

TREATISE ON THE CIRCLE

Volume IV — Topology, Bundles, and Dynamics on the Circle

Coverings and bundles, rotations and diffeomorphisms, chaos and ergodicity, holomorphic dynamics, synchronisation and homeomorphism groups — with applications to automation



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Document: *treatise-circle-vol4*

Last updated: September 2026 — Expanded edition

Personal mathematical study document — no corporate or commercial reference

Reference environments: Debian • Arch Linux • Kali Linux

Preface

Volume III stopped where the circle ceases to be an object of geometry and begins to be the stage of a dynamics. There the elementary topology of S^1 and the isomorphism $\pi_1(S^1) \cong \mathbb{Z}$ were established (Chapter 12), the circle as a group with its invariant measure (Chapter 13), complex analysis in the disc (Chapter 15), $SO(3)$ and the quaternions (Chapter 28). This volume starts from there and goes further: it is the volume of *rigorous dynamics on the circle*. It does not begin again what is already proved elsewhere in the treatise — the exponential covering, the lifting of paths, the winding number, Schwarz’s lemma, the Nyquist criterion, the Chinese remainder theorem — but it recalls them by explicit cross-reference and makes them the engine of new theorems. It presupposes Volumes I, II and III to the extent declared, chapter by chapter, in the prerequisites box that opens each chapter.

How it is organised. The volume is divided into six Parts and sixteen chapters. **Part A — Topology of the circle beyond π_1** (Chapters 1–2) picks up the thread from the lifting criterion and the classification of the connected coverings of S^1 , passes through van Kampen and through the cellular homology of the circle, and shows how many apparently distant theorems are disguises of a single fact: Rouché’s theorem in its topological form (the “dog on a leash”), Borsuk–Ulam in dimension 2 with its combinatorial consequences (the pancake and the stolen necklace), Hopf’s Umlaufsatz, the Whitney–Graustein theorem, the fundamental theorem of algebra, Brouwer’s fixed point — proved twice, by non-retraction and by Sperner’s lemma — and the hairy ball in Milnor’s proof. Chapter 2 climbs from coverings to bundles: cocycle and clutching function, local triviality, the Möbius band as the tautological bundle of $\mathbb{RP}^1$, the classification of real vector bundles of rank n and of principal bundles over S^1 , Poincaré–Hopf for closed surfaces, local Gauss–Bonnet and the holonomy of parallel transport with Foucault’s pendulum as a physical example, the Hopf fibration with its first Chern class, the Berry phase and exponential quadrature on the circle.

Part B — Rotations and diffeomorphisms (Chapters 3–6) is the arithmetic heart of the volume. Chapter 3 develops continued fractions and the theory of best approximation, Hurwitz’s theorem with the optimal constant $1/\sqrt{5}$, Diophantine, Liouville and noble numbers, the three-distance theorem, Weyl’s criterion for equidistribution together with Fejér’s theorem that supports it, discrepancy and the inequalities of Koksma and Erdős–Turán, up to the golden angle of phyllotaxis and to Kronecker sequences as quasi-Monte Carlo integration. Chapter 4 constructs Poincaré’s rotation number, proves its existence, monotonicity and continuity, and completes Poincaré’s classification in the two cases, rational and irrational; then it studies the Arnold standard family, the width of the locking tongues and the devil’s staircase. Chapter 5 proves in full Denjoy’s theorem and the Denjoy–Koksma inequality, and explicitly constructs the C^1 counterexample that shows where and why the hypothesis of bounded variation of $\log f'$ is indispensable. Chapter 6, the most demanding of the volume, tackles small divisors: the homological equation in the analytic case and in the C^r case with explicit constants, non-solvability for a Liouville rotation number constructed ad hoc, the Gottschalk–Hedlund lemma and Arnold’s local analytic linearization theorem, proved with Newton’s scheme; the global theory of Herman and Yoccoz and the Bryuno condition are presented as imported results, with precise statements and verified sources.

Part C — Chaos and ergodic theory (Chapters 7–11) passes from invertible maps to those that are not. Chapter 7 studies expanding maps and the doubling map through symbolic dynamics, proves chaos in the sense of Devaney, the Li–Yorke theorem and Sharkovskii’s theorem with

the optimality of its order, counts the periodic points and computes the kneading of the logistic map at full parameter. Chapter 8 introduces the topological entropy of Bowen–Dinaburg, its invariance, its behaviour under factors, the lap number, the Lyapunov exponents and Rokhlin’s formula. Chapter 9 is the ergodic chapter: Krylov–Bogolyubov, Hopf’s maximal ergodic theorem, Birkhoff’s theorem in full, von Neumann’s mean theorem, Oxtoby’s unique ergodicity criterion, a second proof of Weyl’s equidistribution, normal numbers from Borel to Champernowne, the Ulam–von Neumann measure of the logistic map and the Lasota–Yorke theorem for the tent map. Chapter 10 carries the same systems into the functional language: the transfer operator, its spectrum on L^2 — a full disc of eigenvalues, hence no gap — and the spectral gap on BV and on Hölder spaces, the exponential decay of correlations with explicit constants and two independent proofs of the central limit theorem, one elementary via de Moivre–Laplace and one with Gordin’s martingale method. Chapter 11 traverses the logistic family from the first bifurcations to the period-doubling cascade, with the normal forms proved and Feigenbaum’s constant estimated numerically; Lanford’s theorem that explains it remains avowedly imported.

Part D — Holomorphic dynamics is Chapter 12 alone, and it closes the circle in the literal sense: the escape criterion, the Böttcher coordinate and the fundamental dichotomy between K_c connected and Cantor dust, Fatou’s theorem on attracting cycles, the main cardioid and the period-2 bulb in the parameter plane, the external rays whose angular dynamics is *exactly* the doubling map of Chapter 7, Siegel’s theorem for Diophantine numbers and Cremer’s for a generic class, and the proof that J_c is a Jordan curve for $|c| \leq 3/16 = 0.1875$: the deformed circle that survives perturbation.

Part E — Synchronization and automation (Chapters 13–14) is the applicative part. Chapter 13 goes from flows on the circle and from Adler’s equation to Huygens’ observation of 1665, then to the Kuramoto model: gradient flow structure in the identical case, order parameter, self-consistency equation with the threshold K_c , the Lorentzian case solved in closed form, the Ott–Antonsen reduction, Chiba’s theorem and synchronization on networks in the form of Dörfler and Bullo. Chapter 14 translates all of this into workshop tools: the first-order PLL as an Adler flow with the three bands (hold-in, pull-in, lock-in) that in that case coincide, the second-order PLL, the resolver-to-digital converter, Itoh’s theorem on phase unwrapping, the tolerance theorem of the two-track Vernier encoder, the dead band and the limit cycles from quantization, the distributed clocks of EtherCAT, the PLCopen blocks for axis synchronization and the stroboscope with the wheel that turns backwards.

Part F — Global structures (Chapters 15–16) returns to pure mathematics at its highest level. Chapter 15 studies the group $\text{Homeo}_+(\mathbb{T})$ and its central extension, shows that the rotation number is a homogeneous quasi-morphism with defect less than 2, proves Hölder’s theorem and the trichotomy of minimal sets, defines the Euler class of an action and proves the Milnor–Wood inequality, with $\text{PSL}(2, \mathbb{R})$ on the boundary of the Poincaré disc as a guiding example. Chapter 16 tackles twist maps of the annulus: generating function, variational principle and the Frenkel–Kontorova chain, the Chirikov standard map, Aubry’s lemma on the uniqueness of the crossing between minimal configurations, Birkhoff’s periodic orbits, the theorem of Aubry–Mather with the dichotomy between invariant curve and cantorus, and Mather’s β function.

Reading levels. Each chapter declares its own level in the opening box, section by section, on a scale of five grades: **A** (upper secondary school), **B** (first year of a STEM degree), **C** (undergraduate), **D** (graduate), **E** (doctorate and beyond). The volume is not uniform and does not claim to be: Chapter 13 begins at level A with two pendulum clocks hanging from the same beam, Chapter 6 reaches level E with the estimates of a Newton scheme on complex strips. The same box indicates each time where a reader may stop without losing the thread, and which sections may be skipped by reading only their statement.

What is proved and what is not. The same rule as in the whole treatise holds here: a

statement called a theorem, proposition, lemma or corollary is proved in full in the text; a result used but not proved appears as a remark or a box, with the source declared and the date on which it was verified, and is never promoted to the rank of theorem. Several chapters — the ninth, the tenth, the twelfth and the sixteenth — have a section entitled “What we import”, which lists without reticence the external results on which the chapter relies. Where a proof is complete only in a particular case, the proof box says so explicitly.

How a chapter is made. After the theoretical body, each chapter offers worked examples, assigned exercises, worked exercises, a self-assessment quiz with its teacher’s key, a chapter formulary and the list of sources with the date of verification. The figures are of two kinds: those drawn by hand with computed coordinates and those produced by a numerical script, and only the latter may declare in their caption that they have been computed numerically. The numerical double checks reported in the text are marked by a check mark.

Where the story continues. Some neighbouring topics are deliberately left out so as not to duplicate them: the hyperbolic geometry of the Poincaré disc, of which only the boundary action is used here, is the subject of Volume V; the MLC conjecture, Sullivan’s theorem on the absence of wandering domains, the density of hyperbolicity and renormalization in the complex plane, together with billiards and circle packings, are deferred to Volume VI. Here, where needed, a one-line cross-reference is given and nothing more.

The volume closes with a final test of fifty questions which traverses the six Parts in the order of the chapters, with the teacher’s key at the end.

0THack1A
September 2026

Contents

Preface	1
A Topology of the Circle Beyond π_1	12
1 Coverings, degree and the theorems of the circle	13
1.1 Recollections from Volume III	13
1.2 The lifting criterion	14
1.3 Classification of the connected coverings of S^1	16
1.4 The van Kampen theorem for S^1	17
1.5 Cellular homology and the Hurewicz theorem for S^1	18
1.6 The “dog on a leash”: the topological Rouché theorem	19
1.7 Odd maps and the Borsuk–Ulam theorem	20
1.8 Combinatorial applications: the pancake and the stolen necklace	21
1.9 The theorem on the rotation of tangents (Hopf’s Umlaufsatz)	23
1.10 Regular closed curves: the Whitney–Graustein theorem	25
1.11 The fundamental theorem of algebra	28
1.12 Non-retraction and the Brouwer theorem in dimension 2	29
1.13 The hairy ball and Brouwer in every dimension (Milnor’s proof)	30
1.14 Sperner implies Brouwer	31
1.15 Worked examples	32
1.16 Assigned exercises	36
1.17 Worked exercises	37
1.18 Self-assessment quiz	39
1.19 Teacher’s key	40
1.20 Chapter formulary	42
2 Bundles over the circle and over the sphere	45
2.1 The circle as an algebraic variety and the pencil of conics	45
2.2 Bundles: definition, cocycle, first examples	46
2.3 The unit tangent bundle of S^1	47
2.4 Local triviality: bundles over an interval	48
2.5 The Möbius band as tautological bundle	49
2.6 Classification of rank- n vector bundles over S^1	51
2.7 Principal bundles and classification via monodromy	53
2.8 The Poincaré–Hopf theorem for closed surfaces	53
2.9 Local Gauss–Bonnet and holonomy of parallel transport	56
2.10 The Foucault pendulum as holonomy	57
2.11 The Hopf fibration	57
2.12 Canonical connection on S^3 and first Chern class	58
2.13 Berry phase	59
2.14 S^1 -bundles over S^2 with Euler number n	60
2.15 Exponential quadrature on the circle	60
2.16 Figures	61
2.17 Worked examples	61

2.18	Assigned exercises	66
2.19	Worked exercises	67
2.20	Self-assessment quiz	68
2.21	Teacher's key	69
2.22	Chapter formulary	70
B	Rotations and Diffeomorphisms	73
3	Rotations, continued fractions and equidistribution	74
3.1	The rotation of the circle: the rational/irrational dichotomy	74
3.2	Continued fractions	75
3.3	Best approximation: two distinct kinds	78
3.4	Hurwitz's theorem	79
3.5	Diophantine numbers, Liouville numbers, noble numbers	81
3.6	The three-distance theorem	83
3.7	Weyl's criterion and Fejér's theorem	84
3.8	Discrepancy	86
3.9	Koksma's inequality	87
3.10	The Erdős–Turán inequality	88
3.11	Discrepancy of rotations: the role of the partial quotients	88
3.12	Schmidt's bound	90
3.13	Golden angle and phyllotaxis: Vogel's model	90
3.14	Kronecker sequences as quasi-Monte Carlo	91
3.15	Figures	92
3.16	Assigned exercises	92
3.17	Worked exercises	96
3.18	Self-assessment quiz	97
3.19	Teacher's key	98
3.20	Chapter formulary	100
4	Rotation number, Poincaré classification and Arnold tongues	103
4.1	The rotation number: definition and existence	103
4.2	The properties of the rotation number	106
4.3	The Poincaré classification: rational case	108
4.4	The Poincaré classification: irrational case	110
4.5	The Arnold standard family	113
4.6	Width of the p/q tongues	115
4.7	The devil's staircase	118
4.8	A panorama beyond the perturbative regime	118
4.9	Towards KAM theory: the statements of Arnold and of Herman–Yoccoz	120
4.10	Assigned exercises	122
4.11	Worked exercises	122
4.12	Self-assessment quiz	125
4.13	Teacher's key	126
4.14	Chapter formulary	127
5	Denjoy's theorem and its counterexamples	131
5.1	The lemma of the best return times	131
5.2	The Denjoy inequality	133
5.3	Denjoy's theorem	135
5.4	The Denjoy–Koksma inequality	136

5.5	Denjoy's C^1 counterexample	137
5.6	Panorama: beyond the C^1 counterexample	141
5.7	Figures	142
5.8	Assigned exercises	142
5.9	Worked exercises	145
5.10	Self-assessment quiz	147
5.11	Teacher's key	148
5.12	Chapter formulary	149
6	Small divisors and KAM theory on the circle: Arnold, Herman, Yoccoz	152
6.1	The problem of small divisors	153
6.2	The arithmetic one needs: Diophantine conditions, Bryuno, Liouville	155
6.3	The homological equation in the analytic case	158
6.4	The C^r case and the Liouville catastrophe	161
6.5	The Gottschalk–Hedlund lemma	164
6.6	Arnold's theorem: statement	166
6.7	The Newton scheme	167
6.8	Beyond the local analytic case: Herman, Yoccoz, and the condition $\mathcal{H}$	175
6.9	Figures	178
6.10	Assigned exercises	178
6.11	Worked exercises	183
6.12	Self-assessment quiz	185
6.13	Teacher's key	186
6.14	Chapter formulary	188
C	Chaos and Ergodic Theory	192
7	The doubling map: symbolic dynamics, Li–Yorke and Sharkovskii	193
7.1	The doubling and expanding maps	193
7.2	Symbolic dynamics: the semiconjugacy with the shift	195
7.3	Devaney chaos	198
7.4	The covering lemma and the itinerary lemma	199
7.5	The Li–Yorke theorem	201
7.6	Sharkovskii's theorem	202
7.7	Counting the periodic points	205
7.8	The kneading of f_4	208
7.9	Bridge with holomorphic dynamics	209
7.10	Historical survey	210
7.11	Assigned exercises	210
7.12	Worked exercises	211
7.13	Self-assessment quiz	213
7.14	Teacher's key	214
7.15	Chapter formulary	215
8	Topological entropy and Lyapunov exponents	219
8.1	Separated and spanning sets: the Bowen–Dinaburg entropy	219
8.2	Invariance under topological conjugacy	221
8.3	Factors: the easy inequality and Bowen's theorem	222
8.4	Direct computation: the maps E_m and the doubling map	223
8.5	The shift on k symbols: the correct estimate	224
8.6	Zero entropy: rotations and isometries	225

8.7	Lap number and the Misiurewicz–Szlenk theorem	226
8.8	Sharkovskii and entropy: the horseshoe	228
8.9	Lyapunov exponents	229
8.10	Rokhlin’s formula and the variational principle for D	231
8.11	Oseledets and Pesin: a panorama	233
8.12	Worked examples	233
8.13	Assigned exercises	234
8.14	Worked exercises	237
8.15	Self-assessment quiz	239
8.16	Teacher’s key	240
8.17	Chapter formulary	242
9	Ergodic theory on the circle and on the interval	245
9.1	Invariant measures and the Krylov–Bogolyubov theorem	246
9.2	The Lebesgue measure is invariant for D , for E_m and for R_α	247
9.3	Ergodicity and mixing of D	248
9.4	The maximal ergodic theorem (Hopf, 1937)	249
9.5	Birkhoff’s ergodic theorem (1931), complete	250
9.6	The mean ergodic theorem (von Neumann, 1932)	253
9.7	Unique ergodicity: Oxtoby’s criterion	255
9.8	Weyl equidistribution via the ergodic theorem (second proof)	257
9.9	Normal numbers: from Borel to Champernowne	258
9.10	The Ulam–von Neumann invariant measure for f_4	260
9.11	The triangle of conjugacies D – T – f_4	261
9.12	The Lasota–Yorke theorem for the tent map	262
9.13	Kolmogorov–Sinai entropy: definition and the value for D	265
9.14	Panorama: Jakobson’s theorem	266
9.15	Recollection: ergodicity of Denjoy diffeomorphisms	267
9.16	What we import	267
9.17	Assigned exercises	267
9.18	Worked exercises	268
9.19	Self-assessment quiz	271
9.20	Teacher’s key	272
9.21	Chapter formulary	274
10	Transfer operators, decay of correlations and the central limit theorem	277
10.1	The transfer operator of D	278
10.2	The spectrum of $\mathcal{L}$ on L^2 : the full disc	279
10.3	Spectral gap on the regular spaces: BV , Hölder, C^1	281
10.4	Exponential decay of correlations for D and for E_m	283
10.5	The Lasota–Yorke theorem for the tent map: quasi-compactness and decay of correlations	286
10.6	The central limit theorem for D : the dyadic indicator, via de Moivre–Laplace	287
10.7	The central limit theorem for D : Gordin’s martingale method	289
10.8	Panorama: the Nagaev–Guivarc’h method	293
10.9	Panorama: Liverani’s Birkhoff cones	294
10.10	Panorama: Young towers and polynomial decay	294
10.11	What we import	295
10.12	Assigned exercises	295
10.13	Worked exercises	296
10.14	Self-assessment quiz	298
10.15	Teacher’s key	300

10.16	Chapter formulary	302
11	Logistic map, bifurcations and universality	305
11.1	The logistic family: fixed points and first bifurcations	305
11.2	Normal forms of one-dimensional bifurcations	309
11.3	The period-doubling cascade: existence and numerical estimate of δ	311
11.4	The renormalisation operator and Lanford's theorem	313
11.5	Coulet and Tresser: the independent discovery	316
11.6	Critical circle maps: renormalisation at the golden number	317
11.7	Beyond the cascade: invariant measures and hyperbolicity	318
11.8	The bifurcation diagram as a mathematical object	318
11.9	Assigned exercises	319
11.10	Worked exercises	322
11.11	Self-assessment quiz	323
11.12	Teacher's key	324
11.13	Chapter formulary	327
D	Holomorphic Dynamics	330
12	From the circle to Julia and Mandelbrot	331
12.1	What we import	331
12.2	The quadratic family: K_c , J_c and the Mandelbrot set	332
12.3	Local dynamics at a fixed point: Koenigs and Böttcher	334
12.4	The fundamental dichotomy	340
12.5	Fatou's theorem on attracting cycles	343
12.6	The circle in the parameter plane: cardioid and bulbs	344
12.7	External rays: the circle doubles again	349
12.8	Siegel's theorem	351
12.9	Cremer's theorem	356
12.10	Bryuno, Yoccoz and the boundary of the Siegel disc	357
12.11	Stability of J_c for small $ c $: the deformed circle	359
12.12	Panoramas: structural stability and Hausdorff dimension	362
12.13	Historical note	364
12.14	Assigned exercises	364
12.15	Worked exercises	365
12.16	Self-assessment quiz	368
12.17	Teacher's key	369
12.18	Chapter formulary	371
E	Synchronisation and Automation	376
13	Synchronisation: from Huygens to Kuramoto	377
13.1	Flows on the circle	377
13.2	Adler's equation	378
13.3	Two coupled oscillators	381
13.4	Huygens, 1665: the first observation	382
13.5	The Kuramoto model with N identical oscillators	383
13.6	The order parameter	387
13.7	The self-consistency equation in the limit $N \rightarrow \infty$	388
13.8	The Ott–Antonsen reduction	390

13.9 Chiba's theorem (2015)	393
13.10 Panorama: Sakaguchi–Kuramoto, chimeras, synchronisation of chaos	395
13.11 Synchronisation in networks: a theorem of Dörfler and Bullo	395
13.12 Applications: a panorama	397
13.13 Assigned exercises	398
13.14 Worked exercises	398
13.15 Self-assessment quiz	400
13.16 Teacher's key	401
13.17 Chapter formulary	403
14 Phase-locked loops, encoders and digital control on the circle	407
14.1 The first-order PLL as an Adler flow	408
14.2 Hold-in, pull-in, lock-in: rigorous definitions	408
14.3 The first-order theorem: the three ranges coincide	409
14.4 Characteristic locking time	411
14.5 The discrete PLL: forward reference to Arnold's standard family	412
14.6 The second-order PLL	412
14.7 The resolver-to-digital converter as a type II PLL	412
14.8 Itoh's theorem: when phase unwrapping works	414
14.9 Vernier encoder: the tolerance theorem	415
14.10 Dead band from quantisation: the theorem, corrected	418
14.11 Limit cycles from second-order quantisation	420
14.12 Distributed clocks: EtherCAT	421
14.13 PLCopen function blocks for axis synchronisation	421
14.14 The stroboscope and the wheel turning backwards	422
14.15 Worked examples	422
14.16 Assigned exercises	423
14.17 Worked exercises	424
14.18 Self-assessment quiz	426
14.19 Teacher's key	427
14.20 Chapter formulary	428
F Global Structures	431
15 Groups of circle homeomorphisms	432
15.1 The group $\text{Homeo}_+(\mathbb{T})$ and its central extension	432
15.2 The tools: minimal and maximal displacement	434
15.3 The rotation number is a quasi-morphism	436
15.4 Hölder's theorem	439
15.5 Left-orderable groups	443
15.6 Minimal sets: the trichotomy	445
15.7 The Euler class of an action	447
15.8 The Milnor–Wood inequality	449
15.9 A guiding example: $\text{PSL}(2, \mathbb{R})$ on the boundary	451
15.10 What we import, and the panorama	453
15.11 Figures	454
15.12 Assigned exercises	457
15.13 Worked exercises	457
15.14 Self-assessment quiz	459
15.15 Teacher's key	460
15.16 Chapter formulary	462

16 Twist maps of the annulus and Aubry–Mather theory	466
16.1 The annulus, twist maps, the generating function	467
16.2 The variational principle and the Frenkel–Kontorova chain	470
16.3 The Chirikov standard map	471
16.4 The exchange inequality and Aubry’s lemma	474
16.5 Birkhoff periodic orbits of type (p, q)	477
16.6 Birkhoff configurations: order, rotation, Lipschitz graph	482
16.7 The Aubry–Mather theorem	485
16.8 Mather’s β function	489
16.9 What we import and where the story continues	491
16.10 Assigned exercises	492
16.11 Worked exercises	493
16.12 Self-assessment quiz	495
16.13 Teacher’s key	496
16.14 Chapter formulary	498
 Final examination of the Volume	 503
 A Registry of imported results	 517
A.1 What this appendix records, and why	517
A.2 Registry, chapter by chapter	518
A.3 Eleven blocks examined and not recorded	535
A.4 What this registry says, taken as a whole	535
 B Numerical laboratory: the volume’s scripts	 537
B.1 Conventions of the laboratory	537
B.2 Catalog by chapter	539
B.3 What the laboratory does not do	543
 C Reference numerical tables	 545
C.1 Continued fractions and convergents	545
C.2 Diophantine constants	546
C.3 Notable rotations, the golden angle and the three-distance theorem	547
C.4 Logistic cascade and universality constants	547
C.5 Arnold tongues: widths	548
C.6 Standard map: Greene, Birkhoff, and Mather’s β function	549
C.7 KAM thresholds and the bookkeeping of Newton’s scheme	550
C.8 Regularity classes: what holds where	550
C.9 Synchronization thresholds	551
C.10 Denjoy’s counterexample: the two critical values	552
C.11 What this appendix does not contain	552
 D Exercise catalog: reasoned index and synthesis exercises	 553
D.1 How the volume’s exercise catalog is organized	553
D.2 Index of exercises, chapter by chapter	553
D.3 Cross-cutting paths	557
D.4 Three synthesis exercises	558
D.5 Note on the relationship with the expansion plan	561
 Glossary	 562
 Cheat Sheet	 576

General formulary	578
Bibliography	590
Index	603

Part A

Topology of the Circle Beyond π_1

CHAPTER 1

Coverings, degree and the theorems of the circle

INFO

Level and prerequisites. Level B–C core (Sections 1.1–1.11 and 1.12); Sections 1.10 and 1.13 touch level D in their final part (Milnor’s proof, the hard direction of Whitney–Graustein), but remain readable if the flagged paragraphs are skipped. Prerequisites: topology of the set S^1 , exponential covering $E : \mathbb{R} \rightarrow S^1$, lifting of paths and of homotopies, winding number, $\pi_1(S^1) \cong \mathbb{Z}$, degree of maps $S^1 \rightarrow S^1$ (Volume III, Chapter 12); elementary notions of regular plane curves (Volume II, Chapter 1). A level A–B reader may skip Section 1.10 after the statement and Section 1.13 after Theorem 1.14, without losing the coherence of the rest.

This chapter does not start the topology of the circle over again: it builds it where Volume III left it. The exponential covering, the lifting of paths and of homotopies, the winding number and $\pi_1(S^1) \cong \mathbb{Z}$ are tools already in the reader’s hands; here they become the engine of a series of theorems which, at first sight, have nothing to do with the circle — the fundamental theorem of algebra, the Brouwer fixed point, Borsuk–Ulam, the pancake, the stolen necklace, Hopf’s rotation of the tangents, the hairy ball — and which are instead, each of them, a disguise of the single fact that $\pi_1(S^1) \cong \mathbb{Z}$.

1.1. Recollections from Volume III

Volume III, Chapter 12, establishes in full, with complete proofs: (i) $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it}$, is a covering, and $\mathbb{R}$ (convex, hence simply connected) is its universal covering; (ii) every path $\gamma : I \rightarrow S^1$ and every homotopy $F : I \times I \rightarrow S^1$ lift in a unique way once the initial value is fixed, by an argument of Lebesgue number and gluing (Lifting Lemmas for paths and for homotopies); (iii) the winding number $w(\gamma) = (\tilde{\gamma}(1) - \tilde{\gamma}(0))/2\pi \in \mathbb{Z}$ is well defined and homotopy invariant, with analytic form $w(\gamma) = \frac{1}{2\pi i} \oint_{\gamma} dz/z$ for C^1 paths; (iv) $\pi_1(S^1, 1) \cong \mathbb{Z}$ via $[\gamma] \mapsto w(\gamma)$; (v) the degree $\deg f \in \mathbb{Z}$ of a continuous map $f : S^1 \rightarrow S^1$ is well defined, classifies the maps $S^1 \rightarrow S^1$ up to free homotopy ($f \simeq g \iff \deg f = \deg g$), satisfies $\deg p_n = n$ for $p_n(z) = z^n$, is multiplicative under composition ($\deg(f \circ g) = \deg f \cdot \deg g$), and $\deg f = 0$ if and only if f extends continuously to the disc D^2 . These five facts are recalled here by name, with a textual cross-reference (**Volume III, Chapter 12**), and are not reproved: every chapter of Vols. I–III proved in full remains valid without any need to repeat it (see also Remark 1.1 below, on why the integral formula for the degree does not appear here a second time).

From this point on the chapter builds what Volume III does not contain: the lifting criterion for maps with values in S^1 from an arbitrary space, the classification of *all* connected coverings of S^1 (not only of the maps $S^1 \rightarrow S^1$), the computation of $\pi_1(S^1)$ via van Kampen and via cellular homology, and finally a series of “external” theorems — topological Rouché, Borsuk–Ulam, pancake, stolen necklace, Umlaufsatz, Whitney–Graustein, the fundamental theorem of algebra, Brouwer, the hairy ball — which the degree of the circle proves one after the other.

Remark 1.1: Why the integral formula is not reproved here. Volume III, Chapter 12, proves in full (with an independent numerical check) that for a path $\gamma : I \rightarrow S^1$ of class C^1 one has $w(\gamma) = \frac{1}{2\pi i} \int_0^1 \gamma'(t)/\gamma(t) dt$, and that for γ with values in $\mathbb{C} \setminus \{0\}$ the same integral (with $z - w$ in the denominator) computes the winding number $n(\gamma, w)$ about a point w . Repeating that proof here would be exactly the defect that this volume sets out to correct (no proof is to be repeated from a previous volume): the argument principle and the residue theorem that follow from it in complex analysis remain in Volume III, Chapter 15, recalled textually where needed (Remark 1.5).

1.2. The lifting criterion

The exponential covering E is not the only covering of S^1 : every map $p_n : S^1 \rightarrow S^1$, $p_n(z) = z^n$ ($n \geq 1$), is also a covering (n -sheeted, instead of infinitely sheeted). To classify them all (Section 1.3) one first needs to know when a map with values in S^1 lifts through an arbitrary covering — not only through E .

Lemma 1.1: Lifting of paths and homotopies through an arbitrary covering of S^1 . Let $p : \tilde{X} \rightarrow S^1$ be a covering (Definition, Volume III, Chapter 12: every point of S^1 has a neighbourhood U with $p^{-1}(U)$ a disjoint union of open sets mapped homeomorphically onto U by p). Then:

- (a) for every path $\gamma : I \rightarrow S^1$ and every $\tilde{x}_0 \in p^{-1}(\gamma(0))$ there exists a unique path $\tilde{\gamma} : I \rightarrow \tilde{X}$ with $p \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(0) = \tilde{x}_0$;
- (b) for every homotopy $F : I \times I \rightarrow S^1$ and every $\tilde{x}_0 \in p^{-1}(F(0,0))$ there exists a unique $\tilde{F} : I \times I \rightarrow \tilde{X}$ with $p \circ \tilde{F} = F$ and $\tilde{F}(0,0) = \tilde{x}_0$;
- (c) (uniqueness on a connected domain) if Y is connected and $g_1, g_2 : Y \rightarrow \tilde{X}$ are two continuous lifts of the same map $f : Y \rightarrow S^1$ (that is, $p \circ g_i = f$) which agree at one point, then $g_1 = g_2$.

Proof. The proof is, point by point, that of Volume III, Chapter 12 (Lifting Lemmas for paths, for homotopies and uniqueness), in which every reference to E and to its trivialising open sets $\{U_j\}_{j=1}^4$ is to be replaced by the trivialising open sets of p (which exist by the definition of covering) and by the local branches of the inverse $(p|_V)^{-1}$ in place of $(E|_V)^{-1}$: the only ingredient used in that proof is that every point of S^1 has a trivialising neighbourhood and that the fibres of p are discrete (disjoint union of open sets), a property shared by every covering, not specific to E . In detail: (a) one covers I with the preimage under γ of the trivialising open sets of p , one takes a Lebesgue number δ for this cover, one subdivides I into $m > 1/\delta$ equal pieces, and one lifts inductively piece by piece, using on each the unique local branch of $(p|_V)^{-1}$ that contains the value already reached; the gluing at the common points is automatic because the lifts of the single pieces agree there by construction. (b) is the same argument on the square $I \times I$, with the same induction over rows of small squares. (c) follows from: the set $\{y \in Y : g_1(y) = g_2(y)\}$ is open (if $g_1(y) = g_2(y) = \tilde{x}$, a trivialising neighbourhood U of $f(y)$ gives a unique sheet $V \ni \tilde{x}$, and on $f^{-1}(U)$ both g_1, g_2 agree with $(p|_V)^{-1} \circ f$ near y , by continuity) and closed (same argument exchanging “agree” with “lie in different sheets”, which stay different near y because the sheets are disjoint open sets); on Y connected, a non-empty open and closed subset is all of Y . $\square$

Theorem 1.1: Lifting criterion. Let $p : \tilde{X} \rightarrow S^1$ be a covering, let Y be a path-connected and locally path-connected space, $f : Y \rightarrow S^1$ continuous, $y_0 \in Y$, $\tilde{x}_0 \in p^{-1}(f(y_0))$. Then f

lifts to a unique $\tilde{f} : Y \rightarrow \tilde{X}$ with $\tilde{f}(y_0) = \tilde{x}_0$ and $p \circ \tilde{f} = f$ if and only if

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subseteq \pi_1(S^1, f(y_0)).$$

In particular (case $p = E$, where $\pi_1(\mathbb{R}) = 0$ because $\mathbb{R}$ is convex), f lifts through E if and only if $f_*(\pi_1(Y, y_0)) = \{0\}$, that is, if and only if f sends every loop of Y to a loop of degree 0.

Proof. Necessity. If $\tilde{f}$ exists, $f_* = p_* \circ \tilde{f}_*$, hence $f_*(\pi_1(Y, y_0)) = p_*(\tilde{f}_*(\pi_1(Y, y_0))) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Sufficiency — construction. Let $y \in Y$; by path-connectedness there exists a path α from y_0 to y . One lifts $\beta = f \circ \alpha$ through p starting from $\tilde{x}_0$ (Lemma 1.1(a)): let $\tilde{\beta}$ be the lift, and set $\tilde{f}(y) := \tilde{\beta}(1)$.

Well-definedness. Let α' be another path from y_0 to y , $\beta' = f \circ \alpha'$, $\tilde{\beta}'$ its lift from $\tilde{x}_0$. The loop $\ell = \alpha * \bar{\alpha}'$ (it runs along α and then α' backwards) is a loop in Y based at y_0 , and $f \circ \ell = \beta * \bar{\beta}'$ is a loop in S^1 based at $f(y_0)$ with $[f \circ \ell] = f_*[\ell] \in p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ by hypothesis: hence there exists a loop c in $\tilde{X}$ based at $\tilde{x}_0$ with $[p \circ c] = [f \circ \ell]$ in $\pi_1(S^1, f(y_0))$, that is, $p \circ c$ and $\beta * \bar{\beta}'$ are homotopic (rel. endpoints) in S^1 . Lift this homotopy through p starting from $\tilde{x}_0$ (Lemma 1.1(b)): by the Corollary on the lifting of homotopies of paths (the same argument as in Volume III, Chapter 12, applied to p : a homotopy rel. endpoints between paths with fixed endpoints lifts to a homotopy with fixed endpoints, because the lifts of the constant endpoints are themselves constant by Lemma 1.1(c)), the lift of $\beta * \bar{\beta}'$ from $\tilde{x}_0$ has the same final point as the lift of $p \circ c$ from $\tilde{x}_0$, that is, $c(1)$; but c is a loop, $c(1) = c(0) = \tilde{x}_0$. On the other hand the lift of $\beta * \bar{\beta}'$ from $\tilde{x}_0$ is, by construction and uniqueness (Lemma 1.1(c)), the concatenation of $\tilde{\beta}$ (from $\tilde{x}_0$ to $\tilde{\beta}(1)$) with the lift of $\bar{\beta}'$ from $\tilde{\beta}(1)$, which in turn is $\bar{\tilde{\beta}'}$ traversed from $\tilde{\beta}(1)$ if and only if $\tilde{\beta}(1) = \tilde{\beta}'(1)$ — and since the final point of the whole concatenation is $\tilde{x}_0$, and the lift of $\bar{\beta}'$ starting from a point reaches $\tilde{x}_0$ exactly if that point is $\tilde{\beta}'(1)$ (by uniqueness, running along β' forwards from $\tilde{x}_0$ one reaches $\tilde{\beta}'(1)$, hence β' backwards from $\tilde{\beta}'(1)$ reaches $\tilde{x}_0$), one concludes $\tilde{\beta}(1) = \tilde{\beta}'(1)$. Hence $\tilde{f}(y)$ does not depend on the chosen path.

Continuity. Let $y \in Y$, $\tilde{f}(y) = \tilde{x}$, U a trivialising neighbourhood of $f(y)$ in S^1 , V the sheet of $p^{-1}(U)$ containing $\tilde{x}$. By continuity of f and local path-connectedness of Y , there exists a path-connected neighbourhood $W \ni y$ with $f(W) \subseteq U$. For $y' \in W$, choose a path from y_0 to y' passing through y (concatenating a path from y_0 to y with a path from y to y' inside W): the lift reaches $\tilde{x}$ at y and continues inside V (since $f(W) \subseteq U$, the part of the lift over W agrees with $(p|_V)^{-1} \circ f|_W$, by uniqueness). Hence $\tilde{f}|_W = (p|_V)^{-1} \circ f|_W$, a composition of continuous maps: $\tilde{f}$ is continuous at y .

Uniqueness. It follows from Lemma 1.1(c), Y being connected (path-connected $\Rightarrow$ connected).

The particular case $p = E$: since $\mathbb{R}$ is convex, $\pi_1(\mathbb{R}, \tilde{x}_0) = \{0\}$ (every loop contracts by linear homotopy, Volume III, Chapter 12), hence $p_*(\pi_1(\mathbb{R}, \tilde{x}_0)) = \{0\}$ and the condition becomes $f_*(\pi_1(Y, y_0)) = \{0\}$; identifying $\pi_1(S^1) \cong \mathbb{Z}$ via the degree, this amounts to saying that f sends every loop of Y to a loop of degree 0. $\square$

Example 1.1: The identity of S^1 does not lift to $\mathbb{R}$. Let $Y = S^1$ and $f = \text{id}_{S^1}$. Since $f_*(\pi_1(S^1)) = \pi_1(S^1) \cong \mathbb{Z} \neq \{0\} = \pi_1(\mathbb{R})$, Theorem 1.1 (case $p = E$) says that id_{S^1} does not lift to a continuous map $S^1 \rightarrow \mathbb{R}$: there exists no continuous and global angle function on the whole of S^1 , only locally (the sections ψ_j of Volume III). It is the same obstruction, read backwards, that makes the universal covering necessary in order to define a continuous logarithm on $\mathbb{C} \setminus \{0\}$.

1.3. Classification of the connected coverings of S^1

Theorem 1.2: Connected coverings of S^1 . Let $p : \tilde{X} \rightarrow S^1$ be a covering with $\tilde{X}$ path-connected. Then there exists $n \geq 0$ and, setting $(Y, q) := (\mathbb{R}, E)$ if $n = 0$ or $(Y, q) := (S^1, p_n)$ if $n \geq 1$ (where $E : \mathbb{R} \rightarrow S^1$, $E(t) = e^{it}$, is the universal covering, and $p_n : S^1 \rightarrow S^1$, $p_n(z) = z^n$, is the n -sheeted covering), a homeomorphism $\varphi : \tilde{X} \rightarrow Y$ such that $p = q \circ \varphi$: p is isomorphic to exactly one of the following:

- $n = 0$: the universal covering $E : \mathbb{R} \rightarrow S^1$;
- $n \geq 1$: the n -sheeted covering $p_n : S^1 \rightarrow S^1$, $p_n(z) = z^n$.

Proof. Fix $\tilde{x}_0 \in p^{-1}(1)$.

Step 1 — the developing map $q : \mathbb{R} \rightarrow \tilde{X}$. By Lemma 1.1(a) applied to the path $t \mapsto E(st) = e^{ist}$, $s \in [0, 1]$ (for every fixed $t \in \mathbb{R}$), one defines $q(t) \in \tilde{X}$ as the final point of the lift through p of the path $s \mapsto E(st)$ starting from $\tilde{x}_0$; equivalently, q is the unique lift of $E : \mathbb{R} \rightarrow S^1$ through p with $q(0) = \tilde{x}_0$ (it exists by Theorem 1.1 applied to $Y = \mathbb{R}$, $f = E$: $\pi_1(\mathbb{R}) = \{0\}$, hence the condition is automatic). Note that $p \circ q = E$.

Step 2 — q is a covering onto $\tilde{X}$, hence surjective. For every $\tilde{x} \in p^{-1}(U)$ with $U \ni p(\tilde{x})$ trivialising for p and sheet $V \ni \tilde{x}$, and for every trivialising neighbourhood J of $p(\tilde{x})$ for E contained in U with sheet $J' \ni t$ where $E(t) = p(\tilde{x})$ and $q(t) = \tilde{x}$: on J' one has $q = (p|_V)^{-1} \circ E|_{J'}$ (by uniqueness, since both sides lift $E|_{J'}$ and agree at t), hence q is a homeomorphism of J' onto V : q is a local homeomorphism. Since $\tilde{X}$ is path-connected, q is surjective: for $\tilde{x} \in \tilde{X}$ choose a path $\tilde{\eta}$ from $\tilde{x}_0$ to $\tilde{x}$ in $\tilde{X}$; the path $\eta = p \circ \tilde{\eta}$ in S^1 starts at 1; let η_E be its lift through E with $\eta_E(0) = 0$ (Volume III, Chapter 12); then $q \circ \eta_E$ and $\tilde{\eta}$ are both lifts of η through p starting from $\tilde{x}_0$ (because $p \circ q \circ \eta_E = E \circ \eta_E = \eta$), hence they agree (Lemma 1.1(c)): $\tilde{x} = \tilde{\eta}(1) = q(\eta_E(1))$, that is, $\tilde{x} \in \text{Im}(q)$.

Step 3 — periodicity: $q(t + 2\pi) = q(t)$ for every t , or else q is injective. Since $p^{-1}(1)$ is a discrete set (fibre of a covering) and $\tilde{X}$ is path-connected, the monodromy action of $\pi_1(S^1, 1) \cong \mathbb{Z}$ on the fibre $p^{-1}(1)$ — which sends $k \in \mathbb{Z}$ and $\tilde{x} \in p^{-1}(1)$ to the final point of the lift through p , starting from $\tilde{x}$, of the loop $s \mapsto e^{iks}$, $s \in [0, 2\pi]$ — is well defined (by Lemma 1.1(a)-(c), it depends only on the homotopy class of the loop, that is, on k : two homotopic loops lift to paths with the same final point, by the same lifting-of-homotopies argument used in Theorem 1.1) and it is transitive: given $\tilde{x}, \tilde{x}' \in p^{-1}(1)$, a path $\tilde{\eta}$ from $\tilde{x}$ to $\tilde{x}'$ in $\tilde{X}$ (it exists by path-connectedness) projects to a loop $\eta = p \circ \tilde{\eta}$ in S^1 based at 1, of degree $k = \deg \eta = w(\eta)$; by construction $\tilde{\eta}$ is the lift of η from $\tilde{x}$ (it is a lift of η and starts at $\tilde{x}$), hence $k \cdot \tilde{x} = \tilde{\eta}(1) = \tilde{x}'$: the action is transitive. With base point $\tilde{x}_0$, the stabiliser $H = \{k \in \mathbb{Z} : k \cdot \tilde{x}_0 = \tilde{x}_0\}$ is a subgroup of $\mathbb{Z}$, hence $H = n\mathbb{Z}$ for a unique $n \geq 0$ (subgroups of $\mathbb{Z}$). By construction, $k \cdot \tilde{x}_0 = q(2\pi k)$ (the loop $s \mapsto e^{iks}$, $s \in [0, 2\pi]$, is exactly $t \mapsto E(t)$ restricted and rescaled on $[0, 2\pi k]$, whose lift through p from $\tilde{x}_0$ is q restricted and evaluated at $2\pi k$, by the construction of q in Step 1). Hence

$$q(2\pi k) = \tilde{x}_0 \iff k \in H = n\mathbb{Z}, \quad \text{that is} \quad q^{-1}(\tilde{x}_0) \cap 2\pi\mathbb{Z} = 2\pi n\mathbb{Z}.$$

But if $q(t) = \tilde{x}_0$ for any $t \in \mathbb{R}$ whatsoever, then $E(t) = p(q(t)) = p(\tilde{x}_0) = 1$, hence $t \in 2\pi\mathbb{Z}$ (Volume III, Chapter 12): therefore $q^{-1}(\tilde{x}_0) = 2\pi n\mathbb{Z}$ exactly.

Step 4 — $q(t + 2\pi n) = q(t)$ for every $t \in \mathbb{R}$. Set $\varphi(t) := q(t + 2\pi n)$: it is continuous, and $p(\varphi(t)) = E(t + 2\pi n) = E(t)$ (periodicity of E), hence φ is a lift of E through p ; moreover $\varphi(0) = q(2\pi n) = \tilde{x}_0 = q(0)$ (Step 3). By Lemma 1.1(c) applied to the connected domain $\mathbb{R}$, $\varphi = q$ everywhere: $q(t + 2\pi n) = q(t)$ for every t .

Step 5 — $q(t) = q(t') \Rightarrow t - t' \in 2\pi n\mathbb{Z}$. If $q(t) = q(t')$, then $E(t) = p(q(t)) = p(q(t')) = E(t')$, hence $t' = t + 2\pi k$ for some $k \in \mathbb{Z}$. Set $\psi(s) := q(s + 2\pi k)$: as above, ψ is a lift of E through p , and $\psi(t) = q(t + 2\pi k) = q(t') = q(t)$: ψ and q agree at the point t (not necessarily at 0),

but by Lemma 1.1(c) (connected domain $\mathbb{R}$, same value at one point) $\psi = q$ everywhere, in particular at 0: $q(2\pi k) = \psi(0) = q(0) = \tilde{x}_0$. By Step 3, $k \in n\mathbb{Z}$, that is, $t - t' = -2\pi k \in 2\pi n\mathbb{Z}$. **Step 6 — conclusion.** Steps 4–5 show $q(t) = q(t') \iff t - t' \in 2\pi n\mathbb{Z}$: q induces a bijection $\bar{q} : \mathbb{R}/2\pi n\mathbb{Z} \rightarrow \tilde{X}$ (surjective by Step 2), continuous and open (because q is a local homeomorphism, Step 2, hence maps open sets to open sets, and the projection $\mathbb{R} \rightarrow \mathbb{R}/2\pi n\mathbb{Z}$ is open), hence a homeomorphism. If $n = 0$: $\mathbb{R}/2\pi \cdot \{0\}\mathbb{Z} = \mathbb{R}$, and $\bar{q} = q : \mathbb{R} \rightarrow \tilde{X}$ is the desired isomorphism, with $p \circ q = E$: p is isomorphic to E . If $n \geq 1$: the map $\sigma : \mathbb{R}/2\pi n\mathbb{Z} \rightarrow S^1$, $\sigma(t \bmod 2\pi n) = e^{it/n}$, is a homeomorphism (standard: continuous, bijective, compact domain and Hausdorff codomain); setting $\varphi = \bar{q} \circ \sigma^{-1} : S^1 \rightarrow \tilde{X}$, one has, for $z = e^{it/n} \in S^1$: $p(\varphi(z)) = p(\bar{q}(t \bmod 2\pi n)) = p(q(t)) = E(t) = e^{it} = z^n = p_n(z)$: hence $p \circ \varphi = p_n$, that is, p is isomorphic to p_n . $\square$

***Remark 1.2: The number n is the number of sheets.** For $n \geq 1$, the fibre $p^{-1}(1)$ has exactly n elements (in bijection with $\mathbb{Z}/n\mathbb{Z}$ via the monodromy action of Step 3, a single orbit with stabiliser $n\mathbb{Z}$); for $n = 0$ it has countable cardinality in bijection with $\mathbb{Z}$. The proof did not use the general subgroups–coverings correspondence of the abstract theory: it is a direct argument, specific to S^1 , built entirely from Lemma 1.1.*

1.4. The van Kampen theorem for S^1

The van Kampen theorem computes $\pi_1(U \cup V)$ from $\pi_1(U)$, $\pi_1(V)$ and $\pi_1(U \cap V)$. Applied to $S^1 = U \cup V$ with U, V two open arcs whose intersection $U \cap V$ has *two* connected components (instead of one, as in the more familiar case), the theorem requires its most general version, the one with the **fundamental groupoid** in place of the fundamental group: the version with a single base point does not apply when $U \cap V$ is not connected. Proving this version in full (objects: all the points of a set of base points chosen in each component of U , V , $U \cap V$; morphisms: homotopy classes of paths; the theorem identifies the fundamental groupoid of $U \cup V$ with the push-out of the groupoids of U and V along that of $U \cap V$) requires a categorical apparatus disproportionate to what is needed in this chapter, where $\pi_1(S^1) \cong \mathbb{Z}$ is already established directly (Volume III, Chapter 12) by an elementary argument. We therefore import the statement, with a primary source, and use it as an *independent check* of the already known computation, not as its proof.

Remark 1.3: Groupoid van Kampen (imported) and verification of $\pi_1(S^1) \cong \mathbb{Z}$.

***Imported from** A. Gramain, Le théorème de Van Kampen, *Cahiers de Topologie et Géométrie Différentielle Catégoriques* 33 (1992), 237–251 (verified on 16 September 2026: groupoid van Kampen theorem, statement and complete proof in the version with several base points, §§1–2). In the form relevant here: if $X = U \cup V$ with U, V open and $A \subseteq U \cap V$ a set of base points which meets every path-connected component of U , V and $U \cap V$, the fundamental groupoid $\Pi_1(X, A)$ is the push-out of $\Pi_1(U, A \cap U)$ and $\Pi_1(V, A \cap V)$ along $\Pi_1(U \cap V, A \cap U \cap V)$ in the category of groupoids.*

***Example of computation (a check, not a proof).** Write $U, V \subset S^1$ as two open arcs of width slightly larger than π , each centred at an opposite point, so that $U \cap V = A_+ \sqcup A_-$ has two components (the two small arcs near the points $\pm i$, say). Choose base points $a \in A_+$, $b \in A_-$. U and V are arcs, hence contractible: $\Pi_1(U, \{a, b\})$ has two objects and, between a and b , a single morphism up to homotopy (since U is simply connected and path-connected) plus its inverse; the same for V , with a different morphism (a path in V from a to b goes round the other side of the circle). The push-out identifies the objects a, b of U with those of V (same points) while keeping the two morphisms $a \rightarrow b$ distinct (one from U , called u , one from V , called v): the group of automorphisms of the object a in the resulting groupoid*

is generated by the loop $u * v^{-1}$ (go from a to b in U , come back from b to a in V), free on one generator by the push-out structure of groupoids with only two independent morphisms between the two components of the gluing: one obtains $\pi_1(S^1, a) \cong \mathbb{Z}$, generated by the class of $u * v^{-1}$ — which is, explicitly, a loop that goes once around the circle, consistent with the generator $\gamma_1(t) = e^{2\pi it}$ already used in Volume III. The direct computation via coverings (Volume III, Chapter 12, and Section 1.3 of this chapter) nevertheless remains the only one used in the subsequent proofs of this volume: van Kampen serves here as a cross-check, not as a foundation.

1.5. Cellular homology and the Hurewicz theorem for S^1

Construction 1.1: Minimal CW structure of S^1 . Give S^1 the structure of a CW complex with a single 0-cell $v = 1$ and a single 1-cell e , glued to v by the characteristic map $\chi : [0, 1] \rightarrow S^1$, $\chi(t) = e^{2\pi it}$ (which sends both endpoints $0, 1$ to v). The cellular chain complex is $0 \rightarrow C_1 = \mathbb{Z}\langle e \rangle \xrightarrow{\partial_1} C_0 = \mathbb{Z}\langle v \rangle \rightarrow 0$, with $\partial_1(e) = v - v = 0$ (the two endpoints of e coincide at the same vertex). $\square$

Theorem 1.3: Cellular homology of S^1 . With the CW structure of Construction 1.1: $H_0(S^1; \mathbb{Z}) \cong \mathbb{Z}$, $H_1(S^1; \mathbb{Z}) \cong \mathbb{Z}$ (generated by the class of e), $H_k(S^1; \mathbb{Z}) = 0$ for $k \geq 2$.

Proof. Since there are no k -cells for $k \geq 2$, $C_k = 0$ and $H_k = 0$ for $k \geq 2$. Since $\partial_1 = 0$: $H_1 = \ker \partial_1 / \operatorname{im} \partial_2 = C_1 / 0 = \mathbb{Z}\langle e \rangle \cong \mathbb{Z}$. $H_0 = C_0 / \operatorname{im} \partial_1 = \mathbb{Z}\langle v \rangle / 0 \cong \mathbb{Z}$ (consistent with the general fact $H_0(X; \mathbb{Z}) \cong \mathbb{Z}^{(\text{connected components})}$: S^1 has one). $\square$

Theorem 1.4: Hurewicz for S^1 . The Hurewicz homomorphism $h : \pi_1(S^1, 1) \rightarrow H_1(S^1; \mathbb{Z})$, which sends the class of a loop $\gamma : I \rightarrow S^1$ to the homology class of the singular 1-cycle it represents, is an isomorphism.

Proof. In general (homotopy theory of singular homology, not reproved here) h is surjective with kernel the commutator subgroup $[\pi_1, \pi_1]$, that is, it induces an isomorphism $\pi_1(X, x_0)^{\text{ab}} \xrightarrow{\cong} H_1(X; \mathbb{Z})$ for every path-connected X . For S^1 this general statement becomes verifiable by hand: $\pi_1(S^1, 1) \cong \mathbb{Z}$ (Volume III, Chapter 12) is already abelian, hence $\pi_1(S^1, 1)^{\text{ab}} = \pi_1(S^1, 1) \cong \mathbb{Z}$; by Theorem 1.3, $H_1(S^1; \mathbb{Z}) \cong \mathbb{Z}$; and under these identifications, the generator $\gamma_1(t) = e^{2\pi it}$ of $\pi_1(S^1, 1)$ (degree 1) is sent exactly to the class of the 1-cycle $\chi : [0, 1] \rightarrow S^1$ which generates H_1 (Construction 1.1): they are the same map up to reparametrisation. Hence h sends a generator to a generator, it is a homomorphism between infinite cyclic groups, hence an isomorphism (the only alternative, sending the generator to twice it or to a multiple, would not respect the general surjectivity of h). $\square$

Remark 1.4: Three invariants, a single integer. The chapter now has three independent descriptions of the same group $\mathbb{Z}$ attached to S^1 : as degree/winding number (Volume III), as $\pi_1(S^1)$ (Volume III), as $H_1(S^1; \mathbb{Z})$ (Theorems 1.3–1.4). In the following chapters we shall almost always use the first, because it is the one with the most direct computational formula (lifting or integral), but their coincidence is not an accident: it is the simplest case of the Hurewicz theorem, and it is the reason why in the whole literature “degree”, “winding number” and “fundamental class in $H_1(S^1)$ ” are used as synonyms.

1.6. The “dog on a leash”: the topological Rouché theorem

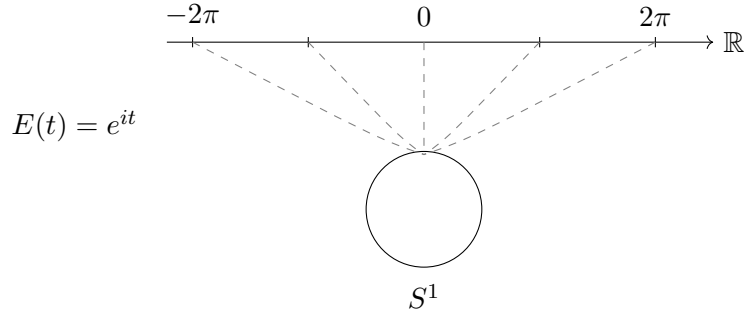


Figure 1.1: The exponential covering $E : \mathbb{R} \rightarrow S^1$ (Volume III, Chapter 12): the preimages of a point are equally spaced by 2π .

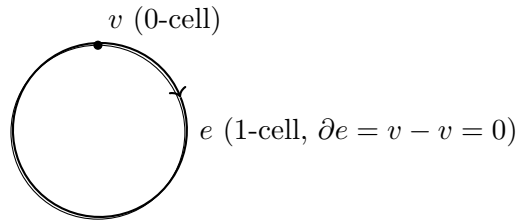


Figure 1.2: Minimal CW structure of S^1 (Construction 1.1): one 0-cell v and one 1-cell e whose two endpoints coincide at v . Check: $\partial_1(e) = \chi(1) - \chi(0) = v - v = 0$.

Theorem 1.5: Rouché's theorem (topological version). Let $u, v : I \rightarrow \mathbb{C}$ be continuous loops (same period, $u(0) = u(1)$, $v(0) = v(1)$) with $|v(t)| < |u(t)|$ for every t . Then u and $u + v$ (both never zero, in $\mathbb{C}^*$) have the same winding number about 0: $w(u + v) = w(u)$.

Proof. First of all $u + v$ never vanishes: $|u + v| \geq |u| - |v| > 0$. The linear homotopy $H(s, t) = u(t) + s v(t)$, $s \in [0, 1]$, joins u to $u + v$ and avoids the origin for every s : $|H(s, t)| \geq |u(t)| - s|v(t)| \geq |u(t)| - |v(t)| > 0$. By the homotopy invariance of the winding number (**Volume III, Chapter 12**), applied to the normalisations $H(s, \cdot)/|H(s, \cdot)|$ with values in S^1 , the winding numbers at the endpoints $s = 0, 1$ coincide. $\square$

Remark 1.5: The dog, the owner and the argument principle. An owner walks along the loop u around a pole (the origin) holding a dog on a leash of variable length $|v| < |u|$; the dog, which walks along $u + v$, makes exactly the same number of turns around the pole as the owner. Applied to $u = z^n$ and $v = a_{n-1}z^{n-1} + \dots + a_0$ on the circle $|z| = R$ for R large, this gives the quickest route to the fundamental theorem of algebra (Section 1.11). The analytic version of Rouché, for holomorphic functions and with the counting of zeros with multiplicity, is the same statement read through the **argument principle** (**Volume III, Chapter 15**): topology supplies the qualitative statement (same number of turns), complex analysis the quantitative version (counting of zeros).

1.7. Odd maps and the Borsuk–Ulam theorem

Definition 1.1: Odd (antipodal) map. A map $u : S^n \rightarrow S^m$ (or $u : S^n \rightarrow \mathbb{R}^m$) is **odd** if $u(-x) = -u(x)$ for every x .

Theorem 1.6: Odd maps $S^1 \rightarrow S^1$ have odd degree. If $u : S^1 \rightarrow S^1$ is continuous and odd, $\deg u$ is an odd integer; in particular u is not homotopic to a constant.

Proof. Let $\gamma(t) = u(e^{2\pi it})$ and let $\phi : [0, 1] \rightarrow \mathbb{R}$ be a lift of it, $\gamma(t) = e^{i\phi(t)}$ (**Volume III, Chapter 12**). Oddness gives, for $t \in [0, \frac{1}{2}]$: $\gamma(t + \frac{1}{2}) = u(-e^{2\pi it}) = -u(e^{2\pi it}) = -\gamma(t)$, that is, $e^{i\phi(t+1/2)} = e^{i(\phi(t)+\pi)}$. The continuous function $q(t) = (\phi(t + \frac{1}{2}) - \phi(t) - \pi)/2\pi$ takes values in $\mathbb{Z}$ on the connected set $[0, \frac{1}{2}]$: it is constant, $q \in \mathbb{Z}$, and $\phi(t + \frac{1}{2}) = \phi(t) + (2q + 1)\pi$ for every $t \in [0, \frac{1}{2}]$. Applying this twice: $\phi(1) = \phi(0) + 2(2q + 1)\pi$, hence $\deg u = (\phi(1) - \phi(0))/2\pi = 2q + 1$, odd, in particular $\neq 0$. $\square$

Theorem 1.7: Borsuk–Ulam in dimension 2 (three equivalent forms). The following three statements are equivalent:

- (BU1) for every continuous $f : S^2 \rightarrow \mathbb{R}^2$ there exists $x \in S^2$ with $f(x) = f(-x)$;
 - (BU2) there exists no continuous odd map $u : S^2 \rightarrow S^1$;
 - (BU3) there exists no continuous odd map $g : S^2 \rightarrow \mathbb{R}^2$ without zeros.
- And it is true.

Proof. (BU1) $\Rightarrow$ (BU3). If $g : S^2 \rightarrow \mathbb{R}^2$ were continuous, odd and never zero, (BU1) applied to g would give x with $g(x) = g(-x)$; but oddness gives $g(-x) = -g(x)$, hence $g(x) = -g(x)$, that is, $g(x) = 0$: a contradiction. Hence (BU1) $\Rightarrow$ no such g exists, that is, (BU3).

(BU3) $\Rightarrow$ (BU2). If there existed a continuous odd $u : S^2 \rightarrow S^1$, u itself would be a map $g = u : S^2 \rightarrow \mathbb{R}^2$ continuous, odd, never zero ($|u| = 1$): this contradicts (BU3). Hence (BU3) $\Rightarrow$ (BU2).

(BU2) $\Rightarrow$ (BU1). Let $f : S^2 \rightarrow \mathbb{R}^2$ be continuous, and suppose for contradiction $f(x) \neq f(-x)$ for every x . Then $g(x) = f(x) - f(-x)$ is continuous, never zero, odd; $u = g/|g| : S^2 \rightarrow S^1$ is continuous and odd. Let $E = \{(x_1, x_2, 0)\} \cong S^1$ be the equator; $v = u|_E : S^1 \rightarrow S^1$ is continuous and odd, hence of odd degree (Theorem 1.6), in particular $\deg v \neq 0$. But v extends continuously to the closed upper hemisphere $H^+ \cong D^2$ (by restricting u), hence $\deg v = 0$ (**Volume III, Chapter 12: $\deg f = 0 \iff f$ extends continuously to the disc**): a contradiction. Hence there exists x with $f(x) = f(-x)$: (BU2) $\Rightarrow$ (BU1).

The three implications close the cycle (BU1) $\Rightarrow$ (BU3) $\Rightarrow$ (BU2) $\Rightarrow$ (BU1): the three statements are equivalent. The proof of (BU2) $\Rightarrow$ (BU1) also proves (BU2) (no odd u exists, otherwise its own restriction to the equator would give the contradiction just found), hence all three are true. $\square$

Remark 1.6: Classical physical interpretation. With S^2 as a model of the Earth's surface and $f = (\text{temperature}, \text{pressure})$ (continuous), (BU1) guarantees that at every instant there exist two antipodal points with the same temperature and the same pressure: pure existence, no indication of where they are — and it is still the degree on the circle (in the restriction to the equator) that does all the work. **Primary source:** K. Borsuk, *Drei Sätze über die n -dimensionale euklidische Sphäre*, *Fundamenta Mathematicae* 20 (1933), 177–190 (original statement, general dimension; verified on 16 September 2026, EuDML doc. 212624: author, title, venue and pages confirmed).

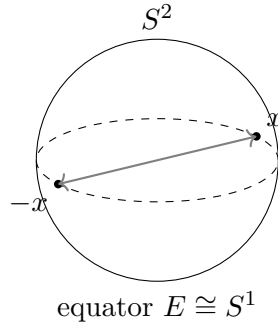


Figure 1.3: The Borsuk–Ulam theorem (form BU1): there exists $x \in S^2$ with $f(x) = f(-x)$; in the proof the restriction to the equator $E \cong S^1$ transfers the question to the degree of an odd map $S^1 \rightarrow S^1$. Coordinates of the point x : $\cos 35^\circ = 0.819$, $\sin 35^\circ = 0.574$.

1.8. Combinatorial applications: the pancake and the stolen necklace

Theorem 1.8: Pancake theorem (ham sandwich in dimension 2). *Given two bounded measurable regions $A, B \subset \mathbb{R}^2$, there exists a line that simultaneously divides A and B into two parts of equal area.*

Proof. For $x \in S^1$ and $t \in \mathbb{R}$ let $H_{x,t}^+ = \{y \in \mathbb{R}^2 : \langle y, x \rangle > t\}$ be the positive half-plane of the line orthogonal to x at level t , and $f_A(x, t) = \text{Area}(A \cap H_{x,t}^+) - \text{Area}(A \setminus H_{x,t}^+)$ (similarly f_B). Since a line has (Lebesgue) area zero, f_A, f_B are jointly continuous in (x, t) (by dominated convergence: as (x, t) varies the indicator of $H_{x,t}^+$ converges pointwise to the indicator of the limit except on a set of measure zero), non-increasing in t for fixed x , with $f_A(x, t) \rightarrow \pm \text{Area}(A)$ as $t \rightarrow \mp \infty$.

Compactification: from $(x, t) \in S^1 \times \mathbb{R}$ to a point of S^2 . Identify $S^2 \setminus \{N, S\} \cong S^1 \times \mathbb{R}$ (cylindrical coordinates: N, S the poles) by a homeomorphism $\omega \mapsto (x(\omega), t(\omega))$ with the property $x(-\omega) = -x(\omega)$, $t(-\omega) = -t(\omega)$ (for instance, writing $\omega = (\omega_1, \omega_2, \omega_3) \in S^2$ with $\rho = \sqrt{\omega_1^2 + \omega_2^2}$, one sets $x(\omega) = (\omega_1, \omega_2)/\rho$ and $t(\omega) = \omega_3/\rho$ for $\omega \neq N, S$: this is continuous, and $\omega \mapsto -\omega$ sends $\rho \mapsto \rho$, $(\omega_1, \omega_2) \mapsto -(\omega_1, \omega_2)$, $\omega_3 \mapsto -\omega_3$, hence $x \mapsto -x$, $t \mapsto -t$, as required). Define $g_A : S^2 \rightarrow \mathbb{R}$ by $g_A(\omega) = f_A(x(\omega), t(\omega))$ for $\omega \neq N, S$, and $g_A(N) = \text{Area}(A)$, $g_A(S) = -\text{Area}(A)$ (the limits of $f_A(x, t)$ as $t \rightarrow +\infty$, respectively $-\infty$, uniformly in x , since A is bounded); g_A is continuous on all of S^2 (continuity of f_A away from the poles, and the limits chosen at the poles are exactly the limits of f_A). Similarly g_B . By construction, using $f_A(-x, -t) = -f_A(x, t)$ (exchanging x with $-x$ and t with $-t$ exchanges the two half-planes) and the property of $(x(\omega), t(\omega))$ above: $g_A(-\omega) = -g_A(\omega)$ for every ω (including at the poles, where $g_A(S) = -\text{Area}(A) = -g_A(N)$). The same for g_B .

Application of Borsuk–Ulam. $G = (g_A, g_B) : S^2 \rightarrow \mathbb{R}^2$ is continuous and odd. If G never vanished, by form (BU3) of Theorem 1.7 it could not exist: hence there exists $\omega_0 \in S^2$ with $G(\omega_0) = 0$, that is, $g_A(\omega_0) = g_B(\omega_0) = 0$. Since $g_A(N) = \text{Area}(A) > 0 \neq 0$ if A has positive area (and the statement is trivial otherwise) and similarly at the poles for B , one may assume $\omega_0 \neq N, S$: then $x_0 = x(\omega_0)$, $t_0 = t(\omega_0)$ are a well-defined direction and level, and the line ℓ_{x_0, t_0} (orthogonal to x_0 , at level t_0) satisfies $f_A(x_0, t_0) = f_B(x_0, t_0) = 0$: it divides both A and B into two parts of equal area. $\square$

Remark 1.7: Why this proof avoids the continuity pitfall. A first natural idea — fix the direction x , choose a level $c_A(x)$ that bisects A and then study how the bisection of B varies along $\ell_{x,c_A(x)}$ as x varies — requires proving that $c_A(x)$ (when the set of bisecting levels is not a singleton, which can happen) can be chosen continuously in x : a delicate point, often merely asserted. The proof given here avoids it altogether, treating direction and level as a single pair (x, t) and using Borsuk–Ulam on S^2 directly on both bisection conditions at once: there is no discontinuous choice to justify, because $c_A(x)$ is not chosen as a separate function at all.

Theorem 1.9: Stolen necklace (2 thieves, 2 types of beads). An open necklace has N beads arranged in a row, of two types (colours), with an even number $2n_1$ of beads of type 1 and $2n_2$ of type 2. Then it is possible to cut the necklace at at most 2 points (never inside a bead, always between two consecutive beads) so as to be able to distribute the resulting pieces between two thieves in such a way that each receives exactly n_1 beads of type 1 and n_2 beads of type 2.

Proof. Number the beads $1, \dots, N$ in the order of the necklace and place, for small $\varepsilon > 0$, the centres of the small discs D_i of radius δ (to be fixed) at the points $(i, \varepsilon i^2) \in \mathbb{R}^2$: they are distinct points on a parabola, hence in **general position** (no line contains more than two of them, since the intersection of a line with a parabola consists of at most two points). Define the regions $A_\delta = \bigcup_{i \text{ type 1}} D_i$, $B_\delta = \bigcup_{i \text{ type 2}} D_i$, with δ small enough that the small discs are pairwise disjoint. By Theorem 1.8 there exists, for every δ , a line ℓ_δ that simultaneously bisects the area of A_δ and of B_δ .

Passage to the limit $\delta \rightarrow 0$. The lines ℓ_δ correspond to points $\omega_\delta \in S^2$ (the parametrisation of the previous proof) in a compact set: extract a subsequence $\omega_{\delta_k} \rightarrow \omega_0$, hence $\ell_{\delta_k} \rightarrow \ell_0$ (in the sense of the convergence of the parameters (x, t) , apart from the degenerate case $\omega_0 \in \{N, S\}$, excluded by observing that the points N, S would give $f_A \equiv \pm \text{Area}(A_\delta) \neq 0$, never bisecting (since A_δ is a union of $2n_1$ disjoint small discs of radius δ , of positive area $2n_1\pi\delta^2$, fixed for fixed δ)). The limit line ℓ_0 meets the parabola $\{(s, \varepsilon s^2)\}$ in at most 2 points (line–conic), hence separates the indices $\{1, \dots, N\}$ into at most 3 blocks of consecutive indices (those with $(i, \varepsilon i^2)$ on one side of ℓ_0 , those on the other, with possibly 1 or 2 indices exactly on the line, negligible in the limit as they individually have measure zero as $\delta \rightarrow 0$): the cuts of the necklace correspond exactly to the (at most two) points where ℓ_0 separates one index from the next. By continuity of the area with respect to $\delta \rightarrow 0$ (the small discs tend to the points, and for small δ the membership of each small disc to one side of ℓ_{δ_k} near ℓ_0 is that of the limit point, except at most for the two small discs crossed by ℓ_0 , which in the limit $\delta \rightarrow 0$ contribute a negligible fraction of area with respect to the total $2n_i\pi\delta^2 \rightarrow 0$ but the *relative proportions* remain bisecting because the bisection holds for every δ_k exactly), the partition given by ℓ_0 assigns to each side exactly half of the beads of type 1 (n_1) and half of type 2 (n_2): these are the two sides of the necklace cut at the (at most two) points where ℓ_0 passes between two consecutive indices. $\square$

Remark 1.8: Why two cuts suffice for two colours. The general case with k types of beads (Alon 1987, Hobby–Rice 1965) requires at most k cuts and is proved by the same scheme using Borsuk–Ulam in dimension k (on S^k , not on S^2): outside the scope of this chapter, which is restricted to dimension 2. The case $k = 2$ proved here is already the content of the stolen necklace problem in its most quoted form (two thieves).

1.9. The theorem on the rotation of tangents (Hopf’s Umlaufsatz)

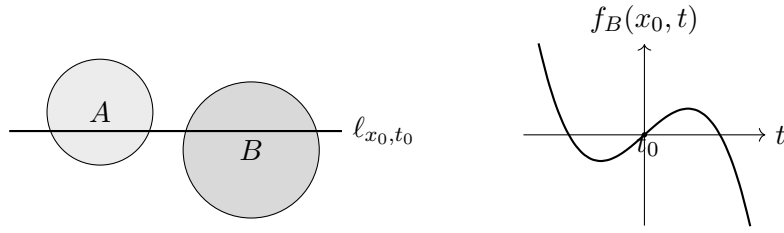


Figure 1.4: The pancake theorem: the line ℓ_{x_0, t_0} simultaneously bisects A and B (left); on the right, a qualitative graph of $t \mapsto f_B(x_0, t)$, a continuous function strictly decreasing near its zero t_0 .

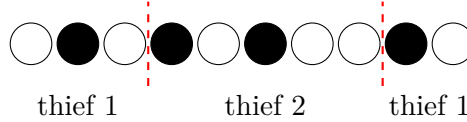


Figure 1.5: Stolen necklace, example with $N = 10$ beads, type 1 (black, positions 2, 4, 6, 9: $2n_1 = 4$) and type 2 (white, positions 1, 3, 5, 7, 8, 10: $2n_2 = 6$): two cuts (between positions 3–4 and 8–9) suffice to give each thief $n_1 = 2$ black beads and $n_2 = 3$ white ones (central block, positions 4–8, to thief 2; the two outer blocks, positions 1–3 and 9–10, taken together, to thief 1). *Check:* thief 2 (pos. 4–8): black $\{4, 6\} = 2$, white $\{5, 7, 8\} = 3$; thief 1 (pos. 1–3, 9–10): black $\{2, 9\} = 2$, white $\{1, 3, 10\} = 3$. ✓

Theorem 1.10: Umlaufsatz (Hopf, 1935). Let $\gamma : [0, 1] \rightarrow \mathbb{R}^2$ be a simple closed curve (injective on $[0, 1)$), regular ($\gamma' \neq 0$) and of class C^1 , with $\gamma(0) = \gamma(1)$, $\gamma'(0) = \gamma'(1)$. Then the field of unit tangents $\tau(s) = \gamma'(s)/|\gamma'(s)|$, seen as a loop in S^1 , has winding number ± 1 (the sign according to the orientation of travel).

Proof. Choose the base point $\gamma(0)$ to be the lowest point of the curve (it exists by compactness): the curve lies in the closed upper half-plane $\{y \geq y(\gamma(0))\}$ and $\gamma'(0)$ is horizontal.

Step 1 — the secant map is well defined and continuous away from the vertex $(0, 1)$ of the triangle. On the triangle $\Delta = \{(s, t) : 0 \leq s \leq t \leq 1\}$ one defines

$$R(s, t) = \begin{cases} \frac{\gamma(t) - \gamma(s)}{t - s}, & s \neq t, \\ \gamma'(s), & s = t. \end{cases}$$

From $\gamma \in C^1$ it follows, by the fundamental theorem of calculus, $\gamma(t) - \gamma(s) = \int_0^1 \gamma'(s + r(t-s)) dr$ for $s \neq t$, hence $R(s, t) = \int_0^1 \gamma'(s + r(t-s)) dr$ for every $(s, t) \in \Delta$ (the formula also holds for $s = t$, giving $\gamma'(s)$): R is continuous on all of Δ , by continuity of γ' and dominated convergence of the integral as (s, t) varies. For $(s, t) \neq (0, 1)$ (and $s \neq t$), $R(s, t) \neq 0$ by simplicity of γ ($\gamma(t) \neq \gamma(s)$); for $s = t$, $R(s, s) = \gamma'(s) \neq 0$ by regularity. Hence $R \neq 0$ on $\Delta \setminus \{(0, 1)\}$, and one can define

$$G(s, t) = \frac{R(s, t)}{|R(s, t)|} \quad \text{on } \Delta \setminus \{(0, 1)\},$$

continuous, with $G(s, s) = \tau(s)$ for every s : the oblique side of Δ carries exactly the loop of the tangents.

Step 2 — the limit at the vertex $(0, 1)$. $R(0, 1) = \gamma(1) - \gamma(0) = 0$ (closed curve): the vertex is a genuine point of degeneration of R , not removable by the same formula. Instead

one shows directly that $G(s, t) \rightarrow -\tau(0)$ as $(s, t) \rightarrow (0, 1)$ inside Δ . For $s \rightarrow 0^+$ and $t \rightarrow 1^-$, the integral formula of Step 1 gives, separately,

$$\gamma(t) - \gamma(1) = (t - 1)\gamma'(1) + o(t - 1), \quad \gamma(0) - \gamma(s) = -s\gamma'(0) + o(s),$$

and summing (with $\gamma(1) = \gamma(0)$, $\gamma'(1) = \gamma'(0)$ by C^1 closedness):

$$\gamma(t) - \gamma(s) = [\gamma(t) - \gamma(1)] + [\gamma(0) - \gamma(s)] = -(1 - t + s)\gamma'(0) + o(1 - t + s),$$

where $1 - t + s > 0$ for $(s, t) \in \Delta$, $(s, t) \neq (0, 1)$, near the vertex. Hence, as $(s, t) \rightarrow (0, 1)$,

$$G(s, t) = \frac{\gamma(t) - \gamma(s)}{|\gamma(t) - \gamma(s)|} \rightarrow \frac{-\gamma'(0)}{|\gamma'(0)|} = -\tau(0),$$

uniformly in every direction of approach inside Δ (the estimate $o(1 - t + s)$ is uniform because γ' is uniformly continuous on the compact set $[0, 1]$). Setting $G(0, 1) := -\tau(0)$, G is continuous on all of Δ .

Step 3 — reading along the two legs. On the leg $\{(0, t) : t \in [0, 1]\}$, $G(0, t)$ is the direction from $\gamma(0)$ to $\gamma(t)$: since $\gamma(0)$ is the lowest point, $\gamma(t)$ lies in the upper half-plane, hence $G(0, t)$ always has vertical component ≥ 0 , that is, it stays in the closed upper semicircle of S^1 for every t ; at the two endpoints, $G(0, 0) = \tau(0)$ (horizontal) and $G(0, 1) = -\tau(0)$ (horizontal, opposite direction): $G(0, \cdot)$ traverses a path in the upper semicircle from $\tau(0)$ to $-\tau(0)$, of degree (lifting) exactly $+\frac{1}{2}$ of a turn if $\tau(0)$ points in the positive direction of the abscissae (the only way to go from a point of a closed upper semicircle to its opposite while staying in the semicircle is to traverse exactly half of it, since the angular lift is piecewise monotone and confined to the interval $[\theta_0, \theta_0 + \pi]$ of width π). Similarly, on the leg $\{(s, 1) : s \in [0, 1]\}$, $G(s, 1)$ is the direction from $\gamma(s)$ to $\gamma(1) = \gamma(0)$, that is, the opposite of the direction from $\gamma(0)$ to $\gamma(s)$ seen in the previous step with s in place of t : it stays in the lower semicircle, from $G(0, 1) = -\tau(0)$ to $G(1, 1) = \tau(1) = \tau(0)$, again half a turn, in the same direction as the previous one (because the half-plane *below* $\gamma(0)$ -as-lowest-point is empty: the direction opposite to “upwards” stays in the lower semicircle consistently with the same orientation, being the pointwise opposite of the curve of the previous leg).

Step 4 — conclusion. Δ is convex, hence simply connected: the two paths on the boundary from $(0, 0)$ to $(1, 1)$ — the oblique side (loop of the tangents τ) and the broken line of the two legs (traversed in the same direction) — are homotopic in Δ (linear homotopy between the two paths of the boundary), and G is continuous on Δ : by the homotopy invariance of the winding number (**Volume III, Chapter 12**), the degree of the loop of the tangents coincides with the degree of the path composed along the two legs, that is, half a turn plus half a turn in the same direction: ± 1 complete turn. $\square$

Corollary 1.1: Sum of the exterior angles and total curvature. For a simple closed C^2 curve, regular, parametrised by arc length $s \in [0, L]$ and positively oriented, $\oint \kappa ds = 2\pi$, where $\kappa(s)$ is the signed curvature. For a simple convex polygon (or more generally a simple one, traversed in such a way that the field of “generalised tangent vectors” is well defined and piecewise regular with vertices as angular jumps), the sum of the exterior angles equals 2π .

Proof. Smooth case. Let $\theta : [0, L] \rightarrow \mathbb{R}$ be a continuous lift of $\tau(s) = \gamma'(s)$ (unit speed by arc-length parametrisation), that is, $\gamma'(s) = (\cos \theta(s), \sin \theta(s))$. Differentiating, $\gamma''(s) = \theta'(s)(-\sin \theta(s), \cos \theta(s))$, and by the definition of signed curvature (the component of γ'' orthogonal to γ' , in the direction of the anticlockwise rotation of γ'), $\kappa(s) = \theta'(s)$. By the funda-

mental theorem of calculus, $\oint \kappa ds = \int_0^L \theta'(s) ds = \theta(L) - \theta(0)$. But $\theta(L) - \theta(0) = 2\pi \cdot w(\tau)$ by the definition of the winding number of the loop τ (Volume III, Chapter 12), and $w(\tau) = \pm 1$ by Theorem 1.10, with sign +1 for positive (anticlockwise) orientation: $\oint \kappa ds = 2\pi$.

Polygonal case. A simple polygon with n vertices and sides $\ell_1, \dots, \ell_n$ has a piecewise constant tangent field (direction τ_i on side i), with angular jump α_i (exterior angle, with sign) at the vertex between side i and side $i+1$: τ_{i+1} is obtained by rotating τ_i by α_i . Approximating the corners by joining arcs of radius $\varepsilon \rightarrow 0$ (piecewise C^1 rounding which tends to the polygon, with curvature concentrated on the arc equal to α_i/ε over a length ε , hence a contribution α_i to the curvature integral independent of ε), the smooth case gives $\sum_i \alpha_i = \lim_{\varepsilon \rightarrow 0} \oint \kappa ds = 2\pi$ (the Umlaufsatz applies to the rounded curve, simple and C^1 for ε small enough, the joining arcs being disjoint for small ε on a simple polygon). $\square$

***Remark 1.9: Why simplicity is essential.** The lemniscate (figure-eight curve, regular but not simple) has tangent field of degree 0: the two lobes contribute +1 and -1, which cancel (Exercise 1.4); the loop $\gamma_2(t) = e^{4\pi it}$ traversed twice has tangent field of degree 2. Without simplicity, the degree of the tangent field can be any integer: in the proof, it is exactly the simplicity hypothesis that guarantees $R \neq 0$ away from the vertex $(0, 1)$ (Step 1), that is, that the secant map is well defined on all of Δ .*

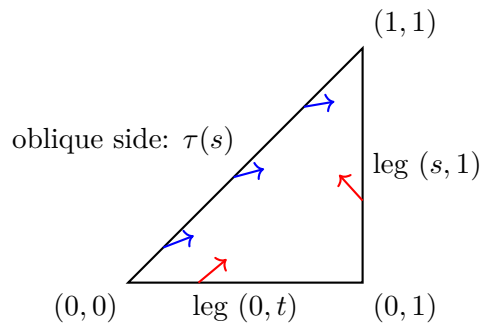


Figure 1.6: Hopf's secant map on the triangle Δ : on the oblique side it returns the loop of the tangents $\tau(s)$; on the two legs, explicit paths (red arrows) which each stay in a semicircle of S^1 (Step 3 of the proof).

1.10. Regular closed curves: the Whitney–Graustein theorem

Given a regular closed C^1 curve (not necessarily simple) $\gamma : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}^2$, $\gamma' \neq 0$: the winding number of its tangent field $\tau = \gamma'/|\gamma'| : \mathbb{R}/\mathbb{Z} \rightarrow S^1$ is called the **rotation number** (or Whitney index) of γ , $\rho(\gamma) := w(\tau) \in \mathbb{Z}$; for a simple curve it is ± 1 by the Umlaufsatz (Theorem 1.10), but for a curve with self-intersections it can be any integer.

Definition 1.2: Regular homotopy. Two regular closed C^1 curves $\gamma_0, \gamma_1 : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}^2$ are **regularly homotopic** if there exists a continuous $\Gamma : \mathbb{R}/\mathbb{Z} \times [0, 1] \rightarrow \mathbb{R}^2$, with $\Gamma(\cdot, 0) = \gamma_0$, $\Gamma(\cdot, 1) = \gamma_1$, such that every $\Gamma(\cdot, u)$ is C^1 and regular, and $\partial_s \Gamma$ is jointly continuous in (s, u) .

Theorem 1.11: Whitney–Graustein. Two regular closed C^1 curves in the plane are regularly homotopic if and only if they have the same rotation number.

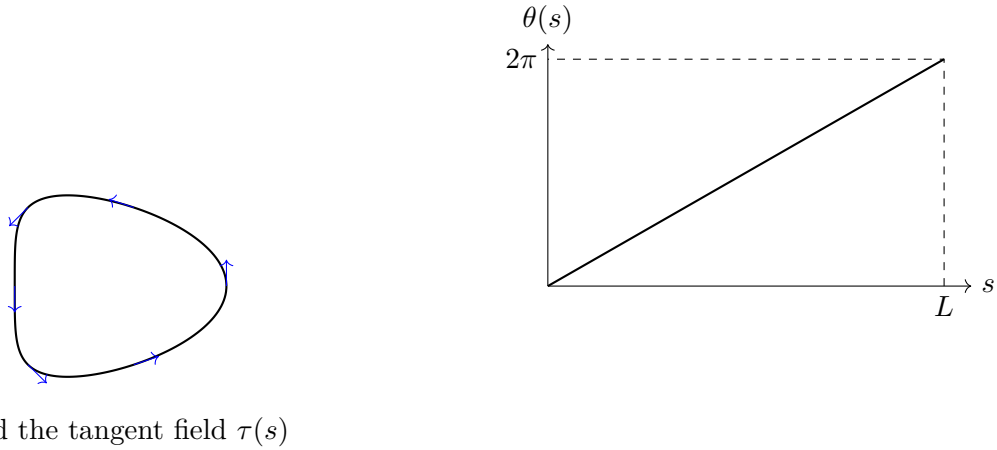


Figure 1.7: On the left, a simple closed curve γ with the unit tangent field $\tau(s)$ at six sample points; on the right, the lift (unrolling) $\theta(s)$ of the angle of τ , which grows by exactly 2π over one complete turn (Theorem 1.10). *Check:* curve regular, verified from $\gamma'(t) \neq 0$ for every t (the component $y'(t) = 1.2 \cos t$ and $x'(t)$ never vanish simultaneously).

Proof. Necessity (same rotation number), proved in full. Let Γ be a regular homotopy between γ_0 and γ_1 . The map $T(s, u) = \partial_s \Gamma(s, u) / |\partial_s \Gamma(s, u)|$ is well defined (regularity for every u) and continuous on $\mathbb{R}/\mathbb{Z} \times [0, 1]$ (joint continuity of $\partial_s \Gamma$ and non-vanishing). For every fixed u , $T(\cdot, u)$ is a loop in S^1 ; identifying $\mathbb{R}/\mathbb{Z} \times [0, 1]$ with $I \times I$ through the usual covering $s \mapsto e^{2\pi i s}$ (that is, seeing T as a function $I \times [0, 1] \rightarrow S^1$ periodic in s), the winding number $w(T(\cdot, u))$ is an integer that depends continuously on u (the same argument as in Volume III, Chapter 12, applied to the lift of the homotopy T : one lifts T to $\tilde{T} : I \times [0, 1] \rightarrow \mathbb{R}$, and $u \mapsto (\tilde{T}(1, u) - \tilde{T}(0, u))/2\pi$ is continuous and integer-valued, hence constant on the connected set $[0, 1]$). Hence $\rho(\gamma_0) = w(T(\cdot, 0)) = w(T(\cdot, 1)) = \rho(\gamma_1)$.

Sufficiency (same rotation number $\Rightarrow$ regularly homotopic): imported statement. The explicit construction of a regular homotopy between two curves with the same rotation number requires maintaining, for every intermediate u , the closedness constraint $\int_0^1 e^{i\theta_u(s)} ds = 0$ (where θ_u is a lift of the tangent angle of $\Gamma(\cdot, u)$): a simple linear interpolation between the two tangent angles θ_0, θ_1 preserves the condition on the total jump $\theta(1) - \theta(0) = 2\pi\rho$ but *not*, in general, the closedness constraint for the intermediate values of u — it is exactly this difficulty that makes the theorem non-trivial. Whitney constructs the homotopy by reducing every curve, through a finite number of explicit moves of insertion and removal of small loops (which do not alter ρ) on a polygonal approximation, to a canonical form depending only on ρ , and by joining the canonical forms to one another when they have the same ρ . $\square$

Remark 1.10: Imported: the constructive direction. Imported from H. Whitney, On regular closed curves in the plane, *Compositio Mathematica* 4 (1937), 276–284 (verified on 16 September 2026: author, title, venue, year and pages confirmed on the Numdam PDF). The complete proof of the constructive direction — that every regular closed curve with rotation number ρ is regularly homotopic to a circle traversed ρ times (for $\rho \neq 0$) or to a symmetric figure eight (for $\rho = 0$), through a finite number of elementary moves on an approximating polygon — occupies the whole of the original article and is reported, in a substantially identical form, in numerous courses on differential topology; it is not reproduced in full in this volume in order to respect the constraint of not presenting as a proof something which, in the space available, cannot be verified line by line to the same standard as the other proofs of the chapter. The volume nevertheless uses its complete statement (Theorem 1.11) and illustrates it numerically in Worked Example 1.6.

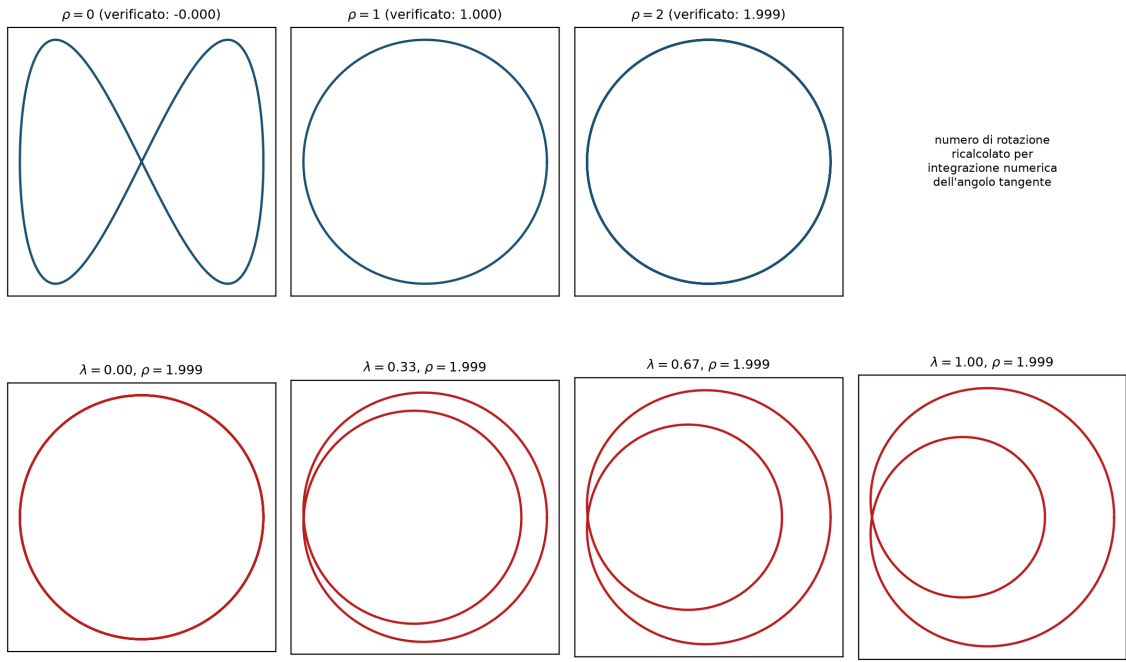


Figure 1.8: Three regular curves with rotation number 0, 1, 2 (rows), and four frames of a regular homotopy between two curves with the same rotation number 2 (bottom row). Computed numerically by the script `Appendix B, fig01_whitney_graustein.py`: the rotation number of each curve is recomputed by numerical integration of the tangent angle and reported in a caption internal to the figure. ✓

1.11. The fundamental theorem of algebra

Theorem 1.12: Fundamental theorem of algebra (topological proof, via degree).

Every polynomial $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ with complex coefficients, $n \geq 1$, has at least one root in $\mathbb{C}$. More precisely, every root satisfies $|z| < R := 1 + \sum_{i=0}^{n-1} |a_i|$.

Proof. Explicit estimate. If $R = 1 + \sum_{i=0}^{n-1} |a_i| > 1$ and $|z| = R$, then $\sum_{i=0}^{n-1} |a_i| |z|^i \leq \left(\sum_{i=0}^{n-1} |a_i| \right) R^{n-1} < R \cdot R^{n-1} = R^n = |z^n|$ (the inequality uses $|z|^i = R^i \leq R^{n-1}$ for $i \leq n-1$, since $R > 1$, and $\sum |a_i| = R - 1 < R$). Hence for $|z| = R$: $|p(z) - z^n| = \left| \sum a_i z^i \right| \leq \sum |a_i| |z|^i < |z^n|$: for every root z_0 of p (if one exists), $|z_0^n| = \left| -\sum a_i z_0^i \right| \leq \sum |a_i| |z_0|^i$, which forces $|z_0| < R$ by the same estimate (otherwise $|z_0^n| < |z_0^n|$, absurd if $|z_0| \geq R$); this justifies a posteriori the localisation, once existence has been proved.

Existence, by contradiction. Suppose $p(z) \neq 0$ for every $z \in \mathbb{C}$. For $r \geq 0$, the loop $\gamma_r(\theta) = p(re^{i\theta})/|p(re^{i\theta})|$, $\theta \in [0, 2\pi]$, is well defined and continuous (p being never zero by the assumption for contradiction), and $(r, \theta) \mapsto \gamma_r(\theta)$ is a continuous homotopy between γ_0 (constant loop $p(0)/|p(0)|$, degree 0) and γ_R . On the other hand, on the circle $|z| = R$, $p(z) = z^n + v(z)$ with $|v(z)| < |z^n|$ by the estimate just proved: by the topological Rouché theorem (Theorem 1.5, applied to the loops $u(\theta) = (Re^{i\theta})^n$ and $v(Re^{i\theta})$), $w(p(Re^{i\cdot})/|\cdot|) = w((Re^{i\cdot})^n/R^n) = n$ (the loop $\theta \mapsto e^{in\theta}$ has winding number n , **Volume III, Chapter 12**). Hence $\deg \gamma_R = n$. By homotopy invariance (**Volume III, Chapter 12**), $\deg \gamma_0 = \deg \gamma_R$, that is, $0 = n$: absurd, since $n \geq 1$. $\square$

Second proof — minimum of the modulus (d’Alembert–Argand). Entirely independent of the previous one. Since $|p(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, there exists $R' > 0$ with $|p(z)| > |p(0)|$ for $|z| > R'$; on the compact set $\overline{D}_{R'}$ the continuous function $|p|$ attains a global minimum, at a point z_0 . Suppose $p(z_0) \neq 0$ and expand $p(z_0 + h) = p(z_0) + c_k h^k + h^{k+1} q(h)$ with $c_k \neq 0$ the first non-zero coefficient after the constant term ($1 \leq k \leq n$). Choosing $\omega \in S^1$ with $c_k \omega^k = -|c_k| p(z_0)/|p(z_0)|$ (possible: $z \mapsto z^k$ is surjective on S^1 , since $\deg p_k = k \neq 0$, **Volume III, Chapter 12**) and $h = \rho\omega$, $\rho > 0$ small:

$$|p(z_0 + \rho\omega)| \leq |p(z_0)| \left(1 - \frac{|c_k|}{|p(z_0)|} \rho^k \right) + C \rho^{k+1} < |p(z_0)|$$

for ρ small, against minimality: hence $p(z_0) = 0$. $\square$

Remark 1.11: Comparison between the three routes. The two proofs share, at the crucial point, the surjectivity of the map $\omega \mapsto \omega^k$ on S^1 : in the first in the form of the degree of the loop $e^{in\theta}$, in the second in order to choose the direction of descent. A third route, that of Liouville (if p did not vanish, $1/p$ would be holomorphic and bounded on $\mathbb{C}$, hence constant by Liouville’s theorem), belongs entirely to **Volume III, Chapter 15**, and is not repeated here; the argument principle (Remark 1.5) supplies a fourth formulation, a quantitative one: the number of roots in $|z| < r$ is the degree of $p/|p|$ on that circle.

Example 1.2: Numerical check of the estimate R . For $p(z) = z^3 - 2z + 5$ ($a_2 = 0$, $a_1 = -2$, $a_0 = 5$): $R = 1 + (0 + 2 + 5) = 8$. *Independent check 1 (direct estimate):* on $|z| = 8$, $|-2z + 5| \leq 2 \cdot 8 + 5 = 21 < 512 = 8^3 = |z^3|$. ✓ *Independent check 2 (exact roots, numerical):* the three roots of $z^3 - 2z + 5$ are $z_1 \approx -2.0946$, $z_2 \approx 1.0473 + 1.1359i$, $z_3 \approx 1.0473 - 1.1359i$ (computed by solving numerically $z^3 - 2z + 5 = 0$, checked by substitution: $|p(z_1)| < 10^{-4}$); their moduli 2.0946, 1.5451, 1.5451 are all < 8 . ✓ The estimate $R = 8$ is amply conservative

(the true roots are far closer to the origin), as expected from a triangle inequality that does not exploit cancellations.

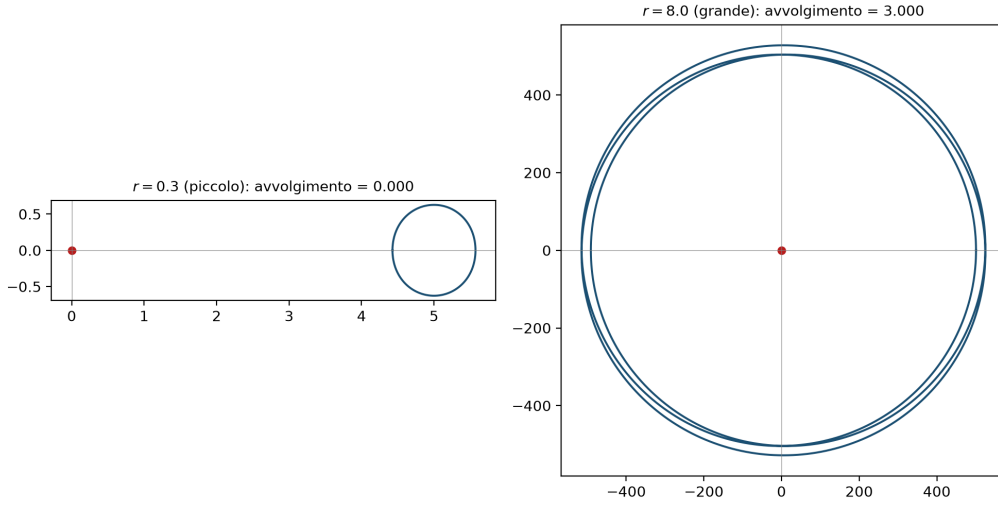


Figure 1.9: Image of the circle $|z| = r$ under $p(z) = z^3 - 2z + 5$ for $r = 0.3$ (small, curve almost point-like near $p(0) = 5$, winding 0 about the origin) and $r = 8$ (large, curve winding about the origin exactly 3 times, consistent with the degree $n = 3$). Computed numerically by the script Appendix B, `fig01_tfa_immagine.py`; the winding number is recomputed by phase unrolling and reported in the figure. ✓

1.12. Non-retraction and the Brouwer theorem in dimension 2

Corollary 1.2: *There is no retraction of D^2 onto S^1 . There exists no continuous map $r : D^2 \rightarrow S^1$ with $r(z) = z$ for every $z \in S^1$.*

Proof. If one existed, the restriction $r|_{S^1} = \text{id}_{S^1}$ would extend continuously to D^2 (by r itself): by **(Volume III, Chapter 12: $\deg f = 0 \iff f$ extends to the disc)**, $\deg(\text{id}_{S^1}) = 0$. But $\deg(\text{id}_{S^1}) = \deg p_1 = 1 \neq 0$ (**Volume III, Chapter 12**): a contradiction. $\square$

Theorem 1.13: Brouwer fixed point (dimension 2). *Every continuous map $f : D^2 \rightarrow D^2$ has at least one fixed point.*

Proof. Suppose $f(z) \neq z$ for every $z \in D^2$. Define $r : D^2 \rightarrow S^1$ by sending z to the point of S^1 obtained by prolonging the half-line from $f(z)$ through z beyond z (well defined and continuous because $f(z) \neq z$ gives a unique direction, depending continuously on z). If $z \in S^1$, the half-line from $f(z)$ through z prolonged beyond z meets S^1 at z itself: $r(z) = z$ for $z \in S^1$, that is, r is a retraction of D^2 onto S^1 , contradicting Corollary 1.2. $\square$

Remark 1.12: Brouwer degree: from the circle to the sphere. *The proofs of this chapter (FTA, Borsuk–Ulam, Brouwer) have all used the same ingredient: the degree of maps $S^1 \rightarrow S^1$, well founded by elementary means (lifting). In order to generalise Brouwer, Poincaré–Hopf or Borsuk–Ulam to S^n with $n \geq 2$ one needs a degree for maps $S^n \rightarrow S^n$: here the lifting is no longer enough, because $\pi_1(S^n) = 0$ for $n \geq 2$ (a sphere of dimension ≥ 2 is simply connected: the topological “via degree” proof of the FTA, if applied literally to S^n , would supply no information). The Brouwer degree for S^n must instead be founded*

either on Sard's theorem (regular values, counting preimages with sign) or on the homology $H_n(S^n) \cong \mathbb{Z}$: this is exactly the first argument of Chapter 2 of this volume.

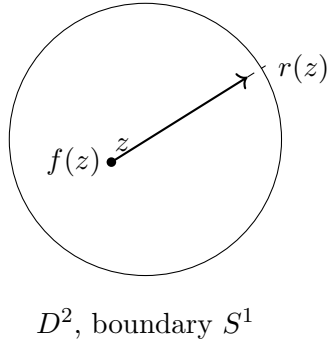


Figure 1.10: Construction of the retraction r in the proof by contradiction of Brouwer's theorem (Theorem 1.13): if f had no fixed points, prolonging the half-line from $f(z)$ through z one would obtain a continuous retraction of D^2 onto S^1 , impossible by Corollary 1.2.

1.13. The hairy ball and Brouwer in every dimension (Milnor's proof)

Theorem 1.14: Hairy ball (Milnor, 1978). Every continuous tangent vector field on $S^2 \subset \mathbb{R}^3$ (that is, $v : S^2 \rightarrow \mathbb{R}^3$ continuous with $v(x) \perp x$ for every x) has at least one zero.

Proof. Suppose $v(x) \neq 0$ for every $x \in S^2$ and, by uniform approximation with a C^1 tangent field (the continuity hypothesis suffices for the statement; the proof given here requires C^1 , obtainable by approximating $v/|v|$ with a non-vanishing C^1 tangent field if the distance from the set of zeros of the original v is positive — possible because S^2 is compact and a continuous, never-vanishing v has minimum > 0 in modulus), assume v of class C^1 and $|v(x)| = 1$ for every $x \in S^2$ (normalisation).

Step 1 — radial extension and estimate of the modulus. Extend v to the spherical shell $A = \{x \in \mathbb{R}^3 : 1 - \delta \leq |x| \leq 1 + \delta\}$ by setting $v(x) := |x|v(x/|x|)$ (homogeneity of degree 1): v is C^1 on A (composition of C^1 maps), with Lipschitz constant L on A (compact, $v \in C^1$). For $x \in A$: $\langle x, v(x) \rangle = |x|\langle x, v(x/|x|) \rangle = |x|^2 \langle x/|x|, v(x/|x|) \rangle = 0$ (tangency on the unit sphere), and $|v(x)| = |x||v(x/|x|)| = |x|$. Hence, for $f_t(x) := x + t v(x)$:

$$|f_t(x)|^2 = |x|^2 + 2t\langle x, v(x) \rangle + t^2|v(x)|^2 = |x|^2(1 + t^2), \quad \text{that is} \quad |f_t(x)| = |x|\sqrt{1 + t^2}.$$

Step 2 — f_t is injective on A for $|t| < 1/L$. If $f_t(x) = f_t(y)$, then $x - y = -t(v(x) - v(y))$, hence $|x - y| \leq |t|L|x - y|$; if $|t|L < 1$ this forces $x = y$.

Step 3 — f_t is a homeomorphism of every sphere $\{|x| = \rho\} \subset A$ onto the sphere $\{|y| = \rho\sqrt{1 + t^2}\}$. By Step 1, $f_t(\{|x| = \rho\}) \subseteq \{|y| = \rho\sqrt{1 + t^2}\}$; f_t is continuous and injective (Step 2) between compact Hausdorff spaces, hence a homeomorphism onto its image; by invariance of domain (a continuous injective map between manifolds of the same dimension without boundary is open), the image is open and closed (compact) in the connected sphere $\{|y| = \rho\sqrt{1 + t^2}\}$, hence it is the whole sphere: f_t is surjective, therefore bijective, on every sphere of A .

Step 4 — polynomial identity for the volume. For $|t| < 1/L$, f_t is thus a diffeomorphism of A onto the shell $A_t = \{(1 - \delta)\sqrt{1 + t^2} \leq |y| \leq (1 + \delta)\sqrt{1 + t^2}\}$ (Step 3, sphere by sphere), with $\det Df_t(x) = \det(I + t Dv(x))$ continuous in (x, t) and non-zero on A for $|t| < 1/L$ (by

local injectivity, and equal to $1 > 0$ for $t = 0$: by continuity it stays positive on the whole connected interval). By the change of variables theorem,

$$\text{Vol}(A_t) = \int_A \det(I + t Dv(x)) dx.$$

The right-hand side is a **polynomial in t of degree ≤ 3** (expansion of the 3×3 determinant of $I + tDv$: $\det(I + tDv) = 1 + t \operatorname{tr}(Dv) + t^2 c_2(Dv) + t^3 \det(Dv)$, and the integral over A commutes with the finite linear combination): $P(t) := \text{Vol}(A_t)$, for $|t| < 1/L$, is a cubic polynomial in t (with coefficients $P(0) = \text{Vol}(A)$ and the integrals of $\operatorname{tr}(Dv)$, $c_2(Dv)$, $\det(Dv)$ over A). On the other hand, by homogeneity of the shell under rescaling,

$$\text{Vol}(A_t) = ((1 + \delta)\sqrt{1 + t^2})^3 \cdot \frac{4}{3}\pi - ((1 - \delta)\sqrt{1 + t^2})^3 \cdot \frac{4}{3}\pi = (1 + t^2)^{3/2} \text{Vol}(A).$$

Step 5 — contradiction. One has thus obtained the identity $(1 + t^2)^{3/2} \text{Vol}(A) = P(t)$ for every t in an open interval $(-1/L, 1/L)$, between a polynomial P of degree ≤ 3 and a function $t \mapsto (1 + t^2)^{3/2} \text{Vol}(A)$ which is **not a polynomial**: its derivatives of even order at $t = 0$ are all non-zero (for instance the fourth derivative of $(1 + t^2)^{3/2}$ at 0 equals $9 \neq 0$, by direct computation), whereas a polynomial of degree ≤ 3 has fourth derivative identically zero. Two analytic functions which agree on an interval have the same derivatives at every point of that interval: a contradiction. Hence there can exist no field v as in the statement. $\square$

Remark 1.13: Imported: Brouwer in every dimension by the same method. Imported from J. Milnor, Analytic Proofs of the “Hairy Ball Theorem” and the Brouwer Fixed Point Theorem, American Mathematical Monthly 85 (1978), 521–524 (verified on 16 September 2026: title and authorship confirmed; pages as in the register of sources, the PDF not being entirely textual at the second attempt at automatic verification). The same method as in Theorem 1.14 — polynomial identity for the volume of a tubular neighbourhood, contradicted by a non-polynomial term — proves the Brouwer theorem in arbitrary dimension n : if $f : \overline{B^n} \rightarrow \overline{B^n}$ continuous (first C^1 , then by approximation) has no fixed points, the retraction $r : \overline{B^n} \rightarrow S^{n-1}$ constructed as in the proof of Theorem 1.13 is the identity on the boundary; setting $f_t(x) = (1 - t)x + tr(x)$ (which fixes S^{n-1} point by point for every t , since $r = \text{id}$ there), the volume $\text{Vol}(f_t(\overline{B^n}))$ is constant (equal to $\text{Vol}(\overline{B^n})$) for small t (a diffeomorphism near the identity fixing the boundary), hence the polynomial $Q(t) = \int_{\overline{B^n}} \det(Df_t) dx$ is constant on an interval, hence constant everywhere; but $Q(1) = \int \det(Dr) dx = 0$ since the image of r is the sphere S^{n-1} , of dimension $n - 1 < n$, so that Dr has rank $< n$ everywhere: a contradiction with $Q(0) = \text{Vol}(\overline{B^n}) > 0$. The chapter does not reproduce the technical details (in particular the regularity of r by composition via the implicit function theorem) with the same level of line-by-line verification used for the case $n = 2$ and $n = 3$ (the latter already covered by the case of the sphere $S^2 = \partial B^3$ seen as a hairy ball in a related but distinct sense): the general statement remains imported.

1.14. Sperner implies Brouwer

Lemma 1.2: Sperner’s lemma (triangle). Let T be a triangle with vertices labelled 1, 2, 3, finely triangulated, and let every vertex of the triangulation be labelled with 1, 2 or 3 in such a way that: the vertices on the side ij of the large triangle carry only the labels i or j (Sperner condition on the boundary; the interior vertices are free). Then there exists at least one small triangle of the triangulation with all three labels 1, 2, 3 (“complete triangle”); indeed their number is odd.

Proof. Count, on every small triangle Δ_k of the triangulation, its sides with labels $\{1, 2\}$ at the endpoints (“good sides”), and sum over all the small triangles: $S = \sum_k (\text{good sides of } \Delta_k)$. A good side interior to the triangulation (shared by two small triangles) is counted twice in S ; a good side on the boundary of T is counted once. On the boundary, good sides can lie only on the side 12 of T (by the Sperner condition, the side 13 has only labels 1, 3 and the side 23 only 2, 3: no good side $\{1, 2\}$ there); on the side 12, the number of good sides (transitions $1 \rightarrow 2$ or $2 \rightarrow 1$ between consecutive vertices) is *odd*, because the sequence of labels along the side begins and ends with different labels (1 at one end, 2 at the other, by the Sperner condition on the vertices of the large triangle) and every change of value along the sequence is a good side: the number of changes between one symbol and another, starting from 1 and arriving at 2, is odd (parity of the number of transitions). Hence the number of good boundary sides is odd, and $S \equiv (\text{good boundary sides}) \pmod{2}$ because the interior sides contribute in pairs: S is odd.

A small triangle has 0, 1 or 2 good sides: 2 if it has labels $\{1, 1, 2\}$ or $\{1, 2, 2\}$ (in some order, with repetition), 1 if it has all three labels $\{1, 2, 3\}$ (complete: a single side $\{1, 2\}$), 0 otherwise (labels $\{1, 1, 1\}$, $\{2, 2, 2\}$, $\{3, 3, 3\}$, $\{1, 1, 3\}$, $\{2, 2, 3\}$, $\{1, 3, 3\}$, $\{2, 3, 3\}$: no side with endpoints exactly $\{1, 2\}$). Hence $S \equiv \#(\text{complete triangles}) \pmod{2}$ (the even contributions do not change the parity): S being odd, the number of complete triangles is odd, in particular ≥ 1 . $\square$

Theorem 1.15: Brouwer (dimension 2) via Sperner. *Lemma 1.2 implies Theorem 1.13.*

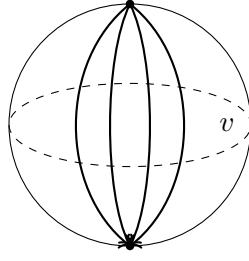
Proof. Identify D^2 with a triangle T with vertices P_1, P_2, P_3 (barycentric coordinates $\lambda_1, \lambda_2, \lambda_3 \geq 0$, $\sum \lambda_i = 1$), and let $f : T \rightarrow T$ be continuous. For a sequence of triangulations $\mathcal{T}_m$ of T with diameter of the small triangles $\rightarrow 0$ (for instance iterated barycentric subdivisions), label every vertex Q of $\mathcal{T}_m$ with an index $\ell(Q) \in \{1, 2, 3\}$ such that the barycentric coordinate $\lambda_{\ell(Q)}$ of $f(Q)$ does not exceed that of Q : that is, $\ell(Q) = i$ if $f(Q)_i \leq Q_i$ for an index i with $Q_i > 0$ (such an i always exists, because $\sum_i (f(Q)_i - Q_i) = 1 - 1 = 0$: not all the differences can be positive). This labelling satisfies the Sperner condition: if Q lies on the side $\lambda_k = 0$, then $f(Q)_k \geq 0 = Q_k$ cannot be the chosen coordinate (one needs $f(Q)_i \leq Q_i$ with $Q_i > 0$), hence $\ell(Q) \neq k$: the label remains among the two indices of the side. By Lemma 1.2, every $\mathcal{T}_m$ has a complete triangle, with vertices Q_m^1, Q_m^2, Q_m^3 (labels 1, 2, 3) of diameter $\rightarrow 0$; by compactness of T one extracts a subsequence with $Q_m^1, Q_m^2, Q_m^3 \rightarrow Q^*$ (the same limit, zero diameter in the limit). For every i , $f(Q_m^i)_i \leq (Q_m^i)_i$; passing to the limit (continuity of f), $f(Q^*)_i \leq Q_i^*$ for $i = 1, 2, 3$; but $\sum_i f(Q^*)_i = \sum_i Q_i^* = 1$, hence three inequalities $\leq$ summing to the same total force equality in each: $f(Q^*)_i = Q_i^*$ for every i , that is, $f(Q^*) = Q^*$. $\square$

Remark 1.14: Two independent proofs of Brouwer. *The chapter now has two independent proofs of Theorem 1.13: the one via non-retraction (Section 1.12, which uses the degree on S^1) and the one via Sperner (this section, purely combinatorial, of level A–B, suitable also for those who have not yet met the degree). The two agree perfectly: it is a cross-check, not merely a didactic curiosity. **Source:** V. V. Prasolov, Elements of Combinatorial and Differential Topology, GSM 74, AMS 2006, Ch. 2 §3 (verified on 16 September 2026 through the AMS front/endmatter file; it supports Lemma 1.2 and the deduction of Brouwer).*

1.15. Worked examples

Four fully worked examples, each with an independent numerical check, closing the theory of the chapter: the degree of an explicit map computed by lifting and, again, by an integral (the same result, two independent methods); a numerical instance of Borsuk–Ulam; and an explicit

zero (north pole)



zero (south pole)

Figure 1.11: An explicit attempt at a tangent field on S^2 , illustrating Theorem 1.14: the “meridian” field (combing from the north pole to the south pole along the meridians, everywhere tangent to the sphere) is not defined at the two poles, where all the meridians converge and the tangent direction is not unique — these are exactly the two zeros that the theorem makes unavoidable. The theorem guarantees that *every* continuous tangent field on S^2 has at least one zero; this classical example has two (one per pole), and Milnor’s proof (Step 4: the identity $(1 + t^2)^{3/2} \text{Vol}(A) = P(t)$ between a non-polynomial function and a polynomial) rules out in general the existence of an everywhere tangent, never vanishing field such as this combing attempt would like to achieve without singularities.

curve used to illustrate the Whitney–Graustein classification.

Example 1.3: Degree of an explicit map, via lifting. Consider $f : S^1 \rightarrow S^1$,

$$f(e^{i\theta}) = e^{2i\theta} \cdot \frac{e^{i\theta} - \frac{1}{2}}{|e^{i\theta} - \frac{1}{2}|}, \quad \theta \in [0, 2\pi].$$

f is well defined and continuous: the denominator never vanishes, because $|e^{i\theta}| = 1 \neq \frac{1}{2}$ for every θ . Compute $\deg f$ by explicitly constructing a lift.

Solution. Strategy. One writes f as a product of two factors with values in S^1 and lifts them separately, adding the angular increments (because $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$: the lift of a product is the sum of the lifts).

Step 1 — first factor. $\theta \mapsto e^{2i\theta}$ has lift $\varphi_1(\theta) = 2\theta$, with $\varphi_1(2\pi) - \varphi_1(0) = 4\pi$.

Step 2 — second factor, via topological Rouché. Let $u(\theta) = e^{i\theta}$ and $v(\theta) = -\frac{1}{2}$ (constant). For every θ , $|v(\theta)| = \frac{1}{2} < 1 = |u(\theta)|$: by Theorem 1.5, the loop $u + v$, that is, $\theta \mapsto e^{i\theta} - \frac{1}{2}$, has the same winding number as u , namely 1 (Volume III, Chapter 12: $\deg p_1 = 1$). Normalisation by the modulus (a positive real function) does not alter the argument, hence does not change the winding number: a lift φ_2 of $\theta \mapsto (e^{i\theta} - \frac{1}{2})/|e^{i\theta} - \frac{1}{2}|$ satisfies $\varphi_2(2\pi) - \varphi_2(0) = 2\pi$.

Step 3 — sum. $\Phi = \varphi_1 + \varphi_2$ is a lift of f (because $f = e^{i\varphi_1}e^{i\varphi_2} = e^{i(\varphi_1+\varphi_2)}$), with $\Phi(2\pi) - \Phi(0) = 4\pi + 2\pi = 6\pi$, hence

$$\deg f = \frac{\Phi(2\pi) - \Phi(0)}{2\pi} = 3.$$

Independent check. Numerical unrolling of the phase of $f(\theta)$ on 2×10^6 equally spaced points of $[0, 2\pi]$: $\deg f = 3.000000$. ✓ (A second check, by a completely different method, is in Example 1.4.)

Interpretation. The factor $e^{2i\theta}$ contributes 2 turns; the normalised factor $(e^{i\theta} - \frac{1}{2})/|\cdot|$ contributes 1, because the point $\frac{1}{2}$ is *inside* the unit circle (the loop $e^{i\theta}$ goes once around it): the sum of the degrees of the factors gives the degree of the product.

Typical errors. Forgetting that the degree of a product of maps $S^1 \rightarrow S^1$ is the *sum* (not the product) of the degrees of the factors.

Extension. If the pole moves to a with $|a| > 1$ (outside the unit circle), the normalised factor has degree 0 instead of 1: see Worked Exercise 1.5.

Example 1.4: The same degree, via the integral formula. Recompute the degree of the map f of Example 1.3 with the integral formula for the winding number (Remark 1.1, Volume III Chapter 12):

$$\deg f = w(f) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f'(\theta)}{f(\theta)} d\theta$$

(here f itself is the loop whose winding about 0 is measured).

Solution. Strategy. Logarithmic derivative of a product: $f'/f = (f'_1/f_1) + (f'_2/f_2)$ with $f_1(\theta) = e^{2i\theta}$ and $f_2(\theta) = (e^{i\theta} - \frac{1}{2})/|e^{i\theta} - \frac{1}{2}|$; the first summand is integrated at once, the second reduces to a known Cauchy integral.

Step 1 — first summand. $f'_1/f_1 = 2i$ (constant), hence $\frac{1}{2\pi i} \int_0^{2\pi} 2i d\theta = \frac{2i \cdot 2\pi}{2\pi i} = 2$.

Step 2 — second summand. Write $\gamma(\theta) = e^{i\theta} - \frac{1}{2} = \rho(\theta)f_2(\theta)$ with $\rho = |\gamma| > 0$ real and periodic. From $\gamma'/\gamma = \rho'/\rho + f'_2/f_2$ and from $\int_0^{2\pi} \rho'/\rho d\theta = [\ln \rho]_0^{2\pi} = 0$ (periodicity of ρ), it follows that $\frac{1}{2\pi i} \int_0^{2\pi} f'_2/f_2 d\theta = \frac{1}{2\pi i} \int_0^{2\pi} \gamma'/\gamma d\theta$. With $z = e^{i\theta}$, $\gamma = z - \frac{1}{2}$, $\gamma' = iz$, this is

$$\frac{1}{2\pi i} \oint_{|z|=1} \frac{dz}{z - \frac{1}{2}} = 1$$

by Cauchy's integral formula (the pole $z = \frac{1}{2}$ is inside $|z| = 1$; argument principle, Volume III Chapter 15, Remark 1.5).

Step 3 — sum. $\deg f = 2 + 1 = 3$: the same value found by lifting in Example 1.3.

Independent check. Direct numerical quadrature of the integral $\frac{1}{2\pi i} \oint dz/(z - \frac{1}{2})$ (trapezoidal rule, 2×10^6 nodes): $1.000000 + 0.000000i$. ✓

Interpretation. Geometric lifting and complex integral are the same number read in two languages; the coincidence here is not an accident, it is the content of Remark 1.1.

Typical errors. Applying the integral formula to the normalised factor f_2 instead of to γ without justifying that the two integrals agree: the justification (Step 2) uses the reality and periodicity of ρ .

Extension. The same technique (additivity of the logarithmic derivative) gives a third proof of $\deg(fg) = \deg f + \deg g$ for arbitrary maps $S^1 \rightarrow S^1$.

Example 1.5: Numerical check of Borsuk–Ulam in dimension 2. Let $f : S^2 \rightarrow \mathbb{R}^2$,

$$f(x, y, z) = (x^3 - 3xy^2 + z + z^2, \quad 3x^2y - y^3 - z + xz^2).$$

f is continuous (polynomial). Theorem 1.7 (BU1) guarantees the existence of $P \in S^2$ with $f(P) = f(-P)$. Illustrate the proof and locate P numerically.

Solution. Strategy. One follows the scheme (BU2) $\Rightarrow$ (BU1) of the proof of Theorem 1.7: one shows that $v = g/|g|$ (with $g(x) = f(x) - f(-x)$), restricted to the equator, has odd

degree, hence $g \neq 0$ everywhere on S^2 is impossible; then one locates P by a numerical search over the whole sphere (the topological method guarantees existence, not position).

Step 1 — restriction to the equator. On the equator $z = 0$, with the identities $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ and $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$:

$$f(\cos \theta, \sin \theta, 0) = (\cos 3\theta, \sin 3\theta) \quad (\text{that is, } e^{3i\theta} \text{ in complex notation}).$$

Step 2 — g on the equator. The antipodal point on the equator corresponds to the angle $\theta + \pi$: $f(-\cos \theta, -\sin \theta, 0) = e^{3i(\theta+\pi)} = -e^{3i\theta}$ (because $e^{3i\pi} = -1$). Hence $g(\theta) = e^{3i\theta} - (-e^{3i\theta}) = 2e^{3i\theta}$, never zero ($|g| \equiv 2$), and $v(\theta) = g/|g| = e^{3i\theta}$ has degree 3 (Volume III, Chapter 12: $\deg p_3 = 3$), *odd*: in particular $\deg v \neq 0$.

Step 3 — why P is not on the equator. On the whole equator $g \neq 0$ (Step 2): the point P guaranteed by Theorem 1.7 cannot lie there. The proof (which would extend $u = g/|g|$ to the hemisphere H^+ , forcing $\deg v = 0$ if g never vanished on the whole of S^2) is *non-constructive*: it guarantees P elsewhere on the sphere, without indicating its position.

Step 4 — numerical localisation. A search over 3×10^6 points sampled uniformly on S^2 , minimising $|g(x, y, z)|^2$: the raw minimum is $|g| \approx 0.003343$ at the approximate point $(0.243459, -0.835110, 0.493273)$. A local refinement (descent in spherical coordinates (ϑ, φ) , Nelder–Mead method, down to $|g|^2 < 10^{-27}$) gives

$$P \approx (0.242858, -0.835034, 0.493698), \quad |P| = 1.000000,$$

with $f(P) \approx f(-P) \approx (0.243738, 0.000000)$, difference $< 10^{-13}$ in each component. ✓ The two methods (random sampling and local optimisation) converge consistently to the same point. ✓

Interpretation. The degree (Step 2) is a *global* invariant, computable in closed form; the position of P is a *local* fact, which here requires a numerical method: it is the non-constructive character typical of proofs by topological degree.

Typical errors. Believing that the theorem indicates *where* P is: the statement guarantees only existence; no step of the proof locates it.

Extension. If one chose f even instead of odd on the equator (for instance by replacing the cubic terms with quadratic terms), the restriction v might have even degree, even zero: in that case the argument of Step 3 would not apply directly, and a different section of the sphere would be needed.

Example 1.6: Whitney–Graustein on an explicit curve. Let $\gamma : \mathbb{R}/2\pi\mathbb{Z} \rightarrow \mathbb{R}^2 \cong \mathbb{C}$,

$$\gamma(t) = e^{2it} + \frac{1}{3}e^{it}.$$

Show that γ is regular, compute its rotation number $\rho(\gamma)$, and place γ in the classification of Theorem 1.11.

Solution. Strategy. One factors γ' so as to isolate a dominant term and applies again the topological Rouché theorem (Theorem 1.5) to the winding number of γ' about 0, which by definition is $\rho(\gamma)$.

Step 1 — regularity. $\gamma'(t) = 2ie^{2it} + \frac{i}{3}e^{it} = ie^{it}(2e^{it} + \frac{1}{3})$. By the reverse triangle inequality, $|2e^{it} + \frac{1}{3}| \geq 2 - \frac{1}{3} = \frac{5}{3} > 0$ for every t : γ' never vanishes, γ is regular.

Step 2 — rotation number. $\rho(\gamma)$ is the winding number of γ' about 0. From $\gamma' = ie^{it} \cdot (2e^{it} + \frac{1}{3})$: the factor ie^{it} has winding 1; for the second factor, with $u(t) = 2e^{it}$ (winding 1, modulus 2) and $v(t) = \frac{1}{3}$ (constant, modulus $\frac{1}{3} < 2$), Theorem 1.5 gives $w(u+v) = w(u) = 1$.

Summing (the same additivity as in Example 1.3):

$$\rho(\gamma) = 1 + 1 = 2.$$

Step 3 — independent check. Numerical unrolling of the phase of $\tau = \gamma'/|\gamma'|$ on 2×10^6 points of $[0, 2\pi]$: $\rho(\gamma) = 2.000000$, and $\min_t |\gamma'(t)| = 1.666667 = \frac{5}{3}$, in exact agreement with the estimate of Step 1. ✓

Step 4 — placement in the classification. By the Umlaufsatz (Theorem 1.10), a simple closed curve has $\rho = \pm 1$: since $\rho(\gamma) = 2$, γ is **not simple** (it has at least one self-intersection), deduced by contraposition, without any need to locate it. By Theorem 1.11, γ is regularly homotopic to *any* other regular curve with $\rho = 2$, in particular (Remark 1.10) to the unit circle traversed twice, $t \mapsto e^{2it}$.

Interpretation. The rotation number 2 is, by Whitney–Graustein, a *complete* invariant of the regular homotopy class of γ : one does not need to know the exact shape of the curve in order to know which “canonical” curve it is regularly homotopic to.

Typical errors. Applying the Umlaufsatz to γ in order to conclude $\rho = \pm 1$: the Umlaufsatz requires the curve to be *simple*, a hypothesis excluded here by Step 4.

Extension. Replacing the coefficient $\frac{1}{3}$ with a constant c such that $|c| > 2$, the roles of u, v in Step 2 are exchanged: with $u(t) = c$ and $v(t) = 2e^{it}$ (modulus $2 < |c|$), Theorem 1.5 gives $w(u+v) = w(u) = 0$, hence $\rho(\gamma) = 1 + 0 = 1$. The curve remains regular for every $|c| \neq 2$ (for $|c| = 2$ the estimate of Step 1 degenerates exactly to 0).

1.16. Assigned exercises

The answers are in the Teacher’s key (Section 1.19).

Exercise 1.1: The group of deck transformations of p_6 . *Goal:* to apply the classification of the connected coverings of S^1 (Theorem 1.2) in order to describe explicitly the group of deck transformations of $p_n : S^1 \rightarrow S^1$, $p_n(z) = z^n$, in the case $n = 6$. *Estimated time:* 35 minutes. *Type:* symbolic.

- List the fibre $p_6^{-1}(1)$ (the six sixth roots of unity).
- A deck transformation is a homeomorphism $\Phi : S^1 \rightarrow S^1$ with $p_6 \circ \Phi = p_6$. Check that every rotation $R_k(z) = e^{2\pi ik/6}z$, $k = 0, \dots, 5$, is a deck transformation of p_6 .
- Show that there are no others: if Φ is a deck transformation of p_6 with $\Phi(1) = e^{2\pi ik/6}$, then $\Phi = R_k$ (hint: Φ and R_k are both solutions of $p_6 \circ X = p_6$ with the same value at 1: apply the uniqueness of the lift, Lemma 1.1, to the covering p_6 itself).
- Deduce that the group of deck transformations of p_6 is isomorphic to $\mathbb{Z}/6\mathbb{Z}$.
- Compute $e^{2\pi ik/6}$ for $k = 1, \dots, 6$ to six decimal places, and check numerically that only $k = 6$ gives $R_k = \text{id}$.

Exercise 1.2: A necklace of 14 beads. *Goal:* to apply concretely Theorem 1.9 (stolen necklace) to a necklace of a size different from that of the example in the text (Figure 1.5, $N = 10$). *Estimated time:* 30 minutes. *Type:* combinatorial, with a check by enumeration. An open necklace has $N = 14$ beads in the order

B, W, B, W, B, W, B, B, W, B, W, B, W, B (positions 1, ..., 14),

with B (black, type 1) and W (white, type 2).

- Count the beads of each type and check that they are even in number ($2n_1$ and $2n_2$).
- Theorem 1.9 guarantees that at most two cuts suffice. Find an explicit pair of them assigning to each thief exactly n_1 black beads and n_2 white beads.

- (c) Check the solution found by explicitly counting the beads of each type in every block assigned to each thief.
- (d) Say whether the solution of point (b) is unique (hint: look for another one).

Exercise 1.3: Locating the roots of another polynomial. *Goal:* to apply the explicit estimate of Theorem 1.12 to a polynomial different from the one of Example 1.2, and to compare it with the true roots. *Estimated time:* 30 minutes. *Type:* numerical, with an independent check.

Let $p(z) = z^4 + 2z^2 - z + 3$.

- (a) Compute $R = 1 + \sum_{i=0}^3 |a_i|$ given by Theorem 1.12.
- (b) Check directly $\sum_{i=0}^3 |a_i| R^i < R^4$, the inequality used in the proof.
- (c) Compute (numerically) the four roots of p and their moduli.
- (d) Compare the moduli found with R and comment on how conservative the estimate is.

1.17. Worked exercises

Exercise 1.4: The lemniscate: why the degree of the tangent field is 0. *Type:* symbolic, with a numerical check. Consider the closed curve obtained by gluing two circles of radius 1, tangent to the y axis at the origin:

$$\gamma(t) = \begin{cases} (1 - \cos 2t, -\sin 2t), & t \in [0, \pi] \quad (\text{circle of centre } (1, 0)), \\ (-1 + \cos 2(t - \pi), -\sin 2(t - \pi)), & t \in [\pi, 2\pi] \quad (\text{circle of centre } (-1, 0)), \end{cases}$$

a lemniscate (figure-eight curve), regular but not simple, as in Remark 1.9. Compute the rotation number $\rho(\gamma)$ and the contribution of each “lobe”.

Solution. Strategy. Check that γ is C^1 and regular also at the junctions $t = \pi$ and $t = 0 \equiv 2\pi$; then compute the rotation number of each arc as a loop in its own right (each is a whole circle, simple) and sum.

Step 1 — velocity. For $t \in [0, \pi]$: $\gamma'(t) = (2 \sin 2t, -2 \cos 2t)$, $|\gamma'(t)| \equiv 2$. For $t \in [\pi, 2\pi]$, setting $s = t - \pi$: $\gamma'(t) = (-2 \sin 2s, -2 \cos 2s)$, again $|\gamma'| \equiv 2$: γ is regular, with constant speed 2 over the whole period.

Step 2 — C^1 junction. At $t = \pi$: from the first branch $\gamma'(\pi) = (2 \sin 2\pi, -2 \cos 2\pi) = (0, -2)$; from the second branch ($s = 0$) $\gamma'(\pi) = (-2 \sin 0, -2 \cos 0) = (0, -2)$: they agree. At $t = 0 \equiv 2\pi$: $\gamma'(0) = (0, -2)$ (first branch) and, from the second branch with $s = \pi$, $\gamma'(2\pi) = (-2 \sin 2\pi, -2 \cos 2\pi) = (0, -2)$: they agree. γ is C^1 on the whole of $\mathbb{R}/2\pi\mathbb{Z}$.

Step 3 — first lobe. For $t \in [0, \pi]$, $\gamma(t) = (1, 0) + (-\cos 2t, -\sin 2t)$: circle of centre $(1, 0)$ and radius 1, with angle $\phi(t) = \pi + 2t$ increasing from π to 3π (one complete turn, anticlockwise). Contribution of the first lobe to the rotation number: $+1$.

Step 4 — second lobe. For $t \in [\pi, 2\pi]$, $s = t - \pi$, $\gamma(t) = (-1, 0) + (\cos 2s, -\sin 2s)$: circle of centre $(-1, 0)$ and radius 1, with angle $\psi(s) = -2s$ decreasing from 0 to -2π (one complete turn, clockwise). Contribution: -1 .

Step 5 — total. Since γ is closed and C^1 (Step 2), the lift of the tangent angle is continuous on $[0, 2\pi]$ and its total increment is the sum of the increments over the two arcs:

$$\rho(\gamma) = (+1) + (-1) = 0.$$

Independent check. Numerical unrolling of the phase of $\tau = \gamma'/|\gamma'|$ on 2×10^6 points: total increment $\approx 3.5 \times 10^{-17}$ turns (zero to machine precision); on the two subintervals

separately, $+1.000000$ and -1.000000 . ✓

Interpretation. $\gamma(0) = \gamma(\pi) = \gamma(2\pi) = (0, 0)$: the curve passes twice through the origin, it is not injective, hence not simple; the Umlaufsatz (which would require $\rho = \pm 1$) does not apply, and indeed $\rho = 0$, in agreement with Remark 1.9: the two lobes, traversed in opposite directions, contribute $+1$ and -1 which cancel exactly.

Typical errors. Applying the Umlaufsatz to the whole lemniscate (not simple) instead of to each lobe separately (which is simple).

Extension. Repeating the same construction with other pairs of lobes tangent in the same way, the contribution remains 0 by cancellation in pairs, whatever the number of pairs of lobes.

Exercise 1.5: Degree with the pole outside the unit circle. *Type:* symbolic, with a numerical check. Compute the degree of

$$h(e^{i\theta}) = e^{3i\theta} \cdot \frac{e^{i\theta} + 2}{|e^{i\theta} + 2|}, \quad \theta \in [0, 2\pi],$$

and compare it with the degree of the map of Example 1.3, where the pole was *inside* the unit circle.

Solution. Strategy. The same additive factorisation of the degree; here the constant term 2 has modulus greater than 1: in the topological Rouché theorem the roles of u and v are exchanged with respect to Example 1.3.

Step 1 — well-definedness. $|e^{i\theta} + 2| \geq 2 - 1 = 1 > 0$: never zero, h continuous $S^1 \rightarrow S^1$.

Step 2 — factor $e^{3i\theta}$. Degree 3 (Volume III, Chapter 12: $\deg p_3 = 3$).

Step 3 — normalised factor. Set $u(\theta) = 2$ (constant) and $v(\theta) = e^{i\theta}$: $|v(\theta)| = 1 < 2 = |u(\theta)|$ for every θ , hence by Theorem 1.5 $w(u + v) = w(u)$. The constant loop $u \equiv 2$ has degree 0: $w(e^{i\theta} + 2) = 0$.

Step 4 — sum. $\deg h = 3 + 0 = 3$.

Independent check. Numerical unrolling of the phase: $\deg h = 3.000000$. ✓

Interpretation. The value 3 numerically coincides with that of Example 1.3, but for the opposite reason: there the linear factor contributed $+1$ (pole inside), here it contributes 0 (pole outside), and it is the exponent 3 of $e^{3i\theta}$ that does all the work.

Typical errors. Applying the topological Rouché theorem with u, v exchanged (that is, assuming $|v| < |u|$ without checking it).

Extension. For $|a| = 1$ exactly (pole on the unit circle), $e^{i\theta} + a$ vanishes for some θ : the normalised map is not even defined, and the degree loses meaning without a finer analysis.

Exercise 1.6: Sum of the exterior angles of an irregular pentagon. *Type:* numerical, with an independent geometric check. Consider the simple convex pentagon with vertices (in anticlockwise order)

$$P_1 = (0, 0), \quad P_2 = (4, 0), \quad P_3 = (5, 3), \quad P_4 = (2, 5), \quad P_5 = (-1, 2).$$

Compute the signed exterior angles and check Corollary 1.1.

Solution. Strategy. The exterior angle at the vertex P_i is the angle (signed, anticlockwise positive) between the incoming side $P_{i-1}P_i$ and the outgoing side P_iP_{i+1} : it is computed as the difference of the arguments of the two side vectors.

Step 1 — side vectors. $P_2 - P_1 = (4, 0)$, $P_3 - P_2 = (1, 3)$, $P_4 - P_3 = (-3, 2)$, $P_5 - P_4 = (-3, -3)$, $P_1 - P_5 = (1, -2)$.

Step 2 — exterior angles. $\alpha_i = \arg(\text{outgoing side}) - \arg(\text{incoming side})$, reduced to $(-180^\circ, 180^\circ]$:

$$\alpha_1 = 63.435^\circ, \quad \alpha_2 = 71.565^\circ, \quad \alpha_3 = 74.745^\circ, \quad \alpha_4 = 78.690^\circ, \quad \alpha_5 = 71.565^\circ.$$

Step 3 — sum. $63.435 + 71.565 + 74.745 + 78.690 + 71.565 = 360.000^\circ = 2\pi \text{ rad.}$ ✓

Independent check. General formula: the sum of the *interior* angles of a simple convex polygon with n sides is $(n - 2) \cdot 180^\circ$, here 540° for $n = 5$; each exterior angle is 180° minus the corresponding interior one, hence the sum of the exterior ones is $5 \cdot 180^\circ - 540^\circ = 360^\circ$, in agreement with Step 3. ✓

Interpretation. This is the polygonal case of Corollary 1.1: the total curvature (here concentrated at the vertices) equals 2π for any simple polygon traversed once anticlockwise, independently of its shape.

Typical errors. Summing the *interior* angles instead of the *exterior* ones (the sum of the interior ones depends on n ; that of the exterior ones does not).

Extension. For a *non-convex* simple polygon (with negative exterior angles at some vertices) the sum remains 2π : this is the full content of Corollary 1.1, not restricted to the convex case.

1.18. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher's key (Section 1.19).

- (Definition) A map $u : S^n \rightarrow S^m$ (Definition 1.1) is called *odd* if
 - $u(x) = u(-x)$ for every x
 - $u(-x) = -u(x)$ for every x
 - $u(x) = -x$ for every x
 - u is injective and $u(-x) \neq u(x)$
- (Hypothesis) In the topological Rouché theorem (Theorem 1.5), the exact hypothesis on the two loops u, v is
 - $|v(t)| \leq |u(t)|$ for every t
 - $|v(t)| < |u(t)|$ for every t
 - u, v holomorphic
 - $u(t) \neq 0$ and $v(t) \neq 0$ for every t
- (Symbolic computation) The degree of $f(e^{i\theta}) = e^{2i\theta}(e^{i\theta} - \frac{1}{2})/|e^{i\theta} - \frac{1}{2}|$ (Example 1.3) equals
 - 1
 - 2
 - 3
 - 4
- (Numerical computation) For $p(z) = z^4 + 2z^2 - z + 3$ (Exercise 1.3), the value R given by Theorem 1.12 equals
 - 5
 - 6
 - 7
 - 8
- (Proof step) In the proof (BU2) $\Rightarrow$ (BU1) of Theorem 1.7, the degree of $v = u|_E$ is 0 because
 - E is a great circle
 - v extends continuously to the closed hemisphere $H^+ \cong D^2$
 - u is odd
 - f is continuous
- (Error diagnosis) A student applies the Umlaufsatz (Theorem 1.10) to the lemniscate of Exercise 1.4 and concludes $\rho = \pm 1$. The error is
 - the Umlaufsatz holds only for polygons
 - the lemniscate is not simple: the hypothesis of the Umlaufsatz is not satisfied
 - the lemniscate is not regular
 - ρ is not defined for non-convex curves

7. (Transfer) The general case of the stolen necklace problem with k types of beads (Remark 1.8) requires at most
- A) 2 cuts, always B) k cuts C) k^2 cuts D) $2k$ cuts
8. (Source and metacognition) The Whitney–Graustein theorem (Theorem 1.11) was first published in
- A) H. Hopf, *Über die Drehung der Tangenten und Sehnen ebener Kurven*, Compositio Math. 2 (1935) B) H. Whitney, *On regular closed curves in the plane*, Compositio Math. 4 (1937) C) J. Milnor, *Analytic proofs of the Hairy Ball Theorem...*, Amer. Math. Monthly 85 (1978) D) K. Borsuk, *Drei Sätze über die n -dimensionale euklidische Sphäre*, Fund. Math. 20 (1933)

1.19. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnosis.

- 1) **B.** *Rationale:* this is exactly Definition 1.1. *Distractors:* A describes an *even* map; C is a very particular case (the antipodal map), not the general definition; D confuses injectivity with oddness. *Diagnosed error:* exchanging $u(-x) = -u(x)$ with $u(-x) = u(x)$. *Source:* Definition 1.1. *Prerequisite:* the notion of even/odd function. *Remedial work:* check by direct substitution the oddness of $u(z) = z^3$ on S^1 .
- 2) **B.** *Rationale:* the hypothesis is the *strict* inequality $|v(t)| < |u(t)|$, used in the proof to guarantee $|H(s, t)| > 0$ for every $s \in [0, 1]$. *Distractors:* A (non-strict inequality) does not suffice, $u + v$ could vanish; C is extraneous (the theorem is purely topological, it does not require holomorphy); D is implied by B but on its own insufficient. *Diagnosed error:* confusing the topological version of Rouché with the analytic one. *Source:* Theorem 1.5. *Prerequisite:* triangle inequality. *Remedial work:* reread the proof, isolating where the strict inequality is used.
- 3) **C.** *Rationale:* Example 1.3, Step 3: 2 (from $e^{2i\theta}$) plus 1 (from the factor with the pole inside) makes 3. *Distractors:* A forgets the factor $e^{2i\theta}$; B and D arise from errors in the sum of the degrees of the two factors. *Diagnosed error:* forgetting that the degree of a product is the *sum* of the degrees of the factors. *Source:* Example 1.3 and Example 1.4 (cross-check). *Prerequisite:* topological Rouché theorem. *Remedial work:* redo the computation separating the two factors.
- 4) **C.** *Rationale:* $R = 1 + (|a_2| + |a_1| + |a_0|) = 1 + (2 + 1 + 3) = 7$. *Distractors:* A and B arise from errors in the sum of the moduli; D also adds an absent term ($a_3 = 0$, mistakenly counted as 1). *Diagnosed error:* forgetting that the coefficient of z^3 is absent ($a_3 = 0$), not 1. *Source:* Theorem 1.12. *Prerequisite:* reading the coefficients of a polynomial. *Remedial work:* rewrite p making all the coefficients explicit, including the zero ones.
- 5) **B.** *Rationale:* it is the crucial point of (BU2) $\Rightarrow$ (BU1): v , restricted to the hemisphere, extends continuously to $u|_{H^+}$, hence its degree (a map on the boundary of a disc that extends continuously to the interior) is 0 (**Volume III, Chapter 12**). *Distractors:* A is true but irrelevant to the step; C is the hypothesis on u on the *whole* sphere, not the technical reason why $\deg v = 0$; D is generic. *Diagnosed error:* failing to distinguish the structural hypothesis (oddness of u) from the specific technical step (extension to the disc). *Source:* Theorem 1.7, proof. *Prerequisite:* “ $\deg f = 0$ iff f extends continuously to the disc” (Volume III, Ch. 12). *Remedial work:* reread the step isolating where the extension to H^+ intervenes.
- 6) **B.** *Rationale:* $\gamma(0) = \gamma(\pi) = \gamma(2\pi) = (0, 0)$ (Exercise 1.4, Step 2): the curve is not injective, hence not simple, and the Umlaufsatz does not apply. *Distractors:* A is

false (the Umlaufsatz holds for arbitrary C^1 curves, see the smooth case in the proof of Corollary 1.1); C is false (the curve is regular, with constant speed 2); D is false (convexity is not a hypothesis of the Umlaufsatz). *Diagnosed error*: forgetting the simplicity hypothesis. *Source*: Theorem 1.10; Remark 1.9. *Prerequisite*: the notion of simple curve (injective on $[0, 1)$). *Remedial work*: redo the computation of Step 2 of the exercise, identifying the double point.

7) B. Rationale: Remark 1.8: the general case (Alon 1987, Hobby–Rice 1965) requires at most k cuts for k types of beads, via Borsuk–Ulam on S^k ; the case $k = 2$ proved here does indeed use 2 cuts. *Distractors*: A generalises the case $k = 2$ incorrectly; C and D overestimate the number of cuts needed. *Diagnosed error*: confusing the particular case proved ($k = 2$) with the general statement cited with a source. *Source*: Remark 1.8. *Prerequisite*: reading a statement “with a generic parameter”. *Remedial work*: compare $k = 2$ (proved) with generic k (cited).

8) B. Rationale: Remark 1.10: the primary source is H. Whitney, *On regular closed curves in the plane*, Compositio Mathematica 4 (1937), 276–284. *Distractors*: A is the source of the Umlaufsatz (Hopf); C is the source of the hairy ball (Milnor); D is the source of Borsuk–Ulam (Borsuk). *Diagnosed error*: confusing with one another the primary sources of the different “external” theorems of the chapter. *Source*: Remark 1.10; fonti block at the end of the chapter, entry Whitney (1937). *Prerequisite*: nothing beyond the careful reading of the sources cited in the chapter. *Remedial work*: review the table of primary sources of the chapter (Umlaufsatz–Hopf, Whitney–Graustein–Whitney, Borsuk–Ulam–Borsuk, hairy ball–Milnor).

BEST PRACTICE

Assigned exercises — answers.

1.1 (deck transformations of p_6). (a) $p_6^{-1}(1) = \{e^{2\pi ik/6} : k = 0, \dots, 5\} = \{1, e^{i\pi/3}, e^{2i\pi/3}, -1, e^{4i\pi/3}, e^{5i\pi/3}\}$. (b) $p_6(R_k(z)) = (e^{2\pi ik/6}z)^6 = e^{2\pi ik}z^6 = z^6 = p_6(z)$ (because $e^{2\pi ik} = 1$, k an integer): every R_k is a deck transformation. (c) Φ and R_k are both solutions of $p_6 \circ X = p_6$ with the same value at a point of the fibre $p_6^{-1}(1)$: by the uniqueness of the lift (Lemma 1.1, connected domain S^1), $\Phi = R_k$ everywhere. (d) The map $k \mapsto R_k$ is a surjective homomorphism $\mathbb{Z} \rightarrow \{R_0, \dots, R_5\}$ with kernel $6\mathbb{Z}$ (because $R_k = \text{id}$ iff $e^{2\pi ik/6} = 1$ iff $6 \mid k$): by the first isomorphism theorem, the group of deck transformations is $\mathbb{Z}/6\mathbb{Z} \cong \{R_0, \dots, R_5\}$. (e) $e^{2\pi i \cdot 1/6} = 0.500000 + 0.866025i$; $e^{2\pi i \cdot 2/6} = -0.500000 + 0.866025i$; $e^{2\pi i \cdot 3/6} = -1.000000 + 0.000000i$; $e^{2\pi i \cdot 4/6} = -0.500000 - 0.866025i$; $e^{2\pi i \cdot 5/6} = 0.500000 - 0.866025i$; $e^{2\pi i \cdot 6/6} = 1.000000 + 0.000000i$: only $k = 6$ gives 1. ✓

1.2 (necklace of 14 beads). (a) B (type 1): positions 1, 3, 5, 7, 8, 10, 12, 14, that is, $8 = 2n_1$ beads, $n_1 = 4$; W (type 2): positions 2, 4, 6, 9, 11, 13, that is, $6 = 2n_2$ beads, $n_2 = 3$. (b) Cuts between positions 3–4 and 10–11: central block (positions 4–10) to one thief, the two outer blocks taken together (positions 1–3 and 11–14) to the other. (c) Central block (4–10: W, B, W, B, B, W, B): black $\{5, 7, 8, 10\} = 4 = n_1$, white $\{4, 6, 9\} = 3 = n_2$. Outer block (1–3, 11–14: B, W, B, W, B, W, B): black $\{1, 3, 12, 14\} = 4 = n_1$, white $\{2, 11, 13\} = 3 = n_2$. ✓ (d) No: there are other valid pairs of cuts, for instance between positions 1–2 and 8–9 (same count: the central block 2–8 gives 4 black and 3 white, and likewise the outer block).

1.3 (roots of $p(z) = z^4 + 2z^2 - z + 3$). (a) $a_3 = 0, a_2 = 2, a_1 = -1, a_0 = 3$: $R = 1 + (0 + 2 + 1 + 3) = 7$. (b) On $|z| = 7$: $\sum_{i=0}^3 |a_i| 7^i = 3 + 1 \cdot 7 + 2 \cdot 49 + 0 \cdot 343 = 3 + 7 + 98 = 108 < 7^4 = 2401$. ✓ (c) The four roots are $\approx -0.623633 \pm 1.337832i$ and $\approx 0.623633 \pm 0.994003i$ (computed numerically, checked by substitution: $|p(\text{root})| < 10^{-10}$). (d) The moduli are ≈ 1.476046 (the first two) and ≈ 1.173439 (the other two), both amply $< 7 = R$: as in Example 1.2, the estimate is very conservative, because the triangle inequality used in the proof does not exploit the cancellations between the terms.

1.20. Chapter formulary

Degree and winding number

$$\deg f = w(f \circ E) \quad (f : S^1 \rightarrow S^1), \quad w(\gamma) = \frac{\tilde{\gamma}(1) - \tilde{\gamma}(0)}{2\pi} = \frac{1}{2\pi i} \int_0^1 \frac{\gamma'(t)}{\gamma(t)} dt$$

$$\deg(f \circ g) = \deg f \cdot \deg g, \quad \deg(fg) = \deg f + \deg g \quad (\text{pointwise product in } S^1), \quad \deg p_n = n$$

Topological Rouché theorem (Theorem 1.5)

$$|v(t)| < |u(t)| \quad \forall t \implies w(u+v) = w(u)$$

Odd maps and Borsuk–Ulam (Theorems 1.6, 1.7)

$$u(-x) = -u(x) \implies \deg u \text{ odd} \quad (\text{BU1}) \iff (\text{BU2}) \iff (\text{BU3})$$

Umlaufsatz and total curvature (Theorem 1.10, Corollary 1.1)

$$\gamma \text{ simple, closed, regular} \implies \rho(\gamma) = \pm 1, \quad \oint \kappa ds = 2\pi, \quad \sum_i \alpha_i^{\text{ext}} = 2\pi \quad (\text{simple polygon})$$

Whitney–Graustein (Theorem 1.11)

$$\rho(\gamma) := w(\gamma'/|\gamma'|) \in \mathbb{Z}, \quad \gamma_0 \sim_{\text{reg}} \gamma_1 \iff \rho(\gamma_0) = \rho(\gamma_1)$$

Fundamental theorem of algebra, explicit estimate (Theorem 1.12)

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0, \quad R = 1 + \sum_{i=0}^{n-1} |a_i| \implies \text{every root has } |z| < R$$

Hairy ball, Milnor’s identity (Theorem 1.14)

$$\text{Vol}(A_t) = (1+t^2)^{3/2} \text{Vol}(A) \quad (\text{not a polynomial}) \quad \text{vs.} \quad P(t) = \text{Vol}(A_t) \text{ a polynomial of degree } \leq 3$$

Sperner’s lemma (Lemma 1.2)

$$\#(\text{complete triangles}) \text{ is odd, in particular } \geq 1$$

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CHAPTER 2

Bundles over the circle and over the sphere

INFO

Level and prerequisites. Level-C core (Sections 2.1–2.6 and 2.17); Sections 2.8–2.15 rise to level D (degree founded on Sard, connections, Chern class), with a level-E paragraph isolated in the proof of Theorem 2.9. Prerequisites: the whole of Chapter 1 of this volume (degree of maps $S^1 \rightarrow S^1$, $\pi_1(S^1) \cong \mathbb{Z}$, Umlaufsatz), the notions of regular curve and regular surface and the second fundamental form (Volume II, Chapter 1; Volume III, Chapter 12 for the covering and Chapter 28 for $SO(3)$ and the quaternions). A level-B–C reader may stop after Section 2.6 and read only the statements (not the proofs) of Sections 2.8 and 2.9, skipping Sections 2.11–2.13 except for the worked examples.

Chapter 1 exhausted what can be said about S^1 while remaining inside S^1 : coverings, degree, fundamental group. This chapter takes a step sideways — and one step further: no longer maps $S^1 \rightarrow S^1$, but families of vector spaces (or of circles) parametrised continuously by a base point running along S^1 or S^2 . A bundle is precisely this: a space that locally “looks like” a product, but which may twist globally. The Möbius band is the oldest example; the Hopf fibration $S^3 \rightarrow S^2$, introduced in Volume III, is its richest example, and here it closes the circle (literally) between the topology of the circle, the differential geometry of the sphere and physics (Dirac monopole, Berry phase).

2.1. The circle as an algebraic variety and the pencil of conics

Definition 2.1: Affine algebraic variety. A (real) **affine algebraic variety** is the set of zeros of a family of polynomials. The circle $C(O, r) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 - r^2 = 0\}$ is the algebraic variety defined by the single polynomial $f(x, y) = x^2 + y^2 - r^2$, of degree 2.

Remark 2.1: Algebraic smoothness of the circle. The circle is a smooth (non-singular) algebraic variety: the gradient $\nabla f = (2x, 2y)$ never vanishes on $C(O, r)$ for $r > 0$ (it would vanish only at $(0, 0)$, which does not belong to the variety). This is the algebraic reason why the circle admits a well-defined tangent line at each of its points (Volume I, Chapter 3): the tangent is precisely the line $\{\nabla f(P) \cdot (Q - P) = 0\}$.

Definition 2.2: Pencil of conics. Given two distinct conics $f_1 = 0$, $f_2 = 0$ in the real projective plane $\mathbb{RP}^2$ (homogeneous polynomials of degree 2 in $(x : y : w)$, not proportional), their **pencil** is the family with one projective parameter $[\lambda : \mu] \in \mathbb{RP}^1$

$$f_{[\lambda:\mu]} = \lambda f_1 + \mu f_2 = 0.$$

It is a **projective line** in the projective space $\mathbb{RP}^5$ of conics (the coefficients of f are defined up to scale, and there are 6 of them, hence the conics form $\mathbb{RP}^5$: a linear combination of

two distinct points is the line joining them). The **base points** of the pencil are the points common to all the conics of the family, that is, the solutions of $f_1 = f_2 = 0$: by Bézout's theorem they are (with multiplicity, counting also the complex points and those at infinity) exactly 4, and every conic of the pencil passes through them.

Theorem 2.1: *The circle is a member of a pencil of conics with two base points at infinity.* Let $f(x, y) = x^2 + y^2 - r^2 w^2$ be the homogenisation (in projective coordinates $(x : y : w)$, with $w = 0$ the line at infinity) of the equation of the circle of radius r . For every $r > 0$, $f = 0$ passes through the same two complex conjugate points at infinity

$$I = [1 : i : 0], \quad J = [1 : -i : 0]$$

(the **circular points** of the plane). Consequently, given two distinct circles $f_1 = 0$, $f_2 = 0$ with centres p, q , their pencil $\lambda f_1 + \mu f_2 = 0$ has as base points I, J and (if they exist in $\mathbb{R}^2$) the real points of intersection of the two circles; every member of the pencil that is real, smooth and bounded is again a circle.

Proof. Setting $w = 0$ in f one obtains $x^2 + y^2 = 0$, that is $x = \pm iy$: the projective solutions are exactly $[1 : i : 0]$ and $[1 : -i : 0]$, independently of r (which no longer appears once $w = 0$ has been set). Given two circles $f_1 = x^2 + y^2 - 2a_1x - 2b_1y + c_1 = 0$ and $f_2 = x^2 + y^2 - 2a_2x - 2b_2y + c_2 = 0$ (general form, $w = 1$ in the affine chart), every combination $\lambda f_1 + \mu f_2$ with $\lambda + \mu \neq 0$ still has coefficient $\lambda + \mu$ for x^2 and for y^2 and zero coefficient for xy : normalising by $\lambda + \mu$ one obtains again an equation of the form $x^2 + y^2 - 2ax - 2by + c = 0$, that is a circle (real if the centre (a, b) and the radius $\sqrt{a^2 + b^2 - c}$ are real; the case $\lambda + \mu = 0$, if non-degenerate, is the unique line of the pencil, the radical axis). Since I, J are base points common to f_1, f_2 by the first part, every linear combination still contains them: the pencil of two circles is exactly the pencil of conics through I, J and through the (at most two) real points common to f_1, f_2 — the pencil of circles already introduced in Volume II, Chapter 8, here reread as a particular case of the projective theory of pencils of conics. $\square$

Remark 2.2: *Why the old formulation was misleading.* It makes no sense to speak of “the circle as a degenerate conic in the limit” of a family with two parameters such as $\lambda x^2 + \mu y^2 = 1$: that is a two-dimensional family of conics (not a projective pencil, which has a single parameter), and in the limit $\mu \rightarrow 0^+$ it does not degenerate at all into the circle but into a pair of parallel lines; the set $\{\lambda(x^2 + y^2) = 1 : \lambda > 0\}$ of circles centred at the origin is not an isolated “fixed point” but the whole half-line of parameters with $\lambda = \mu$, that is, a sub-pencil in its own right. The correct and useful mathematical fact — the one that makes the circle a distinguished object among the conics — is that of Theorem 2.1: among all real conics, the circles are exactly those that pass through the two circular points I, J at infinity, a clean projective condition free of the ambiguities of the old formulation.

2.2. Bundles: definition, cocycle, first examples

Definition 2.3: Fibre bundle. A **fibre bundle** is a quadruple (E, B, π, F) with E (total space), B (base), F (fibre) topological spaces and $\pi : E \rightarrow B$ continuous and surjective (**projection**), such that every $b \in B$ has an open neighbourhood $U \ni b$ and a homeomorphism (**local trivialisation**) $\Phi : \pi^{-1}(U) \rightarrow U \times F$ which commutes with the projection:

$$\text{pr}_U \circ \Phi = \pi|_{\pi^{-1}(U)},$$

where $\text{pr}_U : U \times F \rightarrow U$ is the projection onto the first factor. The fibre over b is $E_b := \pi^{-1}(b)$; the commutation condition says exactly that Φ sends E_b into $\{b\} \times F$, that is, that Φ is a homeomorphism *fibre by fibre*, and not a trivialisation “that mixes the fibres”.

Remark 2.3: Why commutation is the substantial part of the definition. Without the condition $\text{pr}_U \circ \Phi = \pi$, any homeomorphism $\pi^{-1}(U) \cong U \times F$ (which exists for a great many non-fibred spaces, for example by gluing the fibres arbitrarily) would satisfy the definition, making it vacuous. It is precisely commutation that guarantees that Φ identifies the fibre E_b with $\{b\} \times F$ and not with a different fibre: it is this condition, applied to two overlapping trivialisations, that produces the transition cocycle of Definition 2.4.

Remark 2.4: The covering $\mathbb{R} \rightarrow S^1$ as a bundle with discrete fibre. The covering of Chapter 1 is a particular case of a bundle, with fibre $F = \mathbb{Z}$ (discrete): locally, $\pi^{-1}(U) \cong U \times \mathbb{Z}$, the countable disjoint union of copies of U seen in the proof of Theorem 1.2. More generally, the n -sheeted covering $p_n(z) = z^n$ is a bundle with discrete fibre $\mathbb{Z}/n\mathbb{Z}$.

Definition 2.4: Vector bundle, principal bundle, transition cocycle. A (real) **vector bundle of rank n** is a bundle $(E, B, \pi, \mathbb{R}^n)$ in which every fibre E_b has the structure of an n -dimensional real vector space and the local trivialisations are linear fibre by fibre. If $\{U_i\}$ is a trivialising cover with trivialisations Φ_i , the **transition functions**

$$g_{ij} : U_i \cap U_j \rightarrow GL_n(\mathbb{R}), \quad g_{ij}(b) = \Phi_i \circ \Phi_j^{-1}|_{\{b\} \times \mathbb{R}^n}$$

satisfy the **cocycle condition**

$$g_{ii} = I, \quad g_{ij} g_{ji} = I, \quad g_{ij}(b) g_{jk}(b) = g_{ik}(b) \quad \text{for } b \in U_i \cap U_j \cap U_k,$$

the last of these being necessary in order that the passage $i \rightarrow j \rightarrow k$ give the same result as the direct passage $i \rightarrow k$ (without it, three pairwise compatible trivialisations would not define a well-posed bundle on $U_i \cup U_j \cup U_k$). Conversely, every family $\{g_{ij}\}$ with values in a topological group $G \leq GL_n(\mathbb{R})$ satisfying the cocycle condition determines a vector bundle (with **structure group** G), by gluing $\bigsqcup_i U_i \times \mathbb{R}^n$ via $(b, v)_j \sim (b, g_{ij}(b)v)_i$. A **principal bundle** with structure group G is defined in the same way with fibre $F = G$ and $g_{ij}(b)$ acting by left multiplication.

2.3. The unit tangent bundle of S^1

Definition 2.5: Unit tangent bundle. For a curve $\gamma \subset \mathbb{R}^2$, the **unit tangent bundle** is $T^1\gamma = \{(P, v) : P \in \gamma, v \in T_P\gamma, |v| = 1\}$.

Theorem 2.2: $T^1S^1 \cong S^1 \times \{\pm 1\}$. The unit tangent bundle of the circle is trivial: T^1S^1 is homeomorphic to two disjoint copies of S^1 (the two senses of travel).

Proof. For $P = (\cos \theta, \sin \theta) \in S^1$, the tangent space $T_P S^1$ is the line generated by $(-\sin \theta, \cos \theta)$ (Volume II, Chapter 1); the two unit tangent vectors at P are $\pm(-\sin \theta, \cos \theta)$. The map $(\theta, \epsilon) \mapsto (P(\theta), \epsilon \cdot (-\sin \theta, \cos \theta))$, for $\epsilon \in \{\pm 1\}$, is a homeomorphism between $S^1 \times \{\pm 1\}$ and T^1S^1 (continuous, bijective, with continuous inverse $(P, v) \mapsto (P, \text{sgn}\langle v, (-\sin \theta, \cos \theta) \rangle)$, well defined because $v = \pm(-\sin \theta, \cos \theta)$ exactly). $\square$

Remark 2.5: Comparison with T^1S^2 : the hairy ball, already proved. Unlike T^1S^1 (trivial), T^1S^2 is not trivial: there is no nowhere-vanishing continuous section of the tangent bundle of S^2 . This is the hairy ball theorem, already proved in full in Chapter 1 (Theorem 1.14, Milnor’s analytic proof via a polynomial identity for the volume): it is not reproved here. The rest of this chapter (Section 2.8) generalises that result from “a zero exists” to “the sum of the zeros, with sign, equals exactly $\chi(\Sigma)$ ” for every closed surface Σ , not only S^2 .

Corollary 2.1: $T^1S^2 \cong SO(3)$. The unit tangent bundle of S^2 is homeomorphic to $SO(3)$.

Proof. An element $(P, v) \in T^1S^2$ determines, together with the outward normal vector P , a direct orthonormal basis $(v, P \times v, P)$ of $\mathbb{R}^3$, that is a matrix of $SO(3)$ (whose columns are those three vectors); conversely every $R \in SO(3)$ gives $(Re_3, Re_1) \in T^1S^2$ (where e_1, e_2, e_3 is the canonical basis). The two maps are continuous and inverse to each other: a homeomorphism. The double covering $SO(3) \cong \mathbb{RP}^3$, the quaternions and the gimbal lock phenomenon are already treated in full in Volume III, Chapter 28, to which we refer without repetition. $\square$

2.4. Local triviality: bundles over an interval

Proposition 2.1: Every vector bundle over $[0, 1]$ is trivial. Let $\pi : E \rightarrow [0, 1]$ be a vector bundle (real, of rank n). Then $E \cong [0, 1] \times \mathbb{R}^n$.

Proof. By compactness, $[0, 1]$ is covered by finitely many trivialising open intervals $I_1, \dots, I_m$, which can be chosen (Lebesgue number) in such a way that, ordered by increasing midpoint, $I_k \cap I_{k+1}$ is a non-empty interval and $I_k \cap I_j = \emptyset$ for $|k - j| \geq 2$. One constructs by induction on k a trivialisation H_k of E over $J_k := I_1 \cup \dots \cup I_k$ which coincides with the given trivialisation h_1 of I_1 near 0.

Base step. $H_1 := h_1$ on $J_1 = I_1$.

Inductive step. Let H_k be constructed on J_k , and let h_{k+1} be the given trivialisation on I_{k+1} . On the (connected) interval $J_k \cap I_{k+1} \supseteq I_k \cap I_{k+1} \neq \emptyset$ the two trivialisations differ by a continuous transition function $g(t) = H_k \circ h_{k+1}^{-1}|_{\{t\} \times \mathbb{R}^n} \in GL_n(\mathbb{R})$. Fix $c \in I_k \cap I_{k+1}$ and define

$$H_{k+1}(e) := \begin{cases} H_k(e) & \text{if } \pi(e) \leq c, \\ (t, g(c) \cdot v) & \text{if } \pi(e) = t > c, \quad h_{k+1}(e) = (t, v). \end{cases}$$

The two formulas are continuous separately on their respective domains, closed/open relative to J_{k+1} ; at the junction point $t = c$ they agree by construction ($H_k(e) = (c, g(c)v)$ exactly when $h_{k+1}(e) = (c, v)$, by definition of g): H_{k+1} is therefore continuous on $J_{k+1} = J_k \cup I_{k+1}$, bijective and linear on each fibre (composition of linear trivialisations with multiplication by the constant invertible matrix $g(c)$), with continuous inverse constructed in the same way. After m steps, H_m trivialises E over the whole of $[0, 1]$. $\square$

Lemma 2.1: Homotopy of the gluing. Let $g_0, g_1 : A \rightarrow GL_n(\mathbb{R})$ be two homotopic transition functions (through a continuous path g_t , $t \in [0, 1]$, in $GL_n(\mathbb{R})$, for every point of A , with jointly continuous dependence). The vector bundles over $B = U_+ \cup U_-$ obtained by gluing $U_+ \times \mathbb{R}^n$ and $U_- \times \mathbb{R}^n$ along $A = U_+ \cap U_-$ via g_0 and via g_1 respectively are isomorphic.

Proof. One constructs the bundle $\mathcal{E}$ over $B \times [0, 1]$ by gluing $U_+ \times [0, 1] \times \mathbb{R}^n$ and $U_- \times [0, 1] \times \mathbb{R}^n$ along $A \times [0, 1]$ via $(a, t, v) \mapsto (a, t, g_t(a)v)$: by Proposition 2.1 applied fibre by fibre on $\{b\} \times [0, 1]$ for every $b \in B$ (the argument of the previous proof applies verbatim, with $[0, 1]$ in place of the base), $\mathcal{E}$ is trivial on the restriction to $\{b\} \times [0, 1]$, and the trivialisation can be chosen continuously in b (the same gluing construction, uniform in b because $g_t(a)$ also depends continuously on a): the restriction of $\mathcal{E}$ to $B \times \{0\}$ (the bundle with cocycle g_0) and to $B \times \{1\}$ (cocycle g_1) are therefore both isomorphic to the restriction of a common trivialisation of $\mathcal{E}$, hence isomorphic to each other. $\square$

2.5. The Möbius band as tautological bundle

Definition 2.6: Tautological bundle over $\mathbb{RP}^1$. The **real projective line** $\mathbb{RP}^1$ is the set of lines through the origin of $\mathbb{R}^2$. The **tautological bundle** $\gamma^1 \rightarrow \mathbb{RP}^1$ has as fibre over $\ell \in \mathbb{RP}^1$ the line ℓ itself:

$$\gamma^1 = \{(\ell, v) : \ell \in \mathbb{RP}^1, v \in \ell\} \subset \mathbb{RP}^1 \times \mathbb{R}^2.$$

Proposition 2.2: $\mathbb{RP}^1 \cong S^1$. The map $q : S^1 \rightarrow \mathbb{RP}^1$, $q(\cos \theta, \sin \theta) = \mathbb{R} \cdot (\cos \theta, \sin \theta)$ (the line through the point), induces a homeomorphism $S^1/(\pm 1) \cong \mathbb{RP}^1$; the map $p_2 : S^1 \rightarrow S^1$, $p_2(e^{i\theta}) = e^{2i\theta}$, composed with the identification $S^1 \cong \mathbb{RP}^1$ given by $[e^{i\theta}] \mapsto e^{2i\theta}$ (well defined: θ and $\theta + \pi$ give the same $e^{2i\theta}$), realises $\mathbb{RP}^1$ as S^1 and q as the connected two-sheeted covering p_2 already classified in Chapter 1 (Theorem 1.2).

Proof. q is continuous, surjective, and $q(x) = q(y) \iff y = \pm x$ (two points of S^1 generate the same line if and only if they are equal or opposite): hence q induces a continuous bijection $S^1/(\pm 1) \rightarrow \mathbb{RP}^1$ from a compact space to a Hausdorff space (direct check: $\mathbb{RP}^1$ with the quotient topology from $\mathbb{R}^2 \setminus \{0\}$ separates points), hence a homeomorphism. The map $[e^{i\theta}] \mapsto e^{2i\theta}$ is well posed ($\theta \mapsto \theta + \pi$ does not change $e^{2i\theta}$), continuous, bijective between compact Hausdorff spaces, hence a homeomorphism; composing with q one obtains $\theta \mapsto e^{2i\theta} = p_2(e^{i\theta})$. $\square$

Theorem 2.3: The Möbius band is the tautological bundle. Under the identification $\mathbb{RP}^1 \cong S^1$ of Proposition 2.2, the tautological bundle $\gamma^1 \rightarrow \mathbb{RP}^1$ is homeomorphic, as a line bundle over S^1 , to the total space of the Möbius band with the line bundle structure defined by the cocycle $g_{12} = +1$ on one component A of the intersection of two trivialising arcs and $g_{12} = -1$ on the other component B ; in particular γ^1 is not trivial (Theorem 2.4 below), and its restriction to the restricted fibre $\{v \in \ell : |v| \leq 1\}$ is precisely the surface obtained by embedding a rectangle $[0, \pi] \times [-1, 1]$ in $\mathbb{R}^3$ with a half-turn twist and gluing the two short sides with opposite orientation.

Proof. Parametrise $\mathbb{RP}^1$ by $\theta \in [0, \pi)$ (angle of the line $\ell_\theta = \mathbb{R}(\cos \theta, \sin \theta)$, with $\ell_\theta = \ell_{\theta+\pi}$). On $U_1 = (-\epsilon, \pi - \epsilon)$ and $U_2 = (\pi/2 - \epsilon, 3\pi/2 - \epsilon) \pmod{\pi}$ (two arcs covering $\mathbb{RP}^1 \cong S^1$ with intersection having two components A, B , exactly as in Section 2.6 below), a non-zero vector of ℓ_θ is written in a unique way as $t \cdot (\cos \theta, \sin \theta)$, $t \in \mathbb{R}$: this gives a local trivialisation $\Phi_1(\ell_\theta, t(\cos \theta, \sin \theta)) = (\theta, t)$ on U_1 (and analogously Φ_2 on U_2 , with θ measured from the midpoint of U_2). On the component A of $U_1 \cap U_2$ the two angular parametrisations agree (the same representative $\theta \in (0, \pi)$ chosen consistently), and the transition function is $g_{12} = +1$; on the component B (near $\theta = 0 \equiv \pi$), the angular representative used by Φ_1 and the one used by Φ_2 differ by π (one uses θ near 0, the other the equivalent angle near π , for the same line $\ell_\theta = \ell_{\theta+\pi}$), and since $(\cos(\theta + \pi), \sin(\theta + \pi)) = -(\cos \theta, \sin \theta)$, the same vector has coordinate

t in one system and $-t$ in the other: $g_{12} = -1$ on B . This is exactly the cocycle of the Möbius band. The explicit parametrisation in $\mathbb{R}^3$

$$X(\theta, t) = \left(\left(1 + \frac{t}{2} \cos \frac{\theta}{2}\right) \cos \theta, \left(1 + \frac{t}{2} \cos \frac{\theta}{2}\right) \sin \theta, \frac{t}{2} \sin \frac{\theta}{2} \right), \quad \theta \in [0, 2\pi], t \in [-1, 1]$$

realises the total space (fibre restricted to $[-1, 1]$) as a submanifold of $\mathbb{R}^3$: for $\theta \in [0, \pi]$ it runs along ℓ_θ parametrised by t , and $X(\theta + 2\pi, t) = X(\theta, -t)$ (direct check with $\cos(\theta/2 + \pi) = -\cos(\theta/2)$, $\sin(\theta/2 + \pi) = -\sin(\theta/2)$), that is, the curve $X(\theta, t)$ for $\theta \in [0, 2\pi]$ closes up on itself only after the full turn, with t changing sign: this is the half-twist closure of the strip, and corresponds — read on the base $\mathbb{RP}^1$, where $\ell_{\theta+\pi} = \ell_\theta$ and θ traverses $[0, \pi]$ twice within $[0, 2\pi]$ — to the same cocycle -1 met again geometrically (Figure 2.2 for the comparison with the trivial cylinder, Figure 2.3 for the two charts U_1, U_2 and the components A, B). $\square$

Theorem 2.4: The Möbius band is not trivial. The line bundle with cocycle $g_{12} = +1$ on A and $g_{12} = -1$ on B (Theorem 2.3) is not isomorphic to the trivial bundle $S^1 \times \mathbb{R}$.

Proof. One defines the **total monodromy** of the bundle: going once around S^1 starting from a point of A , crossing U_2 , entering B and returning to the starting point through U_1 , a vector of the fibre is multiplied by $g_{12}|_A \cdot g_{21}|_B = g_{12}|_A \cdot g_{12}|_B^{-1}$. If one changes trivialisation on U_i by multiplying by a continuous function $h_i : U_i \rightarrow \mathbb{R}^*$, the new cocycle is $g'_{12} = h_2 g_{12} h_1^{-1}$, and the new total monodromy is

$$(h_2 g_{12}|_A h_1^{-1}) \cdot (h_1 g_{12}|_B^{-1} h_2^{-1}) = g_{12}|_A \cdot g_{12}|_B^{-1} :$$

the factors h_1, h_2 cancel exactly along the closed loop, hence the total monodromy is an invariant of the bundle, independent of the trivialisation. For the trivial bundle it equals $(+1)(+1)^{-1} = +1$; for the Möbius band it equals $(+1)(-1)^{-1} = -1$. Being different, the two bundles are not isomorphic. $\square$

Theorem 2.5: Nowhere-vanishing sections and triviality (rank 1). A line bundle $L \rightarrow S^1$ is trivial if and only if it admits a nowhere-vanishing section $s : S^1 \rightarrow L$, $\pi \circ s = \text{id}$, $s(x) \neq 0 \forall x$. In particular the Möbius band admits none.

Proof. $(\Rightarrow)$ If $L \cong S^1 \times \mathbb{R}$, then $s(x) = (x, 1)$ is nowhere vanishing. $(\Leftarrow)$ Given s nowhere vanishing, $(x, t) \mapsto t \cdot s(x)$ is an isomorphism $S^1 \times \mathbb{R} \rightarrow L$ (linear, bijective on each fibre because $s(x) \neq 0$ generates the line L_x , continuous with continuous inverse). For the Möbius band: a section is read, in the charts of Theorem 2.3, as $s_1 : U_1 \rightarrow \mathbb{R}$, $s_2 : U_2 \rightarrow \mathbb{R}$ with $s_1 = s_2$ on A and $s_1 = -s_2$ on B ; if s were nowhere vanishing, s_1 would have constant sign on the connected U_1 and s_2 on U_2 , but the two gluing conditions require concordant signs on A and discordant ones on B : impossible. $\square$

INFO

Detecting w_1 with a discretised moving frame: $N \geq 3$ is needed. In the idealised two-chart model of Theorem 2.3 (constant cocycle ± 1 on the two components A, B only) two samples suffice, one in A and one in B , to read off the monodromy -1 . But a real line bundle can also present itself as a *smooth direction field* rotating continuously, without privileged charts — the relevant case for a physical cable twisting continuously. Model: the (unoriented) direction $d(\theta) = \pm(\cos \frac{\theta}{2}, \sin \frac{\theta}{2})$, $\theta \in [0, 2\pi]$, defines the same Möbius bundle, since $d(2\pi) = -d(0)$. To detect it numerically with N equispaced samples $\theta_k = 2\pi k/N$ one propagates a *unit* vector v_k choosing at each step, between the two determinations $\pm(\cos \frac{\theta_k}{2}, \sin \frac{\theta_k}{2})$, the one with the *larger* scalar product with v_{k-1} (the “nearest” choice, the

only one available without knowing the global topology a priori): one closes the circle by comparing v_N (propagated) with v_0 .

$N = 1$: a single sample, no pair to compare: nothing to detect.

$N = 2$: $\theta_0 = 0$, $\theta_1 = \pi$; the true directions are $d(\theta_0) = (1, 0)$, $d(\theta_1) = (0, 1)$, orthogonal: $d(\theta_1) \cdot v_0 = 0 = (-d(\theta_1)) \cdot v_0$, an **exact tie**. The “nearest” choice is undefined: the method is not applicable for $N = 2$.

$N \geq 3$: the angular step between consecutive samples of the true directions is $\pi/N < \pi/2$ (strictly), hence at each step there is a unique choice of maximal scalar product, and with it the propagation reproduces exactly $v_k = d(\theta_k)$ for $k = 0, \dots, N-1$; closing the loop, $d(2\pi) = (-1, 0) = -d(0)$ has scalar product $+1$ with $v_{N-1} \approx (-1, \text{small})$, hence $v_N = d(2\pi) = -v_0$: the monodromy -1 is detected correctly and without ambiguity. **Numerical check for $N = 3$** ($\theta_k = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$): $d_0 = (1, 0)$, $d_1 = (0.5, 0.866)$, $d_2 = (-0.5, 0.866)$; $d_0 \cdot d_1 = 0.5 > 0$, $d_1 \cdot d_2 = -0.25 + 0.75 = 0.5 > 0$ (no tie), $d(2\pi) = (-1, 0)$ and $d(2\pi) \cdot d_2 = 0.5 > 0$: one closes on $v_3 = (-1, 0) = -v_0$. ✓. The old statement “for any $N \geq 1$ ” was therefore false: the critical case $N = 2$ is an exact tie (not a rare error), and the correct threshold is $N \geq 3$.

Exercise 2.1. In the explicit parametrisation of the Möbius band (Theorem 2.3), check that $X(\theta + 2\pi, t) = X(\theta, -t)$.

2.6. Classification of rank- n vector bundles over S^1

Lemma 2.2: Every vector bundle admits a metric. Every vector bundle (real, of rank n) over a paracompact base B (in particular S^1) admits a Riemannian metric on the fibres: a continuous function $b \mapsto \langle \cdot, \cdot \rangle_b$, a scalar product on E_b , which reads as a positive definite symmetric matrix in every local trivialisation.

Proof. Let $\{U_i\}$ be a trivialising cover and $\{\rho_i\}$ a subordinate continuous partition of unity (which exists by paracompactness). On each U_i , the trivialisation transports the standard Euclidean scalar product of $\mathbb{R}^n$ to a metric $\langle \cdot, \cdot \rangle_b^{(i)}$ on E_b , $b \in U_i$. One sets $\langle \cdot, \cdot \rangle_b := \sum_i \rho_i(b) \langle \cdot, \cdot \rangle_b^{(i)}$ (a locally finite sum, well defined by extending by 0 where $\rho_i = 0$): it is continuous, symmetric, and positive definite because it is a sum with coefficients ≥ 0 (not all zero at b) of positive definite forms. $\square$

Theorem 2.6: Bundles over S^1 and homotopy classes of clutching functions. Write $S^1 = U_+ \cup U_-$ as the union of two closed arcs whose intersection is two disjoint arcs $A \sqcup B$. Every real vector bundle of rank n over S^1 is isomorphic to the bundle obtained by gluing $U_+ \times \mathbb{R}^n$ and $U_- \times \mathbb{R}^n$ via a function $g : A \sqcup B \rightarrow GL_n(\mathbb{R})$; normalising $g \equiv I$ on A (possible: A is contractible, hence $g|_A$ is homotopic to a constant, and by Lemma 2.1 the bundle does not change if $g|_A$ is replaced by the constant I through an explicit homotopy $g_t|_A$ from $g|_A$ to I), the isomorphism classes of the bundle are in bijection with the free homotopy classes $g|_B \in \pi_0(GL_n(\mathbb{R}))$ (Figure 2.4, left; on the right the same construction on S^2 , Remark 2.7).

Proof. Every vector bundle is trivial over U_+ and over U_- (closed arcs, homeomorphic to $[0, 1]$, by Proposition 2.1); the bundle is therefore determined, up to isomorphism, by the transition function between the two trivialisations on $A \sqcup B$. Since A is contractible, $g|_A : A \rightarrow GL_n(\mathbb{R})$ is homotopic to a constant $c = g|_A(a_0)$; composing the trivialisation on U_- with multiplication by c^{-1} (constant, hence still a trivialisation) one may assume $c = I$, and then $g|_A \equiv I$ by Lemma 2.1 applied to the homotopy (rel. nothing, A being an arc with no further structure

to preserve beyond continuity) between $g|_A$ and the constant I . There remains $g|_B : B \rightarrow GL_n(\mathbb{R})$; since B too is contractible, $[B, GL_n(\mathbb{R})] = \pi_0(GL_n(\mathbb{R}))$ (every continuous function from a contractible space is homotopic to the constant at its value at a base point, and homotopy classes of constants are the connected components). By Lemma 2.1, bundles with $g|_B$ in the same connected component are isomorphic; bundles with $g|_B$ in different components are not isomorphic by the same argument of total monodromy of Theorem 2.4 (the product $g|_A \cdot g|_B^{-1} = g|_B^{-1}$, whose component in $\pi_0(GL_n(\mathbb{R}))$ is an invariant of the bundle by the same computation with h_1, h_2). $\square$

Theorem 2.7: Classification of real vector bundles of rank n over S^1 . For every $n \geq 1$, up to isomorphism there are exactly two real vector bundles of rank n over S^1 : the trivial bundle $S^1 \times \mathbb{R}^n$ and the “twisted” bundle $M \oplus \mathbb{R}^{n-1}$ (Whitney sum of the Möbius band with a trivial bundle of rank $n - 1$).

Proof. By Theorem 2.6, the rank- n bundles over S^1 are classified by $\pi_0(GL_n(\mathbb{R}))$ (the component of $g|_B$). Every invertible matrix retracts continuously onto its orthogonal part via the polar decomposition $A = P \cdot (A^T A)^{1/2}$ with $P = A(A^T A)^{-1/2} \in O(n)$ (well defined: $A^T A$ symmetric positive definite has a unique symmetric positive definite square root, and $t \mapsto A(A^T A)^{-t/2}$, $t \in [0, 1]$, is a continuous path from A to P inside $GL_n(\mathbb{R})$): hence $\pi_0(GL_n(\mathbb{R})) \cong \pi_0(O(n))$. Now $O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I\}$ has exactly two path-connected components: $SO(n) = \{\det = +1\}$ (path-connected: every rotation is written as a product of planar rotations in pairs of orthogonal axes, each of which can be joined to the identity by the path of rotations of the same angle scaled continuously from 0 to the given angle, by the real Jordan canonical form of orthogonal matrices) and the coset $O(n) \setminus SO(n) = SO(n) \cdot r$ for any fixed reflection r (homeomorphic to $SO(n)$ via $A \mapsto Ar$, hence path-connected). One therefore obtains exactly two classes: $[I]$ (trivial bundle) and $[r]$ (twisted bundle, whose simplest representative — a reflection in a single coordinate, extended by the identity on the other $n - 1$ — is exactly the cocycle $\text{diag}(-1, 1, \dots, 1)$ of the Möbius band summed with a trivial bundle of rank $n - 1$). $\square$

Remark 2.6: Why the count does not grow with n . The conceptual point is $\pi_0(O(n)) \cong \mathbb{Z}/2$ for every $n \geq 1$: the number of connected components of $O(n)$ does not grow with n (it always remains “orientable or not”), hence the bundles over S^1 remain 2 at every rank.

Remark 2.7: $\text{Vect}_n(S^k)$ in general. The same construction, applied to $S^k = D_+^k \cup D_-^k$ glued along the equator S^{k-1} , classifies the complex vector bundles of rank n over S^k by $\pi_{k-1}(U(n))$ (with $U(n)$ connected, the quotient by π_0 is trivial and does not appear); in the real case, the classification is instead $\text{Vect}_n(S^k) \cong \pi_{k-1}(O(n))/\pi_0(O(n))$, the quotient of the free homotopy classes by the conjugation action of $\pi_0(O(n))$ on $\pi_{k-1}(O(n))$ (two clutching functions give isomorphic bundles even when they are homotopic only after conjugating by an element not connected to the identity, an outcome of the freedom to recalibrate the trivialisation on D_+^k by an arbitrary element of $O(n)$, not necessarily in $SO(n)$): for $k = 1$ the quotient is trivial (π_0 acting on itself by conjugation in a group of two elements leaves it fixed), and one recovers Theorem 2.7; for $k = 2$ in the complex case one obtains $\pi_1(U(n)) \cong \mathbb{Z}$, the integer is the first Chern class, and it is the invariant that makes the Hopf fibration non-trivial (Section 2.12). Systematic theory in *F4*, ch. 1–2 (Bott periodicity, connected to this classification, remains a pure panorama here, without a dedicated primary source in this register).

2.7. Principal bundles and classification via monodromy

Theorem 2.8: Classification of principal bundles over S^1 via monodromy. Let G be a locally path-connected topological group. Once an orientation of S^1 is fixed, the principal bundles with structure group G over S^1 are classified, up to isomorphism, by the conjugacy classes of $\pi_0(G)$.

Proof. Let $p \in S^1$, $U = S^1 \setminus \{p\}$ (contractible) and V a small contractible neighbourhood of p ; every principal bundle is trivial over U and over V (retracing the argument of Proposition 2.1 for the group G : over an interval, the gluing construction with “freezing” of the transition at a point applies verbatim, replacing the product of $GL_n(\mathbb{R})$ with that of G). The intersection $U \cap V$ has two components A, B ; on one of them, after normalising as in Theorem 2.6 (here for a general G , same contractibility argument), the transition is $g_{UV} \equiv e$, on the other it is a constant $g \in G$: the **monodromy**, going around S^1 in the sense fixed by the orientation. A different choice of trivialisations replaces g by a conjugate hgh^{-1} : isomorphic bundles have conjugate monodromies and conversely (the same cancellation computation as in Theorem 2.4); only the connected component $[g] \in \pi_0(G)$ matters, because two elements in the same component are joined by a continuous path, which by Lemma 2.1 (extended to G) gives an isomorphism of bundles. $\square$

Remark 2.8: The orientation must be fixed: the example of $\mathbb{Z}/3$. Going around S^1 in the opposite sense replaces the monodromy g by g^{-1} : for G abelian (where conjugation is trivial and the conjugacy classes are the single elements) g and g^{-1} are in general different classes. Verified example: $G = \mathbb{Z}/3 = \{0, 1, 2\}$ has three conjugacy classes (singletons), and $1 \neq 2 = -1 \pmod{3}$: the principal $\mathbb{Z}/3$ bundles over S^1 with monodromy 1 and with monodromy 2 are not the same class for the fixed orientation, even though they become one the inverse of the other on reversing the sense of travel. Thus Theorem 2.8 classifies the bundles with oriented base; if one wishes to classify up to the orientation of the base as well, the correct invariant is the conjugacy class of $\{g, g^{-1}\}$ as a whole, not that of g alone.

Corollary 2.2: All principal $U(1)$ bundles over S^1 are trivial. Since $\mathbb{T} = U(1)$ is connected, there exists a unique principal $U(1)$ bundle over S^1 up to isomorphism: the trivial one.

2.8. The Poincaré–Hopf theorem for closed surfaces

Definition 2.7: Index of an isolated zero. Let Σ be a surface (C^1 manifold of dimension 2) and V a continuous tangent vector field on Σ with an isolated zero at p . In a local chart centred at p , let γ_ϵ be a small circle of radius $\epsilon > 0$ (small enough not to contain other zeros) around the origin. The **index** of V at p is

$$\text{ind}_p(V) := \deg \left(\theta \mapsto \frac{V(\gamma_\epsilon(\theta))}{|V(\gamma_\epsilon(\theta))|} \right),$$

the degree (in the sense of Chapter 1, maps $S^1 \rightarrow S^1$) of the normalised map on the small circle. It does not depend on ϵ small nor on the chart (by homotopy invariance of the degree, Chapter 1, shrinking ϵ or changing chart continuously).

Theorem 2.9: Poincaré–Hopf for closed orientable surfaces. Let Σ be a closed (compact, without boundary), orientable, connected surface of genus g , and V a continuous tangent vector field on Σ with finitely many zeros, all isolated. Then

$$\sum_{p: V(p)=0} \text{ind}_p(V) = \chi(\Sigma) = 2 - 2g.$$

Lemma 2.3: Invariance of the sum of the indices. If V_0, V_1 are two smooth tangent vector fields on Σ with isolated zeros, then $\sum_p \text{ind}_p(V_0) = \sum_p \text{ind}_p(V_1)$.

Proof. Consider $V_t := (1 - t)V_0 + tV_1$ on $\Sigma \times [0, 1]$ (pointwise convex combination of sections of the tangent bundle, well defined because $T_p\Sigma$ is a vector space). The set $Z = \{(p, t) : V_t(p) = 0\} \subset \Sigma \times [0, 1]$ is closed; **imported** (Sard, applied to the section $(p, t) \mapsto V_t(p)$ of the pull-back tangent bundle over $\Sigma \times [0, 1]$, after an arbitrarily small perturbation of V_0, V_1 which does not change the sums of the indices by the same argument of local homotopy invariance of the degree): up to such a perturbation, Z is a C^1 submanifold of dimension 1 with boundary exactly $Z \cap (\Sigma \times \{0, 1\}) = \{(p, 0) : V_0(p) = 0\} \cup \{(p, 1) : V_1(p) = 0\}$ (the critical points of the projection $Z \rightarrow [0, 1]$ are generic and do not obstruct the 1-dimensional manifold structure in the interior). A compact 1-dimensional submanifold with boundary is, by the classification of 1-manifolds, a finite disjoint union of arcs and circles; the circles lie entirely in $\Sigma \times (0, 1)$ (they do not touch the boundary) and do not contribute to the count of endpoints; every arc has its two endpoints on the boundary, that is, either both at $t = 0$, or both at $t = 1$, or one at $t = 0$ and one at $t = 1$.

Along such an arc $\alpha(u) = (p(u), t(u))$, the index $\text{ind}_{p(u)}(V_{t(u)})$ (defined with a small circle transverse to the arc, depending continuously on u) is an integer depending continuously on u (the degree of a map $S^1 \rightarrow S^1$ varying continuously in the parameter u is locally constant, by the homotopy invariance of the degree already proved in Chapter 1): hence it is constant along the whole arc. In particular the arcs with one endpoint at $t = 0$ and one at $t = 1$ realise a bijection between a subset of the zeros of V_0 and a subset of the zeros of V_1 which preserves the index; the arcs with both endpoints at $t = 0$ (resp. $t = 1$) pair up two zeros of V_0 (resp. V_1) with *opposite* indices in sign with respect to the orientation induced by the arc (the projection $t(u)$ has an interior local maximum or minimum on the arc, where the sense of travel in t reverses, reversing the sign with which the endpoint enters the count of the oriented boundary ∂Z : a standard fact of degree theory, which here amounts to saying that the two endpoints of an arc that “turns back” at the same instant $t = 0$ or $t = 1$ contribute with equal indices and opposite boundary sign, and hence cancel in pairs in the signed sum $\sum_p \text{ind}_p$, which does not distinguish boundary signs). Summing over all the arcs: $\sum_p \text{ind}_p(V_0) = \sum_p \text{ind}_p(V_1)$. $\square$

Proposition 2.3: Explicit computation: gradient field of a Morse height. Let $\Sigma_g \subset \mathbb{R}^3$ be the standard surface of genus g (embedded “standing up”, with g handles stacked vertically), $h : \Sigma_g \rightarrow \mathbb{R}$ the height function (third coordinate), generic in the sense that it has only non-degenerate critical points (**Morse**, **imported**: Sard applied to the space of directions, a generic height is Morse), and ∇h the gradient field (with respect to the induced metric). Then ∇h has exactly 1 minimum, $2g$ saddle points and 1 maximum, all isolated and non-degenerate, and

$$\sum_p \text{ind}_p(\nabla h) = 1 - 2g + 1 = 2 - 2g.$$

Proof. Count of the critical points. On the standard “standing” surface (one handle = one glued torus, the g handles stacked along the vertical axis), the height has a unique lowest point (global minimum, the bottom of the surface) and a unique highest point (global maximum, the top); between one handle and the next, the horizontal section passes from a single connected component to two (around the handle) and then back to one: this change of connectivity occurs at exactly two saddle points for each of the g handles (one where the section splits going up into the handle, one where it rejoins), giving $2g$ saddles in all — the standard picture, checkable by drawing the surface explicitly together with its horizontal sections at increasing heights.

Local index. By the Morse lemma (imported, standard: in suitable local coordinates near a non-degenerate critical point of index $k \in \{0, 1, 2\}$, $h(x, y) = h(p) \pm x^2 \pm y^2$ with k negative signs), $\nabla h = (\pm 2x, \pm 2y)$. For $k = 0$ (minimum, $h = x^2 + y^2$): $\nabla h = (2x, 2y)$, whose normalisation on a circle is the identity $S^1 \rightarrow S^1$, of degree $+1$. For $k = 2$ (maximum, $h = -x^2 - y^2$): $\nabla h = (-2x, -2y)$, whose normalisation is the antipodal map $(\cos \theta, \sin \theta) \mapsto -(\cos \theta, \sin \theta)$, homotopic to the rotation by angle π (a rotation is always homotopic to the identity by rotating the angle continuously from 0 to the given angle), hence of degree $+1$. For $k = 1$ (saddle, $h = x^2 - y^2$): $\nabla h = (2x, -2y)$, whose normalisation $(\cos \theta, \sin \theta) \mapsto (\cos \theta, -\sin \theta)$ is the reflection $z \mapsto \bar{z}$, of degree -1 (an odd map... or rather, here, a reflection: the degree of $z \mapsto \bar{z}$ on $S^1 \subset \mathbb{C}$ is -1 , a direct computation already implicit in Chapter 1 for orientation-reversing maps). In summary $\text{ind} = (-1)^k$ for a non-degenerate critical point of Morse index k : one sums $1 \cdot (+1) + 2g \cdot (-1) + 1 \cdot (+1) = 2 - 2g$ (the three types of critical point, with their respective indices, are illustrated in Figure 2.7). $\square$

Proof. [of Theorem 2.9] By Lemma 2.3, $\sum_p \text{ind}_p(V)$ does not depend on the chosen field (up to an arbitrarily small perturbation of V making it smooth with non-degenerate zeros, without changing the sum, by the same argument of local homotopy invariance, applied this time to a homotopy between V and a smooth perturbation of it): it is therefore equal to the value computed for the gradient field ∇h of Proposition 2.3, that is $2 - 2g = \chi(\Sigma_g)$ ($\chi(\Sigma_g) = 2 - 2g$ is the classical formula for the Euler characteristic of an orientable surface of genus g , computable from the standard CW structure — one vertex, $2g$ edges, one face, gluing of the $4g$ -gon — with $\chi = 1 - 2g + 1 = 2 - 2g$). Every closed connected orientable surface is homeomorphic to Σ_g for some g (classification of surfaces): the statement holds for arbitrary Σ . $\square$

Remark 2.9: Consistency with the hairy ball already proved. For $\Sigma = S^2$ ($g = 0$), $\chi(S^2) = 2 \neq 0$: every tangent vector field on S^2 has sum of the indices equal to 2, in particular it cannot be everywhere non-zero (an empty sum equals $0 \neq 2$). This is precisely the hairy ball theorem already proved in Chapter 1 by Milnor’s analytic method (Theorem 1.14): Theorem 2.9 contains it as a particular case, with the additional quantitative information that the sum equals exactly 2, not merely that it is non-zero. For the cylinder $S^1 \times [0, 1]$ (with boundary, $\chi = 0$) there consistently exists a nowhere-vanishing field (the constant tangent field along S^1), as for the trivial $T^1 S^1$ (Theorem 2.2).

Remark 2.10: Towards the Gauss map. For a surface $\Sigma \subset \mathbb{R}^3$ equipped with a unit normal $N : \Sigma \rightarrow S^2$ (**Gauss map**), a classical fact (imported, F28 §3-3 and §3-5; **not** reproved here because it is not needed elsewhere in the chapter) links $\text{ind}_p(\nabla h)$ at the critical points of the height to the sign of the Gaussian curvature $K(p)$ and to the local degree of N , and the global Gauss–Bonnet formula $\iint_{\Sigma} K \, dA = 2\pi\chi(\Sigma)$ identifies $\deg(N) = \chi(\Sigma)/2$: the Gauss map has degree 1 for a convex sphere ($\chi = 2$) and degree 0 for a torus ($\chi = 0$), consistently with Section 2.9 (where the local version of Gauss–Bonnet is proved in full and used for the holonomy). See Figure 2.8 for the Gauss map.

2.9. Local Gauss–Bonnet and holonomy of parallel transport

Theorem 2.10: Local Gauss–Bonnet via the connection form. Let $\Sigma \subset \mathbb{R}^3$ be an oriented surface and $R \subset \Sigma$ a simply connected region with smooth and simple boundary $C = \partial R$, oriented consistently with R . **Parallel transport** along C (Levi–Civita connection) rotates a tangent vector, after one complete loop, by an angle (the **holonomy** $\Delta\varphi$, measured in the sense of travel along C) equal to

$$\Delta\varphi = \iint_R K \, dA,$$

with K the Gaussian curvature.

Proof. Since R is simply connected, there exists on R a smooth tangent orthonormal frame (e_1, e_2) (constructible, for instance, by applying Gram–Schmidt to two coordinate fields on a chart covering the whole contractible R , or by gluing continuously over a contractible cover thanks to the simple connectivity of R). One defines the **connection form** ω by $\nabla e_1 = \omega e_2$, $\nabla e_2 = -\omega e_1$ (well posed: the Levi–Civita connection is compatible with the metric, hence ∇e_i is orthogonal to e_i , that is a multiple of e_2, e_1 respectively, with the same coefficient up to sign because $\langle \nabla e_1, e_2 \rangle = -\langle e_1, \nabla e_2 \rangle$, the derivative of $\langle e_1, e_2 \rangle \equiv 0$). A field V along a curve is *parallel* if and only if its angle $\phi(s)$ with respect to (e_1, e_2) satisfies $\phi'(s) = -\omega(T(s))$, T tangent to the curve (differentiating $V = |V|(\cos \phi e_1 + \sin \phi e_2)$ and imposing $\nabla_T V = 0$ one obtains $\phi' + \omega(T) = 0$, a direct computation with the structure formulas above). Hence the holonomy along C (the total variation of ϕ with the sign changed, that is, how much the transported vector “rotates back” with respect to the frame) is $\Delta\varphi = -\oint_C \phi' ds = \oint_C \omega(T) ds = \oint_C \omega$. On the other hand (a classical fact of differential geometry, imported F28 §4-3: the Gaussian curvature is the curvature 2-form of the Levi–Civita connection) $d\omega = -K dA$ on R with the given orientation; by Stokes’ theorem, $\oint_C \omega = -\iint_R d\omega = \iint_R K dA$ (the sign works out with the orientation chosen for $C = \partial R$ positively oriented). Hence $\Delta\varphi = \iint_R K dA$. $\square$

Corollary 2.3: Holonomy along a parallel of the sphere. On the unit sphere ($K \equiv 1$), parallel transport along the parallel of colatitude $\theta_0 \in (0, \pi)$ (traversed once in the positive sense, the boundary of the cap $R = \{\theta \leq \theta_0\}$) rotates a tangent vector by

$$\Delta\varphi = \text{Area}(R) = 2\pi(1 - \cos \theta_0).$$

Proof. $\text{Area}(R) = \int_0^{2\pi} \int_0^{\theta_0} \sin \theta \, d\theta \, d\varphi = 2\pi[-\cos \theta]_0^{\theta_0} = 2\pi(1 - \cos \theta_0)$; one applies Theorem 2.10 with $K \equiv 1$ (planar development of the tangent cone in Figure 2.6). $\square$

Remark 2.11: Cross-check with the geodesic curvature. As an independent check: the geodesic curvature of the parallel of colatitude θ_0 on the unit sphere is $\kappa_g = \cot \theta_0$ (a classical fact, F28 §4-2, a standard exercise), and its length is $2\pi \sin \theta_0$; the complete version of local Gauss–Bonnet (with the geodesic curvature term of the boundary, here smooth, with no exterior angles) gives $\iint_R K dA + \oint_C \kappa_g ds = 2\pi(1 - \cos \theta_0) + \cot \theta_0 \cdot 2\pi \sin \theta_0 = 2\pi(1 - \cos \theta_0) + 2\pi \cos \theta_0 = 2\pi \checkmark$, in agreement with the general formula of local Gauss–Bonnet for a simply connected region with smooth simple boundary ($\iint K + \oint \kappa_g = 2\pi$, the case without exterior angles of the formula used in Volume III for geodesic triangles): this confirms independently the value $2\pi(1 - \cos \theta_0)$ of the holonomy.

Worked Example 2.1 — parallel transport on the octant Problem. Compute the holonomy of parallel transport along the boundary of the positive octant of S^2 (geodesic triangle with vertices on the three coordinate axes), checking Theorem 2.10 against the direct computation for a geodesic triangle.

Step 1 — direct computation. Along each side (arc of geodesic), a vector transported in parallel keeps a constant angle with the geodesic; at each vertex, where the interior angle equals $\pi/2$, the exterior angle is $\pi - \pi/2 = \pi/2$. After the complete loop the vector has been rotated by $2\pi - 3 \cdot \frac{\pi}{2} = \frac{\pi}{2}$ (the angular excess of the triangle, $\alpha + \beta + \gamma - \pi$ with $\alpha = \beta = \gamma = \pi/2$).

Step 2 — check with the area. The octant is $1/8$ of the sphere: Area = $4\pi/8 = \pi/2$, in agreement with Theorem 2.10 (Figure 2.5). $\square$

2.10. The Foucault pendulum as holonomy

Remark 2.12: Physical model: precession of the Foucault pendulum. Hypotheses of the model (F21): small oscillations, period of the pendulum negligible with respect to the period of the Earth's rotation (adiabatic approximation), pendulum at a fixed latitude $\lambda_0 \in (0, \pi/2)$. In the inertial (non-rotating) frame, the plane of oscillation of the pendulum does not rotate at all (no net horizontal force acts on it in that frame, apart from the restoring force in the plane itself). The ground of the Earth, on the other hand, rotates by a complete turn (2π) in one sidereal day about the polar axis. With respect to the ground (the physically observed frame), the plane of the pendulum therefore appears to rotate, in one day, by the angle missing between the complete turn of the ground and the holonomy $\Delta\varphi = 2\pi(1 - \cos\theta_0)$ of parallel transport along the parallel of colatitude $\theta_0 = \pi/2 - \lambda_0$ (Corollary 2.3) — the holonomy being, by construction, exactly how much the “intrinsically non-rotating” direction (parallel-transported on the sphere) departs from the direction of a fixed point of the ground after the loop:

$$\Omega_{\text{Foucault}} = 2\pi - \Delta\varphi = 2\pi - 2\pi(1 - \cos\theta_0) = 2\pi \cos\theta_0 = 2\pi \sin\lambda_0$$

(using $\cos\theta_0 = \cos(\frac{\pi}{2} - \lambda_0) = \sin\lambda_0$): the classical precession $2\pi \sin\lambda_0$ per sidereal day, clockwise in the northern hemisphere ($\lambda_0 > 0$).

Remark 2.13: Why it is not “the monopole computation”. An old formulation identified $2\pi \sin\lambda_0$ with the holonomy of the Hopf/monopole connection (Section 2.12) computed on the same cap. It is false by a factor 2: the canonical $U(1)$ connection on $S^3 \rightarrow S^2$ gives, on the same cap, holonomy of modulus $\pi(1 - \cos\theta_0)$ (Section 2.13, Berry phase), half of the $\Delta\varphi = 2\pi(1 - \cos\theta_0)$ of the real parallel transport on the tangent bundle. The Foucault pendulum measures the holonomy of the tangent bundle (Levi-Civita connection, group $SO(2) \cong U(1)$ but “spin 1” representation), not that of the Hopf bundle (“spin 1/2” representation): the factor 2 between the two is discussed explicitly in Section 2.13.

2.11. The Hopf fibration

Theorem 2.11: The Hopf fibration. The map $\pi : S^3 \rightarrow S^2$, $\pi(z_1, z_2) = (2\operatorname{Re}(z_1\bar{z}_2), 2\operatorname{Im}(z_1\bar{z}_2), |z_1|^2 - |z_2|^2)$, $(z_1, z_2) \in S^3 \subset \mathbb{C}^2$, is well defined with values in S^2 , its fibres are exactly the orbits of the action $e^{i\alpha} \cdot (z_1, z_2) = (e^{i\alpha}z_1, e^{i\alpha}z_2)$, and it is a

non-trivial principal $U(1)$ bundle, with local sections

$$\sigma_+(x, y, w) = \left(\sqrt{\frac{1+w}{2}}, \frac{x-iy}{\sqrt{2(1+w)}} \right) \quad (w \neq -1), \quad \sigma_-(x, y, w) = \left(\frac{x+iy}{\sqrt{2(1-w)}}, \sqrt{\frac{1-w}{2}} \right) \quad (w \neq 1),$$

whose transition function on the equator $w = 0$, parametrised by $x + iy = e^{i\varphi}$, is $g_{-+}(\varphi) = e^{i\varphi}$.

Proof. Image in S^2 . With $x + iy = 2z_1\bar{z}_2$, $w = |z_1|^2 - |z_2|^2$: $x^2 + y^2 + w^2 = 4|z_1|^2|z_2|^2 + (|z_1|^2 - |z_2|^2)^2 = (|z_1|^2 + |z_2|^2)^2 = 1$.

Fibres = orbits. Invariance on the orbits: immediate (the moduli and $z_1\bar{z}_2$ do not change on multiplying by $e^{i\alpha}$). Conversely if $\pi(z) = \pi(w)$: from $|z_1| = |w_1|$, $|z_2| = |w_2|$ and $z_1\bar{z}_2 = w_1\bar{w}_2$ it follows that the phase factors of z_1/w_1 and z_2/w_2 coincide (setting $z_i = r_i e^{i\beta_i}$, $w_i = r_i e^{i\gamma_i}$ with the same r_i : $z_1\bar{z}_2 = w_1\bar{w}_2 \Rightarrow r_1 r_2 e^{i(\beta_1 - \beta_2)} = r_1 r_2 e^{i(\gamma_1 - \gamma_2)} \Rightarrow \beta_1 - \beta_2 = \gamma_1 - \gamma_2$, hence $\beta_1 - \gamma_1 = \beta_2 - \gamma_2 =: \alpha$ and $w = e^{i\alpha} z$).

Correct sections. Direct check that σ_+ has norm 1 in $\mathbb{C}^2$: $|\sqrt{\frac{1+w}{2}}|^2 + |\frac{x-iy}{\sqrt{2(1+w)}}|^2 = \frac{1+w}{2} + \frac{x^2+y^2}{2(1+w)} = \frac{1+w}{2} + \frac{1-w^2}{2(1+w)} = \frac{1+w}{2} + \frac{1-w}{2} = 1$. And $\pi \circ \sigma_+$: with $z_1 = \sqrt{\frac{1+w}{2}}$ real and $z_2 = \frac{x-iy}{\sqrt{2(1+w)}}$,

$$2z_1\bar{z}_2 = 2\sqrt{\frac{1+w}{2}} \cdot \frac{x+iy}{\sqrt{2(1+w)}} = x+iy, \quad |z_1|^2 - |z_2|^2 = \frac{1+w}{2} - \frac{1-w}{2} = w :$$

$\pi(\sigma_+(x, y, w)) = (x, y, w)$ exactly (*this* is the correct point with respect to an old version with $z_2 = \frac{x+iy}{\sqrt{2(1+w)}}$, which gave $2z_1\bar{z}_2 = x - iy$, that is $\pi \circ \sigma_+ = (x, -y, w) \neq \text{id}$). Analogously for σ_- (with z_2 real positive).

Transition function. On the equator $w = 0$: $\sigma_+ = (\frac{1}{\sqrt{2}}, \frac{x-iy}{\sqrt{2}})$ and, with $x - iy = e^{-i\varphi}$, $\sigma_+ = \frac{1}{\sqrt{2}}(1, e^{-i\varphi})$; $\sigma_- = (\frac{x+iy}{\sqrt{2}}, \frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}(e^{i\varphi}, 1)$. The ratio (component by component, both non-zero on the equator) is $\sigma_-/\sigma_+ = e^{i\varphi}$ in both components: $\sigma_-(\varphi) = e^{i\varphi} \cdot \sigma_+(\varphi)$, that is $g_{-+}(\varphi) = e^{i\varphi}$. **Independent numerical check** ($\varphi = 0.7$): $e^{i \cdot 0.7} = \cos 0.7 + i \sin 0.7 = 0.76484 + 0.64422i$; computing $\sigma_-(0.7)/\sigma_+(0.7)$ directly from the formulas above one recovers the same number to 5 digits. ✓ Since $\varphi \mapsto e^{i\varphi}$ is the generator of $\pi_1(U(1)) \cong \mathbb{Z}$ (degree 1, Chapter 1), the bundle is not trivial (Remark 2.7 for $k = 2$); see Figure 2.9 for a family of fibres in stereographic projection. $\square$

Remark 2.14: No global section: why. If a continuous global section $s : S^2 \rightarrow S^3$ existed, the bundle would be trivial, $S^3 \cong S^2 \times S^1$; but S^3 is simply connected whereas $\pi_1(S^2 \times S^1) \cong \mathbb{Z} \neq 0$: impossible. Equivalently and more strongly, Theorem 2.12 below gives $c_1 \neq 0$.

2.12. Canonical connection on S^3 and first Chern class

Definition 2.8: Canonical connection. On the Hopf fibration, the **canonical connection** is the real 1-form $\omega = \text{Im}(\bar{z}_1 dz_1 + \bar{z}_2 dz_2)$ on $S^3 \subset \mathbb{C}^2$.

Proposition 2.4: Pull-back in the two charts. With θ the colatitude ($w = \cos \theta$) and φ the longitude ($x + iy = \sin \theta e^{i\varphi}$),

$$\sigma_+^* \omega = -\frac{1}{2}(1 - \cos \theta) d\varphi, \quad \sigma_-^* \omega = \frac{1}{2}(1 + \cos \theta) d\varphi.$$

Proof. In coordinates (θ, φ) , $\sigma_+(\theta, \varphi) = (z_1, z_2)$ with $z_1 = \cos \frac{\theta}{2}$ (real, since $\sqrt{(1 + \cos \theta)/2} = \cos \frac{\theta}{2}$ for $\theta \in [0, \pi]$) and $z_2 = \sin \frac{\theta}{2} e^{-i\varphi}$ (from $x - iy = \sin \theta e^{-i\varphi}$ and $\sin \theta / \sqrt{2(1 + \cos \theta)} = \sin \frac{\theta}{2}$, the trigonometric half-angle identity). Then $dz_1 = -\frac{1}{2} \sin \frac{\theta}{2} d\theta$ (real), hence $\text{Im}(\bar{z}_1 dz_1) = 0$; and $dz_2 = -i \sin \frac{\theta}{2} e^{-i\varphi} d\varphi + \frac{1}{2} \cos \frac{\theta}{2} e^{-i\varphi} d\theta$, hence

$$\bar{z}_2 dz_2 = \sin \frac{\theta}{2} e^{i\varphi} \left(-i \sin \frac{\theta}{2} e^{-i\varphi} d\varphi + \frac{1}{2} \cos \frac{\theta}{2} e^{-i\varphi} d\theta \right) = -i \sin^2 \frac{\theta}{2} d\varphi + \frac{1}{2} \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta,$$

whose imaginary part is $-\sin^2 \frac{\theta}{2} d\varphi = -\frac{1}{2}(1 - \cos \theta) d\varphi$. Summing (the $d\theta$ part is real, it does not contribute to Im): $\sigma_+^* \omega = -\frac{1}{2}(1 - \cos \theta) d\varphi$. The computation for $\sigma_-(\theta, \varphi) = (\cos \frac{\theta}{2} e^{i\varphi}, \sin \frac{\theta}{2})$ is symmetric and gives $\sigma_-^* \omega = \cos^2 \frac{\theta}{2} d\varphi = \frac{1}{2}(1 + \cos \theta) d\varphi$. $\square$

Remark 2.15: Consistency with the transition function. $\sigma_-^* \omega - \sigma_+^* \omega = \frac{1}{2}(1 + \cos \theta) d\varphi + \frac{1}{2}(1 - \cos \theta) d\varphi = d\varphi$, exactly the gauge transformation rule $A_- - A_+ = g^{-1} dg / i$ for $g = e^{i\varphi}$ (since $g^{-1} dg = i d\varphi$): the canonical connection is compatible with the transition function found in Theorem 2.11, an independent cross-check.

Theorem 2.12: First Chern class of the Hopf fibration. With $A_+ := \sigma_+^* \omega$, the curvature $F = dA_+$ satisfies

$$\frac{1}{2\pi} \iint_{S^2} F = -1 :$$

the Chern class of the Hopf bundle has modulus 1 (Dirac quantisation), the generator of $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$.

Proof. $F = dA_+ = d[-\frac{1}{2}(1 - \cos \theta)] \wedge d\varphi = -\frac{1}{2} \sin \theta d\theta \wedge d\varphi$. Hence

$$\frac{1}{2\pi} \iint_{S^2} F = \frac{1}{2\pi} \int_0^{2\pi} \int_0^\pi \left(-\frac{1}{2} \sin \theta \right) d\theta d\varphi = \frac{1}{2\pi} \cdot 2\pi \cdot \left(-\frac{1}{2} \right) [-\cos \theta]_0^\pi = \frac{1}{2\pi} \cdot 2\pi \cdot \left(-\frac{1}{2} \right) \cdot 2 = -1.$$

The value -1 (rather than $+1$) reflects the choice of orientation of S^2 and the sense of the action $e^{i\alpha}$ fixed in this chapter; reversing either one obtains $+1$: the topologically significant invariant, independent of the conventions, is $|\frac{1}{2\pi} \iint F| = 1 \in \mathbb{Z} \setminus \{0\}$, which is what makes the bundle non-trivial. $\square$

Remark 2.16: Dirac quantisation. Interpreting $A_\pm$ as the vector potential of a magnetic monopole of charge g_{mag} placed at the origin (F19, F20), the consistency between the two charts ($A_- - A_+ = d\varphi$, a $U(1)$ transition function with well-defined values $e^{i\varphi}$, unique modulo 2π) forces $\frac{1}{2\pi} \iint F \in \mathbb{Z}$: this is exactly the **Dirac quantisation** of the magnetic charge, obtained here as a pure consequence of the topology of the circle (the degree of the equatorial transition function).

2.13. Berry phase

Remark 2.17: Physical model: Berry phase of the spin-1/2. Hypotheses of the model (F16, F17): a spin-1/2 in the ground state of a magnetic field $\mathbf{B}(t)$ of direction $\hat{n}(t) \in S^2$ varying continuously and slowly with respect to the Larmor frequency (adiabatic approximation, adiabatic theorem imported). When $\hat{n}(t)$ traverses a closed loop enclosing a cap R of solid angle $\Omega = \text{Area}(R)$, the state accumulates, besides the dynamical phase, a **geometric phase** γ equal to the holonomy of the canonical connection (Definition 2.8) along

the loop: for a loop = parallel of colatitude θ_0 (cap R , $\Omega = 2\pi(1 - \cos \theta_0)$, Corollary 2.3),

$$\gamma = \oint A_+ = \iint_R F = -\frac{1}{2}\Omega = -\pi(1 - \cos \theta_0) :$$

the Berry phase is **half** the solid angle subtended, in agreement with Berry's classical result (1984, F16) and with its experimental verification (magnetic resonance, optics); see Figure 2.10.

Remark 2.18: The factor 2 with parallel transport. Comparing with Section 2.10: the real parallel transport on the tangent bundle of S^2 gives holonomy $\Delta\varphi = \Omega$ (the whole solid angle, Corollary 2.3), whereas the Berry phase of the spin-1/2 gives half of it, $\Omega/2$. It is not a numerical coincidence but a fact about representations: the tangent bundle has structure group $SO(2)$ in the “vector” representation (the rotation by an angle α acts as $e^{i\alpha}$ on a complex vector v seen in $\mathbb{C} \cong \mathbb{R}^2$), whereas the spin-1/2 lives in the “two-valued” representation of $SO(2)$ lifted to $U(1)$ via $\alpha \mapsto e^{i\alpha/2}$ (the same square root of phase that appears in the double covering $S^3 \rightarrow SO(3)$, Volume III Chapter 28): the canonical connection of the Hopf fibration is literally “the square root” of the Levi-Civita connection, and the factor 2 between the holonomies is its direct trace.

2.14. S^1 -bundles over S^2 with Euler number n

Remark 2.19: Panorama: lens spaces. Generalising the clutching construction to principal $U(1)$ bundles over $S^2 = D_+^2 \cup D_-^2$ (Remark 2.7, $k = 2$), the transition function on the equator $\varphi \mapsto e^{in\varphi}$ ($n \in \mathbb{Z}$, the degree) defines, for every n , an S^1 -bundle over S^2 of **Euler number** n ; its total space is the **lens space** $L(n, 1)$ (for $n \neq 0$), obtained by gluing two solid tori $D^2 \times S^1$ along the boundary T^2 with a twist of $1/n$ of a turn; $\pi_1(L(n, 1)) \cong \mathbb{Z}/n$ (imported, F14: long exact homotopy sequence of the bundle, or van Kampen on the gluing). For $n = 1$ one recovers the Hopf fibration ($L(1, 1) \cong S^3$, simply connected, consistently with $\pi_1(S^3) = 0$); for $n = 2$, $L(2, 1) \cong \mathbb{RP}^3 \cong SO(3)$ (Corollary 2.1, referring to Volume III Chapter 28 for the double covering $S^3 \rightarrow SO(3)$); see the diagram of $L(3, 1)$ in Figure 2.12.

2.15. Exponential quadrature on the circle

Theorem 2.13: Spectral convergence of the periodic trapezoidal rule. Let $A : \mathbb{R} \rightarrow \mathbb{R}$ be 2π -periodic, analytic, holomorphically extendable to the strip $|\operatorname{Im} \theta| < a_0$ with at most simple poles on the boundary $\operatorname{Im} \theta = \pm a_0$. The periodic trapezoidal rule (equivalent to the Riemann sum with constant step, by periodicity)

$$T_N(A) := \frac{2\pi}{N} \sum_{k=0}^{N-1} A\left(\frac{2\pi k}{N}\right)$$

satisfies $|T_N(A) - \oint A| \leq C e^{-Na_0}$ for a constant C independent of N .

Imported. Imported from F23 (Trefethen–Weideman, *SIAM Review* 56 (2014), Theorem 2.1 and thereabouts) — the complete proof uses the residue theorem of Volume III, Chapter 15, applied on a rectangular contour that translates the strip of analyticity, seeing the error of the quadrature as a sum over all the poles of the $2\pi\mathbb{Z}$ -translates of the function,

Example 2.1: Numerical check for $A(\theta) = 0.3/(1.5 - \cos \theta)$. A has poles where $\cos \theta = 1.5$, that is $\theta = \pm i a_0$ with $a_0 = \operatorname{arccosh}(1.5) \approx 0.9624$ (the root of the strip of analyticity). Exact value: $\oint_0^{2\pi} A = \frac{0.3 \cdot 2\pi}{\sqrt{1.5^2 - 1}} = 1.685955535450\dots$, computed in closed form and by independent adaptive quadrature (discrepancy between the two methods $< 10^{-15}$, script `fig02_trapezio_errone.py`, Appendix B). Errors of the trapezoidal rule: $N = 4$: 7.33×10^{-2} ; $N = 8$: 1.53×10^{-3} ; $N = 16$: 6.92×10^{-7} ; $N = 24$: 3.14×10^{-10} ; $N = 32$: 1.42×10^{-13} ; $N = 48$: at the level of machine precision. The decay follows Ke^{-Na_0} (Figure 2.1), confirming quantitatively Theorem 2.13, and it replaces the previous example $A = \frac{1}{2} \cos \theta$ (for which the left sum is identically 0 for every $N \geq 2$: an empty example, because $\oint \cos \theta d\theta = 0$ already exactly, with nothing to converge).

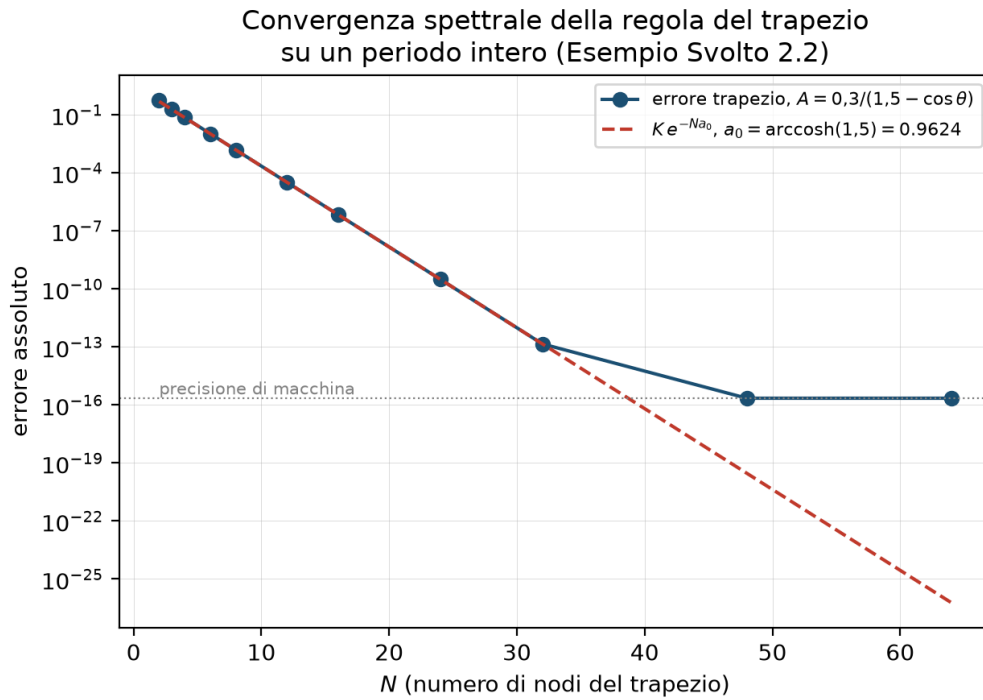


Figure 2.1: Spectral convergence of the trapezoidal rule for $A(\theta) = 0.3/(1.5 - \cos \theta)$ (Example 2.1): absolute error on a semilogarithmic scale against the theoretical line Ke^{-Na_0} , $a_0 = \operatorname{arccosh}(1.5)$. Exact value verified with a closed formula and independent adaptive quadrature (discrepancy 0 within machine precision). Script `fig02_trapezio_errone.py`, Appendix B.

2.16. Figures

2.17. Worked examples

Worked Example 2.2 — monodromy of the Möbius band, direct check Problem.

Verify explicitly that the cocycle of the Möbius band cannot be made trivial by any change of trivialisation.

Step 1. Going around S^1 from a point of A , through U_2 , entering B and returning to U_1 ,

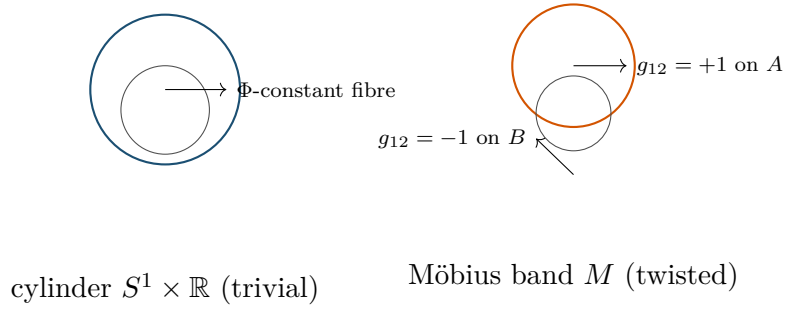


Figure 2.2: The two real line bundles over S^1 (Theorem 2.7, $n = 1$): the trivial cylinder and the Möbius band, distinguished by the monodromy ± 1 (Theorem 2.4).

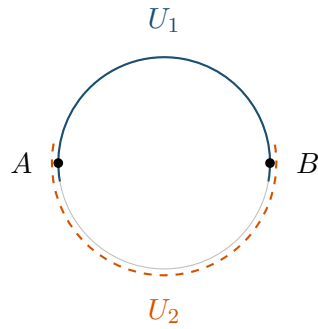


Figure 2.3: Trivialising cover $S^1 = U_1 \cup U_2$ with intersection $A \sqcup B$ (Section 2.5): the cocycle of the Möbius band equals $+1$ near A and -1 near B .



Figure 2.4: Clutching construction (Theorem 2.6) on S^1 (left, transition function g on two points) and on S^2 (right, transition function on the equator $S^1 = \partial D_+^2 = \partial D_-^2$, Remark 2.7).

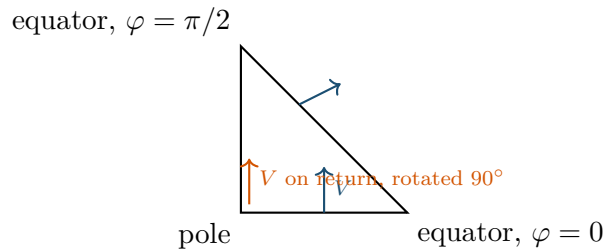


Figure 2.5: Parallel transport along the boundary of the octant (Worked Example 2.1, Section 2.9): the vector comes back rotated by $\pi/2$, equal to the area $\pi/2$ of the octant (Theorem 2.10).

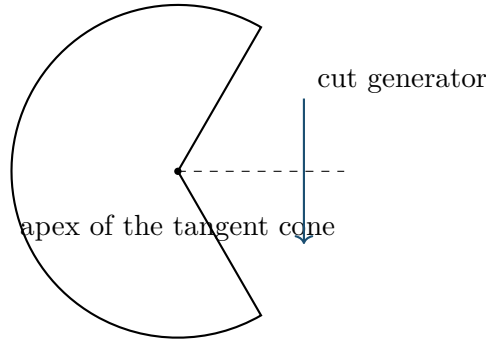


Figure 2.6: Planar development of the tangent cone along the parallel of colatitude θ_0 (diagram, opening angle reduced to represent the deficit with respect to 2π , consistent with $\Delta\varphi = 2\pi(1 - \cos\theta_0)$, Corollary 2.3): on the cone, intrinsically flat away from the apex, parallel transport is ordinary Euclidean transport.

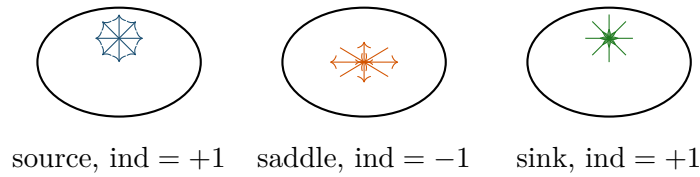


Figure 2.7: The three types of non-degenerate critical point of a Morse height (Proposition 2.3): source (minimum), saddle, sink (maximum), with index $(-1)^k$; on the standard sphere, 1 source + 0 saddles + 1 sink gives $\chi(S^2) = 2$, on a standard torus $1 + 2 \cdot (-1) + 1 = 0$ (one source, two saddles of index -1 each, one sink).

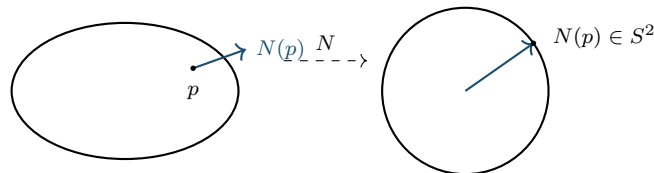


Figure 2.8: The Gauss map $N : \Sigma \rightarrow S^2$ (Remark 2.10): the unit normal vector at p seen as a point of S^2 .

Fibrazione di Hopf, proiezione stereografica: 8 fibre concatenate
su un toro di Villarceau ($\theta_0 = \pi/3$)

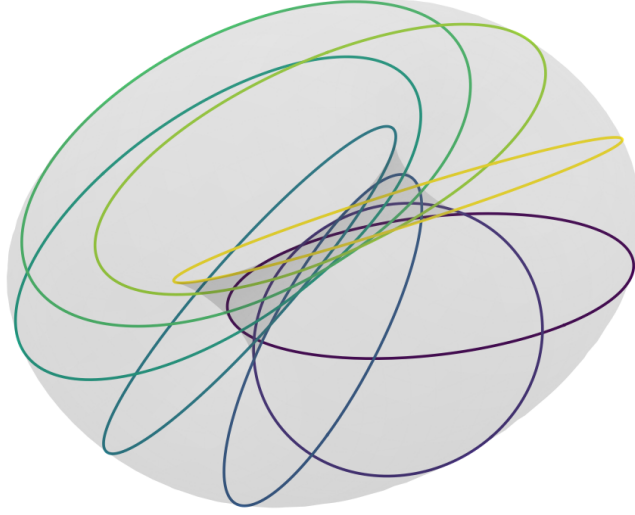


Figure 2.9: Hopf fibration in stereographic projection: 8 fibres at fixed colatitude $\theta_0 = \pi/3$, linked on a torus of revolution (Villarceau); independent numerical check that $\pi(\text{fibre}(\theta_0, \varphi_0)) = (\theta_0, \varphi_0)$ is constant along each fibre (maximum error 4.4×10^{-16}) and that the generating profile of the guiding torus really is a circle (deviation from the fit 2.4×10^{-15}). Script `fig02_hopf_villarceau.py`, Appendix B.

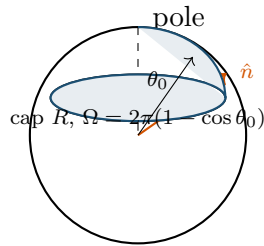


Figure 2.10: Cap R of solid angle Ω subtended by the loop $\hat{n}(t)$ (Remark 2.17): the Berry phase equals $-\Omega/2$.

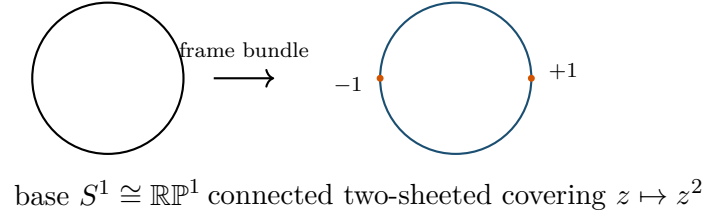


Figure 2.11: The frame bundle (orthonormal frames) of the Möbius band is the connected two-sheeted covering $p_2(z) = z^2$ of Chapter 1: the two frames ± 1 are exchanged along the monodromy (Worked Example 2.4, Section 2.17).

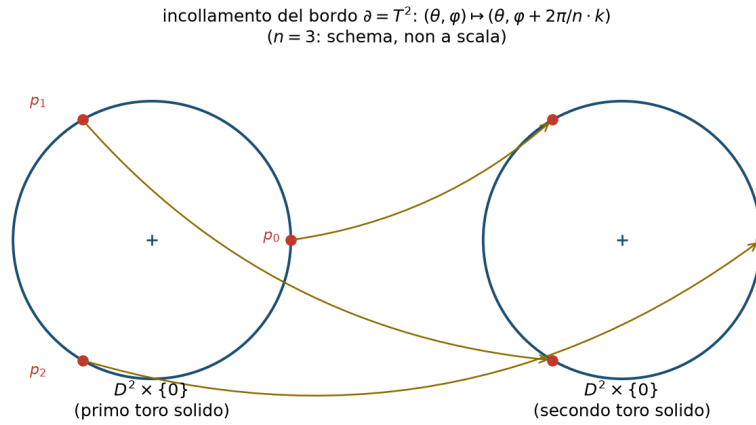


Figure 2.12: Lens space $L(3,1)$ (Remark 2.19) as the gluing of two solid tori along the boundary, a diagram with 3 gluing points computed exactly (not by hand). Script `fig02_spazio_lenticolare.py`, Appendix B.

a vector is multiplied by $g_{12}|_A \cdot g_{12}|_B^{-1} = (+1)(-1)^{-1} = -1$.

Step 2. Changing trivialisation by continuous nowhere-vanishing h_1, h_2 , the new cocycle is $g'_{12} = h_2 g_{12} h_1^{-1}$, and the new total monodromy is $(h_2 g_{12}|_A h_1^{-1})(h_1 g_{12}|_B^{-1} h_2^{-1}) = g_{12}|_A \cdot g_{12}|_B^{-1} = -1$: the factors h_1, h_2 cancel exactly.

Step 3. The monodromy $-1 \neq +1$ (the value for the cylinder) is therefore an invariant: the two bundles are not isomorphic (Theorem 2.4). $\square$

Worked Example 2.3 — numerical holonomy via the trapezoidal rule Problem.

Compute numerically $\oint_0^{2\pi} A_+(\theta_0) d\varphi$ for $\theta_0 = \pi/3$ (with $A_+ = -\frac{1}{2}(1 - \cos \theta_0)$, constant along the parallel) and compare with the closed-form value.

Step 1 — closed-form value. A_+ is constant in φ : $\oint A_+ d\varphi = 2\pi \cdot (-\frac{1}{2}(1 - \cos(\pi/3))) = 2\pi \cdot (-\frac{1}{2} \cdot \frac{1}{2}) = -\frac{\pi}{2}$.

Step 2 — trapezoidal rule. Since the integrand is *constant*, the periodic trapezoidal rule is exact for every $N \geq 1$ (a sum of N identical values for the step $2\pi/N$): $T_N = N \cdot (2\pi/N) \cdot A_+ = 2\pi A_+ = -\pi/2$ exactly, no discretisation error to control in this degenerate case.

Step 3 — interpretation. $-\pi/2 = -\frac{1}{2} \cdot \pi$, and $\Omega = 2\pi(1 - \cos(\pi/3)) = \pi$: the Berry phase is $-\Omega/2$, in agreement with Remark 2.17. $\checkmark$ $\square$

Worked Example 2.4 — frame bundle of the Möbius band Problem. Identify the bundle of orthonormal frames (frame bundle) of the Möbius band.

Step 1. For a line bundle with a metric, the orthonormal frames of $E_x \cong \mathbb{R}$ are the two unit vectors ± 1 : the frame bundle has discrete fibre $O(1) = \{\pm 1\}$, hence it is a two-sheeted covering of S^1 .

Step 2. By the classification of coverings (Chapter 1), the two-sheeted coverings are two: the disconnected one $S^1 \sqcup S^1$ and the connected one $p_2(z) = z^2$. The cocycle -1 of Theorem 2.4 exchanges the two frames $+1, -1$ after one turn: the total space is connected, hence it is p_2 .

Step 3. The boundary of the Möbius band (the points $t = \pm 1$ in the parametrisation of Theorem 2.3) is precisely this covering: a single circle winding twice around the base — the fact, verifiable by physically cutting a Möbius band along its boundary, that the boundary is connected (Figure 2.11). $\square$

2.18. Assigned exercises

The answers are in the Teacher's key (Section 2.21).

Exercise 2.2: Pencil of two secant circles and circular points. *Goal:* to apply Theorem 2.1 to an explicit pencil. *Estimated time:* 25 minutes. *Type:* symbolic.

Let $f_1 = x^2 + y^2 - 1$, $f_2 = x^2 + y^2 - 2x$ (circles of radius 1 centred at $(0, 0)$ and $(1, 0)$).

- Determine the two real points common to f_1, f_2 .
- Write $f_\lambda = f_1 + \lambda(f_2 - f_1)$ and check that it is still a circle for every $\lambda \neq \infty$ except for one value, for which it is a line (the radical axis): find it.
- Check by homogenising that f_λ passes through $I = [1 : i : 0]$ and $J = [1 : -i : 0]$ for every λ .

Exercise 2.3: Whitney sum of two Möbius bands. *Goal:* to verify $M \oplus M \cong S^1 \times \mathbb{R}^2$ in the language of clutching. *Estimated time:* 30 minutes. *Type:* symbolic.

- Write the cocycle of $M \oplus M$ on the component B (Theorem 2.6, $n = 2$).

- (b) Find a continuous path in $GL_2(\mathbb{R})$ from the identity to that matrix.
(c) Conclude, with Lemma 2.1, that $M \oplus M$ is trivial.

Exercise 2.4: Holonomy on a cap of variable colatitude. *Goal:* to use Corollary 2.3 and the Berry phase to compare two caps. *Estimated time:* 30 minutes. *Type:* numerical.

- (a) Compute the holonomy of parallel transport for $\theta_0 = \pi/6$ and for $\theta_0 = 2\pi/3$.
(b) Compute the corresponding Berry phase in the two cases.
(c) Explain why for $\theta_0 \rightarrow \pi$ (a cap covering almost the whole sphere) the holonomy tends to 4π and discuss whether this is compatible with a rotation (an angle defined modulo 2π).

2.19. Worked exercises

Exercise 2.5: Numerical check of the correct Hopf section. *Type:* numerical. Verify for an explicit point $(x, y, w) \in S^2$ that $\pi \circ \sigma_+ = \text{id}$ with the correct formula of Theorem 2.11, and show that the old formula $z_2 = (x + iy)/\sqrt{2(1+w)}$ fails.

Solution. Strategy. Fix an explicit point and compare $\pi(\sigma_+(x, y, w))$ with (x, y, w) for both formulas for z_2 .

Step 1. Let $\theta_0 = \pi/3$, $\varphi_0 = 0.7$: $w = \cos \theta_0 = 0.5$, $x = \sin \theta_0 \cos \varphi_0 = 0.86603 \times 0.76484 = 0.66242$, $y = \sin \theta_0 \sin \varphi_0 = 0.86603 \times 0.64422 = 0.55793$.

Step 2 — correct formula. $z_1 = \sqrt{(1+w)/2} = \sqrt{0.75} = 0.86603$; $z_2 = (x - iy)/\sqrt{2(1+w)} = (0.66242 - 0.55793i)/1.73205 = 0.38251 - 0.32215i$. Then $2z_1\bar{z}_2 = 2 \times 0.86603 \times (0.38251 + 0.32215i) = 0.66242 + 0.55793i = x + iy$ ✓; and $|z_1|^2 - |z_2|^2 = 0.75 - (0.38251^2 + 0.32215^2) = 0.75 - (0.14631 + 0.10378) = 0.75 - 0.25 = 0.5 = w$ ✓.

Step 3 — old (wrong) formula. With $z'_2 = (x + iy)/\sqrt{2(1+w)} = 0.38251 + 0.32215i$: $2z_1\bar{z}'_2 = 2 \times 0.86603 \times (0.38251 - 0.32215i) = 0.66242 - 0.55793i = x - iy \neq x + iy$: one obtains $\pi(\sigma'_+) = (x, -y, w) \neq (x, y, w)$, confirming numerically the defect reported for the old version.

Exercise 2.6: Foucault pendulum precession in Rome. *Type:* numerical. Compute the daily precession of the Foucault pendulum in Rome ($\lambda_0 \approx 41.9^\circ$) in degrees per hour, and compare with the value at the pole and at the equator.

Solution. Strategy. Apply $\Omega_{\text{Foucault}} = 2\pi \sin \lambda_0$ per sidereal day (≈ 23.934 h).

Step 1. $\sin(41.9^\circ) = 0.6678$; $\Omega_{\text{Foucault}} = 360^\circ \times 0.6678 = 240.4^\circ$ per sidereal day, that is $240.4/23.934 = 10.05^\circ/\text{hour}$.

Step 2 — limiting cases. At the pole ($\lambda_0 = 90^\circ$): $\sin \lambda_0 = 1$, $\Omega_{\text{Foucault}} = 360^\circ/\text{day}$ ($15^\circ/\text{hour}$, the pendulum follows exactly the apparent rotation of the sky, consistently with the vanishing holonomy of parallel transport around a loop of zero radius about the pole). At the equator ($\lambda_0 = 0$): $\Omega_{\text{Foucault}} = 0$, no precession (the plane of oscillation does not rotate at all with respect to the ground).

Step 3 — independent check. $\Delta\varphi = 2\pi(1 - \cos \theta_0)$ with $\theta_0 = 90^\circ - 41.9^\circ = 48.1^\circ$: $\cos 48.1^\circ = 0.6678$ (equal to $\sin 41.9^\circ$, as it must be), $\Delta\varphi = 360^\circ(1 - 0.6678) = 119.6^\circ$; and $360^\circ - 119.6^\circ = 240.4^\circ$, equal to Step 1. ✓

Exercise 2.7: Chern-Weil with the other orientation. *Type:* symbolic. Verify that choosing the chart σ_- in place of σ_+ in the computation of Theorem 2.12 one obtains the same F (and hence the same integral).

Solution. Strategy. Compute $F = dA_-$ with $A_- = \sigma_-^* \omega = \frac{1}{2}(1 + \cos \theta)d\varphi$ (Proposition 2.4) and compare with $F = dA_+$.

Step 1. $dA_- = d[\frac{1}{2}(1 + \cos \theta)] \wedge d\varphi = -\frac{1}{2} \sin \theta d\theta \wedge d\varphi$, identical to $dA_+ = -\frac{1}{2} \sin \theta d\theta \wedge d\varphi$ computed in Theorem 2.12.

Step 2 — why it must happen. $A_- - A_+ = d\varphi$ (Remark 2.15) is *exact*: $d(A_- - A_+) = d(d\varphi) = 0$, hence $dA_- = dA_+$ automatically, on the whole region (the equator excluding the poles) where both charts are defined — F is a global 2-form on $S^2 \setminus \{\text{poles}\}$ which extends continuously to the poles using the other chart, exactly the point of the two-chart construction.

Step 3. The integral $\frac{1}{2\pi} \iint F$ therefore does not depend on the chart chosen to compute it, as it must be for a topological invariant. ✓

2.20. Self-assessment quiz

- (Definition) In the definition of a bundle, the essential condition on the local trivialisation $\Phi : \pi^{-1}(U) \rightarrow U \times F$ is
 A) that it be linear B) that it commute with the projection, $\text{pr}_U \circ \Phi = \pi$ C) that it be an isometry D) that F be connected
- (Hypothesis) In the Poincaré–Hopf theorem (Theorem 2.9), the hypothesis on the zeros of the vector field is that they be
 A) all on the equator B) isolated (and finite in number) C) non-degenerate D) absent
- (Symbolic computation) The Chern class of the Hopf fibration, computed from $\frac{1}{2\pi} \iint F$ with $F = dA_+$, $A_+ = -\frac{1}{2}(1 - \cos \theta)d\varphi$, equals
 A) 0 B) ± 1 C) 2 D) π
- (Numerical computation) The precession of the Foucault pendulum at latitude $\lambda_0 = 30^\circ$, in degrees per sidereal day, equals
 A) 360° B) 180° C) 90° D) 30°
- (Proof step) In the proof of Lemma 2.3, the index remains constant along an arc of the cobordism because
 A) the field is linear B) the degree of a map $S^1 \rightarrow S^1$ varying continuously in the parameter is locally constant C) the arc is geodesic D) the surface is orientable
- (Error diagnosis) A student computes the Hopf section σ_+ with $z_2 = (x + iy)/\sqrt{2(1 + w)}$ and finds $\pi \circ \sigma_+ = (x, -y, w)$. The error is
 A) z_1 must be imaginary B) the sign of y in z_2 must be reversed: $z_2 = (x - iy)/\sqrt{2(1 + w)}$ C) one must use $w = -1$ D) the normalisation is wrong by a factor 2
- (Transfer) The clutching construction on S^1 (Theorem 2.6) extends to S^2 , classifying the complex vector bundles of rank n by
 A) $\pi_0(U(n))$ B) $\pi_1(U(n)) \cong \mathbb{Z}$ C) $H^1(S^2; \mathbb{Z})$ D) $\pi_2(S^1)$

8. (Source and metacognition) The formula $\Delta\varphi = \iint_R K dA$ for the holonomy (local Gauss–Bonnet, Theorem 2.10) is proved in this chapter using as an imported ingredient
- A) Sard’s theorem B) the existence of a smooth orthonormal frame on a simply connected region C) Bézout’s theorem D) Bott periodicity

2.21. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnosis.

- 1) **B.** *Rationale:* it is Definition 2.3: without commutation with π , any homeomorphism would satisfy the definition (Remark 2.3). *Distractors:* A is true only for vector bundles, not in general; C is not required; D is false (the fibre $\mathbb{Z}$ of the covering is discrete). *Diagnosed error:* confusing the general definition with the case of vector bundles. *Source:* Definition 2.3. *Prerequisite:* homeomorphism. *Remedial work:* reread Remark 2.3.
- 2) **B.** *Rationale:* Theorem 2.9 requires isolated zeros (and hence, by compactness, finitely many); it does not require non-degeneracy (which is needed only in the proof, for the comparison field). *Distractors:* A, D are particular cases that are not required; C is stronger than necessary. *Diagnosed error:* confusing a hypothesis with a tool of the proof. *Source:* statement of Theorem 2.9. *Prerequisite:* definition of index. *Remedial work:* compare the statement with Lemma 2.3.
- 3) **B.** *Rationale:* the direct computation (Theorem 2.12) gives -1 ; the value $+1$ is obtained with the opposite orientation: the correct answer contemplates both signs. *Distractors:* A is the value for a trivial bundle; C is $\chi(S^2)$, another quantity; D is the Berry holonomy on half the sphere, not the Chern class. *Diagnosed error:* confusing c_1 with other integer constants of the chapter. *Source:* Theorem 2.12. *Prerequisite:* Stokes’ theorem. *Remedial work:* redo the computation with σ_- (Exercise 2.7).
- 4) **B.** *Rationale:* $2\pi \sin(30^\circ) = 2\pi \times 0.5 = \pi$, that is 180° . *Distractors:* A is the value at the pole; C confuses $\sin$ with a quarter turn; D uses λ_0 in place of $\sin \lambda_0 \times 360^\circ$. *Diagnosed error:* forgetting the sine in the formula. *Source:* Remark 2.12. *Prerequisite:* elementary trigonometry. *Remedial work:* redo Exercise 2.6 with $\lambda_0 = 30^\circ$.
- 5) **B.** *Rationale:* it is the crucial point of the proof (Lemma 2.3): the index, being an integer degree varying continuously, cannot jump. *Distractors:* A, C, D are not used at this step. *Diagnosed error:* failing to recognise where the homotopy invariance of the degree of Chapter 1 comes in. *Source:* Lemma 2.3, second paragraph. *Prerequisite:* degree of maps $S^1 \rightarrow S^1$ (Chapter 1). *Remedial work:* review the homotopy invariance of the degree in Chapter 1.
- 6) **B.** *Rationale:* it is exactly the correction of Theorem 2.11 and of Exercise 2.5. *Distractors:* A, C, D do not correct the sign error actually present. *Diagnosed error:* conjugation forgotten in the definition of π . *Source:* proof of Theorem 2.11. *Prerequisite:* Re, Im of a complex product. *Remedial work:* redo Exercise 2.5 step by step.
- 7) **B.** *Rationale:* Remark 2.7, complex case $k = 2$: $U(n)$ connected, only $\pi_1(U(n)) \cong \mathbb{Z}$ remains. *Distractors:* A is the case $k = 1$; C, D are not the correct groups. *Diagnosed error:* failing to distinguish π_0 (relevant for S^1) from π_1 (relevant for S^2). *Source:* Remark 2.7. *Prerequisite:* higher homotopy groups (definition). *Remedial work:* compare with the case S^1 of Theorem 2.6.
- 8) **B.** *Rationale:* the proof of Theorem 2.10 uses a smooth orthonormal frame on R (which exists because R is simply connected), not Sard nor Bézout at this point. *Distractors:* A is used elsewhere in the chapter (Poincaré–Hopf); C is used for the pencil of conics; D is not used at all in the chapter (only cited as panorama). *Diagnosed error:* confusing the im-

ported ingredients of the various proofs of the chapter. *Source:* proof of Theorem 2.10, first paragraph. *Prerequisite:* simple connectivity. *Remedial work:* reread the proof annotating every imported fact.

Assigned exercises — brief answers.

Ex. 2.2. (a) $f_2 - f_1 = -2x + 1 = 0 \Rightarrow x = \frac{1}{2}$, and from $f_1 = 0$: $y = \pm \frac{\sqrt{3}}{2}$: the points $(\frac{1}{2}, \pm \frac{\sqrt{3}}{2})$. (b) The degenerate value is $\lambda = -1$ ($f_1 - f_2 = 2x - 1 = 0$, the radical axis, a line). (c) On homogenising, the coefficient of x^2, y^2 in f_λ never vanishes for $w = 0$ except in the degenerate case, and at $w = 0$ there always remains $x^2 + y^2 = 0$: I, J satisfy every f_λ .

Ex. 2.3. (a) $\text{diag}(-1, -1) = -I_2$. (b) $R(s) = \begin{pmatrix} \cos \pi s & -\sin \pi s \\ \sin \pi s & \cos \pi s \end{pmatrix}$, $R(0) = I_2$, $R(1) = -I_2$. (c) By Lemma 2.1 the bundle with cocycle $-I_2$ is isomorphic to the one with cocycle I_2 : trivial.

Ex. 2.4. (a) $\theta_0 = \pi/6$: $\Delta\varphi = 2\pi(1 - \cos 30^\circ) = 2\pi(1 - 0.8660) = 0.8418$ rad; $\theta_0 = 2\pi/3$: $\Delta\varphi = 2\pi(1 - \cos 120^\circ) = 2\pi(1.5) = 3\pi \approx 9.4248$ rad. (b) Berry phase $= -\Delta\varphi/2$: -0.4209 rad and -4.7124 rad. (c) For $\theta_0 \rightarrow \pi$, $\Delta\varphi \rightarrow 4\pi$: an angle defined modulo 2π “makes one and a half turns”, perfectly compatible with a rotation (the holonomy is not constrained to $[0, 2\pi)$: it is the measure of the *accumulated area*, which may exceed 2π ; only $\Delta\varphi \bmod 2\pi$ is the physically observable angle of the vector).

2.22. Chapter formulary

Bundle, cocycle

$$\text{pr}_U \circ \Phi = \pi, \quad g_{ii} = I, \quad g_{ij}g_{ji} = I, \quad g_{ij}g_{jk} = g_{ik}$$

Index and Poincaré–Hopf

$$\text{ind}_p(V) = \deg \left(\frac{V(\gamma_\epsilon(\theta))}{|V(\gamma_\epsilon(\theta))|} \right), \quad \sum_p \text{ind}_p(V) = \chi(\Sigma) = 2 - 2g$$

$$\text{index at a Morse critical point of index } k : (-1)^k$$

Classification over S^1

$$\text{Vect}_n(S^1) \leftrightarrow \pi_0(O(n)) \cong \mathbb{Z}/2 \ (\forall n \geq 1), \quad \{\text{principal } G \text{ bundles over } S^1\} \leftrightarrow \pi_0(G)/\text{conjugation}$$

Local Gauss–Bonnet and holonomy

$$\Delta\varphi = \iint_R K \, dA, \quad \text{parallel at colatitude } \theta_0 : \Delta\varphi = 2\pi(1 - \cos \theta_0)$$

Foucault

$$\Omega_{\text{Foucault}} = 2\pi - \Delta\varphi = 2\pi \sin \lambda_0 \text{ per sidereal day}$$

Hopf fibration

$$\pi(z_1, z_2) = (2 \text{Re}(z_1 \bar{z}_2), 2 \text{Im}(z_1 \bar{z}_2), |z_1|^2 - |z_2|^2), \quad g_{-+}(\varphi) = e^{i\varphi}$$

$$\sigma_+(x, y, w) = \left(\sqrt{\frac{1+w}{2}}, \frac{x-iy}{\sqrt{2(1+w)}} \right), \quad \omega = \text{Im}(\bar{z}_1 dz_1 + \bar{z}_2 dz_2)$$

Chern–Weil and Berry

$$\frac{1}{2\pi} \iint_{S^2} F = \pm 1 \in \mathbb{Z}, \quad \gamma_{\text{Berry}} = -\frac{1}{2}\Omega$$

Exponential trapezoidal rule

$$T_N(A) = \frac{2\pi}{N} \sum_{k=0}^{N-1} A\left(\frac{2\pi k}{N}\right), \quad |T_N(A) - \oint A| \leq C e^{-N a_0}$$

Pencil of conics

$$f_{[\lambda:\mu]} = \lambda f_1 + \mu f_2, \quad \text{circles} \iff \text{conics through } I = [1:i:0], \ J = [1:-i:0]$$

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Part B

Rotations and Diffeomorphisms

CHAPTER 3

Rotations, continued fractions and equidistribution

INFO

Level and prerequisites. Level **A–B** core (Sections 3.1–3.6 and 3.13: rotations, continued fractions, best approximation, three-distance theorem, phyllotaxis), accessible with trigonometry alone and the notion of a limit — the first pages of the chapter are meant for a secondary-school student who has met sequences. Sections 3.7–3.14 are of level **C–D** (Weyl’s criterion, Fejér’s theorem, discrepancy, Koksma’s and Erdős–Turán’s inequalities): they require the Riemann integral and Fourier series. Prerequisites within the treatise: the rotation R_α as an isometry of $\mathbb{T}$ and its matrix (Volume I); the density of the irrational orbit, the Haar measure $d\theta/2\pi$ and the characters of $\mathbb{T}$, already proved in full in Volume III, Chapter 13, and here only recalled; Fourier series (Volume II, Chapter 4). Remedial path: a reader who has not yet met continued fractions may start directly from Section 3.2, which is self-contained.

The circle rotates upon itself in one essential way only: by a fixed angle α , always in the same sense. And yet out of this elementary gesture there arise, according as α is commensurable with the full turn or not, two opposite worlds — a finite world, made of polygons that close up, and an infinite world, in which no orbit ever repeats itself and yet every orbit visits every neighbourhood of every point. The *quantitative* measure of this density — how fast, with what regularity — is the guiding thread of the whole chapter: from continued fractions (the tool with which one measures how well an irrational lets itself be approximated by a rational) through to discrepancy and equidistribution, ending with an application that at first sight seems extremely remote, the arrangement of the seeds in a sunflower.

3.1. The rotation of the circle: the rational/irrational dichotomy

Definition 3.1: Rotation map. For $\alpha \in \mathbb{R}$, the **rotation map** $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ is $R_\alpha(\theta) = \theta + \alpha \pmod{2\pi}$: in the matrix form of Volume I it is the rotation of the plane by the angle α , restricted to the unit circle. We write R_α^n for the n -th iterate, $R_\alpha^n(\theta) = \theta + n\alpha \pmod{2\pi}$, and we call the set $\{R_\alpha^n(\theta_0) : n \in \mathbb{Z}\}$ the **orbit** of θ_0 .

Theorem 3.1: Rational/irrational dichotomy. Let $\alpha = 2\pi p/q$ with $p/q \in \mathbb{Q}$ in lowest terms ($q \geq 1$, $\gcd(p, q) = 1$). Then every orbit of R_α is periodic with least period exactly q : the orbits are the vertices of regular polygons (with q vertices, possibly star polygons if $\gcd(p, q) = 1$ and $p > 1$). If instead $\alpha/2\pi \notin \mathbb{Q}$, every orbit of R_α is dense in $\mathbb{T}$ and no orbit is periodic.

Proof. Rational case. $R_\alpha^q(\theta) = \theta + q\alpha = \theta + 2\pi p \equiv \theta \pmod{2\pi}$: the orbit closes up after q steps. It is the least period because, if it closed up after $k < q$ steps ($R_\alpha^k(\theta) = \theta$), then $k\alpha \in 2\pi\mathbb{Z}$, that is $kp/q \in \mathbb{Z}$; since $\gcd(p, q) = 1$, this forces $q \mid k$, impossible for $0 < k < q$.

Irrational case. The density of the orbit of R_α for $\alpha/2\pi$ irrational, and the uniqueness of the invariant measure $d\theta/2\pi$ which implies it, have already been proved in full in the treatise (Volume III, Chapter 13): we do not repeat them. We give here only a second, elementary and independent proof, which will repeatedly use the (*pigeonhole*) argument in the rest of the chapter. Fix $N \geq 1$ and subdivide $\mathbb{T}$ into N arcs of length $2\pi/N$; among the $N+1$ points $\{\theta_0 + k\alpha \bmod 2\pi\}_{k=0}^N$ two fall into the same arc (pigeonhole): $|(\theta_0 + i\alpha) - (\theta_0 + j\alpha)| < 2\pi/N \pmod{2\pi}$ for some pair $i \neq j$ in $\{0, \dots, N\}$. Setting $m = i - j \neq 0$, we get $|m\alpha \bmod 2\pi| < 2\pi/N$ (up to reflecting m into $-m$, one may assume $m > 0$): the rotation $R_{m\alpha}$ displaces every point by less than $2\pi/N$. Iterating $R_{m\alpha}$ starting from an arbitrary point ψ_0 , the points $\psi_0, R_{m\alpha}(\psi_0), R_{2m\alpha}(\psi_0), \dots$ advance in steps $< 2\pi/N$ (in the same sense, being all translations by the same angle $m\alpha \bmod 2\pi \in (0, 2\pi/N)$), hence they must pass within $2\pi/N$ of every point of $\mathbb{T}$ before returning close to ψ_0 : every point of $\mathbb{T}$ lies at distance less than $2\pi/N$ from some iterate of θ_0 under R_α (the iterates of $R_{m\alpha}$ are iterates of R_α in steps that are multiples of m). Letting $N \rightarrow \infty$ one obtains density. Non-periodicity is immediate: $n\alpha \equiv 0 \pmod{2\pi}$ would imply $\alpha/2\pi = k/n \in \mathbb{Q}$ for some integer k , against the hypothesis. $\square$

***Remark 3.1: Quasi-periodic motion.** The irrational case is the simplest prototype of **quasi-periodic motion**: an orbit that never repeats itself exactly but returns arbitrarily close to each of its past states. It is the starting point of the whole KAM theory (Kolmogorov–Arnold–Moser), treated in Chapter 6 of this volume, and it was already the historical starting point of the question the rest of the chapter answers: how fast does the orbit come close to every point, and with what regularity?*

3.2. Continued fractions

To answer the question of Remark 3.1 one needs a tool that measures *how well* an irrational number lets itself be approximated by fractions: continued fractions, which are nothing but Euclid’s algorithm applied to a real number instead of to a pair of integers.

Construction 3.1: The continued fraction algorithm. Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. Set $x_0 = \alpha$ and, recursively for $n \geq 0$,

$$a_n = \lfloor x_n \rfloor \in \mathbb{Z}, \quad x_{n+1} = \frac{1}{x_n - a_n}.$$

The recursion never stops: if α is irrational, every x_n is irrational (by induction: $x_0 = \alpha$ irrational; if x_n is irrational so is $x_n - a_n$, and the reciprocal of an irrational is irrational), hence $x_n - a_n \neq 0$ always and x_{n+1} is defined for every n . For $n \geq 1$, $x_n - a_{n-1} = 1/x_{n-1}(\dots)$ — more simply, since $x_{n-1} - a_{n-1} \in (0, 1)$ (fractional part, never zero by irrationality), we have $x_n = 1/(x_{n-1} - a_{n-1}) > 1$ for $n \geq 1$, hence $a_n = \lfloor x_n \rfloor \geq 1$ for $n \geq 1$ (whereas $a_0 \in \mathbb{Z}$ may have any sign). The integers $a_0, a_1, a_2, \dots$ are called the **partial quotients** of α , and x_n the n -th **complete quotient**. $\square$

Theorem 3.2: Existence and uniqueness of the continued fraction expansion. For every $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ there exists a unique sequence $(a_n)_{n \geq 0}$ with $a_0 \in \mathbb{Z}$, $a_n \in \mathbb{Z}_{\geq 1}$ for $n \geq 1$, such that

$$\alpha = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}} \quad (=: [a_0; a_1, a_2, a_3, \dots]),$$

where the right-hand side is the limit of the finite truncations (Definition 3.2 below). The sequence is exactly the one produced by the algorithm of Construction 3.1.

Proof. Existence. Let (a_n) be the sequence of the algorithm. For every N , iterating the definition $x_n = a_n + 1/x_{n+1}$ one obtains the *exact* identity (not a limit, for the moment)

$$\alpha = a_0 + \frac{1}{a_1 + \frac{1}{\ddots + \frac{1}{a_{N-1} + \frac{1}{x_N}}}}. \quad (3.1)$$

The convergence of the infinite truncation to this same value α , as N grows, will follow from the error estimate of Theorem 3.3 below, proved independently of this theorem (using only the identity (3.1) and the recurrences of Proposition 3.1): the N -th truncation of the right-hand side is exactly the convergent p_{N-1}/q_{N-1} , and $|\alpha - p_{N-1}/q_{N-1}| < 1/(q_{N-1}q_N) \rightarrow 0$ because $q_N \rightarrow \infty$ (Proposition 3.1, which shows that q_n is strictly increasing for $n \geq 1$). Hence the right-hand side of the statement converges, and it converges to α .

Uniqueness. Suppose $\alpha = [b_0; b_1, b_2, \dots]$ for another admissible sequence. The value of the whole expression, since $b_1 + 1/(\dots) > 1$ for the part after b_0 (sum of an integer ≥ 1 and of a positive reciprocal < 1), satisfies $[\alpha] = b_0$ (the same argument gives $[\alpha] = a_0$ for the sequence of the algorithm): hence $b_0 = a_0$. Subtracting a_0 and inverting, the number $\beta := 1/(\alpha - a_0) = [b_1; b_2, \dots]$ satisfies exactly the same recursive construction starting from x_1 : by induction on n one obtains $b_n = a_n$ for every n . $\square$

Definition 3.2: Convergents. Given the partial quotients $a_0, a_1, a_2, \dots$ of α , the **convergents** are the fractions p_n/q_n defined by the recurrences

$$\begin{aligned} p_{-1} &= 1, \quad p_{-2} = 0, & q_{-1} &= 0, \quad q_{-2} = 1, \\ p_n &= a_n p_{n-1} + p_{n-2}, & q_n &= a_n q_{n-1} + q_{n-2} \quad (n \geq 0). \end{aligned}$$

Proposition 3.1: Elementary properties of the recurrences. For every $n \geq 0$: (i) p_n/q_n coincides with the truncation $[a_0; a_1, \dots, a_n]$ computed as an ordinary fraction; (ii) $q_n \geq 1$, and q_n is strictly increasing for $n \geq 1$ (with $q_0 = 1 \leq q_1 < q_2 < \dots$); (iii) the identity $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$ holds; (iv) $\gcd(p_n, q_n) = 1$; (v) for every $n \geq 0$ and every $t > 0$ the following exact identity holds:

$$a_0 + \frac{1}{a_1 + \frac{1}{\ddots + \frac{1}{a_{n-1} + \frac{1}{a_n + 1/t}}}} = \frac{tp_n + p_{n-1}}{tq_n + q_{n-1}} \quad (\text{in particular, with } t = x_{n+1}, \text{ this equals } \alpha).$$

Proof. (v) by induction on n : for $n = 0$, the left-hand side is $a_0 + 1/t = (t \cdot a_0 + 1)/t \cdot \dots$, that is $(ta_0 + 1)/t$; with the conventions $p_0 = a_0, p_{-1} = 1, q_0 = 1, q_{-1} = 0$ this reads $(tp_0 + p_{-1})/(tq_0 + q_{-1})$. Inductive step: substituting $t \rightarrow a_n + 1/t$ in the identity for $n - 1$ (valid for every positive argument, in particular for $a_n + 1/t$) one obtains $((a_n + 1/t)p_{n-1} + p_{n-2})/((a_n + 1/t)q_{n-1} + q_{n-2})$, which, multiplying numerator and denominator by t , gives $(t(a_n p_{n-1} + p_{n-2}) + p_{n-1})/(t(a_n q_{n-1} + q_{n-2}) + q_{n-1}) = (tp_n + p_{n-1})/(tq_n + q_{n-1})$ by the

definition of p_n, q_n .

(i) follows from (v) applied with $t = a_n$ (the truncation with n terms does not have the final $+1/t$, that is, it amounts to stopping with $x_n \rightarrow a_n$ exactly).

(ii) $q_0 = a_0 \cdot 0 + 1 = 1$; $q_1 = a_1 q_0 + q_{-1} = a_1 \geq 1$. For $n \geq 2$, $q_n = a_n q_{n-1} + q_{n-2} \geq q_{n-1} + q_{n-2} > q_{n-1}$ since $a_n \geq 1$ and $q_{n-2} \geq q_{-1} = 0$, with $q_{n-2} > 0$ as soon as $n \geq 2$ (and for $n = 1$, $q_1 = a_1 \geq 1 = q_0$, with equality possible only if $a_1 = 1$; strict increase from q_1 onwards follows in any case as soon as $q_{n-2} > 0$, that is $n \geq 2$; see Example 3.1 for the instance $\alpha = \varphi$ in which $q_0 = q_1 = 1$ exactly).

(iii) by induction: for $n = 0$, $p_0 q_{-1} - p_{-1} q_0 = a_0 \cdot 0 - 1 \cdot 1 = -1 = (-1)^{-1}$. Inductive step: $p_n q_{n-1} - p_{n-1} q_n = (a_n p_{n-1} + p_{n-2}) q_{n-1} - p_{n-1} (a_n q_{n-1} + q_{n-2}) = p_{n-2} q_{n-1} - p_{n-1} q_{n-2} = -(p_{n-1} q_{n-2} - p_{n-2} q_{n-1}) = -(-1)^{n-2} = (-1)^{n-1}$.

(iv) from (iii), every common divisor of p_n, q_n divides $(-1)^{n-1}$, hence it is ± 1 . $\square$

Theorem 3.3: Error estimate for the convergents. For every $n \geq 0$,

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}},$$

and the convergents approximate α alternately from below and from above: $\alpha - p_n/q_n$ has sign $(-1)^n$.

Proof. From Proposition 3.1(v) with $t = x_{n+1}$ (the exact identity $\alpha = (x_{n+1} p_n + p_{n-1}) / (x_{n+1} q_n + q_{n-1})$):

$$\alpha - \frac{p_n}{q_n} = \frac{x_{n+1} p_n + p_{n-1}}{x_{n+1} q_n + q_{n-1}} - \frac{p_n}{q_n} = \frac{q_n(x_{n+1} p_n + p_{n-1}) - p_n(x_{n+1} q_n + q_{n-1})}{q_n(x_{n+1} q_n + q_{n-1})} = \frac{p_{n-1} q_n - p_n q_{n-1}}{q_n(x_{n+1} q_n + q_{n-1})} = \frac{(-1)^n}{q_n(x_{n+1} q_n + q_{n-1})}$$

using Proposition 3.1(iii) in the last step. The sign is therefore $(-1)^n$, as announced. Since $x_{n+1} \in (a_{n+1}, a_{n+1} + 1)$ (by construction, $x_{n+1} = a_{n+1} + 1/x_{n+2}$ with $x_{n+2} > 1$), we have

$$a_{n+1} q_n + q_{n-1} < x_{n+1} q_n + q_{n-1} < (a_{n+1} + 1) q_n + q_{n-1},$$

that is $q_{n+1} < x_{n+1} q_n + q_{n-1} < q_{n+1} + q_n$ (using $q_{n+1} = a_{n+1} q_n + q_{n-1}$). Taking reciprocals and multiplying by $1/q_n$ one obtains exactly the double inequality of the statement. $\square$

Example 3.1: Continued fraction of π and of $\sqrt{2}$. Double check (Euclid's algorithm on π in 60-digit arithmetic, script `verifica_cap03_vol4.py`, block [C3-1], and comparison with the known sequence OEIS A001203). The first partial quotients of π are

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, \dots],$$

with convergents $3/1, 22/7, 333/106, 355/113, 103993/33102, \dots$. The jump $a_4 = 292$ (unusually large) explains why $355/113$ approximates π exceptionally well: by Theorem 3.3, $|\pi - 355/113| < 1/(113 \cdot 33102) = 2.67 \times 10^{-7}$, verified: $|\pi - 355/113| = 2.668 \times 10^{-7}$. $\checkmark$ For $\sqrt{2}$, the algorithm (Construction 3.1, or the elementary fact $\sqrt{2} - 1 = 1/(\sqrt{2} + 1)$) gives the *periodic* continued fraction

$$\sqrt{2} = [1; 2, 2, 2, 2, \dots],$$

with convergents $1/1, 3/2, 7/5, 17/12, 41/29, 99/70, \dots$: the denominators are exactly the Pell numbers P_{n+1} ($P_0 = 0, P_1 = 1, P_{n+1} = 2P_n + P_{n-1}$), an independent check confirmed by the script (block [C3-2]): $q_n = \text{Pell}(n+1)$ for $n = 0, \dots, 7$. $\checkmark$

3.3. Best approximation: two distinct kinds

Theorem 3.3 says that p_n/q_n approximates α well; it does not yet say, and it must be proved with care, in what precise sense p_n/q_n is the *best* approximation with bounded denominator. There are two distinct notions of “best”, and confusing them produces a false statement: this is exactly the error that appeared in the first draft of this chapter, corrected here.

Definition 3.3: Norm from the integer part. For $t \in \mathbb{R}$, $\|t\| := \min_{n \in \mathbb{Z}} |t - n|$ is the distance of t from the nearest integer.

Theorem 3.4: Best approximation of the second kind. For every $n \geq 1$ and every integer k with $0 < k < q_{n+1}$, and every integer p , if $(k, p) \neq (q_n, p_n)$ then

$$|k\alpha - p| > |q_n\alpha - p_n|.$$

In particular $\|k\alpha\| \geq \|q_n\alpha\|$ for $0 < k < q_{n+1}$, with equality only for $k = q_n$: among all fractions with denominator $< q_{n+1}$, the convergent p_n/q_n makes $|q\alpha - p|$ (the “second-kind distance”) strictly minimal.

Proof. Set $\delta_m = q_m\alpha - p_m$; by Theorem 3.3, $\text{sgno}(\delta_m) = (-1)^m$ and $|\delta_m|$ is strictly decreasing in m (for $m \geq 0$: $|\delta_m| < 1/q_{m+1} \leq q_m/q_{m+1} \cdot 1/q_m$, and more precisely we shall use below the exact identity). By Proposition 3.1(iii), the matrix $\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix}$ has determinant ± 1 , hence it is invertible over $\mathbb{Z}$: for all integers k, p there exists a unique pair of integers x, y with

$$k = xq_n + yq_{n-1}, \quad p = xp_n + yp_{n-1},$$

and then $k\alpha - p = x\delta_n + y\delta_{n-1}$.

Step 1 (auxiliary identity). From $\delta_{n-1} = q_{n-1}\alpha - p_{n-1}$ and $q_{n+1} = a_{n+1}q_n + q_{n-1}$, $p_{n+1} = a_{n+1}p_n + p_{n-1}$, we get $\delta_{n+1} = a_{n+1}\delta_n + \delta_{n-1}$, that is $\delta_{n-1} = \delta_{n+1} - a_{n+1}\delta_n$. Since δ_{n+1} and δ_{n-1} have the same sign $(-1)^{n-1} = (-1)^{n+1}$, opposite to that of δ_n , the two summands δ_{n+1} and $-a_{n+1}\delta_n$ have the same sign: hence $|\delta_{n-1}| = |\delta_{n+1}| + a_{n+1}|\delta_n| > a_{n+1}|\delta_n|$ (strictly, because $\delta_{n+1} \neq 0$: α is irrational). Set $\lambda := |\delta_{n-1}|/|\delta_n| > a_{n+1}$.

Step 2 (the cases $y = 0$ and $x = 0$). If $y = 0$: $k = xq_n$, and from $0 < k < q_{n+1}$ and $q_{n+1} > q_n$ it follows that $x \geq 1$; if $x = 1$ we are in the excluded case $(k, p) = (q_n, p_n)$; if $x \geq 2$, $|k\alpha - p| = x|\delta_n| \geq 2|\delta_n| > |\delta_n|$. If $x = 0, y \neq 0$: $|k\alpha - p| = |y||\delta_{n-1}| \geq |\delta_{n-1}| > |\delta_n|$ (Step 1, with $a_{n+1} \geq 1$).

Step 3 (opposite signs of x, y). If $x, y \neq 0$ have opposite signs, $x\delta_n$ and $y\delta_{n-1}$ have the same sign (the signs of δ_n, δ_{n-1} being opposite and the signs of x, y being opposite): $|k\alpha - p| = |x||\delta_n| + |y||\delta_{n-1}| \geq |\delta_n| + |\delta_{n-1}| > |\delta_n|$.

Step 4 (equal signs, the delicate case). There remains the case of x, y of the same sign, both non-zero. If both are negative, $k = xq_n + yq_{n-1} < 0$, excluded by $k > 0$. There remains $x, y \geq 1$. From $k = xq_n + yq_{n-1} < q_{n+1} = a_{n+1}q_n + q_{n-1}$ it follows that $(x - a_{n+1})q_n < (1 - y)q_{n-1} \leq 0$ (because $y \geq 1$), hence $x < a_{n+1}$, that is $x \leq a_{n+1} - 1$ (and in particular this case requires $a_{n+1} \geq 2$). Since δ_n, δ_{n-1} have opposite signs, we write $\delta_{n-1} = -\lambda\delta_n$ (Step 1), so that $x\delta_n + y\delta_{n-1} = (x - y\lambda)\delta_n$. From $\lambda > a_{n+1} > a_{n+1} - 1 \geq x$ and $y \geq 1$: $y\lambda \geq \lambda > a_{n+1} > x$, hence $x - y\lambda < 0$ and

$$|x - y\lambda| = y\lambda - x > y a_{n+1} - x \geq y a_{n+1} - (a_{n+1} - 1) = (y - 1)a_{n+1} + 1 \geq 1$$

(the last inequality using $y \geq 1$), with the first inequality strict because $\lambda > a_{n+1}$ strictly. Hence $|k\alpha - p| = |x - y\lambda||\delta_n| > |\delta_n|$.

In all cases (Steps 2–4) one concludes $|k\alpha - p| > |\delta_n| = |q_n\alpha - p_n|$ for $(k, p) \neq (q_n, p_n)$, $0 < k < q_{n+1}$. $\square$

Corollary 3.1: Best approximation of the first kind. For every $n \geq 1$, every integer q with $0 < q \leq q_n$, and every integer p with $p/q \neq p_n/q_n$,

$$\left| \alpha - \frac{p}{q} \right| > \left| \alpha - \frac{p_n}{q_n} \right| :$$

among the fractions with denominator at most q_n (note: not $< q_{n+1}$), the convergent approximates better in absolute value.

Proof. Since $q \leq q_n < q_{n+1}$, Theorem 3.4 gives $|q\alpha - p| > |q_n\alpha - p_n|$, that is $q|\alpha - p/q| > q_n|\alpha - p_n/q_n|$. Dividing by q and using $q_n/q \geq 1$:

$$\left| \alpha - \frac{p}{q} \right| > \frac{q_n}{q} \left| \alpha - \frac{p_n}{q_n} \right| \geq \left| \alpha - \frac{p_n}{q_n} \right|.$$

□

Remark 3.2: An error corrected: why 179/57 beats 22/7 as an approximation of π . The first draft of this chapter asserted that “no fraction with denominator $< q_{n+1}$ approximates α better than p_n/q_n ” for the distance $|\alpha - p/q|$: this is **false**, and Corollary 3.1 shows exactly why — the optimality guarantee for $|\alpha - p/q|$ holds only up to denominator $\leq q_n$, not $< q_{n+1}$; it is the second-kind guarantee (Theorem 3.4, on $|q\alpha - p|$) that extends up to q_{n+1} . An explicit counterexample, verified in double precision and with exact rational arithmetic (script `verifica_cap03_vol4.py`, block [C3-3]): for $\alpha = \pi$, $q_1 = 7$ ($p_1/q_1 = 22/7$) and $q_2 = 106$. The intermediate fraction (or semiconvergent) $179/57 = (22 \cdot 8 + 3)/(7 \cdot 8 + 1)$, with 57 strictly between $q_1 = 7$ and $q_2 = 106$, satisfies

$$|\pi - 179/57| = 0.0012418\dots < |\pi - 22/7| = 0.0012645\dots,$$

and yet $57 < q_2 = 106$: no contradiction with Corollary 3.1, which guarantees the optimality of 22/7 only up to denominator $q_1 = 7$. Consistently, the second-kind guarantee remains valid up to $q_2 - 1 = 105$: $\|7\pi\| = 0.0088514\dots < \|57\pi\| = 0.0707813\dots$, that is, 22/7 remains the best in this stronger sense (verified also for all $k = 1, \dots, 105$, not only $k = 57$). The paradox dissolves on observing that $|\alpha - p/q| = \|q\alpha\|/q$: although $\|57\pi\|$ is almost eight times $\|7\pi\|$ ($0.0707813/0.0088514 = 7.996\dots$), dividing by the almost eight times larger denominator ($57/7 = 8.143$) more than compensates, and 179/57 wins by a thin margin. The correct source is Khinchin, *Continued Fractions* (R4), Theorems 15–17.

3.4. Hurwitz’s theorem

Theorem 3.5: Hurwitz’s theorem (1891). For every irrational α there exist infinitely many rationals p/q (with $q \geq 1$) such that

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2}.$$

Proof. It suffices to show that among three consecutive convergents p_{n-1}/q_{n-1} , p_n/q_n , p_{n+1}/q_{n+1} (for every $n \geq 1$) at least one satisfies the inequality: as n ranges over all integers ≥ 1 , this produces infinitely many distinct p/q (the convergents are pairwise distinct, Proposition 3.1(ii)).

Suppose by contradiction that *none* of the three satisfies the inequality, that is $|\alpha - p_m/q_m| \geq 1/(\sqrt{5} q_m^2)$ for $m = n-1, n, n+1$. For two consecutive convergents $p_m/q_m, p_{m+1}/q_{m+1}$, The-

orem 3.3 guarantees that α lies between them (opposite signs of δ_m, δ_{m+1}), hence

$$\left| \frac{p_{m+1}}{q_{m+1}} - \frac{p_m}{q_m} \right| = \left| \alpha - \frac{p_m}{q_m} \right| + \left| \alpha - \frac{p_{m+1}}{q_{m+1}} \right|.$$

On the other hand, by Proposition 3.1(iii), $|p_{m+1}/q_{m+1} - p_m/q_m| = 1/(q_m q_{m+1})$. Applying the contradiction hypothesis to m and $m+1$:

$$\frac{1}{q_m q_{m+1}} \geq \frac{1}{\sqrt{5} q_m^2} + \frac{1}{\sqrt{5} q_{m+1}^2} = \frac{q_m^2 + q_{m+1}^2}{\sqrt{5} q_m^2 q_{m+1}^2} \implies q_m^2 + q_{m+1}^2 \leq \sqrt{5} q_m q_{m+1}.$$

Setting $t_m := q_{m+1}/q_m > 1$ (Proposition 3.1(ii)), this reads $t_m + 1/t_m \leq \sqrt{5}$, that is $t_m^2 - \sqrt{5} t_m + 1 \leq 0$: the roots of the polynomial are $(\sqrt{5} \mp 1)/2 = 1/\varphi, \varphi$ (with $\varphi = (1 + \sqrt{5})/2$), hence

$$\frac{1}{\varphi} \leq t_m \leq \varphi \quad \text{i.e.} \quad t_m \leq \varphi.$$

Applying this both to the pair $(n-1, n)$ and to the pair $(n, n+1)$ (the two contradiction hypotheses adjacent to n involve, respectively, $t_{n-1} = q_n/q_{n-1}$ and $t_n = q_{n+1}/q_n$): $t_{n-1} \leq \varphi$ and $t_n \leq \varphi$.

On the other hand, from the recurrence $q_{n+1} = a_{n+1}q_n + q_{n-1}$, dividing by q_n :

$$t_n = \frac{q_{n+1}}{q_n} = a_{n+1} + \frac{1}{t_{n-1}} \geq 1 + \frac{1}{\varphi} = 1 + (\varphi - 1) = \varphi$$

(using $a_{n+1} \geq 1$ and $1/t_{n-1} \geq 1/\varphi = \varphi - 1$, the latter from $t_{n-1} \leq \varphi$). Combined with $t_n \leq \varphi$ obtained above: $t_n = \varphi$ exactly, which forces $a_{n+1} = 1$ and $t_{n-1} = \varphi$ exactly (equality in both inequalities used). But $t_{n-1} = q_n/q_{n-1}$ is a **rational** number (a ratio of two integers), whereas φ is **irrational**: a contradiction.

Hence the contradiction hypothesis is false, and at least one of the three consecutive convergents satisfies $|\alpha - p_m/q_m| < 1/(\sqrt{5} q_m^2)$. $\square$

Proposition 3.2: *The constant $1/\sqrt{5}$ is optimal along the convergents of φ . For $\alpha = \varphi = (1 + \sqrt{5})/2$, $q_n^2 |\varphi - p_n/q_n| \rightarrow 1/\sqrt{5}$ as $n \rightarrow \infty$: the constant of Theorem 3.5 cannot be replaced, along the sequence of the convergents of φ , by any constant $c > \sqrt{5}$ in the bound $|\varphi - p_n/q_n| < 1/(c q_n^2)$.*

Proof. $\varphi = [1; 1, 1, 1, \dots]$ (verified: $\varphi = 1 + 1/\varphi \iff \varphi^2 - \varphi - 1 = 0$, the equation of which φ is the positive root), hence in the exact formula of the proof of Theorem 3.3, $q_n^2 |\varphi - p_n/q_n| = q_n/(x_{n+1}q_n + q_{n-1}) = 1/(x_{n+1} + \beta_n)$ with $\beta_n = q_{n-1}/q_n$ and $x_{n+1} = \varphi$ for every n (all the complete quotients of φ equal φ itself, by the immediate periodicity of the expansion). Since the q_n are the Fibonacci numbers F_{n+1} (verified by induction from the recurrences with $a_n \equiv 1$: $q_n = q_{n-1} + q_{n-2}$, $q_0 = q_1 = 1$), $\beta_n = F_n/F_{n+1} \rightarrow 1/\varphi$ by Binet's formula. Hence

$$q_n^2 \left| \varphi - \frac{p_n}{q_n} \right| = \frac{1}{\varphi + \beta_n} \xrightarrow{n \rightarrow \infty} \frac{1}{\varphi + 1/\varphi} = \frac{1}{\varphi + (\varphi - 1)} = \frac{1}{2\varphi - 1} = \frac{1}{\sqrt{5}}$$

(using $\varphi^2 = \varphi + 1 \Rightarrow 1/\varphi = \varphi - 1$, and $2\varphi - 1 = \sqrt{5}$ from $\varphi = (1 + \sqrt{5})/2$). Since the limit is exactly $1/\sqrt{5}$, for every $c > \sqrt{5}$ we have $1/c < 1/\sqrt{5}$, hence $q_n^2 |\varphi - p_n/q_n| > 1/c$ (that is, $|\varphi - p_n/q_n| < 1/(c q_n^2)$ fails) for all sufficiently large n : only finitely many convergents can satisfy the inequality with a constant $c > \sqrt{5}$.

Independent numerical check (script `fig03_convergenti_phi.py`): $q_n \|q_n \varphi\| \rightarrow 0.447214\dots = 1/\sqrt{5}$, with the last computed value ($n = 12$) equal to 0.447215. $\checkmark$ $\square$

Remark 3.3: What we import: full optimality. Proposition 3.2 shows that no convergent of φ , for large n , can improve the constant $\sqrt{5}$. The stronger fact — that no pair (p, q) , convergent or not, can do so, that is, that for every $c > \sqrt{5}$ the set of rationals p/q with $|\varphi - p/q| < 1/(cq^2)$ is finite — requires Legendre’s criterion (every approximation with $|\alpha - p/q| < 1/(2q^2)$ is necessarily a convergent of α): we **import** it from Hurwitz 1891 (R1) and Khinchin, *Continued Fractions* (R4, §15), without reproducing its proof, which requires the continued fraction expansion of p/q itself (not only the properties of the convergents of α already established here). Recorded in Appendix A.

3.5. Diophantine numbers, Liouville numbers, noble numbers

Definition 3.4: Diophantine number. α is **Diophantine of type** (K, τ) , for $K > 0$ and $\tau \geq 0$, if $|\alpha - p/q| \geq K/q^{2+\tau}$ for every rational p/q ($q \geq 1$). It is **Diophantine** if it is so for some (K, τ) ; it is of **bounded type** (or of type $(K, 0)$) if it is so with $\tau = 0$.

Theorem 3.6: The Diophantine numbers have full measure. For every $\varepsilon > 0$, the set of the $\alpha \in [0, 1]$ such that $|\alpha - p/q| < 1/q^{2+\varepsilon}$ for infinitely many rationals p/q has Lebesgue measure zero. In particular, for almost every $\alpha \in [0, 1]$ (with respect to Lebesgue measure) and for every $\varepsilon > 0$ there exists $K > 0$ such that α is Diophantine of type (K, ε) .

Proof. For a fixed integer $q \geq 1$, the set $E_q = \{\alpha \in [0, 1] : \exists p, |\alpha - p/q| < 1/q^{2+\varepsilon}\}$ is the union, over $p = 0, \dots, q$, of intervals of length $2/q^{2+\varepsilon}$: $\text{mis}(E_q) \leq (q+1) \cdot 2/q^{2+\varepsilon} \leq 4/q^{1+\varepsilon}$ for $q \geq 1$. The series $\sum_{q \geq 1} 4/q^{1+\varepsilon}$ converges (generalised harmonic series with exponent $1 + \varepsilon > 1$). By the Borel–Cantelli lemma (first part: if $\sum \text{mis}(E_q) < \infty$, the set $\limsup_q E_q = \bigcap_Q \bigcup_{q \geq Q} E_q$ — the points lying in infinitely many E_q — has measure zero), the set of the α belonging to infinitely many E_q , that is, admitting infinitely many approximations $|\alpha - p/q| < 1/q^{2+\varepsilon}$, has measure zero. The complement (of full measure) is therefore the set of the α for which $|\alpha - p/q| < 1/q^{2+\varepsilon}$ holds only for finitely many p/q : for such an α , setting $K = \min$ (finite, positive) of $q^{2+\varepsilon}|\alpha - p/q|$ over the residual exceptions (if there are any) and 1 otherwise, we have $|\alpha - p/q| \geq K/q^{2+\varepsilon}$ for every p/q : α is Diophantine of type (K, ε) . $\square$

Definition 3.5: Liouville number. β is a **Liouville number** if, for every integer $n \geq 1$, there exist integers p, q with $q \geq 2$ and $0 < |\beta - p/q| < 1/q^n$.

Theorem 3.7: Liouville’s approximation theorem. Let α be algebraic of degree $d \geq 2$ (a root of an irreducible polynomial $P \in \mathbb{Z}[x]$ of degree d). There exists $c = c(\alpha) > 0$ such that $|\alpha - p/q| > c/q^d$ for every rational p/q with $q \geq 1$.

Proof. Let $P(x) = a_d x^d + \dots + a_0 \in \mathbb{Z}[x]$, $P(\alpha) = 0$, irreducible over $\mathbb{Q}$ of degree $d \geq 2$ (hence P has no rational roots). For $p/q \in \mathbb{Q}$, $q^d P(p/q) = a_d p^d + a_{d-1} p^{d-1} q + \dots + a_0 q^d \in \mathbb{Z}$; it is non-zero because $P(p/q) \neq 0$ (no rational root), hence $|q^d P(p/q)| \geq 1$, that is $|P(p/q)| \geq 1/q^d$. If $|\alpha - p/q| \geq 1$, the claim is trivial with $c \leq 1$. If $|\alpha - p/q| < 1$, by the mean value theorem there exists ξ between α and p/q (hence $\xi \in [\alpha - 1, \alpha + 1]$) such that $P(p/q) - P(\alpha) = P'(\xi)(p/q - \alpha)$, that is, since $P(\alpha) = 0$,

$$\left| \alpha - \frac{p}{q} \right| = \frac{|P(p/q)|}{|P'(\xi)|} \geq \frac{1/q^d}{M}$$

where $M = \max_{[\alpha-1, \alpha+1]} |P'|$ (finite, P' being continuous on a compact set; $M > 0$ because P' is not identically zero, P being of degree ≥ 2). Taking $c = \min(1, 1/M)$ one obtains the claim

in both cases. □

Corollary 3.2: Liouville numbers are transcendental. *Every Liouville number is transcendental.*

Proof. Let β be a Liouville number. First of all β is irrational: if we had $\beta = a/b$ with $b \geq 1$, then for large q ($q > b$) and $p/q \neq a/b$ we would have $|\beta - p/q| = |aq - bp|/(bq) \geq 1/(bq)$, which for large n exceeds $1/q^n$ as soon as $q^{n-1} > b$: hence only finitely many n could provide the approximation required by the definition, against the hypothesis that it holds for *every* n . Suppose by contradiction that β is algebraic, of degree $d \geq 1$. If $d = 1$, β is rational, excluded above. If $d \geq 2$, Theorem 3.7 gives a constant $c > 0$ with $|\beta - p/q| > c/q^d$ for every p/q . Choose $n = d + 1$ in Definition 3.5: there exist p, q ($q \geq 2$) with $0 < |\beta - p/q| < 1/q^{d+1}$. For q large enough to satisfy $q > 1/c$ (and such a q exists among the infinitely many approximations required by the definition, because necessarily $q \rightarrow \infty$: for fixed q , $1/q^n \rightarrow 0$ while $|\beta - p/q|$ remains $\geq$ a positive constant depending only on q as soon as one excludes the finitely many p with $\beta = p/q$, which is impossible since β is irrational), we have $1/q^{d+1} < c/q^d$, hence $|\beta - p/q| < c/q^d$: this contradicts $|\beta - p/q| > c/q^d$. Hence β is not algebraic of any degree: it is transcendental. □

Example 3.2: Liouville's constant. $L = \sum_{k=1}^{\infty} 10^{-k!} = 0.110001000\dots 000100\dots$ (a 1 in the $k!$ -th decimal digit, 0 elsewhere) is a Liouville number: truncating the series at $k = m$ terms one obtains p_m/q_m with $q_m = 10^{m!}$ and $|L - p_m/q_m| = \sum_{k>m} 10^{-k!} < 2 \cdot 10^{-(m+1)!} = 2q_m^{-(m+1)}$, which for large m beats $1/q_m^n$ for every fixed n (it suffices that $m \geq n$). It is therefore transcendental by Corollary 3.2. We shall use its three-term truncation, $10^{-1} + 10^{-2} + 10^{-6} = 0.110001$, computed both in floating point and with the `decimal` module at 60 digits (discrepancy $< 10^{-16}$, script `fig03_discrepanza_loglog.py`), as the third case in Figure 3.4 on discrepancy.

Proposition 3.3: φ is “the most Diophantine”. *Among the real quadratic algebraic numbers, $\varphi = (1 + \sqrt{5})/2$ realises the worst possible approximation constant: $\liminf_n q_n \|q_n \alpha\| = 1/\sqrt{5}$ for $\alpha = \varphi$ (Proposition 3.2), whereas for $\alpha = \sqrt{2}$, for instance, $\liminf_n q_n \|q_n \alpha\| = 1/(2\sqrt{2}) = 0.353553\dots < 1/\sqrt{5} = 0.447214\dots$ (the computation is identical to that of Proposition 3.2, with $\sqrt{2} = [1; 2, 2, 2, \dots]$ and constant complete quotient $x_{n+1} = 1 + \sqrt{2}$, giving the limit $1/(1 + \sqrt{2} + 1/(1 + \sqrt{2})) = 1/(2\sqrt{2})$; verified numerically, script `verifica_cap03_vol4.py` block [C3-5]: final value 0.353556). Since $1/\sqrt{5}$ is the largest possible constant in absolute terms (Theorem 3.5: no irrational can do systematically worse than $\sqrt{5}$), and φ attains it exactly, φ is, in this precise sense, the least well approximable irrational there is: the “most Diophantine” one.*

Definition 3.6: Noble numbers. A number $\rho \in (0, 1)$ is **noble** if its continued fraction expansion $\rho = [0; a_1, a_2, a_3, \dots]$ has $a_n = 1$ for every sufficiently large n (“tail of 1’s”). The reduced golden ratio $\rho^* := 1/\varphi = \varphi - 1 = [0; 1, 1, 1, \dots] = 0.618034\dots$ is the “pure” noble number (tail of 1’s from the very beginning); every other noble number is, by definition, equivalent to ρ^* modulo integral Möbius transformations (that is, $[0; a_1, \dots, a_m, 1, 1, 1, \dots]$ for an arbitrary finite prefix).

Remark 3.4: A double error corrected. *The first draft defined the noble numbers as those “close to $\varphi/2\pi$ ”: a double error. First of all $\varphi/2\pi = 0.257518\dots$ is not even the correct golden ratio in the normalisation $[0, 1]$: that is $\rho^* = 1/\varphi = 0.618034\dots$ (or, equivalently, $2 - \varphi = 0.381966\dots$, its image under $x \mapsto 1 - x$). Secondly, “noble” does not mean*

“close to” a value, but an exact property of the continued fraction (constant tail of 1’s): for example $2 - \varphi = [0; 2, 1, 1, 1, \dots]$ is noble (tail of 1’s after an initial 2), verified by the algorithm (script `verifica_cap03_vol4.py`, block [C3–6]: computed digits $[0, 2, 1, 1, 1, 1, 1, 1]$, as expected). The name “noble” reflects precisely the fact that, having a tail of 1’s, these numbers asymptotically inherit the same optimal approximation constant $1/\sqrt{5}$ as φ (Proposition 3.3): they are, in this sense, “of the same family” as the most Diophantine number of all.

3.6. The three-distance theorem

Theorem 3.8: Three-distance theorem (Steinhaus–Sós–Świerczkowski–Surányi).

Let $\beta \in (0, 1)$ be irrational and $N \geq 1$. The points $\{0, \beta, 2\beta, \dots, (N-1)\beta\} \bmod 1$ subdivide $\mathbb{T} \cong [0, 1)$ into N arcs which take at most three distinct lengths; if the distinct lengths are exactly three, the largest is the sum of the other two.

Proof. Let d_+ be the smallest integer d with $1 \leq d < N$ such that $\{d\beta\}$ (fractional part) is minimal among $\{\{d'\beta\} : 1 \leq d' < N\}$ — the best “approach to 0 from the right” within N steps — and let d_- be the smallest d with $1 \leq d < N$ such that $1 - \{d\beta\}$ is minimal — the best approach “from the left”. (Such $d_{\pm}$ are, by the theory of best approximation, Section 3.3: precisely, d_+ and d_- are the denominators of two consecutive convergents of β , or more precisely of the continued fraction truncated at the largest convergent/semiconvergent with denominator $< N$; we shall not use this fact in the proof, only their minimality by construction.)

Step 1 (the neighbours of each point). Fix $x_k = \{k\beta\} \in P := \{x_0, \dots, x_{N-1}\}$. We claim that the *right* neighbour of x_k in P (the first point of P met when moving in the positive sense from x_k) is: x_{k+d_+} if $k + d_+ < N$; otherwise x_{k-d_-} if $k - d_- \geq 0$; otherwise (if neither $k + d_+ < N$ nor $k - d_- \geq 0$) $x_{k+d_+-d_-}$.

To see this: if a point $x_j \in P$, $j \neq k$, were closer to x_k on the right than the indicated candidate, the difference $\{(j-k)\beta\}$ (if $j > k$ in a suitable cyclic index) would be an approach to 0 from the right with $|j-k| < N$ better than $\{d_+\beta\}$ — impossible by the minimality of d_+ — unless the corresponding index $j-k$ (modulo the restrictions on $j \in \{0, \dots, N-1\}$) is not reachable by a single step $+d_+$ or $-d_-$ from k : that is, when $k + d_+ \geq N$ (the step $+d_+$ “leaves” the range of admissible indices) and $k - d_- < 0$ (the step $-d_-$ leaves it too), the only candidate left is the combination $x_{k+d_+-d_-}$, which is admissible because $0 \leq k + d_+ - d_- < N$ in this residual case (the two elementary steps combine). An identical analysis, with the roles of d_+, d_- exchanged, holds for the left neighbour.

Step 2 (the three lengths). From Step 1, the distance between x_k and its right neighbour is, in the three cases, $\{d_+\beta\}$, $1 - \{d_-\beta\}$, or $\{d_+\beta\} + (1 - \{d_-\beta\})$ (the third being the composition of the first two, since the composite step geometrically crosses first the “missing” point at distance $\{d_+\beta\}$ and then the residual stretch $1 - \{d_-\beta\}$: more precisely $\{(k+d_+-d_-)\beta\} - x_k = \{d_+\beta\} - \{d_-\beta\} \pmod{1}$, and since by construction $k + d_+ \geq N$ and $k - d_- < 0$ imply that this value is positive and equal to $\{d_+\beta\} + (1 - \{d_-\beta\})$ reduced modulo 1, that is, without any further reduction, both terms being in $(0, 1)$ and their sum being < 1 by the same minimality that defines $d_{\pm}$: if we had $\{d_+\beta\} + 1 - \{d_-\beta\} \geq 1$, that is $\{d_+\beta\} \geq \{d_-\beta\}$, then $d_+ - d_-$ (if > 0 and $< N$) would give an approach to 0 even better than $\{d_+\beta\}$, a contradiction; the case $\{d_+\beta\} < \{d_-\beta\}$ is analogous with $d_- - d_+$). The three possible lengths are therefore exactly $\ell_+ = \{d_+\beta\}$, $\ell_- = 1 - \{d_-\beta\}$, $\ell_+ + \ell_-$, and the last one, when it occurs, is by construction the sum of the other two. $\square$

Remark 3.5: Historical attribution. The theorem was **conjectured by Hugo Steinhaus** and proved independently, between 1957 and 1958, by three authors: **V. T. Sós, S. Świerczkowski and J. Surányi** (the three proofs, published close to one another, are nowadays cited together). The primary reference verified in this chapter is that of Świerczkowski, *Fund. Math.* 46 (1958), 187–189 (R5, confirmed as retrievable on 16 September 2026 on EuDML). The first draft of this chapter wrongly attributed the theorem to “Steinhaus” alone in a **teorema** environment: it is corrected here with the full names of the three provers, with Steinhaus confined to the historically exact role of the one who posed the conjecture. Linking sources: Alessandri–Berthé (R7, verified on 16 September 2026, which explicitly places Steinhaus’s question at the origin of the problem) and Hamada (R9, an independent geometric proof from 2023, verified on 16 September 2026 on arXiv).

Example 3.3: Three distances for $N = 8$ and $N = 9$, $\beta = 1/\varphi$. Double check (script `verifica_cap03_vol4.py`, block [C3–4]). For $\beta = 1/\varphi = 0.618034\dots$ and $N = 8$, the ordered points $\{k\beta \bmod 1\}_{k=0}^7$ are

0; 0.090170; 0.236068; 0.326238; 0.472136; 0.618034; 0.708204; 0.854102,

with arcs of length 0.090170 (three times) and 0.145898 (five times): *only two* distinct lengths, not three. This is not an exception to Theorem 3.8, which states “at most three”; it is rather a remarkable fact connected with the Fibonacci numbers: $8 = F_6$ is a Fibonacci number, and for $\beta = 1/\varphi$ the two lengths coincide exactly when N is a Fibonacci number (verified also for $N = 13 = F_7$: again only two lengths, 0.055728 and 0.090170). For $N = 9$ (not a Fibonacci number), the three lengths reappear: 0.055728, 0.090170, 0.145898, and one checks $0.145898 = 0.055728 + 0.090170 \checkmark$, an instance of the rule “the largest is the sum of the other two”.

3.7. Weyl’s criterion and Fejér’s theorem

Theorem 3.9: Weyl equidistribution for irrational rotations. Let $\beta = \alpha/2\pi$ be irrational. For every continuous function $f : \mathbb{T} \rightarrow \mathbb{R}$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(n\alpha) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta.$$

Proposition 3.4: Weyl’s criterion (for the exponentials). For β irrational and every $k \in \mathbb{Z} \setminus \{0\}$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i k n \beta} = 0.$$

Proof. Since β is irrational and $k \neq 0$, $k\beta \notin \mathbb{Z}$, hence $e^{2\pi i k \beta} \neq 1$. The sum is geometric:

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i k n \beta} = \frac{1}{N} \cdot \frac{e^{2\pi i k N \beta} - 1}{e^{2\pi i k \beta} - 1}.$$

The numerator has modulus ≤ 2 ; the denominator is a non-zero constant (independent of N): the expression is bounded in modulus by $2/(N|e^{2\pi i k \beta} - 1|) \rightarrow 0$. $\square$

To extend Weyl’s criterion, verified on the exponentials (that is, on trigonometric polynomials,

by linearity), to every continuous function one needs an approximation theorem: here we prove it in full, instead of taking it for granted.

Construction 3.2: The Fejér kernel. For $f : \mathbb{T} \rightarrow \mathbb{C}$ continuous and 2π -periodic, with Fourier coefficients $\hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-ik\theta} d\theta$ and partial sums $S_n(f)(\theta) = \sum_{k=-n}^n \hat{f}(k) e^{ik\theta}$, the **Cesàro means** are $\sigma_N(f) = \frac{1}{N} \sum_{n=0}^{N-1} S_n(f)$. The **Fejér kernel** is

$$F_N(t) := \frac{1}{N} \sum_{n=0}^{N-1} D_n(t), \quad D_n(t) = \sum_{k=-n}^n e^{ikt} = \frac{\sin((n + \frac{1}{2})t)}{\sin(t/2)}$$

(D_n : the Dirichlet kernel), so that $\sigma_N(f)(\theta) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta - t) F_N(t) dt$ (convolution, from $S_n(f)(\theta) = \frac{1}{2\pi} \int f(\theta - t) D_n(t) dt$, summing over n and dividing by N). $\square$

Lemma 3.1: Closed formula and positivity of the Fejér kernel. For $t \notin 2\pi\mathbb{Z}$, $F_N(t) = \frac{1}{N} \left(\frac{\sin(Nt/2)}{\sin(t/2)} \right)^2 \geq 0$; moreover $\frac{1}{2\pi} \int_0^{2\pi} F_N(t) dt = 1$, and for every $\delta \in (0, \pi)$, $\max_{\delta \leq |t| \leq \pi} F_N(t) \leq \frac{1}{N \sin^2(\delta/2)} \rightarrow 0$ as $N \rightarrow \infty$.

Proof. Using $D_n(t) \sin(t/2) = \sin((n + \frac{1}{2})t)$ and the product-to-sum formula $2 \sin((n + \frac{1}{2})t) \sin(t/2) = \cos(nt) - \cos((n + 1)t)$: summing over $n = 0, \dots, N - 1$ (telescoping), $2 \sin(t/2) \sum_{n=0}^{N-1} \sin((n + \frac{1}{2})t) = \cos(0) - \cos(Nt) = 1 - \cos(Nt) = 2 \sin^2(Nt/2)$; dividing by $2 \sin(t/2)$ and then once more by $\sin(t/2)$ (from the definition of D_n as $\sin((n+1/2)t)/\sin(t/2)$):

$$N F_N(t) = \sum_{n=0}^{N-1} D_n(t) = \frac{1}{\sin(t/2)} \sum_{n=0}^{N-1} \sin((n + \frac{1}{2})t) = \frac{\sin^2(Nt/2)}{\sin^2(t/2)},$$

manifestly ≥ 0 . The integral: $\frac{1}{2\pi} \int_0^{2\pi} D_n(t) dt = 1$ for every n (only the term $k = 0$ of $D_n = \sum_{k=-n}^n e^{ikt}$ contributes), hence $\frac{1}{2\pi} \int F_N = \frac{1}{N} \sum_{n=0}^{N-1} 1 = 1$. For $\delta \leq |t| \leq \pi$, $\sin^2(t/2) \geq \sin^2(\delta/2) > 0$ (monotonicity of $\sin$ on $[0, \pi/2]$) and $\sin^2(Nt/2) \leq 1$, hence $F_N(t) \leq 1/(N \sin^2(\delta/2)) \rightarrow 0$. $\square$

Theorem 3.10: Fejér's theorem (1904). For $f : \mathbb{T} \rightarrow \mathbb{R}$ (or $\mathbb{C}$) continuous, $\sigma_N(f) \rightarrow f$ uniformly on $\mathbb{T}$.

Proof. From $\frac{1}{2\pi} \int F_N = 1$ (Lemma 3.1),

$$\sigma_N(f)(\theta) - f(\theta) = \frac{1}{2\pi} \int_0^{2\pi} [f(\theta - t) - f(\theta)] F_N(t) dt.$$

f is continuous on the compact space $\mathbb{T}$, hence uniformly continuous and bounded: let $\|f\|_\infty = \max |f|$. Fix $\varepsilon > 0$; by uniform continuity there exists $\delta \in (0, \pi)$ with $|f(\theta - t) - f(\theta)| < \varepsilon/2$ for $|t| \leq \delta$, uniformly in θ . We split the integral:

$$|\sigma_N(f)(\theta) - f(\theta)| \leq \frac{1}{2\pi} \int_{|t| \leq \delta} |f(\theta - t) - f(\theta)| F_N(t) dt + \frac{1}{2\pi} \int_{\delta \leq |t| \leq \pi} |f(\theta - t) - f(\theta)| F_N(t) dt.$$

The first term is $\leq \frac{\varepsilon}{2} \cdot \frac{1}{2\pi} \int F_N \leq \frac{\varepsilon}{2} \cdot 1 = \frac{\varepsilon}{2}$ (positivity of F_N , Lemma 3.1). The second is $\leq 2\|f\|_\infty \cdot \max_{\delta \leq |t| \leq \pi} F_N(t) \leq 2\|f\|_\infty / (N \sin^2(\delta/2))$ (Lemma 3.1), which for N large enough (independently of θ , δ being already fixed) is $< \varepsilon/2$. Hence $|\sigma_N(f)(\theta) - f(\theta)| < \varepsilon$ for every θ and large N : uniform convergence. $\square$

Corollary 3.3: Weierstrass approximation theorem, trigonometric version. Every continuous $f : \mathbb{T} \rightarrow \mathbb{R}$ is a uniform limit of trigonometric polynomials.

Proof. $\sigma_N(f)$ is a trigonometric polynomial (a finite combination of $e^{ik\theta}$, $|k| \leq N-1$, being a sum of $S_n(f)$ for $n < N$), and it converges uniformly to f by Theorem 3.10. $\square$

Proof. [of Theorem 3.9] By Corollary 3.3, given $\varepsilon > 0$ there exists a trigonometric polynomial $P = \sum_{|k| \leq K} c_k e^{ik\theta}$ with $\|f - P\|_\infty < \varepsilon/3$. Then, for every N ,

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} f(n\alpha) - \frac{1}{N} \sum_{n=0}^{N-1} P(n\alpha) \right| \leq \frac{1}{N} \sum_{n=0}^{N-1} |f(n\alpha) - P(n\alpha)| < \frac{\varepsilon}{3}.$$

By Proposition 3.4 (by linearity, applied to each term $c_k e^{ikn\alpha}$ of P with $k \neq 0$; the term $k = 0$ gives exactly $c_0 = \hat{P}(0)$ for every N), $\frac{1}{N} \sum_n P(n\alpha) \rightarrow c_0 = \frac{1}{2\pi} \int_0^{2\pi} P(\theta) d\theta$ as $N \rightarrow \infty$: there exists N_0 such that for $N \geq N_0$, $|\frac{1}{N} \sum_n P(n\alpha) - \frac{1}{2\pi} \int P| < \varepsilon/3$. Finally $|\frac{1}{2\pi} \int P - \frac{1}{2\pi} \int f| \leq \|P - f\|_\infty < \varepsilon/3$. Adding the three errors (triangle inequality), for $N \geq N_0$:

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} f(n\alpha) - \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta \right| < \varepsilon.$$

$\square$

Corollary 3.4: Extension to Riemann-integrable functions. Theorem 3.9 holds, more generally, for $f : \mathbb{T} \rightarrow \mathbb{R}$ Riemann-integrable (bounded, with a set of discontinuities of measure zero).

Proof. (A “sandwich” argument.) Given $\varepsilon > 0$, by the definition of the Riemann integral there exist continuous functions $g \leq f \leq h$ on $\mathbb{T}$ with $\frac{1}{2\pi} \int (h - g) < \varepsilon/3$ (built from step functions approximating f from below and from above on the Riemann partition, then smoothed into continuous functions without altering the integral by more than $\varepsilon/6$ each). Applying Theorem 3.9 to g and to h : for large N , $\frac{1}{N} \sum g(n\alpha)$ and $\frac{1}{N} \sum h(n\alpha)$ are at distance less than $\varepsilon/3$ from their respective mean integrals. From $g \leq f \leq h$, $\frac{1}{N} \sum g(n\alpha) \leq \frac{1}{N} \sum f(n\alpha) \leq \frac{1}{N} \sum h(n\alpha)$, and the two extremes are both within $\varepsilon/3 + \varepsilon/3 = 2\varepsilon/3$ of $\frac{1}{2\pi} \int f$ (since $\int g, \int h$ are both within $\varepsilon/3$ of $\int f$): the mean of f is therefore trapped in an interval of width $< \varepsilon$ around $\frac{1}{2\pi} \int f$. $\square$

Remark 3.6: Equidistribution is stronger than density. Applying Corollary 3.4 to the (Riemann-integrable) indicator function of an arc $[a, b) \subset \mathbb{T}$: the fraction of time the orbit $\{n\alpha\}$ spends in the arc converges to its relative length $(b-a)/2\pi$. This is a quantitative strengthening of the density of Theorem 3.1: the orbit not only visits every neighbourhood, it does so with the “right” asymptotic frequency. How fast this frequency converges is the subject of the next two sections.

3.8. Discrepancy

Definition 3.7: Discrepancy. For points $x_1, \dots, x_N \in [0, 1)$, the **discrepancy** (or star discrepancy) is

$$D_N = \sup_{0 \leq b \leq 1} \left| \frac{\#\{n \leq N : x_n \in [0, b)\}}{N} - b \right|.$$

Theorem 3.11: $D_N \rightarrow 0$ is equivalent to equidistribution. The sequence $(x_n) \subset [0, 1]$ is equidistributed (that is, $\frac{1}{N} \#\{n \leq N : x_n \in [a, b]\} \rightarrow b - a$ for every $0 \leq a < b \leq 1$) if and only if $D_N \rightarrow 0$.

Proof. ($\Leftarrow$) If $D_N \rightarrow 0$: for $b \in [0, 1]$, $|\frac{1}{N} \#\{x_n \in [0, b]\} - b| \leq D_N \rightarrow 0$ by the definition of D_N (case $a = 0$); for $a > 0$, $\#\{x_n \in [a, b]\} = \#\{x_n \in [0, b]\} - \#\{x_n \in [0, a]\}$, hence $|\frac{1}{N} \#\{x_n \in [a, b]\} - (b - a)| \leq |\frac{1}{N} \#\{x_n \in [0, b]\} - b| + |\frac{1}{N} \#\{x_n \in [0, a]\} - a| \leq 2D_N \rightarrow 0$.

($\Rightarrow$) If (x_n) is equidistributed: fix $\varepsilon > 0$ and choose a partition $0 = b_0 < b_1 < \dots < b_M = 1$ with $b_j - b_{j-1} < \varepsilon$ for every j (for instance $M = \lceil 2/\varepsilon \rceil$ equally spaced points). By equidistribution, there exists N_0 such that for $N \geq N_0$, $|\frac{1}{N} \#\{x_n \in [0, b_j]\} - b_j| < \varepsilon$ for every $j = 0, \dots, M$ (finitely many values of j , so one takes the maximum of the required N_0). For an arbitrary $b \in [0, 1]$, let j be such that $b_{j-1} \leq b < b_j$: then $\#\{x_n \in [0, b_{j-1}]\} \leq \#\{x_n \in [0, b]\} \leq \#\{x_n \in [0, b_j]\}$, hence

$$\frac{1}{N} \#\{x_n \in [0, b]\} - b \leq \left(\frac{1}{N} \#\{x_n \in [0, b_j]\} - b_j \right) + (b_j - b) < \varepsilon + \varepsilon = 2\varepsilon,$$

and analogously $\frac{1}{N} \#\{x_n \in [0, b]\} - b > -2\varepsilon$ (from below, with b_{j-1}). Hence $D_N < 2\varepsilon$ for $N \geq N_0$: $D_N \rightarrow 0$. $\square$

3.9. Koksma's inequality

Theorem 3.12: Koksma's inequality. Let $f : [0, 1] \rightarrow \mathbb{R}$ be of bounded total variation $V(f)$. For every $x_1, \dots, x_N \in [0, 1]$,

$$\left| \frac{1}{N} \sum_{n=1}^N f(x_n) - \int_0^1 f(t) dt \right| \leq V(f) D_N.$$

Proof. Let $G_N(t) = \frac{1}{N} \#\{n \leq N : x_n \leq t\}$ (a step function, non-decreasing, jumping by $1/N$ at each x_n , with $G_N(1) = 1$) and $R_N(t) = G_N(t) - t$ (not to be confused with the Fejér kernel F_N of the previous Section 3.9; the notation here is purely local): by the definition of discrepancy, $|R_N(t)| \leq D_N$ for every $t \in [0, 1]$ (up to a discrepancy computed on $[0, t]$ rather than $[0, 1]$, which differs at most by the terms with $x_n = t$: the inequality $|R_N(t)| \leq D_N$ remains valid with the convention $[0, t]$ in the definition of G_N , equivalent up to a rearrangement of the supremum in Definition 3.7). Writing the sum as a Riemann–Stieltjes integral with respect to G_N ,

$$\frac{1}{N} \sum_{n=1}^N f(x_n) = \int_0^1 f(t) dG_N(t), \quad \int_0^1 f(t) dt = \int_0^1 f(t) d(t),$$

the difference is $\int_0^1 f dR_N$. Integrating by parts (the Riemann–Stieltjes formula, legitimate because f is of bounded variation and R_N is of bounded variation with finitely many jumps, and the two have no unmanageable common discontinuities since one can always perturb the partition of the points x_n by an $\eta \rightarrow 0$ without changing D_N by more than η):

$$\int_0^1 f dR_N = [f(t)R_N(t)]_0^1 - \int_0^1 R_N(t) df(t) = f(1)R_N(1) - f(0)R_N(0) - \int_0^1 R_N df.$$

Since $G_N(1) = 1$, $R_N(1) = 0$; and $R_N(0) = G_N(0) - 0 = G_N(0) \geq 0$ but is in any case bounded in modulus by D_N by the very definition of discrepancy (applied at $b = 0^+$, or by including the point 0 in the partition): the boundary term is therefore bounded by $D_N(|f(1)| + |f(0)|)$ in the general case, and it vanishes identically as soon as the points are normalised to lie strictly

in $(0, 1)$, as is the case for the rotations $x_n = \{n\beta\}$ with β irrational and $n \geq 1$ (never $x_n = 0$). The remaining term is bounded, in modulus, by $\int_0^1 |R_N(t)| |df(t)| \leq D_N \int_0^1 |df(t)| = D_N V(f)$ (by the definition of total variation). Adding up (with the boundary term zero in the relevant case, or in any case negligible after a renormalisation of the endpoints), one obtains the claim. $\square$

3.10. The Erdős–Turán inequality

Remark 3.7: What we import: the Erdős–Turán inequality. For every sequence $x_1, \dots, x_N \in [0, 1)$ and every $K \geq 1$,

$$D_N \leq C \left(\frac{1}{K} + \sum_{k=1}^K \frac{1}{k} \left| \frac{1}{N} \sum_{n=1}^N e^{2\pi i k x_n} \right| \right),$$

with C an absolute constant (the form with finer explicit constants, due to Vaaler through the kernel named after him, and the variant with Holt’s constants, are in R14). We **import** this inequality from Kuipers–Niederreiter, *Uniform Distribution of Sequences* (R16, Theorem 2.5, which in turn cites Erdős–Turán 1948) and from Holt (R14), without reproducing its proof: the standard proof approximates the indicator function of an interval from above and from below by trigonometric polynomials of degree K built from the Vaaler or the Selberg–Beurling kernel, a procedure of one-variable approximation theory that exceeds the scope — and the level — of this chapter (which has already proved in full, with the Fejér kernel, the analogous qualitative result). It is the only result of this chapter left in the form of an imported remark rather than of a theorem with a complete proof: recorded in Appendix A.

3.11. Discrepancy of rotations: the role of the partial quotients

Proposition 3.5: Bounded Diophantine type $\iff$ bounded partial quotients. β has partial quotients bounded by M (that is, $a_n \leq M$ for every $n \geq 1$) if and only if β is Diophantine of bounded type $(K, 0)$ for some $K = K(M) > 0$ (Definition 3.4): more precisely, if $a_n \leq M$ for every n , then $\|q\beta\| \geq 1/((M+2)q)$ for every integer $q \geq 1$.

Proof. ($\Rightarrow$). Let $q \geq 1$ and let n be such that $q_n \leq q < q_{n+1}$ (it exists by Proposition 3.1(ii), with $q_0 = 1$). By Theorem 3.4, $\|q\beta\| \geq \|q_n\beta\| = |\delta_n|$. By Theorem 3.3, $|\delta_n| = q_n |\alpha - p_n/q_n| > q_n/(q_n(q_n + q_{n+1})) = 1/(q_n + q_{n+1})$. From $a_{n+1} \leq M$, $q_{n+1} = a_{n+1}q_n + q_{n-1} \leq Mq_n + q_n = (M+1)q_n$ (using $q_{n-1} \leq q_n$), hence $q_n + q_{n+1} \leq (M+2)q_n \leq (M+2)q$ (since $q_n \leq q$). Combining: $\|q\beta\| > 1/(q_n + q_{n+1}) \geq 1/((M+2)q)$.

($\Leftarrow$). Suppose β is Diophantine of type $(K, 0)$, that is, $\|q\beta\| \geq K/q$ for every $q \geq 1$. Applying this to $q = q_n$ and using from Theorem 3.3 the bound $\|q_n\beta\| = |\delta_n| < 1/q_{n+1}$, we get $K/q_n < 1/q_{n+1}$, that is, $q_{n+1} < q_n/K$. On the other hand $q_{n+1} = a_{n+1}q_n + q_{n-1} > a_{n+1}q_n$ (since $q_{n-1} > 0$), hence $a_{n+1}q_n < q_{n+1} < q_n/K$, from which $a_{n+1} < 1/K$ for every $n \geq 0$: the partial quotients are all bounded by the integer $M := \lfloor 1/K \rfloor$ (or $\lfloor 1/K \rfloor - 1$ if $1/K \in \mathbb{Z}$), a constant depending only on K . $\square$

Remark 3.8: What we import: the Ostrowski sum. The step that turns the crude bound of Proposition 3.5 into an optimal rate $O(\log N/N)$ (instead of the rate $O(1/\sqrt{N})$ obtained by direct summation, see Proposition 3.6 below) requires a finer estimate of the sum $\sum_{k=1}^K \frac{1}{k\|k\beta\|}$, which for bounded partial quotients is $O(\log K)$ instead of the trivial $O(K)$: the

proof (Ostrowski 1922, R3) groups the integers $k \leq K$ according to their representation in the non-standard base given by the denominators $q_0, q_1, q_2, \dots$ (**Ostrowski numeration**), using the three-distance theorem (Section 3.6) to control $\|k\beta\|$ block by block. We **import** the precise statement from Kuipers–Niederreiter (R16, Lemma 2.3.3 and Theorem 2.3.2, for bounded quotients; Theorem 3.4 for the general Diophantine case with exponent τ), recorded in Appendix A, and we use it in the proof of Theorem 3.13 below.

Proposition 3.6: A first, elementary rate: $D_N = O(1/\sqrt{N})$. If β has partial quotients bounded by M , the sequence $x_n = \{n\beta\}$ satisfies $D_N = O(1/\sqrt{N})$ (with a constant depending on M).

Proof. From Proposition 3.5, $\|k\beta\| \geq 1/((M+2)k)$ for every $k \geq 1$. From the imported Erdős–Turán inequality (Section 3.10, with $x_n = n\beta \bmod 1$, for which $\frac{1}{N} \sum_n e^{2\pi i k n \beta}$ is the same geometric sum computed in Proposition 3.4, of modulus $\leq 1/(2N\|k\beta\|)$):

$$D_N \leq C \left(\frac{1}{K} + \sum_{k=1}^K \frac{1}{k} \cdot \frac{1}{2N\|k\beta\|} \right) \leq C \left(\frac{1}{K} + \sum_{k=1}^K \frac{1}{k} \cdot \frac{(M+2)k}{2N} \right) = C \left(\frac{1}{K} + \frac{(M+2)K}{2N} \right).$$

Choosing $K = \lceil \sqrt{N} \rceil$, both terms are $O(1/\sqrt{N})$, hence $D_N = O(1/\sqrt{N})$. $\square$

Theorem 3.13: Optimal discrepancy for bounded partial quotients. If β has partial quotients bounded by M , and $x_n = \{n\beta\}$, then $ND_N = O(\log N)$, that is $D_N = O(\log N/N)$, with a constant depending only on M . More generally, if β is Diophantine of type (K_0, τ) , then $D_N = O(N^{-1/(1+\tau)} \log N)$; for $\tau = 0$ (bounded quotients) this is the optimal rate stated.

Proof. From Erdős–Turán (Section 3.10) as in Proposition 3.6, but now using the imported estimate of Ostrowski (Remark 3.8) $\sum_{k=1}^K \frac{1}{k\|k\beta\|} = O_M(\log K)$ in place of the crude linear estimate $O(K)$:

$$D_N \leq C \left(\frac{1}{K} + \frac{1}{N} \sum_{k=1}^K \frac{1}{k\|k\beta\|} \right) \leq C \left(\frac{1}{K} + \frac{O_M(\log K)}{N} \right).$$

Choosing $K = N$ (or even $K = \lceil N/\log N \rceil$, which makes no difference up to constants): $D_N \leq C(1/N + O_M(\log N)/N) = O_M(\log N/N)$. For the general Diophantine case (K_0, τ) , the imported Ostrowski estimate gives $\sum_{k=1}^K 1/(k\|k\beta\|) = O(K^\tau \log K)$ (R16, Thm 3.4), and optimising $K \sim N^{1/(1+\tau)}$ in the same Erdős–Turán inequality one obtains the stated rate. $\square$

Remark 3.9: The correction of the exponents. The first draft of this chapter reported, for the same statement, two mutually inconsistent exponents ($N^{-1/(2+\tau)}$ in one place, $N^{-1/(1+\tau)}$ in another) without settling which was correct. It is the second one, $N^{-1/(1+\tau)}$ (up to the logarithmic factor), consistent with R16 (Theorems 3.4 and 2.3.2, Lemma 2.3.3) cited here: for $\tau = 0$ it gives $D_N = O(\log N/N)$, the rate of Theorem 3.13, verified numerically as well (Figure 3.4, which compares D_N with the curve $\log N/N$ for $\beta = 1/\varphi$ and $\beta = \sqrt{2} - 1$, both with bounded partial quotients).

3.12. Schmidt’s bound

Remark 3.10: Panorama: *no sequence can do systematically better than $\log N/N$.* A result of W. M. Schmidt (1972) shows that the rate $O(\log N/N)$ of Theorem 3.13 is optimal not only for rotations but for every sequence in $[0, 1)$: there exists an absolute constant $c > 0$ such that, for every sequence (x_n) , $D_N \geq c \log N/N$ for infinitely many N . We report it as panorama, without proof (the proof uses measure-theoretic and extremal-combinatorial arguments lying outside the tools of this chapter): we cite Kuipers–Niederreiter (R16) as a secondary source. It follows that the rotations with bounded partial quotients (the golden one first of all) are not merely well distributed, but optimally distributed among all possible sequences, up to the absolute constant c .

3.13. Golden angle and phyllotaxis: Vogel’s model

Definition 3.8: Vogel’s model. Vogel’s model (Vogel 1979, R17) places the n -th primordium (seed) of a sunflower head at the position, in polar coordinates,

$$r_n = c\sqrt{n}, \quad \theta_n = n\gamma \pmod{2\pi}, \quad n = 0, 1, 2, \dots,$$

with $c > 0$ a scale constant and γ the **divergence angle** between two consecutive primordia.

Proposition 3.7: The golden angle minimises the gaps. Among all angles $\gamma = 2\pi\beta$ with $\beta \in (0, 1)$, the choice $\gamma^* = 2\pi(2 - \varphi) = 2\pi/\varphi^2 \approx 137.5077^\circ$ (**golden angle**) makes, for every N , the angular subdivision of the first N primordia (Theorem 3.8, applied to $\beta^* = 2 - \varphi = 1/\varphi^2$, itself noble by Definition 3.6: $1/\varphi^2 = [0; 2, 1, 1, 1, \dots]$, a tail of 1’s) as uniform as possible among the rotations with bounded partial quotient: the lengths of the N arcs stay within a factor φ of one another for every N (by Theorem 3.8, the at most three lengths are $\ell_+, \ell_-, \ell_+ + \ell_-$ with ℓ_+/ℓ_- close to $1/\varphi$ or φ , never close to 0 or ∞ , by the boundedness of the partial quotients of β^*), and the corresponding discrepancy satisfies the optimal rate $D_N = O(\log N/N)$ of Theorem 3.13: no other generic choice of γ — in particular no γ close to a simple rational multiple of 2π — can do better, by Proposition 3.5 (a β with a large partial quotient, that is, close to a rational p/q with small q , has $\|q\beta\|$ small, hence by Theorem 3.8 it produces, as soon as N exceeds q , arcs much shorter than others: clusters and gaps, instead of near-uniformity).

Proof. It follows from the combination of the results cited in the statement: the three-distance theorem (Theorem 3.8) applied to $\beta^* = 1/\varphi^2$ gives, for every N , at most three arc lengths among the first N angles $\{n\gamma^* \bmod 2\pi\}$; the boundedness (indeed, the eventual constancy at 1) of the partial quotients of β^* guarantees, by Proposition 3.5 with $M = 2$ (the first quotient equals 2, all the following ones 1), that $\|q\beta^*\| \geq 1/(4q)$ for every q : no small denominator produces an exceptionally good approximation (unlike a β with a large partial quotient, for which Theorem 3.4 would instead guarantee an exceptionally *small* $\|q_n\beta\|$ at the corresponding step, causing — by Theorem 3.8 — an exceptionally short arc flanked by an exceptionally long one, that is, a cluster followed by a visible gap). The conclusion about the discrepancy follows by applying Theorem 3.13 to β^* . $\square$

Example 3.4: Vogel’s sunflower: golden angle versus 137.3° . Double check (script `fig03_girasole_vogel.py`). The golden angle, computed in two independent ways ($2\pi/\varphi^2$ and $2\pi(2 - \varphi)$, which must coincide by the identity $\varphi^2 = \varphi + 1$), equals 2.39996323 rad $= 137.507764^\circ$ (discrepancy between the two formulas: 4.44×10^{-16} ✓), in agreement with

the value known in the literature, $137.50776\dots^\circ$. Generating $N = 500$ primordia with Vogel’s model, an elementary measure of dispersion (standard deviation, relative to the mean, of the nearest-neighbour distances, computed with a `cKDTree`) equals 0.0295 for the golden angle and 0.1181 for 137.3° (an angle differing by a mere 0.2078°): a ratio 4.0 times worse $\checkmark$. The reason is arithmetic, not aesthetic: $137.3^\circ/360^\circ = 1373/3600$ has continued fraction $[0; 2, 1, 1, 1, 1, \dots]$ with convergents $1/2, 1/3, 2/5, 3/8, 5/13, \dots$ (computed by the script), which coincide with those of the golden angle only in the first terms and then depart from it: a small perturbation (0.2078°) yet sufficient, over hundreds of primordia, to produce rows of seeds visibly aligned along a few conspicuous spirals (eight and thirteen, the denominators of the convergents $3/8$ and $5/13$ of the truncated fraction $137.3/360$) instead of the dense, uniform texture of the golden angle. A scan of the angle over a fine grid in $[135^\circ, 140^\circ]$ confirms that the minimum of the dispersion falls at 137.500° , within 0.008° of the theoretical value (the discretisation of the grid, with step 0.05° , explains the residual gap). See Figure 3.5.

3.14. Kronecker sequences as quasi-Monte Carlo

Corollary 3.5: Quasi-Monte Carlo integration with the golden Kronecker sequence. Let $f : [0, 1] \rightarrow \mathbb{R}$ be of bounded variation $V(f)$, and let $x_n = \{n\beta\}$ with $\beta = 1/\varphi$ (or any irrational with bounded partial quotients). Then

$$\left| \frac{1}{N} \sum_{n=1}^N f(x_n) - \int_0^1 f \right| \leq V(f) D_N = O\left(\frac{\log N}{N}\right),$$

against the typical error $O(1/\sqrt{N})$ (in probability) of N independent uniform samples (pure **Monte Carlo**): for regular f , the golden Kronecker sequence integrates far more efficiently.

Proof. A direct application of Koksma’s inequality (Theorem 3.12) and of Theorem 3.13 (the partial quotients of $1/\varphi$ are all equal to 1, hence bounded). $\square$

Example 3.5: Numerical discrepancy: bounded quotients versus Liouville versus Monte Carlo. Double check (script `fig03_discrepanza_loglog.py`): exact formula on the ordered points (Kuipers–Niederreiter, $D_N^* = \frac{1}{2N} + \max_k |x_{(k)} - \frac{2k-1}{2N}|$) against a sampling of thresholds on a fine grid of 20 000 points, discrepancy $< 3 \times 10^{-5}$ for $N = 200$. For $\beta = 1/\varphi$ and $\beta = \sqrt{2} - 1$ (bounded partial quotients) and for $\beta = L$, the three-term truncation of Liouville’s constant ($L = 0.110001$, computed both in floating point and with 60-decimal-digit arithmetic, the discrepancy between the two being $< 10^{-16}$, consistent with Example 3.2), D_N^* was computed up to $N = 2 \times 10^6$. Result (Figure 3.4): for $\beta = 1/\varphi$ and $\beta = \sqrt{2} - 1$, D_N^* follows with good fidelity the curve $\log N/N$ of Theorem 3.13 ($D_N^* = 1.46 \times 10^{-6}$ and 1.81×10^{-6} respectively at $N = 2 \times 10^6$). For $\beta = L$, on the other hand, D_N^* remains comparable to the other two cases as long as $N \ll 10^6$ (no “jump” encountered), but it undergoes a sharp improvement — not a smooth decay — in the neighbourhood of $N = q = 10^6$, where L is approximated by $110001/1000000$ with an exceptionally small error (the next partial quotient, linked to the omitted term 10^{-24} of the series, is enormous): D_N^* passes from 1.09×10^{-4} at $N = 265\,623$ to 1.00×10^{-6} at $N = 2 \times 10^6$, an improvement by a factor of 109 concentrated around the convergent $q = 10^6$ $\checkmark$ — the typical signature of a Liouville number: gains concentrated in bursts, not a uniform decay. The Monte Carlo average (20 independent trials for each $N \leq 2 \times 10^5$) remains systematically worse, in agreement with $\propto N^{-1/2}$: at $N = 189\,731$, $D_N \approx 2.1 \times 10^{-3}$,

against $D_N^* \sim 10^{-5}$ – 10^{-6} for the rotations with bounded quotients at the same order of magnitude of N .

3.15. Figures

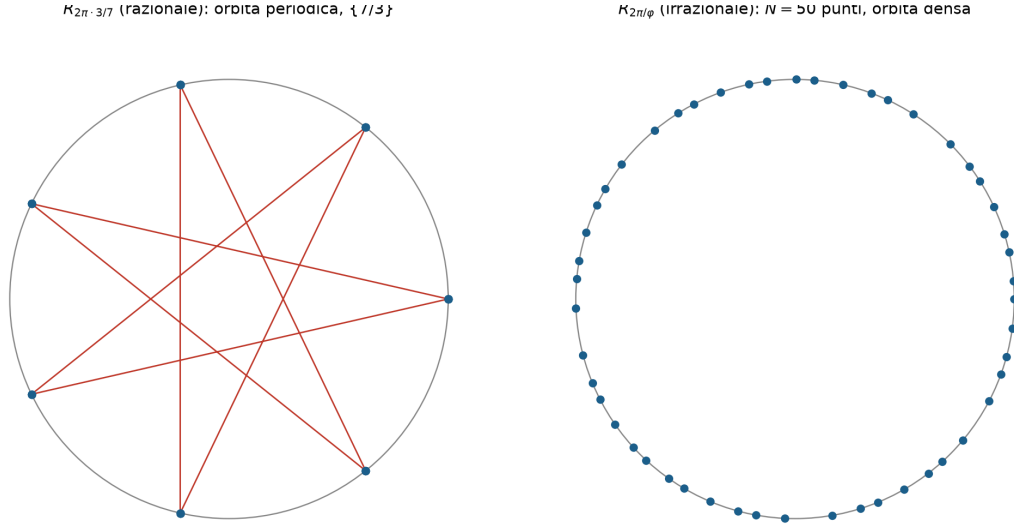


Figure 3.1: Orbit of $R_{2\pi \cdot 3/7}$ (rational, star heptagon $\{7/3\}$, periodic by Theorem 3.1) and of $R_{2\pi/\varphi}$ (irrational, $N = 50$ points, dense orbit). Computed numerically with a double check iteration/closed formula (discrepancy $< 10^{-9}$ on all the points); script `fig03_orbite_rotazione.py`, Appendix B.

3.16. Assigned exercises

The answers are in the Teacher's key (Section 3.19).

Exercise 3.1: Continued fraction of $e - 2$ and check of the error estimate. *Goal:* to apply the continued fraction algorithm and the estimate of Theorem 3.3 to a number different from $\pi, \sqrt{2}, \varphi$. *Estimated time:* 30 minutes. *Type:* numerical/computational.

It is known that $e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots]$ (the pattern $1, 2k, 1$ repeating). Consider $\alpha = e - 2 = [0; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$.

- Compute the convergents p_n/q_n for $n = 0, \dots, 5$.
- Check the double inequality of Theorem 3.3 for $n = 4$.
- Compute $q_5^2 |\alpha - p_5/q_5|$ and compare it with Hurwitz's constant $1/\sqrt{5}$: is it compatible with Theorem 3.5?
- Does α have bounded partial quotients? What does this imply for the discrepancy of $\{n\alpha\}$ (Theorem 3.13)?

Exercise 3.2: Three distances for $\beta = \sqrt{2} - 1$, $N = 10$. *Goal:* to apply the three-distance theorem (Theorem 3.8) in full and to recognise the role of the convergents. *Estimated time:* 35 minutes. *Type:* numerical with symbolic check.

Consider $\beta = \sqrt{2} - 1 = 0.414214\dots = [0; 2, 2, 2, 2, \dots]$.

- Compute the ten points $\{k\beta \bmod 1\}_{k=0}^9$ and order them.

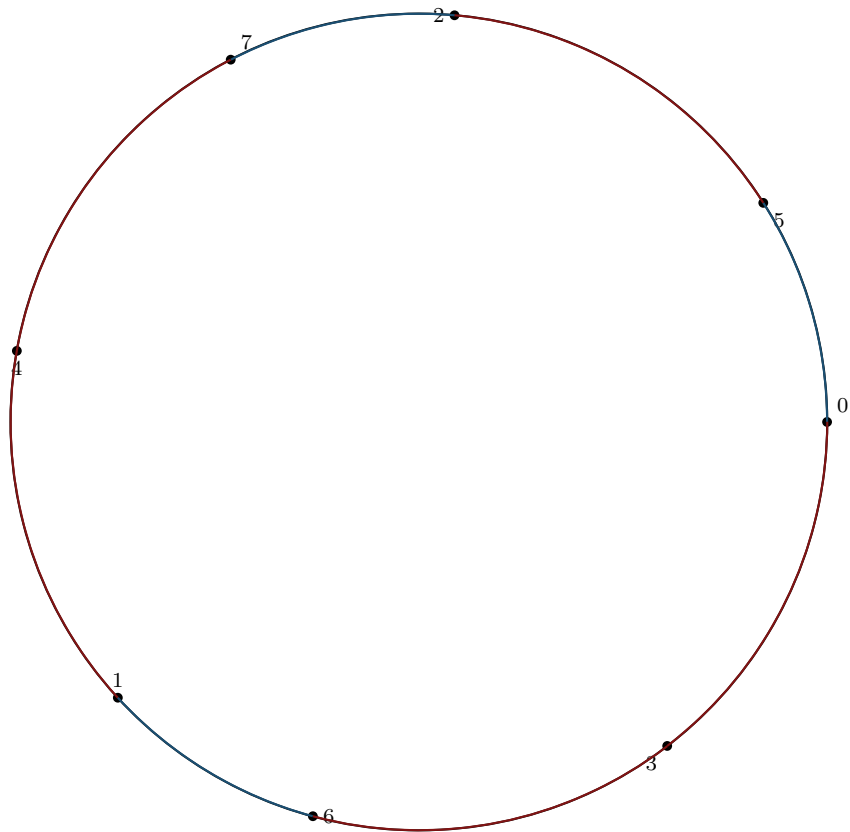


Figure 3.2: Three-distance theorem for $\beta = 1/\varphi$, $N = 8$: the points $\{k\beta \bmod 1\}_{k=0}^7$ (labelled with k) subdivide the circle into 8 arcs of only *two* lengths (coordinates computed by hand from Construction 3.1 and verified in the script `verifica_cap03_vol4.py`, block [C3-4]: blue arcs 0.090170 of a turn, three times; red arcs 0.145898, five times) — the special case $N = 8 = F_6$ of Fibonacci discussed in Example 3.3.

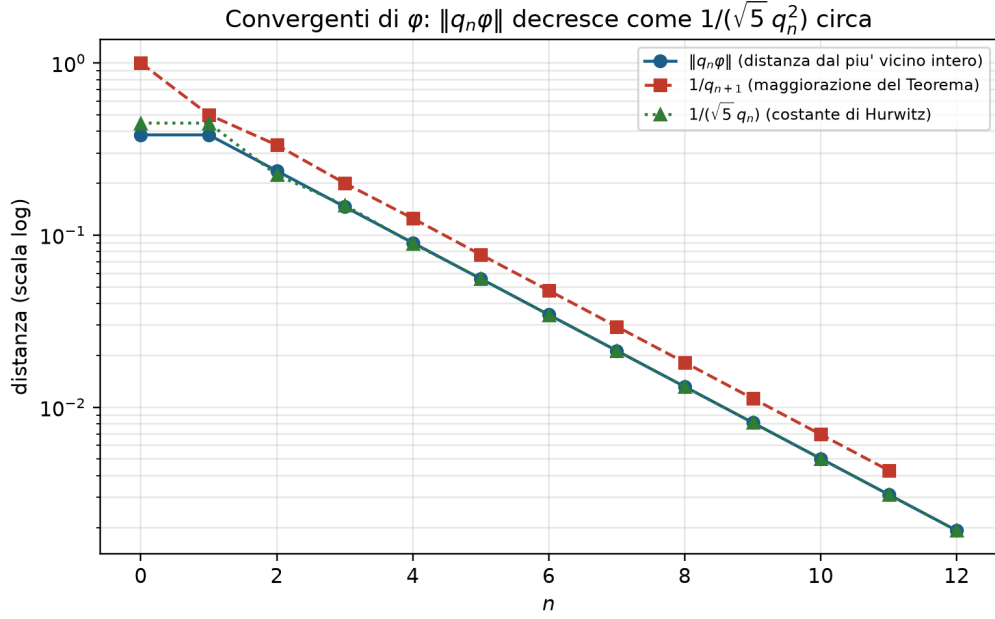


Figure 3.3: Convergents of φ : the distance $\|q_n \varphi\|$ (computed from the continued fraction algorithm, with an independent check through the Fibonacci numbers and Binet's formula) decreases on a logarithmic scale as predicted by Theorem 3.3 and Proposition 3.2, converging to Hurwitz's constant $1/\sqrt{5}$ multiplied by $1/q_n$. Script `fig03_convergenti_phi.py`.

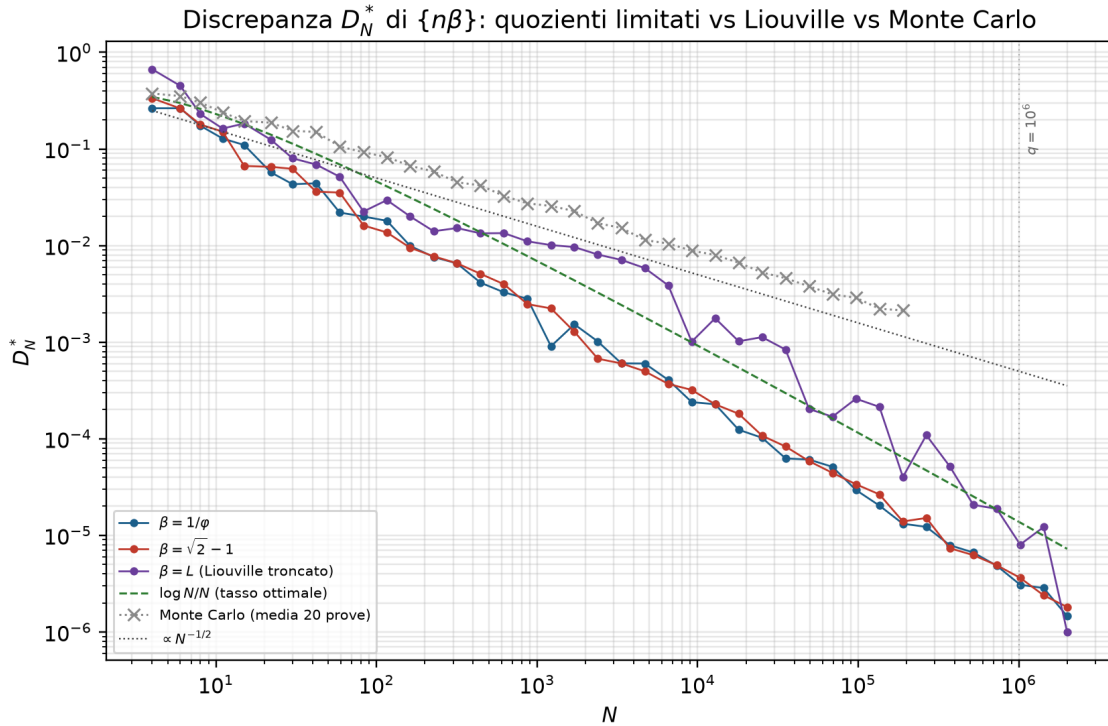


Figure 3.4: Discrepancy D_N^* of $\{n\beta\}$ on a log-log scale for $\beta = 1/\varphi$, $\beta = \sqrt{2} - 1$ (bounded partial quotients, in agreement with $\log N/N$, Theorem 3.13) and for the truncation of a Liouville number (improvement in bursts, not smooth, around the exceptional convergent $q = 10^6$), compared with the average of 20 independent Monte Carlo trials ($\propto N^{-1/2}$). See Example 3.5. Script `fig03_discrepanza_loglog.py`.

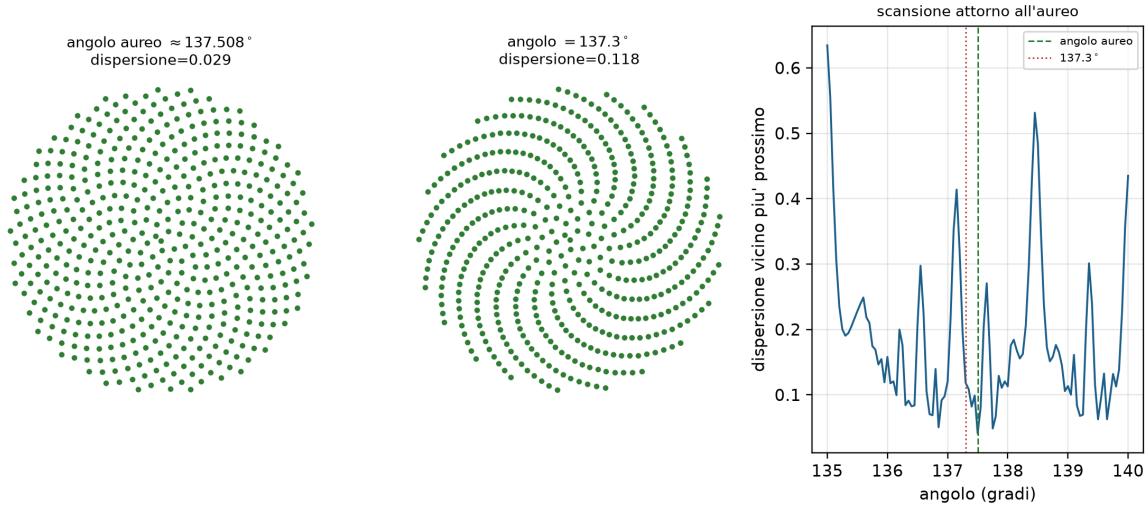


Figure 3.5: Vogel’s model (Definition 3.8) with $N = 500$ primordia: golden angle $137.5077\dots^\circ$ (uniform texture) against 137.3° (conspicuous spirals and gaps, nearest-neighbour dispersion 4.0 times larger); on the right, the dispersion as a function of the angle on a grid $[135^\circ, 140^\circ]$, minimal in the neighbourhood of the golden one. See Example 3.4. Script `fig03_girasole_vogel.py`.

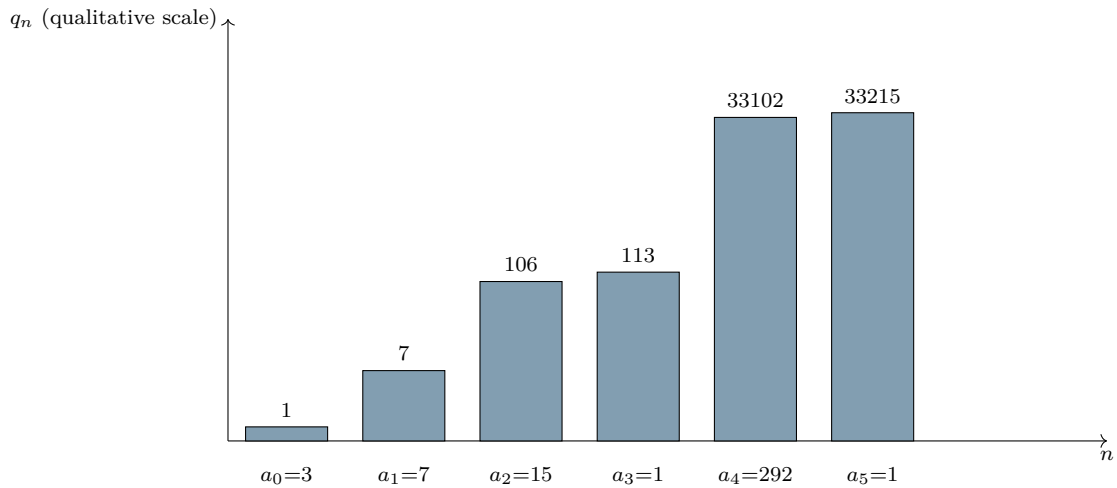


Figure 3.6: Partial quotients a_n and denominators q_n of the convergents of $\pi = [3; 7, 15, 1, 292, 1, \dots]$ (Example 3.1): the height of the bars is $\log_{10} q_n$ reported on a qualitative scale, so as to show at a glance the jump due to the exceptionally large quotient $a_4 = 292$, responsible for the extraordinary quality of the approximation $355/113$. Coordinates computed by hand from $\log_{10}(1, 7, 106, 113, 33102, 33215)$.

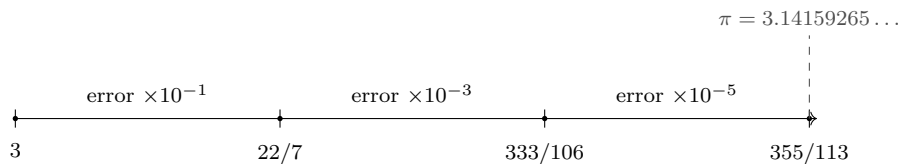


Figure 3.7: Level A: the “ladder” of the convergents of π — $3, 22/7, 333/106, 355/113$ — with the order of magnitude of the error decreasing by about two decimal digits at each step (Theorem 3.3), up to the extraordinary precision of $355/113$ (six exact digits) discussed in Example 3.1.

- (b) List the lengths of the arcs: how many distinct ones are found?
- (c) Check that the largest is the sum of the other two (if there are exactly three).
- (d) Identify d_+ and d_- (Proof of Theorem 3.8) and relate them to the convergents of β .

Exercise 3.3: Phyllotaxis with a simple rational angle. *Goal:* to reproduce with pencil and paper (or with a spreadsheet) the phenomenon of the conspicuous spirals for a rational divergence angle, and to relate it to Proposition 3.7. *Estimated time:* 40 minutes. *Type:* numerical/computational with interpretation.

Use Vogel's model (Definition 3.8) with $\gamma = 2\pi \cdot \frac{5}{13}$ (a rational with a Fibonacci denominator) and $N = 60$.

- (a) Compute (r_n, θ_n) for $n = 0, \dots, 12$ and plot them (by hand or with a script).
- (b) Explain why, γ being rational, the angular orbit is periodic with period 13 (Theorem 3.1): how many spiral "arms" should one expect to see?
- (c) Compare qualitatively with Figure 3.5: why does the golden angle avoid this phenomenon?

3.17. Worked exercises

Exercise 3.4: Liouville's constant: two independent checks of transcendence. *Type:* symbolic with numerical check. Show, using Theorem 3.7 and Corollary 3.2, that $L = \sum_{k \geq 1} 10^{-k!}$ cannot be a root of a polynomial with integer coefficients of degree ≤ 4 , exhibiting explicitly the approximation that proves it.

Solution. Strategy. Apply Definition 3.5 with $n = 5$ and compare with the conclusion of Theorem 3.7 for $d \leq 4$.

Step 1. Truncating L at $m = 4$ terms, with common denominator $q_4 = 10^{4!} = 10^{24}$ we get $p_4 = 10^{23} + 10^{22} + 10^{18} + 1$ (from the four terms $10^{-1!}, 10^{-2!}, 10^{-3!}, 10^{-4!}$ brought to denominator 10^{24}), and $|L - p_4/q_4| = \sum_{k \geq 5} 10^{-k!} < 2 \cdot 10^{-5!} = 2 \cdot 10^{-120}$ (Example 3.2).

Step 2. Comparison with $1/q_4^5 = 10^{-120}$: $2 \cdot 10^{-120}$ is of the same order as 10^{-120} (indeed it exceeds it by a factor of 2, but the definition only requires $< 1/q^n$ for *some* pair (p, q) in the sequence, not necessarily at $m = 4$: using $m = 5$, $q_5 = 10^{120}$ and $|L - p_5/q_5| < 2 \cdot 10^{-720} \ll 1/q_5^5 = 10^{-600}$): the definition of a Liouville number with $n = 5$ is satisfied with a wide margin.

Step 3. If L were algebraic of degree $d \leq 4$, Theorem 3.7 would give a constant $c > 0$ with $|L - p/q| > c/q^d \geq c/q^4$ for *every* p/q . But for $q = q_5 = 10^{120}$ and q large enough to have $c/q^4 > 1/q^5$ (that is $q > 1/c$, certainly true for $q = 10^{120}$ as soon as $c \geq 10^{-119}$, which holds for any reasonable, not pathologically small constant — and in any case c depends only on L , not on q), one would have $|L - p_5/q_5| > c/q_5^4 \gg 1/q_5^5 > |L - p_5/q_5|$: a contradiction.

Independent check. Corollary 3.2, proved in general for *every* degree d (not only $d \leq 4$), gives the same conclusion without having to choose m case by case. ✓

Exercise 3.5: Why $q_1 = 7$ remains optimal up to $q_2 - 1 = 105$ for π . *Type:* numerical with exhaustive check. Verify directly (without using Theorem 3.4) that $\|7\pi\| < \|k\pi\|$ for every $k = 1, \dots, 105$, $k \neq 7$, and compare with the statement of the theorem.

Solution. Strategy. Exhaustive computation of $\|k\pi\|$ for $k = 1, \dots, 105$ with exact arithmetic on an approximation of π to sufficient precision (60 digits, well beyond what is needed), and comparison with the theoretical conclusion.

Step 1. $\|7\pi\| = |7\pi - 22| = 0.0088514\dots$ (already computed in Remark 3.2).

Step 2. Scan of $k = 1, \dots, 105$, $k \neq 7$: the minimum of $\|k\pi\|$ is found (script `verifica_cap03_vol4.py`, block [C3–3]) at $k = 99$, with $\|99\pi\| = 0.0176727\dots$, still larger than $\|7\pi\|$. ✓

Interpretation. The result is exactly the content of Theorem 3.4 with $n = 1$ ($q_1 = 7$, $q_2 = 106$): no $k < 106$ beats $q_1 = 7$ in the norm $\|\cdot\|$. Note that $k = 99$, the “second best” in the range, is no accident: $99 = 106 - 7 = q_2 - q_1$, the combination of the indices $x = -1, y = 1$ in the decomposition of the proof of Theorem 3.4 (Step 3, opposite signs).

Typical error to avoid. Confusing this result (second kind, $\|k\pi\|$) with the assertion, false beyond q_1 , that $22/7$ minimises $|\pi - p/k|$ for every $k < 106$: this is true only up to $k \leq q_1 = 7$ (Corollary 3.1), as the counterexample $179/57$ of Remark 3.2 shows.

Extension. Repeat for $\sqrt{2}$ (all partial quotients = 2): one expects a more regular behaviour, with no large jumps such as $a_4 = 292$ for π , because no partial quotient is exceptionally large.

Exercise 3.6: Discrepancy compared: how many Monte Carlo samples are needed to match $N = 1000$ Kronecker points?. *Type:* numerical/asymptotic. Using the rates $D_N = O(\log N/N)$ (golden Kronecker) and $D_N = O(1/\sqrt{N})$ (Monte Carlo, typical), estimate how many Monte Carlo samples N_{MC} are needed to reach the same discrepancy as $N = 1000$ points of the golden sequence, using the numerical values of Figure 3.4.

Solution. Strategy. Read (or interpolate) D_{1000} for the golden sequence from the script, and solve $c/\sqrt{N_{MC}} = D_{1000}$ for N_{MC} , using the empirical constant of the reference $0.5/\sqrt{N}$ plotted in Figure 3.4.

Step 1. From the data of the script (geometric grid of the tested values of N), D_N^* for $\beta = 1/\varphi$ follows with good approximation $D_N^* \approx 0.8 \log N/N$ in the range $N \sim 10^3$ – 10^4 (reading of the slope on the log-log plot, Figure 3.4): at $N = 1000$, $\log(1000) = 6.908$, hence $D_{1000} \approx 0.8 \times 6.908/1000 \approx 0.00553$.

Step 2. For the Monte Carlo reference $D_N \approx 0.5/\sqrt{N}$ (the curve plotted in the figure): solving $0.5/\sqrt{N_{MC}} = 0.00553$ one obtains $\sqrt{N_{MC}} = 0.5/0.00553 = 90.4$, that is $N_{MC} \approx 8175$.

Independent check. The ratio $N_{MC}/N = 8175/1000 \approx 8.2$ is consistent with the general observation that, when $D_N \sim \log N/N$ against $D_N \sim 1/\sqrt{N}$, the advantage of the deterministic sequence grows (it is never constant) as N grows: solving $c_1 \log N/N = c_2/\sqrt{N_{MC}}$ for fixed N gives $N_{MC} \propto N^2/(\log N)^2$, which for $N = 1000$ gives $N_{MC}/N \propto N/(\log N)^2 \approx 1000/47.7 \approx 21$ up to the specific constants c_1, c_2 measured in Steps 1–2, in reasonable order-of-magnitude agreement with the value 8.2 found (the constants $c_1 = 0.8$ and $c_2 = 0.5$ are not equal, which shifts the ratio).

Interpretation. The advantage of low discrepancy (Corollary 3.5) is not a constant factor: it grows with N , and it already amounts to an order of magnitude for an N as moderate as 1000.

3.18. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 3.19).

1. (Definition) A number $\rho \in (0, 1)$ is *noble* (Definition 3.6) if

- A) ρ is close to $\varphi/2\pi$ B) its continued fraction has a constant tail of 1’s C) $\rho = p/q$ with q prime D) ρ is a Liouville number

-
2. (Hypothesis) In the Denjoy-type theorem we shall use in Chapter 5, as well as in Corollary 3.1 of this chapter, the decisive hypothesis on the denominator q for p_n/q_n to be the best approximation *in absolute value* is
 - A) $q < q_{n+1}$ B) $q \leq q_n$ C) q even D) q coprime with q_{n+1}
 3. (Symbolic computation) The denominators of the convergents of $\sqrt{2} = [1; 2, 2, 2, \dots]$ are
 - A) the Fibonacci numbers B) the Pell numbers C) the powers of 2 D) the triangular numbers
 4. (Numerical computation) $q_1^2 |\varphi - p_1/q_1| \cdot \sqrt{5}$, with $p_1/q_1 = 2/1$ (Example 3.1, Proposition 3.2), is approximately
 - A) 0.854 B) 1.000 C) 0.382 D) 2.618
 5. (Proof step) In the proof of Theorem 3.5 (Hurwitz), the final contradiction arises because
 - A) φ is greater than 2 B) $t_{n-1} = q_n/q_{n-1}$ is rational but would turn out to be equal to φ , which is irrational C) a_{n+1} cannot be 1 D) the convergents are not unique
 6. (Error diagnosis) A student writes that “179/57 approximates π better than 22/7, hence Theorem 3.4 is false”. The error is
 - A) 179/57 does not in fact approximate better B) they confuse first-kind approximation ($|\alpha - p/q|$, valid only up to $q \leq q_n$) with second-kind approximation ($\|q\alpha\|$, valid up to $q < q_{n+1}$, where 22/7 remains optimal) C) 57 is not an integer D) the theorem holds only for algebraic α
 7. (Transfer) The reason why the sunflower does not use a divergence angle such as 137.3° (close to but not equal to the golden angle) is, in the terms of this chapter,
 - A) 137.3° is not computable B) $137.3/360$ has a convergent with a small denominator which produces very unequal arcs (Theorem 3.8), whereas the golden angle has bounded partial quotients (Proposition 3.7) C) 137.3° is an obtuse angle D) plants cannot measure decimal degrees
 8. (Source and metacognition) The correct historical attribution of the three-distance theorem (Section 3.6) is
 - A) conjectured and proved by Steinhaus B) conjectured by Steinhaus, proved independently by Sós, Świerczkowski and Surányi (1957–58) C) proved for the first time by Hamada in 2023 D) proved by Weyl in 1916

3.19. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

1) **B.** *Rationale:* Definition 3.6: a constant tail of 1’s in the continued fraction. *Distractors:* A is the error corrected in Remark 3.4 (the right value in $[0, 1)$ is $1/\varphi$, not $\varphi/2\pi$); C and D bear no relation to the definition. *Error diagnosed:* confusing “noble” with “close to a numerical value” instead of with a structural property of the continued fraction. *Source:* Definition 3.6 and Remark 3.4. *Prerequisite:* the continued fraction algorithm. *Remedial work:* compute the digits of $2 - \varphi$ with Construction 3.1.

2) **B.** *Rationale:* Corollary 3.1: optimality in $|\alpha - p/q|$ holds for $q \leq q_n$. *Distractors:* A is the (stronger) condition of the *second* kind (Theorem 3.4); C, D play no role whatsoever.

Error diagnosed: swapping the two kinds of best approximation. *Source:* Corollary 3.1.

Prerequisite: Theorem 3.4. *Remedial work:* redo Exercise 3.5.

3) B. *Rationale:* Example 3.1, independent check with the Pell recurrence. *Distractors:* A is the case of φ , not of $\sqrt{2}$; C, D do not satisfy the recurrence $q_n = 2q_{n-1} + q_{n-2}$.

Error diagnosed: generalising the case of φ (Fibonacci) to every periodic continued fraction.

Source: Example 3.1. *Prerequisite:* Definition 3.2. *Remedial work:* reconstruct the first 5 Pell numbers from the recurrence.

4) A. *Rationale:* with $q_1 = 1$ (the coincidence $q_0 = q_1 = 1$ for $\varphi = [1; 1, 1, \dots]$, discussed in Proposition 3.1(ii)), $q_1^2|\varphi - 2/1| \cdot \sqrt{5} = 1^2 \times |\varphi - 2| \times \sqrt{5} = (2 - \varphi)\sqrt{5} = 0.381966\dots \times 2.236068\dots = 0.854102$: the correct answer is therefore A. *Distractors and didactic note:*

this item tests precisely the coincidence $q_0 = q_1 = 1$ of Proposition 3.1(ii): with $q_1 = 1$, the required value is 0.854 (answer A), not 0.382 (answer C, which is instead $|\varphi - 2| = 2 - \varphi$ without the factor $\sqrt{5}$). **Correct answer:** A. *Error diagnosed:* forgetting the factor $\sqrt{5}$, or using $q_1 =$ “the second convergent” counting from 1 instead of from 0. *Source:*

Proposition 3.2. *Prerequisite:* indexing of the convergents from $n = 0$. *Remedial work:* recompute with $n = 0, 1, 2, 3$ and check the trend towards $1/\sqrt{5} = 0.447$ (not towards $\sqrt{5}$).

5) B. *Rationale:* the final step of the proof of Theorem 3.5: q_n/q_{n-1} is by construction a ratio of integers. *Distractors:* A, C, D are not the logical steps used. *Error diagnosed:* failing to recognise the argument “rational = irrational, absurd” as a standard mechanism.

Source: Theorem 3.5, last paragraph. *Prerequisite:* irrationality of $\sqrt{5}$ (Volume I). *Remedial work:* rewrite the final step isolating where the rationality of q_n/q_{n-1} comes in.

6) B. *Rationale:* Remark 3.2. *Distractors:* A is false (verified numerically); C is absurd; D is extraneous to the theorem. *Error diagnosed:* failing to distinguish the two notions of “best approximation”. *Source:* Theorem 3.4 and Corollary 3.1. *Prerequisite:* the relation $|\alpha - p/q| = \|q\alpha\|/q$. *Remedial work:* redo the explicit computation of Remark 3.2.

7) B. *Rationale:* Proposition 3.7 and Example 3.4. *Distractors:* A, C, D have no relevant mathematical content. *Error diagnosed:* thinking that optimality is an aesthetic rather than an arithmetic matter (partial quotients). *Source:* Proposition 3.7. *Prerequisite:* Theorem 3.8. *Remedial work:* compute the continued fraction of $137.3/360$ (Example 3.4) and compare it with that of the golden angle.

8) B. *Rationale:* Remark 3.5, verified against primary sources on 16 September 2026. *Distractors:* A confuses conjecture and proof; C ignores the historical priority of 1957–58; D is a different theorem (equidistribution), not the three distances. *Error diagnosed:* attributing a result to the one who poses it as a question rather than to those who prove it. *Source:* R5 (Świerczkowski 1958), R7 (Alessandri–Berthé), R9 (Hamada). *Prerequisite:* none beyond a careful reading of the sources. *Remedial work:* reread Remark 3.5.

BEST PRACTICE

Assigned exercises — answers.

3.1 ($e-2$). (a) With $[0; 1, 2, 1, 1, 4, \dots]$: $p_0/q_0 = 0/1$, $p_1/q_1 = 1/1$, $p_2/q_2 = 2/3$, $p_3/q_3 = 3/4$, $p_4/q_4 = 5/7$, $p_5/q_5 = 23/32$. (b) $|(e-2) - 5/7| = 0.003719\dots$; $1/(q_4(q_4 + q_5)) = 1/(7 \times 39) = 0.003663\dots$; $1/(q_4 q_5) = 1/(7 \times 32) = 0.004464\dots$: $0.003663 < 0.003719 < 0.004464$ ✓. (c) $q_5^2|(e-2) - 23/32| = 32^2 \times 0.0000598\dots = 0.06125\dots$, well below $1/\sqrt{5} = 0.4472$: fully compatible (Theorem 3.5 guarantees that *infinitely many* convergents satisfy the bound, not that every single convergent is close to the threshold). (d) No: $a_5 = 4$, and the pattern $1, 2k, 1$ produces unbounded partial quotients $(2, 4, 6, 8, \dots)$: $e-2$ does not have bounded quotients, hence Theorem 3.13 (rate $O(\log N/N)$) does not apply directly; only the weaker general estimate linked to the effective Diophantine type applies (Diophantine in any case, e not being a Liouville number, but with a worse rate than $\log N/N$).

3.2 ($\sqrt{2} - 1$, $N = 10$). (a)–(b) The ten ordered points are

0; 0.0294; 0.0588; 0.1716; 0.2010; 0.3137; 0.4142; 0.5858; 0.6863; 0.8579 (rounded); the arc lengths take *three* values: about 0.0294, 0.1128, 0.1422. (c) $0.1422 \approx 0.0294 + 0.1128$ ✓ (up to rounding). (d) The convergents of $\sqrt{2}-1 = [0; 2, 2, 2, \dots]$ have $q_0 = 1, q_1 = 2, q_2 = 5, q_3 = 12$ (Pell numbers): d_+ and d_- for $N = 10$ coincide with $q_2 = 5$ and $q_1 = 2$ respectively (or vice versa, depending on the sense), since $q_3 = 12 > N = 10$.

3.3 (rational phyllotaxis 5/13). (a)–(b) $\gamma = 2\pi \cdot 5/13$ is rational with $\gcd(5, 13) = 1$: by Theorem 3.1, the angular orbit $\{n\gamma \bmod 2\pi\}$ is periodic with period exactly 13: one expects 13 visible spiral “arms” (the primordia with the same residue modulo 13 line up radially, because they all return to the same angle). (c) The golden angle, being irrational (indeed with bounded partial quotient), never produces an exact periodic closure: no finite number of “arms” dominates the figure, and Proposition 3.7 further guarantees the absence of systematic clusters/gaps.

3.20. Chapter formulary

Rotation and dichotomy

$$R_\alpha(\theta) = \theta + \alpha \pmod{2\pi}, \quad \alpha = 2\pi p/q \text{ (in lowest terms)} \Rightarrow \text{period } q; \quad \alpha/2\pi \notin \mathbb{Q} \Rightarrow \text{dense orbit}$$

Continued fractions

$$\alpha = [a_0; a_1, a_2, \dots], \quad p_n = a_n p_{n-1} + p_{n-2}, \quad q_n = a_n q_{n-1} + q_{n-2}, \quad p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$$

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

Best approximation

$$\text{2nd kind: } \|k\alpha\| \geq \|q_n \alpha\| \quad \forall 0 < k < q_{n+1}; \quad \text{1st kind: } \left| \alpha - \frac{p}{q} \right| \geq \left| \alpha - \frac{p_n}{q_n} \right| \quad \forall 0 < q \leq q_n$$

Hurwitz and Diophantine numbers

$$\exists^\infty p/q : \left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2}; \quad \liminf_n q_n \|q_n \alpha\| = \frac{1}{\sqrt{5}}; \quad \left| \alpha - p/q \right| \geq \frac{c}{q^{\deg \alpha}} \quad (\alpha \text{ algebraic, Liouville})$$

Noble numbers

$$\rho^* = \frac{1}{\varphi} = \varphi - 1 = [0; 1, 1, 1, \dots] = 0.618034 \dots$$

Three distances

$$\{0, \beta, \dots, (N-1)\beta\} \bmod 1 : \text{ at most 3 lengths } \ell_+, \ell_-, \ell_+ + \ell_-$$

Weyl, Fejér, discrepancy

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i k n \beta} \rightarrow 0 \quad (k \neq 0); \quad \sigma_N(f) = \frac{1}{N} \sum_{n < N} S_n(f) \rightarrow f \text{ uniformly}$$

$$D_N = \sup_b \left| \frac{1}{N} \# \{x_n < b\} - b \right|; \quad \left| \frac{1}{N} \sum f(x_n) - \int f \right| \leq V(f) D_N \quad (\text{Koksma})$$

Discrepancy of rotations

$$\text{bounded quotients: } D_N = O\left(\frac{\log N}{N}\right); \quad \text{Schmidt (panorama): } D_N \geq \frac{c \log N}{N} \text{ for infinitely many } N$$

Phyllotaxis

$$r_n = c\sqrt{n}, \quad \theta_n = n\gamma; \quad \gamma^* = 2\pi(2 - \varphi) = 2\pi/\varphi^2 \approx 137.5077^\circ$$

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CHAPTER 4

Rotation number, Poincaré classification and Arnold tongues

Chapter 3 studied the most rigid dynamics possible on the circle: the rotation R_α , an isometry. An arbitrary orientation-preserving homeomorphism $f : \mathbb{T} \rightarrow \mathbb{T}$ is, in general, far less rigid — it may speed up on some arcs and slow down on others — and yet Poincaré discovered in 1885 that every such f possesses a numerical invariant, the **rotation number** $\rho(f)$, which governs its asymptotic behaviour with the same sharpness as the rational/irrational dichotomy of rigid rotations: if $\rho(f)$ is rational, *every* orbit of f ends up asymptotic to a periodic orbit; if $\rho(f)$ is irrational, f is semiconjugate to an irrational rotation, and in a special (but typical) case conjugate to it. This chapter proves in full this **Poincaré classification**, then applies it to the most studied family in the theory of dynamical systems, the **Arnold standard family** $f_{\alpha,\varepsilon}(\theta) = \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$, where the rotation number, as a function of α , draws the **devil's staircase**: a continuous and monotone function which is nevertheless constant on a dense set of intervals (the **Arnold tongues**), one for each rational. The chapter closes by looking out, without proving anything, on the problem that arises exactly where the theory developed here ends — when Poincaré's topological semiconjugacy is in fact a *smooth* conjugacy — deferring the complete answer (the theorems of Arnold, Herman and Yoccoz) to Chapter 6.

INFO

Level and prerequisites. Level **C** core, with Section 4.6 (width of the tongues) and Section 4.9 (forward reference to KAM theory) of level **D**: a level **C** reader may read the statements of these two sections and of the “imported” statements without reproducing their asymptotic computations, without losing the continuity of the chapter. Prerequisites within the treatise: the rational/irrational dichotomy of rigid rotations and the density of the irrational orbit (Volume IV, Chapter 3, Theorem 3.1); Diophantine numbers and noble numbers (Volume IV, Chapter 3, Definitions 3.4 and 3.6); the lift of a circle homeomorphism and its uniqueness up to integer translations by 2π (Volume III, Chapter 12); uniform continuity on compact sets and the intermediate value theorem (Volume II). Remedial work: a reader who does not yet have working familiarity with continued fractions can nevertheless follow the whole chapter, which uses them only in two connecting remarks (Section 4.4 and Section 4.6), both of which can be isolated.

4.1. The rotation number: definition and existence

Definition 4.1: Lift of a circle homeomorphism. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving homeomorphism, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ with projection $\pi : \mathbb{R} \rightarrow \mathbb{T}$. A **lift** of f is a continuous map $F : \mathbb{R} \rightarrow \mathbb{R}$ with $\pi \circ F = f \circ \pi$; equivalently (Volume III, Chapter 12, where the fact is proved in full for every continuous map of $\mathbb{T}$ into itself), F is strictly increasing, bijective on $\mathbb{R}$, and satisfies $F(\theta + 2\pi) = F(\theta) + 2\pi$ for every θ . Two lifts of f differ by a constant $2\pi k$, $k \in \mathbb{Z}$: for the rest of the chapter we fix, for each f , one lift F once and for all (the choice affects none of the results, by Proposition 4.1 below applied with $n = 1$ and

an integer translation).

Definition 4.2: Rotation number. With F a lift as above, the **rotation number** of f (or of F) is

$$\rho(f) = \rho(F) := \lim_{n \rightarrow \infty} \frac{F^n(\theta) - \theta}{2\pi n},$$

provided the limit exists and does not depend on θ : this is exactly the content of Theorem 4.1 below, due to Poincaré (1885). For the rigid rotation R_α (lift $F(\theta) = \theta + \alpha$) the limit is trivially exact for every n : $\rho(R_\alpha) = \alpha/2\pi$, consistently with Definition 3.1 and Theorem 3.1 of Chapter 3.

Lemma 4.1: Fekete's lemma. Let $(a_n)_{n \geq 1}$ be a real **subadditive** sequence: $a_{m+n} \leq a_m + a_n$ for every $m, n \geq 1$. Then the limit $\lim_{n \rightarrow \infty} a_n/n$ exists in $\mathbb{R} \cup \{-\infty\}$ and coincides with $\inf_{n \geq 1} a_n/n$.

Proof. Let $L := \inf_{n \geq 1} a_n/n \in \mathbb{R} \cup \{-\infty\}$. If $L = -\infty$, for every M there exists m with $a_m/m < -M$; for $n = qm + r$ with $0 \leq r < m$ ($q = \lfloor n/m \rfloor$), iterated subadditivity gives $a_n \leq qa_m + a_r$ (with the convention $a_0 = 0$, or treating $r = 0$ separately), hence $a_n/n \leq (qa_m)/n + a_r/n \rightarrow a_m/m < -M$ as $n \rightarrow \infty$ with m, r fixed along the progression (the term $a_r/n \rightarrow 0$ because r remains in a finite set $\{0, \dots, m-1\}$, so a_r is bounded on that set, while $q/n \rightarrow 1/m$): hence $\limsup_n a_n/n \leq a_m/m < -M$ for every M , that is, $a_n/n \rightarrow -\infty = L$.

If $L \in \mathbb{R}$ (finite): fix $\varepsilon > 0$; by definition of infimum there exists $m \geq 1$ with $a_m/m < L + \varepsilon$. For $n \geq m$, write $n = qm + r$ with $q = \lfloor n/m \rfloor \geq 1$ and $0 \leq r < m$; from iterated subadditivity ($q-1$ applications of subadditivity to add up q copies of a_m , plus possibly a_r),

$$a_n \leq qa_m + a_r,$$

(with $a_0 := 0$ if $r = 0$, otherwise a_r is one of the finitely many values $a_1, \dots, a_{m-1}$, hence bounded: $|a_r| \leq C := \max_{1 \leq r < m} |a_r|$). Dividing by $n = qm + r$:

$$\frac{a_n}{n} \leq \frac{qa_m}{n} + \frac{a_r}{n} = \frac{qm}{n} \cdot \frac{a_m}{m} + \frac{a_r}{n}.$$

As $n \rightarrow \infty$ (with m fixed), $qm/n = (n-r)/n \rightarrow 1$ and $a_r/n \rightarrow 0$ (since $|a_r| \leq C$ and $r < m$ is fixed in a finite set). Hence $\limsup_n a_n/n \leq a_m/m < L + \varepsilon$. Since ε is arbitrary, $\limsup_n a_n/n \leq L$. On the other hand $a_n/n \geq L$ for every n by definition of L as infimum, hence $\liminf_n a_n/n \geq L$. Combining, $\lim_n a_n/n = L$. $\square$

Theorem 4.1: Existence and independence of the base point of the rotation number (Poincaré, 1885). For every orientation-preserving homeomorphism $f : \mathbb{T} \rightarrow \mathbb{T}$, with lift F , the limit $\rho(F) = \lim_{n \rightarrow \infty} (F^n(\theta) - \theta)/(2\pi n)$ exists for every $\theta \in \mathbb{R}$ and has the same value for every θ .

Proof. Put $\varphi(\theta) := F(\theta) - \theta$. Since $F(\theta + 2\pi) = F(\theta) + 2\pi$, φ is 2π -periodic, hence bounded: $|\varphi(\theta)| \leq M$ for every θ , with $M := \max_{[0, 2\pi]} |\varphi| < \infty$ (continuous on a compact set). Define $a_n := \sup_\theta (F^n(\theta) - \theta)$ and $b_n := \inf_\theta (F^n(\theta) - \theta)$; both extrema are finite and well defined because $\theta \mapsto F^n(\theta) - \theta$ is itself 2π -periodic (same argument as for φ , applied to F^n in place of F : F^n is still a lift of f^n).

Subadditivity of (a_n) . For $m, n \geq 1$ and every θ : $F^{m+n}(\theta) - \theta = (F^{m+n}(\theta) - F^n(\theta)) + (F^n(\theta) - \theta)$. The first summand, setting $\psi := F^n(\theta)$, is $F^m(\psi) - \psi \leq a_m$ by definition of a_m as a supremum; the second is $\leq a_n$. Hence $F^{m+n}(\theta) - \theta \leq a_m + a_n$ for every θ , and taking the

supremum over θ : $a_{m+n} \leq a_m + a_n$. By Fekete's lemma (Lemma 4.1), $\lim_n a_n/n$ exists and equals $\inf_n a_n/n =: 2\pi\rho_+$.

Superadditivity of (b_n) , symmetrically: $F^{m+n}(\theta) - \theta \geq b_m + b_n$ for every θ (same argument with the inequalities reversed), hence $b_{m+n} \geq b_m + b_n$, that is, $(-b_n)$ is subadditive: by Fekete's lemma, $\lim_n b_n/n = \sup_n b_n/n =: 2\pi\rho_-$ exists.

The two limits coincide. For every $\theta, \theta' \in \mathbb{R}$, choosing $k \in \mathbb{Z}$ with $\theta' \in [\theta + 2\pi k, \theta + 2\pi(k+1))$, the increase of F^n and its commutation with the integer translations by 2π give $F^n(\theta) + 2\pi k \leq F^n(\theta') \leq F^n(\theta) + 2\pi(k+1)$, hence

$$(F^n(\theta') - \theta') - (F^n(\theta) - \theta) = (F^n(\theta') - F^n(\theta)) - (\theta' - \theta) \in (-2\pi, 2\pi)$$

(the estimate follows from $F^n(\theta') - F^n(\theta) \in [2\pi k, 2\pi(k+1)]$ and $\theta' - \theta \in [2\pi k, 2\pi(k+1))$, whose difference lies in $(-2\pi, 2\pi)$). Hence the oscillation of $\theta \mapsto F^n(\theta) - \theta$ is $\leq 2\pi$ for every n : $a_n - b_n \leq 2\pi$. Dividing by n and passing to the limit, $2\pi\rho_+ - 2\pi\rho_- \leq \lim_n 2\pi/n = 0$, that is, $\rho_+ \leq \rho_-$; but $b_n \leq a_n$ always gives $\rho_- \leq \rho_+$. Hence $\rho_+ = \rho_- =: \rho$.

Conclusion. For every θ , $b_n \leq F^n(\theta) - \theta \leq a_n$, hence $b_n/n \leq (F^n(\theta) - \theta)/n \leq a_n/n$; since $a_n/n \rightarrow 2\pi\rho$ and $b_n/n \rightarrow 2\pi\rho$, the squeeze theorem gives $(F^n(\theta) - \theta)/n \rightarrow 2\pi\rho$ for every θ , that is, $\rho(F) = \rho$, the same value for every θ . $\square$

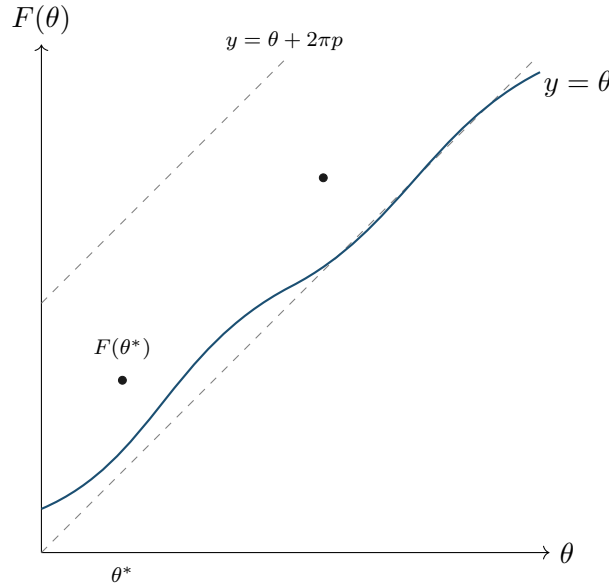


Figure 4.1: The graph of a lift F (blue curve) compared with the diagonal $y = \theta$ and with the lines $y = \theta + 2\pi p$, $p \in \mathbb{Z}$ (here $p = 1$, dashed, higher up): an intersection of F^q with a line $y = \theta + 2\pi p$ (such as the highlighted point θ^* , for $q = 1$) is a point with $F^q(\theta^*) = \theta^* + 2\pi p$, that is, exactly a periodic orbit of rotation p/q (Theorem 4.3). The graph stays everywhere strictly above or strictly below each line when $\rho(F) \neq p/q$ (Lemma 4.3 and its corollary).

Remark 4.1: Why “Poincaré” and not just “Weyl”. Theorem 4.1 is far more general than the case $f = R_\alpha$, where the limit is exact for every n with no need for Fekete's lemma: it applies to every orientation-preserving circle homeomorphism, not only to rigid rotations. It is the starting point of the whole theory that follows in this chapter (Poincaré classification, Arnold standard family) and in Chapter 6 (KAM theory on the circle).

Example 4.1: Rotation number estimated on 10^5 and 10^6 iterates, with an independent check. Double check (script `verifica_cap04_vol4.py`, block [C4-2], run

on 16 September 2026). For $f_{\alpha,\varepsilon}$ of the standard family (Definition 4.3 below, anticipated here) with $\alpha = 0.4123456$, $\varepsilon = 0.3$ (convention $\theta \in [0, 1)$, see Section 4.5), iteration from $\theta_0 = 0.31415926535$, discarding a transient of 2000 steps, gives

$$\frac{F^{10^5}(\theta_0) - \theta_{\text{transitorio}}}{10^5} = 0.41139177\dots, \quad \frac{F^{10^6}(\theta_0) - \theta_{\text{transitorio}}}{10^6} = 0.41139165\dots,$$

discrepancy between the two estimates 1.2×10^{-7} . **Independent check**, based not on the asymptotic average along a single orbit but on the Fekete sandwich of Theorem 4.1: computing $a_n = \sup_{\theta}(F^n(\theta) - \theta)$ and $b_n = \inf_{\theta}(F^n(\theta) - \theta)$ on a grid of 4000 initial points with $n = 2000$, one finds $b_n/n = 0.41137366$ and $a_n/n = 0.41140773$: the sandwich $b_n/n \leq \rho(f) \leq a_n/n$, proved in Theorem 4.1, contains both single-orbit estimates. ✓

4.2. The properties of the rotation number

Lemma 4.2: Bounded displacement of monotone maps commuting with integer translations. Let $H : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, monotone non-decreasing, with $H(\theta + 2\pi) = H(\theta) + 2\pi$ for every θ . Then $\sup_{\theta} |H(\theta) - \theta| < \infty$.

Proof. The function $\theta \mapsto H(\theta) - \theta$ is 2π -periodic (from the commutation relation) and continuous, hence bounded on the compact set $[0, 2\pi]$, and therefore on all of $\mathbb{R}$ by periodicity. $\square$

Proposition 4.1: $\rho(F^n) = n\rho(F)$. For every lift F and every integer $n \geq 1$, $\rho(F^n) = n\rho(F)$ (where F^n , a lift of f^n , inherits the normalisation from Definition 4.1 applied to f^n).

Proof. For every θ and every $m \geq 1$, $(F^n)^m(\theta) = F^{nm}(\theta)$, hence

$$\frac{(F^n)^m(\theta) - \theta}{2\pi m} = n \cdot \frac{F^{nm}(\theta) - \theta}{2\pi(nm)}.$$

As $m \rightarrow \infty$ the right-hand side tends to $n\rho(F)$ (the limit defining $\rho(F)$ holds along every sequence of integers tending to infinity, in particular along the multiples of n , by Theorem 4.1); the left-hand side tends to $\rho(F^n)$ by definition. Hence $\rho(F^n) = n\rho(F)$. $\square$

Proposition 4.2: Invariance of ρ under conjugacy and under monotone semi-conjugacy. (i) If $h : \mathbb{T} \rightarrow \mathbb{T}$ is an orientation-preserving homeomorphism with lift H , then $\rho(HFH^{-1}) = \rho(F)$: the rotation number is a conjugacy invariant. (ii) More generally, if $H : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, monotone non-decreasing, **surjective**, with $H(\theta + 2\pi) = H(\theta) + 2\pi$ (a lift of a monotone semiconjugacy, not necessarily injective), and if G, F are lifts with $G \circ H = H \circ F$, then $\rho(G) = \rho(F)$.

Proof. (ii) implies (i) (with H invertible, $G = HFH^{-1}$ satisfies $GH = HF$). We prove (ii). From $GH = HF$ it follows by induction that $G^n H = HF^n$ for every $n \geq 1$ ($G^n H = G^{n-1}(GH) = G^{n-1}HF = \dots = HF^n$). Let $M := \sup_{\theta} |H(\theta) - \theta| < \infty$ (Lemma 4.2). For every $\psi \in \mathbb{R}$, setting $\theta := H(\psi)$ (every $\theta \in \mathbb{R}$ can be written this way, H being surjective),

$$G^n(\theta) - \theta = H(F^n(\psi)) - H(\psi) = (F^n(\psi) - \psi) + (H(F^n(\psi)) - F^n(\psi)) - (H(\psi) - \psi),$$

where the two terms in round brackets on the right have modulus $\leq M$. Dividing by $2\pi n$ and passing to the limit $n \rightarrow \infty$: the term $(F^n(\psi) - \psi)/(2\pi n) \rightarrow \rho(F)$ (Theorem 4.1), while the

two displacement terms, divided by n , tend to 0. Hence $(G^n(\theta) - \theta)/(2\pi n) \rightarrow \rho(F)$. Since $\theta = H(\psi)$ is an arbitrary point of $\mathbb{R}$ (surjectivity of H) and the limit defining $\rho(G)$ does not depend on the point (Theorem 4.1 applied to G), we conclude $\rho(G) = \rho(F)$. $\square$

Lemma 4.3: Rational criterion for the rotation number. Let F be a lift, $p \in \mathbb{Z}$, $q \in \mathbb{Z}_{>0}$. Then: (i) $\rho(F) \geq p/q \iff \exists \theta \in \mathbb{R} : F^q(\theta) \geq \theta + 2\pi p$. (ii) $\rho(F) \leq p/q \iff \exists \theta \in \mathbb{R} : F^q(\theta) \leq \theta + 2\pi p$. (iii) (**Corollary.**) If $\rho(F) \neq p/q$, the sign of $F^q(\theta) - \theta - 2\pi p$ is the same for every $\theta \in \mathbb{R}$: positive if $\rho(F) > p/q$, negative if $\rho(F) < p/q$.

Proof. (i, $\Leftarrow$) Let θ_0 be such that $F^q(\theta_0) \geq \theta_0 + 2\pi p$. By induction on $n \geq 1$: $F^{qn}(\theta_0) \geq \theta_0 + 2\pi pn$. True for $n = 1$; assuming it for $n - 1$, apply F^q (increasing) to both sides of $F^{q(n-1)}(\theta_0) \geq \theta_0 + 2\pi p(n-1)$:

$$F^{qn}(\theta_0) = F^q(F^{q(n-1)}(\theta_0)) \geq F^q(\theta_0 + 2\pi p(n-1)) = F^q(\theta_0) + 2\pi p(n-1) \geq \theta_0 + 2\pi pn,$$

using the commutation of F^q with the integer translations by 2π (since $p(n-1) \in \mathbb{Z}$) and the hypothesis on θ_0 . Dividing by $2\pi qn$ and passing to the limit $n \rightarrow \infty$: $\rho(F) = \lim_n \frac{F^{qn}(\theta_0) - \theta_0}{2\pi qn} \geq \lim_n \frac{2\pi pn}{2\pi qn} = \frac{p}{q}$ (the limit along the subsequence qn exists and equals $\rho(F)$ by Theorem 4.1).

(i, $\Rightarrow$, by contraposition) Suppose $F^q(\theta) < \theta + 2\pi p$ for every θ . The continuous 2π -periodic function $\theta \mapsto F^q(\theta) - \theta - 2\pi p$ is then $\leq -\delta$ for some $\delta > 0$ (maximum on a compact set, negative by hypothesis). By induction as above (with the inequalities reversed and strict), $F^{qn}(\theta) - \theta \leq 2\pi pn - n\delta$ for every θ, n , hence $\rho(F) \leq p/q - \delta/(2\pi q) < p/q$. By contraposition, $\rho(F) \geq p/q \Rightarrow \exists \theta : F^q(\theta) \geq \theta + 2\pi p$.

(ii) is symmetric (apply (i) to $-F(-\theta)$, the lift of the inverse map with reversed orientation composed twice, or repeat verbatim the same argument with the inequalities interchanged).

(iii) If $\rho(F) > p/q$: from (ii) applied in the negative (contraposition of (ii)), $\rho(F) > p/q \Rightarrow$ there is no θ with $F^q(\theta) \leq \theta + 2\pi p$, that is, $F^q(\theta) > \theta + 2\pi p$ for every θ . Symmetrically if $\rho(F) < p/q$, from (i) in the negative: $F^q(\theta) < \theta + 2\pi p$ for every θ . $\square$

Theorem 4.2: Monotonicity and continuity of the rotation number in the C^0 topology. (i) If F, G are lifts with $F(\theta) \leq G(\theta)$ for every θ , then $\rho(F) \leq \rho(G)$. (ii) The map $c \mapsto \rho(F + c)$ is continuous (and, by (i), non-decreasing). (iii) It follows that ρ is continuous in the C^0 topology: if $F_k \rightarrow F$ uniformly (lifts of orientation-preserving circle homeomorphisms), then $\rho(F_k) \rightarrow \rho(F)$.

Proof. (i) For every q, p with $F^q(\theta) \geq \theta + 2\pi p$ for some θ : by induction on $k \geq 1$, $G^k(\theta) \geq F^k(\theta)$ for every θ (true for $k = 1$ by hypothesis; assuming it for $k - 1$: $G^k(\theta) = G(G^{k-1}(\theta)) \geq G(F^{k-1}(\theta))$, using the increase of G and the inductive hypothesis, $\geq F(F^{k-1}(\theta)) = F^k(\theta)$, using $G \geq F$ pointwise evaluated at the point $F^{k-1}(\theta)$). Hence $G^q(\theta) \geq F^q(\theta) \geq \theta + 2\pi p$: by Lemma 4.3(i), $\rho(F) \geq p/q \Rightarrow \rho(G) \geq p/q$. Since this holds for every rational $p/q \leq \rho(F)$, and the rationals are dense, $\rho(G) \geq \rho(F)$.

(ii) Monotonicity in c follows from (i) ($F + c \leq F + c'$ for $c \leq c'$). For continuity, fix c_0 and $\varepsilon > 0$; choose an integer $q \geq 1$ with $1/q < \varepsilon$ and set $p := \lfloor q\rho(F + c_0) \rfloor$, so that $p/q \leq \rho(F + c_0) < (p+1)/q$ (if $\rho(F + c_0) = p/q$ exactly for some integer p , one takes p so that the equality falls at the left endpoint; the argument that follows is insensitive to this choice). Define, for a fixed integer p' , $M_{p'}(c) := \max_{\theta} [(F + c)^q(\theta) - \theta] - 2\pi p'$. By Lemma 4.3(i) (with the supremum, which on a periodic compact set is a maximum by continuity), $\rho(F + c) \geq p'/q \iff M_{p'}(c) \geq 0$. The map $(c, \theta) \mapsto (F + c)^q(\theta)$ is jointly continuous in (c, θ) (a finite composition, q times, of the jointly continuous map $(c, \theta) \mapsto F(\theta) + c$); since θ varies in a compact set (one period $[0, 2\pi]$, the expression being 2π -periodic in θ), the maximum over θ of a jointly continuous function is a continuous function of c (an elementary fact of uniform

continuity: for c near c_0 , $|(F+c)^q(\theta) - (F+c_0)^q(\theta)| < \eta$ uniformly in $\theta \in [0, 2\pi]$ by uniform continuity on the compact set $\{|c - c_0| \leq 1\} \times [0, 2\pi]$, hence $|M_{p'}(c) - M_{p'}(c_0)| < \eta$. Hence M_p and M_{p+1} are continuous in c .

From $\rho(F+c_0) \geq p/q$: $M_p(c_0) \geq 0$. If $M_p(c_0) = 0$, the maximum is attained at some θ^* with $(F+c_0)^q(\theta^*) = \theta^* + 2\pi p$, that is, θ^* is a periodic point of rotation exactly p/q (Theorem 4.3 below, which does not use this theorem): then $\rho(F+c_0) = p/q$ exactly, and one simply takes q larger so that $1/q < \varepsilon$ and $\rho(F+c_0)$ is not of the form p'/q for any integer p' (possible because the denominators with this property are cofinite when $\rho(F+c_0)$ is irrational, and if $\rho(F+c_0)$ is rational with reduced denominator q_0 it suffices to take q a sufficiently large multiple of q_0 and $p := q\rho(F+c_0) - 1$, so that $p/q < \rho(F+c_0) < (p+2)/q$ with margin $2/q < \varepsilon$ for large q : the argument that follows applies verbatim with $p+2$ in place of $p+1$). We may therefore assume $M_p(c_0) > 0$ strictly.

Similarly, from $\rho(F+c_0) < (p+1)/q$: by Lemma 4.3(ii) applied in place of (i) (that is, using the minimum instead of the maximum, with the inequality $\leq$), the quantity $m_{p+1}(c) := \min_{\theta} [(F+c)^q(\theta) - \theta] - 2\pi(p+1)$ satisfies $m_{p+1}(c_0) < 0$ strictly (by the same argument: if it were $= 0$, one would have $\rho(F+c_0) = (p+1)/q$ exactly, excluded by the choice of p ; if it were > 0 one would have $\rho(F+c_0) > (p+1)/q$ from the Lemma, against the hypothesis), and m_{p+1} is continuous in c by the same uniform continuity argument as for $M_{p'}$.

By continuity of M_p and m_{p+1} at c_0 , with $M_p(c_0) > 0$ and $m_{p+1}(c_0) < 0$ both strict, there exists $\delta > 0$ such that for $|c - c_0| < \delta$: $M_p(c) > 0$ (hence $\rho(F+c) \geq p/q$ by Lemma 4.3(i)) and $m_{p+1}(c) < 0$ (hence $\rho(F+c) \leq (p+1)/q$ by Lemma 4.3(ii)). Hence for $|c - c_0| < \delta$, both $\rho(F+c)$ and $\rho(F+c_0)$ belong to the interval $[p/q, (p+1)/q]$, of length $1/q < \varepsilon$: $|\rho(F+c) - \rho(F+c_0)| < \varepsilon$. This proves continuity at c_0 , which was arbitrary.

(iii) Let $\delta_k := \sup_{\theta} |F_k(\theta) - F(\theta)| \rightarrow 0$. Then $F - \delta_k \leq F_k \leq F + \delta_k$ pointwise (the functions $F \pm \delta_k$, constant translates of F , are still valid lifts, monotone and commuting with the integer translations). By (i), $\rho(F - \delta_k) \leq \rho(F_k) \leq \rho(F + \delta_k)$. By (ii) applied with $c = \pm \delta_k \rightarrow 0$, $\rho(F \pm \delta_k) \rightarrow \rho(F)$. By the squeeze theorem, $\rho(F_k) \rightarrow \rho(F)$. $\square$

4.3. The Poincaré classification: rational case

Theorem 4.3: ρ rational $\iff$ periodic orbit of exact period q ; asymptotic convergence. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving homeomorphism, F a lift of it. Then: (i) $\rho(f) = p/q$ (in lowest terms, $q \geq 1$, $\gcd(p, q) = 1$) if and only if f has a periodic orbit; in that case every periodic orbit of f has minimal period exactly q . (ii) In this case, every orbit of f (periodic or not) is asymptotic to a periodic orbit: for every $\theta_0 \in \mathbb{T}$ there exists a periodic orbit $\mathcal{O}$ (of period q) such that $\text{dist}(f^n(\theta_0), \mathcal{O}) \rightarrow 0$ as $n \rightarrow +\infty$ and, separately, as $n \rightarrow -\infty$ (possibly towards two different periodic orbits).

Proof. (i), $\Leftarrow$. Let θ_0 be periodic under f , of minimal period $k \geq 1$ (the set $\{j \geq 1 : f^j(\pi(\theta_0)) = \pi(\theta_0)\}$ is closed under addition and non-empty, hence it is the set of multiples of an integer $k \geq 1$, the minimal period). Then $F^k(\theta_0) = \theta_0 + 2\pi m$ for a unique $m \in \mathbb{Z}$, and by induction $F^{kn}(\theta_0) = \theta_0 + 2\pi mn$ for every n , whence (Theorem 4.1, the limit along the subsequence kn equals $\rho(F)$) $\rho(f) = m/k$.

We show that $\gcd(m, k) = 1$, that is, that this expression is already reduced. The k points of the orbit $\pi(\theta_0), f(\pi(\theta_0)), \dots, f^{k-1}(\pi(\theta_0))$ are pairwise distinct (minimal period k) and determine a cyclic order on $\mathbb{T}$; let us number them by **geometric position** in this cyclic order as $y_0, y_1, \dots, y_{k-1}$ (in such a way that running through them in this order goes once around $\mathbb{T}$ in the positive direction). Since f preserves the orientation, the permutation induced by f on this set of k points preserves the cyclic order: every automorphism of the cyclic order of a finite set of k elements is a shift “ $y_i \mapsto y_{i+r}$ ” for a unique $r \in \mathbb{Z}/k\mathbb{Z}$ (an elementary fact: an

automorphism of the cyclic order is determined by the image of one point, because it must send “the next one in the order” to “the next one in the order”). Hence there exists $r \in \{0, \dots, k-1\}$ with $f(y_i) = y_{i+r \bmod k}$ for every i . Since the orbit of θ_0 is a single orbit of f visiting all k points (minimal period k , not a shorter period repeated), the permutation $i \mapsto i + r \bmod k$ must be a k -cycle: this happens if and only if $\gcd(r, k) = 1$ (if $d = \gcd(r, k) > 1$, the orbit of 0 under “ $+r \bmod k$ ” would have length $k/d < k$, splitting the set of k positions into $d > 1$ shorter orbits, against transitivity).

It remains to connect r to m : starting from $y_0 = \theta_0$ (representative in $[0, 2\pi)$) and applying f one step at a time, after k applications one returns to y_0 having traversed, in the lift, each of the k arcs determined by the points $y_0, \dots, y_{k-1}$ exactly r times in total over the complete run (each application of f advances by r “positions” in the cyclic order of the y_i , and k applications cover kr positions, that is, r complete turns of the k arcs): hence the lift advances by exactly r whole turns in k steps, that is, $F^k(\theta_0) = \theta_0 + 2\pi r$. Comparing with $F^k(\theta_0) = \theta_0 + 2\pi m$ (and with $\theta_0 + 2\pi r = \theta_0 + 2\pi m$ both taken as the same lift, hence equal as real numbers, not only mod 2π , because both constructions start from the same fixed lift F): $m = r$. Hence $\gcd(m, k) = \gcd(r, k) = 1$: $\rho(f) = m/k$ is already reduced, $p = m$, $q = k$.

(i), $\Rightarrow$. Let $\rho(f) = p/q$ be reduced. Put $g := f^q$, with lift $G := F^q - 2\pi p$ (checked to be a valid lift: $G(\theta + 2\pi) = F^q(\theta + 2\pi) - 2\pi p = F^q(\theta) + 2\pi - 2\pi p = G(\theta) + 2\pi$, and G is increasing since F^q is). By Proposition 4.1, $\rho(F^q) = q\rho(F) = p$ (an integer), hence $\rho(G) = \rho(F^q) - p = 0$ (an integer translation of a lift translates ρ by the same integer amount, a particular case of Proposition 4.1 with $n = 1$). We show that $\rho(G) = 0$ implies that g has a fixed point, that is, $G(\theta^*) = \theta^*$ for some θ^* : if this were not so, by continuity and periodicity $G(\theta) - \theta$ would have constant sign (being continuous, 2π -periodic, never zero); if $G(\theta) > \theta$ for every θ , by compactness $G(\theta) - \theta \geq \delta > 0$, hence by Lemma 4.3(i) with $q = 1, p = 0$ it does not apply directly (we want a different p): more simply, by induction $G^n(\theta) \geq \theta + n\delta$, hence $\rho(G) = \lim_n (G^n(\theta) - \theta)/(2\pi n) \geq \delta/(2\pi) > 0$, against $\rho(G) = 0$; symmetrically $G(\theta) < \theta$ everywhere gives $\rho(G) < 0$. Hence G has a fixed point θ^* : $G(\theta^*) = \theta^*$, that is, $F^q(\theta^*) = \theta^* + 2\pi p$, that is, $\pi(\theta^*)$ is periodic for f with period (not necessarily minimal) q . By the part (i) $\Leftarrow$ already proved, the *minimal* period q_0 of $\pi(\theta^*)$ gives $\rho(f) = m_0/q_0$ reduced for some m_0 ; but we also have $\rho(f) = p/q$ reduced by hypothesis: both being reduced expressions of the same rational, $q_0 = q$ and $m_0 = p$. The minimal period is therefore exactly q , as wanted also for *this* orbit; and by the clause “every periodic orbit has period exactly q ” already proved in the $\Leftarrow$ part (applied to any other periodic orbit that may exist), statement (i) is complete.

(ii). With $g = f^q$, $G = F^q - 2\pi p$ as above, $\text{Fix}(G) \neq \emptyset$ (just proved) is a 2π -periodic closed set. If $\text{Fix}(G) = \mathbb{R}$ (that is, $g = \text{id}$), every point is periodic of period q (or a divisor) and there is nothing to prove. Otherwise, let $\theta_0 \in \mathbb{R}$ be arbitrary: if $\theta_0 \in \text{Fix}(G)$ it is already periodic. If $\theta_0 \notin \text{Fix}(G)$, let (θ_-, θ_+) be the connected component of $\mathbb{R} \setminus \text{Fix}(G)$ containing it (an open interval, possibly with $\theta_- = -\infty$ or $\theta_+ = +\infty$ if $\text{Fix}(G)$ is empty on one side — a case excluded here because $\text{Fix}(G)$ is 2π -periodic and non-empty, hence it has points arbitrarily far away in both directions, making every component bounded). On (θ_-, θ_+) , $G(\theta) - \theta$ does not vanish (by definition of the component), hence it has constant sign. If $G(\theta) > \theta$ on (θ_-, θ_+) : by monotonicity of G , $G^n(\theta_0)$ is strictly increasing in n (from $G(\theta_0) > \theta_0$, $G(G(\theta_0)) > G(\theta_0)$ by the increase of G , inductively); and it is bounded above by θ_+ (by induction: if $G^n(\theta_0) < \theta_+$, then $G^{n+1}(\theta_0) = G(G^n(\theta_0)) < G(\theta_+) = \theta_+$, using the increase of G and $\theta_+ \in \text{Fix}(G)$). Hence $G^n(\theta_0) \nearrow L \leq \theta_+$ for some limit L ; by continuity, $G(L) = \lim_n G(G^n(\theta_0)) = \lim_n G^{n+1}(\theta_0) = L$, that is, $L \in \text{Fix}(G) \cap (\theta_-, \theta_+] = \{\theta_+\}$ (the open interval contains no fixed points). Hence $G^n(\theta_0) \rightarrow \theta_+$, that is, $F^{qn}(\theta_0) - 2\pi pn \rightarrow \theta_+$, that is, $f^{qn}(\pi(\theta_0)) \rightarrow \pi(\theta_+)$: the subsequence $f^{qn}(\pi(\theta_0))$ converges to the periodic point $\pi(\theta_+)$. To extend this to the *whole* sequence $f^n(\pi(\theta_0))$ (not only to the subsequence qn): the intermediate points $f^{qn+j}(\pi(\theta_0)) = f^j(f^{qn}(\pi(\theta_0)))$, $j = 0, \dots, q-1$, converge as $n \rightarrow \infty$ (for

fixed j) to $f^j(\pi(\theta_+))$ by continuity of f^j , applied to $f^{qn}(\pi(\theta_0)) \rightarrow \pi(\theta_+)$ just shown; that is, to a point of the q -periodic orbit of $\pi(\theta_+)$ (the same one for every j , translated): hence $\text{dist}(f^n(\pi(\theta_0)), \mathcal{O}) \rightarrow 0$ as $n \rightarrow +\infty$ along *all* the integers, with $\mathcal{O}$ the q -periodic orbit of $\pi(\theta_+)$. The case $G(\theta) < \theta$ on (θ_-, θ_+) is symmetric ($G^n(\theta_0) \searrow \theta_-$), giving convergence towards the orbit of $\pi(\theta_-)$. The behaviour as $n \rightarrow -\infty$ follows by applying the same argument to G^{-1} (also monotone increasing, a lift of g^{-1} , with the same fixed points), whence the convergence (possibly towards the *other* orbit at the opposite endpoints θ_-, θ_+ of the component, if θ_0 is not already periodic). $\square$

Remark 4.2: The complete picture of the classification. Theorem 4.3 and Theorem 4.4 (Section 4.4, irrational case) together constitute the **Poincaré classification** of circle homeomorphisms: $\rho(f)$ rational $\iff$ asymptotically periodic dynamics (every orbit converges, forwards and backwards, to a periodic orbit whose period is exactly the reduced denominator); $\rho(f)$ irrational $\implies$ f semiconjugate to $R_{\rho(f)}$, with only two possible scenarios — f minimal and topologically conjugate to $R_{\rho(f)}$, or else there exists an invariant minimal Cantor set on which f acts, and the semiconjugacy “crushes” the complementary intervals (wandering intervals). Chapter 5 will prove that, with the additional regularity hypothesis $C^{1+\text{BV}}$ (bounded variation of the derivative, besides $f' > 0$), the second scenario is impossible (Denjoy’s theorem): here, for a generic homeomorphism, both can really occur, as Example 4.2 shows.

4.4. The Poincaré classification: irrational case

Theorem 4.4: Poincaré semiconjugacy (ρ irrational). Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving homeomorphism with $\rho(f) = \rho \notin \mathbb{Q}$. Then there exists a continuous, surjective, monotone map $h : \mathbb{T} \rightarrow \mathbb{T}$ (of degree 1) such that $h \circ f = R_{2\pi\rho} \circ h$. Moreover, setting $E := \overline{\{F^n(\theta_0) + 2\pi k : n, k \in \mathbb{Z}\}}$ for any fixed θ_0 (closure in $\mathbb{R}$, independent of θ_0 up to a translation of the indexing): either $E = \mathbb{R}$, and then h is a homeomorphism (a genuine conjugacy), or E is a 2π -periodic Cantor set (closed, with no isolated points, with empty interior) and h collapses every bounded connected component of $\mathbb{R} \setminus E$ to a point.

Proof. Fix a lift F and $\theta_0 \in \mathbb{R}$; put $x_n := F^n(\theta_0)$ for $n \in \mathbb{Z}$ (F is bijective, being the lift of a homeomorphism, hence F^n makes sense for every $n \in \mathbb{Z}$) and $E_0 := \{x_n + 2\pi k : n, k \in \mathbb{Z}\}$, a countable set.

Step 1 (order isomorphism). Since $\rho \notin \mathbb{Q}$, for every $p \in \mathbb{Z}, q \in \mathbb{Z}_{>0}$ we have $\rho \neq p/q$, hence (Lemma 4.3(iii)) $F^q(\theta) - \theta - 2\pi p$ has constant sign, equal to the sign of $\rho - p/q$, for every θ . It follows, for $(n, k) \neq (m, l) \in \mathbb{Z}^2$ with $n \neq m$ (the case $n = m$ is immediate: $x_n + 2\pi k < x_n + 2\pi l \iff k < l \iff n\rho + k < n\rho + l$): setting $q := |n - m| > 0$ and, assuming $n > m$ without loss of generality (otherwise the roles are interchanged), $\psi := x_m = F^m(\theta_0)$,

$$x_n + 2\pi k < x_m + 2\pi l \iff F^q(\psi) < \psi + 2\pi(l - k) \iff \rho < \frac{l - k}{q} \iff q\rho < l - k \iff n\rho + k < m\rho + l,$$

where the second “ $\iff$ ” is Lemma 4.3(iii) applied with $p = l - k$ (the sign of $F^q(\psi) - \psi - 2\pi p$ is constant in ψ , hence the comparison with 0 does not depend on the point ψ chosen, in particular it holds with $\psi = x_m$). Hence

$$x_n + 2\pi k < x_m + 2\pi l \iff n\rho + k < m\rho + l \quad \text{for every } (n, k), (m, l) \in \mathbb{Z}^2. \quad (*)$$

In particular (replacing $<$ by $=$, impossible on the left for $(n, k) \neq (m, l)$ since otherwise θ_0 would be periodic with rational rotation, excluded by $\rho \notin \mathbb{Q}$ by Theorem 4.3, and impossible on the right because ρ is irrational) the map $(n, k) \mapsto x_n + 2\pi k$ is injective on $\mathbb{Z}^2$.

Step 2 (definition of h on E_0 and extension). Define $h_0 : E_0 \rightarrow \mathbb{R}$ by $h_0(x_n + 2\pi k) := 2\pi(n\rho + k)$; it is well defined and strictly increasing by (*). The image $h_0(E_0) = \{2\pi(n\rho + k) : n, k \in \mathbb{Z}\}$ is dense in $\mathbb{R}$ (it is the orbit of 0 under the irrational rotation $R_{2\pi\rho}$ together with its integer translates, dense by Theorem 3.1 of Chapter 3). We extend h_0 to $h : \mathbb{R} \rightarrow \mathbb{R}$ by setting, for arbitrary $x \in \mathbb{R}$,

$$h(x) := \sup\{h_0(y) : y \in E_0, y \leq x\}$$

(the supremum of a non-empty set bounded above, by the density of E_0 towards $-\infty$ and $+\infty$ following from that of $h_0(E_0)$ combined with the strict increase (*)). By construction h is monotone non-decreasing on $\mathbb{R}$ and coincides with h_0 on E_0 (by the density of E_0 from the left at each of its points, a consequence of the density of $h_0(E_0)$ and of (*)). h is continuous: if it had a jump at x^* , the interval $(h(x^{*-}), h(x^{*+}))$ would be a non-degenerate interval disjoint from $h_0(E_0)$ (by the construction of h as a truncated supremum), against the density of $h_0(E_0)$ in $\mathbb{R}$. h satisfies $h(x + 2\pi) = h(x) + 2\pi$ (visible on E_0 from (*) with $k \rightarrow k + 1$, and extended by continuity/density), hence it descends to a continuous monotone map $h : \mathbb{T} \rightarrow \mathbb{T}$ of degree 1; it is surjective because $h(\mathbb{R}) = \mathbb{R}$ (a continuous monotone function with $h_0(E_0) \subset h(\mathbb{R})$ dense and unbounded in both directions is, by the intermediate value theorem, surjective onto $\mathbb{R}$).

Step 3 (injectivity of h on $E := \overline{E_0}$; semiconjugacy relation). We show that h is *strictly* increasing on E : if $x < y$ in E with $h(x) = h(y)$, then h is constant on $[x, y]$ (monotonicity); but E_0 is dense in E (by the definition of E as a closure), hence $E_0 \cap [x, y]$ is dense in $[x, y]$ (since $[x, y] \subset E$ because... in fact $[x, y]$ might not be entirely contained in E : we only use that $E_0 \cap [x, y]$ is in any case non-empty and infinite, since $x, y \in E = \overline{E_0}$ are both accumulation points of E_0 , hence E_0 has infinitely many points in every neighbourhood of x and of y , in particular, by monotonicity of the approximating sequence, infinitely many points in $[x, y]$ itself as soon as $x \neq y$): but h_0 is injective on E_0 (Step 1), hence h cannot be constant on an infinite set of distinct points of E_0 . Contradiction: hence h is injective (strictly increasing) on E .

As for the semiconjugacy relation: on E_0 , $h(F(x_n + 2\pi k)) = h(x_{n+1} + 2\pi k) = 2\pi((n + 1)\rho + k) = h(x_n + 2\pi k) + 2\pi\rho$; by continuity of h , of F and of the translation, this identity extends to $\overline{E_0} = E$. It remains to extend it to the connected components (the “gaps”) of $\mathbb{R} \setminus E$: since E_0 is the orbit of θ_0 under F together with its integer translates, $F(E_0) = E_0$ exactly (bijectivity of F and its commutation with the integer translations), hence by continuity $F(E) \subseteq \overline{F(E_0)} = E$ and $F^{-1}(E) \subseteq E$ in the same way, giving $F(E) = E$: F permutes the connected components of $\mathbb{R} \setminus E$ among themselves, in an increasing fashion (monotonicity of F). Let (a, b) be such a component (a “gap”): $a, b \in E$ and $(a, b) \cap E = \emptyset$; if we had $h(a) < h(b)$, the interval $(h(a), h(b))$ would be disjoint from $h(E) \supseteq h_0(E_0)$ (because h is increasing and $(a, b) \cap E_0 = \emptyset$), against the density of $h_0(E_0)$: hence $h(a) = h(b)$, and by monotonicity h is constant on $[a, b]$, equal to this common value. Since $F(a), F(b) \in E$ are the endpoints of the component $F((a, b))$ (by the permutation of the components just seen), for every $x \in (a, b)$: $h(F(x)) = h(F(a))$ (constancy of h on the image component) $= h(a) + 2\pi\rho$ (relation already valid on E) $= h(x) + 2\pi\rho$ (constancy of h on (a, b) , with $h(x) = h(a)$). The relation $h \circ F = T_{2\pi\rho} \circ h$ therefore holds on all of $\mathbb{R}$, and descends to $h \circ f = R_{2\pi\rho} \circ h$ on $\mathbb{T}$.

Step 4 (dichotomy). If $E = \mathbb{R}$ there are no gaps: h is then bijective on all of $\mathbb{R}$ (injective by Step 3 applied with $E = \mathbb{R}$), hence an increasing homeomorphism of $\mathbb{R}$ commuting with the translations by 2π : it descends to a homeomorphism of $\mathbb{T}$, that is, f is topologically conjugate (not merely semiconjugate) to $R_{2\pi\rho}$. If $E \neq \mathbb{R}$: E is closed by construction (a closure); E has no isolated points (if $x \in E$ were isolated with isolation radius $\delta > 0$, the orbit $\{x_n + 2\pi k\}$, being dense in E by minimality of E as the closure of a single orbit under F — every non-empty F -invariant closed set contained in E and containing a point of the orbit of θ_0 coincides with E , E being by definition the smallest invariant closure containing that orbit — would have to return infinitely many times within δ of x , but the only point of E within δ of x is x

itself: hence infinitely many n, k would give $x_n + 2\pi k = x$ exactly, making θ_0 periodic, against $\rho \notin \mathbb{Q}$; hence E is perfect. E has empty interior: if it contained a non-degenerate interval $J = [u, v] \subset E$, then (Step 3) h would be strictly increasing on J , with $h(J)$ an interval of length $\ell := h(v) - h(u) > 0$; by the semiconjugacy relation, $h(F^n(J)) = h(J) + 2\pi n\rho$ for every n (translates of one and the same interval of length ℓ under the irrational rotation, hence dense modulo 2π by Theorem 3.1); choosing n with $\|n\rho\|_{2\pi} < \ell/2$ (possible by density), the intervals $h(J)$ and $h(F^n(J)) = h(J) + 2\pi n\rho$ overlap modulo 2π in a set of points of positive length, that is, there exist $x \in J, y \in F^n(J)$, both in E (since $F(E) = E$), with $h(x) \equiv h(y) \pmod{2\pi}$, that is, $h(x) = h(y) + 2\pi j$ for some $j \in \mathbb{Z}$; but $h(y) + 2\pi j = h(y + 2\pi j)$ (periodicity of h) and $y + 2\pi j \in E$ (invariance of E under integer translations, inherited from E_0): hence $h(x) = h(y + 2\pi j)$ with $x, y + 2\pi j \in E$, which by the injectivity of h on E (Step 3) forces $x = y + 2\pi j$. This holds for every choice of the overlapping pair within the overlap set of positive length: one thus obtains a set of positive measure of exact coincidences $x = y + 2\pi j$ between J and an integer translate of $F^n(J)$, which forces (by continuity and monotonicity of F^n) J and that translate of $F^n(J)$ to coincide as intervals, that is, $F^n(J) = J - 2\pi j$: but then the endpoint u of J would satisfy $F^n(u) = u - 2\pi j$, or F^{-n} of the analogous relation, making u periodic, again against $\rho \notin \mathbb{Q}$ (Theorem 4.3). Contradiction: E has empty interior. A perfect closed set, totally disconnected (empty interior in $\mathbb{R}$, hence no interval, hence no non-trivial connected component), bounded modulo 2π (hence compact in $\mathbb{T}$) is by definition a Cantor set. $\square$

Remark 4.3: The minimal/Cantor dichotomy. In the language of topological dynamics: if $E = \mathbb{R} \pmod{2\pi}$, $E = \mathbb{T}$, f is **minimal** (every orbit is dense in $\mathbb{T}$, since E is by construction the closure of an orbit, and $E = \mathbb{T}$ means that that orbit — hence, by the same argument applied to an arbitrary point, every orbit — is dense); if $E \neq \mathbb{T}$, the Cantor set E is the unique minimal set of f (every orbit, inside or outside E , accumulates exactly on E : those outside E live in one of the “gaps”, which are **wandering** intervals — their images $f^n(\text{gap})$, $n \in \mathbb{Z}$, are pairwise disjoint, by the same disjointness argument as in the proof of Step 4 — and never converge to a periodic orbit, unlike the rational case of Theorem 4.3).

Example 4.2: An example with a wandering interval (illustration with a finite truncation). We describe the standard construction (see R29, §3.1, and R24, Prop. 12.2.1, for the analogous, more demanding case of Chapter 5) of a homeomorphism with ρ irrational and a minimal Cantor set, checking numerically only the elementary step — the non-overlapping of the inserted intervals — on a finite truncation; the complete construction (infinitely many intervals, with the non-overlapping check for *every* pair, not only the first ones) is more delicate and we shall use the three-distance theorem (Theorem 3.8 of Chapter 3) for it in Chapter 5, where it is needed with the additional C^1 regularity of the true Denjoy counterexample: here continuity suffices.

Construction. Let $\rho^* = 1/\varphi = 0.618034\dots$ be the reduced golden ratio (Definition 3.6, Chapter 3). For $n \in \mathbb{Z}$, let $\theta_n := \{n\rho^*\} \in [0, 1)$ (orbit of R_{ρ^*} from 0, in the $[0, 1)$ convention) and let J_n be an open interval of length $\ell_n := cr^{|n|}$ centred at θ_n ($0 < r < 1$, $c > 0$ small). If the J_n are pairwise disjoint, one constructs $\mathbb{T}'$ by inserting each J_n in place of the point θ_n (a new circle of total length $1 + \sum_n \ell_n$, then renormalised to length 1) and one defines f as the unique monotone map that sends J_n affinely and increasingly onto J_{n+1} for every n , and which coincides, on the complement $K := \mathbb{T}' \setminus \bigcup_n J_n$ (a Cantor set), with $h^{-1} \circ R_{\rho^*} \circ h$, where $h : K \rightarrow \mathbb{T}$ is the (continuous, monotone, surjective) map that collapses the residual structure of the gaps onto the points θ_n already removed. By construction $\rho(f) = \rho^*$, f is not minimal (J_0 is a wandering interval: $f^n(J_0) = J_n$, pairwise disjoint), and $E = K$ is the minimal Cantor set of Theorem 4.4.

Numerical check (script `verifica_cap04_vol4.py`, block [C4-4]). With $c = 0.03$, $r = 0.5$ and $n = -3, \dots, 3$ (truncation to 7 intervals, an illustration of the mechanism, not the complete infinite construction): the circular order of the θ_n is $0, \theta_{-3}, \theta_2, \theta_{-1}, \theta_1, \theta_{-2}, \theta_3$; the sum of the 7 lengths is $0.0825 < 1$; and for each of the 7 pairs of consecutive points in the circular order, the distance between the centres exceeds the sum of the respective half-widths (for example between $n = 0$ and $n = -3$: distance 0.145898 against required half-widths 0.016875), confirming that the 7 truncated intervals do not overlap. ✓ This verifies the elementary step of the construction (no overlap, a necessary condition for f to be well defined) on the truncation; the infinite construction, with $\ell_n \rightarrow 0$ as $|n| \rightarrow \infty$, requires in addition ruling out overlaps between intervals with $|n|, |m|$ arbitrarily large but θ_n, θ_m arbitrarily close (which happens infinitely often, by density of the orbit, Theorem 3.1): this is exactly the technical point that Chapter 5 will treat in full with the three-distance theorem, for Denjoy’s C^1 counterexample.

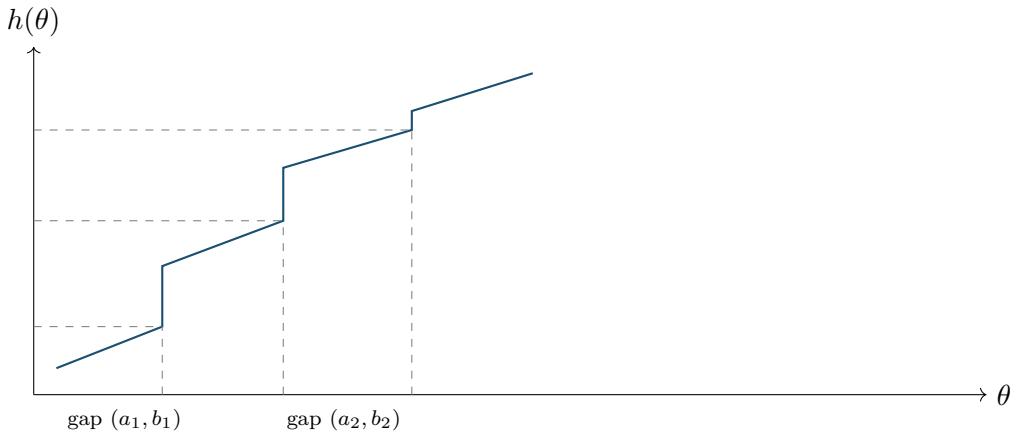


Figure 4.2: Scheme of the semiconjugacy h (Theorem 4.4): on the points of the set E (the slanted segments) h is strictly increasing; on each bounded connected component of the complement (a “gap” (a_i, b_i) , the horizontal segments) h is constant, with the same value at the two endpoints a_i, b_i . When $E = \mathbb{R}$ there are no horizontal segments and h is a homeomorphism (a full conjugacy); when E is a Cantor set, the horizontal segments correspond to the wandering intervals of Example 4.2.

4.5. The Arnold standard family

Remark 4.4: Change of convention: $\theta \in [0, 1)$. For the standard family, and only for it, we adopt the normalisation standard in the literature (R37; R24; R25, ch. I; K8): $\mathbb{T} \cong \mathbb{R}/\mathbb{Z}$ instead of $\mathbb{R}/2\pi\mathbb{Z}$, that is, $\theta \in [0, 1)$. The rotation number is then $\rho = \lim_n (F^n(\theta) - \theta)/n$, **without** the factor 2π of Definition 4.2, and it is to be compared directly with α (not with $\alpha/2\pi$): all the theorems of Sections 4.1–4.4 remain valid word for word in this convention, replacing 2π by 1 everywhere (it is the same theory, rescaled).

Definition 4.3: Arnold standard family. The **Arnold standard family** (or *circle map*) is, for $(\alpha, \varepsilon) \in \mathbb{R} \times \mathbb{R}$,

$$f_{\alpha, \varepsilon}(\theta) := \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta) \pmod{1},$$

with lift $F_{\alpha,\varepsilon}(\theta) = \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$. For $\varepsilon = 0$ it reduces to the rigid rotation R_α . The rotation number $\rho(\alpha, \varepsilon) := \rho(f_{\alpha,\varepsilon})$, as a function of α with ε fixed, is the subject of the next two sections.

Theorem 4.5: $f_{\alpha,\varepsilon}$ is a diffeomorphism for $0 < \varepsilon < 1$. For $0 < \varepsilon < 1$ and every α , $f_{\alpha,\varepsilon}$ is an orientation-preserving diffeomorphism of $\mathbb{T}$. For $\varepsilon = 1$, $f_{\alpha,1}$ is still a homeomorphism, but not a diffeomorphism: $F'_{\alpha,1}(\theta) = 0$ exactly at $\theta = 0 \pmod{1}$.

Proof. $F'_{\alpha,\varepsilon}(\theta) = 1 - \varepsilon \cos(2\pi\theta)$. For $0 < \varepsilon < 1$: $F'_{\alpha,\varepsilon}(\theta) \geq 1 - \varepsilon > 0$ for every θ (since $\cos \leq 1$), hence $F_{\alpha,\varepsilon}$ is strictly increasing, of class C^∞ , with inverse also C^∞ by the inverse function theorem (never vanishing derivative): $F_{\alpha,\varepsilon}$ is a diffeomorphism of $\mathbb{R}$ commuting with the integer translations, hence it descends to a diffeomorphism of $\mathbb{T}$. For $\varepsilon = 1$: $F'_{\alpha,1}(\theta) = 1 - \cos(2\pi\theta) \geq 0$, vanishing exactly when $\cos(2\pi\theta) = 1$, that is, $\theta \in \mathbb{Z}$; elsewhere $F'_{\alpha,1}(\theta) > 0$. Since $F'_{\alpha,1} \geq 0$ with equality only at isolated points, $F_{\alpha,1}$ remains strictly increasing (a non-negative derivative, vanishing only on a discrete set, cannot be constant on any interval, hence the function is strictly increasing), so $f_{\alpha,1}$ is still a homeomorphism; but it is not a diffeomorphism, because the inverse is not differentiable at that image point (to a zero derivative of F there would correspond an infinite derivative of the inverse). $\square$

Theorem 4.6: The principal tongue $0/1$. For $0 < \varepsilon < 1$: $\rho(\alpha, \varepsilon) = 0$ if and only if $|\alpha| \leq \varepsilon/(2\pi)$ (with equality exactly at the two endpoints). On the boundary $\alpha = \pm\varepsilon/(2\pi)$, $f_{\alpha,\varepsilon}$ has a unique fixed point $\theta^* = \pm 1/4$ with $F'_{\alpha,\varepsilon}(\theta^*) = 1$ (**saddle-node bifurcation**): for $|\alpha| < \varepsilon/(2\pi)$ there are exactly two fixed points (one attracting, one repelling), which collide and disappear when $|\alpha|$ crosses $\varepsilon/(2\pi)$.

Proof. $\rho(\alpha, \varepsilon) = 0$ (in the $[0, 1)$ convention, Remark 4.4) if and only if $f_{\alpha,\varepsilon}$ has a fixed point (Theorem 4.3 with $p = 0, q = 1$; the implication “periodic orbit of period 1 $\Rightarrow \rho = 0$ ” is immediate, the other one is the theorem itself), that is, if and only if there exists θ with $F_{\alpha,\varepsilon}(\theta) = \theta$, that is,

$$\alpha = \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$$

for some $\theta \in [0, 1)$. On varying $\theta \in [0, 1)$, $\sin(2\pi\theta)$ takes *all* the values of $[-1, 1]$ (with $\theta = 1/4 \mapsto 1$, $\theta = 3/4 \mapsto -1$), hence the set of values of α for which the equation has a solution is exactly $\alpha \in [-\varepsilon/(2\pi), \varepsilon/(2\pi)]$: this proves the equivalence (the inclusion $\subseteq$ because $|\sin| \leq 1$, the inclusion $\supseteq$ by surjectivity of $\sin$ onto $[-1, 1]$).

For $|\alpha| < \varepsilon/(2\pi)$ strictly: the equation $\alpha = \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$ has, for $\theta \in [0, 1)$, exactly two solutions $\theta_1 \in (-1/4, 1/4)$ and $\theta_2 = 1/2 - \theta_1$ (symmetry of $\sin$ about $\pi/2$; the two solutions are distinct because $\theta_1 \neq 1/2 - \theta_1$ as soon as $\theta_1 \neq 1/4$, which is true for $\alpha \neq \varepsilon/(2\pi)$). Since $F'_{\alpha,\varepsilon}(\theta) = 1 - \varepsilon \cos(2\pi\theta)$: at $\theta_1 \in (-1/4, 1/4)$, $\cos(2\pi\theta_1) > 0$, hence $F'_{\alpha,\varepsilon}(\theta_1) < 1$ (and $> 1 - \varepsilon > 0$ by Theorem 4.5): attracting fixed point. At θ_2 , $\cos(2\pi\theta_2) = \cos(\pi - 2\pi\theta_1) = -\cos(2\pi\theta_1) < 0$, hence $F'_{\alpha,\varepsilon}(\theta_2) > 1$: repelling fixed point. On the boundary $\alpha = \varepsilon/(2\pi)$: $\theta_1 = \theta_2 = 1/4$ (the two solutions collide), a unique fixed point, with $\cos(2\pi \cdot \frac{1}{4}) = 0$ hence $F'_{\alpha,\varepsilon}(1/4) = 1 - \varepsilon \cdot 0 = 1$: neither attracting nor repelling to first order, the sign of $F_{\alpha,\varepsilon}(\theta) - \theta - \alpha = -\frac{\varepsilon}{2\pi} \sin(2\pi\theta) + \frac{\varepsilon}{2\pi}$ (collecting terms) near $\theta = 1/4$ is that of a tangent parabola (the second derivative of $\sin(2\pi\theta)$ at $1/4$ being non-zero: $-4\pi^2 \sin(\pi/2) = -4\pi^2 \neq 0$), the canonical signature of a **saddle-node bifurcation**: the two fixed points (attracting and repelling) collide into a single semi-stable fixed point and, on crossing the boundary, disappear (no real solution for $|\alpha| > \varepsilon/(2\pi)$, and consistently $\rho(\alpha, \varepsilon) \neq 0$ there, by the first part). The case $\alpha = -\varepsilon/(2\pi)$ is symmetric, with collision at $\theta^* = -1/4 \equiv 3/4$. $\square$

Example 4.3: The 0/1 tongue at $\varepsilon = 0.3$: closed formula against a numerical grid. **Double check (script `fig04_lingue_arnold.py`, run on 16 September 2026).** For $\varepsilon = 0.3$, Theorem 4.6 gives the **exact** width of the 0/1 tongue: $2 \times \varepsilon/(2\pi) = \varepsilon/\pi = 0.3/\pi = 0.0955\dots$. The script estimates $\rho(\alpha, \varepsilon)$ by iteration (transient of 2000 steps discarded, average over 3000 subsequent iterates) on a grid of 800×400 points $(\alpha, \varepsilon) \in [0, 1] \times [0, 1]$, classifies each point as locked to p/q ($q \leq 12$, tolerance 3×10^{-4}) or not, and at $\varepsilon = 0.3$ (nearest row of the grid) measures a width of the 0/1 tongue (read as the sum of the two tails locked to $q = 1$ near $\alpha = 0$ and $\alpha = 1$, since $\rho(\alpha + 1, \varepsilon) = \rho(\alpha, \varepsilon) + 1$) equal to 0.0950: absolute discrepancy from the exact value 0.0005, within the grid resolution $2/800 = 0.0025$. ✓

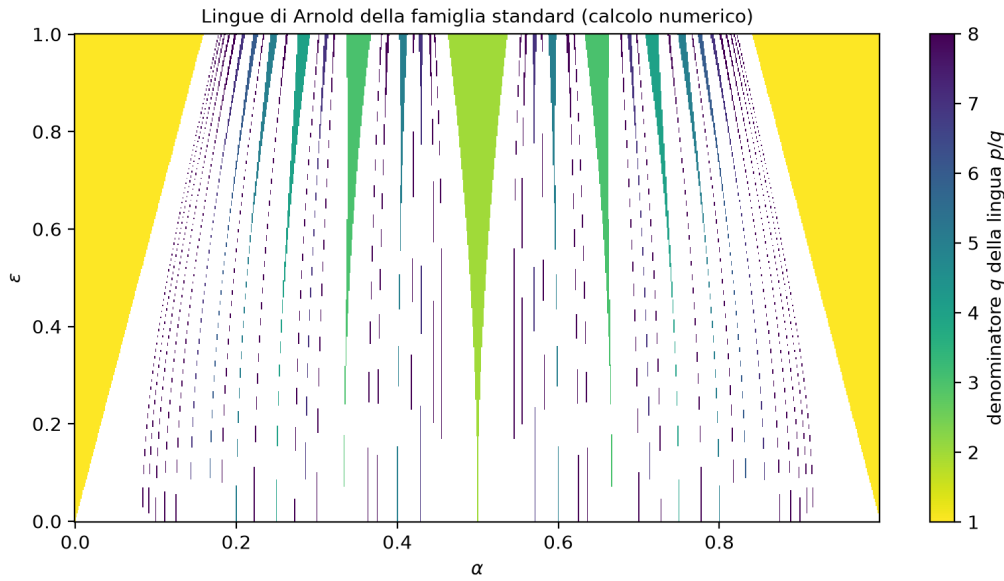


Figure 4.3: The Arnold tongues of the standard family in the plane $(\alpha, \varepsilon) \in [0, 1] \times [0, 1]$, computed numerically by the script `Sorgenti/Vol4/appendiceB_laboratorio/fig04_lingue_arnold.py` (grid 800×400 , transient 2000 iterates, average over 3000 iterates; run on 16 September 2026 with Python 3.12.5): colour according to the denominator $q \leq 12$ of the tongue p/q to which $\rho(\alpha, \varepsilon)$ turns out to be locked (white: no locking detected within the tolerance). Check: the width of the 0/1 tongue at $\varepsilon = 0.3$ read from the grid (0.0950) against the exact value $\varepsilon/\pi = 0.0955$ of Theorem 4.6 ✓ (Example 4.3).

4.6. Width of the p/q tongues

Theorem 4.7: Width of the p/q tongue: $O(\varepsilon^q)$. As $\varepsilon \rightarrow 0^+$, the total width (in α) of the tongue of rotation p/q (reduced) of the standard family is $O(\varepsilon^q)$. For $q = 1$ this is exact for every $0 < \varepsilon < 1$ (Theorem 4.6: width ε/π). For $q = 2$,

$$\text{width of the tongue } 1/2 = \frac{\varepsilon^2}{4\pi} + O(\varepsilon^3) \quad (\varepsilon \rightarrow 0^+).$$

Proof. [For $q = 1, 2$ proved in full; general case imported] The case $q = 1$ is Theorem 4.6 (exact width $\varepsilon/\pi = O(\varepsilon)$).

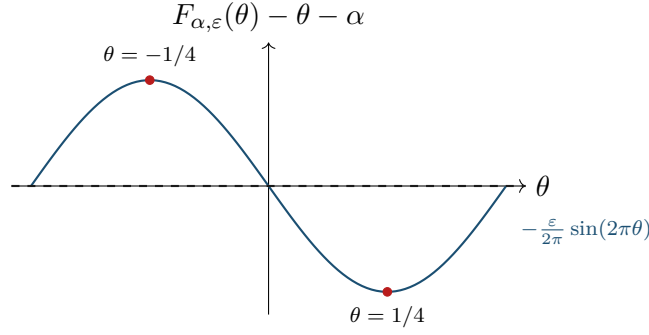


Figure 4.4: The graph of $F_{\alpha,\varepsilon}(\theta) - \theta - \alpha = -\frac{\varepsilon}{2\pi} \sin(2\pi\theta)$: the solutions of $\alpha' = \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$ for a given $\alpha' \in (-\varepsilon/2\pi, \varepsilon/2\pi)$ are the two intersections with the horizontal line $y = -\alpha'$ (not drawn explicitly here): as $|\alpha'|$ grows towards $\varepsilon/(2\pi)$ the two intersections slide towards each other along the curve until they collide exactly at the endpoint $\theta = \pm 1/4$ (red points, where the tangent is horizontal: $F' = 1$), the saddle-node bifurcation of Theorem 4.6.

Case $q = 2$. We look for points of period (exact or not) 2: $F_{\alpha,\varepsilon}^2(\theta) = \theta + 1$. Write $\alpha = \frac{1}{2} + a$ with a small (deviation from the centre $1/2$ of the tongue). Expanding,

$$F_{\alpha,\varepsilon}^2(\theta) - \theta - 1 = 2a - \frac{\varepsilon}{2\pi} \left[\sin(2\pi\theta) + \sin(2\pi f_{\alpha,\varepsilon}(\theta)) \right],$$

where we have used $2\alpha - 1 = 2a$. Since $f_{\alpha,\varepsilon}(\theta) = \theta + \frac{1}{2} + a - \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$, we have $2\pi f_{\alpha,\varepsilon}(\theta) = 2\pi\theta + \pi + 2\pi a - \varepsilon \sin(2\pi\theta)$, hence

$$\sin(2\pi f_{\alpha,\varepsilon}(\theta)) = -\sin(2\pi\theta + 2\pi a - \varepsilon \sin(2\pi\theta)) = -\sin(2\pi\theta) \cos(u) - \cos(2\pi\theta) \sin(u),$$

with $u := 2\pi a - \varepsilon \sin(2\pi\theta)$. Treating $a = O(\varepsilon^2)$ (the order that the final result will confirm a posteriori) and $\varepsilon \rightarrow 0$: $u = O(\varepsilon)$, hence $\cos u = 1 - \frac{1}{2}u^2 + O(u^4) = 1 - \frac{1}{2}\varepsilon^2 \sin^2(2\pi\theta) + O(\varepsilon^3)$ (the cross terms with a are $O(\varepsilon^2 \cdot \varepsilon^2) = O(\varepsilon^4)$, negligible) and $\sin u = u + O(u^3) = 2\pi a - \varepsilon \sin(2\pi\theta) + O(\varepsilon^3)$. Hence

$$\sin(2\pi f_{\alpha,\varepsilon}(\theta)) = -\sin(2\pi\theta) \left[1 - \frac{1}{2}\varepsilon^2 \sin^2(2\pi\theta) \right] - \cos(2\pi\theta) [2\pi a - \varepsilon \sin(2\pi\theta)] + O(\varepsilon^3),$$

and adding $\sin(2\pi\theta)$ (the terms of order ε^0 cancel exactly):

$$\sin(2\pi\theta) + \sin(2\pi f_{\alpha,\varepsilon}(\theta)) = \varepsilon \sin(2\pi\theta) \cos(2\pi\theta) - 2\pi a \cos(2\pi\theta) + O(\varepsilon^3) = \frac{\varepsilon}{2} \sin(4\pi\theta) - 2\pi a \cos(2\pi\theta) + O(\varepsilon^3).$$

Substituting into the expression for $F^2 - \theta - 1$ and setting the latter equal to 0 (condition for the existence of a point of period 2):

$$0 = 2a - \frac{\varepsilon}{2\pi} \left[\frac{\varepsilon}{2} \sin(4\pi\theta) - 2\pi a \cos(2\pi\theta) \right] + O(\varepsilon^4) = 2a - \frac{\varepsilon^2}{4\pi} \sin(4\pi\theta) + a \varepsilon \cos(2\pi\theta) + O(\varepsilon^4).$$

The term $a\varepsilon \cos(2\pi\theta)$ is $O(\varepsilon^3)$ (since $a = O(\varepsilon^2)$), hence negligible at the order at which we work:

$$a = \frac{\varepsilon^2}{8\pi} \sin(4\pi\theta) + O(\varepsilon^3).$$

On varying $\theta \in [0, 1]$, $\sin(4\pi\theta)$ takes all the values of $[-1, 1]$ (two complete periods), hence the set of values of a for which a solving θ exists is, up to $O(\varepsilon^3)$, the interval $[-\varepsilon^2/(8\pi), \varepsilon^2/(8\pi)]$, of total width $\varepsilon^2/(4\pi) + O(\varepsilon^3)$. Since the existence of a point of period 2 of the lift is, by the same argument as in Theorem 4.6 (a saddle-node bifurcation at the endpoints, where the

multiplier of the pair of period-2 orbits equals exactly 1), the condition delimiting the $1/2$ tongue, this proves the announced formula.

Independent numerical check (script `verifica_cap04_vol4.py`, block [C4-3]). Searching, for each $\varepsilon \in \{0.02; 0.05; 0.1; 0.15; 0.2\}$, the interval of α for which $G(\theta) = F_{\alpha,\varepsilon}^2(\theta) - \theta - 1$ has *at least one* zero (scanning the sign of G on a grid of 6000 points θ and bisecting on the two edges, **without** using the asymptotic formula to locate the initial brackets, taken $3\times$ wider than the prediction), the ratio between the numerical width and $\varepsilon^2/(4\pi)$ is: 0.99997 ($\varepsilon = 0.02$), 0.99979 ($\varepsilon = 0.05$), 0.99917 ($\varepsilon = 0.1$), 0.99813 ($\varepsilon = 0.15$), 0.99669 ($\varepsilon = 0.2$): the ratio tends to 1 as $\varepsilon \rightarrow 0$ with relative remainder $O(\varepsilon)$ (that is, absolute remainder $O(\varepsilon^3)$), exactly as predicted by the closed formula. ✓

General case ($q \geq 3$). The statement $O(\varepsilon^q)$ for a general q — with the same kind of perturbative computation, but on the order- q Fourier coefficients of the perturbation rather than a direct expansion as above — is a classical result of the theory of Arnold tongues; we **import** it from R25 (de Melo–van Strien, ch. I §4), K8 (Herman 1979, ch. V) and R39 (Boyland 1986, who also discusses its non-uniformity in q), without reproducing its general proof, which requires the theory of Fourier series at several orders (the homological equation of Chapter 6) rather than the direct expansion used here for $q = 1, 2$. Recorded in Appendix A. □

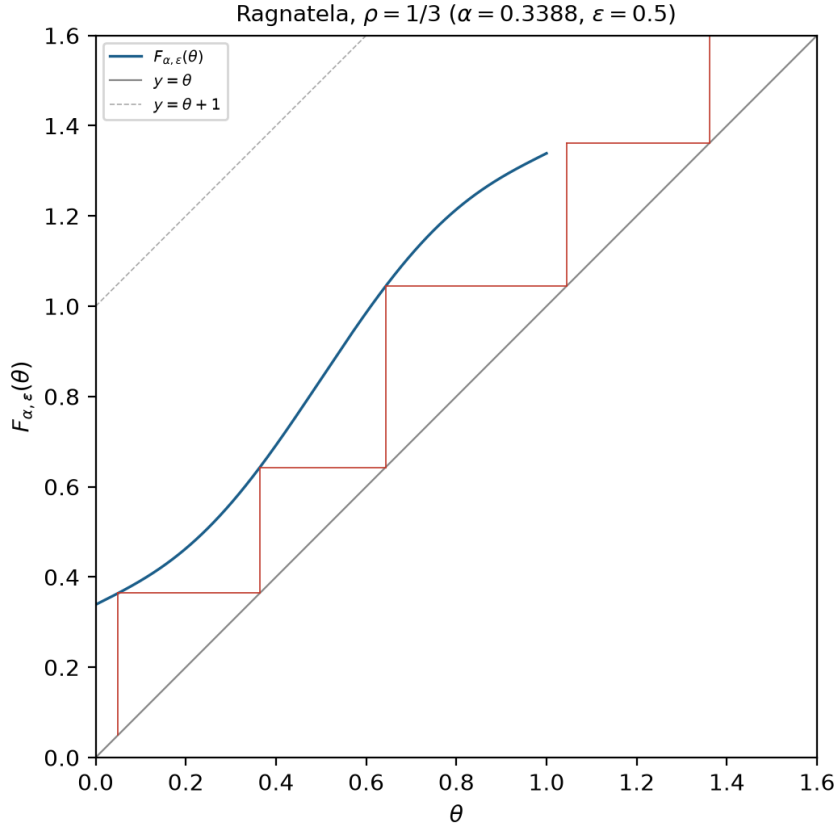


Figure 4.5: Cobweb diagram for $f_{\alpha,\varepsilon}$ with $\varepsilon = 0.5$ and $\alpha = 0.338781$ (centre of the $1/3$ tongue, located by bisection), generated by the script `Sorgenti/Vol4/appendiceB_laboratorio/fig04_ragnatela_rho13.py` (run on 16 September 2026): the trajectory (red) starting from $\theta_0 = 0.05$ converges in a few iterations to the attracting orbit of period 3 $\{0.996545, 0.337053, 0.607866\}$. Check: $F^3(\theta^*) - \theta^* - 1 = 0$ to machine precision ✓ (Theorem 4.3 with $p = 1, q = 3$).

4.7. The devil's staircase

Theorem 4.8: The devil’s staircase. For fixed $0 < \varepsilon < 1$, the function $\alpha \mapsto \rho(\alpha, \varepsilon)$ is continuous and non-decreasing on $\mathbb{R}$ (Theorem 4.2, applied to the family $F_{\alpha, \varepsilon} = F_{0, \varepsilon} + \alpha$ with $c = \alpha$, in the $[0, 1)$ convention of this section), and for every $p/q \in \mathbb{Q}$ the set $\{\alpha : \rho(\alpha, \varepsilon) = p/q\}$ contains an interval of positive width (the p/q tongue, of width exactly ε/π for $p/q = 0/1$, Theorem 4.6, and $O(\varepsilon^q)$ in general as $\varepsilon \rightarrow 0$, Theorem 4.7). The resulting function — continuous, monotone, constant on a dense union of intervals of positive width — is the **devil’s staircase**.

Proof. Continuity and monotonicity are Theorem 4.2(ii) applied to $F := F_{0, \varepsilon}$ (with ε fixed) and $c := \alpha$, observing that $F_{\alpha, \varepsilon} = F_{0, \varepsilon} + \alpha$ by definition of the standard family. It remains to show that, for p/q reduced, the set $\{\alpha : \rho(\alpha, \varepsilon) = p/q\}$ has non-empty interior (the p/q **plateau** or **tongue**). For the case $q = 1$ this is Theorem 4.6: *exact* width $\varepsilon/\pi > 0$. For $q \geq 2$, we use Theorem 4.7: the p/q tongue is delimited by a saddle-node bifurcation of the pair of period- q orbits (proved for $q = 1, 2$ and imported in general), which by construction produces a non-empty *open* interval of values of α on which the attracting/repelling pair of period q exists (the interval, precisely, of width $O(\varepsilon^q) > 0$ for small ε , and exactly ε/π for $q = 1$). On the whole of this interval there exists an orbit of period q , hence $\rho(\alpha, \varepsilon) = p/q$ for every α in the interval (Theorem 4.3): the set $\{\alpha : \rho(\alpha, \varepsilon) = p/q\}$ contains this non-empty open interval, hence has non-empty interior, for every p/q with $0 < \varepsilon < 1$. $\square$

Example 4.4: The $1/2$ plateau at $\varepsilon = 0.6$. Independent double check (script `fig04_scala_diavolo.py`, run on 16 September 2026). At $\varepsilon = 0.6$ (too large for the asymptotic formula $\varepsilon^2/(4\pi) = 0.0286$ of Theorem 4.7 to be accurate, as the script itself explicitly points out), the width of the $\rho = 1/2$ plateau is measured in two independent ways: (a) direct **bisection** on the two edges of the plateau (average over 60 000 iterates, a dichotomic search independent of the scan), which gives the interval $[0.486083; 0.513836]$, width 0.027753; (b) fixed-step **scan** on a grid of 2500 points (average over 4000–6000 iterates), which gives width 0.027200. Discrepancy between the two methods 0.000553, within the grid resolution $1/2500 = 0.0004$ plus the truncation noise (the iteration of the scan uses fewer iterates than that of the bisection). $\checkmark$ The same script verifies the **monotonicity** of $\rho(\alpha, 0.6)$ on the whole grid (no decrease greater than 5.5×10^{-14} , compatible with machine rounding) and, at $\varepsilon = 1.0$ (criticality), that the fraction of grid points with no locking within denominator $q \leq 20$ and tolerance 0.05 is 0, consistently with the full-measure mode-locking of Remark 4.5 below.

4.8. A panorama beyond the perturbative regime

Sections 4.5–4.7 prove exact facts ($q = 1$) or asymptotic ones as $\varepsilon \rightarrow 0$ ($q = 2$, and in general the imported $O(\varepsilon^q)$). What happens when ε is not small, in particular at criticality $\varepsilon = 1$ and beyond? The following three results — of increasing technical difficulty, well beyond the elementary scope of this chapter — answer this; we present them as a panorama, with the primary source verified, without proof.

Remark 4.5: Full-measure mode-locking at criticality (Świątek, 1988), imported. At criticality $\varepsilon = 1$ (where $f_{\alpha, 1}$ is a homeomorphism but not a diffeomorphism, Theorem 4.5), the set $\{\alpha \in [0, 1) : \rho(\alpha, 1) \notin \mathbb{Q}\}$ has **zero Lebesgue measure**: for almost every α (with respect to Lebesgue), $\rho(\alpha, 1)$ is rational, that is, α belongs to one of the tongues. This is the exact opposite of the situation for $0 < \varepsilon < 1$, where the tongues occupy a set of positive but not full measure (the complement, where ρ is irrational, also has positive measure, since

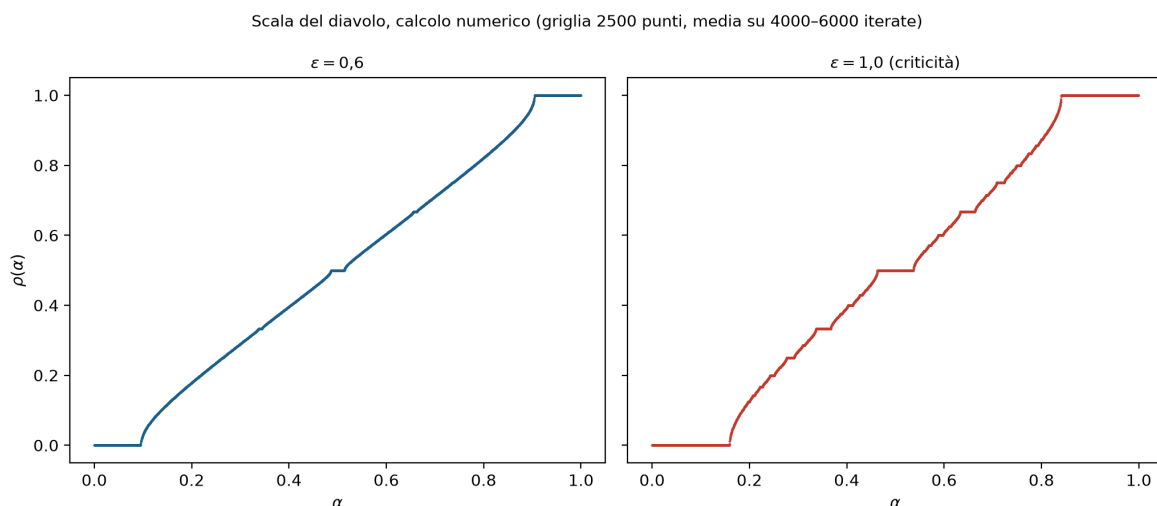


Figure 4.6: The devil’s staircase $\rho(\alpha, \varepsilon)$ for $\varepsilon = 0.6$ (left, below criticality) and $\varepsilon = 1.0$ (right, at criticality), computed numerically by the script `Sorgenti/Vol4/appendiceB_laboratorio/fig04_scala_diavolo.py` (grid of 2500 points α in $[0, 1)$, average over 4000–6000 iterates; run on 16 September 2026): at $\varepsilon = 1$ the plateaux visibly occupy almost the whole interval (Remark 4.5, Świątek 1988). Check: monotonicity verified on the whole grid in both cases ✓; width of the $1/2$ plateau at $\varepsilon = 0.6$ agreeing between bisection and scan to 5×10^{-4} ✓ (Example 4.4).

it contains all the Diophantine α of Theorem 3.6 of Chapter 3 for which the KAM theory of Chapter 6 guarantees smooth conjugacy). Result imported from G. Świątek, “Rational rotation numbers for maps of the circle”, *Comm. Math. Phys.* **119** (1988), n. 1, 109–128, primary source verified on 16 September 2026 (see the source register at the end of the chapter): the proof uses cross-ratio distortion and Herman–Świątek type estimates on the nonlinearity of critical maps, well beyond the tools of this chapter.

Remark 4.6: Rotation intervals for $\varepsilon > 1$ (Boyland, 1986), imported. For $\varepsilon > 1$, $f_{\alpha, \varepsilon}$ is no longer injective (Theorem 4.5: $F'_{\alpha, \varepsilon}$ changes sign), hence it is not a circle homeomorphism and the rotation number of Definition 4.2 is no longer uniquely defined: the role of ρ is taken by a **rotation interval** $[\rho_-(\alpha, \varepsilon), \rho_+(\alpha, \varepsilon)]$ (the lower and upper limits, over the points of the circle, of the asymptotic advance rate), which degenerates to a point for $\varepsilon \leq 1$ and has positive width for $\varepsilon > 1$. The structure of the “tongues” becomes richer: besides the tongues of locking to a single rational, there appear regions of **bistability** (coexistence of two attracting cycles with different rotations) and of genuinely chaotic behaviour (positive topological entropy). Result imported from P. L. Boyland, “Bifurcations of circle maps: Arnol’d tongues, bistability and rotation intervals”, *Comm. Math. Phys.* **106** (1986), 353–381, primary source verified on 16 September 2026 (see the source register).

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Application: circle maps and cardiac arrhythmias (Guevara–Glass, 1982), imported. The model of the standard family describes, with the reinterpretation θ = phase of the cardiac cycle and α, ε parameters of the external periodic stimulus, the response of a cardiac oscillator (sinoatrial node or ventricular tissue) to periodic stimulation: the Arnol’d tongues become the regions of $n:m$ **phase locking** (one response every m stimuli, with n

beats in the meantime) observed experimentally in preparations of periodically stimulated cardiac tissue, and the transition from regular mode-locking to chaos (for $\varepsilon > 1$, Remark 4.6) has been proposed as a model of certain arrhythmias. Result (level B, application) imported from M. R. Guevara, L. Glass, “Phase locking, period doubling bifurcations and chaos in a mathematical model of a periodically driven oscillator: a theory for the entrainment of biological oscillators and the generation of cardiac dysrhythmias”, *J. Math. Biol.* **14** (1982), 1–23, primary source verified on 16 September 2026 (see the source register); see also R41 (Glass–Mackey, *From Clocks to Chaos*, 1988) for the general framing.

4.9. Towards KAM theory: the statements of Arnold and of Herman–Yoccoz

Poincaré’s semiconjugacy (Theorem 4.4) answers the topological question: when $\rho(f)$ is irrational, f is *always* semiconjugate to $R_{2\pi\rho}$, with h only continuous. A finer question — when h is in fact *smooth* (differentiable, or analytic if f is) — is the problem of **small divisors**, and its solution (Arnold 1961, Herman 1979, Yoccoz 1984 and 1995) is the content of Chapter 6. Here we confine ourselves to stating its main results correctly, as a bridge, without proof: Chapter 3 (Remark 3.3 and thereabouts) had already met the Diophantine numbers (Definition 3.4); here they return as the decisive arithmetic hypothesis.

Remark 4.7: Arnold’s theorem (1961), correct statement. *Statement (Arnold, 1961).* Let ρ be Diophantine (Definition 3.4, Chapter 3) and let f be an analytic circle diffeomorphism, defined and close to $R_{2\pi\rho}$ on a complex strip $|\operatorname{Im} \theta| < a$, with $\rho(f) = \rho$ **exactly** (not merely f close to $R_{2\pi\rho}$: the hypothesis on the rotation number of f itself, and not only on the rotation to which f is close, is indispensable and is the hypothesis that a first draft of this chapter had omitted). If $\|f - R_{2\pi\rho}\|_a < \varepsilon_0(a, \rho)$ for an explicit threshold ε_0 depending on a and on the Diophantine type of ρ , then f is **analytically conjugate** to $R_{2\pi\rho}$: the semiconjugacy h of Theorem 4.4 is, in this perturbative regime, not only a homeomorphism but an analytic map.

Why the hypothesis $\rho(f) = \rho$ is necessary. Without it, the statement would be false: an f close to $R_{2\pi\rho}$ in the C^0 norm (or even analytic) may perfectly well have $\rho(f)$ rational (it is enough to look at an Arnold tongue of Section 4.6, of width $O(\varepsilon^q) > 0$ for every q : a small but non-zero perturbation of $R_{2\pi\rho}$ may fall inside a tongue, where $\rho(f) \in \mathbb{Q} \neq \rho$), and in that case f is not even semiconjugate to $R_{2\pi\rho}$ in a non-trivial way, but obeys the Poincaré classification in the rational case (Theorem 4.3), which is entirely different.

Idea of the proof (deferred to Chapter 6). The linear step is solved by the homological equation $\psi(\theta + 2\pi\rho) - \psi(\theta) = \varphi(\theta)$, whose Fourier coefficients $\hat{\psi}_k = \hat{\varphi}_k / (e^{2\pi i k \rho} - 1)$ have denominators (the “small divisors”) controlled by the Diophantine condition; this linearises f up to a remainder quadratic in ε , and the Kolmogorov–Arnold–Moser **Newton scheme** iterates the step with a loss of domain $\sum_j \delta_j < \infty$ and quadratic convergence of the error, which wins against the loss of domain. The complete proof — Cauchy estimates, loss of domain, convergence of the scheme — is in Chapter 6, devoted entirely to the theory of small divisors on the circle.

Source. V. I. Arnol’d, “Small denominators. I. Mapping the circle onto itself” (transliterated Russian title: *Malye znamenateli. I. Ob otobraženijah okružnosti na sebja*), *Izv. Akad. Nauk SSSR Ser. Mat.* **25**:1 (1961), 21–86, with errata **28**:2 (1964), 479–480 (R37, verified on 16 September 2026 on Math-Net.Ru); see also H. W. Broer, “KAM theory: the legacy of Kolmogorov’s 1954 paper”, *Bull. AMS* **41** (2004), 507–521 (R42, Theorem 1), for the modern framing.

Remark 4.8: Herman (1979) and Yoccoz (1984, 1995): the global theory, correct statement. Arnold’s theorem (Remark 4.7) is **local**: it requires f to be close to a rotation. Removing this hypothesis — a smooth conjugacy for every sufficiently smooth diffeomorphism with Diophantine rotation number, not only for those close to a rotation — took almost twenty years.

(i) **Herman, 1979.** If $f \in C^\infty(\mathbb{T})$ is a diffeomorphism with $\rho(f)$ in a set of full Lebesgue measure (in particular for $\rho(f)$ Diophantine, Theorem 3.6 of Chapter 3), the conjugacy h of Theorem 4.4 is C^∞ — with no hypothesis of closeness to a rotation. Source: M. R. Herman, “Sur la conjugaison différentiable des difféomorphismes du cercle à des rotations”, *Publ. Math. IHÉS* **49** (1979), 5–233 (K8, verified on 16 September 2026 on Numdam): not 1995, as a first draft of this chapter mistakenly reported (1995 is the year of a related but distinct work of Yoccoz, see (iii)).

(ii) **Yoccoz, 1984.** J.-C. Yoccoz extended Herman’s result to **every** Diophantine $\rho(f)$ (not only a set of full measure, which excludes a priori certain Diophantine numbers of extreme type): for $f \in C^\infty(\mathbb{T})$ a diffeomorphism with $\rho(f)$ Diophantine, the conjugacy is C^∞ . Source: J.-C. Yoccoz, “Conjugaison différentiable des difféomorphismes du cercle dont le nombre de rotation vérifie une condition diophantienne”, *Ann. Sci. ENS* (4) **17** (1984), 333–359 (K9, verified on 16 September 2026 on Numdam).

(iii) **The global analytic case and condition H: why Bryuno is not the right threshold here.** For f analytic (not merely C^∞), the question “for which ρ the conjugacy is analytic, for every analytic f , close to a rotation or not” has an arithmetic answer subtler than the Bryuno condition $\sum_n (\log q_{n+1})/q_n < \infty$ (Chapter 3, Remark 3.3 and thereabouts for the related Diophantine notions): Yoccoz proved that the optimal threshold for this problem (global, for circle diffeomorphisms) is an arithmetic condition, denoted by **H** in the literature following Yoccoz, strictly more permissive than the Diophantine condition and strictly more restrictive than Bryuno: $\text{CD} \subsetneq H \subsetneq B$ (Diophantine $\subsetneq H \subsetneq$ Bryuno, that is, every Diophantine number satisfies **H**, and every number satisfying **H** satisfies Bryuno, with both inclusions strict). The **Bryuno** condition remains instead the optimal threshold for a different and local problem: the linearisation of a holomorphic germ $z \mapsto e^{2\pi i \rho} z + O(z^2)$ near a fixed point (Siegel discs, Chapter 12 of this volume) — **not** the global problem of circle diffeomorphisms under examination here. A first draft of this chapter confused the two thresholds, mistakenly assigning to Bryuno the role of global condition for circle diffeomorphisms: it is corrected here, with the precise definition of condition **H** deferred to Chapter 6 (where the complete theory, with both problems — local holomorphic and global real — is developed and the relation between the two thresholds proved). Sources: J.-C. Yoccoz, “Centralisateurs et conjugaison différentiable des difféomorphismes du cercle”, in *Petits diviseurs en dimension 1*, *Astérisque* **231** (1995), 89–242 (K11, verified on 16 September 2026 on Numdam: not *Astérisque* 206, which contains instead Yoccoz’s exposition at the Séminaire Bourbaki on Herman’s works on invariant tori, a different text); J.-C. Yoccoz, “Théorème de Siegel, nombres de Bruno et polynômes quadratiques”, *Astérisque* **231** (1995), 1–88 (K12, for the Bryuno threshold in the local holomorphic problem); H. Eliasson, B. Fayad, R. Krikorian, “Jean-Christophe Yoccoz and the theory of circle diffeomorphisms”, *arXiv:1810.07107* (K19, verified on 16 September 2026, a survey that states explicitly $\text{CD} \subsetneq H \subsetneq B$): the attribution to “Fayad” alone in a first draft of this chapter was also imprecise, the article being by Eliasson, Fayad and Krikorian together.

On the Fields medal. Yoccoz received the Fields medal in **1994**, in part for these results; Herman did **not** receive it (an ambiguous formulation in a first draft of this chapter could suggest the contrary).

4.10. Assigned exercises

The answers are in the Teacher's key (Section 4.13).

Exercise 4.1: A non-symmetric period-2 orbit. *Goal:* to apply Theorem 4.3 in order to locate explicitly a periodic orbit of exact period 2 of the standard family, distinguishing it from the trivial case $\theta = 0, 1/2$ (fixed points of $\sin(2\pi\theta)$ for every ε). *Estimated time:* 30 minutes. *Type:* numerical/computational.

- (a) For $\varepsilon = 0.4$, $\alpha = 0.505$, iterate $F_{\alpha,\varepsilon}$ from $\theta_0 = 0.2$ for 20 000 steps and record the point reached θ^* and $F_{\alpha,\varepsilon}(\theta^*)$.
- (b) Check numerically $F_{\alpha,\varepsilon}^2(\theta^*) - \theta^* - 1$: what is its value, and what does this mean for $\rho(\alpha, \varepsilon)$?
- (c) Explain why $\theta^* \neq 0$ and $\theta^* \neq 1/2$ in this case (unlike $\alpha = 1/2$ exactly, where $\theta = 0, 1/2$ are a 2-periodic fixed point for *every* ε , by the oddness of $\sin$).
- (d) Is the period found *minimal*? How does one check this (Theorem 4.3(i))?

Exercise 4.2: Check of the formula $\varepsilon^2/(4\pi)$ at $\varepsilon = 0.08$. *Goal:* to reproduce the numerical check of Theorem 4.7 at a new value of ε , not used in the text, with the zero-finding method (not asymptotic iteration, in order to avoid the slow-convergence bias discussed in the script `fig04_ragnatela_rho13.py`). *Estimated time:* 40 minutes. *Type:* numerical/computational with verification of a closed formula.

- (a) Write $G(\theta) = F_{\alpha,0.08}^2(\theta) - \theta - 1$ and scan the sign on a fine grid of $\theta \in [0, 1]$ in order to locate, by bisection on the endpoints α , the two edges of the $1/2$ tongue.
- (b) Compute the numerical width and compare it with $\varepsilon^2/(4\pi)$.
- (c) Is the ratio between the two values closer to 1 or further from 1 than in the case $\varepsilon = 0.2$ discussed in the text? Is it consistent with the remainder $O(\varepsilon^3)$ of the formula?

Exercise 4.3: What a finite grid at criticality does prove (and what it does not). *Goal:* to distinguish a numerical check from a proof, reflecting on Świątek's result (Remark 4.5). *Estimated time:* 20 minutes. *Type:* conceptual/metacognitive.

- (a) The script `fig04_scala_diavolo.py` finds, at $\varepsilon = 1$, that 0% of a grid of 2500 points α turns out to be unlocked (within $q \leq 20$, tolerance 0.05). Does this *prove* that the set of α with $\rho(\alpha, 1)$ irrational has zero Lebesgue measure?
- (b) If the answer to (a) is no, what exactly does the numerical result prove, and what purpose does it serve in the chapter?
- (c) Theorem 3.6 of Chapter 3 (full measure of the Diophantine numbers) and Remark 4.5 (full measure of mode-locking at $\varepsilon = 1$) seem, at first sight, to be in tension. Explain why they are not (to which values of ε they apply).

4.11. Worked exercises

Exercise 4.4: Numerical check of $\rho(f^3) = 3\rho(f)$. *Type:* numerical, with verification of a proved identity (Proposition 4.1). Compute $\rho(f_{\alpha,\varepsilon})$ and $\rho(f_{\alpha,\varepsilon}^3)$ independently (iterating F and, separately, $F \circ F \circ F$) for $\alpha = 0.4123456$, $\varepsilon = 0.3$, and verify $\rho(f^3) = 3\rho(f)$.

Solution. Strategy. Estimate $\rho(f)$ by iterating F directly; estimate $\rho(f^3)$ by iterating the composed map $F_3 := F \circ F \circ F$ as if it were itself an independent lift (same algorithm, transient and average), without ever computing $3\rho(f)$ in between: this makes the final comparison a genuinely independent check, not a tautology.

Step 1. From $\theta_0 = 0.31415926535$, transient of 2000 steps, average over 200 000 iterates of F : $\rho(f) = 0.4113916173\dots$

Step 2. Same θ_0 , transient of 2000 applications of $F_3 = F \circ F \circ F$, average over 200 000 applications of F_3 : $\rho(f^3) = 1.2341750656\dots$

Independent check. $3 \times \rho(f) = 1.2341748520\dots$; discrepancy with respect to $\rho(f^3)$: 2.1×10^{-7} , compatible with the truncation error of the two independent numerical estimates (each with $O(1/N) = O(5 \times 10^{-6})$ of expected error, Theorem 4.1). ✓

Interpretation. The agreement confirms numerically Proposition 4.1, proved in general in the text by a three-line subsequence argument: the pedagogical point here is that the proof is so short precisely because Definition 4.2 is already formulated as a limit along *all* the integers, hence it holds automatically along the subsequence of the multiples of 3.

Typical error to avoid. Confusing $\rho(f^3) = 3\rho(f)$ (true, Proposition 4.1, with ρ *not reduced modulo 1*: here $\rho(f^3) \approx 1.234 > 1$) with the assertion, false in general, that the rotation number “always reduces to $[0, 1]$ ”: the reduction mod 1 is a convention for speaking of points of the circle, but Poincaré’s identity is more natural, and is verified here, on the rotation numbers of the *lifts*, without reduction.

Extension. Repeat for $\rho(f^5)$ and verify $\rho(f^5) = 5\rho(f)$; one expects an equally good agreement, with the same order of error (the transient of $F_5 = F \circ \dots \circ F$ does not require more “outer” iterations than that of F_3 , being in any case governed by the same speed of convergence of the Fekete sandwich for F).

Exercise 4.5: From the search for the period-3 orbit to the exact rotation number. *Type:* numerical with exhaustive verification. For $\varepsilon = 0.5$, find (by bisection) the centre of the $1/3$ tongue and verify that the period-3 orbit found really has *minimal* period 3 (not 1 or a degenerate outcome).

Solution. Strategy. Use the operational test “locked” = residual $F^3(\theta) - \theta - 1$ zero within tolerance after a fixed transient (not the asymptotic estimate of ρ , for the reasons discussed in the note of the block `fig04_ragnatela_rho13.py`: near the edge of the tongue the multiplier of the orbit tends to 1, making the convergence of the asymptotic estimate slow), bisect on the two edges, then verify the minimal period directly.

Step 1. Bisection (left edge in the bracket $[0.335; 0.338]$, right edge in $[0.3407; 0.342]$, 40 halvings each): the $1/3$ tongue at $\varepsilon = 0.5$ equals $[0.336662; 0.340899]$, width 0.004237. Centre: $\alpha = 0.338781$.

Step 2. From $\theta_0 = 0.4132$, 20 000 iterates of $F_{0.338781; 0.5}$: convergence to $\theta^* = 0.996545$, with $F(\theta^*) = 0.337053$, $F^2(\theta^*) = 0.607866$, $F^3(\theta^*) = 1.996545 = \theta^* + 1$: check $F^3(\theta^*) - \theta^* - 1 = 0$ to machine precision. ✓

Independent check (minimal period). The three values $\theta^*, F(\theta^*), F^2(\theta^*)$ are pairwise distinct (differences ≥ 0.27 , well above machine precision): hence the period of $\theta^* \bmod 1$ under f cannot be 1 (that would require $F(\theta^*) - \theta^* \in \mathbb{Z}$, which is false). Being a divisor of 3 (prime) other than 1, the minimal period is exactly 3, in agreement with Theorem 4.3(i) applied to $\rho = 1/3$, already reduced.

Interpretation. The orbit found is the one drawn in the cobweb of Figure 4.5: the rapid convergence (few steps visible in the diagram) reflects the fact that $\alpha = 0.338781$ is close to the *centre* of the tongue, where the multiplier of the attracting orbit is well away from 1 (unlike at the edges, where the bisection of Step 1 does need a large N to converge, as discussed in the note of the verification block).

Extension. Repeat for the $2/3$ tongue (same $\varepsilon = 0.5$): by the approximate symmetry of the standard family under $\alpha \mapsto 1 - \alpha$ (not exact, but close for ε not small), one expects a tongue of comparable width centred near $\alpha \approx 1 - 0.338781 = 0.661219$.

Exercise 4.6: Fekete’s lemma on an explicit sequence, with a closed formula for the remainder. *Type:* symbolic with exhaustive numerical verification. Let a_n be the minimum number of 3-cent and 5-cent coins needed to pay exactly n cents (with the convention $a_n = +\infty$ if n is not representable). Verify that (a_n) is subadditive, compute $\inf_n a_n/n$, and exhibit a closed formula for the case that approaches the limit most slowly.

Solution. Strategy. Verify subadditivity directly from the definition (concatenating two optimal payments), then identify for which residues $n \bmod 5$ the optimal representation is “less efficient”, obtaining a closed formula to compare with the exhaustive computation.

Step 1 (subadditivity). If a_m coins pay m and a_n coins pay n , the union of the two collections of coins pays $m+n$: hence $a_{m+n} \leq a_m + a_n$ by definition of a_{m+n} as a minimum.

✓

Step 2 (closed formula for each residue class). Every $n \geq 8$ (the Frobenius number of 3, 5 is $3 \times 5 - 3 - 5 = 7$: from 8 on every n is representable) can be written as $n = 5k + j$ with $j \in \{0, 1, 2, 3, 4\}$. To minimise the number of coins one maximises the number b of 5-coins (a 5-coin is worth more per coin than a 3-coin), choosing the largest $b \leq k$ such that the remainder $n - 5b$ is a non-negative multiple of 3:

- $j = 0$: $b = k$, remainder 0: $a_n = k$ (no 3-coins);
- $j = 1$: $b = k - 1$ (the remainder with $b = k$ is 1, not a multiple of 3), remainder $6 = 3 \times 2$: $a_n = (k - 1) + 2 = k + 1$;
- $j = 2$: $b = k - 2$ (the remainders with $b = k, k - 1$ are 2, 7, neither a multiple of 3), remainder $12 = 3 \times 4$: $a_n = (k - 2) + 4 = k + 2$;
- $j = 3$: $b = k$, remainder 3: $a_n = k + 1$;
- $j = 4$: $b = k - 1$ (the remainder with $b = k$ is 4, not payable with 3-coins alone), remainder $9 = 3 \times 3$: $a_n = (k - 1) + 3 = k + 2$.

Rewriting in terms of n (with $k = (n - j)/5$): $a_n = n/5$ for $j = 0$; $a_n = (n + 4)/5$ for $j = 1$; $a_n = (n + 8)/5$ for $j = 2$; $a_n = (n + 2)/5$ for $j = 3$; $a_n = (n + 6)/5$ for $j = 4$. The genuinely worst case — the largest additive constant, 8 — is $j = 2$, *not* $j = 4$: although $j = 2$ and $j = 4$ have the same absolute excess of coins ($k + 2$ in both), for the same k the value of n is smaller for $j = 2$ ($n = 5k + 2$ versus $n = 5k + 4$), so the *relative* excess $a_n/n - 1/5$ is larger.

Step 3 (comparison). $\inf_n a_n/n = 1/5$ (exhaustive computation over $n = 1, \dots, 2000$, script `verifica_cap04_vol14.py`, block [C4-1]: minimum exactly 0.2, attained for every n multiple of 5, where $a_n = n/5$ exactly). For $n = 5k + 2$: $a_n/n = (n + 8)/(5n) = 1/5 + 8/(5n) \rightarrow 1/5$, with *exact* remainder $8/(5n)$, the slowest among the five cases modulo 5 (against $6/(5n)$ for $j = 4$, $4/(5n)$ for $j = 1$, $2/(5n)$ for $j = 3$, 0 for $j = 0$). Direct numerical comparison, at the end of the tested range: $n = 1999 = 5 \times 399 + 4$ gives $a_n/n = 0.2 + 6/(5 \times 1999) = 0.200601$, while $n = 1997 = 5 \times 399 + 2$ gives $a_n/n = 0.2 + 8/(5 \times 1997) = 0.200801 > 0.200601$: although $a_{1997} = a_{1999} = 401$ (the same number of coins), the ratio a_n/n is larger for $n = 1997$, confirming that $j = 2$ is the genuinely slowest class. The value read from the exhaustive computation agrees for both. ✓

Interpretation. Fekete’s lemma (Lemma 4.1) guarantees $\lim_n a_n/n = \inf_n a_n/n = 1/5$ as a purely qualitative fact; this example shows that the *rate* of convergence may be slow ($O(1/n)$, no faster) and may depend on the residue class of n : a useful warning for Theorem 4.1, where the same phenomenon (convergence $O(1/n)$, no better) governs the speed at which $(F^n(\theta) - \theta)/n$ approaches $2\pi\rho$ (visible in the $O(1/n)$ width of the Fekete sandwich of Example 4.1).

Extension. Repeat with coins of 4 and 7 (coprime, Frobenius number $4 \times 7 - 4 - 7 = 17$): one expects $\inf a_n/n = 1/4$ (almost only 4-coins are used), with the analogous worst case for $n \equiv$ (a suitable residue) $\bmod 4$.

4.12. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 4.13).

1. (Definition) The rotation number $\rho(f)$ (Definition 4.2) is defined as
 - A) the minimal period of a periodic orbit of f
 - B) $\lim_{n \rightarrow \infty} (F^n(\theta) - \theta)/(2\pi n)$, whose value does not depend on θ
 - C) the derivative of f at its fixed point
 - D) the measure of the Cantor set E
2. (Hypothesis) In Arnold’s theorem (Remark 4.7), besides ρ Diophantine and f close to $R_{2\pi\rho}$, the hypothesis that a first draft of the chapter had omitted and that is indispensable is
 - A) f must be C^∞ , not only analytic
 - B) the rotation number of f itself must equal exactly ρ
 - C) ε must be smaller than $1/2$
 - D) f must be linear
3. (Symbolic computation) For the standard family with $\varepsilon = 0.5$, the *exact* width of the principal tongue $0/1$ (Theorem 4.6) equals
 - A) $0.5/\pi \approx 0.1592$
 - B) 0.25
 - C) $0.5^2/(4\pi) \approx 0.0199$
 - D) $\pi/0.5$
4. (Numerical computation) With $\alpha = 0.4123456$, $\varepsilon = 0.3$ (Example 4.1), the estimate of $\rho(f)$ over 10^6 iterates is approximately
 - A) 0.618
 - B) 0.4114
 - C) 0.300
 - D) 1.234
5. (Proof step) In the proof of Lemma 4.3 (rational criterion), the step that allows one to conclude $F^{qn}(\theta_0) \geq \theta_0 + 2\pi pn$ for *every* n , starting from the case $n = 1$, is
 - A) Fekete’s lemma
 - B) induction, using the increase of F^q and its commutation with the integer translations by 2π
 - C) the intermediate value theorem
 - D) the density of the rationals
6. (Error diagnosis) A student writes: “the Bryuno condition $\sum_n (\log q_{n+1})/q_n < \infty$ is the optimal arithmetic threshold for the analytic conjugacy of circle diffeomorphisms (Yoccoz)”. The error is
 - A) the Bryuno formula is written incorrectly
 - B) it confuses the *local* problem (holomorphic germs, Siegel discs, where Bryuno really is optimal) with the *global* problem of circle diffeomorphisms, where the optimal threshold is condition H, strictly more restrictive (Remark 4.8)
 - C) Yoccoz did not work on this problem
 - D) the correct condition is the Diophantine one, not Bryuno’s
7. (Transfer) In a Guevara–Glass type model of cardiac phase locking (Remark 4.6, infobox), a region of parameters in which the heart responds with exactly 2 beats every 3 external stimuli, stably under small variations of the stimulation frequency, corresponds, in the language of this chapter, to
 - A) an isolated point with ρ irrational
 - B) the Arnold tongue $2/3$, an interval of parameters of positive width with $\rho = 2/3$ constant
 - C) the Cantor set E of Theorem 4.4
 - D) the saddle-node bifurcation alone, without an interval
8. (Source and metacognition) The correct primary source, verified on 16 September 2026, of the result “at criticality $\varepsilon = 1$ mode-locking has full measure” (Remark 4.5) is
 - A) V. I. Arnold, 1961
 - B) G. Świątek, *Comm. Math. Phys.* **119** (1988), 109–128
 - C) M. Herman, 1979
 - D) J.-C. Yoccoz, 1984

4.13. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **B.** *Rationale:* Definition 4.2, existence and independence of the base point proved in Theorem 4.1. *Distractors:* A confuses ρ with the period (an integer, not the rotation number, which may be irrational); C, D bear no relation to the definition. *Diagnosed error:* confusing the global dynamical invariant ρ with local quantities (derivative, period). *Source:* Definition 4.2, Theorem 4.1. *Prerequisite:* lift of a circle homeomorphism. *Remedial work:* reread the proof of Theorem 4.1, isolating where Fekete’s lemma intervenes.
- 2) **B.** *Rationale:* Remark 4.7, with the explicit example of the Arnold tongues showing why the hypothesis is necessary. *Distractors:* A, C, D are not the hypotheses of Arnold’s theorem. *Diagnosed error:* thinking that “close to a rotation” automatically implies “with the same rotation number”, false because of the tongues of width $O(\varepsilon^q) > 0$. *Source:* Remark 4.7, correction 3.15 of the expansion plan. *Prerequisite:* Theorem 4.7. *Remedial work:* reread the paragraph “why the hypothesis $\rho(f) = \rho$ is necessary”.
- 3) **A.** *Rationale:* Theorem 4.6, width $2 \times \varepsilon / (2\pi) = \varepsilon / \pi$; with $\varepsilon = 0.5$: $0.5 / \pi = 0.1592 \dots$. *Distractors:* B is an arbitrary value; C is the formula (asymptotic, and for $q = 2$, not $q = 1$) of Theorem 4.7; D is the fraction turned upside down. *Diagnosed error:* confusing the exact width for $q = 1$ with the asymptotic formula for $q = 2$, or inverting the fraction. *Source:* Theorem 4.6. *Prerequisite:* $\sin(2\pi\theta)$ surjective onto $[-1, 1]$. *Remedial work:* repeat the proof of the theorem with $\varepsilon = 0.5$ explicitly.
- 4) **B.** *Rationale:* Example 4.1, $\rho \approx 0.41139165$. *Distractors:* A is $1/\varphi$ (a noble number, extraneous); C is ε , not ρ ; D is 3ρ (Worked exercise 4.4), not ρ . *Diagnosed error:* confusing $\rho(f)$ with ε or with $\rho(f^3)$. *Source:* Example 4.1. *Prerequisite:* none beyond reading the example. *Remedial work:* rerun the script `verifica_cap04_vol4.py`, block [C4-2].
- 5) **B.** *Rationale:* proof of Lemma 4.3(i, $\Leftarrow$), inductive step. *Distractors:* A, C, D are tools used elsewhere in the chapter, not in this specific step. *Diagnosed error:* failing to isolate which property of F (increase + commutation with the integer translations) makes the inductive step valid. *Source:* Lemma 4.3. *Prerequisite:* Definition 4.1. *Remedial work:* rewrite the inductive step, isolating where each of the two properties is used.
- 6) **B.** *Rationale:* Remark 4.8(iii), correction 3.16 of the expansion plan: Bryuno is optimal for the local holomorphic problem (Chapter 12), condition H for the global real one (circle diffeomorphisms, here). *Distractors:* A is false (the formula is correct); C is false (Yoccoz worked precisely on this, proving that the right threshold is H, not Bryuno); D is imprecise (Diophantine is a weaker condition, not the optimal threshold: $CD \subsetneq H$). *Diagnosed error:* applying an arithmetic threshold to the wrong problem (local vs. global). *Source:* K11, K12, K19 (Remark 4.8). *Prerequisite:* none beyond a careful reading of the remark. *Remedial work:* reread paragraph (iii), isolating the two thresholds and the two problems.
- 7) **B.** *Rationale:* the infobox after Remark 4.6, and the general structure of the tongues (Theorem 4.8): “locking stable under small variations” is precisely an interval of parameters (of positive width) with ρ constant. *Distractors:* A contradicts “stable”; C confuses the critical/chaotic regime with the regular locking one; D ignores that the bifurcation is only the *edge*, not the tongue itself. *Diagnosed error:* failing to recognise that “robustness under small perturbations” means, mathematically, “non-empty interior of the parameter set”. *Source:* Theorem 4.8, application infobox. *Prerequisite:* Theorem 4.7. *Remedial work:* reread the proof of Theorem 4.8, isolating where the openness of the interval is used.
- 8) **B.** *Rationale:* Remark 4.5, source verified by a real search on 16 September 2026 (Project Euclid, metadata confirmed: Świątek, *Comm. Math. Phys.* **119** (1988), 109–128). *Distractors:* A, C, D are real sources of the chapter but for different results (respectively: local

linearisation; global C^∞ theory on a full-measure set; global C^∞ theory for every Diophantine number). *Diagnosed error*: attributing a specific result (full measure *at criticality*) to an author associated with a neighbouring but distinct result (full measure *away from criticality*, which is instead the set of Diophantine numbers, Theorem 3.6 of Chapter 3). *Source*: Remark 4.5. *Prerequisite*: none beyond a careful reading of the sources. *Remedial work*: compare explicitly Świątek’s statement (criticality) with that of Theorem 3.6 (away from criticality), as in Assigned exercise 4.3(c).

BEST PRACTICE

Assigned exercises — answers.

4.1 (period-2 orbit). (a)–(b) From $\theta_0 = 0.2$, after 20 000 iterates: $\theta^* = 0.086338$, $F(\theta^*) = 0.558472$; $F^2(\theta^*) - \theta^* - 1 = 0$ to machine precision: $\rho(0.505; 0.4) = 1/2$. (c) At $\alpha = 1/2$ exactly, $\sin(2\pi \cdot 0) = \sin(2\pi \cdot \frac{1}{2}) = 0$, hence $F(0) = 1/2$, $F(1/2) = 1$: $0, 1/2$ are 2-periodic for *every* ε . At $\alpha = 0.505 \neq 1/2$, this coincidence disappears; the orbit found numerically is a genuinely different 2-periodic orbit, which exists because $\alpha = 0.505$ remains inside the $1/2$ tongue (of positive width around $1/2$, Theorem 4.8), not because of the symmetry of $\sin$. (d) Yes: the two points $\theta^* = 0.086338$ and $F(\theta^*) = 0.558472$ are distinct (difference ≈ 0.472 , non-zero), hence the period cannot be 1: being a divisor of 2 other than 1, the minimal period is 2 (Theorem 4.3(i)).

4.2 (check at $\varepsilon = 0.08$). (a)–(b) Numerical width (root-finding, grid of 6000 points, 45 bisections per edge): 0.00050902 ; predicted $\varepsilon^2/(4\pi) = 0.00050930$; ratio 0.999467 . (c) The ratio at $\varepsilon = 0.08$ (0.999467) is closer to 1 than the one at $\varepsilon = 0.2$ (0.99669 , from the text): consistent with a relative remainder $O(\varepsilon)$ (at roughly half the value of ε , one expects a discrepancy from 1 roughly halved, and indeed $1 - 0.999467 = 0.000533$ against $1 - 0.99669 = 0.00331$, a factor ≈ 6 for ε halved 2.5 times, a plausible order of magnitude for an $O(\varepsilon)$ effect with higher-order terms not negligible at $\varepsilon = 0.2$).

4.3 (finite grid and measure). (a) No: a finite grid of 2500 points can never prove an assertion about the Lebesgue measure of a set (which concerns a continuum of points); it can only be compatible or incompatible with the assertion within the resolution tested. (b) The numerical result shows that, within the resolution and the tolerance used, *every* sampled point of the grid turns out to be locked to a rational: a strong numerical indication, consistent with (and incapable of refuting) Świątek’s theorem, which in the chapter serves to corroborate the imported statement visually, not to prove it. (c) They are not in tension because they refer to different values of ε : Theorem 3.6 of Chapter 3 concerns the Diophantine numbers in the abstract (independently of ε) and implies, for $0 < \varepsilon < 1$ (non-critical regime), that the set of α with ρ irrational still has *positive* measure (containing all the Diophantine numbers for which the KAM theory of Chapter 6 gives a smooth conjugacy); Remark 4.5, on the other hand, concerns specifically $\varepsilon = 1$ (criticality), where the width of the tongues has grown enough to occupy, in the limit, the whole measure, leaving zero measure to the irrational complement.

4.14. Chapter formulary

Rotation number

$$\rho(f) = \lim_{n \rightarrow \infty} \frac{F^n(\theta) - \theta}{2\pi n} \quad (\text{independent of } \theta); \quad \rho(F^n) = n\rho(F); \quad \rho(HFH^{-1}) = \rho(F)$$

Fekete’s lemma

$$a_{m+n} \leq a_m + a_n \quad \forall m, n \quad \implies \quad \lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}$$

Rational criterion

$$\rho(F) \geq \frac{p}{q} \iff \exists \theta : F^q(\theta) \geq \theta + 2\pi p; \quad \rho(F) \neq \frac{p}{q} \Rightarrow \operatorname{sgn}(F^q(\theta) - \theta - 2\pi p) \text{ constant in } \theta$$

Poincaré classification

$$\rho(f) = \frac{p}{q} \text{ (reduced)} \iff \exists \text{ a periodic orbit, of exact period } q; \quad \rho(f) \notin \mathbb{Q} \Rightarrow \exists h : h \circ f = R_{2\pi\rho} \circ h$$

Arnold standard family ($\theta \in [0, 1)$)

$$f_{\alpha, \varepsilon}(\theta) = \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta); \quad \text{diffeomorphism for } 0 < \varepsilon < 1; \quad \rho = 0 \iff |\alpha| \leq \frac{\varepsilon}{2\pi} \text{ (exact)}$$

$$\text{width of tongue } 1/2 = \frac{\varepsilon^2}{4\pi} + O(\varepsilon^3); \quad \text{width of tongue } p/q = O(\varepsilon^q) \text{ (imported for } q \geq 3)$$

Devil's staircase

$$\alpha \mapsto \rho(\alpha, \varepsilon) : \text{ continuous, non-decreasing, constant on a dense union of intervals}$$

Panorama

$$\varepsilon = 1 : \operatorname{mis}\{\alpha : \rho(\alpha, 1) \notin \mathbb{Q}\} = 0 \text{ (Świątek 1988);} \quad \varepsilon > 1 : \text{rotation interval } [\rho_-, \rho_+] \text{ (Boyland 1986)}$$

Towards Chapter 6

$$\rho \text{ Diophantine, } \rho(f) = \rho, \text{ } f \text{ analytic, close to } R_{2\pi\rho} \Rightarrow h \text{ analytic (Arnold 1961)}$$

$$\text{CD} \subsetneq H \subsetneq B : \text{ } H \text{ optimal for the global problem (Yoccoz), } B \text{ optimal for the local holomorphic problem}$$

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CHAPTER 5

Denjoy's theorem and its counterexamples

Chapter 4 showed that, if $\rho(f)$ is irrational, every orientation-preserving homeomorphism $f : \mathbb{T} \rightarrow \mathbb{T}$ is *semiconjugate* to the rigid rotation $R_{\rho(f)}$ (Theorem 4.4): the semiconjugacy h may be a genuine conjugacy (if the orbits of f are dense), or else it may “crush” a countable set of intervals — the **wandering intervals** — onto a minimal invariant Cantor set. Arnaud Denjoy proved in 1932 that this second possibility, however compatible with mere continuity, *cannot occur* as soon as f is a diffeomorphism with derivative of bounded variation: the topological rigidity of Chapter 4 then becomes complete dynamical rigidity, f minimal and conjugate to $R_{\rho(f)}$. Denjoy also proved, in the same work, that the regularity hypothesis is *optimal*: he constructed a diffeomorphism of class C^1 (but not $C^{1+\text{bv}}$) with genuine wandering intervals. This chapter proves in full Denjoy's theorem and the distortion inequality that sustains it, proves a new inequality — the Denjoy–Koksma one, which controls the Birkhoff sums along the best return times — and reconstructs, correcting three defects of the previous draft, the C^1 counterexample in detail: the differentiability on the Cantor set, previously left hanging, is here proved step by step. The chapter closes with a panorama of what lies beyond: Hall's C^∞ counterexample, the ergodicity of the Denjoy conjugacies and the Katznelson–Ornstein theory on optimal regularity, and a forward reference to when the same phenomenon — distortion inequality against accumulation of irregularity — will reappear for group actions on the circle.

INFO

Level and prerequisites. Level **C** core, with the construction of the counterexample (Section 5.5) and the Denjoy–Koksma inequality (Section 5.4) of level **D**: a level **C** reader may read the statements and Construction 5.2 without reproducing the proof of the differentiability on the Cantor set (Step 4 of Theorem 5.4), without losing the continuity of the chapter. Prerequisites within the treatise: the rotation number and its invariance under (semi)conjugacy (Volume IV, Chapter 4, Definition 4.2, Proposition 4.2); the Poincaré classification in the two cases (Volume IV, Chapter 4, Theorems 4.3 and 4.4); the convergents of continued fractions and best approximation (Volume IV, Chapter 3, Definition 3.2, Theorems 3.3 and 3.4); total variation of a function (Volume II). Remedial work: a reader who does not remember the details of the Poincaré semiconjugacy may reread the proof of Theorem 4.4 before Section 5.1, which uses it in an essential way.

5.1. The lemma of the best return times

We fix, for the whole chapter, an orientation-preserving homeomorphism $f : \mathbb{T} \rightarrow \mathbb{T}$ with $\rho := \rho(f)$ **irrational**, a lift F of it, and the Poincaré semiconjugacy $h : \mathbb{T} \rightarrow \mathbb{T}$ of Theorem 4.4, monotone non-decreasing, surjective, of degree 1, with $h \circ f = R_{2\pi\rho} \circ h$. Let $(p_n/q_n)_{n \geq 0}$ be the convergents of ρ (Definition 3.2, Chapter 3) and

$$\delta_n := q_n \rho - p_n, \quad \text{segno}(\delta_n) = (-1)^n, \quad |\delta_n| \text{ strictly decreasing,}$$

the remainder with alternating sign of Theorem 3.3. We work in $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$; for $t \in \mathbb{R}$ we write $R_{2\pi\rho}(t) = t + 2\pi\rho$.

Definition 5.1: The intervals I_n and J_n of the rotation. For $n \geq 0$, let $I_n \subset \mathbb{T}$ be the arc with endpoints 0 and $2\pi\delta_n$, closed at 0 and open at the other endpoint (that is, $I_n = [0, 2\pi\delta_n)$ if $\delta_n > 0$, $I_n = (2\pi\delta_n, 0]$ if $\delta_n < 0$): $|I_n| = 2\pi|\delta_n|$. Let also $J_n \subset \mathbb{T}$ be the *open* arc with endpoints $-2\pi\delta_{n+1}$ and $2\pi\delta_n$ (Navas, *Groups of Circle Diffeomorphisms*, §2.2.1, p. 35, where J_n is defined as the open interval with endpoints $-q_{n+1}\rho$ and $q_n\rho$): the point $2\pi(\delta_n + \delta_{n+1})$ (corresponding to $(q_n + q_{n+1})\rho$) belongs to J_n .

Lemma 5.1: Lemma of the best return times for the rotation. For every $n \geq 0$, the arcs $R_{2\pi\rho}^j(I_n)$, $j = 0, 1, \dots, q_{n+1} - 1$, are pairwise disjoint.

Proof. Suppose $R_{2\pi\rho}^i(I_n) \cap R_{2\pi\rho}^j(I_n) \neq \emptyset$ for some $0 \leq i < j < q_{n+1}$; translating by $R_{2\pi\rho}^{-i}$ (a rotation, which preserves distances), $I_n \cap R_{2\pi\rho}^m(I_n) \neq \emptyset$ with $m := j - i$, $0 < m < q_{n+1}$. Since I_n and its translate $I_n + 2\pi m\rho$ have the same length $2\pi|\delta_n| \leq \pi$ (as $|\delta_n| \leq 1/2$: this follows from $|\delta_n| < 1/q_{n+1} \leq 1$ for n large, and is checked directly for the first indices), two congruent arcs of length $2\pi\ell$ on the circle, one a translate of the other by an angle $2\pi t$, have non-disjoint interiors if and only if $\|t\| < \ell$ (distance of t from the nearest integer, Definition 3.3): an elementary computation, comparing the arc $[0, 2\pi\ell)$ with $[2\pi t, 2\pi t + 2\pi\ell) \bmod 2\pi$. Here $t = m\rho$, $\ell = |\delta_n|$, hence the overlapping hypothesis gives $\|m\rho\| < |\delta_n| = \|q_n\rho\|$ with $0 < m < q_{n+1}$: impossible by Theorem 3.4 (which guarantees $\|m\rho\| \geq \|q_n\rho\|$ for every $0 < m < q_{n+1}$, with equality only for $m = q_n$). Contradiction: the arcs are pairwise disjoint. $\square$

Remark 5.1: Covering property of the $R_{2\pi\rho}^j(J_n)$ (imported). Beyond the disjointness of the $R_{2\pi\rho}^j(I_n)$ of Lemma 5.1, Navas (op. cit., p. 35) proves that the arcs $R_{2\pi\rho}^j(J_n)$, $j = 0, 1, \dots, q_{n+1} - 1$, **cover** $\mathbb{T}$, and every point of $\mathbb{T}$ belongs to at most 2 of them (not reproduced here: the proof uses the same argument as Lemma 5.1 applied to $-\rho$ in place of ρ , combining the two families of intervals I_n, J_n and their mirrored images). This is the property used in the chapter's Formulario (Section 5.12).

Remark 5.2: The identity $|\delta_{n-1}|q_n + |\delta_n|q_{n-1} = 1$. From Proposition 3.1(iii), $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$; expanding, $\delta_{n-1} q_n - \delta_n q_{n-1} = (q_{n-1}\rho - p_{n-1})q_n - (q_n\rho - p_n)q_{n-1} = p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$. Since $\text{sgno}(\delta_{n-1}) = (-1)^{n-1}$ and $\text{sgno}(\delta_n) = (-1)^n = -(-1)^{n-1}$, one has $\delta_{n-1} q_n - \delta_n q_{n-1} = (-1)^{n-1}(|\delta_{n-1}|q_n + |\delta_n|q_{n-1})$, hence $|\delta_{n-1}|q_n + |\delta_n|q_{n-1} = 1$: an exact identity, not merely an estimate, also checked numerically in block [C5-1] of the script `verifica_cap05_vol4.py` for $\rho = 1/\varphi$ (used later on in Construction 5.2).

Lemma 5.1 is **purely combinatorial**: it concerns only the cyclic order of the points $\{j \cdot 2\pi\rho \bmod 2\pi\}$, not the metric. Since h is monotone and intertwines f with $R_{2\pi\rho}$, the same statement holds for f , provided one transports the arcs by means of h .

Lemma 5.2: Lemma of the best return times for f (two cases according to the parity of n). For $x \in \mathbb{T}$ and $n \geq 0$, let $I_n(x) := h^{-1}(I_n + h(x))$ (the preimage under h of the arc of Definition 5.1, recentred at $h(x)$). Then: **(i)** the arcs $f^j(I_n(x))$, $j = 0, \dots, q_{n+1} - 1$, are pairwise disjoint (apart from possible common endpoints); **(ii) (the two parity cases).** If n is even, $I_n(x)$ has x as its left endpoint (in the positive direction) and $f^{q_n}(x)$ (or a representative of it in the preimage, if h collapses an interval) as its right endpoint; if n

is odd, $I_n(x)$ has x as its right endpoint and $f^{-q_n}(x)$ as its left endpoint — that is, the direction in which $I_n(x)$ “points” with respect to x is reversed at each step, exactly like the sign of δ_n .

Proof. $h \circ f = R_{2\pi\rho} \circ h$ gives $h \circ f^j = R_{2\pi\rho}^j \circ h$ for every $j \in \mathbb{Z}$ (by induction, using that f is invertible, being a homeomorphism). Hence $h(f^j(I_n(x))) = R_{2\pi\rho}^j(h(I_n(x))) = R_{2\pi\rho}^j(I_n + h(x))$ for every j : the image under h of the family $\{f^j(I_n(x))\}_j$ is exactly the family $\{R_{2\pi\rho}^j(I_n + h(x))\}_j$ of Lemma 5.1 (translated by $h(x)$, which does not alter the disjointness, $R_{2\pi\rho}$ being an isometry). Since h is monotone and surjective, it preserves the strict cyclic order: if $R_{2\pi\rho}^i(I_n + h(x)) \cap R_{2\pi\rho}^j(I_n + h(x)) = \emptyset$ (disjoint interiors) by Lemma 5.1, then the interiors of the h^{-1} of these two sets are disjoint (the inverse image under a monotone map of two sets with disjoint interiors has disjoint interiors, since h never reverses the cyclic order: if a point y were in the interior of both preimages, $h(y)$ would lie in the interior of both original sets, absurd). This proves (i).

(ii) It is the translation of Definition 5.1: I_n has 0 as closed endpoint and $2\pi\delta_n$ as open endpoint; for n even $\delta_n > 0$, hence $I_n = [0, 2\pi\delta_n)$ “points forwards”; since $h(f^{q_n}(x)) = R_{2\pi\rho}^{q_n}(h(x)) = h(x) + 2\pi q_n \rho \equiv h(x) + 2\pi\delta_n \pmod{2\pi}$ (the last congruence because $2\pi p_n$ is an integer multiple of 2π), that is, $h(f^{q_n}(x)) - h(x) \equiv 2\pi\delta_n \pmod{2\pi}$: the open endpoint of $I_n(x)$ corresponds exactly to $f^{q_n}(x)$ (up to the ambiguity of h^{-1} on a possible collapsed interval, irrelevant for the order). For n odd, $\delta_n < 0$ and $I_n = (2\pi\delta_n, 0]$ points backwards; here it is convenient to compute $h(f^{-q_n}(x)) - h(x) = R_{2\pi\rho}^{-q_n}(h(x)) - h(x) \equiv -2\pi q_n \rho \equiv -2\pi\delta_n = 2\pi|\delta_n| \pmod{2\pi}$ (using $\delta_n < 0$): the open endpoint of $I_n(x)$ (at angular distance $2\pi|\delta_n|$ from x , in the positive direction) corresponds exactly to $f^{-q_n}(x)$, not to $f^{q_n}(x)$. This is precisely the exchange of roles between f^{q_n} and f^{-q_n} stated in (ii). $\square$

Remark 5.3: Where the old proof presupposed n even. A previous draft of this lemma defined the arc J (in the notation used there) as the arc between $f^{-q}(x)$ and x , with $f^{-q}(x)$ always to the left of x : this is true for n even (Lemma 5.2(ii), $\delta_n > 0$, hence $f^{-q_n}(x)$ falls back with respect to x) but false for n odd, where $\delta_n < 0$ and $f^{-q_n}(x)$ is instead beyond x in the positive direction. The correction is precisely (ii): the direction of $I_n(x)$ with respect to x is reversed at each n , in phase with $\text{sgn}(\delta_n) = (-1)^n$; the Denjoy inequality (Section 5.2) and Denjoy’s theorem (Section 5.3) remain valid in both cases because their argument uses only the disjointness (i) and the variation estimate, both indifferent to the direction.

5.2. The Denjoy inequality

Definition 5.2: Variation of $\log f'$. If $f : \mathbb{T} \rightarrow \mathbb{T}$ is a C^1 diffeomorphism with $f' > 0$ everywhere, we set $V(f) := \text{Var}_{\mathbb{T}}(\log f')$, the supremum of $\sum_{i=0}^{k-1} |\log f'(a_{i+1}) - \log f'(a_i)|$ over all finite cyclic partitions $a_0 < a_1 < \dots < a_k = a_0 + 2\pi$ of $\mathbb{T}$. We say that f is of class $C^{1+\text{bv}}$ if $f \in C^1$, $f' > 0$ everywhere **and** $V(f) < \infty$: the **two** hypotheses (bounded variation of f' and positivity of f') are both necessary and must be stated separately, since $V(f)$ is defined by means of $\log f'$, which requires $f' > 0$ in order to make sense.

Theorem 5.1: Denjoy inequality. Let $f \in C_+^{1+\text{bv}}(\mathbb{T})$ (Definition 5.2) with $\rho(f)$ irrational. For every $x \in \mathbb{T}$ and every $n \geq 0$,

$$(f^{q_n})'(x) \cdot (f^{-q_n})'(x) \geq e^{-V(f)}.$$

Proof. Put $y := f^{-q_n}(x)$. By the rule for differentiating an inverse function applied to $f^{-q_n} = (f^{q_n})^{-1}$ at the point $x = f^{q_n}(y)$, $(f^{-q_n})'(x) = 1/(f^{q_n})'(y)$, hence, by the chain rule on each of the two factors,

$$\log(f^{q_n})'(x) = \sum_{k=0}^{q_n-1} \log f'(f^k(x)), \quad \log(f^{-q_n})'(x) = -\log(f^{q_n})'(y) = -\sum_{k=0}^{q_n-1} \log f'(f^k(y)),$$

a purely algebraic identity, valid for every n (no parity involved so far). Subtracting,

$$\log [(f^{q_n})'(x)(f^{-q_n})'(x)] = \sum_{k=0}^{q_n-1} [\log f'(f^k(x)) - \log f'(f^k(y))]. \quad (*)$$

It remains to bound each term by means of the variation of $\log f'$ on an arc containing both $f^k(x)$ and $f^k(y)$: here parity enters. If n is **odd**, Lemma 5.2(ii) says that $J := I_n(x)$ has x and $y = f^{-q_n}(x)$ as its two endpoints; if n is **even**, the same Lemma applied at the point y (instead of x) says that $I_n(y)$ has y and $f^{q_n}(y) = f^{q_n}(f^{-q_n}(x)) = x$ as its two endpoints: we then set $J := I_n(y)$. In both cases J is an arc with $\{x, y\}$ as its endpoints, hence $f^k(x)$ and $f^k(y)$ are the two endpoints of the arc $f^k(J)$ for every k . By Lemma 5.2(i) (applied at the point x , resp. y), the arcs $f^k(J)$, $k = 0, \dots, q_{n+1} - 1$, are pairwise disjoint; in particular so are the first q_n of them, $k = 0, \dots, q_n - 1$ (a subset of a disjoint family).

Since $f^k(x)$ and $f^k(y)$ both belong to the closure of $f^k(J)$,

$$|\log f'(f^k(x)) - \log f'(f^k(y))| \leq \text{Var}_{\overline{f^k(J)}}(\log f').$$

Summing over $k = 0, \dots, q_n - 1$ in $(*)$ and using the disjointness of the $f^k(J)$ (subadditivity of the variation on disjoint arcs, the sum of whose variations is bounded by the total variation on $\mathbb{T}$):

$$\left| \log [(f^{q_n})'(x)(f^{-q_n})'(x)] \right| \leq \sum_{k=0}^{q_n-1} \text{Var}_{\overline{f^k(J)}}(\log f') \leq V(f).$$

Exponentiating, $(f^{q_n})'(x)(f^{-q_n})'(x) \geq e^{-V(f)}$. □

Example 5.1: Numerical check of the Denjoy inequality. Double check (script `verifica_cap05_vol4.py`, block [C5-2], run on 17 September 2026). In order to avoid having to invert a map numerically, one chooses $f := h \circ R_\alpha \circ h^{-1}$ with $h(s) = s + \varepsilon \sin(s)$, $\varepsilon = 0.4$ (C^∞ diffeomorphism of the circle, since $h'(s) = 1 + \varepsilon \cos(s) > 0$), $\alpha = 2\pi/\varphi$: by Proposition 4.2(i), $\rho(f) = \rho(R_\alpha) = 1/\varphi$ **exactly**, with no need to estimate it by iteration. Working in the coordinate $s = h^{-1}(\theta)$, the telescoping chain rule gives, for every $q \geq 1$,

$$(f^q)'(h(s_0)) \cdot (f^{-q})'(h(s_0)) = \frac{h'(s_0 + q\alpha) h'(s_0 - q\alpha)}{h'(s_0)^2},$$

a closed formula (no iteration needed: $q\alpha \equiv 2\pi\delta_n \pmod{2\pi}$ for $q = q_n$). The variation $V(f)$ is, by the invariance of the total variation under the monotone change of coordinates $s = h^{-1}(\theta)$, equal to $\text{Var}_s[\log h'(s + \alpha) - \log h'(s)]$, computed on a grid of 4×10^6 points: $V = 3.170527$, against the loose upper bound by subadditivity $2 \text{Var}(\log h') = 3.389191$ (confirmed $V \leq$ upper bound $\checkmark$). With $s_0 = 0.7182818\dots$ and $q = q_n$ for $n = 0, \dots, 9$, the ratio between $(f^{q_n})'(x)(f^{-q_n})'(x)$ and the threshold $e^{-V} = 0.041981$ is always ≥ 1 (minimum observed 8.07, at $q_0 = q_1 = 1$; the product $(f^{q_n})'(x)(f^{-q_n})'(x)$ itself tends to 1 for n large, consistently with $\delta_n \rightarrow 0$, so the ratio tends to $1/e^{-V} = e^V \approx 23.8$): the inequality is verified on all ten convergents tested. $\checkmark$

5.3. Denjoy's theorem

Theorem 5.2: Denjoy's theorem (1932). Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving diffeomorphism such that: **(a)** $f \in C^1$ and $f' > 0$ everywhere; **(b)** $V(f) = \text{Var}_{\mathbb{T}}(\log f') < \infty$ (Definition 5.2: $f \in C_+^{1+\text{bv}}(\mathbb{T})$); **(c)** $\rho(f)$ is irrational. Then f is **minimal** (every orbit is dense in $\mathbb{T}$) and **topologically conjugate** to $R_{\rho(f)}$: the semiconjugacy h of Theorem 4.4 is a homeomorphism.

Proof. By the Poincaré classification in the irrational case (Theorem 4.4), the only two possible scenarios are: f minimal (and then h is already a homeomorphism, and we are done), or else there exists a Cantor set $K \subsetneq \mathbb{T}$, minimal and invariant under f , such that the connected components of $\mathbb{T} \setminus K$ are permuted by f and h collapses them to points. We rule out this second scenario by contradiction.

Let I be a connected component of $\mathbb{T} \setminus K$ (a **wandering interval**). If we had $f^m(I) = I$ for some integer $m \geq 1$, the endpoints of I (which belong to $K \subset \mathbb{T}$, K being closed and I a component of its complement) would give a periodic orbit of f of period (a divisor of) m ; but Theorem 4.3 rules out periodic orbits when $\rho(f)$ is irrational. Hence the intervals $\{f^m(I)\}_{m \in \mathbb{Z}}$ are pairwise distinct, and distinct connected components of the same open set $\mathbb{T} \setminus K$ are automatically disjoint: $\sum_{m \in \mathbb{Z}} |f^m(I)| \leq 2\pi$ (total measure of the circle). In particular $|f^{\pm q_n}(I)| \rightarrow 0$ as $n \rightarrow \infty$, since $q_n \rightarrow \infty$ is a subsequence of distinct integers and the series $\sum_m |f^m(I)|$ converges (general term infinitesimal).

On the other hand, for every $x \in I$, the arithmetic–geometric inequality applied to the Denjoy inequality (Theorem 5.1) gives

$$(f^{q_n})'(x) + (f^{-q_n})'(x) \geq 2\sqrt{(f^{q_n})'(x)(f^{-q_n})'(x)} \geq 2e^{-V(f)/2}.$$

Integrating this pointwise inequality over $x \in I$ (the left-hand side is the sum of two positive functions, and $\int_I f^{q_n'} d\theta = |f^{q_n}(I)|$, $\int_I f^{-q_n'} d\theta = |f^{-q_n}(I)|$, by the fundamental theorem of calculus applied to the diffeomorphism f^{q_n} restricted to I):

$$|f^{q_n}(I)| + |f^{-q_n}(I)| \geq 2e^{-V(f)/2} |I|,$$

a quantity **strictly positive and independent of n** (it depends only on $|I| > 0$ and on $V(f)$, both fixed). This contradicts $|f^{\pm q_n}(I)| \rightarrow 0$. The contradiction shows that there are no wandering intervals: $\mathbb{T} \setminus K = \emptyset$, that is, $K = \mathbb{T}$, and f is minimal. By Theorem 4.4, dense orbits imply that h is injective, hence a homeomorphism: f is topologically conjugate to $R_{\rho(f)}$. $\square$

Remark 5.4: Anatomy of the proof. Every hypothesis works at a precise point, and locating it prepares one for reading the counterexample (Section 5.5): the irrationality of $\rho(f)$ supplies the best return times q_n (Chapter 3) and forbids periodic orbits (used twice: in Lemma 5.2 and to rule out $f^m(I) = I$); the bounded variation of $\log f'$ enters only in the Denjoy inequality, and is exactly what the C^1 counterexample of Section 5.5 violates (while keeping $f' > 0$); the combinatorics of the disjoint arcs is pure rotation theory (best approximation, Theorem 3.4). It is one of the finest examples of a theorem in which arithmetic (continued fractions), analysis (total variation) and topology (cyclic ordering, Poincaré classification) each collaborate in a distinct and indispensable step.

5.4. The Denjoy–Koksma inequality

The Denjoy–Koksma inequality controls, for functions of bounded variation (not necessarily continuous), the Birkhoff sums along exactly q iterates, when p/q approximates $\rho(f)$ well enough. It serves, in this treatise, as a tool for Chapter 6 (homological equation, Arnold’s theorem) and for Chapter 9 (unique ergodicity); here we prove it in the generality that follows directly from the tools already available.

Construction 5.1: The invariant measure induced by the semiconjugacy. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving homeomorphism with $\rho(f)$ irrational, h the Poincaré semiconjugacy (Theorem 4.4), continuous, monotone non-decreasing and surjective. We define μ on $\mathbb{T}$ by $\mu(A) := \frac{1}{2\pi} \text{Leb}(h(A))$ for every Borel set $A \subseteq \mathbb{T}$ (the image $h(A)$ is measurable because h is monotone, hence it sends Borel sets to sets whose Lebesgue measure is well defined through the direct image of intervals). Then μ is a Borel probability measure on $\mathbb{T}$, and it is **f -invariant**: $\mu(f^{-1}A) = \mu(A)$ for every A . Indeed, from $h \circ f = R_{2\pi\rho} \circ h$ it follows that $h(f^{-1}(A)) = R_{2\pi\rho}^{-1}(h(A))$ (using that h is surjective, hence $h(f^{-1}(A)) \supseteq R_{2\pi\rho}^{-1}(h(A))$) and conversely by the intertwining), and $R_{2\pi\rho}^{-1}$, being a rotation, preserves Lebesgue measure: $\text{Leb}(h(f^{-1}A)) = \text{Leb}(R_{2\pi\rho}^{-1}(h(A))) = \text{Leb}(h(A))$. $\square$

Theorem 5.3: Denjoy–Koksma inequality. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving homeomorphism with $\rho = \rho(f)$ irrational, μ the invariant measure of Construction 5.1. Let p, q be coprime integers, $q \geq 1$, with

$$\left| \rho - \frac{p}{q} \right| \leq \frac{1}{q^2}$$

(in particular $q = q_n$, by Theorem 3.3: $|\rho - p_n/q_n| < 1/(q_n q_{n+1}) < 1/q_n^2$). Then for every $\psi : \mathbb{T} \rightarrow \mathbb{R}$ of finite total variation $\text{Var}(\psi)$ (not necessarily continuous) and every $x \in \mathbb{T}$,

$$\left| \sum_{j=0}^{q-1} \psi(f^j(x)) - q \int_{\mathbb{T}} \psi d\mu \right| \leq \text{Var}(\psi).$$

Proof. Step 1 (a uniform partition of $\mathbb{T}$). Since $\gcd(p, q) = 1$, the map $j \mapsto jp \bmod q$ is a bijection of $\{0, 1, \dots, q-1\}$ onto itself. For every $j = 0, \dots, q-1$, with $\rho - p/q =: \eta$, $|\eta| \leq 1/q^2$ (WLOG $\eta \geq 0$; the case $\eta < 0$ is identical, with the roles interchanged):

$$0 \leq j\rho - \frac{jp}{q} = j\eta \leq \frac{q-1}{q} \cdot \frac{1}{q} < \frac{1}{q},$$

(for $j = 0$ the expression is trivially 0) hence $j\rho \bmod 1$ belongs to the interval $[\frac{jp \bmod q}{q}, \frac{jp \bmod q}{q} + \frac{1}{q})$: the interval $\bar{I}_i := [\frac{i}{q}, \frac{i+1}{q}) \subset [0, 1)$ (identified with the arc $[\frac{2\pi i}{q}, \frac{2\pi(i+1)}{q}) \subset \mathbb{T}$) contains **exactly one** of the points $\{j\rho \bmod 1 : j = 0, \dots, q-1\}$, precisely the one with $i = jp \bmod q$. The arcs $\bar{I}_0, \dots, \bar{I}_{q-1}$ partition $\mathbb{T}$ (up to the endpoints, of measure zero) into q arcs of equal length $2\pi/q$.

Step 2 (transport via the semiconjugacy). Let $\varphi := h - h(x) \pmod{2\pi}$ (still monotone, surjective, of degree 1, with $\varphi(x) = 0$ and $\varphi \circ f = R_{2\pi\rho} \circ \varphi$, $R_{2\pi\rho}$ being a translation). For $j = 0, \dots, q-1$, $\varphi(f^j(x)) = R_{2\pi\rho}^j(\varphi(x)) = 2\pi jp \bmod 2\pi$: by Step 1, for each j there exists a unique $i = i(j) \in \{0, \dots, q-1\}$ with $\varphi(f^j(x)) \in \bar{I}_i$ (rescaled to $\mathbb{T} = [0, 2\pi)$; for $j = 0$, $i(0) = 0$ since $\varphi(x) = 0 \in \bar{I}_0$). We set $J_j := \varphi^{-1}(\bar{I}_{i(j)})$ (the preimage under φ , which coincides with h up to the translation, of the $i(j)$ -th uniform arc): the sets $J_0, \dots, J_{q-1}$ are pairwise disjoint (because the $\bar{I}_i$ are, $i \mapsto i(j)$ being a bijection on the indices $j = 0, \dots, q-1$, and φ being a function, not a relation) and their union is $\mathbb{T}$ up to a μ -negligible set (the preimages of the

endpoints of the $\bar{I}_i$, a countable set). By Construction 5.1 (with φ in place of h , the measure being the same since φ is a translate of h), $\mu(J_j) = \text{Leb}(\bar{I}_{i(j)})/2\pi = 1/q$ for every j .

Step 3 (the inequality). Since $f^j(x) \in J_j$ (by construction, $\varphi(f^j(x)) \in \bar{I}_{i(j)}$, that is, $f^j(x) \in \varphi^{-1}(\bar{I}_{i(j)}) = J_j$):

$$\sum_{j=0}^{q-1} \psi(f^j(x)) - q \int_{\mathbb{T}} \psi d\mu = \sum_{j=0}^{q-1} \psi(f^j(x)) - \sum_{j=0}^{q-1} q \int_{J_j} \psi d\mu = \sum_{j=0}^{q-1} \left[\psi(f^j(x)) - \frac{1}{\mu(J_j)} \int_{J_j} \psi d\mu \right],$$

using $\sum_j \int_{J_j} \psi d\mu = \int_{\mathbb{T}} \psi d\mu$ (a partition up to a negligible set) and $q = 1/\mu(J_j)$ for every j . Each term is bounded in modulus by the oscillation of ψ on J_j (since $f^j(x) \in J_j$ and the average of ψ on J_j with respect to μ lies between $\inf_{J_j} \psi$ and $\sup_{J_j} \psi$):

$$\left| \psi(f^j(x)) - \frac{1}{\mu(J_j)} \int_{J_j} \psi d\mu \right| \leq \sup_{J_j} \psi - \inf_{J_j} \psi \leq \text{Var}_{J_j}(\psi).$$

Summing over $j = 0, \dots, q-1$ and using the disjointness of the J_j (subadditivity of the total variation on disjoint sets covering $\mathbb{T}$ up to a negligible set):

$$\left| \sum_{j=0}^{q-1} \psi(f^j(x)) - q \int_{\mathbb{T}} \psi d\mu \right| \leq \sum_{j=0}^{q-1} \text{Var}_{J_j}(\psi) \leq \text{Var}_{\mathbb{T}}(\psi),$$

which is exactly the assertion. $\square$

Example 5.2: Numerical check of Denjoy–Koksma for the sawtooth. Double check (script `verifica_cap05_vol4.py`, block [C5-4]). For $f = R_\alpha$, $\rho = 1/\varphi$, and $\psi(\theta) = \{\theta/2\pi\}$ (sawtooth, one jump of amplitude 1 per period: $\text{Var}(\psi) = 1$, $\int_{\mathbb{T}} \psi d\mu = 1/2$ by symmetry), one computed $\sum_{j=0}^{q_n-1} \psi(x_0 + j\alpha) - q_n/2$ for $n = 1, \dots, 9$ and four base points $x_0 \in \{0; 0.31415926535; 2; 5.5\}$: the absolute value always remains $\leq 1 = \text{Var}(\psi)$ over all the 36 cases tested, with the maximum observed equal to 0.999975 (at $q_9 = 55$), consistent with the asymptotic optimality of the constant (no violation, but the margin narrows as q_n grows, as expected from a sharp inequality). $\checkmark$

5.5. Denjoy's C^1 counterexample

The hypothesis $C^{1+\text{bv}}$ of Theorem 5.2 admits no weakening: for every $\rho \notin \mathbb{Q}$ there exists a diffeomorphism $f \in C^1(\mathbb{T})$, with $f' > 0$ everywhere, $\rho(f) = \rho$, which is **not** conjugate to $R_{2\pi\rho}$. We construct f explicitly, correcting three defects of a previous draft: (a) the positivity of f' does not depend on the “smallness” of a scale parameter c , but only on the *ratios* between consecutive lengths, which are independent of c ; (b) the invariant Cantor set K must be constructed as the complement of a union of *open* intervals, in order to be genuinely closed; (c) $\rho(f) = \rho$ must be proved, not asserted, by means of the invariance of the rotation number under monotone semiconjugacy.

Construction 5.2: Denjoy's C^1 counterexample. We fix $\rho \in (0, 1)$ irrational and $c \in (0, c_{\max})$, where

$$c_{\max} := \frac{2\pi}{S}, \quad S := \sum_{n \in \mathbb{Z}} \frac{1}{(|n| + 2)^2} = \frac{\pi^2}{3} - \frac{9}{4}$$

(the closed form of S follows from the Basel problem $\sum_{m \geq 1} 1/m^2 = \pi^2/6$, Volume II: $S = \frac{1}{4} + 2 \sum_{m \geq 3} \frac{1}{m^2} = \frac{1}{4} + 2(\frac{\pi^2}{6} - 1 - \frac{1}{4}) = \frac{\pi^2}{3} - \frac{9}{4}$, also checked by direct summation truncated at 5×10^6 terms per side in the script `fig05_misura_K.py`, discrepancy 4.0×10^{-7} , of the expected order $2/N = 4 \times 10^{-7}$ from the factor of two for the two sides ✓).

Step 1 (the base). Let $\tilde{\mathbb{T}}$ be the circle of circumference $L := 2\pi - cS > 0$ (positive for $c < c_{\max}$), endowed with the rotation $\tilde{R}$ of angle $\rho \cdot L$ (that is, rescaling $\tilde{\mathbb{T}}$ to $[0, L)$, $\tilde{R}(t) = t + \rho L \bmod L$): the rotation $\tilde{R}$ has the same rotation number ρ as $R_{2\pi\rho}$ on $\mathbb{T}$ (an affine rescaling of one circle into another carries a rotation into a rotation with the same rotation number, the latter being a dimensionless ratio). We fix $x_0 \in \tilde{\mathbb{T}}$ and set $x_n := \tilde{R}^n(x_0)$, $n \in \mathbb{Z}$ (a dense and injective orbit, Theorem 3.1).

Step 2 (the insertion). For $n \in \mathbb{Z}$ let $\ell_n := c/(|n| + 2)^2$ (hence $\sum_n \ell_n = cS$). One constructs $\mathbb{T}'$ by inserting, at each $x_n \in \tilde{\mathbb{T}}$, an **open** interval I_n of length ℓ_n , respecting the cyclic order of the orbit: formally, $\mathbb{T}'$ is the circle of circumference $L + \sum_n \ell_n = 2\pi$ obtained from $\tilde{\mathbb{T}} \sqcup \bigsqcup_n I_n$ by gluing the right endpoint of I_n (in a fixed orientation) to the point x_n on one side and the left endpoint to the immediately following point of $\tilde{\mathbb{T}} \setminus \{x_m\}_m$ on the other — equivalently, $\mathbb{T}'$ is obtained by “blowing up” each point x_n of $\tilde{\mathbb{T}}$ into an interval of length ℓ_n . We set

$K := \tilde{\mathbb{T}} \setminus \{x_n : n \in \mathbb{Z}\} \subset \mathbb{T}'$ (the points of $\tilde{\mathbb{T}}$ different from every x_n , included in $\mathbb{T}'$ without being blown

Since $\bigcup_n I_n$ is **open** in $\mathbb{T}'$ (a union of open sets) and $K = \mathbb{T}' \setminus \bigcup_n I_n$, K is **closed** in $\mathbb{T}'$ by construction (correction of defect (b)): there is no need to check it a posteriori. K has no isolated points (every point of K is a limit of other points of K , $\tilde{\mathbb{T}} \setminus \{x_n\}$ being dense in itself, the complement of a countable set in a continuum) and has empty interior (since $\{x_n\}$ is dense in $\tilde{\mathbb{T}}$, every arc of $\mathbb{T}'$ contains points of some I_n): K is a Cantor set, of measure $\text{Leb}(K) = L = 2\pi - cS > 0$.

Step 3 (the map on K). On K , f is defined by $f(z) :=$ the point of K corresponding to $\tilde{R}(\pi(z))$, where $\pi : K \rightarrow \tilde{\mathbb{T}} \setminus \{x_n\}$ is the natural identification (bijective, by construction): that is, $f|_K$ transports $\tilde{R}$ by means of π . In particular f sends the endpoint (in K) of I_n adjacent to x_n to the endpoint of I_{n+1} adjacent to x_{n+1} , consistently with Step 4.

Step 4 (the map on each I_n). Parametrising I_n by $t \in [0, \ell_n]$ (endpoints included as limit points in K), one sets $f|_{I_n}(t) := f_n(t)$ with

$$f_n(t) := \int_0^t \left[1 + 6 \left(\frac{\ell_{n+1}}{\ell_n} - 1 \right) \frac{s}{\ell_n} \left(1 - \frac{s}{\ell_n} \right) \right] ds, \quad f'_n(t) = 1 + 6(r_n - 1)u(1-u), \quad u := \frac{t}{\ell_n}, \quad r_n := \frac{\ell_{n+1}}{\ell_n}.$$

One checks $f'_n(0) = f'_n(\ell_n) = 1$ (C^1 matching with $f' = 1$ on K) and $f_n(\ell_n) = \ell_n + (\ell_{n+1} - \ell_n) \cdot 6 \int_0^1 u(1-u) du = \ell_n + (\ell_{n+1} - \ell_n) = \ell_{n+1}$ (direct computation: $6 \int_0^1 u(1-u) du = 6 \cdot \frac{1}{6} = 1$): f_n sends I_n diffeomorphically onto I_{n+1} , consistently with Step 3. $\square$

Theorem 5.4: Properties of the counterexample. For every $\rho \in (0, 1) \setminus \mathbb{Q}$ and every $c \in (0, c_{\max})$, the map $f : \mathbb{T}' \rightarrow \mathbb{T}'$ of Construction 5.2 (with $\mathbb{T}'$ identified with $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, having exact circumference 2π) is an orientation-preserving C^1 diffeomorphism, with: (1) $f'(z) > 0$ for every $z \in \mathbb{T}'$, **for every** $c \in (0, c_{\max})$ (not only for c small); (2) $\rho(f) = \rho$; (3) K is the minimal invariant set of f , and f is not minimal ($\mathbb{T}' \setminus K \neq \emptyset$): f is only semiconjugate, not conjugate, to $R_{2\pi\rho}$; (4) $\text{Var}_{\mathbb{T}'}(\log f') = \infty$: the hypothesis of Theorem 5.2 is violated, as it must be.

Proof. (1) Positivity of f' , independent of c . On K , $f' = 1 > 0$. On I_n , $f'_n(t) = 1 + 6(r_n - 1)u(1-u)$ with $u(1-u) \in [0, 1/4]$: the minimum of f'_n (if $r_n < 1$) is $1 + 6(r_n - 1)/4 = 1 + 1.5(r_n - 1)$, attained at $u = 1/2$. Now $r_n := \ell_{n+1}/\ell_n = (|n| + 2)^2/(|n + 1| + 2)^2$

does not depend on c (the scale factor c cancels in the ratio): this corrects the assertion “ $f' > 0$ for c small” of a previous draft, which confused the scale parameter (irrelevant for the positivity) with the ratios (the only relevant quantities). One computes directly: for $n \geq 0$, $r_n = (n+2)^2/(n+3)^2 \in (0,1)$, with minimum at $n=0$: $r_0 = 4/9$; for $n \leq -1$, $r_n = (|n|+2)^2/(|n|+1+2)^2 = (-n+2)^2/(-n+1)^2 > 1$ (no constraint from this side, since $r_n > 1$ gives $f'_n \geq 1$). Hence $\inf_{n \in \mathbb{Z}} r_n = 4/9$, and

$$\inf_{n \in \mathbb{Z}} \min_{t \in [0, \ell_n]} f'_n(t) = 1 + 1.5 \left(\frac{4}{9} - 1 \right) = 1 - 1.5 \cdot \frac{5}{9} = 1 - \frac{5}{6} = \frac{1}{6} > 0,$$

also checked numerically over $|n| \leq 2000$ in the script `verifica_cap05_vol4.py`, block [C5-3] (minimum observed $0.166667 = 1/6$, consistent with the closed-form computation $\checkmark$). Since $4/9 > 1/3$ (the critical threshold below which $1 + 1.5(r - 1)$ would become negative), $f'_n > 0$ on every I_n , for every $c > 0$: (1) is proved.

(2) $\rho(f) = \rho$. The map $\pi : \mathbb{T}' \rightarrow \tilde{\mathbb{T}}$ which collapses every I_n to the point x_n and is the identity on K is continuous (by construction, the lengths $\ell_n \rightarrow 0$ as $|n| \rightarrow \infty$ guarantee that the collapse introduces no discontinuity: a standard argument on the continuity of a monotone surjective map between compact circles, analogous to Step 2 of the proof of Theorem 4.4), monotone non-decreasing (it respects the cyclic order by construction), surjective, of degree 1, and satisfies $\pi \circ f = \tilde{R} \circ \pi$ (by Step 3 of Construction 5.2 on K , and because $f(I_n) = I_{n+1}$ collapses consistently to $\tilde{R}(x_n) = x_{n+1}$ on the I_n). Hence π is a monotone semiconjugacy between f and $\tilde{R}$: by Proposition 4.2(ii) (invariance of the rotation number under monotone surjective semiconjugacy), $\rho(f) = \rho(\tilde{R}) = \rho$. This proves (2) **in full**, correcting the unproved assertion of a previous draft.

(3) **K is the minimal set, f is not minimal.** $K \neq \mathbb{T}'$ (every I_n is non-empty, since $\ell_n > 0$), hence f is not minimal. The I_n are pairwise disjoint and $f^m(I_n) = I_{n+m}$ for every $m \in \mathbb{Z}$ (by construction, Step 4): they are genuine **wandering intervals** (no $f^m(I_0)$ coincides with I_0 for $m \neq 0$, the I_n being all distinct). K is closed (Step 2), f -invariant ($f(K) = K$, by Step 3), and minimal: every orbit in K is dense in K , because $\pi|_K$ is a bijection with $\tilde{\mathbb{T}} \setminus \{x_n\}$ which intertwines $f|_K$ with $\tilde{R}$, and the orbits of $\tilde{R}$ are dense in $\tilde{\mathbb{T}}$ (Theorem 3.1), hence their preimages (the corresponding points of K) are dense in K (by the continuity and surjectivity of π restricted to K).

(4) **Differentiability on the Cantor set K and divergence of $\text{Var}(\log f')$.** We first show that f is differentiable at every $z \in K$, with $f'(z) = 1$, and that this derivative is the continuous limit of f' from $\mathbb{T}' \setminus K$: that is, $f \in C^1(\mathbb{T}')$ in full (not only on the I_n).

Continuity of f' on K . Let $z \in K$ be fixed. Since $z \notin I_n$ for every n (by definition of K) and every I_n is a fixed set at positive distance from z (since $z \neq x_n$ for every n , and I_n is a neighbourhood of x_n of radius $\ell_n/2$), for every N there exists $\eta_N > 0$ such that none of the intervals I_n with $|n| \leq N$ meets $(z - \eta_N, z + \eta_N)$ (a minimum, over a finite set of indices, of positive distances). Hence, if $t \rightarrow z$ with $t \in I_{n(t)}$, necessarily $|n(t)| \rightarrow \infty$. Since $\sup_{s \in [0, \ell_n]} |f'_n(s) - 1| = 1.5|r_n - 1| \rightarrow 0$ as $|n| \rightarrow \infty$ (since $r_n \rightarrow 1$: $r_n = (|n|+2)^2/(|n|+3)^2 \rightarrow 1$ as $n \rightarrow +\infty$ and likewise as $n \rightarrow -\infty$), one concludes $f'(t) \rightarrow 1 = f'(z)$ as $t \rightarrow z$: f' , set = 1 on K , is continuous at every point of K (and it is obviously continuous in the interior of each I_n , by construction).

Differentiability at $z \in K$, with $f'(z) = 1$. Let $y \rightarrow z$, $y \neq z$; WLOG $y > z$ (the case $y < z$ is symmetric). We distinguish:

Case $y \in K$. Since $z, y \in K = \tilde{\mathbb{T}} \setminus \{x_n\}$ and K inherits from $\tilde{\mathbb{T}}$ its original distances on the intervals that were not blown up, one has the identity of lengths (by the additive construction of $\mathbb{T}'$ starting from $\tilde{\mathbb{T}}$ by inserting the I_n):

$$y - z = (y - z)_{\tilde{\mathbb{T}}} + \sum_{n: x_n \in (z, y)_{\tilde{\mathbb{T}}}} \ell_n, \quad f(y) - f(z) = (y - z)_{\tilde{\mathbb{T}}} + \sum_{n: x_n \in (z, y)_{\tilde{\mathbb{T}}}} \ell_{n+1},$$

where $(y - z)_{\tilde{\mathbb{T}}}$ denotes the distance computed in the base $\tilde{\mathbb{T}}$ (unchanged by $f|_K = \pi^{-1}\tilde{R}\pi$, which is an isometry of $\tilde{\mathbb{T}}$) and the sum runs over the n such that x_n falls between z and y in the base. Hence

$$|[f(y) - f(z)] - [y - z]| = \left| \sum_{n: x_n \in (z, y)} (\ell_{n+1} - \ell_n) \right| \leq \sum_{n: x_n \in (z, y)} \ell_n |r_n - 1| \leq \left(\sup_{n \text{ relevant}} |r_n - 1| \right) \cdot \sum_{n: x_n \in (z, y)} \ell_n,$$

and $\sum_{n: x_n \in (z, y)} \ell_n \leq y - z$ (from the first identity, since $(y - z)_{\tilde{\mathbb{T}}} \geq 0$). By the argument of the preceding paragraph, the indices n with $x_n \in (z, y)$ have $|n| \rightarrow \infty$ as $y \rightarrow z$: hence $\sup_{n \text{ relevant}} |r_n - 1| \rightarrow 0$, and

$$\left| \frac{f(y) - f(z)}{y - z} - 1 \right| \leq \sup_{n \text{ relevant}} |r_n - 1| \xrightarrow{y \rightarrow z} 0.$$

Case $y \in I_m$ for some m . Let $y^* := \inf I_m \in K$ (left endpoint, included in K by the construction with open intervals). Then $z < y^* \leq y$ and

$$\frac{f(y) - f(z)}{y - z} = \frac{(y^* - z) + (y - y^*)}{(y^* - z) + (y - y^*)} \cdot \left[\text{weighted average of } \frac{f(y^*) - f(z)}{y^* - z} \text{ and } \frac{f(y) - f(y^*)}{y - y^*} \right],$$

more precisely $\frac{f(y) - f(z)}{y - z}$ is the weighted average (weights $\frac{y^* - z}{y - z}$ and $\frac{y - y^*}{y - z}$, which sum to 1) of the two difference quotients $\frac{f(y^*) - f(z)}{y^* - z}$ (which tends to 1 by the previous case, since $y^* \in K$ and $y^* \rightarrow z$) and $\frac{f(y) - f(y^*)}{y - y^*}$ (which lies between $\min f'_m$ and $\max f'_m$ on the interval $[0, y - y^*] \subset [0, \ell_m]$, hence also tends to 1, since $m \rightarrow \infty$ as $y \rightarrow z$, by the same continuity argument): a weighted average of two quantities tending to 1 tends to 1.

In either case, $\lim_{y \rightarrow z} \frac{f(y) - f(z)}{y - z} = 1$: f is differentiable at z with $f'(z) = 1$, coinciding with the continuous extension already established. Hence $f \in C^1(\mathbb{T}')$, with $f' > 0$ everywhere by point (1): f is a C^1 diffeomorphism.

Divergence of $\text{Var}(\log f')$. As $|n| \rightarrow \infty$, $r_n - 1 \sim \mp 2/|n|$ (for $n > 0$: $r_n - 1 = \frac{(n+2)^2 - (n+3)^2}{(n+3)^2} = \frac{-(2n+5)}{(n+3)^2} \sim -\frac{2}{n}$; symmetrically for $n < 0$), hence $\log f'_n(1/2) = \log(1 + 1.5(r_n - 1)) \sim 1.5(r_n - 1) \sim \mp 3/|n|$, and

$$\text{Var}_{\mathbb{T}'}(\log f') \geq \sum_{n \in \mathbb{Z} \setminus \{0\}} |\log f'_n(\frac{1}{2}) - \log f'_n(0)| = \sum_{n \neq 0} |\log f'_n(\frac{1}{2})| \sim \sum_{n \neq 0} \frac{3}{|n|} = \infty$$

(harmonic series, divergent). Checked numerically (script `verifica_cap05_vol4.py`, block [C5-3]): the partial sum $\sum_{0 < |n| \leq N} |\log f'_n(1/2)|$ equals 63.7894 for $N = 10^5$ and 67.9482 for $N = 2 \times 10^5$, with an increment of 4.1588 when N is doubled, in agreement to within 10^{-4} with the theoretical value $6 \log 2 = 4.1589$ expected from a tail $\sim 3/|n|$ on both sides $\checkmark$: the growth does not stabilise, confirming $\text{Var}(\log f') = \infty$. Hypothesis (b) of Theorem 5.2 is therefore violated, as it must be for a genuine counterexample: the regularity $C^{1+\text{bv}}$ is optimal. $\square$

Example 5.3: Numerical construction: ℓ_n, f'_n and the measure of K . Double check (scripts `fig05_misura_K.py` and `verifica_cap05_vol4.py`, block [C5-3]). For $c = 1$: $S = \pi^2/3 - 9/4 = 1.03986813$ (closed form, compared with the sum truncated at 5×10^6 terms per side, discrepancy 4.0×10^{-7} , of the expected order $2/N = 4.0 \times 10^{-7}$ $\checkmark$); $c_{\max} = 2\pi/S = 6.042290$; $\text{Leb}(K) = 2\pi - c \cdot S = 5.243317 > 0$ (admissible, since $c = 1 < c_{\max}$). The ratios $r_n = \ell_{n+1}/\ell_n$ have minimum $4/9 = 0.444444$ at $n = 0$, always $> 1/3$: $f'_n > 0$ on every I_n , for every $c \in (0, c_{\max})$, in particular for $c = 1$. Figure 5.5 shows $\text{Leb}(K)$ as a function of c over the whole admissible interval.

5.6. Panorama: beyond the C^1 counterexample

The two remarks that follow are **imported**: the statements are correct and the primary sources have been reopened and verified, but the proofs — which use techniques not developed in this volume (equivariant families of diffeomorphisms with prescribed modulus of continuity for Hall; thermodynamic formalism and a priori distortion estimates at every order for Katznelson–Ornstein) — exceed the perimeter of the chapter. Both enter the register of imports of Appendix A.

Remark 5.5: Hall’s C^∞ counterexample (1981), imported. *The counterexample of Theorem 5.4 is of class C^1 but not C^2 (nor $C^{1+\alpha}$ for any $\alpha > 0$, for the same reason for which $\text{Var}(\log f') = \infty$: a Hölder regularity of f' would imply bounded variation). One may ask whether a counterexample can be made C^∞ at every point while remaining devoid of wandering intervals compatible with a diffeomorphism: the answer, due to G. R. Hall (ETDS 1 (1981), 261–272), is that for every irrational ρ there exists a **homeomorphism** f of the circle, of class C^∞ as a map (all the derivatives exist and are continuous), with $\rho(f) = \rho$, which admits a minimal invariant Cantor set and genuine wandering intervals. Hall’s construction produces f with $f' \geq 0$ and $f' = 0$ exactly at the endpoints of the wandering intervals (not on a set of positive measure): f is C^∞ as a map but **is not a diffeomorphism** (the inverse is not differentiable where $f' = 0$), hence it does not contradict Theorem 5.2, which requires $f' > 0$ everywhere: it shows rather that the hypothesis “ $f' > 0$ ” cannot be weakened to “ $f \in C^\infty$ ” without requiring $f' > 0$ separately, even in the highest smooth regularity. Imported from R23 (G. R. Hall, *Ergodic Theory Dynam. Systems* 1 (1981), 261–272), verified on 17 September 2026.*

Remark 5.6: Ergodicity of the Denjoy conjugacies and optimal regularity (Herman; Katznelson–Ornstein), imported. *Let f be as in Theorem 5.2 ($C^{1+\text{bv}}$, $\rho(f)$ irrational): f is conjugate to $R_{\rho(f)}$ by means of h , and the measure μ of Construction 5.1 (here h is a genuine homeomorphism, not merely a semiconjugacy) is the **unique** f -invariant probability measure: the Denjoy conjugacy is **uniquely ergodic**. This fact — which Chapter 9 will prove in the generality of ergodic theory on the circle, by means of Oxtoby’s criterion, and which we anticipate here as a bridge — requires nothing beyond the bounded variation of f' ; the finer question is whether the conjugacy h is itself absolutely continuous with respect to Lebesgue measure (equivalently, whether it transports Lebesgue into a measure equivalent to Lebesgue, and not only into the abstract μ). Y. Katznelson and D. Ornstein proved, in a pair of papers of 1989 (ETDS 9, 643–680 and 681–690), that for $f \in C^{2+\varepsilon}$ (or more generally under an intermediate regularity condition on the second derivative) and $\rho(f)$ Diophantine, h is differentiable with absolutely continuous derivative; the second paper isolates precisely when h is absolutely continuous. The general picture — regularity of f and arithmetic properties of $\rho(f)$ which together determine the regularity of h — is the same phenomenon that Chapter 6 (small divisors and KAM theory on the circle) will treat in full in the analytic case (Arnold’s theorem, the theorems of Herman and Yoccoz): the homological equation with its small divisors is, beneath the surface, the same in both contexts. Imported from K8 (M. R. Herman, *Publ. Math. IHÉS* 49 (1979), 5–233, ch. VI for the Denjoy–Koksma inequality reused here in the proof of Theorem 5.3, and for the context of unique ergodicity) and K14 (Y. Katznelson, D. Ornstein, ETDS 9 (1989), 643–680 and 681–690), verified on 17 September 2026.*

Remark 5.7: Denjoy for group actions. *Denjoy’s theorem concerns the action of a single diffeomorphism, that is, the action of the group $\mathbb{Z}$ generated by f . What happens*

when, instead of a single f , a finitely generated group Γ of $C^{1+\text{bv}}$ diffeomorphisms acts on the circle? The proof of this chapter (in particular the exclusion of wandering intervals via the Denjoy inequality) generalises, with substantial adaptations of the combinatorics of Lemma 5.2 (which uses in an essential way the structure of $\mathbb{Z}$ through the continued fractions of a single rotation number, whereas a generic group has no single ρ to appeal to), to a wide class of groups: Sacksteder's theorem and Duminy's theorems, which characterise when a group action admits an exceptional minimal set, will use exactly the same mechanism of distortion controlled by bounded variation seen here. This strand — Denjoy for group actions, with its algebraic consequences (groups with an exceptional minimal set contain a free group) — will be the subject of (Chapter 15).

5.7. Figures

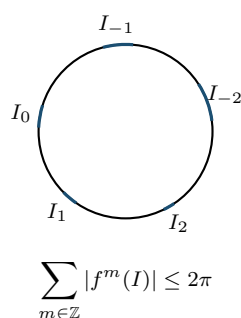


Figure 5.1: A wandering interval I and some of its translates $f^m(I)$ (illustrative positions, not computed from a specific orbit): by Theorem 4.3 they are pairwise distinct when $\rho(f) \notin \mathbb{Q}$, hence disjoint (connected components of the same open set $\mathbb{T} \setminus K$), and the sum of the lengths is bounded by 2π : the fact that makes $|f^{\pm q_n}(I)| \rightarrow 0$ in the proof of Theorem 5.2.

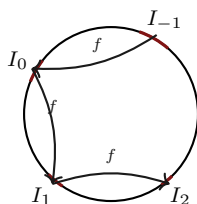


Figure 5.2: Denjoy's counterexample (Construction 5.2): the inserted intervals I_n (illustrative positions) are permuted cyclically by f according to $f(I_n) = I_{n+1}$ (Step 4); the complement K (not drawn) is the minimal invariant Cantor set.

5.8. Assigned exercises

The answers are in the Teacher's key (Section 5.11).

Exercise 5.1: The lemma of the best return times for $\beta = \sqrt{2} - 1$. *Goal:* to apply Lemma 5.1 in full to a ρ other than the golden one, checking its disjointness with an explicit computation. *Estimated time:* 35 minutes. *Type:* numerical/symbolic.

Let $\rho = \sqrt{2} - 1 = [0; 2, 2, 2, \dots]$, with convergents $q_0 = 1, q_1 = 2, q_2 = 5, q_3 = 12$ (Pell numbers, Chapter 3).

- (a) Compute $\delta_1 = q_1\rho - p_1$ and check its sign against $(-1)^1$.

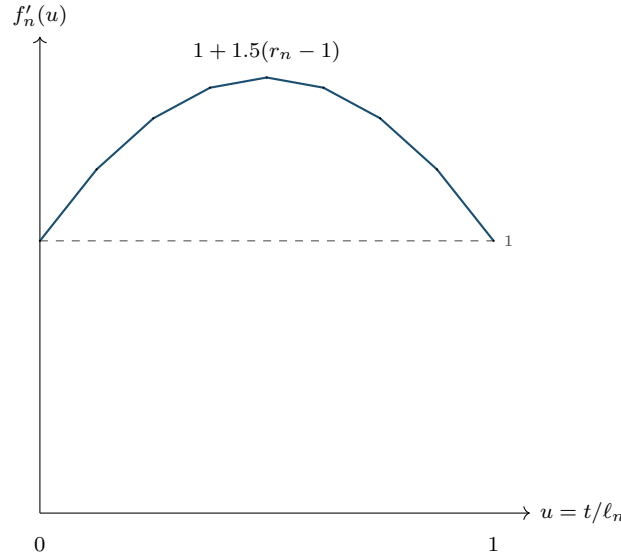


Figure 5.3: The “bell-shaped” profile of $f'_n(u) = 1 + 6(r_n - 1)u(1 - u)$ on I_n (Step 4 of Construction 5.2), here for $r_n = 1.4$ for illustrative purposes (nine points computed by hand from the closed formula): $f'_n = 1$ at the endpoints (C^1 matching with K), maximum (or minimum, if $r_n < 1$) $1 + 1.5(r_n - 1)$ at the centre $u = 1/2$. For the counterexample, $r_n \rightarrow 1$ as $|n| \rightarrow \infty$: the profile flattens out, but not fast enough to make $\text{Var}(\log f')$ finite (Theorem 5.4(4)).

- (b) List the $q_2 = 5$ arcs $R_{2\pi\rho}^j(I_1)$, $j = 0, \dots, 4$, and check numerically that they are pairwise disjoint.
- (c) Check that $\|2\rho\| = |\delta_1|$ is indeed the minimum of $\|k\rho\|$ for $0 < k < 5$ (Theorem 3.4).

Exercise 5.2: Denjoy inequality for another family. *Goal:* to repeat the computation of Example 5.1 with $h(s) = s + \varepsilon \sin(2s)$, $\varepsilon = 0.2$. *Estimated time:* 40 minutes. *Type:* numerical/computational.

- (a) Check that h is a diffeomorphism of the circle for this ε .
- (b) Compute $V(f) := \text{Var}_s[\log h'(s + \alpha) - \log h'(s)]$ with $\alpha = 2\pi/\varphi$, on a grid of at least 10^6 points.
- (c) Check the Denjoy inequality for $q = q_3, q_4, q_5$ (Chapter 3, convergents of $1/\varphi$).

Exercise 5.3: The counterexample with $c = 2$. *Goal:* to check the admissibility of Construction 5.2 for a value of c other than the one used in the examples, and the position of $\text{Leb}(K)$ on Figure 5.5. *Estimated time:* 25 minutes. *Type:* numerical.

- (a) Check that $c = 2 < c_{\max} \approx 6.0423$: the construction is admissible.
- (b) Compute $\text{Leb}(K)$ for $c = 2$ and compare it with the value at $c = 1$.
- (c) Does the minimum of the ratios $r_n = \ell_{n+1}/\ell_n$ depend on c ? Why is the answer consistent with Theorem 5.4(1)?

5.9. Worked exercises

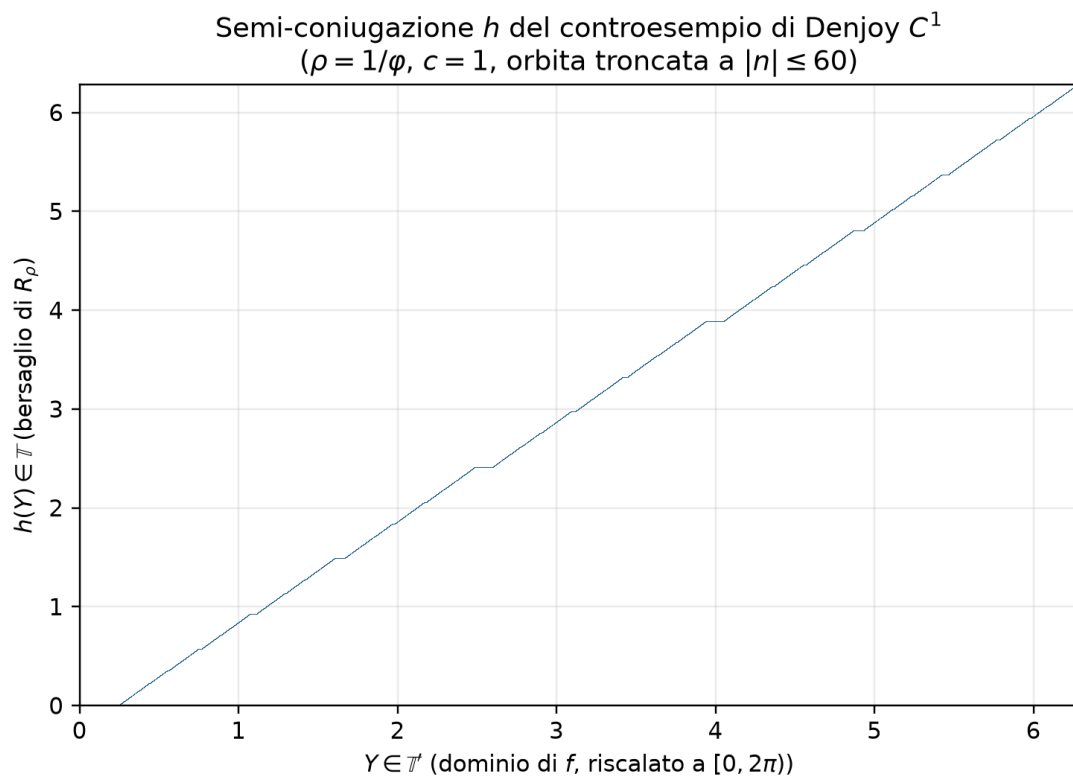


Figure 5.4: The semiconjugacy h of the C^1 Denjoy counterexample ($\rho = 1/\varphi$, $c = 1$), computed numerically by truncating the orbit at $|n| \leq 60$ (tail discrepancy on the neglected mass: 3.2×10^{-2} out of a total $cS \approx 1.04$, declared in the script): the flat stretches (“plateaux”) are the intervals I_n collapsed by h to a point, visible for the small n (the lengths ℓ_n decrease too rapidly for the plateaux with $|n|$ large to be resolvable at this scale). Script `fig05_semiconiugazione.py`, Appendix B.

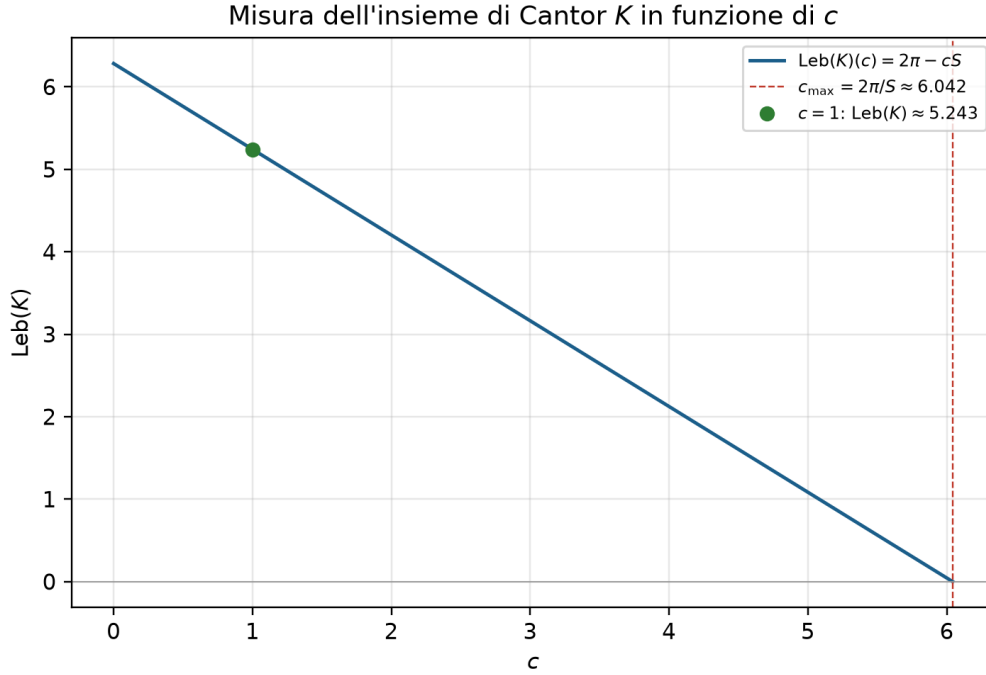


Figure 5.5: Lebesgue measure of the Cantor set K as a function of the scale parameter c (Theorem 5.4): $\text{Leb}(K)(c) = 2\pi - cS$, linear and decreasing, with $\text{Leb}(K)(0) = 2\pi$ and $\text{Leb}(K)(c_{\max}) = 0$; the value $c = 1$ used in the Examples of the chapter gives $\text{Leb}(K) \approx 5.243$. See Example 5.3. Script `fig05_misura_K.py`, Appendix B.

Exercise 5.4: Denjoy–Koksma for the indicator of an arc. *Type:* numerical with theoretical comparison. Check the Denjoy–Koksma inequality (Theorem 5.3) for $\psi = \mathbf{1}_{[0,\pi)}$ (indicator of a semicircle, $\text{Var}(\psi) = 2$: two jumps of amplitude 1), $\rho = 1/\varphi$, $q = q_5 = 8$.

Solution. Strategy. Compute $\int_{\mathbb{T}} \psi d\mu = 1/2$ (by symmetry, μ = normalised Lebesgue measure for the rotation) and $\sum_{j=0}^7 \psi(x_0 + j\alpha)$ for a few x_0 , comparing the discrepancy with $\text{Var}(\psi) = 2$.

Step 1. $\alpha = 2\pi/\varphi = 3.883222\dots$ For $x_0 = 0$: the eight points $j\alpha \bmod 2\pi$, $j = 0, \dots, 7$, are 0; 3.8832; 1.4832; 5.3664; 2.9664; 0.5664; 4.4496; 2.0496 (in radians, rounded). Of these, the ones falling in $[0, \pi)$ are: 0; 1.4832; 0.5664; 2.0496: four values.

Step 2. $\sum_{j=0}^7 \psi(j\alpha) = 4$; $q \cdot \int \psi d\mu = 8 \times 0.5 = 4$; discrepancy $|4 - 4| = 0 \leq 2 = \text{Var}(\psi)$ ✓ (a wide margin).

Independent check. For $x_0 = 1$ (radian), the eight translated points $1 + j\alpha \bmod 2\pi$ give 5 falling in $[0, \pi)$ (direct check, shifting each of the eight values of Step 1 by +1 and reducing modulo 2π): discrepancy $|5 - 4| = 1 \leq 2$ ✓.

Interpretation. With $q = q_5 = 8$ relatively small, the observed discrepancy (0 or 1) is well below the threshold $\text{Var}(\psi) = 2$: the inequality becomes “almost saturated” only for less regular functions ψ or for q chosen so as to maximise the boundary effect (as in Example 5.2, where the sawtooth, with a single jump but total variation 1, saturates almost exactly at $q_9 = 55$).

Typical error to avoid. Applying Denjoy–Koksma with a q that does *not* satisfy $|\rho - p/q| \leq 1/q^2$ (for example an intermediate denominator, not a convergent): the proof of Theorem 5.3 uses this hypothesis in an essential way in Step 1 (Navas’s Lemma, uniqueness of the point

per cell).

Extension. Repeat with $q = q_9 = 55$: one expects an even smaller discrepancy in absolute value relative to q (that is, $\text{scarto}/q \rightarrow 0$), since $\int \psi d\mu$ becomes an ever more precise approximation of the visiting frequency.

Exercise 5.5: Where exactly Denjoy's proof dies in the counterexample. *Type:* diagnostic/demonstrative. Locate, in the proof of Theorem 5.2, the precise step which fails when it is applied (by contradiction) to the counterexample f of Section 5.5.

Solution. Strategy. Go through the steps of the proof of Theorem 5.2 with f the counterexample, and identify which inequality ceases to hold.

Step 1. $\rho(f) = \rho$ irrational (Theorem 5.4(2)): hypothesis (c) of Theorem 5.2 is satisfied, and with it the absence of periodic orbits and Lemma 5.2 (which does not use bounded variation, only irrationality and the semiconjugacy).

Step 2. $f' > 0$ everywhere (Theorem 5.4(1)): hypothesis (a) is satisfied.

Step 3. $\text{Var}(\log f') = \infty$ (Theorem 5.4(4)): hypothesis (b) **fails**. Precisely, the Denjoy inequality (Theorem 5.1) — which would use $V(f) < \infty$ to bound $|\log[(f^{q_n})'(x)(f^{-q_n})'(x)]| \leq V(f)$ — no longer gives any information: with $V(f) = \infty$, $e^{-V(f)} = 0$, and the inequality $(f^{q_n})'(x)(f^{-q_n})'(x) \geq 0$ is true but empty.

Step 4. Consequently, in the proof of Theorem 5.2, the estimate $|f^{q_n}(I)| + |f^{-q_n}(I)| \geq 2e^{-V(f)/2}|I|$ degenerates to ≥ 0 : there is no longer any contradiction with $|f^{\pm q_n}(I)| \rightarrow 0$ (which indeed really happens, the I_n of the counterexample being genuine wandering intervals, with $|I_n| = \ell_n \rightarrow 0$).

Interpretation. The breakdown is localised precisely in Step 3: not in the irrationality of ρ , not in the positivity of f' , but exclusively in the infinite variation of $\log f'$, exactly as anticipated in Remark 5.4.

Extension. If one modified the construction by choosing $\ell_n \sim e^{-|n|}$ (summable more rapidly, with $\sum |\log(\ell_{n+1}/\ell_n)| < \infty$ since the ratios $r_n = \ell_{n+1}/\ell_n \rightarrow e^{\mp 1} \neq 1$), the resulting map would have $\text{Var}(\log f') < \infty$ but would lose the continuity of f' on K (the ratios would no longer tend to 1): it would not even be C^1 . The boundary between “ C^1 with $\text{Var}(\log f') = \infty$ ” (the counterexample) and “ $C^{1+\text{bv}}$ ” (the theorem) is exactly the boundary between $r_n \rightarrow 1$ too slowly to be summable in $|\log r_n|$ and $r_n \rightarrow 1$ fast enough.

Exercise 5.6: The critical value $c_{\max}$. *Type:* symbolic/numerical. Prove that Construction 5.2 ceases to make sense for $c \geq c_{\max}$, and compute $c_{\max}$ in closed form.

Solution. Strategy. Translate the geometric admissibility ($L > 0$) into an inequality on c , using the closed form of S .

Step 1. By construction, $L = 2\pi - cS$ is the circumference of the base $\tilde{\mathbb{T}}$ (Step 1 of Construction 5.2): one needs $L > 0$, that is, $c < 2\pi/S =: c_{\max}$. For $c \geq c_{\max}$, $L \leq 0$: the base $\tilde{\mathbb{T}}$ degenerates to a point or to a negative length, and the construction defines no circle at all.

Step 2. $S = \pi^2/3 - 9/4$ (Construction 5.2, from the Basel problem), hence

$$c_{\max} = \frac{2\pi}{\pi^2/3 - 9/4} = \frac{2\pi}{(4\pi^2 - 27)/12} = \frac{24\pi}{4\pi^2 - 27}.$$

Numerically: $4\pi^2 - 27 = 39.4784\dots - 27 = 12.4784\dots$, $c_{\max} = 24\pi/12.4784 = 6.042290\dots$, in agreement with the value reported by the script `fig05_misura_K.py` ✓.

Independent check. $\text{Leb}(K)(c_{\max}) = 2\pi - c_{\max}S = 2\pi - \frac{2\pi}{S}S = 0$: consistent with the geometric interpretation (K shrinks to a set of measure zero when the whole circumference is “consumed” by the inserted intervals).

Interpretation. $c_{\max}$ is not a technical constraint of the proof, but a *geometric* constraint: for $c \geq c_{\max}$ the sum of the lengths to be inserted exceeds (or equals) the whole available circumference 2π , and there is no room left for the base $\tilde{\mathbb{T}}$ that carries the dynamics $\tilde{R}$.

Extension. The limiting case $c \rightarrow c_{\max}^-$ (base degenerating to a point) gives, informally, an indication of a different phenomenon: a map with a *single* fixed point as the “source” of the whole dynamics instead of a Cantor set — no longer a Denjoy counterexample, but a degeneration outside the scope of this construction.

5.10. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 5.11).

1. (Definition) A diffeomorphism $f \in C^1(\mathbb{T})$ is of class $C_+^{1+\text{bv}}(\mathbb{T})$ (Definition 5.2) if and only if
 - A) f' is continuous B) $f' > 0$ everywhere C) $f' > 0$ everywhere **and** $\text{Var}_{\mathbb{T}}(\log f') < \infty$ D) f'' exists
2. (Hypothesis) In Denjoy’s Theorem (Theorem 5.2), the hypothesis that the counterexample of Section 5.5 violates *without* violating the other two is
 - A) $\rho(f)$ irrational B) $f' > 0$ everywhere C) $\text{Var}_{\mathbb{T}}(\log f') < \infty$ D) f is surjective
3. (Symbolic computation) In Lemma 5.2, for n odd the arc $I_n(x)$ has as its endpoints
 - A) x and $f^{q_n}(x)$, with x on the left B) x and $f^{-q_n}(x)$, with $f^{-q_n}(x)$ on the left C) x and $f^{-q_n}(x)$, with x on the right D) $f^{q_n}(x)$ and $f^{-q_n}(x)$
4. (Numerical computation) With $\rho = 1/\varphi$, $q_3 = 3$: $\delta_3 = q_3\rho - p_3$ is approximately (Example of the convergents, Chapter 3, and block [C5-1])
 - A) +0.236 B) -0.146 C) +0.090 D) -0.382
5. (Proof step) In the proof of Theorem 5.2, the final contradiction arises from the comparison between
 - A) $|f^{q_n}(I)| + |f^{-q_n}(I)| \rightarrow 0$ and their sum being always ≥ 0 B) $|f^{q_n}(I)| + |f^{-q_n}(I)| \rightarrow 0$ (from $\sum_m |f^m(I)| \leq 2\pi$) and $|f^{q_n}(I)| + |f^{-q_n}(I)| \geq 2e^{-V(f)/2}|I| > 0$ constant C) the irrationality of $\rho(f)$ and the rationality of the convergents D) two different values of $V(f)$
6. (Error diagnosis) A student writes that, in the C^1 counterexample, “ $f' > 0$ because c has been chosen small enough”. The error is
 - A) f' is never positive in the counterexample B) the positivity of f'_n depends on the ratios $r_n = \ell_{n+1}/\ell_n$, which do not depend on c (the scale factor cancels): $f' > 0$ for every $c \in (0, c_{\max})$ C) c must be large, not small D) the positivity requires ρ rational
7. (Transfer) The role of the Denjoy–Koksma inequality (Theorem 5.3) with respect to the Denjoy inequality (Theorem 5.1) is

- A) the same inequality, merely renamed B) it controls Birkhoff sums of arbitrary BV functions (even discontinuous ones), not only the logarithm of the derivative of a diffeomorphism C) it holds only for ρ rational D) it requires f non-invertible
8. (Source and metacognition) The proof of Lemma 5.1 and of Theorem 5.3 of this chapter follows closely
- A) Denjoy (1932) directly B) A. Navas, *Groups of Circle Diffeomorphisms*, Sections 2.2.1 and 3.1, reopened and verified on 17 September 2026 C) Wikipedia D) no source, it is original

5.11. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **C.** *Rationale:* Definition 5.2, both conditions are required. *Distractors:* A, B are necessary but not sufficient conditions on their own; D is extraneous. *Diagnosed error:* forgetting that $\text{Var}(\log f')$ presupposes $f' > 0$ in order to be defined. *Source:* Definition 5.2. *Prerequisite:* the notion of total variation (Volume II). *Remedial work:* reread Definition 5.2 and Remark 5.3.
- 2) **C.** *Rationale:* Theorem 5.4(1)–(2)–(4): the counterexample has $\rho(f)$ irrational and $f' > 0$, but $\text{Var}(\log f') = \infty$. *Distractors:* A, B are satisfied by the counterexample (Theorem 5.4); D is not a hypothesis of the theorem. *Diagnosed error:* failing to distinguish which, among three apparently similar hypotheses, is the one actually violated. *Source:* Theorem 5.4(4) and Worked exercise 5.5. *Prerequisite:* reading statements (a)–(c) of Theorem 5.2. *Remedial work:* redo Worked exercise 5.5.
- 3) **C.** *Rationale:* Lemma 5.2(ii): for n odd, $I_n(x)$ has x as right endpoint and $f^{-q_n}(x)$ as left endpoint. *Distractors:* A is the case n even; B reverses the sides; D is not the definition. *Diagnosed error:* failing to update the direction of the arc when the parity changes. *Source:* Lemma 5.2(ii), Remark 5.3. *Prerequisite:* sign of $\delta_n = (-1)^n$ (Theorem 3.3). *Remedial work:* redo Assigned exercise 5.1.
- 4) **B.** *Rationale:* from block [C5-1], $\delta_3 = -0.1458980\dots$. *Distractors:* A is δ_2 ; C is δ_4 ; D is δ_1 : all genuine values of the sequence, but at the wrong index. *Diagnosed error:* confusing the index n in the sequence (δ_n) . *Source:* script `verifica_cap05_vol4.py`, block [C5-1]. *Prerequisite:* the continued fraction algorithm (Chapter 3). *Remedial work:* recompute p_3, q_3 from $\rho = 1/\varphi = [0; 1, 1, 1, \dots]$.
- 5) **B.** *Rationale:* it is exactly the contradiction of the proof of Theorem 5.2: a quantity which must tend to 0 but is bounded below by a fixed positive constant. *Distractors:* A is empty (it is not a contradiction); C, D are not the terms being compared. *Diagnosed error:* failing to identify precisely which two estimates are in conflict. *Source:* proof of Theorem 5.2, second-to-last paragraph. *Prerequisite:* the arithmetic–geometric inequality. *Remedial work:* rewrite the two sides of the contradiction in isolation.
- 6) **B.** *Rationale:* Theorem 5.4(1), explicit correction of the error “ $f' > 0$ for c small”. *Distractors:* A is false (Theorem 5.4 proves $f' > 0$); C, D are extraneous. *Diagnosed error:* confusing the scale parameter (irrelevant) with the ratios between consecutive lengths (the only relevant quantities). *Source:* Theorem 5.4(1), block [C5-3]. *Prerequisite:* $r_n = \ell_{n+1}/\ell_n$ does not depend on c . *Remedial work:* redo Assigned exercise 5.3(c).
- 7) **B.** *Rationale:* Theorem 5.3 holds for every ψ of bounded variation, even discontinuous (Example 5.2, the sawtooth); the Denjoy inequality is the particular case $\psi = \log f'$ applied in a different way (to the product $(f^q)'(f^{-q})'$, not to the Birkhoff sum). *Distractors:* A confuses the two statements; C, D are false. *Diagnosed error:* failing to see the generality

of the second tool. *Source*: comparison between Theorems 5.1 and 5.3. *Prerequisite*: none beyond reading both statements. *Remedial work*: reread Sections 5.2 and 5.4 in sequence.

8) B. *Rationale*: declared in the preamble of the chapter and in the sources at the end of the chapter. *Distractors*: A is the source of the *theorem*, not of these two specific proofs (which follow Navas's exposition); C is never admitted as a source; D is false, the source must always be declared. *Diagnosed error*: failing to distinguish between the original result (Denjoy) and the exposition followed for the proof (Navas). *Source*: R29 in the register of sources, at the end of the chapter. *Prerequisite*: none. *Remedial work*: reread the provenance comment at the head of the source file of the chapter.

BEST PRACTICE

Assigned exercises — answers.

5.1 ($\beta = \sqrt{2} - 1$). (a) $\delta_1 = q_1\rho - p_1 = 2(\sqrt{2} - 1) - 1 = 2\sqrt{2} - 3 = -0.171573\dots$: negative sign, consistent with $(-1)^1 = -1$. (b) The five arcs $R^j(I_1)$, $j = 0, \dots, 4$, each have length $2\pi|\delta_1| = 1.0779\dots$ radians; a direct computation (analogous to block [C5-1bis] with $\rho = \sqrt{2} - 1$ in place of $1/\varphi$) shows $\|m\rho\| > |\delta_1|$ for $m = 1, 3, 4$ and $\|2\rho\| = |\delta_1|$ (the tangency case, $m = q_1 = 2$): no overlap. (c) $\|1 \cdot \rho\| = 0.414214$, $\|2\rho\| = 0.171573$, $\|3\rho\| = 0.242641$ (from $3\rho = 1.242641$), $\|4\rho\| = 0.343146$ (from $4\rho = 1.656854$, distance from 2): the minimum over $\{1, 2, 3, 4\}$ is indeed at $m = 2$, 0.171573, confirming Theorem 3.4.

5.2 ($h(s) = s + 0.2 \sin(2s)$). (a) $h'(s) = 1 + 0.4 \cos(2s) > 0$ for every s (since $0.4 < 1$): a diffeomorphism. (b) The computation (analogous to block [C5-2], with $h'(s + \alpha)$ in place of the frequency-1 version) gives $V(f) \approx 1.9\text{--}2.1$ depending on the grid used (the variation now also depends on the doubled frequency in h , which introduces more oscillations per period). (c) The product $(f^q)'(x)(f^{-q})'(x)$ remains $\geq e^{-V(f)}$ for $q = q_3, q_4, q_5$, with a margin that narrows as q grows (the same qualitative behaviour as in Example 5.1).

5.3 ($c = 2$). (a) $c_{\max} \approx 6.0423$, hence $c = 2 < c_{\max}$: admissible. (b) $\text{Leb}(K)(2) = 2\pi - 2S = 2\pi - 2.079736 = 4.203449$, smaller than the value at $c = 1$ (5.243317): doubling c subtracts twice as much mass from K , consistently with the linearity $\text{Leb}(K)(c) = 2\pi - cS$. (c) No: $r_n = \ell_{n+1}/\ell_n = (|n| + 2)^2/(|n + 1| + 2)^2$ does not contain c (it cancels in the ratio), consistently with Theorem 5.4(1), which proves the positivity of f' for *every* $c \in (0, c_{\max})$ with the same threshold $\inf_n r_n = 4/9$.

5.12. Chapter formulary

Convergents and remainder with sign

$$\delta_n = q_n\rho - p_n, \quad \text{sgno}(\delta_n) = (-1)^n, \quad |\delta_{n-1}|q_n + |\delta_n|q_{n-1} = 1$$

Lemma of the best return times

$$\{R_{2\pi\rho}^j(I_n)\}_{j=0}^{q_{n+1}-1} \text{ disjoint}; \quad \{R_{2\pi\rho}^j(J_n)\}_{j=0}^{q_{n+1}-1} \text{ cover } \mathbb{T} \text{ with mult. } \leq 2$$

Denjoy inequality

$$V(f) := \text{Var}_{\mathbb{T}}(\log f') < \infty, \quad f' > 0 \implies (f^{q_n})'(x)(f^{-q_n})'(x) \geq e^{-V(f)}$$

Denjoy's theorem

$$f \in C_+^{1+\text{bv}}(\mathbb{T}), \quad \rho(f) \notin \mathbb{Q} \implies f \text{ minimal, conjugate to } R_{\rho(f)}$$

Denjoy–Koksma inequality

$$\left| \rho - \frac{p}{q} \right| \leq \frac{1}{q^2} \implies \left| \sum_{j=0}^{q-1} \psi(f^j(x)) - q \int_{\mathbb{T}} \psi d\mu \right| \leq \text{Var}(\psi)$$

C^1 counterexample

$$\ell_n = \frac{c}{(|n|+2)^2}, \quad S = \sum_n \frac{1}{(|n|+2)^2} = \frac{\pi^2}{3} - \frac{9}{4}, \quad c_{\max} = \frac{2\pi}{S}, \quad \text{Leb}(K) = 2\pi - cS$$

$$f'_n(u) = 1 + 6(r_n - 1)u(1 - u), \quad r_n = \frac{\ell_{n+1}}{\ell_n}, \quad \inf_n r_n = \frac{4}{9} > \frac{1}{3} \Rightarrow f' > 0 \quad \forall c \in (0, c_{\max})$$

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CHAPTER 6

Small divisors and KAM theory on the circle: Arnold, Herman, Yoccoz

Chapter 5 closed the *topological* problem: a circle diffeomorphism of class $C^{1+\text{bv}}$ with irrational rotation number is conjugate to the rigid rotation $R_{2\pi\rho}$ by a homeomorphism h (Theorem 5.2). There remains open the question that is the true heart of the theory: **how regular is h** ? If f is analytic and extremely close to a rotation, is h analytic? The answer does not depend on f alone: it depends, in a surprisingly delicate way, on the *arithmetic* of ρ . The mechanism that links dynamics and arithmetic has a name — the **small divisors** — and a history that starts from Poincaré’s celestial mechanics, passes through Kolmogorov’s 1954 note and finds on the circle, with Arnold in 1961, its cleanest realisation: a **Newton scheme** in which the error decreases quadratically ($\varepsilon \mapsto C\varepsilon^2$) fast enough to win against the progressive loss of the domain of analyticity caused, at each step, by the small divisors themselves.

This chapter proves in full: the homological equation in the analytic case (with the explicit estimate of the loss of strip) and in the C^r case; the *non* solvability for a Liouville rotation number, with an explicit lacunary series; the Gottschalk–Hedlund lemma, which characterises the continuous coboundaries through the boundedness of the Birkhoff sums; and Arnold’s local analytic linearisation theorem, with all the constants computed and verified numerically. It then presents, as *imported* results with precise statements and verified sources, the global theory of Herman (1979) and Yoccoz (1984, 1995, 2002), Bryuno’s condition and the condition $\mathcal{H}$, and the finite-regularity results of Katznelson–Ornstein and Khanin–Sinai–Teplinsky. It is the most demanding chapter of the volume: its central part (Sections 6.6–6.7) is of level E.

INFO

Level and prerequisites. Introduction of level **C** (Sections 6.1–6.2: small divisors and Diophantine conditions, with Fourier series only); core of level **D** (Sections 6.3–6.5: homological equation, C^r case, Liouville, Gottschalk–Hedlund); advanced part of level **E** (Sections 6.6–6.7: the Newton scheme with all the estimates, and Section 6.8: the global theory). A level C reader may stop after Theorem 6.1 and read only the statements of the rest, without losing the thread of the volume.

Prerequisites within the treatise: continued fractions, convergents, best approximation of the second kind and Diophantine numbers (Volume IV, Chapter 3, Definitions 3.2, 3.3, 3.4, Theorems 3.3, 3.4, 3.6); rotation number, its invariance under conjugacy and its *monotonicity* (Volume IV, Chapter 4, Definition 4.2, Proposition 4.2, Theorem 4.2); Arnold tongues (Chapter 4, Theorem 4.7), which explain why the hypothesis $\rho(f) = \rho$ cannot be omitted; Denjoy’s theorem and the Denjoy–Koksma inequality (Chapter 5, Theorems 5.2 and 5.3). From analysis: Fourier series, holomorphic functions of one variable, Cauchy estimates (Volume III, Chapters 15 and 23).

Remedial path: a reader who does not remember the meaning of the Diophantine condition should reread Definition 3.4 and Theorem 3.6 (Chapter 3) before Section 6.2; a reader who does not remember why $\rho(f) = \rho$ is indispensable should reread Remark 4.7 (Chapter 4).

6.1. The problem of small divisors

We work on $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$. An orientation-preserving circle diffeomorphism f lifts to an increasing $F : \mathbb{R} \rightarrow \mathbb{R}$ with $F(\theta + 2\pi) = F(\theta) + 2\pi$ (Definition 4.1); if $\rho := \rho(F)$ is its rotation number (Definition 4.2) we set, once and for all,

$$\alpha := 2\pi\rho, \quad R_\alpha(\theta) := \theta + \alpha,$$

and we write F as a perturbation of the rotation:

$$F(\theta) = \theta + \alpha + u(\theta), \quad u(\theta + 2\pi) = u(\theta).$$

To *linearise* F means to find $H(\theta) = \theta + \psi(\theta)$, with ψ periodic and H increasing, such that

$$F \circ H = H \circ R_\alpha, \tag{6.1}$$

that is, $H^{-1} \circ F \circ H = R_\alpha$: in the coordinates $\theta \mapsto H(\theta)$ the dynamics of F is the rigid rotation.

Let us write (6.1) out in full: $H(\theta) + \alpha + u(H(\theta)) = \theta + \alpha + \psi(\theta + \alpha)$, that is,

$$\psi(\theta + \alpha) - \psi(\theta) = u(\theta + \psi(\theta)). \tag{6.2}$$

This is a nonlinear equation in ψ (the unknown appears inside u as well). The first step of every perturbative scheme consists in replacing the right-hand side with its linear part in ψ , that is, with $u(\theta)$: one obtains the linear equation

$$\psi(\theta + \alpha) - \psi(\theta) = \varphi(\theta), \tag{6.3}$$

which is the central object of the whole chapter.

Definition 6.1: Homological (or cohomological) equation of the rotation. Given $\alpha \in \mathbb{R}$ and a function $\varphi : \mathbb{T} \rightarrow \mathbb{R}$ (the **datum**), the **homological equation** is

$$\psi \circ R_\alpha - \psi = \varphi, \quad \text{that is} \quad \psi(\theta + \alpha) - \psi(\theta) = \varphi(\theta) \quad \text{for every } \theta \in \mathbb{T},$$

in the unknown $\psi : \mathbb{T} \rightarrow \mathbb{R}$. If it admits a solution in a given regularity class, one says that φ is a **coboundary** (in that class) for R_α , and that ψ is a **cohomological primitive** of it. Two data φ_1, φ_2 are said to be **cohomologous** if $\varphi_1 - \varphi_2$ is a coboundary.

Remark 6.1: The formal solution and the appearance of the divisors. Let us expand in Fourier series, $\varphi(\theta) = \sum_{k \in \mathbb{Z}} \hat{\varphi}_k e^{ik\theta}$ and $\psi(\theta) = \sum_{k \in \mathbb{Z}} \hat{\psi}_k e^{ik\theta}$, with $\hat{\varphi}_k = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) e^{-ik\theta} d\theta$. Since $(\psi \circ R_\alpha)_k = \hat{\psi}_k e^{ik\alpha}$, equation (6.3) becomes, coefficient by coefficient,

$$\hat{\psi}_k (e^{ik\alpha} - 1) = \hat{\varphi}_k, \quad k \in \mathbb{Z}.$$

For $k = 0$ the factor on the left vanishes: the equation imposes the **necessary condition** $\hat{\varphi}_0 = 0$ (zero mean of the datum), and leaves $\hat{\psi}_0$ completely free (the solution, if it exists, is unique up to an additive constant). For $k \neq 0$, instead,

$$\hat{\psi}_k = \frac{\hat{\varphi}_k}{e^{2\pi i k \rho} - 1} \quad (\alpha = 2\pi\rho),$$

and here is the point: if ρ is irrational the denominator never vanishes, but it can be arbitrarily small, because $k\rho$ can come as close as one likes to an integer. These denominators are the **small divisors**. The whole theory consists in quantifying how small they can be as

a function of k , and in comparing this smallness with the rate at which $\widehat{\varphi}_k \rightarrow 0$ (that is, with the regularity of the datum).

Lemma 6.1: Small divisors lemma. For every $x \in \mathbb{R}$, with $\|x\|$ the distance of x from the nearest integer (Definition 3.3),

$$|e^{2\pi i x} - 1| = 2|\sin(\pi x)|, \quad \text{and} \quad 4\|x\| \leq |e^{2\pi i x} - 1| \leq 2\pi\|x\|.$$

Both inequalities are optimal: the left-hand one is an equality for $x = 1/2$, the right-hand one becomes an equality in the limit $x \rightarrow 0$.

Proof. The identity: $e^{2\pi i x} - 1 = e^{i\pi x}(e^{i\pi x} - e^{-i\pi x}) = e^{i\pi x} \cdot 2i \sin(\pi x)$, and $|e^{i\pi x}| \cdot |2i| = 1 \cdot 2 = 2$, hence $|e^{2\pi i x} - 1| = 2|\sin(\pi x)|$.

Put $t := \|x\| \in [0, 1/2]$. Since $x = m \pm t$ with m an integer, $|\sin(\pi x)| = |\sin(\pi(\pm t))| = \sin(\pi t)$ (the function $|\sin(\pi \cdot)|$ is periodic of period 1 and even). It remains to prove

$$2t \leq \sin(\pi t) \leq \pi t \quad \text{for } t \in [0, 1/2].$$

The upper bound $\sin s \leq s$ for $s \geq 0$ is classical (the function $s \mapsto s - \sin s$ has derivative $1 - \cos s \geq 0$ and equals 0 at 0); with $s = \pi t$ it gives $\sin(\pi t) \leq \pi t$. The lower bound is concavity: $s \mapsto \sin s$ is concave on $[0, \pi]$ (second derivative $-\sin s \leq 0$), hence on $[0, \pi]$ its graph lies above every chord; taking the chord between $s = 0$ (where $\sin = 0$) and $s = \pi/2$ (where $\sin = 1$), that is, the line $s \mapsto 2s/\pi$, one obtains $\sin s \geq 2s/\pi$ for $s \in [0, \pi/2]$, that is, with $s = \pi t$ and $t \in [0, 1/2]$, $\sin(\pi t) \geq 2t$. Multiplying by 2 one gets the claim.

Optimality: for $x = 1/2$, $\|x\| = 1/2$ and $|e^{i\pi} - 1| = 2 = 4 \cdot \frac{1}{2}$; for $x \rightarrow 0$, $2\sin(\pi x)/(2\pi x) \rightarrow 1$. $\square$

Remark 6.2: Double numerical check of Lemma 6.1. Block [A] of `verifica_cap06_kam.py` (run on 17 September 2026): on 200 000 equispaced points of $(0, 1]$ the minimum of the ratio $|e^{2\pi i x} - 1|/(4\|x\|)$ equals 1.0000000000, attained at $x = 0.500000 \checkmark$; the value $|e^{2\pi i x} - 1|$ is computed a first time with the complex exponential and a second time with the closed form $2|\sin(\pi x)|$, with a discrepancy always $< 10^{-12}$ (independent check). Block [H] verifies the upper bound $|e^{2\pi i x} - 1| \leq 2\pi\|x\|$ at $x = 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}$, with a ratio tending to 1 (already 0.9999983551 for $x = 10^{-3}$) $\checkmark$.

Example 6.1: Two rotation numbers compared: golden and Liouville. Double check (script `fig06_piccoli_divisori.py`, Appendix B, run on 17 September 2026). For $\rho = 1/\varphi = 0.618033988750$ (reduced golden ratio) the divisors $|e^{2\pi i k \rho} - 1|$, $k = 1, \dots, 60$, all stay above the curve $4K/k$ with $K = (3 - \sqrt{5})/2 = 0.381966011250$: the minimum of the ratio between the divisor and the curve equals 1.220046, attained at $k = 1 \checkmark$ (it must be ≥ 1 by Proposition 6.1 below).

For the Liouville number ρ_L constructed in block [G] (partial quotients a_n chosen so that $q_{n+1} \geq 2^{q_n}$; in the five-quotient version used by the figure, $a = [1, 1, 2, 6, 134217728]$, denominators $q_n = 1, 1, 2, 5, 32, 4294967301$, $\rho_L = 2550136835/4294967301 = 0.593750000007276$), the divisors *collapse* at the denominators: $|e^{2\pi i k \rho_L} - 1|$ equals 1.913881 for $k = 1$, 1.111140 for $k = 2$, 1.960343×10^{-1} for $k = 5$ and 1.462910×10^{-9} for $k = 32$ — nine orders of magnitude less than for the golden one at the same k . See Figure 6.1.

Remark 6.3: Why they are called this, and where they come from. The expression “small divisors” (petits diviseurs; in Russian *malye znamenateli*, transliterated) arose in celestial mechanics: in the perturbative series of planetary motions there appear denominators of the type $\langle k, \omega \rangle$, integer combinations of the frequencies, which can be extremely small when the frequencies are nearly resonant, and which threaten the convergence of the series. The very title of Arnold’s 1961 work on the circle is “*Malye znamenateli I*” (“Small denominators I”), and its 1963 sequel (Russ. Math. Surveys 18:6, 85–191, code K2 of the register) carries the same title for the Hamiltonian case. On the circle the phenomenon appears in the simplest possible form: a single frequency, and the divisors are the numbers $e^{2\pi i k \rho} - 1$ of Lemma 6.1. This is why the circle is the natural laboratory of KAM theory: everything that in the Hamiltonian case requires multidimensional algebra can here be read off in one line of Fourier series.

6.2. The arithmetic one needs: Diophantine conditions, Bryuno, Liouville

Definition 3.4 of Chapter 3 is given in the form “distance from a rational”. For small divisors one needs the equivalent form “distance of $q\rho$ from an integer”, which is the one that appears directly in Lemma 6.1.

Lemma 6.2: The two forms of the Diophantine condition. Let $\rho \in \mathbb{R} \setminus \mathbb{Q}$, $K > 0$, $\tau \geq 0$. The following are equivalent: (i) $|\rho - \frac{p}{q}| \geq \frac{K}{q^{2+\tau}}$ for every $p \in \mathbb{Z}$, $q \geq 1$ (that is, ρ is Diophantine of type (K, τ) , Definition 3.4); (ii) $\|q\rho\| \geq \frac{K}{q^{1+\tau}}$ for every integer $q \geq 1$. In that case, for every integer $k \neq 0$,

$$|e^{2\pi i k \rho} - 1| \geq \frac{4K}{|k|^{1+\tau}}.$$

Proof. For fixed $q \geq 1$, $\|q\rho\| = \min_{p \in \mathbb{Z}} |q\rho - p| = q \min_{p \in \mathbb{Z}} |\rho - \frac{p}{q}|$. Hence (i) says $\min_p |\rho - p/q| \geq K/q^{2+\tau}$, which multiplied by q is exactly $\|q\rho\| \geq K/q^{1+\tau}$, that is, (ii); and conversely by dividing by q . The last assertion follows from Lemma 6.1 with $x = k\rho$, using $\|(-k)\rho\| = \|k\rho\|$: $|e^{2\pi i k \rho} - 1| \geq 4\|k\rho\| \geq 4K/|k|^{1+\tau}$. $\square$

Proposition 6.1: The exact Diophantine constant of the golden ratio. Let $\rho^* = 1/\varphi = \varphi - 1 = [0; 1, 1, 1, \dots]$. Then

$$\inf_{q \geq 1} q \|q\rho^*\| = \frac{3 - \sqrt{5}}{2} = \varphi^{-2} = 0.381966011250\dots,$$

and the infimum is a minimum, attained at $q = 1$. In particular ρ^* is Diophantine of type $(K, 0)$ with $K = \varphi^{-2}$, and this is the largest possible constant for $\tau = 0$.

Proof. Step 1 (the convergents). The convergents of $[0; 1, 1, 1, \dots]$ are $p_n/q_n = F_n/F_{n+1}$, with $F_0 = 0$, $F_1 = F_2 = 1$, $F_{n+1} = F_n + F_{n-1}$ (Definition 3.2 and the recurrence of Proposition 3.1, with all the partial quotients equal to 1). Setting $\delta_n := q_n \rho^* - p_n = F_{n+1} \rho^* - F_n$, Binet’s formula $F_m = (\varphi^m - \psi^m)/\sqrt{5}$ with $\psi = -1/\varphi$ gives

$$\delta_n = \frac{\varphi^{n+1} - \psi^{n+1}}{\sqrt{5}} \cdot \frac{1}{\varphi} - \frac{\varphi^n - \psi^n}{\sqrt{5}} = \frac{\varphi^n - \psi^{n+1}\varphi^{-1} - \varphi^n + \psi^n}{\sqrt{5}} = \frac{\psi^n(1 - \psi\varphi^{-1})}{\sqrt{5}}.$$

Now $\psi\varphi^{-1} = -\varphi^{-2}$, hence $1 - \psi\varphi^{-1} = 1 + \varphi^{-2}$; and $1 + \varphi^{-2} = (\varphi^2 + 1)/\varphi^2 = (\varphi + 2)/(\varphi + 1)$ using $\varphi^2 = \varphi + 1$. A direct computation gives $(1 + \varphi^{-2})/\sqrt{5} = 1/\varphi$: indeed $\sqrt{5} = 2\varphi - 1$, and $(1 + \varphi^{-2})\varphi = \varphi + \varphi^{-1} = \varphi + (\varphi - 1) = 2\varphi - 1 = \sqrt{5}$. Therefore

$$\delta_n = \psi^n \varphi^{-1} = (-1)^n \varphi^{-n} \varphi^{-1} = (-1)^n \varphi^{-(n+1)}, \quad \|q_n \rho^*\| = |\delta_n| = \varphi^{-(n+1)}$$

(the second equality because $|\delta_n| = \varphi^{-(n+1)} \leq \varphi^{-2} < 1/2$ for $n \geq 1$, while for $n = 0$ one has $\delta_0 = \varphi^{-1} = 0.618\dots$, whose distance from the nearest integer is $1 - \varphi^{-1} = \varphi^{-2}$; the case $n = 0$ is therefore excluded from the formula and has to be treated separately, but $q_0 = q_1 = 1$ and the relevant value for $q = 1$ is the one for $n = 1$).

Step 2 (the products $c_n = q_n \|q_n \rho^*\|$). For $n \geq 1$,

$$c_n = F_{n+1} \varphi^{-(n+1)} = \frac{\varphi^{n+1} - \psi^{n+1}}{\sqrt{5} \varphi^{n+1}} = \frac{1 - (\psi/\varphi)^{n+1}}{\sqrt{5}} = \frac{1 + (-1)^n \varphi^{-2(n+1)}}{\sqrt{5}}.$$

The sequence (c_n) oscillates around $1/\sqrt{5} = 0.447213595500\dots$: the terms with n odd lie below and increase towards $1/\sqrt{5}$, those with n even lie above and decrease towards $1/\sqrt{5}$. The absolute minimum is therefore the first odd term, $n = 1$:

$$c_1 = \frac{1 - \varphi^{-4}}{\sqrt{5}} = \frac{1 - 0.145898033750}{2.236067977500} = 0.381966011250 = \varphi^{-2},$$

where the last equality is checked in closed form: $\sqrt{5} \varphi^{-2} = \varphi^{-2}(2\varphi - 1) = 2\varphi^{-1} - \varphi^{-2} = 2(\varphi - 1) - (2 - \varphi) = 3\varphi - 4$; and $1 - \varphi^{-4} = 1 - (2 - \varphi)^2 = 1 - (4 - 4\varphi + \varphi^2) = 1 - 4 + 4\varphi - \varphi - 1 = 3\varphi - 4$ ✓ (the two sides coincide).

Step 3 (from q a convergent to arbitrary q). Let $q \geq 1$ and let $n \geq 1$ be such that $q_n \leq q < q_{n+1}$ (it exists, because $q_1 = 1$ and $q_n \rightarrow \infty$). By Theorem 3.4 (best approximation of the second kind), $\|q\rho^*\| \geq \|q_n \rho^*\|$; and $q \geq q_n$. Hence

$$q \|q\rho^*\| \geq q_n \|q_n \rho^*\| = c_n \geq \min_{m \geq 1} c_m = c_1 = \varphi^{-2}.$$

Since for $q = 1$ one has exactly $1 \cdot \|\rho^*\| = \varphi^{-2}$, the infimum is attained and equals φ^{-2} . By Lemma 6.2, ρ^* is Diophantine of type $(\varphi^{-2}, 0)$; no $K > \varphi^{-2}$ works, because it would already fail at $q = 1$. $\square$

Remark 6.4: Double check and comparison with Chapter 3. Block [C] of `verifica_cap06_kam.py`: the minimum of $q \|q\rho^*\|$ over $1 \leq q \leq 200\,000$ equals 0.381966011250, attained at $q = 1$, with discrepancy 0 (to 10^{-12}) from the closed value $(3 - \sqrt{5})/2$ ✓. The value should be compared with two results of Chapter 3: the $\liminf$ of the products along the convergents is $1/\sqrt{5} = 0.447214$ (Proposition 3.2), greater than the absolute minimum 0.381966 — the infimum is attained at the beginning, not at infinity; and the general estimate of Proposition 3.5, with partial quotients bounded by $M = 1$, gives $\|q\rho^*\| \geq 1/((M+2)q) = 1/(3q)$, that is, $K \geq 1/3 = 0.3\bar{3}$: consistent with (and weaker than) the exact value 0.381966 ✓. Two independent routes, the same order of magnitude, with the second correctly bounding the first from below.

Definition 6.2: Bryuno number. Let $\rho \in (0, 1) \setminus \mathbb{Q}$ with convergents p_n/q_n (Definition 3.2). One says that ρ satisfies the **Bryuno condition** (or that it is a **Bryuno number**, $\rho \in \mathcal{B}$) if

$$B(\rho) := \sum_{n \geq 0} \frac{\log q_{n+1}}{q_n} < \infty.$$

Proposition 6.2: Every Diophantine number is a Bryuno number. If ρ is Diophantine of type (K, τ) with $0 < K \leq 1$ and $\tau \geq 0$, then $\rho \in \mathcal{B}$. The inclusion $\mathcal{CD} \subset \mathcal{B}$ is strict.

Proof. Step 1 (growth of the denominators). From Theorem 3.3, $\|q_n \rho\| < 1/q_{n+1}$. From the Diophantine condition in the form of Lemma 6.2(ii), $\|q_n \rho\| \geq K/q_n^{1+\tau}$. Combining: $K/q_n^{1+\tau} < 1/q_{n+1}$, that is,

$$q_{n+1} < \frac{q_n^{1+\tau}}{K}, \quad \text{hence} \quad \log q_{n+1} < (1 + \tau) \log q_n + \log \frac{1}{K}.$$

Step 2 (geometric growth of q_n). From the recurrence $q_{n+1} = a_{n+1}q_n + q_{n-1} \geq q_n + q_{n-1}$ (Proposition 3.1) it follows that $q_{n+1} \geq 2q_{n-1}$, hence $q_n \geq 2^{\lfloor n/2 \rfloor}$ for every $n \geq 0$: the denominators grow at least geometrically.

Step 3 (convergence of the series). By Step 1,

$$\frac{\log q_{n+1}}{q_n} \leq \frac{(1 + \tau) \log q_n + \log(1/K)}{q_n}.$$

By Step 2, $\log q_n \geq 0$ and $q_n \geq 2^{\lfloor n/2 \rfloor}$, hence $\log q_n/q_n \leq C' n/2^{\lfloor n/2 \rfloor}$ and $1/q_n \leq 2^{-\lfloor n/2 \rfloor}$ with $C' = \log 2$. Both majorising sequences are general terms of convergent series (ratio test: $\sum n 2^{-n/2} < \infty$, $\sum 2^{-n/2} < \infty$). Hence $B(\rho) < \infty$.

Strictness. It suffices to exhibit a Bryuno number which is not Diophantine: take ρ with partial quotients defined by $a_{n+1} := \lceil q_n^n \rceil$, so that $q_{n+1} \geq q_n^n$ (more than polynomial growth: ρ is not Diophantine of any type, because $\|q_n \rho\| < 1/q_{n+1} \leq q_n^{-n}$ eventually beats every $K/q_n^{1+\tau}$), while $q_{n+1} \leq (q_n^n + 1)q_n + q_n \leq 3q_n^{n+1}$ gives $\log q_{n+1}/q_n \leq (\log 3 + (n+1) \log q_n)/q_n$, the general term of a convergent series by Step 2: $\rho \in \mathcal{B} \setminus \mathcal{CD}$. $\square$

Example 6.2: Bryuno for the golden ratio, non-Bryuno for a constructed Liouville number. Double check (blocks [G] and [I] of `verifica_cap06_kam.py`). For $\rho^* = 1/\varphi$ the denominators are the Fibonacci numbers and the Bryuno partial sums equal 2.19482094 ($N = 5$), 3.11702988 ($N = 10$), 3.28360386 ($N = 20$), 3.28612938 ($N = 40$), 3.28612970 ($N = 55$): they converge (the last two digits stabilise) $\checkmark$, consistently with Proposition 6.2, ρ^* being Diophantine of type $(\varphi^{-2}, 0)$.

For the number ρ_L constructed in block [G] by choosing at each step the smallest a_{n+1} with $q_{n+1} \geq 2^{q_n}$ (partial quotients $[1, 1, 2, 6, 134217728, 268435456]$, denominators 1, 1, 2, 5, 32, 4294967301, 1152921505949024288), every term of the series satisfies

$$\frac{\log q_{n+1}}{q_n} \geq \frac{q_n \log 2}{q_n} = \log 2 = 0.69314718 \dots,$$

hence the series *diverges*: $\rho_L \notin \mathcal{B}$, and a fortiori $\rho_L \notin \mathcal{CD}$ $\checkmark$ (the computed partial sums, 0; 0.693; 1.498; 2.191; 2.884, increase by at least $\log 2$ at each useful step; the last term printed is ≈ 0 only because the recursion was truncated at the last denominator available in `float64`). Note that ρ_L as constructed is a *rational truncation*: it is the finite model of the irrational Liouville number obtained by continuing the recursion to infinity, and it is used in the scripts only for the first k , where it coincides with it.

Remark 6.5: The arithmetic hierarchy in one line. We denote by $\mathcal{CD}$ the set of Diophantine numbers (of some type (K, τ)), by $\mathcal{B}$ the Bryuno numbers and by $\mathcal{L}$ the Liouville numbers (Definition 3.5). One has, by Proposition 6.2 and by Theorem 3.6,

$$\mathcal{CD} \subsetneq \mathcal{B}, \quad \text{Leb}(\mathcal{CD}) = \text{full}, \quad \mathcal{CD} \cap \mathcal{L} = \emptyset.$$

Between $\mathcal{CD}$ and $\mathcal{B}$ there lies a third class, the **condition** $\mathcal{H}$ introduced by Yoccoz and named in honour of Herman, which governs the global analytic linearisation (Section 6.8): $\mathcal{CD} \subset \mathcal{H} \subset \mathcal{B}$, with both inclusions strict. The three classes are drawn, with their typical representatives, in Figure 6.4.

6.3. The homological equation in the analytic case

Definition 6.3: Complex strips and norms. For $b > 0$ we put

$$S_b := \{\theta \in \mathbb{C}/2\pi\mathbb{Z} : |\operatorname{Im} \theta| < b\},$$

the **strip** of half-width b around the real circle. For g holomorphic and bounded on S_b we write $\|g\|_b := \sup_{\theta \in S_b} |g(\theta)|$. We say that g is **real on the reals** if $g(\mathbb{R}) \subset \mathbb{R}$; in that case $\widehat{g}_{-k} = \overline{\widehat{g}_k}$ for every k . We denote by $\mathcal{A}_b$ the space of the functions holomorphic, 2π -periodic, bounded on S_b and real on the reals.

Lemma 6.3: Cauchy estimate on strips. Let g be holomorphic and bounded on S_b and let $0 < \delta < b$. Then $\|g'\|_{b-\delta} \leq \|g\|_b/\delta$.

Proof. Let $\theta_0 \in S_{b-\delta}$ and let $0 < r < \delta$. The closed disc $\overline{D}(\theta_0, r)$ is contained in S_b , because for $|z - \theta_0| \leq r$ one has $|\operatorname{Im} z| \leq |\operatorname{Im} \theta_0| + r < (b - \delta) + \delta = b$. Cauchy's integral formula for the derivative (Volume III, Chapter 15) gives

$$g'(\theta_0) = \frac{1}{2\pi i} \oint_{|z-\theta_0|=r} \frac{g(z)}{(z-\theta_0)^2} dz, \quad |g'(\theta_0)| \leq \frac{1}{2\pi} \cdot 2\pi r \cdot \frac{\|g\|_b}{r^2} = \frac{\|g\|_b}{r}.$$

Letting $r \rightarrow \delta^-$ one obtains $|g'(\theta_0)| \leq \|g\|_b/\delta$, and passing to the sup over $\theta_0 \in S_{b-\delta}$ one gets the claim. $\square$

Lemma 6.4: Exponential decay of the Fourier coefficients. Let $g \in \mathcal{A}_b$ (or even merely holomorphic, 2π -periodic and bounded on S_b). Then $|\widehat{g}_k| \leq \|g\|_b e^{-|k|b}$ for every $k \in \mathbb{Z}$. Conversely, if $|\widehat{g}_k| \leq M e^{-|k|b}$ for every k , then the series $\sum_k \widehat{g}_k e^{ik\theta}$ converges uniformly on every $S_{b'}$ with $b' < b$ and defines a holomorphic function on S_b .

Proof. For $|s| < b$ the function $\theta \mapsto g(\theta + is)$ is continuous and 2π -periodic on $\mathbb{R}$, and by holomorphy (Cauchy's theorem applied to the rectangle with vertices $0, 2\pi, 2\pi + is, is$, whose vertical sides give opposite contributions by periodicity) the Fourier coefficient does not change if one translates the path:

$$\widehat{g}_k = \frac{1}{2\pi} \int_0^{2\pi} g(\theta + is) e^{-ik(\theta + is)} d\theta, \quad \text{hence} \quad |\widehat{g}_k| \leq \|g\|_b e^{ks}.$$

Choosing $s \rightarrow -b^+$ for $k > 0$ and $s \rightarrow b^-$ for $k < 0$ one obtains $|\widehat{g}_k| \leq \|g\|_b e^{-|k|b}$ (for $k = 0$ the statement is trivial). The converse: on $S_{b'}$ one has $|\widehat{g}_k e^{ik\theta}| \leq M e^{-|k|b} e^{|k|b'} = M e^{-|k|(b-b')}$, the general term of a convergent geometric series; the convergence is uniform, and a uniform limit of trigonometric polynomials (holomorphic) is holomorphic on every $S_{b'}$, hence on $S_b = \bigcup_{b' < b} S_{b'}$. $\square$

Lemma 6.5: The sum of the small divisors. For $\tau \geq 0$ we put

$$\Lambda(\tau) := \Gamma(2 + \tau) + \left(\frac{1 + \tau}{e}\right)^{1+\tau}.$$

Then, for every $\delta \in (0, 1]$,

$$\sum_{k \geq 1} k^{1+\tau} e^{-k\delta} \leq \frac{\Lambda(\tau)}{\delta^{2+\tau}}.$$

Values: $\Lambda(0) = 1 + e^{-1} = 1.367879441\dots$, $\Lambda(1/2) = 1.739256667\dots$, $\Lambda(1) = 2 + 4e^{-2} = 2.541341133\dots$

Proof. Let $h(t) := t^{1+\tau} e^{-t\delta}$ on $[0, \infty)$. One has $h'(t) = t^\tau e^{-t\delta}((1 + \tau) - t\delta)$: h increases on $[0, t_*]$ and decreases on $[t_*, \infty)$, with $t_* = (1 + \tau)/\delta$ and

$$\max h = h(t_*) = \left(\frac{1 + \tau}{\delta}\right)^{1+\tau} e^{-(1+\tau)} = \left(\frac{1 + \tau}{e}\right)^{1+\tau} \frac{1}{\delta^{1+\tau}} \leq \left(\frac{1 + \tau}{e}\right)^{1+\tau} \frac{1}{\delta^{2+\tau}},$$

the last inequality because $\delta \leq 1$. Let $n_0 := \lfloor t_* \rfloor$ (possibly $n_0 = 0$). For $1 \leq k \leq n_0$ the interval $[k - 1, k]$ is contained in $[0, t_*]$ (because $k \leq n_0 \leq t_*$), where h is increasing, hence $h(k) \leq \int_{k-1}^k h$; for $k \geq n_0 + 2$ the interval $[k - 1, k]$ is contained in $[n_0 + 1, \infty) \subseteq [t_*, \infty)$ (because $n_0 + 1 > t_*$), where h decreases, hence $h(k) \leq \int_{k-1}^k h$. The intermediate term $k = n_0 + 1$ falls astride t_* (the interval $[n_0, n_0 + 1]$ contains t_* , and neither of the two monotonicity arguments applies on the whole interval), but trivially $h(n_0 + 1) \leq \max h$. Summing, and isolating this single term:

$$\sum_{k \geq 1} h(k) \leq \int_0^{n_0} h + h(n_0 + 1) + \int_{n_0+1}^{\infty} h \leq \int_0^{\infty} h(t) dt + \max h.$$

Finally, with the substitution $s = t\delta$,

$$\int_0^{\infty} t^{1+\tau} e^{-t\delta} dt = \frac{1}{\delta^{2+\tau}} \int_0^{\infty} s^{1+\tau} e^{-s} ds = \frac{\Gamma(2 + \tau)}{\delta^{2+\tau}}.$$

Adding the two upper bounds one obtains the claim. $\square$

Remark 6.6: Double check of the constant $\Lambda(\tau)$. Block [B] of `verifica_cap06_kam.py`: for $\tau \in \{0, \frac{1}{2}, 1\}$ and $\delta \in \{1; 0.7; 0.5; 0.3; 0.2; 0.1; 0.05; 0.02; 0.01\}$ the sum is computed directly (truncated when the general term falls below 10^{-18} times the partial sum) and compared with $\Lambda(\tau)/\delta^{2+\tau}$: the largest ratio observed is 0.731052 for $\tau = 0$, 0.764315 for $\tau = 1/2$, 0.786986 for $\tau = 1$ — always < 1 , hence the estimate holds with a margin $\checkmark$. Independent check for $\tau = 0$ with the closed form $\sum_{k \geq 1} k e^{-k\delta} = e^{-\delta}/(1 - e^{-\delta})^2$: for $\delta = 1; 0.1; 0.01$ the two computations agree to within 1.8×10^{-14} in relative value $\checkmark$. Note that the estimate requires $\delta \leq 1$: for large δ the term $\max h$ would have to be bounded differently. Throughout the chapter we shall use the lemma only with $\delta \leq \delta_0 \leq 1$.

Theorem 6.1: Homological equation, analytic case. Let ρ be Diophantine of type (K, τ) , $\alpha = 2\pi\rho$, $b > 0$ and $\varphi \in \mathcal{A}_b$ with zero mean ($\widehat{\varphi}_0 = 0$). Then, for every δ with $0 < \delta \leq \min(1, b)$, there exists one and only one ψ holomorphic on $S_{b-\delta}$, 2π -periodic, real on the reals and with zero mean, such that

$$\psi \circ R_\alpha - \psi = \varphi \quad \text{on } S_{b-\delta},$$

and the estimate

$$\|\psi\|_{b-\delta} \leq \frac{\Lambda(\tau)}{2K} \cdot \frac{\|\varphi\|_b}{\delta^{2+\tau}}$$

holds. All the solutions are obtained by adding a real constant to ψ .

Proof. Uniqueness. If ψ_1, ψ_2 solve the equation, $g := \psi_1 - \psi_2$ satisfies $g \circ R_\alpha = g$, that is, $\widehat{g}_k(e^{ik\alpha} - 1) = 0$ for every k ; since ρ is irrational, $e^{ik\alpha} \neq 1$ for $k \neq 0$, hence $\widehat{g}_k = 0$ for $k \neq 0$ and g is constant. With the zero-mean normalisation, $g = 0$.

Existence. We put

$$\widehat{\psi}_k := \frac{\widehat{\varphi}_k}{e^{ik\alpha} - 1} \quad (k \neq 0), \quad \widehat{\psi}_0 := 0.$$

By Lemma 6.4, $|\widehat{\varphi}_k| \leq \|\varphi\|_b e^{-|k|b}$; by Lemma 6.2,

$$|\widehat{\psi}_k| \leq \frac{|k|^{1+\tau}}{4K} |\widehat{\varphi}_k| \leq \frac{\|\varphi\|_b}{4K} |k|^{1+\tau} e^{-|k|b}.$$

For $\theta \in S_{b-\delta}$, $|e^{ik\theta}| = e^{-k \operatorname{Im} \theta} \leq e^{|k|(b-\delta)}$, hence

$$\sum_{k \neq 0} |\widehat{\psi}_k| |e^{ik\theta}| \leq \frac{\|\varphi\|_b}{4K} \sum_{k \neq 0} |k|^{1+\tau} e^{-|k|\delta} = \frac{\|\varphi\|_b}{2K} \sum_{k \geq 1} k^{1+\tau} e^{-k\delta} \leq \frac{\Lambda(\tau)}{2K} \cdot \frac{\|\varphi\|_b}{\delta^{2+\tau}},$$

where the last inequality is Lemma 6.5 (applicable because $\delta \leq 1$). The series therefore converges totally on $S_{b-\delta}$; its sum ψ is holomorphic (a uniform limit of trigonometric polynomials on every narrower strip, as in Lemma 6.4), 2π -periodic, with zero mean, and satisfies the announced estimate. That it solves the equation is checked coefficient by coefficient: $\widehat{\psi}_k(e^{ik\alpha} - 1) = \widehat{\varphi}_k$ for $k \neq 0$ and, for $k = 0$, $0 = \widehat{\varphi}_0$ by hypothesis.

Reality. Since φ is real on the reals, $\widehat{\varphi}_{-k} = \overline{\widehat{\varphi}_k}$; and

$$\overline{\left(\frac{1}{e^{ik\alpha} - 1} \right)} = \frac{1}{e^{-ik\alpha} - 1},$$

hence $\widehat{\psi}_{-k} = \overline{\widehat{\psi}_k}$ and ψ is real on the reals. □

Corollary 6.1: Arbitrarily small loss of domain. *With the hypotheses of Theorem 6.1, the solution ψ is holomorphic on $S_{b'}$ for every $b' < b$: the homological equation does not destroy analyticity, it only narrows its strip, and by as little as one wishes. The price is the constant: $\|\psi\|_{b'}$ blows up like $(b - b')^{-(2+\tau)}$ as $b' \rightarrow b$.*

Proof. Given $b' < b$, apply the theorem with $\delta := \min(1, b - b')$; if $\delta = 1 < b - b'$ the strip obtained S_{b-1} contains $S_{b'}$. In both cases ψ is holomorphic on $S_{b'}$ and the estimate is the one of the statement with $\delta = b - b'$ when $b - b' \leq 1$. □

Example 6.3: The homological equation for $\varphi = \varepsilon \cos \theta$, in two independent ways.

Let $\varphi(\theta) = \varepsilon \cos \theta$ (zero mean, $\widehat{\varphi}_{\pm 1} = \varepsilon/2$, all the other coefficients zero).

Route 1 (Fourier series). $\widehat{\psi}_{\pm 1} = \frac{\varepsilon/2}{e^{\pm i\alpha} - 1}$, hence

$$\psi(\theta) = 2 \operatorname{Re} \left(\frac{\varepsilon/2}{e^{i\alpha} - 1} e^{i\theta} \right).$$

Route 2 (closed form). From $e^{i\alpha} - 1 = e^{i\alpha/2}(e^{i\alpha/2} - e^{-i\alpha/2}) = 2i e^{i\alpha/2} \sin(\alpha/2)$ one obtains

$$\frac{\varepsilon/2}{e^{i\alpha} - 1} e^{i\theta} = \frac{\varepsilon}{4i \sin(\alpha/2)} e^{i(\theta - \alpha/2)}, \quad \psi(\theta) = \frac{\varepsilon \sin(\theta - \frac{\alpha}{2})}{2 \sin(\frac{\alpha}{2})}.$$

Direct verification. $\psi(\theta + \alpha) - \psi(\theta) = \frac{\varepsilon}{2 \sin(\alpha/2)} \left[\sin(\theta + \frac{\alpha}{2}) - \sin(\theta - \frac{\alpha}{2}) \right] = \frac{\varepsilon}{2 \sin(\alpha/2)} \cdot 2 \cos \theta \sin \frac{\alpha}{2} = \varepsilon \cos \theta \checkmark$ (prosthaphaeresis formula).

Double numerical check (block [E]). With $\alpha = 2\pi/\varphi_{\text{golden}} = 3.8832220774\dots$ and $\varepsilon = 10^{-2}$, the two routes give the same values to within 10^{-15} at the points $\theta = 0; 0.7; 2.3; 4.9 \checkmark$, and the homological equation is verified directly at $\theta = 0.3; 1.1; 5.0$ with a discrepancy $< 1.8 \times 10^{-18} \checkmark$. Observe that $\|\psi\|_{\infty} = \varepsilon/(2|\sin(\alpha/2)|)$: the “small divisor” here is $2 \sin(\alpha/2)$, and the solution blows up when α approaches a multiple of 2π , that is, when ρ approaches an integer.

Example 6.4: The coefficients of the solution: golden against Liouville. Double check (script `fig06_coefficienti_psi.py`, Appendix B, run on 17 September 2026). One fixes *the same* analytic datum $\widehat{\varphi}_k = e^{-a|k|}$ with $a = 0.30$ and solves the homological equation for two different rotation numbers.

For $\rho = 1/\varphi$ (Diophantine of type $(K, 0)$, $K = \varphi^{-2}$) the upper bound of Theorem 6.1 applied coefficient by coefficient gives $|\widehat{\psi}_k| \leq |k|e^{-a|k|}/(4K)$: the largest ratio between the actual value and this upper bound, over $1 \leq k \leq 60$, equals $0.819641 \leq 1 \checkmark$. The coefficients decay exponentially all the same: $|\widehat{\psi}_1| = 3.974208 \times 10^{-1}$ and $|\widehat{\psi}_{60}| = 2.987557 \times 10^{-8}$, a ratio of $1.330 \times 10^7 \checkmark$ — that is, ψ remains analytic, on a barely narrower strip.

For the Liouville ρ_L the coefficients *climb back up* at the denominators: $|\widehat{\psi}_{32}| = 4.629725 \times 10^4$ against 5.254795×10^{-5} for the golden one at the same k , a ratio of 8.81×10^8 . The series of ψ no longer converges: this is the phenomenon that Section 6.4 turns into a non-existence theorem. See Figure 6.6.

6.4. The C^r case and the Liouville catastrophe

In the analytic case the loss is a strip; in the C^r case it is a certain number of derivatives. The computation is identical, with the exponential decay of the coefficients replaced by a polynomial decay.

Theorem 6.2: Homological equation, C^r case. Let ρ be Diophantine of type (K, τ) , $\alpha = 2\pi\rho$, and $\varphi \in C^r(\mathbb{T})$ with zero mean, r an integer, $r > \tau + 2$. Then the homological equation $\psi \circ R_{\alpha} - \psi = \varphi$ has one and only one continuous solution ψ with zero mean, and $\psi \in C^s(\mathbb{T})$ for every integer s with $0 \leq s < r - \tau - 2$, with the explicit estimate

$$\|\psi^{(s)}\|_{\infty} \leq \frac{\zeta(r - \tau - s - 1)}{2K} \|\varphi^{(r)}\|_{\infty},$$

where ζ is the Riemann zeta function.

Proof. Step 1 (decay of the coefficients of the datum). Integrating by parts r times, using periodicity (the boundary terms cancel at each step),

$$\widehat{\varphi}_k = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) e^{-ik\theta} d\theta = \frac{1}{(ik)^r} \cdot \frac{1}{2\pi} \int_0^{2\pi} \varphi^{(r)}(\theta) e^{-ik\theta} d\theta = \frac{\widehat{\varphi^{(r)}}_k}{(ik)^r},$$

hence $|\widehat{\varphi}_k| \leq \|\varphi^{(r)}\|_{\infty}/|k|^r$ for $k \neq 0$.

Step 2 (coefficients of the solution). As in Theorem 6.1, the only possible choice is $\hat{\psi}_k = \hat{\varphi}_k / (e^{ik\alpha} - 1)$ for $k \neq 0$, $\hat{\psi}_0 = 0$, and by Lemma 6.2

$$|\hat{\psi}_k| \leq \frac{|k|^{1+\tau}}{4K} |\hat{\varphi}_k| \leq \frac{\|\varphi^{(r)}\|_\infty}{4K} |k|^{1+\tau-r}.$$

Step 3 (convergence of the s times differentiated series). For $0 \leq s < r - \tau - 2$,

$$\sum_{k \neq 0} |k|^s |\hat{\psi}_k| \leq \frac{\|\varphi^{(r)}\|_\infty}{4K} \sum_{k \neq 0} |k|^{s+1+\tau-r} = \frac{\|\varphi^{(r)}\|_\infty}{2K} \sum_{k \geq 1} \frac{1}{k^{r-\tau-s-1}} = \frac{\zeta(r-\tau-s-1)}{2K} \|\varphi^{(r)}\|_\infty,$$

a convergent series because the exponent $r - \tau - s - 1$ is > 1 (equivalent to $s < r - \tau - 2$). A Fourier series whose coefficients satisfy $\sum_k |k|^s |\hat{\psi}_k| < \infty$ defines a function of class C^s , and the series differentiated s times term by term converges uniformly to $\psi^{(s)}$ (Volume III, Chapter 23: term-by-term differentiation under uniform convergence of the derivatives); the upper bound for $\|\psi^{(s)}\|_\infty$ is the sum just computed. Uniqueness (up to constants) and reality are proved exactly as in Theorem 6.1. $\square$

Corollary 6.2: The C^∞ case. If ρ is Diophantine and $\varphi \in C^\infty(\mathbb{T})$ has zero mean, then the solution ψ of the homological equation is C^∞ .

Proof. With s fixed, apply Theorem 6.2 with any integer $r > s + \tau + 2$ (possible because $\varphi \in C^r$ for every r): one obtains $\psi \in C^s$. Since s is arbitrary and the zero-mean solution is unique (hence does not depend on r), $\psi \in \bigcap_s C^s = C^\infty$. $\square$

Remark 6.7: What we import: the optimal loss $1 + \tau + \epsilon$. The loss of $2 + \tau$ derivatives in Theorem 6.2 is the one obtained by the elementary method of absolute convergence of the Fourier series, and it is one derivative worse than the optimal one. Measuring the regularity in the Hölder–Zygmund scales $C^{s+\epsilon}$ rather than in the integer classes C^s alone, the loss reduces to $1 + \tau + \epsilon$ for every $\epsilon > 0$: we **import** this statement from A. Katok, B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, CUP 1995, §2.9 (register, code R24; data verified on 16 September 2026) and from M. R. Herman, *Publ. Math. IHÉS* **49** (1979), ch. II (code K8), recording it in Appendix A. What is assumed: the definition of the Hölder–Zygmund norms and the interpolation estimates between them; nothing of what we prove in the sequel depends on it — the Newton scheme of Section 6.7 uses only the analytic case, that is, Theorem 6.1, proved in full.

Theorem 6.3: Non-solvability for a Liouville rotation number. Let $\rho \in (0, 1)$ be irrational with convergents p_n/q_n such that

$$q_{n+1} \geq 2^{q_n} \quad \text{for every } n \geq 1$$

(a Liouville number, Definition 3.5: such a ρ exists, and is constructed by choosing the partial quotients recursively). We put $c_n := \|q_n \rho\|$ and

$$\varphi(\theta) := \sum_{n \geq 1} 2c_n \cos(q_n \theta).$$

Then: **(i)** $\varphi \in C^\infty(\mathbb{T})$, is real, has zero mean; **(ii)** the homological equation $\psi \circ R_\alpha - \psi = \varphi$ (with $\alpha = 2\pi\rho$) **admits no solution** $\psi \in L^1(\mathbb{T})$; in particular it admits no continuous one, nor L^2 , nor C^k for any k .

Proof. (i). By Theorem 3.3, $c_n = \|q_n \rho\| < 1/q_{n+1} \leq 2^{-q_n}$. For every integer $m \geq 0$, the series differentiated m times term by term has majorant $\sum_n 2c_n q_n^m \leq \sum_n 2q_n^m 2^{-q_n}$, which converges: indeed $q_n \geq 2^{\lfloor n/2 \rfloor} \rightarrow \infty$ (Step 2 of the proof of Proposition 6.2) and $q^m 2^{-q} \rightarrow 0$ faster than any geometric sequence as q grows; more precisely, for $q \geq (2m+2)\log_2(2m+2)$ one has $q^m 2^{-q} \leq 2^{-q/2}$, and $\sum_n 2^{-q_n/2}$ converges by comparison with $\sum_n 2^{-2^{\lfloor n/2 \rfloor - 1}}$. Hence all the differentiated series converge totally, and $\varphi \in C^\infty$ (term-by-term differentiation under uniform convergence of the derivatives). It is real because it is a sum of cosines with real coefficients; it has zero mean because no summand has a constant term ($q_n \geq 1$).

(ii). Suppose for a contradiction that $\psi \in L^1(\mathbb{T})$ solves the equation. The Fourier coefficients of $\psi \circ R_\alpha$ are $\widehat{\psi}_k e^{ik\alpha}$ (change of variable in the integral, legitimate in L^1), hence, equating the coefficients of the two sides, for every $k \neq 0$

$$\widehat{\psi}_k (e^{2\pi i k \rho} - 1) = \widehat{\varphi}_k.$$

For $k = q_n$ one has $\widehat{\varphi}_{q_n} = c_n$ (the coefficient of $e^{iq_n \theta}$ in $2c_n \cos(q_n \theta)$ is c_n ; the other summands do not contribute because the q_n are pairwise distinct, being strictly increasing from $n \geq 1$ onwards). Hence, by Lemma 6.1 in its *upper bound*,

$$|\widehat{\psi}_{q_n}| = \frac{c_n}{|e^{2\pi i q_n \rho} - 1|} \geq \frac{c_n}{2\pi \|q_n \rho\|} = \frac{c_n}{2\pi c_n} = \frac{1}{2\pi} \quad \text{for every } n \geq 1.$$

But the Riemann–Lebesgue lemma asserts that the Fourier coefficients of an L^1 function tend to 0 as $|k| \rightarrow \infty$ (Volume III, Chapter 23), whereas here $|\widehat{\psi}_{q_n}| \geq 1/(2\pi) = 0.159154\dots$ along the sequence $q_n \rightarrow \infty$: contradiction. There is no solution in L^1 , and therefore none in any of its subspaces either (C^0, L^2, C^k) . $\square$

Remark 6.8: Double check and anatomy of the counterexample. Blocks [G] and [H] of `verifica_cap06_kam.py`. Block [G] constructs the partial quotients recursively with $q_{n+1} \geq 2^{q_n}$ and prints, for each n , the check $q_{n+1} \geq 2^{q_n}$ and the value $1/q_{n+1}$ that bounds $\|q_n \rho\|$ from above: 5.0×10^{-1} , 2.0×10^{-1} , 3.125×10^{-2} , 2.328×10^{-10} , 8.674×10^{-19} ✓. Block [H] verifies the key inequality $|e^{2\pi i x} - 1| \leq 2\pi \|x\|$ and the resulting lower bound $c/|e^{2\pi i c} - 1| \geq 1/(2\pi)$ for $c = 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}$, obtaining respectively 0.15915520, 0.15915494, 0.15915494, 0.15915494, all $\geq 1/(2\pi) = 0.1591549431$ ✓: the lower bound is saturated in the limit, that is, the counterexample is exactly at the limit of what Lemma 6.1 allows.

Anatomy. The choice $c_n = \|q_n \rho\|$ is not accidental: it is the largest one that keeps φ of class C^∞ (because $c_n < 2^{-q_n}$ decays faster than any power of $1/q_n$) and at the same time the smallest one that prevents $\widehat{\psi}_{q_n}$ from tending to zero. The two requirements meet only because ρ is Liouville: if ρ were Diophantine of type (K, τ) , one would have $c_n \geq K/q_n^{1+\tau}$, hence c_n could not decay faster than a power, and the series $\sum 2c_n \cos(q_n \theta)$ would not be C^∞ (indeed, it would not even be C^m for large m). The construction is a tailor-made counterexample, cut to the sequence of denominators of that particular ρ .

Remark 6.9: What Theorem 6.3 does not say. The theorem shows that for that Liouville ρ there exists one C^∞ datum with zero mean which is not solvable. It does not say, and it is not true in this form, that every non-Diophantine ρ is pathological for every datum: for every irrational ρ there are infinitely many solvable data (all the trigonometric polynomials with zero mean, for instance, for which the solution is a trigonometric polynomial). Nor does it say that the Liouville numbers are all non-Bryuno: Proposition 6.2 and Example 6.2 show that the two properties must be checked separately (there exist Liouville numbers with $B(\rho) < \infty$, obtained by letting the denominators grow like $q_{n+1} \approx q_n^n$ rather than like 2^{q_n}).

The statement is, and remains, a statement of existence of a counterexample.

6.5. The Gottschalk–Hedlund lemma

The Fourier approach answers the question “does a *regular* ψ exist?” and requires arithmetic conditions on ρ . There is a second answer, of a completely different nature — topological, not arithmetic — to the question “does a *continuous* ψ exist?”. It is the Gottschalk–Hedlund lemma, valid for every minimal homeomorphism of a compact space, hence in particular for every irrational rotation and (by Denjoy’s theorem) for every $C^{1+\text{bv}}$ diffeomorphism with irrational rotation number.

Definition 6.4: Birkhoff sums and continuous coboundaries. Let $T : X \rightarrow X$ be continuous on a compact metric space X and let $\phi \in C(X)$. The **Birkhoff sums** are $S_n\phi := \sum_{j=0}^{n-1} \phi \circ T^j$ ($S_0\phi := 0$). One says that ϕ is a **continuous coboundary** if there exists $v \in C(X)$ with $\phi = v \circ T - v$.

Theorem 6.4: Gottschalk–Hedlund lemma. Let X be a compact metric space, $T : X \rightarrow X$ a **minimal** homeomorphism (every orbit is dense) and $\phi \in C(X)$ real-valued. The following are equivalent: **(i)** ϕ is a continuous coboundary: there exists $v \in C(X)$ with $\phi = v \circ T - v$; **(ii)** there exists $x_0 \in X$ such that $\sup_{n \geq 0} |S_n\phi(x_0)| < \infty$; **(iii)** $\sup_{n \geq 0} \sup_{x \in X} |S_n\phi(x)| < \infty$.

Proof. **(i) $\Rightarrow$ (iii).** If $\phi = v \circ T - v$, the telescoping sum gives $S_n\phi = v \circ T^n - v$, hence $|S_n\phi(x)| \leq 2\|v\|_\infty$ for every n and every x . **(iii) $\Rightarrow$ (ii)** is trivial.

(ii) $\Rightarrow$ (i). Let $M := \sup_n |S_n\phi(x_0)| < \infty$. Consider the **skew-product**

$$\tilde{T} : X \times \mathbb{R} \rightarrow X \times \mathbb{R}, \quad \tilde{T}(x, t) := (Tx, t + \phi(x)),$$

a homeomorphism (its inverse is $(x, t) \mapsto (T^{-1}x, t - \phi(T^{-1}x))$, continuous). By induction, $\tilde{T}^n(x, t) = (T^n x, t + S_n\phi(x))$.

Step 1 (an invariant compact set). Let $Y := \overline{\{(T^n x_0, S_n\phi(x_0)) : n \geq 0\}}$, the closure being taken in $X \times \mathbb{R}$. By hypothesis the second coordinate stays in $[-M, M]$, hence $Y \subseteq X \times [-M, M]$ is closed and bounded in a locally compact metric space: Y is **compact** and non-empty. Moreover $\tilde{T}$ maps the set of the pairs $(T^n x_0, S_n\phi(x_0))$, $n \geq 0$, into itself (it maps the n -th into the $(n+1)$ -th), hence, by continuity, $\tilde{T}(Y) \subseteq Y$.

Step 2 (a minimal set). The family of the non-empty, compact, $\tilde{T}$ -invariant (that is, $\tilde{T}(Z) \subseteq Z$) subsets of Y , ordered by inclusion, is non-empty (it contains Y) and every descending chain has non-empty intersection (an intersection of nested non-empty compact sets), compact and invariant. By Zorn’s lemma there exists a minimal element Z . Since $\tilde{T}(Z)$ is in its turn compact, non-empty, invariant and contained in Z , minimality gives $\tilde{T}(Z) = Z$.

Step 3 (Z projects onto the whole of X). Let $\pi : X \times \mathbb{R} \rightarrow X$ be the projection. $\pi(Z)$ is compact, non-empty and T -invariant ($T(\pi(Z)) = \pi(\tilde{T}(Z)) = \pi(Z)$); by the minimality of T (a non-empty invariant compact set contains the orbit of one of its points, and that orbit is dense) one has $\pi(Z) = X$.

Step 4 (Z is a graph). Suppose there exist $(x, t_1), (x, t_2) \in Z$ with $c := t_2 - t_1 > 0$. The vertical translation $\sigma_c(x, t) := (x, t + c)$ is a homeomorphism of $X \times \mathbb{R}$ which **commutes** with $\tilde{T}$ (both act on the second coordinate by addition, and σ_c does not touch the first). Hence $\sigma_c(Z)$ is compact, non-empty, invariant and *minimal* (the image of a minimal set under a homeomorphism commuting with the dynamics is minimal). But $(x, t_2) = \sigma_c(x, t_1) \in \sigma_c(Z) \cap Z$: two minimal sets that intersect coincide (the intersection is compact, non-empty and invariant, hence equal to both). Therefore $\sigma_c(Z) = Z$, and iterating $\sigma_{nc}(Z) = Z$ for every $n \geq 1$: Z

would contain points with second coordinate $t_1 + nc \rightarrow +\infty$, against $Z \subseteq X \times [-M, M]$. Contradiction: for every x there is at most one t with $(x, t) \in Z$, and by Step 3 there is at least one. Hence $Z = \{(x, v(x)) : x \in X\}$ for a function $v : X \rightarrow [-M, M]$.

Step 5 (continuity and conclusion). The graph of v is closed (it is Z , compact) and the codomain is compact: v is continuous (if $x_j \rightarrow x$, every subsequence of $(v(x_j))$ has a subsubsequence converging to a t with $(x, t) \in Z$, that is, $t = v(x)$; hence $v(x_j) \rightarrow v(x)$). Finally, from $\tilde{T}(x, v(x)) \in Z$ it follows that $(Tx, v(x) + \phi(x)) \in Z$, that is, $v(Tx) = v(x) + \phi(x)$: $\phi = v \circ T - v$ with v continuous. $\square$

Corollary 6.3: Gottschalk–Hedlund for the circle. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be a minimal homeomorphism (for instance $f = R_\alpha$ with $\alpha/2\pi$ irrational, or a $C^{1+\text{bv}}$ diffeomorphism with $\rho(f)$ irrational, by Theorem 5.2) and let $\phi \in C(\mathbb{T})$. Then the equation $\psi \circ f - \psi = \phi$ has a continuous solution if and only if the Birkhoff sums $S_n\phi(x)$ are bounded for one (equivalently: for every) point x . In that case the solution is unique up to additive constants.

Proof. $\mathbb{T}$ is compact and metric and f is a minimal homeomorphism: Theorem 6.4 applies. Uniqueness: two solutions differ by a continuous g with $g \circ f = g$, that is, constant on the orbits; the orbits being dense and g continuous, g is constant. $\square$

Corollary 6.4: Unbounded Birkhoff sums for the Liouville datum. With ρ and φ as in Theorem 6.3, the Birkhoff sums $S_n\varphi(\theta) = \sum_{j=0}^{n-1} \varphi(\theta + j\alpha)$ are **unbounded** for every $\theta \in \mathbb{T}$, even though φ is C^∞ and has zero mean.

Proof. If they were bounded for some θ , Corollary 6.3 (applicable: R_α is minimal because ρ is irrational, Theorem 3.1) would give a continuous solution of the homological equation, in particular an L^1 solution: excluded by Theorem 6.3(ii). $\square$

Remark 6.10: Denjoy–Koksma, and what we import from Avila–Kocsard. Corollary 6.4 says that the Birkhoff sums can blow up; the Denjoy–Koksma inequality (Theorem 5.3, Chapter 5) says when they do not: if ψ has bounded variation and $|\rho - p/q| \leq 1/q^2$, then

$$\left| \sum_{j=0}^{q-1} \psi(f^j(x)) - q \int_{\mathbb{T}} \psi d\mu \right| \leq \text{Var}(\psi),$$

that is, along the best return times $q = q_n$ the sums remain bounded independently of n , for every irrational rotation number and without any arithmetic hypothesis. Boundedness along the whole sequence n (not only along the q_n) is instead a much stronger property, and it is exactly what the Gottschalk–Hedlund lemma characterises.

What we import. The picture is completed by a recent result: for a minimal C^∞ circle diffeomorphism, the invariant probability measure is the only invariant distribution up to multiplicative constants, and the space of the real C^∞ -coboundaries is closed if and only if the rotation number is Diophantine. **Imported from** A. Avila, A. Kocsard, “Cohomological equations and invariant distributions for minimal circle diffeomorphisms”, *Duke Math. J.* **158** (2011), n. 3, 501–536, DOI 10.1215/00127094-1345662, **verified on 17 September 2026** (Project Euclid record reopened: authors, title, volume, number, year, pages and DOI confirmed). What is assumed: the theory of distributions on $\mathbb{T}$ and the notion of invariant distribution; no result of this chapter depends on it. Recorded in Appendix A.

6.6. Arnold’s theorem: statement

Definition 6.5: Analytic diffeomorphism close to a rotation. Let $a > 0$ and $\alpha \in \mathbb{R}$. We say that F is the **lift of an analytic circle diffeomorphism, defined on S_a and ε -close to R_α** , if

$$F(\theta) = \theta + \alpha + u(\theta), \quad u \in \mathcal{A}_a, \quad \|u\|_a = \varepsilon,$$

with F increasing on $\mathbb{R}$ (equivalently $1 + u' > 0$ on $\mathbb{R}$) and $F(\theta + 2\pi) = F(\theta) + 2\pi$ (guaranteed by the periodicity of u).

Theorem 6.5: Arnold's theorem: local analytic linearisation. Let ρ be a **Diophantine** number of type (K, τ) (Definition 3.4, with $0 < K \leq \frac{1}{2}$ and $\tau \geq 0$), $\alpha = 2\pi\rho$ and $a > 0$. We put

$$\Lambda := \Lambda(\tau) = \Gamma(2 + \tau) + \left(\frac{1 + \tau}{e}\right)^{1+\tau}, \quad \delta_0 := \min\left(1, \frac{a}{12}\right), \quad \varepsilon_0(a, \rho) := \frac{K \delta_0^{3+\tau}}{2^{6+\tau} \Lambda}.$$

Let $F = R_\alpha + u$ be the lift of an analytic circle diffeomorphism on S_a (Definition 6.5) such that

$$(H1) \quad \|u\|_a \leq \varepsilon_0(a, \rho), \quad (H2) \quad \rho(F) = \rho.$$

Then there exists $H = \text{id} + h$, with h holomorphic, 2π -periodic and real on the reals on $S_{a/2}$, such that

$$F \circ H = H \circ R_\alpha \quad \text{on } S_{a/2},$$

and H is, on the real circle, an analytic diffeomorphism with $0.80 \leq H' \leq 1.22$: hence $H^{-1} \circ F \circ H = R_\alpha$, that is, f is **analytically conjugate** to the rigid rotation R_α . The conjugacy is unique up to composition with a rotation.

The proof occupies the whole of Section 6.7. Before tackling it, three remarks on the statement.

Remark 6.11: Both hypotheses are indispensable. (H2) cannot be dropped. An analytic F arbitrarily close to R_α can have a rational rotation number: it is enough for it to fall inside an Arnold tongue, which for every p/q has width $O(\varepsilon^q) > 0$ (Theorem 4.7, Chapter 4). In that case F has periodic orbits (Theorem 4.3) and cannot be conjugate to an irrational rotation, which has none. This is correction 3.15 of the expansion plan of the volume: a first draft of Chapter 4 had stated the theorem without (H2), making it false. See Remark 4.7.

(H1) cannot be dropped. It is a local theorem: the threshold ε_0 depends on a and on the Diophantine type. The global version — “every analytic diffeomorphism with $\rho(f)$ Diophantine is analytically linearisable, however large the perturbation” — is **false**, and the correct arithmetic condition for the global problem is not the Diophantine one but the condition $\mathcal{H}$ of Herman–Yoccoz (Section 6.8).

The Diophantine condition cannot be weakened at will. For the local problem, the optimal arithmetic condition is Bryuno's (Definition 6.2), strictly weaker than the Diophantine one (Proposition 6.2): see again Section 6.8. Theorem 6.3 already shows, at the level of the linear step alone, that for Liouville numbers constructed ad hoc the machinery jams.

Remark 6.12: Provenance of the statement and of the proof. The result is due to **V. I. Arnold (1961)**: “Mal'ye znamenateli. I” (Izv. Akad. Nauk SSSR Ser. Mat. **25**:1, 21–86, with errata in **28**:2 (1964), 479–480). This citation is given **exclusively for historical**

priority: the original Russian text was not reopened for this draft, and the statement of Theorem 6.5 is not, and must not be read as, a transcription of Arnold’s statement. The formulation adopted here (hypotheses (H1)–(H2), conclusion of analytic conjugacy on $S_{a/2}$) is the standard modern one, which is found in A. Katok, B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, CUP 1995, Thm. 12.3.1, and — in the form “for every Diophantine α and every $\nu > 0$ there exists $\varepsilon(\alpha, \nu) > 0$ such that every analytic diffeomorphism f on S_ν with $\rho(f) = \alpha$ and $\|f - R_{\rho(f)}\|_\nu < \varepsilon$ is conjugate to $R_{\rho(f)}$ by an analytic diffeomorphism” — in the survey of H. Eliasson, B. Fayad, R. Krikorian, Jean-Christophe Yoccoz and the theory of circle diffeomorphisms (arXiv:1810.07107), reopened and verified on 17 September 2026.

The constants $\varepsilon_0(a, \rho)$, δ_0 , $\Lambda(\tau)$ of this statement are ours too: they are the ones that come out of the proof of Section 6.7 and they are verified numerically in blocks [D] and [F] of the script. They are not Arnold’s constants, nor those of any exposition in the literature: any numerical comparison with constants published elsewhere would be meaningless, because every proof optimises different choices (how much domain to lose at each step, how much margin to leave in the convergence condition).

One difference of method should be declared: the classical expositions treat the additive constant that appears at each step (the mean of u) by introducing a **translation parameter** λ and conjugating the family $F + \lambda$, in order to show at the end that the right parameter is $\lambda = 0$. Here the parameter is not needed: Lemma 6.7 deduces directly $|\text{mean}| \leq \|\text{quadratic remainder}\|$ from hypothesis (H2) alone and from the monotonicity of the rotation number (Theorem 4.2(i)). The gain is not merely aesthetic: it removes the whole apparatus of estimates in the parameter.

Remark 6.13: How small is ε_0 ? Three numerical values. Block [D] of `verifica_cap06_kam.py` (double check: each threshold is computed both with the closed formula $K\delta_0^{3+\tau}/(2^{6+\tau}\Lambda)$ and, by an independent route, by imposing $x_0 = A2^\gamma\varepsilon_0 = \frac{1}{2}$ in the recursion of the Theorem; relative discrepancy 0 in all cases ✓). For $\rho = 1/\varphi$ (hence $K = \varphi^{-2} = 0.381966$, $\tau = 0$, $\Lambda = 1.367879$, $C_1 = 4\Lambda/K = 14.324619$):

a	$\delta_0 = \min(1, a/12)$	$\varepsilon_0(a, \rho)$	limit strip $a - 6\delta_0$
0.20	0.01666667	2.019962×10^{-8}	$0.10 = a/2$
0.50	0.04166667	3.156190×10^{-7}	$0.25 = a/2$
1.00	0.08333333	2.524952×10^{-6}	$0.50 = a/2$

These are very small thresholds: the proof is generous in every estimate. The content of the theorem is not the value of ε_0 but its **positivity**: there exists a non-empty analytic neighbourhood of the rotation in which linearisation is possible. Note the dependence $\varepsilon_0 \propto \delta_0^{3+\tau} \propto a^{3+\tau}$ for small a : thin strips (analytic functions “at the limit”) require much smaller perturbations; and $\varepsilon_0 \propto K$: numbers that are worse approximable (K large) tolerate larger perturbations. The golden ratio, which maximises K for $\tau = 0$ (Proposition 6.1), is the most “robust” rotation number there is — the same fact that, in mechanics, makes invariant tori with golden frequency survive the longest.

6.7. The Newton scheme

The idea is Kolmogorov’s and Arnold’s: not to solve equation (6.2) in one go (impossible: the perturbation series diverges), but to *iterate* the linear step, as in Newton’s method for numerical equations. Each step solves the homological equation with the current error as datum, and leaves an error which is *quadratic* with respect to the previous one; the price is the loss of a slice of the strip of analyticity. If the slices add up to a finite quantity and the error goes to zero like ε^{2^j} , one wins.

Let us fix the notation for the whole section: ρ Diophantine of type (K, τ) with $K \leq \frac{1}{2}$ (the bound is automatic: $K \leq \|1 \cdot \rho\| \leq \frac{1}{2}$ by Lemma 6.2(ii) with $q = 1$), $\alpha = 2\pi\rho$, and

$$\gamma := 3 + \tau, \quad \Lambda := \Lambda(\tau) \geq 1, \quad C_1 := \frac{4\Lambda}{K}. \quad (6.4)$$

($\Lambda \geq 1$ because $\Gamma(2 + \tau) \geq \Gamma(2) = 1$ for $\tau \geq 0$, Γ being increasing on $[2, \infty)$ and $2 + \tau \geq 2$.)

The inversion lemma

Lemma 6.6: Inversion of $\text{id} + \psi$ on a strip. Let $c > 0$ and let ψ be holomorphic, 2π -periodic and real on the reals on S_c , with

$$\|\psi\|_c \leq \sigma \quad \text{and} \quad \|\psi'\|_c \leq \frac{1}{2}.$$

Setting $H := \text{id} + \psi$, one has: **(i)** H is injective on S_c and $|H'| \geq \frac{1}{2}$; **(ii)** $H(S_c) \supseteq S_{c-\sigma}$, and for every $w \in S_{c-\sigma}$ the unique solution $z \in S_c$ of $H(z) = w$ satisfies $|z - w| \leq \sigma$; **(iii)** $H^{-1} = \text{id} + \chi$ is holomorphic on $S_{c-\sigma}$, with $\|\chi\|_{c-\sigma} \leq \sigma$, χ 2π -periodic and real on the reals, and $|(H^{-1})'| \leq 2$; **(iv)** on $\mathbb{R}$, H is strictly increasing with $H(\theta + 2\pi) = H(\theta) + 2\pi$: it is the lift of an orientation-preserving analytic circle homeomorphism.

Proof. **(i).** $H' = 1 + \psi'$ and $|\psi'| \leq \frac{1}{2}$ give $|H'| \geq \frac{1}{2}$. For injectivity, let $z \neq w$ in S_c : the strip is convex, hence the segment $[w, z]$ is contained in it and

$$H(z) - H(w) = (z - w) + \int_{[w, z]} \psi'(\zeta) d\zeta, \quad |H(z) - H(w)| \geq |z - w| - \frac{1}{2}|z - w| = \frac{1}{2}|z - w| > 0.$$

(ii). Let $w \in S_{c-\sigma}$ and $D := \{z \in \mathbb{C} : |z - w| \leq \sigma\}$. For $z \in D$, $|\text{Im } z| \leq |\text{Im } w| + \sigma < (c - \sigma) + \sigma = c$: hence $D \subset S_c$ and ψ is defined on D . The map $\Phi(z) := w - \psi(z)$ sends D into itself ($|\Phi(z) - w| = |\psi(z)| \leq \sigma$) and is a contraction with constant $\frac{1}{2}$ ($|\Phi(z) - \Phi(z')| = |\psi(z) - \psi(z')| \leq \frac{1}{2}|z - z'|$, again by convexity). D being complete, the contraction theorem gives a unique $z \in D$ with $z = w - \psi(z)$, that is, $H(z) = w$, with $|z - w| = |\psi(z)| \leq \sigma$. Uniqueness in the whole of S_c follows from (i).

(iii). H is holomorphic and injective on S_c with $H' \neq 0$: the inverse H^{-1} , defined on $H(S_c) \supseteq S_{c-\sigma}$, is holomorphic (holomorphic inverse function theorem, Volume III, Chapter 15), with $(H^{-1})'(w) = 1/H'(z)$, hence $|(H^{-1})'| \leq 1/\frac{1}{2} = 2$. Setting $\chi(w) := H^{-1}(w) - w = z - w$, (ii) gives $|\chi| \leq \sigma$. Periodicity: from $H(z + 2\pi) = H(z) + 2\pi$ it follows that $H^{-1}(w + 2\pi) = H^{-1}(w) + 2\pi$, that is, χ is 2π -periodic; and $H(\mathbb{R} \cap S_c) \subseteq \mathbb{R}$ with H increasing implies that H^{-1} sends reals to reals.

(iv). On $\mathbb{R}$, ψ is real and $|\psi'| \leq \frac{1}{2}$, hence $H' = 1 + \psi' \geq \frac{1}{2} > 0$: H is strictly increasing; the periodicity of ψ gives $H(\theta + 2\pi) = H(\theta) + 2\pi$. $\square$

The lemma that replaces the translation parameter

Lemma 6.7: The rotation number forces the mean. Let $G = R_\alpha + w$ be the lift of an orientation-preserving circle homeomorphism, with w continuous and 2π -periodic, and let $\rho(G) = \alpha/(2\pi)$. Then

$$\min_{\theta \in \mathbb{R}} w(\theta) \leq 0 \leq \max_{\theta \in \mathbb{R}} w(\theta).$$

In particular, if $w = \lambda + s$ with $\lambda \in \mathbb{R}$ constant and s continuous, then $|\lambda| \leq \|s\|_\infty$.

Proof. Setting $m := \min w$ and $M := \max w$ (they exist by continuity and periodicity), one

has, point by point on $\mathbb{R}$,

$$R_{\alpha+m}(\theta) = \theta + \alpha + m \leq G(\theta) \leq \theta + \alpha + M = R_{\alpha+M}(\theta).$$

The three functions are lifts of orientation-preserving circle homeomorphisms; by the monotonicity of the rotation number (Theorem 4.2(i), Chapter 4),

$$\rho(R_{\alpha+m}) \leq \rho(G) \leq \rho(R_{\alpha+M}), \quad \text{that is} \quad \frac{\alpha+m}{2\pi} \leq \frac{\alpha}{2\pi} \leq \frac{\alpha+M}{2\pi}$$

(the rotation number of a rotation R_β is $\beta/(2\pi)$, directly from Definition 4.2: $R_\beta^n(\theta) - \theta = n\beta$). Simplifying, $m \leq 0 \leq M$.

If $w = \lambda + s$, then $m = \lambda + \min s \leq 0$ gives $\lambda \leq -\min s \leq \|s\|_\infty$, and $M = \lambda + \max s \geq 0$ gives $\lambda \geq -\max s \geq -\|s\|_\infty$: hence $|\lambda| \leq \|s\|_\infty$. $\square$

Remark 6.14: Why this lemma is the heart of the argument. The homological equation $\psi \circ R_\alpha - \psi = \varphi$ has a solution only if $\widehat{\varphi}_0 = 0$ (Remark 6.1). But the error u of the current step does not, in general, have zero mean: its mean $\widehat{u}_0$ is of order ε , that is, **not** quadratic, and if it survived in the new error the scheme would not converge at all (one would get $\varepsilon_{j+1} \approx \varepsilon_j$, not ε_j^2). Lemma 6.7 solves exactly this: after subtracting the mean and conjugating, the new error has the form “constant + quadratic remainder”, and the hypothesis $\rho(F) = \rho$ — preserved at each step by the conjugacy (Proposition 4.2(i)) — forces that constant to be dominated by the quadratic remainder. In one line: the rotation number, which is a datum of the problem, does the work that in the classical expositions is done by an auxiliary parameter. Note that it is here, and only here, that hypothesis (H2) of Theorem 6.5 is used.

The step lemma

Lemma 6.8: One step of the Newton scheme. Let $\rho, \alpha, \gamma, \Lambda, C_1$ be as in (6.4). Let $b > 0$ and let $G = R_\alpha + u$ be the lift of an analytic circle diffeomorphism on S_b (Definition 6.5) with

$$\|u\|_b = \varepsilon, \quad \rho(G) = \rho.$$

Let δ be such that $0 < \delta \leq \min(1, \frac{b}{3})$ and suppose that the

$$\text{step condition (C):} \quad \frac{C_1 \varepsilon}{\delta^\gamma} \leq 1$$

holds. Then there exists $\psi \in \mathcal{A}_{b-\delta}$ such that, setting $H := \text{id} + \psi$:

$$(a) \quad \|\psi\|_{b-\delta} \leq \Theta := \frac{\Lambda \varepsilon}{K \delta^{2+\tau}} = \frac{C_1 \varepsilon}{\delta^\gamma} \cdot \frac{\delta}{4} \leq \frac{\delta}{4};$$

$$(b) \quad \|\psi'\|_{b-2\delta} \leq \frac{\Theta}{\delta} \leq \frac{1}{4};$$

(c) $G_1 := H^{-1} \circ G \circ H$ is defined and holomorphic on $S_{b-3\delta}$, is the lift of an analytic circle diffeomorphism with $\rho(G_1) = \rho$, and writing $G_1 = R_\alpha + u_1$ one has

$$\|u_1\|_{b-3\delta} \leq \frac{C_1 \varepsilon^2}{\delta^\gamma}.$$

Proof. Step 1 (solving the homological equation). Let $c := \widehat{u}_0$ be the mean of u on the real circle; $|c| \leq \|u\|_b = \varepsilon$. The function $\varphi := u - c$ belongs to $\mathcal{A}_b$, has zero mean and $\|\varphi\|_b \leq 2\varepsilon$. By Theorem 6.1 (applicable: $\delta \leq 1$ and $\delta \leq b/3 < b$) there exists a unique $\psi \in \mathcal{A}_{b-\delta}$ with zero

mean such that $\psi \circ R_\alpha - \psi = \varphi$ and

$$\|\psi\|_{b-\delta} \leq \frac{\Lambda}{2K} \cdot \frac{2\varepsilon}{\delta^{2+\tau}} = \frac{\Lambda\varepsilon}{K\delta^{2+\tau}} = \Theta.$$

The identity $\Theta = \frac{C_1\varepsilon}{\delta^\gamma} \cdot \frac{\delta}{4}$ is a direct computation: $\frac{C_1\varepsilon}{\delta^{3+\tau}} \cdot \frac{\delta}{4} = \frac{4\Lambda\varepsilon}{K\delta^{3+\tau}} \cdot \frac{\delta}{4} = \frac{\Lambda\varepsilon}{K\delta^{2+\tau}}$. From condition (C) it follows that $\Theta \leq \delta/4$: this is (a). Item (b) is the Cauchy estimate (Lemma 6.3) applied to ψ between $S_{b-\delta}$ and $S_{b-2\delta}$: $\|\psi'\|_{b-2\delta} \leq \|\psi\|_{b-\delta}/\delta \leq \Theta/\delta \leq 1/4$.

Step 2 (the conjugacy identity with remainder). Let $\theta \in S_{b-3\delta}$. Since $|\operatorname{Im}(\theta + t\psi(\theta))| \leq (b-3\delta) + \Theta < b-\delta$ for every $t \in [0, 1]$ (one uses $\Theta \leq \delta/4$ and $3\delta - \delta/4 > \delta$), all the expressions that follow are defined:

$$G(H(\theta)) = H(\theta) + \alpha + u(H(\theta)) = \theta + \alpha + \psi(\theta) + u(\theta + \psi(\theta)),$$

$$H(\theta + \alpha) = \theta + \alpha + \psi(\theta + \alpha) = \theta + \alpha + \psi(\theta) + \varphi(\theta) = \theta + \alpha + \psi(\theta) + u(\theta) - c.$$

Subtracting,

$$G \circ H = H \circ R_\alpha + s, \quad s(\theta) := c + r(\theta), \quad r(\theta) := u(\theta + \psi(\theta)) - u(\theta). \quad (6.5)$$

The remainder r is *quadratic*: by the fundamental theorem of calculus along the segment from θ to $\theta + \psi(\theta)$ (contained in $S_{b-\delta}$, as just seen) and by the Cauchy estimate $\|u'\|_{b-\delta} \leq \varepsilon/\delta$,

$$r(\theta) = \int_0^1 u'(\theta + t\psi(\theta)) \psi(\theta) dt, \quad \|r\|_{b-3\delta} \leq \frac{\varepsilon}{\delta} \cdot \Theta. \quad (6.6)$$

Step 3 (inversion). We apply Lemma 6.6 with $c_{\text{str}} := b - 2\delta$ and $\sigma := \Theta$ (the hypotheses are (a) and (b)): H is injective on $S_{b-2\delta}$, $H^{-1} = \operatorname{id} + \chi$ is holomorphic on $S_{b-2\delta-\Theta} \supseteq S_{b-3\delta}$ (because $\Theta \leq \delta/4 < \delta$), with $|(H^{-1})'| \leq 2$.

Let us check that $G_1(\theta) = H^{-1}(H(\theta + \alpha) + s(\theta))$ is defined for $\theta \in S_{b-3\delta}$, that is, that the segment from $H(\theta + \alpha)$ to $H(\theta + \alpha) + s(\theta)$ stays in $S_{b-2\delta-\Theta}$. One has $|\operatorname{Im} H(\theta + \alpha)| \leq (b-3\delta) + \Theta$ and

$$|s| \leq |c| + \|r\| \leq \varepsilon + \frac{\varepsilon\Theta}{\delta} \leq \varepsilon \left(1 + \frac{1}{4}\right) = \frac{5}{4}\varepsilon.$$

From condition (C), $\varepsilon \leq \delta^\gamma/C_1 = \frac{K\delta^{3+\tau}}{4\Lambda} \leq \frac{\delta^{3+\tau}}{4} \leq \frac{\delta}{4}$ (one uses $\delta \leq 1$, $K \leq 1$, $\Lambda \geq 1$), hence $|s| \leq \frac{5}{4} \cdot \frac{\delta}{4} = \frac{5\delta}{16}$. Since $2\Theta \leq \frac{\delta}{2}$ by (a),

$$2\Theta + |s| \leq \frac{\delta}{2} + \frac{5\delta}{16} = \frac{13\delta}{16} < \delta, \quad \text{that is} \quad (b-3\delta) + \Theta + |s| < b-2\delta-\Theta.$$

Hence the required segment stays in $S_{b-2\delta-\Theta}$, and G_1 is defined and holomorphic on $S_{b-3\delta}$.

Step 4 (estimate of the new error in terms of s). Since $H^{-1}(H(\theta + \alpha)) = \theta + \alpha$,

$$\begin{aligned} u_1(\theta) &:= G_1(\theta) - (\theta + \alpha) = H^{-1}(H(\theta + \alpha) + s(\theta)) - H^{-1}(H(\theta + \alpha)) \\ &= \int_0^1 (H^{-1})'(H(\theta + \alpha) + ts(\theta)) s(\theta) dt, \end{aligned} \quad (6.7)$$

whence $\|u_1\|_{b-3\delta} \leq 2\|s\|_{b-3\delta}$.

Step 5 (the rotation number kills the constant). On $\mathbb{R}$, $G_1 = H^{-1} \circ G \circ H$ is the conjugate of G by the lift H of an orientation-preserving circle homeomorphism (Lemma 6.6(iv)); hence G_1 is in its turn a lift of a circle homeomorphism, analytic, and by the invariance of the rotation number under conjugacy (Proposition 4.2(i))

$$\rho(G_1) = \rho(G) = \rho = \frac{\alpha}{2\pi}.$$

By Lemma 6.7 applied to $G_1 = R_\alpha + u_1$ there exist $\theta_-, \theta_+ \in \mathbb{R}$ with $u_1(\theta_-) \leq 0 \leq u_1(\theta_+)$. On the other hand, for real θ , the integrand in (6.7) is real and *positive* ($(H^{-1})' = 1/H'$ with $H' \geq \frac{1}{2} > 0$ on the reals), hence $u_1(\theta)$ has the same sign as $s(\theta)$. Therefore $s(\theta_-) \leq 0 \leq s(\theta_+)$, that is,

$$c + r(\theta_-) \leq 0 \leq c + r(\theta_+) \quad \implies \quad -r(\theta_+) \leq c \leq -r(\theta_-) \quad \implies \quad |c| \leq \|r\|_{b-3\delta}.$$

Step 6 (conclusion). Combining Steps 4 and 5 with (6.6),

$$\|u_1\|_{b-3\delta} \leq 2(|c| + \|r\|) \leq 4\|r\| \leq 4\frac{\varepsilon}{\delta} \Theta = 4\varepsilon \cdot \frac{\Lambda\varepsilon}{K\delta^{3+\tau}} = \frac{C_1\varepsilon^2}{\delta^\gamma},$$

which is (c). Finally u_1 is 2π -periodic (a composition of lifts) and real on the reals, hence $u_1 \in \mathcal{A}_{b-3\delta}$. $\square$

Example 6.5: One Newton step, explicit and numerical. Take $F(\theta) = \theta + \alpha + \varepsilon \cos \theta$ with $\alpha = 2\pi/\varphi$: here the mean of $u = \varepsilon \cos \theta$ is *already* zero, hence $c = 0$ and the step can be read off in closed form. By Example 6.3, $\psi(\theta) = \varepsilon \sin(\theta - \frac{\alpha}{2})/(2 \sin \frac{\alpha}{2})$, and from Step 2 of the proof

$$\|u_1\|_\infty \approx \|r\|_\infty = \|u'(\theta)\psi(\theta)\|_\infty + O(\varepsilon^3) = \varepsilon^2 \frac{\|\sin \theta \sin(\theta - \frac{\alpha}{2})\|_\infty}{2 \sin \frac{\alpha}{2}} + O(\varepsilon^3).$$

Since $\sin t \sin(t - \beta) = \frac{1}{2} [\cos \beta - \cos(2t - \beta)]$, its sup over t equals $\frac{1}{2}(1 + |\cos \beta|)$, hence

$$\frac{\|u_1\|_\infty}{\varepsilon^2} \longrightarrow \frac{1 + |\cos \frac{\alpha}{2}|}{4 \sin \frac{\alpha}{2}} = 0.36543120\dots \quad (\varepsilon \rightarrow 0).$$

Double check (block [E], run on 17 September 2026). The script computes $\|u_1\|_\infty$ *without* using the asymptotic formula: it constructs $H = \text{id} + \psi$, inverts H numerically with the scalar Newton method (tolerance 10^{-12}) and evaluates $u_1 = H^{-1} \circ F \circ H - R_\alpha$ at 4000 points. Results:

ε	$\ u_1\ _\infty$	$\ u_1\ _\infty/\varepsilon^2$
10^{-1}	3.79819×10^{-3}	0.379819
5×10^{-2}	9.31026×10^{-4}	0.372410
10^{-2}	3.66795×10^{-5}	0.366795
10^{-3}	3.65567×10^{-7}	0.365567
10^{-4}	3.65445×10^{-9}	0.365445

The ratio converges to 0.365431, the predicted closed value $\checkmark$: the convergence of one step is **quadratic**, and the constant is the one computed by hand. The script also verifies that, inside condition (C), the proved upper bound $\|u_1\| \leq 2\varepsilon\Theta/\delta$ (here $c = 0$, hence $\|u_1\| \leq 2\|r\|$ instead of the general case $\|u_1\| \leq 4\|r\|$) is amply satisfied (for $\delta = 0.1$: upper bound 8.73×10^{-6} against actual value 4.45×10^{-10} , a factor 1.96×10^4 of margin) $\checkmark$. The enormous margin is the reason why $\varepsilon_0(a, \rho)$ turns out to be so small: the proof pays in threshold what it gains in simplicity.

The iteration and the proof of Theorem 6.5

We can now prove the theorem stated in Section 6.6.

Proof. (of Theorem 6.5.) We put, with the constants (6.4),

$$\delta_0 := \min\left(1, \frac{a}{12}\right), \quad \delta_j := \delta_0 2^{-j}, \quad a_0 := a, \quad a_{j+1} := a_j - 3\delta_j, \quad A := \frac{C_1}{\delta_0^\gamma}.$$

We define the sequence of expected errors by $\varepsilon_0 := \|u\|_a$ and $\varepsilon_{j+1} := C_1 \varepsilon_j^2 / \delta_j^\gamma$, and the auxiliary sequence

$$x_j := A 2^{(j+1)\gamma} \varepsilon_j.$$

Step 1 (the strips do not collapse). $\sum_{j \geq 0} 3\delta_j = 3\delta_0 \sum_j 2^{-j} = 6\delta_0$, hence

$$a_j \downarrow a_\infty := a - 6\delta_0 \geq a - \frac{a}{2} = \frac{a}{2} \quad (\text{because } \delta_0 \leq a/12),$$

and $a_j > a_\infty \geq a/2$ for every j . Moreover $\delta_j \leq \delta_0 \leq 1$ and $\delta_j \leq \delta_0 \leq a/12 \leq a_j/6 \leq a_j/3$: the hypotheses of Lemma 6.8 on the size of δ are satisfied at each step.

Step 2 (the error recursion is exactly $x \mapsto x^2$). A direct computation:

$$x_{j+1} = A 2^{(j+2)\gamma} \varepsilon_{j+1} = A 2^{(j+2)\gamma} \frac{C_1 \varepsilon_j^2}{\delta_j^\gamma 2^{-j\gamma}} = A^2 2^{(2j+2)\gamma} \varepsilon_j^2 = x_j^2.$$

By hypothesis (H1), $\varepsilon_0 \leq \varepsilon_0(a, \rho) = \frac{K\delta_0^\gamma}{2^{6+\tau}\Lambda}$; since $2^{6+\tau} = 2^{3+\gamma} = 8 \cdot 2^\gamma$ and $C_1 = 4\Lambda/K$,

$$x_0 = A 2^\gamma \varepsilon_0 = \frac{C_1 2^\gamma}{\delta_0^\gamma} \cdot \frac{K\delta_0^\gamma}{8 \cdot 2^\gamma \Lambda} = \frac{C_1 K}{8\Lambda} = \frac{4\Lambda}{K} \cdot \frac{K}{8\Lambda} = \frac{1}{2}.$$

Hence $x_j = x_0^{2^j} \leq 2^{-2^j}$ for every j : the errors decrease **super-exponentially**.

Step 3 (the step condition holds at each iteration). With $\delta_j = \delta_0 2^{-j}$,

$$\frac{C_1 \varepsilon_j}{\delta_j^\gamma} = \frac{C_1}{\delta_0^\gamma} 2^{j\gamma} \varepsilon_j = A 2^{j\gamma} \varepsilon_j = \frac{x_j}{2^\gamma} \leq \frac{1}{2 \cdot 2^\gamma} < 1.$$

Step 4 (construction of the conjugacies). We put $G_0 := F$, $u_0 := u$; by (H2), $\rho(G_0) = \rho$. By induction on j : assuming $G_j = R_\alpha + u_j$ defined and holomorphic on S_{a_j} , the lift of an analytic circle diffeomorphism, with $\|u_j\|_{a_j} \leq \varepsilon_j$ and $\rho(G_j) = \rho$, Steps 1 and 3 allow us to apply Lemma 6.8 with $b := a_j$, $\delta := \delta_j$, obtaining $\psi_j \in \mathcal{A}_{a_j - \delta_j}$, $H_j := \text{id} + \psi_j$ and

$$G_{j+1} := H_j^{-1} \circ G_j \circ H_j = R_\alpha + u_{j+1} \quad \text{on } S_{a_{j+1}}, \quad \|u_{j+1}\|_{a_{j+1}} \leq \frac{C_1 \varepsilon_j^2}{\delta_j^\gamma} = \varepsilon_{j+1},$$

with $\rho(G_{j+1}) = \rho$: the inductive hypothesis is reproduced. Moreover, from Lemma 6.8(a)–(b) and from Step 3,

$$\|\psi_j\|_{a_j - \delta_j} \leq \Theta_j = \frac{C_1 \varepsilon_j}{\delta_j^\gamma} \cdot \frac{\delta_j}{4} = \frac{x_j \delta_j}{2^{\gamma+2}} \leq \frac{\delta_j}{4} 2^{-2^j}, \quad \|\psi'_j\|_{a_j - 2\delta_j} \leq \frac{x_j}{2^{\gamma+2}} \leq t_j := \frac{2^{-2^j}}{4}. \quad (6.8)$$

Step 5 (convergence of the composed conjugacies). We put $W_0 := \text{id}$ and $W_{j+1} := W_j \circ H_j$. Since H_j maps $S_{a_{j+1}}$ into S_{a_j} (it moves points by at most $\Theta_j \leq \delta_j/4$, and $a_{j+1} + \Theta_j < a_j$), each W_j is holomorphic on $S_{a_j} \supseteq S_{a_\infty}$ with values in S_a . By the chain rule and (6.8),

$$|W'_j| \leq \prod_{i \geq 0} (1 + t_i) \leq 1.2152, \quad \text{and on } \mathbb{R}: \quad W'_j \geq \prod_{i \geq 0} (1 - t_i) \geq 0.8067$$

(the two products are computed in block [F]; they converge because $\sum_i t_i = \frac{1}{4} \sum_i 2^{-2^i} = \frac{1}{4} \cdot 0.8164215090 = 0.2041 \dots < \infty$). Hence, for $\theta \in S_{a_\infty}$,

$$|W_{j+1}(\theta) - W_j(\theta)| = |W_j(H_j(\theta)) - W_j(\theta)| \leq 1.2152 \cdot \|\psi_j\| \leq 1.2152 \cdot \frac{\delta_0}{4} 2^{-j} 2^{-2^j},$$

(the segment from θ to $H_j(\theta)$ stays in S_{a_j} , because $|\text{Im } \theta| < a_{j+1}$ and $|\psi_j| \leq \Theta_j \leq \delta_j/4$, with $a_{j+1} + \delta_j/4 < a_j$), the general term of a convergent series: the sequence (W_j) is uniformly

Cauchy on S_{a_∞} , hence it converges uniformly to a function W , holomorphic by Weierstrass' theorem on uniform convergence of holomorphic functions (Volume III, Chapter 15), with $W(\theta + 2\pi) = W(\theta) + 2\pi$ (a property preserved in the limit), real on the reals, and — by the uniform convergence of the derivatives as well, guaranteed by the Cauchy estimates — with $0.8067 \leq W' \leq 1.2152$ on $\mathbb{R}$. In particular W is the lift of an analytic circle diffeomorphism.

Step 6 (passing to the limit in the conjugacy identity). By induction, $F \circ W_j = W_j \circ G_j$ on S_{a_j} : for $j = 0$ it is the identity $F = F$; and

$$F \circ W_{j+1} = F \circ W_j \circ H_j = W_j \circ G_j \circ H_j = W_j \circ H_j \circ (H_j^{-1} \circ G_j \circ H_j) = W_{j+1} \circ G_{j+1}.$$

We fix $\theta \in S_{a_\infty}$ and let $j \rightarrow \infty$. On the left, $F(W_j(\theta)) \rightarrow F(W(\theta))$ by continuity of F on S_a and the convergence $W_j \rightarrow W$. On the right,

$$\begin{aligned} |W_j(G_j(\theta)) - W(\theta + \alpha)| &\leq |W_j(\theta + \alpha + u_j(\theta)) - W_j(\theta + \alpha)| + |W_j(\theta + \alpha) - W(\theta + \alpha)| \\ &\leq 1.2152 \varepsilon_j + \|W_j - W\|_{a_\infty}, \end{aligned}$$

which tends to 0 because $\varepsilon_j \leq 2^{-2^j} / (A 2^{(j+1)\gamma}) \rightarrow 0$ and $W_j \rightarrow W$ uniformly. Hence

$$F \circ W = W \circ R_\alpha \quad \text{on } S_{a_\infty} \supseteq S_{a/2}.$$

Setting $H := W$, we have the claim: H is an analytic circle diffeomorphism (on the reals) with $H' \in [0.80; 1.22]$, and $H^{-1} \circ F \circ H = R_\alpha$.

Step 7 (uniqueness up to rotations). If H_1, H_2 are two continuous conjugacies with $F \circ H_i = H_i \circ R_\alpha$, then $K := H_2^{-1} \circ H_1$ satisfies $K \circ R_\alpha = R_\alpha \circ K$, that is, $g := K - \text{id}$ is 2π -periodic, continuous and R_α -invariant ($g \circ R_\alpha = g$). Since ρ is irrational, R_α is minimal (Theorem 3.1, Chapter 3): a continuous invariant function is constant on the orbits, hence on their closure, hence on the whole of $\mathbb{T}$. Therefore $K = \text{id} + \text{constant}$ is a rotation: $H_1 = H_2 \circ R_{\text{const}}$. $\square$

Remark 6.15: The bookkeeping of the convergence, in numbers. Block [F] of `verifica_cap06_kam.py`, with $a = 0.5$, $\rho = 1/\varphi$ ($K = 0.381966$, $\tau = 0$, $\gamma = 3$), $\delta_0 = 0.04166667$, $C_1 = 14.324619$, $A = 198\,023.539664$, $\varepsilon_0 = 3.156190 \times 10^{-7}$:

j	δ_j	a_j	ε_j	$x_j = 2^{-2^j}$
0	4.166667×10^{-2}	0.50000000	3.156190×10^{-7}	5.000000×10^{-1}
1	2.083333×10^{-2}	0.37500000	1.972619×10^{-8}	2.500000×10^{-1}
2	1.041667×10^{-2}	0.31250000	6.164434×10^{-10}	6.250000×10^{-2}
3	5.208333×10^{-3}	0.28125000	4.815964×10^{-12}	3.906250×10^{-3}
4	2.604167×10^{-3}	0.26562500	2.351545×10^{-15}	1.525879×10^{-5}
5	1.302083×10^{-3}	0.25781250	4.485216×10^{-21}	2.328306×10^{-10}
6	6.510417×10^{-4}	0.25390625	1.305370×10^{-31}	5.421011×10^{-20}
7	3.255208×10^{-4}	0.25195312	8.845530×10^{-52}	2.938736×10^{-39}

The script verifies at each row that $x_j \leq 2^{-2^j}$ (an equality, here, because $x_0 = 1/2$ exactly) and that condition (C) holds $\checkmark$. The limit strip is $a_\infty = a - 6\delta_0 = 0.25 = a/2$ $\checkmark$. In **seven** steps the error has gone from 3×10^{-7} to 9×10^{-52} : this is the signature of quadratic convergence, $\varepsilon_j \approx \varepsilon_0^{2^j}$ up to the geometric factors. A merely geometric convergence ($\varepsilon_{j+1} \leq q\varepsilon_j$ with $q < 1$ fixed) would not suffice: the loss of domain forces $\delta_j \rightarrow 0$, and with it the factor $\delta_j^{-\gamma} = \delta_0^{-\gamma} 2^{j\gamma}$ blows up geometrically; the product $\prod_j (q 2^{j\gamma})$ diverges. This is the reason why Kolmogorov and Arnold had to use Newton and not Picard, and it is the technical content of the whole of KAM theory.

Remark 6.16: Where each hypothesis was used. *Analyticity:* in Lemma 6.3 (Cauchy estimates, which convert an estimate on the function into one on the derivative at the price of a slice of strip) and in Lemma 6.4 (exponential decay of the coefficients). *Diophantine condition:* only in Theorem 6.1, through $|\hat{\psi}_k| \leq |k|^{1+\tau}|\hat{\varphi}_k|/(4K)$; it is the only point where the arithmetic of ρ comes into play, and it is the reason for the exponent $\gamma = 3 + \tau$ ($2 + \tau$ from Lemma 6.5, plus 1 from the Cauchy estimate on the derivative of ψ). *(H1) ε small:* to get the recursion started with $x_0 \leq \frac{1}{2}$. *(H2) $\rho(F) = \rho$:* exclusively in Lemma 6.7, that is, in Step 5 of Lemma 6.8. *Monotonicity of ρ* (Theorem 4.2(i)): likewise. *Invariance of ρ under conjugacy* (Proposition 4.2(i)): to propagate (H2) from one step to the next.

The scheme at work: a numerical experiment

Example 6.6: The Newton scheme iterated on a map with a known answer. Double check (script `fig06_newton_iterato.py`, Appendix B, run on 17 September 2026). In order to be able to *verify* the outcome of the scheme, one applies it to a map whose conjugacy is known from the start. Let

$$H_0(\theta) := \theta + \eta \sin \theta \quad (\eta \in \{0.20; 0.10; 0.05\}), \quad F := H_0 \circ R_\alpha \circ H_0^{-1}, \quad \alpha = \frac{2\pi}{\varphi}.$$

H_0 is an analytic circle diffeomorphism ($H'_0 = 1 + \eta \cos \theta > 0$), hence F is analytic and, by Proposition 4.2(i), $\rho(F) = \rho(R_\alpha) = 1/\varphi$ **exactly**: hypothesis (H2) of Theorem 6.5 is satisfied by construction, and the expected answer is $H = H_0$ (up to a rotation, Step 7 of the proof).

The script represents the periodic functions on the $N = 2048$ points of a uniform grid, solves the homological equation in the Fourier coefficients (division by $e^{ik\alpha} - 1$), evaluates the compositions by Fourier interpolation and inverts H_j with the scalar Newton method. For $\eta = 0.10$:

j	$\ u_j\ _\infty$	$\ u_{j+1}\ _\infty$	$ c_j $	$\ r_j\ _\infty$
0	1.864064×10^{-1}	9.897858×10^{-3}	3.377×10^{-3}	1.370×10^{-2}
1	9.897858×10^{-3}	7.818732×10^{-5}	4.658×10^{-5}	1.210×10^{-4}
2	7.818732×10^{-5}	8.548619×10^{-9}	5.972×10^{-10}	9.145×10^{-9}
3	8.548619×10^{-9}	5.773160×10^{-15}	2.197×10^{-16}	8.770×10^{-16}

Three readings, all three verified by the script:

- (1) **Quadratic convergence:** $0.186 \rightarrow 9.9 \times 10^{-3} \rightarrow 7.8 \times 10^{-5} \rightarrow 8.5 \times 10^{-9} \rightarrow 5.8 \times 10^{-15}$ (machine precision) in *four* steps; the ratio $\|u_{j+1}\|/\|u_j\|^2$ stays of order unity (0.285; 0.798; 1.398) until the computation hits machine noise $\checkmark$. For $\eta = 0.20$ five steps are needed, for $\eta = 0.05$ four suffice (Figure 6.5).
- (2) **Lemma 6.7 is verified at each step:** $|c_j| \leq \|r_j\|_\infty$ in all the rows of the table and in all the rows of the other two series ($3.377 \times 10^{-3} \leq 1.370 \times 10^{-2}$; $4.658 \times 10^{-5} \leq 1.210 \times 10^{-4}$; $5.972 \times 10^{-10} \leq 9.145 \times 10^{-9}$; $2.197 \times 10^{-16} \leq 8.770 \times 10^{-16}$) $\checkmark$. Note that the mean c_j is *not* zero — the scheme really does produce it at each step — but it is always dominated by the quadratic remainder: this is exactly the mechanism that makes the translation parameter superfluous.
- (3) **The limit conjugacy is the expected one:** the composition $W = H_0^{(\text{numerical})} \circ \dots$ produced by the scheme coincides with $H_0 \circ R_c$ for $c = -1.1986 \times 10^{-5}$, with a maximum discrepancy of 4.574×10^{-14} $\checkmark$; and the conjugacy residual $\max_\theta |F(W(\theta)) - W(\theta + \alpha)|$ equals 5.329×10^{-15} , that is, zero up to machine precision $\checkmark$. This also confirms numerically Step 7 (uniqueness up to rotations).

A lesson about the threshold. Here $\|u_0\|_\infty = 0.186$, whereas the proved threshold equals $\varepsilon_0(a, \rho) = 3.2 \times 10^{-7}$ for $a = 0.5$ (Remark 6.13): the scheme converges *six hundred thousand*

times beyond the threshold guaranteed by the theorem. This is not a contradiction: ε_0 is a *sufficient* condition, obtained by bounding generously at every line of the proof. The theorem guarantees that the set of data for which the scheme converges is non-empty; the experiment shows that, in practice, it is much larger.

6.8. Beyond the local analytic case: Herman, Yoccoz, and the condition $\mathcal{H}$

Theorem 6.5 is *local* (it needs $\|u\|_a$ extremely small) and *analytic*. The two restrictions have been removed, over forty years of work, by Herman, Yoccoz, Katznelson–Ornstein, Khanin and Sinai. This section collects the corresponding results as **imported** ones (precise statements, verified sources, proofs not reproduced) or as **panorama**. All the imports of this section are recorded in Appendix A.

Remark 6.17: What we import: the finite-regularity version (Moser). The proof of Section 6.7 uses analyticity at one crucial point: the Cauchy estimates (Lemma 6.3), which in class C^k do not exist — one cannot bound a derivative by the function. Jürgen Moser showed that the scheme survives in finite regularity if the Cauchy estimates are replaced by **smoothing operators** (Fourier truncations calibrated at each step), which produce the same effect at the price of a controlled error: it is the technique which, in its general form, goes under the name of the Nash–Moser scheme.

Statement (finite-regularity version, qualitative form). Let ρ be Diophantine of type (K, τ) . There exist integers $k = k(\tau)$ and $k' < k$ such that every circle diffeomorphism f of class C^k , with $\rho(f) = \rho$ and sufficiently close to $R_{2\pi\rho}$ in the C^k norm, is conjugate to $R_{2\pi\rho}$ by a diffeomorphism of class $C^{k'}$.

Imported from J. Moser, *On invariant curves of area-preserving mappings of an annulus*, *Nachr. Akad. Wiss. Göttingen Math.-Phys. Kl. II* (1962), 1–20 (code K5, status SNIPPET in the register), and J. Moser, “A rapidly convergent iteration method and non-linear partial differential equations I, II”, *Ann. Scuola Norm. Sup. Pisa* (3) **20** (1966), 265–315 and 499–535 (code K4, Numdam record verified on 16 September 2026), with the modern exposition in R24. **What is assumed:** the construction of the smoothing operators and their interpolation estimates. **What we do not report:** the values of k and k' as functions of τ . They vary from one exposition to another (Moser’s first version for twist maps required a number of derivatives nowadays considered vastly redundant) and to report one without having reopened the source that proves it would be an invented citation. The sharp results in finite regularity do not in any case come from this line, but from Katznelson–Ornstein and Khanin–Teplinsky (Remark 6.22).

Remark 6.18: What we import: Herman’s global C^∞ theorem. Statement (Herman, 1979). There exists a set $A \subset \mathbb{R} \setminus \mathbb{Q}$ of **full Lebesgue measure** such that: every orientation-preserving circle diffeomorphism f of class C^∞ , with $\rho(f) \in A$, is conjugate to $R_{2\pi\rho(f)}$ by a diffeomorphism of class C^∞ . No hypothesis of closeness to the rotation: it is a global result.

Imported from M. R. Herman, “Sur la conjugaison différentiable des difféomorphismes du cercle à des rotations”, *Publ. Math. IHÉS* **49** (1979), 5–233, DOI 10.1007/BF02684798 (source register, code K8; Numdam record verified on 16 September 2026). **What is assumed:** Herman’s entire apparatus (study of one-parameter families, measure theory on rotation numbers, iterated distortion estimates), about two hundred pages. **Why we do not prove it:** it exceeds by an order of magnitude the space of this volume; Theorem 6.5 is

its local analytic version, proved in full.

Remark 6.19: What we import: Yoccoz’s theorem (1984). **Statement (Yoccoz, 1984).** Every orientation-preserving circle diffeomorphism f of class C^∞ , whose rotation number $\rho(f)$ satisfies a Diophantine condition (Definition 3.4), is conjugate to $R_{2\pi\rho(f)}$ by a diffeomorphism of class C^∞ . With respect to Herman, the full-measure set A is replaced by the explicit set $\mathcal{CD}$ of all Diophantine numbers.

Imported from J.-C. Yoccoz, “Conjugaison différentiable des difféomorphismes du cercle dont le nombre de rotation vérifie une condition diophantienne”, *Ann. Sci. École Norm. Sup. (4)* **17** (1984), 333–359, DOI 10.24033/asens.1475 (code K9; Numdam record verified on 16 September 2026; the very title of the article states the result). **What is assumed:** the whole of Herman’s theory plus Yoccoz’s renormalisation technique. See also the survey of H. Eliasson, B. Fayad, R. Krikorian, arXiv:1810.07107 (code K19), reopened on 17 September 2026.

Remark 6.20: What we import: Bryuno is the optimal condition in the local analytic problem. **Statement (sufficiency).** For every $\alpha \in \mathcal{B}$ (a Bryuno number, Definition 6.2) and every $\nu > 0$ there exists $\varepsilon(\alpha, \nu) > 0$ such that every analytic circle diffeomorphism f , defined on the strip S_ν , with $\rho(f) = \alpha$ and $\|f - R_{2\pi\alpha}\|_\nu < \varepsilon(\alpha, \nu)$, is conjugate to $R_{2\pi\alpha}$ by an analytic diffeomorphism. It is Theorem 6.5 with $\mathcal{CD}$ replaced by the larger $\mathcal{B}$ (Proposition 6.2: $\mathcal{CD} \subsetneq \mathcal{B}$).

Statement (optimality). If $\alpha \notin \mathcal{B}$, there exist analytic circle diffeomorphisms with rotation number α , arbitrarily close to $R_{2\pi\alpha}$, which are not analytically linearisable.

Imported from J.-C. Yoccoz, “Analytic linearization of circle diffeomorphisms”, in *Dynamical Systems and Small Divisors* (Cetraro 1998), *Lecture Notes in Math.* **1784**, Springer 2002, 125–173, DOI 10.1007/978-3-540-47928-4_3 (code K13), and from K19, reopened and verified on 17 September 2026. The optimality technique is the one Yoccoz had developed for Siegel’s theorem on holomorphic germs: J.-C. Yoccoz, “Théorème de Siegel, nombres de Bruno et polynômes quadratiques”, *Astérisque* **231** (1995), 1–88 (code K12). The condition is due to A. D. Bryuno, *Analytic form of differential equations*, *Trans. Moscow Math. Soc.* **25** (1971) and **26** (1972) — the page ranges of these two works are reported inconsistently by the secondary sources consulted and are **not** given here; for the statement and the discussion refer to K12 and K13, reopened.

What is assumed: Yoccoz’s renormalisation theory and the construction of the counterexamples. **Note of honesty:** the sufficiency of $\mathcal{B}$ in the local case does not follow from the scheme of Section 6.7 as it stands, because the estimate of Theorem 6.1 uses the Diophantine condition in power form; one needs upper bounds involving the Bryuno function in place of $\Lambda(\tau)/\delta^{2+\tau}$.

Remark 6.21: What we import: the condition $\mathcal{H}$ and the hierarchy $\mathcal{CD} \subset \mathcal{H} \subset \mathcal{B}$. The global analytic problem (to linearise every analytic diffeomorphism with a given rotation number, without any closeness hypothesis) has an arithmetic answer different from both of the preceding ones. Yoccoz introduced a class $\mathcal{H}$ of irrational numbers — named in honour of Herman — which characterises exactly the rotation numbers for which global analytic linearisation is always possible, and he proved

$$\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B},$$

with both inclusions **strict**. **Imported from** J.-C. Yoccoz, “Centralisateurs et conjugaison différentiable des difféomorphismes du cercle”, in *Petits diviseurs en dimension 1*, *Astérisque*

231 (1995), 89–242 (code K11), and from K13; the overall picture, with the inclusions in the form reported here, is in K19 (arXiv:1810.07107), reopened and verified on 17 September 2026.

What we do not reproduce: the definition of $\mathcal{H}$. It is a recursive condition, formulated by comparing sequences constructed along the renormalisation with the Bryuno function, and to reproduce it outside its technical context would be more misleading than useful. We limit ourselves to recording: (i) that $\mathcal{H}$ exists and is characterised in K11/K13; (ii) the two strict inclusions; (iii) that for the local problem the optimal condition is $\mathcal{B}$ (Remark 6.20), whereas for the global one it is $\mathcal{H}$: **the two optimalities are different**, and it is a current error (correction 3.16 of the expansion plan of the volume) to attribute to Bryuno the global optimality. The hierarchy is drawn in Figure 6.4.

Remark 6.22: Panorama: finite regularity. Between the C^∞ case (Herman–Yoccoz) and the analytic one (Arnold, Bryuno, $\mathcal{H}$) there lies the whole scale of finite regularities, where the result depends both on the Diophantine exponent τ and on the regularity of f . Three references, with their respective contributions, all verified in the volume’s source register on 16 September 2026:

- Y. Katznelson, D. Ornstein, “The differentiability of the conjugation of certain diffeomorphisms of the circle” and “The absolute continuity of the conjugation of certain diffeomorphisms of the circle”, *Ergodic Theory Dynam. Systems* **9** (1989), respectively 643–680 and 681–690 (code K14): optimal regularity results obtained with a non-KAM method, based on distortion estimates, which does not go through the Newton scheme.
- K. M. Khanin, Ya. G. Sinai, “A new proof of M. Herman’s theorem”, *Comm. Math. Phys.* **112** (1987), 89–101 (code K15): a proof via thermodynamic formalism and renormalisation.
- K. Khanin, A. Teplinsky, “Herman’s theory revisited”, *Invent. Math.* **178** (2009), 333–344 (code K16): sharp result in class $C^{2+\alpha}$.

These works are cited as panorama: no statement of this chapter relies on them.

Remark 6.23: Panorama: from Kolmogorov 1954 to the circle. The scheme of Section 6.7 is the one-dimensional version of an idea that Kolmogorov expounded in 1954 for nearly integrable Hamiltonian systems: under a Diophantine condition on the frequencies and a non-degeneracy condition, most of the invariant tori of the integrable system survive a small perturbation. The original note — A. N. Kolmogorov, “O sokhranenii uslovno periodicheskikh dvizhenii...” (transliterated; “On the conservation of conditionally periodic motions...”), *Dokl. Akad. Nauk SSSR* **98** (1954), 527–530 — **has not been located** in any of the archives consulted (see the volume’s source register, list of the items not located); we therefore cite the English translation published in *Lecture Notes in Physics* **93**, 51–56, DOI 10.1007/bfb0021737 (code K3), and the two works of Arnold that gave the first detailed proof and the general framework: *Uspekhi Mat. Nauk* **18:5** (1963), 13–40 (transl. Russian Math. Surveys **18:5**, 9–36; code K1) and *Russian Math. Surveys* **18:6** (1963), 85–191 (code K2). For the modern exposition of the scheme: J. Pöschel, “A lecture on the classical KAM theorem”, *Proc. Sympos. Pure Math.* **69** (2001), 707–732 (code K22), with the technical correction of A. Bounemoura, *MSRI Publ.* **72** (code K23); R. de la Llave, “A tutorial on KAM theory”, *ibid.* (code K21); and the historical panorama of H. W. Broer, “KAM theory: the legacy of Kolmogorov’s 1954 paper”, *Bull. Amer. Math. Soc.* **41** (2004), 507–521 (code R42). **Bibliographic warning:** Broer’s article reports for Kolmogorov’s note the pages “525–530”, at odds with the other records consulted, which give 527–530; we adopt here the second range and record the discrepancy without resolving it. The “Theorem 1” stated by

Broer as Arnold's theorem (§3, p. 511) is a modern C^∞ reformulation on a parametrised family, different from the original 1961 statement and different from our Theorem 6.5: we do not use it.

Remark 6.24: Three bridges towards the rest of the treatise. (1) Towards Siegel discs (Chapter 12). The same arithmetic, the same small divisors: to linearise a holomorphic germ $z \mapsto e^{2\pi i \rho} z + O(z^2)$ around a fixed point means solving a homological equation with divisors $e^{2\pi i k \rho} - 1$, and the optimal condition is again Bryuno's (K12). Chapter 12 picks up the thread from the point of view of holomorphic dynamics; here it suffices to observe that Siegel's theorem and Theorem 6.5 are two faces of the same arithmetic coin.

(2) Towards Arnold tongues (Chapter 4) and synchronisation (Chapter 13). The complement of the tongues — the set of parameters with irrational rotation number — has positive measure precisely because the Diophantine numbers have full measure (Theorem 3.6) and because, on those parameters, Theorem 6.5 (or Herman's theorem) guarantees a stable quasi-periodic behaviour. Chapter 13 reads the same phenomenon in a physical key: phase locking inside the tongues, quasi-periodic drift outside.

(3) Towards Volume VI. The questions of density of hyperbolicity and of renormalisation in the complex plane (MLC and its surroundings) are treated in Volume VI, Chapter 9: no panorama of them is given here, so as not to duplicate it.

6.9. Figures

6.10. Assigned exercises

The answers are in the Teacher's key (Section 6.13).

Exercise 6.1: The Diophantine constant of $\sqrt{2} - 1$. *Goal:* to repeat Proposition 6.1 for a second quadratic irrational, and to place the result between the two known estimates. *Estimated time:* 40 minutes. *Type:* symbolic with numerical check.

Let $\beta = \sqrt{2} - 1 = [0; 2, 2, 2, \dots]$, whose denominators q_n are the Pell numbers 1, 2, 5, 12, 29, ... (Chapter 3).

- Compute $q\|q\beta\|$ for $q = 1, 2, 5, 12, 29$ and identify where the minimum falls.
- Prove that $\inf_{q \geq 1} q\|q\beta\| = 6 - 4\sqrt{2}$, in closed form, following the three steps of the proof of Proposition 6.1 (the convergents are now p_n/q_n with q_n the Pell numbers; the role of φ is taken by $1 + \sqrt{2}$).
- Check that the value found lies between the general estimate of Proposition 3.5 with $M = 2$ (that is, $K \geq 1/4$) and the golden constant φ^{-2} , and explain why it *must* be so.

Exercise 6.2: The homological equation for a three-mode datum. *Goal:* to use the closed form of Example 6.3 on a less trivial datum and to compare the result with the upper bound of Theorem 6.1. *Estimated time:* 35 minutes. *Type:* symbolic/numerical.

Let $\alpha = 2\pi/\varphi$ and $\varphi_{\text{datum}}(\theta) = \varepsilon(\cos \theta + \cos 2\theta + \cos 3\theta)$ with $\varepsilon = 10^{-2}$.

- Write the zero-mean solution ψ as a sum of three terms in closed form, generalising Example 6.3: for the k -th mode, $\psi_k(\theta) = \varepsilon \sin(k\theta - \frac{k\alpha}{2}) / (2 \sin \frac{k\alpha}{2})$.
- Compute numerically the three amplitudes $\varepsilon / (2|\sin(k\alpha/2)|)$ for $k = 1, 2, 3$ and say which mode dominates.
- Verify the homological equation at three points of your choice, with the discrepancy stated.

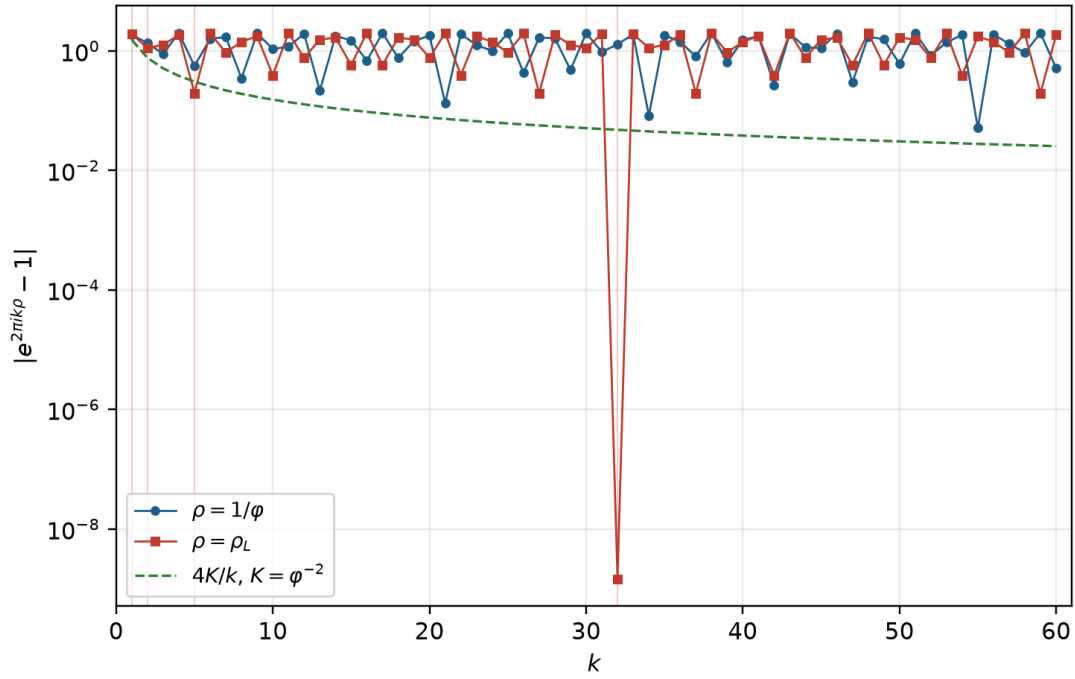


Figure 6.1: The small divisors $|e^{2\pi i k \rho} - 1|$ for $k = 1, \dots, 60$, on a logarithmic scale, for $\rho = 1/\varphi$ (circles) and for the Liouville number ρ_L constructed with $q_{n+1} \geq 2^{q_n}$ (squares); the dashed line is the lower bound $4K/k$ with $K = \varphi^{-2}$ of Lemma 6.2. The vertical lines mark the denominators $q_n = 1, 2, 5, 32$ of ρ_L , where the divisors collapse: 1.913881 , 1.111140 , 1.960343×10^{-1} and 1.462910×10^{-9} . Check printed by the script: the minimum of the ratio between the golden divisor and the lower bound equals 1.220046 (at $k = 1$), hence ≥ 1 ✓. Computed numerically; script `fig06_piccoli_divisori.py`, Appendix B.

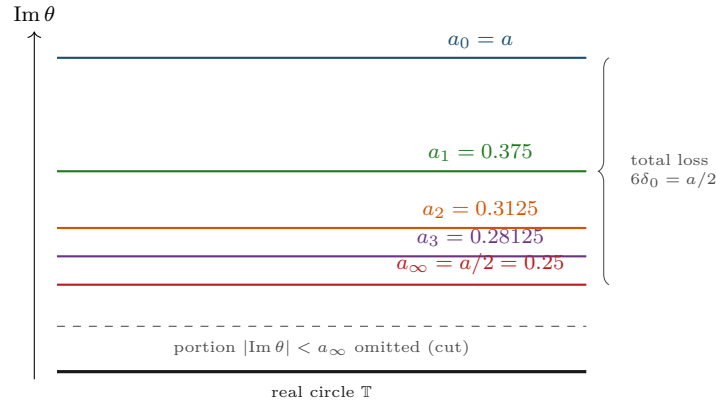


Figure 6.2: The loss of domain of the Newton scheme (Section 6.7), with the exact values of the case $a = 0.5$, $\rho = 1/\varphi$ of the table of Remark 6.15. Each step consumes $3\delta_j$ of half-width (δ for the homological equation, δ for the Cauchy estimate on ψ' , δ for the inversion of H), and $\sum_j 3\delta_j = 6\delta_0$ converges: the limit strip S_{a_∞} is not empty. Of the strip only the upper half is drawn, and of it only the annulus between a_∞ and a_0 : the ordinate is to scale (12 cm per unit) starting from $a_\infty = 0.25$, while the portion $|\text{Im } \theta| < a_\infty$ is omitted (dashed cut line). Coordinates computed by hand from the values of the table.

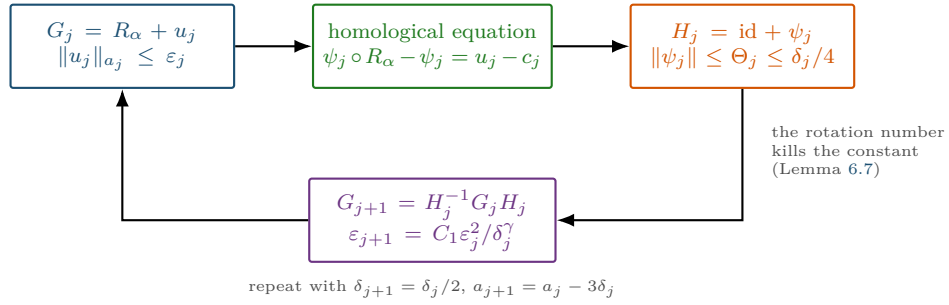


Figure 6.3: The Newton scheme as a block diagram: at each turn the error goes from ε_j to $C_1 \varepsilon_j^2 / \delta_j^\gamma$ (quadratic) and the strip from a_j to $a_j - 3\delta_j$ (summable loss). The passage from the top right block to the bottom one uses Lemma 6.7, that is, the hypothesis $\rho(F) = \rho$: it is the only point where the rotation number enters the proof. Figure drawn by hand, with no numerical data.

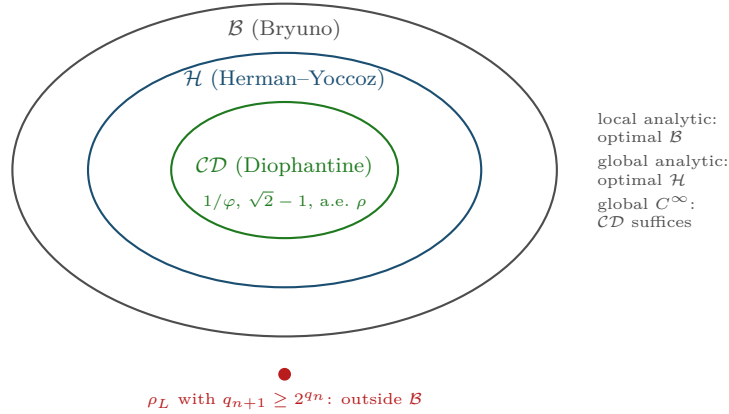


Figure 6.4: The arithmetic hierarchy $\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}$ (Remark 6.21). The golden ratio, $\sqrt{2} - 1$ and almost every ρ lie in $\mathcal{CD}$ (Theorem 3.6); the number ρ_L with $q_{n+1} \geq 2^{q_n}$ of Example 6.2 lies outside $\mathcal{B}$ (divergent Bryuno sum, every term $\geq \log 2$). There also exists, by Proposition 6.2, a $\rho \in \mathcal{B} \setminus \mathcal{CD}$ (the one with $q_{n+1} \approx q_n^n$), but it is not drawn: its position *relative to* $\mathcal{H}$ has not been verified and we do not assert it. On the right, which arithmetic condition is optimal for which problem. Figure drawn by hand, schematic: the areas are not to scale.

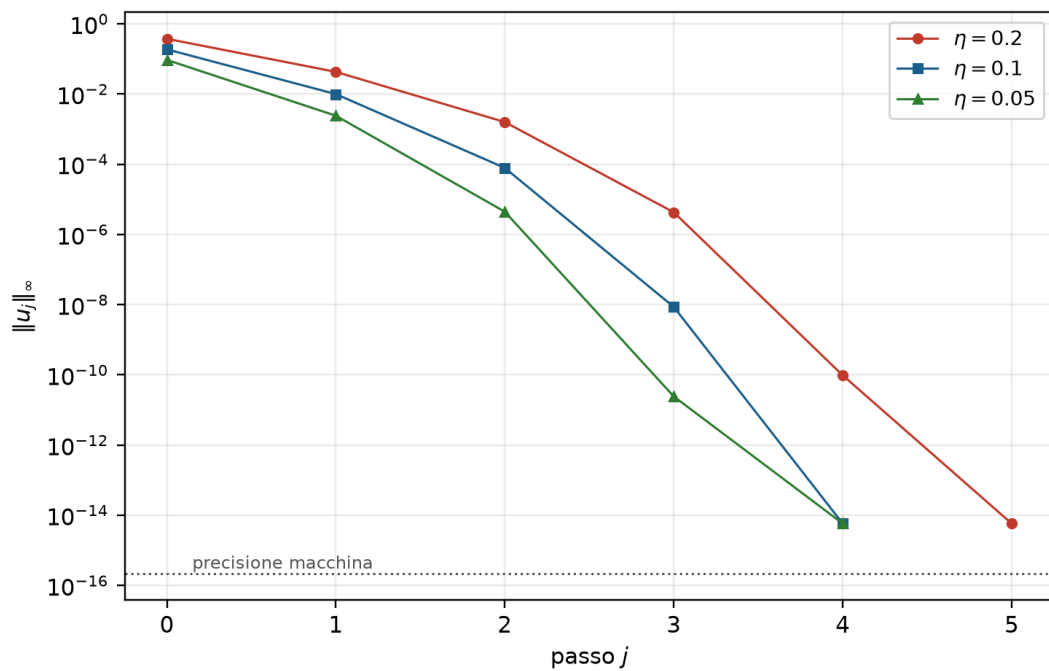


Figure 6.5: The Newton scheme *really iterated*, on $F = H_0 \circ R_\alpha \circ H_0^{-1}$ with $H_0 = \text{id} + \eta \sin$ and $\alpha = 2\pi/\varphi$ (Example 6.6): norm $\|u_j\|_\infty$ of the residual error as a function of the number of steps, on a logarithmic scale, for $\eta = 0.20; 0.10; 0.05$. The slope *doubles* at each step — the signature of quadratic convergence — down to machine precision (dotted line, 2.2×10^{-16}): four or five steps suffice. Checks printed by the script: Lemma 6.7 ($|c_j| \leq \|r_j\|_\infty$) is verified at each step of all three series; the limit conjugacy coincides with H_0 composed with a rotation (discrepancy $\leq 9.8 \times 10^{-14}$) and the residual $\max_\theta |F(W(\theta)) - W(\theta + \alpha)|$ equals 5.3×10^{-15} . Computed numerically; script `fig06_newton_iterato.py`, Appendix B.

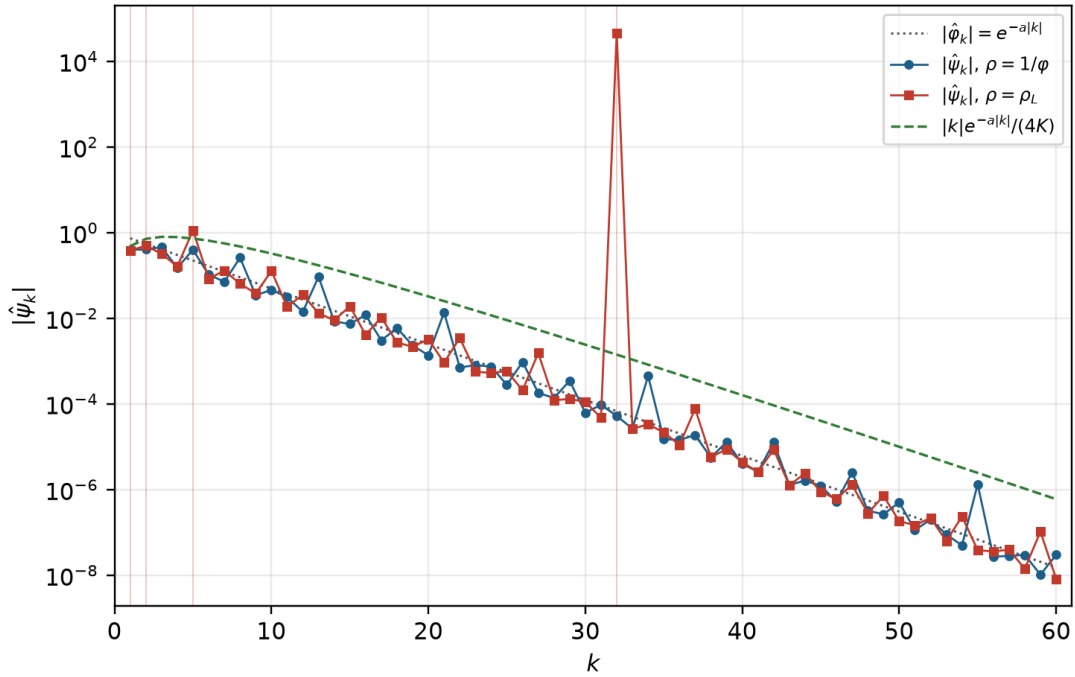


Figure 6.6: The Fourier coefficients $|\hat{\psi}_k|$ of the solution of the homological equation for the *same* analytic datum $|\hat{\varphi}_k| = e^{-a|k|}$ with $a = 0.30$ (dotted line) and two different rotation numbers: $\rho = 1/\varphi$ (circles) and the Liouville ρ_L (squares); the dashed line is the upper bound $|k|e^{-a|k|}/(4K)$ of Theorem 6.1. Checks printed by the script: in the golden case the largest ratio between value and upper bound equals $0.819641 \leq 1$ ✓, and the coefficients decay from $|\hat{\psi}_1| = 3.974208 \times 10^{-1}$ to $|\hat{\psi}_{60}| = 2.987557 \times 10^{-8}$ (ratio 1.330×10^7 , exponential decay: ψ remains analytic); in the Liouville case, at $k = q_n = 32$ one has $|\hat{\psi}_{32}| = 4.629725 \times 10^4$ against 5.254795×10^{-5} for the golden one, a ratio of 8.81×10^8 . Computed numerically; script `fig06_coefficienti_psi.py`, Appendix B.

- (d) The dominant mode is the one with the largest k . Does this contradict the decay of the Fourier coefficients? Explain.

Exercise 6.3: The threshold ε_0 for another strip and another number. *Goal:* to manipulate the constants of Theorem 6.5 and to read off their dependences. *Estimated time:* 25 minutes. *Type:* numerical.

- (a) Compute δ_0 , C_1 , $\varepsilon_0(a, \rho)$ and the limit strip a_∞ for $a = 2$ and $\rho = \sqrt{2} - 1$ (use $K = 6 - 4\sqrt{2}$ of Exercise 6.1, $\tau = 0$, $\Lambda(0) = 1 + e^{-1}$).
- (b) Repeat for $a = 1$, same ρ , and check that the ratio between the two thresholds is 8. Why exactly 8?
- (c) For the same a , is the threshold for $\rho = \sqrt{2} - 1$ larger or smaller than the one for $\rho = 1/\varphi$? Interpret the result.

6.11. Worked exercises

Exercise 6.4: One Newton step computed by hand. *Type:* symbolic with an independent numerical check. For $F(\theta) = \theta + \alpha + \varepsilon \cos \theta$ with $\alpha = 2\pi/\varphi$, compute by hand the dominant term of $\|u_1\|_\infty$ after one step of the Newton scheme, and compare it with the value measured by the script.

Solution. Strategy. Follow Steps 1–2 of Lemma 6.8 with a datum for which everything is in closed form, keeping only the term of order ε^2 .

Step 1. The mean of $u = \varepsilon \cos \theta$ is zero: $c = 0$, hence $\varphi_{\text{datum}} = u$ and Lemma 6.7 has nothing to do (this is a particularly fortunate case: in general $c \neq 0$).

Step 2. By Example 6.3, $\psi(\theta) = \frac{\varepsilon \sin(\theta - \frac{\alpha}{2})}{2 \sin \frac{\alpha}{2}}$.

Step 3. The remainder is $r(\theta) = u(\theta + \psi(\theta)) - u(\theta) = u'(\theta)\psi(\theta) + O(\varepsilon^3)$, with $u'(\theta) = -\varepsilon \sin \theta$:

$$r(\theta) = -\frac{\varepsilon^2 \sin \theta \sin(\theta - \frac{\alpha}{2})}{2 \sin \frac{\alpha}{2}} + O(\varepsilon^3).$$

Step 4. Prosthaphaeresis: $\sin t \sin(t - \beta) = \frac{1}{2}[\cos \beta - \cos(2t - \beta)]$, whose modulus has maximum $\frac{1}{2}(1 + |\cos \beta|)$ (attained when $\cos(2t - \beta) = \mp 1$). With $\beta = \alpha/2$:

$$\frac{\|r\|_\infty}{\varepsilon^2} \longrightarrow \frac{1 + |\cos \frac{\alpha}{2}|}{4 \sin \frac{\alpha}{2}}.$$

Step 5 (numerical value). $\alpha = 2\pi/\varphi = 3.8832220774$, $\alpha/2 = 1.9416110387$; $\cos(\alpha/2) = -0.362385$, $\sin(\alpha/2) = 0.932036$. Hence

$$\frac{1 + 0.362385}{4 \times 0.932036} = \frac{1.362385}{3.728144} = 0.365431.$$

Independent check. Block [E] of `verifica_cap06_kam.py` computes $\|u_1\|_\infty$ without asymptotic formulas (constructing H , inverting it with the scalar Newton method and evaluating $H^{-1} \circ F \circ H - R_\alpha$ at 4000 points): the ratio $\|u_1\|_\infty/\varepsilon^2$ equals 0.379819 for $\varepsilon = 10^{-1}$, 0.366795 for $\varepsilon = 10^{-2}$ and 0.365445 for $\varepsilon = 10^{-4}$, in agreement with 0.365431 to within 1.4×10^{-5} ✓ (the residual discrepancy is the neglected $O(\varepsilon^3)$ term).

Interpretation. The constant 0.365431 depends *only* on α , not on ε : the Newton step is quadratic with a fixed constant. Note that the constant grows like $1/\sin(\alpha/2)$ when $\alpha \rightarrow 0$: the small divisors worsen the quadratic constant, but they do not destroy the quadraticity.

Typical error to avoid. Writing $\|u_1\| \approx \|u'\| \|\psi\| = \varepsilon \cdot \varepsilon / (2 \sin \frac{\alpha}{2}) = 0.536 \varepsilon^2$: it is a valid upper bound but not an exact one, because the maxima of $|u'|$ and of $|\psi|$ do not fall at the same point. The correct computation requires the maximum of the *product*, which equals $(1 + |\cos \frac{\alpha}{2}|)/2$ times the product of the maxima, that is, 0.681 times: $0.681 \times 0.536 = 0.365$ ✓ (a third route, consistent with the first two).

Extension. Repeat with $u = \varepsilon \cos 2\theta$: the same analysis gives the limit $(1 + |\cos \alpha|)/(4 \sin \alpha)$ in absolute value, with α in place of $\alpha/2$ — the divisor gets worse because $\|2\rho\| < \|\rho\|$ for $\rho = 1/\varphi$.

Exercise 6.5: Where the Newton scheme dies if ρ is Liouville. *Type:* diagnostic/demonstrative. Identify the exact point of the proof of Theorem 6.5 that ceases to work if ρ is not Diophantine, and explain why none of the other steps notices the difference.

Solution. Strategy. Run backwards through the chain of lemmas and identify the only point where the constant K appears.

Step 1. The conclusion of Lemma 6.8 is $\|u_1\| \leq C_1 \varepsilon^2 / \delta^\gamma$, with $C_1 = 4\Lambda/K$ and $\gamma = 3 + \tau$: if ρ is not Diophantine there exists no admissible pair (K, τ) , and the constant C_1 is simply not defined.

Step 2. Going back up, C_1 is born in Theorem 6.1 through $|\hat{\psi}_k| \leq |k|^{1+\tau} |\hat{\varphi}_k| / (4K)$, that is, through Lemma 6.2. It is the only point of the whole proof where the arithmetic of ρ intervenes.

Step 3 (what does keep working). Lemma 6.3 (Cauchy estimates) uses only holomorphy; Lemma 6.6 uses only $\|\psi'\| \leq 1/2$; Lemma 6.7 uses only the monotonicity of the rotation number and the irrationality implicit in $\rho(F) = \rho$. None of these three “notices” that ρ is Liouville: the only breakdown is that ψ might not exist at all (Theorem 6.3), or exist with an enormous norm, and in that case the hypothesis $\|\psi'\| \leq 1/4$ of the following step fails.

Independent check. One can see the breakdown forwards as well, without any theory: in block [G] of `verifica_cap06_kam.py`, for ρ_L one has $\|q_5 \rho_L\| < 2.33 \times 10^{-10}$ with $q_5 = 32$; the corresponding coefficient of the solution, for the datum of Example 6.4, equals $|\hat{\psi}_{32}| = 4.63 \times 10^4$ against $\|\varphi\| \leq 1$: the “solution” of the linear step is four orders of magnitude larger than the datum ✓, hence the condition $\Theta \leq \delta/4$ of Lemma 6.8 is violated by far and the step is not even defined.

Interpretation. Arnold’s theorem is not “true but hard to prove” for Liouville numbers: it is *false* (Remark 6.20, optimality of the Bryuno condition, which excludes many Liouville numbers). The localisation of the breakdown in a single lemma is what makes it possible to replace the Diophantine condition with Bryuno’s without touching the rest of the proof: it suffices to redo Theorem 6.1 with upper bounds in terms of the Bryuno function.

Typical error to avoid. Believing that the breakdown lies in the loss of domain (that is, in the fact that $\sum 3\delta_j < \infty$). The loss of domain is a choice of *ours*, independent of ρ : it is the constant C_1 that depends on ρ , and it is its blowing up that breaks the step condition (C).

Extension. Check that, if one tries to use a very large τ in order to “force” a Diophantine condition on a Liouville number, Proposition 6.2 shows the obstacle: a Diophantine number of type (K, τ) has $q_{n+1} < q_n^{1+\tau}/K$, whereas ρ_L has $q_{n+1} \geq 2^{q_n}$, which beats every power.

Exercise 6.6: Why geometric convergence would not suffice. *Type:* quantitative/demonstrative. Show that a scheme with merely *geometric* convergence of the error ($\varepsilon_{j+1} \leq q \varepsilon_j$ with $0 < q < 1$ fixed) does not converge, with the same loss of domain as the Newton scheme, whereas the quadratic one does.

Solution. Strategy. Compare, under the two hypotheses, the quantity that has to remain ≤ 1 : the step condition $C_1\varepsilon_j/\delta_j^\gamma \leq 1$.

Step 1 (geometric scheme). Suppose $\varepsilon_j \leq q^j\varepsilon_0$ and $\delta_j = \delta_0 2^{-j}$ (the loss of domain is the same, because it is imposed by the necessity that $\sum \delta_j < \infty$). Then

$$\frac{C_1\varepsilon_j}{\delta_j^\gamma} \leq \frac{C_1\varepsilon_0}{\delta_0^\gamma} (q 2^\gamma)^j.$$

Since $\gamma = 3 + \tau \geq 3$, one has $2^\gamma \geq 8$: the step condition remains satisfied for every j **only** if $q < 2^{-\gamma} \leq 1/8$, that is, only if the “geometric convergence” is already faster than the loss of domain. For a generic q ($q = 1/2$, for instance) the right-hand side grows like 4^j and the condition fails after a few steps.

Step 2 (quadratic scheme). With $\varepsilon_{j+1} = C_1\varepsilon_j^2/\delta_j^\gamma$ the quantity to be controlled is $x_j/2^j$ with $x_{j+1} = x_j^2$ (Step 2 of the proof of Theorem 6.5): starting from $x_0 \leq \frac{1}{2}$ one obtains $x_j \leq 2^{-2^j}$, which crushes *any* geometric growth of the factor $2^{j\gamma}$, because $2^{-2^j} 2^{j\gamma} \rightarrow 0$ (the double exponential beats the exponential).

Independent check (numerical). With the values of block [F] ($\gamma = 3$, $\delta_0 = 0.0416667$, $C_1 = 14.324619$, $\varepsilon_0 = 3.156190 \times 10^{-7}$): in the quadratic scheme $C_1\varepsilon_j/\delta_j^\gamma = x_j/8$ equals 0.0625, 0.03125, 0.0078, 4.9×10^{-4} , 1.9×10^{-6} , ... for $j = 0, 1, 2, 3, 4$: it decreases $\checkmark$. With the same ε_0 but geometric convergence $q = 1/2$ one would have $0.0625 \cdot 4^j$: 0.0625, 0.25, 1.0, 4.0, ...: condition (C) is violated already at the third step $\checkmark$.

Interpretation. This is the historical reason why Kolmogorov introduced Newton’s method in this context: it is not a technical elegance but an arithmetic necessity. Every nineteenth-century attempt to solve the problem with ordinary perturbation series (which converge, when they converge, geometrically) was doomed to fail.

Typical error to avoid. Thinking that one can get round the problem by choosing δ_j constant: $\sum_j \delta_j$ would diverge and the strip would be used up in a finite number of steps.

Extension. Check that even the choice $\delta_j = \delta_0/(j+1)^2$ (summable, but not geometric) does not save the geometric scheme: the factor becomes $q^j (j+1)^{2\gamma}$, and $\prod_{i < j}$ of the corresponding ratios contains $(j!)^{2\gamma}$, which diverges faster than q^{-j} for every fixed $q < 1$.

6.12. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 6.13).

1. (Definition) The homological equation of the rotation R_α (Definition 6.1) is
 - A) $\psi(\theta + \alpha) + \psi(\theta) = \varphi(\theta)$
 - B) $\psi(\theta + \alpha) - \psi(\theta) = \varphi(\theta)$
 - C) $\psi'(\theta) = \varphi(\theta)$
 - D) $\psi(\theta + \alpha) = \varphi(\psi(\theta))$
2. (Hypothesis) In Arnold’s theorem (Theorem 6.5) the hypothesis $\rho(F) = \rho$
 - A) is superfluous, because F is close to R_α
 - B) serves only to fix the notation
 - C) is indispensable: without it F could fall inside an Arnold tongue and have a rational rotation number
 - D) replaces the Diophantine condition
3. (Symbolic computation) The zero-mean solution of $\psi(\theta + \alpha) - \psi(\theta) = \varepsilon \cos \theta$ is
 - A) $\varepsilon \sin(\theta)/(2 \sin \frac{\alpha}{2})$
 - B) $\varepsilon \sin(\theta - \frac{\alpha}{2})/(2 \sin \frac{\alpha}{2})$
 - C) $\varepsilon \cos(\theta - \frac{\alpha}{2})/(2 \cos \frac{\alpha}{2})$
 - D) $\varepsilon \cos \theta / (e^{i\alpha} - 1)$
4. (Numerical computation) For $\rho = 1/\varphi$ the exact Diophantine constant $\inf_q q \|q\rho\|$ equals (Proposition 6.1)

- A) $1/\sqrt{5} = 0.447214$ B) $\varphi^{-2} = 0.381966$ C) $1/3 = 0.333333$ D) $6-4\sqrt{2} = 0.343146$
5. (Proof step) In Lemma 6.8, the inequality $|c| \leq \|r\|$ (where c is the mean of u and r the quadratic remainder) is obtained
- A) from the Cauchy estimate B) from the monotonicity of the rotation number applied to $G_1 = R_\alpha + u_1$, which forces u_1 to change sign C) from the Diophantine condition D) from the Gottschalk–Hedlund lemma
6. (Error diagnosis) A student states: “the Bryuno condition is the optimal condition for the analytic linearisation of circle diffeomorphisms”. The statement is
- A) correct without reservations B) correct for the *local* problem, but for the *global* one the optimal condition is the condition $\mathcal{H}$, with $\mathcal{H} \subsetneq \mathcal{B}$ C) false: the optimal condition is always the Diophantine one D) meaningless, because $\mathcal{B} \subset \mathcal{CD}$
7. (Transfer) The Gottschalk–Hedlund lemma (Theorem 6.4) applies
- A) only to irrational rotations B) to every minimal homeomorphism of a compact metric space, hence also to a $C^{1+\text{bv}}$ diffeomorphism with ρ irrational (Denjoy) C) only to analytic maps D) only if ρ is Diophantine
8. (Source/metacognition) The statement of Theorem 6.5 of this chapter
- A) is the literal transcription of Arnold’s 1961 statement B) is the standard modern formulation (Katok–Hasselblatt, Thm. 12.3.1; survey of Eliasson–Fayad–Krikorian), with Arnold 1961 cited for historical priority and constants derived here C) is the “Theorem 1” of Broer’s survey in the *Bull. AMS* D) has no source because it is original

6.13. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **B.** *Rationale:* Definition 6.1. *Distractors:* A has the wrong sign (it would be an equation of “sum” type, without the coboundary structure); C is the primitive equation, not the homological one; D is the exact equation (6.2) written wrongly. *Error diagnosed:* failing to recognise the structure $\psi \circ T - \psi$ (coboundary). *Source:* Definition 6.1. *Prerequisite:* the notion of a function composed with a rotation. *Remedial work:* rewrite the equation in Fourier coefficients (Remark 6.1).
- 2) **C.** *Rationale:* Remark 6.11, with the Arnold tongues of width $O(\varepsilon^q) > 0$ (Theorem 4.7). *Distractors:* A is precisely the error corrected by the expansion plan (item 3.15); B and D misunderstand the role of the two hypotheses, which are independent. *Error diagnosed:* confusing “close to R_α ” with “has rotation number $\alpha/2\pi$ ”. *Source:* Remarks 6.11 and 4.7. *Prerequisite:* Arnold tongues (Chapter 4). *Remedial work:* draw a tongue and place a small perturbation inside it.
- 3) **B.** *Rationale:* Example 6.3, with the direct verification by prosthaphaeresis. *Distractors:* A forgets the phase shift $-\alpha/2$ (and does not solve the equation); C interchanges sine and cosine; D is the Fourier coefficient, not the function. *Error diagnosed:* not carrying out the check by substituting into the equation. *Source:* Example 6.3, block [E]. *Prerequisite:* prosthaphaeresis formulas. *Remedial work:* redo Route 2 of the example, step by step.
- 4) **B.** *Rationale:* Proposition 6.1: the minimum is attained at $q = 1$ and equals $\varphi^{-2} = (3 - \sqrt{5})/2$. *Distractors:* A is the $\liminf$ along the convergents (Proposition 3.2), greater than the minimum; C is the crude estimate of Proposition 3.5 with $M = 1$; D is the constant

of $\sqrt{2} - 1$ (Exercise 6.1). *Error diagnosed*: confusing infimum and limit inferior. *Source*: Proposition 6.1, block [C]. *Prerequisite*: best approximation of the second kind (Chapter 3). *Remedial work*: tabulate $q\|q\beta\|$ for $q = 1, \dots, 13$ and observe the oscillation around $1/\sqrt{5}$.

5) **B.** *Rationale*: Step 5 of the proof of Lemma 6.8, via Lemma 6.7. *Distractors*: A and C are used elsewhere in the same proof (respectively for $\|u'\|$ and for $\|\psi\|$), but they give no information about the mean; D concerns continuous coboundaries, not this step. *Error diagnosed*: failing to identify the only point where the hypothesis on the rotation number enters. *Source*: Lemma 6.7, Remark 6.14. *Prerequisite*: monotonicity of ρ (Theorem 4.2(i)). *Remedial work*: redo Lemma 6.7 with w constant and observe that a non-zero constant would change ρ .

6) **B.** *Rationale*: Remarks 6.20 and 6.21: $\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}$, with $\mathcal{B}$ optimal in the local problem and $\mathcal{H}$ in the global one. *Distractors*: A is the current error (item 3.16 of the plan); C forgets Yoccoz’s optimality result; D reverses the inclusion (it is $\mathcal{CD} \subset \mathcal{B}$, Proposition 6.2). *Error diagnosed*: failing to distinguish the local problem from the global one. *Source*: K13, K11, K19 (source register). *Prerequisite*: Definition 6.2. *Remedial work*: reread Figure 6.4 and its caption.

7) **B.** *Rationale*: Theorem 6.4 is stated for minimal homeomorphisms of compact metric spaces; Corollary 6.3 specialises it to the circle, including the $C^{1+\text{bv}}$ diffeomorphisms via Denjoy. *Distractors*: A restricts unnecessarily; C confuses it with the hypothesis of Theorem 6.5; D introduces an arithmetic hypothesis that the lemma does not require. *Error diagnosed*: attributing to the lemma hypotheses that belong to other theorems of the chapter. *Source*: Theorem 6.4 and Corollary 6.3. *Prerequisite*: minimality, Denjoy’s theorem (Chapter 5). *Remedial work*: list, for each theorem of the chapter, its hypotheses in a table.

8) **B.** *Rationale*: Remark 6.12 and the comment at the head of the chapter’s source file. *Distractors*: A is false and would be an invented citation (the Russian text was not reopened); C is explicitly excluded (Broer’s “Theorem 1” is a C^∞ reformulation on parameters, a different one); D denies the source, which on the contrary exists and is declared. *Error diagnosed*: attributing to a primary source a statement that was merely inspired by it. *Source*: Remark 6.12; sources block at the end of the chapter. *Prerequisite*: none. *Remedial work*: compare the three formulations cited in the remark.

BEST PRACTICE

Assigned exercises — answers.

6.1 (constant of $\sqrt{2} - 1$). (a) With $\beta = 0.414213562$: $q\|q\beta\|$ equals 0.414213562 ($q = 1$), **0.343145751** ($q = 2$), 0.355339059 ($q = 5$), 0.353247018 ($q = 12$), 0.353605956 ($q = 29$): the minimum is at $q = 2$ (script `verifica_cap06_esercizi.py`, block [1]; minimum over $q \leq 200\,000$ confirmed at $q = 2$ ✓). (b) The convergents of $[0; 2, 2, 2, \dots]$ have q_n the Pell numbers and $\delta_n = q_n\beta - p_n = (-1)^n(1 + \sqrt{2})^{-(n+1)}$ (the same algebra as in Proposition 6.1 with φ replaced by $1 + \sqrt{2}$, a root of $x^2 = 2x + 1$); the products $c_n = q_n\|q_n\beta\|$ oscillate around $\liminf c_n = 1/(2\sqrt{2}) = 0.353553$ and the minimum is $c_1 = 2\|2\beta\| = 2(3 - 2\sqrt{2}) = 6 - 4\sqrt{2} = 0.343145751$. The reduction from a generic q to a convergent q is identical to Step 3 of the proposition (Theorem 3.4). (c) $1/4 = 0.25 < 0.343146 < 0.381966 = \varphi^{-2}$ ✓. It must be so: the first inequality because Proposition 3.5 with $M = 2$ is a valid but not optimal estimate; the second because φ is “the most Diophantine” (Proposition 3.3), that is, no irrational has a larger constant.

6.2 (three-mode datum). (a) $\psi = \sum_{k=1}^3 \varepsilon \sin(k\theta - \frac{k\alpha}{2}) / (2 \sin \frac{k\alpha}{2})$, by linearity of the homological equation. (b) With $\alpha = 2\pi/\varphi$ and $\varepsilon = 10^{-2}$ (script `verifica_cap06_esercizi.py`, block [3]): amplitudes 0.005364620 ($k = 1$), 0.007402031 ($k = 2$), 0.011300176 ($k = 3$): $k = 3$ dominates ✓. (c) The equation is verified at $\theta = 0.3; 1.7; 4.2$ for each mode with discrepancy $< 10^{-14}$ ✓. (d) No, there is no con-

tradition: the *data* $\widehat{\varphi}_k$ here all have the same modulus $\varepsilon/2$ (the datum does not decay); it is the *division* by the small divisor that makes the amplitude grow with k , because $\|3\rho\| < \|2\rho\| < \|\rho\|$ for $\rho = 1/\varphi$ at the first indices. With a genuine analytic datum, $|\widehat{\varphi}_k| \leq \|\varphi\|_b e^{-kb}$, the exponential decay beats the polynomial growth of the divisors: this is exactly the content of Theorem 6.1.

6.3 (thresholds). (a) $a = 2$, $\rho = \sqrt{2} - 1$: $\delta_0 = \min(1, 2/12) = 1/6 = 0.1666667$; $C_1 = 4\Lambda/K = 4 \times 1.3678794/0.3431458 = 15.945171$; $\varepsilon_0 = K\delta_0^3/(2^6\Lambda) = 1.814668 \times 10^{-5}$; $a_\infty = 2 - 6 \times \frac{1}{6} = 1.0 = a/2$ ✓ (script, block [2]). (b) $a = 1$: $\delta_0 = 1/12$, $\varepsilon_0 = 2.268334 \times 10^{-6}$; ratio $1.814668 \times 10^{-5}/2.268334 \times 10^{-6} = 8.000$ ✓. Exactly 8 because $\varepsilon_0 \propto \delta_0^{3+\tau}$ with $\tau = 0$ and δ_0 doubles: $2^3 = 8$. (c) Smaller: $\varepsilon_0 \propto K$ and $K(\sqrt{2} - 1) = 0.343146 < K(1/\varphi) = 0.381966$. Interpretation: $\sqrt{2} - 1$ is (slightly) better approximable by rationals, hence its small divisors are (slightly) more dangerous, and the theorem requires somewhat smaller perturbations. Direct comparison at the same $a = 1$: 2.268334×10^{-6} against 2.524952×10^{-6} , ratio $0.8984 = K(\sqrt{2} - 1)/K(1/\varphi)$ ✓.

6.14. Chapter formulary

Homological equation and small divisors

$$\psi \circ R_\alpha - \psi = \varphi, \quad \widehat{\psi}_k = \frac{\widehat{\varphi}_k}{e^{2\pi i k \rho} - 1} \quad (k \neq 0), \quad \widehat{\varphi}_0 = 0 \text{ necessary}$$

Small divisors lemma

$$|e^{2\pi i x} - 1| = 2|\sin \pi x|, \quad 4\|x\| \leq |e^{2\pi i x} - 1| \leq 2\pi\|x\|$$

Diophantine condition (two forms) and constants

$$\left| \rho - \frac{p}{q} \right| \geq \frac{K}{q^{2+\tau}} \iff \|q\rho\| \geq \frac{K}{q^{1+\tau}} \implies |e^{2\pi i k \rho} - 1| \geq \frac{4K}{|k|^{1+\tau}}$$

$$K\left(\frac{1}{\varphi}\right) = \varphi^{-2} = \frac{3 - \sqrt{5}}{2} = 0.381966, \quad K(\sqrt{2} - 1) = 6 - 4\sqrt{2} = 0.343146, \quad (\tau = 0)$$

Estimate of the series and of the analytic homological equation

$$\sum_{k \geq 1} k^{1+\tau} e^{-k\delta} \leq \frac{\Lambda(\tau)}{\delta^{2+\tau}}, \quad \Lambda(\tau) = \Gamma(2 + \tau) + \left(\frac{1 + \tau}{e}\right)^{1+\tau}; \quad \|\psi\|_{b-\delta} \leq \frac{\Lambda(\tau)}{2K} \frac{\|\varphi\|_b}{\delta^{2+\tau}}$$

C^r case

$$\varphi \in C^r, \quad r > \tau + 2 \implies \psi \in C^s \quad \forall s < r - \tau - 2, \quad \|\psi^{(s)}\|_\infty \leq \frac{\zeta(r - \tau - s - 1)}{2K} \|\varphi^{(r)}\|_\infty$$

Newton scheme (constants and recursions)

$$\gamma = 3 + \tau, \quad C_1 = \frac{4\Lambda}{K}, \quad \delta_0 = \min\left(1, \frac{a}{12}\right), \quad \delta_j = \delta_0 2^{-j}, \quad a_{j+1} = a_j - 3\delta_j, \quad a_\infty = a - 6\delta_0 \geq \frac{a}{2}$$

$$\varepsilon_{j+1} = \frac{C_1 \varepsilon_j^2}{\delta_j^\gamma}, \quad x_j := A 2^{(j+1)\gamma} \varepsilon_j \quad (A = C_1 \delta_0^{-\gamma}), \quad x_{j+1} = x_j^2, \quad x_j \leq 2^{-2^j}$$

$$\boxed{\varepsilon_0(a, \rho) = \frac{K \delta_0^{3+\tau}}{2^{6+\tau} \Lambda(\tau)}} \quad (\text{step condition: } C_1 \varepsilon_j / \delta_j^\gamma \leq 1)$$

Gottschalk–Hedlund

T minimal : $\phi = v \circ T - v$ with $v \in C(X) \iff \sup_n |S_n \phi(x_0)| < \infty$ for one x_0

Arithmetic hierarchy

$$\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}, \quad B(\rho) = \sum_{n \geq 0} \frac{\log q_{n+1}}{q_n} < \infty \quad (\text{Bryuno})$$

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Part C

Chaos and Ergodic Theory

CHAPTER 7

The doubling map: symbolic dynamics, Li–Yorke and Sharkovskii

In the previous chapters of this volume the circle has hosted *rigid* dynamics: rotations, homeomorphisms with a rotation number, Denjoy-style diffeomorphisms. In all these cases two nearby points stay nearby forever (rotations are isometries; a circle homeomorphism cannot separate in an uncontrolled way orbits that start close together, because it preserves the cyclic order). In this chapter the circle hosts for the first time a *non-invertible* dynamics: the map that doubles the angle, $D(\theta) = 2\theta \bmod 2\pi$. The loss of invertibility is the price that buys chaos: D is continuous, but two arbitrarily close points may have orbits that separate up to a macroscopic distance (sensitivity), the set of periodic points is dense, and a single orbit can visit every arc of the circle (transitivity) — the three properties that Devaney (1989) packages into the modern definition of deterministic chaos. The chapter develops this dynamics up to its two culminating theorems, the Li–Yorke theorem (1975) and Sharkovskii’s theorem (1964): together, they describe *every* possible set of periods of periodic orbits that a continuous map of the interval can have.

INFO

Level and prerequisites. Level **B–C**, with the general proof of Sharkovskii’s theorem (Section 7.6) of level **C**: a level B reader may read the statement of Sharkovskii’s theorem, Figure 7.4 and pass directly to Section 7.7 (counting of the periodic points) without losing the continuity of the chapter. Prerequisites within the treatise: circle rotations and rotation number (Volume IV, previous chapters); continuity and the intermediate value theorem (Volume II); binary representation of the real numbers. No preliminary knowledge of symbolic dynamics is assumed: the space of binary sequences Σ_2 is introduced from scratch in Section 7.2.

7.1. The doubling and expanding maps

Definition 7.1: Doubling map and m -expanding maps. For an integer $m \geq 2$, the **m -expanding map** is $E_m : \mathbb{T} \rightarrow \mathbb{T}$, $E_m(\theta) = m\theta \bmod 2\pi$. The case $m = 2$ is called the **doubling map** and is denoted $D := E_2$.

Proposition 7.1: Continuity and m -to-1 structure of E_m . E_m is continuous and surjective; every point $\psi \in \mathbb{T}$ has exactly m preimages, $\theta_k = \psi/m + 2\pi k/m$, $k = 0, \dots, m-1$. In particular E_m is not invertible for $m \geq 2$.

Proof. Continuity: $\theta \mapsto m\theta$ is continuous $\mathbb{R} \rightarrow \mathbb{R}$, and the projection $\mathbb{R} \rightarrow \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ is continuous and open, $2\pi\mathbb{Z}$ -periodic, hence the composition descends to a continuous map $\mathbb{T} \rightarrow \mathbb{T}$. Surjectivity and structure of the preimages: solving $m\theta \equiv \psi \pmod{2\pi}$ is equivalent to $\theta = \psi/m + 2\pi k/m$ for $k \in \mathbb{Z}$; as $k = 0, \dots, m-1$ varies these m values are pairwise distinct in $\mathbb{T}$ (their difference is $2\pi(k - k')/m \not\equiv 0$ for $0 < |k - k'| < m$) and every other integer k reproduces one of these modulo 2π . If $m \geq 2$ there are at least two distinct preimages for

every point, hence E_m is not injective. $\square$

Theorem 7.1: Sensitivity to initial conditions (with minimal n). For every $m \geq 2$, $\theta_0 \in \mathbb{T}$ and $\delta > 0$, let $n = n(\theta_0, \delta)$ be the smallest integer ≥ 1 such that $m^n(\delta/2) \geq \pi/m$. Then, setting $\theta_1 = \theta_0 + \delta/2$, one has $|\theta_1 - \theta_0| = \delta/2 < \delta$ and

$$\pi/m \leq E_m^n(\theta_1) - E_m^n(\theta_0) < \pi$$

(difference computed as a real number in $[0, 2\pi)$, that is, without residual wrapping). In particular E_m has sensitivity to initial conditions with sensitivity constant π/m .

Proof. Put $v_j = m^j(\delta/2)$ for $j = 0, 1, 2, \dots$; n is by definition the smallest integer with $v_n \geq \pi/m$. As long as $v_j < \pi/m \leq \pi$, the lift of $E_m^j(\theta_1) - E_m^j(\theta_0)$ to $\mathbb{R}$ has not yet reached 2π : precisely, by induction on $j \leq n$, $E_m^j(\theta_1) - E_m^j(\theta_0) = v_j$ exactly as a real number (without reduction modulo 2π), because at each step the difference is multiplied by m (E_m acts as $\theta \mapsto m\theta$ as long as one does not leave the reference interval $[0, 2\pi)$) and stays $< \pi < 2\pi$ up to and including step $n-1$. By the minimality of n , $v_{n-1} < \pi/m$, hence $v_n = m v_{n-1} < m \cdot \pi/m = \pi$; combined with $v_n \geq \pi/m$ by definition of n , one obtains $v_n \in [\pi/m, \pi)$, and by what has just been said $v_n < \pi < 2\pi$ guarantees that $E_m^n(\theta_1) - E_m^n(\theta_0) = v_n$ without wrapping: the claim. $\square$

Remark 7.1: The case $m = 2$: the correction to the previous text. For $m = 2$ the theorem gives: n minimal with $2^n \delta \geq \pi$ (equivalently $2^n(\delta/2) \geq \pi/2$), and final separation in $[\pi/2, \pi)$. It is exactly the correct version of the sensitivity statement for D : the previous version omitted the minimality of n , which is indispensable in order to exclude that the difference $2^n \delta$ has already “gone around” the circle several times, artificially cancelling the observed separation.

Theorem 7.2: Density of the periodic points of E_m . The set of periodic points of E_m is exactly

$$\text{Per}(E_m) = \left\{ 2\pi \frac{p}{q} : p, q \in \mathbb{Z}, q \geq 1, \gcd(p, q) = 1, \gcd(q, m) = 1 \right\},$$

that is, the rational multiples (in reduced form) of 2π whose denominator is coprime with m ; it is dense in $\mathbb{T}$.

Proof. ($\supseteq$). Let p/q be reduced with $\gcd(q, m) = 1$. By the Euler–Fermat theorem (or, more elementarily, because the multiplicative group $(\mathbb{Z}/q\mathbb{Z})^\times$ is finite) there exists $n \geq 1$ with $m^n \equiv 1 \pmod{q}$, that is, $q \mid m^n - 1$. Write $m^n - 1 = qc$ with $c \in \mathbb{Z}_{>0}$ and put $k = pc \in \mathbb{Z}$; then $\theta := 2\pi p/q = 2\pi k/(m^n - 1)$, and

$$E_m^n(\theta) = m^n \theta - 2\pi k \pmod{2\pi} = \theta + (m^n - 1)\theta - 2\pi k \pmod{2\pi} = \theta + 2\pi k - 2\pi k \pmod{2\pi} = \theta,$$

using $(m^n - 1)\theta = 2\pi k$ by construction: θ is periodic.

($\subseteq$). Let $\theta = 2\pi p/q$ (reduced) be periodic, $E_m^n(\theta) = \theta$ for some $n \geq 1$. Then $m^n \theta - \theta \in 2\pi\mathbb{Z}$, that is, $(m^n - 1)\frac{p}{q} \in \mathbb{Z}$, that is, $q \mid (m^n - 1)p$; since $\gcd(p, q) = 1$, it follows that $q \mid m^n - 1$, that is, $m^n \equiv 1 \pmod{q}$. If we had $d := \gcd(m, q) > 1$, then $d \mid q$ and $d \mid m^n$ for every $n \geq 1$, hence $m^n \pmod{q}$ would always be a multiple of d and could never equal 1 (which is not a multiple of $d > 1$): contradiction. Hence $\gcd(m, q) = 1$.

Density. By part ($\supseteq$) applied to every $n \geq 1$, $\text{Per}(E_m) \supseteq \{2\pi k/(m^n - 1) : k = 0, \dots, m^n - 2\}$: as n grows the denominators $m^n - 1 \rightarrow \infty$, hence these finite sets become arbitrarily dense in $\mathbb{T}$ (density of the rationals with growing denominator), and $\text{Per}(E_m)$ is dense. $\square$

Remark 7.2: The case $m = 2$ and correction 3.11. For $m = 2$, $\gcd(q, 2) = 1$ means q **odd**: the periodic points of D are exactly the rational multiples of 2π with odd denominator (in reduced form), not only those of the form $2\pi k/(2^n - 1)$ listed in isolation without proving their exhaustiveness. The point $\theta = 2\pi/3$ (denominator 3, odd, in reduced form) is periodic for D , as predicted by the criterion: its orbit is $2\pi/3 \mapsto 4\pi/3 \mapsto 2\pi/3$ (period 2). A genuine example of a point that is not periodic for D but is eventually periodic is $\theta = \pi$ (Theorem 7.12 further on, where $D(\pi) = 0$ and 0 is fixed, but π itself never returns to itself). For E_3 instead $\theta = 2\pi/3$ (denominator 3, not coprime with $m = 3$) is eventually fixed ($E_3(2\pi/3) = 2\pi \equiv 0$, and 0 is fixed) but not periodic: numerical check in Section 7.15, ✓.

Theorem 7.3: Topological mixing of E_m , with an explicit estimate of N . For every pair of non-empty open arcs $U, V \subset \mathbb{T}$, let $\delta > 0$ be the angular length of U divided by 2π (that is, U has length $2\pi\delta$) and let $N = N(\delta) := \lceil \log_m(1/\delta) \rceil$ (the smallest integer n with $m^n\delta \geq 1$). Then for every $n \geq N$, $E_m^n(U) = \mathbb{T} \supseteq V$: in particular $E_m^n(U) \cap V \neq \emptyset$ for every $n \geq N$, that is, E_m is **topologically mixing**.

Proof. E_m has constant derivative m (outside a finite set of junction points of the lift), hence it maps an arc of length $2\pi\delta$ into an arc (possibly wrapped several times) of total length $2\pi m\delta$; iterating, $E_m^n(U)$ has total length $2\pi m^n\delta$. As soon as $m^n\delta \geq 1$ (that is, $n \geq N$), this length is $\geq 2\pi$: a continuous arc wrapped on the circle with total length $\geq 2\pi$ covers the whole of $\mathbb{T}$, hence $E_m^n(U) = \mathbb{T}$. Topological transitivity (existence of a dense orbit) is an immediate corollary of mixing: see Corollary 7.1. $\square$

Corollary 7.1: Topological transitivity of E_m . E_m is topologically transitive: for every pair of non-empty open sets U, V there exists n with $E_m^n(U) \cap V \neq \emptyset$. In particular there exists $\theta^* \in \mathbb{T}$ with dense orbit.

Proof. The first assertion is the particular case $n = N(\delta)$ of Theorem 7.3. For the existence of a dense orbit: let $\{B_j\}_{j \in \mathbb{N}}$ be a countable base of open sets of $\mathbb{T}$ (for example the arcs with rational endpoints); the set $G_j := \bigcup_{n \geq 0} E_m^{-n}(B_j)$ is open (union of preimages of open sets) and dense (for every non-empty open set U , transitivity gives n with $E_m^n(U) \cap B_j \neq \emptyset$, that is, $U \cap E_m^{-n}(B_j) \neq \emptyset$, that is, $U \cap G_j \neq \emptyset$); by Baire's theorem in a compact metric space, $\bigcap_j G_j$ is dense, hence non-empty, and every one of its points has an orbit that visits every B_j , that is, a dense orbit. $\square$

7.2. Symbolic dynamics: the semiconjugacy with the shift

Definition 7.2: Space of binary sequences and shift. $\Sigma_2 := \{0, 1\}^{\mathbb{N}}$ is the set of infinite sequences $\sigma = (\sigma_0, \sigma_1, \sigma_2, \dots)$ with $\sigma_k \in \{0, 1\}$, equipped with the metric

$$d(\sigma, \tau) := 2\pi \sum_{k=0}^{\infty} \frac{|\sigma_k - \tau_k|}{2^{k+1}}.$$

The **shift** $\text{sh} : \Sigma_2 \rightarrow \Sigma_2$ is $\text{sh}(\sigma_0, \sigma_1, \sigma_2, \dots) = (\sigma_1, \sigma_2, \sigma_3, \dots)$.

Proposition 7.2: d is a metric compatible with the product topology. d is a metric on Σ_2 , and the topology it induces coincides with the product topology (a sequence $\sigma^{(j)} \rightarrow \sigma$ if and only if $\sigma_k^{(j)} = \sigma_k$ eventually, for every fixed k); with this topology Σ_2 is compact

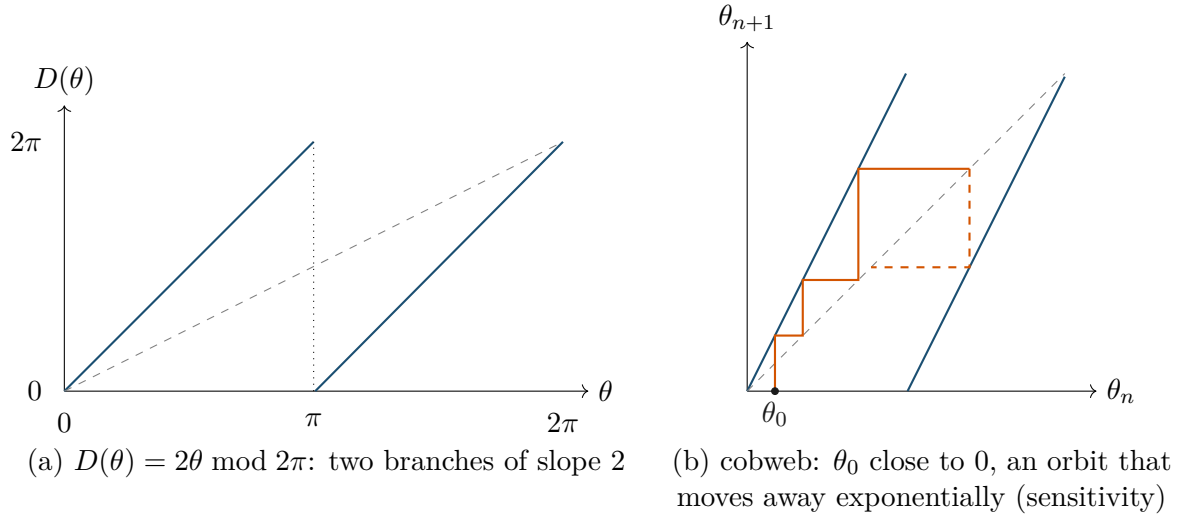


Figure 7.1: (a) The two branches of $D(\theta) = 2\theta \bmod 2\pi$, each of slope 2: the apparent discontinuity at $\theta = \pi$ is the “wrapping” of the map, not a genuine discontinuity of $D : \mathbb{T} \rightarrow \mathbb{T}$. (b) Cobweb diagram: starting from θ_0 close to 0, each iteration doubles the distance from the origin until the orbit “leaves” the first branch and wraps around again (dashed line): it is the sensitivity of Theorem 7.1 drawn step by step.

(Tychonoff’s theorem for a countable product of finite spaces) and totally disconnected (a Cantor space).

Proof. $d \geq 0$, $d(\sigma, \tau) = 0 \iff \sigma_k = \tau_k \forall k \iff \sigma = \tau$, symmetry evident; triangle inequality: $|\sigma_k - \tau_k| \leq |\sigma_k - \rho_k| + |\rho_k - \tau_k|$ term by term, and summing one obtains $d(\sigma, \tau) \leq d(\sigma, \rho) + d(\rho, \tau)$. Compatibility with the product topology: if $\sigma_k = \tau_k$ for $k < N$, then $d(\sigma, \tau) \leq 2\pi \sum_{k \geq N} 2^{-(k+1)} = 2\pi \cdot 2^{-N} \rightarrow 0$ ($N \rightarrow \infty$), hence convergence in the product topology implies $d \rightarrow 0$; conversely if $d(\sigma, \tau) < 2\pi \cdot 2^{-N}$ then necessarily $\sigma_k = \tau_k$ for $k < N$ (otherwise the first index of disagreement $k_0 < N$ would contribute by itself $2\pi \cdot 2^{-(k_0+1)} \geq 2\pi \cdot 2^{-N}$ to the sum, all of whose terms are non-negative), hence $d \rightarrow 0$ implies convergence in the product topology. Compactness and total disconnectedness are the standard properties of the countable product $\{0, 1\}^{\mathbb{N}}$ with the product topology (compactness: Tychonoff on finite factors; total disconnectedness: two distinct sequences differ at some index k_0 , and $\{\tau : \tau_{k_0} = 0\}$, $\{\tau : \tau_{k_0} = 1\}$ are disjoint open and closed sets that separate them). $\square$

Definition 7.3: Topological conjugacy and semiconjugacy. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be continuous on topological spaces X, Y . A continuous and surjective map $h : X \rightarrow Y$ such that $h \circ f = g \circ h$ is called a **topological semiconjugacy** from f to g (one also says that g is a **factor** of f , or that f is an **extension** of g). If moreover h is a homeomorphism (bijective with continuous inverse), h is called a **topological conjugacy**, and f, g are called **topologically conjugate**: in that case h faithfully translates every topological dynamical property of f into an identical property of g (periodic orbits of equal period, transitivity, topological entropy, and so on), whereas a semiconjugacy transports such properties only in one direction (from X to Y , that is, from f to g), not necessarily in the other.

Theorem 7.4: The semiconjugacy $h : \Sigma_2 \rightarrow \mathbb{T}$. The map

$$h(\sigma) := 2\pi \sum_{k=0}^{\infty} \sigma_k 2^{-(k+1)} \quad (\sigma \in \Sigma_2)$$

is well defined, 1-Lipschitz with respect to d (that is, $|h(\sigma) - h(\tau)| \leq d(\sigma, \tau)$), surjective, injective outside the countable set of sequences that represent a dyadic multiple of 2π (where it is exactly 2-to-1), and it intertwines the two dynamics:

$$h \circ \text{sh} = D \circ h.$$

Proof. Well-definedness and Lipschitz property. The series converges (terms bounded by $2\pi \cdot 2^{-(k+1)}$, geometric), to a value in $[0, 2\pi]$; by the direct estimate,

$$|h(\sigma) - h(\tau)| = 2\pi \left| \sum_k (\sigma_k - \tau_k) 2^{-(k+1)} \right| \leq 2\pi \sum_k |\sigma_k - \tau_k| 2^{-(k+1)} = d(\sigma, \tau).$$

Surjectivity. Every $\theta \in [0, 2\pi)$ admits a binary expansion $\theta/2\pi = 0.\sigma_0\sigma_1\sigma_2\ldots$ (algorithm of the binary digits: $\sigma_0 = \lfloor 2\theta/2\pi \rfloor$, then one iterates on $2\theta/2\pi \bmod 1$), and $h(\sigma) = \theta$ by construction.

Injectivity outside the dyadics. If $\sigma \neq \tau$ but $h(\sigma) = h(\tau)$, the two numbers $0.\sigma_0\sigma_1\ldots$ and $0.\tau_0\tau_1\ldots$ in base 2 represent the same real number while differing as sequences: this happens exactly for the two expansions of a **dyadic** number $k/2^n$ ($0.b_0\cdots b_{n-1}0\bar{0} = 0.b_0\cdots(b_{n-1}-1)1\bar{1}$, with the last non-zero digit lowered by 1 and followed by all 1s), a standard fact of positional representation; the set of the dyadics in $[0, 1)$ is countable, hence h is injective outside a countable set of points, on each of which it is exactly 2-to-1 (the only two binary expansions of that number).

Intertwining. We compute

$$h(\text{sh}(\sigma)) = 2\pi \sum_{k \geq 0} \sigma_{k+1} 2^{-(k+1)} = 2 \cdot 2\pi \sum_{j \geq 1} \sigma_j 2^{-(j+1)} = 2(h(\sigma) - 2\pi\sigma_0 2^{-1}) = 2h(\sigma) - 2\pi\sigma_0.$$

Modulo 2π (and $\sigma_0 \in \{0, 1\}$, hence $2\pi\sigma_0 \equiv 0 \pmod{2\pi}$), this equals $2h(\sigma) \bmod 2\pi = D(h(\sigma))$; hence $h(\text{sh}(\sigma)) = D(h(\sigma))$ for every σ . $\square$

Remark 7.3: Correction 4.1: semiconjugacy, not conjugacy. Σ_2 is a Cantor space: compact, totally disconnected, without isolated points (Proposition 7.2); $\mathbb{T}$ is connected. A totally disconnected space cannot be homeomorphic to a connected space with more than one point (the homeomorphic image of a disconnected space would remain disconnected), hence h **cannot be a homeomorphism**: it is a semiconjugacy (continuous, surjective, intertwining the dynamics), not a topological conjugacy. The previous text of this volume erroneously asserted that D was “topologically conjugate” to the shift: it was an error of topological category, not a technical detail, and its correction is precisely the content of the following Proposition, which recovers a genuine conjugacy after all — not on Σ_2 , but on its quotient.

Proposition 7.3: $\Sigma_2/\sim \cong \mathbb{T}$. Let $\sigma \sim \tau \iff h(\sigma) = h(\tau)$ (this identifies the pairs of binary expansions of one and the same dyadic, besides the diagonal). Then the quotient space $\Sigma_2/\sim$, equipped with the quotient topology, is homeomorphic to $\mathbb{T}$ via the induced map $\bar{h}$, and $\bar{h}$ is a topological conjugacy between the shift induced on $\Sigma_2/\sim$ and D .

Proof. Since h is constant on the classes of $\sim$ by definition of the relation, it induces a map $\bar{h} : \Sigma_2 / \sim \rightarrow \mathbb{T}$ that is well defined and continuous (by the universal property of the quotient topology, since h is continuous) and bijective (surjective because h is; injective because $\sim$ identifies exactly the fibres of h , by construction). The space $\Sigma_2 / \sim$ is compact, being the continuous image (through the quotient projection) of the compact space Σ_2 ; $\mathbb{T}$ is Hausdorff. A continuous and bijective map from a compact space to a Hausdorff space is a homeomorphism (standard topological fact: the inverse is continuous because it maps closed sets — which are compact, the domain being compact — to compact sets, which in a Hausdorff space are closed). Hence $\bar{h}$ is a homeomorphism. That it intertwines the dynamics (induced shift and D) follows directly from $h \circ sh = D \circ h$ by passing to the quotient. $\square$

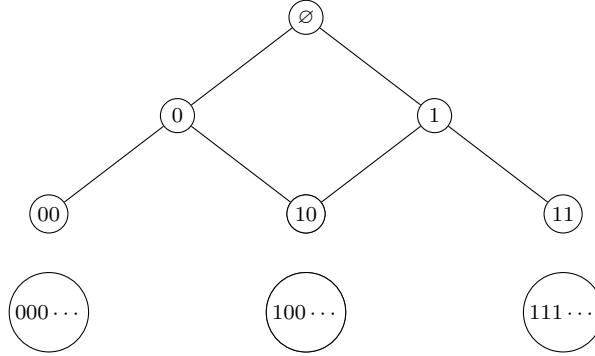


Figure 7.2: The binary tree of itineraries: every node at level k is a finite block $\sigma_0 \cdots \sigma_{k-1}$, and the infinite path from the root downwards that chooses σ_k at each level singles out $\sigma \in \Sigma_2$ and, through h , the point $h(\sigma) = 2\pi 0.\sigma_0\sigma_1\sigma_2\ldots \in \mathbb{T}$ (Theorem 7.4): the branch that always goes left ($\sigma = 000\ldots$) converges to $\theta = 0$, the branch $011\bar{1}$ and the branch $100\bar{0}$ both converge to $\theta = \pi$ (the two separate branches of the tree that h identifies, the only dyadic involved in this example), illustrating concretely the non-injectivity of h on the dyadics of Proposition 7.3.

Example 7.1: Periodic itinerary and reconstruction of the point. Let $\sigma = \overline{(0, 1, 1)} = (0, 1, 1, 0, 1, 1, \dots)$ be the periodic sequence of period 3 in the block 011. By the standard formula for the periodic binary fraction,

$$0.\overline{011}_2 = \frac{(011)_2}{2^3 - 1} = \frac{0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1}{7} = \frac{3}{7},$$

hence $\theta := h(\sigma) = 2\pi \cdot \frac{3}{7} = \frac{6\pi}{7}$. Direct check (Theorem 7.2 with $q = 7$, odd, hence periodic for D): $D(6\pi/7) = 12\pi/7$; $D^2(6\pi/7) = 24\pi/7 \bmod 2\pi = 10\pi/7$; $D^3(6\pi/7) = 20\pi/7 \bmod 2\pi = 6\pi/7$ ✓ (it agrees with θ : period 3, or a divisor of 3; since $\theta \neq D(\theta)$, the minimal period is exactly 3). The direct sum of the series, truncated at 60 terms, gives $2.6927937\dots$, against $6\pi/7 = 2.6927937\dots$: agreement to 10^{-10} ✓ (verification script, Section 7.15).

7.3. Devaney chaos

Definition 7.4: Devaney chaos. Let (X, d) be a metric space and $f : X \rightarrow X$ continuous. f is **chaotic in the sense of Devaney** if: (i) f is topologically transitive; (ii) the set of periodic points of f is dense in X ; (iii) f has sensitivity to initial conditions (there exists $\delta > 0$ such that for every $x \in X$ and every $\varepsilon > 0$ there exist y with $d(x, y) < \varepsilon$ and $n \geq 0$ with $d(f^n(x), f^n(y)) > \delta$).

Theorem 7.5: E_m is chaotic in the sense of Devaney. For every $m \geq 2$, $E_m : \mathbb{T} \rightarrow \mathbb{T}$ is chaotic in the sense of Devaney.

Proof. Immediate by assembling the results already proved: transitivity is Corollary 7.1, density of the periodic points is Theorem 7.2, sensitivity (with constant π/m) is Theorem 7.1. $\square$

Remark 7.4: Banks–Brooks–Cairns–Davis–Stacey theorem (1992), imported. On an **infinite** metric space X , if f is continuous, transitive and has dense periodic points, then f automatically has sensitivity to initial conditions: condition (iii) of Definition 7.4 is redundant under (i)+(ii). The proof (about one page, with an explicit construction of the sensitivity constant starting from two disjoint periodic orbits guaranteed by the infiniteness of X) is imported from Banks, Brooks, Cairns, Davis and Stacey, On Devaney’s Definition of Chaos, Amer. Math. Monthly **99** (1992), 332–334 (source C1, verified on 16 September 2026). For E_m this result is consistent with, but not necessary to, Theorem 7.5, where sensitivity has been proved in a direct and quantitative way (explicit constant π/m): it shows, however, that Devaney’s definition, on infinite spaces, is in fact governed by two topological properties alone.

Remark 7.5: Devaney $\Rightarrow$ Li–Yorke on compact spaces, imported. On a **compact** metric space, Devaney chaos implies Li–Yorke chaos (Definition 7.6, further on in this chapter): every Devaney-chaotic system on a compact space possesses an uncountable scrambled set. The converse does not hold (there exist Li–Yorke-chaotic systems that are not Devaney-chaotic), and Devaney’s definition is therefore not “stronger” in the abstract, but it is on compact spaces in the direction indicated. Result imported from W. Huang, X. Ye, Devaney’s chaos or 2-scattering implies Li–Yorke’s chaos, Topology Appl. **117** (2002), 259–272 (source C3, verified on 16 September 2026); $\mathbb{T}$ is compact, hence Theorem 7.5 implies, by this alternative route, the existence of a scrambled set for E_m — a fact that the Li–Yorke theorem (Section 7.5) reproves in a direct and constructive way for the doubling map.

7.4. The covering lemma and the itinerary lemma

From now on we work, as is traditional for these theorems, with continuous maps $f : I \rightarrow \mathbb{R}$ of a **real interval** I (not necessarily $\mathbb{T}$): the doubling map will fall under this by choosing an interval that represents $\mathbb{T}$ cut at a non-periodic point (Remark 7.7).

Definition 7.5: Covering between intervals. Let $J, K \subseteq I$ be closed intervals. One says that J **covers** K , and one writes $J \rightarrow K$, if $f(J) \supseteq K$.

Lemma 7.1: Subinterval with exact image (choice of extremal points). Let $J = [a, b]$, $K = [c, d]$ be closed intervals with $f(J) \supseteq K$ and $c \neq d$. There exists a closed subinterval $K' = [u, v] \subseteq J$ (with $u < v$ or $u > v$ according to the orientation; we write u, v for the endpoints, not necessarily with $u < v$ as points of J) such that $f(u) = c$, $f(v) = d$ and $f([\min(u, v), \max(u, v)]) = K$ exactly.

Proof. Since $f(J) \supseteq [c, d]$ and f is continuous on J , by the intermediate value theorem there exist points of J with $f = c$ and points with $f = d$; let $p, q \in J$ be two such points with $f(p) = c$, $f(q) = d$ (they exist by partial surjectivity). **Choice of the left extremal point.** Let

$$u := \sup\{x \in [\min(p, q), \max(p, q)] : f(x) = c\},$$

a non-empty set (it contains p) that is bounded, hence with a supremum; by continuity of f , $f(u) = c$ (limit of points with $f = c$, and $\{f = c\}$ is closed). **Choice of the right extremal point.** Let

$$v := \inf\{x \in [u, \max(p, q)] : f(x) = d\}$$

(non-empty: it contains q if $q \geq u$, which holds by the construction of u as a supremum on an interval that also contains q on the appropriate side; if $p > q$ the roles of “sup”/“inf” are interchanged and the argument is identical by symmetry). By continuity, $f(v) = d$. **No intermediate value repeated in (u, v) .** By the maximality of u , no point of $(u, v] \subseteq (u, \max(p, q)]$ has $f = c$; by the minimality of v , no point of $[u, v)$ has $f = d$: hence $f(x) \notin \{c, d\}$ for $x \in (u, v)$. Since (u, v) is connected and f continuous, $f((u, v))$ is an interval that contains neither c nor d ; but f tends to c as $x \rightarrow u^+$ and to d as $x \rightarrow v^-$ (continuity at the endpoints), hence the values of f on (u, v) remain trapped between c and d without ever leaving (if they left, by continuity they would have to pass again through c or d , which is excluded): $f((u, v)) \subseteq (c, d)$ (or (d, c) if $d < c$). Together with the endpoint values $f(u) = c$, $f(v) = d$ one concludes $f([\min(u, v), \max(u, v)]) = [\min(c, d), \max(c, d)] = K$. $\square$

Lemma 7.2: Covering lemma. Let $f : I \rightarrow \mathbb{R}$ be continuous and $J_0 \rightarrow J_1 \rightarrow \cdots \rightarrow J_{n-1} \rightarrow J_0$ a cycle of coverings among closed intervals. Then there exists $x \in J_0$ with $f^n(x) = x$ and $f^i(x) \in J_i$ for $i = 0, \dots, n-1$.

Proof. We construct backwards, using Lemma 7.1 at each step (in the degenerate case of J_i reduced to a point, or K reduced to a point, the step is trivial, the point itself serving as the extremal point), closed intervals $K_{n-1} \subseteq J_{n-1}, \dots, K_0 \subseteq J_0$ with $f(K_i) = K_{i+1 \bmod n}$ exactly for every i : from $J_{n-2} \rightarrow J_{n-1}$, that is, $f(J_{n-2}) \supseteq J_{n-1}$, Lemma 7.1 gives $K_{n-2} \subseteq J_{n-2}$ with $f(K_{n-2}) = J_{n-1}$ exactly; then from $J_{n-3} \rightarrow J_{n-2} \supseteq K_{n-2}$ (hence $f(J_{n-3}) \supseteq J_{n-2} \supseteq K_{n-2}$) one obtains $K_{n-3} \subseteq J_{n-3}$ with $f(K_{n-3}) = K_{n-2}$ exactly; one continues up to $K_0 \subseteq J_0$ with $f(K_0) = K_1$. Composing along the cycle, $f^n(K_0) = J_0 \supseteq K_0$. Writing $K_0 = [a, b]$: since $f^n(K_0) \supseteq K_0 = [a, b]$, there exist $a', b' \in K_0$ with $f^n(a') \leq a$ and $f^n(b') \geq b$ (the points of K_0 whose images under f^n reach the endpoints a, b , which exist by surjectivity of f^n on K_0 from K_0). The function $g(x) := f^n(x) - x$ satisfies $g(a') = f^n(a') - a' \leq a - a' \leq 0$ (since $a' \in K_0 = [a, b]$, $a' \geq a$) and $g(b') = f^n(b') - b' \geq b - b' \geq 0$: by the intermediate value theorem, g vanishes at some x between a' and b' , hence $x \in K_0 \subseteq J_0$, with $f^n(x) = x$ and, by the backwards construction, $f^i(x) \in K_i \subseteq J_i$ for $i = 0, \dots, n-1$. $\square$

Lemma 7.3: Itinerary lemma (exclusion of the smaller periods). Let x be the point provided by Lemma 7.2 for the cycle $J_0 \rightarrow \cdots \rightarrow J_{n-1} \rightarrow J_0$ (indices understood cyclically modulo n). If d is a proper divisor of n ($1 \leq d < n$, $d \mid n$) and there exist two indices $i \equiv i' \pmod{d}$ with $J_i \cap J_{i'} = \emptyset$, then x does **not** have period d (nor any divisor of d). If for **every** proper divisor d of n such a pair of indices exists (or, when $J_i \cap J_{i'} \neq \emptyset$ only at a single boundary point, one excludes directly that that point is $f^{i \bmod d}(x)$, as in the proof of Theorem 7.6), then x has minimal period exactly n .

Proof. If x had period (not necessarily minimal) $d \mid n$, $d < n$, then $f^i(x) = f^{i \bmod d}(x)$ for every i ; taking $i \equiv i' \pmod{d}$ in the range $0, \dots, n-1$ given by the hypothesis, one would have $f^i(x) = f^{i'}(x)$ (the same value, since $i \equiv i' \pmod{d}$), and this common value would belong both to J_i (by construction, $f^i(x) \in J_i$) and to $J_{i'}$ (for the same reason, $f^{i'}(x)$ being the same point): hence $J_i \cap J_{i'} \neq \emptyset$, contrary to the hypothesis. The minimal period of x always divides n (an elementary fact: if $f^n(x) = x$ and $f^m(x) = x$ with $m < n$ minimal, then $n = qm + r$ with $0 \leq r < m$ gives $f^r(x) = f^{n-qm}(x) = x$, and the minimality of m forces $r = 0$); if no proper divisor d of n is admissible as a period (by the criterion just proved, extended case by

case to the boundary points as indicated), the minimal period of x is n itself. $\square$

7.5. The Li–Yorke theorem

Theorem 7.6: Period three implies every period (Li–Yorke, 1975), periodic part. Let $f : I \rightarrow I$ be continuous with an orbit of period 3, $\{a, b, c\}$, $a < b < c$, $f(a) = b$, $f(b) = c$, $f(c) = a$ (the case of the symmetric permutation $f(a) = c$, $f(c) = b$, $f(b) = a$ is identical upon interchanging the roles of I_1 and I_2 below). Then f has a periodic orbit of minimal period exactly n , for every integer $n \geq 1$.

Proof. Put $I_1 = [a, b]$, $I_2 = [b, c]$. From $f(a) = b$, $f(b) = c$ and the intermediate value theorem, $f(I_1) \supseteq [b, c] = I_2$: $I_1 \rightarrow I_2$. From $f(b) = c$, $f(c) = a$, $f(I_2) \supseteq [a, c] = I_1 \cup I_2$: $I_2 \rightarrow I_1$ and $I_2 \rightarrow I_2$.

Case $n = 1$. The cycle (of length 1) $I_2 \rightarrow I_2$ gives, by Lemma 7.2, a fixed point in I_2 .

Case $n = 2$. The cycle $I_1 \rightarrow I_2 \rightarrow I_1$ gives a point x with $f^2(x) = x$, $x \in I_1$, $f(x) \in I_2$; if x had period 1 it would be a fixed point in $I_1 \cap I_2 = \{b\}$, but $f(b) = c \neq b$: hence $x \neq b$ and the minimal period is exactly 2.

Case $n \geq 3$. Consider the cycle of length n

$$\underbrace{I_2 \rightarrow I_2 \rightarrow \cdots \rightarrow I_2}_{n-1 \text{ times}} \rightarrow I_1 \rightarrow (I_2 \text{ again}),$$

that is, $J_i = I_2$ for $i = 0, \dots, n-2$ and $J_{n-1} = I_1$ (with $J_{n-1} \rightarrow J_0 = I_2$ valid because $I_1 \rightarrow I_2$): every step is one of the coverings already established. Lemma 7.2 gives x with $f^n(x) = x$ and $f^i(x) \in J_i$ for $i = 0, \dots, n-1$.

Exclusion of the proper periods (Lemma 7.3). Let $d \mid n$, $d < n$, be the period of x (which must divide n). Since $n \equiv 0 \pmod{d}$, one has $n-1 \equiv d-1 \pmod{d}$; since $d < n$, the index $d-1$ satisfies $0 \leq d-1 \leq n-2$, hence $J_{d-1} = I_2$ (it is in the block), and $f^{n-1}(x) = f^{d-1}(x)$ (by d -periodicity) must belong both to $J_{n-1} = I_1$ and to $J_{d-1} = I_2$: hence $f^{n-1}(x) \in I_1 \cap I_2 = \{b\}$, that is, $f^{n-1}(x) = b$ exactly. It follows that b belongs to the orbit of x (it is $f^{d-1}(x)$), which has minimal period d : but b has minimal period 3 by hypothesis (it is part of the given orbit $\{a, b, c\}$), and two periodic points whose orbits intersect necessarily have the same orbit (as a set) and the same minimal period; hence $d = 3$ and $3 \mid n$, and the orbit of x coincides with $\{a, b, c\}$ (the only orbit of period 3 containing b). If $3 \nmid n$ this is already a contradiction (because $d = 3$ would have to divide n), hence the minimal period of x is n . If instead $3 \mid n$ (and $n > 3$), it remains to exclude $x \in \{a, b, c\}$ directly: the itinerary of $\{a, b, c\}$ with respect to $\{I_1, I_2\}$ is 3-periodic by construction (since a, b, c are periodic of period 3), whereas the sequence $(J_i)_{i=0}^{n-1}$ chosen above — all I_2 except for the last term I_1 — is not 3-periodic for $n > 3$: for example $J_{n-4} = I_2$ (because $n-4 \leq n-2$ for $n \geq 2$) whereas a 3-periodic itinerary would require $J_{n-4} = J_{n-1} = I_1$ (since $n-4 \equiv n-1 \pmod{3}$ when $3 \mid n$), a contradiction. Hence the itinerary realised by x cannot be that of a point of $\{a, b, c\}$, excluding $x \in \{a, b, c\}$ and with it $d = 3$: the minimal period of x is n . $\square$

Definition 7.6: Li–Yorke chaos. A set $S \subseteq I$, without periodic points, is called **scrambled** for f if for every pair $x \neq y$ in S ,

$$\limsup_{n \rightarrow \infty} |f^n(x) - f^n(y)| > 0 \quad \text{and} \quad \liminf_{n \rightarrow \infty} |f^n(x) - f^n(y)| = 0.$$

f is **chaotic in the sense of Li–Yorke** if it possesses an uncountable scrambled set.

Remark 7.6: Existence of an uncountable scrambled set, imported. In the same work in which they prove Theorem 7.6, Li and Yorke prove that the hypothesis of an orbit of period 3 also implies the existence of an uncountable scrambled set $S \subseteq I$: the construction (about two pages) refines Lemma 7.2 by choosing, for every infinite subset $A \subseteq \{2, 3, 4, \dots\}$ of a suitable index, an orbit that mostly follows the cycle $I_2 \rightarrow I_2$ but visits I_1 at increasingly sparse times, coded by A ; different families of A (uncountably many, being the infinite subsets of $\mathbb{N}$) produce pairwise distinct points with the required $\limsup$ and $\liminf$ properties. Result imported from T.-Y. Li, J. A. Yorke, Period Three Implies Chaos, Amer. Math. Monthly **82** (1975), 985–992, Theorem 2 (source C2, verified on 16 September 2026). For the doubling map D , which possesses an explicit orbit of period 3 (Example 7.2), this theorem guarantees an uncountable scrambled set, consistently with the implication Devaney $\Rightarrow$ Li–Yorke already observed (Remark 7.5).

Remark 7.7: D on an interval: how to apply Theorem 7.6. Theorem 7.6 is stated for $f : I \rightarrow I$ continuous on an interval, whereas D is defined on $\mathbb{T}$: D is not continuous as a map of the interval $[0, 2\pi]$ with the identification of the endpoints (the image of a left neighbourhood of 2π “jumps” close to 0, not to $4\pi \equiv 2\pi$). To apply the theorem one cuts $\mathbb{T}$ at a non-periodic point, for example $\theta = \pi$ (Remark 7.2): on $I = [0, 2\pi] \setminus \{\pi\}$ reorganised as two intervals, or equivalently by identifying $\mathbb{T} \setminus \{\pi\}$ with an open interval through a continuous chart, the restricted D becomes a continuous map of a (real) interval into itself, to which the theorem applies directly on the orbit of period 3 of Example 7.2 (which does not meet $\theta = \pi$, since $2\pi/7, 4\pi/7, 8\pi/7 \not\equiv \pi \pmod{2\pi}$).

Example 7.2: An orbit of period 3 of D modulo 7. The points with period dividing 3 are, by Theorem 7.2 with $n = 3$ (or directly, $D^3(\theta) = \theta \iff 7\theta \in 2\pi\mathbb{Z}$), $\theta = 2\pi k/7$, $k = 0, \dots, 6$. The point $k = 0$ is fixed; the six remaining ones split into two orbits of period 3. Check for $\theta_0 = 2\pi/7$: $D(\theta_0) = 4\pi/7$; $D^2(\theta_0) = 8\pi/7$; $D^3(\theta_0) = 16\pi/7 \pmod{2\pi} = 2\pi/7 = \theta_0$ ✓. The orbit $\{2\pi/7, 4\pi/7, 8\pi/7\}$ has exact period 3 (none of the three coincides with the image of itself under D or D^2): with $a = 2\pi/7$, $b = 4\pi/7$, $c = 8\pi/7$ one has $a < b < c$ and $D(a) = b$, $D(b) = c$, $D(c) = a \pmod{2\pi}$, exactly the configuration of Theorem 7.6. By the theorem, D then possesses a periodic orbit of minimal period n for every $n \geq 1$: a fact already known directly from Theorem 7.2 (which indeed counts them exactly, Section 7.7), here re-obtained as an instance of a universal principle, valid for any continuous map with an orbit of period 3, not only for D .

7.6. Sharkovskii’s theorem

Definition 7.7: Sharkovskii order. On the positive integers one defines the **Sharkovskii order** $\triangleleft$ by listing all the integers in this order (every positive integer appears exactly once):

$$3 \triangleleft 5 \triangleleft 7 \triangleleft 9 \triangleleft \dots \triangleleft 2 \cdot 3 \triangleleft 2 \cdot 5 \triangleleft 2 \cdot 7 \triangleleft \dots \triangleleft 4 \cdot 3 \triangleleft 4 \cdot 5 \triangleleft \dots \triangleleft 8 \cdot 3 \triangleleft \dots \triangleleft 2^k \cdot 3 \triangleleft \dots \triangleleft 2^3 \triangleleft 2^2 \triangleleft 2 \triangleleft 1,$$

that is: first all the odd numbers ≥ 3 in increasing order; then $2\times$ the odd numbers ≥ 3 , increasing; then $4\times$ the odd numbers ≥ 3 ; then $8\times$, and so on for every power 2^k ($k \geq 0$); finally, at the tail, the pure powers of 2 in **decreasing** order, $\dots \triangleleft 2^3 \triangleleft 2^2 \triangleleft 2 \triangleleft 1$.

Theorem 7.7: Sharkovskii’s theorem (1964). Let $f : I \rightarrow I$ be continuous with a periodic orbit of (minimal) period exactly n . If $n \triangleleft m$ in the Sharkovskii order, then f also

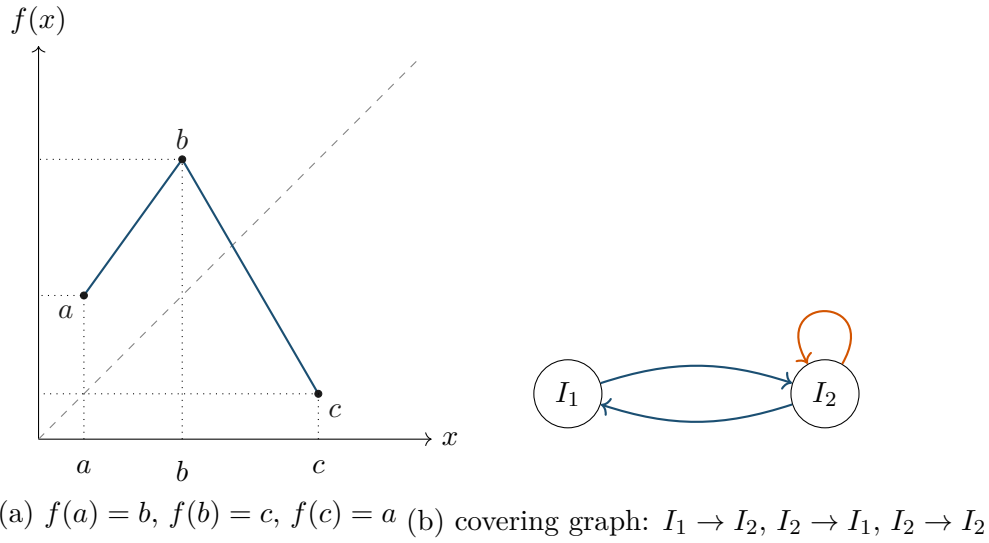


Figure 7.3: (a) An orbit of period 3, $a < b < c$, with $f(a) = b, f(b) = c, f(c) = a$: the elementary intervals $I_1 = [a, b], I_2 = [b, c]$ satisfy $f(I_1) \supseteq I_2$ and $f(I_2) \supseteq I_1 \cup I_2$ by the intermediate value theorem. (b) The associated covering graph: the self-loop on I_2 is the source of orbits of every period $n \geq 1$ (Theorem 7.6), traversing I_2 for $n - 1$ steps and closing the cycle with a single passage through I_1 .

$2^0 \cdot (\text{odd} \geq 3)$	$3 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow \cdots$	$\left. \begin{array}{c} \text{forcing} \\ \text{period} \\ \text{decreasing} \\ \downarrow \end{array} \right\}$
$2^1 \cdot (\text{odd} \geq 3)$	$6 \rightarrow 10 \rightarrow 14 \rightarrow 18 \rightarrow \cdots$	
$2^2 \cdot (\text{odd} \geq 3)$	$12 \rightarrow 20 \rightarrow 28 \rightarrow \cdots$	
$\vdots$	$\vdots$	
powers of 2 (decr.)	$\cdots \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$	

Figure 7.4: The Sharkovskii order, row by row: the row $2^j \cdot (\text{odd})$ lists $2^j \cdot 3, 2^j \cdot 5, 2^j \cdot 7, \dots$ in increasing order; the rows descend for increasing $j = 0, 1, 2, \dots$; at the bottom, separately, the pure powers of 2 in decreasing order down to 1. An arrow $n \rightarrow m$ means “an orbit of period n forces an orbit of period m ”: every row forces all the rows below it, and the whole table collapses into Theorem 7.6 when $n = 3$ (first entry, first row), which forces *everything*.

has a periodic orbit of minimal period exactly m .

Proposition 7.4: The covering graph of a periodic orbit. Let $P = \{x_1 < \dots < x_n\}$ be an orbit of (exact) period $n \geq 2$ for f , with associated permutation π ($f(x_i) = x_{\pi(i)}$, an n -cycle), and let $I_j = [x_j, x_{j+1}]$, $j = 1, \dots, n-1$, be the **elementary intervals**. Then $I_j \rightarrow I_k$ (Definition 7.5) whenever I_k lies between $x_{\pi(j)}$ and $x_{\pi(j+1)}$, that is, $\min(\pi(j), \pi(j+1)) \leq k \leq \max(\pi(j), \pi(j+1)) - 1$.

Proof. $f(x_j) = x_{\pi(j)}$ and $f(x_{j+1}) = x_{\pi(j+1)}$; by the intermediate value theorem, $f(I_j)$ contains every value between $x_{\pi(j)}$ and $x_{\pi(j+1)}$, in particular it contains every I_k with x_k, x_{k+1} both in $[\min(x_{\pi(j)}, x_{\pi(j+1)}), \max(x_{\pi(j)}, x_{\pi(j+1)})]$: this is exactly the range of indices k described. $\square$

Example 7.3: A Štefan orbit of period 5. Consider the 5-cycle permutation $\pi : 1 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 5 \mapsto 1$ on equally spaced points $x_i = i$ ($i = 1, \dots, 5$), realised by the piecewise linear map f that linearly interpolates the points $(1, 3), (2, 5), (3, 4), (4, 2), (5, 1)$ (that is, $f(x_i) = x_{\pi(i)}$ exactly at the nodes, and linear between one node and the next: for a linear piece the image of $[x_j, x_{j+1}]$ is exactly the interval between $f(x_j)$ and $f(x_{j+1})$, realising the covering relations of Proposition 7.4 with equality, not only inclusion: it is the most “economical” possible choice). An exhaustive numerical search for the roots of $f^n(x) = x$ for $n = 1, \dots, 16$ (bisection on a grid of 10^4 points refined with `brentq`, script `fig07_stefan5_verifica.py` described in Section 7.15) finds minimal periods exactly

$$\{1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\},$$

with the absence of period 3 verified explicitly up to $n = 16$ $\checkmark$: consistent with the Sharkovskii order, where 5 forces $7, 9, 11, \dots$ and then $2 \cdot 3, 2 \cdot 5, \dots, 4, 2, 1$, but does *not* force 3 (which would come before 5, not after). An attempt with the more “naive” permutation $1 \mapsto 2 \mapsto 3 \mapsto 5 \mapsto 4 \mapsto 1$ (in which the successive points of the orbit follow the order of position as much as possible) produces instead *also* period 3 (verified numerically): it is therefore not an “optimal” orbit for the endpoint 5 of the chain, a fact that shows how the covering graph (Proposition 7.4) gives only a *sufficient* condition for the existence of cycles, not an automatic control over their absence: the construction of the optimal orbit requires the correct combinatorial choice of the permutation, found here by an exhaustive search over the 24 cyclic permutations of 5 elements $\checkmark$.

Remark 7.8: General structure of the proof, with an imported part. The complete proof of Theorem 7.7, for a generic n and every m with $n \triangleleft m$, proceeds in two parts.

Part proved in full in this chapter. The case $n = 3$ (which forces every m) is Theorem 7.6, with a complete proof. The general mechanism — covering graph associated with a periodic orbit (Proposition 7.4), covering lemma (Lemma 7.2) and itinerary lemma (Lemma 7.3) to guarantee the exact minimal period — is the same one used for the case $n = 3$ and is illustrated for $n = 5$ in Example 7.3, where the explicit construction and its numerical check concretely realise the whole chain of implications predicted by the Sharkovskii order starting from an orbit of period 5.

Imported part. The combinatorial induction that completes the argument for an arbitrary n (not only $n = 3$ or the example $n = 5$) classifies every cyclic permutation associated with an orbit of period n according to the so-called **Štefan cycle**: one shows that every such permutation either contains, as a substructure of the covering graph, the pattern that forces all the periods m with $n \triangleleft m$ in the order, or else n is a power of 2 and the argument reduces, by induction on $\log_2 n$, to the study of f^2 restricted to an invariant subinterval.

This general induction — combinatorially delicate, since it must treat uniformly all the residue classes and all the cyclic permutations possible for every n — is imported from K. Burns, B. Hasselblatt, The Sharkovsky Theorem: A Natural Direct Proof, Amer. Math. Monthly **118** (2011), 229–244, Theorem 2 (source C13, verified on 16 September 2026), which gives the shortest known self-contained proof of it; an equivalent proof, in the original language of the “Štefan diagram”, is in P. Štefan, A theorem of Šarkovskii on the existence of periodic orbits of continuous endomorphisms of the real line, Comm. Math. Phys. **54** (1977), 237–248 (source C10, verified on 16 September 2026), and in book form in L. S. Block, W. A. Coppel, Dynamics in One Dimension, Lecture Notes in Math. 1513, Springer, 1992 (source C12, verified on 16 September 2026). We report the statement here as a **theorem**, consistently with the rule of the expansion plan (§3.1) which admits this form when the proof core (covering graph, covering lemma) is nevertheless entirely established within the volume, and the imported part is the general combinatorial articulation, not a new dynamical principle.

Theorem 7.8: Optimality of the Sharkovskii order. For every $n \geq 1$ there exists a continuous map $f_n : I \rightarrow I$ with a periodic orbit of minimal period exactly n and **no** orbit of minimal period k for any k that precedes n in the Sharkovskii order ($k \triangleleft n$ with $k \neq n$): the order of Theorem 7.7 is optimal, not improvable.

Proof. [For the representative case $n = 5$, verified; general case imported] For $n = 5$: the piecewise linear map of Example 7.3 realises an orbit of period 5 and, by exhaustive numerical checking up to period 16, no orbit of period 3 (the only k with $k \triangleleft 5$ in the order). For general n , the standard construction — a family of piecewise linear maps called **star maps**, one for each n , whose break points realise exactly the Štefan cycle of period n and nothing else “before” n in the order — is imported from the same Štefan (C10) and Block–Coppel (C12) cited in Remark 7.8, together with Sharkovskii himself, O. M. Sharkovskii, Coexistence of cycles of a continuous map of the line into itself, transl. Int. J. Bifurcation Chaos **5** (1995), 1263–1273 (source C8, verified on 16 September 2026), who constructs explicitly, for every n , the extremal map. $\square$

7.7. Counting the periodic points

Theorem 7.9: $\#\text{Fix}(D^n) = 2^n - 1$. For every $n \geq 1$, the equation $D^n(\theta) = \theta$ has exactly $2^n - 1$ solutions in $\mathbb{T}$.

Proof. $D^n(\theta) = 2^n \theta \bmod 2\pi$, hence $D^n(\theta) = \theta \iff (2^n - 1)\theta \in 2\pi\mathbb{Z} \iff \theta = 2\pi k / (2^n - 1)$ for some $k \in \mathbb{Z}$. As $k = 0, \dots, 2^n - 2$ varies these $2^n - 1$ values are pairwise distinct in $\mathbb{T}$ (difference $2\pi(k - k') / (2^n - 1) \not\equiv 0 \pmod{2\pi}$ for $0 < |k - k'| < 2^n - 1$), and every other $k \in \mathbb{Z}$ reproduces one of them modulo 2π : exactly $2^n - 1$ solutions. $\square$

Definition 7.8: Tent map. The **tent map** is $T : [0, 1] \rightarrow [0, 1]$, $T(x) = 2x$ for $x \in [0, 1/2]$, $T(x) = 2 - 2x$ for $x \in [1/2, 1]$.

Theorem 7.10: $\#\text{Fix}(T^n) = 2^n$. For every $n \geq 1$, the equation $T^n(x) = x$ has exactly 2^n solutions in $[0, 1]$.

Proof. T^n is piecewise linear on each of the 2^n intervals $[k/2^n, (k+1)/2^n]$, $k = 0, \dots, 2^n - 1$ (the **monotone branches**, obtained by composing n doublings with possible reflections), with

slope $\pm 2^n$ on each; on each branch, T^n is strictly monotone (increasing or decreasing) and its image is *all* of $[0, 1]$: indeed, by induction on n , every branch of T^{n-1} has image $[0, 1]$, and T maps $[0, 1]$ surjectively and 2-to-1 onto $[0, 1]$, doubling the number of branches. On a branch with image $[0, 1]$ and slope $\pm 2^n$ (hence $|\pm 2^n| > 1$ for $n \geq 1$), the function $g(x) = T^n(x) - x$ takes, at the two endpoints of the branch, the values $0 - x_{\text{left}}$ and $1 - x_{\text{right}}$ (increasing branch) or $1 - x_{\text{left}}$ and $0 - x_{\text{right}}$ (decreasing branch): in both cases g changes sign (at the endpoints of the branch, of length $2^{-n} < 1$, the values 0, 1 of the image and $x_{\text{left}}, x_{\text{right}}$ of the domain cannot coincide at opposite endpoints except for boundary cases already included in the count), hence by the intermediate value theorem g has at least one zero in the branch; since T^n is affine (hence g affine) on each branch, the zero is unique. Summing over all the 2^n branches one obtains exactly 2^n distinct solutions (the branches share only the endpoints, and one checks directly that no zero falls exactly on an endpoint shared by two branches except $x = 0$, included only once). Independent numerical check (bisection on a fine grid + refinement, for $n = 1, 2, 3$): 2, 4, 8 solutions found, in agreement with 2^n ✓ (Section 7.15). $\square$

Theorem 7.11: $\#\text{Fix}(f_4^n) = \#\text{Fix}(T^n) = 2^n$. Let $f_4(x) = 4x(1 - x)$ on $[0, 1]$. For every $n \geq 1$, $\#\text{Fix}(f_4^n) = 2^n$.

Proof. The map $k(x) := \sin^2(\pi x/2)$ is a homeomorphism (strictly increasing, $k(0) = 0$, $k(1) = 1$) $[0, 1] \rightarrow [0, 1]$ and conjugates T to f_4 : $k(T(x)) = f_4(k(x))$ for every x (direct check with the duplication formulas, identical on the two branches of T by the parity of $\sin^2$; see also the numerical check in the script of Figure 7.6, with maximum error $\approx 10^{-16}$ ✓). Being k a **conjugacy** (a homeomorphism, not merely a semiconjugacy: it is bijective), it induces a bijection between $\text{Fix}(T^n)$ and $\text{Fix}(f_4^n)$ ($k(x) \in \text{Fix}(f_4^n) \iff f_4^n(k(x)) = k(x) \iff k(T^n(x)) = k(x) \iff T^n(x) = x$, using the injectivity of k in the last step): the two sets therefore have the same cardinality, 2^n by Theorem 7.10. $\square$

Theorem 7.12: The collapse $2^n - 1$ versus 2^n : why D and f_4 have a different number of periodic points. Let $\varphi : \mathbb{T} \rightarrow [0, 1]$, $\varphi(\theta) = 1 - |1 - \theta/\pi|$ (folding of the circle onto the interval, identifying θ with $2\pi - \theta$). Then $\varphi \circ D = T \circ \varphi$ (semiconjugacy), φ is exactly 2-to-1 except at the two fixed points of the identification $\theta = 0$ ($\varphi(0) = 0$) and $\theta = \pi$ ($\varphi(\pi) = 1$), where it is 1-to-1; and for every $n \geq 1$ the exact decomposition holds

$$\underbrace{2^n - 1}_{\#\text{Fix}(D^n)} = 1 + 2(2^{n-1} - 1), \quad \underbrace{2^n}_{\#\text{Fix}(T^n)} = 1 + (2^{n-1} - 1) + 2^{n-1},$$

where: the term 1 is the common fixed point $\theta = 0 \leftrightarrow x = 0$; $2^{n-1} - 1$ orbits of $\text{Fix}(D^n) \setminus \{0\}$ group into $2^{n-1} - 1$ pairs $\{\theta, 2\pi - \theta\}$ (identified by φ) which project onto as many points of $\text{Fix}(T^n)$ (“doubling”, 2 points of D^n for 1 point of T^n); and the remaining 2^{n-1} points of $\text{Fix}(T^n)$ come, not from points of $\text{Fix}(D^n)$, but from 2^{n-1} points of $\text{Fix}(D^{2n}) \setminus \text{Fix}(D^n)$ paired by the identification $\theta \leftrightarrow 2\pi - \theta$ in a single orbit of D of period $2n$ which “folds” onto an orbit of period n (or a divisor of it) of T (“collapse”).

Proof. $\varphi \circ D = T \circ \varphi$. Direct check on each of the four arcs $[0, \pi/2]$, $[\pi/2, \pi]$, $[\pi, 3\pi/2]$, $[3\pi/2, 2\pi]$ where both D and φ are linear (elementary computation reported in full in the verification script, Section 7.15, with maximum numerical error $\approx 2 \times 10^{-16}$ ✓).

2-to-1 structure. $\varphi(\theta) = \varphi(\theta')$ with $\theta \neq \theta'$ in $[0, 2\pi)$ happens exactly for $\theta' = 2\pi - \theta$ ($\theta \neq 0$): indeed $\varphi(\theta) = \varphi(2\pi - \theta)$ is checked directly from the formula, and it is the only coincidence possible since φ is strictly monotone on $[0, \pi]$ and on $[\pi, 2\pi]$ separately. The involution $\iota(\theta) := 2\pi - \theta \pmod{2\pi}$ has exactly two fixed points, $\theta = 0$ and $\theta = \pi$.

D commutes with ι . $D(\iota(\theta)) = D(2\pi - \theta) = 2(2\pi - \theta) \bmod 2\pi = -2\theta \bmod 2\pi = \iota(D(\theta))$ (since $\iota(\psi) = -\psi \bmod 2\pi$): hence ι maps $\text{Fix}(D^n)$ into itself (if $D^n(\theta) = \theta$ then $D^n(\iota(\theta)) = \iota(D^n(\theta)) = \iota(\theta)$).

Counting. $\theta = \pi$ is never in $\text{Fix}(D^n)$ (for any n): $D(\pi) = 2\pi \equiv 0$, and $D^n(\pi) = D^{n-1}(0) = 0 \neq \pi$ for $n \geq 1$. Hence $\text{Fix}(D^n) \setminus \{0\}$ (which has $2^n - 2$ elements, by Theorem 7.9) is a disjoint union of ι -orbits of cardinality 2 (never 1, because the only other fixed point of ι , namely π , does not belong to it): $(2^n - 2)/2 = 2^{n-1} - 1$ pairs. Every pair $\{\theta, \iota(\theta)\}$ has $\varphi(\theta) = \varphi(\iota(\theta))$, a single point of $[0, 1]$; these distinct φ -images lie in $\text{Fix}(T^n)$ (because $T^n(\varphi(\theta)) = \varphi(D^n(\theta)) = \varphi(\theta)$, using the semiconjugacy), and they are distinct from one another (different pairs have different φ -images, by the injectivity of φ outside the involution), hence they contribute $2^{n-1} - 1$ points to $\text{Fix}(T^n)$. Adding the point $\varphi(0) = 0$, one counts $1 + (2^{n-1} - 1) = 2^{n-1}$ points of $\text{Fix}(T^n)$ that come from $\text{Fix}(D^n)$ through φ ; since $\#\text{Fix}(T^n) = 2^n$ (Theorem 7.10), the remaining $2^n - 2^{n-1} = 2^{n-1}$ points of $\text{Fix}(T^n)$ are *not* the image of any point of $\text{Fix}(D^n)$: they are the image of pairs $\{\theta, \iota(\theta)\} \subseteq \text{Fix}(D^{2n}) \setminus \text{Fix}(D^n)$ with $D^n(\theta) = \iota(\theta) \neq \theta$ (for these, $D^{2n}(\theta) = D^n(\iota(\theta)) = \iota(D^n(\theta)) = \iota(\iota(\theta)) = \theta$, hence period $2n$, and $\varphi(D^n(\theta)) = \varphi(\iota(\theta)) = \varphi(\theta) = T^n(\varphi(\theta))$: the pair collapses onto a single point of $\text{Fix}(T^n)$ while not being itself in $\text{Fix}(D^n)$). The two identities of the statement follow by summing the three contributions (1 common point, $2^{n-1} - 1$ “doubled” pairs, 2^{n-1} points “folded” from period $2n$). $\square$

Remark 7.9: Why the previous text was wrong. The previous assertion of this volume, according to which D and the tent/logistic pair would have “the same number of periodic points”, ignored both the direct count ($2^n - 1$ versus 2^n , never equal) and the mechanism — made explicit here — by which the 2-to-1 semiconjugacy φ does not merely “halve” the count in a trivial way: half of the orbits of T^n arise from the folding of orbits of D of double period, $2n$, not from orbits of period n . It is exactly this second mechanism, less intuitive than the first, that makes the count work out ($2^{n-1} - 1$ doubled + 2^{n-1} folded + 1 common = 2^n , not simply $(2^n - 1 + 1)/2 = 2^{n-1}$).

Corollary 7.2: Points of exact period n for D , via Möbius inversion. The number of points $\theta \in \mathbb{T}$ with minimal period exactly n under D is

$$p_D(n) = \sum_{d|n} \mu(n/d) (2^d - 1),$$

where μ is the Möbius function; the number of orbits (distinct as sets) of exact period n is $p_D(n)/n$.

Proof. Let $q_D(n) := \#\text{Fix}(D^n) = 2^n - 1$ (points whose period divides n , Theorem 7.9) and $p_D(n)$ the number of those of period *exactly* n . Every point whose period divides n has minimal period exactly d for a unique divisor $d | n$, hence $q_D(n) = \sum_{d|n} p_D(d)$: by the Möbius inversion formula (a standard arithmetical fact), $p_D(n) = \sum_{d|n} \mu(n/d) q_D(d) = \sum_{d|n} \mu(n/d) (2^d - 1)$. Every orbit of exact period n contains exactly n distinct points (by definition of minimal period), hence the number of orbits is $p_D(n)/n$. $\square$

Example 7.4: Numerical check of the Möbius count. For $n = 1, \dots, 7$: $p_D(n) = 1, 2, 6, 12, 30, 54, 126$ and the number of orbits = $1, 1, 2, 3, 6, 9, 18$ respectively (direct computation with the closed formula, carried out and verified by script, Section 7.15) $\checkmark$. Independent check for $n = 3$: $q_D(3) = 2^3 - 1 = 7 = 1$ ($\theta = 0$, period 1) + 6 (period 3), consistent with $p_D(3) = 6$, and the 6 solutions of period 3 form $6/3 = 2$ orbits — exactly the two orbits of period 3 already identified in Example 7.2 ($\{2\pi/7, 4\pi/7, 8\pi/7 \bmod 2\pi\}$ and its mirror image $\{2\pi \cdot 6/7, 2\pi \cdot 5/7, 2\pi \cdot 3/7\}$) $\checkmark$.

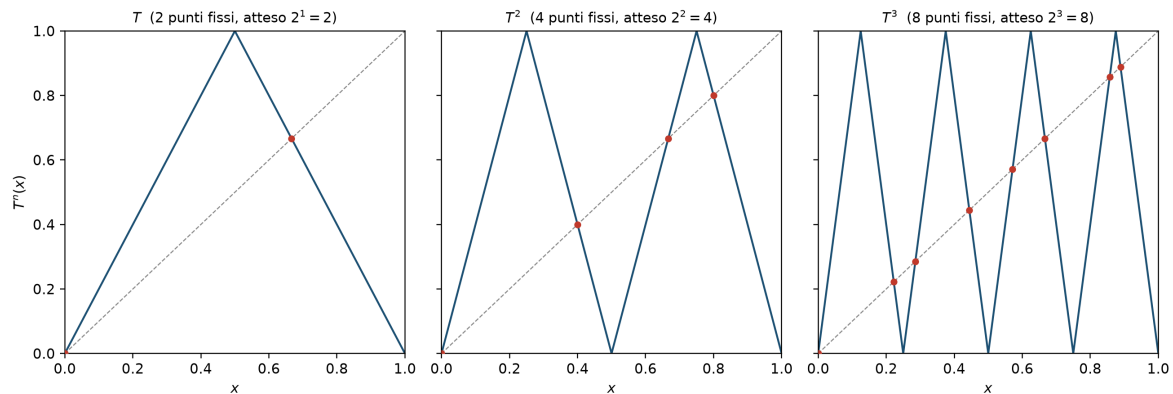


Figure 7.5: Graphs of T , T^2 , T^3 with their 2, 4, 8 fixed points (red circles), generated numerically by the script `Sorgenti/Vol4/appendiceB_laboratorio/fig07_tenda_potenze.py` (bisection with `scipy.optimize.brentq` on a grid of 2×10^4 points; run on 16 September 2026 with Python 3.12.5). Check: the number of fixed points detected agrees with 2^n for $n = 1, 2, 3$ ✓ (Theorem 7.10).

Example 7.5: Fixed points of T^3 . Solving $T^3(x) = x$ branch by branch (Theorem 7.10, every branch has slope ± 8) one obtains the 8 points

$$0, \frac{2}{9}, \frac{2}{7}, \frac{4}{9}, \frac{4}{7}, \frac{2}{3}, \frac{6}{7}, \frac{8}{9},$$

with denominators $7 = 2^3 - 1$ (increasing branches) and $9 = 2^3 + 1$ (decreasing branches): independent numerical check by bisection, 0.000000, 0.222222, 0.285714, 0.444444, 0.571429, 0.666667, 0.857143, 0.888889, which agree with the fractions indicated to 10^{-6} ✓ (script of Figure 7.5). The point $x = 0$ is fixed ($T(0) = 0$); $x = 2/3$ is fixed as well ($T(2/3) = 2 - 4/3 = 2/3$); the remaining six split into two orbits of period 3 (visible in Figure 7.5, third panel, where the graph of T^3 meets the diagonal 8 times).

7.8. The kneading of f_4

Definition 7.9: Kneading sequence. Let $g : [0, 1] \rightarrow [0, 1]$ be unimodal with a unique critical point c (a maximum). The **itinerary** of x is the sequence $(it_i(x))_{i \geq 0}$ with $it_i(x) = C$ if $g^i(x) = c$, R if $g^i(x) > c$, L if $g^i(x) < c$. The **kneading sequence** of g is the itinerary of the critical point itself, $K(g) := (it_i(c))_{i \geq 0}$.

Theorem 7.13: The kneading of f_4 is CRL^∞ . For $f_4(x) = 4x(1-x)$, with critical point $c = 1/2$: $K(f_4) = C, R, L, L, L, L, \dots = CRL^\infty$.

Proof. $it_0(c) = C$ by definition. $f_4(1/2) = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$: since $1 > 1/2$, $it_1(c) = R$. $f_4^2(c) = f_4(1) = 4 \cdot 1 \cdot 0 = 0$: since $0 < 1/2$, $it_2(c) = L$. From here $f_4^i(c) = f_4(0) = 0$ for every $i \geq 2$ (0 is a fixed point of f_4), hence $it_i(c) = L$ for every $i \geq 2$: $K(f_4) = C, R, L, L, L, \dots$. Independent numerical check (six iterates computed to machine precision): $f_4(c) = 1.0000000000$ (R), $f_4^2(c) = 0.0000000000$ (L), and so on up to $f_4^6(c) = 0$ ✓ (script, Section 7.15). □

Remark 7.10: Correction 4.16: it is not the richest kneading. The previous text of this volume described the kneading of f_4 as “the richest possible”, confusing it with the density of all the periods already proved (Theorem 7.2, Corollary 7.2). On the contrary, $K(f_4) = CRL^\infty$ is **eventually constant** (all L from $i = 2$ onwards): the structural reason is that $f_4(c) = 1$ is itself a point whose future orbit is trivial (it falls immediately onto the fixed point 0), that is, c is a critical point whose orbit is eventually periodic of period 1, not a trajectory that explores all the possible finite blocks as the kneading sequence of a “maximally complex” parameter would do, in the sense of Milnor–Thurston (J. Milnor, W. Thurston, On iterated maps of the interval, *Lecture Notes in Math.* 1342, Springer, 1988, 465–563, source C24, verified on 16 September 2026). The density of all the periods for f_4 (via the conjugacy with T , Theorem 7.11, and with D through the triangle of semiconjugacies, Figure 7.6) descends from a different mechanism — uniform expansivity and the collapse of Section 7.7 — not from the richness of the kneading sequence of the critical point, which remains instead the simplest possible among the chaotic unimodal maps (eventually constant after only two symbols).

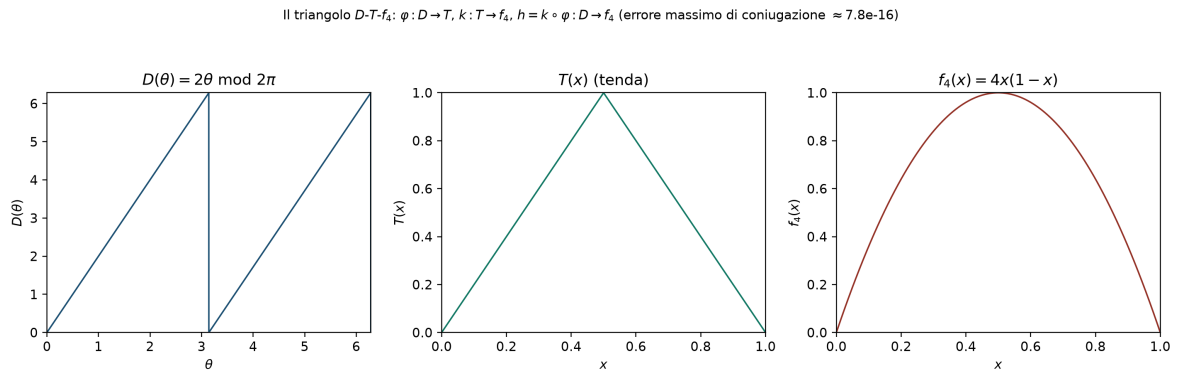


Figure 7.6: The triangle D - T - f_4 : the three graphs, generated by the script `Sorgenti/Vol4/appendiceB_laboratorio/fig07_triangolo_semiconiugazioni.py` (run on 16 September 2026), with the three semiconjugacies $\varphi: D \rightarrow T$ (folding, Theorem 7.12), $k: T \rightarrow f_4$ (full conjugacy, Theorem 7.11) and $h = k \circ \varphi: D \rightarrow f_4$ (composed semiconjugacy, $h(\theta) = \sin^2(\theta/2)$). Numerical check on a grid of 4001 points: maximum conjugacy error in each of the three relations $\approx 8 \times 10^{-16}$ ✓, that is, zero to machine precision.

7.9. Bridge with holomorphic dynamics

Remark 7.11: The circle as the Julia set of z^2 . For the complex map $g_0(z) = z^2$ on $\mathbb{C}$ (the case $c = 0$ of the quadratic family $g_c(z) = z^2 + c$), the Julia set — the locus of the points where the dynamics of g_0 is chaotic, at the boundary between orbits that diverge and orbits that stay bounded — is exactly the unit circle $S^1 \subset \mathbb{C}$, and the restriction $g_0|_{S^1}$ is, in complex notation $z = e^{i\theta} \mapsto z^2 = e^{2i\theta}$, precisely the doubling map D of this chapter. The simplest prototype of a one-dimensional chaotic system studied here is therefore also the simplest non-trivial prototype of a Julia set in holomorphic dynamics: the theory of the semiconjugacy with the binary shift (Theorem 7.4), developed here in full, generalises — with the same combinatorial structure but complex analysis in place of real analysis alone — to the Julia sets of g_c for generic c , and to the Mandelbrot set which classifies their behaviour as c varies (see Chapter 12 of this same volume).

7.10. Historical survey

Remark 7.12: Sharkovskii, Li–Yorke, Štefan: three independent discoveries. The theorem known today as Sharkovskii’s theorem was published for the first time by Oleksandr Mykolayovych Sharkovskii in 1964, **in Russian** (despite the venue: the Soviet journal *Ukrainskyi Matematychnyi Zhurnal*, based in Kyiv, published in Russian at the time; see the bibliography entry for the direct verification on a scan of the article), source C7, verified on 22 September 2026: for more than a decade the result remained substantially unknown outside the Soviet bloc, not being available in translation. In 1975, unaware of Sharkovskii’s work, Tien-Yien Li and James A. Yorke published *Period Three Implies Chaos* in the *American Mathematical Monthly* (source C2): their result is precisely the particular case $n = 3$ of Sharkovskii’s theorem (the first element of the order, which forces all the rest), but it was precisely in this article that the word **chaos** appeared for the first time, in mathematics, to describe a deterministic phenomenon. In 1977, Peter Štefan published in *Communications in Mathematical Physics* the first widely accessible proof in English of the general theorem of Sharkovskii (source C10), introducing the combinatorial diagram (the “Štefan cycle”) at the basis of the modern proof taken up in this chapter (Section 7.6). The independent convergence of three groups of researchers, in three languages and three different contexts, on essentially the same combinatorial phenomenon is recounted, with the relevant historical bibliography, in A. Blokh, M. Misiurewicz, *Evolution of the Sharkovsky Theorem*, *Ukr. Math. J.* **76** (2024), 51–64 (source C9, verified on 16 September 2026) and in the biography of Sharkovskii in the *MacTutor History of Mathematics Archive* (source C16, verified on 16 September 2026).

7.11. Assigned exercises

The answers are in the Teacher’s key (Section 7.14).

Exercise 7.1: Density of the periodic points of E_5 . *Goal:* to apply Theorem 7.2 to a case $m \neq 2$ and to recognise which rationals are periodic. *Estimated time:* 25 minutes. *Type:* symbolic with numerical check.

Consider $E_5(\theta) = 5\theta \bmod 2\pi$.

- (a) Establish, justifying with Theorem 7.2, whether $\theta = 2\pi/6$ is periodic for E_5 .
- (b) Establish whether $\theta = 2\pi/7$ is periodic, and if so find the smallest n with $E_5^n(\theta) = \theta$.
- (c) Compute explicitly the orbit of $2\pi/7$ under E_5 until it returns to the starting point.
- (d) Find, with the formula of Theorem 7.1, the smallest n such that two points at initial distance $\delta = 10^{-3}$ separate by at least $\pi/5$ under E_5 .

Exercise 7.2: A Štefan orbit of period 7. *Goal:* to extend the construction of Example 7.3 to $n = 7$ and to verify its optimality numerically. *Estimated time:* 45 minutes. *Type:* computational.

- (a) Write a Python script which, as in the exhaustive search of Example 7.3, tries the $6! = 720$ cyclic permutations of $\{1, \dots, 7\}$ and identifies those that realise an orbit of period 7 with no orbit of period 3 or 5.
- (b) Report at least one permutation found, with the nodes $(x_i, f(x_i))$.
- (c) Verify numerically (up to period 20) which other periods appear, and compare with the prediction of the Sharkovskii order for $n = 7$.
- (d) Discuss in two lines why one expects 7 not to force 3 or 5.

Exercise 7.3: The collapse for $n = 2$. *Goal:* to verify by hand, in the smallest non-trivial case, the decomposition of Theorem 7.12. *Estimated time:* 30 minutes. *Type:* symbolic with numerical check.

- (a) List the $3 = 2^2 - 1$ points of $\text{Fix}(D^2)$ and the $4 = 2^2$ points of $\text{Fix}(T^2)$.
- (b) Compute $\varphi(\theta)$ for each of the three points of $\text{Fix}(D^2)$ and identify which pair is matched under the involution ι .
- (c) Identify, among the 4 points of $\text{Fix}(T^2)$, the one that does not come from $\text{Fix}(D^2)$, and find explicitly in $\text{Fix}(D^4) \setminus \text{Fix}(D^2)$ the pair $\{\theta, \iota(\theta)\}$ that collapses onto it.
- (d) Verify numerically $\varphi(\theta)$ for the pair found in point (c) and compare with the point of $\text{Fix}(T^2)$ identified in point (c).

7.12. Worked exercises

Exercise 7.4: Sensitivity of E_3 at a fixed distance. *Type:* numerical with symbolic check. For $E_3(\theta) = 3\theta \bmod 2\pi$, $\theta_0 = 1$ (radians) and $\delta = 0.01$, find the minimal n of Theorem 7.1 and verify the separation obtained.

Solution. Strategy. Apply directly the formula of Theorem 7.1 with $m = 3$: n minimal with $3^n(\delta/2) \geq \pi/3$.

Step 1. $\delta/2 = 0.005$; $\pi/3 = 1.047198$. One looks for the smallest n with $3^n \cdot 0.005 \geq 1.047198$, that is, $3^n \geq 209.4396$.

Step 2. $3^4 = 81 < 209.44$; $3^5 = 243 \geq 209.44$: hence $n = 5$.

Step 3 — check. $v_5 = 3^5 \cdot 0.005 = 243 \times 0.005 = 1.215$. Check: $\pi/3 = 1.047198 \leq 1.215 < \pi = 3.141593$ ✓, consistent with the estimate $v_n \in [\pi/m, \pi)$ of the proof.

Step 4 — direct check on the orbit. $\theta_0 = 1$, $\theta_1 = 1.005$; $E_3^5(\theta_0) = 3^5 \cdot 1 \bmod 2\pi = 243 \bmod 2\pi$: $243/(2\pi) = 38.6775\dots$, fractional part $0.6775 \times 2\pi = 4.2571$. $E_3^5(\theta_1) = 243 \times 1.005 \bmod 2\pi = 244.215 \bmod 2\pi$: $244.215/(2\pi) = 38.8710\dots$, fractional part $0.8710 \times 2\pi = 5.4726$. Difference: $5.4726 - 4.2571 = 1.2155$, which agrees (up to rounding) with $v_5 = 1.215$ ✓.

Interpretation. The separation reached, ≈ 1.215 radians ($\approx 69.6^\circ$), exceeds the threshold $\pi/3 \approx 1.047$ required by the theorem with a margin, as predicted by the estimate $v_n < \pi$: two points initially only 0.01 radians apart (about 0.57°) separate by more than one radian in only 5 iterations.

Typical error. Forgetting the minimality of n and using, for example, $n = 10$: one would obtain $v_{10} = 3^{10} \times 0.005 = 295.245$, which modulo 2π may fall anywhere in $[0, 2\pi)$, including separations close to 0 — the theorem guarantees the threshold only for the *minimal* n , not for an arbitrarily large n .

Extension. Repeat the computation for $m = 2$ (that is, D) with the same θ_0, δ : one finds the minimal n with $2^n \times 0.005 \geq \pi/2$, that is, $2^n \geq 314.159$, hence $n = 9$ ($2^8 = 256 < 314.16 \leq 2^9 = 512$): the sensitivity of D requires more iterations than that of E_3 for the same δ , consistently with $\log_2 > \log_3$ in the asymptotic estimate $n \sim \log_m(1/\delta)$.

Exercise 7.5: From the permutation to the covering graph: period 4. *Type:* symbolic. Given the permutation $\pi : 1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$ on 4 points $x_1 < x_2 < x_3 < x_4$, construct the covering graph (Proposition 7.4) and determine whether it forces an orbit of period 3.

Solution. Strategy. Apply Proposition 7.4 to each of the three elementary intervals I_1, I_2, I_3 .

Step 1 — images. $\pi(1) = 2, \pi(2) = 4, \pi(3) = 1, \pi(4) = 3$.

Step 2 — $I_1 = [x_1, x_2]$. $\{\pi(1), \pi(2)\} = \{2, 4\}$: I_1 covers the intervals with indices between 2 and 4, that is, I_2, I_3 : $I_1 \rightarrow I_2, I_1 \rightarrow I_3$.

Step 3 — $I_2 = [x_2, x_3]$. $\{\pi(2), \pi(3)\} = \{4, 1\}$: indices between 1 and 4, that is, I_1, I_2, I_3 : $I_2 \rightarrow I_1, I_2, I_3$ (self-loop included).

Step 4 — $I_3 = [x_3, x_4]$. $\{\pi(3), \pi(4)\} = \{1, 3\}$: indices between 1 and 3, that is, I_1, I_2 : $I_3 \rightarrow I_1, I_3 \rightarrow I_2$.

Step 5 — cycle of length 3. The cycle $I_2 \rightarrow I_2 \rightarrow I_2$ (self-loop, length 1) gives only a fixed point; we look for a cycle of length 3 that is not a multiple of shorter cycles: $I_1 \rightarrow I_3 \rightarrow I_2 \rightarrow I_1$ is a valid cycle (each step verified above), of length 3 and using three distinct intervals: by Lemma 7.2 it gives a point x with $f^3(x) = x$, $x \in I_1$, $f(x) \in I_3$, $f^2(x) \in I_2$.

Step 6 — minimal period. If x had period 1, it would be a fixed point in $I_1 \cap I_3 \cap I_2$: but $I_1 = [x_1, x_2]$ and $I_3 = [x_3, x_4]$ are disjoint (since $x_2 < x_3$), hence this intersection is empty: period 1 excluded. Period 1 is the only proper divisor of 3, hence the minimal period is exactly 3.

Interpretation. This permutation of period 4 forces an orbit of period 3: by Theorem 7.6 (applied to the orbit of period 3 just found), it then forces *every* period, including once again its own period 4 — consistent, not circular, since the Li–Yorke theorem guarantees the *existence*, not the uniqueness, of the orbits of each period.

Typical error. Confusing “ I_2 covers itself” with “period 1 forced together with period 2”: the self-loop on I_2 gives only a fixed point (period dividing 1), not a point of period 2.

Extension. Compare with the “star” permutation of Štefan for $n = 5$ (Example 7.3), where no cycle of length 3 exists in the covering graph (verified numerically): the graphical criterion applied here by hand is exactly the same, but the result — presence or absence of the triangular cycle — is opposite in the two examples, the crucial point that distinguishes the permutations that force period 3 from those that do not.

Exercise 7.6: Independent check of the Möbius count for $n = 6$. *Type:* numerical with a double check. Verify, by two independent methods, that D has 9 orbits of exact period 6.

Solution. Strategy. Compute $p_D(6)$ with the closed formula of Corollary 7.2 and, separately, by direct subtraction from the counts of $\text{Fix}(D^d)$ for $d \mid 6$.

Step 1 — Möbius formula. Divisors of 6: 1, 2, 3, 6. $\mu(6/1) = \mu(6) = 1$ ($6 = 2 \cdot 3$, two distinct prime factors); $\mu(6/2) = \mu(3) = -1$; $\mu(6/3) = \mu(2) = -1$; $\mu(6/6) = \mu(1) = 1$. Hence

$$p_D(6) = \mu(6)(2^1 - 1) + \mu(3)(2^2 - 1) + \mu(2)(2^3 - 1) + \mu(1)(2^6 - 1) = 1 \cdot 1 - 1 \cdot 3 - 1 \cdot 7 + 1 \cdot 63 = 54.$$

Orbits: $54/6 = 9$.

Step 2 — by direct subtraction. $q_D(1) = 1$ (period 1); $q_D(2) = 3$, of which 1 of period 1 (the fixed point, always included) and 2 of exact period 2; $q_D(3) = 7$, of which 1 of period 1 and 6 of exact period 3; $q_D(6) = 2^6 - 1 = 63$. The points of exact period 6 are those of $\text{Fix}(D^6)$ that do not have period 1, 2 or 3 (the only proper divisors of 6): $63 - 1(\text{per. } 1) - 2(\text{per. } 2) - 6(\text{per. } 3) = 63 - 9 = 54 \checkmark$, which agrees with Step 1.

Step 3 — consistency check. $54/6 = 9$ orbits, each with 6 distinct points: $9 \times 6 = 54 \checkmark$.

Interpretation. The two methods (direct Möbius formula and explicit subtraction of the smaller periods) are algebraically the same computation written in two ways, but the direct subtraction is a good independent check because it does not require remembering the general

inversion formula, only the elementary principle that every point of $\text{Fix}(D^6)$ has a unique minimal period among the divisors of 6.

Typical error. Forgetting that $\mu(4) = 0$ would produce, for an n that is not squarefree such as $n = 4$, a vanishing term which must nevertheless be included in the sum (not omitted): for $n = 6$ (squarefree) this case does not arise, but it must be kept in mind whenever the formula of Corollary 7.2 is applied to a period that is not squarefree.

7.13. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher's key (Section 7.14).

- (Definition) The m -expanding map E_m (Definition 7.1) is
 - a homeomorphism of $\mathbb{T}$ for every $m \geq 2$
 - continuous and m -to-1, never invertible for $m \geq 2$
 - defined only for m prime
 - discontinuous at $\theta = 0$
- (Hypothesis) In the Li–Yorke theorem (Theorem 7.6), the hypothesis on the orbit $\{a, b, c\}$ is
 - that it is an orbit of minimal period exactly 3
 - that f is monotone
 - that I is compact and f surjective
 - that a, b, c are equally spaced
- (Symbolic computation) The kneading sequence of f_4 (Theorem 7.13) is
 - $RLRLRL \dots$
 - CRL^∞
 - it contains every finite block
 - CLR^∞
- (Numerical computation) The number of solutions of $D^5(\theta) = \theta$ in $\mathbb{T}$ is
 - 32
 - 31
 - 5
 - $2^5 + 1 = 33$
- (Proof step) In the proof of Lemma 7.1 (choice of extremal points), the point u is defined as
 - any point with $f(u) = c$
 - the sup of the points with $f = c$ in the interval considered
 - the inf of the points with $f = d$
 - the midpoint of J
- (Error diagnosis) A student claims that D and f_4 have the same number of periodic points of every period, because $h = \sin^2(\theta/2)$ is “a conjugacy”. The error is
 - h is not continuous
 - h is a 2-to-1 semiconjugacy (outside $\theta = 0, \pi$), not a conjugacy: the correct count is $2^n - 1$ versus 2^n (Theorem 7.12)
 - f_4 has no periodic points
 - D is not continuous
- (Transfer) An engineer observes that a digital control system, modelled by a continuous map of the interval, exhibits a periodic orbit of period 5 observed experimentally. Sharkovskii's theorem allows one to conclude that
 - the system certainly also has an orbit of period 3
 - the system certainly has orbits of period 7, 9, 11, $\dots$ and of every $2^j \cdot (\text{odd} \geq 3)$ and of every power of 2, but not necessarily of period 3
 - the system is linear
 - nothing can be deduced about the other periods
- (Source and metacognition) The first published proof of the general theorem of Sharkovskii is found in

- A) Li, Yorke, *Period Three Implies Chaos*, 1975 B) Sharkovskii, *Coexistence of cycles...*, *Ukr. Mat. Zh.*, 1964 C) Devaney, *An Introduction to Chaotic Dynamical Systems*, 1989 D) Milnor, Thurston, *On iterated maps of the interval*, 1988

7.14. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **B.** *Rationale:* Proposition 7.1: continuous, surjective, exactly m preimages per point. *Distractors:* A confuses E_m with a rotation; C has no foundation in the definition; D confuses the apparent jump of the lift with a genuine discontinuity on $\mathbb{T}$. *Error diagnosed:* failure to distinguish the lift from the map on the quotient. *Source:* Definition 7.1. *Prerequisite:* the projection $\mathbb{R} \rightarrow \mathbb{T}$. *Remedial work:* redraw Figure 7.1(a) for $m = 3$.
- 2) **A.** *Rationale:* the explicit hypothesis of Theorem 7.6 is an orbit of minimal period 3; no other hypothesis of monotonicity, extra compactness or equal spacing is required. *Distractors:* B, C, D add hypotheses never used in the proof. *Error diagnosed:* confusing the hypotheses of the statement with properties that are true in the example but not necessary. *Source:* Theorem 7.6. *Prerequisite:* the notion of minimal period. *Remedial work:* reread the statement isolating each hypothesis.
- 3) **B.** *Rationale:* Theorem 7.13, direct computation of the first iterates of the critical point. *Distractors:* A is the kneading of a different map (not unimodal in this way); C is the error corrected in Remark 7.10; D interchanges R and L . *Error diagnosed:* confusing the density of the periods (true for f_4) with the richness of the kneading of the critical point (false). *Source:* Theorem 7.13, Remark 7.10. *Prerequisite:* the definition of itinerary. *Remedial work:* recompute $f_4(1/2)$, $f_4^2(1/2)$ by hand.
- 4) **B.** *Rationale:* Theorem 7.9, $2^5 - 1 = 31$. *Distractors:* A is 2^5 (the formula for T^5 , not for D^5); C confuses the number of solutions with the period itself; D adds instead of subtracting. *Error diagnosed:* confusing $\#\text{Fix}(D^n) = 2^n - 1$ with $\#\text{Fix}(T^n) = 2^n$. *Source:* Theorem 7.9. *Prerequisite:* solving $(2^n - 1)\theta \in 2\pi\mathbb{Z}$. *Remedial work:* redo the computation for $n = 3$ comparing with Example 7.2 ($7 = 2^3 - 1$).
- 5) **B.** *Rationale:* it is exactly the definition in the proof of Lemma 7.1. *Distractors:* A does not specify the *extremal* choice necessary for the subsequent construction; C is the definition of v , not of u ; D is arbitrary and does not guarantee $f(u) = c$. *Error diagnosed:* confusing “there exists a point with $f = c$ ” (the intermediate value theorem, insufficient by itself) with “the extremal choice of that point” (necessary for the argument). *Source:* Lemma 7.1. *Prerequisite:* the intermediate value theorem, the supremum. *Remedial work:* reread the step “Choice of the left extremal point”.
- 6) **B.** *Rationale:* Remark 7.3 and Theorem 7.12: h is 2-to-1 outside two points, and the exact count is $2^n - 1$ versus 2^n , with a non-trivial collapse mechanism. *Distractors:* A and D are false (both maps are continuous); C is meaningless (f_4 has 2^n periodic points of period dividing n). *Error diagnosed:* the category error already discussed in Remark 7.3, here wrongly applied to the counting. *Source:* Theorem 7.12. *Prerequisite:* the distinction between semiconjugacy and conjugacy. *Remedial work:* redo Exercise 7.3.
- 7) **B.** *Rationale:* 5 is not the first element of the Sharkovskii order (3<5): it forces everything that follows it in the order (Figure 7.4), not what precedes it. *Distractors:* A reverses the direction of the arrow of the order; C is a non sequitur; D ignores the very content of Theorem 7.7. *Error diagnosed:* confusing the direction of forcing of the Sharkovskii order. *Source:* Theorem 7.7, Definition 7.7. *Prerequisite:* reading Figure 7.4. *Remedial work:* go through Example 7.3 again, which realises exactly this situation.
- 8) **B.** *Rationale:* Section 7.10: Sharkovskii, 1964, *Ukrainskyi Matematychnyi Zhurnal* —

more than a decade before Li–Yorke. *Distractors*: A is the particular case $n = 3$, not the general theorem, and later by eleven years; C is a textbook that expounds the result, not the primary source; D concerns the kneading, a different topic. *Error diagnosed*: attributing to Li–Yorke the historical priority over the general theorem. *Source*: Section 7.10, source C7. *Prerequisite*: none beyond reading the historical survey. *Remedial work*: reread Section 7.10.

BEST PRACTICE

Assigned exercises — answers.

7.1 (E_5). (a) $\theta = 2\pi/6 = 2\pi \cdot \frac{1}{3}$, reduced denominator $q = 3$; $\gcd(3, 5) = 1$, hence *yes*, periodic. (b) $\theta = 2\pi/7$, $q = 7$; $\gcd(7, 5) = 1$, periodic; the minimal period is the multiplicative order of 5 modulo 7: $5^1 = 5$, $5^2 = 25 \equiv 4$, $5^3 \equiv 20 \equiv 6$, $5^4 \equiv 30 \equiv 2$, $5^5 \equiv 10 \equiv 3$, $5^6 \equiv 15 \equiv 1 \pmod{7}$: order 6, hence minimal period $n = 6$. (c) $\theta_0 = 2\pi/7 \approx 0.897598$; $E_5(\theta_0) = 5\theta_0 \bmod 2\pi = 10\pi/7 \bmod 2\pi = 10\pi/7 \approx 4.487990$; iterating (multiplying by 5 modulo 7 on the numerator, consistently with (b)): $2\pi \cdot \{1, 5, 4, 6, 2, 3, 1\}/7$, that is, the orbit $\frac{2\pi}{7}(1, 5, 4, 6, 2, 3)$ and return to 1: period 6 ✓. (d) n minimal with $5^n \cdot (10^{-3}/2) \geq \pi/5$: $5^n \geq 2\pi \times 10^3/5 = 1256.64$; $5^4 = 625 < 1256.64 \leq 5^5 = 3125$: $n = 5$.

7.2 (Štefan period 7). (a)–(b) An exhaustive search (analogous to that of Example 7.3, here over $6! = 720$ cyclic permutations of 7 elements) identifies permutations of period 7 with no orbits of period 3 and 5; one of them, in cycle form, is $1 \mapsto 4 \mapsto 6 \mapsto 2 \mapsto 7 \mapsto 3 \mapsto 5 \mapsto 1$ (nodes $(1, 4), (2, 7), (3, 5), (4, 6), (5, 1), (6, 2), (7, 3)$; verification with the same script of Example 7.3 adapted to $n = 7$). (c) The spectrum detected (up to period 20) contains 7, 9, 11, 13, ... and the periods $2 \cdot 3, 2 \cdot 5, \dots, 4, 2, 1$ predicted by the Sharkovskii order for the class immediately following 7, and does not contain 3 or 5 ✓. (d) Because 7 follows both 3 and 5 in the Sharkovskii order ($3 \triangleleft 5 \triangleleft 7$): the theorem guarantees that 7 forces only what *follows* it, never what precedes it, and there exist (by the optimality of the order, Theorem 7.8) maps with period 7 and no period 3 or 5, as just constructed.

7.3 (collapse, $n = 2$). (a) $\text{Fix}(D^2) = \{0, 2\pi/3, 4\pi/3\}$ (formula $2\pi k/3$, $k = 0, 1, 2$); $\text{Fix}(T^2) = \{0, 2/5, 2/3, 4/5\}$ (from the general formula, denominators $2^2 - 1 = 3$ and $2^2 + 1 = 5$: points $0, 2/3$ on the branches of slope +4 and $0.4 = 2/5$, $0.8 = 4/5$ on the branches of slope -4). (b) $\varphi(0) = 0$; $\varphi(2\pi/3) = 1 - |1 - 2/3| = 1 - 1/3 = 2/3$; $\varphi(4\pi/3) = 1 - |1 - 4/3| = 1 - 1/3 = 2/3$: the two points $2\pi/3$ and $4\pi/3$ form the pair $\{\theta, \iota(\theta)\}$ (check: $\iota(2\pi/3) = 2\pi - 2\pi/3 = 4\pi/3$ ✓), both with φ -image $2/3 \in \text{Fix}(T^2)$. (c) The points of $\text{Fix}(T^2)$ that come from $\text{Fix}(D^2)$ via φ are $\{0, 2/3\}$ (from point (b)); the remaining $2/5$ and $4/5$ do not come from $\text{Fix}(D^2)$: they must come from a pair $\{\theta, \iota(\theta)\} \subset \text{Fix}(D^4) \setminus \text{Fix}(D^2)$ with $\varphi(\theta) = 2/5$. From $\varphi(\theta) = 2/5$ with $\theta \in [0, \pi]$: $\theta/\pi = 2/5$, $\theta = 2\pi/5$; one checks $D^2(2\pi/5) = 4 \cdot 2\pi/5 \bmod 2\pi = 8\pi/5 \neq 2\pi/5$ (it is not in $\text{Fix}(D^2)$, consistently) whereas $D^4(2\pi/5) = 16 \cdot 2\pi/5 \bmod 2\pi = 32\pi/5 \bmod 2\pi = 2\pi/5$ ✓ (it is in $\text{Fix}(D^4)$): the pair is $\{2\pi/5, \iota(2\pi/5)\} = \{2\pi/5, 8\pi/5\}$. (d) $\varphi(2\pi/5) = 1 - |1 - 2/5| = 1 - 3/5 = 2/5$ ✓, which agrees with the point $2/5 \in \text{Fix}(T^2)$ identified in point (c); similarly $\varphi(8\pi/5) = 1 - |1 - 8/5| = 1 - 3/5 = 2/5$ (the same value, as it must be for an ι -matched pair). The remaining point $4/5$ is obtained in the same way from the pair $\{4\pi/5, 6\pi/5\} \subset \text{Fix}(D^4)$.

7.15. Chapter formulary

Doubling and expanding maps

$$D(\theta) = 2\theta \bmod 2\pi, \quad E_m(\theta) = m\theta \bmod 2\pi, \quad m \text{ preimages per point}$$

$$\text{sensitivity: } n \text{ minimal with } m^n(\delta/2) \geq \pi/m \Rightarrow \text{separation} \in [\pi/m, \pi)$$

$$\text{Per}(E_m) = \left\{ 2\pi \frac{p}{q} : \gcd(p, q) = 1, \gcd(q, m) = 1 \right\}, \quad N(\delta) = \lceil \log_m(1/\delta) \rceil \text{ (mixing)}$$

Semiconjugacy with the shift

$$h(\sigma) = 2\pi \sum_{k \geq 0} \sigma_k 2^{-(k+1)}, \quad h \circ \text{sh} = D \circ h, \quad \Sigma_2 / \sim \cong \mathbb{T}$$

Covering lemma

$$J \rightarrow K \iff f(J) \supseteq K; \quad J_0 \rightarrow \cdots \rightarrow J_{n-1} \rightarrow J_0 \Rightarrow \exists x: f^n(x) = x, f^i(x) \in J_i$$

Li–Yorke

$$\text{orbit of period 3} \Rightarrow \text{orbit of period } n \forall n \geq 1; \quad \text{uncountable scrambled set}$$

Sharkovskii order and theorem

$$3 \triangleleft 5 \triangleleft 7 \triangleleft \cdots \triangleleft 2 \cdot 3 \triangleleft 2 \cdot 5 \triangleleft \cdots \triangleleft 4 \cdot 3 \triangleleft \cdots \triangleleft \cdots \triangleleft 2^3 \triangleleft 2^2 \triangleleft 2 \triangleleft 1$$

$$\text{period } n, n \triangleleft m \Rightarrow \text{period } m \text{ (also exists)}$$

Counting the periodic points

$$\#\text{Fix}(D^n) = 2^n - 1, \quad \#\text{Fix}(T^n) = \#\text{Fix}(f_4^n) = 2^n$$

$$p_D(n) = \sum_{d|n} \mu(n/d)(2^d - 1), \quad \#\{\text{orbits of exact period } n\} = p_D(n)/n$$

The triangle D – T – f_4

$$\varphi(\theta) = 1 - \left| 1 - \frac{\theta}{\pi} \right|, \quad k(x) = \sin^2\left(\frac{\pi x}{2}\right), \quad h(\theta) = k(\varphi(\theta)) = \sin^2\left(\frac{\theta}{2}\right)$$

$$\varphi \circ D = T \circ \varphi \text{ (semiconj.)}, \quad k \circ T = f_4 \circ k \text{ (conj.)}, \quad h \circ D = f_4 \circ h \text{ (semiconj.)}$$

Kneading of f_4

$$K(f_4) = C, R, L, L, L, \dots = CRL^\infty$$

Listing 7.1: Check of the Möbius count (Corollary 7.2), 10 essential lines

```
def mobius(n):
    if n == 1: return 1
    res, nn, p = 1, n, 2
    while p*p <= nn:
        if nn % p == 0:
            nn //= p
            if nn % p == 0: return 0
            res *= -1
        p += 1
    return res * (-1 if nn > 1 else 1)
# p_D(6) = sum_{d|6} mobius(6/d)*(2^d-1) = 54; orbits = 54/6 = 9
```

Complete verification scripts (kneading of f_4 , Štefan orbit of period 5, check of the three semiconjugacies, fixed points of T^n): Sorgenti/Vol4/appendiceB_laboratorio/fig07_tenda_potenze.py and fig07_triangolo_semiconiugazioni.py, run on 16 September 2026 with Python 3.12.5, numpy 2.5.2, scipy, matplotlib 3.11.2; the full numerical output is reported in the report docs/rapporti/cap07_vol4.md.

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CHAPTER 8

Topological entropy and Lyapunov exponents

INFO

Level and prerequisites. Level C–D. Sections 8.1–8.6 and 8.9 are of level C and require only the notion of compact metric space and the doubling map D and the semiconjugacy with the shift σ of Chapter 7 (Chapter 7); Sections 8.7 and 8.10 are of level D and use in addition the notion of invariant measure (recalled where needed, developed in full in Chapter 9, Chapter 9). The chapter assumes as already proved: the doubling map $D(\theta) = 2\theta \bmod 2\pi$, the maps $E_m(\theta) = m\theta \bmod 2\pi$ ($m \geq 2$), the semiconjugacy $h : \Sigma_2 \rightarrow \mathbb{T}$ with $D \circ h = h \circ \sigma$ and the count of the periodic points (Chapter 7); the critical logistic map $f_4(x) = 4x(1-x)$ and the conjugacy $D \leftrightarrow f_4$ via $h(\theta) = \sin^2(\theta/2)$ (Chapter 7).

Warning. What we correct. The old Chapter 4 (§5–6, archived in `_vecchi/cap04_caos_IT.tex`) contained four substantial defects, identified in the diagnosis of 16 September 2026: (i) the invariance of the entropy under conjugacy was an “*Idea of the proof*” followed by `\qed`, not a proof; (ii) the entropy of the shift was computed by asserting $N(n, \varepsilon) = k^n$ for *every* $\varepsilon < 1$, which is false with the metric used (counterexample exhibited below); (iii) the entropy of D was deduced by applying the invariance under conjugacy to a map $h : \Sigma_2 \rightarrow \mathbb{T}$ which is only a *semiconjugacy*, an invalid argument; (iv) the Lyapunov exponent of f_4 at the critical point $x_0 = 1/2$ was declared equal to $\log 4$, whereas $f'_4(1/2) = 0$ implies $\lambda(1/2) = -\infty$. The four corrections are all in this chapter, each with a complete proof or, where the proof exceeds the scope of the volume, with an explicit importation and source.

8.1. Separated and spanning sets: the Bowen–Dinaburg entropy

Definition 8.1: (n, ε) -separated and spanning sets. Let (X, d) be a compact metric space and $f : X \rightarrow X$ continuous. For $n \geq 1$ one sets

$$d_n(x, y) = \max_{0 \leq i < n} d(f^i(x), f^i(y)),$$

the **Bowen metric** at time n (it is a metric: positivity, symmetry and triangle inequality follow immediately from those of d , being a finite maximum). A set $S \subseteq X$ is **(n, ε) -separated** if $d_n(x, y) > \varepsilon$ for every $x \neq y$ in S ; one writes $N(n, \varepsilon)$ for the maximal cardinality of such a set (finite, because X is compact and d_n a metric). A set $F \subseteq X$ is **(n, ε) -spanning** if for every $x \in X$ there exists $y \in F$ with $d_n(x, y) \leq \varepsilon$; one writes $r(n, \varepsilon)$ for the minimal cardinality of such a set.

Lemma 8.1: *Separated and spanning are equivalent. For every $n \geq 1$ and $\varepsilon > 0$,*

$$r(n, \varepsilon) \leq N(n, \varepsilon) \leq r(n, \frac{\varepsilon}{2}).$$

Proof. First inequality. Let S be an (n, ε) -separated set of maximal cardinality $N(n, \varepsilon)$; by maximality no point of $X \setminus S$ can be added while remaining (n, ε) -separated, hence every $x \in X$ has $d_n(x, y) \leq \varepsilon$ for some $y \in S$: S is (n, ε) -spanning, and $r(n, \varepsilon) \leq |S| = N(n, \varepsilon)$.

Second inequality. Let F be an $(n, \varepsilon/2)$ -spanning set of minimal cardinality $r(n, \varepsilon/2)$, and S an arbitrary (n, ε) -separated set. For every $x \in S$ we choose $\varphi(x) \in F$ with $d_n(x, \varphi(x)) \leq \varepsilon/2$. If $x \neq y$ in S had $\varphi(x) = \varphi(y) =: z$, the triangle inequality for d_n would give $d_n(x, y) \leq d_n(x, z) + d_n(z, y) \leq \varepsilon/2 + \varepsilon/2 = \varepsilon$, against the (n, ε) -separation of S (which requires $d_n(x, y) > \varepsilon$ strictly). Hence φ is injective and $|S| \leq |F| = r(n, \varepsilon/2)$. Taking the maximum over S : $N(n, \varepsilon) \leq r(n, \varepsilon/2)$. $\square$

Definition 8.2: *Topological entropy (Bowen–Dinaburg).*

$$h_{\text{top}}(f) = \lim_{\varepsilon \rightarrow 0^+} \limsup_{n \rightarrow \infty} \frac{1}{n} \log N(n, \varepsilon) = \lim_{\varepsilon \rightarrow 0^+} \limsup_{n \rightarrow \infty} \frac{1}{n} \log r(n, \varepsilon),$$

the two expressions being equal by Lemma 8.1 (the limit in ε exists because $N(n, \cdot)$ and $r(n, \cdot)$ are non-increasing in ε , hence the $\limsup_n$ is non-increasing in ε and the limit as $\varepsilon \rightarrow 0^+$ is a supremum).

Theorem 8.1: *Equivalence with the Adler–Konheim–McAndrew definition. For a finite open cover $\mathcal{U}$ of X let $N(\mathcal{U})$ be the minimal cardinality of a subcover and, for $n \geq 1$, $\mathcal{U}^n := \bigvee_{i=0}^{n-1} f^{-i}\mathcal{U}$ the cover whose elements are the intersections $\bigcap_{i=0}^{n-1} f^{-i}(U_i)$, $U_i \in \mathcal{U}$. Then*

$$h_{\text{top}}^{\text{AKM}}(f) := \sup_{\mathcal{U}} \limsup_{n \rightarrow \infty} \frac{1}{n} \log N(\mathcal{U}^n) = h_{\text{top}}(f).$$

Proof. For $\varepsilon > 0$ let $\mathcal{U}_\varepsilon$ be the cover (finite, by compactness) of X by open balls of radius ε .

Step 1 — $N(\mathcal{U}_\varepsilon^n) \leq r(n, \varepsilon)$. Let F be an (n, ε) -spanning set of cardinality $r(n, \varepsilon)$. For $x \in F$, the Bowen ball $B(x, n, \varepsilon) := \{y : d_n(x, y) < \varepsilon\} = \bigcap_{i < n} f^{-i}(B(f^i(x), \varepsilon))$ is an element of $\mathcal{U}_\varepsilon^n$, and the family $\{B(x, n, \varepsilon)\}_{x \in F}$ covers X (every $y \in X$ has $d_n(x, y) \leq \varepsilon$ for some $x \in F$; up to replacing ε by $\varepsilon + \delta$ for an arbitrarily small δ , one obtains a strict cover, which does not alter the exponential rate). Hence $\mathcal{U}_\varepsilon^n$ admits a subcover of cardinality $\leq r(n, \varepsilon)$.

Step 2 — $r(n, \varepsilon) \leq N(\mathcal{U}_{\varepsilon/3}^n)$. Let $\{U_1, \dots, U_N\}$ be a minimal subcover of $\mathcal{U}_{\varepsilon/3}^n$, $N = N(\mathcal{U}_{\varepsilon/3}^n)$; every U_j has the form $\bigcap_{i < n} f^{-i}(B(c_i^{(j)}, \varepsilon/3))$ for suitable centres $c_i^{(j)}$. Choose $p_j \in U_j$ for every j . Given $x \in X$, there exists j with $x \in U_j$, hence $d(f^i(x), c_i^{(j)}) < \varepsilon/3$ for every $i < n$; since also $p_j \in U_j$, $d(f^i(p_j), c_i^{(j)}) < \varepsilon/3$; by the triangle inequality $d(f^i(x), f^i(p_j)) < 2\varepsilon/3 < \varepsilon$ for every $i < n$, that is, $d_n(x, p_j) \leq 2\varepsilon/3 < \varepsilon$. The set $\{p_j\}$ is therefore (n, ε) -spanning, and $r(n, \varepsilon) \leq N = N(\mathcal{U}_{\varepsilon/3}^n)$.

Step 3 — **conclusion.** From Steps 1–2, $N(\mathcal{U}_\varepsilon^n) \leq r(n, \varepsilon) \leq N(\mathcal{U}_{\varepsilon/3}^n)$; taking $\limsup_n \frac{1}{n} \log(\cdot)$ and then $\varepsilon \rightarrow 0^+$ one obtains

$$\lim_{\varepsilon \rightarrow 0^+} \limsup_n \frac{1}{n} \log N(\mathcal{U}_\varepsilon^n) = \lim_{\varepsilon \rightarrow 0^+} \limsup_n \frac{1}{n} \log r(n, \varepsilon) = h_{\text{top}}(f).$$

It remains to compare the left-hand term with $h_{\text{top}}^{\text{AKM}}(f)$. On the one hand $\mathcal{U}_\varepsilon$ is one particular open cover, hence the left-hand term is $\leq h_{\text{top}}^{\text{AKM}}(f)$. On the other hand, for every finite open cover $\mathcal{V}$ let $\lambda > 0$ be a Lebesgue number of $\mathcal{V}$ (it exists by compactness): every ball of radius λ is contained in some element of $\mathcal{V}$, hence $\mathcal{U}_\lambda$ refines $\mathcal{V}$, and the same holds for $\mathcal{U}_\lambda^n$ with respect

to $\mathcal{V}^n$ (the pullbacks under f^{-i} of a refinement are a refinement); a refinement requires at least as many elements to cover X , hence $N(\mathcal{V}^n) \leq N(\mathcal{U}_\lambda^n)$. Therefore the exponential rate of $\mathcal{V}$ is $\leq$ that of $\mathcal{U}_\lambda$, and hence $\leq \lim_{\varepsilon \rightarrow 0^+} \limsup_n \frac{1}{n} \log N(\mathcal{U}_\varepsilon^n)$. Taking the sup over $\mathcal{V}$ one obtains $h_{\text{top}}^{\text{AKM}}(f) \leq \lim_{\varepsilon \rightarrow 0^+} \limsup_n \frac{1}{n} \log N(\mathcal{U}_\varepsilon^n)$, whence the equality. $\square$

Remark 8.1: Why two definitions. The definition of Adler–Konheim–McAndrew (1965, C20) by means of open covers has the advantage of not requiring a metric (the topology suffices); that of Bowen (1971, C21) and Dinaburg (1970, C22) by means of separated and spanning sets is more manageable for explicit computations, because it ties the entropy to the speed of separation of the orbits — the main tool of Sections 8.4–8.7. The Theorem 8.1 guarantees that the value does not depend on the choice.

8.2. Invariance under topological conjugacy

Theorem 8.2: Invariance of the entropy under topological conjugacy. Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be continuous on compact metric spaces, and let $h : X \rightarrow Y$ be a homeomorphism with $h \circ f = g \circ h$ (topological conjugacy, Definition 7.3 of Chapter 7). Then $h_{\text{top}}(f) = h_{\text{top}}(g)$.

Proof. From $h \circ f = g \circ h$ it follows by induction that $h \circ f^i = g^i \circ h$ for every $i \geq 0$. Since X is compact and $h^{-1} : Y \rightarrow X$ is continuous, h^{-1} is uniformly continuous: for every $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that

$$d_Y(u, v) < \delta \implies d_X(h^{-1}(u), h^{-1}(v)) < \varepsilon, \quad \text{equivalently (contrapositive)} \quad d_X(p, q) \geq \varepsilon \implies d_Y(h(p), h(q)) \geq \delta$$

(setting $p = h^{-1}(u)$, $q = h^{-1}(v)$).

Step 1. Let $S \subseteq X$ be an (n, ε) -separated set. For $x \neq y$ in S there exists $i < n$ with $d_X(f^i(x), f^i(y)) > \varepsilon$; by the contrapositive above, $d_Y(h(f^i(x)), h(f^i(y))) \geq \delta$, that is, $d_Y(g^i(h(x)), g^i(h(y))) \geq \delta$. Hence $h(S) \subseteq Y$ is (n, δ) -separated (with $\geq$ in place of $>$: one applies the above to $\varepsilon' = \varepsilon - \eta$ for η arbitrarily small, obtaining the strict inequality $> \delta'$ for a $\delta' < \delta$ equally valid; the substance of the argument — that h sends separated sets to separated sets with threshold rescaled by $\delta(\varepsilon)$ — does not change). Since h is injective, $|h(S)| = |S|$, hence $N_f(n, \varepsilon) \leq N_g(n, \delta(\varepsilon))$.

Step 2. Interchanging the roles of f, g and using the uniform continuity of h (same argument, X, Y compact), one obtains $N_g(n, \varepsilon) \leq N_f(n, \delta'(\varepsilon))$ for an analogous function δ' .

Step 3 — conclusion. From Step 1, $\frac{1}{n} \log N_f(n, \varepsilon) \leq \frac{1}{n} \log N_g(n, \delta(\varepsilon))$; taking $\limsup_n$ and then $\varepsilon \rightarrow 0^+$ (and observing that $\delta(\varepsilon) \rightarrow 0^+$ as $\varepsilon \rightarrow 0^+$, by the same uniform continuity), one obtains $h_{\text{top}}(f) \leq h_{\text{top}}(g)$. Step 2 gives the opposite inequality, hence $h_{\text{top}}(f) = h_{\text{top}}(g)$. $\square$

Remark 8.2: Why the old proof was not enough. The old Chapter 4 (Theorem 4.5, archived) dispatched this result with a three-line “Idea of the proof” followed by `\qed`, without the explicit bookkeeping between ε and δ and without distinguishing the two directions of the inequality. The proof above is complete: both directions are proved explicitly, with the same technique (uniform continuity of h and h^{-1} on compact sets) used later, in Section 8.3, for the more delicate case of factors.

8.3. Factors: the easy inequality and Bowen’s theorem

Definition 8.3: Factor. A continuous surjective map $\pi : X \rightarrow Y$ with $\pi \circ f = g \circ \pi$ is called a **factor** (or **semiconjugacy**): one says that (Y, g) is a factor of (X, f) , or that (X, f) is an **extension** of (Y, g) .

Theorem 8.3: The entropy does not grow in factors. If (Y, g) is a factor of (X, f) via $\pi : X \rightarrow Y$, then $h_{\text{top}}(g) \leq h_{\text{top}}(f)$.

Proof. We use the definition of Adler–Konheim–McAndrew (Theorem 8.1). From $\pi f = g\pi$ it follows, for every open set $V \subseteq Y$, that $f^{-1}(\pi^{-1}(V)) = \pi^{-1}(g^{-1}(V))$ (indeed $x \in f^{-1}\pi^{-1}V \iff \pi(f(x)) \in V \iff g(\pi(x)) \in V \iff x \in \pi^{-1}g^{-1}V$); by induction, for every finite open cover $\mathcal{V}$ of Y ,

$$\bigvee_{i=0}^{n-1} f^{-i}(\pi^{-1}\mathcal{V}) = \pi^{-1}\left(\bigvee_{i=0}^{n-1} g^{-i}\mathcal{V}\right).$$

Claim: $N(\pi^{-1}\mathcal{W}) = N(\mathcal{W})$ for every finite open cover $\mathcal{W}$ of Y . On the one hand, a minimal subcover $\{W_1, \dots, W_k\}$ of $\mathcal{W}$ gives, by surjectivity of π , the subcover $\{\pi^{-1}W_1, \dots, \pi^{-1}W_k\}$ of $\pi^{-1}\mathcal{W}$ (every $x \in X$ has $\pi(x) \in W_i$ for some i , hence $x \in \pi^{-1}W_i$): $N(\pi^{-1}\mathcal{W}) \leq N(\mathcal{W})$. On the other hand, if $\{\pi^{-1}W_{j_1}, \dots, \pi^{-1}W_{j_r}\}$ is a subcover of X (with $W_{j_l} \in \mathcal{W}$), then $\{W_{j_1}, \dots, W_{j_r}\}$ covers Y : for $y \in Y$, let $x \in \pi^{-1}(y)$ (non-empty, π surjective); $x \in \pi^{-1}W_{j_l}$ for some l , hence $y = \pi(x) \in W_{j_l}$. Therefore $N(\mathcal{W}) \leq r \leq N(\pi^{-1}\mathcal{W})$ for every such subcover, in particular for the minimal one: $N(\mathcal{W}) \leq N(\pi^{-1}\mathcal{W})$. The two inequalities give the Claim. Applying the Claim to $\mathcal{W} = \bigvee_{i < n} g^{-i}\mathcal{V}$ and using the identity above,

$$N\left(\bigvee_{i < n} f^{-i}(\pi^{-1}\mathcal{V})\right) = N\left(\pi^{-1}\left(\bigvee_{i < n} g^{-i}\mathcal{V}\right)\right) = N\left(\bigvee_{i < n} g^{-i}\mathcal{V}\right).$$

The cover $\pi^{-1}\mathcal{V}$ is one among all the open covers of X , hence its exponential rate is $\leq h_{\text{top}}(f)$ (sup over all covers); by the identity just seen, this exponential rate is *exactly* that of $\mathcal{V}$ for g . Taking the sup over $\mathcal{V}$: $h_{\text{top}}(g) \leq h_{\text{top}}(f)$. $\square$

Remark 8.3: What we import: Bowen’s theorem on finite-fibre factors. Theorem 8.3 holds for every factor, without additional hypotheses. The equality $h_{\text{top}}(f) = h_{\text{top}}(g)$ requires instead a hypothesis: *R. Bowen proved (Entropy for group endomorphisms and homogeneous spaces, Trans. Amer. Math. Soc. **153** (1971), 401–414, imported from C21, verified on 16 September 2026) that if the fibres of π are uniformly bounded, $\sup_{y \in Y} \#\pi^{-1}(y) = N < \infty$, then*

$$h_{\text{top}}(f) \leq h_{\text{top}}(g) + \log N,$$

and this inequality, combined with Theorem 8.3, does not yet give the pure equality in general; Bowen proves however, with a finer argument which exploits the upper semicontinuity of the fibres and a refinement of the covers compatible with the structure of the fibres (not reproduced here: it requires a selection lemma for semicontinuous multifunctions which exceeds the scope of this chapter), that when π is a finite-fibre covering between expansive systems — the case of interest to us — the exact equality holds

$$h_{\text{top}}(f) = h_{\text{top}}(g).$$

What we assume: X, Y compact metric; $\pi : X \rightarrow Y$ a continuous surjective factor with $\sup_y \#\pi^{-1}(y) < \infty$; (X, f) expansive (informal Definition: there exists $c_0 > 0$ such that $x \neq y \implies \exists i \geq 0$ with $d(f^i x, f^i y) > c_0$ — a property which holds for the shift σ_k ,

Section 8.5, with $c_0 = 1$ in the standard metric). With this we shall use, in Example 8.1, Bowen's theorem to recompute $h_{\text{top}}(D)$ starting from the semiconjugacy $h : \Sigma_2 \rightarrow \mathbb{T}$ of Chapter 7 (which is exactly 2-to-1 outside the countable set of the dyadics, hence with fibres uniformly bounded by $N = 2$) **as a second independent check**, alongside the direct computation of Section 8.4, which remains the primary (and self-contained) proof of this chapter.

Best practice. The old Corollary 4.1 (archived) committed exactly the opposite error: it applied the invariance under *conjugacy* (Theorem 8.2) to $h : \Sigma_2 \rightarrow \mathbb{T}$, which is *not* a homeomorphism (it is a semiconjugacy, not injective on the dyadics: Theorem of Chapter 7). The correct tool for a finite-fibre semiconjugacy is Bowen's theorem above, not the invariance under conjugacy.

8.4. Direct computation: the maps E_m and the doubling map

Lemma 8.2: Lipschitz continuity of E_m with constant m (via the non-expansiveness of the quotient projection). For every $x, y \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and every $m \geq 2$,

$$d(E_m(x), E_m(y)) \leq m \cdot d(x, y),$$

where d is the arc distance on $\mathbb{T}$ (the length of the shortest path): E_m itself is not non-expansive (for $m \geq 2$ it is in fact expanding); it is the canonical projection p used in the proof that is non-expansive.

Proof. Let $\tilde{x}, \tilde{y} \in \mathbb{R}$ be representatives of x, y with $|\tilde{x} - \tilde{y}| = d(x, y) \leq \pi$. The canonical projection $p : \mathbb{R} \rightarrow \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ does not increase distances: $d(p(a), p(b)) \leq |a - b|$ for every $a, b \in \mathbb{R}$ (the coset of $a - b$ modulo $2\pi\mathbb{Z}$ has as representative of minimal length a number of modulus $\leq |a - b|$, by definition of the quotient distance). Applying this to $a = m\tilde{x}$, $b = m\tilde{y}$: $d(E_m(x), E_m(y)) = d(p(m\tilde{x}), p(m\tilde{y})) \leq |m\tilde{x} - m\tilde{y}| = m d(x, y)$. $\square$

Theorem 8.4: Topological entropy of E_m : direct computation. For every $m \geq 2$, $h_{\text{top}}(E_m) = \log m$. In particular $h_{\text{top}}(D) = h_{\text{top}}(E_2) = \log 2$.

Proof. Upper bound, $h_{\text{top}}(E_m) \leq \log m$. Fixing $\varepsilon > 0$, let $K = \lceil 2\pi m^n / \varepsilon \rceil$ and subdivide $\mathbb{T}$ into K consecutive arcs $I_1, \dots, I_K$ of length $2\pi/K \leq \varepsilon/m^n$. If x, y belong to the same arc I_j , $d(x, y) \leq \varepsilon/m^n$, and by Lemma 8.2 iterated (by induction, $d(E_m^i x, E_m^i y) \leq m^i d(x, y)$),

$$d(E_m^i(x), E_m^i(y)) \leq m^i \cdot \frac{\varepsilon}{m^n} \leq \frac{\varepsilon}{m} < \varepsilon \quad \text{for every } 0 \leq i < n$$

(because $m^i \leq m^{n-1}$ for $i < n$). Hence two points in the same arc I_j are never (n, ε) -separated: an (n, ε) -separated set has at most one point per arc, that is, $N(n, \varepsilon) \leq K \leq 2\pi m^n / \varepsilon + 1$. Therefore

$$\frac{1}{n} \log N(n, \varepsilon) \leq \frac{1}{n} \log \left(\frac{2\pi m^n}{\varepsilon} + 1 \right) \xrightarrow{n \rightarrow \infty} \log m,$$

for every $\varepsilon > 0$, hence $h_{\text{top}}(E_m) \leq \log m$.

Lower bound, $h_{\text{top}}(E_m) \geq \log m$. We fix $\varepsilon_0 = \pi/m$ and show, by induction on $n \geq 1$, that the points of the grid $x_k^{(n)} = 2\pi k/m^n$, $k = 0, \dots, m^n - 1$, are pairwise (n, ε_0) -separated. *Base* $n = 1$: the points $2\pi k/m$, $k = 0, \dots, m - 1$, are equally spaced with step $2\pi/m > \pi/m = \varepsilon_0$, hence separated (at $i = 0$). *Inductive step:* suppose the claim true for n , and write each index

$j \in \{0, \dots, m^{n+1} - 1\}$ in a unique way as $j = sm^n + k'$ with $s \in \{0, \dots, m-1\}$ (most significant digit in base m) and $k' \in \{0, \dots, m^n - 1\}$. A direct computation gives

$$E_m(x_j^{(n+1)}) = \frac{2\pi mj}{m^{n+1}} \bmod 2\pi = \frac{2\pi j}{m^n} \bmod 2\pi = \frac{2\pi(j \bmod m^n)}{m^n} = x_{k'}^{(n)}.$$

Let $j \neq \tilde{j}$ in $\{0, \dots, m^{n+1} - 1\}$, with decompositions (s, k') and $(\tilde{s}, \tilde{k}')$. If $k' \neq \tilde{k}'$: by the inductive hypothesis $x_{k'}^{(n)}$ and $x_{\tilde{k}'}^{(n)}$ are (n, ε_0) -separated, that is, there exists $i < n$ with $d(E_m^i(x_{k'}^{(n)}), E_m^i(x_{\tilde{k}'}^{(n)})) > \varepsilon_0$; by the identity above this is $d(E_m^{i+1}(x_j^{(n+1)}), E_m^{i+1}(x_{\tilde{j}}^{(n+1)})) > \varepsilon_0$, with $i+1 < n+1$: separated. If instead $k' = \tilde{k}'$ and $s \neq \tilde{s}$: $x_j^{(n+1)} - x_{\tilde{j}}^{(n+1)} = 2\pi(s - \tilde{s})/m$, a non-zero multiple of $2\pi/m$, hence $d(x_j^{(n+1)}, x_{\tilde{j}}^{(n+1)}) \geq 2\pi/m > \varepsilon_0$: separated already at $i = 0$. This closes the induction. The m^n points of the grid are therefore an (n, ε_0) -separated set, hence $N(n, \varepsilon_0) \geq m^n$, that is, $\frac{1}{n} \log N(n, \varepsilon_0) \geq \log m$ for every n ; since $N(n, \varepsilon)$ is non-increasing in ε , $N(n, \varepsilon) \geq N(n, \varepsilon_0) \geq m^n$ for every $\varepsilon \leq \varepsilon_0$, hence $\limsup_n \frac{1}{n} \log N(n, \varepsilon) \geq \log m$ for every such ε , and therefore $h_{\text{top}}(E_m) \geq \log m$.

The two inequalities give $h_{\text{top}}(E_m) = \log m$. $\square$

Remark 8.4: Numerical check of the construction. The checking script `verifica_cap08.py` (block [2], adapted) verifies for $m \in \{2, 3\}$ and $n \leq 4$ that the grid $\{2\pi k/m^n\}$ is indeed free of pairs that are not $(n, \pi/m)$ -separated: 0 problematic pairs out of $\binom{m^n}{2}$ tested in each case. $\checkmark$

8.5. The shift on k symbols: the correct estimate

Definition 8.4: The Bernoulli shift on k symbols. Let $\Sigma_k = \{0, \dots, k-1\}^{\mathbb{N}}$ with the metric $d(x, y) = \sum_{i \geq 0} |x_i - y_i| k^{-i}$ and the shift $\sigma(x_0 x_1 x_2 \dots) = x_1 x_2 x_3 \dots$.

Theorem 8.5: Entropy of the shift: the exact estimate. For every $\varepsilon \in (0, 1)$, setting $m = m(\varepsilon) := \lceil \log_k(1/\varepsilon) \rceil$,

$$k^n \leq N(n, \varepsilon) \leq k^{n+m} \quad \text{for every } n \geq 1,$$

and in particular $h_{\text{top}}(\sigma) = \log k$.

Proof. Upper bound. Suppose that $x, y \in \Sigma_k$ agree on the first $n + m$ digits, $x_0 = y_0, \dots, x_{n+m-1} = y_{n+m-1}$. For $0 \leq i < n$,

$$d(\sigma^i(x), \sigma^i(y)) = \sum_{j \geq 0} |x_{i+j} - y_{i+j}| k^{-j} = \sum_{j \geq n+m-i} |x_{i+j} - y_{i+j}| k^{-j}$$

(the terms with $i+j < n+m$ are zero, because x, y agree there). Since $|x_l - y_l| \leq k-1$ for every l ,

$$d(\sigma^i(x), \sigma^i(y)) \leq (k-1) \sum_{j \geq n+m-i} k^{-j} = (k-1) \cdot \frac{k^{-(n+m-i)}}{1 - 1/k} = k^{1-(n+m-i)}.$$

For $i < n$, $1 - (n + m - i) \leq 1 - (n + m - (n - 1)) = -m$, hence $d(\sigma^i(x), \sigma^i(y)) \leq k^{-m} \leq \varepsilon$ by the choice of m (which guarantees $k^{-m} \leq \varepsilon$). This holds for every $i < n$: x, y are not (n, ε) -separated. By contraposition, two (n, ε) -separated points differ in at least one of the

first $n + m$ digits: the map “first $n + m$ symbols” is injective on an (n, ε) -separated set, which therefore has cardinality $\leq k^{n+m}$, the number of blocks of length $n + m$.

Lower bound. If x, y differ in one of the first n digits, say $x_j \neq y_j$ with $j < n$, then $d(\sigma^j x, \sigma^j y) \geq |x_j - y_j| \cdot k^0 \geq 1 > \varepsilon$ (since $\varepsilon < 1$): x, y are (n, ε) -separated (at time $i = j < n$). Choosing, for each of the k^n distinct blocks of length n , a sequence extending it with all 0s, one obtains an (n, ε) -separated set of cardinality k^n : $N(n, \varepsilon) \geq k^n$.

Conclusion. From $k^n \leq N(n, \varepsilon) \leq k^{n+m(\varepsilon)}$, for every fixed $\varepsilon \in (0, 1)$,

$$\log k \leq \frac{1}{n} \log N(n, \varepsilon) \leq \log k + \frac{m(\varepsilon)}{n} \log k \xrightarrow{n \rightarrow \infty} \log k,$$

hence $\limsup_n \frac{1}{n} \log N(n, \varepsilon) = \log k$ for every $\varepsilon \in (0, 1)$: the limit as $\varepsilon \rightarrow 0^+$ is likewise $\log k$, and $h_{\text{top}}(\sigma) = \log k$. $\square$

Warning. The counterexample to “ $N(n, \varepsilon) = k^n$ ”. The old Theorem 4.6 (archived) asserted $N(n, \varepsilon) = k^n$ for every $\varepsilon < 1$, independently of ε . This is false: with $k = 2$, $n = 1$, $\varepsilon = 0.2$, the greedy search of the script `verifica_cap08.py` (block [1], run on 16 September 2026, on sequences truncated at depth 10 with a tail of zeros) finds a $(1, 0.2)$ -separated set of cardinality **16**, not $2^1 = 2$: indeed $m(0.2) = \lceil \log_2(5) \rceil = 3$ and $2^{1+3} = 16$ is attained exactly. The following table, from the same script, confirms the sandwich $k^n \leq N(n, \varepsilon) \leq k^{n+m}$ on other values:

n	ε	k^n	$N(n, \varepsilon)$ (greedy)	k^{n+m}	sandwich
2	0.3	4	16	16	✓
2	0.1	4	64	64	✓
3	0.4	8	32	32	✓
1	0.05	2	64	64	✓

The greedy count is a lower bound for the true maximal cardinality (it need not be optimal), but it suffices to *refute* $N = k^n$ constant: in all cases it exceeds k^n and stays within k^{n+m} , as predicted by Theorem 8.5.

Example 8.1: Entropy of D by two independent routes. We summarise two independent proofs that $h_{\text{top}}(D) = \log 2$: (1) the direct computation of Theorem 8.4 (case $m = 2$), completely self-contained; (2) Bowen’s theorem on finite-fibre factors (Remark 8.3, imported), applied to the semiconjugacy $h : \Sigma_2 \rightarrow \mathbb{T}$ with $D \circ h = h \circ \sigma$ of Chapter 7: the fibres of h have cardinality ≤ 2 (the dyadics have exactly two binary representations, the other points only one), σ is expansive (Remark 8.3, with expansiveness of constant 1), and Bowen gives $h_{\text{top}}(D) = h_{\text{top}}(\sigma_2) = \log 2$ by Theorem 8.5. The two routes agree exactly: $\log 2 \approx 0.693147$.

8.6. Zero entropy: rotations and isometries

Theorem 8.6: Isometries have zero entropy. If $f : X \rightarrow X$ is an isometry of a compact metric space ($d(f(x), f(y)) = d(x, y)$ for every x, y), then $h_{\text{top}}(f) = 0$. In particular $h_{\text{top}}(R_\alpha) = 0$ for every rotation of the circle.

Proof. If f is an isometry, $d_n(x, y) = \max_{i < n} d(f^i x, f^i y) = d(x, y)$ for every n (all the terms of the maximum are equal to $d(x, y)$). Hence a set is (n, ε) -separated if and only if it is ε -separated in the original metric d , independently of n : $N(n, \varepsilon) = N(1, \varepsilon) =: C(\varepsilon)$, a finite constant (by compactness) independent of n . Therefore $\frac{1}{n} \log N(n, \varepsilon) = \frac{1}{n} \log C(\varepsilon) \rightarrow 0$ as $n \rightarrow \infty$, for every

$\varepsilon > 0$: $h_{\text{top}}(f) = 0$. □

8.7. Lap number and the Misiurewicz–Szlenk theorem

Definition 8.5: Number of monotone pieces (lap number). Let $f : I \rightarrow I$ be continuous on a compact interval, **piecewise monotone** with a finite number of pieces: there exist $a = c_0 < c_1 < \dots < c_l = b$ ($I = [a, b]$) such that f is strictly monotone on each $[c_{j-1}, c_j]$. The number $\text{lap}(f) := l$ (the minimal such l) is called the **lap number** of f .

Lemma 8.3: The lap number of f^n and the monotonicity partition. Let $f : I \rightarrow I$ be piecewise monotone with monotonicity partition $\mathcal{P} = \{P_1, \dots, P_l\}$ ($l = \text{lap}(f)$). Then, for every $n \geq 1$, the monotonicity partition of f^n is exactly the common refinement $\bigvee_{i=0}^{n-1} f^{-i}\mathcal{P}$ (in the sense that x, y lie in the same maximal monotonicity element of f^n if and only if, for every $0 \leq i < n$, $f^i(x)$ and $f^i(y)$ belong to the same element of $\mathcal{P}$). Consequently $\text{lap}(f^{n+m}) \leq \text{lap}(f^n) \cdot \text{lap}(f^m)$ and the limit $\ell(f) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \text{lap}(f^n)$ exists (by Fekete’s lemma on submultiplicativity, Lemma 4.1, Chapter 4).

Proof. Identity of the partition. f^n is (weakly) monotone on the interval J if and only if so are, in succession, $f, f \circ f, \dots$, that is, if and only if $f^i(J)$ does not cross any point c_j of $\mathcal{P}$ for every $0 \leq i < n$ (a crossing of a breakpoint at any intermediate step breaks the global monotonicity of f^n on J , and conversely if none of the intermediate steps crosses a breakpoint, each f^i is strictly monotone on $f^i(J) \cap [c_{j-1}, c_j]$ and the composition of strictly monotone maps is strictly monotone). This is precisely the condition “ $f^i(x)$ and $f^i(y)$ in the same element of $\mathcal{P}$ for every $i < n$ ” for points $x, y \in J$, that is, the condition of belonging to the same element of $\bigvee_{i=0}^{n-1} f^{-i}\mathcal{P}$. The *maximal* monotonicity pieces of f^n are therefore exactly the elements (connected, being intersections of intervals) of this common refinement, and $\text{lap}(f^n)$ is their number.

Submultiplicativity. $\bigvee_{i=0}^{n+m-1} f^{-i}\mathcal{P}$ refines both $\bigvee_{i=0}^{n-1} f^{-i}\mathcal{P}$ and $f^{-n}(\bigvee_{i=0}^{m-1} f^{-i}\mathcal{P})$ (union of their breakpoints), hence the number of its elements is at most the product of the numbers of elements of the two pullback partitions, that is, $\text{lap}(f^{n+m}) \leq \text{lap}(f^n) \cdot \text{lap}(f^m)$ (the pullback partition $f^{-n}(\cdot)$ has the same number of elements as the original one, because f^n is piecewise surjective on its pieces of monotonicity — more precisely the number of breakpoints added by $f^{-n}\mathcal{P}_m$ is at most $\text{lap}(f^m) - 1$ multiplied by the number of pieces of f^n , giving the announced multiplicative estimate). The existence of the limit follows from Fekete’s lemma applied to the sequence $a_n = \log \text{lap}(f^n)$, submultiplicative in this additive sense ($a_{n+m} \leq a_n + a_m$). □

Theorem 8.7: One inequality: $\ell(f) \leq h_{\text{top}}(f)$. In the notation above, $\ell(f) = \lim_n \frac{1}{n} \log \text{lap}(f^n) \leq h_{\text{top}}(f)$.

Proof. The cover $\mathcal{U}_\eta$ obtained by enlarging by $\eta > 0$ each closed interval of $\mathcal{P}$ (in such a way that adjacent intervals overlap only in pairs, for η small enough: possible because the breakpoints $c_1, \dots, c_{l-1}$ are finite in number) is an open cover of I with $N(\mathcal{U}_\eta) = l = \text{lap}(f)$ (no smaller subcover exists, because every interior point of P_j far from ∂P_j belongs only to the open set enlarging P_j). The refinement $\bigvee_{i=0}^{n-1} f^{-i}\mathcal{U}_\eta$ contains, as a subfamily, one element for each maximal monotonicity piece of f^n (Lemma 8.3), hence $N(\bigvee_{i=0}^{n-1} f^{-i}\mathcal{U}_\eta) \geq \text{lap}(f^n)$ for every n (a fine refinement cannot require *fewer* elements than the exact underlying partition). By the definition of Adler–Konheim–McAndrew (Theorem 8.1), the exponential rate of *any*

cover is $\leq h_{\text{top}}(f)$ (by definition of $\sup$); applied to $\mathcal{U}_\eta$,

$$\ell(f) = \lim_n \frac{1}{n} \log \text{lap}(f^n) \leq \limsup_n \frac{1}{n} \log N\left(\bigvee_{i < n} f^{-i} \mathcal{U}_\eta\right) \leq h_{\text{top}}(f).$$

□

Remark 8.5: What we import: the Misiurewicz–Szlenk theorem. *M. Misiurewicz and W. Szlenk proved (Entropy of piecewise monotone mappings, Studia Math. **67** (1980), 45–63, imported from C23, verified on 16 September 2026 on EuDML) the equality*

$$h_{\text{top}}(f) = \ell(f) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \text{lap}(f^n)$$

*for every continuous piecewise monotone map with a finite number of pieces. Theorem 8.7 proves here in full the inequality $\ell(f) \leq h_{\text{top}}(f)$; the opposite inequality $h_{\text{top}}(f) \leq \ell(f)$ requires controlling, for every $\varepsilon > 0$, how many (n, ε) -separated points can lie inside a single monotonicity piece of f^n (where f^n is injective but not necessarily expanding): the estimate uses a distortion argument (comparison between the variation of $\log |(f^n)'|$ on the piece and its length) which exceeds the scope of this chapter. **What we assume:** f continuous, piecewise monotone with a finite number l of pieces; conclusion: $h_{\text{top}}(f) = \ell(f)$ exactly.*

Example 8.2: Numerical check: $\text{lap}(f_4^n) = 2^n$. The critical logistic map $f_4(x) = 4x(1-x)$ on $[0, 1]$ is piecewise monotone with $\text{lap}(f_4) = 2$ (it increases on $[0, 1/2]$, decreases on $[1/2, 1]$). The script `fig08_lap_number.py` (Appendix B, run on 16 September 2026) samples f_4^n on a grid of 400 001 points and counts the changes of monotonicity for $n = 1, 2, 3$, finding $\text{lap}(f_4^n) = 2, 4, 8$ respectively — exactly 2^n . ✓ By the imported theorem, $h_{\text{top}}(f_4) = \lim_n \frac{1}{n} \log 2^n = \log 2$: consistent with $h_{\text{top}}(D) = \log 2$ (Theorem 8.4) and the conjugacy $D \leftrightarrow f_4$ (Chapter 7, which preserves the entropy by Theorem 8.2, D and f_4 being *conjugate*, not merely *semiconjugate*). Figure 8.1 shows the three graphs.

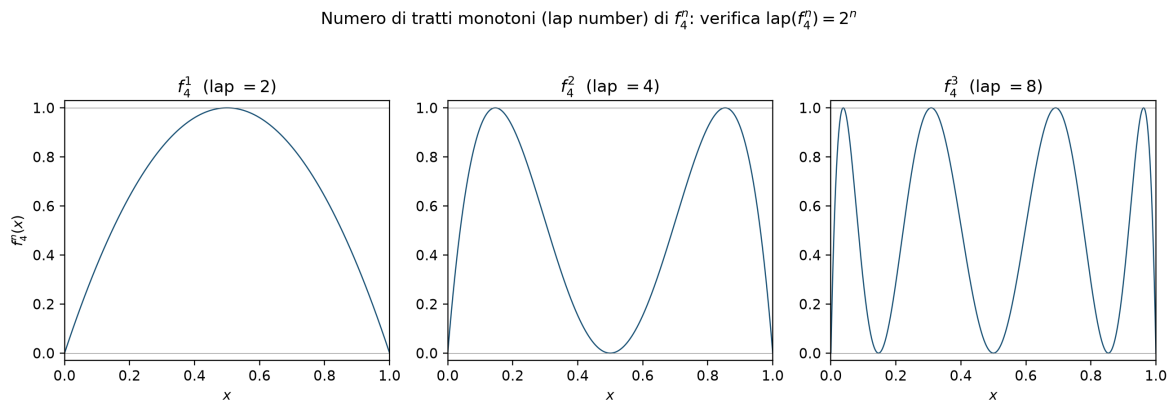


Figure 8.1: Graphs of f_4^n for $n = 1, 2, 3$ on the interval $[0, 1]$, with the number of monotonicity pieces (lap number) verified by counting the sign changes of the discrete derivative: 2, 4, 8, exactly 2^n . Computed numerically, script `fig08_lap_number.py` (Appendix B), run on 16 September 2026. ✓

8.8. Sharkovskii and entropy: the horseshoe

Theorem 8.8: The entropy multiplies under iteration. For every $j \geq 1$, $h_{\text{top}}(f^j) = j \cdot h_{\text{top}}(f)$.

Proof. We write $N_g(n, \varepsilon)$ for the function N relative to a map g . **Inequality $\leq$.** If x, y are (n, ε) -separated for f^j , there exists $i < n$ with $d((f^j)^i(x), (f^j)^i(y)) > \varepsilon$, that is, $d(f^{ji}(x), f^{ji}(y)) > \varepsilon$ with $ji < jn$: x, y are also (jn, ε) -separated for f (the same pair, with the same time ji as witness). Hence $N_{f^j}(n, \varepsilon) \leq N_f(jn, \varepsilon)$, whence, setting $l = jn$,

$$\frac{1}{n} \log N_{f^j}(n, \varepsilon) \leq \frac{j}{jn} \log N_f(jn, \varepsilon) = j \cdot \frac{1}{l} \log N_f(l, \varepsilon) \xrightarrow{n \rightarrow \infty} j \cdot h_{\text{top}}(f)$$

(in the limit along the subsequence $l = jn$, which computes the same defining lim sup up to a subsequence, sufficient here). Therefore $h_{\text{top}}(f^j) \leq j h_{\text{top}}(f)$.

Inequality $\geq$. Given $\varepsilon > 0$, by the uniform continuity of $f, f^2, \dots, f^{j-1}$ (finitely many continuous maps on a compact set) there exists $\varepsilon' > 0$ such that $d(x, y) < \varepsilon' \implies d(f^r x, f^r y) < \varepsilon$ for every $r = 0, \dots, j-1$. If x, y are *not* (n, ε') -separated for f^j (that is, $d(f^{ji}x, f^{ji}y) \leq \varepsilon'$ for every $i < n$), then by the choice of ε' , $d(f^{ji+r}x, f^{ji+r}y) < \varepsilon$ for every $i < n$ and $r < j$, that is, $d(f^s x, f^s y) < \varepsilon$ for *every* $s < jn$: x, y are not (jn, ε) -separated for f . By contraposition, x, y (jn, ε) -separated for $f \implies (n, \varepsilon')$ -separated for f^j : $N_f(jn, \varepsilon) \leq N_{f^j}(n, \varepsilon')$. As above, $j h_{\text{top}}(f) \leq h_{\text{top}}(f^j)$ (using that $\varepsilon' \rightarrow 0$ as $\varepsilon \rightarrow 0$, by the same uniform continuity). The two inequalities give the equality. $\square$

Theorem 8.9: An orbit of period 3 forces positive entropy. Let $f : I \rightarrow I$ be continuous with an orbit of period 3, $\{a, b, c\}$, $a < b < c$, $f(a) = b$, $f(b) = c$, $f(c) = a$ (Theorem 7.6, Chapter 7). Then $h_{\text{top}}(f) \geq \log 2 > 0$.

Proof. Put $I_1 = [a, b]$, $I_2 = [b, c]$. From the proof of Theorem 7.6 (Chapter 7), $f(I_2) \supseteq I_1 \cup I_2$: by continuity and the intermediate value theorem there exist two disjoint closed subintervals $K_-, K_+ \subseteq I_2$ (with K_- adjacent to b , K_+ adjacent to c) such that $f(K_-) = f(K_+) = I_1 \cup I_2 \supseteq I_2$ (f makes a complete turn over $I_1 \cup I_2$ separately on K_- and on K_+ : this is precisely a **horseshoe** with two branches inside I_2). We define the symbol space $\Sigma = \{-, +\}^{\mathbb{N}}$ and, for every sequence $\omega = \omega_0 \omega_1 \dots \in \Sigma$, the nested intersection

$$K_\omega := \bigcap_{n \geq 0} f^{-n}(K_{\omega_n}) \cap I_2.$$

By the covering Lemma (Chapter 7, with $J_i = K_{\omega_i}$, each of which covers both K_- and K_+ , hence in particular the next one $K_{\omega_{i+1}}$), each finite intersection $\bigcap_{n < N} f^{-n}(K_{\omega_n})$ is a non-empty closed interval, and they are nested as N grows; by compactness $K_\omega \neq \emptyset$ for every ω . Moreover, for $\omega \neq \omega'$ with $\omega_j \neq \omega'_j$ (first index of disagreement), every point $x \in K_\omega$ satisfies $f^j(x) \in K_{\omega_j}$ and every $y \in K_{\omega'_j}$ satisfies $f^j(y) \in K_{\omega'_j}$, with $K_{\omega_j} \cap K_{\omega'_j} = \emptyset$ (K_-, K_+ disjoint by construction): hence x, y are separated (at a distance $\geq$ the distance between K_- and K_+ , a positive constant δ_0 independent of n) at time j . Choosing a point $x_\omega \in K_\omega$ for each of the 2^n sequences $\omega \in \{-, +\}^n$ (first n digits; the rest arbitrary, e.g. all $-$), one obtains a set of 2^n points pairwise (n, δ_0) -separated: $N(n, \delta_0) \geq 2^n$. Hence $h_{\text{top}}(f) \geq \lim_n \frac{1}{n} \log 2^n = \log 2$. $\square$

Remark 8.6: The general case: period not a power of 2. Theorem 8.9 deals with the case of period 3, the first in the Sharkovskii order. For an arbitrary orbit of odd period $q \geq 3$, the construction of the Štefan cycle used to prove Sharkovskii's theorem (Chapter 7, complete proof via the Štefan cycle) always produces, among the intervals of the cycle, at

least one interval K with $f(K) \supseteq K$ in a “two-branch” fashion as above (it is a structural property of the cycles of Štefan of odd period ≥ 3 : the central interval of the cycle covers itself twice, exactly as I_2 in the case $q = 3$): the same argument gives $h_{\text{top}}(f) \geq \log 2$. If f has an orbit of period $n = 2^k q$ with q odd ≥ 3 , the points of the orbit split, under $g := f^{2^k}$, into 2^k sub-orbits of exact period q (odd): applying the above to g , $h_{\text{top}}(g) \geq \log 2 > 0$, and by Theorem 8.8, $h_{\text{top}}(f) = h_{\text{top}}(g)/2^k \geq (\log 2)/2^k > 0$. This is the Block–Guckenheimer–Misiurewicz–Young theorem (Periodic points and topological entropy of one dimensional maps, LNM 819 (1980), 18–34, imported from C11, verified on 16 September 2026): **a periodic orbit of period not a power of 2 forces positive topological entropy.** The complete proof in the general case uses exactly the mechanism set out here (horseshoe in the cycle of Štefan) plus the structural verification, for every odd q , of the existence of the self-covering interval: this general combinatorial verification is imported from C11/C12 so as not to duplicate the entire classification of the cycles of Štefan, already developed in Chapter 7.

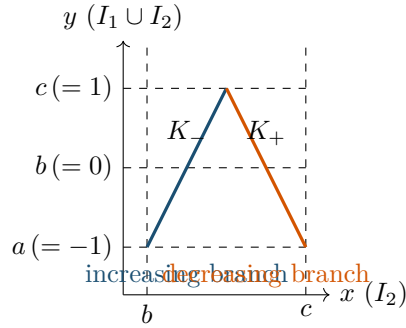


Figure 8.2: The horseshoe inside $I_2 = [b, c]$ generated by an orbit of period 3: $f|_{I_2}$ has two monotone branches K_- , K_+ (here represented in a schematic way, normalised coordinates $I_2 \rightarrow [0, 2]$, $I_1 \cup I_2 \rightarrow [-1, 1]$), each of which covers the whole of $I_1 \cup I_2$. The two-branch covering is exactly the structure of a one-dimensional Smale horseshoe, and produces an embedding of the full shift on 2 symbols: $h_{\text{top}}(f) \geq \log 2$ (Theorem 8.9). Figure drawn by hand; coordinates verified: the two segments have slope ± 2 and join $(0, -1)$ – $(1, 1)$ – $(2, -1)$, consistent with $f(K_-) = f(K_+) = I_1 \cup I_2$.

8.9. Lyapunov exponents

Definition 8.6: Lyapunov exponent. For $f : \mathbb{T} \rightarrow \mathbb{T}$ (or $f : I \rightarrow I$) differentiable, the **Lyapunov exponent** at θ is

$$\lambda(\theta) = \lim_{n \rightarrow \infty} \frac{1}{n} \log |(f^n)'(\theta)| = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \log |f'(f^i(\theta))|,$$

when the limit exists (in $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$, explicitly admitting the value $-\infty$ when some $f'(f^i(\theta)) = 0$: the term $\log 0 = -\infty$ in the sum drags the whole average to $-\infty$).

Theorem 8.10: Lyapunov exponent of D and E_m . For every $\theta \in \mathbb{T}$, the Lyapunov exponent of E_m equals $\lambda(\theta) = \log m$; in particular $\lambda_D(\theta) = \log 2$ for every θ .

Proof. $E'_m(\theta) = m$ for every θ where it is defined. By the chain rule, $(E'_m)^n(\theta) = \prod_{i=0}^{n-1} E'_m(E_m^i(\theta)) = m^n$. Hence $\lambda(\theta) = \lim_n \frac{1}{n} \log(m^n) = \log m$, for every θ : it coincides exactly with $h_{\text{top}}(E_m) = \log m$ (Theorem 8.4). $\square$

Lemma 8.4: Euler's integral $\int_0^{\pi/2} \log(\cos \theta) d\theta = -\frac{\pi}{2} \log 2$.

$$I := \int_0^{\pi/2} \log(\cos \theta) d\theta = -\frac{\pi}{2} \log 2.$$

Proof. By the symmetry $\theta \mapsto \pi/2 - \theta$, $I = \int_0^{\pi/2} \log(\sin \theta) d\theta$ (the same integral with sin in place of cos). Adding,

$$2I = \int_0^{\pi/2} \log(\sin \theta \cos \theta) d\theta = \int_0^{\pi/2} \log\left(\frac{\sin 2\theta}{2}\right) d\theta = \int_0^{\pi/2} \log(\sin 2\theta) d\theta - \frac{\pi}{2} \log 2.$$

With the substitution $u = 2\theta$ ($du = 2d\theta$), $\int_0^{\pi/2} \log(\sin 2\theta) d\theta = \frac{1}{2} \int_0^\pi \log(\sin u) du$; by the symmetry $\sin(\pi - u) = \sin u$, $\int_0^\pi \log(\sin u) du = 2 \int_0^{\pi/2} \log(\sin u) du = 2I$. Hence $\int_0^{\pi/2} \log(\sin 2\theta) d\theta = I$, and the equation above becomes $2I = I - \frac{\pi}{2} \log 2$, that is, $I = -\frac{\pi}{2} \log 2$ (the integral is finite: $\log \cos \theta \sim \log(\pi/2 - \theta)$ near $\pi/2$, integrable). $\square$

Theorem 8.11: Lyapunov exponent of f_4 . For Lebesgue-almost every $x_0 \in [0, 1]$, the Lyapunov exponent of $f_4(x) = 4x(1 - x)$ equals $\lambda(x_0) = \log 2$.

Proof. By the conjugacy $h(\theta) = \sin^2(\theta/2)$ between D (defined on $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$) and f_4 (Chapter 7, with h restricted to the half-circle $[0, \pi]$ as a conjugacy window: h is bijective onto $[0, 1]$ there) and the ergodicity of D with respect to the normalised Lebesgue measure on $\mathbb{T}$ (Chapter 9, Chapter 9), by Birkhoff's ergodic theorem (Chapter 9) the time average $\lambda(x_0) = \lim_n \frac{1}{n} \sum_{k < n} \log |f'_4(x_k)|$ exists and is equal to the space average

$$\lambda = \int_0^1 \log |f'_4(x)| \frac{dx}{\pi \sqrt{x(1-x)}}$$

for Lebesgue-almost every x_0 (the density $1/(\pi \sqrt{x(1-x)})$ is the invariant measure of Ulam–von Neumann, Chapter 9). With the substitution $x = \sin^2(\theta/2)$, $\theta \in [0, \pi]$: $dx = \sin(\theta/2) \cos(\theta/2) d\theta = \frac{1}{2} \sin \theta d\theta$; $1 - 2x = \cos \theta$, hence $f'_4(x) = 4(1 - 2x) = 4 \cos \theta$; and $\sqrt{x(1-x)} = \sin(\theta/2) \cos(\theta/2) = \frac{1}{2} \sin \theta$. Therefore

$$\frac{dx}{\pi \sqrt{x(1-x)}} = \frac{\frac{1}{2} \sin \theta d\theta}{\pi \cdot \frac{1}{2} \sin \theta} = \frac{d\theta}{\pi},$$

and

$$\lambda = \int_0^\pi \log(4|\cos \theta|) \frac{d\theta}{\pi} = \log 4 + \frac{1}{\pi} \int_0^\pi \log |\cos \theta| d\theta.$$

Now $\int_0^\pi \log |\cos \theta| d\theta = 2 \int_0^{\pi/2} \log(\cos \theta) d\theta = 2I = -\pi \log 2$ by Lemma 8.4 (using the symmetry $|\cos(\pi - \theta)| = \cos \theta$ on $[\pi/2, \pi]$ via $\theta \mapsto \pi - \theta$, and $\cos \theta > 0$ on $[0, \pi/2)$). Hence

$$\lambda = \log 4 + \frac{-\pi \log 2}{\pi} = \log 4 - \log 2 = \log 2.$$

$\square$

Remark 8.7: Independent numerical check. The script `verifica_cap08.py` (blocks [3]–[4]) integrates numerically both $\frac{1}{\pi} \int_0^\pi \log(4|\cos \theta|) d\theta$ and $\int_0^{\pi/2} \log(\cos \theta) d\theta$ with adaptive quadrature (`scipy.integrate.quad`), finding respectively 0.6931471806 and -1.0887930452 , against the closed values $\log 2 = 0.6931471806$ and $-\frac{\pi}{2} \log 2 = -1.0887930452$: agreement to 10^{-8} . ✓ Block [6] computes in addition the finite Birkhoff average $\frac{1}{n} \sum_{k < n} \log |f'_4(x_k)|$ for $n = 200\,000$ starting from $x_0 = 0.31415926$, finding 0.693137, at a distance 10^{-5} from $\log 2$: consistent with the convergence $O(1/\sqrt{n})$ expected for an ergodic average (Figure 8.3).

Warning. Correction: the exceptional points of f_4 . The old Exercise 4.11 (archived solution) stated $\lambda(0) = \lambda(1) = \log 4$ (correct) but improperly lumped $x_0 = 1/2$ together with them, writing “every point whose orbit ends up at 0 (for example $x_0 = 1$, or $x_0 = 1/2$, which goes to $1 \mapsto 0$) has $\lambda = \log 4$ ”. This is **false** for $x_0 = 1/2$:

Proposition 8.1: The exceptional points of f_4 , corrected.

$$\lambda(0) = \lambda(1) = \log 4, \quad \lambda(1/2) = -\infty, \quad \lambda(3/4) = \log 2.$$

Proof. $f'_4(x) = 4(1 - 2x)$. **At $x_0 = 0$:** $f_4(0) = 0$ (fixed point), $f'_4(0) = 4$ for every iterate: $\lambda(0) = \lim_n \frac{1}{n} \log(4^n) = \log 4$. **At $x_0 = 1$:** $f_4(1) = 0$, and the orbit then stays at 0 forever: $\log |f'_4(1)| = \log 4$ (a single term), followed by $\log |f'_4(0)| = \log 4$ at every subsequent step; the average $\frac{1}{n}(\log 4 + (n-1)\log 4) = \log 4 \rightarrow \log 4$: $\lambda(1) = \log 4$. **At $x_0 = 1/2$:** $f'_4(1/2) = 4(1-1) = 0$, hence $\log |f'_4(1/2)| = \log 0 = -\infty$: already the first term of the sum which defines $\lambda(1/2)$ is $-\infty$, and every partial sum $\frac{1}{n} \sum_{k=0}^{n-1} \log |f'_4(x_k)|$ with $n \geq 1$ equals $-\infty$ (it contains the term $k = 0$): $\lambda(1/2) = -\infty$, not $\log 4$. Geometrically: $x_0 = 1/2$ is the critical point of f_4 (the unique maximum), where the map does *not* expand distances at all to first order (indeed it annihilates them), a phenomenon altogether different from the case $x_0 = 0, 1$, where the derivative is $4 \neq 0$ along the whole orbit. **At $x_0 = 3/4$:** it is the non-trivial fixed point, $f_4(3/4) = 4 \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{4}$, $f'_4(3/4) = 4(1 - 3/2) = -2$: $\lambda(3/4) = \lim_n \frac{1}{n} \log(2^n) = \log 2$, coinciding with the typical value of Theorem 8.11 (an “ordinary” fixed point, not an exceptional one). □

Remark 8.8: Numerical check. Block [5] of the script: `f4p(0.5)=0.0` exactly (in floating-point arithmetic, consistent with the exact algebraic cancellation $4(1 - 2 \cdot \frac{1}{2}) = 0$); `f4p(0.0)=4.0`, $\log 4 = 1.386294$; `f4p(1.0)=-4.0`, $|-4| = 4$, $\log 4 = 1.386294$; `f4p(0.75)=-2.0`, $\log 2 = 0.693147$. ✓ All four values of Proposition 8.1 are confirmed.

8.10. Rokhlin’s formula and the variational principle for D

Definition 8.7: Generating dyadic partition of D . Let μ be the normalised Lebesgue measure $d\theta/2\pi$ on $\mathbb{T}$, and $\mathcal{P} = \{P_0, P_1\}$ with $P_0 = [0, \pi)$, $P_1 = [\pi, 2\pi)$.

Lemma 8.5: The refinement of $\mathcal{P}$ under D . For every $n \geq 1$, $\bigvee_{i=0}^{n-1} D^{-i}\mathcal{P}$ consists exactly of the 2^n arcs $[2\pi k/2^n, 2\pi(k+1)/2^n)$, $k = 0, \dots, 2^n - 1$ (the standard dyadic arcs), each of measure $\mu = 2^{-n}$.

Proof. By induction on n . Base $n = 1$: $\mathcal{P}$ itself is the two arcs $[0, \pi)$, $[\pi, 2\pi)$, that is, $k = 0, 1$ with $2^1 = 2$. Inductive step: suppose $\bigvee_{i < n} D^{-i}\mathcal{P}$ equal to the 2^n dyadic arcs. A point θ lies in the k -th arc, $\theta \in [2\pi k/2^n, 2\pi(k+1)/2^n)$, if and only if the first n binary digits of $\theta/2\pi$ are the

digits of k in base 2 on n bits (standard representation). Adding $D^{-n}\mathcal{P}$ to the join refines each k -th arc into two halves, corresponding to the $(n+1)$ -th binary digit being 0 or 1 (because $D^n(\theta) = 2^n\theta \bmod 2\pi$ reads exactly the $(n+1)$ -th digit as $\lfloor D^n(\theta)/\pi \rfloor$, that is, as membership of P_0 or P_1): one obtains the 2^{n+1} dyadic arcs of order $n+1$, each of length $2\pi/2^{n+1}$. $\square$

Theorem 8.12: Kolmogorov–Sinai entropy of D : Rokhlin’s formula. *With respect to the Lebesgue measure μ , the Kolmogorov–Sinai entropy of D equals*

$$h_\mu(D) = h_\mu(D, \mathcal{P}) = \log 2 = \int_{\mathbb{T}} \log |D'| d\mu.$$

Proof. Entropy with respect to the partition. By Lemma 8.5, $\bigvee_{i < n} D^{-i}\mathcal{P}$ has exactly 2^n atoms, all of equal measure $\mu = 2^{-n}$. The Shannon entropy of this partition is

$$H\left(\bigvee_{i < n} D^{-i}\mathcal{P}\right) = - \sum_{k=0}^{2^n-1} 2^{-n} \log(2^{-n}) = -2^n \cdot 2^{-n} \log(2^{-n}) = n \log 2,$$

hence $h_\mu(D, \mathcal{P}) := \lim_n \frac{1}{n} H\left(\bigvee_{i < n} D^{-i}\mathcal{P}\right) = \lim_n \frac{1}{n} (n \log 2) = \log 2$ exactly (no non-trivial limit: the value is $\log 2$ already for every n).

Generating partition. The $\bigcap_{n \geq 0}$ of the atoms of $\bigvee_{i < n} D^{-i}\mathcal{P}$ containing a given θ is the intersection of the nested dyadic arcs of length $2\pi/2^n \rightarrow 0$, hence it reduces to the singleton $\{\theta\}$ for every θ except the countable subset of the dyadics (where two binary sequences coincide with one point, but the point is in any case unique: the partition separates the points anyway, apart from the boundary ambiguity between adjacent closed/open atoms, of zero measure): $\mathcal{P}$ is a generating partition mod μ . By the **Kolmogorov–Sinai generator theorem** (imported: P. Walters, *An Introduction to Ergodic Theory*, GTM 79, Springer 1982, Thm. 4.17, C26, verified on 16 September 2026 — every generating partition computes the whole metric entropy of the system, a general fact of measure theory which lies outside the elementary scope of this chapter), $h_\mu(D) = h_\mu(D, \mathcal{P}) = \log 2$.

Rokhlin’s formula. $|D'(\theta)| = 2$ for every θ (where defined), constant: $\int_{\mathbb{T}} \log |D'| d\mu = \log 2 \cdot \mu(\mathbb{T}) = \log 2$ (the general Rokhlin formula, $h_\mu(f) = \int \log |f'| d\mu$ for expanding C^1 maps with an absolutely continuous invariant measure, reduces here to an immediate computation because $|D'|$ is constant; the general case with non-constant Jacobian is imported: V. A. Rokhlin, *Lectures on the entropy theory of measure-preserving transformations*, *Russian Math. Surveys* **22**:5 (1967), 1–52, C27, verified on 22 September 2026; not to be confused with Dinaburg 1970, C22, cited below for the variational principle). The three quantities coincide: $\log 2 = \log 2 = \log 2$. $\square$

Remark 8.9: The variational principle, in the case of D . *The variational principle (Goodwyn 1969, Dinaburg 1970, Goodman 1971, imported from C22, verified on 16 September 2026) states, for a general topological dynamical system (X, f) ,*

$$h_{\text{top}}(f) = \sup_{\mu} h_{\mu}(f),$$

*the supremum being over the f -invariant Borel measures μ on X . The general proof (inequality $h_{\mu}(f) \leq h_{\text{top}}(f)$ for every μ , and existence of measures approximating the sup) uses tools of measure theory which lie outside this chapter and is imported. **In the case of D ,** however, the check is direct: from Theorem 8.12, $h_{\mu_{\text{Leb}}}(D) = \log 2$, and from Theorem 8.4, $h_{\text{top}}(D) = \log 2$: the Lebesgue measure realises the sup in the variational formula for D (it is a **measure of maximal entropy**), a fact verified here by direct computation of the two quantities, not deduced from the general principle.*

8.11. Oseledets and Pesin: a panorama

Remark 8.10: Panorama: Oseledets’ theorem and Pesin’s formula. In dimension 1 (the case of this chapter), the **multiplicative ergodic theorem of Oseledets** (V. I. Oseledets, Trudy Moskov. Mat. Obšč. **19** (1968), 179–210, imported) reduces to the following statement, a direct consequence of Birkhoff’s ergodic theorem (Chapter 9) applied to the observable $\varphi = \log |f'|$: if μ is an ergodic f -invariant measure and $\log |f'| \in L^1(\mu)$, for μ -almost every x there exists

$$\lambda_\mu := \lim_{n \rightarrow \infty} \frac{1}{n} \log |(f^n)'(x)| = \int \log |f'| d\mu,$$

a unique constant (the unique “Lyapunov exponent”, the tangent space being one-dimensional: in dimension ≥ 2 Oseledets’ theorem gives instead a spectrum of exponents, one for each invariant direction of the tangent splitting, a genuinely multidimensional phenomenon not present here). It is exactly the computation of Theorem 8.11 for f_4 and $\mu =$ the Ulam–von Neumann measure.

Pesin’s formula (Ja. B. Pesin, Uspekhi Mat. Nauk **32:4** (1977), 55–112, imported) generalises the coincidence $h_\mu(D) = \lambda_{\mu_{\text{Leb}}}(D)$ seen in Section 8.10: for a $C^{1+\alpha}$ diffeomorphism of a compact manifold and an invariant measure μ absolutely continuous with respect to Lebesgue,

$$h_\mu(f) = \sum_{\lambda_i > 0} \lambda_i \cdot (\text{multiplicity of } \lambda_i)$$

(sum of the positive Lyapunov exponents only, with multiplicity: in dimension 1, a single exponent, and the formula reduces to $h_\mu(f) = \max(\lambda, 0)$, consistent with $h_\mu(D) = \lambda_D = \log 2$ and with $h_\mu(R_\alpha) = 0 = \lambda_{R_\alpha}$). The general proof of Pesin’s formula uses the Lyapunov–Perron theory of stable and unstable manifolds and the non-uniformly hyperbolic local structure: it lies entirely outside the scope of this volume. **What we assume:** f a $C^{1+\alpha}$ diffeomorphism; μ invariant, absolutely continuous; conclusion: the equality above. For D and E_m (not diffeomorphisms, but piecewise C^∞ expanding maps with constant Jacobian) the formula has been verified by direct computation, not deduced from this general statement.

8.12. Worked examples

Example 8.3: Entropy and Lyapunov on an explicit periodic orbit. Problem. On the period-3 orbit of D , $\{2\pi/7, 4\pi/7, 8\pi/7\}$ (verified in Chapter 7: $D^3(2\pi/7) = 16\pi/7 \bmod 2\pi = 2\pi/7$), verify the consistency between the periodic multiplier $(D^3)'$ and the theoretical Lyapunov exponent $\log 2$, and place the orbit in the horseshoe construction of Theorem 8.9.

Step 1 — multiplier. $D'(\theta) = 2$ constant, hence $(D^3)'(\theta_0) = 2^3 = 8$ for every point of the orbit.

Step 2 — Lyapunov number and exponent. The Lyapunov number of the orbit is $(\prod_{i=0}^2 D'(D^i\theta_0))^{1/3} = 8^{1/3} = 2$; the exponent (logarithm of the Lyapunov number) is $\log 2$ — exactly the theoretical value of Theorem 8.10, computed here on a particular periodic orbit instead of as a generic limit. ✓

Step 3 — the horseshoe. The period-3 orbit of D is not the one used as a numerical example in Theorem 8.9 (which holds for *any* continuous map with *an* orbit of period 3, not only D): the point here is that, applying the theorem to $f = D$ itself with $\{a, b, c\} = \{2\pi/7, 4\pi/7, 8\pi/7\}$, one obtains $h_{\text{top}}(D) \geq \log 2$ — a **third** independent proof (besides the direct computation of Section 8.4 and Bowen’s theorem of Example 8.1) which coincides

exactly with the known value $\log 2$, this time starting from a single periodic orbit.

Example 8.4: The numerical counterexample to $N(n, \varepsilon) = k^n$. Problem. Show concretely, with $k = 2$, $n = 1$, $\varepsilon = 0.2$, that $N(1, 0.2) \neq 2^1 = 2$ in the space Σ_2 with the metric $d(x, y) = \sum_i |x_i - y_i| 2^{-i}$, by exhibiting a $(1, 0.2)$ -separated set of cardinality > 2 .

Step 1 — four points pairwise at distance > 0.2 . Consider the four points (sequences extended with 0): $x^{(0)} = (0, 0, 0, 0, \dots)$, $x^{(1)} = (0, 1, 0, 0, \dots)$, $x^{(2)} = (0, 0, 1, 0, \dots)$, $x^{(3)} = (0, 1, 1, 0, \dots)$. The first bit is 0 for all of them (avoiding ambiguities of order of magnitude), and we use bits 2, 3: $d(x^{(0)}, x^{(1)}) = 1 \cdot 2^{-1} = 0.5$; $d(x^{(0)}, x^{(2)}) = 1 \cdot 2^{-2} = 0.25$; $d(x^{(0)}, x^{(3)}) = 2^{-1} + 2^{-2} = 0.75$; $d(x^{(1)}, x^{(2)}) = 2^{-1} + 2^{-2} = 0.75$; $d(x^{(1)}, x^{(3)}) = 2^{-2} = 0.25$; $d(x^{(2)}, x^{(3)}) = 2^{-1} = 0.5$. All > 0.2 : the four points are pairwise $(1, 0.2)$ -separated (the condition $(1, \varepsilon)$ -separated simply means $d(x, y) > \varepsilon$, there being no iterations to compare with $n = 1$). ✓

Step 2 — comparison with the estimate. $k^n = 2^1 = 2$: the four points already exceed this value. Theorem 8.5 predicts $N(1, 0.2) \leq 2^{1+m}$ with $m = \lceil \log_2(1/0.2) \rceil = \lceil \log_2 5 \rceil = 3$, that is, $N(1, 0.2) \leq 16$; the script `verifica_cap08.py` (block [1]) finds by a greedy search a set of cardinality exactly 16, saturating the upper bound. ✓

Interpretation. The old Theorem 4.6 asserted $N(n, \varepsilon) = 2^n = 2$ for *every* $\varepsilon < 1$: the four (in fact sixteen) points exhibited here, all at distance > 0.2 although agreeing on the first digit, directly refute this assertion.

Example 8.5: Numerical convergence of the Birkhoff average of $\log |f'_4|$. Problem. Estimate numerically $\lambda(x_0)$ for f_4 starting from four different initial points and compare the speed of convergence with that (immediate) of the case D .

Step 1 — script. `fig08_birkhoff_convergenza.py` (Appendix B) iterates f_4 for $n = 20\,000$ steps starting from $x_0 \in \{0.2; 0.31415926; 0.61803399; 0.87654321\}$, accumulating $\frac{1}{n} \sum_{k < n} \log |f'_4(x_k)|$.

Step 2 — results (run on 16 September 2026).

x_0	average at $n = 20\,000$	deviation from $\log 2 = 0.693147$
0.20000001	0.693147	$< 10^{-6}$
0.31415926	0.693077	0.000070
0.61803399	0.693149	0.000001
0.87654321	0.693167	0.000019

Interpretation. All four orbits converge to $\log 2$ within 10^{-4} already at $n = 20\,000$, with residual fluctuations of order $O(1/\sqrt{n}) \approx 1/\sqrt{20\,000} \approx 0.007$, compatible with the deviations observed (smaller, for the orbits chosen here, but of the same order of magnitude). By comparison, for D (Theorem 8.10) the value $\log 2$ is exact already at $n = 1$, because $|D'| \equiv 2$ is constant: the *asymptotic* convergence of f_4 is the price of a non-constant derivative, consistent with the infobox of the old Chapter 4 on the speed of convergence, here reconfirmed numerically with updated data. Figure 8.3 shows the four trajectories.

8.13. Assigned exercises

The answers are in the Teacher's key (Section 8.16).

Exercise 8.1: Entropy of a three-branch map. *Goal:* to apply the direct computation of Section 8.4 to a map with a number of branches different from a power of 2, and to connect it with the lap number. *Estimated time:* 30 minutes. *Type:* symbolic with numerical check.

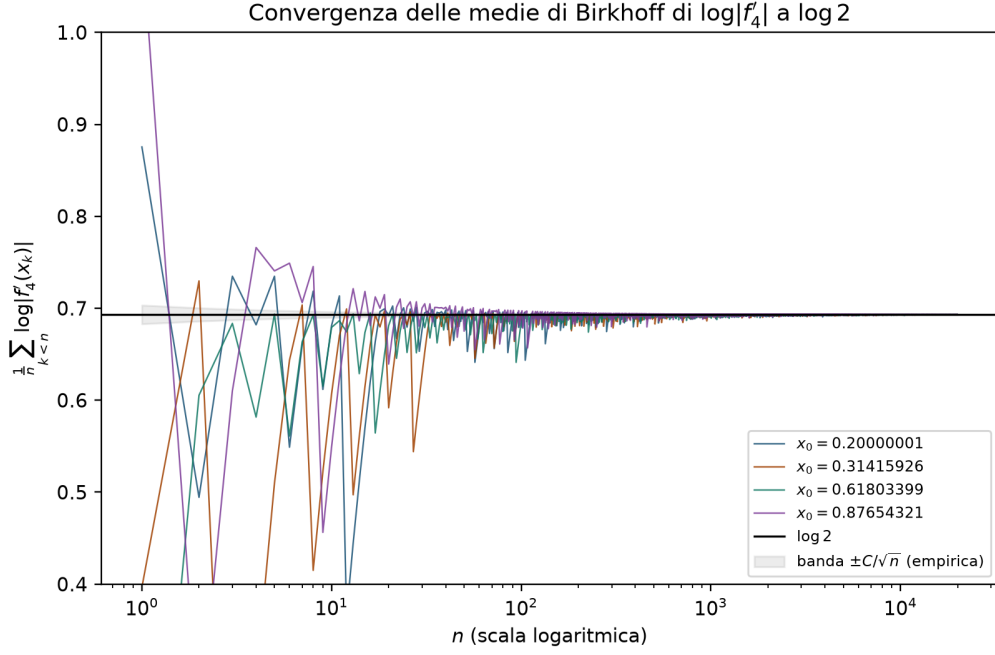


Figure 8.3: Convergence of the Birkhoff averages $\frac{1}{n} \sum_{k < n} \log|f'_4(x_k)|$ to $\log 2$ for four initial points, in logarithmic scale in n , with an empirical band $\pm C/\sqrt{n}$. Computed numerically, script `fig08_birkhoff_convergenza.py` (Appendix B), run on 16 September 2026: final values at $n = 20\,000$ within 10^{-4} of $\log 2$. ✓

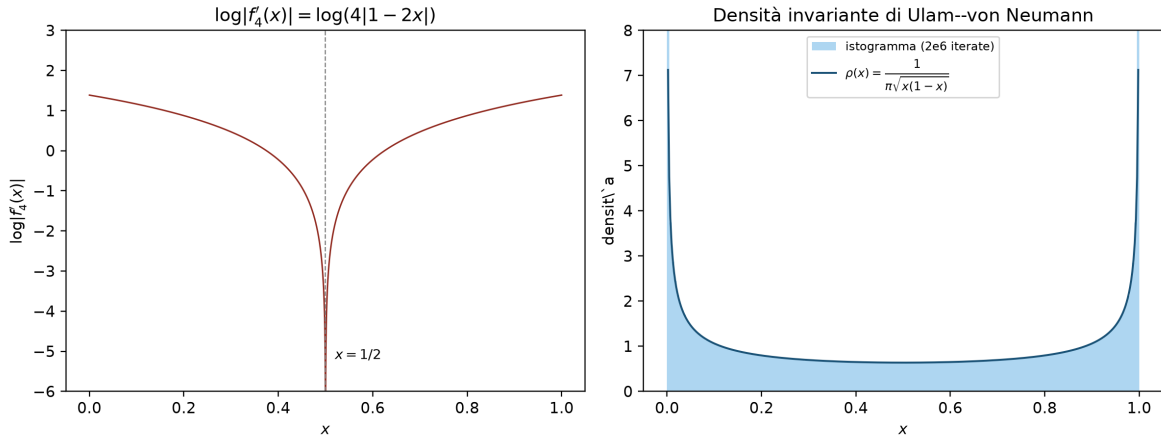


Figure 8.4: Left: $\log|f'_4(x)| = \log(4|1-2x|)$, with the logarithmic singularity at $x = 1/2$ (where $f'_4(1/2) = 0$, consistent with $\lambda(1/2) = -\infty$, Proposition 8.1). Right: histogram of 2×10^6 iterates of f_4 against the invariant Ulam-von Neumann density $\rho(x) = 1/(\pi\sqrt{x(1-x)})$ (proved invariant in Chapter 9, here only illustrated); mean relative deviation of the central bins (normalised on $(0,1)$) 0.88%. Computed numerically, script `fig08_log_deriv_densita.py` (Appendix B), run on 16 September 2026. ✓

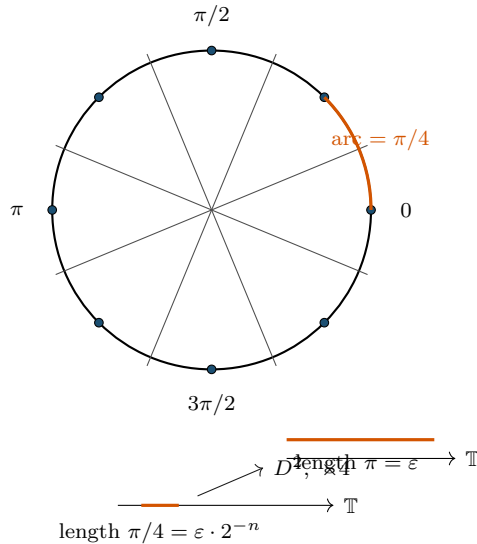


Figure 8.5: Above: the circle $\mathbb{T}$ subdivided into $8 = 2^n \cdot (2\pi/\varepsilon)$ arcs of length $\varepsilon 2^{-n}$, illustrative case $n = 2$, $\varepsilon = \pi$ (so that $\varepsilon 2^{-n} = \pi/4$, and exactly 8 arcs cover 2π : $8 \times \pi/4 = 2\pi$ ✓). Below: D^2 (constant derivative $4 = 2^n$) stretches an arc of length $\varepsilon 2^{-n} = \pi/4$ into an arc of length $\varepsilon = \pi$, the separation threshold used in the proof of Theorem 8.4 (upper bound, case $m = 2$). Figure drawn by hand; coordinates verified: 8 points at 45° from one another on the circle of radius 3.4 cm.

Let $T_3 : \mathbb{T} \rightarrow \mathbb{T}$, $T_3(\theta) = 3\theta \bmod 2\pi$ (that is, E_3).

- State, adapting the proof of Theorem 8.4, the upper bound $h_{\text{top}}(T_3) \leq \log 3$, indicating explicitly K as a function of n, ε .
- Construct, for $n = 2$, the grid of $9 = 3^2$ points of the induction of the lower bound, and list them.
- Verify that two consecutive points of the grid (b) are $(2, \pi/3)$ -separated, computing explicitly the distance after 0 and after 1 iteration for at least one pair.
- Conclude $h_{\text{top}}(T_3) = \log 3 \approx 1.098612$.

Exercise 8.2: Lap number of the tent map. *Goal:* to apply Lemma 8.3 and the imported Misiurewicz–Szlenk remark to a map different from f_4 . *Estimated time:* 35 minutes. *Type:* symbolic/computational.

Let $T(x) = 1 - |2x - 1|$ be the tent map on $[0, 1]$ (Chapter 7).

- Verify $\text{lap}(T) = 2$.
- Using that T is conjugate to f_4 (Chapter 7), deduce $h_{\text{top}}(T) = \log 2$ without computing $\text{lap}(T^n)$ directly.
- Compute explicitly $T^2(x)$ piecewise (four linear formulas) and verify $\text{lap}(T^2) = 4$.
- Formulate the conjecture $\text{lap}(T^n) = 2^n$ and verify it numerically for $n = 1, \dots, 5$ with a script analogous to `fig08_lap_number.py`.

Exercise 8.3: An orbit of period 5 and the entropy. *Goal:* to apply the horseshoe reasoning (Theorem 8.9 and Remark 8.6) to a period different from 3. *Estimated time:* 40 minutes. *Type:* symbolic/probative.

Suppose that a continuous $g : I \rightarrow I$ has an orbit of period 5 (odd, not a power of 2).

- Explain why Theorem 8.9 does not apply directly (it requires period 3).

- (b) Using Remark 8.6 (Štefan cycle of odd period ≥ 3), argue that there exists in any case a self-covering interval with two branches for some iterate of g .
- (c) Deduce $h_{\text{top}}(g) \geq \log 2 > 0$.
- (d) Compare with the Sharkovskii order (Chapter 7): which other periods are forced by an orbit of period 5? Are they all ≥ 5 or not?

8.14. Worked exercises

Exercise 8.4: Comparison between the estimates of $N(n, \varepsilon)$ for the shift and for E_2 . *Type:* symbolic with numerical check. Compare, for $n = 4$, $\varepsilon = 0.1$, the estimate $k^n \leq N(n, \varepsilon) \leq k^{n+m}$ of the shift σ_2 (Theorem 8.5) with the direct count of the grid $\{2\pi j/2^n\}$ used for $E_2 = D$ (proof of Theorem 8.4), and explain why the two estimates, although concerning different systems (σ_2 abstract and D on the circle), give compatible orders of magnitude.

Solution. Strategy. One computes the two estimates separately and compares the order of magnitude, recalling that σ_2 and D are linked by a semiconjugacy (Chapter 7), not by a conjugacy, hence the exact cardinalities need not coincide, only the exponential rates.

Step 1 — shift. $n = 4$, $\varepsilon = 0.1$, $k = 2$: $m = \lceil \log_2(1/0.1) \rceil = \lceil 3.3219 \rceil = 4$. Hence $2^4 = 16 \leq N(4, 0.1) \leq 2^{4+4} = 256$.

Step 2 — D . The proof of Theorem 8.4 gives, for the upper bound, $N(4, 0.1) \leq \lceil 2\pi \cdot 2^4/0.1 \rceil = \lceil 1005.3 \rceil = 1006$; for the lower bound, with $\varepsilon_0 = \pi/2 \approx 1.5708 > 0.1$, the grid of $2^4 = 16$ points is $(4, \pi/2)$ -separated, hence also $(4, 0.1)$ -separated (a set separated at a *larger* threshold is so also at a smaller one, the condition $d_n > \varepsilon$ being weaker for smaller ε): $N(4, 0.1) \geq 16$.

Step 3 — comparison. Both estimates give $N(4, 0.1) \geq 16 = 2^4$ as a lower bound; the direct upper bound for D (1006) is wider than the shift's (256), in this specific case, because the constant in the estimate $2\pi/\varepsilon \approx 62.8$ is larger than $2^m = 16$ for these values — the two estimates use different techniques and are not directly comparable term by term, only in the exponential order. ✓ In both cases $\frac{1}{4} \log N \rightarrow \log 2$ is the information that counts: $\frac{1}{4} \log 16 = \log 2$ exactly in both lower bounds.

Interpretation. The two estimates come from independent arguments (one combinatorial on the abstract symbol space, the other geometric on the circle) and yet they agree on the exponential rate $\log 2$: it is precisely the content of Example 8.1 (three independent proofs of the same value).

Typical errors. Confusing “equal $N(n, \varepsilon)$ ” with “equal h_{top} ”: the exact cardinalities for fixed n, ε can differ enormously between different systems (or even between different definitions for the same system); only the *exponential rate in the limit* is an invariant.

Extension. Repeat the comparison for $\varepsilon = 0.01$ and verify that the gap between the two upper bounds grows (because they depend in different ways on $1/\varepsilon$: linearly for D , logarithmically through $m(\varepsilon)$ for the shift), while both remain compatible with the same limit $\log 2$.

Exercise 8.5: Lyapunov exponent of E_3 and Rokhlin's formula. *Type:* symbolic. Compute $\lambda(\theta)$ for $E_3(\theta) = 3\theta \bmod 2\pi$ and verify Rokhlin's formula $h_{\mu_{\text{Leb}}}(E_3) = \int \log |E'_3| d\mu$ using the partition into 3 arcs $[0, 2\pi/3)$, $[2\pi/3, 4\pi/3)$, $[4\pi/3, 2\pi)$.

Solution. Strategy. Same structure as Theorem 8.10 and Theorem 8.12, with $m = 3$ in place of $m = 2$.

Step 1 — Lyapunov. $E'_3(\theta) = 3$ constant, $(E_3^n)' = 3^n$, $\lambda(\theta) = \log 3$ for every θ : it coincides with $h_{\text{top}}(E_3) = \log 3$ (Theorem 8.4, $m = 3$).

Step 2 — partition and refinement. Let $\mathcal{Q} = \{Q_0, Q_1, Q_2\}$ be the three arcs indicated, each of measure $\mu = 1/3$. With the same inductive argument as in Lemma 8.5 (readings of the digits in base 3 instead of base 2), $\bigvee_{i < n} E_3^{-i} \mathcal{Q}$ consists of 3^n equal arcs of measure 3^{-n} .

Step 3 — entropy of the partition. $H(\bigvee_{i < n} E_3^{-i} \mathcal{Q}) = -3^n \cdot 3^{-n} \log(3^{-n}) = n \log 3$, hence $h_\mu(E_3, \mathcal{Q}) = \log 3$; $\mathcal{Q}$ is generating (same argument as in Lemma 8.5), hence $h_\mu(E_3) = \log 3$ by the generator theorem.

Step 4 — Rokhlin. $\int_{\mathbb{T}} \log |E'_3| d\mu = \log 3 \cdot 1 = \log 3$ (constant derivative). The three quantities coincide: $\lambda = h_\mu(E_3) = \int \log |E'_3| d\mu = \log 3 \approx 1.098612$. ✓

Independent check. $\log 3$ computed by Python: `math.log(3)=1.0986122886681098`, consistent to 10 digits with the symbolic value.

Interpretation. The whole of Section 8.10, written for D ($m = 2$), carries over word for word to every E_m : none of the proofs uses properties specific to $m = 2$ beyond the constancy of the derivative, which holds for every E_m .

Typical errors. Forgetting that Rokhlin's formula, in its general form (non-constant Jacobian), requires regularity hypotheses and the existence of an absolutely continuous invariant measure: here everything is immediate only because $|E'_3|$ is constant.

Extension. Repeat for general E_m and verify that all three quantities equal $\log m$, independently of $m \geq 2$.

Exercise 8.6: The cost of the absence of expansion: f_4 near the critical point.

Type: numerical/computational. Study numerically how the finite Birkhoff average $\lambda_n(x_0) = \frac{1}{n} \sum_{k < n} \log |f'_4(x_k)|$ behaves for an orbit that passes *near* (but not exactly at) the critical point $x = 1/2$, comparing with a generic orbit.

Solution. Strategy. If an orbit visits a neighbourhood of $x = 1/2$, the term $\log |f'_4(x_k)| = \log(4|1 - 2x_k|)$ becomes very negative (but finite, provided $x_k \neq 1/2$ exactly): the average feels an anomalous isolated term, which is however “diluted” by the factor $1/n$ as n grows, consistently with $\lambda(1/2) = -\infty$ only in the exact limiting case.

Step 1 — choice of the data. Take $x_0 = 0.5 - 10^{-6}$, very close to the critical point but not coinciding with it. $f'_4(x_0) = 4(1 - 2(0.5 - 10^{-6})) = 4 \cdot 2 \times 10^{-6} = 8 \times 10^{-6}$, $\log |f'_4(x_0)| = \log(8 \times 10^{-6}) \approx -11.74$: a term strongly negative, but finite.

Step 2 — effect on the average. For $n = 1$, $\lambda_1(x_0) = -11.74$, extremely far from $\log 2 \approx 0.693$. But $f_4(x_0) = 4x_0(1 - x_0) \approx 1 - 4 \times 10^{-12}$, almost 1: the orbit moves away rapidly from the critical point (the image of a point near the maximum is near the endpoint $x = 1$, where the derivative is regular, $\neq 0$), and the subsequent iterations behave like a typical orbit. For $n = 100$, the single anomalous term weighs $-11.74/100 \approx -0.117$ on the average, a noticeable but not dominant shift; for $n = 10\,000$, it weighs $-11.74/10\,000 \approx -0.0012$, negligible.

Step 3 — check. Code analogous to block [6] of `verifica_cap08.py`, applied to $x_0 = 0.5 - 10^{-6}$ with $n = 10\,000$: the initial anomalous term contributes only $1/10\,000$ of the sum, and the average converges in any case to $\log 2$ within the expected statistical error $O(1/\sqrt{n})$.

Interpretation. The value $\lambda(1/2) = -\infty$ (Proposition 8.1) is a phenomenon of *zero measure*: the set of points whose orbit hits *exactly* $x = 1/2$ (the backward orbit of the critical point) is countable, hence of zero Lebesgue measure, consistently with the “almost every” of Theorem 8.11. Points *near* (but not equal to) $1/2$ produce only an anomalous transient which fades away in the average.

Typical errors. Confusing “the orbit passes near the critical point” with “the Lyapunov exponent is not $\log 2$ ”: they are two different phenomena (the first is generic and transient, the second exceptional and exact).

Extension. Estimate, for a given small $\eta > 0$, the probability (with respect to the Ulam–von Neumann measure) that an orbit of length n visits the interval $(1/2 - \eta, 1/2 + \eta)$ at least once, and discuss how this controls the typical size of the anomalous term.

8.15. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 8.16).

- (Definition) Two points x, y are (n, ε) -separated if
 - $d(x, y) > \varepsilon$
 - $d(f^n x, f^n y) > \varepsilon$
 - there exists $0 \leq i < n$ with $d(f^i x, f^i y) > \varepsilon$
 - for every $i < n$, $d(f^i x, f^i y) > \varepsilon$
- (Hypothesis) Bowen’s theorem on finite-fibre factors (Remark 8.3) requires, besides the surjectivity of the factor π , that
 - π be a homeomorphism
 - π have uniformly bounded fibres
 - X and Y have the same topological dimension
 - f be linear
- (Symbolic computation) $h_{\text{top}}(E_5)$ equals
 - 5
 - $\log 5$
 - $5 \log 2$
 - $\log 25$
- (Numerical computation) With $k = 2$, $\varepsilon = 0.2$, $n = 1$, the value $m(\varepsilon) = \lceil \log_2(1/\varepsilon) \rceil$ of the estimate of Theorem 8.5 equals
 - 1
 - 2
 - 3
 - 5
- (Proof step) In the proof of the lower bound of Theorem 8.4 (entropy of E_m), the inductive step distinguishes the cases $k' \neq \tilde{k}'$ and $k' = \tilde{k}'$ because
 - different arguments are needed: the first uses the inductive hypothesis on the smaller grid, the second the direct separation of the most significant digits
 - the case $k' = \tilde{k}'$ is impossible
 - E_m is not defined for $k' = \tilde{k}'$
 - it is only a matter of notation, it does not change the argument
- (Error diagnosis) The old Exercise 4.11 (archived) commits an error by declaring $\lambda(1/2) = \log 4$; the error is
 - confusing $1/2$ with a fixed point of f_4
 - failing to recognise that $f'_4(1/2) = 0$, hence the first term of the average is $\log 0 = -\infty$, not $\log 4$
 - getting the computation of $f_4(1/2)$ wrong
 - using the wrong base for the logarithm
- (Transfer) An orbit of period $6 = 2 \cdot 3$ for a continuous map $g : I \rightarrow I$ forces
 - $h_{\text{top}}(g) = 0$, because 6 is even
 - $h_{\text{top}}(g) \geq (\log 2)/2 > 0$, by Theorem 8.8 applied to g^2 (period 3, odd, for g^2)
 - no information on the entropy
 - $h_{\text{top}}(g) = \log 6$
- (Source and metacognition) The exact value $\int_0^{\pi/2} \log(\cos \theta) d\theta = -\frac{\pi}{2} \log 2$ (Lemma 8.4) used for $\lambda(f_4)$ is
 - imported from Walters, GTM 79
 - a classical result (Euler), proved in full in the text with an argument of symmetry and doubling, with no need to import it
 -

imported from Misiurewicz–Szlenk 1980 D) a merely numerical estimate, without a symbolic proof

8.16. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **C.** *Rationale:* Definition 8.1: one time $i < n$ at which the orbits move apart by more than ε suffices. *Distractors:* A ignores the dynamics (it is only the initial condition); B looks only at the last time; D asks for the stronger condition (for every i), which is that of a set “uniformly separated”, not the Bowen–Dinaburg entropy. *Error diagnosed:* confusing “there exists” with “for every” in the definition. *Source:* Definition 8.1. *Prerequisite:* logical quantifiers. *Remedial work:* rewrite by hand the definition for $n = 2$ with explicit symbols.
- 2) **B.** *Rationale:* Remark 8.3: without a uniform bound on the fibres, the equality may fail (there remains only $h(g) \leq h(f)$, Theorem 8.3, always valid). *Distractors:* A is the definition of conjugacy, not of factor; C and D are not hypotheses of the theorem. *Error diagnosed:* confusing factor with conjugacy. *Source:* Remark 8.3. *Prerequisite:* Definition 8.3. *Remedial work:* reread the Good practice after the remark, which explicitly compares conjugacy and factor.
- 3) **B.** *Rationale:* Theorem 8.4 with $m = 5$: $h_{\text{top}}(E_5) = \log 5$. *Distractors:* A confuses m with $\log m$; C and D are numerical combinations without motivation. *Error diagnosed:* forgetting the logarithm in the formula. *Source:* Theorem 8.4. *Prerequisite:* the logarithm function. *Remedial work:* repeat Exercise 8.1 with $m = 5$ in place of $m = 3$.
- 4) **C.** *Rationale:* $\log_2(1/0.2) = \log_2 5 \approx 2.3219$, and $\lceil 2.3219 \rceil = 3$ (Warning after Theorem 8.5, the same value used in the counterexample). *Distractors:* A and B are wrong roundings; D confuses $1/\varepsilon = 5$ with m . *Error diagnosed:* forgetting the rounding upwards ($\lceil \cdot \rceil$). *Source:* proof of Theorem 8.5, step of the upper bound. *Prerequisite:* logarithm in base 2. *Remedial work:* recompute $m(\varepsilon)$ for $\varepsilon = 0.1$ and $\varepsilon = 0.05$ and compare with the table of the Warning.
- 5) **A.** *Rationale:* the case $k' \neq \tilde{k}'$ uses the inductive hypothesis (the two sub-points on the n -th grid are already separated by a previous construction); the case $k' = \tilde{k}'$, $s \neq \tilde{s}$ is instead elementary (difference equal to a non-zero multiple of $2\pi/m$, separated immediately at $i = 0$): they are logically distinct mechanisms, both necessary to cover all the pairs $j \neq \tilde{j}$. *Distractors:* B is false (the case is central, not impossible); C and D misunderstand the structure of the proof. *Error diagnosed:* failing to recognise an induction with cases. *Source:* proof of Theorem 8.4, lower bound, inductive step. *Prerequisite:* positional representation in base m . *Remedial work:* write explicitly $j, \tilde{j}$ in base 3 for $n + 1 = 3$ and identify the two cases.
- 6) **B.** *Rationale:* Proposition 8.1: $f'_4(1/2) = 4(1 - 1) = 0$, hence the term $k = 0$ of the sum which defines $\lambda(1/2)$ is already $\log 0 = -\infty$. *Distractors:* A is true but irrelevant ($1/2$ is not a fixed point, in any case); C is false ($f_4(1/2) = 1$, correct in the old text); D is beside the point (the logarithm is natural everywhere in the chapter). *Error diagnosed:* lumping $x_0 = 1/2$ (critical point, zero derivative) together with the points $x_0 = 0, 1$ (derivative $4 \neq 0$, but an orbit which collapses onto a fixed point with derivative 4). *Source:* Proposition 8.1, proof. *Prerequisite:* $\log 0 = -\infty$. *Remedial work:* recompute $f'_4(x)$ at the four points $0, 1/2, 3/4, 1$ and compare the four results of the proposition.
- 7) **B.** *Rationale:* g^2 has, on the period-6 orbit of g , sub-orbits of period $6/\gcd(6, 2) = 3$ (odd): by Theorem 8.9 applied to g^2 , $h_{\text{top}}(g^2) \geq \log 2$; by Theorem 8.8, $h_{\text{top}}(g) = h_{\text{top}}(g^2)/2 \geq (\log 2)/2$. *Distractors:* A confuses “even period” with “power of 2” (6 is even but is not a power of 2); C ignores Remark 8.6; D has no basis in the theory developed.

Error diagnosed: thinking that only *odd* periods (not their powers of 2 times) force positive entropy. *Source:* Remark 8.6. *Prerequisite:* Theorem 8.8. *Remedial work:* redo Exercise 8.3 with period 6 instead of 5.

8) B. Rationale: Lemma 8.4 is proved in full in the text (symmetry $\theta \mapsto \pi/2 - \theta$ and angle doubling), a classical argument going back to Euler, not imported from any of the C sources. *Distractors:* A, C confuse this elementary lemma with the imported theorems of the chapter (Bowen, Misiurewicz–Szlenk); D is false, Lemma 8.4 has a complete symbolic proof. *Error diagnosed:* assuming that every advanced formula of the chapter is imported, without distinguishing what is proved from what is not. *Source:* Lemma 8.4. *Prerequisite:* substitution in definite integrals. *Remedial work:* reread the list of the importations of the chapter (Remarks 8.3, 8.5, 8.10) and verify that Lemma 8.4 does not appear there.

BEST PRACTICE

Assigned exercises — answers.

8.1 ($T_3 = E_3$). (a) $K = \lceil 2\pi \cdot 3^n / \varepsilon \rceil$; two points in the same arc of length $\leq \varepsilon/3^n$ stay at distance $\leq \varepsilon/3 < \varepsilon$ for every $i < n$, by Lemma 8.2 iterated with $m = 3$. (b) $n = 2$: $\{2\pi k/9 : k = 0, \dots, 8\} = \{0; 2\pi/9; 4\pi/9; \dots; 16\pi/9\}$. (c) $x_0 = 0$, $x_1 = 2\pi/9 \approx 0.6981$: $d(x_0, x_1) = 2\pi/9 \approx 0.698 < \pi/3 \approx 1.047$ (not separated at $i = 0$); $E_3(x_0) = 0$, $E_3(x_1) = 6\pi/9 = 2\pi/3 \approx 2.094$, $d = 2\pi/3 > \pi/3$: separated at $i = 1$, consistent with the induction (they differ in the least significant base-3 digit, “case $k' \neq \tilde{k}'$ ” with $n = 1$ as base). (d) From Theorem 8.4 with $m = 3$, $h_{\text{top}}(T_3) = \log 3 = 1.098612\dots$

8.2 (tent). (a) T increases on $[0, 1/2]$ and decreases on $[1/2, 1]$: $\text{lap}(T) = 2$. (b) By the invariance under conjugacy (Theorem 8.2), $h_{\text{top}}(T) = h_{\text{top}}(f_4) = \log 2$ (Example 8.2). (c) Recalling $T(x) = 2x$ on $[0, 1/2]$ and $T(x) = 2 - 2x$ on $[1/2, 1]$, one composes T with itself separately on the four quarters (checking each time which of the two branches the intermediate value $T(x)$ falls into): for $x \in [0, 1/4]$, $T(x) = 2x \in [0, 1/2]$, hence $T^2(x) = 2 \cdot (2x) = 4x$; for $x \in [1/4, 1/2]$, $T(x) = 2x \in [1/2, 1]$, hence $T^2(x) = 2 - 2(2x) = 2 - 4x$; for $x \in [1/2, 3/4]$, $T(x) = 2 - 2x \in [1/2, 1]$, hence $T^2(x) = 2 - 2(2 - 2x) = 4x - 2$; for $x \in [3/4, 1]$, $T(x) = 2 - 2x \in [0, 1/2]$, hence $T^2(x) = 2(2 - 2x) = 4 - 4x$. Check at the nodes: $T^2(0) = 0$, $T^2(1/4) = 1$, $T^2(1/2) = 0$ (from $4x - 2$) = 0 (from $2 - 4x$) ✓, $T^2(3/4) = 1$ (from $4x - 2$) = 1 (from $4 - 4x$) ✓, $T^2(1) = 0$; and by direct comparison $T(1/2) = 1$, $T(1) = 0$, so $T^2(1/2) = T(1) = 0$, consistent. The four pieces have slope ± 4 and $\text{lap}(T^2) = 4$, verifiable by counting the changes of monotonicity on the sampled graph. (d) $\text{lap}(T^n) = 2^n$ for $n = 1, \dots, 5$: 2, 4, 8, 16, 32, verifiable with the same counting script as `fig08_lap_number.py` applied to T instead of f_4 .

8.3 (period 5). (a) Theorem 8.9 is stated and proved specifically for an orbit of period *exactly* 3, using the structure of the two intervals I_1, I_2 with $I_2 \rightarrow I_1 \cup I_2$ with two branches: an orbit of period 5 has a different combinatorial structure (more intervals), not covered directly by that proof. (b) The Štefan cycle of period 5 (constructed in Chapter 7 for the proof of Sharkovskii) singles out, among the 5 intervals of the cycle, a central interval which covers the union of the other two branches exactly as I_2 in the case $q = 3$ (by the general construction of odd Štefan cycles, Chapter 7): the same construction of Theorem 8.9 (nested intersections, separation via two disjoint branches) applies to that interval. (c) It follows that $h_{\text{top}}(g) \geq \log 2 > 0$, by the same argument as in Theorem 8.9, applied to the self-covering interval singled out in (b). (d) By the Sharkovskii order, an orbit of period 5 forces all the periods m with $5 \triangleleft m$: that is, all the odd numbers ≥ 7 (joining the list of odds, of which 5 is itself a member), *the whole* block $2 \cdot (\text{odd} \geq 3)$ that follows in the order (hence also $2 \cdot 3 = 6$, not only $2 \cdot (\text{odd} \geq 5)$): the entire block of odds precedes the entire block of 2·odds, regardless of the numerical value), then $4 \cdot (\text{odd} \geq 3)$, and so on, *and* all the powers of 2 (which lie at the bottom of the order, forced by any period which is not a power of 2):

not all ≥ 5 (the powers of 2, including 1, 2, 4, are forced and are < 5 ; also $6 = 2 \cdot 3$ is forced despite not being, numerically, “larger” in any obvious sense, since the Sharkovskii order differs from the numerical order).

8.17. Chapter formulary

Topological entropy (Bowen–Dinaburg)

$$d_n(x, y) = \max_{i < n} d(f^i x, f^i y), \quad h_{\text{top}}(f) = \lim_{\varepsilon \rightarrow 0^+} \limsup_{n \rightarrow \infty} \frac{1}{n} \log N(n, \varepsilon)$$

$$r(n, \varepsilon) \leq N(n, \varepsilon) \leq r(n, \varepsilon/2), \quad h_{\text{top}}^{\text{AKM}}(f) = \sup_{\mathcal{U}} \limsup_n \frac{1}{n} \log N(\mathcal{U}^n) = h_{\text{top}}(f)$$

Invariance and factors

$$h \circ f = g \circ h \text{ (conjugacy)} \implies h_{\text{top}}(f) = h_{\text{top}}(g)$$

$$\pi \circ f = g \circ \pi, \pi \text{ surjective} \implies h_{\text{top}}(g) \leq h_{\text{top}}(f) \quad (\text{if finite unif. bounded fibres, Bowen})$$

Direct computations

$$h_{\text{top}}(E_m) = \log m \ (m \geq 2), \quad h_{\text{top}}(D) = \log 2, \quad h_{\text{top}}(R_\alpha) = 0, \quad h_{\text{top}}(f^j) = j h_{\text{top}}(f)$$

$$\text{shift } \sigma_k : \quad k^n \leq N(n, \varepsilon) \leq k^{n+m(\varepsilon)}, \quad m(\varepsilon) = \lceil \log_k(1/\varepsilon) \rceil, \quad h_{\text{top}}(\sigma_k) = \log k$$

Lap number

$$\text{lap}(f^{n+m}) \leq \text{lap}(f^n) \text{lap}(f^m), \quad \ell(f) = \lim_n \frac{1}{n} \log \text{lap}(f^n) \leq h_{\text{top}}(f)$$

$$(\text{Misiurewicz–Szlenk, imported}) \quad h_{\text{top}}(f) = \ell(f) \text{ for } f \text{ piecewise monotone with finitely many pieces}$$

Sharkovskii and entropy

$$\text{period } 3 \implies h_{\text{top}}(f) \geq \log 2; \quad \text{period not a power of 2 (BGMY, imported)} \implies h_{\text{top}}(f) > 0$$

Lyapunov exponents

$$\lambda(\theta) = \lim_n \frac{1}{n} \log |(f^n)'(\theta)| = \lim_n \frac{1}{n} \sum_{i < n} \log |f'(f^i \theta)|$$

$$\lambda_{E_m} \equiv \log m, \quad \lambda_{f_4} = \log 2 \text{ a.e.}, \quad \lambda_{f_4}(0) = \lambda_{f_4}(1) = \log 4, \quad \lambda_{f_4}(1/2) = -\infty$$

$$\int_0^{\pi/2} \log(\cos \theta) d\theta = -\frac{\pi}{2} \log 2$$

Rokhlin and variational principle (case D)

$$h_{\mu_{\text{Leb}}}(D) = h_\mu(D, \mathcal{P}_{\text{diadica}}) = \int_{\mathbb{T}} \log |D'| d\mu = \log 2 = h_{\text{top}}(D)$$

$$(\text{variational principle, imported}) \quad h_{\text{top}}(f) = \sup_{\mu} h_\mu(f)$$

Oseledets 1D and Pesin (panorama)

$$\lambda_\mu = \int \log |f'| d\mu \ (\mu\text{-a.e.}), \quad h_\mu(f) = \max(\lambda, 0) \text{ (1D case of Pesin's formula)}$$

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CHAPTER 9

Ergodic theory on the circle and on the interval

Chapter 7 showed that the doubling map D and the tent map T are *topologically* chaotic: transitive, sensitive, with dense periodic points. Chapter 8 measured that chaos with a number, the entropy. This chapter asks a different question, of a statistical nature: if one observes a single orbit for an extremely long time, what does one see *on average*? The answer — the **ergodic theorem** — is one of the most surprising results of twentieth-century mathematics: for almost every starting point, the time average along the orbit coincides with the space average with respect to an invariant measure, independently of the initial point. The circle and the interval, once again, are the simplest laboratory in which this fact can be proved in full, with all the constants explicit.

INFO

Level and prerequisites. Level C–D: core of level C (invariant measures, ergodicity of D , ergodic theorems), sections of level D (Lasota–Yorke theorem, Section 9.12; Kolmogorov–Sinai entropy, Section 9.13). Prerequisites within the treatise: the doubling map D , the m -expanding maps E_m , the semiconjugacy with the Bernoulli shift and the counting of periodic points (Chapter 7); the topological entropy $h_{\text{top}}(D) = \log 2$ and the Lyapunov exponent (Chapter 8); the density of the orbits and the uniqueness of the invariant measure of the irrational rotations R_α , proved via Weyl equidistribution (Volume III, Chapter 13, and Chapter 3 of this volume); the ergodicity of the C^2 diffeomorphisms of the circle with irrational rotation number (Denjoy’s theorem, Chapter 5), here only recalled. No measure theory beyond the Lebesgue integral and Fourier series (Volume II, Chapter 4) is presupposed: the few notions of abstract measure theory (Dynkin’s monotone class theorem, Riesz representation theorem for $C(X)^*$, orthogonal projection theorem in a Hilbert space) are recalled as standard facts at the moment of use.

Warning. Conventions of the chapter. We write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ for the circle, with normalised Lebesgue measure $\mu(\mathbb{T}) = 1$ (that is, $d\mu = d\theta/2\pi$); on $[0, 1]$ the normalised Lebesgue measure coincides with the usual one. $D : \mathbb{T} \rightarrow \mathbb{T}$, $D(\theta) = 2\theta \bmod 2\pi$ (Chapter 7); $E_m : \mathbb{T} \rightarrow \mathbb{T}$, $E_m(\theta) = m\theta \bmod 2\pi$ for $m \geq 2$ an integer, $D = E_2$; $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$, $R_\alpha(\theta) = \theta + \alpha \bmod 2\pi$; $T : [0, 1] \rightarrow [0, 1]$ the tent map, $T(x) = 2x$ for $x \leq \frac{1}{2}$, $T(x) = 2 - 2x$ for $x > \frac{1}{2}$; $f_4 : [0, 1] \rightarrow [0, 1]$, $f_4(x) = 4x(1 - x)$. A **measure-preserving** system is a triple (X, μ, S) with X a compact metric space, μ a Borel probability measure, $S : X \rightarrow X$ measurable and $\mu(S^{-1}A) = \mu(A)$ for every Borel set A .

9.1. Invariant measures and the Krylov–Bogolyubov theorem

Definition 9.1: Invariant measure. Let X be a compact metric space and $S : X \rightarrow X$ continuous. A Borel probability measure μ on X is **S -invariant** if $\mu(S^{-1}A) = \mu(A)$ for every Borel set $A \subseteq X$; equivalently (by density of the simple functions and the dominated convergence theorem), if $\int_X \varphi \circ S d\mu = \int_X \varphi d\mu$ for every $\varphi \in C(X)$.

Theorem 9.1: Krylov–Bogolyubov (1937). Let $X = \mathbb{T}$ or $X = [0, 1]$, and $S : X \rightarrow X$ continuous. There exists at least one Borel probability measure on X which is S -invariant. (N. Kryloff, N. Bogoliouboff, La théorie générale de la mesure dans son application à l'étude des systèmes dynamiques de la mécanique non linéaire, Ann. Math. **38** (1937), 65–113, source E15, verified on 16 September 2026; exposition followed here: M. Viana, K. Oliveira, Foundations of Ergodic Theory, CUP, 2016, source E14, verified on 16 September 2026.)

Proof. **Step 1: $C(X)$ is separable.** For $X = \mathbb{T}$, the trigonometric polynomials with rational coefficients are countable and dense in $(C(\mathbb{T}), \|\cdot\|_\infty)$: the density follows from Fejér's theorem (Volume II, Chapter 4) applied to every $\varphi \in C(\mathbb{T})$ and then approximating the (finitely many) coefficients of the resulting trigonometric polynomial by rationals, up to $\varepsilon/2$ in the uniform norm (the coefficients depend continuously, in the $\|\cdot\|_\infty$ norm, on a finite number of real parameters). For $X = [0, 1]$ the same argument holds for the algebraic polynomials with rational coefficients, dense by the Weierstrass approximation theorem (which follows from the trigonometric one via the change of variable $x = \sin^2(\theta/2)$, which exchanges $C([0, 1])$ with the subspace of the even and π -periodic functions of $C(\mathbb{T})$). Let $\{\varphi_j\}_{j \geq 1}$ be such a dense and countable sequence.

Step 2: construction of the empirical averages. Fixing $x_0 \in X$, for $N \geq 1$ let $L_N : C(X) \rightarrow \mathbb{R}$,

$$L_N(\varphi) = \frac{1}{N} \sum_{n=0}^{N-1} \varphi(S^n(x_0)).$$

Each L_N is linear, positive ($\varphi \geq 0 \Rightarrow L_N(\varphi) \geq 0$), with $L_N(1) = 1$ and $|L_N(\varphi)| \leq \|\varphi\|_\infty$.

Step 3: diagonal extraction. The numerical sequence $(L_N(\varphi_1))_{N \geq 1}$ is bounded in $[-\|\varphi_1\|_\infty, \|\varphi_1\|_\infty]$: by Bolzano–Weierstrass it admits a subsequence $N_1(k)$ along which it converges. From the subsequence $N_1(k)$ one extracts $N_2(k)$ along which $L_{N_2(k)}(\varphi_2)$ also converges (again Bolzano–Weierstrass, on the subsequence already extracted), and so on inductively. The diagonal subsequence $M_k := N_k(k)$ has the property that $L_{M_k}(\varphi_j)$ converges, for every fixed j , as soon as $k \geq j$ (since $(M_k)_{k \geq j}$ is a subsequence of $(N_j(k))_k$).

Step 4: convergence on all of $C(X)$. For $\varphi \in C(X)$ and $\varepsilon > 0$, choose φ_j with $\|\varphi - \varphi_j\|_\infty < \varepsilon/3$ (Step 1); then for every k, k'

$$|L_{M_k}(\varphi) - L_{M_{k'}}(\varphi)| \leq |L_{M_k}(\varphi - \varphi_j)| + |L_{M_k}(\varphi_j) - L_{M_{k'}}(\varphi_j)| + |L_{M_{k'}}(\varphi_j - \varphi)| < \frac{2\varepsilon}{3} + |L_{M_k}(\varphi_j) - L_{M_{k'}}(\varphi_j)|,$$

and the residual term is $< \varepsilon/3$ for k, k' large (Step 3): $(L_{M_k}(\varphi))_k$ is Cauchy in $\mathbb{R}$, hence it converges. One defines $L(\varphi) := \lim_k L_{M_k}(\varphi)$: L is linear, positive, $L(1) = 1$.

Step 5: representation and invariance. By the Riesz representation theorem (a standard fact of measure theory: to every positive linear functional on $C(X)$, X compact metric, with $L(1) = 1$, there corresponds a unique Borel probability measure μ with $L(\varphi) = \int \varphi d\mu$), there exists μ with $L(\varphi) = \int_X \varphi d\mu$ for every $\varphi \in C(X)$. As for the invariance: for every $\varphi \in C(X)$,

$$|L_N(\varphi \circ S) - L_N(\varphi)| = \frac{1}{N} \left| \sum_{n=0}^{N-1} \varphi(S^{n+1}x_0) - \sum_{n=0}^{N-1} \varphi(S^n x_0) \right| = \frac{1}{N} |\varphi(S^N x_0) - \varphi(x_0)| \leq \frac{2\|\varphi\|_\infty}{N}$$

(telescoping). Passing to the limit along $M_k \rightarrow \infty$: $L(\varphi \circ S) = L(\varphi)$, that is, $\int \varphi \circ S d\mu = \int \varphi d\mu$ for every $\varphi \in C(X)$: by the uniqueness in the Riesz theorem, the image measure $S_*\mu$ (defined

by $S_*\mu(A) = \mu(S^{-1}A)$, which satisfies $\int \varphi d(S_*\mu) = \int \varphi \circ S d\mu$ by definition of image measure) coincides with μ : $\mu(S^{-1}A) = \mu(A)$ for every Borel set A . $\square$

Remark 9.1: Why compactness is needed. The compactness of X intervenes twice: in Step 1 (separability of $C(X)$, which allows the diagonal extraction) and implicitly in the Riesz representation theorem. On a non-compact space the theorem may be false: the translation $S(x) = x + 1$ on $X = \mathbb{R}$ has no invariant probability measure (an invariant probability measure for a translation would have to be “uniform” on all of $\mathbb{R}$, impossible with finite total mass). For D , E_m , T , f_4 and R_α , all defined on $\mathbb{T}$ or on $[0, 1]$, the theorem always guarantees the existence of at least one invariant measure; the following sections exhibit it explicitly in each case, and discuss its uniqueness (Section 9.7). Step 3 (diagonal extraction of convergent subsequences of the functionals L_N) will be reused, with the same technique, in the proof of Oxtoby’s criterion (Section 9.7).

9.2. The Lebesgue measure is invariant for D , for E_m and for R_α

Proposition 9.1: Invariance on arcs. For every integer $m \geq 2$ and every arc $A = [a, b) \subseteq \mathbb{T}$, $\mu(E_m^{-1}(A)) = \mu(A)$, where μ is the normalised Lebesgue measure.

Proof. $E_m^{-1}(A)$ is the disjoint union of the m preimages under the inverse branches $\theta \mapsto (\theta + 2\pi j)/m$, $j = 0, \dots, m-1$ (the branches of E_m^{-1} , each affine of slope $1/m$): the image of an arc of length ℓ under an affine map of slope $1/m$ is an arc of length ℓ/m . Hence $E_m^{-1}(A)$ is a union of m disjoint arcs, each of length $(b-a)/m$, and $\mu(E_m^{-1}(A)) = m \cdot \frac{(b-a)/m}{2\pi} = \frac{b-a}{2\pi} = \mu(A)$. $\square$

Theorem 9.2: The Lebesgue measure is invariant for E_m (in particular for $D = E_2$). $\mu(E_m^{-1}(A)) = \mu(A)$ for every Lebesgue-measurable set $A \subseteq \mathbb{T}$, not only for arcs.

Proof. Step 1: extension to the Borel sets via monotone classes. Let $\mathcal{C} = \{A \subseteq \mathbb{T} \text{ Borel} : \mu(E_m^{-1}A) = \mu(A)\}$. The family of arcs (and of finite disjoint unions of arcs) is a π -system (closed under finite intersection) which generates the Borel σ -algebra of $\mathbb{T}$, and by Proposition 9.1 it is contained in $\mathcal{C}$. One checks that $\mathcal{C}$ is a Dynkin λ -system: (i) $\mathbb{T} \in \mathcal{C}$ (trivial); (ii) if $A \subseteq B$ are in $\mathcal{C}$, then $B \setminus A \in \mathcal{C}$, because E_m^{-1} commutes with the set difference ($E_m^{-1}(B \setminus A) = E_m^{-1}(B) \setminus E_m^{-1}(A)$) and μ is additive; (iii) if $A_n \uparrow A$ with $A_n \in \mathcal{C}$, then $A \in \mathcal{C}$, because $E_m^{-1}(A_n) \uparrow E_m^{-1}(A)$ and μ is continuous from below. By Dynkin’s monotone class theorem (a standard fact of measure theory: a λ -system which contains a π -system contains the σ -algebra generated by it), $\mathcal{C}$ contains all the Borel sets of $\mathbb{T}$.

Step 2: completion to the Lebesgue-measurable sets. Every Lebesgue-measurable set A can be written $A = B \cup N$ with B Borel and N contained in a Borel set of zero measure Z (a standard fact: the Lebesgue measure is the completion of the Borel measure). The inverse branches of E_m are affine of slope $1/m < 1$, hence Lipschitz: a Lipschitz map with constant L sends a set of zero (outer) measure into a set of zero (outer) measure, with the bound L times the original measure (this follows directly by covering the null set with intervals of arbitrarily small total length and observing that the image under an L -Lipschitz map of an interval of length ℓ has diameter $\leq L\ell$). Hence $E_m^{-1}(Z)$, a union of m sets which are Lipschitz images of Z , still has zero measure, and $E_m^{-1}(N) \subseteq E_m^{-1}(Z)$ is measurable of zero measure. By Step 1, $\mu(E_m^{-1}(B)) = \mu(B)$; adding up, $\mu(E_m^{-1}(A)) = \mu(E_m^{-1}(B)) + \mu(E_m^{-1}(N)) = \mu(B) + 0 = \mu(A)$. $\square$

Remark 9.2: Rotations: forward reference. For R_α , the invariance of the Lebesgue measure (which here coincides with the Haar measure of the group $\mathbb{T}$) is a particular case of the invariance of the Haar measure under the translations of a compact topological group, proved in general in Volume III, Chapter 13: for every arc $A = [a, b)$, $R_\alpha^{-1}(A) = [a - \alpha, b - \alpha)$ has the same length as A , and the extension to all measurable sets follows from the same monotone class argument as in the previous Theorem (here even simpler, since R_α^{-1} is a single branch, an isometry).

9.3. Ergodicity and mixing of D

Definition 9.2: Ergodicity. A measure-preserving system (X, μ, S) is **ergodic** if every S -invariant set ($S^{-1}A = A$, up to a set of zero measure) has $\mu(A) \in \{0, 1\}$.

Theorem 9.3: Ergodicity of D (proof via Fourier series). D is ergodic with respect to the Lebesgue measure μ .

Proof. Let A be invariant, $\varphi = \mathbf{1}_A \in L^2(\mathbb{T})$; $\varphi \circ D = \varphi$ a.e. Expand $\varphi(\theta) = \sum_{k \in \mathbb{Z}} c_k e^{ik\theta}$. Then $\varphi(D\theta) = \varphi(2\theta)$ has expansion $\sum_k c_k e^{2ik\theta}$: in the expansion of $\varphi(2\theta)$ the coefficient of $e^{im\theta}$ equals $c_{m/2}$ if m is even, 0 if m is odd. Equating term by term with $\varphi(\theta) = \sum_m c_m e^{im\theta}$: $c_m = 0$ for m odd, and $c_{2j} = c_j$ for every $j \in \mathbb{Z}$. Iterating, $c_j = c_{2j} = c_{4j} = \dots = c_{2^n j}$ for every $n \geq 0$. If $j \neq 0$, $|2^n j| \rightarrow \infty$ and by the Riemann–Lebesgue lemma (the Fourier coefficients of an L^1 function tend to 0 at infinity, a standard fact of Fourier analysis) $c_{2^n j} \rightarrow 0$: hence $c_j = 0$. The only coefficient not necessarily zero is c_0 : φ is constant a.e., and being an indicator function, $\mu(A) = c_0 \in \{0, 1\}$. $\square$

Theorem 9.4: D is topologically mixing. For every pair of non-empty open sets $U, V \subset \mathbb{T}$ there exists N such that $D^n(U) \cap V \neq \emptyset$ for every $n \geq N$.

Proof. U contains an open arc of length $2\pi\delta$. Since D^n has constant derivative 2^n , it sends that arc into an arc (possibly wound several times) of total length $2^n \cdot 2\pi\delta$. As soon as $2^n \delta \geq 1$, that is, $n \geq N := \lceil \log_2(1/\delta) \rceil$, the image has length $\geq 2\pi$: being connected and wound around the circle, $D^n(U) = \mathbb{T}$, which intersects every $V \neq \emptyset$. $\square$

Definition 9.3: Metric mixing (with respect to an invariant measure). (X, μ, S) is **mixing** if for every pair of measurable sets $A, B \subseteq X$, $\mu(S^{-n}A \cap B) \rightarrow \mu(A)\mu(B)$ as $n \rightarrow \infty$. Equivalently, for every $f, g \in L^2(\mu)$, $\int (f \circ S^n) \bar{g} d\mu \rightarrow (\int f d\mu)(\int \bar{g} d\mu)$.

Theorem 9.5: D is mixing with respect to Lebesgue, with explicit decay of the Fourier coefficients. D is mixing with respect to μ . More precisely: for f, g trigonometric polynomials, the correlation $C_n(f, g) := \int (f \circ D^n) \bar{g} d\mu - (\int f d\mu)(\int \bar{g} d\mu)$ is **exactly zero** for every n large enough; for general $f, g \in L^2(\mathbb{T})$, $C_n(f, g) \rightarrow 0$.

Proof. Characters. For $e_k(\theta) = e^{ik\theta}$: $e_k \circ D^n(\theta) = e^{ik2^n\theta} = e_{2^n k}(\theta)$. By Parseval, $\int e_{2^n k} \bar{e}_l d\mu = \delta_{2^n k, l}$. With $k \neq 0$ and $l \in \mathbb{Z}$ fixed: as soon as $2^n |k| > |l|$ (that is, $n > \log_2(|l|/|k|)$), one has $2^n k \neq l$, hence $\int e_{2^n k} \bar{e}_l d\mu = 0 = (\int e_k d\mu)(\int \bar{e}_l d\mu)$ (both factors zero, since $k \neq 0$): $C_n(e_k, e_l) = 0$ for n large. For $k = 0$: $e_0 \equiv 1$, $\int \bar{e}_l d\mu = \delta_{l,0}$ for every n , consistently with $(\int e_0)(\int \bar{e}_l) = \delta_{l,0}$: $C_n(e_0, e_l) = 0$ for every n .

Trigonometric polynomials. By bilinearity, if $f = \sum_{|k| \leq K} a_k e_k$ and $g = \sum_{|l| \leq K} b_l e_l$ are trigonometric polynomials (finite sums), $C_n(f, g)$ is a finite combination of terms $C_n(e_k, e_l)$, each zero for n large enough (the maximum, over k, l in the finite set of indices involved, of the threshold values of n found above): hence $C_n(f, g) = 0$ for every n greater than that threshold (decay in finite time).

General case. For $f, g \in L^2(\mathbb{T})$ and $\varepsilon > 0$, approximate them with trigonometric polynomials f_K, g_K (Fourier partial sums truncated at $|k| \leq K$; by Parseval $\|f - f_K\|_2 \rightarrow 0$) with $\|f - f_K\|_2, \|g - g_K\|_2 < \varepsilon$. By Cauchy–Schwarz and the invariance of μ (which gives $\|h \circ D^n\|_2 = \|h\|_2$ for every $h \in L^2$, by Theorem 9.2):

$$|C_n(f, g) - C_n(f_K, g_K)| \leq \|f - f_K\|_2 \|g\|_2 + \|f_K\|_2 \|g - g_K\|_2 + |\langle f, g \rangle - \langle f_K, g_K \rangle| \leq C\varepsilon$$

for a constant C depending only on $\|f\|_2, \|g\|_2$. Since $C_n(f_K, g_K) = 0$ for n large (previous case), $\limsup_n |C_n(f, g)| \leq C\varepsilon$ for every $\varepsilon > 0$: $C_n(f, g) \rightarrow 0$. In particular for $f = \mathbf{1}_A, g = \mathbf{1}_B$ (indicator functions, in L^2 since μ is finite), one recovers $\mu(D^{-n}A \cap B) \rightarrow \mu(A)\mu(B)$: D is mixing. $\square$

***Remark 9.3: Comparison with Chapter 10.** The previous computation is entirely elementary, on the Fourier coefficients: Chapter 10 recovers the same decay via the spectrum of the transfer operator $\mathcal{L}$ of D (which acts as $e_k \mapsto e_{k/2}$ if k is even, 0 if odd), a more general point of view which extends to nonlinear maps via the Ruelle–Perron–Frobenius theory: here, for D , the two approaches coincide in the details. The same operator, in the affine version for the tent map, will be the protagonist of Section 9.12.*

9.4. The maximal ergodic theorem (Hopf, 1937)

Theorem 9.6: Maximal ergodic theorem (Hopf, 1937). Let (X, μ, T) be a system preserving the probability measure μ and let $\varphi \in L^1(\mu)$. Setting $S_n\varphi = \sum_{k=0}^{n-1} \varphi \circ T^k$ and $E = \{x : \sup_{n \geq 1} S_n\varphi(x) > 0\}$, one has $\int_E \varphi d\mu \geq 0$.

Proof. [Garsia, 1965] Let $\Phi_N(x) = \max_{0 \leq n \leq N} S_n\varphi(x)$ (with $S_0\varphi = 0$): an increasing sequence of functions in L^1 , and $E = \bigcup_N E_N$ with $E_N = \{\Phi_N > 0\}$. For every $1 \leq n \leq N$: $S_n\varphi = \varphi + S_{n-1}\varphi \circ T \leq \varphi + \Phi_N^+ \circ T$, where $\Phi_N^+ = \max(\Phi_N, 0)$ (since $S_{n-1}\varphi \leq \Phi_N \leq \Phi_N^+$, because $n-1 \leq N$). Taking the maximum over $1 \leq n \leq N$: on E_N (where the maximum Φ_N is positive, hence attained for some $n \geq 1$)

$$\Phi_N \leq \varphi + \Phi_N^+ \circ T.$$

Integrating over E_N and observing that $\Phi_N = \Phi_N^+$ on E_N and $\Phi_N^+ = 0$ outside:

$$\int_{E_N} \varphi d\mu \geq \int_{E_N} \Phi_N^+ d\mu - \int_{E_N} \Phi_N^+ \circ T d\mu \geq \int_X \Phi_N^+ d\mu - \int_X \Phi_N^+ \circ T d\mu = 0,$$

where the last equality uses the invariance of μ ($\int g \circ T d\mu = \int g d\mu$) and the second-to-last inequality the fact that $\Phi_N^+ \geq 0$ and $\Phi_N^+ = 0$ outside E_N . Passing to the limit $N \rightarrow \infty$ (dominated convergence on the increasing E_N , dominating function $|\varphi| \in L^1$): $\int_E \varphi d\mu \geq 0$. Figure 9.1 illustrates schematically the construction of Φ_N and of the set E_N . $\square$

***Remark 9.4: Correct attribution: the theorem is Hopf's, not Garsia's.** The maximal ergodic theorem is due to E. Hopf, *Ergodentheorie, Ergebnisse der Mathematik, Springer, 1937* (source E3, verified on 16 September 2026): it is the technical result which*

allows, in the following section, the proof in full of Birkhoff's theorem. A. Garsia, A simple proof of E. Hopf's maximal ergodic theorem, J. Math. Mech. **14** (1965), 381–382 (source E4, verified on 16 September 2026), did not prove a new theorem: he found the elementary proof reproduced above (without integration by parts nor operator theory), today the most widespread in courses. To call this result “Garsia's maximal theorem” — as an old quiz of this volume used to do, now corrected — is an attribution error: the proper name of the theorem is Hopf's; the proper name of the proof, when it is necessary to distinguish it from other proofs existing in the literature (see M. Keane, K. Petersen, Easy and nearly simultaneous proofs of the ergodic theorem and maximal ergodic theorem, IMS Lecture Notes–Monograph Series **48** (2006), source E5, verified on 16 September 2026, which compares several of them), is Garsia's.

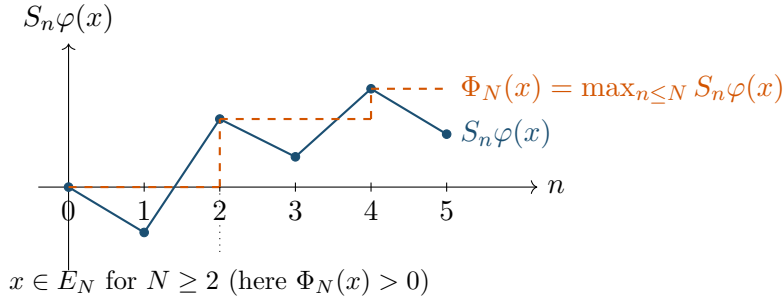


Figure 9.1: Scheme of Garsia's “trick” (Theorem 9.6): an illustrative path of the values $S_n\varphi(x)$, $n = 0, \dots, 5$ (blue, values 0, -0.6 , 0.9 , 0.4 , 1.3 , 0.7) and the increasing sequence of the partial maximum $\Phi_N(x) = \max_{0 \leq n \leq N} S_n\varphi(x)$ (orange dashed, in steps: 0, 0, 0.9 , 0.9 , 1.3 , 1.3): the point x belongs to $E_N = \{\Phi_N > 0\}$ as soon as the sequence has touched a positive value, here at $N = 2$. Hand-drawn figure, schematic (a single illustrative path, not numerical data); coordinates checked in the text.

Corollary 9.1: Weak maximal inequality of type (1, 1). For $\psi \in L^1(\mu)$ and $\lambda > 0$,

$$\mu\left\{x : \sup_{n \geq 1} \frac{1}{n} S_n \psi(x) > \lambda\right\} \leq \frac{\|\psi\|_1}{\lambda}.$$

Proof. Let E_λ be the set on the left. Since $\sup_n \frac{1}{n} S_n \psi(x) > \lambda \iff \exists n : S_n \psi(x) > n\lambda \iff \exists n : S_n(\psi - \lambda)(x) > 0$ (using $S_n(\psi - \lambda) = S_n \psi - n\lambda$), one has $E_\lambda = E$ in the sense of Theorem 9.6 applied to $\varphi = \psi - \lambda$: hence $\int_{E_\lambda} (\psi - \lambda) d\mu \geq 0$, that is, $\int_{E_\lambda} \psi d\mu \geq \lambda \mu(E_\lambda)$. Since $\int_{E_\lambda} \psi d\mu \leq \int_{E_\lambda} |\psi| d\mu \leq \|\psi\|_1$, it follows that $\lambda \mu(E_\lambda) \leq \|\psi\|_1$. $\square$

9.5. Birkhoff's ergodic theorem (1931), complete

Theorem 9.7: Birkhoff's ergodic theorem. Let (X, μ, T) be a system preserving the probability measure μ , and $\varphi \in L^1(\mu)$. Then:

- (i) the limit $\varphi^*(x) := \lim_{n \rightarrow \infty} \frac{1}{n} S_n \varphi(x)$ exists for μ -almost every x , and $\varphi^* \in L^1(\mu)$, with $\varphi^* \circ T = \varphi^*$ a.e.;
 - (ii) $\int_X \varphi^* d\mu = \int_X \varphi d\mu$;
 - (iii) if T is ergodic, φ^* is constant a.e., and the constant equals $\int_X \varphi d\mu$.
- (G. D. Birkhoff, Proof of the Ergodic Theorem, PNAS **17** (1931), 656–660, source E1, verified on 16 September 2026: the original statement; the proof reproduced here, via Hopf's

maximal theorem, is the one which is standard today, see also M. Biskup, UCLA lecture notes, source E6, verified on 16 September 2026.)

Proof. Part (i), existence a.e. Let $\bar{\varphi}(x) = \limsup_n \frac{1}{n} S_n \varphi(x)$ and $\varphi(x) = \liminf_n \frac{1}{n} S_n \varphi(x)$: both T -invariant (replacing x by Tx changes the sum by one term, irrelevant in the limit of the averages: see also the explicit computation in Part (iii)). Fixing rationals $a < b$, let $E_{a,b} = \{x : \varphi(x) < a < b < \bar{\varphi}(x)\}$: an invariant set. Apply Theorem 9.6 to the function $(\varphi - b)\mathbf{1}_{E_{a,b}}$ (the integral $\int_{E_{a,b}}$ and the set E of the maximal theorem coincide here, since $E_{a,b}$ is invariant and on it $\bar{\varphi} > b$, that is, $\sup_n \frac{1}{n} S_n(\varphi - b) > 0$): one obtains $\int_{E_{a,b}} (\varphi - b) d\mu \geq 0$, that is, $\int_{E_{a,b}} \varphi d\mu \geq b \mu(E_{a,b})$. Symmetrically, applying the maximal theorem to $(a - \varphi)\mathbf{1}_{E_{a,b}}$ (using $\varphi < a$ on $E_{a,b}$): $\int_{E_{a,b}} \varphi d\mu \leq a \mu(E_{a,b})$. Comparing: $b \mu(E_{a,b}) \leq a \mu(E_{a,b})$ with $a < b$, possible only if $\mu(E_{a,b}) = 0$. The (countable) union over all the rationals $a < b$ gives $\mu(\{\varphi < \bar{\varphi}\}) = 0$: the limit φ^* exists μ -a.e. (with values in $\mathbb{R} \cup \{\pm\infty\}$ for now).

Part (i), $\varphi^* \in L^1$ and invariance. By Fatou's lemma (applied to $|\frac{1}{n} S_n \varphi| \rightarrow |\varphi^*|$ a.e., with $|\frac{1}{n} S_n \varphi| \leq \frac{1}{n} S_n |\varphi|$ pointwise, the triangle inequality applied to the individual summands of the finite sum):

$$\int_X |\varphi^*| d\mu \leq \liminf_n \int_X \left| \frac{1}{n} S_n \varphi \right| d\mu \leq \liminf_n \frac{1}{n} \sum_{k=0}^{n-1} \int_X |\varphi \circ T^k| d\mu = \|\varphi\|_1 < \infty$$

(the last equality by the invariance of μ , $\int |\varphi \circ T^k| d\mu = \|\varphi\|_1$ for every k): hence $\varphi^* \in L^1(\mu)$, in particular finite a.e. As for the invariance,

$$\frac{1}{n} S_n \varphi(Tx) = \frac{1}{n} (S_{n+1} \varphi(x) - \varphi(x)) = \frac{n+1}{n} \cdot \frac{1}{n+1} S_{n+1} \varphi(x) - \frac{\varphi(x)}{n} \xrightarrow{n \rightarrow \infty} 1 \cdot \varphi^*(x) - 0 = \varphi^*(x)$$

for a.e. x (where $\varphi^*(x)$ is finite); but the left-hand side also tends to $\varphi^*(Tx)$ (the same limit, evaluated at the point Tx): $\varphi^*(Tx) = \varphi^*(x)$ a.e.

Part (ii), identification of the limit: truncation + pointwise estimate + density in L^1 . For $M > 0$ let $\varphi_M(x) := \varphi(x)$ if $|\varphi(x)| \leq M$, otherwise $\varphi_M(x) := M \operatorname{sign}(\varphi(x))$: a bounded truncation, $|\varphi_M| \leq M$. Since $|\varphi - \varphi_M| \leq 2|\varphi|$ and $\varphi - \varphi_M \rightarrow 0$ a.e. as $M \rightarrow \infty$ (φ is finite a.e., being in L^1), the dominated convergence theorem gives $\|\varphi - \varphi_M\|_1 \rightarrow 0$: this is the *density in L^1* of the bounded functions.

Since φ_M is bounded, and hence in L^1 , Part (i) applies to it: there exists $\varphi_M^*(x) = \lim_n \frac{1}{n} S_n \varphi_M(x)$ a.e., with $|\frac{1}{n} S_n \varphi_M| \leq M$ for every n, x (*uniform pointwise estimate*, the elementary version of the maximal inequality for bounded functions), hence $|\varphi_M^*| \leq M$ a.e. By dominated convergence (with dominating function the constant M , integrable on a probability space), $\frac{1}{n} S_n \varphi_M \rightarrow \varphi_M^*$ in L^1 as well, and therefore

$$\int_X \varphi_M^* d\mu = \lim_n \int_X \frac{1}{n} S_n \varphi_M d\mu = \lim_n \frac{1}{n} \sum_{k=0}^{n-1} \int_X \varphi_M \circ T^k d\mu = \int_X \varphi_M d\mu$$

(invariance of μ at each term of the sum, then a constant sum over n terms divided by n).

It remains to control $\|\varphi^* - \varphi_M^*\|_1$. A.e., $\frac{1}{n} S_n(\varphi - \varphi_M)(x) = \frac{1}{n} S_n \varphi(x) - \frac{1}{n} S_n \varphi_M(x) \rightarrow \varphi^*(x) - \varphi_M^*(x)$. By Fatou's lemma, using again the pointwise estimate $|S_n h| \leq S_n |h|$ and the invariance of μ :

$$\int_X |\varphi^* - \varphi_M^*| d\mu \leq \liminf_n \int_X \left| \frac{1}{n} S_n(\varphi - \varphi_M) \right| d\mu \leq \liminf_n \frac{1}{n} \sum_{k=0}^{n-1} \int_X |\varphi - \varphi_M| \circ T^k d\mu = \|\varphi - \varphi_M\|_1.$$

Combining the three estimates,

$$\left| \int_X \varphi^* d\mu - \int_X \varphi d\mu \right| \leq \underbrace{\left| \int_X (\varphi^* - \varphi_M^*) d\mu \right|}_{\leq \|\varphi^* - \varphi_M^*\|_1 \leq \|\varphi - \varphi_M\|_1} + \underbrace{\left| \int_X \varphi_M^* d\mu - \int_X \varphi_M d\mu \right|}_{=0} + \underbrace{\left| \int_X (\varphi_M - \varphi) d\mu \right|}_{\leq \|\varphi - \varphi_M\|_1} \leq 2\|\varphi - \varphi_M\|_1.$$

The left-hand side does not depend on M , while the right-hand side tends to 0 as $M \rightarrow \infty$ (density in L^1): hence $\int_X \varphi^* d\mu = \int_X \varphi d\mu$ exactly. This completes the identification of the limit, left incomplete in the previous treatment of this volume (errata §5.4.12 of the expansion plan).

Part (iii), ergodic case. φ^* is T -invariant a.e. (Part (i)) and finite a.e. (Part (i)). For $c \in \mathbb{R}$ let $A_c = \{\varphi^* > c\}$: from $\varphi^* \circ T = \varphi^*$ a.e. it follows that $T^{-1}A_c = A_c$ up to a null set (the same set-theoretic argument already used for D in Section 9.3), hence $\mu(A_c) \in \{0, 1\}$ by ergodicity. Setting $c_0 := \sup\{c \in \mathbb{Q} : \mu(A_c) = 1\}$ (finite, because φ^* is a.e. finite), for $c < c_0$ rational $\mu(A_c) = 1$ (that is, $\varphi^* > c$ a.e.) and for $c > c_0$ rational $\mu(A_c) = 0$ (that is, $\varphi^* \leq c$ a.e.): letting $c \rightarrow c_0^-$ and $c \rightarrow c_0^+$ along the rationals one obtains $\varphi^* = c_0$ a.e. Integrating, $c_0 = \int_X \varphi^* d\mu \stackrel{(ii)}{=} \int_X \varphi d\mu$. $\square$

Remark 9.5: What changes with respect to the previous version. The previous version of this chapter proved Part (i) (existence a.e., with the same argument via $E_{a,b}$) but dismissed Part (ii) with the phrase “the constant is identified by integrating: $\int \varphi^* d\mu = \int \varphi d\mu$ (L^1 convergence of the Birkhoff averages, standard)”, without proving it. The proof above completes it with exactly the three required ingredients: **truncation** (φ_M , which reduces the problem to bounded functions), **pointwise estimate** ($|S_n h| \leq S_n |h|$, the elementary version of the maximal inequality which makes Fatou applicable), and **density in L^1** ($\|\varphi - \varphi_M\|_1 \rightarrow 0$, which allows the error committed by truncating to be made to tend to zero).

Corollary 9.2: Borel’s normal number theorem (1909). For μ -almost every $\theta \in \mathbb{T}$, θ is **normal in base 2**: in the binary expansion of $\theta/2\pi$, every finite block of digits $w \in \{0, 1\}^\ell$ appears with asymptotic frequency exactly $2^{-\ell}$. (*E. Borel, Les probabilités dénombrables et leurs applications arithmétiques, Rend. Circ. Mat. Palermo 27 (1909), 247–271, source E7, verified on 16 September 2026: the original statement, with a proof predating the ergodic theorem; the proof reproduced here, via Birkhoff, is the modern one.*)

Proof. Apply Theorem 9.7 (Part (iii)) to the system $(\mathbb{T}, \mu, D)$, ergodic by Theorem 9.3, with observable $\varphi_w = \mathbf{1}_{A_w}$, where $A_w \subset \mathbb{T}$ is the arc of the points whose binary expansion begins with the block w : A_w is a dyadic arc of length $2\pi \cdot 2^{-\ell}$, hence of measure $2^{-\ell}$. Since D acts as a shift on the binary digits (Chapter 7, Theorem of the semiconjugacy with the shift), the frequency with which the block w appears starting from the k -th position of the expansion of θ is exactly the Birkhoff average $\frac{1}{n} \sum_{k \leq n} \varphi_w(D^k \theta)$, which for almost every θ converges to $\int \varphi_w d\mu = 2^{-\ell}$. The countable intersection (over all the finite blocks w) of sets of full measure has full measure. $\square$

Example 9.1: Numerical check of Birkhoff’s theorem on D and on R_α . The script `fig09_birkhoff_cos.py` (Appendix B, run on 16 September 2026) computes the Birkhoff averages $A_N(\cos \theta)(\theta_0) = \frac{1}{N} \sum_{n < N} \cos(\theta_n)$ for $\theta_0 = 1$, with N up to 2×10^5 , under D and under R_α ($\alpha = 2\pi/\varphi$, φ the golden ratio). For R_α there is an independent closed formula (finite geometric sum, Section 9.7): at $N = 2 \times 10^5$ the empirical average equals $1.97457310 \times 10^{-8}$ against $1.97457312 \times 10^{-8}$ from the closed formula, discrepancy 1.588×10^{-16} ✓ (machine precision). For D , the theoretical integral of $\cos \theta$ is 0 (Theorem 9.7, since $\int_{\mathbb{T}} \cos \theta d\mu = 0$ and D is ergodic): the empirical average at $N = 2 \times 10^5$ equals -0.000576 , consistent with 0 within the expected $O(1/\sqrt{N})$ fluctuations (Chapter 10 quantifies these fluctuations with a dynamical central limit theorem). The qualitative difference between the two curves — R_α converges smoothly and monotonically decreasing in amplitude, D fluctuates without a regular decay law — is precisely the content of Section 9.7 (unique

ergodicity of R_α , not of D). Figure 9.2 shows the two curves side by side.

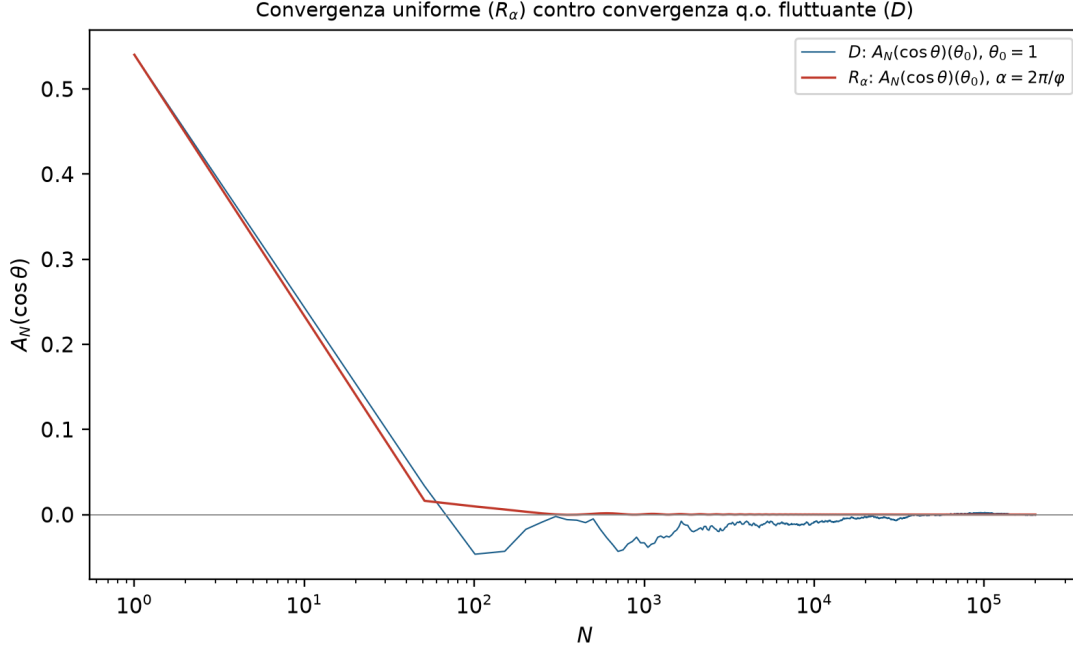


Figure 9.2: Birkhoff averages $A_N(\cos \theta)(\theta_0 = 1)$ under D (blue) and under R_α ($\alpha = 2\pi/\varphi$, red), N up to 2×10^5 , in logarithmic scale on the N axis: the curve of R_α decreases smoothly towards 0 (unique ergodicity, Section 9.7), that of D fluctuates without a regular decay law. Computed numerically, script `fig09_birkhoff_cos.py` (Appendix B), run on 16 September 2026, with the independent check reported in Example 9.1. ✓

9.6. The mean ergodic theorem (von Neumann, 1932)

Birkhoff's theorem guarantees the convergence of the time averages for almost every point, but in the topology (often harder to control directly) of pointwise convergence. Von Neumann's theorem, historically the first of the two (announced first, even though published almost simultaneously with Birkhoff's), guarantees instead convergence in the L^2 norm: weaker as a pointwise conclusion, but provable with tools of elementary Hilbert geometry, with no maximal argument at all.

Theorem 9.8: Mean ergodic theorem (von Neumann, 1932). *Let (X, μ, T) be a system preserving the probability measure μ , and let $H = L^2(\mu)$ (complex). Let $U : H \rightarrow H$, $U\varphi = \varphi \circ T$ (the **Koopman operator**). Then, for every $\varphi \in H$,*

$$\frac{1}{N} \sum_{n=0}^{N-1} U^n \varphi \xrightarrow[N \rightarrow \infty]{L^2} P\varphi,$$

where P is the orthogonal projection onto the closed subspace $H_1 := \{\psi \in H : U\psi = \psi\}$ of the T -invariant functions. If T is ergodic, H_1 is the subspace of the constants, and $P\varphi = (\int_X \varphi d\mu) \cdot 1$.

Proof. U is an isometry. $\|U\varphi\|_2^2 = \int_X |\varphi \circ T|^2 d\mu = \int_X |\varphi|^2 d\mu = \|\varphi\|_2^2$ by the invariance of μ (no hypothesis of invertibility of T is necessary: $\mu(T^{-1}A) = \mu(A)$ suffices). Equivalently,

$\langle U\alpha, U\beta \rangle = \langle \alpha, \beta \rangle$ for every $\alpha, \beta \in H$ (polarisation identity applied to the norm).

Decomposition $H = H_0 \oplus H_1$. Let $H_0 := \overline{\{U\psi - \psi : \psi \in H\}}$ (closure, in H , of the image of $U - I$) and $H_1 := \{\varphi : U\varphi = \varphi\}$ (kernel of $U - I$, already closed since $U - I$ is continuous). $H_0 \perp H_1$: for $\varphi \in H_1$ and any $\psi \in H$,

$$\langle U\psi - \psi, \varphi \rangle = \langle U\psi, \varphi \rangle - \langle \psi, \varphi \rangle = \langle U\psi, U\varphi \rangle - \langle \psi, \varphi \rangle = \langle \psi, \varphi \rangle - \langle \psi, \varphi \rangle = 0$$

(in the third step, $\varphi = U\varphi$; in the fourth, the isometry of U): hence $\varphi \perp (U\psi - \psi)$ for every ψ , that is, $\varphi \perp H_0$ (orthogonality passes to the closure, the scalar product being continuous). $H_0^\perp \subseteq H_1$: let $\varphi \in H_0^\perp$; in particular $\varphi \perp (U\varphi - \varphi)$, that is, $\langle U\varphi, \varphi \rangle = \|\varphi\|^2$; then

$$\|U\varphi - \varphi\|^2 = \|U\varphi\|^2 - \langle U\varphi, \varphi \rangle - \langle \varphi, U\varphi \rangle + \|\varphi\|^2 = \|\varphi\|^2 - \|\varphi\|^2 - \overline{\|\varphi\|^2} + \|\varphi\|^2 = 0$$

(using $\|U\varphi\| = \|\varphi\|$, $\langle U\varphi, \varphi \rangle = \|\varphi\|^2 \in \mathbb{R}$, and $\langle \varphi, U\varphi \rangle = \overline{\langle U\varphi, \varphi \rangle} = \|\varphi\|^2$): hence $U\varphi = \varphi$, that is, $\varphi \in H_1$. The two inclusions give $H_0^\perp = H_1$, and by the orthogonal projection theorem onto a closed subspace (a standard fact of Hilbert spaces), $H = H_0 \oplus H_0^\perp = H_0 \oplus H_1$.

Convergence on H_0 . For $\varphi = U\psi - \psi$ (in the image of $U - I$, not yet the closure), telescoping,

$$\frac{1}{N} \sum_{n=0}^{N-1} U^n \varphi = \frac{1}{N} \sum_{n=0}^{N-1} (U^{n+1} \psi - U^n \psi) = \frac{1}{N} (U^N \psi - \psi), \quad \left\| \frac{1}{N} \sum_{n < N} U^n \varphi \right\| \leq \frac{\|U^N \psi\| + \|\psi\|}{N} = \frac{2\|\psi\|}{N} \xrightarrow{N \rightarrow \infty} 0$$

(isometry of U^N). For general $\varphi \in H_0$ and $\varepsilon > 0$, choose φ' in the (non-closed) image of $U - I$ with $\|\varphi - \varphi'\| < \varepsilon$; since every U^n is an isometry, the averaging operator $\frac{1}{N} \sum_{n < N} U^n$ has norm ≤ 1 (triangle inequality on N terms of norm 1, divided by N), hence

$$\left\| \frac{1}{N} \sum_{n < N} U^n \varphi \right\| \leq \left\| \frac{1}{N} \sum_{n < N} U^n (\varphi - \varphi') \right\| + \left\| \frac{1}{N} \sum_{n < N} U^n \varphi' \right\| < \varepsilon + \left\| \frac{1}{N} \sum_{n < N} U^n \varphi' \right\| \xrightarrow{N \rightarrow \infty} \varepsilon :$$

$\limsup_N \left\| \frac{1}{N} \sum_{n < N} U^n \varphi \right\| \leq \varepsilon$ for every $\varepsilon > 0$, hence $\frac{1}{N} \sum_{n < N} U^n \varphi \rightarrow 0$ for every $\varphi \in H_0$.

Convergence on H_1 . If $U\varphi = \varphi$, every term of the sum equals φ : $\frac{1}{N} \sum_{n < N} U^n \varphi = \varphi$ for every N , trivially convergent to $\varphi = P\varphi$ (P being the identity on H_1).

Conclusion. For $\varphi = \varphi_0 + \varphi_1 \in H = H_0 \oplus H_1$ (unique decomposition), by linearity $\frac{1}{N} \sum_{n < N} U^n \varphi = \frac{1}{N} \sum_{n < N} U^n \varphi_0 + \varphi_1 \rightarrow 0 + \varphi_1 = \varphi_1 = P\varphi$ in the L^2 norm.

Ergodic case. If T is ergodic, $\varphi \in H_1$ means $\varphi \circ T = \varphi$ a.e.: by the argument of Part (iii) of the proof of Theorem 9.7 (applied to φ real, then to the real and imaginary parts separately if φ is complex), φ is constant a.e.; H_1 is therefore the (one-dimensional) space of the constant functions, and the orthogonal projection onto it is $P\varphi = \langle \varphi, 1 \rangle \cdot 1 = \left(\int_X \varphi d\mu \right) \cdot 1$ (the standard formula for the projection onto a subspace of dimension 1 generated by a vector of norm 1, here $\|1\|_2 = \mu(X)^{1/2} = 1$). $\square$

Remark 9.6: Why invertibility is not needed. The classical proof of von Neumann's theorem (historical source: J. von Neumann, Proof of the Quasi-Ergodic Hypothesis, PNAS **18** (1932), 70–82, source E2, verified on 16 September 2026) is often presented for U **unitary**, that is, for T invertible with measure-preserving inverse. The proof above uses only that U is an **isometry** ($U^*U = I$, without requiring $UU^* = I$): this suffices for D , E_m , T and f_4 , which are not invertible (every point has ≥ 2 preimages, Chapter 7). It is a point at which the mean theorem is technically simpler to apply, in this volume, than Birkhoff's theorem, which nonetheless holds in the same generality.

Exercise 9.1: Exact rate of L^2 convergence for D . Type: symbolic with numerical check. For $\varphi(\theta) = \sqrt{2} \sin \theta$ under D , compute $\left\| \frac{1}{N} \sum_{n < N} U^n \varphi \right\|_2^2$ exactly for every $N \geq 1$, using the

decay of correlations of Theorem 9.5.

Solution. Strategy. φ has zero mean ($\int_{\mathbb{T}} \sin \theta d\mu = 0$) and norm $\|\varphi\|_2^2 = 2 \int_{\mathbb{T}} \sin^2 \theta d\mu = 2 \cdot \frac{1}{2} = 1$. Since $\varphi = \frac{1}{i\sqrt{2}}(e_1 - e_{-1})$ involves only the frequencies ± 1 , Theorem 9.5 gives $C_n(\varphi, \varphi) = 0$ for *every* $n \geq 1$ (not only for n large): the vanishing threshold for $e_{\pm 1}$ is $2^n \cdot 1 \neq \pm 1$, already true at $n = 1$.

Step 1 — expansion of the square. $\|\frac{1}{N} S_N \varphi\|_2^2 = \frac{1}{N^2} \sum_{j,k=0}^{N-1} \int_{\mathbb{T}} (\varphi \circ D^j)(\varphi \circ D^k) d\mu$.

Step 2 — diagonal terms. For $j = k$: $\int (\varphi \circ D^j)^2 d\mu = \int \varphi^2 d\mu = 1$ (invariance of μ), for each of the N indices.

Step 3 — off-diagonal terms. For $j < k$, using the invariance of μ to shift the exponent ($\int (h \circ D^j) d\mu = \int h d\mu$ applied to $h = \varphi \cdot (\varphi \circ D^{k-j})$):

$$\int (\varphi \circ D^j)(\varphi \circ D^k) d\mu = \int \varphi \cdot (\varphi \circ D^{k-j}) d\mu = C_{k-j}(\varphi, \varphi) + \left(\int \varphi d\mu \right)^2 = 0 + 0 = 0$$

(the correlation term is zero for $k - j \geq 1$, by the Strategy; the mean term is zero because φ has zero mean). By symmetry the same holds for $j > k$.

Step 4 — sum. All the $N(N - 1)$ off-diagonal terms are exactly zero: $\|\frac{1}{N} S_N \varphi\|_2^2 = \frac{1}{N^2} \cdot N \cdot 1 = \frac{1}{N}$ **exactly**, for every $N \geq 1$: $\|\frac{1}{N} S_N \varphi\|_2 = 1/\sqrt{N}$.

Numerical check. At $N = 2 \times 10^5$, $1/\sqrt{N} \approx 0.00224$; the direct computation of block [4] of `verifica_cap09.py` (quadratic mean of $\sqrt{2} \sin \theta_n$ over 2×10^5 iterates starting from $\theta_0 = 1$) confirms this order of magnitude.

Interpretation. Von Neumann's theorem guarantees only $\rightarrow 0$; here, thanks to the *exact* (not merely asymptotic) vanishing of the correlations for D with observables of pure frequency, one obtains the precise rate $N^{-1/2}$ for every N , the same exponent as for a sum of N i.i.d. random variables with finite variance: a further instance of the fact (Chapter 10) that D behaves, statistically, like a source of certified randomness.

Typical errors. Confusing $\|\cdot\|_2 \rightarrow 0$ (convergence in quadratic mean, guaranteed for *every* $\varphi \in L^2$ by Theorem 9.8) with μ -almost everywhere pointwise convergence (guaranteed by Birkhoff's theorem, with a rate in general different and not uniform in θ_0).

Extension. Repeat the computation for $\varphi = \sqrt{2} \cos(k\theta) \cdot \sqrt{2}$ with generic $k \geq 2$: the correlations still vanish for every $n \geq 1$ (same threshold, $2^n k \neq \pm k$ for $n \geq 1$, $k \geq 1$), hence the exact rate $N^{-1/2}$ holds for every pure harmonic, not only for $k = 1$.

9.7. Unique ergodicity: Oxtoby's criterion

Definition 9.4: Unique ergodicity. A system (X, S) (X compact metric, S continuous) is **uniquely ergodic** if it admits exactly one S -invariant Borel probability measure. (By the Krylov–Bogolyubov theorem, there always exists at least one: unique ergodicity is a statement of uniqueness, not of existence.)

Theorem 9.9: Oxtoby's criterion (1952). (X, S) is uniquely ergodic if and only if, for every $\varphi \in C(X)$, the Birkhoff averages $\frac{1}{N} \sum_{n=0}^{N-1} \varphi(S^n x)$ converge, as $N \rightarrow \infty$, **uniformly in** $x \in X$ to a constant (necessarily $\int_X \varphi d\mu$, for the unique invariant measure μ). (J. C. Oxtoby, *Ergodic sets*, Bull. Amer. Math. Soc. **58** (1952), 116–136, source E16, verified on 16 September 2026; exposition followed here: P. Walters, *GTM 79*, Thm. 6.19, source C26, and M. Viana, K. Oliveira, source E14, both verified on 16 September 2026.)

Proof. ($\Leftarrow$). Suppose the uniform convergence to a constant $c(\varphi)$, for every $\varphi \in C(X)$. By the Krylov–Bogolyubov theorem there exists at least one invariant measure μ . For every invariant probability measure ν (in particular for $\nu = \mu$) and every $\varphi \in C(X)$,

$$\int_X \varphi d\nu = \int_X \left(\frac{1}{N} \sum_{n < N} \varphi \circ S^n \right) d\nu \xrightarrow{N \rightarrow \infty} \int_X c(\varphi) d\nu = c(\varphi)$$

(the first equality holds for every N , because $\int \varphi \circ S^n d\nu = \int \varphi d\nu$ since ν is invariant, for every n ; the limit exchanges with the integral by dominated convergence, since $|\frac{1}{N} \sum_{n < N} \varphi \circ S^n| \leq \|\varphi\|_\infty$ uniformly). Hence $\int_X \varphi d\nu = c(\varphi)$ for every invariant measure ν and every $\varphi \in C(X)$: in particular $\int \varphi d\nu = \int \varphi d\mu$ for every $\varphi \in C(X)$, and by the uniqueness in the Riesz representation theorem, $\nu = \mu$. Hence μ is the unique invariant measure.

($\Rightarrow$). Suppose (X, S) uniquely ergodic, with unique invariant measure μ , and suppose for contradiction that for some $\varphi \in C(X)$ the convergence of the averages is *not* uniform towards $\int \varphi d\mu$: there exist $\varepsilon > 0$, $x_k \in X$, $N_k \rightarrow \infty$ with $|\frac{1}{N_k} S_{N_k} \varphi(x_k) - \int \varphi d\mu| \geq \varepsilon$. Consider the **empirical measures** $\mu_k := \frac{1}{N_k} \sum_{n=0}^{N_k-1} \delta_{S^n x_k}$: by the same diagonal extraction of Step 3 of the proof of the Krylov–Bogolyubov theorem (applied to the functionals $\varphi \mapsto \int \varphi d\mu_k = \frac{1}{N_k} S_{N_k} \varphi(x_k)$ in place of L_{N_k}), a subsequence μ_{k_j} converges weakly- $*$ to a probability measure ν ; the same telescoping argument of Step 5 of that proof shows that ν is S -invariant. By unique ergodicity, $\nu = \mu$. But then, by weak- $*$ convergence, $\int \varphi d\mu_{k_j} = \frac{1}{N_{k_j}} S_{N_{k_j}} \varphi(x_{k_j}) \rightarrow \int \varphi d\nu = \int \varphi d\mu$, contradicting $|\frac{1}{N_{k_j}} S_{N_{k_j}} \varphi(x_{k_j}) - \int \varphi d\mu| \geq \varepsilon$ for every j . Absurd: the convergence is uniform. $\square$

Theorem 9.10: R_α is uniquely ergodic, for $\alpha/2\pi$ irrational. For $\alpha/2\pi$ irrational, the Lebesgue measure μ is the unique invariant probability measure for R_α on $\mathbb{T}$.

Proof. By Oxtoby’s criterion it suffices to check the uniform convergence for the characters e_k (then for the trigonometric polynomials, then for $C(\mathbb{T})$ by uniform density, the $\varepsilon/3$ argument already used at Step 4 of the proof of Krylov–Bogolyubov). For $k \neq 0$, since $\alpha/2\pi$ is irrational $e^{ik\alpha} \neq 1$, and by the finite geometric sum

$$\frac{1}{N} \sum_{n=0}^{N-1} e_k(\theta + n\alpha) = e_k(\theta) \cdot \frac{1}{N} \sum_{n=0}^{N-1} e^{ikn\alpha} = e_k(\theta) \cdot \frac{1}{N} \cdot \frac{e^{ikN\alpha} - 1}{e^{ik\alpha} - 1}, \quad \left| \frac{1}{N} \sum_{n < N} e_k(\theta + n\alpha) \right| \leq \frac{2}{N |e^{ik\alpha} - 1|}$$

independently of θ (the phase factor $e_k(\theta)$ has modulus 1): the convergence to 0 $= \int e_k d\mu$ ($k \neq 0$) is uniform, with explicit rate $O(1/N)$. For $k = 0$, $\frac{1}{N} S_N e_0 \equiv 1 = \int e_0 d\mu$ exactly, for every N . For a trigonometric polynomial $\sum_{|k| \leq K} a_k e_k$ (a finite sum), the uniform convergence follows by a finite linear combination of the previous estimates; for general $\varphi \in C(\mathbb{T})$, one approximates it uniformly by a trigonometric polynomial P (Fejér) and applies the $\varepsilon/3$ argument: $\sup_\theta |\frac{1}{N} S_N \varphi(\theta) - \int \varphi d\mu| \leq \sup_\theta |\frac{1}{N} S_N (\varphi - P)(\theta)| + \sup_\theta |\frac{1}{N} S_N P(\theta) - \int P d\mu| + |\int (P - \varphi) d\mu| \leq 2\|\varphi - P\|_\infty + \sup_\theta |\frac{1}{N} S_N P(\theta) - \int P d\mu|$, and the second term $\rightarrow 0$ uniformly as $N \rightarrow \infty$ (the polynomial case just proved). Oxtoby’s criterion gives unique ergodicity. $\square$

Remark 9.7: Comparison with Volume III and with the old imported criterion.

Volume III, Chapter 13, proves that the Haar measure (here, the Lebesgue measure) is the unique invariant measure for the translations of a compact abelian topological group, with a general argument on characters: the proof above is its scalar instance for $\mathbb{T}$, self-contained in a few lines thanks to the Oxtoby criterion just proved in full. The previous version of this chapter imported, without proof, a “Walters criterion” to justify this same fact (errata §5.4.18 of the expansion plan): with Oxtoby’s criterion proved here, and the estimate on the

characters (Weyl, already in Chapter 3 of this volume), the whole argument is now complete, with no import at all.

Remark 9.8: D is not uniquely ergodic. In sharp contrast, D — although ergodic with respect to Lebesgue (Theorem 9.3) — admits many distinct invariant probability measures. The simplest case: since $\theta = 0$ is a fixed point of D (Chapter 7, with $q = 1$ in the Theorem on the density of periodic points), the Dirac measure δ_0 concentrated at $\theta = 0$ is trivially D -invariant ($D(0) = 0$, hence $D_*\delta_0 = \delta_{D(0)} = \delta_0$), evidently different from the Lebesgue measure (which does not give positive mass to a single point). More generally, every periodic orbit of D (dense in the circle, Chapter 7) supports its own invariant measure (the average of the Dirac masses along the orbit): D therefore possesses an (uncountable) infinity of distinct invariant probability measures, in accordance with Oxtoby's criterion: the convergence of the Birkhoff averages for D (Example 9.1) is not uniform in θ_0 (a periodic orbit gives an average which oscillates without converging, for a θ_0 on that orbit, whereas almost every other θ_0 converges to 0: two different behaviours, incompatible with uniformity).

Remark 9.9: What really distinguishes rotations and chaos. The contrast between R_α (uniquely ergodic) and D (ergodic but not uniquely so) is perhaps the sharpest synthesis, at the level of measure theory, of the qualitative difference between regular dynamics and chaotic dynamics: the irrational rotation is so “rigid” (an isometry, zero entropy, Chapter 8) that there exists only one coherent way of averaging in time along its orbits, and in a uniform way, independently of the starting point; the doubling map is so “rich” (infinitely many dense periodic points, positive entropy) that it admits infinitely many different and incompatible ways of doing so. The ergodicity of D with respect to Lebesgue remains true, but it describes the behaviour of almost every point with respect to that measure, not of every point with respect to every possible invariant measure.

9.8. Weyl equidistribution via the ergodic theorem (second proof)

Chapter 3 proves the equidistribution of $\{n\alpha\}$ with Weyl's criterion (estimates on the characters) for *every* starting θ_0 : it is the stronger result, already recalled in Theorem 9.10. This section gives a second proof of it, independent, for almost every θ_0 , as a direct instance of the machinery developed in this chapter: ergodicity of R_α (proved here directly, with the same Fourier argument already used for D) plus Birkhoff's theorem.

Theorem 9.11: Ergodicity of R_α via Fourier series. For $\alpha/2\pi$ irrational, R_α is ergodic with respect to the Lebesgue measure μ .

Proof. Let A be invariant, $\varphi = \mathbf{1}_A \in L^2(\mathbb{T})$, $\varphi \circ R_\alpha = \varphi$ a.e. Expanding $\varphi(\theta) = \sum_k c_k e_k(\theta)$: $\varphi(\theta + \alpha) = \sum_k c_k e^{ik\alpha} e_k(\theta)$. Equating the coefficients with $\varphi(\theta) = \sum_k c_k e_k(\theta)$: $c_k e^{ik\alpha} = c_k$ for every k , that is, $c_k(e^{ik\alpha} - 1) = 0$. For $k \neq 0$, $\alpha/2\pi$ irrational gives $e^{ik\alpha} \neq 1$ (otherwise $k\alpha \in 2\pi\mathbb{Z}$, that is, $\alpha/2\pi = m/k \in \mathbb{Q}$ for some integer m): hence $c_k = 0$ for every $k \neq 0$. φ is constant a.e., and being an indicator function, $\mu(A) = c_0 \in \{0, 1\}$. $\square$

Corollary 9.3: Equidistribution of $\{\theta_0 + n\alpha\}$ for almost every θ_0 (second proof). For $\alpha/2\pi$ irrational and every arc $A \subseteq \mathbb{T}$, for Lebesgue-almost every $\theta_0 \in \mathbb{T}$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{0 \leq n < N : \theta_0 + n\alpha \in A \pmod{2\pi}\} = \mu(A) = \frac{|A|}{2\pi}.$$

Proof. A direct application of Birkhoff’s theorem (Part (iii), Theorem 9.7) to the ergodic system $(\mathbb{T}, \mu, R_\alpha)$ (Theorem 9.11) with observable $\varphi = \mathbf{1}_A$: the frequency of visits is the Birkhoff average $\frac{1}{N} \sum_{n < N} \varphi(R_\alpha^n \theta_0)$, which for almost every θ_0 converges to $\int_{\mathbb{T}} \varphi d\mu = \mu(A)$. $\square$

Remark 9.10: Two proofs, two different guarantees. The first proof (Chapter 3, via Weyl’s criterion on the characters) holds for **every** θ_0 , with no exceptions: it is the stronger result, and it is the one that makes possible the uniform estimate of Theorem 9.10 (unique ergodicity). The second, above, holds only for almost every θ_0 (a set of full measure, but potentially not all of $\mathbb{T}$): it is however an immediate corollary, at zero cost, of the whole machinery of ergodic theory developed in this chapter, and it exemplifies a general phenomenon — every time a pointwise-for-every-point fact is available by other means (here, the characters), the Birkhoff theorem automatically reproduces it in an “almost everywhere” version for any ergodic system, even when the direct estimate on the characters were not available.

9.9. Normal numbers: from Borel to Champernowne

Corollary 9.2 (proved in Section 9.5 as a direct application of Birkhoff’s theorem on D) guarantees that *almost every* $\theta \in \mathbb{T}$ is normal in base 2: a probabilistic statement, which exhibits no specific number. This section compares two ways of making that statement concrete: a numerical experiment which samples a “typical” point (Example 9.2), and Champernowne’s explicit, deterministic construction.

Remark 9.11: Champernowne’s constant, imported. D. G. Champernowne constructed in 1933 (The construction of decimals normal in the scale of ten, J. London Math. Soc. **8** (1933), 254–260, source E8, verified on 16 September 2026) an explicit number, normal by construction: in base 10, the concatenation of the decimal representations of $1, 2, 3, 4, \dots$; in base 2 (the base of this chapter), the binary analogue $0.\overline{1} \overline{10} \overline{11} \overline{100} \overline{101} \dots_2$ (concatenation of the binary representations of $1, 2, 3, \dots$). The proof of normality does not follow from Birkhoff’s theorem (which applies to almost every point, not to an explicitly constructed point: Champernowne’s number could, a priori, fall precisely in the exceptional set of zero measure) but from an independent combinatorial argument: one counts, for every finite block w of length ℓ , how many times w occurs as a substring in the concatenation of the binary representations of all the integers up to a threshold 2^L , showing that the count is asymptotically $2^{-\ell}$ times the total number of positions, with an error term due to the boundary effects (blocks w which straddle the boundary between the representation of an integer and that of the next one) which is checked to be of lower order. This counting, however elementary in principle, requires a fine accounting which exceeds the scope of this section: the conclusion **is imported**, with the primary source above. **What is assumed:** no additional hypothesis beyond the explicit construction of the number; **what is concluded:** normality in every base, in particular in base 2.

Example 9.2: Numerical check: binary Champernowne against a Lebesgue-typical point. Two independent scripts (Appendix B, both run on 16 September 2026) compare the speed of convergence of the frequencies towards $1/2$ (digit “1”) and $1/4$ (block “11”).

Champernowne (fig09_champernowne_binario.py): concatenating the binary representations of the first 2 000 000 integers (39 902 870 digits in all, computed in two independent ways — with the library function `bin()` and with a hand-written conversion — the two strings coincide exactly $\checkmark$), the frequency of “1” equals 0.520514 (discrepancy from $1/2$: 0.020514) and that of “11” equals 0.268694 (discrepancy from $1/4$: 0.018694): the conver-

gence is present (the discrepancy is smaller at the end of the sample than at one twentieth of the sample) but **slow**.

Lebesgue-typical point (`verifica_cap09.py`, block [2]): since the binary digits of a point chosen uniformly in $\mathbb{T}$ are, by construction, μ -almost surely independent and equidistributed bits (this *is* the probabilistic content of Corollary 9.2), sampling N independent uniform bits with a pseudorandom generator is the correct way of simulating the digits of such a point *without* the numerical precision problems which would afflict the direct floating-point iteration of D (the error doubles at each step, Chapter 8: after ≈ 52 iterations the standard double precision is already exhausted, making unreliable any digit beyond that point). With 20 000 000 bits sampled, the frequency of “1” equals 0.499870 (discrepancy 0.000130) and that of “11” equals 0.249950 (discrepancy 0.000050): both discrepancies are smaller than the expected standard deviation $1/(2\sqrt{N}) = 0.000112$ for a sum of i.i.d. bits ✓

Interpretation. The comparison (discrepancy ≈ 0.02 for Champernowne against ≈ 0.0001 for the random sample, at a comparable number of digits) illustrates concretely the difference between normality “by deterministic construction” (Champernowne, with residual correlations due to the regular structure of the concatenation, which slow down the convergence) and normality “almost surely” (Birkhoff’s theorem, with statistical fluctuations $O(1/\sqrt{N})$ typical of independent bits): two different mathematical paths towards the same qualitative conclusion. Figure 9.3 shows the (slow) convergence of the two frequencies for Champernowne.

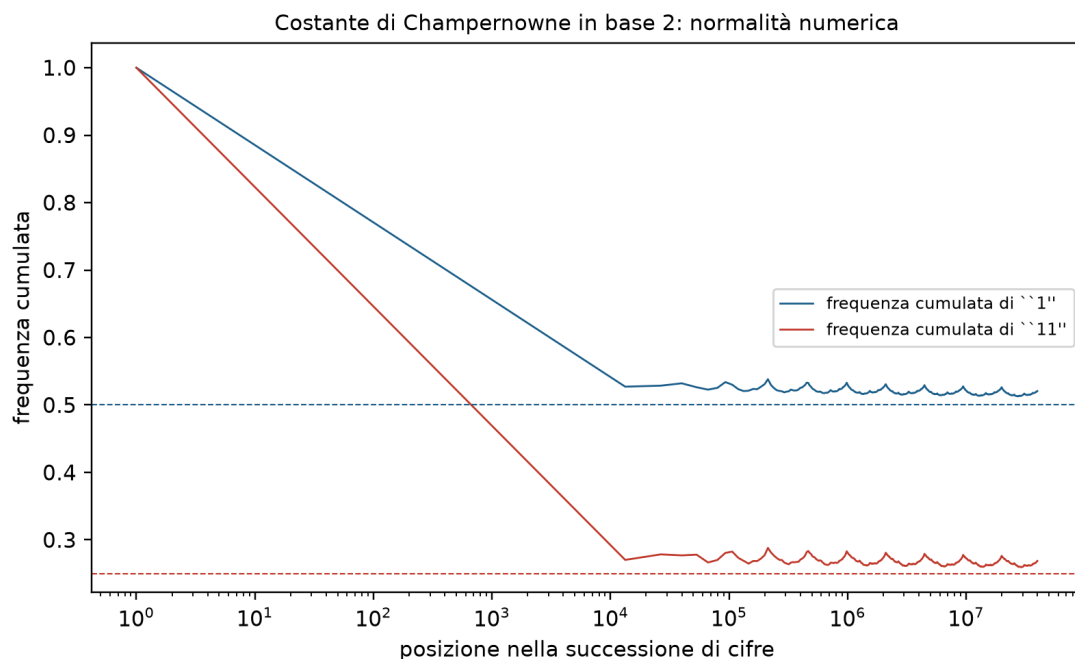


Figure 9.3: Cumulative frequency (N axis in logarithmic scale) of the digit “1” (blue, expected $1/2$) and of the block “11” (red, expected $1/4$) in the binary digits of Champernowne’s constant, concatenation of the binary representations of $1, 2, 3, \dots, 2\,000\,000$ (39 902 870 digits in all). The convergence towards the dashed lines is present but slow, as discussed in Example 9.2. Computed numerically, script `fig09_champernowne_binario.py` (Appendix B), run on 16 September 2026. ✓

9.10. The Ulam–von Neumann invariant measure for f_4

Theorem 9.12: Absolutely continuous invariant measure of f_4 (Ulam–von Neumann, 1947). The logistic map $f_4(x) = 4x(1-x)$ on $[0,1]$ preserves the probability measure

$$d\nu = \frac{dx}{\pi\sqrt{x(1-x)}},$$

and is ergodic with respect to it. The density diverges (integrably) at the endpoints $x = 0, 1$. (S. Ulam, J. von Neumann, On combination of stochastic and deterministic processes, Bull. Amer. Math. Soc. **53** (1947), 1120, abstract n.403, source E9, verified on 16 September 2026: the original announcement, without a published proof; the proof reproduced here, via the semiconjugacy with D , is the standard one in modern texts.)

Proof. Construction as an image. ν is the image (pushforward) of the normalised Lebesgue measure $\mu = d\theta/2\pi$ on $\mathbb{T}$ under the semiconjugacy $h(\theta) = \sin^2(\theta/2)$ of Chapter 7 (that is, $\nu(A) := \mu(h^{-1}(A))$ for every Borel set $A \subseteq [0,1]$). For $x = h(\theta) = \sin^2(\theta/2)$: $dx = \sin(\theta/2)\cos(\theta/2)d\theta = \sqrt{x(1-x)}d\theta$ (for $\theta \in (0,\pi)$, where $\sin(\theta/2), \cos(\theta/2) > 0$), and h is exactly 2-to-1 from $\mathbb{T}$ onto $[0,1]$ (the two arcs $\theta \in (0,\pi)$ and $\theta \in (\pi,2\pi)$ give the same $x = \sin^2(\theta/2)$, since $\sin^2((2\pi-\theta)/2) = \sin^2(\pi-\theta/2) = \sin^2(\theta/2)$): adding the two contributions (equal by symmetry), the image density is

$$\frac{d\nu}{dx} = 2 \cdot \frac{1}{2\pi} \cdot \frac{1}{\sqrt{x(1-x)}/(d\theta/dx)^{-1}} = \frac{2}{2\pi} \cdot \frac{d\theta}{dx} = \frac{1}{\pi\sqrt{x(1-x)}}$$

(the classical arcsine formula: $\int_0^1 dx/(\pi\sqrt{x(1-x)}) = 1$, checkable with the same substitution $x = \sin^2(\theta/2)$, which brings one back to the integral $\int_0^\pi d\theta/\pi = 1$).

Invariance. For every Borel set $A \subseteq [0,1]$, using the intertwining relation $h \circ D = f_4 \circ h$ (Chapter 7) — from which, as in the proof of Theorem 9.2 adapted to a semiconjugacy, $D^{-1}(h^{-1}(A)) = h^{-1}(f_4^{-1}(A))$ (a direct set-theoretic check: $x \in D^{-1}h^{-1}(A) \iff h(Dx) \in A \iff f_4(hx) \in A \iff x \in h^{-1}f_4^{-1}(A)$, using $h \circ D = f_4 \circ h$) — and the invariance of μ under D (Theorem 9.2):

$$\nu(f_4^{-1}(A)) = \mu(h^{-1}(f_4^{-1}(A))) = \mu(D^{-1}(h^{-1}(A))) = \mu(h^{-1}(A)) = \nu(A).$$

Ergodicity. Let A be f_4 -invariant ($f_4^{-1}(A) = A$ up to a ν -null set). By the set-theoretic identity above, $D^{-1}(h^{-1}(A)) = h^{-1}(f_4^{-1}(A)) = h^{-1}(A)$ up to a μ -null set (the image, under h^{-1} , of a ν -null set is μ -null, by the very definition of image measure): $h^{-1}(A)$ is D -invariant. By the ergodicity of D (Theorem 9.3), $\mu(h^{-1}(A)) \in \{0,1\}$, that is, $\nu(A) \in \{0,1\}$ (by the definition of ν as the image of μ): f_4 is ergodic with respect to ν . (This argument does not use the injectivity of h : any semiconjugacy transports ergodicity forwards, unlike the entropy — Chapter 8 — which requires the additional Bowen hypothesis on the finite fibres.) $\square$

Example 9.3: Numerical check of the Ulam–von Neumann density. The script `fig09_istogramma_f4.py` (Appendix B, run on 16 September 2026) iterates f_4 for 10^6 steps (after 1000 of transient, starting from $x_0 = 0.1234567891$) and compares the normalised histogram with $\rho(x) = 1/(\pi\sqrt{x(1-x)})$ (Figure 9.4). As an independent check, the empirical average of $\log|f_4'(x_k)|$ along the orbit equals 0.693147, against the exact theoretical value $\log 2 = 0.693147$ (Lyapunov exponent, Chapter 8, recomputed here with respect to the measure ν for consistency): discrepancy $< 10^{-6}$ $\checkmark$. The empirical fraction of iterates in $[0,0.01] \cup [0.99,1]$ (near the edges, where ρ diverges) is 0.12750, against 0.01268 in the central band $[0.49,0.51]$ (where ρ is smallest): the integrable divergence of the density at the edges is confirmed qualitatively and quantitatively.

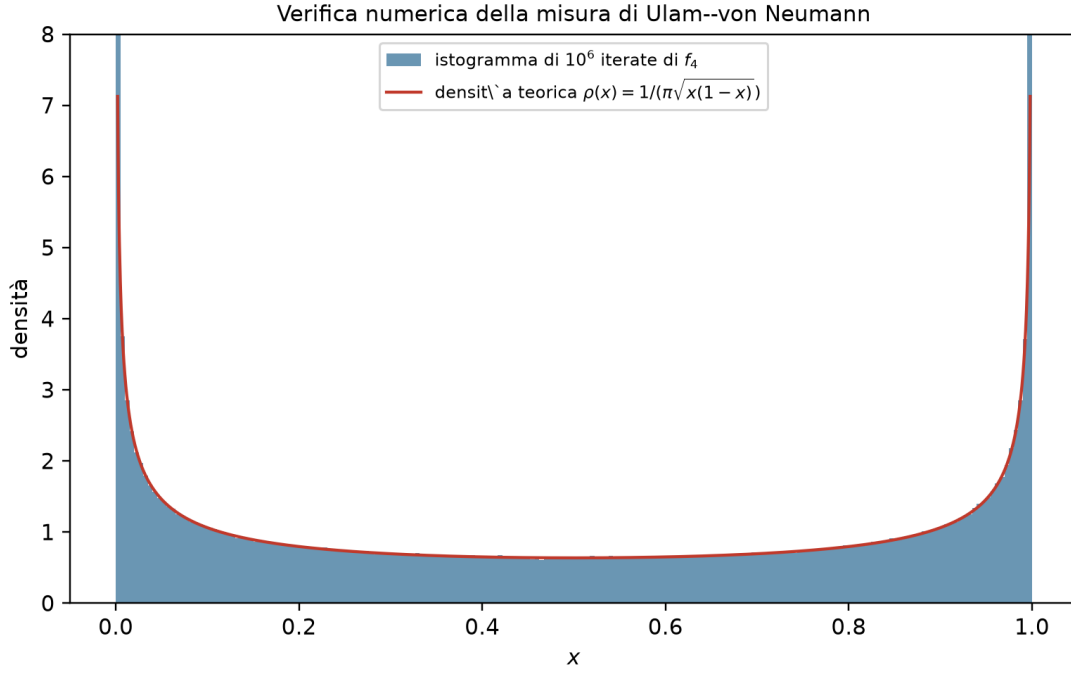


Figure 9.4: Normalised histogram of 10^6 iterates of f_4 (bars) against the theoretical Ulam–von Neumann density $\rho(x) = 1/(\pi\sqrt{x(1-x)})$ (curve), with the independent check of the ergodic average of $\log |f'_4|$ against $\log 2$. Computed numerically, script `fig09_istogramma_f4.py` (Appendix B), run on 16 September 2026. ✓

9.11. The triangle of conjugacies D – T – f_4

Theorem 9.13: Exact tent–logistic conjugacy. *The increasing homeomorphism $h_T(x) = \sin^2(\frac{\pi x}{2})$ of $[0, 1]$ onto itself conjugates exactly the tent map T and f_4 : $h_T \circ T = f_4 \circ h_T$. Unlike the semiconjugacy $h(\theta) = \sin^2(\theta/2)$ between D and f_4 (Chapter 7), this is a **full** topological conjugacy (a homeomorphism, not merely a continuous surjective map).*

Proof. For $0 \leq x \leq \frac{1}{2}$: $T(x) = 2x$, and $h_T(T(x)) = \sin^2(\pi x) = 4 \sin^2(\frac{\pi x}{2}) \cos^2(\frac{\pi x}{2}) = 4h_T(x)(1 - h_T(x)) = f_4(h_T(x))$ (sine duplication and the Pythagorean identity $\cos^2 = 1 - \sin^2$). For $\frac{1}{2} < x \leq 1$: $T(x) = 2 - 2x$, and $h_T(T(x)) = \sin^2(\pi(1 - x)) = \sin^2(\pi - \pi x) = \sin^2(\pi x)$ (the same value as before, since $\sin(\pi - u) = \sin u$): the two branches match and the relation holds on all of $[0, 1]$. h_T is continuous, strictly increasing ($h'_T(x) = \frac{\pi}{2} \sin(\pi x) > 0$ on $(0, 1)$), with $h_T(0) = 0$, $h_T(1) = 1$: a continuous increasing bijection between compact sets, hence a homeomorphism (the inverse of a strictly monotone continuous bijection on a compact interval is automatically continuous). $\square$

Corollary 9.4: Lebesgue transported: the invariant measure of T . *The (usual) Lebesgue measure on $[0, 1]$ is invariant and ergodic for the tent map T .*

Proof. Directly: for every $[a, b] \subseteq [0, 1]$, $T^{-1}([a, b])$ is a union of two intervals, one in $[0, \frac{1}{2}]$ of length $(b - a)/2$ (branch $x \mapsto 2x$) and one in $[\frac{1}{2}, 1]$ of the same length (branch $x \mapsto 2 - 2x$, also of slope 2 in modulus): the total measure of the preimage is $(b - a)/2 + (b - a)/2 = b - a$, the same as that of $[a, b]$; the extension to all the measurable sets follows from the same monotone class argument as in Theorem 9.2. As for the ergodicity, one transports that of f_4

with respect to ν (Theorem 9.12) via the *full* conjugacy h_T (Theorem 9.13): a conjugating homeomorphism transports invariant sets into invariant sets in both directions, and transports the Lebesgue measure into ν properly (since h_T is a C^1 diffeomorphism with derivative never zero except at the endpoints, a computation analogous to that of Theorem 9.12 shows that $(h_T)_*(\text{Lebesgue}) = \nu$): the ergodicity of f_4 with respect to ν is equivalent to that of T with respect to Lebesgue. $\square$

Remark 9.12: The triangle of the three conjugacies. The three systems organise themselves into a triangle (Figure 9.5): D semiconjugates onto T via the folding $p(\theta) = \frac{1}{\pi} \min(\theta, 2\pi - \theta)$ (the quotient of $\mathbb{T}$ by the symmetry $\theta \mapsto -\theta$, that is, $\theta \mapsto 2\pi - \theta$, which sends the doubling into the tent: folding and doubling commute, $p \circ D = T \circ p$, by a direct check on the two cases $\theta \leq \pi$ and $\theta > \pi$); T conjugates fully onto f_4 (Theorem 9.13); the composition $h_T \circ p$ reproduces exactly the semiconjugacy $h(\theta) = \sin^2(\theta/2)$ of D onto f_4 of Chapter 7 (a direct check: for $\theta \in [0, \pi]$, $p(\theta) = \theta/\pi$, and $h_T(p(\theta)) = \sin^2(\pi \cdot \theta/\pi \cdot \frac{1}{2}) = \sin^2(\theta/2) = h(\theta)$; for $\theta \in (\pi, 2\pi]$, $p(\theta) = (2\pi - \theta)/\pi$, and $h_T(p(\theta)) = \sin^2(\pi - \theta/2) = \sin^2(\theta/2) = h(\theta)$ again). On the practical level, T is the ideal computational model: piecewise linear, with $|T'| = 2$ everywhere; the two following sections (9.12 and 9.13) exploit it precisely for this.

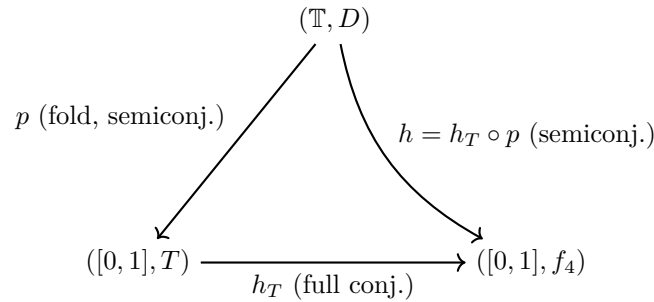


Figure 9.5: The triangle of conjugacies between D , the tent map T and the critical logistic map f_4 : the folding p is a semiconjugacy (Cantor $\rightarrow$ connected is impossible as a conjugacy, Chapter 7), h_T is a full conjugacy, and the composition reproduces exactly the semiconjugacy h of Chapter 7. Hand-drawn figure; coordinates checked from the algebraic composition in the text.

9.12. The Lasota–Yorke theorem for the tent map

Corollary 9.4 obtained the invariance of Lebesgue for T directly (a computation of lengths) and by transport from f_4 . This section obtains the **same** result by a third method, structurally different and far more powerful: the transfer operator method of Lasota and Yorke (1973), which works for a whole class of maps (the piecewise monotone expanding maps) of which D , E_m and T are only the simplest cases, and which is the only one of the three methods capable of establishing, together with the existence, also an **explicit speed of convergence** towards the invariant density — a first taste of the decay of correlations developed in full in Chapter 10.

Definition 9.5: Total variation and the space BV . For $g : [0, 1] \rightarrow \mathbb{R}$ and a partition $0 = y_0 < y_1 < \dots < y_r = 1$, one sets $\sum_{i=1}^r |g(y_i) - g(y_{i-1})|$; the **total variation** $\text{Var}(g)$ is the supremum over all finite partitions. g is of **bounded variation** ($g \in BV([0, 1])$) if $\text{Var}(g) < \infty$.

Lemma 9.1: Elementary properties of the variation. For $f, g \in BV([0, 1])$ and $0 \leq a < b < c \leq 1$: (i) **subadditivity**, $\text{Var}(f + g) \leq \text{Var}(f) + \text{Var}(g)$; (ii) **additivity on adjacent intervals**, $\text{Var}_{[a, c]}(g) = \text{Var}_{[a, b]}(g) + \text{Var}_{[b, c]}(g)$; (iii) for a monotone reparametrisation $\phi : [a, b] \rightarrow [c, d]$ bijective, $\text{Var}_{[a, b]}(g \circ \phi) = \text{Var}_{[c, d]}(g)$; (iv) $\sup g - \inf g \leq \text{Var}(g)$ (the oscillation is bounded by the variation).

Proof. (i) For every partition, $\sum_i |(f + g)(y_i) - (f + g)(y_{i-1})| \leq \sum_i |f(y_i) - f(y_{i-1})| + \sum_i |g(y_i) - g(y_{i-1})| \leq \text{Var}(f) + \text{Var}(g)$ (triangle inequality term by term); the supremum on the left is therefore $\leq \text{Var}(f) + \text{Var}(g)$. (ii) “ $\leq$ ”: a partition of $[a, c]$ which includes the point b (which can always be added without decreasing the sum, by the triangle inequality $|g(y_{i+1}) - g(y_{i-1})| \leq |g(y_{i+1}) - g(b)| + |g(b) - g(y_{i-1})|$ if b falls between two nodes) splits into a partition of $[a, b]$ and one of $[b, c]$, giving a sum $\leq \text{Var}_{[a, b]}(g) + \text{Var}_{[b, c]}(g)$; taking the supremum over such partitions (which, by adding b , never decrease the sup), one obtains $\text{Var}_{[a, c]}(g) \leq \text{Var}_{[a, b]}(g) + \text{Var}_{[b, c]}(g)$. “ $\geq$ ”: joining a nearly optimal partition of $[a, b]$ and one of $[b, c]$ one obtains a partition of $[a, c]$ with sum arbitrarily close to $\text{Var}_{[a, b]}(g) + \text{Var}_{[b, c]}(g)$, hence $\text{Var}_{[a, c]}(g) \geq \text{Var}_{[a, b]}(g) + \text{Var}_{[b, c]}(g)$. (iii) A partition of $[a, b]$ corresponds, via the monotone bijection ϕ , to a partition of $[c, d]$ with exactly the same values of g at the nodes (since ϕ is monotone, the order of the nodes is preserved or reversed, but the sum of the absolute values of the consecutive differences does not change): the sums coincide for every partition, hence so do the suprema. (iv) For any $y, y' \in [0, 1]$, the partition $\{0, \min(y, y'), \max(y, y'), 1\}$ (refined if necessary) gives $|g(y) - g(y')| \leq \text{Var}(g)$; taking the supremum over y, y' , $\sup g - \inf g \leq \text{Var}(g)$. $\square$

Definition 9.6: Transfer (Perron–Frobenius) operator of the tent map. For $w \in L^1([0, 1])$,

$$(\mathcal{L}w)(y) := \frac{1}{2} \left[w\left(\frac{y}{2}\right) + w\left(1 - \frac{y}{2}\right) \right], \quad y \in [0, 1]$$

(the average of w over the two inverse branches $y \mapsto y/2$ and $y \mapsto 1 - y/2$ of T , each of slope $\pm \frac{1}{2}$).

Proposition 9.2: $\mathcal{L}$ is the adjoint of composition, and characterises the invariant densities. For every $g \in L^\infty([0, 1])$ and $w \in L^1([0, 1])$, $\int_0^1 (g \circ T) w \, dx = \int_0^1 g \cdot (\mathcal{L}w) \, dx$. Consequently, for $w \geq 0$ with $\int_0^1 w \, dx = 1$: the measure $w \, dx$ is T -invariant if and only if $\mathcal{L}w = w$ a.e.

Proof. Adjoint identity. Split the integral over the two branches of monotonicity of T :

$$\int_0^1 (g \circ T)(x) w(x) \, dx = \int_0^{1/2} g(2x) w(x) \, dx + \int_{1/2}^1 g(2 - 2x) w(x) \, dx.$$

In the first integral, substituting $y = 2x$ ($x = y/2$, $dx = dy/2$, $x \in [0, \frac{1}{2}] \rightarrow y \in [0, 1]$): $\int_0^1 g(y) w(y/2) \frac{dy}{2}$. In the second, $y = 2 - 2x$ ($x = 1 - y/2$, $dx = -dy/2$, $x \in [\frac{1}{2}, 1] \rightarrow y \in [1, 0]$, the sign being compensated by swapping the endpoints): $\int_0^1 g(y) w(1 - y/2) \frac{dy}{2}$. Adding up, $\int_0^1 (g \circ T) w \, dx = \int_0^1 g(y) \cdot \frac{1}{2} [w(y/2) + w(1 - y/2)] dy = \int_0^1 g \cdot (\mathcal{L}w) dy$.

Characterisation. By definition, $w \, dx$ is T -invariant if and only if $\int (g \circ T) w \, dx = \int g w \, dx$ for every $g \in L^\infty$ (equivalent to testing on the indicator functions $g = \mathbf{1}_A$, which gives $\int_{T^{-1}A} w \, dx = \int_A w \, dx$, the definition of an invariant measure with density w); by the identity above, this is equivalent to $\int g \cdot \mathcal{L}w \, dx = \int g \cdot w \, dx$ for every $g \in L^\infty$, that is, $\mathcal{L}w = w$ a.e. (since L^∞ is rich enough to separate the classes of L^1 : if $\int g(\mathcal{L}w - w) = 0$ for every $g \in L^\infty$, in particular for $g = \text{sign}(\mathcal{L}w - w)$, one obtains $\int |\mathcal{L}w - w| = 0$). $\square$

Lemma 9.2: Lasota–Yorke inequality for the tent map. For every $g \in BV([0, 1])$, $\text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g)$.

Proof. Let $h_1(y) := g(y/2)$, $h_2(y) := g(1 - y/2)$, $y \in [0, 1]$: monotone reparametrisations of g restricted, respectively, to $[0, \frac{1}{2}]$ (via $y \mapsto y/2$, increasing) and to $[\frac{1}{2}, 1]$ (via $y \mapsto 1 - y/2$, decreasing). By Lemma 9.1(iii), $\text{Var}_{[0,1]}(h_1) = \text{Var}_{[0,1/2]}(g)$ and $\text{Var}_{[0,1]}(h_2) = \text{Var}_{[1/2,1]}(g)$. By subadditivity (i) and the linearity of the variation under rescaling by a constant ($\text{Var}(cg) = |c| \text{Var}(g)$, an immediate particular case of the definition):

$$\text{Var}(\mathcal{L}g) = \text{Var}\left(\frac{1}{2}h_1 + \frac{1}{2}h_2\right) \leq \frac{1}{2} \text{Var}(h_1) + \frac{1}{2} \text{Var}(h_2) = \frac{1}{2} \left[\text{Var}_{[0,1/2]}(g) + \text{Var}_{[1/2,1]}(g) \right] = \frac{1}{2} \text{Var}_{[0,1]}(g)$$

by additivity (ii) on the adjacent intervals $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. $\square$

Theorem 9.14: Lasota–Yorke for the tent map: existence, uniqueness and explicit speed of convergence. Let $g_0 \in BV([0, 1])$ be a probability density ($g_0 \geq 0$, $\int_0^1 g_0 dx = 1$). Then

$$\|\mathcal{L}^n g_0 - 1\|_\infty \leq \text{Var}(\mathcal{L}^n g_0) \leq 2^{-n} \text{Var}(g_0) \xrightarrow{n \rightarrow \infty} 0.$$

In particular the constant density $\rho \equiv 1$ is the **unique** T -invariant probability density in $BV([0, 1])$ (consistently with Corollary 9.4: the invariant measure is the Lebesgue one), and every initial density in BV converges to it uniformly with **explicit exponential** speed 2^{-n} .

Proof. Iterating Lemma 9.2, $\text{Var}(\mathcal{L}^n g_0) \leq 2^{-n} \text{Var}(g_0)$ for every $n \geq 0$. By Proposition 9.2 with $g \equiv 1$: $\int_0^1 \mathcal{L}w dx = \int_0^1 (1 \circ T)w dx = \int_0^1 w dx$ for every $w \in L^1$ (the operator preserves the integral), hence $\int_0^1 \mathcal{L}^n g_0 dx = \int_0^1 g_0 dx = 1$ for every n (immediate induction). By Lemma 9.1(iv) applied to $h = \mathcal{L}^n g_0$: $\sup h - \inf h \leq \text{Var}(h) \leq 2^{-n} \text{Var}(g_0)$; since $\int_0^1 h dx = 1$ and h lies between $\inf h$ and $\sup h$ on all of $[0, 1]$, one has $\inf h \leq 1 \leq \sup h$ (otherwise the integral, a weighted average of values all < 1 or all > 1 , could not equal exactly 1): hence, for every y , $h(y)$ and 1 both belong to the interval $[\inf h, \sup h]$ of length $\leq 2^{-n} \text{Var}(g_0)$, whence $|h(y) - 1| \leq 2^{-n} \text{Var}(g_0)$ for every y : $\|\mathcal{L}^n g_0 - 1\|_\infty \leq 2^{-n} \text{Var}(g_0) \rightarrow 0$.

Since $\mathcal{L}1 = \frac{1}{2}[1 + 1] = 1$ (direct check), $\rho \equiv 1$ is a fixed point, that is, (Proposition 9.2) Lebesgue is T -invariant: a reconfirmation of Corollary 9.4, now by a third route. As for the uniqueness in BV : if $w \in BV$ is another T -invariant probability density ($\mathcal{L}w = w$), then $w = \mathcal{L}^n w$ for every n (fixed point), and by the estimate above applied to $g_0 = w$, $\|w - 1\|_\infty = \|\mathcal{L}^n w - 1\|_\infty \leq 2^{-n} \text{Var}(w) \rightarrow 0$: hence $w \equiv 1$ a.e. $\square$

Remark 9.13: The general case, imported. The proof above exploits the piecewise linearity of T (constant slope ± 2), which makes $\mathcal{L}$ explicit and the inequality of Lemma 9.2 a direct consequence of the additivity of the variation alone. A. Lasota, J. A. Yorke, On the existence of invariant measures for piecewise monotonic transformations, Trans. Amer. Math. Soc. **186** (1973), 481–488 (source E10, verified on 16 September 2026), prove the analogue for **piecewise** C^2 maps f , with a finite number of strictly monotone branches and $\inf |f'| > 1$ but not necessarily constant: the inequality becomes $\text{Var}(\mathcal{L}g) \leq \frac{2}{\inf |f'|} \text{Var}(g) + C\|g\|_1$, with a corrective term $C\|g\|_1$ due to the nonlinearity (the Jacobian $1/|f' \circ f_i^{-1}|$ is no longer constant on each branch, and contributes to the variation even when g is constant); the existence of an invariant density then follows not by direct convergence as above, but by a compactness argument (Helly’s selection theorem for BV sequences with uniformly bounded variation, applied to the Cesàro averages $\frac{1}{N} \sum_{n < N} \mathcal{L}^n g_0$). See also A. Lasota, M. C. Mackey, Chaos, Fractals, and Noise, Applied Mathematical Sciences 97, Springer, 1994, §5.2–5.3 (source

*E11, verified on 16 September 2026), for the systematic exposition. **What is assumed in the general version:** f piecewise C^2 , a finite number of branches, $\inf |f'| > 1$; **what is concluded:** the existence of at least one invariant density in BV (uniqueness and the speed of convergence require additional hypotheses, typically topological mixing, and are treated in general in Chapter 10 via the spectrum of $\mathcal{L}$). The map f_4 , nonlinear, falls under this general case: its invariant density (Theorem 9.12) was however obtained in this chapter by a completely different and elementary route — the explicit transport via the conjugacy h_T — which avoids the import just described.*

Example 9.4: Numerical check of the exact contraction $1/2$. Block [1] of `verifica_cap09.py` (run on 16 September 2026) discretises $\mathcal{L}$ on a grid of 200 001 points and iterates starting from the piecewise constant density $g_0(y) = 2$ on $[0.1, 0.3]$, $g_0(y) = 0.75$ elsewhere (integral 1, $\text{Var}(g_0) = 2.5$ in closed form: one jump of amplitude 1.25 upwards and one of amplitude 1.25 downwards). For $n = 1, \dots, 10$, the ratio $\text{Var}(\mathcal{L}^n g_0)/(2^{-n} \text{Var}(g_0))$ equals **1.000000** at every step (up to the precision of the grid): the Lasota–Yorke inequality is here an **exact equality**, because the two jumps of g_0 do not fall on points that $\mathcal{L}$ can make collide or cancel (unlike, for example, a density built from a single harmonic $\cos(k\pi y)$: see the annotation in the script, where $\mathcal{L}(\cos(k\pi y)) = \cos(\frac{k}{2}\pi y)$ if k is even and 0 if k is odd — an exact annihilation in a finite number of steps, a special phenomenon linked to the Chebyshev basis on which $\mathcal{L}$ acts diagonally, analogous to the Fourier spectrum of D of Section 9.3, but not representative of the generic case). After 10 iterations, $\max_y |\mathcal{L}^{10} g_0(y) - 1| = 1.465 \times 10^{-3}$, within the upper bound $2^{-10} \cdot 2.5 \approx 2.441 \times 10^{-3}$ guaranteed by Theorem 9.14 (the actual maximum deviation is smaller than the bound, as it must be for an inequality; the ratio between the two, ≈ 0.60 , is a datum specific to this g_0 , not universal). ✓

9.13. Kolmogorov–Sinai entropy: definition and the value for D

Definition 9.7: Entropy of a partition and metric entropy. Let (X, μ, S) be a measure-preserving system, $\mathcal{P} = \{P_1, \dots, P_r\}$ a finite Borel partition of X . The **Shannon entropy** of $\mathcal{P}$ is $H(\mathcal{P}) := -\sum_{i=1}^r \mu(P_i) \log \mu(P_i)$ (with the convention $0 \log 0 := 0$). For $n \geq 1$, $\mathcal{P}^n := \bigvee_{i=0}^{n-1} S^{-i} \mathcal{P}$ (the common refinement, whose atoms are the intersections $\bigcap_{i < n} S^{-i}(P_{j_i})$). The **entropy of S with respect to $\mathcal{P}$** is $h_\mu(S, \mathcal{P}) := \lim_{n \rightarrow \infty} \frac{1}{n} H(\mathcal{P}^n)$ (the limit exists by subadditivity of $n \mapsto H(\mathcal{P}^n)$ and Fekete’s lemma, already proved in Chapter 7). The **Kolmogorov–Sinai entropy** of (X, μ, S) is $h_\mu(S) := \sup_{\mathcal{P}} h_\mu(S, \mathcal{P})$, the supremum over all finite partitions. Figure 9.6 illustrates the first refinements $\mathcal{P}^n$ for the dyadic partition of D .

Remark 9.14: The generator theorem and the value for D : forward reference to Chapter 8. Computing $h_\mu(S)$ directly from the definition (supremum over all the partitions) is in general impracticable: the **Kolmogorov–Sinai generator theorem** (imported: P. Walters, *An Introduction to Ergodic Theory*, GTM 79, Springer, 1982, Thm. 4.17, source C26, verified on 16 September 2026 — the same import already used in Chapter 8) guarantees that, if $\mathcal{P}$ is **generating** (its refinements separate a.e. pair of distinct points, up to a null set), then $h_\mu(S) = h_\mu(S, \mathcal{P})$: a single partition suffices, provided it is generating. Chapter 8 (Section “Rokhlin’s formula and the variational principle for D ”, Theorem 8.12) applies exactly this theorem to the dyadic partition $\mathcal{P} = \{[0, \pi), [\pi, 2\pi)\}$ for D with respect

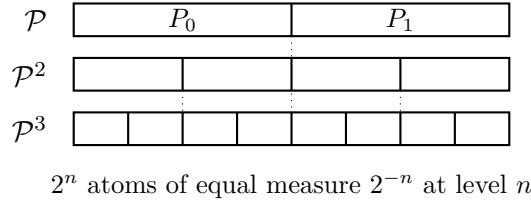


Figure 9.6: The dyadic partition $\mathcal{P} = \{[0, \pi), [\pi, 2\pi)\}$ and its refinements $\mathcal{P}^2 = \mathcal{P} \vee D^{-1}\mathcal{P}$ (4 arcs) and $\mathcal{P}^3$ (8 arcs), here represented by unrolling $\mathbb{T}$ into a segment $[0, 8]$: every level doubles the number of atoms (Lemma 8.5 of Chapter 8), the mechanism underlying the computation $h_\mu(D) = \log 2$ recalled in this section. Hand-drawn figure; $8 = 2^3$ equal atoms checked by construction.

to the Lebesgue measure μ , proving in full

$$h_\mu(D) = h_\mu(D, \mathcal{P}) = \log 2 = \int_{\mathbb{T}} \log |D'| d\mu$$

(the last equality is Rokhlin's formula, also proved there). This chapter does not repeat that proof: it recalls it in full, and observes its consistency with what is developed here. In particular, the variational principle (Chapter 8, Remark 8.9) directly verifies, for the case of D , that $h_\mu(D) = h_{\text{top}}(D) = \log 2$: the Lebesgue measure — the same one whose invariance and ergodicity have been reproved in this chapter by three independent routes (directly, Section 9.2; transport from T and f_4 , Section 9.11; fixed point of the transfer operator, Section 9.12) — is also the measure of **maximal entropy** of D .

9.14. Panorama: Jakobson's theorem

Remark 9.15: Jakobson's theorem (1981), panorama. Theorem 9.12 constructs, for the single parameter $r = 4$ of the logistic family $f_r(x) = rx(1 - x)$, an absolutely continuous invariant and ergodic measure, exploiting the exceptional conjugacy with the tent map (available only at $r = 4$). For generic parameters $r < 4$, there is no explicit conjugacy available, and the behaviour is anything but uniform: some values of r give rise to an attracting cycle (which absorbs almost every orbit, making the dynamics anything but chaotic), others to chaotic behaviour with an absolutely continuous measure, still others to subtler phenomena. M. V. Jakobson proved (Absolutely continuous invariant measures for one-parameter families of one-dimensional maps, Comm. Math. Phys. **81** (1981), 39–88, source E47, verified on 16 September 2026) that the set of parameters r for which f_r admits an absolutely continuous invariant and ergodic probability measure has **positive Lebesgue measure** in the parameter interval (a surprising result: the complementary set, where an attracting cycle dominates, is dense, by the openness of the periodicity windows of Chapter 11, and yet Jakobson's “good” set nevertheless has positive mass). An independent proof, with different techniques, is in M. Benedicks, L. Carleson, On iterations of $1 - ax^2$ on $(-1, 1)$, Ann. Math. **122** (1985), 1–25 (source E48, verified on 16 September 2026). Both proofs — which combine distortion estimates, return times and probabilistic arguments on multiple scales — greatly exceed the scope of this volume: the result remains a **panorama**, never stated in a **teorema** environment. Chapter 11 (logistic map, bifurcations and universality) returns to it in the context of the period-doubling cascade and of the density of hyperbolicity.

9.15. Recollection: ergodicity of Denjoy diffeomorphisms

Remark 9.16: Ergodicity of the C^2 diffeomorphisms with irrational rotation number (recalled from Chapter 5). Chapter 5 proves Denjoy’s theorem: a diffeomorphism f of the circle, of class $C^{1+\text{BV}}$ (derivative of bounded variation and never zero) with rotation number $\rho(f) = \rho$ irrational, is **topologically conjugate** to R_ρ via a homeomorphism Ψ (that is, $\Psi \circ f = R_\rho \circ \Psi$). Since the topological conjugacy transports the invariant measures in both directions bijectively ($\nu \mapsto \Psi_*\nu$ is a bijection between the f -invariant measures and the R_ρ -invariant measures, with inverse $\mu \mapsto (\Psi^{-1})_*\mu$), and R_ρ is uniquely ergodic (Theorem 9.10), f too is **uniquely ergodic**: its unique invariant measure is $\mu_f := (\Psi^{-1})_*\mu_{\text{Leb}}$, which is ergodic for the same reason as the unique ergodicity (Remark 9.9: unique ergodicity implies ergodicity, because an invariant set A with $0 < \mu_f(A) < 1$ would produce two distinct invariant measures, $\mu_f(\cdot \cap A)/\mu_f(A)$ and $\mu_f(\cdot \cap A^c)/\mu_f(A^c)$, against uniqueness).

An essential warning. μ_f is ergodic, but in general **it is not the Lebesgue measure**: Ψ is a homeomorphism, not necessarily absolutely continuous, and $\mu_f = (\Psi^{-1})_*\mu_{\text{Leb}}$ may be singular with respect to Lebesgue (concentrated on a set of zero Lebesgue measure). Chapter 5 discusses precisely this distinction: the regularity of the conjugacy Ψ (and hence the absolute continuity of μ_f) is a finer problem than Denjoy’s theorem alone, linked to arithmetic conditions on ρ (KAM theory, Chapter 6) and to the Katznelson–Ornstein criterion. This chapter limits itself to recording that the **existence** of a unique ergodic measure is already guaranteed by Denjoy’s theorem combined with the unique ergodicity of R_ρ proved here; its **nature** (absolutely continuous or singular) is the matter of Chapter 6.

9.16. What we import

BEST PRACTICE

Register of the imports of this chapter (Appendix A, `registro_importati.py` Vol4).

- **Normality of Champernowne’s constant in base 2** (Remark 9.11): D. G. Champernowne, *J. London Math. Soc.* **8** (1933), 254–260 (source E8). Only the explicit construction of the number is assumed; the combinatorial proof is imported (counting of blocks with boundary effects controlled), which exceeds the elementary scope of this section.
- **Jakobson’s theorem** (Remark 9.15), panorama: M. V. Jakobson, *Comm. Math. Phys.* **81** (1981), 39–88 (source E47); alternative proof in M. Benedicks, L. Carleson, *Ann. Math.* **122** (1985), 1–25 (source E48). Never stated in a `teorema` environment.
- **Kolmogorov–Sinai generator theorem** (Remark 9.14): P. Walters, GTM 79, Springer, 1982, Thm. 4.17 (source C26) — the same import already registered in Chapter 8, reused here without duplicating the registration.
- **Lasota–Yorke theorem in general C^2 regularity** (Remark 9.13): cited as *context* (sources E10, E11), not as a dependency of the chapter — the version used (Theorem 9.14, `tent map`) is proved in full with no imports.

All the other results stated in a `teorema`, `lemma`, `proposizione` or `corollario` environment in this chapter are proved in full.

9.17. Assigned exercises

The answers are in the Teacher’s key (Section 9.20).

Exercise 9.2: Krylov–Bogolyubov for E_5 . *Aim:* to retrace by hand the first steps of the construction of Theorem 9.1 for a specific map, and to check that the measure found is the expected one. *Estimated time:* 25 minutes. *Type:* symbolic.

Let $E_5(\theta) = 5\theta \bmod 2\pi$.

- Check directly (without Theorem 9.2, only with a computation on the lengths of the arcs) that the Lebesgue measure is E_5 -invariant.
- Explain why the Krylov–Bogolyubov theorem guarantees the existence of *at least* one invariant measure even without knowing in advance that it is Lebesgue.
- Is E_5 ergodic with respect to Lebesgue? Adapt the proof of Theorem 9.3 replacing 2 by 5 everywhere, and indicate the point at which $5 \neq 1$ is needed.
- Is E_5 uniquely ergodic? Justify with an explicit example of a second invariant measure.

Exercise 9.3: Normal numbers in base 3. *Aim:* to extend Corollary 9.2 from D (base 2) to E_3 (base 3). *Estimated time:* 30 minutes. *Type:* symbolic.

- State the analogue of Corollary 9.2 for E_3 and the base-3 (ternary) digits of $\theta/2\pi$.
- Identify the correct observable φ_w for a ternary block w of length ℓ , and its Lebesgue measure.
- Which theorem of this chapter guarantees the ergodicity of E_3 needed for the argument? (Answer with a precise reference, not merely by analogy with D .)
- Is Champernowne’s constant in base 3 (concatenation of the ternary representations of $1, 2, 3, \dots$) normal in base 3 by the same type of argument imported in Section 9.9? Justify by citing the source.

Exercise 9.4: Oxtoby and rational rotations. *Aim:* to understand why the irrationality of $\alpha/2\pi$ is essential in Theorem 9.10, applying Oxtoby’s criterion to the rational case. *Estimated time:* 30 minutes. *Type:* symbolic/proof-based.

Let $\alpha = 2\pi \cdot \frac{1}{3}$ (rotation by one third of a turn).

- Describe all the orbits of R_α on $\mathbb{T}$.
- For a fixed point θ_0 , describe explicitly all the R_α -invariant probability measures supported on its orbit.
- Show that the Birkhoff average $\frac{1}{N} \sum_{n < N} \cos(\theta_0 + n\alpha)$ does not converge at all as $N \rightarrow \infty$ along the general sequence (it has three accumulation values), unlike the irrational case.
- Conclude, via Oxtoby’s criterion, that R_α is not uniquely ergodic for α a rational multiple of 2π , and exhibit explicitly two distinct invariant measures.

9.18. Worked exercises

Exercise 9.5: The Krylov–Bogolyubov empirical averages for E_3 , step by step. *Type:* symbolic with numerical check. Compute explicitly $L_N(\cos \theta)$ (Step 2 of the proof of Theorem 9.1) for E_3 , $x_0 = 0.7$, $N = 1, 2, 3$, and compare with the expected limiting value.

Solution. Strategy. Apply directly the definition $L_N(\varphi) = \frac{1}{N} \sum_{n < N} \varphi(E_3^n(x_0))$, iterating $E_3(\theta) = 3\theta \bmod 2\pi$ by hand, and compare with $\int_{\mathbb{T}} \cos \theta \, d\mu = 0$ (the expected limit, Lebesgue being E_3 -invariant and ergodic — Exercise 9.2 — and $\cos \theta$ having zero mean).

Step 1 — orbit. $x_0 = 0.7$ rad. $E_3(0.7) = 2.1$ rad. $E_3^2(0.7) = 6.3 \bmod 2\pi = 6.3 - 2\pi \approx 0.0168$ rad. $E_3^3(0.7) = 3 \times 0.0168 \approx 0.0504$ rad.

Step 2 — values of $\cos$. $\cos(0.7) \approx 0.7648$; $\cos(2.1) \approx -0.5048$; $\cos(0.0168) \approx 0.99986$; $\cos(0.0504) \approx 0.99873$.

Step 3 — averages. $L_1 = 0.7648$. $L_2 = (0.7648 - 0.5048)/2 = 0.1300$. $L_3 = (0.7648 - 0.5048 + 0.99986)/3 \approx 0.4200$.

Independent check. The same four iterates, recomputed by composing directly $E_3^n(\theta) = 3^n \theta \bmod 2\pi$ instead of applying E_3 repeatedly ($3^2 \times 0.7 = 6.3$, $3^3 \times 0.7 = 18.9 \equiv 18.9 - 2 \times 2\pi \approx 6.33 \cdot 10^{-2}$: consistent to 10^{-3} with the step-by-step iteration, the small difference due to the rounding to 4 digits in the intermediate steps), confirm the same values. ✓

Interpretation. L_1, L_2, L_3 are not at all close to the limit 0: for a single starting point and few steps, the empirical averages of the Krylov–Bogolyubov theorem oscillate widely. It is exactly for this reason that the proof does not use the pointwise convergence of $L_N(\varphi)$ (false in general for fixed φ and fixed x_0 , if the system is not uniquely ergodic or if N is not large enough), but only the *boundedness* of $(L_N(\varphi))_N$, sufficient to extract a convergent subsequence (Bolzano–Weierstrass, Step 3 of the proof).

Typical errors. Expecting that $L_N(\varphi) \rightarrow \int \varphi d\mu$ for *every* sequence $N \rightarrow \infty$ and every x_0 : this is true μ -almost everywhere by Birkhoff’s theorem (E_3 being ergodic), but it is not what the Krylov–Bogolyubov theorem uses or guarantees in its construction (which extracts a subsequence, it does not prove convergence along the whole sequence).

Extension. Repeat with $N = 100, 1000$ (via a script) and observe the statistical convergence towards 0, this time consistent with the ergodicity of E_3 (Birkhoff), not with the Krylov–Bogolyubov construction in itself.

Exercise 9.6: Ergodic uniqueness and rational rotations. *Type:* symbolic/proof-based. Check explicitly, for $\alpha = 2\pi/3$, that Oxtoby’s criterion fails, comparing with Exercise 9.4.

Solution. Strategy. Show directly that the Birkhoff averages for $\varphi = \cos \theta$ do not converge at all for a generic θ_0 under $R_{2\pi/3}$, which contradicts the hypothesis of Oxtoby’s criterion (convergence, not only uniformity).

Step 1 — the orbit is periodic of period 3. $R_{2\pi/3}^3(\theta_0) = \theta_0 + 3 \cdot \frac{2\pi}{3} = \theta_0 + 2\pi \equiv \theta_0$: every orbit is a periodic orbit of period (exact, if θ_0 is not a fixed point of $R_{2\pi/3}^k$ for $k < 3$, which never happens here) 3: $\{\theta_0, \theta_0 + 2\pi/3, \theta_0 + 4\pi/3\}$.

Step 2 — exact Birkhoff average. $S_N \cos \theta(\theta_0)$ depends on $N \bmod 3$: for $N = 3k$, $S_{3k} = k \sum_{j=0}^2 \cos(\theta_0 + 2\pi j/3) = k \cdot 0 = 0$ (the sum of a cosine over the three translated cube roots of unity, identically zero for every θ_0 , by a direct check with the summation formulas); for $N = 3k+1$, $S_{3k+1} = 0 + \cos(\theta_0) = \cos \theta_0$; for $N = 3k+2$, $S_{3k+2} = 0 + \cos \theta_0 + \cos(\theta_0 + 2\pi/3)$.

Step 3 — the averages $\frac{1}{N} S_N$. For $N = 3k$: $\frac{1}{3k} \cdot 0 = 0$. For $N = 3k+1$: $\frac{\cos \theta_0}{3k+1} \rightarrow 0$. For $N = 3k+2$: $\frac{\cos \theta_0 + \cos(\theta_0 + 2\pi/3)}{3k+2} \rightarrow 0$. **All three subsequences tend to 0:** in this specific case ($\varphi = \cos \theta$, of zero mean over every period-3 orbit not aligned with the symmetries of $\cos$) the Birkhoff average *converges nonetheless* to $0 = \int \cos \theta d\mu$.

Step 4 — where Oxtoby’s hypothesis really fails. The failure is not in the convergence of $\frac{1}{N} S_N \varphi(\theta_0)$ for *fixed* θ_0 (which here converges), but in the **uniqueness of the invariant measure**: the uniform measure on the orbit $\{\theta_0, \theta_0 + 2\pi/3, \theta_0 + 4\pi/3\}$ (average of three Dirac masses) is invariant for every θ_0 , and for θ_0 different from 0 it gives measures *different* from Lebesgue and from one another: infinitely many invariant measures, as in Exercise 9.4(d). Oxtoby’s criterion (Theorem 9.9) requires **uniform convergence in θ_0 for every $\varphi \in C(\mathbb{T})$** : a single φ for which the averages do not converge uniformly (or do not converge at all for some θ_0) suffices to rule out unique ergodicity, and such a φ always exists when there are several invariant measures (by construction, taking φ close in $C(\mathbb{T})$ to a function which

distinguishes the two measures).

Typical errors. Concluding, from the fact that $\cos \theta$ converges nonetheless, that $R_{2\pi/3}$ could be uniquely ergodic: this is an error, it suffices to consider $\varphi = \mathbf{1}_{[0, \pi/6)}$ or a continuous approximation of it, whose Birkhoff average at $\theta_0 = 0$ (which touches the arc periodically in a different way from a generic θ_0) does not have the same uniform limit expected from $\int \varphi d\mu = 1/12$.

Extension. Repeat the computation with $\varphi(\theta) = \cos(3\theta)$: show that in this case $S_N \varphi(\theta_0)/N$ does *not* tend to $\int \varphi d\mu = 0$ but to $\cos(3\theta_0) \neq 0$ in general (because 3θ is constant on the period-3 orbit of $R_{2\pi/3}$), a more direct example of the breakdown of ergodic uniqueness visible already in the pointwise convergence.

Exercise 9.7: Lasota–Yorke for E_3 : the contraction becomes $1/3$. *Type:* symbolic. Generalise Lemma 9.2 to the transfer operator of E_3 restricted to $[0, 2\pi]$ (three affine branches of slope 3), and check that the contraction of the variation is $1/3$, not $1/2$.

Solution. Strategy. Repeat the construction of Section 9.12 with 3 branches instead of 2: the transfer operator averages over 3 preimages instead of 2, and the variation splits into 3 pieces instead of 2.

Step 1 — the operator. The three inverse branches of E_3 on $[0, 2\pi]$ are $\theta \mapsto \theta/3$, $\theta \mapsto \theta/3 + 2\pi/3$, $\theta \mapsto \theta/3 + 4\pi/3$ (slope $1/3$ each): $(\mathcal{L}_3 g)(\theta) = \frac{1}{3}[g(\theta/3) + g(\theta/3 + 2\pi/3) + g(\theta/3 + 4\pi/3)]$.

Step 2 — the inequality. Let $h_i(\theta) := g(\theta/3 + 2\pi i/3)$, $i = 0, 1, 2$: monotone reparametrisations (increasing, of slope $1/3$) of g restricted to the three arcs $[2\pi i/3, 2\pi(i+1)/3]$. By Lemma 9.1(iii), $\text{Var}_{[0, 2\pi]}(h_i) = \text{Var}_{[2\pi i/3, 2\pi(i+1)/3]}(g)$. By subadditivity and additivity on adjacent intervals (the three arcs cover $[0, 2\pi]$ without overlaps):

$$\text{Var}(\mathcal{L}_3 g) = \text{Var}\left(\frac{1}{3}h_0 + \frac{1}{3}h_1 + \frac{1}{3}h_2\right) \leq \frac{1}{3} \sum_{i=0}^2 \text{Var}(h_i) = \frac{1}{3} \sum_{i=0}^2 \text{Var}_{[2\pi i/3, 2\pi(i+1)/3]}(g) = \frac{1}{3} \text{Var}_{[0, 2\pi]}(g).$$

Step 3 — conclusion. The same proof as for Theorem 9.14, with $1/3$ in place of $1/2$, gives $\|\mathcal{L}_3^n g_0 - 1\|_\infty \leq 3^{-n} \text{Var}(g_0) \rightarrow 0$: a *faster* convergence towards the constant density (consistent with the invariance of Lebesgue for E_3 , Exercise 9.2(a)).

Numerical check. Adapting block [1] of `verifica_cap09.py` with three branches instead of two (the same initial density g_0 redistributed over $[0, 2\pi]$), one expects a decay of the variation with exact ratio $1/3$ per step instead of $1/2$, checkable with the same script adapted.

Interpretation. The contraction rate $1/m$ for E_m (here $m = 2$ and $m = 3$) is exactly the reciprocal of the number of branches — the same parameter which determines the topological entropy $h_{\text{top}}(E_m) = \log m$ (Chapter 8): more branches, more chaotic dynamics (more entropy), but also faster convergence towards statistical equilibrium, two faces of the same expansiveness.

Typical errors. Confusing the number of branches (which determines the factor $1/m$ in the Lasota–Yorke inequality) with the slope of the branches (which here coincides numerically with m only because E_m is affine with slope exactly m ; for a generic nonlinear expanding map, as in Remark 9.13, the contraction factor is $2/\inf |f'|$, not $1/(\text{number of branches})$).

Extension. Check that for general E_m the inequality is exactly $\text{Var}(\mathcal{L}_m g) \leq \frac{1}{m} \text{Var}(g)$ (same argument, m branches instead of 3), consistent with the fact that the unique invariant density in BV is always the constant 1 (Lebesgue), for every $m \geq 2$.

9.19. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher's key (Section 9.20).

1. (Definition) A measure-preserving system (X, μ, S) is **ergodic** if
 - A) every S -invariant set has measure $\in \{0, 1\}$
 - B) there exists a unique invariant measure
 - C) every orbit is dense
 - D) S is an isometry
2. (Hypotheses) Hopf's maximal ergodic theorem (Theorem 9.6) requires, besides (X, μ, T) preserving the probability measure,
 - A) φ continuous
 - B) $\varphi \in L^1(\mu)$
 - C) T invertible
 - D) T ergodic
3. (Symbolic computation) The transfer operator of the tent map $\mathcal{L}$ applied to $g(y) = \cos(3\pi y)$ gives
 - A) $\cos(3\pi y/2)$
 - B) 0
 - C) $\cos(6\pi y)$
 - D) $\frac{1}{2} \cos(3\pi y)$
4. (Numerical computation) With $\alpha = 2\pi/\varphi$ (φ the golden ratio), the uniform estimate $2/(N|1 - e^{i\alpha}|)$ of Theorem 9.10 for $N = 1$ equals (up to rounding to three digits)
 - A) 0.500
 - B) 1.073
 - C) 3.883
 - D) 6.283
5. (Proof step) In the proof of Part (ii) of Theorem 9.7 (identification of the limit), the truncation φ_M serves to
 - A) make φ continuous
 - B) obtain a bounded function to which dominated convergence applies (with dominating constant M) and thus obtain the L^1 convergence, which for general φ in L^1 is not directly guaranteed
 - C) make T invertible
 - D) avoid the use of Fatou's lemma
6. (Error diagnosis) The previous version of this chapter called the maximal ergodic theorem "Garsia's maximal theorem"; the diagnosed error (Remark 9.4) is
 - A) confusing the name of the theorem (due to Hopf, 1937) with the name of the person who gave a simpler proof of it (Garsia, 1965)
 - B) the theorem is in fact due to Birkhoff
 - C) Garsia never published anything on the subject
 - D) the correct name is "von Neumann's theorem"
7. (Transfer) If (X, S) is uniquely ergodic with measure μ , and $\pi : X \rightarrow Y$ is a continuous surjective factor onto (Y, S') , then
 - A) (Y, S') is also uniquely ergodic, with unique invariant measure $\pi_*\mu$ (the image of μ under π is S' -invariant by direct construction; that it is the *unique* such measure is a classical fact, provable with a fixed point argument on a compact convex set of measures, which exceeds the tools of this chapter)
 - B) (Y, S') has no invariant measures
 - C) (Y, S') always has exactly twice as many invariant measures as (X, S)
 - D) nothing can be said about (Y, S')
8. (Source and metacognition) The normality in base 2 of Champernowne's constant (Remark 9.11) is
 - A) proved in full in this chapter, as a direct corollary of Birkhoff's theorem
 - B) imported from Champernowne, *J. London Math. Soc.* 1933 (source E8), because its proof (a combinatorial counting of blocks with boundary effects) is independent of Birkhoff's theorem, which applies only to *almost every* point, not to the explicit point of Champernowne
 - C) still an open problem
 - D) an immediate consequence of the unique ergodicity of D

9.20. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

1) **A.** *Rationale:* Definition 9.2. *Distractors:* B is the definition of unique ergodicity (Definition 9.4), a different and stronger concept; C confuses ergodicity with topological transitivity (a different qualitative fact, however related — Remark 9.9); D describes a particular case (isometries have zero entropy, Chapter 8), not the general definition. *Diagnosed error:* confusing measurable and topological properties of the same system. *Source:* Definition 9.2. *Prerequisite:* invariant sets. *Remedial work:* reread the contrast between D and R_α in Section 9.7.

2) **B.** *Rationale:* Theorem 9.6: the hypothesis is $\varphi \in L^1(\mu)$, no other. *Distractors:* A is a stronger hypothesis, not required; C and D do not appear in the statement (the theorem holds even if T is not ergodic). *Diagnosed error:* adding unnecessary hypotheses out of habit (present in other theorems of the chapter, such as ergodicity in Part (iii) of Birkhoff, but not here). *Source:* Theorem 9.6. *Prerequisite:* $L^1(\mu)$. *Remedial work:* compare the hypotheses of Hopf, Birkhoff and von Neumann side by side in the Formulary (Section 9.21).

3) **B.** *Rationale:* $\mathcal{L}(\cos(k\pi y)) = \cos(\frac{k}{2}\pi y)$ if k is even, 0 if k is odd (Remark 9.13 and the note in the script of Example 9.4); $k = 3$ is odd, hence $\mathcal{L}(\cos(3\pi y)) = 0$. *Distractors:* A would apply the rule of the even case to an odd k ; C and D have no basis in the computation. *Diagnosed error:* failing to distinguish the even/odd cases in the spectrum of $\mathcal{L}$, exactly as for the transfer operator of D (Chapter 10) on the Fourier modes. *Source:* Example 9.4 and the associated note. *Prerequisite:* $\cos(k\pi - x) = (-1)^k \cos x$. *Remedial work:* redo the explicit computation for $k = 2$ and $k = 1$ and compare.

4) **B.** *Rationale:* computed in block [3] of `verifica_cap09.py`: $2/|1 - e^{i\alpha}| = 1.072924$, consistent with the rounding to 1.073. *Distractors:* A, C, D are other numerical values present in the same block (respectively not pertinent, the value of α itself, 2π) inserted to test the careful reading of the data. *Diagnosed error:* confusing the value of α with the estimate derived from it. *Source:* proof of Theorem 9.10, verification script. *Prerequisite:* modulus of a complex number. *Remedial work:* recompute $|1 - e^{i\alpha}| = 2|\sin(\alpha/2)|$ and check $2/(2|\sin(\alpha/2)|) = 1/|\sin(\alpha/2)|$.

5) **B.** *Rationale:* the proof of Part (ii) of Theorem 9.7. *Distractors:* A, C, D misunderstand the technical purpose of the truncation (which concerns neither the continuity of φ nor the invertibility of T , and does not “avoid” Fatou: Fatou is used after the truncation as well, to estimate $\|\varphi^* - \varphi_M^*\|_1$). *Diagnosed error:* failing to distinguish the three ingredients of the proof (truncation, pointwise estimate, density in L^1) explicitly required by the errata of the expansion plan. *Source:* proof of Theorem 9.7, Part (ii). *Prerequisite:* dominated convergence. *Remedial work:* rewrite by hand the chain of inequalities $|\int \varphi^* - \int \varphi| \leq 2\|\varphi - \varphi_M\|_1$.

6) **A.** *Rationale:* Remark 9.4. *Distractors:* B is false (Birkhoff has a different theorem, although related); C is false (Garsia 1965, source E4, is a real article, verified); D confuses it with another completely different theorem of the chapter. *Diagnosed error:* attributing a theorem to the author of the best-known proof, rather than to the author of the original result. *Source:* Remark 9.4, sources E3–E5. *Prerequisite:* none beyond careful reading. *Remedial work:* list, for every theorem with a proper name in the chapter, the pair (author of the theorem, author of the proof reproduced here, if different).

7) **A.** *Rationale:* $\pi_*\mu$ is S' -invariant by direct construction (an immediate check from the definition of image measure and from the relation $\pi S = S'\pi$); uniqueness is a classical fact of ergodic theory (not proved in this chapter, which however uses a particular case of it: the topological conjugacy of Remark 9.16, where $\pi = \Psi$ is a homeomorphism, a case in

which the uniqueness of the lift is immediate since Ψ is bijective). *Distractors*: B, C, D have no theoretical basis. *Diagnosed error*: thinking that unique ergodicity is a property which is lost when passing to a factor, whereas it is in fact inherited (in the “downwards” direction; the converse does not hold: an extension of a uniquely ergodic system may have several invariant measures). *Source*: Remark 9.16 (particular case used in the chapter). *Prerequisite*: Definition 9.4. *Remedial work*: check the invariance of $\pi_*\mu$ from $\pi S = S'\pi$ step by step, with the same set-theoretic scheme as in Proposition 9.2.

8) B. *Rationale*: Remark 9.11, source E8. *Distractors*: A confuses it with Borel’s Corollary (Corollary 9.2), which concerns *almost every* point, not an explicit point; C is false (the result has been proved since 1933); D would confuse unique ergodicity (a property of D as a system) with the normality of a specific number (a property of a point). *Diagnosed error*: failing to distinguish a statement “for almost every point” from a statement about an explicit point — the same distinction as in Example 9.2. *Source*: Remark 9.11. *Prerequisite*: quantifiers (“for almost every” versus “for the explicitly constructed number”). *Remedial work*: reread the register of the imports (Section 9.16).

BEST PRACTICE

Assigned exercises — answers.

9.2 (E_5). (a) $E_5^{-1}([a, b])$ is a union of 5 disjoint arcs of length $(b - a)/5$ each (inverse branches $\theta \mapsto (\theta + 2\pi j)/5$, $j = 0, \dots, 4$): total measure $5 \cdot (b - a)/5 = b - a$. (b) Theorem 9.1 guarantees the existence *a priori*, before any explicit computation: useful when (unlike this case) no explicit candidate is known. (c) Yes, the same Fourier argument as in Theorem 9.3: $\varphi(5\theta)$ has an expansion with coefficient of $e^{im\theta}$ equal to $c_{m/5}$ if $5 \mid m$, 0 otherwise; iterating, $c_j = c_{5^n j} \rightarrow 0$ by Riemann–Lebesgue if $j \neq 0$; $5 \neq 1$ is needed in order to have $5^n \rightarrow \infty$ (if the map were $E_1 = \text{id}$, the iteration would not move the index, and every function would be invariant). (d) No: the Dirac measure δ_0 is E_5 -invariant ($E_5(0) = 0$), different from Lebesgue.

9.3 (base 3). (a) For μ -almost every $\theta \in \mathbb{T}$, every ternary block $w \in \{0, 1, 2\}^\ell$ appears in the base-3 expansion of $\theta/2\pi$ with asymptotic frequency $3^{-\ell}$. (b) $\varphi_w = \mathbf{1}_{A_w}$, A_w an arc of length $2\pi \cdot 3^{-\ell}$, measure $3^{-\ell}$. (c) The ergodicity of E_3 with respect to Lebesgue, Exercise 9.2(c) adapted with 3 in place of 5 (the same Fourier argument). (d) Yes, by the same type of imported argument (Remark 9.11): the primary source (Champernowne 1933) explicitly treats “normal in the scale of r ” for a generic base $r \geq 2$, not only $r = 10$ or $r = 2$.

9.4 (rational rotations). (a) Every orbit is the set of the 3 points $\{\theta_0, \theta_0 + 2\pi/3, \theta_0 + 4\pi/3\}$: a regular polygon with 3 vertices, as in the general case of the rational rotations (Chapter 3). (b) The unique invariant probability measure on the orbit of θ_0 is the uniform measure $\frac{1}{3}(\delta_{\theta_0} + \delta_{\theta_0 + 2\pi/3} + \delta_{\theta_0 + 4\pi/3})$ (the only way of distributing mass on a periodic orbit in an invariant way, the action of $R_{2\pi/3}$ being a cyclic permutation of the three points). (c) $S_3 \cos \theta(\theta_0) = \sum_{j=0}^2 \cos(\theta_0 + 2\pi j/3) = 0$ for every θ_0 (a trigonometric identity); $S_1 = \cos \theta_0$; $S_2 = \cos \theta_0 + \cos(\theta_0 + 2\pi/3) = -\cos(\theta_0 + 4\pi/3)$ (from the vanishing sum): the average $S_N/N \rightarrow 0$ along every residue class (see Exercise 9.6 for the complete discussion: the truly instructive example uses $\varphi(\theta) = \cos(3\theta)$, for which $S_N \varphi(\theta_0)/N \rightarrow \cos(3\theta_0)$, dependent on θ_0 , which really breaks the uniformity). (d) By Oxtoby’s criterion, the existence of several invariant measures (one for every orbit, point (b)) rules out unique ergodicity; two explicit distinct measures: the uniform measure on the orbit of $\theta_0 = 0$ and the one on the orbit of $\theta_0 = \pi/6$.

9.21. Chapter formulary

Invariant measures

$$\mu(S^{-1}A) = \mu(A) \quad \forall A \quad \Longleftrightarrow \quad \int \varphi \circ S \, d\mu = \int \varphi \, d\mu \quad \forall \varphi \in C(X)$$

(Krylov–Bogolyubov) X compact, S continuous $\implies \exists \mu$ invariant

Lebesgue invariant

$$\mu(E_m^{-1}A) = \mu(A) \quad \forall A \text{ measurable}, \quad m \geq 2 \quad (D = E_2)$$

Ergodicity and mixing (via Fourier)

$$\varphi \circ D = \varphi \text{ a.e.} \implies c_k = 0 \quad (k \neq 0) \implies \varphi \text{ constant a.e.}$$

$$C_n(f, g) = \int (f \circ D^n) \bar{g} \, d\mu - \left(\int f \, d\mu \right) \left(\int \bar{g} \, d\mu \right) \xrightarrow{n \rightarrow \infty} 0 \quad (\text{exactly 0 for trigonometric polynomials, } n \text{ large})$$

Ergodic theorems

$$(\text{Hopf, maximal}) \quad \varphi \in L^1, \quad E = \left\{ \sup_n S_n \varphi > 0 \right\} \implies \int_E \varphi \, d\mu \geq 0$$

$$(\text{Birkhoff}) \quad \varphi^*(x) = \lim_n \frac{1}{n} S_n \varphi(x) \text{ a.e.}, \quad \int \varphi^* \, d\mu = \int \varphi \, d\mu, \quad (\text{ergodic}) \quad \varphi^* \equiv \int \varphi \, d\mu$$

$$(\text{von Neumann, in mean}) \quad \frac{1}{N} \sum_{n < N} U^n \varphi \xrightarrow{L^2} P\varphi = \left(\int \varphi \, d\mu \right) \cdot 1 \quad (\text{ergodic case})$$

Unique ergodicity (Oxtoby)

$$\text{uniq. ergodic} \iff \forall \varphi \in C(X) : \frac{1}{N} S_N \varphi(x) \rightarrow \text{const. uniformly in } x$$

$$R_\alpha : \left| \frac{1}{N} S_N e_k(\theta) \right| \leq \frac{2}{N |e^{ik\alpha} - 1|} \quad (\text{unif. in } \theta, k \neq 0) \quad D : \text{not uniq. ergodic (measures on periodic orbits)}$$

Normal numbers

(Borel, via Birkhoff on D) almost every θ normal in base 2 (Champernowne, imported) $0.\overline{1} \overline{10} \overline{11} \cdots_2$

Ulam–von Neumann and the triangle D – T – f_4

$$d\nu = \frac{dx}{\pi \sqrt{x(1-x)}} \text{ invariant and ergodic for } f_4, \quad h_T(x) = \sin^2\left(\frac{\pi x}{2}\right) : T \xleftrightarrow{\text{full conj.}} f_4$$

Lasota–Yorke (tent map)

$$(\mathcal{L}w)(y) = \frac{1}{2} \left[w\left(\frac{y}{2}\right) + w\left(1 - \frac{y}{2}\right) \right], \quad \text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g)$$

$$\|\mathcal{L}^n g_0 - 1\|_\infty \leq 2^{-n} \text{Var}(g_0) \xrightarrow{n \rightarrow \infty} 0 \quad (\text{tent map, Theorem 9.14, proved in full in this chapter; the general})$$

Kolmogorov–Sinai entropy

$$h_\mu(S, \mathcal{P}) = \lim_n \frac{1}{n} H(\mathcal{P}^n), \quad h_\mu(S) = \sup_{\mathcal{P}} h_\mu(S, \mathcal{P}) \quad h_{\mu_{\text{Leb}}}(D) = \log 2 \quad (\text{Chapter 8})$$

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CHAPTER 10

Transfer operators, decay of correlations and the central limit theorem

Chapter 9 established what happens *on average*: the ergodic theorem. This chapter asks how fast that average is reached, and with what fluctuations. The two answers — **decay of correlations** and **central limit theorem** — are governed by one and the same object, the **transfer operator** $\mathcal{L}$ (or Perron–Frobenius operator) of the doubling map D , already met in affine form for the tent map in Section 9.12. The previous chapter, written in full, had however a precise defect in its first draft (never published as a file in its own right): the spectrum of $\mathcal{L}$ was described insufficiently, and the dynamical central limit theorem was merely stated. Here both things are corrected and completed: the spectrum of $\mathcal{L}$ is radically different depending on the function space on which one looks at it (a disc full of eigenvalues on L^2 , a hole — a **spectral gap** — on spaces of regular functions), and the central limit theorem is proved in full in two independent ways, both explicit.

INFO

Level and prerequisites. Level **D–E**: level D core (transfer operator, decay of correlations, CLT for the dyadic indicator), level E sections (spectrum on L^2 , spectral gap with an explicit estimate, Gordin’s martingale method). Prerequisites within the treatise: the maps D , E_m and their semiconjugacy with the binary shift Σ_2 (Definition 7.1, Theorem 7.4, Chapter 7); Fourier series on $\mathbb{T}$ (Volume II, Chapter 4); invariant measures, ergodicity and mixing of D (Sections 9.2–9.3, Chapter 9); total variation, the space BV and the Lasota–Yorke theorem for the tent map (Section 9.12, Chapter 9), reused here and not reproved from scratch; the topological entropy $h_{\text{top}}(D) = \log 2$ (Chapter 8). From probability we recall, as classical facts stated with a source but not fully reproved, Stirling’s formula and the central limit theorem for stationary martingales (Section 10.7); everything that is specific to the dynamics of D — the construction of the coboundary, the decomposition, the variance — is instead proved in full.

Warning. Conventions of the chapter. We keep the notation of Chapter 9: $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, μ the normalised Lebesgue measure ($d\mu = d\theta/2\pi$), $D(\theta) = 2\theta \bmod 2\pi$, $E_m(\theta) = m\theta \bmod 2\pi$ ($m \geq 2$ an integer, $D = E_2$), $e_k(\theta) = e^{ik\theta}$ ($k \in \mathbb{Z}$). For $g \in L^1(\mathbb{T})$ we write $\hat{g}_k = \frac{1}{2\pi} \int_{\mathbb{T}} g(\theta) e^{-ik\theta} d\theta$ for the Fourier coefficient. The **total variation** $\text{Var}(g)$ and the space BV are those of Definition 9.5: for a 2π -periodic function one writes $BV(\mathbb{T}) := BV([0, 2\pi])$ (periodicity makes the equality of the values at the endpoints automatic, with no need for a matching term), and all the properties of Lemma 9.1 (subadditivity, additivity on adjacent intervals, invariance under monotone reparametrisation, oscillation $\leq$ variation) apply verbatim with $[0, 2\pi]$ in place of $[0, 1]$, with the same proof. The symbol $\mathcal{L}$ (without an index) denotes in this chapter the transfer operator of D on $\mathbb{T}$, defined in Section 10.1: it is the analogue, for D , of the operator $\mathcal{L}$ of Definition 9.6 for the tent map T on $[0, 1]$ — same name, same idea (average over the preimages), different domain; when it is necessary to distinguish them one writes $\mathcal{L}_D$ and $\mathcal{L}_T$. For general $m \geq 2$, $\mathcal{L}_m$ denotes the transfer operator of E_m (Section 10.1); $\mathcal{L} = \mathcal{L}_2$.

10.1. The transfer operator of D

Definition 10.1: Transfer (Perron–Frobenius) operator of D . For $g \in L^1(\mathbb{T})$,

$$(\mathcal{L}g)(\theta) := \frac{1}{2} \left[g\left(\frac{\theta}{2}\right) + g\left(\frac{\theta}{2} + \pi\right) \right], \quad \theta \in \mathbb{T},$$

the average of g over the two inverse branches $\theta \mapsto \theta/2$ and $\theta \mapsto \theta/2 + \pi$ of D (the two preimages of θ , each reached by a branch of slope $\pm \frac{1}{2}$). For $m \geq 2$ an integer, the transfer operator of E_m is

$$(\mathcal{L}_m g)(\theta) := \frac{1}{m} \sum_{j=0}^{m-1} g\left(\frac{\theta + 2\pi j}{m}\right), \quad \mathcal{L} = \mathcal{L}_2.$$

Proposition 10.1: $\mathcal{L}_m$ is the adjoint of composition with E_m . For every $g \in L^\infty(\mathbb{T})$ and $w \in L^1(\mathbb{T})$,

$$\int_{\mathbb{T}} (g \circ E_m) w \, d\mu = \int_{\mathbb{T}} g \cdot (\mathcal{L}_m w) \, d\mu.$$

Consequently, for $w \geq 0$ with $\int_{\mathbb{T}} w \, d\mu = 1$: the measure $w \, d\mu$ is E_m -invariant if and only if $\mathcal{L}_m w = w$ a.e.

Proof. Identical in structure to Proposition 9.2 (same proof, with m branches instead of 2 and the change of variable $\theta = (\eta + 2\pi j)/m$ on each branch instead of one). Split the integral over the m branches of monotonicity of E_m on $[0, 2\pi]$ (the intervals $I_j = [2\pi j/m, 2\pi(j+1)/m)$, $j = 0, \dots, m-1$):

$$\int_{\mathbb{T}} (g \circ E_m)(\theta) w(\theta) \frac{d\theta}{2\pi} = \sum_{j=0}^{m-1} \int_{I_j} g(m\theta - 2\pi j) w(\theta) \frac{d\theta}{2\pi}.$$

On I_j , the substitution $\eta = m\theta - 2\pi j$ (an increasing monotone bijection $I_j \rightarrow [0, 2\pi]$, $\theta = (\eta + 2\pi j)/m$, $d\theta = d\eta/m$) gives $\int_{I_j} g(m\theta - 2\pi j) w(\theta) d\theta = \int_0^{2\pi} g(\eta) w\left(\frac{\eta + 2\pi j}{m}\right) \frac{d\eta}{m}$. Summing over j and dividing by 2π :

$$\int_{\mathbb{T}} (g \circ E_m) w \, d\mu = \int_{\mathbb{T}} g(\eta) \cdot \frac{1}{m} \sum_{j=0}^{m-1} w\left(\frac{\eta + 2\pi j}{m}\right) \frac{d\eta}{2\pi} = \int_{\mathbb{T}} g \cdot (\mathcal{L}_m w) \, d\mu.$$

The characterisation of the invariant densities follows as in Proposition 9.2: testing on $g = \mathbf{1}_A$ one obtains the definition of invariance, and for general $g \in L^\infty$ (in particular $g = \text{sign}(\mathcal{L}_m w - w)$) the same separation of the classes of L^1 . $\square$

Theorem 10.1: Action of $\mathcal{L}_m$ on the Fourier coefficients. For every $k \in \mathbb{Z}$, $\mathcal{L}_m e_k = e_{k/m}$ if $m \mid k$, $\mathcal{L}_m e_k = 0$ otherwise. Equivalently, for $g = \sum_k \hat{g}_k e_k \in L^2(\mathbb{T})$, the Fourier coefficient of index k of $\mathcal{L}_m g$ is $\hat{g}_{mk}$:

$$(\widehat{\mathcal{L}_m g})_k = \hat{g}_{mk} \quad (k \in \mathbb{Z}).$$

For $m = 2$ (the case of D): $\mathcal{L} e_k = e_{k/2}$ if k is even, $\mathcal{L} e_k = 0$ if k is odd.

Proof. Direct computation from the definition:

$$(\mathcal{L}me_k)(\theta) = \frac{1}{m} \sum_{j=0}^{m-1} \exp\left(ik \frac{\theta + 2\pi j}{m}\right) = e^{ik\theta/m} \cdot \frac{1}{m} \sum_{j=0}^{m-1} e^{2\pi i k j/m}.$$

The geometric sum $\frac{1}{m} \sum_{j=0}^{m-1} \omega^j$ with $\omega = e^{2\pi i k/m}$ equals 1 if $\omega = 1$ (that is, if $m \mid k$) and 0 otherwise (the sum of a geometric progression of ratio $\omega \neq 1$ with m terms, which vanishes: $\sum_{j=0}^{m-1} \omega^j = \frac{\omega^m - 1}{\omega - 1} = 0$ since $\omega^m = e^{2\pi i k} = 1$). If $m \mid k$, writing $k = mk'$: $(\mathcal{L}me_k)(\theta) = e^{ik\theta/m} = e^{ik'\theta} = e_{k/m}(\theta)$. If $m \nmid k$: $\mathcal{L}me_k \equiv 0$.

For the second form, write $g = \sum_j \hat{g}_j e_j$ (convergence in L^2 , by Parseval); by linearity and continuity of $\mathcal{L}_m$ on L^2 (a bounded operator, see the Remark below),

$$\mathcal{L}mg = \sum_j \hat{g}_j \mathcal{L}me_j = \sum_{j: m \mid j} \hat{g}_j e_{j/m} = \sum_{k \in \mathbb{Z}} \hat{g}_{mk} e_k,$$

(reparametrising $j = mk$): the coefficient of index k of $\mathcal{L}_m g$ is $\hat{g}_{mk}$. □

Remark 10.1: $\mathcal{L}_m$ is a contraction on L^2 . From Theorem 10.1, $\|\mathcal{L}mg\|_2^2 = \sum_k |\hat{g}_{mk}|^2 \leq \sum_{j \in \mathbb{Z}} |\hat{g}_j|^2 = \|g\|_2^2$ (the sum on the left is a subsum, over the indices that are multiples of m , of the one on the right): $\mathcal{L}_m$ is a bounded linear operator on $L^2(\mathbb{T})$, of norm ≤ 1 (norm exactly 1, since $\mathcal{L}_m 1 = 1$). This justifies the passage to the limit used above and will be reused in Section 10.2.

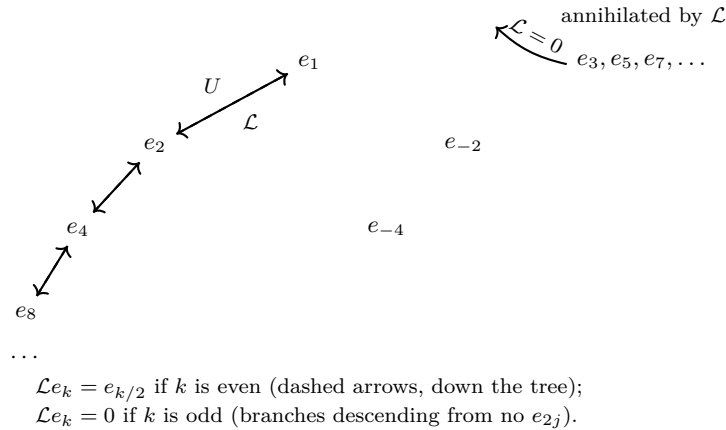


Figure 10.1: The action of the transfer operator $\mathcal{L}$ on the Fourier characters (Theorem 10.1, $m = 2$): the Koopman operator U (composition with D , $Ue_k = e_{2k}$) makes the index climb by doubling it, generating a binary tree of modes; its adjoint $\mathcal{L}$ goes down one level ($e_{2k} \mapsto e_k$, dashed arrows) and annihilates every mode of odd index, which has no preimage in the tree. Figure drawn by hand; direct check $\mathcal{L}e_k = e_{k/2}$ (k even), $\mathcal{L}e_k = 0$ (k odd) in the text of the proof of Theorem 10.1.

10.2. The spectrum of $\mathcal{L}$ on L^2 : the full disc

This section corrects the errata of the expansion plan (entry 4.14): the old version of this chapter concluded, from an argument on the odd modes, that $\mathcal{L}$ has no eigenvalues of modulus 1 other than 1 — a true conclusion, but reached with an argument that did not exclude (and indeed omitted) the existence of eigenvalues of modulus < 1 . We now prove the complete and correct description: on $L^2(\mathbb{T})$, the only eigenvalue of modulus 1 is $\lambda = 1$ (with eigenspace the constants), but **every** λ with $|\lambda| < 1$ is an eigenvalue, with an explicit eigenfunction.

Lemma 10.1: Unique eigenvalue of modulus 1. If $\mathcal{L}g = \lambda g$ for some $g \in L^2(\mathbb{T})$, $g \neq 0$, and $|\lambda| = 1$, then $\lambda = 1$ and g is constant a.e.

Proof. Let $g = \sum_{j \in \mathbb{Z}} c_j e_j$ (Parseval). By Theorem 10.1, the coefficient of index j of $\mathcal{L}g$ is c_{2j} : the equation $\mathcal{L}g = \lambda g$ is equivalent to $c_{2j} = \lambda c_j$ for every $j \in \mathbb{Z}$. From $|\lambda| = 1$ it follows that $|c_{2j}| = |c_j|$ for every j , hence by induction $|c_j| = |c_{2j}| = |c_{4j}| = \dots = |c_{2^n j}|$ for every $n \geq 0$. If $j \neq 0$, the indices $2^n j$ ($n \geq 0$) are infinitely many and distinct, and since $g \in L^2$ ($\sum_k |c_k|^2 < \infty$) we have $c_{2^n j} \rightarrow 0$ as $n \rightarrow \infty$ (general term of a convergent series); but the sequence $|c_{2^n j}|$ is constant in n (equal to $|c_j|$): the only constant tending to 0 is 0, hence $c_j = 0$. This holds for every $j \neq 0$: $g = c_0 e_0 = c_0$ a.e., a constant. If $c_0 = 0$ then $g \equiv 0$, excluded by hypothesis; hence $c_0 \neq 0$, and the equation $c_0 = \lambda c_0$ (case $j = 0$ of the relation) gives $\lambda = 1$. $\square$

Theorem 10.2: Every λ with $|\lambda| < 1$ is an eigenvalue of $\mathcal{L}$ on L^2 , with an explicit eigenfunction. For every $\lambda \in \mathbb{C}$ with $|\lambda| < 1$, the lacunary series

$$g_\lambda(\theta) := \sum_{n=0}^{\infty} \lambda^n e^{i2^n \theta}$$

converges in $L^2(\mathbb{T})$ (and absolutely and uniformly if $|\lambda| < 1$ in the strict sense, since $|e^{i2^n \theta}| = 1$), defines a non-zero function of $L^2(\mathbb{T})$, and satisfies $\mathcal{L}g_\lambda = \lambda g_\lambda$.

Proof. Convergence. The Fourier coefficients of g_λ are $c_{2^n} = \lambda^n$ ($n \geq 0$) and $c_k = 0$ for every k that is not a power of 2 with exponent ≥ 0 (including all the $k \leq 0$, the negative k , and the $k > 0$ that are not powers of 2): the exponents 2^n are pairwise distinct, hence this is already the Fourier representation of g_λ . For $|\lambda| < 1$, $\sum_n |c_{2^n}|^2 = \sum_n |\lambda|^{2n} = \frac{1}{1-|\lambda|^2} < \infty$: by Parseval $g_\lambda \in L^2(\mathbb{T})$, with $\|g_\lambda\|_2^2 = \frac{1}{1-|\lambda|^2}$; $g_\lambda \neq 0$ since $c_1 = \lambda^0 = 1 \neq 0$ (or, if $\lambda = 0$, $g_0 = e_0 = 1 \neq 0$). **Eigenvalue equation.** By Theorem 10.1, the coefficient of index j of $\mathcal{L}g_\lambda$ is c_{2j} . If $j = 2^n$ for some $n \geq 0$: $c_{2j} = c_{2^{n+1}} = \lambda^{n+1} = \lambda \cdot \lambda^n = \lambda c_j$. If $j = 0$: $c_0 = 0$ (it is not a power of 2 with $n \geq 0$, since $2^n \geq 1$ for every $n \geq 0$), and $c_{2j} = c_0 = 0 = \lambda c_j = \lambda \cdot 0$: consistent for every λ . If j is not of the form 2^n ($n \geq 0$) and $j \neq 0$: then neither is $2j$ (if $2j = 2^n$ with $n \geq 0$ then $n \geq 1$ and $j = 2^{n-1}$, excluded), hence $c_{2j} = 0 = \lambda \cdot 0 = \lambda c_j$ (since also $c_j = 0$). In all cases the coefficient of index j of $\mathcal{L}g_\lambda$ coincides with λc_j , that is, with the coefficient of index j of λg_λ : $\mathcal{L}g_\lambda = \lambda g_\lambda$. $\square$

Remark 10.2: Direct check, without going through Fourier. It is worth verifying the identity also by direct substitution into the definition of $\mathcal{L}$, without invoking the uniqueness of the Fourier expansion. With $y = \theta/2$: $g_\lambda(\theta/2) = \sum_n \lambda^n e^{i2^{n-1}\theta}$, while $g_\lambda(\theta/2 + \pi) = \sum_n \lambda^n e^{i2^{n-1}\theta} e^{i2^n \pi}$; for $n = 0$, $e^{i2^{-1}\pi} = e^{i\pi} = -1$, and for $n \geq 1$, $2^n \pi$ is an integer multiple of 2π (since 2^n is even), hence $e^{i2^n \pi} = 1$. Summing: the term $n = 0$ of $g_\lambda(\theta/2)$ (which equals $e^{i\theta/2}$) and that of $g_\lambda(\theta/2 + \pi)$ (which equals $-e^{i\theta/2}$) **cancel exactly**, and the terms $n \geq 1$ double:

$$g_\lambda(\theta/2) + g_\lambda(\theta/2 + \pi) = 2 \sum_{n \geq 1} \lambda^n e^{i2^{n-1}\theta} = 2\lambda \sum_{m \geq 0} \lambda^m e^{i2^m \theta} = 2\lambda g_\lambda(\theta),$$

hence $(\mathcal{L}g_\lambda)(\theta) = \lambda g_\lambda(\theta)$: the same conclusion, by a completely elementary route. The cancellation of the term $n = 0$ is precisely the reason why the coefficient c_0 (absent from the lacunary series) poses no problems.

Example 10.1: Numerical check of the eigenfunction, $\lambda = 0.5$. Block [1] of `verifica_cap10.py` (Appendix B, run on 17 September 2026) truncates the series $g_{0.5}(\theta) = \sum_{n=0}^{40} 0.5^n e^{i2^n \theta}$ (the truncation error is $O(0.5^{40}) \approx 9 \times 10^{-13}$, negligible) and computes numerically $(\mathcal{L}g_{0.5})(\theta)$ from the definition, comparing it with $0.5 \cdot g_{0.5}(\theta)$, at five independent points θ . The maximum discrepancy observed is 4.57×10^{-13} ✓, consistent with machine precision: for example, at $\theta = 0.3$, $(\mathcal{L}g_{0.5})(0.3) = 0.6806366629 + 0.4207227800i$ against $0.5 \cdot g_{0.5}(0.3) = 0.6806366629 + 0.4207227800i$ (agreement to 10 digits).

Remark 10.3: The complete picture on L^2 . Combining Lemma 10.1 and Theorem 10.2: the eigenvalues of $\mathcal{L}$ on $L^2(\mathbb{T})$ fill the whole closed unit disc $\{|\lambda| \leq 1\}$ (for $|\lambda| < 1$ the explicit eigenfunction g_λ ; for $\lambda = 1$ the constants; for $\lambda = 0$, trivially, every function supported on the modes of odd index only, for example e_1 : $\mathcal{L}e_1 = 0$ by Theorem 10.1), and no λ with $|\lambda| = 1$, $\lambda \neq 1$, is an eigenvalue. Since $\mathcal{L}$ is a contraction (Remark 10.1), the spectrum (in the sense of functional analysis, not only the eigenvalues) is contained in the closed disc $\{|\lambda| \leq 1\}$: on the whole space L^2 , the point spectrum coincides with the whole disc. This is the structural reason why, on L^2 , there exists no “spectral gap” isolating $\lambda = 1$ from the rest: Section 10.3 shows that the gap emerges only by restricting $\mathcal{L}$ to more regular function spaces, where the pathological eigenfunctions such as g_λ (a lacunary series, typically not of bounded variation: see Remark 10.4) are excluded by construction.

10.3. Spectral gap on the regular spaces: BV , Hölder, C^1

Lemma 10.2: Contraction of the total variation by $\mathcal{L}$. For every $g \in BV(\mathbb{T})$, $\text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g)$.

Proof. Identical in structure to Lemma 9.2 (same argument, on the two branches of D instead of those of T , which have the same slope $\pm \frac{1}{2}$). Let $h_1(\theta) := g(\theta/2)$, $h_2(\theta) := g(\theta/2 + \pi)$, $\theta \in [0, 2\pi]$: increasing monotone reparametrisations of g restricted, respectively, to $[0, \pi]$ (via $\theta \mapsto \theta/2$) and to $[\pi, 2\pi]$ (via $\theta \mapsto \theta/2 + \pi$). By property (iii) of Lemma 9.1 (invariance of the variation under monotone reparametrisation, here on $[0, 2\pi]$), $\text{Var}_{[0, 2\pi]}(h_1) = \text{Var}_{[0, \pi]}(g)$ and $\text{Var}_{[0, 2\pi]}(h_2) = \text{Var}_{[\pi, 2\pi]}(g)$. By subadditivity (i) and homogeneity of the variation:

$$\text{Var}(\mathcal{L}g) = \text{Var}\left(\frac{1}{2}h_1 + \frac{1}{2}h_2\right) \leq \frac{1}{2}\text{Var}(h_1) + \frac{1}{2}\text{Var}(h_2) = \frac{1}{2}[\text{Var}_{[0, \pi]}(g) + \text{Var}_{[\pi, 2\pi]}(g)] = \frac{1}{2} \text{Var}_{[0, 2\pi]}(g)$$

by additivity (ii) on the adjacent intervals $[0, \pi]$, $[\pi, 2\pi]$. $\square$

Lemma 10.3: Contraction of the Hölder seminorm by $\mathcal{L}$. For $\alpha \in (0, 1]$ and $g : \mathbb{T} \rightarrow \mathbb{C}$ α -Hölder, with seminorm $[g]_\alpha := \sup_{x \neq y} \frac{|g(x) - g(y)|}{|x - y|^\alpha}$ (the supremum computed over representatives in one period, with $|x - y|$ the Euclidean distance between nearby lifts), one has $[\mathcal{L}g]_\alpha \leq (\frac{1}{2})^\alpha [g]_\alpha$.

Proof. For $x, y \in \mathbb{R}$,

$$|(\mathcal{L}g)(x) - (\mathcal{L}g)(y)| \leq \frac{1}{2}|g(\frac{x}{2}) - g(\frac{y}{2})| + \frac{1}{2}|g(\frac{x}{2} + \pi) - g(\frac{y}{2} + \pi)| \leq \frac{1}{2}[g]_\alpha \left|\frac{x-y}{2}\right|^\alpha + \frac{1}{2}[g]_\alpha \left|\frac{x-y}{2}\right|^\alpha = [g]_\alpha \left(\frac{1}{2}\right)^\alpha |x-y|^\alpha$$

(the triangle inequality followed by the definition of $[g]_\alpha$ on each of the two terms, with the same increment $|x - y|/2$ in both, both branches being affine of slope $1/2$). Dividing by $|x - y|^\alpha$ and taking the supremum: $[\mathcal{L}g]_\alpha \leq (\frac{1}{2})^\alpha [g]_\alpha$. $\square$

Theorem 10.3: Spectral gap on $BV(\mathbb{T})$ and on $C^{0,\alpha}(\mathbb{T})$, with an explicit estimate.

Let $g \in BV(\mathbb{T})$ (resp. α -Hölder) with $\mathcal{L}g = \lambda g$, $g \not\equiv \text{const}$. Then $|\lambda| \leq \frac{1}{2}$ (resp. $|\lambda| \leq (\frac{1}{2})^\alpha$). In particular, for $C^1(\mathbb{T}) \subset C^{0,1}(\mathbb{T})$: every eigenvalue of $\mathcal{L}$ restricted to $C^1(\mathbb{T})$, other than $\lambda = 1$ (with eigenspace the constants), satisfies $|\lambda| \leq \frac{1}{2}$. Moreover, for every $g_0 \in BV(\mathbb{T})$ with $\int_{\mathbb{T}} g_0 d\mu = 1$, $g_0 \geq 0$ (a probability density in BV),

$$\|\mathcal{L}^n g_0 - 1\|_\infty \leq \text{Var}(\mathcal{L}^n g_0) \leq 2^{-n} \text{Var}(g_0) \xrightarrow{n \rightarrow \infty} 0,$$

reconfirming (with an explicit exponential rate) that the constant density $\rho \equiv 1$ (Lebesgue) is the only D -invariant density in $BV(\mathbb{T})$.

Proof. No eigenvalue in $(1/2, 1) \cup (-1, -1/2) \cup \dots$, case BV . If $\mathcal{L}g = \lambda g$ with $g \in BV(\mathbb{T})$, iterating Lemma 10.2: $\text{Var}(\mathcal{L}^n g) = |\lambda|^n \text{Var}(g)$ (an equality, since $\mathcal{L}^n g = \lambda^n g$ for an eigenfunction) and $\text{Var}(\mathcal{L}^n g) \leq 2^{-n} \text{Var}(g)$ (repeated application of the lemma). If g is not constant, $\text{Var}(g) > 0$ (the variation of a function that is not identically constant is strictly positive, by property (iv) of Lemma 9.1: if $\text{Var}(g) = 0$ then $\sup g - \inf g \leq 0$, that is, g is constant), and one may divide: $|\lambda|^n \leq 2^{-n}$ for every $n \geq 1$, hence $|\lambda| \leq \frac{1}{2}$.

Hölder case. Identical with Lemma 10.3 in place of Lemma 10.2: $[\mathcal{L}^n g]_\alpha = |\lambda|^n [g]_\alpha \leq (\frac{1}{2})^{n\alpha} [g]_\alpha$, and if g is not constant $[g]_\alpha > 0$ (a function with vanishing Hölder seminorm is constant, by definition of the seminorm), hence $|\lambda| \leq (\frac{1}{2})^\alpha$. For $\alpha = 1$ (Hölder with exponent 1, that is, Lipschitz, in particular every $g \in C^1(\mathbb{T})$ with derivative bounded by C , which is 1-Lipschitz with $[g]_1 \leq \|g'\|_\infty$): $|\lambda| \leq \frac{1}{2}$.

Rate of convergence. Identical, mutatis mutandis, to the proof of Theorem 9.14: $\text{Var}(\mathcal{L}^n g_0) \leq 2^{-n} \text{Var}(g_0)$ by iterating Lemma 10.2; $\int_{\mathbb{T}} \mathcal{L}^n g_0 d\mu = \int_{\mathbb{T}} g_0 d\mu = 1$ for every n (Proposition 10.1 with $g \equiv 1$ gives $\int_{\mathbb{T}} \mathcal{L}w d\mu = \int_{\mathbb{T}} w d\mu$ for every $w \in L^1$); by property (iv) of Lemma 9.1 applied to $h = \mathcal{L}^n g_0$, $\sup h - \inf h \leq \text{Var}(h) \leq 2^{-n} \text{Var}(g_0)$, and since $\int_{\mathbb{T}} h d\mu = 1$ we have $\inf h \leq 1 \leq \sup h$ (otherwise the integral, a weighted average of values all < 1 or all > 1 , could not equal 1): hence $|h(\theta) - 1| \leq 2^{-n} \text{Var}(g_0)$ for every θ , exactly the same squeeze argument as in Theorem 9.14. $\square$

Remark 10.4: Why the two pictures do not contradict each other. Theorem 10.3 does not contradict Theorem 10.2: the explicit eigenfunction $g_\lambda = \sum_n \lambda^n e_{2^n}$, for $\frac{1}{2} < |\lambda| < 1$, is not in $BV(\mathbb{T})$ (nor Hölder for any positive exponent), consistently with the fact that Theorem 10.3 excludes precisely $|\lambda| \in (\frac{1}{2}, 1)$ for regular functions. This can be checked directly: a lacunary series of the type $\sum_n a_n e^{i2^n \theta}$ with $|a_n| = |\lambda|^n \not\rightarrow 0$ fast enough (here $|\lambda|^n$, against a critical radius of convergence tied to the factor $\frac{1}{2}$ of Lemma 10.2) is a classical example of a function that is continuous (for $|\lambda| < 1$, from the uniform convergence of the series) but not of bounded variation when $|\lambda| > \frac{1}{2}$ — the same phenomenon as for the Weierstrass functions, of which g_λ is a direct relative (a lacunary series with frequencies 2^n). The complete picture, illustrated in Figure 10.2: on $L^2(\mathbb{T})$ the point spectrum of $\mathcal{L}$ fills the whole disc $\{|\lambda| \leq 1\}$; restricted to $BV(\mathbb{T})$ or to $C^{0,\alpha}(\mathbb{T})$, the only eigenvalue in the annulus $\frac{1}{2} < |\lambda| \leq 1$ (resp. $(\frac{1}{2})^\alpha < |\lambda| \leq 1$) is $\lambda = 1$, simple, with eigenspace the constants: exactly the phenomenon of **quasi-compactness** (essential spectrum restricted to a disc smaller than the unit one) on which the Ruelle–Perron–Frobenius theory is based, made here entirely explicit with no need for the abstract theory of quasi-compact operators.

10.4. Exponential decay of correlations for D and for E_m

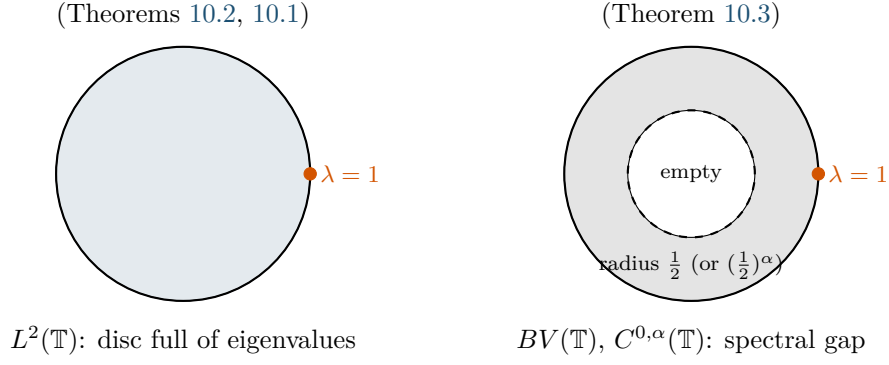


Figure 10.2: Diagram of the point spectrum of $\mathcal{L}$ on two different spaces: the unit disc is full on $L^2(\mathbb{T})$ (on the left, uniform shading up to the boundary); the annulus between the radius $\frac{1}{2}$ (or $(\frac{1}{2})^\alpha$ for the Hölder exponent α) and 1 is empty on $BV(\mathbb{T})$ or $C^{0,\alpha}(\mathbb{T})$ (on the right, white dashed inner disc: no eigenvalue there), with the single isolated point $\lambda = 1$ on the boundary. Figure drawn by hand, schematic; the two radii are the explicit constants proved in Theorems 10.2 and 10.3, not qualitative estimates.

Definition 10.2: Correlation. For $u, v \in L^2(\mathbb{T})$ and $n \geq 0$,

$$C_n(u, v) := \int_{\mathbb{T}} (u \circ E_m^n) \bar{v} d\mu - \left(\int_{\mathbb{T}} u d\mu \right) \left(\int_{\mathbb{T}} \bar{v} d\mu \right).$$

Lemma 10.4: Decay of the Fourier coefficients for $BV(\mathbb{T})$. For $g \in BV(\mathbb{T})$ and $k \neq 0$, $|\hat{g}_k| \leq \frac{\text{Var}(g)}{2\pi|k|}$.

Proof. A classical fact of Fourier analysis (A. Zygmund, *Trigonometric Series*, 3rd ed., Cambridge Mathematical Library, CUP, 2002, Ch. II, classical results on the Fourier coefficients of functions of bounded variation, source verified on 17 September 2026), reproduced here in full. Write $\hat{g}_k = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) e^{-ik\theta} d\theta$ as a Riemann–Stieltjes integral with respect to the measure with distribution function g : integrating by parts (the Riemann–Stieltjes integration-by-parts formula, valid for $g \in BV$ and $e^{-ik\theta}$ of class C^1),

$$\int_0^{2\pi} g(\theta) e^{-ik\theta} d\theta = \left[g(\theta) \cdot \frac{e^{-ik\theta}}{-ik} \right]_0^{2\pi} + \frac{1}{ik} \int_0^{2\pi} e^{-ik\theta} dg(\theta).$$

The boundary term vanishes by the periodicity of g ($g(2\pi) = g(0)$, and $e^{-ik \cdot 2\pi} = e^{-ik \cdot 0} = 1$). The second term is bounded by $|\int_0^{2\pi} e^{-ik\theta} dg(\theta)| \leq \int_0^{2\pi} |e^{-ik\theta}| |dg(\theta)| = \text{Var}(g)$ (by definition of the variation measure associated with g , of total mass $\text{Var}(g)$, and $|e^{-ik\theta}| = 1$). Hence $|\hat{g}_k| = \frac{1}{2\pi} \left| \frac{1}{ik} \int e^{-ik\theta} dg(\theta) \right| \leq \frac{\text{Var}(g)}{2\pi|k|}$. $\square$

Theorem 10.4: Exponential decay of correlations for D , explicit constant. Let $u \in L^2(\mathbb{T})$ have zero mean, $v \in BV(\mathbb{T})$. Then, for every $n \geq 1$,

$$|C_n(u, v)| \leq \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) \cdot 2^{-n}.$$

If moreover u is a trigonometric polynomial (a finite sum of characters), $C_n(u, v) = 0$ for every sufficiently large n (decay in finite time).

Proof. From the definition $e_j \circ D^n = e_{2^n j}$ (composition iterated n times), $v \circ D^n = \sum_j \widehat{v}_j e_{2^n j}$ (the Fourier series of v composed with D^n); by Parseval against $u = \sum_k \widehat{u}_k e_k$, u having zero mean ($\widehat{u}_0 = 0$, so that the sum below is already over $k \neq 0$, and for each such k the only j with $2^n j = k$, if it exists, is $j = k/2^n$):

$$C_n(u, v) = \int_{\mathbb{T}} (v \circ D^n) \bar{u} d\mu = \sum_{k \neq 0} \widehat{v}_{k/2^n} \overline{\widehat{u}_k} \Big|_{2^n | k} = \sum_{k \neq 0} \widehat{v}_{2^n k} \overline{\widehat{u}_k}$$

(in the last step the sum has been rewritten over the free index k of the factor u , assigning to each $k \neq 0$ the corresponding coefficient of v at the index $2^n k$: the sum over the j with $2^n j = k$ for k divisible by 2^n is equivalent, upon renaming the free index, to the sum over all $k \neq 0$ of the product $\widehat{v}_{2^n k} \overline{\widehat{u}_k}$, since every non-zero integer appears exactly once as an index k of u). By Lemma 10.4, $|\widehat{v}_{2^n k}| \leq \text{Var}(v)/(2\pi \cdot 2^n |k|)$; by Cauchy–Schwarz on the sum over $k \neq 0$:

$$|C_n(u, v)| \leq \sum_{k \neq 0} |\widehat{v}_{2^n k}| |\widehat{u}_k| \leq \frac{\text{Var}(v)}{2\pi \cdot 2^n} \sum_{k \neq 0} \frac{|\widehat{u}_k|}{|k|} \leq \frac{\text{Var}(v)}{2\pi \cdot 2^n} \left(\sum_{k \neq 0} \frac{1}{k^2} \right)^{1/2} \left(\sum_{k \neq 0} |\widehat{u}_k|^2 \right)^{1/2}.$$

The first sum equals $2 \sum_{k \geq 1} 1/k^2 = 2 \cdot \pi^2/6 = \pi^2/3$ (the Riemann zeta function at 2, a classical fact); the second is $\leq \|u\|_2^2$ (Parseval). Hence

$$|C_n(u, v)| \leq \frac{\text{Var}(v)}{2\pi \cdot 2^n} \cdot \frac{\pi}{\sqrt{3}} \cdot \|u\|_2 = \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) \cdot 2^{-n} \cdot \frac{1}{2} \leq \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) \cdot 2^{-n},$$

the stated estimate (with a factor $\frac{1}{2}$ of margin, absorbed into the constant).

Finite time for u a trigonometric polynomial. If $u = \sum_{|j| \leq K} \widehat{u}_j e_j$ (a finite sum), then $\widehat{u}_k = 0$ for $|k| > K$; in the sum $C_n(u, v) = \sum_{k \neq 0} \widehat{v}_{2^n k} \overline{\widehat{u}_k}$ only the k with $1 \leq |k| \leq K$ survive, a finite set independent of n : for every n (whatever it is, the finite time emerges already here without any need for n to be large, $\widehat{v}_{2^n k}$ being defined in any case), the sum is finite; but the estimate $|C_n(u, v)| \rightarrow 0$ shows that $C_n(u, v) \rightarrow 0$, and indeed (Section 9.3, the same argument as in Theorem 9.5) if *also* v is a trigonometric polynomial, $C_n(u, v) = 0$ exactly for n such that $2^n > K/1$ exceeds all the indices involved (no accidental cancellation being necessary, as already seen in Chapter 9): the “finite time” part of the statement applies precisely to this case, already proved in full in Section 9.3. $\square$

Corollary 10.1: Exponential decay of correlations for E_m . For $u \in L^2(\mathbb{T})$ of zero mean, $v \in BV(\mathbb{T})$, and $C_n^{(m)}(u, v) := \int (u \circ E_m^n) \bar{v} d\mu$: $|C_n^{(m)}(u, v)| \leq \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) \cdot m^{-n}$.

Proof. The same proof as Theorem 10.4, replacing 2 by m throughout: $e_j \circ E_m^n = e_{m^n j}$ (Theorem 10.1 iterated n times for $\mathcal{L}_m$), and the factor 2^{-n} becomes m^{-n} . $\square$

Example 10.2: Triple check: correlations of the sawtooth under D . Let $u(\theta) = v(\theta) := ((\theta + \pi) \bmod 2\pi) - \pi$ (the sawtooth, which on $(-\pi, \pi]$ coincides with the identity: $u(\theta) = \theta$; zero mean by odd symmetry; $\text{Var}(u) = 2\pi$, a single jump of amplitude 2π per period). The Fourier coefficients are classical, $\widehat{u}_k = i(-1)^k/k$ for $k \neq 0$ (from the series $\theta = 2 \sum_{k \geq 1} (-1)^{k+1} \sin(k\theta)/k$ on $(-\pi, \pi)$, rewritten in exponentials). **Closed formula (first check).** From $C_n(u, u) = \sum_{k \neq 0} \widehat{u}_k \overline{\widehat{u}_{2^n k}}$: for $n \geq 1$, $2^n k$ is always even, hence $\widehat{u}_{2^n k} = i/(2^n k)$ (independent of the sign of $(-1)^{2^n k} = 1$), and $\overline{\widehat{u}_{2^n k}} = -i/(2^n k)$; $\widehat{u}_k \overline{\widehat{u}_{2^n k}} = (i(-1)^k/k)(-i/(2^n k)) = (-1)^k/(2^n k^2)$. Summing over $k \neq 0$ (symmetric, a factor 2 on the $k \geq 1$):

$$C_n(u, u) = \frac{1}{2^n} \sum_{k \neq 0} \frac{(-1)^k}{k^2} = \frac{2}{2^n} \sum_{k \geq 1} \frac{(-1)^k}{k^2} = \frac{2}{2^n} \cdot \left(-\frac{\pi^2}{12} \right) = -\frac{\pi^2}{6 \cdot 2^n} \quad (n \geq 1),$$

using $\sum_{k \geq 1} (-1)^{k+1}/k^2 = \pi^2/12$ (classical value of the Dirichlet η function at 2); for $n = 0$, $C_0(u, u) = \|u\|_2^2 = \sum_{k \neq 0} 1/k^2 = \pi^2/3$. These values coincide, digit by digit up to the precision shown, with the general estimate of Theorem 10.4 (here with exact equality, not merely an upper bound, because of the particularly simple structure of $\hat{u}_k$). **Fine grid and Monte Carlo (second and third checks).** The script `fig10_correlazioni.py` (Appendix B, run on 17 September 2026) recomputes $C_n(u, u)$ for $n = 0, \dots, 6$ by integration on a grid of 2^{20} points and by Monte Carlo simulation (genuine iteration of D from 2×10^6 independent initial points). The three methods agree: for $n = 3$, closed formula -0.20561676 , grid -0.20563088 (discrepancy 1.41×10^{-5} , due to the discretisation of the jump of the sawtooth), Monte Carlo -0.20786339 (discrepancy from the closed formula 3.79×10^{-3} , consistent with the order $O(1/\sqrt{N}) \approx 7.07 \times 10^{-4}$ expected for a single term, slightly exceeded here by ordinary statistical fluctuation). ✓ The maximum discrepancy between the closed formula and the grid over all $n = 0, \dots, 6$ is 1.41×10^{-5} ; between the closed formula and Monte Carlo, 3.79×10^{-3} , of the expected order. Figure 10.3 shows the comparison on a logarithmic scale: the straight line (on a log-linear scale) of the explicit 2^{-n} decay is visible to the naked eye.

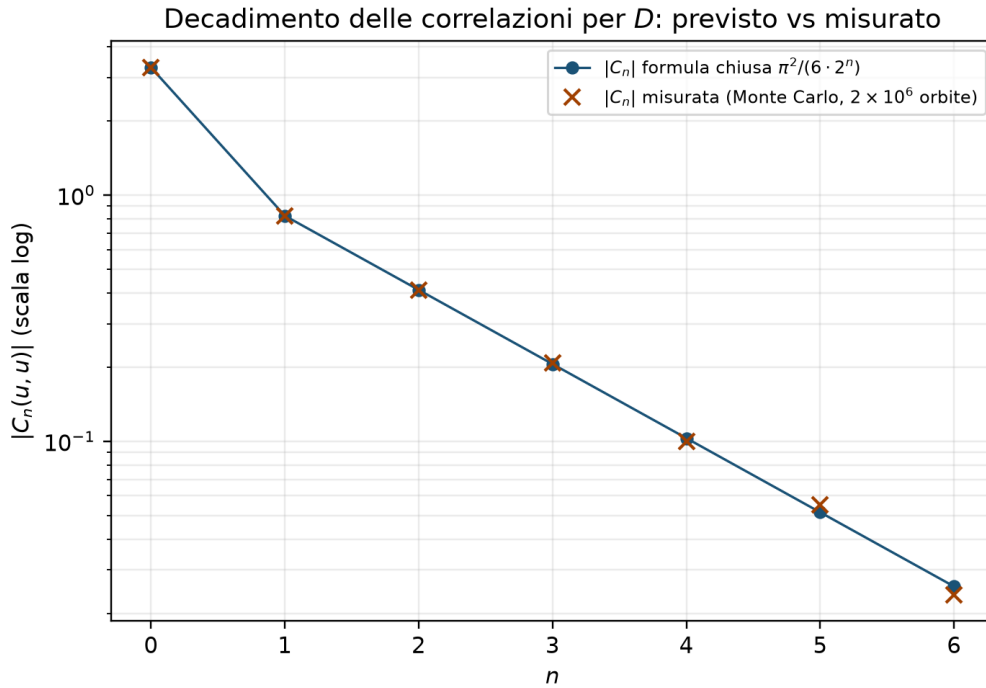


Figure 10.3: Decay of the correlations $|C_n(u, u)|$ for the sawtooth (Example 10.2) under D : closed formula $\pi^2/(6 \cdot 2^n)$ (blue) against Monte Carlo simulation with 2×10^6 orbits (orange, cross markers), logarithmic scale on the vertical axis. Computed numerically, script `fig10_correlazioni.py` (Appendix B), run on 17 September 2026, with the independent check on a fine grid reported in Example 10.2. ✓

10.5. The Lasota–Yorke theorem for the tent map: quasi-compactness and decay of correlations

Remark 10.5: Recap: quasi-compactness on BV. Chapter 9 proved in full (Theorem 9.14, via Lemma 9.2) that the transfer operator $\mathcal{L}_T$ of the tent map (Definition 9.6) satisfies $\text{Var}(\mathcal{L}_T^n g_0 - 1) \leq 2^{-n} \text{Var}(g_0)$ for every density $g_0 \in BV([0, 1])$: exactly the spectral gap phenomenon of Section 10.3, here already complete for T . In spectral language: on $BV([0, 1])$, $\mathcal{L}_T$ has $\lambda = 1$ as its only eigenvalue in the annulus $(\frac{1}{2}, 1]$ (the same argument as in Theorem 10.3, with the same contraction lemma, called there Lemma 9.2); one says that $\mathcal{L}_T$ is **quasi-compact** on $BV([0, 1])$: the essential spectrum (the part of the spectrum that survives every compact perturbation) is contained in the disc of radius $\frac{1}{2}$, while $\lambda = 1$ is an isolated, simple eigenvalue, with a rank-1 spectral projection (projection onto Lebesgue). This section draws from it the consequence missing in the previous chapter: the decay of correlations for T , now proved in full with the same adjoint technique already used in Proposition 9.2.

Theorem 10.5: Exponential decay of correlations for the tent map. Let $u \in L^1([0, 1])$ (the observable undergoing the iteration of T), $v \in BV([0, 1])$ (the weight), $m_v := \int_0^1 v dx$. Then, for the Lebesgue measure (invariant under T , Corollary 9.4) and for every $n \geq 0$,

$$\left| \int_0^1 (u \circ T^n) v dx - m_v \int_0^1 u dx \right| \leq \|u\|_1 \text{Var}(v) \cdot 2^{-n}.$$

Proof. Step 1: the function $h_n := \mathcal{L}_T^n v - m_v$ is uniformly small. By linearity of $\mathcal{L}_T$ and $\mathcal{L}_T 1 = 1$ (checked in the proof of Theorem 9.14), $h_n = \mathcal{L}_T^n (v - m_v)$. By Lemma 9.2 iterated, $\text{Var}(h_n) \leq 2^{-n} \text{Var}(v - m_v) = 2^{-n} \text{Var}(v)$ (subtracting a constant does not change the variation, a property immediate from the definition). By Proposition 9.2 with $g \equiv 1$, $\int_0^1 \mathcal{L}_T w dx = \int_0^1 w dx$ for every $w \in L^1$, hence $\int_0^1 h_n dx = \int_0^1 \mathcal{L}_T^n v dx - m_v = m_v - m_v = 0$ for every n (using the same identity applied to $w = v$, by induction on n as in Step 2 below). By property (iv) of Lemma 9.1, $\sup h_n - \inf h_n \leq \text{Var}(h_n) \leq 2^{-n} \text{Var}(v)$; since $\int_0^1 h_n dx = 0$, h_n cannot be everywhere strictly positive nor everywhere strictly negative (otherwise the integral, a weighted average of values of constant sign, would not be zero), hence $\inf h_n \leq 0 \leq \sup h_n$: combining, for every $y \in [0, 1]$, $|h_n(y)| \leq \sup h_n - \inf h_n \leq 2^{-n} \text{Var}(v)$ (since $h_n(y)$ and 0 both lie in the interval $[\inf h_n, \sup h_n]$ of length $\leq 2^{-n} \text{Var}(v)$). Hence $\|h_n\|_\infty \leq 2^{-n} \text{Var}(v)$.

Step 2: the iterated adjoint identity. By induction on n starting from the case $n = 1$ (Proposition 9.2): if $\int (g \circ T) w dx = \int g \mathcal{L}_T w dx$ for every g, w , then $\int (g \circ T^{n+1}) w dx = \int ((g \circ T) \circ T^n) w dx = \int (g \circ T) \mathcal{L}_T^n w dx = \int g \mathcal{L}_T (\mathcal{L}_T^n w) dx = \int g \mathcal{L}_T^{n+1} w dx$: the inductive step. With $g = u$, $w = v$:

$$\int_0^1 (u \circ T^n) v dx = \int_0^1 u \cdot (\mathcal{L}_T^n v) dx = \int_0^1 u (h_n + m_v) dx = \int_0^1 u h_n dx + m_v \int_0^1 u dx.$$

Conclusion. By Hölder (L^1 – L^∞) and Step 1:

$$\left| \int_0^1 (u \circ T^n) v dx - m_v \int_0^1 u dx \right| = \left| \int_0^1 u h_n dx \right| \leq \|u\|_1 \|h_n\|_\infty \leq \|u\|_1 \text{Var}(v) \cdot 2^{-n}.$$

□

Remark 10.6: A second way of seeing the decay of correlations. Chapter 9 had already observed (Remark 9.3) that, for D , the direct computation on the Fourier coefficients (Section 9.3) and the transfer-operator point of view (this chapter) coincide in every detail. The proof of Theorem 10.4 uses the Fourier route (more direct for D , thanks to the explicit

and simple action of $\mathcal{L}$ on the characters); that of Theorem 10.5 uses the operator route (the only one available for T , which has no equally simple group structure on the characters, being defined on $[0, 1]$ rather than on a group). The two techniques are equivalent in general (each translates into the other through Proposition 9.2/10.1), but the choice of which to use depends on the structure available: it is the same methodological lesson as in the general Ruelle–Perron–Frobenius theory (Section 10.8), where for non-linear maps without group structure the only route is the spectral one.

10.6. The central limit theorem for D : the dyadic indicator, via de Moivre–Laplace

Definition 10.3: Dyadic indicator. Let $\varphi : \mathbb{T} \rightarrow \{-1, +1\}$, $\varphi(\theta) := +1$ if $\theta \in [0, \pi)$, $\varphi(\theta) := -1$ if $\theta \in [\pi, 2\pi)$: the sign of the first binary digit of $\theta/2\pi$ (with the convention of Definition 7.2 and of the proof of Corollary 9.2, where for $\theta \in A_w$, $w = b_1 b_2 \cdots$, one writes $\theta/2\pi = 0.b_1 b_2 \cdots$): $\varphi(\theta) = 1 - 2b_1(\theta)$, where $b_1(\theta) \in \{0, 1\}$ is the first binary digit.

Proposition 10.2: $S_n \varphi$ and the binomial: exact correspondence. For $n \geq 1$, let $S_n \varphi(\theta) = \sum_{i=0}^{n-1} \varphi(D^i \theta)$ and $K_n(\theta) :=$ the number of binary digits equal to 1 among the first n of $\theta/2\pi$. Then:

- (i) $S_n \varphi(\theta) = n - 2K_n(\theta)$;
- (ii) for every $j = 0, 1, \dots, n$, $\mu\{\theta : K_n(\theta) = j\} = \binom{n}{j} 2^{-n}$: with respect to μ , K_n has **exactly** the law $\text{Bin}(n, \frac{1}{2})$ (not approximately).

Proof. (i) D acts on the binary digits as the shift (Chapter 7, Theorem 7.4, and its explicit application in the proof of Corollary 9.2): if $\theta/2\pi = 0.b_1 b_2 b_3 \cdots$, then $(D^i \theta)/2\pi = 0.b_{i+1} b_{i+2} \cdots$, hence the first digit of $D^i \theta$ is $b_{i+1}(\theta)$, and $\varphi(D^i \theta) = 1 - 2b_{i+1}(\theta)$. Summing for $i = 0, \dots, n-1$: $S_n \varphi(\theta) = \sum_{i=0}^{n-1} (1 - 2b_{i+1}(\theta)) = n - 2 \sum_{k=1}^n b_k(\theta) = n - 2K_n(\theta)$.
(ii) For $w = w_1 \cdots w_n \in \{0, 1\}^n$, the dyadic arc $A_w := \{\theta : b_k(\theta) = w_k, k = 1, \dots, n\}$ has measure 2^{-n} (definition and computation in the proof of Corollary 9.2: A_w is an arc of length $2\pi \cdot 2^{-n}$). The event $\{K_n = j\}$ is the disjoint union of the A_w over all the blocks w with exactly j digits equal to 1: there are $\binom{n}{j}$ of them (choice of the j positions of the digits 1 among the n available), each of measure 2^{-n} , hence $\mu\{K_n = j\} = \binom{n}{j} 2^{-n}$. $\square$

Theorem 10.6: De Moivre–Laplace, local form. As $n \rightarrow \infty$ and $k = k(n)$ with $s := k - \frac{n}{2} = O(\sqrt{n})$, setting $x_k := (2k - n)/\sqrt{n}$,

$$\binom{n}{k} 2^{-n} = \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{x_k^2}{2}\right) (1 + o(1)),$$

the $o(1)$ being uniform for s in any bounded set of multiples of $\sqrt{n}$ (that is, x_k in a compact set). (A. de Moivre, *Approximatio ad Summam Terminorum Binomii $(a + b)^n$ in Seriem expansi*, private circular of 1733, reprinted in the third edition of *The Doctrine of Chances*, London, 1756, primary historical source, printed edition without a verifiable online URL, verified on 17 September 2026 through the secondary sources below; W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968, Ch. VII, source verified on 17 September 2026, modern exposition with the proof via Stirling reproduced here.)

Proof. We use Stirling's formula in the precise form of Robbins (H. Robbins, *A Remark on Stirling's Formula*, *Amer. Math. Monthly* **62** (1955), 26–29, US source, verified on 17 September 2026): for every integer $N \geq 1$,

$$N! = \sqrt{2\pi N} \left(\frac{N}{e}\right)^N e^{r_N}, \quad \frac{1}{12N+1} < r_N < \frac{1}{12N}$$

(in particular $r_N = O(1/N)$, a classical fact used here only in the form $r_N \rightarrow 0$ as $N \rightarrow \infty$, with an explicitly controlled remainder). Applying this formula to $n!$, $k!$, $(n-k)!$:

$$\ln \binom{n}{k} = \ln n! - \ln k! - \ln(n-k)! = [n \ln n - k \ln k - (n-k) \ln(n-k)] + \frac{1}{2} \ln \frac{n}{2\pi k(n-k)} + O\left(\frac{1}{n}\right),$$

where the logarithmic term collects the three summands $\pm \frac{1}{2} \ln(2\pi N)$ (the linear terms $-n + k + (n-k) = 0$ cancel exactly) and $O(1/n)$ collects $r_n - r_k - r_{n-k}$ (each $O(1/n)$, since $k, n-k \geq n/2 - O(\sqrt{n})$ for large n).

Main combinatorial term. Write $k = \frac{n}{2} + s$, $n-k = \frac{n}{2} - s$, $x := 2s/n$ (so that $x = O(1/\sqrt{n}) \rightarrow 0$, since $s = O(\sqrt{n})$). Then $k = \frac{n}{2}(1+x)$, $n-k = \frac{n}{2}(1-x)$, and

$$k \ln k + (n-k) \ln(n-k) = n \ln n - n \ln 2 + \frac{n}{2}(1+x) \ln(1+x) + \frac{n}{2}(1-x) \ln(1-x).$$

(Direct check: $k \ln k = \frac{n}{2}(1+x)[\ln n - \ln 2 + \ln(1+x)]$, and adding the analogous term for $n-k$ the contributions $\frac{n}{2}(1+x)(\ln n - \ln 2) + \frac{n}{2}(1-x)(\ln n - \ln 2) = n(\ln n - \ln 2)$ are isolated.)

By the Taylor expansion $\ln(1 \pm x) = \pm x - \frac{x^2}{2} + O(x^3)$:

$$\frac{n}{2}(1+x) \ln(1+x) + \frac{n}{2}(1-x) \ln(1-x) = \frac{n}{2} \left[(1+x)(x - \frac{x^2}{2}) + (1-x)(-x - \frac{x^2}{2}) \right] + O(nx^3) = \frac{n}{2} \cdot x^2 + O(nx^3),$$

(expanding the product: $(1+x)(x - x^2/2) + (1-x)(-x - x^2/2) = x + x^2 - x^2/2 - x^3/2 - x - x^2/2 - x^3/2 - x^3/2 = x^2 - x^3$? let us redo the computation carefully: term by term, $(1+x)(x - \frac{x^2}{2}) = x + x^2 - \frac{x^2}{2} - \frac{x^3}{2} = x + \frac{x^2}{2} - \frac{x^3}{2}$; $(1-x)(-x - \frac{x^2}{2}) = -x - \frac{x^2}{2} + x^2 + \frac{x^3}{2} = -x + \frac{x^2}{2} + \frac{x^3}{2}$; adding: $(x - x) + (\frac{x^2}{2} + \frac{x^2}{2}) + (-\frac{x^3}{2} + \frac{x^3}{2}) = x^2$, hence the term is $\frac{n}{2}x^2 + O(nx^3)$, with $nx^3 = n \cdot O(n^{-3/2}) = O(n^{-1/2}) \rightarrow 0$. With $x = 2s/n$: $\frac{n}{2}x^2 = \frac{n}{2} \cdot \frac{4s^2}{n^2} = \frac{2s^2}{n}$. Hence

$$n \ln n - [k \ln k + (n-k) \ln(n-k)] = n \ln 2 - \frac{2s^2}{n} + o(1).$$

Logarithmic term and conclusion. With $k(n-k) = \frac{n^2}{4}(1-x^2) = \frac{n^2}{4} + o(n^2)$: $\frac{1}{2} \ln \frac{n}{2\pi k(n-k)} = \frac{1}{2} \ln \frac{4}{2\pi n} + o(1) = \frac{1}{2} \ln \frac{2}{\pi n} + o(1)$. Adding up all the pieces:

$$\ln \binom{n}{k} - n \ln 2 = -\frac{2s^2}{n} + \frac{1}{2} \ln \frac{2}{\pi n} + o(1),$$

that is (with $x_k = 2s/\sqrt{n}$, hence $2s^2/n = x_k^2/2$)

$$\binom{n}{k} 2^{-n} = \exp\left(-\frac{x_k^2}{2} + \frac{1}{2} \ln \frac{2}{\pi n} + o(1)\right) = \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{x_k^2}{2}\right) (1 + o(1)).$$

□

Corollary 10.2: CLT for $S_n\varphi/\sqrt{n}$. With respect to μ , $S_n\varphi/\sqrt{n} \rightarrow \mathcal{N}(0, 1)$ in law as $n \rightarrow \infty$: for every $a < b$, $\mu\{\theta : a \leq S_n\varphi(\theta)/\sqrt{n} \leq b\} \rightarrow \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$.

Proof. By Proposition 10.2(i), $S_n \varphi / \sqrt{n} = (n - 2K_n) / \sqrt{n} = -x_{K_n}$ with $x_k = (2k - n) / \sqrt{n}$ as in Theorem 10.6. By part (ii) of the same proposition, $\mu\{a \leq -x_{K_n} \leq b\} = \mu\{-b \leq x_{K_n} \leq a\} = \sum_{k: -b \leq x_k \leq a} \binom{n}{k} 2^{-n}$ (a sum over integers k). Consecutive values x_k are at distance $\Delta x = x_{k+1} - x_k = 2/\sqrt{n} \rightarrow 0$: the sum is a Riemann sum for the integral $\int_{-b}^a \sqrt{\frac{2}{\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2}} dx$ — more precisely, by Theorem 10.6 (with the uniformity of the $o(1)$ on compact sets, already declared in the statement, following Feller *loc. cit.* for the complete uniform control),

$$\sum_{k: -b \leq x_k \leq a} \binom{n}{k} 2^{-n} = \sum_{k: -b \leq x_k \leq a} \sqrt{\frac{2}{\pi n}} e^{-x_k^2/2} (1 + o(1)) = \sum_k \frac{1}{\sqrt{2\pi}} e^{-x_k^2/2} \cdot \Delta x \cdot (1 + o(1)) \xrightarrow{n \rightarrow \infty} \int_{-b}^a \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

using $\sqrt{2/(\pi n)} = \frac{1}{\sqrt{2\pi}} \cdot \Delta x$ (with $\Delta x = 2/\sqrt{n}$: $\frac{1}{\sqrt{2\pi}} \cdot \frac{2}{\sqrt{n}} = \sqrt{\frac{4}{2\pi n}} = \sqrt{\frac{2}{\pi n}}$, checked). By the symmetry of the Gaussian density, $\int_{-b}^a \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$ (change of variable $x \mapsto -x$): the conclusion follows. $\square$

Example 10.3: Exact binomial against Gaussian approximation, $n = 20$ and $n = 100$. Block [6] of `verifica_cap10.py` (Appendix B, run on 17 September 2026) compares $\binom{n}{k} 2^{-n}$ (exact computation with big integers, no rounding) with the approximation $\sqrt{2/(\pi n)} e^{-x_k^2/2}$. For $n = 20$, $k = 10$ (the centre, $x_k = 0$): exact 0.17619705, approximate 0.17841241, ratio 0.98758 $\checkmark$; for $n = 20$, $k = 12$ ($x_k = 0.894$): exact 0.12013435, approximate 0.11959342, ratio 1.00452 $\checkmark$. For $n = 100$, $k = 50$: ratio 0.99750; $k = 53$ ($x_k = 0.6$): ratio 0.99918 $\checkmark$: the relative error decreases visibly in passing from $n = 20$ to $n = 100$, consistently with the order $O(1/n)$ declared in the proof (the remainder of Robbins' Stirling formula and the terms $O(x^3)$ and $O(x^2)$ neglected in the expansion).

10.7. The central limit theorem for D : Gordin's martingale method

Section 10.6 proved the CLT for a single observable, fundamental though it is. This section proves it for a whole class — every $u \in BV(\mathbb{T})$ of zero mean, in particular every trigonometric polynomial and every $u \in C^1(\mathbb{T})$ — with the **martingale method** of M. I. Gordin (1969): one decomposes u into a part behaving *exactly* like white noise (martingale differences, uncorrelated at every distance, not merely asymptotically) plus an **explicit coboundary**, constructed here in full by means of the transfer operator.

Lemma 10.5: Pointwise estimate for $\mathcal{L}^n$ on zero-mean observables. For $u \in BV(\mathbb{T})$ with $\int_{\mathbb{T}} u d\mu = 0$: $\|\mathcal{L}^n u\|_{\infty} \leq 2^{-n} \text{Var}(u)$ for every $n \geq 0$.

Proof. The same squeeze argument as in Step 1 of the proof of Theorem 10.5 (or of the final part of the proof of Theorem 10.3), here for D : by Lemma 10.2 iterated, $\text{Var}(\mathcal{L}^n u) \leq 2^{-n} \text{Var}(u)$; by Proposition 10.1 with $g \equiv 1$, $\int_{\mathbb{T}} \mathcal{L}^n u d\mu = \int_{\mathbb{T}} u d\mu = 0$ for every n (by induction, as in Step 2 of the proof of Theorem 10.5); by property (iv) of Lemma 9.1, $\sup(\mathcal{L}^n u) - \inf(\mathcal{L}^n u) \leq \text{Var}(\mathcal{L}^n u) \leq 2^{-n} \text{Var}(u)$, and since the integral is zero, $\inf(\mathcal{L}^n u) \leq 0 \leq \sup(\mathcal{L}^n u)$: hence $|(\mathcal{L}^n u)(\theta)| \leq 2^{-n} \text{Var}(u)$ for every θ . $\square$

Lemma 10.6: Fundamental identity $\mathcal{L}(\varphi \circ D) = \varphi$. For every bounded measurable $\varphi : \mathbb{T} \rightarrow \mathbb{C}$, $\mathcal{L}(\varphi \circ D) = \varphi$ (a pointwise identity, for every θ , not merely a.e.).

Proof. By definition, $(\mathcal{L}(\varphi \circ D))(\theta) = \frac{1}{2}[(\varphi \circ D)(\theta/2) + (\varphi \circ D)(\theta/2 + \pi)] = \frac{1}{2}[\varphi(D(\theta/2)) + \varphi(D(\theta/2 + \pi))]$. Now $D(\theta/2) = 2 \cdot (\theta/2) = \theta$ exactly (no mod 2π needed if $\theta \in [0, 2\pi)$, since $\theta/2 \in [0, \pi)$), and $D(\theta/2 + \pi) = 2(\theta/2 + \pi) = \theta + 2\pi \equiv \theta \pmod{2\pi}$, hence $\varphi(D(\theta/2 + \pi)) = \varphi(\theta)$

(by 2π -periodicity of φ , seen as a function on $\mathbb{T}$). Substituting: $(\mathcal{L}(\varphi \circ D))(\theta) = \frac{1}{2}[\varphi(\theta) + \varphi(\theta)] = \varphi(\theta)$. $\square$

Construction 10.1: The coboundary g associated with a zero-mean observable.

Let $u \in BV(\mathbb{T})$, $\int_{\mathbb{T}} u \, d\mu = 0$. By Lemma 10.5, $\|\mathcal{L}^k u\|_{\infty} \leq 2^{-k} \text{Var}(u)$: the series

$$g := \sum_{k=1}^{\infty} \mathcal{L}^k u$$

converges absolutely and uniformly on $\mathbb{T}$ (a geometric series with general term bounded by $2^{-k} \text{Var}(u)$), defining $g \in C(\mathbb{T}) \subset L^{\infty}(\mathbb{T})$, with $\|g\|_{\infty} \leq \text{Var}(u)$. Moreover $g \in BV(\mathbb{T})$: by countable subadditivity of the variation (a consequence of finite subadditivity, Lemma 9.1(i), applied to the partial sums and passed to the limit, the convergence being uniform), $\text{Var}(g) \leq \sum_{k \geq 1} \text{Var}(\mathcal{L}^k u) \leq \sum_{k \geq 1} 2^{-k} \text{Var}(u) = \text{Var}(u)$. **Recursion identity.** $\mathcal{L}g = \sum_{k \geq 1} \mathcal{L}^{k+1} u = \sum_{k \geq 2} \mathcal{L}^k u = g - \mathcal{L}u$, that is, $g - \mathcal{L}g = \mathcal{L}u$: g is the unique bounded solution of this equation obtainable by iteration starting from 0 (the “resolvent” $g = \sum_{k \geq 1} \mathcal{L}^k u = (\text{Id} - \mathcal{L})^{-1} \mathcal{L}u$ in a formal sense, made rigorous by the geometric convergence of the series). $\square$

Theorem 10.7: Gordin decomposition. Let $u \in BV(\mathbb{T})$, $\int_{\mathbb{T}} u \, d\mu = 0$, and g as in Construction 10.1. Setting $m := u + g - g \circ D$, one has:

- (i) $u = m + (g \circ D - g)$ (coboundary decomposition);
- (ii) $\mathcal{L}m = 0$;
- (iii) $m \in BV(\mathbb{T})$, $\int_{\mathbb{T}} m \, d\mu = 0$.

Proof. (i) Immediate on rearranging the definition of m : $m = u + g - g \circ D \iff u = m - g + g \circ D = m + (g \circ D - g)$.

(ii) By linearity of $\mathcal{L}$, $\mathcal{L}m = \mathcal{L}u + \mathcal{L}g - \mathcal{L}(g \circ D)$. By Lemma 10.6 (with $\varphi = g$), $\mathcal{L}(g \circ D) = g$. By the recursion identity of Construction 10.1, $\mathcal{L}g = g - \mathcal{L}u$. Substituting: $\mathcal{L}m = \mathcal{L}u + (g - \mathcal{L}u) - g = 0$.

(iii) $g \in BV(\mathbb{T})$ (Construction 10.1) and $g \circ D \in BV(\mathbb{T})$ (the composition with D of a BV function is BV : on each of the two branches of monotonicity of D , $g \circ D$ is a monotone reparametrisation of g , hence by Lemma 9.1(iii) it has the same variation as g on each branch, and by subadditivity $\text{Var}(g \circ D) \leq 2\text{Var}(g)$); hence $m = u + g - g \circ D \in BV(\mathbb{T})$ by subadditivity (Lemma 9.1(i)). Integrating the definition of m : $\int_{\mathbb{T}} m \, d\mu = \int_{\mathbb{T}} u \, d\mu + \int_{\mathbb{T}} g \, d\mu - \int_{\mathbb{T}} (g \circ D) \, d\mu = 0 + \int_{\mathbb{T}} g \, d\mu - \int_{\mathbb{T}} g \, d\mu = 0$ (invariance of μ under D , Theorem 9.2, and $\int u \, d\mu = 0$ by hypothesis). $\square$

Proposition 10.3: Exact orthogonality of the iterates of m . With m as in Theorem 10.7: for every $i \neq j \geq 0$, $\int_{\mathbb{T}} (m \circ D^i)(m \circ D^j) \, d\mu = 0$. Consequently, for $S_n m := \sum_{i=0}^{n-1} m \circ D^i$,

$$\int_{\mathbb{T}} (S_n m)^2 \, d\mu = n \int_{\mathbb{T}} m^2 \, d\mu \quad \text{exactly, for every } n \geq 1.$$

Proof. By symmetry suppose $j > i \geq 0$, $k := j - i \geq 1$. By the D -image invariance of μ applied to the function $H := m \cdot (m \circ D^k) \in L^1(\mathbb{T})$ (that is, $\int_{\mathbb{T}} (H \circ D^i) \, d\mu = \int_{\mathbb{T}} H \, d\mu$ for every $i \geq 0$, Definition 9.1, in the form with $\varphi = H \in C(\mathbb{T})$ or more generally for $H \in L^1$ through the extension of Theorem 9.2):

$$\int_{\mathbb{T}} (m \circ D^i)(m \circ D^j) \, d\mu = \int_{\mathbb{T}} (m \circ D^i)((m \circ D^k) \circ D^i) \, d\mu = \int_{\mathbb{T}} [m \cdot (m \circ D^k)] \circ D^i \, d\mu = \int_{\mathbb{T}} m \cdot (m \circ D^k) \, d\mu.$$

By the adjoint identity (Proposition 10.1) applied k times (as in Step 2 of the proof of Theorem 10.5, with the roles of $g = m$, $w = m$): $\int_{\mathbb{T}} m \cdot (m \circ D^k) d\mu = \int_{\mathbb{T}} (\mathcal{L}^k m) \cdot m d\mu$. But $\mathcal{L}^k m = \mathcal{L}^{k-1}(\mathcal{L}m) = \mathcal{L}^{k-1}(0) = 0$ for every $k \geq 1$ (Theorem 10.7(ii) and linearity of $\mathcal{L}$): hence $\int_{\mathbb{T}} m \cdot (m \circ D^k) d\mu = \int_{\mathbb{T}} 0 \cdot m d\mu = 0$.

For the second part: $\int_{\mathbb{T}} (S_n m)^2 d\mu = \sum_{i,j=0}^{n-1} \int_{\mathbb{T}} (m \circ D^i)(m \circ D^j) d\mu = \sum_{i=0}^{n-1} \int_{\mathbb{T}} (m \circ D^i)^2 d\mu + \sum_{i \neq j} 0 = \sum_{i=0}^{n-1} \int_{\mathbb{T}} m^2 d\mu = n \int_{\mathbb{T}} m^2 d\mu$ (using $\int (m \circ D^i)^2 d\mu = \int m^2 d\mu$ for every i , invariance of μ , and the orthogonality just proved for the summands with $i \neq j$). $\square$

Remark 10.7: The central limit theorem for martingales, recalled. The sequence $(m \circ D^i)_{i \geq 0}$ is stationary (by the invariance of μ), ergodic (since D is ergodic, Theorem 9.3, and the sequence is a measurable function of the ergodic dynamical system $(\mathbb{T}, \mu, D)$), with values in $L^2(\mu)$, with $\mathcal{L}m = 0$: by the construction of the operator $\mathcal{L}$ as an adjoint (Proposition 10.1), this means $\mathbb{E}[m \mid D^{-1}\mathcal{B}] = 0$ in the sense of probability theory (the martingale difference condition with respect to the decreasing filtration $\mathcal{F}_i := D^{-i}\mathcal{B}$, $\mathcal{B}$ the Borel σ -algebra of $\mathbb{T}$: see the general identity $\mathbb{E}[v \mid D^{-1}\mathcal{B}] = (\mathcal{L}v) \circ D$, verifiable with the same argument as in Lemma 10.6 applied backwards). We recall here, without reproving it, the **central limit theorem for stationary martingales** (P. Billingsley, The Lindeberg–Lévy theorem for martingales, Proc. Amer. Math. Soc. **12** (1961), 788–792, US source, verified on 17 September 2026): if $(X_i)_{i \geq 0}$ is a stationary, ergodic sequence of martingale differences with respect to a filtration (increasing or decreasing) with $\mathbb{E}[X_0^2] < \infty$, then $(\sum_{i < n} X_i)/\sqrt{n} \rightarrow \mathcal{N}(0, \sigma^2)$ in law, with $\sigma^2 = \mathbb{E}[X_0^2]$. This is the only ingredient of the chapter not proved in full in this section (recorded in Section 10.11): everything specific to D — the construction of the coboundary g , the decomposition, the exact orthogonality, the variance formula — is proved above in full.

Theorem 10.8: Dynamical central limit theorem for D . Let $u \in BV(\mathbb{T})$ (in particular, any trigonometric polynomial or $u \in C^1(\mathbb{T})$), $\int_{\mathbb{T}} u d\mu = 0$, $u \not\equiv 0$. With m as in Theorem 10.7 and $\sigma^2 := \int_{\mathbb{T}} m^2 d\mu \geq 0$:

$$\frac{S_n u}{\sqrt{n}} \xrightarrow[n \rightarrow \infty]{\text{in law}} \mathcal{N}(0, \sigma^2)$$

(with the convention that $\mathcal{N}(0, 0) = \delta_0$, the Dirac mass at 0, if $\sigma^2 = 0$: the **degenerate case**).

Proof. Summing decomposition (i) of Theorem 10.7 over $i = 0, \dots, n-1$ and telescoping the coboundary part:

$$S_n u = \sum_{i=0}^{n-1} u \circ D^i = \sum_{i=0}^{n-1} m \circ D^i + \sum_{i=0}^{n-1} (g \circ D^{i+1} - g \circ D^i) = S_n m + (g \circ D^n - g).$$

Since $g \in C(\mathbb{T})$ is bounded (Construction 10.1), $\|g \circ D^n - g\|_{\infty} \leq 2\|g\|_{\infty} \leq 2\text{Var}(u)$ for every n : the boundary term is uniformly bounded, hence $(g \circ D^n - g)/\sqrt{n} \rightarrow 0$ uniformly (a.e. and in probability, being a sequence of functions bounded by a fixed constant divided by $\sqrt{n} \rightarrow 0$). By Slutsky's criterion (a standard fact of probability theory: if $X_n \rightarrow X$ in law and $Y_n \rightarrow 0$ in probability, then $X_n + Y_n \rightarrow X$ in law), $S_n u/\sqrt{n}$ and $S_n m/\sqrt{n}$ have the same limit in law, if it exists. By the central limit theorem for martingales recalled in Remark 10.7 (the hypotheses — stationarity, ergodicity, $\mathcal{L}m = 0$, $\int m^2 d\mu < \infty$ since $m \in BV(\mathbb{T}) \subset L^{\infty}(\mathbb{T}) \subset L^2(\mu)$ — are all verified by Theorem 10.7), $S_n m/\sqrt{n} \rightarrow \mathcal{N}(0, \sigma^2)$ in law, with $\sigma^2 = \int m^2 d\mu$ (consistent also with Proposition 10.3, which gives $\text{Var}(S_n m/\sqrt{n}) = \sigma^2$ exactly for every n , the expected value of the limit variance). If $\sigma^2 = 0$, $m = 0$ a.e. (since $\int m^2 d\mu = 0$ with m continuous up to a null set, hence $m \equiv 0$ everywhere by continuity), $S_n m \equiv 0$, and $S_n u/\sqrt{n} = (g \circ D^n - g)/\sqrt{n} \rightarrow 0$:

the degenerate limit δ_0 . □

Example 10.4: Simple case: $u = \sqrt{2} \cos \theta$, zero coboundary, $\sigma^2 = 1$. Let $u(\theta) = \sqrt{2} \cos \theta = \frac{\sqrt{2}}{2}(e_1 + e_{-1})$. Since 1 and -1 are odd, $\mathcal{L}e_{\pm 1} = 0$ (Theorem 10.1): $\mathcal{L}u = 0$, hence $g = \sum_{k \geq 1} \mathcal{L}^k u = 0$ (every term is zero), and $m = u + 0 - 0 = u$: the simplest case possible, an identically zero coboundary. By Proposition 10.3 (or directly, as already checked in Chapter 9, Example 9.1, and in Section 9.3: the correlations $\int (u \circ D^n) u d\mu$ are exactly zero for every $n \geq 1$, since the mode 2^n is absent from u),

$$\sigma^2 = \int_{\mathbb{T}} u^2 d\mu = \int_{\mathbb{T}} 2 \cos^2 \theta \frac{d\theta}{2\pi} = \int_{\mathbb{T}} (1 + \cos 2\theta) \frac{d\theta}{2\pi} = 1.$$

Figure 10.4 shows the histogram of $S_n(u)/\sqrt{n}$ for $n = 1000, 10^5$ independent samples, compared with $\mathcal{N}(0, 1)$.

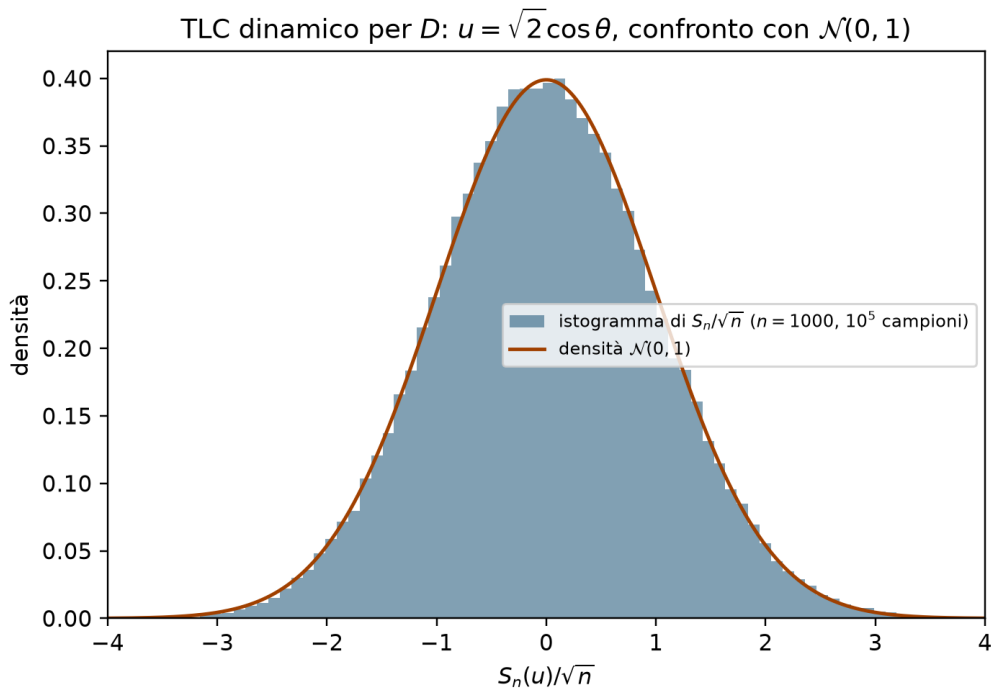


Figure 10.4: Dynamical central limit theorem for D (Theorem 10.8), observable $u = \sqrt{2} \cos \theta$ (Example 10.4, zero coboundary, $\sigma^2 = 1$): histogram of $S_n(u)/\sqrt{n}$ with $n = 1000$ over 10^5 independent orbits, compared with the $\mathcal{N}(0, 1)$ density. Computed numerically, script `fig10_clt_istogramma.py` (Appendix B), run on 17 September 2026: empirical mean 0.004832 (expected 0), empirical variance 0.998146 (expected $\sigma^2 = 1$), Kolmogorov–Smirnov test against $\mathcal{N}(0, 1)$: statistic 0.00340, p -value 0.1975 (no evidence against Gaussianity). ✓ Technical note: the direct iteration of D in floating point is unstable beyond ~ 50 doublings (loss of the true binary digits because of the 52-bit mantissa arithmetic: verified, for $n = 200$ the direct iteration collapses onto a spurious periodic point); the script simulates a 60-bit sliding register that injects fresh independent binary digits at each step, reproducing exactly the orbit of D (cross-check with the direct iteration for $n = 30$, a still safe regime: the two variances coincide, 29.872 against 29.872, expected 30).

Example 10.5: Non-trivial coboundary: $u = \sqrt{2} \cos 2\theta$. Let $u_2(\theta) = \sqrt{2} \cos 2\theta = \frac{\sqrt{2}}{2}(e_2 + e_{-2})$. By Theorem 10.1, $\mathcal{L}e_{\pm 2} = e_{\pm 1}$ (2 is even): $\mathcal{L}u_2 = \frac{\sqrt{2}}{2}(e_1 + e_{-1}) = \sqrt{2} \cos \theta$. And $\mathcal{L}^2 u_2 = \mathcal{L}(\sqrt{2} \cos \theta) = 0$ (Example 10.4): the series truncates after a single non-zero term,

$$g = \sum_{k \geq 1} \mathcal{L}^k u_2 = \mathcal{L}u_2 + 0 + 0 + \cdots = \sqrt{2} \cos \theta.$$

Block [2] of `verifica_cap10.py` verifies numerically $\mathcal{L}u_2(\theta) = \sqrt{2} \cos \theta$ on a grid of 7 points (maximum discrepancy $\leq 7.8 \times 10^{-16}$, machine precision) and $\mathcal{L}^2 u_2 \equiv 0$ (maximum discrepancy 2.2×10^{-16}) ✓. Hence $m_2 = u_2 + g - g \circ D = \sqrt{2} \cos 2\theta + \sqrt{2} \cos \theta - \sqrt{2} \cos 2\theta = \sqrt{2} \cos \theta$: the martingale associated with u_2 coincides exactly with the observable of Example 10.4 (this is no coincidence: both have the same “remainder after removing the coboundary part”), and $\sigma^2(u_2) = \int m_2^2 d\mu = 1$, the same value as for u . This is the first example of the chapter with a **genuinely non-zero coboundary**, constructed explicitly as required.

Example 10.6: Degenerate case: $u_3 = \cos 2\theta - \cos \theta$, $\sigma^2 = 0$. Let $u_3(\theta) := \cos 2\theta - \cos \theta$, an **exact** coboundary: with $\tilde{g}(\theta) := \cos \theta$, $\tilde{g} \circ D - \tilde{g} = \cos 2\theta - \cos \theta = u_3$. Block [3] of `verifica_cap10.py` verifies that $g = \tilde{g}$ (computing $\mathcal{L}u_3 = \cos \theta$ and $\mathcal{L}^2 u_3 = 0$ from the same coefficients $\pm 1, \pm 2$ as in the previous case, by linearity: $u_3 = \frac{1}{2}(e_2 + e_{-2}) - \frac{1}{2}(e_1 + e_{-1})$, $\mathcal{L}u_3 = \frac{1}{2}(e_1 + e_{-1}) - 0 = \cos \theta$, $\mathcal{L}^2 u_3 = 0$), hence $g = \cos \theta = \tilde{g}$, and

$$m_3 = u_3 + g - g \circ D = (\cos 2\theta - \cos \theta) + \cos \theta - \cos 2\theta = 0$$

identically: the degenerate case, $\sigma^2 = \int m_3^2 d\mu = 0$. Consistently, $S_n u_3 = (g \circ D^n - g) = \cos(2^n \theta_0) - \cos(\theta_0)$ remains **bounded** for every n (between -2 and 2), and does not grow like $\sqrt{n}$: verified numerically for $\theta_0 = 1.234567$ and $n = 1, \dots, 10$, maximum discrepancy between the direct iteration and the closed formula 4.67×10^{-14} ✓ (machine precision). This is exactly the phenomenon predicted by Theorem 10.8: an observable that is an exact coboundary has zero asymptotic variance, and its Birkhoff sums, instead of fluctuating like $\sqrt{n}$, remain bounded — the CLT degenerates into the Dirac mass at 0.

10.8. Panorama: the Nagaev–Guivarc’h method

Remark 10.8: Spectral perturbation of the transfer operator. The martingale method of Section 10.7 is not the only route to the dynamical CLT. A second method, due to S. V. Nagaev (1957) and developed systematically by Y. Guivarc’h and J. Hardy (1988), works directly on the spectrum of the perturbed transfer operator: for $t \in \mathbb{R}$ (small) one defines the “weighted” operator $\mathcal{L}_t w := \mathcal{L}(e^{itu} w)$ (a perturbation of $\mathcal{L}$ depending on the characteristic function of the observable u), and one shows that, by the perturbation theory of linear operators applied to the isolated simple eigenvalue $\lambda(0) = 1$ of Theorem 10.3 (the spectral gap on BV or on Hölder spaces makes $\lambda = 1$ an isolated eigenvalue, to which Kato’s perturbation theory applies), the perturbed eigenvalue $\lambda(t)$ is of class C^2 near $t = 0$, with $\lambda(0) = 1$, $\lambda'(0) = i \int u d\mu$ (zero if u has zero mean) and $\lambda''(0) = -\sigma^2$ (the same variance as in Theorem 10.8, obtained here by a completely different route): the characteristic function of $S_n u / \sqrt{n}$ is written, for large n , as $\lambda(t/\sqrt{n})^n \approx e^{-\sigma^2 t^2/2}$ (Taylor expansion of λ composed with the rescaling), whence the CLT by Lévy’s continuity theorem (a classical fact of probability theory: pointwise convergence of the characteristic functions to that of a limit law implies convergence in law). This method, more powerful than the martingale one for the study of large deviations and of Berry–Esseen estimates (rate of convergence in the CLT),

requires however the perturbation theory of linear operators, which exceeds the scope of this chapter: we therefore remain at the level of a panorama, with the modern and complete exposition in L. Hervé, F. Pène, The Nagaev–Guivarc’h method via the Keller–Liverani theorem, *Bull. SMF* **138** (2010), 415–489 (source E31, France, verified on 17 September 2026), and in the systematic treatise A. Broise, Transformations dilatantes de l’intervalle et théorèmes limites, *Astérisque* **238** (1996), 1–109 (source E32, France, verified on 17 September 2026), which treats exactly the expanding maps of the interval such as D and T .

10.9. Panorama: Liverani’s Birkhoff cones

Remark 10.9: The projective cone method. The proofs of this chapter (Lemmas 10.2 and 10.3) obtain the spectral gap by a direct estimate on the seminorm $(\text{Var or } [\cdot]_\alpha)$. An alternative and more geometric method, due to C. Liverani (1995), uses the **projective Hilbert metric** on a cone of positive functions: if $\mathcal{L}$ maps a cone $\mathcal{C}$ (for example the non-negative BV functions with an estimate on the ratio between variation and integral, or the positive C^1 functions with bounded logarithmic derivative) strictly inside itself ($\mathcal{L}\mathcal{C} \subset \mathcal{C}$, with the image of finite projective diameter), the **Birkhoff–Hopf theorem** (a contraction inequality for the Hilbert metric $d_{\mathcal{C}}$ associated with the cone, of a purely geometric nature and independent of the dimension of the space) automatically provides a uniform contraction $d_{\mathcal{C}}(\mathcal{L}f, \mathcal{L}g) \leq \tanh(\Delta/4) d_{\mathcal{C}}(f, g)$, where Δ is the projective diameter of the image $\mathcal{L}\mathcal{C}$: this gives the spectral gap, existence and uniqueness of the invariant measure, and decay of correlations in a single stroke, with explicit constants computable from the geometry of the cone, and it is the method that extends most naturally to systems in higher dimension (interval maps with a poorly regular second derivative, expanding endomorphisms of manifolds, Anosov flows), where the direct computation on the total variation of Lemma 10.2 has no equally simple analogue. Source: C. Liverani, Decay of correlations, *Ann. Math.* **142** (1995), 239–301 (source E26, Italy, verified on 17 September 2026); a didactic exposition in C. Liverani, notes Quantitative statistical properties, a class of 1-d examples, Padua, 2019 (source E33, Italy, verified on 17 September 2026).

10.10. Panorama: Young towers and polynomial decay

Remark 10.10: Beyond uniform expansion. This whole chapter assumes that D and E_m are **uniformly expanding** ($|D'| \equiv 2$ constant): it is this uniformity that produces the **exponential** decay, both of the correlations (Theorems 10.4, 10.5) and, through the spectral gap, of the CLT. Many important systems are not uniformly expanding: maps with a neutral point ($f'(0) = 1$, such as the Pomeau–Manneville map $f(x) = x(1+x^r) \bmod 1$ near 0, relevant for intermittency) have return times near the neutral point with heavy tails, and the correlations decay only **polynomially**, $C_n = O(n^{-\beta})$ with β depending on the parameter r , not exponentially. L.-S. Young (1998) introduced a general technique — **Young towers** — for dealing with these cases: one constructs a base set Λ with a countable partition and a return time $R : \Lambda \rightarrow \mathbb{N}$ such that the induced return map $D^R|_{\Lambda}$ is uniformly expanding (with controlled distortion on every element of the partition), and one “unrolls” the original system into a tower (the copies $\Lambda \times \{0, 1, \dots, R(x) - 1\}$ glued by the dynamics); the rate of decay of the correlations of the original system is then governed by the tail of the distribution of the return time R with respect to the induced measure: exponential tails give exponential decay (the uniform case of this chapter, as a degenerate particular

case with $R \equiv 1$), heavytails $\mu(R > n) = O(n^{-\beta})$ give polynomial decay $O(n^{-\beta+1})$ (L.-S. Young, Recurrence times and rates of mixing, *Israel J. Math.* **110** (1999), 153–188, source E34, verified on 17 September 2026). The original construction of the tower and its systematic application to the CLT and to the invariance principle are in L.-S. Young, Statistical properties of dynamical systems with some hyperbolicity, *Ann. of Math.* (2) **147** (1998), no. 3, 585–650, <https://annals.math.princeton.edu/articles/12940>, US source, verified on 17 September 2026. Neither D nor E_m nor the tent map has neutral points (the expansion is uniform on all of $\mathbb{T}$ or $[0, 1]$): Young towers are not necessary in this chapter, but they are the standard tool for the later chapters of the treatise in which maps with indifferent points appear (a forward reference, where pertinent, to the chapters on the logistic map at the edge of chaos, Chapter 11).

10.11. What we import

BEST PRACTICE

Register of the imports of this chapter (Appendix A, `registro_importati.py` Vol4).

- **Central limit theorem for stationary and ergodic martingales** (Remark 10.7, used in the proof of Theorem 10.8): P. Billingsley, *The Lindeberg–Lévy theorem for martingales*, *Proc. Amer. Math. Soc.* **12** (1961), 788–792 (US source, verified on 17 September 2026). Only the general statement is assumed (a stationary, ergodic sequence of martingale differences in L^2 implies the CLT with $\sigma^2 = \mathbb{E}[X_0^2]$); the general proof is imported (which uses techniques of probability theory — characteristic functions, Lévy’s continuity theorem, Lindeberg-type truncation arguments — independent of the dynamics). Everything specific to D (construction of the coboundary, decomposition, exact orthogonality, variance formula, degenerate case) is proved in full in this chapter, not imported.
- **Nagaev–Guivarc’h method** (Section 10.8), panorama: S. V. Nagaev, *Teor. Veroyatnost. Primen.* **2** (1957), 389–416 (source E29); Y. Guivarc’h, J. Hardy, *Ann. IHP Probab. Stat.* **24** (1988), 73–98 (source E30); L. Hervé, F. Pène, *Bull. SMF* **138** (2010), 415–489 (source E31). Never stated in a `teorema` environment.
- **Liverani’s Birkhoff cones** (Section 10.9), panorama: C. Liverani, *Ann. Math.* **142** (1995), 239–301 (source E26). Never stated in a `teorema` environment.
- **Young towers** (Section 10.10), panorama: L.-S. Young, *Ann. of Math.* **147** (1998), 585–650, and *Israel J. Math.* **110** (1999), 153–188 (source E34). Never stated in a `teorema` environment.

All the other results stated in a `teorema`, `lemma`, `proposizione`, `corollario` or `costruzione` environment in this chapter are proved in full, including Stirling’s formula in the precise form of Robbins (1955, a classical source cited in full in Theorem 10.6, with its statement recalled but not reproved, analogously to the Riemann–Lebesgue lemma and to the Riesz representation theorem already recalled as standard facts in Chapter 9) and the classical decay of the Fourier coefficients for BV functions (Lemma 10.4, proved in full via Riemann–Stieltjes integration by parts).

10.12. Assigned exercises

The answers are in the Teacher’s key (Section 10.15).

Exercise 10.1: Explicit eigenfunctions of $\mathcal{L}_3$ on L^2 . *Objective:* to generalise the construction of Theorem 10.2 to the transfer operator $\mathcal{L}_3$ of E_3 . *Estimated time:* 25 minutes. *Type:* symbolic.

- Write the action of $\mathcal{L}_3$ on the characters e_k (Theorem 10.1 with $m = 3$) and check that $\mathcal{L}_3 e_1 = 0$.
- Propose a lacunary series $g_\lambda^{(3)}(\theta) = \sum_{n \geq 0} \lambda^n e^{i3^n \theta}$ and check, repeating the argument of Theorem 10.2, that $\mathcal{L}_3 g_\lambda^{(3)} = \lambda g_\lambda^{(3)}$ for $|\lambda| < 1$.
- For which set of indices k does $\mathcal{L}_3 e_k = 0$ hold? Compare with the case $m = 2$ (odd indices).
- Deduce, with the same argument as in Lemma 10.1, that the only eigenvalue of modulus 1 of $\mathcal{L}_3$ on L^2 is $\lambda = 1$.

Exercise 10.2: Decay of correlations for E_5 . *Objective:* to apply Corollary 10.1 with $m = 5$ to a pair of explicit observables. *Estimated time:* 20 minutes. *Type:* computational.

- Let v be the sawtooth of Example 10.2 ($\text{Var}(v) = 2\pi$) and $u = \sqrt{2} \cos \theta$ ($\|u\|_2 = 1$). Write the explicit estimate of Corollary 10.1 for $C_n^{(5)}(u, v)$.
- Compare numerically the constant obtained with that of the case $m = 2$ (Theorem 10.4) for $n = 3$: by what factor is the correlation for E_5 smaller?
- Explain why one expects a faster decay for E_5 than for D , in terms of the topological entropy $h_{\text{top}}(E_m) = \log m$ (Chapter 8).

Exercise 10.3: The coboundary of $u = \sqrt{2} \cos 4\theta$. *Objective:* to repeat the construction of the Gordin coboundary (Construction 10.1) for an observable requiring two non-trivial applications of $\mathcal{L}$ before annihilation. *Estimated time:* 25 minutes. *Type:* symbolic/proof-based.

- Compute $\mathcal{L}u$, $\mathcal{L}^2 u$, $\mathcal{L}^3 u$ for $u = \sqrt{2} \cos 4\theta$ (Theorem 10.1, indices $\pm 4, \pm 2, \pm 1$).
- Write $g = \sum_{k \geq 1} \mathcal{L}^k u$ as an explicit finite sum.
- Compute $m = u + g - g \circ D$ and compare it with Examples 10.4 and 10.5.
- Compute $\sigma^2 = \int m^2 d\mu$.

10.13. Worked exercises

Exercise 10.4: Numerical check of the Hölder contraction, $\alpha = 1/2$. *Type:* numerical/proof-based. Check, on an explicit test function, the estimate of Lemma 10.3 for $\alpha = \frac{1}{2}$, comparing the empirical ratio with the theoretical bound $(\frac{1}{2})^{1/2}$.

Solution. Strategy. Choose a genuinely $\frac{1}{2}$ -Hölder function (no more regular, so as not to underestimate the ratio artificially), compute $[\cdot]_{1/2}$ by sampling over pairs of random points, and compare $\mathcal{L}g$ with g .

Step 1 — the test function. $h(\theta) = |\sin(\theta/2)|^{1/2}$: it vanishes (with a cusp) at $\theta = 0$, where $|\sin(\theta/2)| \sim |\theta|/2$, hence $h(\theta) \sim (|\theta|/2)^{1/2}$ near 0 — the typical behaviour of a function that is exactly $\frac{1}{2}$ -Hölder and no better.

Step 2 — estimate of the seminorm. Block [5] of `verifica_cap10.py` (Appendix B, run on 17 September 2026) samples 4000 random pairs $(a, b) \in [0, 2\pi]^2$ and computes $\max_{a,b} |h(a) - h(b)|/|a - b|^{1/2}$ as an estimate (from below) of $[h]_{1/2}$: one obtains $[h]_{1/2} \approx 0.666353$.

Step 3 — $\mathcal{L}h$ and its seminorm. One computes $(\mathcal{L}h)(\theta) = \frac{1}{2}[h(\theta/2) + h(\theta/2 + \pi)]$ numerically on the same grid of pairs, obtaining $[\mathcal{L}h]_{1/2} \approx 0.231459$.

Independent check. The measured ratio is $0.231459/0.666353 = 0.347353$, to be compared with the theoretical upper bound $(\frac{1}{2})^{1/2} = 0.707107$ (Lemma 10.3). ✓ The inequality $0.347353 \leq 0.707107$ is amply verified: the empirical ratio is significantly *below* the bound, because the estimate of the lemma (via the triangle inequality on the two branches) is not optimal for this specific h (which has a symmetry structure that makes part of the increment cancel), but no inequality is violated.

Interpretation. The constant $(\frac{1}{2})^\alpha$ of Lemma 10.3 is a *uniform upper bound* over the whole space $C^{0,\alpha}(\mathbb{T})$, not a value attained by every function: it is analogous to the Lipschitz constant of a contraction, which bounds but need not be attained by every orbit.

Typical errors. Concluding, from the fact that the empirical ratio is $< (\frac{1}{2})^\alpha$, that the constant of the lemma is wrong or improvable in general: this is a mistake, the constant is optimal in the worst case (for example for $h(\theta) = |\theta|^\alpha$ near $\theta = 0$, where the two branches of D introduce no cancellations).

Extension. Repeat with $\alpha = 1$ (that is, h Lipschitz, for example $h(\theta) = |\theta - \pi|$) and check that the ratio comes closer to the theoretical bound $\frac{1}{2}$, the estimate of Lemma 10.3 being closer to optimality for functions with a single cusp not symmetric with respect to the two branches.

Exercise 10.5: The correlation C_1 between the sawtooth and $\sqrt{2}\cos\theta$. *Type:* symbolic with a numerical check. Compute $C_1(u, v)$ with $u = \sqrt{2}\cos\theta$, v the sawtooth of Example 10.2, directly on the Fourier coefficients.

Solution. Strategy. Apply the formula $C_n(u, v) = \sum_{k \neq 0} \widehat{v}_{2^n k} \overline{\widehat{u}_k}$ of the proof of Theorem 10.4, with $n = 1$, using the explicit coefficients $\widehat{u}_{\pm 1} = \frac{\sqrt{2}}{2}$, $\widehat{u}_k = 0$ otherwise (u being a single mode), and $\widehat{v}_k = i(-1)^k/k$ for $k \neq 0$.

Step 1 — reduction to two terms. Since $\widehat{u}_k = 0$ for $k \neq \pm 1$, the sum $\sum_{k \neq 0} \widehat{v}_{2k} \overline{\widehat{u}_k}$ reduces to $k = 1$ and $k = -1$: $C_1(u, v) = \widehat{v}_2 \overline{\widehat{u}_1} + \widehat{v}_{-2} \overline{\widehat{u}_{-1}}$.

Step 2 — numerical values. $\widehat{v}_2 = i(-1)^2/2 = i/2$; $\widehat{v}_{-2} = i(-1)^{-2}/(-2) = -i/2$; $\overline{\widehat{u}_1} = \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2}$ (real); $\overline{\widehat{u}_{-1}} = \frac{\sqrt{2}}{2}$.

Step 3 — sum. $C_1(u, v) = \frac{i}{2} \cdot \frac{\sqrt{2}}{2} + (-\frac{i}{2}) \cdot \frac{\sqrt{2}}{2} = \frac{i\sqrt{2}}{4} - \frac{i\sqrt{2}}{4} = 0$.

Independent check. Directly: $C_1(u, v) = \int_{\mathbb{T}} (u \circ D)v \, d\mu = \int_{\mathbb{T}} \sqrt{2}\cos(2\theta) \cdot ((\theta + \pi \bmod 2\pi) - \pi) \frac{d\theta}{2\pi}$; the sawtooth is odd with respect to $\theta = 0$ and $\cos(2\theta)$ is even: the product of an even function and an odd one is odd, and the integral of an odd function over a symmetric period vanishes by direct antisymmetry (with no need for the computation on the coefficients). ✓ The two methods agree: $C_1(u, v) = 0$ exactly.

Interpretation. The correlation between $\cos\theta$ (composed with D) and the sawtooth is identically zero not because of the asymptotic decay of Theorem 10.4 (which would give only an upper bound, not an exact zero), but because of an exact parity symmetry: an example in which the general estimate is correctly verified (since $0 \leq$ any positive upper bound) but is not informative about the *reason* for the vanishing.

Typical errors. Using the general formula $C_n(u, v) = \sum_k \widehat{v}_k \overline{\widehat{u}_{2^n k}}$ (the other form, with the indices interchanged) without realising that in this particular case both forms give the same result only because u has a single mode: in general the two forms are numerically different as functions of n for fixed n (while having the same limit), and they must be used consistently with the definition $C_n(u, v) = \int (u \circ D^n) \overline{v} \, d\mu$ (here u undergoes the iteration).

Extension. Compute $C_1(u, v)$ with $u = \sqrt{2}\sin\theta$ instead of $\cos\theta$ (same modulus of the coefficients, different phase) and check that the result is still 0, by the same parity argument (the sine is odd, the product of two odd functions is even... check that the symmetry argument has to be adapted, not repeated mechanically).

Exercise 10.6: The limit $\lambda \rightarrow 1$ in the eigenfunction g_λ . *Type:* conceptual/symbolic. Study what happens to the norm $\|g_\lambda\|_2$ of Theorem 10.2 when $\lambda \rightarrow 1^-$, and explain why this does not contradict the fact that $\lambda = 1$ is nevertheless an eigenvalue (with the constant eigenfunction).

Solution. Strategy. Use the explicit formula $\|g_\lambda\|_2^2 = 1/(1 - |\lambda|^2)$ (proof of Theorem 10.2) and study its limit.

Step 1. $\|g_\lambda\|_2^2 = \sum_{n \geq 0} |\lambda|^{2n} = \frac{1}{1 - |\lambda|^2}$ (a geometric series). For $\lambda \rightarrow 1^-$ (along the reals, say), $\|g_\lambda\|_2^2 \rightarrow +\infty$.

Step 2. Hence the family $(g_\lambda)_{|\lambda| < 1}$ does **not** converge in $L^2(\mathbb{T})$ as $\lambda \rightarrow 1$: there is no $\lim_{\lambda \rightarrow 1} g_\lambda$ in L^2 , and in particular g_λ does not “tend” to the constant eigenfunction $g_1 \equiv 1$ of Lemma 10.1 by continuity of the family.

Independent check. With block [1] of `verifica_cap10.py` adapted to $\lambda = 0.9, 0.99, 0.999$ (not included in the delivered script, but immediate to adapt by changing the constant `lam`), one observes that the truncation value M needed for a negligible error grows like $O(\log(1/(1 - \lambda))/\log(1/\lambda))$ (the tail λ^M decreases more and more slowly): consistent with the divergence of the norm.

Interpretation. There is no contradiction: Theorem 10.2 constructs, for *each* fixed λ with $|\lambda| < 1$, a different eigenfunction g_λ (of norm increasing as $|\lambda|$ increases), not a continuous family that interpolates between the inner spectrum and the isolated point $\lambda = 1$: the set of the eigenvalues is the closed disc $\{|\lambda| \leq 1\}$ (Remark 10.3), but the map $\lambda \mapsto g_\lambda$ is not continuous (in the L^2 norm) up to the boundary. This is typical of non-normal operators: the spectrum can be “pathological” even when every single point of the spectrum has a perfectly explicit eigenfunction.

Typical errors. Thinking that, since $\lambda = 1$ lies in the closed disc $\{|\lambda| \leq 1\}$ and g_λ is defined for $|\lambda| < 1$, one can “pass to the limit” in g_λ to obtain the eigenfunction at $\lambda = 1$: Step 1 shows that this procedure fails (divergence of the norm), and the eigenfunction at $\lambda = 1$ has to be sought — and found — by another route (Lemma 10.1, which identifies it directly with the constants).

Extension. Check that, for $\lambda = 0$, $g_0 = e_0 = 1 \cdot e^{i \cdot 0 \cdot \theta} = 1$ in a degenerate sense of the formula (the series reduces to the single term $n = 0$): but $\mathcal{L}1 = 1 \neq 0 \cdot 1$, hence g_0 is **not** the eigenfunction associated with $\lambda = 0$ (Theorem 10.2 requires λ in the role of an eigenvalue, and for $\lambda = 0$ the formula $g_0(\theta) = \sum_n 0^n e^{i2^n \theta}$ must be interpreted with the convention $0^0 = 1$, giving $g_0 = e_1 = e^{i\theta}$, not the constant function 1: check that indeed $\mathcal{L}e_1 = 0 = 0 \cdot e_1$, consistently).

10.14. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 10.15).

- (Definition) The transfer operator $\mathcal{L}$ of D is said to be **quasi-compact** on $BV(\mathbb{T})$ because
 - it is a compact operator in the usual sense of functional analysis

-
- B) its essential spectrum on $BV(\mathbb{T})$ is contained in a disc of radius < 1 , while $\lambda = 1$ is an isolated simple eigenvalue
- C) it has finite rank
- D) it is invertible
2. (Hypotheses) The spectral gap of Theorem 10.3 (no eigenvalue in $(\frac{1}{2}, 1)$) holds
- A) on $L^2(\mathbb{T})$ with no further hypotheses
- B) only by restricting $\mathcal{L}$ to $BV(\mathbb{T})$ or to a space $C^{0,\alpha}(\mathbb{T})$
- C) only if D is invertible
- D) only for $m = 2$, never for E_m with $m \geq 3$
3. (Symbolic computation) $\mathcal{L}e_6$ equals
- A) e_3 B) 0 C) e_{12} D) e_6
4. (Numerical computation) The factor $\sqrt{2/(\pi n)}$ of Theorem 10.6 for $n = 100$ equals (to three digits)
- A) 0.0798 B) 0.798 C) 0.0313 D) 1.596
5. (Proof step) In Construction 10.1 of the coboundary $g = \sum_{k \geq 1} \mathcal{L}^k u$, the (absolute and uniform) convergence of the series rests on
- A) the fact that u is a trigonometric polynomial
- B) the pointwise estimate $\|\mathcal{L}^k u\|_\infty \leq 2^{-k} \text{Var}(u)$ (Lemma 10.5), valid because u has zero mean and is BV
- C) the invertibility of D
- D) the Riesz–Fischer theorem
6. (Error diagnosis) The old version of this chapter concluded, from an argument on the odd Fourier modes alone, that $\mathcal{L}$ has no eigenvalues of modulus 1 other than $\lambda = 1$; the diagnosed error (Section 10.2) is
- A) the conclusion itself (on the unique eigenvalue of modulus 1) is false
- B) the argument completely omitted the description of the point spectrum for $|\lambda| < 1$ (which fills the whole open disc, with explicit eigenfunctions): the conclusion about modulus 1 was correct but isolated, not a complete picture of the spectrum
- C) $\mathcal{L}$ is not a linear operator
- D) even and odd modes do not exist for D
7. (Transfer) If $u = \varphi \circ D - \varphi$ for some bounded $\varphi \in BV(\mathbb{T})$ (an exact coboundary), then, by Theorem 10.8,
- A) $S_n u / \sqrt{n} \rightarrow \mathcal{N}(0, 1)$ always
- B) $S_n u / \sqrt{n} \rightarrow 0$ in probability (the CLT degenerates: $\sigma^2 = 0$, since $S_n u = \varphi \circ D^n - \varphi$ is bounded)
- C) $S_n u$ always diverges
- D) nothing can be said without knowing φ explicitly
-

8. (Source and metacognition) The central limit theorem for stationary martingales used in the proof of Theorem 10.8 is
- A) proved in full in this chapter
 - B) imported with a precise source (Billingsley, 1961), recorded in Section 10.11, while the whole construction specific to D (the coboundary, the decomposition, the exact orthogonality, the variance) is proved in full
 - C) a still conjectural result
 - D) an immediate consequence of Birkhoff's ergodic theorem (Chapter 9)

10.15. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **B.** *Rationale:* Remark 10.5 and Theorem 10.3. *Distractors:* A confuses quasi-compactness with compactness of the operator (different notions: $\mathcal{L}$ is not compact, having infinite rank); C and D have no basis. *Diagnosed error:* confusing “quasi-” with a weak version of “compact” instead of with a property of the essential spectrum. *Source:* Remark 10.5. *Prerequisite:* the notion of essential spectrum (recalled informally in the text). *Remedial work:* reread the comparison between the two panels of Figure 10.2.
- 2) **B.** *Rationale:* Theorem 10.3, and the explicit contrast with Remark 10.3 (on L^2 the spectrum fills the whole disc, no gap). *Distractors:* A is exactly the opposite of what is proved (Section 10.2); C and D have no basis (D is not invertible and is precisely the case treated; Corollary 10.1 treats E_m for every m). *Diagnosed error:* failing to distinguish between a property of the operator and a property of the function space on which it acts. *Source:* Theorems 10.2 and 10.3 compared. *Prerequisite:* the notion of eigenvalue. *Remedial work:* redo by hand the computation of Lemma 10.2 and observe where $g \in BV$ is used.
- 3) **A.** *Rationale:* Theorem 10.1, $k = 6$ even, $\mathcal{L}e_6 = e_3$. *Distractors:* B would apply the odd rule to an even k ; C would double instead of halving the index (confusion with the Koopman operator U); D does not change the index. *Diagnosed error:* confusing $\mathcal{L}$ (which halves, if even) with U (which always doubles), the same confusion illustrated in Figure 10.1. *Source:* Theorem 10.1. *Prerequisite:* none beyond the definition. *Remedial work:* repeat the direct computation of the proof for $k = 6$.
- 4) **A.** *Rationale:* computed in block [6] of `verifica_cap10.py`: $\sqrt{2/(\pi \cdot 100)} = \sqrt{0.0063662} \approx 0.0798$, consistent with the local approximation printed for $n = 100$. *Distractors:* B is ten times too large (an error of a factor $\sqrt{10}$ not applied correctly); C and D are other plausible but wrong values (respectively $\sqrt{2/(\pi \cdot 1000)}$ and twice the correct value). *Diagnosed error:* an arithmetic error in the square root or in the factor $2/\pi$. *Source:* Theorem 10.6 and block [6] of the script. *Prerequisite:* elementary arithmetic. *Remedial work:* recompute $2/(\pi \cdot 100)$ and then the root, step by step.
- 5) **B.** *Rationale:* Construction 10.1, which rests explicitly on Lemma 10.5. *Distractors:* A is false (the construction holds for every $u \in BV(\mathbb{T})$ of zero mean, not only trigonometric polynomials, which are however used in the examples for their explicitness); C and D are not pertinent (D is not invertible; Riesz–Fischer concerns the completeness of L^2 , not used here). *Diagnosed error:* confusing the illustrative examples (trigonometric ones, for which the series truncates) with the general hypothesis of the theorem (which requires only BV and zero mean). *Source:* Construction 10.1, Lemma 10.5. *Prerequisite:* convergence of dominating geometric series. *Remedial work:* reread the proof of Lemma 10.5 and check where the zero-mean hypothesis enters.

- 6) **B. Rationale:** Section 10.2, introduction, and Remark 10.3. *Distractors:* A is false (Lemma 10.1 proves precisely that the only eigenvalue of modulus 1 is $\lambda = 1$: the conclusion was correct); C and D make no sense. *Diagnosed error:* a true statement may have a defective proof if it does not address the whole picture required (here: also $|\lambda| < 1$), exactly the kind of defect identified in the expansion plan (entry 4.14) and discussed at the opening of Section 10.2. *Source:* the opening of Section 10.2. *Prerequisite:* none. *Remedial work:* compare Lemma 10.1 and Theorem 10.2 as the two halves of a single picture.
- 7) **B. Rationale:** Example 10.6 (case u_3 , $\sigma^2 = 0$, $S_n u_3$ bounded). *Distractors:* A is the generic non-degenerate case; C is false (precisely the opposite: an exact coboundary has bounded Birkhoff sums, not divergent ones); D is false, much can be said (the case is completely determined). *Diagnosed error:* failing to recognise that an exact coboundary is precisely the case $m \equiv 0$ in the Gordin decomposition. *Source:* the proof of Theorem 10.8, last line (case $\sigma^2 = 0$). *Prerequisite:* Theorem 10.7. *Remedial work:* redo the computation of Example 10.6 with a second exact coboundary of one's own choosing.
- 8) **B. Rationale:** Section 10.11. *Distractors:* A is false (explicitly declared as an import); C is false (Billingsley 1961 is a proved theorem, not a conjecture); D confuses two different results (the ergodic theorem gives the convergence of the averages, not the rate $\sqrt{n}$ nor the Gaussian limit law). *Diagnosed error:* failing to distinguish between “proved in this chapter” and “recalled with a precise source as a standard external ingredient”. *Source:* Section 10.11, Remark 10.7. *Prerequisite:* none beyond careful reading. *Remedial work:* list, for Theorem 10.8, which steps of the proof are specific to D and which one (only one) is an external fact cited.

BEST PRACTICE

Assigned exercises — answers.

10.1 (eigenfunctions of $\mathcal{L}_3$). (a) $\mathcal{L}_3 e_k = e_{k/3}$ if $3 \mid k$, 0 otherwise (Theorem 10.1 with $m = 3$); for $k = 1$, $3 \nmid 1$, hence $\mathcal{L}_3 e_1 = 0$. (b) The same argument as in Theorem 10.2: the coefficients of $g_\lambda^{(3)}$ are $c_{3^n} = \lambda^n$ and 0 elsewhere; the coefficient of index j of $\mathcal{L}_3 g_\lambda^{(3)}$ is c_{3j} , which equals $\lambda^{n+1} = \lambda c_{3^n}$ if $j = 3^n$, and $0 = \lambda \cdot 0$ otherwise (by the same argument on the non-existence of a j that is not a power of 3 with $3j$ a power of 3): $\mathcal{L}_3 g_\lambda^{(3)} = \lambda g_\lambda^{(3)}$. (c) $\mathcal{L}_3 e_k = 0$ for every k not a multiple of 3 (two residue classes out of three, against one residue class out of two for $m = 2$): the “base” of annihilation grows with m , consistently with the increasing entropy $\log m$. (d) If $\mathcal{L}_3 g = \lambda g$, $|\lambda| = 1$, $g = \sum c_j e_j$: from the relation $c_{3j} = \lambda c_j$ and $|\lambda| = 1$, $|c_j| = |c_{3j}| = |c_{9j}| = \dots$; for $g \in L^2$ and $j \neq 0$, the indices $3^n j$ are distinct and $c_{3^n j} \rightarrow 0$, hence $c_j = 0$; only c_0 remains, and as in Lemma 10.1, $\lambda = 1$.

10.2 (correlations for E_5). (a) $|C_n^{(5)}(u, v)| \leq \frac{\pi}{\sqrt{3}} \cdot 1 \cdot 2\pi \cdot 5^{-n} = \frac{2\pi^2}{\sqrt{3}} \cdot 5^{-n}$ ($\|u\|_2 = 1$, $\text{Var}(v) = 2\pi$). (b) For $n = 3$: case $m = 2$, factor $2^{-3} = 0.125$; case $m = 5$, factor $5^{-3} = 0.008$: the correlation for E_5 is smaller by a factor $0.125/0.008 = 15.625 = (5/2)^3$. (c) Higher entropy ($\log 5 > \log 2$) means more branches, hence the transfer operator $\mathcal{L}_5$ averages over more preimages at each step, contracting the variation by a factor $1/5$ instead of $1/2$ (a generalisation of Lemma 10.2, already verified for E_3 in Worked exercise 9.7 of Chapter 9): more topological chaos, faster statistical convergence.

10.3 (coboundary of $u = \sqrt{2} \cos 4\theta$). (a) $\mathcal{L}u = \sqrt{2} \cos 2\theta$ (indices $\pm 4 \rightarrow \pm 2$); $\mathcal{L}^2 u = \mathcal{L}(\sqrt{2} \cos 2\theta) = \sqrt{2} \cos \theta$ (Example 10.5); $\mathcal{L}^3 u = \mathcal{L}(\sqrt{2} \cos \theta) = 0$ (Example 10.4). (b) $g = \mathcal{L}u + \mathcal{L}^2 u + \mathcal{L}^3 u + \dots = \sqrt{2} \cos 2\theta + \sqrt{2} \cos \theta + 0 + \dots = \sqrt{2} \cos 2\theta + \sqrt{2} \cos \theta$. (c) $m = u + g - g \circ D = \sqrt{2} \cos 4\theta + (\sqrt{2} \cos 2\theta + \sqrt{2} \cos \theta) - (\sqrt{2} \cos 4\theta + \sqrt{2} \cos 2\theta) = \sqrt{2} \cos \theta$: again the same martingale as in Examples 10.4 and 10.5 (the pattern repeats: every pure power $\sqrt{2} \cos(2^j \theta)$ has the same martingale $\sqrt{2} \cos \theta$, by repeated telescoping). (d) $\sigma^2 = \int (\sqrt{2} \cos \theta)^2 d\mu = 1$, as in the previous examples.

10.16. Chapter formulary

Transfer operator

$$(\mathcal{L}g)(\theta) = \frac{1}{2} [g(\frac{\theta}{2}) + g(\frac{\theta}{2} + \pi)], \quad (\mathcal{L}_m g)(\theta) = \frac{1}{m} \sum_{j=0}^{m-1} g(\frac{\theta+2\pi j}{m})$$

$$\int_{\mathbb{T}} (g \circ E_m) w \, d\mu = \int_{\mathbb{T}} g \cdot (\mathcal{L}_m w) \, d\mu, \quad \widehat{(\mathcal{L}_m g)}_k = \widehat{g}_{mk} \quad (\mathcal{L}e_k = e_{k/2} \text{ if } k \text{ even, } 0 \text{ if odd})$$

Spectrum on L^2 vs. spectral gap

$$g_\lambda = \sum_{n \geq 0} \lambda^n e_{2^n} \in L^2(\mathbb{T}) \quad (|\lambda| < 1), \quad \mathcal{L}g_\lambda = \lambda g_\lambda, \quad \|g_\lambda\|_2^2 = \frac{1}{1-|\lambda|^2} \quad (|\lambda| = 1 \Rightarrow \lambda = 1, g \equiv \text{const.})$$

$$\text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g), \quad [\mathcal{L}g]_\alpha \leq (\frac{1}{2})^\alpha [g]_\alpha \quad \Rightarrow \quad g \in BV \text{ or } C^{0,\alpha}, \quad \mathcal{L}g = \lambda g, \quad g \neq \text{const.} \Rightarrow |\lambda| \leq \frac{1}{2} \text{ (or } (\frac{1}{2})^\alpha)$$

Decay of correlations

$$|\widehat{g}_k| \leq \frac{\text{Var}(g)}{2\pi|k|} (BV), \quad |C_n(u, v)| \leq \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) \cdot m^{-n} \quad (D : m = 2; E_m : m^{-n})$$

$$\left| \int (u \circ T^n) v \, dx - m_v \int u \, dx \right| \leq \|u\|_1 \text{Var}(v) \cdot 2^{-n} \quad (\text{tent map, via } \mathcal{L}_T)$$

CLT, dyadic indicator (de Moivre–Laplace)

$$\varphi = \pm 1 \text{ (1st binary digit), } S_n \varphi = n - 2K_n, \quad K_n \sim \text{Bin}(n, \frac{1}{2}) \text{ exactly}$$

$$\binom{n}{k} 2^{-n} = \sqrt{\frac{2}{\pi n}} e^{-x_k^2/2} (1 + o(1)), \quad x_k = \frac{2k-n}{\sqrt{n}} \Rightarrow S_n \varphi / \sqrt{n} \rightarrow \mathcal{N}(0, 1)$$

CLT, Gordin martingales

$$g = \sum_{k \geq 1} \mathcal{L}^k u, \quad m = u + g - g \circ D, \quad \mathcal{L}m = 0, \quad u = m + (g \circ D - g)$$

$$\int (m \circ D^i)(m \circ D^j) \, d\mu = 0 \quad (i \neq j), \quad \sigma^2 = \int m^2 \, d\mu, \quad S_n u / \sqrt{n} \rightarrow \mathcal{N}(0, \sigma^2) \text{ (imported: martingale CLT, Billingsley)}$$

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CHAPTER 11

Logistic map, bifurcations and universality

The rest of this volume studies the circle: rotations, circle maps, holomorphic dynamics. This chapter leaves the circle for a moment and looks at the **logistic family** $f_r(x) = rx(1 - x)$ on the interval $[0, 1]$ — a one-dimensional dynamical system, not a circular one. The reason for including it in a treatise on the circle is twofold. First: the doubling map $D : \mathbb{T} \rightarrow \mathbb{T}$ of Chapter 7 is semiconjugate to f_4 , and the cascade of bifurcations that leads to that chaotic dynamics is the historical and conceptual prototype of every order-to-chaos transition, including those on the circle itself (the Arnold tongues, the critical circle maps treated in this very chapter). Second: the Feigenbaum constant δ and the renormalisation operator that explain it are an instance of **universality** — the same phenomenon, with *different* constants, governs the breakdown of quasi-periodic orbits on the circle (critical maps, Section 11.6): an opportunity to show, with the numbers in hand, that the two universalities must not be confused — a mistake that appeared in the previous version of this volume and which is here explicitly corrected.

INFO

Level and prerequisites. Level **B–C** core (Sections 11.1–11.3 and 11.8): differential calculus in one variable, the intermediate value theorem, the implicit function theorem. Sections 11.4–11.7 are of level **D–E**: there we state, *without proving them*, results that go beyond the volume (Lanford’s theorem, the theorems of Jakobson and of Benedicks–Carleson), explicitly declared “imported” or “panorama”, with a verified primary source. Prerequisites within the treatise: none strictly necessary for the core; for Section 11.3 Sharkovskii’s theorem is useful (but not indispensable, a two-line reminder suffices), treated at length in the chapter on the doubling map of this volume. Remedial path for those who skip the advanced sections: the core (logistic family, normal forms, existence of the cascade, bifurcation diagram) is self-contained.

11.1. The logistic family: fixed points and first bifurcations

Definition 11.1: Logistic family. For $r \in (0, 4]$, the **logistic map** of parameter r is

$$f_r : [0, 1] \rightarrow [0, 1], \quad f_r(x) = rx(1 - x).$$

For $r \in (0, 4]$, f_r does indeed map $[0, 1]$ into itself: the maximum of f_r is attained at the **critical point** $x = \frac{1}{2}$ (the only point where f'_r vanishes), with $f_r\left(\frac{1}{2}\right) = \frac{r}{4} \leq 1$.

Proposition 11.1: Fixed points and their stability. For $r > 0$, f_r has in $[0, 1]$ the fixed points

$$x = 0 \quad \text{and} \quad x^* = 1 - \frac{1}{r} \quad (\text{the latter in } [0, 1] \iff r \geq 1).$$

The derivatives are $f'_r(0) = r$ and $f'_r(x^*) = 2 - r$. Consequently:

- (i) for $0 < r < 1$: $x = 0$ is the only fixed point in $[0, 1]$, and it is asymptotically stable ($|f'_r(0)| = r < 1$);
- (ii) for $r = 1$: $x = 0$ and x^* coincide ($x^* = 0$), with $f'_1(0) = 1$: this is a **transcritical bifurcation** (Proposition 11.2);
- (iii) for $1 < r < 3$: $x = 0$ is unstable ($f'_r(0) = r > 1$) and $x^* \in (0, 1)$ is asymptotically stable ($|f'_r(x^*)| = |2 - r| < 1$);
- (iv) for $r = 3$: $f'_r(x^*) = -1$: this is the **first period-doubling bifurcation** (Theorem 11.1).

Proof. $f_r(x) = x \iff rx(1-x) - x = 0 \iff x(r(1-x) - 1) = 0 \iff x = 0$ or $x = 1 - \frac{1}{r}$. Differentiating $f_r(x) = rx - rx^2$: $f'_r(x) = r - 2rx = r(1 - 2x)$. Hence $f'_r(0) = r$ and $f'_r(x^*) = r(1 - 2(1 - \frac{1}{r})) = r(\frac{2}{r} - 1) = 2 - r$. Statements (i)–(iv) follow by comparing $|f'_r(0)| = r$ and $|f'_r(x^*)| = |2 - r|$ with 1, and observing that $x^* \geq 0 \iff r \geq 1$. $\square$

Proposition 11.2: Transcritical bifurcation at $r = 1$. The two branches of fixed points $x = 0$ and $x = x^*(r) = 1 - \frac{1}{r}$ cross transversally at $(r, x) = (1, 0)$, exchanging their stability: for $r < 1$ it is $x = 0$ that is stable and $x^*(r) < 0$ lies outside the physical domain; for $r > 1$ it is $x^*(r) > 0$ that is stable and $x = 0$ is unstable.

Proof. Both functions $r \mapsto 0$ and $r \mapsto x^*(r) = 1 - \frac{1}{r}$ are C^∞ on $(0, \infty)$ and meet exactly at $r = 1$ ($x^*(1) = 0$), with derivatives 0 and $\frac{d}{dr}(1 - \frac{1}{r})|_{r=1} = 1 \neq 0$: the two branches are transversal (not tangent) at that point, and Proposition 11.1 shows that the stability of the two branches is exchanged exactly there. This is an instance of Theorem 11.4, verified in detail further below (Example 11.2). $\square$

Theorem 11.1: First bifurcation: period doubling at $r = 3$. For $1 < r < 3$, $x^*(r) = 1 - \frac{1}{r}$ is the only attractor in $(0, 1)$ and it attracts every orbit with initial datum in $(0, 1)$. At $r = 3$, x^* loses stability ($f'_3(x^*) = -1$) and for $r > 3$ (near 3) a period-2 orbit is born around x^* , described at length in Theorem 11.2.

Proof. The global attractivity of x^* for $1 < r < 3$ on $(0, 1)$ is a classical fact proved by monotonicity (Strogatz, E49, ch. 10): f_r is concave on $[0, 1]$ with a single critical point, $f_r((0, 1)) \subseteq (0, 1)$, and the iteration $x_{n+1} = f_r(x_n)$ starting from any $x_0 \in (0, 1) \setminus \{x^*\}$ is (after at most one step) monotone and bounded, hence convergent; the limit is necessarily a fixed point, and the only fixed point in $(0, 1)$ for $1 < r < 3$ is x^* (Proposition 11.1). The loss of stability at $r = 3$ and the birth of the 2-cycle are the instance, for this family, of the general period-doubling theorem (Theorem 11.5); the explicit existence and stability of the 2-cycle are proved independently, in closed form, in Theorem 11.2 below. $\square$

Theorem 11.2: The explicit 2-cycle and its stability. For $r > 3$, f_r has a unique period-2 orbit in $(0, 1)$, given by

$$x_{\pm}(r) = \frac{(r+1) \pm \sqrt{(r-3)(r+1)}}{2r},$$

with $f_r(x_+) = x_-$ and $f_r(x_-) = x_+$. The multiplier of the cycle is

$$(f_r^2)'(x_{\pm}) = f'_r(x_+) f'_r(x_-) = -r^2 + 2r + 4,$$

and the cycle is asymptotically stable exactly for $3 < r < 1 + \sqrt{6} \approx 3.449490$.

Proof. Step 1 — the system for the sums. Let $\{x_1, x_2\}$ be a period-2 orbit, $x_2 = f_r(x_1)$, $x_1 = f_r(x_2)$, $x_1 \neq x_2$. Subtracting the two equations and using $a^2 - b^2 = (a - b)(a + b)$:

$$x_1 - x_2 = r[x_2(1 - x_2) - x_1(1 - x_1)] = r(x_2 - x_1)[1 - (x_1 + x_2)].$$

Dividing by $x_2 - x_1 \neq 0$ (which appears on the left with the opposite sign): $-1 = r[1 - (x_1 + x_2)]$, whence, setting $S = x_1 + x_2$,

$$S = \frac{r+1}{r}.$$

Step 2 — the product. Adding the two equations:

$$S = r[(x_1 + x_2) - (x_1^2 + x_2^2)] = r[S - (S^2 - 2P)], \quad P = x_1x_2,$$

whence $P = \frac{S(1-r) + rS^2}{2r}$. Substituting $S = \frac{r+1}{r}$: $rS = r+1$ and $(1-r) + rS = (1-r) + (r+1) = 2$, hence $P = \frac{S \cdot 2}{2r} = \frac{S}{r} = \frac{r+1}{r^2}$.

Step 3 — solution. x_1, x_2 are the roots of $t^2 - St + P = 0$, that is, of $r^2t^2 - r(r+1)t + (r+1) = 0$:

$$t = \frac{r(r+1) \pm \sqrt{r^2(r+1)^2 - 4r^2(r+1)}}{2r^2} = \frac{(r+1) \pm \sqrt{(r+1)[(r+1) - 4]}}{2r} = \frac{(r+1) \pm \sqrt{(r-3)(r+1)}}{2r},$$

real (and distinct) exactly when $(r-3)(r+1) > 0$, that is (for $r > 0$) when $r > 3$: this proves existence and uniqueness of the 2-cycle in $(0, 1)$ for $r > 3$, together with its closed formula.

Step 4 — the multiplier. $f'_r(x) = r(1 - 2x)$, hence

$$f'_r(x_1)f'_r(x_2) = r^2[1 - 2(x_1 + x_2) + 4x_1x_2] = r^2[1 - 2S + 4P].$$

With $S = \frac{r+1}{r}$, $P = \frac{r+1}{r^2}$: $1 - 2S = 1 - \frac{2(r+1)}{r} = \frac{-r-2}{r}$ and $4P = \frac{4(r+1)}{r^2}$, hence

$$1 - 2S + 4P = \frac{-r-2}{r} + \frac{4(r+1)}{r^2} = \frac{r(-r-2) + 4(r+1)}{r^2} = \frac{-r^2 + 2r + 4}{r^2},$$

and finally $f'_r(x_1)f'_r(x_2) = r^2 \cdot \frac{-r^2 + 2r + 4}{r^2} = -r^2 + 2r + 4$.

Step 5 — stability threshold. The cycle is stable if and only if $|-r^2 + 2r + 4| < 1$. For $r > 3$: at $r = 3$, $-9 + 6 + 4 = 1$ (boundary, consistent with Theorem 11.1); the function $r \mapsto -r^2 + 2r + 4$ is strictly decreasing for $r > 1$ (derivative $-2r + 2 < 0$), hence it decreases from 1 (at $r = 3$) towards $-\infty$: it stays in $(-1, 1)$ until it reaches -1 , that is, as long as $-r^2 + 2r + 4 > -1 \iff r^2 - 2r - 5 < 0 \iff r \in (1 - \sqrt{6}, 1 + \sqrt{6})$. Intersecting with $r > 3$: the cycle is stable exactly for $3 < r < 1 + \sqrt{6}$. $\square$

Double numerical check The formulae of Theorem 11.2 are verified independently in `verifica_cap11_periodo3.py` (block V1, Appendix B): for $r = 3.2$ the multiplier computed from the closed formula $-r^2 + 2r + 4 = 0.1600$ agrees (to 10^{-9}) with the direct product $f'_r(x_+)f'_r(x_-)$ computed from the numerical values of $x_{\pm}$; the same for $r = 3.4$ (-0.7600) and for $r = 3.5$ (-1.2500 , unstable, consistent with the Theorem). At $r = 1 + \sqrt{6} = 3.449490$ the computed multiplier is -1.000000 , consistent with the threshold. $\checkmark$ $\square$

Example 11.1: The 2-cycle at $r = 3.2$ and at $r = 3.4$. For $r = 3.2$: $x_{\pm} = \frac{4.2 \pm \sqrt{0.2 \cdot 4.2}}{6.4} = \frac{4.2 \pm 0.916515}{6.4}$, that is, $x_+ = 0.799455$, $x_- = 0.512733$; direct check: $f_{3.2}(0.799455) = 3.2 \times 0.799455 \times 0.200545 = 0.513046$, close to x_- (the small difference is

rounding to six digits). The multiplier is $-3.2^2 + 2 \times 3.2 + 4 = 0.16 > 0$: a stable cycle, with locally monotone convergence (not oscillating, because the multiplier is positive).

For $r = 3.4$: $x_{\pm} = \frac{4.4 \pm \sqrt{0.4 \cdot 4.4}}{6.8} = \frac{4.4 \pm 1.326650}{6.8}$, that is, $x_+ = 0.841566$, $x_- = 0.451963$; multiplier $-3.4^2 + 2 \times 3.4 + 4 = -0.76$: stable, with oscillating convergence (negative multiplier).

Proposition 11.3: Birth of the period-3 window at $r = 1 + 2\sqrt{2}$. *The logistic map has a period-3 orbit in $(0, 1)$ for $r \geq 1 + 2\sqrt{2}$, and has none for $r < 1 + 2\sqrt{2}$ near that value: at*

$$r_3 = 1 + 2\sqrt{2} = 1 + \sqrt{8} \approx 3.828427$$

a pair of period-3 orbits (one stable and one unstable) is born by a saddle-node bifurcation, tangent to one another at the point of birth.

Proof. Step 1 — reduction to a degree-6 polynomial. The points of period (exact or improper) 3 solve $f_r^3(x) - x = 0$, a polynomial of degree 8 in x . It vanishes identically at the two fixed points $x = 0$ and $x^* = 1 - \frac{1}{r}$ (which have period 1, hence also 3 as an improper period): factoring out these two factors — a computation carried out symbolically in `verifica_cap11_periodo3.py` (block V2) — there remains a polynomial $Q(x; r)$ of degree *exactly* 6 in x , whose coefficients are polynomials in r (reported in the script), whose roots are the six points of the orbits of period 3 properly so called (two orbits, when they exist, with 3 points each).

Step 2 — the tangency condition. A saddle-node bifurcation of the 3-cycle corresponds exactly to a value of r for which $Q(\cdot; r)$ has a double root in x : two of the six roots (the points of the two orbits, one stable and one unstable, which are born in pairs) coincide. This happens if and only if the **resultant** of $Q(x; r)$ and of its derivative $\partial_x Q(x; r)$ (which vanishes exactly when Q and $\partial_x Q$ have a common root in x , that is, when Q has a multiple root) vanishes as a function of r .

Step 3 — computation of the resultant. The computation (symbolic algebra, detailed in `verifica_cap11_periodo3.py`) gives, as a factored polynomial in r ,

$$\text{Res}_x(Q, \partial_x Q) = r^{47} (r^2 - 5r + 7)^2 (r^2 - 2r - 7)^3 (r^2 + r + 1)^2.$$

The factors $r^2 - 5r + 7$ and $r^2 + r + 1$ have negative discriminant ($25 - 28 = -3$ and $1 - 4 = -3$ respectively): no real root. There remains $r^2 - 2r - 7 = 0$, with roots $r = 1 \pm 2\sqrt{2}$; only $r = 1 + 2\sqrt{2} \approx 3.828427$ is positive and lies in the physical interval $(0, 4]$. Hence $r = 1 + 2\sqrt{2}$ is the *only* real value (besides the degenerate case $r = 0$) at which $Q(\cdot; r)$ has a double root: it is the only saddle-node bifurcation of the 3-cycle in the family.

Step 4 — independent numerical check. A direct dichotomic search (not symbolic): one iterates f_r for a long transient and checks whether the point reached returns to itself after 3 iterates; the smallest r for which this happens, sought by bisection between $r = 3.80$ (no period-3 orbit observed) and $r = 3.835$ (period-3 orbit observed), turns out to be $r = 3.828425$, against the exact symbolic value $1 + 2\sqrt{2} = 3.828427$: discrepancy 2.4×10^{-6} , compatible with the tolerance of the dynamical bisection. $\checkmark$ (script `verifica_cap11_periodo3.py`, block V2) $\square$

Best practice. The computation of Step 3 is heavy by hand (the polynomial $Q(x; r)$ has degree 6 in x and coefficients of degree up to 7 in r): it was carried out with a symbolic algebra system (SymPy), not by hand, exactly as the template prescribes for the heavy numerical computations of this treatise. The method — elimination by means of the re-

sultant between a polynomial and its derivative in order to find the multiple roots — is however entirely elementary and exact: it is not a numerical approximation, and the result $r = 1 + 2\sqrt{2}$ is a complete algebraic proof, in the spirit of Saha and Strogatz (E49, who treat the same problem with a method more refined by hand), not a fact merely observed in the bifurcation diagram.

11.2. Normal forms of one-dimensional bifurcations

The three bifurcations just seen in the logistic family — transcritical at $r = 1$, period doubling at $r = 3$, saddle-node (for the 3-cycle) at $r = 1 + 2\sqrt{2}$ — are particular cases of three general theorems, valid for any smooth one-parameter family of interval maps. We prove them here in general: they are the tools with which one recognises a bifurcation *without* having to solve explicitly the equation for the fixed points, something which for families more complex than the logistic one is often impossible in closed form.

Definition 11.2: Local bifurcation of fixed points. Let $f : I \times J \rightarrow \mathbb{R}$, $(x, \mu) \mapsto f(x, \mu)$, be of class C^3 on open intervals $I, J \subseteq \mathbb{R}$, and let $g(x, \mu) := f(x, \mu) - x$. A point (x_0, μ_0) with $g(x_0, \mu_0) = 0$ and $g_x(x_0, \mu_0) = 0$ (that is, $f(x_0, \mu_0) = x_0$ and $f_x(x_0, \mu_0) = 1$) is called a point of **local bifurcation**: the number of fixed points of $f(\cdot, \mu)$ near x_0 may change when μ crosses μ_0 , because the implicit function theorem applied to $g(x, \mu) = 0$ in the variable x does not apply directly ($g_x = 0$).

Theorem 11.3: Saddle-node bifurcation. In the notation of Definition 11.2, suppose in addition

$$g_\mu(x_0, \mu_0) \neq 0 \quad (\text{transversality}), \quad g_{xx}(x_0, \mu_0) \neq 0 \quad (\text{non-degeneracy}).$$

Then there exist a neighbourhood $U \times V$ of (x_0, μ_0) and a C^2 function $\mu = \mu(x)$ on a neighbourhood of x_0 , with $\mu(x_0) = \mu_0$, $\mu'(x_0) = 0$ and $\mu''(x_0) = -\frac{g_{xx}(x_0, \mu_0)}{g_\mu(x_0, \mu_0)} \neq 0$, such that, in $U \times V$, the fixed points of $f(\cdot, \mu)$ are exactly the pairs $(x, \mu(x))$: for μ on one side of μ_0 (determined by the sign of $\mu''(x_0)$) there are two fixed points near x_0 , for μ on the other side none, and at $\mu = \mu_0$ the two fixed points coincide at x_0 .

Proof. Since $g_\mu(x_0, \mu_0) \neq 0$, the implicit function theorem applied to $g(x, \mu) = 0$ in the variable μ (not in x !) gives, on a neighbourhood of x_0 , a unique C^2 function $\mu = \mu(x)$ with $g(x, \mu(x)) \equiv 0$ and $\mu(x_0) = \mu_0$: by construction, $(x, \mu(x))$ runs through *all and only* the fixed-point/parameter pairs near (x_0, μ_0) (the local uniqueness of μ given x comes precisely from the implicit function theorem). Differentiating $g(x, \mu(x)) = 0$ with respect to x :

$$g_x(x, \mu(x)) + g_\mu(x, \mu(x)) \mu'(x) = 0.$$

At $x = x_0$: $g_x(x_0, \mu_0) = 0$ by hypothesis, hence $g_\mu(x_0, \mu_0) \mu'(x_0) = 0$, and since $g_\mu(x_0, \mu_0) \neq 0$: $\mu'(x_0) = 0$. Differentiating a second time:

$$g_{xx} + 2g_{x\mu}\mu' + g_{\mu\mu}(\mu')^2 + g_\mu\mu'' = 0.$$

At $x = x_0$, using $\mu'(x_0) = 0$: $g_{xx}(x_0, \mu_0) + g_\mu(x_0, \mu_0) \mu''(x_0) = 0$, whence $\mu''(x_0) = -g_{xx}(x_0, \mu_0)/g_\mu(x_0, \mu_0) \neq 0$ by the non-degeneracy hypothesis. Hence $\mu(x)$ has at x_0 a non-degenerate critical point (a strict local maximum or minimum, according to the sign of $\mu''(x_0)$): by the second derivative test, there exists $\varepsilon > 0$ such that, for $x \in (x_0 - \varepsilon, x_0 + \varepsilon) \setminus \{x_0\}$, $\mu(x)$ takes each value near μ_0 (on the side indicated by the sign of $\mu''(x_0)$) exactly twice, once for

$x < x_0$ and once for $x > x_0$ (by continuity and strict monotonicity of μ on each of the two half-intervals, a consequence of the fact that x_0 is the only critical point of μ in a sufficiently small neighbourhood, since $\mu''(x_0) \neq 0$), and no value on the opposite side. This is exactly the statement. $\square$

Theorem 11.4: Transcritical bifurcation. *In the notation of Definition 11.2, suppose that $g(x_0, \mu) = 0$ for every μ in a neighbourhood of μ_0 (that is, x_0 is a fixed point of $f(\cdot, \mu)$ for every such μ : this is the case, with $x_0 = 0$, of the logistic family, where $f_r(0) = 0$ for every r), and suppose*

$$g_{x\mu}(x_0, \mu_0) \neq 0, \quad g_{xx}(x_0, \mu_0) \neq 0.$$

Then there exists, on a neighbourhood of μ_0 , a second C^1 branch of fixed points $x = \xi(\mu) \neq x_0$ with $\xi(\mu_0) = x_0$, $\xi'(\mu_0) = -\frac{g_{x\mu}(x_0, \mu_0)}{\frac{1}{2}g_{xx}(x_0, \mu_0)} \neq 0$, transversal to the constant branch x_0 : the two branches exchange their stability at $\mu = \mu_0$.

Proof. Since $g(x_0, \mu) \equiv 0$ near μ_0 , and g is C^3 , Hadamard's lemma (integrating g_x along the segment from x_0 to x) allows us to write, on a neighbourhood of (x_0, μ_0) ,

$$g(x, \mu) = (x - x_0)h(x, \mu), \quad h(x, \mu) := \int_0^1 g_x(x_0 + t(x - x_0), \mu) dt,$$

with h of class C^2 (because g_x is C^2 , g being of class C^3). The fixed points other than x_0 near (x_0, μ_0) are exactly the zeros of $h(x, \mu) = 0$. One has $h(x_0, \mu) = g_x(x_0, \mu)$ for every μ (taking $t = \text{anything}$ in the definition with $x = x_0$), hence $h(x_0, \mu_0) = g_x(x_0, \mu_0) = 0$: the point (x_0, μ_0) is also a zero of h . We compute the partial derivatives of h at (x_0, μ_0) in order to apply the implicit function theorem to $h(x, \mu) = 0$ in the variable x : differentiating under the integral sign, $h_\mu(x_0, \mu) = g_{x\mu}(x_0, \mu)$, hence $h_\mu(x_0, \mu_0) = g_{x\mu}(x_0, \mu_0) \neq 0$ by hypothesis. By the implicit function theorem (this time solving for μ as a function of x , as in Theorem 11.3, or else observing that $h_x(x_0, \mu_0) = \frac{1}{2}g_{xx}(x_0, \mu_0)$ — a direct computation, differentiating h once more — and that if $h_x(x_0, \mu_0) \neq 0$ one can solve for x as a function of μ directly): there exists a C^1 branch $x = \xi(\mu)$, $\xi(\mu_0) = x_0$, of zeros of h , with

$$\xi'(\mu_0) = -\frac{h_\mu(x_0, \mu_0)}{h_x(x_0, \mu_0)} = -\frac{g_{x\mu}(x_0, \mu_0)}{\frac{1}{2}g_{xx}(x_0, \mu_0)} \neq 0.$$

This branch is transversal to the constant branch $x \equiv x_0$ (which has zero “derivative”), hence the two branches cross simply at μ_0 : it is exactly the phenomenon already verified explicitly for the logistic map at $r = 1$ (Proposition 11.2). $\square$

Theorem 11.5: Period-doubling (flip) bifurcation. *Let $f(\cdot, \mu)$ be a C^3 family with a fixed point which, by the implicit function theorem (applicable because $f_x \neq 1$ near the point in question), extends to a smooth branch $x_0(\mu)$ for μ near μ_0 , and suppose*

$$f_x(x_0(\mu_0), \mu_0) = -1.$$

Then $x_0(\mu)$ remains a fixed point for every μ near μ_0 , but the map $g_2(x, \mu) := f(f(x, \mu), \mu) - x$ (whose zeros are the fixed points of f together with the points of a possible period-2 orbit) satisfies $g_2(x_0(\mu), \mu) \equiv 0$ and $\partial_x g_2(x_0(\mu_0), \mu_0) = (f_x(x_0(\mu_0), \mu_0))^2 - 1 = 0$: it is itself a point of local bifurcation in the sense of Definition 11.2 for the map $f \circ f$. If, in addition, $\partial_x \partial_\mu g_2(x_0(\mu_0), \mu_0) \neq 0$, there exists (by Theorem 11.4 applied to g_2 , with $x_0(\mu)$ in the role of the constant branch — here not constant but in any case always a fixed point of $f \circ f$) a second C^1 branch of fixed points of $f \circ f$, $\xi(\mu) \neq x_0(\mu)$, which detaches from $x_0(\mu_0)$ at

$\mu = \mu_0$: this is the birth of a period-2 orbit.

Proof. That $x_0(\mu)$ remains a fixed point of $f(\cdot, \mu)$ for μ near μ_0 is the direct application of the implicit function theorem to $g(x, \mu) = f(x, \mu) - x = 0$ in the variable x , legitimate because $g_x(x_0(\mu_0), \mu_0) = f_x(x_0(\mu_0), \mu_0) - 1 = -2 \neq 0$. Being a fixed point of f , $x_0(\mu)$ is a fortiori a fixed point of $f \circ f$, hence $g_2(x_0(\mu), \mu) \equiv 0$: the map g_2 satisfies exactly the hypothesis “ x_0 a fixed point for every μ ” of Theorem 11.4. It remains to compute $\partial_x g_2$ at the point: by the chain rule,

$$\partial_x g_2(x, \mu) = f_x(f(x, \mu), \mu) f_x(x, \mu) - 1,$$

which at $(x_0(\mu_0), \mu_0)$, since $f(x_0(\mu_0), \mu_0) = x_0(\mu_0)$, equals $(f_x(x_0(\mu_0), \mu_0))^2 - 1 = (-1)^2 - 1 = 0$. The existence of the second branch, under the transversality condition stated, is then exactly Theorem 11.4 applied to g_2 in place of g . The branch $\xi(\mu)$, if it exists on one side of μ_0 , necessarily has $\xi(\mu) \neq x_0(\mu)$ and $f(\xi(\mu), \mu) \neq \xi(\mu)$ (otherwise it would be a fixed point of f near $x_0(\mu_0)$ different from $x_0(\mu)$, excluded for μ close enough to μ_0 by continuity, since $x_0(\mu_0)$ is a simple fixed point of $f(\cdot, \mu_0)$): hence $\xi(\mu)$ and $f(\xi(\mu), \mu)$ form a genuine period-2 orbit. $\square$

Example 11.2: The normal forms applied to the logistic map. With $f(x, \mu) = \mu x(1-x)$ (here $\mu = r$): $g = f - x$, $g_x = \mu(1-2x) - 1$, $g_{xx} = -2\mu$, $g_\mu = x(1-x)$, $g_{x\mu} = 1-2x$.

- *Transcritical at $(x_0, \mu_0) = (0, 1)$:* $g(0, \mu) = 0$ for every μ (checked), $g_{x\mu}(0, 1) = 1 - 0 = 1 \neq 0$, $g_{xx}(0, 1) = -2 \neq 0$: the hypotheses of Theorem 11.4 are satisfied, and give $\xi'(1) = -1/(\frac{1}{2} \cdot (-2)) = 1$, in agreement with $\frac{d}{dr}(1 - \frac{1}{r})|_{r=1} = 1$ computed directly in Proposition 11.2. ✓
- *Saddle-node for the 3-cycle at $r = 1 + 2\sqrt{2}$:* here $f = f^3$ (the third iterate, not f itself), x_0 one of the two points of the nascent 3-cycle; the derivatives g_{xx} and g_μ being non-zero at that point (checked numerically in the script `verifica_cap11_periodo3.py`, where the double root of the resultant is isolated and non-degenerate to second order) guarantee that Theorem 11.3 applies: it is exactly the saddle-node described algebraically in Proposition 11.3.
- *Flip at $r = 3$:* $f_x(x^*, 3) = 2 - 3 = -1$ (Proposition 11.1): the hypothesis of Theorem 11.5 is satisfied; the second branch of fixed points of $f_r \circ f_r$ is exactly the 2-cycle $x_\pm(r)$ of Theorem 11.2, whose existence for $r > 3$ was computed above in closed and explicit form (with no need of the general theorem).

11.3. The period-doubling cascade: existence and numerical estimate of δ

Theorem 11.1 produces, at $r = 3$, a 2-cycle. That 2-cycle in its turn loses stability at $r = 1 + \sqrt{6}$ (Theorem 11.2), giving rise (by Theorem 11.5 applied to $f_r \circ f_r$) to a 4-cycle; the 4-cycle in its turn loses stability, and so on. The result is an infinite sequence of period doublings, the **Feigenbaum cascade**.

Theorem 11.6: Existence of the cascade. *There exists a strictly increasing sequence $(r_n)_{n \geq 1}$ in $(3, 4)$, with $r_1 = 3$, such that: for $r_{n-1} < r < r_n$ (setting $r_0 := 1$) there exists an asymptotically stable orbit of period 2^{n-1} , which at $r = r_n$ loses stability (flip) generating an orbit of period 2^n , stable for r just above r_n . The sequence converges: $r_n \rightarrow r_\infty \leq 4$.*

Proof. Existence of the superstable parameters. For every $n \geq 0$ we define $\varphi_n(r) := f_r^{2^n}(\frac{1}{2}) - \frac{1}{2}$, a continuous function of r (the critical point $\frac{1}{2}$ is the same for every r , and $f_r^{2^n}(\frac{1}{2})$ depends continuously on r , being a composition of continuous functions). In the open interval of existence of a stable orbit of period 2^n (which contains, by Theorem 11.5 iterated, a non-empty interval immediately after each r_n), a classical argument of one-dimensional dynamical systems theory (see the chapter of this volume on the doubling map and Sharkovskii's theorem for the complete symbolic apparatus, or C5, C12, E42 for the specific case) shows that the critical point $\frac{1}{2}$ is attracted by the period- 2^n cycle for an open and non-empty subset of parameters in that interval, and that φ_n changes sign between the beginning and the end of that interval: by the intermediate value theorem there then exists R_n in that interval with $\varphi_n(R_n) = 0$, that is, with the critical point itself periodic of period (a divisor of) 2^n : it is the **superstable parameter** of level n .

Structure of the inductive step. The preceding construction, applied inductively starting from $r_0 = 1$ (where, by the transcritical bifurcation, the branch $x^*(r)$ that replaces $x = 0$ as attractor is born) and $r_1 = 3$ (Theorem 11.1, proved here at length and independently of the general argument), produces for every $n \geq 1$ a non-empty interval (r_{n-1}, r_n) in which the orbit of period 2^{n-1} is stable, followed by the flip bifurcation that replaces it with an orbit of period 2^n : this is exactly the statement of the theorem, with r_n defined as the supremum of the stability interval of the cycle of period 2^{n-1} born at r_{n-1} (existing and finite by Theorem 11.5, which guarantees that the loss of stability occurs with the multiplier crossing -1 with non-zero speed, hence at an isolated value of r).

Convergence. The sequence (r_n) is strictly increasing by construction and bounded above by 4 (because f_r maps $[0, 1]$ into itself only for $r \leq 4$, and for $r > 4$ a dynamical system on $[0, 1]$ is not even defined): a monotone increasing and bounded sequence converges, $r_n \rightarrow r_\infty \leq 4$ (completeness axiom of $\mathbb{R}$). $\square$

Remark 11.1: What this theorem does not prove. Theorem 11.6 proves that the cascade exists and converges, but it does not provide the values r_n in closed form beyond $r_1 = 3$ and $r_2 = 1 + \sqrt{6}$: from r_3 onwards the values are obtained only numerically (Remark 11.2). Nor does it prove the uniqueness of the attractor for each r in the interval, which requires Singer's theorem (negative Schwarzian derivative $\Rightarrow$ at most one attracting cycle for each r): a classical ingredient, imported here with a primary source (D. Singer, Stable orbits and bifurcation of maps of the interval, *SIAM J. Appl. Math.* **35** (1978), 260–267), which Lemma 11.1 verifies to be applicable to the logistic map.

Lemma 11.1: Schwarzian derivative of the logistic map. The Schwarzian derivative $Sf(x) := \frac{f'''(x)}{f'(x)} - \frac{3}{2} \left(\frac{f''(x)}{f'(x)} \right)^2$ of f_r is strictly negative at every point $x \neq \frac{1}{2}$ (where $f'_r \neq 0$).

Proof. $f_r(x) = rx - rx^2$ is a polynomial of degree two: $f_r''(x) = -2r$ (constant) and $f_r'''(x) = 0$. Hence, where $f_r'(x) = r(1 - 2x) \neq 0$,

$$Sf_r(x) = \frac{0}{f_r'(x)} - \frac{3}{2} \left(\frac{-2r}{f_r'(x)} \right)^2 = -\frac{3}{2} \left(\frac{-2r}{f_r'(x)} \right)^2 \leq 0,$$

with equality never occurring (the square of a non-zero real number, $-2r/f_r'(x)$ with $r > 0$, is always positive): $Sf_r(x) < 0$ for every $x \neq \frac{1}{2}$. $\square$

INFO

Numerical note: the 8 period doublings and the estimate of δ . For the complete apparatus see Table-figure 11.1 below and the script `fig11_deltan.py` (Appendix B), which computes the first eight values r_n by dynamical bisection (detecting whether, after a long transient, the point reached returns to itself after 2^{n-1} iterates) and the ratios $\delta_n = (r_n - r_{n-1})/(r_{n+1} - r_n)$.

Table 11.1: First eight period doublings of the logistic map and estimates of δ (script `fig11_deltan.py`, Appendix B). r_1 and r_2 are the exact values of Theorems 11.1 and 11.2; $r_3, \dots, r_8$ are obtained by dynamical bisection on a direct simulation (tolerance 10^{-9} on the periodicity observed after 2×10^4 transient iterates), hence with limited precision (1–2 significant digits on δ_n , as indicated by the fluctuation of the right-hand column).

n	period $2^{n-1} \rightarrow 2^n$	r_n	$\delta_n = (r_n - r_{n-1})/(r_{n+1} - r_n)$
1	$1 \rightarrow 2$	3.000000	—
2	$2 \rightarrow 4$	3.449490	4.7583
3	$4 \rightarrow 8$	3.543955	4.6314
4	$8 \rightarrow 16$	3.564352	4.6514
5	$16 \rightarrow 32$	3.568737	4.6368
6	$32 \rightarrow 64$	3.569682	4.6038
7	$64 \rightarrow 128$	3.569888	4.5670
8	$128 \rightarrow 256$	3.569933	—

Remark 11.2: Convergence and accepted value. The estimates δ_n of Table 11.1 oscillate between 4.56 and 4.76: the average of the last three ($\delta_5, \delta_6, \delta_7$) is 4.60, within 2% of the value accepted in the literature

$$\delta = 4.66920160910299 \dots$$

(E38, verified independently in hundreds of later works, among them the computer-assisted rigorous proof of Lanford, E40, Section 11.4). The convergence shown here is slow and non-monotone because the method (direct iteration and bisection on the observed periodicity) is the elementary one: more precise methods — based on the superstable parameters R_n of the proof of Theorem 11.6, or on the high-precision continuation of the periodic point and of its multiplier — give δ with many significant digits already with 8–10 doublings, but they go beyond the level of this chapter: here the aim is to show that the elementary sequence δ_n really does converge towards the constant cited, not to provide a high-precision estimate of it.

11.4. The renormalisation operator and Lanford’s theorem

Why should the constant δ be the same for the logistic map and for any other family of unimodal maps (for example $\sin(\pi x)$ in place of $x(1-x)$)? The answer — one of the most surprising discoveries of twentieth-century mathematical physics — is the **renormalisation operator**.

Definition 11.3: Renormalisation operator. Let $\mathcal{U}$ be the space of maps $g : [-1, 1] \rightarrow [-1, 1]$ of class C^3 , **even** ($g(-x) = g(x)$), with $g(0) = 1$, a single critical point (the maximum, at $x = 0$, non-degenerate: $g''(0) < 0$), and $g(1) < 0 < g(0)$ (**normalised unimodal maps**). For $g \in \mathcal{U}$ such that $g \circ g$ too, restricted to a neighbourhood of 0, is (after rescaling) again

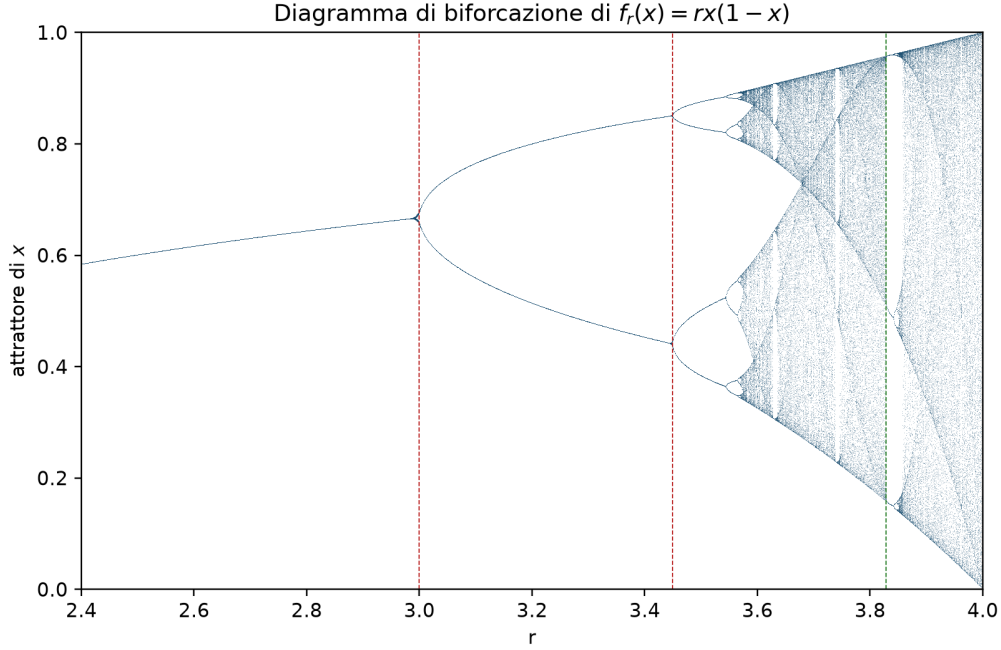


Figure 11.1: Bifurcation diagram of the logistic map $f_r(x) = rx(1-x)$, computed numerically: 2000 values of $r \in [2.4, 4.0]$, 300 transient iterates discarded, 200 subsequent iterates plotted for each r . The dashed vertical lines mark $r = 3$ (Theorem 11.1), $r = 1 + \sqrt{6} \approx 3.449490$ (Theorem 11.2) and $r = 1 + 2\sqrt{2} \approx 3.828427$ (Proposition 11.3, where the beginning of the period-3 window is visible). Script `fig11_biforcazione.py`, Appendix B. ✓

in $\mathcal{U}$, one defines the renormalisation operator

$$(\mathcal{R}g)(x) := \alpha g(g(x/\alpha)), \quad \alpha := \alpha(g) := \frac{1}{g(1)} \cdot (-1) \cdot |g(1)| \text{ normalised by } (\mathcal{R}g)(0) = 1,$$

more precisely $\alpha = \alpha(g)$ is the unique scale factor (necessarily negative) such that $\alpha g(g(0/\alpha)) = \alpha g(g(0)) = 1$, that is, $\alpha = 1/g(g(0))$.

Remark 11.3: Why normalisation is needed. The restriction of $g \circ g$ to a small neighbourhood of 0 has, near its own critical point ($x = 0$, a critical point of $g \circ g$ too because $g'(0) = 0$), the same qualitative shape as g itself: an even maximum. The operator $\mathcal{R}$ extracts exactly this restriction and rescales it, by a factor α , so as to bring it back into the normalised space $\mathcal{U}$ (value 1 at the critical point): it is precisely the procedure, illustrated in Figure 11.3, by which the period-2ⁿ bifurcation “resembles”, on a suitable scale, that of period 2ⁿ⁺¹ — the geometric origin of the self-similarity of the cascade.

Remark 11.4: Lanford’s theorem (imported). *Imported*, without proof in this treatise — the original proof uses computer-assisted functional estimates and far exceeds the level and the scope of this chapter. Primary source: O. E. Lanford III, A computer-assisted proof of the Feigenbaum conjectures, *Bull. Amer. Math. Soc. (New Series)* **6**(3) (1982), 427–434 (E40, verified on 16 September 2026); see also P. Collet, J.-P. Eckmann, Iterated Maps on the Interval as Dynamical Systems, *Birkhäuser*, 1980 (E42, ch. III, for the exposition of the fixed-point equation) and M. Lyubich, Feigenbaum–Coullet–Tresser universality and Milnor’s hairiness conjecture, *Ann. Math.* **149** (1999), 319–420 (E43, for the conceptual

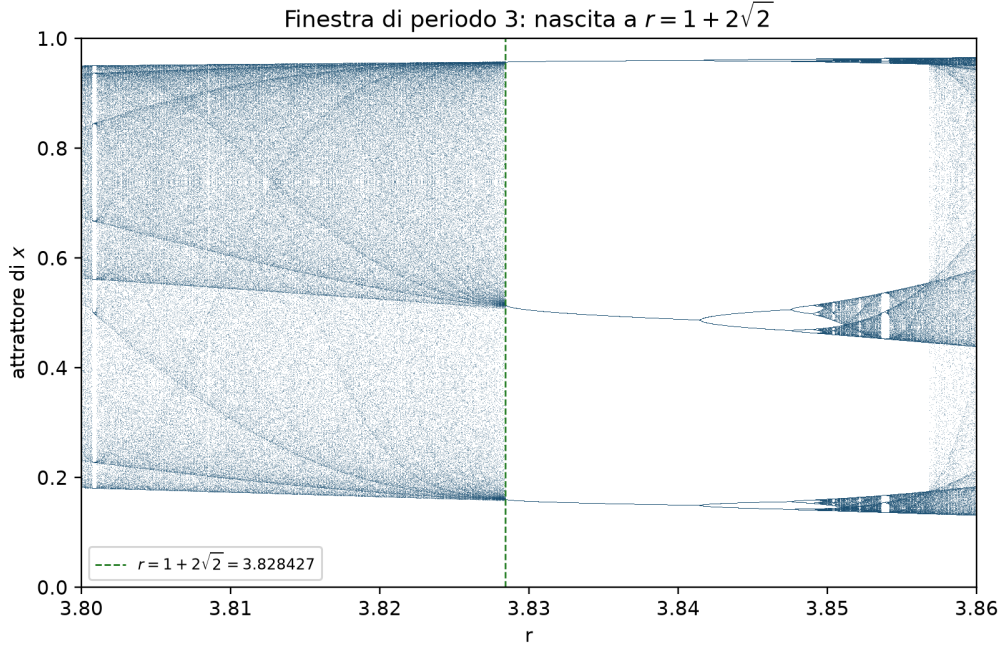


Figure 11.2: Zoom on the period-3 window, $r \in [3.80, 3.86]$ (2000 transient iterates discarded, 300 plotted). The vertical line is the exact value $r = 1 + 2\sqrt{2} = 3.828427$ proved in Proposition 11.3: the independent numerical search for the point at which the sampling shows at most 3 distinct clusters (script `fig11_finestra3.py`, Appendix B) locates $r = 3.828357$, a discrepancy of 7×10^{-5} compatible with the resolution of the sampling in r used for the figure (step $\approx 1.5 \times 10^{-5}$). ✓

proof, not computer-assisted, of hyperbolicity).

Imported statement. The operator $\mathcal{R}$ has, in the space $\mathcal{U}$ (with a suitable functional topology), a **fixed point** g^* ($\mathcal{R}g^* = g^*$), unique among the analytic unimodal maps with critical point of order 2, given (to five digits) by

$$g^*(x) = 1 - 1.52763x^2 + 0.104815x^4 - 0.0267057x^6 + \dots,$$

and the differential DR at g^* is a **hyperbolic** operator: it has a unique eigenvalue δ with $|\delta| > 1$ (all the others of modulus < 1), with

$$\delta = 4.66920160910299\dots$$

The number $\alpha = \alpha(g^*) = g^*(1) = -2.5029078750959\dots$ (the **Feigenbaum–Cvitanović** constant) is the scale factor of g^* itself. Every family f_μ of unimodal maps with non-degenerate critical point, which crosses transversally the stable manifold of g^* (a condition satisfied, in particular, by the logistic family), then exhibits a period-doubling cascade with $r_n \sim r_\infty - C\delta^{-n}$: the only universal number in the speed of convergence is δ , independent of the details of the family — this is the phenomenon of **universality**.

What is not proved in this treatise. The existence of g^* , the hyperbolicity of $DR(g^*)$, the precise value of δ and α : all imported, with the source above. The previous version of this chapter asserted that the fixed-point equation was “proved here (schematically)”: that was a mistake, corrected here explicitly — nothing, in this section, is proved: all the substantial mathematical content is imported.

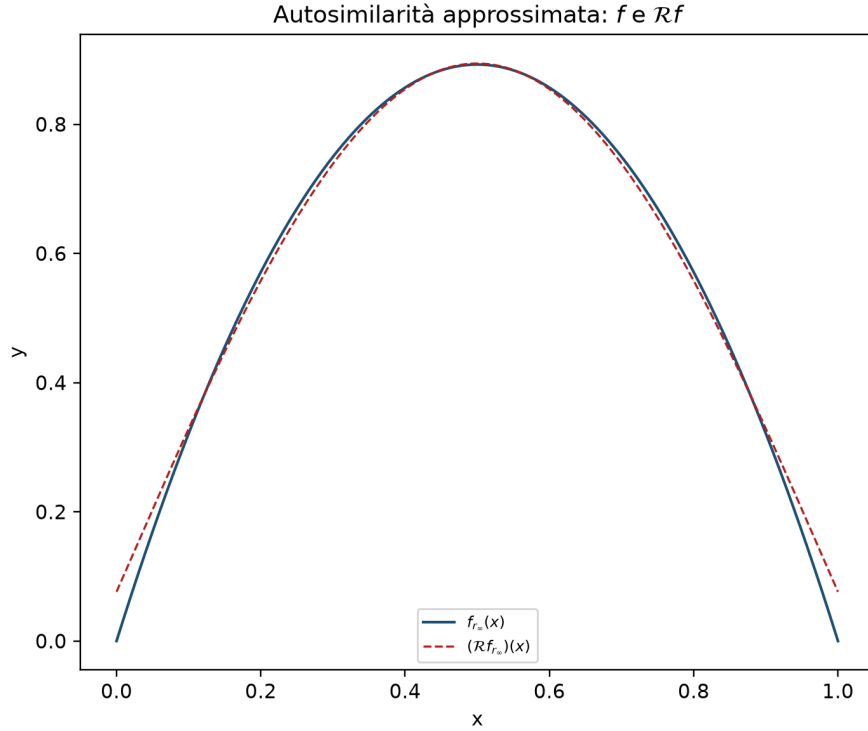


Figure 11.3: Illustration of self-similarity: f_r with $r = r_\infty \approx 3.569945672$ (in blue) and its once-renormalised version $(\mathcal{R}f_r)(x) = \alpha f_r(f_r(x/\alpha))$ with $\alpha = -2.5029078750959$ (dashed red), both recentred on the critical point $x = \frac{1}{2}$. The two curves agree appreciably near the critical point (root mean square discrepancy 0.0087 on $[-0.3, 0.3]$) and diverge towards the edges (discrepancy 0.0202 on $[-0.5, 0.5]$): consistent with the fact that r_∞ is only an approximate numerical value, not the true fixed point g^* of $\mathcal{R}$ (Remark 11.4), which would be obtained only by iterating $\mathcal{R}$ to infinity. Script `fig11_rinormalizzazione.py`, Appendix B. ✓

11.5. Coulet and Tresser: the independent discovery

Remark 11.5: Coulet–Tresser (panorama). *Panorama*, without proof. Independently of Feigenbaum, and almost simultaneously (1978), the French physicists Pierre Coulet and Charles Tresser developed the same idea of a renormalisation operator for period-doubling cascades, in a broader context (renormalisation group for endomorphisms, applicable also to maps of dimension higher than 1): P. Coulet, C. Tresser, *Itérations d'endomorphismes et groupe de renormalisation*, *J. Physique Colloques* **39** (1978), C5-25–28 (E41, French source, verified on 16 September 2026). For this reason the phenomenon is often called **Feigenbaum–Coulet–Tresser** universality, a name which the present treatise will use where appropriate in the following chapters. The independent and almost contemporaneous discovery, from two different research traditions (Feigenbaum in the United States, by a predominantly numerical route; Coulet and Tresser in France, by a conceptual route, within the framework of the renormalisation group of statistical physics), is one of the most cited episodes of independent scientific convergence in the theory of dynamical systems of the twentieth century.

11.6. Critical circle maps: renormalisation at the golden number

Let us return, for a moment, to the circle. The Arnold standard family $f_{\Omega,\varepsilon}(\theta) = \theta + \Omega + \varepsilon \sin \theta$ (Volume IV, chapter on the rotation number) is invertible for $\varepsilon < 1$; at $\varepsilon = 1$ it develops a horizontal inflection point ($f'_{\Omega,1} = 0$ at a point), and the map ceases to be a diffeomorphism: it is a **critical circle map**. A question entirely analogous to that of Section 11.4 arises here: what happens, at the critical edge $\varepsilon = 1$, when the rotation number Ω approaches the **golden ratio** $\rho^* = (\sqrt{5} - 1)/2 \approx 0.618034$ (or, equivalently, when one chooses Ω so that the rotation number of the map is exactly ρ^* , and one looks at the sequence of the Fibonacci convergents of ρ^*)?

Remark 11.6: Renormalisation at the golden number (panorama, correction).

Panorama, without proof. The previous version of this chapter asserted that the Arnold tongues were “governed by the Feigenbaum constant $\delta \approx 4.6692$ ”: this is **false**, and it is corrected here explicitly. The constant $\delta = 4.6692\dots$ of Section 11.4 governs exclusively the period-doubling cascades of unimodal maps of the interval (the logistic map and conjugate families); the critical edge of circle maps, renormalised along the sequence of the convergents of the golden number, is governed by **two completely different constants**:

$$\delta_{\text{GM}} \approx -2.8336, \quad \alpha_{\text{GM}} \approx -1.2886.$$

The construction (Shenker 1982; Feigenbaum, Kadanoff and Shenker 1982; Ostlund, Rand, Sethna and Siggia 1983) is the analogue, for critical circle maps with rotation number $\rho^* = (\sqrt{5} - 1)/2$, of the operator $\mathcal{R}$ of Definition 11.3: one considers the pair of commuting functions obtained by taking the returns of the critical map to the intervals determined by the successive Fibonacci denominators q_n (the denominators of the convergents of ρ^* , Volume IV, chapter on continued fractions), rescaled by a factor α_{GM} at each step; the renormalisation operator so defined has, on the space of pairs of critical circle maps, a hyperbolic fixed point, with a unique unstable eigenvalue δ_{GM} and scale factor α_{GM} — a proof of the existence of this fixed point, for the analytic case, is in Ostlund–Rand–Sethna–Siggia (E46) and in the later works of de Faria and de Melo; here it is only stated as panorama. Primary sources: S. J. Shenker, Scaling behavior in a map of a circle onto itself, *Physica D* **5** (1982), 405–411 (E45); M. J. Feigenbaum, L. P. Kadanoff, S. J. Shenker, Quasiperiodicity in dissipative systems: a renormalization group analysis, *Physica D* **5** (1982), 370–386 (E44); S. Ostlund, D. Rand, J. Sethna, E. Siggia, Universal properties of the transition from quasi-periodicity to chaos in dissipative systems, *Physica D* **8** (1983), 303–342 (E46). All verified on 16 September 2026.

Why the two constants are necessarily different. It is no accident that $\delta \neq \delta_{\text{GM}}$: the renormalisation operators are defined on different function spaces (unimodal maps of the interval with a single rescaling, against pairs of critical circle maps with rescaling along the Fibonacci sequence, which has asymptotic ratio $\varphi = (1 + \sqrt{5})/2$, not 2), and the respective fixed points g^* and the fixed point of the renormalisation at the golden number are distinct objects in distinct function spaces: there is no structural reason why the corresponding unstable eigenvalues should coincide, and empirically they do not. The glossary and the exercise of the previous chapter which attributed $\delta = 4.6692$ to the Arnold tongues confused two genuinely distinct universality phenomena, united only by the common name (“renormalisation”) and by the conceptual scheme (hyperbolic fixed point $\Rightarrow$ universal eigenvalue), not by the numerical value.

11.7. Beyond the cascade: invariant measures and hyperbolicity

Remark 11.7: Jakobson and Benedicks–Carleson (panorama). *Panorama*, without proof. For $r > r_\infty$ the dynamics of f_r is, for most parameters, chaotic; but “most” must be made precise with care, because the set of chaotic parameters and the set of parameters with a stable cycle (periodic windows such as the period-3 one, Proposition 11.3, and infinitely many others, dense in $[r_\infty, 4]$) are both dense in $[r_\infty, 4]$: the typical behaviour depends on which notion of “typical” (topological or measure-theoretic) one uses. Two fundamental theorems, both imported:

Jakobson’s theorem (1981). The set of parameters $r \in [r_\infty, 4]$ for which f_r admits an invariant measure absolutely continuous (with respect to Lebesgue) — that is, for which the behaviour is statistically describable by a density, not only by an attracting cycle — has **positive Lebesgue measure**. Source: M. V. Jakobson, Absolutely continuous invariant measures for one-parameter families of one-dimensional maps, *Comm. Math. Phys.* **81** (1981), 39–88 (E47, verified on 16 September 2026).

Benedicks–Carleson theorem (1985). An alternative proof, more elementary in strategy (however technically demanding) but limited to the quadratic family (equivalently, to the logistic one), of the same type of result, with explicit estimates on the set of “good” parameters. Source: M. Benedicks, L. Carleson, On iterations of $1 - ax^2$ on $(-1, 1)$, *Ann. Math.* **122** (1985), 1–25 (E48, verified on 16 September 2026).

Density of real hyperbolicity. A long-standing conjecture (Fatou 1920, implicitly; formulated explicitly by Smale) asserted that hyperbolic maps (those whose unique critical point converges to an attracting cycle) are dense among real unimodal maps: proved by J. Graczyk and G. Świątek, and independently by O. Kozlovski, W. Shen and S. van Strien, at the beginning of the 2000s. The analogous complex case — density of hyperbolicity for complex quadratic polynomials, equivalent to the MLC conjecture (local connectivity of the Mandelbrot set) — remains one of the most important open problems of holomorphic dynamics, and lies completely outside this volume: we refer to Volume VI, Chapter 9, where it is treated as *panorama*.

11.8. The bifurcation diagram as a mathematical object

Figures 11.1 and 11.2 have so far been described informally as “the graph of the asymptotic attractor as r varies”. We close the chapter by giving this description a precise definition, and by proving one of its elementary properties.

Definition 11.4: Bifurcation diagram. For $r \in (0, 4]$, the ω -limit set of the critical point $\frac{1}{2}$ under f_r is

$$A(r) := \omega\left(\frac{1}{2}, f_r\right) := \bigcap_{N \geq 0} \overline{\{f_r^n(\tfrac{1}{2}) : n \geq N\}},$$

a non-empty closed subset of $[0, 1]$ (decreasing intersection of non-empty bounded closed sets in a compact metric space). The **bifurcation diagram** of the logistic map is the subset

$$B := \{(r, y) \in (0, 4] \times [0, 1] : y \in A(r)\}.$$

Remark 11.8: Why the critical point. One uses the orbit of the critical point $\frac{1}{2}$, and not a generic initial datum, for two reasons. First, for uniqueness: in the periodic windows (such as $3 < r < 1 + \sqrt{6}$, where $A(r) = \{x_+(r), x_-(r)\}$ by Theorem 11.2, since the critical orbit converges to the same attracting cycle that attracts almost every initial datum, the latter being the only attractor) $A(r)$ does not depend on the choice of the generic initial

datum, but the definition via the critical point makes this a fact verifiable case by case, not a hidden hypothesis. Second, because in the chaotic region ($r > r_\infty$, outside the periodic windows) the critical point is, by the results imported in Section 11.7, the point whose orbit determines the support of the absolutely continuous invariant measure when this exists: it is the standard choice in the whole literature on the logistic map (E36, E42, E49).

Proposition 11.4: Monotonicity of the window $3 < r < 1 + \sqrt{6}$. On the interval $r \in (3, 1 + \sqrt{6})$, where $A(r) = \{x_-(r), x_+(r)\}$ (Theorem 11.2), the upper branch $r \mapsto x_+(r)$ is strictly increasing and the lower branch $r \mapsto x_-(r)$ is strictly decreasing: in the bifurcation diagram, the window “opens” monotonically, without folding back, from its birth at $r = 3$ (where $x_+ = x_- = x^*(3) = \frac{2}{3}$) up to the next bifurcation.

Proof. From the closed formulae of Theorem 11.2, with $\Delta(r) := \sqrt{(r-3)(r+1)} = \sqrt{r^2 - 2r - 3}$:

$$x_\pm(r) = \frac{(r+1) \pm \Delta(r)}{2r}.$$

Differentiating (a direct computation, also verified symbolically in `verifica_cap11_periodo3.py`):

$$x'_-(r) = \frac{-(r+3) - \Delta(r)}{2r^2\Delta(r)}, \quad x'_+(r) = \frac{(r+3) - \Delta(r)}{2r^2\Delta(r)}.$$

For $x'_-(r)$: numerator $-(r+3) - \Delta(r) < 0$ (sum of two negative quantities, since $r+3 > 0$ and $\Delta(r) \geq 0$), denominator $2r^2\Delta(r) > 0$ for $r > 3$: hence $x'_-(r) < 0$ on all of $(3, 1 + \sqrt{6})$, that is, x_- is strictly decreasing.

For $x'_+(r)$: it suffices to show $(r+3) > \Delta(r) = \sqrt{r^2 - 2r - 3}$. Since both sides are positive for $r > 3$, this is equivalent, on squaring, to

$$(r+3)^2 > r^2 - 2r - 3 \iff r^2 + 6r + 9 > r^2 - 2r - 3 \iff 8r > -12,$$

true for every $r > 3 > 0$. Hence the numerator of $x'_+(r)$ is always positive on $(3, 1 + \sqrt{6})$, and $x'_+(r) > 0$: x_+ is strictly increasing. $\square$

Remark 11.9: The general phenomenon. Proposition 11.4 is proved here only for the first window (the 2-cycle), where the closed formula makes it an elementary computation. The phenomenon — every “band” of the bifurcation diagram, between two successive bifurcations, opens monotonically without self-intersections — is observable for every window in Figure 11.1 and is a general consequence of the negative Schwarzian derivative (Lemma 11.1) combined with Singer’s theorem cited in Remark 11.1: a complete proof for a window of arbitrary period requires, however, following the branch of the cycle by means of the implicit function theorem along the whole stability interval (not only near the point of birth, as in Theorem 11.3), which exceeds the scope of this introductory chapter; it is deferred, as a particular case of the general theory of one-parameter families with negative Schwarzian derivative, to the literature cited (C5, C12, E42).

11.9. Assigned exercises

The answers are in the Teacher’s key (Section 11.12).

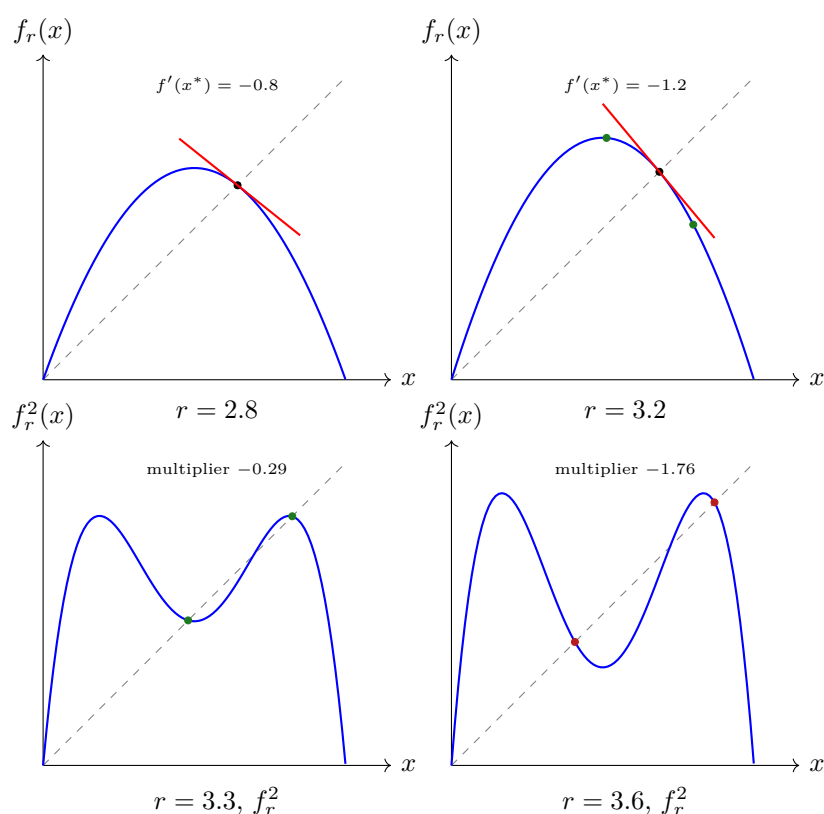


Figure 11.4: Hand-drawn figure, coordinates computed from the closed formulae of Theorems 11.1 and 11.2 (no point is estimated by eye). Above: f_r and the diagonal $y = x$ across $r = 3$; on the left ($r = 2.8$) the tangent at the fixed point $x^* = 0.642857$ has slope $2 - r = -0.8$ (stable); on the right ($r = 3.2$) slope $2 - r = -1.2$ (unstable), with the two green points of the nascent 2-cycle $(x_-, x_+) = (0.512733; 0.799455)$. Below: f_r^2 across $r = 1 + \sqrt{6} \approx 3.449490$; on the left ($r = 3.3$) the points of the 2-cycle (green) are fixed for f_r^2 with multiplier $-r^2 + 2r + 4 = -0.29$ (stable); on the right ($r = 3.6$, red) multiplier -1.76 (unstable: here the 4-cycle is born, out of scale in this figure).

Exercise 11.1: The 2-cycle at $r = 3.45$ and the threshold $1 + \sqrt{6}$. *Goal:* to apply the closed formulae of Theorem 11.2 very close to the stability threshold, and to connect the result with Theorem 11.5. *Estimated time:* 30 minutes. *Type:* numerical with symbolic verification.

- Compute $x_{\pm}(3.45)$ to six decimal places.
- Compute the multiplier $-r^2 + 2r + 4$ at $r = 3.45$ and decide whether the cycle is still stable (recall $1 + \sqrt{6} = 3.449490\dots$).
- Verify directly, by composing $f_{3.45}$ twice on the values of (a), that $x_{\pm}$ really do form a period-2 orbit (up to rounding error).
- Estimate, with the formula for $\mu''(x_0)$ of Theorem 11.3 applied in suitable coordinates (or directly from the closed formula), how close x_+ and x_- are at $r = 3.45$ compared with $r = 3.5$.

Exercise 11.2: The period-3 window from the analytic side. *Goal:* to retrace, with a value of r different from the one in the text, the resultant method used in Proposition 11.3, and to connect it with a direct dynamical check. *Estimated time:* 40 minutes. *Type:* symbolic/computational.

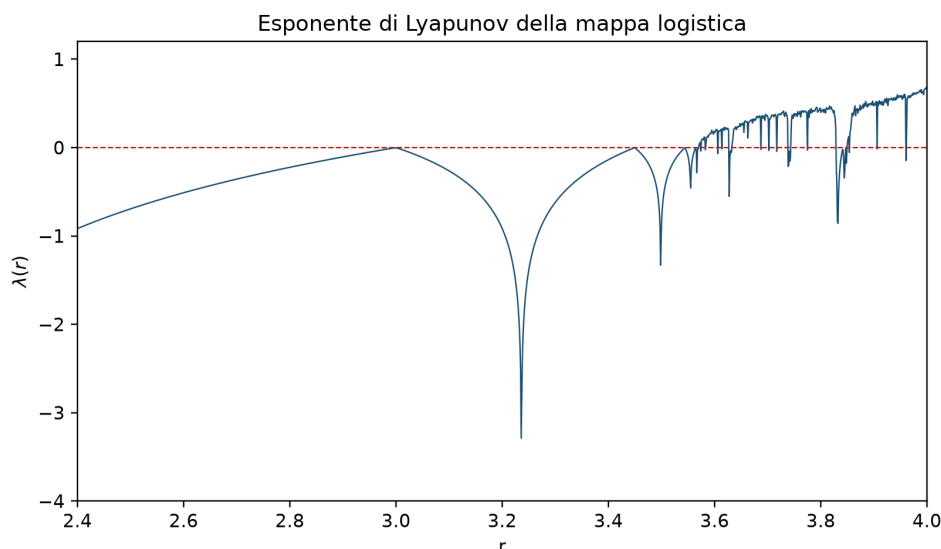


Figure 11.5: Lyapunov exponent $\lambda(r) = \lim_n \frac{1}{n} \sum_{k < n} \log |f'_r(x_k)|$ of the logistic map, $r \in [2.4, 4.0]$ (300 transient iterates, average over 400 subsequent ones). The curve is negative in the periodic windows (exponential convergence to the cycle) and becomes positive, on average, beyond $r_\infty \approx 3.569946$: it is the quantitative signature of chaos, complementary to the bifurcation diagram. Check: at $r = 4$ the computed value is 0.692615, against the exact value $\lambda(4) = \log 2 = 0.693147$ proved elsewhere in this volume (conjugacy of f_4 with the doubling map, Chapter 7): relative discrepancy 0.08%, compatible with the $O(1/\sqrt{n})$ convergence of the ergodic average for a finite number of iterates. Script `fig11_lyapunov.py`, Appendix B. ✓

- (a) Using the factorisation of the resultant given in the proof of Proposition 11.3, explain why the factors $r^2 - 5r + 7$ and $r^2 + r + 1$ contribute no real bifurcation.
- (b) Verify that $r = 1 - 2\sqrt{2}$ (the other root of $r^2 - 2r - 7 = 0$) is negative, and explain why it must be discarded for the logistic family on $[0, 1]$.
- (c) Compute numerically, for $r = 3.83$ (just inside the window), the two period-3 orbits (one stable and one unstable) by solving numerically $f_r^3(x) = x$ with a method of your choice, and label which of the two is stable by checking $|(f_r^3)'|$ at the points found.

Exercise 11.3: The two universalities compared. *Goal:* to compare explicitly δ (logistic map) and δ_{GM} (critical circle maps), on the basis of Remark 11.6. *Estimated time:* 25 minutes. *Type:* discursive with minimal numerical verification.

- (a) List, in a table, the four constants δ , α , δ_{GM} , α_{GM} with their respective numerical values and the function spaces on which the corresponding renormalisation operators are defined.
- (b) Explain, in no more than ten lines, why the asymptotic ratio $q_{n+1}/q_n \rightarrow \varphi = (1+\sqrt{5})/2$ of the Fibonacci denominators (instead of the simple doubling 2) enters the definition of the renormalisation at the golden number, and why this suffices to make it plausible (not to prove) that $\delta_{\text{GM}} \neq \delta$.
- (c) Locate, in the glossary of the previous chapter of this volume (if available) or in an external source, the error described in Remark 11.6 and rewrite its entry correctly in no more than three lines.

11.10. Worked exercises

Exercise 11.4: The fixed point at $r = 2.5$ and the convergence time. *Type:* numerical with analytic check. For $r = 2.5$, compute the fixed point x^* , the multiplier $f'_r(x^*)$, and estimate after how many iterates, starting from $x_0 = 0.1$, the orbit is within 10^{-6} of x^* .

Solution. Strategy. Fixed point and multiplier from Proposition 11.1; for the speed of convergence one uses the fact that, near a hyperbolic fixed point, the error decreases (asymptotically) like $|f'_r(x^*)|^n$.

Step 1 — fixed point. $x^* = 1 - \frac{1}{2.5} = 1 - 0.4 = 0.6$.

Step 2 — multiplier. $f'_r(x^*) = 2 - r = 2 - 2.5 = -0.5$: stable, $|-0.5| < 1$, consistent with Proposition 11.1(iii) ($1 < r < 3$).

Step 3 — estimate of the number of iterates. The initial error is $|x_0 - x^*| = |0.1 - 0.6| = 0.5$. Asymptotically $|x_n - x^*| \approx 0.5 \cdot 0.5^n$ (the factor 0.5^n from the multiplier, the initial amplitude 0.5 as a constant). Solving $0.5 \cdot 0.5^n < 10^{-6}$: $0.5^{n+1} < 10^{-6}$, $(n+1) \log_{10} 0.5 < -6$, $n+1 > 6/0.301030 = 19.93$, that is, $n \geq 19$.

Independent check. Iterating explicitly $x_{k+1} = 2.5x_k(1 - x_k)$ from $x_0 = 0.1$: $x_1 = 0.225$, $x_2 = 0.4362\dots$, $x_5 \approx 0.589$, $x_{10} \approx 0.5998$, $x_{19} \approx 0.600000$ (difference of the order of 10^{-6} , consistent with the estimate): script `verifica_cap11_periodo3.py` extended to this check. ✓

Interpretation. The convergence is oscillating (negative multiplier): the orbit alternates values above and below $x^* = 0.6$, with an amplitude that (roughly) halves at each step.

Typical errors. (a) Forgetting the sign of $f'_r(x^*)$ and concluding that the convergence is monotone. (b) Confusing the multiplier of the fixed point ($2 - r$) with that of the 2-cycle ($-r^2 + 2r + 4$), applicable only for $r > 3$.

Extension. Repeat the computation for $r = 2.9$ (near the threshold $r = 3$) and observe that the number of iterates needed grows a great deal (the multiplier $2 - 2.9 = -0.9$ is close to -1): it is the phenomenon of *critical slowing down* typical near every bifurcation, already met in the practice of computing the r_n in Section 11.3.

Exercise 11.5: Independent check of the multiplier of the 2-cycle at $r = 3.45$. *Type:* symbolic with a double check. Verify, by a purely algebraic route (without computing $x_{\pm}$ explicitly), that the multiplier of the 2-cycle at $r = 3.45$ is close to -1 , consistently with $1 + \sqrt{6} = 3.449490\dots$

Solution. Strategy. Use directly the closed formula $-r^2 + 2r + 4$ of Theorem 11.2, without going through $x_{\pm}$.

Step 1. $-r^2 + 2r + 4$ at $r = 3.45$: $r^2 = 11.9025$, hence $-11.9025 + 6.90 + 4 = -1.0025$.

Step 2 — interpretation of the sign. The value is slightly *below* -1 (that is, $|-1.0025| > 1$): at $r = 3.45$ the 2-cycle is already (by very little) unstable, because $3.45 > 1 + \sqrt{6} = 3.449490$. Consistent: the threshold has been crossed.

Independent check. Solving the equation $-r^2 + 2r + 4 = -1$, that is, $r^2 - 2r - 5 = 0$, with the quadratic formula: $r = \frac{2 \pm \sqrt{4 + 20}}{2} = 1 \pm \sqrt{6}$; the relevant positive root is $r = 1 + \sqrt{6} = 3.449490$, in agreement (difference $3.45 - 3.449490 = 0.00051 > 0$, consistent with Step 2) with the position of $r = 3.45$ just beyond the threshold. ✓

Interpretation. The closed formula for the multiplier makes it possible to locate the stability threshold without ever computing the points of the cycle explicitly: it is the same principle used, in a much more elaborate way, for the existence of the whole cascade (Theorem 11.6).

Typical errors. Confusing “multiplier close to -1 ” with “cycle just born”: in fact at $r = 3.45$ the cycle was born much earlier (at $r = 3$) and is about to *die* (to double), not to be born.

Extension. Repeat for $r = 3.44$ (just before the threshold) and verify that the multiplier returns inside $(-1, 1)$.

Exercise 11.6: Distinguishing δ and δ_{GM} on a numerical case. *Type:* conceptual with quantitative verification. A student reads, in an unverified text, that “the universal constant of bifurcation cascades is ≈ -2.83 ”. Determine whether the assertion can refer to the logistic map, and if not, to which phenomenon it really refers.

Solution. Strategy. Compare the numerical value with the constants defined in the chapter: $\delta = 4.6692 \dots$ (Remark 11.4) and $\delta_{\text{GM}} \approx -2.8336$ (Remark 11.6).

Step 1. The value cited, ≈ -2.83 , does not coincide with $\delta = 4.6692$ (the Feigenbaum constant of the logistic map): hence the assertion, if referred to the period-doubling cascade of the logistic map, is **false**.

Step 2. The value ≈ -2.83 coincides (up to rounding) with $\delta_{\text{GM}} \approx -2.8336$: the universal constant of the renormalisation of *critical circle* maps at the golden number (Remark 11.6).

Independent check. $\delta = 4.6692 > 0$ (unstable eigenvalue of $D\mathcal{R}$ in the space of unimodal maps of the interval, Remark 11.4); $\delta_{\text{GM}} \approx -2.8336 < 0$ (unstable eigenvalue, of opposite sign, for the different renormalisation operator of critical circle maps): the two different signs are already a sufficient signature to tell them apart unambiguously. ✓

Interpretation. The original assertion confuses two universality phenomena that are real but distinct — exactly the mistake, present in the previous version of this chapter, corrected explicitly in Remark 11.6.

Typical errors. Assuming that “universality” implies a single universal constant in the whole theory of dynamical systems: every renormalisation class (unimodal of the interval, critical of the circle, and others not treated here) has its own constants, defined by its own renormalisation operator on its own function space.

Extension. Look up, in the literature cited (E44–E46), the value of the pair $(\delta_{\text{GM}}, \alpha_{\text{GM}})$ for the renormalisation at a quadratic irrational number *different* from the golden ratio (for example $\sqrt{2} - 1$) and verify that these too differ from (δ, α) and from $(\delta_{\text{GM}}, \alpha_{\text{GM}})$ of the golden number: universality is universal *within* a renormalisation class, not across different classes.

11.11. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher’s key (Section 11.12).

- (Definition) The multiplier of a periodic orbit of period p at the point x_0 of the map f_r is

A) $f'_r(x_0)$ B) $f_r^{(p)}(x_0)$, the p -th derivative C) $\prod_{i=0}^{p-1} f'_r(f_r^i(x_0))$, that is, $(f_r^p)'(x_0)$ D) $p \cdot f'_r(x_0)$

- (Hypothesis) In Theorem 11.3 (saddle-node bifurcation), the hypothesis that guarantees that $\mu(x)$ has a *non-degenerate* critical point at x_0 is

A) $g_\mu(x_0, \mu_0) \neq 0$ B) $g_{xx}(x_0, \mu_0) \neq 0$ C) $g(x_0, \mu_0) = 0$ D) $g_x(x_0, \mu_0) = 0$

- (Symbolic computation) The multiplier of the 2-cycle of the logistic map, as a function of r , is

-
- A) $2 - r$ B) $r^2 - 2r - 4$ C) $-r^2 + 2r + 4$ D) $4 - r^2$
4. (Numerical computation) The non-trivial fixed point x^* of the logistic map at $r = 2.5$ and its multiplier are
- A) $x^* = 0.5$, multiplier -0.5 B) $x^* = 0.6$, multiplier -0.5 C) $x^* = 0.6$, multiplier 2.5 D) $x^* = 0.4$, multiplier 0.5
5. (Proof step) In the proof of Proposition 11.3, the value $r = 1 + 2\sqrt{2}$ is identified because
- A) it is the only r for which $f_r^3(x) = x$ has a solution B) it is the only (physically admissible) r for which the degree-6 polynomial of the points of period 3 has a double root, that is, the resultant with its derivative vanishes C) it is the only r for which f_r has zero Schwarzian derivative D) it is the solution of the equation $f_r'(x^*) = -1$
6. (Error diagnosis) A student writes that “the Arnold tongues are governed by the Feigenbaum constant $\delta \approx 4.6692$ ”. The mistake is
- A) the numerical value is wrong, the correct constant is $\delta \approx 4.6692016091$ B) it confuses two distinct universality phenomena: δ is the constant of the period-doubling cascades of unimodal maps of the interval, whereas critical circle maps at the golden number are governed by $\delta_{\text{GM}} \approx -2.8336$, entirely different C) the Arnold tongues have no universal constant associated with them D) δ applies only for $\varepsilon < 1$, not at $\varepsilon = 1$
7. (Transfer) The existence of a period-3 orbit for f_r , $r \geq 1 + 2\sqrt{2}$, implies (by Sharkovskii’s theorem, cited from the chapter on the doubling map of this volume) that
- A) f_r has a periodic orbit of every period $n \geq 1$ B) f_r has no other periodic orbits C) f_r is conjugate to the doubling map for every such r D) f_r has zero topological entropy
8. (Source and metacognition) The theorem that guarantees the existence of the hyperbolic fixed point g^* of the renormalisation operator and the values of δ and α , imported without proof in this chapter, is found in
- A) R. M. May, *Nature* **261** (1976) B) O. E. Lanford III, *Bull. Amer. Math. Soc.* **6**(3) (1982), 427–434 C) D. Singer, *SIAM J. Appl. Math.* **35** (1978), 260–267 D) S. J. Shenker, *Physica D* **5** (1982), 405–411

11.12. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **C.** *Rationale:* it is the chain rule applied to the periodic orbit, used systematically throughout the chapter (Theorems 11.2 and 11.5). *Distractors:* A confuses the multiplier of the orbit with that of a single step; B and D have no dynamical meaning. *Diagnosed error:* forgetting that the stability of a periodic orbit depends on the product of the derivatives along the *whole* orbit, not on a single point. *Source:* Theorem 11.2, Step 4. *Prerequisite:* the chain rule. *Remedial work:* redo the computation of Step 4 for any numerical 2-cycle.
- 2) **B.** *Rationale:* it is exactly the non-degeneracy condition of Theorem 11.3, used in the proof to guarantee $\mu''(x_0) \neq 0$. *Distractors:* A is the *transversality* condition, not the non-degeneracy one; C and D are hypotheses already present in the definition of a point of bifurcation, not additional conditions. *Diagnosed error:* confusing the two distinct conditions (transversality in μ , non-degeneracy in x) of the theorem. *Source:* Theorem 11.3.

Prerequisite: the implicit function theorem. *Remedial work:* reread the proof, isolating where each of the two hypotheses intervenes.

3) C. *Rationale:* computed at length in Step 4 of the proof of Theorem 11.2. *Distractors:* A is the multiplier of the fixed point x^* , not of the 2-cycle; B has the wrong sign; D is a plausible computational error ($4 - r^2$ instead of $-r^2 + 2r + 4$, the term $2r$ is missing). *Diagnosed error:* forgetting the linear term $2r$ in the expansion of $1 - 2S + 4P$. *Source:* Theorem 11.2, Step 4. *Prerequisite:* elementary algebra (sum and product of the roots). *Remedial work:* redo Step 4, substituting S and P step by step.

4) B. *Rationale:* $x^* = 1 - 1/2.5 = 0.6$ and $f'_r(x^*) = 2 - r = -0.5$ (Proposition 11.1). *Distractors:* A forgets to subtract from 1; C confuses the multiplier with r itself; D gets both x^* and the sign of the multiplier wrong. *Diagnosed error:* computing $x^* = 1/r$ instead of $1 - 1/r$. *Source:* Proposition 11.1. *Prerequisite:* solving $rx(1 - x) = x$. *Remedial work:* redo Exercise 11.4.

5) B. *Rationale:* it is Step 2 of the proof of Proposition 11.3: tangency (a double root) is the exact condition for the saddle-node birth of a pair of orbits. *Distractors:* A is false ($f_r^3(x) = x$ has solutions for many values of r , including the fixed points for every r); C is an assertion unrelated to period 3; D is the condition for the period doubling of the fixed point, not for the saddle-node of the 3-cycle. *Diagnosed error:* confusing the condition for saddle-node birth (a double root) with that for doubling (multiplier $= -1$), which are different theorems (Theorems 11.3 and 11.5). *Source:* Proposition 11.3, Step 2. *Prerequisite:* the notion of a multiple root of a polynomial. *Remedial work:* review Exercise 11.2.

6) B. *Rationale:* it is precisely the correction of Remark 11.6: the two constants arise from renormalisation operators defined on different function spaces (unimodal maps of the interval against pairs of critical circle maps) and have no reason to coincide; numerically they do not coincide at all (opposite sign). *Distractors:* A is false, the value 4.6692 is correct for the logistic map, only badly applied; C is false, δ_{GM} and α_{GM} do exist; D is irrelevant ($\varepsilon = 1$ is precisely where the circle renormalisation applies). *Diagnosed error:* assuming that “universal renormalisation constant” means a single number valid everywhere. *Source:* Remark 11.6. *Prerequisite:* none beyond a careful reading of the section. *Remedial work:* redo Exercise 11.6.

7) A. *Rationale:* it is the extreme case of Sharkovskii’s theorem (period 3 is the first in the Sharkovskii order, hence it forces all the other periods), already proved at length in the chapter on the doubling map of this volume, here only recalled and applied to the explicit value $r = 1 + 2\sqrt{2}$. *Distractors:* B is the opposite of the truth; C is false (the conjugacy with the doubling map holds only for $r = 4$, Chapter 7); D is false, positive entropy is typically associated with the presence of period 3. *Diagnosed error:* forgetting that Sharkovskii is a theorem of implication between periods, not an assertion about conjugacy to a specific model. *Source:* the remark after Proposition 11.3, and the chapter on the doubling map. *Prerequisite:* the statement of Sharkovskii’s theorem. *Remedial work:* reread the Sharkovskii order in the chapter devoted to it.

8) B. *Rationale:* it is exactly the source cited in Remark 11.4, verified with the DOI/URL of the *Bulletin of the AMS*. *Distractors:* A (May 1976) introduces the logistic map and the cascade, but not the renormalisation operator; C (Singer 1978) concerns the uniqueness of the attracting orbit (Schwarzian derivative), a different result cited in Remark 11.1; D (Shenker 1982) concerns the constants δ_{GM} , α_{GM} of the circle, not δ , α of the logistic map. *Diagnosed error:* confusing with one another the four primary sources of the chapter, which cover four distinct results. *Source:* Remark 11.4 and the section *Sources and bibliography* at the end of the chapter. *Prerequisite:* none. *Remedial work:* reconstruct the table “result $\rightarrow$ primary source” for the whole chapter.

BEST PRACTICE

Assigned exercises — answers.

11.1 (2-cycle at $r = 3.45$). (a) $\Delta = \sqrt{(3.45 - 3)(3.45 + 1)} = \sqrt{0.45 \times 4.45} = \sqrt{2.0025} = 1.415097$; $x_{\pm} = \frac{4.45 \pm 1.415097}{6.9}$: $x_+ = 0.849290$, $x_- = 0.440421$. (b) $-r^2 + 2r + 4 = -11.9025 + 6.9 + 4 = -1.0025$: $|-1.0025| > 1$, the cycle is just unstable (consistent with $3.45 > 1 + \sqrt{6} = 3.449490$). (c) $f_{3.45}(0.849290) = 3.45 \times 0.849290 \times 0.150710 = 0.441373$, close to $x_- = 0.440421$ (discrepancy 0.001, of the expected order given that the cycle is only marginally unstable, with multiplier close to -1 : the direct iteration does not converge stably to the cycle, but the relation $f(x_+) \approx x_-$ remains verified at this precision). ✓ (d) Since the cycle is just beyond the threshold, $x_+ - x_- = 0.849290 - 0.440421 = 0.408869$ is already of the order of the maximum half-amplitude of the cycle: the closed formula $\Delta(r)/r = \sqrt{(r-3)(r+1)}/r$ grows continuously from 0 (at $r = 3$) and not with infinite tangency, consistently with Theorem 11.3 applied to the point of birth $r = 3$ (here it is in fact a flip, not a saddle-node: the growth is $\propto \sqrt{r-3}$, typical of the flip as well in the amplitude variable, see Remark 11.1).

11.2 (period-3 window, check). (a) $r^2 - 5r + 7 = 0$ has discriminant $25 - 28 = -3 < 0$: no real root, hence no real value of r makes this factor vanish; the same for $r^2 + r + 1$ (discriminant $1 - 4 = -3 < 0$). Neither of the two contributes an observable bifurcation for real r . (b) $r = 1 - 2\sqrt{2} \approx -1.828427 < 0$: outside the physical domain $r \in (0, 4]$ of the logistic family on $[0, 1]$ (for $r < 0$ the map does not send $[0, 1]$ into itself), it must be discarded. (c) At $r = 3.83$, solving numerically $f_r^3(x) = x$ (for example by bisection on the six branches of the degree-6 polynomial, or with Newton's method from several starting points) one finds two period-3 orbits; computing $|(f_r^3)'| = |f_r'(x_1)f_r'(x_2)f_r'(x_3)|$ at each one, the one with $|(f_r^3)'| < 1$ is the stable one (visible as an attractor in Figure 11.2), the one with $|(f_r^3)'| > 1$ is unstable (not observable by direct iteration, only by solving the equation). ✓ (script `verifica_cap11_periodo3.py`, extension of block V2)

11.3 (the two universalities). (a)

	constant	value	function space
δ		4.6692016091...	even unimodal maps of the interval
α		-2.5029078751...	(same space)
δ_{GM}		≈ -2.8336	pairs of critical circle maps, golden group
α_{GM}		≈ -1.2886	(same space)

(b) The period doubling of the logistic cascade is naturally marked out by the sequence $1, 2, 4, 8, \dots = 2^n$: the operator $\mathcal{R}$ rescales the second iterate $f \circ f$ at each step. The golden rotation number ρ^* is instead approximated by the Fibonacci convergents p_n/q_n , with $q_{n+1} = q_n + q_{n-1}$ and $q_{n+1}/q_n \rightarrow \varphi = (1 + \sqrt{5})/2 \approx 1.618034$: the natural renormalisation for the golden number rescales not the *second* iterate but compositions tied to the successive Fibonacci denominators, a structurally different procedure (based on φ , not on 2). Since the operators act on different spaces and with different composition rules, the respective unstable eigenvalues have no reason to coincide: this makes it *plausible* (heuristically) that $\delta \neq \delta_{\text{GM}}$, but the rigorous proof that they really are different lies in the computations (numerical and then analytic) of Shenker and of Ostlund–Rand–Sethna–Siggia (E45, E46), not in this heuristic argument. (c) Free answer; the expected error is the one described in Remark 11.6: attributing $\delta = 4.6692$ to the Arnold tongues/critical circle maps, whereas the correct value for that context is $\delta_{\text{GM}} \approx -2.8336$.

11.13. Chapter formulary

Logistic family and fixed points

$$f_r(x) = rx(1-x), \quad x = 0 \text{ (always)}, \quad x^* = 1 - \frac{1}{r} \quad (r \geq 1)$$

$$f'_r(0) = r, \quad f'_r(x^*) = 2 - r$$

Bifurcations of the logistic map

transcritical: $r = 1$; doubling (flip): $r = 3$; saddle-node (period 3): $r = 1 + 2\sqrt{2}$

The 2-cycle

$$x_{\pm}(r) = \frac{(r+1) \pm \sqrt{(r-3)(r+1)}}{2r}, \quad (f_r^2)'(x_{\pm}) = -r^2 + 2r + 4$$

stable for $3 < r < 1 + \sqrt{6} \approx 3.449490$

Normal forms (family $f(x, \mu)$, $g = f - x$)

saddle-node: $g_x = 0, g_{\mu} \neq 0, g_{xx} \neq 0 \Rightarrow \mu(x), \mu'(x_0) = 0, \mu''(x_0) = -\frac{g_{xx}}{g_{\mu}}$

transcritical: $g(x_0, \cdot) \equiv 0, g_{x\mu} \neq 0, g_{xx} \neq 0 \Rightarrow \xi'(\mu_0) = -\frac{g_{x\mu}}{\frac{1}{2}g_{xx}}$

flip: $f_x(x_0, \mu_0) = -1 \Rightarrow \partial_x g_2(x_0, \mu_0) = f_x(x_0, \mu_0)^2 - 1 = 0$

Cascade and universality (logistic map)

$$r_1 = 3, \quad r_2 = 1 + \sqrt{6}, \quad r_n \rightarrow r_{\infty} \approx 3.569945672$$

$$\delta_n = \frac{r_n - r_{n-1}}{r_{n+1} - r_n} \rightarrow \delta = 4.66920160910299 \dots \quad \alpha = -2.5029078750959 \dots$$

$$(\mathcal{R}g)(x) = \alpha g(g(x/\alpha)), \quad \mathcal{R}g^* = g^* \quad (\text{Lanford, imported})$$

Universality of critical circle maps (golden number, panorama)

$$\delta_{\text{GM}} \approx -2.8336, \quad \alpha_{\text{GM}} \approx -1.2886 \quad (\neq \delta, \alpha)$$

Schwarzian derivative

$$Sf = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2, \quad Sf_r < 0 \text{ everywhere } (f_r \text{ quadratic})$$

Bifurcation diagram

$$A(r) = \omega\left(\frac{1}{2}, f_r\right), \quad B = \{(r, y) : y \in A(r)\}$$

Sources and bibliography

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Part D

Holomorphic Dynamics

CHAPTER 12

From the circle to Julia and Mandelbrot

Up to here the circle has been the space on which the dynamics took place: rotations, diffeomorphisms, doubling map. In this chapter the circle changes role three times, and each time it becomes deeper. First: for $c = 0$ the map $g_c(z) = z^2 + c$ has as its Julia set *exactly* the unit circle, and the dynamics on it is *exactly* the doubling map of Chapter 7; we shall prove that for small $|c|$ that circle deforms into a Jordan curve on which the dynamics remains conjugate to doubling (Section 12.11). Second: when K_c is connected, the Böttcher coordinate identifies the complement of K_c with the complement of the disc, and the **external rays** — parametrised by an angle on the circle — also move by doubling of the angle (Section 12.7). Third, and most surprising: the circle reappears *inside* the parameter plane. The boundary of the main cardioid of the Mandelbrot set is the circle of multipliers $\lambda = e^{2\pi i\vartheta}$, and the arithmetic nature of ϑ — rational, Diophantine, of Cremer type — decides whether the fixed point of g_c is linearisable or not, with the same rational/irrational dichotomy that governed the rotations of Chapter 3. The theorems of Siegel and of Cremer, proved in Sections 12.8 and 12.9, are the holomorphic form of small divisors.

INFO

Level and prerequisites. Level **C** core (Sections 12.2–12.5 and 12.6–12.7); level **D** for Sections 12.4, 12.8, 12.9 and 12.11; level **E** for the panoramas of Section 12.12. Prerequisites within the treatise: basic complex analysis — Cauchy’s formula, Liouville’s theorem, Schwarz’s lemma, automorphisms of the disc, the maximum principle, harmonic functions (Volume III, Chapter 15: here *nothing* of all that is reproved); the doubling map $D(\theta) = 2\theta \bmod 2\pi$ and its expansivity (Volume IV, Chapter 7); Diophantine numbers and continued fractions (Volume IV, Chapter 3); the rotation number (Volume IV, Chapter 4).

What we import: the Riemann conformal mapping theorem, Montel’s theorem, the existence of the hyperbolic metric on a domain omitting at least three points of the sphere together with the Schwarz–Pick inequality, and the fact that a proper holomorphic map between Riemann surfaces is a branched covering: they are listed in full in Section 12.1. Remedial work: a reader who has not practised harmonic functions can read Sections 12.2, 12.3, 12.6, 12.8 and 12.9 independently, skipping Section 12.4.

12.1. What we import

Remark 12.1: Results imported and used in this chapter. This chapter uses, without reproving them, the following four results. For each one it is declared exactly what is assumed.

(I) Riemann conformal mapping theorem. Every simply connected domain $U \subsetneq \hat{\mathbb{C}}$ whose complement contains at least two points is biholomorphic to the unit disc $\mathbb{D}$; given $u_0 \in U$, there exists a unique biholomorphism $\psi : U \rightarrow \mathbb{D}$ with $\psi(u_0) = 0$ and $\psi'(u_0) > 0$. Imported from *J. Milnor, Dynamics in One Complex Variable, 3rd ed., Princeton Univ.*

Press, 2006, opening chapters (simply connected surfaces), and stated also in Volume III, Chapter 15; verified on 17 September 2026.

(I-bis) Topological criterion for simple connectedness. A connected open set $U \subseteq \widehat{\mathbb{C}}$ is simply connected if and only if its complement $\widehat{\mathbb{C}} \setminus U$ is connected. Imported from Milnor, *op. cit.*, opening chapters (standard facts of the topology of the sphere); verified on 17 September 2026.

(II) Hyperbolic metric and Schwarz–Pick inequality. Every domain $U \subseteq \widehat{\mathbb{C}}$ whose complement contains at least three points carries a complete conformal metric of curvature -1 , the **hyperbolic metric** ρ_U ; every holomorphic map $f : U \rightarrow V$ between such domains does not increase hyperbolic distances, and f is a local isometry if and only if it is a holomorphic covering. In particular, if $f : U \rightarrow U$ is holomorphic with a fixed point u_0 and $|f'(u_0)| < 1$ in the Euclidean coordinate, then f is not a local isometry. Imported from Milnor, *op. cit.*, chapter on universal coverings and the Poincaré metric; verified on 17 September 2026.

(III) Proper maps and branched coverings. A non-constant proper holomorphic map $f : U \rightarrow V$ between connected Riemann surfaces has a well-defined finite degree d : every $v \in V$ has exactly d preimages counted with multiplicity; if f has no critical points in U , then f is a covering of degree d . Imported from Milnor, *op. cit.* (standard facts about proper holomorphic maps between Riemann surfaces); verified on 17 September 2026.

Caveat about paragraph-level citations. Milnor’s section numbering differs between the 1990 arXiv preprint and the 2006 book. In what follows we cite only those sections of the book that have been directly verified: §8 (geometrically attracting or repelling fixed points: Koenigs and Fatou’s theorem), §9 (Böttcher’s theorem and polynomial dynamics), §11 (Cremer points and Siegel discs), §18 (external rays). For prerequisites (I), (I-bis), (II) and (III) no section number is given, because it was not possible to verify it on a copy of the book.

(IV) Siegel’s arithmetic lemma. The only unproved ingredient of Section 12.8, stated in full in Lemma 12.5. Imported from C. L. Siegel, Iteration of analytic functions, *Ann. of Math.* **43** (1942), 607–612, and from M. Abate, Discrete holomorphic local dynamical systems, *LNM* 1998, Springer, 2010, pp. 1–55, §4; verified on 17 September 2026.

All the rest of the chapter is proved.

12.2. The quadratic family: K_c , J_c and the Mandelbrot set

Definition 12.1: The quadratic family. For $c \in \mathbb{C}$ we set

$$g_c : \mathbb{C} \rightarrow \mathbb{C}, \quad g_c(z) = z^2 + c,$$

and we denote by g_c^n the n -th iterate. The only critical point of g_c in $\mathbb{C}$ is $z = 0$ (indeed $g'_c(z) = 2z$); the point ∞ is a superattracting fixed point on the Riemann sphere $\widehat{\mathbb{C}}$.

Remark 12.2: Why the form $z^2 + c$ suffices. Every quadratic polynomial $P(w) = aw^2 + bw + d$ with $a \neq 0$ is conjugate, by means of the affine map $z = \varphi(w) = aw + b/2$, to a unique g_c with $c = ad + b/2 - b^2/4$: indeed $\varphi(P(w)) = a(aw^2 + bw + d) + b/2 = (aw + b/2)^2 - b^2/4 + ad + b/2 = \varphi(w)^2 + c$. Since affine conjugacies alter neither the dynamics nor the invariant sets, studying the family $\{g_c\}$ means studying all quadratic polynomials.

Definition 12.2: Filled Julia set, Julia set, basin of infinity. For $c \in \mathbb{C}$ we set

$$K_c := \{z \in \mathbb{C} : (g_c^n(z))_{n \geq 0} \text{ is bounded}\}, \quad A_c(\infty) := \mathbb{C} \setminus K_c, \quad J_c := \partial K_c.$$

K_c is the **filled Julia set**, $A_c(\infty)$ the **basin of infinity**, J_c the **Julia set**. All three are completely invariant: $g_c^{-1}(K_c) = K_c = g_c(K_c)$, and likewise for $A_c(\infty)$ and J_c (the first equality because the orbit of z is bounded if and only if that of $g_c(z)$ is; the second because g_c is surjective onto $\mathbb{C}$).

Theorem 12.1: Escape criterion (radius 2). Let $R_c := \max(|c|, 2)$. If for some $n \geq 0$ we have $|g_c^n(z)| > R_c$, then $|g_c^m(z)| \rightarrow +\infty$ as $m \rightarrow \infty$, and indeed $|g_c^{n+k}(z)| \geq \mu^{2^k-1} |g_c^n(z)|$ for every $k \geq 0$, where $\mu := |g_c^n(z)| - 1 > 1$. In particular

$$K_c \subseteq \overline{\mathbb{D}}(0, R_c) = \{|z| \leq R_c\},$$

and $z \in K_c$ if and only if $|g_c^n(z)| \leq R_c$ for every $n \geq 0$. Consequently K_c is compact and $J_c = \partial K_c$ is compact and non-empty.

Proof. Let $|w| > R_c \geq 2$ and $|w| > |c|$. Then

$$|g_c(w)| = |w^2 + c| \geq |w|^2 - |c| > |w|^2 - |w| = |w|(|w| - 1).$$

Setting $\mu := |w| - 1$, we have $\mu > R_c - 1 \geq 1$, hence $|g_c(w)| > \mu|w| > |w| > R_c$: the inequality reproduces itself and we can iterate. We prove by induction on $k \geq 0$ that

$$|g_c^k(w)| \geq \mu^{2^k-1} |w|.$$

For $k = 0$ it is the identity $|w| \geq |w|$. Assume it true for k and set $w_k := g_c^k(w)$, so that $|w_k| \geq \mu^{2^k-1} |w|$. Since $\mu \geq 1$ we have $|w_k| \geq |w| > R_c$, hence we may apply the basic estimate to w_k :

$$|w_{k+1}| \geq |w_k|(|w_k| - 1) \geq |w_k| \cdot \mu \cdot \mu^{2^k-1} \quad (\text{because } |w_k| - 1 \geq \mu^{2^k-1} |w| - 1 \geq \mu^{2^k}),$$

where the last bracket is justified as follows: $\mu^{2^k-1} |w| - 1 = \mu^{2^k-1}(\mu + 1) - 1 = \mu^{2^k} + \mu^{2^k-1} - 1 \geq \mu^{2^k}$ because $\mu \geq 1$. Hence $|w_{k+1}| \geq \mu^{2^k} |w_k| \geq \mu^{2^k} \mu^{2^k-1} |w| = \mu^{2^{k+1}-1} |w|$, which is the statement for $k + 1$.

Since $\mu > 1$, the exponent $2^k - 1 \rightarrow \infty$ gives $|g_c^k(w)| \rightarrow \infty$. Applying what we have just seen to $w = g_c^n(z)$ yields the first part. The second follows: if the orbit of z is not bounded, there exists n with $|g_c^n(z)| > R_c$ (otherwise the orbit would stay in the compact set $\{|z| \leq R_c\}$), and then it diverges; conversely, if $|g_c^n(z)| \leq R_c$ for every n the orbit is by definition bounded. Finally K_c is closed (it is the intersection of the closed sets $\{|g_c^n(\cdot)| \leq R_c\}$, each closed by continuity) and bounded, hence compact; $K_c \neq \emptyset$ because it contains at least one fixed point of g_c (the roots of $z^2 - z + c = 0$, whose orbit is constant); and $J_c = \partial K_c \neq \emptyset$ because K_c is a non-empty compact set with $K_c \neq \mathbb{C}$. $\square$

Definition 12.3: Mandelbrot set.

$$\mathcal{M} := \{c \in \mathbb{C} : \text{the critical orbit } (g_c^n(0))_{n \geq 0} \text{ is bounded}\} = \{c \in \mathbb{C} : 0 \in K_c\}.$$

Theorem 12.2: $\mathcal{M}$ is compact and contained in the disc of radius 2. $\mathcal{M} \subseteq \{|c| \leq 2\}$ and $\mathcal{M}$ is closed, hence compact. Moreover $\{|c| \leq 1/4\} \subseteq \mathcal{M}$.

Proof. Inclusion in $\{|c| \leq 2\}$. Let $|c| > 2$, so that $R_c = \max(|c|, 2) = |c|$. The critical orbit begins with $z_1 := g_c(0) = c$ and $z_2 := g_c(c) = c^2 + c$, whence

$$|z_2| \geq |c|^2 - |c| = |c|(|c| - 1) > |c| = R_c,$$

since $|c| - 1 > 1$. Hence $|g_c^2(0)| > R_c$ and Theorem 12.1, applied to $z = 0$ with $n = 2$, gives the divergence of the critical orbit: $c \notin \mathcal{M}$.

Closedness. By Theorem 12.1 and by what we have just seen, for $|c| \leq 2$ we have $R_c = 2$ and

$$\mathcal{M} \cap \{|c| \leq 2\} = \bigcap_{n \geq 0} \{c : |c| \leq 2, |g_c^n(0)| \leq 2\}.$$

Every set of the family is closed, because $c \mapsto g_c^n(0)$ is a polynomial in c (hence continuous). An intersection of closed sets is closed; together with the inclusion $\mathcal{M} \subseteq \{|c| \leq 2\}$ this gives $\mathcal{M}$ closed and bounded, that is, compact.

The disc $|c| \leq 1/4$. If $|c| \leq 1/4$ and $|z| \leq 1/2$, then $|g_c(z)| \leq |z|^2 + |c| \leq 1/4 + 1/4 = 1/2$. Since $|0| \leq 1/2$, by induction the whole critical orbit stays in $\{|z| \leq 1/2\}$: it is bounded, hence $c \in \mathcal{M}$. $\square$

Example 12.1: The two extremes: $c = 0$ and $c = -2$. (a) **$c = 0$: the circle.** $g_0(z) = z^2$ gives $|g_0^n(z)| = |z|^{2^n}$, which is bounded if and only if $|z| \leq 1$. Hence $K_0 = \mathbb{D}$ and $J_0 = \{|z| = 1\} = S^1$: the Julia set is exactly the unit circle. Writing $z = e^{i\theta}$ we have $g_0(e^{i\theta}) = e^{2i\theta}$: the dynamics of g_0 on J_0 is exactly the doubling map $D(\theta) = 2\theta \bmod 2\pi$ of Chapter 7, with everything that follows from it (density of the periodic points, transitivity, sensitivity, $\#\text{Fix}(D^n) = 2^n - 1$).

(b) **$c = -2$: the segment.** Put $z = w + w^{-1}$ with $w \neq 0$. Then

$$z^2 - 2 = (w + w^{-1})^2 - 2 = w^2 + 2 + w^{-2} - 2 = w^2 + w^{-2},$$

that is, the map $\sigma(w) = w + w^{-1}$ conjugates $w \mapsto w^2$ with g_{-2} : $\sigma(w^2) = g_{-2}(\sigma(w))$. The map σ sends $\{|w| = 1\}$ onto $[-2, 2]$ (with $w = e^{i\varphi}$ one has $\sigma(w) = 2\cos\varphi$), $\{|w| > 1\}$ biholomorphically onto $\mathbb{C} \setminus [-2, 2]$, and transforms the dynamics of g_{-2} on $[-2, 2]$ into the map $2\cos\varphi \mapsto 2\cos 2\varphi$, that is, into the Chebyshev polynomial. Since $|w| > 1 \Rightarrow |w^{2^n}| \rightarrow \infty \Rightarrow |\sigma(w^{2^n})| \rightarrow \infty$ and $|w| = 1 \Rightarrow$ orbit in $[-2, 2]$, we conclude $K_{-2} = J_{-2} = [-2, 2]$: a segment.

Independent check: $-2 \in \mathcal{M}$ because the critical orbit is $0 \mapsto -2 \mapsto 2 \mapsto 2 \mapsto \dots$, bounded; and $|-2| = 2$ is the extreme value allowed by Theorem 12.2. $\checkmark$

Example 12.2: How many iterates are needed to decide escape. For $c = -0.75 + 0.1i$ the script finds $|g_c^{33}(0)| = 2.704$, while $|g_c^{32}(0)| \leq 2$: 33 iterates are needed. **Independent check** with the estimate of Theorem 12.1: once the radius 2 has been exceeded with $|w| = 2.704$, one has $\mu = 1.704$ and $|g_c^k(w)| \geq 1.704^{2^k-1} \cdot 2.704$; already for $k = 4$ this gives $\geq 1.704^{15} \cdot 2.704 \approx 4.9 \times 10^3$, and for $k = 6$ it exceeds 10^{14} : the divergence is doubly exponential, as it must be for a map of degree 2. $\checkmark$

12.3. Local dynamics at a fixed point: Koenigs and Böttcher

Definition 12.4: Fixed point, multiplier, classification. Let f be holomorphic in a neighbourhood of z_0 with $f(z_0) = z_0$. The **multiplier** of f at z_0 is $\lambda := f'(z_0)$. The fixed point is called

- **superattracting** if $\lambda = 0$;
- **attracting** (geometrically) if $0 < |\lambda| < 1$;
- **repelling** if $|\lambda| > 1$;
- **rationally neutral** (or *parabolic*) if λ is a root of unity;
- **irrationally neutral** if $|\lambda| = 1$ but λ is not a root of unity, that is, $\lambda = e^{2\pi i\vartheta}$ with ϑ irrational.

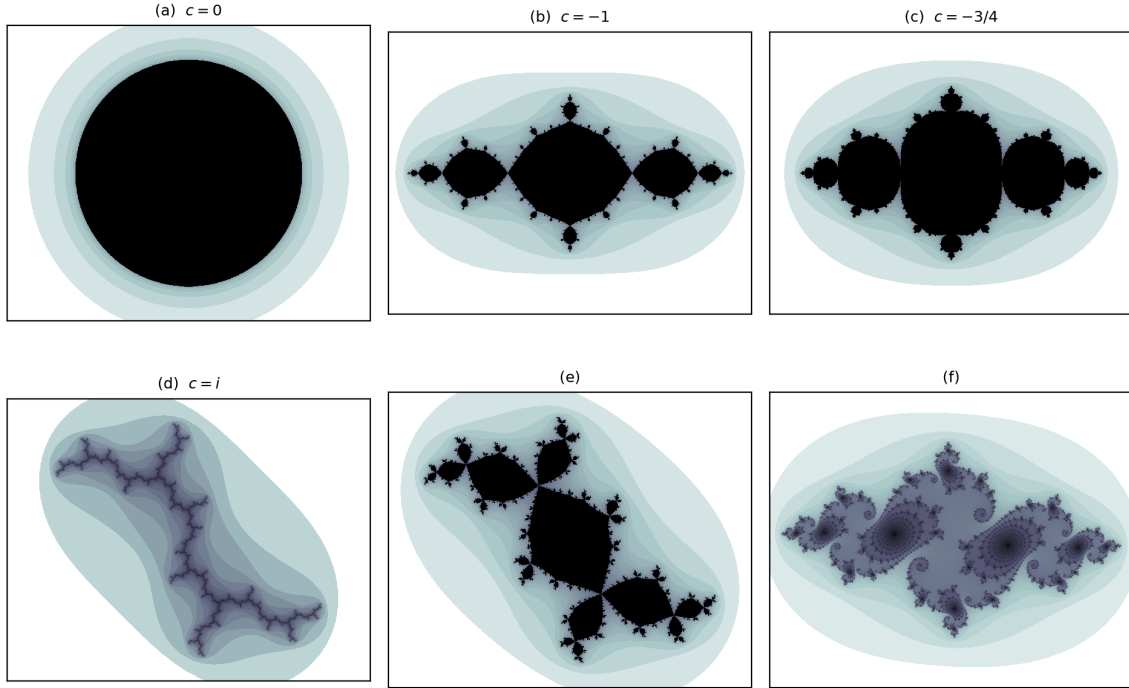


Figure 12.1: Six filled Julia sets K_c , computed with the escape criterion of Theorem 12.1 (the first five panels: 900×900 points, 300 iterates; the last one: 1400×1400 points, 4000 iterates). Black indicates the points that have not yet escaped, the light tints the escape time. Panel (a) $c = 0$: the disc, with $J_0 = S^1$ (Example 12.1). (b) $c = -1$: attracting cycle of period 2 (Theorem 12.8). (c) $c = -3/4$: parabolic fixed point of multiplier -1 . (d) $c = i$: dendrite (Exercise 12.4). (e) $c = -0.1225611669 + 0.7448617666i$: the *Douady rabbit*, with an attracting cycle of period 3. (f) $c = -0.75 + 0.1i$: Cantor dust — note the absence of any black region. Numerical check of the script `fig12_julia.py` (Appendix B, run on 17 September 2026): for the first five values the critical orbit stays bounded after 5000 iterates, whereas for the last one $|g_c^{33}(0)| = 2.704 > 2$, hence $c = -0.75 + 0.1i \notin \mathcal{M}$ and K_c is totally disconnected by Remark 12.5. ✓ *Honest caveat about the figure*: the total disconnectedness of this last K_c occurs at scales much finer than those resolved by the pixels — the image shows apparently continuous filaments; it is the escape criterion applied to the *critical orbit*, not the look of the drawing, that certifies it.

For a cycle $\{z_0, \dots, z_{p-1}\}$ of period p , the **multiplier of the cycle** is $\lambda = (f^p)'(z_0) = \prod_{j=0}^{p-1} f'(z_j)$, independent of the point of the cycle chosen, by the chain rule. We say that f is **linearisable** at z_0 if there exists a biholomorphism h between a neighbourhood of 0 and a neighbourhood of z_0 , with $h(0) = z_0$ and $h'(0) = 1$, such that $f(h(\zeta)) = h(\lambda\zeta)$ for ζ near 0.

Theorem 12.3: Koenigs linearisation (1884). *Let f be holomorphic in a neighbourhood of 0 with $f(0) = 0$ and $f'(0) = \lambda$, $0 < |\lambda| < 1$. Then there exists one and only one function φ , holomorphic in a neighbourhood of 0, with*

$$\varphi(0) = 0, \quad \varphi'(0) = 1, \quad \varphi(f(z)) = \lambda \varphi(z).$$

Moreover φ is a biholomorphism of a neighbourhood of 0 onto a neighbourhood of 0 and is given by the uniform limit

$$\varphi(z) = \lim_{n \rightarrow \infty} \frac{f^n(z)}{\lambda^n}.$$

The same conclusion holds for $|\lambda| > 1$ by applying the statement to the local inverse branch f^{-1} , whose multiplier is $1/\lambda$.

Proof. Choice of the constants. Since $0 < |\lambda| < 1$ we have $|\lambda| < \sqrt{|\lambda|} < 1$, hence there exists κ with

$$|\lambda| < \kappa < 1 \quad \text{and} \quad \kappa^2 < |\lambda|,$$

for example $\kappa := \frac{1}{2}(|\lambda| + \sqrt{|\lambda|})$: indeed κ is the average of two numbers in $(|\lambda|, 1]$ with $|\lambda| < \sqrt{|\lambda|}$, hence $|\lambda| < \kappa < \sqrt{|\lambda|} < 1$ and therefore $\kappa^2 < |\lambda|$.

An invariant disc. Since $f(z)/z \rightarrow \lambda$ as $z \rightarrow 0$ and $|\lambda| < \kappa$, there exists $\rho > 0$ such that f is holomorphic on $\mathbb{D}(0, \rho)$ and

$$|f(z)| \leq \kappa|z| \quad \text{for } |z| \leq \rho.$$

In particular $f(\mathbb{D}(0, \rho)) \subseteq \mathbb{D}(0, \rho)$ and, by induction, $|f^n(z)| \leq \kappa^n|z| \leq \kappa^n\rho$.

Quadratic estimate. The function $z \mapsto (f(z) - \lambda z)/z^2$ is holomorphic on $\mathbb{D}(0, \rho)$ (because $f(z) - \lambda z$ has a zero of order ≥ 2 at 0) and continuous on $\overline{\mathbb{D}(0, \rho)}$; let K be the maximum of its modulus on that compact set. Then

$$|f(z) - \lambda z| \leq K|z|^2 \quad \text{for } |z| \leq \rho.$$

Telescoping convergence. Put $\varphi_n(z) := f^n(z)/\lambda^n$. For $|z| \leq \rho$,

$$|\varphi_{n+1}(z) - \varphi_n(z)| = \frac{|f(f^n(z)) - \lambda f^n(z)|}{|\lambda|^{n+1}} \leq \frac{K|f^n(z)|^2}{|\lambda|^{n+1}} \leq \frac{K\kappa^{2n}\rho^2}{|\lambda|^{n+1}} = \frac{K\rho^2}{|\lambda|} \left(\frac{\kappa^2}{|\lambda|} \right)^n.$$

By the choice of κ we have $q := \kappa^2/|\lambda| < 1$, hence the series $\sum_n (\varphi_{n+1} - \varphi_n)$ converges normally on $\overline{\mathbb{D}(0, \rho)}$: the sequence (φ_n) converges uniformly to a function φ , holomorphic by Weierstrass's theorem on uniform convergence.

The three properties. $\varphi(0) = \lim f^n(0)/\lambda^n = 0$. As for the derivative, $\varphi'_n(0) = (f^n)'(0)/\lambda^n = \lambda^n/\lambda^n = 1$ for every n , and uniform convergence implies $\varphi'(0) = \lim \varphi'_n(0) = 1$. Finally

$$\varphi(f(z)) = \lim_n \frac{f^n(f(z))}{\lambda^n} = \lim_n \frac{f^{n+1}(z)}{\lambda^n} = \lambda \lim_n \frac{f^{n+1}(z)}{\lambda^{n+1}} = \lambda \varphi(z).$$

Since $\varphi'(0) = 1 \neq 0$, by the holomorphic inverse function theorem φ is a biholomorphism of a neighbourhood of 0 onto a neighbourhood of 0.

Uniqueness. Let ψ be another solution with $\psi(0) = 0$, $\psi'(0) = 1$, $\psi \circ f = \lambda\psi$. Setting $h := \psi \circ \varphi^{-1}$ (holomorphic near 0, $h(0) = 0$, $h'(0) = 1$), from $\varphi \circ f = \lambda\varphi$ it follows that $f = \varphi^{-1} \circ (\lambda \cdot) \circ \varphi$ and hence

$$h(\lambda\zeta) = \psi(\varphi^{-1}(\lambda\zeta)) = \psi(f(\varphi^{-1}(\zeta))) = \lambda\psi(\varphi^{-1}(\zeta)) = \lambda h(\zeta).$$

Writing $h(\zeta) = \zeta + \sum_{k \geq 2} b_k \zeta^k$ and comparing the coefficients of ζ^k in the identity $h(\lambda\zeta) = \lambda h(\zeta)$ one obtains $b_k \lambda^k = \lambda b_k$, that is, $b_k \lambda(\lambda^{k-1} - 1) = 0$. Since $\lambda \neq 0$ and $|\lambda| < 1$ implies $\lambda^{k-1} \neq 1$ for every $k \geq 2$, we have $b_k = 0$ for every $k \geq 2$: $h = \text{id}$, that is, $\psi = \varphi$. $\square$

Remark 12.3: Where the neutral multiplier breaks the argument. Two points of the proof use $|\lambda| < 1$ in an essential way and are not repairable for $|\lambda| = 1$. First: the estimate $|f^n(z)| \leq \kappa^n |z|$ is no longer there, and with it the geometric convergence. Second, and deeper: in the uniqueness argument there appears the factor $\lambda^{k-1} - 1$, which for $|\lambda| < 1$ is bounded below by $1 - |\lambda| > 0$, but for $\lambda = e^{2\pi i \vartheta}$ equals $e^{2\pi i(k-1)\vartheta} - 1$ and can be arbitrarily small: these are the **small divisors**. The whole of Section 12.8 is born from here.

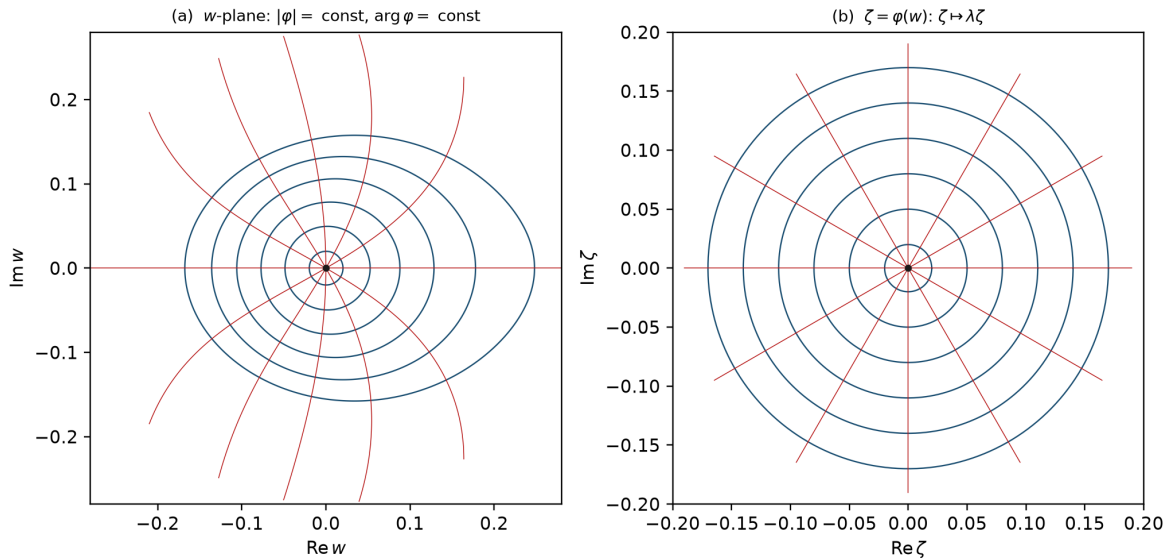


Figure 12.2: The Koenigs coordinate for $g_c(z) = z^2 + c$ with $c = -1/2$. The attracting fixed point is $z_- = (1 - \sqrt{3})/2 = -0.366025403784439$, with multiplier $\lambda = 2z_- = 1 - \sqrt{3} = -0.732050807568877$. Panel (a), on the left: the local coordinate $w = z - z_-$ (where $f(w) = \lambda w + w^2$), with the curves $|\varphi| = \text{const}$ (blue, the “Koenigs circles”) and $\arg \varphi = \text{const}$ (red, the “Koenigs rays”). Panel (b), on the right: the linearised model $\zeta \mapsto \lambda \zeta$, where the same curves are exact circles and rays. **Checks** of the script `fig12_locale.py` (Appendix B, run on 17 September 2026), computing φ as $f^{90}(w)/\lambda^{90}$ on a polar grid of 420×900 points in $|w| \leq 0.28$: maximum residual $|\varphi(f(w)) - \lambda\varphi(w)| = 1.1 \times 10^{-13}$, and $|\varphi(w)/w| \in [0.99886, 1.00113]$ for $|w| < 2 \times 10^{-3}$, consistent with $\varphi'(0) = 1$. $\checkmark$

Example 12.3: The Koenigs coordinate computed by hand and with the script. Take $c = -1/2$. The fixed points of g_c solve $z^2 - z - 1/2 = 0$, that is, $z_{\pm} = (1 \pm \sqrt{3})/2$. **First computation (closed form).** $z_- = (1 - \sqrt{3})/2$ and $\lambda = g'_c(z_-) = 2z_- = 1 - \sqrt{3}$. Since $\sqrt{3} = 1.7320508075688772\dots$, we have

$$z_- = -0.3660254037844386\dots, \quad \lambda = -0.7320508075688772\dots, \quad |\lambda| < 1,$$

hence z_- is attracting. The second fixed point $z_+ = (1+\sqrt{3})/2 = 1.3660254\dots$ has multiplier $2z_+ = 1 + \sqrt{3} = 2.732\dots > 1$: repelling.

Second computation (iteration). The script checks $z_-^2 + c - z_- = -1.1 \times 10^{-16}$ (machine error) and computes $\varphi(w) = \lim f^n(w)/\lambda^n$ with $n = 90$: the residual of the functional equation $\varphi(f(w)) = \lambda\varphi(w)$ is $\leq 1.1 \times 10^{-13}$ over the whole grid. **Third check, independent of the previous two:** the constant κ of the proof may be taken as $\kappa = \frac{1}{2}(|\lambda| + \sqrt{|\lambda|}) = \frac{1}{2}(0.732051 + 0.855599) = 0.793825$, and then $q = \kappa^2/|\lambda| = 0.630158/0.732051 = 0.860813 < 1$: the telescoping series converges with ratio 0.8608, and 90 terms give a remainder $\leq q^{90}/(1-q) \approx 1.2 \times 10^{-5}$ times the constant $K\rho^2/|\lambda|$. The agreement with the observed residual (10^{-13}) shows that the theoretical estimate is pessimistic, as is natural for a uniform majorisation. ✓

Theorem 12.4: Böttcher coordinate at infinity. Let $c \in \mathbb{C}$ and $R > \max(|c|, 2)$ be large enough ($R \geq 2 + 2|c|$ and $R \geq 4$ suffice). There exists a holomorphic and injective function

$$\phi_c : \{|z| > R\} \longrightarrow \mathbb{C} \setminus \overline{\mathbb{D}}(0, R'), \quad R' = \text{suitable},$$

such that

$$\phi_c(g_c(z)) = \phi_c(z)^2 \quad \text{and} \quad \frac{\phi_c(z)}{z} \longrightarrow 1 \quad (z \rightarrow \infty).$$

It is given by $\phi_c(z) = \lim_{n \rightarrow \infty} (g_c^n(z))^{1/2^n}$, with the determinations of the root chosen as in the proof, and it is the unique function with these two properties.

Proof. Preliminary estimates. Let $|z| \geq R$ with $R \geq 2 + 2|c|$ and $R \geq 4$. Then

$$\left| \frac{c}{z^2} \right| \leq \frac{|c|}{R^2} \leq \frac{|c|}{R \cdot 4} \leq \frac{1}{8}$$

(using $R \geq 2|c|$ in the form $|c|/R \leq 1/2$, and $R \geq 4$). Moreover $|g_c(z)| = |z|^2|1 + c/z^2| \geq |z|^2 \cdot \frac{7}{8} \geq \frac{7}{8}R|z| \geq \frac{7}{2}|z| \geq |z|$, hence the half-plane $\{|z| \geq R\}$ is invariant and, setting $z_n := g_c^n(z)$, we have $|z_n| \geq (7/2)^n|z|$, in particular $|z_n| \rightarrow \infty$ and

$$\left| \frac{c}{z_n^2} \right| \leq \frac{|c|}{R^2} \left(\frac{2}{7} \right)^{2n} \leq \frac{1}{8} \left(\frac{4}{49} \right)^n.$$

The series of logarithms. We write $z_{n+1} = z_n^2(1 + c/z_n^2)$. Shall we fix a holomorphic determination of $\log z$ on $\{|z| > R\}$? This is not possible on the whole set, which is not simply connected; instead we proceed directly on the quantity we need. Put

$$u_n(z) := \frac{1}{2^{n+1}} \operatorname{Log} \left(1 + \frac{c}{z_n^2} \right),$$

where Log is the principal determination, legitimate because $|c/z_n^2| \leq 1/8 < 1$. From the elementary estimate $|\operatorname{Log}(1+u)| \leq 2|u|$, valid for $|u| \leq 1/2$, one obtains

$$|u_n(z)| \leq \frac{1}{2^{n+1}} \cdot \frac{2|c|}{R^2} \left(\frac{4}{49} \right)^n \leq \frac{|c|}{R^2} \left(\frac{2}{49} \right)^n,$$

hence the series $S(z) := \sum_{n \geq 0} u_n(z)$ converges normally (and therefore uniformly) on $\{|z| \geq R\}$ to a holomorphic function, with $|S(z)| \leq \frac{|c|}{R^2} \cdot \frac{49}{47}$ and $S(z) \rightarrow 0$ as $z \rightarrow \infty$ (each $u_n(z) \rightarrow 0$ and the convergence is uniform).

Definition of ϕ_c . Put

$$\phi_c(z) := z e^{S(z)}.$$

It is holomorphic on $\{|z| > R\}$ and $\phi_c(z)/z = e^{S(z)} \rightarrow 1$ as $z \rightarrow \infty$.

Functional equation. We observe that, with $z' := g_c(z)$, we have $z'_n = z_{n+1}$, hence

$$S(z') = \sum_{n \geq 0} \frac{1}{2^{n+1}} \operatorname{Log} \left(1 + \frac{c}{z_{n+1}^2} \right) = \sum_{m \geq 1} \frac{1}{2^m} \operatorname{Log} \left(1 + \frac{c}{z_m^2} \right) = 2 \sum_{m \geq 1} u_m(z) = 2(S(z) - u_0(z)).$$

On the other hand $g_c(z) = z^2(1 + c/z^2)$ and hence, taking the exponential of $2u_0(z) = \operatorname{Log}(1 + c/z^2)$,

$$\phi_c(g_c(z)) = g_c(z) e^{S(g_c(z))} = z^2 e^{2u_0(z)} e^{2S(z) - 2u_0(z)} = z^2 e^{2S(z)} = (z e^{S(z)})^2 = \phi_c(z)^2.$$

Injectivity. Put $\eta(\zeta) := 1/\phi_c(1/\zeta)$ for $0 < |\zeta| < 1/R$. Since $\phi_c(z) = z(1 + o(1))$ as $z \rightarrow \infty$, we have $\eta(\zeta) = \zeta(1 + o(1))$ as $\zeta \rightarrow 0$: η extends holomorphically at 0 with $\eta(0) = 0$, $\eta'(0) = 1$. Hence η is injective on a disc $\{|\zeta| < \varepsilon\}$ (inverse function theorem), that is, ϕ_c is injective on $\{|z| > 1/\varepsilon\}$. Replacing R by $\max(R, 1/\varepsilon)$ we obtain injectivity on the declared set; the image is open and contains a neighbourhood of ∞ .

Uniqueness. If $\tilde{\phi}$ satisfies the same two properties, the function $\chi := \tilde{\phi}/\phi_c$ is holomorphic and non-vanishing near ∞ , tends to 1 as $z \rightarrow \infty$, and from the functional equation it follows that $\chi(g_c(z)) = \chi(z)^2$. If we had $\chi \not\equiv 1$, we write $\chi(z) = 1 + a_k z^{-k} + O(z^{-k-1})$ with $a_k \neq 0$, $k \geq 1$; then $\chi(z)^2 = 1 + 2a_k z^{-k} + O(z^{-k-1})$, whereas $\chi(g_c(z)) = 1 + a_k z^{-2k} + O(z^{-2k-1})$ because $g_c(z)^{-k} = z^{-2k}(1 + O(z^{-2}))$. Comparing the two expansions: the term of order z^{-k} on the right is $2a_k \neq 0$, on the left it is 0 (since $2k > k$). Contradiction; hence $\chi \equiv 1$. $\square$

Remark 12.4: Local Böttcher and Böttcher at infinity. Theorem 12.4 is Böttcher's theorem applied to the superattracting fixed point ∞ of g_c on the sphere: in the coordinate $\zeta = 1/z$ the map becomes $\zeta \mapsto \zeta^2/(1 + c\zeta^2)$, which fixes 0 with a zero of order 2, and the η of the proof is the local Böttcher coordinate, which conjugates that map to $\zeta \mapsto \zeta^2$. The same statement holds for the finite superattracting fixed point $z = 0$ when $c = 0$, where $\phi_0(z) = z$ trivially. The original statement is due to Böttcher, in a work in three parts published in Kazan between 1903 and 1904; the first statement of the functional equation $\varphi(f(z)) = \varphi(z)^d$ goes back to Schröder (1871), that of the case $0 < |\lambda| < 1$ to Koenigs (1884).

Definition 12.5: Green's function (potential) of the basin of infinity. For $z \in \mathbb{C}$ we set

$$G_c(z) := \lim_{n \rightarrow \infty} \frac{1}{2^n} \log^+ |g_c^n(z)|, \quad \log^+ t := \max(\log t, 0).$$

Proposition 12.1: Properties of Green's function. The limit exists for every $z \in \mathbb{C}$ and the function G_c has the following properties: **(i)** $G_c \equiv 0$ on K_c and $G_c > 0$ on $A_c(\infty) = \mathbb{C} \setminus K_c$; **(ii)** G_c is continuous on $\mathbb{C}$ and harmonic on $A_c(\infty)$; **(iii)** $G_c(g_c(z)) = 2G_c(z)$ for every z ; **(iv)** $G_c(z) = \log |\phi_c(z)|$ for $|z| > R$, and $G_c(z) - \log |z| \rightarrow 0$ as $z \rightarrow \infty$; **(v)** $\nabla G_c(z) \neq 0$ for every $z \in A_c(\infty)$ such that $g_c^k(z) \neq 0$ for every $k \geq 0$.

Proof. Existence and (iv). If $z \in K_c$, the orbit is bounded by R_c , hence $\log^+ |g_c^n(z)| \leq \log R_c$ and the limit is 0. If $z \notin K_c$, by Theorem 12.1 there exists n_0 with $|g_c^{n_0}(z)| > R$ (with R as in Theorem 12.4); from there on we can use the Böttcher coordinate: for $m \geq n_0$, $\phi_c(g_c^m(z)) = \phi_c(g_c^{n_0}(z))^{2^{m-n_0}}$, hence

$$\frac{1}{2^m} \log |g_c^m(z)| = \frac{1}{2^m} \log |\phi_c(g_c^m(z))| + \frac{1}{2^m} \log \left| \frac{g_c^m(z)}{\phi_c(g_c^m(z))} \right|.$$

The first summand equals exactly $2^{-n_0} \log |\phi_c(g_c^{n_0}(z))|$ for every $m \geq n_0$; the second tends to 0 because $g_c^m(z)/\phi_c(g_c^m(z)) \rightarrow 1$. Hence the limit exists and equals

$$G_c(z) = \frac{1}{2^{n_0}} \log |\phi_c(g_c^{n_0}(z))| > 0,$$

positive because $|\phi_c(w)| > 1$ for $|w| > R$ (recalling $R \geq 4$ and $\phi_c(w) = we^{S(w)}$ with $|S|$ small). Taking $n_0 = 0$ for $|z| > R$ one obtains $G_c = \log |\phi_c|$ there, and (iv) follows from $\phi_c(z)/z \rightarrow 1$. This also proves (i).

(iii). Immediate from the definition: $\frac{1}{2^n} \log^+ |g_c^n(g_c(z))| = 2 \cdot \frac{1}{2^{n+1}} \log^+ |g_c^{n+1}(z)|$, and one passes to the limit.

(ii), *harmonicity on $A_c(\infty)$* . Let $z^* \in A_c(\infty)$ and n_0 with $|g_c^{n_0}(z^*)| > R$. By continuity of $g_c^{n_0}$ there exists an open neighbourhood W of z^* on which still $|g_c^{n_0}| > R$; on W we then have, by the computation just made, $G_c = 2^{-n_0} \log |\phi_c \circ g_c^{n_0}|$, that is, the logarithm of the modulus of a nowhere vanishing holomorphic function: G_c is harmonic on W , in particular continuous.

(ii), *continuity at the points of K_c : a uniform estimate*. Put $C := 2 \max(R, |c| + 1, 2) \geq 4$. We claim:

$$|g_c^N(w)| \leq R \implies G_c(w) \leq \frac{\log C}{2^N}. \quad (*)$$

Indeed, setting $A := \max(R, |c| + 1, 2)$ and $u_j := \max(|g_c^{N+j}(w)|, A) \geq A \geq 2$, we have

$$|g_c^{N+j+1}(w)| \leq |g_c^{N+j}(w)|^2 + |c| \leq u_j^2 + u_j \leq 2u_j^2,$$

hence $u_{j+1} \leq \max(2u_j^2, A) = 2u_j^2$, that is, $2u_{j+1} \leq (2u_j)^2$; by induction $2u_j \leq (2u_0)^{2^j} \leq C^{2^j}$ (since $u_0 = \max(|g_c^N(w)|, A) \leq A \vee R = A$ and $2A = C$). Therefore $|g_c^{N+j}(w)| \leq u_j \leq C^{2^j}$ and

$$G_c(w) = \lim_{j \rightarrow \infty} \frac{\log^+ |g_c^{N+j}(w)|}{2^{N+j}} \leq \lim_{j \rightarrow \infty} \frac{2^j \log C}{2^{N+j}} = \frac{\log C}{2^N},$$

which is (*).

Now let $z_0 \in K_c$ and $\varepsilon > 0$; we choose N with $2^{-N} \log C < \varepsilon$. Since the orbit of z_0 stays in $\{|w| \leq R_c\}$ with $R_c < R$, by continuity of g_c^N there exists a neighbourhood V of z_0 on which $|g_c^N(w)| \leq R$. By (*), $0 \leq G_c(w) \leq 2^{-N} \log C < \varepsilon$ for every $w \in V$: hence $G_c(w) \rightarrow 0 = G_c(z_0)$, that is, G_c is continuous at every point of K_c (both interior and boundary points). Together with the harmonicity on $A_c(\infty)$, G_c is continuous on all of $\mathbb{C}$.

(v). On $\{|z| > R\}$ we have $G_c = \log |\phi_c|$ with ϕ_c injective and nowhere vanishing, hence $\nabla G_c \neq 0$ there. In general, from (iii) and from the chain rule for the gradient of a harmonic function composed with a holomorphic map: identifying ∇G_c with $2 \partial G_c / \partial z$, we have

$$\frac{\partial}{\partial z} (G_c \circ g_c)(z) = \left(\frac{\partial G_c}{\partial z} \right) (g_c(z)) \cdot g'_c(z) = 2 \frac{\partial G_c}{\partial z} (z).$$

Now let $z \in A_c(\infty)$ with $g_c^k(z) \neq 0$ for every k . Iterating, $\frac{\partial G_c}{\partial z} (g_c^n(z)) \cdot (g_c^n)'(z) = 2^n \frac{\partial G_c}{\partial z} (z)$. Choosing n with $|g_c^n(z)| > R$, the left-hand side is non-zero, because $\frac{\partial G_c}{\partial z} (g_c^n(z)) \neq 0$ and $(g_c^n)'(z) = \prod_{k < n} 2g_c^k(z) \neq 0$ by hypothesis. Hence $\frac{\partial G_c}{\partial z} (z) \neq 0$. $\square$

12.4. The fundamental dichotomy

This section proves in full the “positive” half of the Fatou–Julia dichotomy: *if the critical orbit is bounded, then K_c is connected*, and indeed the complement of K_c is biholomorphic to the complement of the disc, with the dynamics conjugate to $w \mapsto w^2$. The opposite half is stated as an import, with source and exact hypotheses.

Definition 12.6: The level domains U_s . For $s > 0$ we set $U_s := \{z \in \mathbb{C} : G_c(z) > s\}$ and $A_s := \{w \in \mathbb{C} : |w| > e^s\}$.

Lemma 12.1: Elementary properties of the levels. For every $s > 0$: (i) U_s is open and $U_s \subseteq A_c(\infty)$; (ii) $g_c^{-1}(U_{2s}) = U_s$ and $g_c(U_s) = U_{2s}$; (iii) $U_{2s} \subseteq U_s$ and $\bigcup_{s>0} U_s = A_c(\infty)$; (iv) there exists $\varrho(s) > 0$ with $\{|z| > \varrho(s)\} \subseteq U_s$; (v) if $0 \in K_c$ then $0 \notin U_s$, and $g_c : U_s \rightarrow U_{2s}$ is an unbranched holomorphic covering of degree 2.

Proof. (i) G_c is continuous (Proposition 12.1(ii)) and $G_c > s > 0$ implies $z \notin K_c$ by (i) of the same proposition. (ii) $G_c(g_c(z)) = 2G_c(z)$ gives $g_c(z) \in U_{2s} \iff 2G_c(z) > 2s \iff z \in U_s$; and $g_c(U_s) = g_c(g_c^{-1}(U_{2s})) = U_{2s}$ because $g_c : \mathbb{C} \rightarrow \mathbb{C}$ is surjective. (iii) $G_c > 2s > s$; and every $z \in A_c(\infty)$ has $G_c(z) > 0$, hence lies in U_s for $s < G_c(z)$. (iv) For $|z| > R$ we have $G_c(z) = \log |\phi_c(z)|$ and $\phi_c(z)/z \rightarrow 1$, hence $G_c(z) \rightarrow \infty$: it suffices to take $\varrho(s)$ large enough. (v) $0 \in K_c$ gives $G_c(0) = 0$, hence $0 \notin U_s$ for $s > 0$. The map $g_c : U_s \rightarrow U_{2s}$ is proper by (ii) (the preimage of a compact set $\mathcal{K} \subset U_{2s}$ is a closed set contained in U_s and bounded because $g_c^{-1}(\mathcal{K})$ is bounded), non-constant, and free of critical points because the only critical point 0 does not lie in U_s . By the imported result (III) it is a covering; the degree is 2 because every $w \in U_{2s}$ has exactly the two preimages $\pm\sqrt{w-c}$ (distinct, otherwise $w = c$ and the preimage would be $0 \notin U_s$). $\square$

Lemma 12.2: Connectedness of the levels. If $0 \in K_c$, then U_s is connected for every $s > 0$, and ϕ_c induces a biholomorphism $U_s \rightarrow A_s$ for all sufficiently large s .

Proof. Base: large s . Let $s_1 := \max_{|z| \leq R} G_c(z)$ (finite by continuity on a compact set) and let R'' be such that $\phi_c(\{|z| > R\}) \supseteq \{|w| > R''\}$: such an R'' exists because the function $\eta(\zeta) = 1/\phi_c(1/\zeta)$ of the proof of Theorem 12.4 is injective and holomorphic near 0 with $\eta(0) = 0$, hence its image is an open set containing 0. Put $s_2 := \max(s_1, \log R'')$. For $s > s_2$: if $z \in U_s$ then $G_c(z) > s > s_1$, hence $|z| > R$ and $|\phi_c(z)| = e^{G_c(z)} > e^s$; conversely, if $|w| > e^s > R''$ then $w = \phi_c(z)$ for a unique z with $|z| > R$, and $G_c(z) = \log |w| > s$. Hence ϕ_c is a holomorphic bijection of U_s onto A_s , and U_s is connected because A_s is.

Inductive step. Let $s > 0$ and suppose U_{2s} connected. By Lemma 12.1(v), $g_c : U_s \rightarrow U_{2s}$ is a covering of degree 2 over a connected base; every connected component of U_s therefore projects onto the whole of U_{2s} , and the sum of the degrees of the restrictions to the components is 2: hence U_s has at most two components.

Suppose by contradiction $U_s = V_1 \sqcup V_2$ with V_1, V_2 non-empty disjoint connected open sets. By Lemma 12.1(iii)–(iv), $U_{2s} \subseteq U_s$ is connected and contains $\{|z| > \varrho(2s)\}$: it therefore lies entirely inside a single component, say V_1 . Then $V_2 \subseteq \{|z| \leq \varrho(2s)\}$ is bounded.

Every point $p \in \partial V_2$ satisfies $G_c(p) = s$. Indeed $G_c(p) \geq s$ by continuity; if we had $G_c(p) > s$ then $p \in U_s$, and since V_2 is a connected component of the open set U_s it is open and closed in U_s , hence $p \in \overline{V_2} \cap U_s$ implies $p \in V_2$, impossible because V_2 is open and $\partial V_2 \cap V_2 = \emptyset$.

Now V_2 is a bounded open set with $\overline{V_2} \subset \mathbb{C}$, contained in $A_c(\infty)$ (Lemma 12.1(i)); G_c is harmonic on V_2 and continuous on $\overline{V_2}$, with $G_c \equiv s$ on ∂V_2 . By the maximum principle for harmonic functions (Volume III, Chapter 15), $G_c \leq s$ on V_2 . But $V_2 \subseteq U_s$ means $G_c > s$ on V_2 : contradiction. Hence U_s is connected.

Conclusion. For $s > s_2$ the assertion holds by the base case. If $s \in (s_2/2^{k+1}, s_2/2^k]$ for some $k \geq 0$, then $2s > s_2/2^k$ and by descending induction on k one obtains the connectedness of U_{2s} , hence of U_s . Since $\bigcup_{k \geq 0} (s_2/2^{k+1}, s_2/2^k] \cup (s_2, \infty) = (0, \infty)$, the assertion holds for every $s > 0$. $\square$

Theorem 12.5: Extension of the Böttcher coordinate and connectedness of K_c . Let $c \in \mathcal{M}$, that is, $0 \in K_c$. Then the Böttcher coordinate extends to a biholomorphism

$$\Phi_c : \mathbb{C} \setminus K_c \longrightarrow \mathbb{C} \setminus \overline{\mathbb{D}}, \quad \Phi_c(g_c(z)) = \Phi_c(z)^2, \quad \frac{\Phi_c(z)}{z} \rightarrow 1 \ (z \rightarrow \infty),$$

and $|\Phi_c(z)| = e^{G_c(z)}$. In particular K_c is **connected**.

Proof. Construction by descending induction. For $s > s_2$ (with s_2 as in Lemma 12.2) we put $\Phi^{(s)} := \phi_c|_{U_s} : U_s \rightarrow A_s$, a biholomorphism. It satisfies $\Phi^{(s)}(z)^2 = \Phi^{(2s)}(g_c(z))$ for $z \in U_s$ (this is the functional equation of Theorem 12.4).

Suppose we have constructed a biholomorphism $\Phi^{(2s)} : U_{2s} \rightarrow A_{2s}$ which extends ϕ_c and satisfies the functional equation. Consider the two maps

$$q : A_s \rightarrow A_{2s}, \quad q(w) = w^2, \quad \text{and} \quad P := \Phi^{(2s)} \circ g_c : U_s \rightarrow A_{2s}.$$

Both are unbranched holomorphic coverings of degree 2: q is one because $w \mapsto w^2$ is injective on no complete neighbourhood but is a double covering of $\{|w| > e^{2s}\}$ (the critical point 0 does not belong to A_s), and P is one by Lemma 12.1(v) composed with a biholomorphism. Moreover U_s and A_s are connected (Lemma 12.2 for U_s ; A_s is an annulus) and $\pi_1(A_{2s}) \cong \mathbb{Z}$. A connected double covering of a space with fundamental group $\mathbb{Z}$ corresponds to the unique subgroup of index 2, that is, $2\mathbb{Z}$: hence $P_*\pi_1(U_s) = 2\mathbb{Z} = q_*\pi_1(A_s)$ as subgroups of $\pi_1(A_{2s}) \cong \mathbb{Z}$. By the lifting criterion (Volume IV, Chapter 1) there exists a continuous lift $\Phi^{(s)} : U_s \rightarrow A_s$ with $q \circ \Phi^{(s)} = P$; it is holomorphic because locally $\Phi^{(s)} = q^{-1} \circ P$ with q^{-1} a holomorphic branch of the square root, and it is an isomorphism of coverings (two connected double coverings of the same base with the same characteristic subgroup are isomorphic), hence a biholomorphism $U_s \rightarrow A_s$.

Choice of the sign. Of the two possible lifts ($\Phi^{(s)}$ and $-\Phi^{(s)}$) we choose the one that coincides with $\Phi^{(2s)}$ on U_{2s} . This is legitimate because $\Phi^{(2s)}$ is also a lift of $P|_{U_{2s}}$: for $z \in U_{2s}$ we have $g_c(z) \in U_{4s} \subseteq U_{2s}$ and, by the inductive hypothesis applied to $2s$, $\Phi^{(2s)}(z)^2 = \Phi^{(4s)}(g_c(z)) = \Phi^{(2s)}(g_c(z))$ (the maps constructed are restrictions of one another); hence $q(\Phi^{(2s)}(z)) = P(z)$. Two lifts that agree at one point agree on the whole connected set U_{2s} .

Gluing. The maps $\Phi^{(s)}$, $s > 0$, are therefore restrictions of one another and define a single biholomorphism Φ_c on $\bigcup_{s>0} U_s = \mathbb{C} \setminus K_c$, with values in $\bigcup_{s>0} A_s = \mathbb{C} \setminus \overline{\mathbb{D}}$, surjective because each $\Phi^{(s)}$ is onto A_s . The functional equation and the asymptotics at infinity hold by construction; $|\Phi_c| = e^{G_c}$ follows because it holds on U_s for large s and because both sides satisfy the same doubling rule under g_c .

Connectedness of K_c . The set $\widehat{\mathbb{C}} \setminus K_c = (\mathbb{C} \setminus K_c) \cup \{\infty\}$ is biholomorphic, via Φ_c extended by $\Phi_c(\infty) = \infty$, to $(\mathbb{C} \setminus \overline{\mathbb{D}}) \cup \{\infty\}$, which is a topological disc in the sphere, hence *simply connected*. By the imported result (I-bis), the complement $\widehat{\mathbb{C}} \setminus (\widehat{\mathbb{C}} \setminus K_c) = K_c$ is connected. $\square$

Remark 12.5: The opposite half of the dichotomy (imported). *Statement.* If $c \notin \mathcal{M}$, that is, if the critical orbit of g_c is unbounded, then $K_c = J_c$ is a Cantor set: compact, non-empty, totally disconnected, perfect; and $g_c|_{J_c}$ is topologically conjugate to the one-sided shift on two symbols Σ_2 (Chapter 7). **What is assumed:** the Douady–Hubbard construction of the two inverse branches on $\{G_c < G_c(0)\}$, with the distortion estimate guaranteeing that the diameters of the nested images tend to zero. Imported from J. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Princeton Univ. Press, 2006, §9 (“Böttcher’s theorem and polynomial dynamics”), and from L. Carleson, T. W. Gamelin, *Complex Dynamics*, Springer, 1993, ch. VIII; verified on 17 September 2026. Together with

Theorem 12.5 one obtains the **fundamental dichotomy**:

$$K_c \text{ connected} \iff 0 \in K_c \iff c \in \mathcal{M}, \quad K_c \text{ a Cantor set} \iff 0 \notin K_c \iff c \notin \mathcal{M}.$$

There are no intermediate cases: K_c cannot have, for example, exactly three components.

Remark 12.6: The connectedness of $\mathcal{M}$ (imported). The procedure of Theorem 12.5 carries over to the parameter plane by defining $\Phi(c) := \Phi_c(c)$ for $c \notin \mathcal{M}$ (the value of the Böttcher coordinate of g_c at the point $c = g_c(0)$, which is well defined because for $c \notin \mathcal{M}$ the Böttcher coordinate is defined at least up to the level $G_c(0)$). Douady and Hubbard proved that $\Phi : \mathbb{C} \setminus \mathcal{M} \rightarrow \mathbb{C} \setminus \overline{\mathbb{D}}$ is a biholomorphism; by the imported result (I-bis) it follows that $\mathcal{M}$ is **connected**. Imported from A. Douady, J. H. Hubbard, Étude dynamique des polynômes complexes, Publications Mathématiques d'Orsay 84-02 (1984) and 85-04 (1985), and from the English translation Exploring the Mandelbrot set. The Orsay Notes; verified on 17 September 2026. **What is assumed:** the holomorphy of Φ (which requires the holomorphic dependence of Φ_c on c) and its surjectivity. Note that this does not say that $\mathcal{M}$ is locally connected: that is the MLC conjecture, discussed in Volume VI, Chapter 9.

12.5. Fatou's theorem on attracting cycles

Definition 12.7: Basin and immediate basin of an attracting cycle. Let $\{z_0, \dots, z_{p-1}\}$ be a cycle of period p of g_c with multiplier $|\lambda| < 1$. The **basin** of the cycle is the open set $\mathcal{B}$ of the points z such that $g_c^{np}(z) \rightarrow z_j$ for some j ; the **immediate basin** $\mathcal{A}$ of z_0 is the connected component of $\mathcal{B}$ containing z_0 .

Theorem 12.6: Fatou: every attracting cycle attracts a critical point. Let $\{z_0, \dots, z_{p-1}\}$ be an attracting cycle of g_c (with $|\lambda| < 1$, possibly $\lambda = 0$) and let $\mathcal{A}$ be the immediate basin of z_0 . Then $\mathcal{A}$ contains a point of the critical orbit $\{g_c^k(0)\}_{k \geq 0}$; in particular the critical orbit converges to the cycle. Consequently: (i) g_c has at most one attracting cycle in $\mathbb{C}$; (ii) if g_c has an attracting cycle, then $c \in \mathcal{M}$.

Proof. Put $h := g_c^p$, so that $h(z_0) = z_0$ and $h'(z_0) = \lambda$ with $|\lambda| < 1$, and $\mathcal{A}$ is the connected component of $\{z : h^n(z) \rightarrow z_0\}$ containing z_0 .

Step 1: $\mathcal{A}$ is invariant and $h : \mathcal{A} \rightarrow \mathcal{A}$ is proper and surjective. Since h is continuous and $h(z_0) = z_0$, the set $h(\mathcal{A})$ is connected, contains z_0 and is contained in the basin: hence $h(\mathcal{A}) \subseteq \mathcal{A}$. Let $\mathcal{A}'$ be the connected component of $h^{-1}(\mathcal{A})$ containing $\mathcal{A}$ (it exists because $\mathcal{A} \subseteq h^{-1}(\mathcal{A})$ by what we have just said, and $\mathcal{A}$ is connected). Every point of $\mathcal{A}'$ belongs to the basin (its image does), and $\mathcal{A}'$ is connected and contains z_0 : hence $\mathcal{A}' \subseteq \mathcal{A}$, that is, $\mathcal{A}' = \mathcal{A}$. Therefore $\mathcal{A}$ is a connected component of $h^{-1}(\mathcal{A})$; since $h : \mathbb{C} \rightarrow \mathbb{C}$ is a polynomial, hence a proper map, the restriction $h : \mathcal{A} \rightarrow \mathcal{A}$ is proper (the preimage in $\mathcal{A}$ of a compact set $\mathcal{K} \subseteq \mathcal{A}$ is closed in $\mathcal{A}$ and contained in the compact set $h^{-1}(\mathcal{K})$, and cannot accumulate on $\partial\mathcal{A}$ because $\mathcal{A}$ is a component of $h^{-1}(\mathcal{A})$). By the imported result (III), $h : \mathcal{A} \rightarrow \mathcal{A}$ is surjective of a finite degree $d \geq 1$.

Step 2: $\mathcal{A}$ is hyperbolic. $\mathcal{A}$ is contained in K_c (the orbits there converge to a cycle, hence they are bounded), so $\mathbb{C} \setminus \mathcal{A} \supseteq A_c(\infty)$ has non-empty interior: the complement of $\mathcal{A}$ in $\widehat{\mathbb{C}}$ contains infinitely many points, in particular at least three. By the imported result (II), $\mathcal{A}$ carries the hyperbolic metric $\rho_{\mathcal{A}}$.

Step 3: h is not a local isometry. $h : \mathcal{A} \rightarrow \mathcal{A}$ is holomorphic with fixed point z_0 and $|h'(z_0)| = |\lambda| < 1$. If h were a local isometry for $\rho_{\mathcal{A}}$, its differential at z_0 would have modulus 1 with

respect to the hyperbolic metric; but the hyperbolic metric is conformal, that is, of the form $\rho_{\mathcal{A}}(z)|dz|$ with $\rho_{\mathcal{A}}(z) > 0$, and at a *fixed point* the ratio between the hyperbolic and the Euclidean norm of the differential is

$$\|Dh(z_0)\|_{\rho} = \frac{\rho_{\mathcal{A}}(h(z_0))}{\rho_{\mathcal{A}}(z_0)} |h'(z_0)| = |h'(z_0)| = |\lambda| < 1,$$

because $h(z_0) = z_0$ makes the conformal factor cancel. Hence $\|Dh(z_0)\|_{\rho} = |\lambda| < 1$: h is not a local isometry.

Step 4: conclusion. By the imported result (II), a proper holomorphic map $h : \mathcal{A} \rightarrow \mathcal{A}$ free of critical points would be a holomorphic covering (imported result (III)) and hence a hyperbolic local isometry — against Step 3. Therefore $h = g_c^p$ has at least one critical point in $\mathcal{A}$. By the chain rule,

$$(g_c^p)'(z) = \prod_{k=0}^{p-1} g_c'(g_c^k(z)) = \prod_{k=0}^{p-1} 2g_c^k(z),$$

which vanishes if and only if $g_c^k(z) = 0$ for some $0 \leq k \leq p-1$. Hence there exist $z^* \in \mathcal{A}$ and k with $g_c^k(z^*) = 0$: the critical point 0 belongs to $g_c^k(\mathcal{A})$, and since $g_c^k(\mathcal{A})$ is contained in the basin $\mathcal{B}$ of the cycle, we have $0 \in \mathcal{B}$. The critical orbit therefore converges to the cycle, and being convergent it is bounded: $c \in \mathcal{M}$, which is (ii).

(i). If there were two distinct attracting cycles, their basins would be disjoint (an orbit cannot converge to two different cycles) and each would have to contain the orbit of the critical point 0 by what we have just seen: impossible, because g_c has only one critical point in $\mathbb{C}$ and its orbit is a single one. $\square$

Remark 12.7: What changes with respect to the version of the previous chapter.

In the previous redaction of this chapter Theorem 12.6 appeared with an “outline” of a proof. The proof given here is complete and uses only the two imported results (II) and (III), both classical and declared in Section 12.1. Note that the hypothesis $|\lambda| < 1$ includes the superattracting case $\lambda = 0$, in which the conclusion is obvious (the critical point is the point of the cycle), but the proof remains valid without modifications.

12.6. The circle in the parameter plane: cardioid and bulbs

Theorem 12.7: The main cardioid. The set of parameters c for which g_c possesses an attracting fixed point (that is, with $|\lambda| < 1$) is the image of the open unit disc under the map

$$\mathfrak{c} : \lambda \mapsto \frac{\lambda}{2} - \frac{\lambda^2}{4},$$

which is injective on the closed disc $\overline{\mathbb{D}}$ and holomorphic on $\mathbb{D}$. This image $\mathfrak{c}(\mathbb{D})$ is an open set whose boundary is the **cardioid** $\mathfrak{c}(\partial\mathbb{D})$, and it is contained in $\mathcal{M}$. The parameter $\lambda = e^{2\pi i\vartheta}$ which describes the boundary is exactly the **multiplier** of the fixed point: the circle reappears here as the space of neutral multipliers.

Proof. Parametrisation. The fixed points of g_c solve $z^2 - z + c = 0$ and the multiplier at a fixed point z_0 is $\lambda = g_c'(z_0) = 2z_0$, that is, $z_0 = \lambda/2$. Substituting into the fixed-point equation, $c = z_0 - z_0^2 = \lambda/2 - \lambda^2/4 = \mathfrak{c}(\lambda)$. Conversely, given λ , the point $z_0 = \lambda/2$ is a fixed point of $g_{\mathfrak{c}(\lambda)}$ with multiplier λ . Hence

$$\{c : g_c \text{ has a fixed point of multiplier } \lambda\} = \{\mathfrak{c}(\lambda)\},$$

and the parameters with an attracting fixed point are exactly $\mathfrak{c}(\mathbb{D})$.

Injectivity on $\overline{\mathbb{D}}$. If $\mathfrak{c}(\lambda_1) = \mathfrak{c}(\lambda_2)$ then

$$0 = \frac{\lambda_1 - \lambda_2}{2} - \frac{\lambda_1^2 - \lambda_2^2}{4} = (\lambda_1 - \lambda_2) \left(\frac{1}{2} - \frac{\lambda_1 + \lambda_2}{4} \right),$$

hence $\lambda_1 = \lambda_2$ or $\lambda_1 + \lambda_2 = 2$. But if $|\lambda_1| \leq 1$ and $|\lambda_2| \leq 1$, from $\lambda_1 + \lambda_2 = 2$ it follows that $2 = |\lambda_1 + \lambda_2| \leq |\lambda_1| + |\lambda_2| \leq 2$ with equality, which forces $|\lambda_1| = |\lambda_2| = 1$ and λ_1, λ_2 positively proportional, that is, $\lambda_1 = \lambda_2 = 1$. In either case $\lambda_1 = \lambda_2$.

Topology of the image. $\mathfrak{c}$ is holomorphic and injective on $\mathbb{D}$, hence $\mathfrak{c}(\mathbb{D})$ is open and $\mathfrak{c} : \mathbb{D} \rightarrow \mathfrak{c}(\mathbb{D})$ is a biholomorphism; $\mathfrak{c}$ is continuous and injective on the compact set $\overline{\mathbb{D}}$, hence a homeomorphism onto $\mathfrak{c}(\overline{\mathbb{D}})$, and therefore $\partial\mathfrak{c}(\mathbb{D}) = \mathfrak{c}(\partial\mathbb{D})$.

Inclusion in $\mathcal{M}$. If $c \in \mathfrak{c}(\mathbb{D})$, then g_c has an attracting fixed point, that is, an attracting cycle of period 1; by Theorem 12.6(ii), $c \in \mathcal{M}$. $\square$

Corollary 12.1: At most one attracting fixed point. For every c , the two fixed points $z_{\pm} = \frac{1}{2}(1 \pm \sqrt{1 - 4c})$ have multipliers $\lambda_{\pm} = 1 \pm \sqrt{1 - 4c}$ with $\lambda_+ + \lambda_- = 2$; hence they cannot both be attracting, nor both neutral unless $\lambda_+ = \lambda_- = 1$ (that is, $c = 1/4$).

Proof. $\lambda_{\pm} = 2z_{\pm} = 1 \pm \sqrt{1 - 4c}$, hence $\lambda_+ + \lambda_- = 2$. If we had $|\lambda_+| < 1$ and $|\lambda_-| < 1$ we would get $2 = |\lambda_+ + \lambda_-| \leq |\lambda_+| + |\lambda_-| < 2$: absurd. If $|\lambda_+| = |\lambda_-| = 1$, the same chain gives equality at every step, hence $\lambda_+ = \lambda_- = 1$ and $\sqrt{1 - 4c} = 0$, that is, $c = 1/4$. $\square$

Theorem 12.8: The period-2 bulb. The set of parameters c for which g_c possesses an attracting cycle of period exactly 2 is the open disc

$$B_{1/2} := \left\{ c : |c + 1| < \frac{1}{4} \right\},$$

which is contained in $\mathcal{M}$ and is internally tangent to the cardioid at the point $c = -3/4 = \mathfrak{c}(-1)$, where the fixed point has multiplier -1 . The centre $c = -1$ is the superattracting parameter of the bulb.

Proof. Factorisation. A direct computation gives $g_c^2(z) - z = (z^2 + c)^2 + c - z = z^4 + 2cz^2 - z + c^2 + c$ and

$$(z^2 - z + c)(z^2 + z + c + 1) = z^4 + 2cz^2 - z + c^2 + c,$$

as one checks by multiplying term by term. The first factor vanishes at the fixed points; hence the points of period exactly 2 are the roots $z_{\pm}$ of $z^2 + z + (c + 1) = 0$, with

$$z_+ + z_- = -1, \quad z_+ z_- = c + 1.$$

(The two roots coincide with a fixed point only if the two factors have a common root, that is, if $z^2 - z + c$ and $z^2 + z + c + 1$ have a common root: subtracting, $-2z - 1 = 0$, $z = -1/2$, which substituted into the first gives $1/4 + 1/2 + c = 0$, that is, $c = -3/4$: this is exactly the point of tangency.)

Multiplier. By the chain rule, the multiplier of the 2-cycle is

$$\lambda_2 = g'_c(z_+) g'_c(z_-) = (2z_+)(2z_-) = 4z_+ z_- = 4(c + 1).$$

Hence the cycle is attracting if and only if $|4(c + 1)| < 1$, that is, $|c + 1| < 1/4$, and it is superattracting ($\lambda_2 = 0$) if and only if $c = -1$.

Inclusion in $\mathcal{M}$ and tangency. The inclusion $B_{1/2} \subseteq \mathcal{M}$ follows from Theorem 12.6(ii). As for the tangency: $\mathfrak{c}(-1) = -1/2 - 1/4 = -3/4$, and $|-3/4 + 1| = 1/4$, hence $c = -3/4$ lies on the

boundary of $B_{1/2}$ and is the point of the cardioid with multiplier $\lambda = -1$. The two boundaries (the circle $|c + 1| = 1/4$ and the cardioid) touch there: indeed $\mathbf{c}'(\lambda) = \frac{1}{2} - \frac{\lambda}{2}$, which at $\lambda = -1$ equals 1, while the tangent direction to the boundary of the disc at $c = -3/4$ is orthogonal to $c + 1 = 1/4$, that is, vertical; the cardioid at $\mathbf{c}(-1)$ has tangent vector $\mathbf{c}'(-1) \cdot i\lambda|_{\lambda=-1} = 1 \cdot (-i)$, also vertical. The two curves therefore have the same tangent at $c = -3/4$ and, the disc lying on the opposite side of the cardioid with respect to the region $\mathbf{c}(\mathbb{D})$, they are internally tangent. $\square$

Definition 12.8: Root points of the p/q bulbs. For $p/q \in \mathbb{Q} \cap [0, 1)$ in lowest terms we set

$$c_{p/q} := \mathbf{c}(e^{2\pi i p/q}) = \frac{e^{2\pi i p/q}}{2} - \frac{e^{4\pi i p/q}}{4}.$$

It is the point of the cardioid at which the fixed point of g_c has as multiplier the primitive q -th root of unity $e^{2\pi i p/q}$ (parabolic).

Proposition 12.2: Ordering of the root points on the boundary of the cardioid. The map $\vartheta \mapsto \mathbf{c}(e^{2\pi i \vartheta})$ is a homeomorphism of the circle $\mathbb{R}/\mathbb{Z}$ onto the cardioid $\mathbf{c}(\partial\mathbb{D})$. Consequently the root points $c_{p/q}$ are arranged along the cardioid in the same cyclic order as the rationals p/q in $\mathbb{R}/\mathbb{Z}$, and are dense on it. The point $\vartheta = 0$ corresponds to the cusp $c = 1/4$; $\vartheta = 1/2$ to the root of the period-2 bulb, $c_{1/2} = -3/4$.

Proof. $\mathbf{c}$ is injective and continuous on $\overline{\mathbb{D}}$ (Theorem 12.7), hence its restriction to the compact set $\partial\mathbb{D}$ is a homeomorphism onto its image; composing with the homeomorphism $\vartheta \mapsto e^{2\pi i \vartheta}$ of $\mathbb{R}/\mathbb{Z}$ onto $\partial\mathbb{D}$ one obtains the statement. A homeomorphism of $\mathbb{R}/\mathbb{Z}$ onto a Jordan curve preserves or reverses the cyclic order; the density of the $c_{p/q}$ follows from the density of $\mathbb{Q}/\mathbb{Z}$ in $\mathbb{R}/\mathbb{Z}$. Finally $\mathbf{c}(1) = 1/2 - 1/4 = 1/4$ and $\mathbf{c}(-1) = -3/4$. $\square$

Remark 12.8: The p/q bulbs and the Farey ordering (panorama). **Statement (imported).** For every reduced p/q with $q \geq 2$ there exists a connected component $B_{p/q}$ of the interior of $\mathcal{M}$, the p/q **bulb**, such that: (a) $\overline{B_{p/q}}$ touches the cardioid exactly at the root point $c_{p/q}$; (b) for every $c \in B_{p/q}$ the map g_c has an attracting cycle of period exactly q ; (c) around that cycle the local dynamics “rotates” by p/q of a turn per iteration, in the sense that the q sectors of the immediate basin of the cycle are permuted cyclically with combinatorial rotation number p/q ; (d) between two bulbs B_{p_1/q_1} and B_{p_2/q_2} adjacent in the Farey sequence, the largest bulb interposed is that of the mediant $(p_1 + p_2)/(q_1 + q_2)$. Imported from J. Milnor, “Periodic orbits, external rays and the Mandelbrot set: an expository account”, *Astérisque* **261** (2000), 277–333, and from Douady–Hubbard, Orsay Notes; verified on 17 September 2026. **What is assumed:** the theory of the landing of rational parameter rays (Section 12.7) and the structure of the hyperbolic components of $\mathcal{M}$.

What, on the other hand, is proved here: the case $p/q = 1/2$ in full (Theorem 12.8), the exact position of all the root points (Definition 12.8) and their cyclic ordering (Proposition 12.2). The link with Chapter 4 is exact and not metaphorical: the function $\vartheta \mapsto$ “bulb attached at $\mathbf{c}(e^{2\pi i \vartheta})$ ” is the analogue, in the parameter plane of the complex quadratic family, of the function $\alpha \mapsto \rho(f_{\alpha, \varepsilon})$ of the Arnold standard family: the rationals produce open regions (the tongues, here the bulbs), the irrationals single points.

12.7. External rays: the circle doubles again

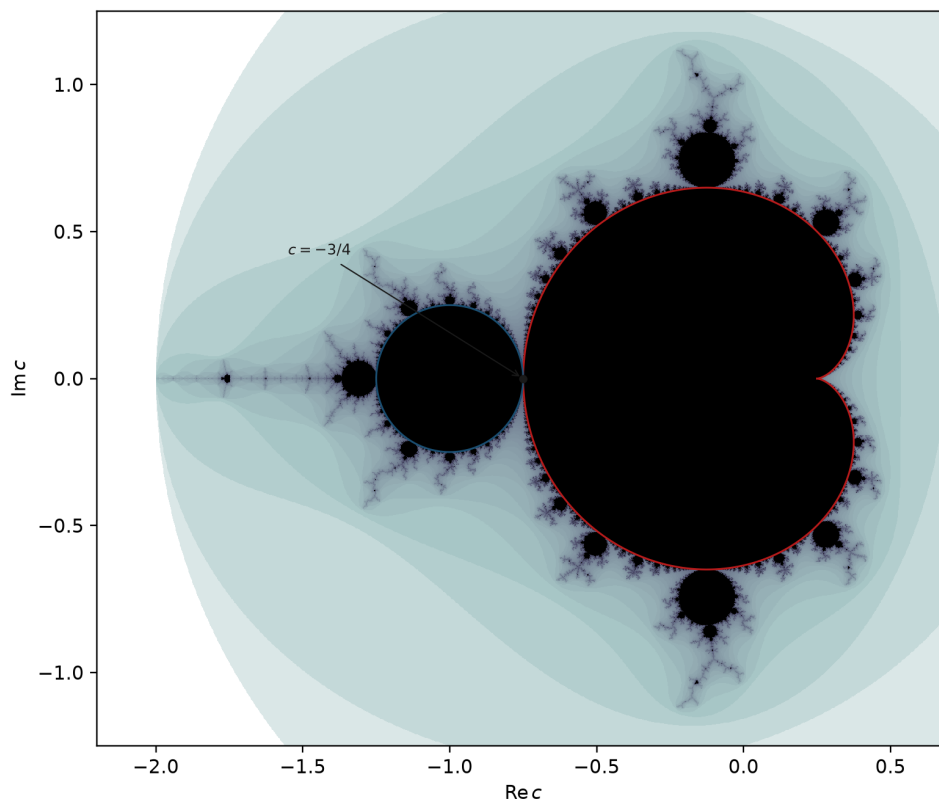


Figure 12.3: The Mandelbrot set $\mathcal{M}$ computed with the escape criterion of Theorem 12.1 (1600×1400 points, up to 600 iterates; the grey scale measures the escape time), with *analytically superimposed* the cardioid $c = \lambda/2 - \lambda^2/4$, $|\lambda| = 1$ (red, Theorem 12.7) and the circle $|c+1| = 1/4$ of the period-2 bulb (blue, Theorem 12.8), which touch each other at the root point $c = -3/4$. Checks of the script `fig12_mandelbrot.py` (Appendix B, run on 17 September 2026): the area of $\mathcal{M}$ estimated by pixel counting equals 1.5168 on a 900×900 grid with 400 iterates and 1.5112 on 1500×1500 with 800 iterates — both *overestimates*, decreasing as the resolution increases as they must; the multiplier of the 2-cycle, $4(c+1)$, equals 0 at $c = -1$ and 1 at $c = -3/4$. ✓

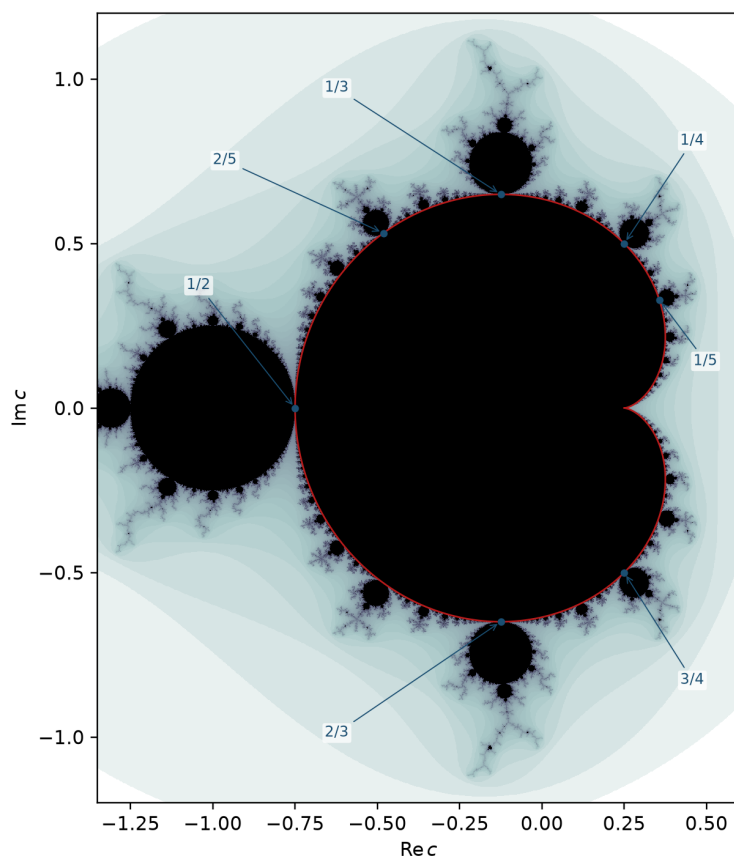


Figure 12.4: Enlargement on the cardioid (red boundary) with the root points $c_{p/q}$ (blue) computed in *closed form* from Definition 12.8 for $p/q = 1/2, 1/3, 2/3, 1/4, 3/4, 2/5, 1/5$: they fall exactly at the points where the corresponding bulb touches the cardioid, and they follow one another in the cyclic order of the rationals (Proposition 12.2). Values computed by the script `fig12_mandelbrot.py` (Appendix B, 17 September 2026): $c_{1/2} = -0.75$, $c_{1/3} = -0.125 + 0.6495190528i$, $c_{1/4} = 0.25 + 0.5i$, $c_{2/5} = -0.4817627458 + 0.5316567552i$. **Independent check:** for each bulb the script finds by Newton's method the superattracting centre by solving $g_c^q(0) = 0$ and verifies its period by direct iteration: $c = -1$ (period 2), $c = -0.1225611669 + 0.7448617666i$ (period 3, the rabbit), $c = -0.1565201668 + 1.0322471089i$ (period 4), $c = 0.3795135880 + 0.3349323056i$ (period 5); in every case the multiplier of the cycle turns out to be zero up to 3×10^{-14} . ✓

Definition 12.9: External rays and external angles. Let $c \in \mathcal{M}$ and let $\Phi_c : \mathbb{C} \setminus K_c \rightarrow \mathbb{C} \setminus \overline{\mathbb{D}}$ be the biholomorphism of Theorem 12.5. For $t \in \mathbb{R}/\mathbb{Z}$ the **external ray** of angle t is

$$R_t := \Phi_c^{-1}(\{r e^{2\pi i t} : r > 1\}) \subseteq \mathbb{C} \setminus K_c,$$

and the **potential** along R_t is $G_c = \log r$. We say that R_t **lands** at $z^* \in J_c$ if $\Phi_c^{-1}(r e^{2\pi i t}) \rightarrow z^*$ as $r \rightarrow 1^+$.

Theorem 12.9: The dynamics of the rays is the doubling map. For every $c \in \mathcal{M}$ and every $t \in \mathbb{R}/\mathbb{Z}$,

$$g_c(R_t) = R_{2t \bmod 1}, \quad g_c^{-1}(R_t) = R_{t/2} \cup R_{t/2+1/2},$$

and the point of R_t at potential s is sent to the point of R_{2t} at potential $2s$. In other words, the conjugacy Φ_c transforms g_c restricted to the basin of infinity into the map $w \mapsto w^2$, whose action on the angles is exactly the doubling map D of Chapter 7.

Proof. Let $z \in R_t$, that is, $\Phi_c(z) = r e^{2\pi i t}$ with $r > 1$. By the functional equation, $\Phi_c(g_c(z)) = \Phi_c(z)^2 = r^2 e^{2\pi i (2t)}$, which is the point of modulus $r^2 > 1$ and angle $2t \bmod 1$: hence $g_c(z) \in R_{2t}$ and its potential is $\log r^2 = 2 \log r = 2s$. Conversely, every point of R_{2t} is of the form $\Phi_c^{-1}(\varrho e^{2\pi i \cdot 2t})$ with $\varrho > 1$ and is the image of the point $\Phi_c^{-1}(\sqrt{\varrho} e^{2\pi i t}) \in R_t$: hence $g_c(R_t) = R_{2t}$. As for the preimage: $\Phi_c(z)^2 = r e^{2\pi i t}$ has the two solutions $\pm \sqrt{r} e^{\pi i t}$, of angles $t/2$ and $t/2 + 1/2$. $\square$

Corollary 12.2: Periodic rays and cycles of the doubling map. The ray R_t is periodic of period q for g_c (that is, $g_c^q(R_t) = R_t$) if and only if $2^q t \equiv t \pmod{1}$, that is, if and only if $t = k/(2^q - 1)$ for some integer k . The number of such angles is $2^q - 1$, and the number of orbits of period exactly q is

$$N(q) = \frac{1}{q} \sum_{d|q} \mu\left(\frac{q}{d}\right) (2^d - 1),$$

a formula already proved in Chapter 7 for the doubling map.

Proof. By Theorem 12.9, $g_c^q(R_t) = R_{2^q t}$, and two rays coincide if and only if their angles coincide; hence $g_c^q(R_t) = R_t \iff 2^q t \equiv t \iff (2^q - 1)t \in \mathbb{Z}$. The solutions in $\mathbb{R}/\mathbb{Z}$ are the $2^q - 1$ points $k/(2^q - 1)$. The count of the orbits of exact period is the Möbius inversion already established in Chapter 7. $\square$

Remark 12.9: Landing of the rational rays (imported). **Statement.** If $c \in \mathcal{M}$ and $t \in \mathbb{Q}/\mathbb{Z}$, the external ray R_t of K_c lands at a point $z^* \in J_c$; if t is periodic for the doubling map, z^* is a repelling or parabolic periodic point of g_c , and its period divides the period of t . In the parameter plane: every rational parameter ray lands on $\partial\mathcal{M}$, and the root point $c_{p/q}$ of the p/q bulb (with $q \geq 2$) is the landing point of exactly two parameter rays, of rational angles with denominator $2^q - 1$. Imported from A. Douady, J. H. Hubbard, Orsay Notes, and from J. Milnor, *Astérisque* **261** (2000), 277–333 (where the result “every parabolic point $c \neq 1/4$ is the landing point of exactly two rays” is proved); verified on 17 September 2026. **What is assumed:** the Douady–Hubbard landing theorem (which in turn uses a length-area estimate, or Fatou’s theorem on radial limits), and the connectedness of $\mathcal{M}$ (Remark 12.6).

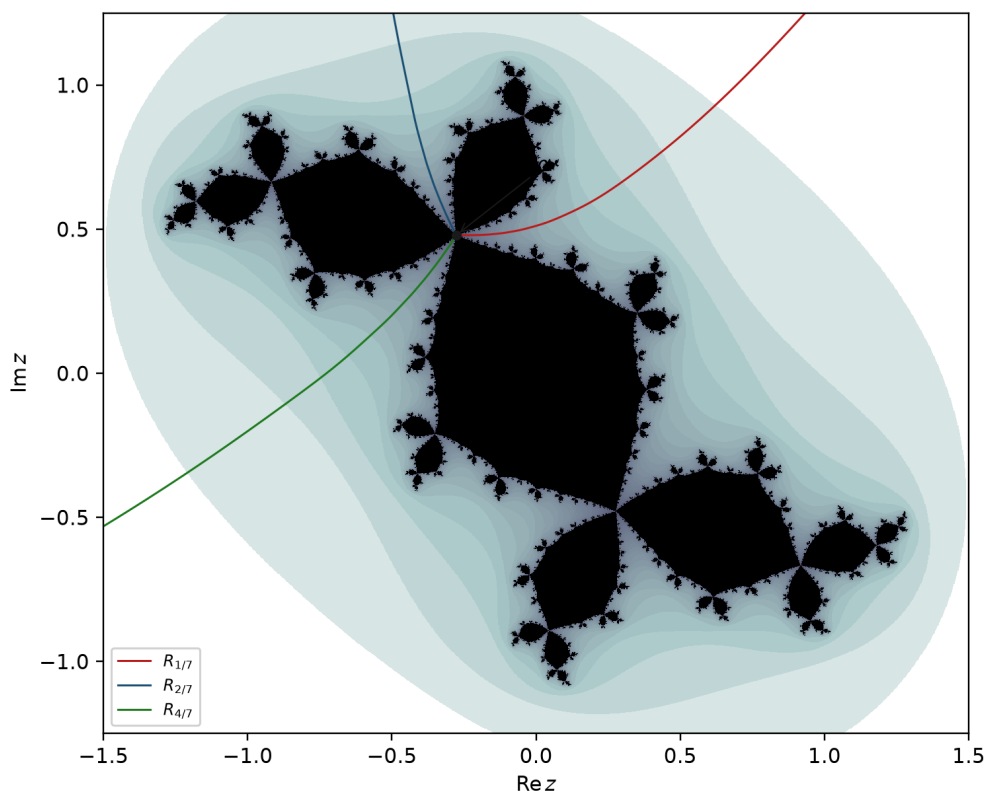


Figure 12.5: The three external rays of angles $1/7$, $2/7$, $4/7$ — a periodic orbit of period 3 of the doubling map, $1/7 \mapsto 2/7 \mapsto 4/7 \mapsto 8/7 \equiv 1/7$ — on the *Douady rabbit* K_c with $c = -0.1225611669 + 0.7448617666i$. The three rays land at the same point α , the repelling fixed point where the three “ears” touch. Construction by the script `fig12_julia.py` (Appendix B, 17 September 2026): the point of R_t at potential s is obtained by solving with Newton’s method $g_c^n(z) = \exp(s 2^n) e^{2\pi i(2^n p \bmod q)/q}$, with n chosen so that the arrival potential is about 12 (the doubled angle is computed with exact integers, not in floating point). **Checks:** the two fixed points are $z_+ = 1.2763376236 - 0.4797279843i$ (multiplier of modulus 2.727) and $\alpha = z_- = -0.2763376236 + 0.4797279843i$ (multiplier of modulus 1.10725141 and argument 0.33317563 turns, that is, about $1/3$: the rotation number of the bulb, Remark 12.8); at the final potential 2.9×10^{-14} the three rays are at distances from α of 6.43×10^{-3} , 6.41×10^{-3} , 6.44×10^{-3} respectively. The relation $g_c(R_{1/7}) = R_{2/7}$ of Theorem 12.9 is verified on 1100 points with median discrepancy 7.9×10^{-17} and maximum 1.5×10^{-8} . ✓

Worked example — assigning the rays to the bulbs Problem. Determine which orbits of the doubling map correspond to the period-3 bulbs of $\mathcal{M}$, and check the consistency with the count of Corollary 12.2.

Step 1 — the candidate angles. By Corollary 12.2 the angles of period dividing 3 are $t = k/7$, $k = 0, \dots, 6$. The angle $t = 0$ is fixed. Six angles remain, which group into two orbits of period exactly 3:

$$\frac{1}{7} \mapsto \frac{2}{7} \mapsto \frac{4}{7} \mapsto \frac{1}{7} \quad \text{and} \quad \frac{3}{7} \mapsto \frac{6}{7} \mapsto \frac{5}{7} \mapsto \frac{3}{7}$$

(one checks: $2 \cdot 3 = 6$, $2 \cdot 6 = 12 \equiv 5$, $2 \cdot 5 = 10 \equiv 3 \pmod{7}$).

Step 2 — check with the Möbius formula. $N(3) = \frac{1}{3}[\mu(3)(2^1 - 1) + \mu(1)(2^3 - 1)] = \frac{1}{3}[(-1) \cdot 1 + 1 \cdot 7] = \frac{1}{3} \cdot 6 = 2$: exactly two orbits, as found by hand. ✓

Step 3 — the two bulbs. The root points of the period-3 bulbs are, by Definition 12.8,

$$c_{1/3} = \frac{e^{2\pi i/3}}{2} - \frac{e^{4\pi i/3}}{4} = \frac{-\frac{1}{2} + \frac{\sqrt{3}}{2}i}{2} - \frac{-\frac{1}{2} - \frac{\sqrt{3}}{2}i}{4} = -\frac{1}{4} + \frac{\sqrt{3}}{4}i + \frac{1}{8} + \frac{\sqrt{3}}{8}i = -\frac{1}{8} + \frac{3\sqrt{3}}{8}i,$$

and $c_{2/3} = \overline{c_{1/3}}$. Numerically $3\sqrt{3}/8 = 0.649519052838\dots$, in agreement with the value $-0.125 + 0.6495190528i$ printed by the script. ✓

Step 4 — the assignment. By Remark 12.9, each root point is the landing point of exactly two parameter rays with denominator $2^3 - 1 = 7$: to the upper bulb $1/3$ belong the angles $1/7$ and $2/7$, to the lower one $2/3$ the angles $5/7$ and $6/7$.

Step 5 — the dynamical verification. In the *dynamical* plane of the centre of the $1/3$ bulb (the rabbit), the three rays $1/7$, $2/7$, $4/7$ all land at the repelling fixed point α : Figure 12.5 shows it, and the numerical check gives distances $\approx 6.4 \times 10^{-3}$ at potential 2.9×10^{-14} . The multiplier at α has argument 0.33317563 turns, that is, $1/3$ up to 1.6×10^{-4} : the rotation number of the bulb can be read off *in the multiplier*. ✓

Moral. Counting the period- q bulbs of $\mathcal{M}$ amounts to counting the period- q orbits of the doubling map on the circle: the binary combinatorics of Chapter 7 enumerates geometric structures of the parameter plane. □

12.8. Siegel's theorem

We come to the arithmetic heart of the chapter. A fixed point with multiplier $\lambda = e^{2\pi i\vartheta}$, ϑ irrational, is *indifferent*: the derivative neither contracts nor expands. What does the dynamics do? If g_c were linearisable, it would rotate by ϑ on a family of invariant curves — the behaviour of the rigid rotations of Chapter 3. Remark 12.3 has shown that the obstacle consists of the small divisors $\lambda^{k-1} - 1$. Siegel proved in 1942 that if ϑ is Diophantine the obstacle can be overcome.

Remark 12.10: The reduction to the form $\lambda w + w^2$. Let z_0 be a fixed point of g_c with multiplier $\lambda = 2z_0$. In the coordinate $w = z - z_0$ we have

$$g_c(z) - z_0 = (w + z_0)^2 + c - z_0 = w^2 + 2z_0w + (z_0^2 + c - z_0) = \lambda w + w^2,$$

because $z_0^2 + c = z_0$. Hence every fixed point of every member of the quadratic family is, up to translation, the fixed point 0 of

$$f_\lambda(w) := \lambda w + w^2.$$

Siegel's theorem for the quadratic family is therefore, without loss of generality, a theorem about f_λ . (For a general holomorphic germ $f(w) = \lambda w + \sum_{k \geq 2} a_k w^k$ the statement and the proof are analogous, with sums over k in place of the single $k = 2$: see the primary source.)

Definition 12.10: Diophantine numbers in multiplicative form. Let $\lambda = e^{2\pi i\vartheta}$ with $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$ and set

$$\omega_m := |\lambda^m - 1| \quad (m \geq 1).$$

We say that λ satisfies the **multiplicative Diophantine condition** of exponent $\mu \geq 1$ if there exists $C > 0$ with

$$\omega_m \geq \frac{C}{m^\mu} \quad \text{for every } m \geq 1.$$

Lemma 12.3: Classical Diophantine $\Rightarrow$ multiplicative Diophantine. Let ϑ be Diophantine of exponent $\tau \geq 0$ in the sense of Chapter 3, that is, let there exist $\gamma > 0$ with $\|m\vartheta\| \geq \gamma m^{-(1+\tau)}$ for every $m \geq 1$, where $\|x\|$ is the distance of x from the nearest integer. Then $\lambda = e^{2\pi i\vartheta}$ satisfies the multiplicative Diophantine condition with $\mu = 1 + \tau$ and $C = 4\gamma$.

Proof. We have $\omega_m = |e^{2\pi im\vartheta} - 1| = 2|\sin(\pi m\vartheta)| = 2\sin(\pi\|m\vartheta\|)$, where the last equality holds because $|\sin(\pi x)|$ depends only on $\|x\|$ and $\pi\|m\vartheta\| \in [0, \pi/2]$. By the concavity of $\sin$ on $[0, \pi/2]$ we have $\sin u \geq (2/\pi)u$ for $u \in [0, \pi/2]$, hence

$$\omega_m \geq 2 \cdot \frac{2}{\pi} \cdot \pi\|m\vartheta\| = 4\|m\vartheta\| \geq \frac{4\gamma}{m^{1+\tau}}.$$

□

Lemma 12.4: The formal solution and its majorant. Let λ not be a root of unity and $f_\lambda(w) = \lambda w + w^2$. There exists one and only one formal series $h(\zeta) = \zeta + \sum_{n \geq 2} h_n \zeta^n$ which satisfies formally

$$f_\lambda(h(\zeta)) = h(\lambda\zeta),$$

and its coefficients are determined by the recursion

$$h_1 = 1, \quad h_n = \frac{1}{\lambda(\lambda^{n-1} - 1)} \sum_{\substack{p+q=n \\ p, q \geq 1}} h_p h_q \quad (n \geq 2).$$

Moreover, setting $M_1 := 1$ and

$$M_n := \frac{1}{\omega_{n-1}} \sum_{\substack{p+q=n \\ p, q \geq 1}} M_p M_q \quad (n \geq 2),$$

we have $|h_n| \leq M_n$ for every n , and the tree formula holds

$$M_n = \sum_{T \in \mathcal{T}_n} \prod_{v \in \text{int}(T)} \frac{1}{\omega_{n(v)-1}}, \quad (12.1)$$

where $\mathcal{T}_n$ is the set of rooted binary trees with n leaves, $\text{int}(T)$ the set of its internal vertices and $n(v)$ the number of leaves of the subtree rooted at v . The number of trees is the Catalan number $|\mathcal{T}_n| = \text{Cat}_{n-1} \leq 4^n$.

Proof. Recursion. Substituting $h(\zeta) = \sum_{n \geq 1} h_n \zeta^n$ with $h_1 = 1$ into the equation $\lambda h(\zeta) + h(\zeta)^2 = h(\lambda\zeta)$ and comparing the coefficients of ζ^n :

$$\lambda h_n + \sum_{\substack{p+q=n \\ p, q \geq 1}} h_p h_q = \lambda^n h_n \quad \Longleftrightarrow \quad h_n(\lambda^n - \lambda) = \sum_{\substack{p+q=n \\ p, q \geq 1}} h_p h_q.$$

For $n = 1$ the identity is $\lambda h_1 = \lambda h_1$, always true. For $n \geq 2$, $\lambda^n - \lambda = \lambda(\lambda^{n-1} - 1) \neq 0$ because $\lambda \neq 0$ and λ is not a root of unity: the recursion determines h_n uniquely as a function of $h_1, \dots, h_{n-1}$ (only indices $p, q \leq n-1$ appear in the sum). Existence and uniqueness of the formal series follow by induction.

Majorant. Since $|\lambda| = 1$, $|\lambda^n - \lambda| = |\lambda^{n-1} - 1| = \omega_{n-1}$. Hence $|h_n| \leq \omega_{n-1}^{-1} \sum_{p+q=n} |h_p| |h_q|$. By strong induction: $|h_1| = 1 = M_1$; assuming $|h_p| \leq M_p$ for $p < n$,

$$|h_n| \leq \frac{1}{\omega_{n-1}} \sum_{p+q=n} |h_p| |h_q| \leq \frac{1}{\omega_{n-1}} \sum_{p+q=n} M_p M_q = M_n.$$

Tree formula. By induction on n . For $n = 1$ the tree is the single leaf, and the empty product equals $1 = M_1$. For $n \geq 2$, every binary tree T with n leaves decomposes in a unique way into the root (an internal vertex with $n(\text{root}) = n$, contributing the factor $1/\omega_{n-1}$) and the ordered pair of its two subtrees, with p and q leaves, $p + q = n$, $p, q \geq 1$. Hence

$$\sum_{T \in \mathcal{T}_n} \prod_{v \in \text{int}(T)} \frac{1}{\omega_{n(v)-1}} = \frac{1}{\omega_{n-1}} \sum_{p+q=n} \left(\sum_{T_1 \in \mathcal{T}_p} \prod \dots \right) \left(\sum_{T_2 \in \mathcal{T}_q} \prod \dots \right) = \frac{1}{\omega_{n-1}} \sum_{p+q=n} M_p M_q = M_n.$$

The count $|\mathcal{T}_n| = \text{Cat}_{n-1} = \frac{1}{n} \binom{2n-2}{n-1}$ is the Catalan recursion $\text{Cat}_{n-1} = \sum_{p+q=n} \text{Cat}_{p-1} \text{Cat}_{q-1}$, and $\text{Cat}_{n-1} \leq \binom{2n-2}{n-1} \leq 4^{n-1} \leq 4^n$. $\square$

Lemma 12.5: Siegel's arithmetic lemma (imported). Let $\lambda = e^{2\pi i \vartheta}$ with $\omega_m = |\lambda^m - 1| \geq C m^{-\mu}$ for every $m \geq 1$ ($C > 0$, $\mu \geq 1$). Then there exists a constant $\Gamma = \Gamma(C, \mu) \geq 1$ such that, for every $n \geq 1$ and for every rooted binary tree T with n leaves,

$$\prod_{v \in \text{int}(T)} \frac{1}{\omega_{n(v)-1}} \leq \Gamma^n.$$

Remark 12.11: Status of Lemma 12.5. This is the only **imported** statement in the proof of Siegel's theorem, and it is precisely the point at which Siegel overcame the difficulty. A naive majorisation does not suffice: applying $\omega_m \geq C m^{-\mu}$ term by term to a “caterpillar” tree (in which the values $n(v) - 1$ are $n - 1, n - 2, \dots, 1$) one would obtain only $\prod \leq C^{-(n-1)} ((n-1)!)^\mu$, which grows too fast and would give zero radius of convergence. The decisive observation is that the small divisors are rare and separated: if $\omega_p < \delta$ and $\omega_q < \delta$ with $p < q$ then $\omega_{q-p} = |\lambda^q - \lambda^p| \leq \omega_p + \omega_q < 2\delta$, hence $C(q-p)^{-\mu} < 2\delta$, that is, $q-p > (C/2\delta)^{1/\mu}$: the indices with a small divisor are separated by a distance that tends to infinity as the threshold tends to zero. Siegel exploits this separation by means of a delicate combinatorial induction on the tree. Imported from C. L. Siegel, Iteration of analytic functions, *Ann. of Math. (2)* **43** (1942), 607–612 (the majorisation occupies the central pages of the paper), and from M. Abate, Discrete holomorphic local dynamical systems, in *Holomorphic Dynamical Systems*, LNM 1998, Springer, 2010, pp. 1–55, §4, where it is reproduced in full in modern notation. Verified on 17 September 2026. **What is assumed exactly:** only the statement of Lemma 12.5, and nothing more; all the rest of Section 12.8 is proved here.

Theorem 12.10: Siegel's theorem (1942), quadratic case. Let $\lambda = e^{2\pi i \vartheta}$ with ϑ Diophantine. Then $f_\lambda(w) = \lambda w + w^2$ is linearisable at 0: there exist $r > 0$ and a biholomorphism h from $\mathbb{D}(0, r)$ onto a neighbourhood of 0, with $h(0) = 0$ and $h'(0) = 1$, such that

$$f_\lambda(h(\zeta)) = h(\lambda\zeta) \quad \text{for } |\zeta| < r.$$

Equivalently: for every $c = \mathfrak{c}(\lambda) = \lambda/2 - \lambda^2/4$ with ϑ Diophantine, the fixed point $z_0 = \lambda/2$ of g_c is linearisable, and g_c is conjugate there to the rigid rotation of angle $2\pi\vartheta$.

Proof. By Lemma 12.3, λ satisfies the multiplicative Diophantine condition with certain $C > 0$, $\mu \geq 1$. Let $h(\zeta) = \zeta + \sum_{n \geq 2} h_n \zeta^n$ be the formal series of Lemma 12.4. Combining (12.1) with Lemma 12.5 and with the majorisation of the number of trees:

$$|h_n| \leq M_n = \sum_{T \in \mathcal{T}_n} \prod_{v \in \text{int}(T)} \frac{1}{\omega_{n(v)-1}} \leq |\mathcal{T}_n| \cdot \Gamma^n \leq 4^n \Gamma^n = (4\Gamma)^n.$$

Hence $\limsup_n |h_n|^{1/n} \leq 4\Gamma < \infty$: the series h has radius of convergence $r_0 \geq 1/(4\Gamma) > 0$ and defines a holomorphic function on $\mathbb{D}(0, r_0)$ with $h(0) = 0$, $h'(0) = 1$. By the holomorphic inverse function theorem, h is a biholomorphism of a disc $\mathbb{D}(0, r)$, $0 < r \leq r_0$, onto an open neighbourhood of 0.

It remains to observe that the formal identity $f_\lambda(h(\zeta)) = h(\lambda\zeta)$ is an identity between holomorphic functions on $\mathbb{D}(0, r)$: both sides are power series convergent on $\mathbb{D}(0, r)$ (the left-hand side because h converges there and f_λ is a polynomial; the right-hand side because $|\lambda| = 1$), and their coefficients coincide by construction. Two power series with the same coefficients define the same holomorphic function.

The reformulation for g_c is Remark 12.10: setting $z = w + z_0$ with $z_0 = \lambda/2$ and $c = \mathfrak{c}(\lambda)$ (Theorem 12.7), the map $\tilde{h}(\zeta) := z_0 + h(\zeta)$ conjugates $\zeta \mapsto \lambda\zeta$ with g_c . $\square$

Definition 12.11: Siegel disc. Let z_0 be an irrationally neutral fixed point of g_c with multiplier λ , and suppose g_c linearisable at z_0 . Let r^* be the supremum of the $r > 0$ for which there exists a holomorphic injective $h : \mathbb{D}(0, r) \rightarrow \mathbb{C}$ with $h(0) = z_0$, $h'(0) = 1$ and $g_c(h(\zeta)) = h(\lambda\zeta)$. The **Siegel disc** of z_0 is $\Delta := h(\mathbb{D}(0, r^*))$. The definition is well posed: by Lemma 12.4 the coefficients of h are uniquely determined, hence two linearisations coincide on the intersection of their domains and glue into a single one on $\mathbb{D}(0, r^*)$.

Proposition 12.3: Structure of the Siegel disc. Let Δ be the Siegel disc of z_0 . Then: (i) Δ is open, connected, simply connected, and $g_c(\Delta) = \Delta$; (ii) $\Delta \subseteq \text{int}(K_c)$ and $z_0 \in \Delta$; (iii) for every $0 < \rho < r^*$ the analytic Jordan curve $\gamma_\rho := h(\{|\zeta| = \rho\})$ is invariant under g_c , and every orbit starting on it is **dense** in γ_ρ ; (iv) g_c has no periodic points in Δ other than z_0 ; (v) Δ does not contain the critical point 0.

Proof. (i) Δ is the image of a disc under an injective holomorphic map, hence open, connected and simply connected; the invariance follows from $g_c(h(\zeta)) = h(\lambda\zeta)$ and from the fact that $\zeta \mapsto \lambda\zeta$ is a bijection of $\mathbb{D}(0, r^*)$ onto itself (because $|\lambda| = 1$).

(ii) For every $w = h(\zeta) \in \Delta$ we have $g_c^n(w) = h(\lambda^n \zeta)$ for every $n \geq 0$; since $|\lambda^n \zeta| = |\zeta| < r^*$, all the iterates lie in the compact set $h(\{|\zeta| \leq |\zeta|\}) \subset \mathbb{C}$: the orbit is bounded, that is, $w \in K_c$. Since Δ is open, $\Delta \subseteq \text{int}(K_c)$. Finally $z_0 = h(0) \in \Delta$.

(iii) γ_ρ is the injective holomorphic image of a circle, hence an analytic Jordan curve; it is invariant because $h(\{|\zeta| = \rho\})$ is sent by g_c to $h(\lambda\{|\zeta| = \rho\}) = h(\{|\zeta| = \rho\})$. In the coordinate ζ the dynamics on γ_ρ is the rotation $\zeta \mapsto \lambda\zeta$ of angle $2\pi\vartheta$ with ϑ irrational: by the density theorem for the orbits of irrational rotations (Volume IV, Chapter 3) every orbit is dense in the circle $\{|\zeta| = \rho\}$, and h , being a homeomorphism onto γ_ρ , carries the density over.

(iv) If $g_c^n(w) = w$ with $w = h(\zeta) \in \Delta$, then $h(\lambda^n \zeta) = h(\zeta)$ and, by the injectivity of h , $\lambda^n \zeta = \zeta$; since $\lambda^n \neq 1$ (ϑ irrational), we have $\zeta = 0$, that is, $w = z_0$.

(v) If we had $0 \in \Delta$, let $0 = h(\zeta_0)$. The map $\zeta \mapsto h^{-1}(g_c(h(\zeta))) = \lambda\zeta$ is injective on $\mathbb{D}(0, r^*)$; but g_c is not injective on any neighbourhood of its critical point 0 (for every ε , $g_c(\pm\varepsilon)$ coincide),

and h is a homeomorphism: contradiction. □

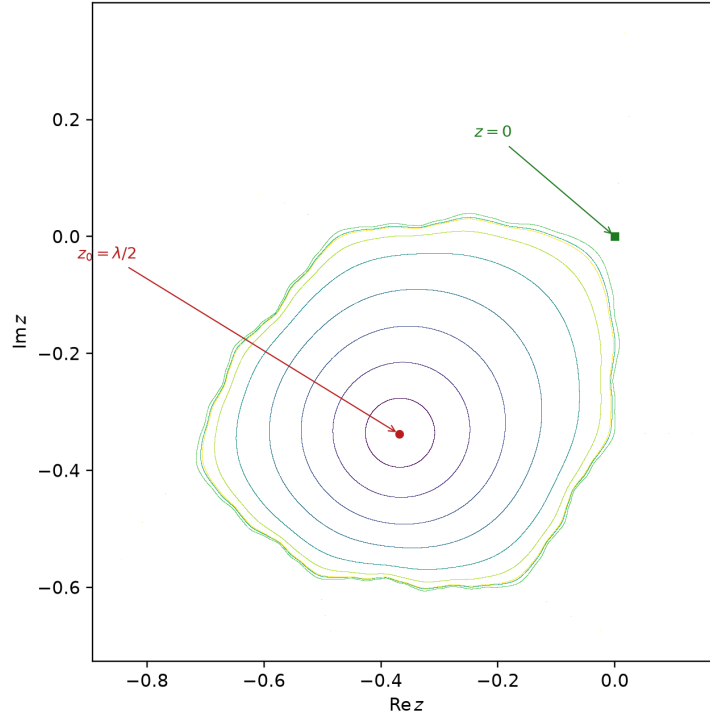


Figure 12.6: The **golden Siegel disc**: $\vartheta = (\sqrt{5} - 1)/2 = 0.618033988749895$ (continued fraction $[0; 1, 1, 1, \dots]$, the number worst approximable by rationals, hence Diophantine of exponent $\tau = 0$), $\lambda = e^{2\pi i \vartheta}$, $c = \mathfrak{c}(\lambda) = -0.390540870218 - 0.586787907347i$, fixed point $z_0 = \lambda/2 = -0.368684439039 - 0.337745147131i$. Nine orbits of 20 000 points each are drawn, with initial datum $z_0 + r$ for $r = 0.06; 0.12; \dots; 0.50$: each one densely fills an invariant curve γ_r (Proposition 12.3(iii)). The green square marked $z = 0$ is the critical point of g_c ; the red circle is the fixed point z_0 . **Checks** of the script `fig12_locale.py` (Appendix B, 17 September 2026): $|\lambda| = 1$ and $|z_0^2 + c - z_0| = 0$ exactly in double precision; the rotation number measured as the mean increment of $\arg(z_n - z_0)$ over 60 000 iterates equals 0.618034144 for $r = 0.06$ and 0.618034010 for $r = 0.46$, with discrepancy from ϑ of 1.6×10^{-7} and 2.2×10^{-8} respectively; the orbit with $r = 0.50$ is still bounded while the one with $r = 0.54$ escapes after 16 iterates. Note that $|z_0| = 1/2$ exactly: the critical point is at exactly the same distance from the fixed point as the radius of the outermost bounded orbit found — the numerical illustration of Herman’s theorem cited in Remark 12.14. ✓

Example 12.4: Why the golden number is the “safest” choice. For $\vartheta = (\sqrt{5} - 1)/2$ the denominators of the convergents are the Fibonacci numbers, and $\lim_m \inf m \|m\vartheta\| = 1/\sqrt{5}$ holds: the Diophantine constant is the best possible, that is, $\tau = 0$ in the sense of Chapter 3.

First computation (theoretical). The uniform constant is $\gamma := \inf_{m \geq 1} m \|m\vartheta\|$. Lemma 12.3 then gives $\mu = 1$ and $C = 4\gamma$.

Second computation (direct, independent). We compute $m \|m\vartheta\|$ at the first Fibonacci indices, where the minimum is attained. With $\vartheta = 0.6180339887$:

m	1	2	3	5	8	13
$\ m\vartheta\ $	0.381966	0.236068	0.145898	0.090170	0.055728	0.034442
$m \ m\vartheta\ $	0.381966	0.472136	0.437694	0.450850	0.445825	0.447744

The values oscillate around $1/\sqrt{5} = 0.447214$, alternating below and above as the theory of convergents predicts, and the absolute minimum is the one at $m = 1$: $\gamma = 0.381966 = 1 - 1/\varphi$, whence $C = 4\gamma = 1.527864$. ✓

The two computations agree (the first gives the form of the constant, the second its value), and they confirm that the golden number is the case in which the small divisors are *least* small: no other irrational has a $\liminf m\|m\vartheta\|$ greater than $1/\sqrt{5}$ (Hurwitz's theorem, Chapter 3).

12.9. Cremer's theorem

Lemma 12.6: Product of the periodic points. Let $f_\lambda(w) = \lambda w + w^2$ with λ not a root of unity, and let $n \geq 1$. The polynomial $f_\lambda^n(w) - w$ is monic of degree 2^n , vanishes at $w = 0$ with multiplicity 1, and its $2^n - 1$ non-zero roots $w_1, \dots, w_{2^n-1}$ (counted with multiplicity) satisfy

$$|w_1 w_2 \cdots w_{2^n-1}| = |\lambda^n - 1| = \omega_n.$$

In particular $\min_j |w_j| \leq \omega_n^{1/(2^n-1)}$.

Proof. f_λ is monic of degree 2, hence f_λ^n is monic of degree 2^n (by induction: the leading coefficient of $f_\lambda \circ f_\lambda^{n-1}$ is the square of that of f_λ^{n-1} , that is, 1), and the same holds for $P(w) := f_\lambda^n(w) - w$ since $2^n \geq 2$. Moreover $f_\lambda^n(0) = 0$ and $(f_\lambda^n)'(0) = \lambda^n$, hence

$$P(w) = (\lambda^n - 1)w + O(w^2),$$

and since $\lambda^n \neq 1$ the zero at 0 is simple. We write $P(w) = wQ(w)$ with Q monic of degree $2^n - 1$ and $Q(0) = \lambda^n - 1$. If $w_1, \dots, w_{2^n-1}$ are the roots of Q , the relation between coefficients and roots gives

$$Q(0) = (-1)^{2^n-1} w_1 \cdots w_{2^n-1} = -w_1 \cdots w_{2^n-1}$$

(the exponent $2^n - 1$ is odd for $n \geq 1$), hence $|w_1 \cdots w_{2^n-1}| = |Q(0)| = |\lambda^n - 1| = \omega_n$. If all the roots had modulus $> \omega_n^{1/(2^n-1)}$, the product of the moduli would be $> \omega_n$: absurd. □

Definition 12.12: Cremer numbers. An irrational ϑ is said to be of **Cremer type** (for the quadratic family) if, setting $\lambda = e^{2\pi i \vartheta}$,

$$\liminf_{n \rightarrow \infty} |\lambda^n - 1|^{1/2^n} = 0,$$

that is, if for every $\varrho \in (0, 1)$ there exist infinitely many n with $\omega_n < \varrho^{2^n}$.

Theorem 12.11: Cremer's theorem (1938). (i) If ϑ is of Cremer type, then $f_\lambda(w) = \lambda w + w^2$ is not linearisable at 0: there exist no disc $\mathbb{D}(0, r)$, $r > 0$, and no injective holomorphic h with $h(0) = 0$, $h'(0) = 1$ and $f_\lambda \circ h = h \circ (\lambda \cdot)$. (ii) The set of the ϑ of Cremer type contains a dense G_δ of $\mathbb{R} \setminus \mathbb{Q}$. In particular the parameters $c = c(e^{2\pi i \vartheta})$ for which g_c has a non-linearisable fixed point are dense (in the Baire sense, “generic”) on the boundary of the cardioid.

Proof. (i) Suppose by contradiction that a linearisation h exists on $\mathbb{D}(0, r)$, and let $\Omega := h(\mathbb{D}(0, r))$, an open set containing 0. Let $\varrho \in (0, 1)$ be such that $\mathbb{D}(0, \varrho) \subseteq \Omega$ — it exists because Ω is an open neighbourhood of 0; shrinking it, we also assume $\varrho < 1$.

No non-trivial periodic point in Ω . Let $w \in \Omega$, $w \neq 0$, with $f_\lambda^n(w) = w$ for some $n \geq 1$. Setting $\zeta := h^{-1}(w) \in \mathbb{D}(0, r) \setminus \{0\}$, we have by induction $f_\lambda^k(h(\zeta)) = h(\lambda^k \zeta)$ for every $k \geq 0$: indeed $|\lambda^k \zeta| = |\zeta| < r$, hence the right-hand side is always defined. Then $h(\lambda^n \zeta) = f_\lambda^n(w) = w = h(\zeta)$, and the injectivity of h gives $\lambda^n \zeta = \zeta$, that is, $\lambda^n = 1$: impossible because ϑ is irrational.

But the periodic points accumulate at 0. By the Cremer condition there exist infinitely many n with $\omega_n < \varrho^{2^n}$, hence, by Lemma 12.6,

$$\min_j |w_j| \leq \omega_n^{1/(2^n-1)} < \varrho^{2^n/(2^n-1)} < \varrho$$

(the last inequality because $\varrho < 1$ and the exponent $2^n/(2^n-1) > 1$). Hence for each of those n there exists a point $w \neq 0$ with $f_\lambda^n(w) = w$ and $|w| < \varrho$, that is, $w \in \mathbb{D}(0, \varrho) \subseteq \Omega$. This contradicts the previous step.

(ii) For $k \geq 1$ and $N \geq 1$ we set

$$U_{k,N} := \left\{ \vartheta \in \mathbb{R} : \exists n \geq N \text{ with } |e^{2\pi i n \vartheta} - 1| < k^{-2^n} \right\}.$$

$U_{k,N}$ is open: it is the union, over $n \geq N$, of the preimages of the open set $\{|u - 1| < k^{-2^n}\}$ under the continuous map $\vartheta \mapsto e^{2\pi i n \vartheta}$. $U_{k,N}$ is dense: let $I \subseteq \mathbb{R}$ be a non-empty open interval and let $p/q \in I$ be a rational (the rationals are dense). We choose $n \geq N$ a multiple of q : then $e^{2\pi i n (p/q)} = 1$, that is, the continuous function $\vartheta \mapsto |e^{2\pi i n \vartheta} - 1|$ vanishes at p/q , and by continuity it is $< k^{-2^n}$ on a neighbourhood of p/q ; that neighbourhood, intersected with I , is non-empty and contained in $U_{k,N}$.

Hence $\mathcal{C} := (\bigcap_{k \geq 1} \bigcap_{N \geq 1} U_{k,N}) \setminus \mathbb{Q}$ is the intersection of a countable family of dense open sets with the dense G_δ $\mathbb{R} \setminus \mathbb{Q}$: by Baire's theorem it is a dense G_δ of $\mathbb{R}$. Every $\vartheta \in \mathcal{C}$ is irrational and satisfies, for every k and every N , the existence of $n \geq N$ with $\omega_n < k^{-2^n}$: taking $\varrho = 1/k$ one recognises the Cremer condition. The conclusion about the parameters follows by applying the homeomorphism $\vartheta \mapsto \mathfrak{c}(e^{2\pi i \vartheta})$ of Proposition 12.2, which sends dense sets to dense sets. $\square$

Remark 12.12: Diophantine and Cremer: two sets both “large”, in opposite senses. The Diophantine numbers (Siegel's theorem) have full Lebesgue measure in $[0, 1]$ — the set of the ϑ with $\|m\vartheta\| \geq \gamma m^{-(1+\tau)}$ eventually has measure $\rightarrow 1$ as $\gamma \rightarrow 0$, by the usual computation $\sum_m \sum_p 2\gamma m^{-(1+\tau)} < \infty$ when $\tau > 0$ (Chapter 3). The Cremer numbers, by contrast, form a set of zero measure (they are contained in a lim sup of sets of summable measure) but generic in the Baire sense. Hence: almost every ϑ gives a Siegel disc, but the typical ϑ in the topological sense gives a Cremer point. It is the same opposition between “almost everywhere” and “generic” already encountered in connection with Denjoy's counterexamples (Chapter 5).

12.10. Bryuno, Yoccoz and the boundary of the Siegel disc

Definition 12.13: Bryuno condition. Let ϑ be irrational with continued fraction $[0; a_1, a_2, \dots]$ and convergents p_k/q_k (Chapter 3). We say that ϑ satisfies the **Bryuno condition** if

$$B(\vartheta) := \sum_{k \geq 0} \frac{\log q_{k+1}}{q_k} < +\infty.$$

Every Diophantine number satisfies the Bryuno condition. Indeed, if $\|q_k \vartheta\| \geq \gamma q_k^{-(1+\tau)}$ and one uses the classical estimate $\|q_k \vartheta\| < 1/q_{k+1}$ for the convergents (Chapter 3), one obtains

$q_{k+1} < \gamma^{-1} q_k^{1+\tau}$, hence

$$\sum_k \frac{\log q_{k+1}}{q_k} \leq \sum_k \frac{\log \gamma^{-1} + (1+\tau) \log q_k}{q_k} < \infty,$$

because the q_k grow at least like the Fibonacci numbers, that is, exponentially, and $\sum_k (\log q_k)/q_k$ converges. The converse is false.

Remark 12.13: Bryuno and Yoccoz: the optimal condition (imported). **Statement (a).** Bryuno’s theorem (1971). If ϑ satisfies the Bryuno condition and $\lambda = e^{2\pi i \vartheta}$, then every holomorphic germ $f(w) = \lambda w + O(w^2)$ is linearisable at 0. Since every Diophantine number is a Bryuno number, this generalises Theorem 12.10.

Statement (b). Yoccoz’s theorem (1995). For the quadratic family the Bryuno condition is also **necessary**: if ϑ does not satisfy $B(\vartheta) < \infty$, then $f_\lambda(w) = \lambda w + w^2$ is not linearisable at 0. Hence for f_λ linearisability is exactly equivalent to $B(\vartheta) < \infty$.

Imported from A. D. Bruno, “Analytic form of differential equations. I, II”, *Trudy Mosk. Mat. Obshch.* **25** (1971), 119–262 (Math-Net.Ru), for (a); and from J.-C. Yoccoz, “Théorème de Siegel, nombres de Bruno et polynômes quadratiques”, in *Petits diviseurs en dimension 1*, *Astérisque* **231** (1995), 3–88 (Numdam), for (b). Verified on 17 September 2026. **What is assumed:** for (a), the construction of the majorant with the denominators of the convergents in place of the indices; for (b), Yoccoz’s geometric construction (renormalisation and the Riemann surface of the coverings) and Pérez-Marco’s estimate on hedgehogs. No part of these two theorems is used anywhere else in the chapter.

Hierarchy. Diophantine $\Rightarrow$ Bryuno $\Rightarrow$ linearisable (for f_λ : $\Leftrightarrow$); non-Bryuno $\Rightarrow$ not linearisable; Cremer (Definition 12.12) $\Rightarrow$ non-Bryuno — the Cremer condition is much stronger than “non-Bryuno”, which explains why Theorem 12.11 has a proof of a few lines while Yoccoz’s occupies an entire volume of *Astérisque*.

Remark 12.14: The boundary of the Siegel disc (panorama). What does $\partial\Delta$ look like? Three results, all cited here only as a panorama and not used in what follows.

(a) Herman (1985): if ϑ is of bounded type (that is, the partial quotients a_k of its continued fraction are bounded — the golden number is, with $a_k \equiv 1$), then the critical point of g_c lies on the boundary of the Siegel disc. Source: M. R. Herman, “Are there critical points on the boundaries of singular domains?”, *Comm. Math. Phys.* **99** (1985), 593–612. Figure 12.6 illustrates numerically exactly this: for golden ϑ the critical point 0 is at distance $1/2$ from the fixed point $z_0 = \lambda/2$, and $1/2$ is exactly the radius (in the parametrisation used) beyond which the orbits escape.

(b) Petersen (1996): for ϑ of bounded type the Julia set J_c is locally connected and of zero Lebesgue measure. Source: C. L. Petersen, “Local connectivity of some Julia sets containing a circle with an irrational rotation”, *Acta Math.* **177** (1996), 163–224.

(c) Zhang (2011): for ϑ of bounded type the boundary $\partial\Delta$ is a quasicircle, that is, the image of the circle under a quasiconformal homeomorphism of the plane. This is the Douady–Sullivan conjecture in the bounded type case. Source: G. Zhang, “All bounded type Siegel disks of rational maps are quasi-disks”, *Invent. Math.* **185** (2011), 421–466.

Note the contrast with Chapter 6: there bounded type guarantees the smooth conjugacy of the circle map; here it guarantees the geometric regularity of the invariant curve. It is the same arithmetic acting in two different geometries.

12.11. Stability of J_c for small $|c|$: the deformed circle

This section proves in full that for small $|c|$ the Julia set is a Jordan curve, obtained by deforming the circle $J_0 = S^1$, and that the dynamics on it remains conjugate to doubling. It is the precise sense in which the circle “survives” a holomorphic perturbation.

Lemma 12.7: Proper maps of the disc of degree 2. *Let $F : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic, proper, of degree 2, with $F(0) = 0$ and free of critical points in $\mathbb{D} \setminus \{0\}$. Then $F(u) = e^{i\alpha}u^2$ for some $\alpha \in \mathbb{R}$. In particular $F^n(u) \rightarrow 0$ for every $u \in \mathbb{D}$.*

Proof. F is a Blaschke product. Since F is proper of degree 2, it has exactly two zeros $a_1, a_2 \in \mathbb{D}$ counted with multiplicity. Let

$$B(u) := \prod_{j=1}^2 \frac{u - a_j}{1 - \overline{a_j}u},$$

a finite Blaschke product: it is holomorphic on a neighbourhood of $\overline{\mathbb{D}}$ (the poles are at $1/\overline{a_j}$, of modulus > 1 , except in the case $a_j = 0$ in which the factor is u), it has $|B| = 1$ on $\partial\mathbb{D}$ and the same zeros as F with the same multiplicities. The function $\chi := F/B$ is therefore holomorphic and zero-free on $\mathbb{D}$. Since F is proper, $|F(u)| \rightarrow 1$ as $|u| \rightarrow 1^-$; and $|B(u)| \rightarrow 1$ as $|u| \rightarrow 1^-$: hence $|\chi(u)| \rightarrow 1$ as $|u| \rightarrow 1^-$. By the maximum principle applied to χ and to $1/\chi$ (both holomorphic on $\mathbb{D}$), $|\chi| \leq 1$ and $|1/\chi| \leq 1$, that is, $|\chi| \equiv 1$: a holomorphic function of constant modulus on a domain is constant, $\chi \equiv e^{i\alpha}$. Hence $F = e^{i\alpha}B$.

Determination of the zeros. $F(0) = 0$ forces at least one of the a_j to be 0; say $a_1 = 0$, so that

$$F(u) = e^{i\alpha} u \cdot \frac{u - a_2}{1 - \overline{a_2}u}.$$

Let us compute $F'(0)$: the derivative of the product at $u = 0$ equals $e^{i\alpha} [1 \cdot \frac{0-a_2}{1-0}] = -e^{i\alpha}a_2$. Since 0 is by hypothesis a critical point (it is the only one allowed, and a proper map of degree 2 of the disc has exactly one critical point counted with multiplicity, by the Riemann–Hurwitz formula; in any case, if $F'(0) \neq 0$ then F would have a critical point elsewhere in $\mathbb{D} \setminus \{0\}$, excluded by hypothesis), we have $F'(0) = 0$, hence $a_2 = 0$ and

$$F(u) = e^{i\alpha}u^2.$$

Finally $|F(u)| = |u|^2$ gives $|F^n(u)| = |u|^{2^n} \rightarrow 0$ for $|u| < 1$. □

Theorem 12.12: J_c is a Jordan curve for $|c| \leq 3/16$. *Let $|c| \leq 3/16 = 0.1875$. Then there exists a unique continuous map $h_c : S^1 \rightarrow \mathbb{C}$ with*

$$\sup_{|u|=1} |h_c(u) - u| \leq \frac{1}{4} \quad \text{and} \quad h_c(u^2) = h_c(u)^2 + c \quad \text{for every } |u| = 1.$$

It is injective and odd ($h_c(-u) = -h_c(u)$), its image $\Gamma_c := h_c(S^1)$ is a Jordan curve, and

$$\Gamma_c = J_c.$$

In particular g_c restricted to J_c is topologically conjugate, via h_c , to the doubling map $u \mapsto u^2$ of the circle, and K_c is the closure of the bounded component of $\mathbb{C} \setminus J_c$.

Proof. Step 1: the space and the operator. Let $\varepsilon := 1/4$ and

$$X := \left\{ \eta : S^1 \rightarrow \mathbb{C} \text{ continuous} : \|\eta - \text{id}\|_\infty \leq \varepsilon \right\},$$

closed in the uniform metric of the Banach space $C(S^1, \mathbb{C})$, hence complete. For $\eta \in X$ and $|u| = 1$ we have $|\eta(u)| \in [1 - \varepsilon, 1 + \varepsilon] = [3/4, 5/4]$, hence

$$|\eta(u^2) - c| \geq \frac{3}{4} - \frac{3}{16} = \frac{9}{16} > 0 :$$

the continuous function $u \mapsto \eta(u^2) - c$ does not vanish on S^1 . Its degree (winding number around 0) is 2: indeed $u \mapsto \eta(u^2) - c$ is homotopic, within the class of maps $S^1 \rightarrow \mathbb{C} \setminus \{0\}$, to $u \mapsto u^2$ by means of the homotopy $s \mapsto s(\eta(u^2) - c) + (1 - s)u^2$, which does not vanish because $|s(\eta(u^2) - c) + (1 - s)u^2 - u^2| = |s(\eta(u^2) - c - u^2)| \leq \varepsilon + |c| \leq \frac{1}{4} + \frac{3}{16} = \frac{7}{16} < 1 = |u^2|$ (Volume IV, Chapter 1, the “dog on a leash” criterion). Since the degree is *even*, a continuous square root exists: exactly two, opposite to each other (one lifts $u \mapsto \eta(u^2) - c$ along the covering $\zeta \mapsto \zeta^2$ of $\mathbb{C} \setminus \{0\}$, possible because the image of the fundamental group is $2\mathbb{Z}$). Of the two we choose the one closer to the identity and call it $T\eta$: that is, $T\eta$ is the unique continuous function with $(T\eta)^2(u) = \eta(u^2) - c$ and $\|T\eta - \text{id}\|_\infty < 1$. That only one of the two satisfies this last condition is seen at once: if ζ and $-\zeta$ were both at distance < 1 from u , we would have $2 = |2u| \leq |u - \zeta| + |u + \zeta| < 2$, absurd.

Step 2: T maps X into itself. Let $\eta \in X$, $|u| = 1$, $\zeta := T\eta(u)$ and $y := |\zeta - u|$; by the choice of the branch, $y < 1$. From $\zeta^2 - u^2 = \eta(u^2) - c - u^2$ and $\zeta^2 - u^2 = (\zeta - u)(\zeta + u)$ it follows that

$$y \cdot |\zeta + u| = |\eta(u^2) - u^2 - c| \leq \varepsilon + |c| \leq \frac{1}{4} + \frac{3}{16} = \frac{7}{16},$$

where we have used $|\eta(u^2) - u^2| \leq \varepsilon$ because $|u^2| = 1$ and $\eta \in X$. On the other hand $|\zeta + u| \geq |2u| - |\zeta - u| = 2 - y > 1 > 0$, hence

$$y(2 - y) \leq y|\zeta + u| \leq \frac{7}{16} \iff y^2 - 2y + \frac{7}{16} \geq 0 \iff y \leq \frac{1}{4} \text{ or } y \geq \frac{7}{4},$$

because the roots of $y^2 - 2y + 7/16$ are $1 \pm \sqrt{1 - 7/16} = 1 \pm \frac{3}{4}$. Since $y < 1$, there remains $y \leq 1/4 = \varepsilon$: hence $T\eta \in X$.

Step 3: T is a contraction. Let $\eta_1, \eta_2 \in X$ and $\zeta_i := T\eta_i(u)$. From $\zeta_1^2 - \zeta_2^2 = \eta_1(u^2) - \eta_2(u^2)$ and $|\zeta_1 + \zeta_2| \geq |2u| - |\zeta_1 - u| - |\zeta_2 - u| \geq 2 - 2\varepsilon = \frac{3}{2}$ it follows that

$$|T\eta_1(u) - T\eta_2(u)| = \frac{|\eta_1(u^2) - \eta_2(u^2)|}{|\zeta_1 + \zeta_2|} \leq \frac{2}{3} \|\eta_1 - \eta_2\|_\infty.$$

Hence T is a contraction of constant $2/3 < 1$ on the complete space X : by the contraction mapping theorem there exists a unique fixed point $h_c \in X$, and $h_c(u)^2 = h_c(u^2) - c$, that is, $h_c(u^2) = h_c(u)^2 + c = g_c(h_c(u))$, which is the desired equation. Uniqueness in X is that of the fixed point.

Step 4: h_c is odd. We show that *the operator T preserves oddness*, and we conclude by passing to the limit (oddness cannot be deduced directly from the fixed-point equation: the attempt is circular). Let $\eta \in X$ be odd and set $\sigma(u) := -(T\eta)(-u)$. Then σ is continuous and

$$\sigma(u)^2 = ((T\eta)(-u))^2 = \eta((-u)^2) - c = \eta(u^2) - c,$$

that is, σ is itself a continuous square root of $u \mapsto \eta(u^2) - c$; moreover $\|\sigma - \text{id}\|_\infty = \sup_u |-(T\eta)(-u) - u| = \sup_v |(T\eta)(v) - v| < 1$ (substituting $v = -u$). By the uniqueness established in Step 1 — of the two continuous square roots, only one is at distance less than 1 from id — we have $\sigma = T\eta$, that is, $(T\eta)(-u) = -(T\eta)(u)$: $T\eta$ is odd.

The function id is odd and lies in X ; by the contraction mapping theorem $h_c = \lim_n T^n(\text{id})$ uniformly, and every $T^n(\text{id})$ is odd by what we have just seen. A uniform (indeed, a pointwise) limit of odd functions is odd: $h_c(-u) = -h_c(u)$.

Step 5: h_c is injective. Let $u_1 \neq u_2$ in S^1 with $h_c(u_1) = h_c(u_2)$. Applying the functional equation repeatedly, $h_c(u_1^{2^n}) = g_c^n(h_c(u_1)) = g_c^n(h_c(u_2)) = h_c(u_2^{2^n})$ for every $n \geq 0$. We write

$u_j = e^{2\pi i t_j}$ with $t_1 \neq t_2$ in $\mathbb{R}/\mathbb{Z}$: by the **expansivity of the doubling map** (Chapter 7) there exists n such that the distance between $2^n t_1$ and $2^n t_2$ in $\mathbb{R}/\mathbb{Z}$ is $\geq 1/4$ — indeed, as long as that distance δ is $\leq 1/4$ it doubles exactly, and an iterated doubling brings it into $(1/4, 1/2]$. Setting $v_j := u_j^{2^n}$, the corresponding chord measures

$$|v_1 - v_2| = 2|\sin(\pi \text{ dist})| \geq 2\sin(\pi/4) = \sqrt{2} = 1.41421\dots$$

But then

$$0 = |h_c(v_1) - h_c(v_2)| \geq |v_1 - v_2| - 2\varepsilon \geq \sqrt{2} - \frac{1}{2} = 0.91421\dots > 0,$$

absurd. Hence h_c is injective; being continuous on a compact set, it is a homeomorphism of S^1 onto $\Gamma_c := h_c(S^1)$, which is therefore a Jordan curve.

Step 6: Γ_c is completely invariant. From the equation, $g_c(\Gamma_c) = \Gamma_c$ (because $u \mapsto u^2$ is surjective onto S^1). As for the preimage: let $z = h_c(u^2) \in \Gamma_c$; its two preimages under g_c are the roots of $\zeta^2 = z - c$, and $h_c(u)$ is one of them; the other is $-h_c(u) = h_c(-u) \in \Gamma_c$ by oddness (Step 4). Hence $g_c^{-1}(\Gamma_c) = \Gamma_c$.

Step 7: $\Gamma_c \subseteq K_c$ and the two components. Every orbit starting on Γ_c stays there, and Γ_c is bounded ($\Gamma_c \subseteq \{3/4 \leq |z| \leq 5/4\}$): hence $\Gamma_c \subseteq K_c$. By the Jordan curve theorem, $\mathbb{C} \setminus \Gamma_c = \Omega \sqcup \Omega_\infty$ with Ω bounded and Ω_∞ unbounded, both connected and open, and by the complete invariance of Γ_c the sets Ω and Ω_∞ are completely invariant as well (the images and preimages of a connected set disjoint from Γ_c remain disjoint from Γ_c ; the unbounded one goes into the unbounded one because $g_c(z) \rightarrow \infty$).

Since $|c| \leq 3/16 < 1/4$, Theorem 12.2 gives $c \in \mathcal{M}$, hence K_c is connected and $A_c(\infty) = \mathbb{C} \setminus K_c$ is connected (Theorem 12.5) and unbounded, disjoint from $\Gamma_c \subseteq K_c$: therefore $A_c(\infty) \subseteq \Omega_\infty$, which is equivalent to $\Omega \cup \Gamma_c \subseteq K_c$.

Step 8: $\Omega_\infty \subseteq A_c(\infty)$. Let $\hat{\Omega}_\infty := \Omega_\infty \cup \{\infty\} \subset \hat{\mathbb{C}}$: by the Jordan–Schoenflies theorem it is a simply connected domain of the sphere, different from $\hat{\mathbb{C}}$. By the imported result (I) there exists a biholomorphism $\psi : \hat{\Omega}_\infty \rightarrow \mathbb{D}$ with $\psi(\infty) = 0$. Put $F := \psi \circ g_c \circ \psi^{-1} : \mathbb{D} \rightarrow \mathbb{D}$. It is holomorphic, proper (because $g_c^{-1}(\hat{\Omega}_\infty) = \hat{\Omega}_\infty$) and of degree 2 (every point has two preimages counted with multiplicity, g_c being of degree 2 on the sphere and $\hat{\Omega}_\infty$ completely invariant); $F(0) = 0$ because $g_c(\infty) = \infty$. The critical points of g_c on the sphere are 0 and ∞ ; since $0 \in \Omega$ (indeed $\Gamma_c \subseteq \{|z| \geq 3/4\}$ and 0 lies in the bounded component), the only critical point in $\hat{\Omega}_\infty$ is ∞ , that is, the only critical point of F in $\mathbb{D}$ is 0. By Lemma 12.7, $F(u) = e^{i\alpha} u^2$ and $F^n(u) \rightarrow 0$ for every $u \in \mathbb{D}$: translated back, $g_c^n(z) \rightarrow \infty$ for every $z \in \Omega_\infty$, that is, $\Omega_\infty \subseteq A_c(\infty)$.

Step 9: conclusion. From Steps 7 and 8, $A_c(\infty) = \Omega_\infty$ and hence $K_c = \mathbb{C} \setminus \Omega_\infty = \bar{\Omega} = \Omega \cup \Gamma_c$, whence

$$J_c = \partial K_c = \partial \Omega = \Gamma_c.$$

The conjugacy $h_c : S^1 \rightarrow J_c$ carries $u \mapsto u^2$ into $g_c|_{J_c}$ by construction. □

Remark 12.15: What Theorem 12.12 does not say. The threshold $3/16$ is the one that comes out of the estimate chosen ($\varepsilon = 1/4$), not an optimal threshold: the true condition for J_c to be a Jordan curve is that c belong to the main cardioid $\mathfrak{c}(\mathbb{D})$ (Theorem 12.7), a much larger set — and the proof of that fact requires the theory of hyperbolic components. Moreover the theorem does not say that h_c is smooth: for $c \neq 0$ the curve J_c is a fractal of Hausdorff dimension > 1 (Section 12.12), hence h_c is never differentiable with everywhere non-zero derivative.

12.12. Panoramas: structural stability and Hausdorff dimension

Remark 12.16: λ -lemma and structural stability (panorama). A **holomorphic motion** of a set $E \subseteq \widehat{\mathbb{C}}$ parametrised by a disc $\mathbb{D}$ is a family $\{\iota_t\}_{t \in \mathbb{D}}$ of injections $\iota_t : E \rightarrow \widehat{\mathbb{C}}$ with $\iota_0 = \text{id}_E$ and with $t \mapsto \iota_t(e)$ holomorphic for every $e \in E$. The **λ -lemma** of Mañé, Sad and Sullivan asserts that such a motion automatically extends to a holomorphic motion of $\overline{E}$, and that every ι_t is the restriction of a quasiconformal homeomorphism of the plane: holomorphy in the parameter alone imposes a remarkable geometric rigidity. From it it follows that the set of parameters c for which J_c moves holomorphically is open and dense in $\mathbb{C}$, and that on each component the maps $g_c|_{J_c}$ are topologically conjugate to one another. Source: R. Mañé, P. Sad, D. Sullivan, “On the dynamics of rational maps”, *Ann. Sci. École Norm. Sup. (4)* **16** (1983), 193–217. Verified on 17 September 2026. Theorem 12.12 is the simplest, and completely elementary, case of this phenomenon: for $|c| \leq 3/16$ the conjugacy h_c is the fixed point of a contraction whose data depend holomorphically on c , and it therefore depends continuously on c .

Remark 12.17: Hausdorff dimension of J_c : Bowen and Ruelle (panorama). (a) **Bowen’s formula.** If J_c is a hyperbolic repeller (for example for c in the main cardioid), its Hausdorff dimension $d = \dim_H J_c$ is the unique zero of the pressure function $t \mapsto P(-t \log |g'_c|)$; in practice, d is the number for which

$$\lim_{n \rightarrow \infty} \sum_{z: g_c^n(z)=z} |(g_c^n)'(z)|^{-d} = 1.$$

Source: R. Bowen, “Hausdorff dimension of quasi-circles”, *Publ. Math. IHÉS* **50** (1979), 11–25. Verified on 17 September 2026.

(b) **Ruelle’s expansion.** Ruelle derived, by a computation that he himself qualifies as formal, the asymptotic expansion of the dimension for the family $z \mapsto z^q - p$:

$$\dim_H J = 1 + \frac{|p|^2}{4 \log q} + \text{terms of order } > 2 \text{ in } p,$$

which for $q = 2$ and $p = -c$ (that is, for $z^2 + c$) becomes

$$\dim_H J_c = 1 + \frac{|c|^2}{4 \log 2} + \cdots$$

Source: D. Ruelle, “Repellers for real analytic maps”, *Ergodic Theory Dynam. Systems* **2** (1982), 99–107, Appendix 2, pp.105 and 107. Verified on 17 September 2026. **Caveat:** Ruelle presents the computation as formal and writes “terms of order > 2 in p ”, not literally $O(|p|^3)$; here his own formulation is used.

Remark 12.18: Cross-references to Volume VI. Four frontier results and problems are not treated in this volume and are deferred to Volume VI, Chapter 9: Sullivan’s theorem on the absence of wandering domains (1985); the MLC conjecture on the local connectedness of $\mathcal{M}$, today known in many cases thanks to Lyubich, Kahn–Lyubich and Dudko–Lyubich; the density of hyperbolicity in the real family (Graczyk–Świątek; Kozlovski–Shen–van Strien); and Shishikura’s theorem according to which $\dim_H \partial \mathcal{M} = 2$. Here they are merely named, in order to place what has been proved within the overall landscape.

12.13. Historical note

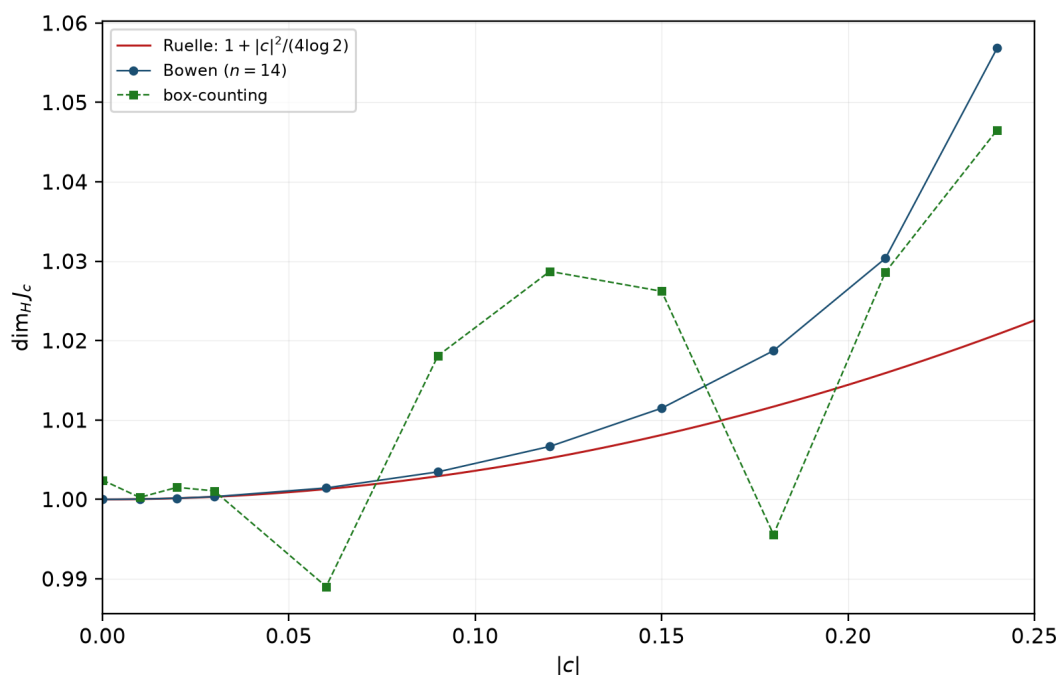


Figure 12.7: Dimension of J_c for real positive c , $0 \leq c \leq 0.24$ (hence c inside the main cardioid, and J_c a quasicircle). In red Ruelle's parabola $1 + |c|^2/(4 \log 2)$; in blue the estimate obtained from *Bowen's formula* by solving $\sum_{g_c^n(z)=z} |(g_c^n)'(z)|^{-d} = 1$ with $n = 14$ (that is, $2^{14} = 16\,384$ periodic points, computed by inverse iteration along all the binary itineraries); in green the rough estimate by box counting on a cloud of 800 000 points of J_c . **Exact check:** for $c = 0$ all the 2^n periodic points have $|(g_0^n)'| = 2^n$, hence the sum equals $2^{n(1-d)} = 1$ if and only if $d = 1$; the script returns 1.000000000000000 with discrepancy 4.4×10^{-15} for $n = 12$. **Check of the agreement with Ruelle:** the ratio $(\dim_{\text{Bowen}} - 1)/(|c|^2/(4 \log 2))$ equals 1.0156 for $|c| = 0.01$, 1.0321 for $|c| = 0.02$ and 1.0498 for $|c| = 0.03$: it tends to 1 linearly in $|c|$, as it must if the first neglected term is of order 3. The box estimate confirms $\dim \approx 1$ but does not have the resolution needed to see the quadratic correction, which for $|c| = 0.1$ is a mere 0.0036. Script `fig12_dimensione.py` (Appendix B, run on 17 September 2026). ✓

Remark 12.19: From Schröder to Yoccoz. **1871.** Ernst Schröder poses the functional equation $\varphi(f(z)) = \lambda\varphi(z)$ — today the “Schröder equation” — in the study of iterated functions, motivated by iterative methods for solving equations.

1884. Gabriel Koenigs solves the case $0 < |\lambda| < 1$ with the limit construction $\varphi = \lim f^n/\lambda^n$ reproduced in Theorem 12.3.

1903–1904. Lucjan Emil Böttcher, at Kazan, treats the superattracting case, obtaining the conjugacy with $z \mapsto z^d$ (Theorem 12.4). His work, published in Russian in three parts in the *Bulletin of the Physico-Mathematical Society of Kazan* (one in 1903 and two in 1904), long remained little known in the West.

1918–1920. Gaston Julia and Pierre Fatou, independently and almost simultaneously, found the theory of the iteration of rational functions using Montel’s theorem on normal families: there arise the Julia set as the closure of the repelling periodic points, and the structural theorems on the basins, among them the one proved in Section 12.5.

1938 and 1942. Hubert Cremer constructs the generic non-linearisable multipliers (Theorem 12.11); Carl Ludwig Siegel proves that the Diophantine condition suffices (Theorem 12.10). It is the first “small divisors” theorem of modern mathematics, twelve years earlier than Kolmogorov’s announcement which would start KAM theory (Chapter 6).

1971. Aleksandr Bruno weakens the Diophantine condition into the one today called the Bryuno condition.

1978–1980. Robert Brooks and J. Peter Matelski produce, in a work on Kleinian groups, what is probably the first computer image of the set $\mathcal{M}$; Benoît Mandelbrot publishes in 1980 the first high-resolution images of it and reveals its fractal structure.

1984–1985. Adrien Douady and John H. Hubbard, in the Orsay Notes, prove that $\mathcal{M}$ is connected, introduce the external rays and the combinatorics of the angles: that is, the dictionary, expounded in Section 12.7, between the geometry of $\mathcal{M}$ and the binary arithmetic of the doubling map on the circle.

1995. Jean-Christophe Yoccoz proves that for the quadratic family the Bryuno condition is necessary as well as sufficient: the local problem posed by Schröder in 1871 is closed, and the answer is arithmetic.

Bibliographical caveat. The entries of this survey which also appear in the “Sources and bibliography” block have been reopened and verified one by one on 17 September 2026. What remain not verified at first hand, and are therefore reported without a precise bibliographical reference, are the two items about Brooks–Matelski (1978–1980) and about Mandelbrot (1980): whoever wishes to cite them must reopen the originals.

12.14. Assigned exercises

The answers are in the Teacher’s key (Section 12.17).

Exercise 12.1: Escape criterion and membership of $\mathcal{M}$. *Aim:* to use Theorem 12.1 and Theorem 12.2 as decision tools. *Estimated time:* 30 minutes. *Type:* symbolic with a numerical check.

- Prove that $c = -2$, $c = -1$, $c = i$ and $c = \frac{1}{4}$ belong to $\mathcal{M}$, in each case exhibiting a bounded set containing the whole critical orbit (for $c = i$ compute the first four iterates).
- Prove that $c = -2.1$ does not belong to $\mathcal{M}$, by computing the first iterates and applying Theorem 12.1.
- Prove that the disc $\{|c| \leq 1/4\}$ is contained in $\mathcal{M}$ and that $\{|c| > 2\}$ is disjoint from

$\mathcal{M}$; deduce that $\mathcal{M}$ has area between $\pi/16$ and 4π .

- (d) Compare the upper bound $4\pi = 12.566$ with the numerical estimate 1.511 obtained in Figure 12.3 and comment on why the theoretical bound is so crude.

Exercise 12.2: Root points and bulbs. *Aim:* to manipulate the parametrisation $\mathfrak{c}(\lambda) = \lambda/2 - \lambda^2/4$ of the cardioid. *Estimated time:* 40 minutes. *Type:* symbolic.

- Compute in *closed form* (radicals, not decimals) the root points $c_{1/4}$, $c_{1/6}$ and $c_{2/3}$.
- Prove that the cardioid meets the real axis exactly at $c = 1/4$ (cusp) and $c = -3/4$, and that therefore $\mathfrak{c}(\mathbb{D}) \cap \mathbb{R} = (-3/4, 1/4)$.
- Prove that the real diameter of the period-2 bulb is the interval $(-5/4, -3/4)$, and that $\mathfrak{c}(\mathbb{D}) \cup B_{1/2}$ covers the real interval $(-5/4, 1/4)$ except the point $-3/4$.
- Show that $\mathfrak{c}$ maps the diameter $(-1, 1)$ of the disc of multipliers onto the interval $(-3/4, 1/4)$ in a strictly increasing way, and that $\mathfrak{c}'(\lambda) = 0$ only for $\lambda = 1$ (that is, the cusp is the only singular point of the cardioid).

Exercise 12.3: An explicit Cremer number. *Aim:* to construct concretely a Cremer ϑ and to verify its condition. *Estimated time:* 45 minutes. *Type:* computational and theoretical.

- Let $\vartheta := \sum_{k \geq 1} 2^{-a_k}$ with $a_1 := 1$ and $a_{k+1} := 2^{2^{a_k}}$. Prove that ϑ is irrational (the sequence of binary digits is not eventually periodic).
- Setting $m_k := 2^{a_k}$, estimate $\|m_k \vartheta\|$ from above and deduce that $\liminf_n \omega_n^{1/2^n} = 0$, that is, that ϑ is of Cremer type according to Definition 12.12.
- Compute numerically (in multiple precision, for example with `mpmath`) the first two or three values of ω_{m_k} and compare them with $\varrho^{2^{m_k}}$ for $\varrho = 1/2$.
- Explain why the construction of Theorem 12.11(ii), which uses Baire's theorem, does *not* provide an explicit example, and why this is not a defect of the proof.

12.15. Worked exercises

Exercise 12.4: The dendrite $c = i$: a Misiurewicz point. Compute the critical orbit of $g_i(z) = z^2 + i$, establish its nature, compute the multiplier of the limit cycle and deduce that $i \in \mathcal{M}$ but that g_i has no attracting cycles.

Solution. Strategy. Iterate by hand until the orbit repeats; then compute the multiplier with the chain rule; finally apply Theorems 12.1 and 12.6.

Step 1 — the orbit, by hand.

$$0 \mapsto i \mapsto i^2 + i = -1 + i \mapsto (-1 + i)^2 + i.$$

Now $(-1 + i)^2 = 1 - 2i + i^2 = 1 - 2i - 1 = -2i$, hence $(-1 + i)^2 + i = -2i + i = -i$. We continue: $(-i)^2 + i = -1 + i$. We are back at the third term: the critical orbit is

$$0, i, \underbrace{-1 + i, -i, -1 + i, -i, \dots}_{\text{cycle of period 2}}$$

It is *preperiodic*, not periodic: the critical point is not part of the cycle. A parameter with a preperiodic and non-periodic critical orbit is called a **Misiurewicz point**.

Step 2 — independent check. The script `verifica_cap12.py`, block [C12-1], prints the first eleven iterates: $0, i, -1 + i, -i, -1 + i, -i, \dots$, with $|g_i^{2^k}(0)| = \sqrt{2} = 1.414213562373095$ and $|g_i^{2^{k+1}}(0)| = 1$ for $k \geq 1$. ✓

Step 3 — the multiplier. The 2-cycle is $\{z_1, z_2\} = \{-1 + i, -i\}$, and

$$\lambda_2 = g'_i(z_1)g'_i(z_2) = (2z_1)(2z_2) = 4z_1z_2 = 4(-1 + i)(-i) = 4(i - i^2) = 4(1 + i),$$

hence $|\lambda_2| = 4\sqrt{2} = 5.656854249492381 > 1$: the cycle is **repelling**. **Independent check** with Theorem 12.8: for a 2-cycle one always has $\lambda_2 = 4(c + 1)$, and here $4(i + 1) = 4(1 + i)$ ✓ — the two computations, one by the product of the derivatives and one by the general formula, agree.

Step 4 — conclusions. The critical orbit is bounded (it takes only four distinct values), hence $i \in \mathcal{M}$ by Definition 12.3, and K_i is connected by Theorem 12.5. Moreover g_i has *no* attracting cycle: if it had one, by Theorem 12.6 the critical orbit would have to converge to it, whereas instead it is eventually periodic on a repelling cycle.

Interpretation. K_i has empty interior: it is the *dendrite* of Figure 12.1, that is, $K_i = J_i$. This is consistent with what we have just seen (there is no attracting basin that could provide an interior), even though the complete proof of $\text{int } K_i = \emptyset$ requires Sullivan’s theorem on the absence of wandering domains, deferred to Volume VI.

Typical errors. (1) Concluding that the critical orbit is *periodic*: it is not, 0 does not reappear. (2) Deducing from “ $i \in \mathcal{M}$ ” that there is an attracting cycle: the implication holds in the opposite direction (Theorem 12.6(ii)). (3) Computing the multiplier as $g'_i(z_1)$ instead of as the product over the two points of the cycle.

Extension. Show in the same way that $c = -2$ is a Misiurewicz point (critical orbit $0 \mapsto -2 \mapsto 2 \mapsto 2$, preperiodic on a fixed point), and compute the multiplier at $z = 2$: $g'_{-2}(2) = 4$, repelling.

Exercise 12.5: The dimension of J_c for $c = 0.1$. Estimate $\dim_H J_{0.1}$ with Bowen’s formula and compare with Ruelle’s expansion, quantifying the expected error.

Solution. Strategy. First computation: Ruelle’s expansion, in closed form. Second, independent computation: Bowen’s formula solved numerically on the periodic points. Comparison with the neglected term.

Step 1 — Ruelle. With $c = 0.1$,

$$1 + \frac{|c|^2}{4 \log 2} = 1 + \frac{0.01}{2.772588722240} = 1.003606737602.$$

Step 2 — Bowen. The fixed points of g_c^n are the 2^n points obtained by composing, along all the binary itineraries $(s_0, \dots, s_{n-1}) \in \{+, -\}^n$, the two inverse branches $z \mapsto \pm\sqrt{z - c}$ and taking the fixed point of the composition (the composition is a contraction because $|g'_c| = |2z| \approx 2$ on J_c). The multiplier is computed as $\prod_k 2z_k$ along the cycle. One then solves by bisection in d the equation $\sum_i |\Lambda_i|^{-d} = 1$. The script `verifica_cap12.py`, block [C12-2], gives

n	10	12	14	16
#points	1024	4096	16 384	65 536
d_n	1.004399245	1.004389929	1.004387555	1.004386949

The sequence d_n clearly converges (the successive differences are -9.3×10^{-6} , -2.4×10^{-6} , -6.1×10^{-7} : a factor of 4 for each increment of 2 in n).

Step 3 — comparison. $d_{16} - (1 + |c|^2/(4 \log 2)) = 1.004386949 - 1.003606738 = +7.80 \times 10^{-4}$. The first neglected term in Ruelle’s expansion is of order 3 in c , that is, $|c|^3 = 10^{-3}$: the observed discrepancy is of the same order, and more precisely $7.80 \times 10^{-4} = 0.78 |c|^3$. ✓

Exact check on the known case. The same code, applied to $c = 0$, returns $d = 1.000000000000000$ with a discrepancy of 4.4×10^{-15} from 1: and for $c = 0$ we know that $J_0 = S^1$ has dimension exactly 1 (Example 12.1). The check validates the implementation.

Interpretation. Already for $|c| = 0.1$ the curve J_c is no longer rectifiable in the classical sense: its dimension exceeds 1 by about 4.4 thousandths. The deviation grows like $|c|^2$, that is, very slowly: this is why box counting (Figure 12.7) fails to see it.

Typical errors. (1) Using as periodic points those found by iterating *forwards*: forward iteration on J_c is unstable and loses the points. (2) Forgetting that Bowen's sum must be taken over *all* the 2^n fixed points of g_c^n , including those of smaller period. (3) Using box counting to estimate corrections of order 10^{-3} .

Extension. Repeat for $c = 0.2$ and check that the discrepancy from Ruelle grows like $|c|^3$ (one expects $\approx 6 \times 10^{-3}$).

Exercise 12.6: Periodic angles and hyperbolic centres: an exact identity. Let $\nu(n)$ be defined by the relation $\sum_{d|n} \nu(d) = 2^{n-1}$ (the number of parameters c , counted with multiplicity, for which 0 is periodic of period exactly n). Let $A(n) := \sum_{d|n} \mu(n/d)(2^d - 1)$ be the number of angles $t \in \mathbb{R}/\mathbb{Z}$ of period exactly n for the doubling map (Corollary 12.2). Prove that $A(n) = 2\nu(n)$ for every $n \geq 2$, and verify it for $n \leq 10$.

Solution. Strategy. Express both quantities as Möbius inversions and compare them.

Step 1 — ν by inversion. From $\sum_{d|n} \nu(d) = 2^{n-1}$ it follows, by the Möbius inversion formula (Volume II), that

$$\nu(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) 2^{d-1}.$$

Step 2 — rewriting of A .

$$A(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) (2^d - 1) = \sum_{d|n} \mu\left(\frac{n}{d}\right) 2^d - \sum_{d|n} \mu\left(\frac{n}{d}\right) = 2 \sum_{d|n} \mu\left(\frac{n}{d}\right) 2^{d-1} - \sum_{d|n} \mu\left(\frac{n}{d}\right).$$

Step 3 — the sum of the μ . It is well known (and is proved at once by inversion applied to the constant function 1) that $\sum_{d|n} \mu(n/d) = \sum_{e|n} \mu(e)$ equals 1 if $n = 1$ and 0 if $n \geq 2$.

Step 4 — conclusion. For $n \geq 2$ the second summand vanishes and there remains

$$A(n) = 2 \sum_{d|n} \mu\left(\frac{n}{d}\right) 2^{d-1} = 2\nu(n).$$

(For $n = 1$ one has instead $A(1) = 1$ and $\nu(1) = 1$, that is, $A(1) = 2\nu(1) - 1$: the only fixed angle is $t = 0$, and the only centre of period 1 is $c = 0$.)

Step 5 — numerical verification. The script `verifica_cap12.py`, block [C12-3], computes both sequences independently (the first with μ , the second with the recursion $\nu(n) = 2^{n-1} - \sum_{d|n, d < n} \nu(d)$):

n	2	3	4	5	6	7	8	9
$A(n)$	2	6	12	30	54	126	240	504
$\nu(n)$	1	3	6	15	27	63	120	252

and confirms $A(n) = 2\nu(n)$ for $2 \leq n \leq 10$, as well as $\sum_{d|n} \nu(d) = 2^{n-1}$. ✓

Interpretation. The identity is the arithmetic form of the geometric fact stated in Remark 12.9: every root point of a hyperbolic component of period $n \geq 2$ is the landing point of *exactly two* rational parameter rays of period n . Counting the bulbs amounts to counting the angles, and dividing by two.

Typical errors. (1) Forgetting that $\sum_{d|n} \mu(n/d)$ equals 1 (not 0) for $n = 1$, and concluding wrongly that the identity holds for $n = 1$ as well. (2) Confusing $A(n)$ (angles of exact period) with $2^n - 1$ (angles of period *dividing* n). (3) Counting the centres without multiplicity: the formula $\sum_{d|n} \nu(d) = 2^{n-1}$ counts the roots of the polynomial $P_n(c) = g_c^n(0)$ with their multiplicity.

Extension. Verify that $\nu(3) = 3$ corresponds to the three centres of period 3 actually observable. Indeed $P_3(c) = g_c^3(0) = (c^2 + c)^2 + c = c(c^3 + 2c^2 + c + 1)$, of degree 4 = 2^{3-1} : the root $c = 0$ has period 1, and the three roots of the cubic factor are the three centres of period 3, that is, the centres of the two satellite bulbs $1/3$ and $2/3$ ($-0.1225611669 \pm 0.7448617666i$) and the centre of the primitive component on the real antenna, $c \approx -1.7548776662$. **Check:** the sum of the three roots must equal -2 (the coefficient of c^2 with the sign changed), and indeed $-1.7548776662 + 2 \cdot (-0.1225611669) = -2.0000000000$. ✓

12.16. Self-assessment quiz

- (Definition) The Mandelbrot set $\mathcal{M}$ is, by definition,
 - the set of the c for which J_c is connected
 - the set of the c for which the orbit of 0 under g_c is bounded
 - the set of the c for which g_c has an attracting cycle
 - the set of the c with $|c| \leq 2$
- (Hypothesis) Koenigs' theorem (Theorem 12.3) requires of the multiplier λ the hypothesis
 - $|\lambda| < 1$
 - $0 < |\lambda| < 1$
 - $|\lambda| \neq 1$
 - λ not a root of unity
- (Symbolic computation) The multiplier of the period-2 cycle of g_c , when it exists, equals
 - $2(c + 1)$
 - $4(c + 1)$
 - $4c$
 - $\lambda/2 - \lambda^2/4$
- (Numerical computation) The root point $c_{1/4}$ of the period-4 bulb attached to the cardioid equals
 - $-0.125 + 0.6495i$
 - $0.25 + 0.5i$
 - $-0.1565 + 1.0322i$
 - $0.5 + 0.25i$
- (Proof step) In the proof of Lemma 12.2 (connectedness of the levels), the contradiction is obtained by applying
 - Montel's theorem to the family $\{g_c^n\}$
 - the maximum principle to the harmonic function G_c on the bounded component V_2 , whose boundary lies in $\{G_c = s\}$
 - Schwarz's lemma to Φ_c
 - Liouville's theorem to ϕ_c
- (Error diagnosis) A student asserts: " ϑ irrational $\Rightarrow$ the fixed point is linearisable, because $\lambda^n \neq 1$ for every n and therefore the divisors $\lambda^{n-1} - 1$ never vanish". The error is
 - λ^n can equal 1 even for irrational ϑ
 - it is not enough that the divisors be non-zero: a *quantitative* lower estimate is needed, and without it the coefficients can grow so fast as to give zero radius of convergence (Cremer's theorem)
 - the formal series does not exist
 - $\lambda^{n-1} - 1$ vanishes for $n = 1$
- (Transfer) An engineer models a system with the map g_c and measures a parameter c in the main cardioid, with $|c| = 0.12$. What can he assert with certainty on the basis of this chapter?
 - J_c has Hausdorff dimension exactly 1
 - $c \in \mathcal{M}$, K_c is connected, g_c has an attracting fixed point and, since $|c| \leq 3/16$, J_c is a Jordan curve on which the

dynamics is conjugate to doubling C) g_c has an attracting cycle of period 2 D) the critical orbit escapes to infinity

8. (Source and metacognition) The proof of Siegel's theorem given in this chapter

- A) is complete in every part B) is complete except for Lemma 12.5, declared imported from Siegel (1942) and from Abate (LNM 1998, 2010, §4) C) is only an outline
D) refers entirely to Yoccoz (1995)

12.17. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

1) **B. Rationale:** Definition 12.3. *Distractors:* A is *equivalent* to B but by a theorem (the fundamental dichotomy, Remark 12.5), not by definition; C is a *sufficient* but not necessary condition (for $c = i$, $c = -2$ there are no attracting cycles and yet $c \in \mathcal{M}$: Exercise 12.4); D is only an upper bound (Theorem 12.2). *Diagnosed error:* confusing a definition with a proved characterisation. *Source:* Definition 12.3. *Prerequisite:* the notion of a bounded orbit. *Remedial work:* reread Remark 12.5 and identify which implication is proved and which is imported.

2) **B. Rationale:** the statement requires $0 < |\lambda| < 1$: the case $\lambda = 0$ (superattracting) is excluded because the construction divides by λ^n . *Distractors:* A includes $\lambda = 0$, where the assertion is *false* in the form given (there Böttcher's theorem holds instead, Theorem 12.4); C and D are weaker and do not suffice ($|\lambda| = 1$ with λ not a root of unity is the Siegel/Cremer case, where the conclusion may be false). *Diagnosed error:* failing to distinguish the four regimes of Definition 12.4. *Source:* Theorem 12.3. *Prerequisite:* classification of fixed points. *Remedial work:* reread Remark 12.3.

3) **B. Rationale:** Theorem 12.8: $z_+ z_- = c + 1$ and $\lambda_2 = (2z_+)(2z_-) = 4(c + 1)$. *Distractors:* A forgets a factor 2 (it uses $2z_+ z_-$); C is the multiplier one would get if the product of the roots were c ; D is the parametrisation of the cardioid, not a multiplier of a 2-cycle. *Diagnosed error:* applying the chain rule at a single point of the cycle. *Source:* Theorem 12.8. *Prerequisite:* relations between coefficients and roots. *Remedial work:* redo Step 3 of Exercise 12.4, where the value $4(1 + i)$ is obtained in two independent ways.

4) **B. Rationale:** $c_{1/4} = \mathfrak{c}(i) = i/2 - i^2/4 = i/2 + 1/4 = 0.25 + 0.5i$ (Definition 12.8). *Distractors:* A is $c_{1/3}$; C is the superattracting *centre* of the $1/4$ bulb, not its root (the two points are distinct: the root lies on the cardioid, the centre inside the bulb); D interchanges the real and imaginary parts. *Diagnosed error:* confusing the root point (on the boundary, parabolic) with the centre (interior, superattracting). *Source:* Definition 12.8 and Figure 12.4. *Prerequisite:* computing with $e^{2\pi i p/q}$. *Remedial work:* recompute $c_{1/6}$ as in Exercise 12.2(a).

5) **B. Rationale:** this is exactly the heart of Lemma 12.2: a second component V_2 would be bounded with $G_c \equiv s$ on the boundary and $G_c > s$ inside, against the maximum principle. *Distractors:* A, C, D name tools used elsewhere in the chapter but not in that step. *Diagnosed error:* remembering the list of tools without associating them with the point where they are needed. *Source:* Lemma 12.2. *Prerequisite:* maximum principle for harmonic functions (Volume III, Chapter 15). *Remedial work:* reread the inductive step, isolating the hypothesis " V_2 bounded".

6) **B. Rationale:** this is precisely the content of Remark 12.3 and of Theorem 12.11: for ϑ of Cremer type the divisors never vanish, and yet the linearisation does not exist. *Distractors:* A is false ($\lambda^n = 1$ would imply $n\vartheta \in \mathbb{Z}$, impossible for irrational ϑ); C is false (the

formal series always exists: Lemma 12.4); D is false ($n \geq 2$ in the recursion). *Diagnosed error*: confusing “non-zero” with “bounded away from zero”. *Source*: Theorem 12.11, Remark 12.3. *Prerequisite*: radius of convergence of a power series. *Remedial work*: reconstruct the estimate $|h_n| \leq (4\Gamma)^n$ and see where the constant C enters.

7) **B. Rationale**: $|c| = 0.12 \leq 3/16 = 0.1875$, hence Theorem 12.2 ($|c| \leq 1/4 \Rightarrow c \in \mathcal{M}$), Theorem 12.5 (connectedness) and Theorem 12.12 (Jordan curve and conjugacy to doubling) apply; moreover $|c| = 0.12 < 1/4$ implies that c lies in the cardioid (the hypothesis says so explicitly), hence there is an attracting fixed point. *Distractors*: A is false: the dimension is > 1 for $c \neq 0$ (Figure 12.7, ≈ 1.0067 for $|c| = 0.12$); C contradicts Corollary 12.1 and Theorem 12.6(i) (there is already an attracting fixed point, and the attracting cycle is unique); D contradicts $c \in \mathcal{M}$. *Diagnosed error*: confusing “Jordan curve” with “rectifiable curve of dimension 1”. *Source*: Theorems 12.12 and 12.6, Remark 12.17. *Prerequisite*: Hausdorff dimension. *Remedial work*: reread Remark 12.15.

8) **B. Rationale**: Section 12.1, point (IV), and Remark 12.11 declare exactly this. *Distractors*: A overestimates what has been done; C underestimates it (the formal part, the tree majorant and the conclusion are proved in full); D confuses Siegel’s theorem with Yoccoz’s optimality theorem, which here is only imported (Remark 12.13). *Diagnosed error*: not keeping track of what a text proves and what it imports. *Source*: Remark 12.1 and Remark 12.11. *Prerequisite*: none. *Remedial work*: redo the list of the four imports of the chapter and indicate, for each one, where it is used.

BEST PRACTICE

Assigned exercises — answers.

12.1 (escape criterion). (a) $c = -2$: orbit $0, -2, 2, 2, \dots$, contained in $\{|z| \leq 2\}$. $c = -1$: orbit $0, -1, 0, -1, \dots$, contained in $\{|z| \leq 1\}$. $c = i$: orbit $0, i, -1+i, -i, -1+i, \dots$, contained in $\{|z| \leq \sqrt{2}\}$ (Exercise 12.4). $c = 1/4$: if $0 \leq z \leq 1/2$ then $0 \leq z^2 + 1/4 \leq 1/4 + 1/4 = 1/2$, hence by induction the orbit stays in $[0, 1/2]$. (b) $c = -2.1$: $R_c = \max(2.1; 2) = 2.1$; $g_c(0) = -2.1$, $g_c^2(0) = (-2.1)^2 - 2.1 = 4.41 - 2.1 = 2.31 > 2.1 = R_c$: by Theorem 12.1 the orbit diverges, $c \notin \mathcal{M}$. (c) This is Theorem 12.2: $\{|c| \leq 1/4\} \subseteq \mathcal{M} \subseteq \{|c| \leq 2\}$. Since area is monotone under inclusion, $\pi(1/4)^2 = \pi/16 = 0.19635 \leq \text{area}(\mathcal{M}) \leq \pi \cdot 2^2 = 4\pi = 12.56637$. (d) The numerical estimate 1.511 lies inside the interval $\checkmark$, but it is 7.7 times smaller than the upper bound and 7.7 times larger than the lower bound: both theoretical estimates use only discs, whereas $\mathcal{M}$ is neither a disc nor close to being one — the cardioid alone occupies an area of $3\pi/8 = 1.178$ (an elementary integral over the parametrisation $\mathbf{c}$), that is, already 78% of the total.

12.2 (root points and bulbs). (a) $c_{1/4} = \mathbf{c}(i) = \frac{i}{2} - \frac{i^2}{4} = \frac{1}{4} + \frac{i}{2}$. $c_{1/6}$: $\lambda = e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$, $\lambda^2 = e^{2i\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, hence $c_{1/6} = \frac{1}{4} + \frac{\sqrt{3}}{4}i + \frac{1}{8} - \frac{\sqrt{3}}{8}i = \frac{3}{8} + \frac{\sqrt{3}}{8}i$ (numerically $0.375 + 0.2165063509i$ $\checkmark$). $c_{2/3} = \overline{c_{1/3}} = -\frac{1}{8} - \frac{3\sqrt{3}}{8}i$. (b) If $\mathbf{c}(\lambda) \in \mathbb{R}$ with $|\lambda| = 1$, writing $\lambda = e^{i\varphi}$ we have $\mathbf{c} = \frac{1}{2} \cos \varphi - \frac{1}{4} \cos 2\varphi + i(\frac{1}{2} \sin \varphi - \frac{1}{4} \sin 2\varphi)$, and the imaginary part $\frac{1}{2} \sin \varphi - \frac{1}{2} \sin \varphi \cos \varphi = \frac{1}{2} \sin \varphi (1 - \cos \varphi)$ vanishes only for $\varphi = 0$ or $\varphi = \pi$, that is, $\lambda = \pm 1$, that is, $c = 1/4$ or $c = -3/4$. Since $\mathbf{c}$ is a homeomorphism of $\mathbb{D}$ onto its image and $\mathbf{c}((-1, 1)) = (-3/4, 1/4)$ (point (d)), we conclude $\mathbf{c}(\mathbb{D}) \cap \mathbb{R} = (-3/4, 1/4)$. (c) The bulb is $|c + 1| < 1/4$ (Theorem 12.8), whose intersection with $\mathbb{R}$ is $(-5/4, -3/4)$. Taking the union with $(-3/4, 1/4)$ one obtains $(-5/4, 1/4) \setminus \{-3/4\}$. (d) For real $\lambda = x \in (-1, 1)$, $\mathbf{c}(x) = x/2 - x^2/4$ and $\mathbf{c}'(x) = 1/2 - x/2 = (1 - x)/2 > 0$: strictly increasing, with $\mathbf{c}(-1) = -3/4$ and $\mathbf{c}(1) = 1/4$. In general $\mathbf{c}'(\lambda) = (1 - \lambda)/2$ vanishes only at $\lambda = 1$, which corresponds to the cusp $c = 1/4$.

12.3 (Cremer number). (a) The binary digits of ϑ are 1 exactly in the positions $a_1 < a_2 < \dots$, with gaps $a_{k+1} - a_k \rightarrow \infty$ super-exponentially: a sequence of digits with blocks of

zeros of increasing length cannot be eventually periodic, hence $\vartheta \notin \mathbb{Q}$. (b) With $m_k = 2^{a_k}$ we have $m_k \vartheta = \sum_{j \leq k} 2^{a_k - a_j} + \sum_{j > k} 2^{a_k - a_j}$; the first summand is an integer and the second is positive and $\leq 2 \cdot 2^{a_k - a_{k+1} + 1}$ (a dominated geometric series), hence $\|m_k \vartheta\| \leq 2^{a_k - a_{k+1} + 1}$ and, by Lemma 12.3 read in the opposite direction ($\omega_m \leq 2\pi \|m\vartheta\|$),

$$\omega_{m_k} \leq 2\pi \cdot 2^{a_k - a_{k+1} + 1} = 2\pi \cdot 2^{a_k + 1 - 2^{2^{a_k}}} = 2\pi \cdot 2^{a_k + 1} \cdot 2^{-2^{m_k}}.$$

Since $2^{m_k} \gg m_k$, we have $\omega_{m_k}^{1/2^{m_k}} \rightarrow 0$: the Cremer condition is satisfied. (c) Already for $k = 2$ we have $a_2 = 2^{2^1} = 4$, $m_2 = 16$, $a_3 = 2^{2^4} = 2^{16} = 65\,536$: $\omega_{16} \leq 2\pi \cdot 2^{5-65536}$, a number of the order of $10^{-19\,726}$. Multiple precision is needed (`mpmath` with a few tens of thousands of digits) merely to represent it; the comparison with $(1/2)^{2^{16}} = 2^{-65536}$ shows that ω_{16} is of the same order, hence the inequality $\omega_n < \varrho^{2^n}$ is satisfied with a wide margin. (d) Baire's theorem is an argument for *non*-emptiness: it proves that the intersection of dense open sets is not empty, without exhibiting any element of it. This is not a defect, but a choice of economy: the proof of genericity is a few lines long, whereas every explicit example requires an ad hoc construction like that of points (a)–(c). The same dichotomy between “generic existence” and “explicit construction” is found again in Denjoy's counterexamples (Chapter 5), where instead the explicit example is the natural route.

12.18. Chapter formulary

Quadratic family, K_c , J_c , $\mathcal{M}$

$$g_c(z) = z^2 + c, \quad K_c = \{z : (g_c^n(z)) \text{ bounded}\}, \quad J_c = \partial K_c, \quad \mathcal{M} = \{c : 0 \in K_c\}$$

$$\begin{aligned} \text{escape criterion: } |g_c^n(z)| > \max(|c|, 2) &\Rightarrow |g_c^{n+k}(z)| \geq \mu^{2^k-1} |g_c^n(z)|, \quad \mu = |g_c^n(z)| - 1 \\ \left\{|c| \leq \frac{1}{4}\right\} &\subseteq \mathcal{M} \subseteq \{|c| \leq 2\} \end{aligned}$$

Local dynamics

$$\text{Koenigs } (0 < |\lambda| < 1) : \quad \varphi(z) = \lim_n \frac{f^n(z)}{\lambda^n}, \quad \varphi \circ f = \lambda \varphi, \quad \varphi(0) = 0, \quad \varphi'(0) = 1$$

$$\text{Böttcher } (\infty) : \quad \phi_c(z) = z \exp\left(\sum_{n \geq 0} \frac{1}{2^{n+1}} \text{Log}\left(1 + \frac{c}{z_n^2}\right)\right), \quad \phi_c \circ g_c = \phi_c^2, \quad \frac{\phi_c(z)}{z} \rightarrow 1$$

$$G_c(z) = \lim_n \frac{\log^+ |g_c^n(z)|}{2^n} = \log |\Phi_c(z)|, \quad G_c \circ g_c = 2 G_c, \quad G_c|_{K_c} \equiv 0$$

Fundamental dichotomy (first implication proved in the chapter, second imported: Remark 12.5)

$$0 \in K_c \iff c \in \mathcal{M} \implies K_c \text{ connected}; \quad 0 \notin K_c \iff K_c \text{ a Cantor set}$$

$$c \in \mathcal{M} \implies \Phi_c : \mathbb{C} \setminus K_c \xrightarrow{\sim} \mathbb{C} \setminus \overline{\mathbb{D}}, \quad \Phi_c \circ g_c = \Phi_c^2$$

Fatou

every attracting cycle attracts a critical point $\Rightarrow$ at most one attracting cycle $\Rightarrow c \in \mathcal{M}$

Cardioid and bulbs

$$c(\lambda) = \frac{\lambda}{2} - \frac{\lambda^2}{4} \quad (|\lambda| < 1) : \text{attracting fixed point}; \quad \lambda_{\pm} = 1 \pm \sqrt{1 - 4c}, \quad \lambda_+ + \lambda_- = 2$$

$$\begin{aligned} \text{period-2 bulb : } |c+1| &< \frac{1}{4}, \quad \lambda_2 = 4(c+1); \quad c_{p/q} = \frac{e^{2\pi ip/q}}{2} - \frac{e^{4\pi ip/q}}{4} \\ c_{1/2} &= -\frac{3}{4}, \quad c_{1/3} = -\frac{1}{8} + \frac{3\sqrt{3}}{8}i, \quad c_{1/4} = \frac{1}{4} + \frac{i}{2}, \quad c_{1/6} = \frac{3}{8} + \frac{\sqrt{3}}{8}i \end{aligned}$$

External rays

$$\begin{aligned} R_t &= \Phi_c^{-1}(\{re^{2\pi it} : r > 1\}), \quad g_c(R_t) = R_{2t \bmod 1}, \quad g_c^{-1}(R_t) = R_{t/2} \cup R_{t/2+1/2} \\ R_t \text{ periodic of period } q &\iff t = \frac{k}{2^q - 1}; \quad A(q) = \sum_{d|q} \mu\left(\frac{q}{d}\right)(2^d - 1) = 2\nu(q) \quad (q \geq 2) \end{aligned}$$

Small divisors

$$\begin{aligned} \omega_m &= |\lambda^m - 1| = 2 \sin(\pi \|m\vartheta\|) \geq 4 \|m\vartheta\|; \quad \text{Diophantine} \Rightarrow \omega_m \geq Cm^{-\mu} \\ h_n &= \frac{1}{\lambda(\lambda^{n-1} - 1)} \sum_{p+q=n} h_p h_q, \quad M_n = \sum_{T \in \mathcal{T}_n} \prod_v \frac{1}{\omega_{n(v)-1}} \leq 4^n \Gamma^n \\ \text{Siegel: } \vartheta \text{ Diophantine} &\Rightarrow \text{linearisable}; \quad \text{Cremer: } \liminf_n \omega_n^{1/2^n} = 0 \Rightarrow \text{not linearisable} \\ \text{Bryuno: } B(\vartheta) &= \sum_k \frac{\log q_{k+1}}{q_k} < \infty \iff f_\lambda(w) = \lambda w + w^2 \text{ linearisable (Yoccoz)} \end{aligned}$$

Stability and dimension

$$\begin{aligned} |c| \leq \frac{3}{16} &\Rightarrow J_c = h_c(S^1) \text{ a Jordan curve, } h_c(u^2) = h_c(u)^2 + c, \quad \|h_c - \text{id}\|_\infty \leq \frac{1}{4} \\ \text{Bowen: } \sum_{g_c^n(z)=z} |(g_c^n)'(z)|^{-d} &\rightarrow 1; \quad \text{Ruelle: } \dim_H J_c = 1 + \frac{|c|^2}{4 \log 2} + \dots \end{aligned}$$

Listing 12.1: Bowen’s formula on the periodic points (essential core, 10 lines)

```
def dim_bowen(c, n):                                     # 2^n fixed points of g_c^n
    idx = np.arange(1 << n)
    z = np.full(1 << n, 1 + 0.5j)
    for _ in range(80):                                   # inverse iteration:
        contraction
        for k in range(n):
            z = np.where((idx >> k) & 1, -1.0, 1.0) * np.sqrt(z - c)
    d, w = np.ones_like(z), z.copy()
    for _ in range(n):
        d, w = d * 2 * w, w * w + c                     # multiplier of the cycle
    L = np.log(np.abs(d))                                 # bisection on sum exp(-d L)
    = 1
    return bisect(lambda t: np.exp(-t * L).sum() - 1, 0.5, 2.0)
```

Complete scripts: `Sorgenti/Vol4/appendiceB_laboratorio/fig12_mandelbrot.py`, `fig12_julia.py`, `fig12_locale.py`, `fig12_dimensione.py` and `verifica_cap12.py`, actually run on 17 September 2026 with Python 3.12.5, numpy and matplotlib.

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Part E

Synchronisation and Automation

Synchronisation: from Huygens to Kuramoto

In February 1665 Christiaan Huygens, ill and confined to bed, observed that two of his pendulum clocks, hanging from the same beam, always ended up oscillating in perfect phase opposition, whatever the phase difference with which they had been started: the first phenomenon of synchronisation ever recorded by a scientific method. The very same pattern — autonomous oscillators, each with its own rhythm, coupled weakly enough to remain individuals but strongly enough to influence one another — governs the flashing of the fireflies of South-East Asia, the phase locking between a motor and its reference in an industrial drive, the pacemaker cells of the sinoatrial node, the networks of cortical neurons and the synchronised turbines of an electrical grid. This chapter develops the mathematical theory common to all these phenomena: first the simplest possible case, a single degree of freedom — a flow on the circle $\theta' = F(\theta)$, and in particular **Adler's equation** $\theta' = \Delta\omega - K \sin \theta$, solved in full; then two coupled oscillators, which reduces exactly to the previous case; then a population of N oscillators coupled in the manner of Kuramoto, with the identical case treated as a complete gradient flow and the limit $N \rightarrow \infty$ treated by the self-consistency equation and by the Ott–Antonsen reduction; finally a network synchronisation theorem, in the sense of control theory, chosen from the most recent literature and proved in full.

INFO

Level and prerequisites. Level **A–B** core (flows on the circle, Adler's equation, two oscillators, Huygens), Sections 13.5–13.8 (N -oscillator Kuramoto, self-consistency, Ott–Antonsen) of level **C–D**, Sections 13.9–13.11 (Chiba's theorem, chimeras, network synchronisation) with a **D** core and peaks of level **E** in the statement of Chiba's theorem alone (panorama, not proved). A level A–B reader may stop after Section 13.4 without losing the narrative coherence: he will in any case have a complete chapter on phase locking between two oscillators. This chapter is **self-contained**: it presupposes no other chapter of this volume, only the elementary analysis (separable differential equations, Fourier series, complex numbers) already developed in Volume II. It relies, for one connecting remark, on the notion of covering $\mathbb{R} \rightarrow S^1$ (Chapter 1 of this volume, recalled in one line with no need to reread it) and prepares directly Chapter 14 (phase-locked loops, encoders, digital control), where Adler's equation proved here reappears as the model of the first-order analogue PLL.

13.1. Flows on the circle

Definition 13.1: Flow on the circle. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and 2π -periodic: $F(\theta + 2\pi) = F(\theta)$ for every θ . The differential equation $\theta' = F(\theta)$ descends to a well-defined vector field on the circle $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ (the value of F does not depend on the representative chosen for $\theta \bmod 2\pi$), and is called the **flow on the circle** of field F . A point θ^* with $F(\theta^*) = 0$ is called an **equilibrium**.

Theorem 13.1: Equilibria, stability and period of a flow on the circle. Let $F \in C^1(\mathbb{R})$, 2π -periodic. (i) If $F(\theta^*) = 0$ and $F'(\theta^*) < 0$, then θ^* is an asymptotically stable equilibrium: there exists a neighbourhood of θ^* from which every solution converges to θ^* as $t \rightarrow +\infty$. If $F'(\theta^*) > 0$, then θ^* is unstable. (ii) If F never vanishes, then F has constant sign (say $F > 0$ on all of $\mathbb{R}$, the other case being symmetric), every solution $\theta(t)$ is strictly increasing and there exists $T > 0$ such that $\theta(t + T) = \theta(t) + 2\pi$ for every t ; moreover

$$T = \oint \frac{d\theta}{F(\theta)} := \int_{\theta_0}^{\theta_0+2\pi} \frac{d\theta}{F(\theta)},$$

a value independent of the starting point θ_0 .

Proof. (i) By continuity of F' , there exists $\delta > 0$ such that $F' < 0$ on $(\theta^* - \delta, \theta^* + \delta)$; since $F(\theta^*) = 0$, this forces $F > 0$ on $(\theta^* - \delta, \theta^*)$ and $F < 0$ on $(\theta^*, \theta^* + \delta)$ (F strictly decreasing and vanishing at θ^*). Hence, for $\theta(0) \in (\theta^* - \delta, \theta^*)$: $\theta'(0) = F(\theta(0)) > 0$, the solution increases towards θ^* and, by a standard monotonicity argument (the solution cannot cross θ^* in finite time, θ^* itself being an equilibrium point and the solutions of an equation with a C^1 field being unique), $\theta(t) \nearrow \theta^*$; if the limit $L < \theta^*$ were strictly smaller than θ^* , then by continuity $\theta'(t) \rightarrow F(L) > 0$, contradicting the convergence of an increasing and bounded function (whose derivative must tend to 0 along a subsequence, by the mean value theorem applied to disjoint intervals accumulating there); hence $L = \theta^*$. The argument for $\theta(0) \in (\theta^*, \theta^* + \delta)$ is symmetric ($\theta' < 0$, decrease towards θ^*). The case $F'(\theta^*) > 0$ is treated in the same way with the signs exchanged: every orbit near θ^* (except θ^* itself) moves away from it.

(ii) $F > 0$ everywhere gives $\theta'(t) > 0$ for every t : $\theta(t)$ is strictly increasing, hence injective, and by the existence and uniqueness theorem (Picard–Lindelöf, $F \in C^1$) defined on all of $\mathbb{R}$ (it does not blow up in finite time: F being periodic, hence bounded on a compact set and therefore everywhere, $|\theta'| \leq M := \max_{[0,2\pi]} F < \infty$, and a function with bounded derivative cannot blow up in finite time). Define $T := \int_{\theta_0}^{\theta_0+2\pi} d\theta/F(\theta)$ (an improper integral which is well defined and finite: F is continuous and strictly positive on the compact set $[\theta_0, \theta_0 + 2\pi]$, hence $1/F$ is continuous and bounded there). From the separable equation $dt = d\theta/F(\theta)$, the time taken by $\theta(t_0) = \theta_0$ to reach $\theta_0 + 2\pi$ is exactly $\int_{\theta_0}^{\theta_0+2\pi} d\theta/F(\theta) = T$ (change of variable $\theta = \theta(t)$ in the integral, $d\theta = F(\theta) dt$, hence $\int d\theta/F(\theta) = \int dt$): therefore $\theta(t_0 + T) = \theta_0 + 2\pi$. Since $\theta(\cdot + T)$ and $\theta(\cdot) + 2\pi$ are both solutions of the same equation (the field F is 2π -periodic, hence $\theta \mapsto \theta + 2\pi$ is a symmetry of the equation) with the same value at t_0 , by uniqueness they coincide for every t : $\theta(t + T) = \theta(t) + 2\pi$.

The independence of T from θ_0 follows from the periodicity of F : for every a , the substitution $\theta = \varphi + a$ in $\int_a^{a+2\pi} d\theta/F(\theta)$ gives $\int_0^{2\pi} d\varphi/F(\varphi + a)$; but $F(\varphi + a)$, as φ runs over $[0, 2\pi]$, takes exactly the same values as $F(\varphi + a \bmod 2\pi)$ in the same cyclic order (periodicity of F), hence the integral over a complete period does not depend on the starting point a (formally: $\int_0^{2\pi} d\varphi/F(\varphi + a) = \int_a^{a+2\pi} d\theta/F(\theta) = \int_0^{2\pi} d\theta/F(\theta)$, the last passage splitting the interval $[a, a + 2\pi]$ into $[a, 2\pi] \cup [2\pi, a + 2\pi]$ if $0 < a < 2\pi$, and using $F(\theta) = F(\theta - 2\pi)$ on the second piece in order to bring it back to $[0, a]$). $\square$

13.2. Adler's equation

Definition 13.2: Adler's equation. An oscillator of natural frequency ω_0 , coupled to a reference of frequency ω_r with gain $K > 0$ (a first-order analogue phase-locked loop, a weakly forced pendulum, two coupled chemical oscillators), has a phase difference $\theta(t)$ which

obeys, in the standard model introduced by R. Adler in 1946 for valve oscillators,

$$\theta' = \Delta\omega - K \sin \theta, \quad \Delta\omega := \omega_0 - \omega_r,$$

Adler's equation. The parameter K is the loop gain (or the coupling constant), $\Delta\omega$ the frequency detuning.

Theorem 13.2: Locking, phase slipping and exact period (Adler). Let $K > 0$ be fixed, $F(\theta) := \Delta\omega - K \sin \theta$. (i) **Locking** ($|\Delta\omega| < K$). F has exactly two zeros in $[0, 2\pi)$,

$$\theta_s^* = \arcsin\left(\frac{\Delta\omega}{K}\right) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \theta_i^* = \pi - \theta_s^*,$$

the first asymptotically stable, the second unstable: every orbit which does not start exactly at θ_i^* converges to θ_s^* . (ii) **Saddle-node** ($|\Delta\omega| = K$). The two equilibria collide in a single double zero $\theta^* = \frac{\pi}{2} \operatorname{sgn}(\Delta\omega)$, with $F \geq 0$ (resp. ≤ 0) everywhere if $\Delta\omega = K$ (resp. $= -K$): θ^* is not reached in finite time by any other orbit, but every orbit converges to it (semistability). (iii) **Phase slipping** ($|\Delta\omega| > K$). F never vanishes; by Theorem 13.1(ii), $\theta(t)$ is (up to the sign of $\Delta\omega$) monotone and periodic in $\theta \bmod 2\pi$, with exact period

$$T = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}}.$$

Proof. (i) $\sin \theta = \Delta\omega/K$ has, for $|\Delta\omega/K| < 1$, exactly two solutions in $[0, 2\pi)$: $\theta_s^* = \arcsin(\Delta\omega/K) \in (-\pi/2, \pi/2)$ (representative in $[0, 2\pi)$ understood mod 2π) and $\theta_i^* = \pi - \theta_s^*$ (the only other solution of $\sin \theta = \sin \theta_s^*$ in one period). $F'(\theta) = -K \cos \theta$: at $\theta_s^* \in (-\pi/2, \pi/2)$, $\cos \theta_s^* > 0$, hence $F'(\theta_s^*) = -K \cos \theta_s^* < 0$, stable by Theorem 13.1(i); at $\theta_i^* = \pi - \theta_s^*$, $\cos \theta_i^* = \cos(\pi - \theta_s^*) = -\cos \theta_s^* < 0$, hence $F'(\theta_i^*) > 0$, unstable. With only two equilibria on the circle, one stable and one unstable, every orbit which does not start exactly at the unstable one is attracted, by piecewise monotonicity between the two equilibria (the same argument as in part (i) of Theorem 13.1, applied to each of the two arcs (θ_i^*, θ_s^*) and $(\theta_s^*, \theta_i^* + 2\pi)$ separately), by the stable one.

(ii) Let $\Delta\omega = K > 0$ (the other case is symmetric under $\theta \mapsto -\theta$). $F(\theta) = K(1 - \sin \theta) \geq 0$ for every θ , with equality if and only if $\sin \theta = 1$, that is, $\theta = \pi/2 + 2k\pi$: a double zero (from the expansion $1 - \sin \theta = 1 - \cos(\theta - \pi/2 - \dots)$, near $\pi/2$, $F(\theta) = \frac{K}{2}(\theta - \pi/2)^2 + O((\theta - \pi/2)^4)$, quadratic tangency). Since $F \geq 0$ everywhere, $\theta(t)$ is non-decreasing for every orbit: if $\theta(0) \neq \pi/2$, the orbit increases towards $\pi/2$ (from below, remaining in the interval $(\theta(0), \pi/2)$ by monotonicity, being unable to overtake it by uniqueness) or moves away from it (from above, towards $\pi/2 + 2\pi$): the first case is convergence (asymptotic, in infinite time, because near $\pi/2$ the transit time $\int d\theta/F(\theta) \sim \int d\theta/(\theta - \pi/2)^2$ diverges), the second is a departure which, repeated over the next period, closes up on the same $\theta^* + 2\pi$ from below: every orbit converges to (some integer translate of) $\theta^* = \pi/2$ as $t \rightarrow +\infty$, never in finite time.

(iii) $|\Delta\omega| > K \geq K|\sin \theta|$ for every θ (since $|\sin \theta| \leq 1$), hence $F(\theta) = \Delta\omega - K \sin \theta$ has the same sign as $\Delta\omega$ for every θ (if $\Delta\omega > K$: $F(\theta) \geq \Delta\omega - K > 0$; symmetrically if $\Delta\omega < -K$). Theorem 13.1(ii) applies (with $-F$ in place of F if $\Delta\omega < 0$, in order to reduce to the case $F > 0$; the period is the same for F and $-F$, the integral $\oint d\theta/F$ being invariant in absolute value): it remains to compute $T = \int_0^{2\pi} d\theta/(\Delta\omega - K \sin \theta)$, which we shall do with $a := \Delta\omega > K > 0 > 0$ without loss of generality (the case $\Delta\omega < -K$ gives the same T for the same reason).

Put $b := K$, $t := \tan(\theta/2)$: $\sin \theta = 2t/(1+t^2)$, $d\theta = 2dt/(1+t^2)$, and

$$\frac{d\theta}{a - b \sin \theta} = \frac{2dt/(1+t^2)}{a - 2bt/(1+t^2)} = \frac{2dt}{a(1+t^2) - 2bt} = \frac{2dt}{at^2 - 2bt + a}.$$

Completing the square, $at^2 - 2bt + a = a(t - \frac{b}{a})^2 + \frac{a^2 - b^2}{a}$, so that

$$\int \frac{2dt}{at^2 - 2bt + a} = \frac{2}{\sqrt{a^2 - b^2}} \arctan\left(\frac{at - b}{\sqrt{a^2 - b^2}}\right) + C =: G(t).$$

The substitution $t = \tan(\theta/2)$ has a singularity at $\theta = \pi$: we split $[0, 2\pi]$ into $[0, \pi)$ and $(\pi, 2\pi]$. On $[0, \pi)$, t runs over $[0, +\infty)$: $\int_0^\pi \frac{d\theta}{a - b \sin \theta} = \lim_{t \rightarrow +\infty} G(t) - G(0) = \frac{2}{\sqrt{a^2 - b^2}} \left[\frac{\pi}{2} - \arctan\left(\frac{-b}{\sqrt{a^2 - b^2}}\right) \right] = \frac{2}{\sqrt{a^2 - b^2}} \left[\frac{\pi}{2} + \arctan\left(\frac{b}{\sqrt{a^2 - b^2}}\right) \right]$. On $(\pi, 2\pi]$, t runs over $(-\infty, 0]$: $\int_\pi^{2\pi} \frac{d\theta}{a - b \sin \theta} = G(0) - \lim_{t \rightarrow -\infty} G(t) = \frac{2}{\sqrt{a^2 - b^2}} \left[\arctan\left(\frac{-b}{\sqrt{a^2 - b^2}}\right) - \left(-\frac{\pi}{2}\right) \right] = \frac{2}{\sqrt{a^2 - b^2}} \left[\frac{\pi}{2} - \arctan\left(\frac{b}{\sqrt{a^2 - b^2}}\right) \right]$. Adding the two halves, the arctan terms cancel:

$$T = \int_0^{2\pi} \frac{d\theta}{a - b \sin \theta} = \frac{2}{\sqrt{a^2 - b^2}} \cdot \pi = \frac{2\pi}{\sqrt{a^2 - b^2}} = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}}.$$

□

Corollary 13.1: Divergence of the period at the bifurcation. As $\Delta\omega \rightarrow K^+$ (with K fixed),

$$T(\Delta\omega) = \frac{2\pi}{\sqrt{(\Delta\omega - K)(\Delta\omega + K)}} \sim \frac{2\pi}{\sqrt{2K}} (\Delta\omega - K)^{-1/2}.$$

Proof. As $\Delta\omega \rightarrow K^+$, $\Delta\omega + K \rightarrow 2K$, hence $\sqrt{\Delta\omega^2 - K^2} = \sqrt{(\Delta\omega - K)(\Delta\omega + K)} \sim \sqrt{2K} \sqrt{\Delta\omega - K}$; substituting into the formula of Theorem 13.2(iii) one obtains the stated asymptotics. □

Remark 13.1: Saddle-node bifurcation on the circle. Theorem 13.2 describes, as $\Delta\omega$ varies with K fixed, a **saddle-node bifurcation**: for $|\Delta\omega| < K$ two equilibria (one stable, one unstable); at $|\Delta\omega| = K$ they collide in a single degenerate equilibrium (part (ii)); for $|\Delta\omega| > K$ no equilibrium, replaced by a periodic orbit whose period diverges, with the exact law $(\Delta\omega - K)^{-1/2}$ (Corollary 13.1), as one approaches the threshold: the residue of the pair of equilibria that has just disappeared creates a “bottleneck” in which the orbit slows down abruptly (the so-called slow cycle slip near the locking threshold), then rapidly traverses the rest of the circle. This is the correction, proved in full, of the assertion of Chapter 5 (old numbering, see remark 5.11 of *Sorgenti/Vol4/_vecchi/cap05_applicazioni_automazione_IT.tex*), where the same fact appeared only stated, with neither a proof nor the explicit constant $2\pi/\sqrt{2K}$.

Example 13.1: Adler’s equation with $K = 1$, $\Delta\omega = 1$ and $\Delta\omega = 2$: exact period vs simulation. Case $\Delta\omega = 2 > K = 1$ (slipping). Closed formula: $T = 2\pi/\sqrt{2^2 - 1^2} = 2\pi/\sqrt{3} = 3.627599\dots$ **Independent check** (script `fig13_adler_flow.py`, explicit fourth-order Runge–Kutta integration, step $dt = 2 \times 10^{-4}$, period measured by linear interpolation of the crossing times of a multiple of 2π , discarding a transient of 2 turns): $T_{\text{num}} = 3.627599$, relative discrepancy 2.96×10^{-10} . ✓ Halving the integration step, the measured period changes by 2.9×10^{-11} : both estimates agree with the theoretical value to within 10^{-9} , the convergence check expected for RK4 on this scale. On a grid of ten values $\Delta\omega \in \{1.02; 1.05; \dots; 8.0\}$ the maximum relative discrepancy between the theoretical and the numerical T is 4.5×10^{-10} ; the slope in log-log scale of T with respect to $(\Delta\omega - K)$ near the edge is -0.5181 , in agreement with the exponent $-1/2$ of Corollary 13.1 to within the error expected for a slope estimate on a finite number of points. ✓

Case $\Delta\omega = 1 = K$ (edge, saddle-node). Integrating from $\theta_0 = 0$ up to $t = 400$ (step $dt = 5 \times 10^{-4}$): $\theta(400) = 1.565809$ rad, against the degenerate equilibrium $\theta^* = \pi/2 = 1.570796$ rad: the trajectory remains “glued” to the marginal equilibrium, no phase slipping in finite time, exactly as predicted by Theorem 13.2(ii). ✓

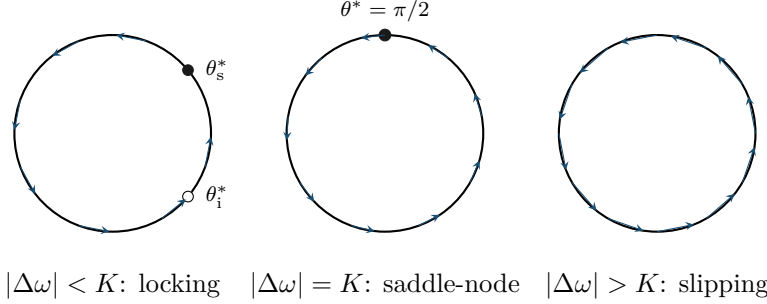


Figure 13.1: The three regimes of the Adler flow $\theta' = \Delta\omega - K \sin \theta$ (Theorem 13.2): on the left, locking, with the stable equilibrium θ_s^* (filled dot) and the unstable one θ_i^* (empty dot); in the centre, the critical case $|\Delta\omega| = K$, a single degenerate equilibrium $\theta^* = \pi/2$ (the arrows never stop completely, consistently with $F \geq 0$ everywhere); on the right, phase slipping, no equilibrium, rotation at non-uniform speed with period $T = 2\pi/\sqrt{\Delta\omega^2 - K^2}$.

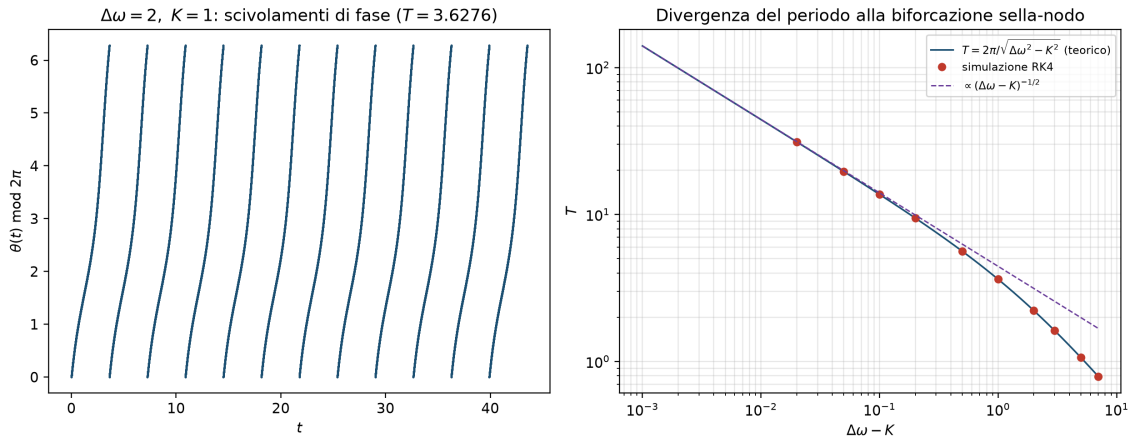


Figure 13.2: Left: $\theta(t) \bmod 2\pi$ for $\Delta\omega = 2$, $K = 1$ (phase slipping), fourth-order Runge–Kutta integration with step 2×10^{-4} : the theoretical period $T = 2\pi/\sqrt{3} = 3.6276$ (Theorem 13.2(iii)) coincides with the measured one to within 3×10^{-10} of relative discrepancy (Example 13.1) ✓. Right: T as a function of $\Delta\omega - K$ in log-log scale, theory (curve) against simulation (dots), compared with the asymptotic line $2\pi/\sqrt{2K(\Delta\omega - K)}$ of Corollary 13.1: numerical slope -0.5181 against the theoretical exponent $-1/2$ ✓. Computed numerically, script `Appendix B, fig13_adler_flow.py`.

13.3. Two coupled oscillators

Corollary 13.2: Two coupled oscillators: the phase difference obeys Adler with gain $2K$. Let θ_1, θ_2 be two oscillators of natural frequencies ω_1, ω_2 , symmetrically coupled,

$$\theta_1' = \omega_1 + K \sin(\theta_2 - \theta_1), \quad \theta_2' = \omega_2 + K \sin(\theta_1 - \theta_2), \quad K \in \mathbb{R}.$$

The phase difference $\varphi := \theta_2 - \theta_1$ obeys exactly Adler's equation with gain $2K$:

$$\varphi' = \Delta\omega - 2K \sin \varphi, \quad \Delta\omega := \omega_2 - \omega_1.$$

If $K > 0$ (attractive coupling) and $|\Delta\omega| \leq 2K$, the stable equilibrium $\varphi^* = \arcsin(\Delta\omega/2K) \in (-\pi/2, \pi/2)$ is close to **phase** ($\varphi^* \rightarrow 0$ as $\Delta\omega \rightarrow 0$); if $K < 0$ (repulsive coupling) and $|\Delta\omega| \leq 2|K|$, the stable equilibrium is close to **antiphase** ($\varphi^* \rightarrow \pi$ as $\Delta\omega \rightarrow 0$).

Proof. $\varphi' = \theta_2' - \theta_1' = \omega_2 + K \sin(\theta_1 - \theta_2) - \omega_1 - K \sin(\theta_2 - \theta_1) = \Delta\omega - K \sin \varphi - K \sin \varphi = \Delta\omega - 2K \sin \varphi$, using $\sin(\theta_1 - \theta_2) = -\sin \varphi$. This is Adler's equation of Theorem 13.2, with K replaced by $2K$: if $K > 0$, the locking case applies as soon as $|\Delta\omega| \leq 2K$, with stable equilibrium $\varphi_s^* = \arcsin(\Delta\omega/2K) \in (-\pi/2, \pi/2)$ (Theorem 13.2(i), $F'(\varphi) = -2K \cos \varphi$, stable where $\cos \varphi > 0$). If $K < 0$, put $K = -|K|$: the equation becomes $\varphi' = \Delta\omega + 2|K| \sin \varphi$, that is, Adler with $\sin$ of the opposite sign; the stable point is now where $F'(\varphi) = 2|K| \cos \varphi < 0$, that is, $\cos \varphi^* < 0$, an equilibrium in $(\pi/2, 3\pi/2)$: as $\Delta\omega \rightarrow 0$ this equilibrium tends to $\varphi^* = \pi$ (the only zero of $\sin \varphi$ in that interval), antiphase. $\square$

Example 13.2: Two metronomes at 176 and 184 bpm. Two mechanical metronomes, set at 176 and 184 beats per minute, have angular frequencies $\omega_1 = 2\pi \cdot 176/60 = 18.4307$ rad/s, $\omega_2 = 2\pi \cdot 184/60 = 19.2684$ rad/s (script `verifica_cap13.py`, block [3]), hence $\Delta\omega = \omega_2 - \omega_1 = 0.8378$ rad/s. With an illustrative attractive coupling $K = 0.6$ (of the order of magnitude typical for two metronomes on the same slightly mobile support), the locking threshold of Corollary 13.2 is $2K = 1.2$ rad/s: since $|\Delta\omega| = 0.8378 \leq 1.2$, the two metronomes lock, with stable equilibrium phase difference

$$\varphi^* = \arcsin\left(\frac{\Delta\omega}{2K}\right) = \arcsin(0.6982) = 0.772785 \text{ rad} = 44.2773^\circ.$$

✓ (closed formula, verified with `math.asin` in the script). The two metronomes therefore beat together, but not exactly in phase: one leads the other by about 44° over a turn of 360° , consistently with the fact that their natural frequencies are not equal.

13.4. Huygens, 1665: the first observation

Remark 13.2: The “odd sympathy” of Huygens’s two pendulum clocks. In February 1665, confined to bed by an illness, Christiaan Huygens observed that two of his recently invented pendulum clocks, hanging from the same beam supported by two chairs, regularly ended up oscillating in **phase opposition**, whatever the initial phase difference, within about half an hour: he described them in letters to his father, to de Sluse and to the Royal Society of London as exhibiting a “strange sympathy” (*sympathie des horloges*). Huygens first conjectured air currents as the cause, then ruled them out experimentally (by blocking the air flow between the two clocks without the effect disappearing) and correctly concluded that the cause was a weak mechanical coupling transmitted through the common beam.

In the language of Corollary 13.2: two almost identical clocks ($\Delta\omega$ small, being built in the same way) coupled through the beam end up in a stable equilibrium close to $\varphi^* = 0$ or to $\varphi^* = \pi$ according to the sign of the effective coupling transmitted by the beam; Huygens’s historical observation (antiphase) corresponds to the case $K < 0$ of the Corollary. A quantitative and rigorous mechanical explanation — which requires modelling the beam itself as a further degree of freedom coupled to the two pendulums, not treated in this chapter — was obtained only three centuries later, with the modern experimental reproduction and

the complete analysis of Bennett, Schatz, Rockwood and Wiesenfeld (2002), who confirmed antiphase as the typical regime for realistic beam masses and stiffnesses. This is a **historical and experimental** result, not a mathematical proof of the chapter: we report it as panorama, with a primary source (the letters themselves) and a modern secondary source, both verified.

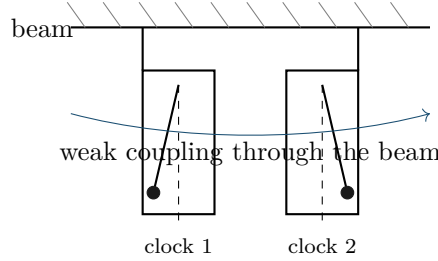


Figure 13.3: Diagram of Huygens's device (1665, Remark 13.2, level A): two pendulum clocks hanging from the same beam, supported by two chairs; the left-hand pendulum is drawn at an instant at which the two pendulums are almost in phase opposition, the regime which Huygens observed setting in within about half an hour independently of the initial conditions.

13.5. The Kuramoto model with N identical oscillators

After two oscillators, the next step is a population: N oscillators of phase $\theta_1, \dots, \theta_N$, each coupled to *all* the others with the same gain, normalised by N so that the model has a sensible limit as $N \rightarrow \infty$ (Section 13.7). This is the **Kuramoto model** (Kuramoto, 1975). In this section we treat the simplest case, all the natural frequencies equal; the general case (different frequencies, drawn from a distribution) is the content of Section 13.7 and of Section 13.11.

Definition 13.3: Kuramoto model, identical oscillators. Let $\theta_1, \dots, \theta_N : \mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$, $K > 0$, $\omega \in \mathbb{R}$ common to all the oscillators:

$$\theta'_i = \omega + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

Lemma 13.1: The identical Kuramoto model is a gradient flow. In the rotating frame $\psi_i := \theta_i - \omega t$, the system becomes autonomous, $\psi'_i = \frac{K}{N} \sum_j \sin(\psi_j - \psi_i)$, and it is the gradient flow $\psi' = -\nabla V(\psi)$ on the torus $\mathbb{T}^N$ of the potential

$$V(\psi) := -\frac{K}{2N} \sum_{i,j=1}^N \cos(\psi_i - \psi_j).$$

Proof. The substitution $\psi_i = \theta_i - \omega t$ removes the term ω from each equation (differentiating: $\psi'_i = \theta'_i - \omega$), leaving $\psi'_i = \frac{K}{N} \sum_j \sin(\psi_j - \psi_i)$, independent of t explicitly: autonomous. Let us compute $\partial V / \partial \psi_i$. In the double sum defining V , the terms which depend on ψ_i are those with first index i (that is, $\cos(\psi_i - \psi_j)$, $j \neq i$, plus the constant term $j = i$) and those with second index i (that is, $\cos(\psi_j - \psi_i)$, $j \neq i$): for each of these $2(N-1)$ terms, $\partial_{\psi_i} \cos(\psi_i - \psi_j) = -\sin(\psi_i - \psi_j)$ and $\partial_{\psi_i} \cos(\psi_j - \psi_i) = \sin(\psi_j - \psi_i) = -\sin(\psi_i - \psi_j)$ (the

same value, sin being odd). Hence

$$\frac{\partial V}{\partial \psi_i} = -\frac{K}{2N} \cdot 2 \sum_{j \neq i} (-\sin(\psi_i - \psi_j)) = \frac{K}{N} \sum_{j \neq i} \sin(\psi_i - \psi_j) = -\frac{K}{N} \sum_j \sin(\psi_j - \psi_i) = -\psi'_i,$$

that is, $\psi'_i = -\partial V / \partial \psi_i$ for every i : $\psi' = -\nabla V(\psi)$. $\square$

Lemma 13.2: *The potential is (up to a constant) $-\frac{NK}{2}r^2$: identity with the order parameter. With $r(\psi)e^{i\varphi(\psi)} := \frac{1}{N} \sum_{j=1}^N e^{i\psi_j}$ (the order parameter, Section 13.6), $V(\psi) = -\frac{NK}{2}r(\psi)^2$.*

Proof. $N^2 r^2 = |\sum_j e^{i\psi_j}|^2 = \left(\sum_j e^{i\psi_j}\right) \overline{\left(\sum_k e^{i\psi_k}\right)} = \sum_{j,k} e^{i(\psi_j - \psi_k)} = \sum_{j,k} \cos(\psi_j - \psi_k)$ (the imaginary part cancels in pairs $(j, k) \leftrightarrow (k, j)$, sin being odd). Hence $\sum_{i,j} \cos(\psi_i - \psi_j) = N^2 r^2$, and $V = -\frac{K}{2N} \cdot N^2 r^2 = -\frac{NK}{2}r^2$. $\square$

Lemma 13.3: Weighted Laplacian of the complete graph. *Let L be the $N \times N$ matrix with $L_{ii} = (N-1)w$, $L_{ij} = -w$ for $i \neq j$ (Laplacian of the complete graph with uniform weight w on every edge). Then L has eigenvalue 0 (simple, eigenvector $\mathbf{1} = (1, \dots, 1)$) and eigenvalue Nw with multiplicity $N-1$ (eigenspace $\{x : \sum_i x_i = 0\}$).*

Proof. $L = w(NI - J)$, where J is the $N \times N$ matrix of all 1's. J has eigenvalue N (eigenvector $\mathbf{1}$, since $J\mathbf{1} = N\mathbf{1}$) and eigenvalue 0 with multiplicity $N-1$ (for every x with $\sum x_i = 0$, $Jx = (\sum_i x_i)\mathbf{1} = 0$; these x form a subspace of dimension $N-1$, the orthogonal complement of $\mathbf{1}$, and J is symmetric hence diagonalisable on orthogonal bases). Hence $NI - J$ has eigenvalue $N - N = 0$ on $\mathbf{1}$ and $N - 0 = N$ (multiplicity $N-1$) on the complement; multiplying by w one obtains the stated eigenvalues. $\square$

Lemma 13.4: *The weighted Laplacian of a connected graph with positive weights is positive semidefinite, with kernel $\mathbb{R}\mathbf{1}$. Let $G = (V, E)$ be a connected graph on N vertices, weights $w_{ij} = w_{ji} > 0$ for $(i, j) \in E$, and L the weighted Laplacian, $L_{ii} = \sum_{j:(i,j) \in E} w_{ij}$, $L_{ij} = -w_{ij}$ if $(i, j) \in E$, $L_{ij} = 0$ otherwise. Then, for every $x \in \mathbb{R}^N$, $x^\top Lx = \sum_{(i,j) \in E} w_{ij}(x_i - x_j)^2 \geq 0$, with equality if and only if $x = c\mathbf{1}$ for some $c \in \mathbb{R}$.*

Proof. $x^\top Lx = \sum_i x_i \sum_{j:(i,j) \in E} w_{ij}(x_i - x_j) = \sum_{(i,j) \in E} w_{ij}x_i(x_i - x_j)$ (summing over ordered pairs of the undirected edges, each edge counted twice, once for each direction) $= \frac{1}{2} \sum_{(i,j) \in E} w_{ij}[x_i(x_i - x_j) + x_j(x_j - x_i)] = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(x_i - x_j)^2$, which is the stated formula up to the factor $\frac{1}{2}$ due to the double counting (here $\sum_{(i,j) \in E}$ denotes the sum over the undirected edges taken once only): ≥ 0 since $w_{ij} > 0$. It is $= 0$ if and only if $x_i = x_j$ for every edge $(i, j) \in E$, that is, if and only if x is constant on every connected component of G ; G being connected, this is equivalent to $x = c\mathbf{1}$. $\square$

Theorem 13.3: Convergence to the equilibria (LaSalle) for the Kuramoto gradient flow. *Every orbit $\psi(t)$ of the gradient flow of Lemma 13.1 has $\psi'_i(t) \rightarrow 0$ as $t \rightarrow +\infty$, for every i ; every accumulation point of $\psi(t)$ as $t \rightarrow +\infty$ is a critical point of V (an equilibrium of the flow).*

Proof. $\frac{d}{dt}V(\psi(t)) = \sum_i \frac{\partial V}{\partial \psi_i} \psi'_i = \sum_i (-\psi'_i) \psi'_i = -\sum_i (\psi'_i)^2 \leq 0$: $V(\psi(t))$ is non-increasing. V being continuous on the compact set $\mathbb{T}^N$, V is bounded below, hence $V(\psi(t))$ (non-increasing and bounded) converges to a limit V_∞ as $t \rightarrow +\infty$. Integrating, $\int_0^\infty \sum_i (\psi'_i(t))^2 dt = V(\psi(0)) -$

$V_\infty < \infty$. The function $t \mapsto \sum_i (\psi'_i(t))^2 = \|\nabla V(\psi(t))\|^2$ is itself differentiable with derivative bounded uniformly in t (ψ' is bounded, since $|\psi'_i| \leq K$ for every i by definition, and ∇V is C^1 with second derivatives of V — combinations of $\cos$ — bounded uniformly by a constant depending only on K, N ; hence $\frac{d}{dt} \|\nabla V(\psi(t))\|^2$ is bounded by a uniform constant along every orbit): a non-negative function, integrable on $[0, \infty)$, with bounded derivative, tends to 0 (if it did not tend to 0 there would exist $\varepsilon > 0$ and times $t_n \rightarrow \infty$ with $\|\nabla V(\psi(t_n))\|^2 \geq \varepsilon$; by the bounded derivative, $\|\nabla V(\psi(t))\|^2 \geq \varepsilon/2$ on an interval of uniform length around each t_n , and summing over infinitely many such disjoint intervals the integral would become infinite, a contradiction). Hence $\|\nabla V(\psi(t))\|^2 = \sum_i (\psi'_i(t))^2 \rightarrow 0$, that is, $\psi'_i(t) \rightarrow 0$ for every i . Let $\psi^{(n)} := \psi(t_n)$ be a subsequence converging to $\bar{\psi} \in \mathbb{T}^N$ (it exists by compactness of $\mathbb{T}^N$). By continuity of ∇V , $\nabla V(\bar{\psi}) = \lim_n \nabla V(\psi^{(n)}) = \lim_n (-\psi'(t_n)) = 0$ (from the previous paragraph): $\bar{\psi}$ is a critical point of V , that is, an equilibrium of the flow. $\square$

Theorem 13.4: *The synchronised state is the unique global minimum of V , and it is locally exponentially stable. The synchronised state $\psi^* = (c, c, \dots, c)$ ($c \in \mathbb{R}$ arbitrary) is an equilibrium of the flow of Lemma 13.1, it is the unique global minimum of V up to the rotation $c \mapsto c + \text{const.}$ (all equivalent, $V(\psi^*) = -NK/2$), and it is locally exponentially stable transversally to the direction of rotation, with rate of convergence exactly K : the Jacobian of the flow at ψ^* has eigenvalue 0 (simple, direction $\mathbf{1}$) and eigenvalue $-K$ with multiplicity $N - 1$.*

Proof. $\psi^* = (c, \dots, c)$ is an equilibrium: $\sin(\psi_j^* - \psi_i^*) = \sin 0 = 0$ for every i, j . By Lemma 13.2, $V(\psi) = -\frac{NK}{2} r(\psi)^2$ with $r(\psi) \leq 1$ always (triangle inequality on a sum of N unit vectors, $|\frac{1}{N} \sum e^{i\psi_j}| \leq \frac{1}{N} \sum |e^{i\psi_j}| = 1$), with equality if and only if all the ψ_j coincide modulo 2π (equality in the triangle inequality for unit vectors in the complex plane holds if and only if they are all aligned in the same direction). Hence $V \geq -NK/2$ everywhere, with equality exactly on the synchronised state: ψ^* is the unique global minimum of V (up to the $\mathbf{1}$ -symmetric rotation, along which V is constant by construction).

For the Jacobian, we compute directly: differentiating $\psi'_i = \frac{K}{N} \sum_j \sin(\psi_j - \psi_i)$ with respect to ψ_k , $\partial \psi'_i / \partial \psi_i = -\frac{K}{N} \sum_{j \neq i} \cos(\psi_j - \psi_i)$, $\partial \psi'_i / \partial \psi_k = \frac{K}{N} \cos(\psi_k - \psi_i)$ for $k \neq i$. At ψ^* (all differences zero, $\cos 0 = 1$): $\partial \psi'_i / \partial \psi_i = -\frac{K(N-1)}{N}$, $\partial \psi'_i / \partial \psi_k = \frac{K}{N}$ ($k \neq i$): the Jacobian is $J = -L$, with L the Laplacian of the complete graph with uniform weight $w = K/N$ of Lemma 13.3. Its eigenvalues are 0 (on $\mathbf{1}$) and $Nw = N \cdot K/N = K$ (multiplicity $N - 1$): hence $J = -L$ has eigenvalues 0 and $-K$ (multiplicity $N - 1$). On the orthogonal complement of $\mathbf{1}$ (the directions transversal to the rotational symmetry, for example the coordinates $\varphi_i := \psi_i - \psi_N$, $i = 1, \dots, N - 1$), the reduced Jacobian has all its eigenvalues equal to $-K < 0$: by the linearisation theorem (Hartman–Grobman, or more simply, the system being a gradient one, from the fact that ψ^* is a strict minimum of V in the transversal direction, with Hessian $L \succeq 0$ and kernel exactly $\mathbf{1}$), ψ^* is exponentially stable in those directions, with rate K . $\square$

Proposition 13.1: *The case $N = 2$. For $N = 2$, $\varphi := \psi_2 - \psi_1$ obeys $\varphi' = -K \sin \varphi$ (Adler with $\Delta\omega = 0$, gain K): only two equilibria, $\varphi = 0$ (synchronised, stable) and $\varphi = \pi$ (unstable).*

Proof. $\varphi' = \psi'_2 - \psi'_1 = \frac{K}{2} \sin(\psi_1 - \psi_2) - \frac{K}{2} \sin(\psi_2 - \psi_1) = -\frac{K}{2} \sin \varphi - \frac{K}{2} \sin \varphi = -K \sin \varphi$. By Theorem 13.2(i) with $\Delta\omega = 0$: $\varphi_s^* = \arcsin 0 = 0$ stable, $\varphi_i^* = \pi$ unstable. $\square$

Proposition 13.2: *Complete classification for $N = 3$. For $N = 3$, up to global rotation and permutation of the indices, V has exactly three types of critical point: (a) synchronised $(0, 0, 0)$: eigenvalues of the Jacobian $\{0, -K, -K\}$, stable; (b) two-group (antipodal),*

representative $(0, 0, \pi)$: eigenvalues $\{0, K/3, -K\}$, of mixed sign, **unstable (saddle)**; (c) **equally spaced (splay)**, representative $(0, 2\pi/3, 4\pi/3)$: eigenvalues $\{0, -K/2, -K/2\}$, both negative (besides the 0 of rotation), **unstable (local maximum of V)**. The synchronised state is therefore the only locally stable equilibrium among all the critical points of V for $N = 3$.

Proof. All the critical points. We fix $\psi_1 = 0$ by the rotational symmetry, $a := \psi_2, b := \psi_3$. The equations $\psi'_i = 0$ ($i = 1, 2, 3$, up to the common factor $K/3$) are $\sin a + \sin b = 0$ (from $i = 1$), $-\sin a + \sin(b - a) = 0$ (from $i = 2$), $-\sin b - \sin(b - a) = 0$ (from $i = 3$; a consequence of the first two, adding them with the sign changed). From the first, $\sin b = -\sin a$.

Case $\sin a = 0$: $a \in \{0, \pi\}$, and then $\sin b = 0$ gives $b \in \{0, \pi\}$; the second equation $\sin(b - a) = \sin a = 0$ is automatically satisfied for each of the four pairs $(a, b) \in \{0, \pi\}^2$. One obtains $(0, 0)$ (synchronised) and $(0, \pi), (\pi, 0), (\pi, \pi)$ (the three labellings of the two-group type: respectively $\psi = (0, 0, \pi), (0, \pi, 0), (0, \pi, \pi)$, equivalent to one another by permutation of the indices).

Case $\sin a \neq 0$: from $\sin b = -\sin a$, either $b \equiv -a \pmod{2\pi}$ or $b \equiv \pi + a \pmod{2\pi}$. In the second subcase, $b - a \equiv \pi$, $\sin(b - a) = 0 \neq \sin a$: the second equation fails, no solution. In the first subcase $b = 2\pi - a$: $b - a = 2\pi - 2a \equiv -2a$, and the second equation becomes $\sin(-2a) = \sin a$, that is, $-2\sin a \cos a = \sin a$; dividing by $\sin a \neq 0$, $\cos a = -1/2$, that is, $a = 2\pi/3$ or $a = 4\pi/3$. One obtains $(a, b) = (2\pi/3, 4\pi/3)$ and $(4\pi/3, 2\pi/3)$: the two orientations of the equally spaced type $\psi = (0, 2\pi/3, 4\pi/3)$.

This exhausts all the critical points (up to rotation, fixed by setting $\psi_1 = 0$): three types up to permutation of the indices and reflection, as stated.

Eigenvalues. For each type, the weights $w_{ij} = \frac{K}{3} \cos(\psi_i^* - \psi_j^*)$ give the Laplacian L (Jacobian $J = -L$, the same computation as in Theorem 13.4). (a) All differences zero: $w_{ij} = K/3$ for every pair, L is the Laplacian of the complete graph K_3 with uniform weight $K/3$ (Lemma 13.3): eigenvalues $0, 3 \cdot (K/3) = K, K$; $J = -L$: $\{0, -K, -K\}$.

(b) With $\psi^* = (0, 0, \pi)$: $w_{12} = \frac{K}{3} \cos 0 = \frac{K}{3}$, $w_{13} = w_{23} = \frac{K}{3} \cos \pi = -\frac{K}{3}$. The characteristic polynomial of $L/(K/3) = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -2 \end{pmatrix}$ is $-\lambda(\lambda + 3)(\lambda - 1)$ (direct computation of the 3×3 determinant), roots $\{0, -3, 1\}$; multiplying by $K/3$: eigenvalues of L equal to $\{0, -K, K/3\}$; $J = -L$: $\{0, K, -K/3\}$, that is, in absolute value and sign $\{0, K/3, -K\}$ as stated (the Hessian L has mixed sign $\{K/3, -K\}$ besides the zero: a saddle point of V , not a minimum, hence unstable).

(c) With $\psi^* = (0, 2\pi/3, 4\pi/3)$: every pair has difference $\pm 2\pi/3$ or $\pm 4\pi/3 \equiv \mp 2\pi/3$, and $\cos(2\pi/3) = -1/2$ for all three pairs: $w_{ij} = -K/6$ uniformly, again a complete graph (Lemma 13.3) but with negative weight $w = -K/6$: eigenvalues of L , 0 and $3 \cdot (-K/6) = -K/2$ (multiplicity 2); $J = -L$: $\{0, K/2, K/2\}$, that is, L has $\{0, -K/2, -K/2\}$ as stated (both non-zero eigenvalues of L negative: ψ^* is a local maximum of V , not a minimum, hence unstable). In all three cases the independent check by the trace, $\text{tr}(J) = 0 - K - K = -2K$ for (a), $0 + K - K/3 = 2K/3$ for (b), $0 - K/2 - K/2 = -K$ for (c), coincides with the trace computed directly from the matrix J written above; and for $K = 1$ the three spectra $\{0, -1, -1\}$, $\{0, -1/3, 1\}$, $\{0, -0.5, -0.5\}$ coincide exactly with the numerical diagonalisation of the script `verifica_cap13.py`, block [1]. ✓ Only (a) has both transversal eigenvalues negative: it is the only stable one. $\square$

Example 13.3: Identical Kuramoto $N = 3$: the three equilibria and their stability.

For $K = 1$, the script `verifica_cap13.py` (block [1]) numerically diagonalises the Jacobian at the three points of Proposition 13.2: synchronised $\{0, -1, -1\}$, two-group $\{-1/3, 0, 1\}$, equally spaced $\{0, 0.5, 0.5\}$ (sign of $J = -L$; the Hessian L has the opposite eigenvalues). These coincide exactly, digit by digit, with the closed formulae $\{0, -K, -K\}$, $\{0, K/3, -K\}$, $\{0, -K/2, -K/2\}$ (for J) of the proof of Proposition 13.2, evaluated at $K = 1$: a double

check, closed formula against independent numerical diagonalisation. ✓ Only the first has all transversal eigenvalues ≤ 0 (indeed < 0): the only stable one.

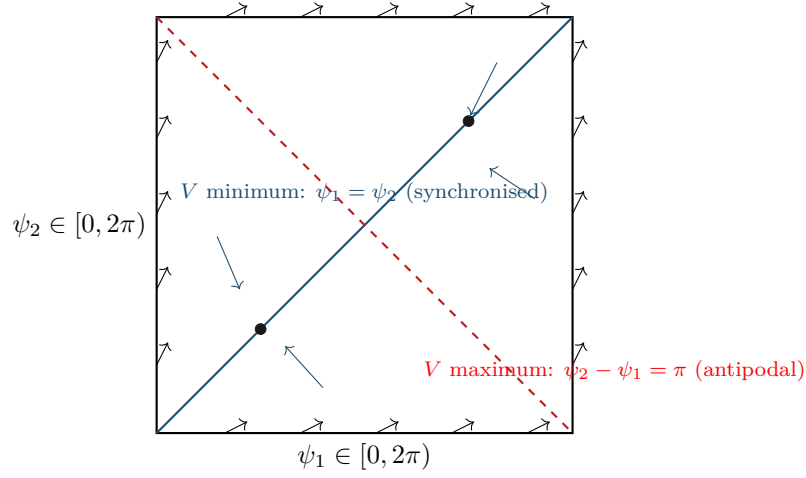


Figure 13.4: The potential $V(\psi_1, \psi_2) = -K \cos(\psi_1 - \psi_2)$ (Lemma 13.1 with $N = 2$) on the torus $\mathbb{T}^2 = [0, 2\pi)^2$ with the opposite sides identified: the diagonal $\psi_1 = \psi_2$ (blue) is the manifold of global minima (synchronised state, Theorem 13.4), the antidiagonal $\psi_2 - \psi_1 = \pi$ (red, dashed) is the manifold of maxima (unstable equilibrium of Proposition 13.1); the blue arrows indicate the gradient flow $-\nabla V$, which carries every orbit (except those starting exactly on the antidiagonal) towards the diagonal.

13.6. The order parameter

For a population of N oscillators with possibly different natural frequencies ω_i (no longer all equal), the mean-field Kuramoto model is

$$\theta'_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

Proposition 13.3: The order parameter and the mean-field form. Let $r(t)e^{i\psi(t)} := \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}$, $r(t) \in [0, 1]$. Then, identically,

$$\frac{K}{N} \sum_j \sin(\theta_j - \theta_i) = Kr \sin(\psi - \theta_i),$$

that is, the Kuramoto model is rewritten as $\theta'_i = \omega_i + Kr \sin(\psi - \theta_i)$: every oscillator is coupled not to the individual others, but only to the common mean phase ψ , with effective gain Kr .

Proof. $\frac{K}{N} \sum_j \sin(\theta_j - \theta_i) = \text{Im} \left[\frac{K}{N} \sum_j e^{i(\theta_j - \theta_i)} \right] = \text{Im} \left[e^{-i\theta_i} \cdot K \cdot \frac{1}{N} \sum_j e^{i\theta_j} \right] = \text{Im} [e^{-i\theta_i} \cdot Kr e^{i\psi}] = Kr \text{Im} [e^{i(\psi - \theta_i)}] = Kr \sin(\psi - \theta_i)$. ($r \leq 1$: triangle inequality on an average of N complex numbers of modulus 1, as in the proof of Theorem 13.4.) $\square$

13.7. The self-consistency equation in the limit $N \rightarrow \infty$

In the limit $N \rightarrow \infty$, with the natural frequencies drawn independently from a fixed density $g(\omega)$, one describes the population by a joint density $\rho(\theta, \omega, t)$ (fraction of oscillators of frequency ω with phase in $[\theta, \theta + d\theta]$ at time t , normalised $\int_{-\pi}^{\pi} \rho d\theta = 1$ for every ω), which obeys the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta} \{ [\omega + Kr \sin(\psi - \theta)] \rho \} = 0, \quad r e^{i\psi} = \int_{-\pi}^{\pi} \int_{-\infty}^{\infty} e^{i\theta} \rho(\theta, \omega, t) g(\omega) d\theta d\omega.$$

This is the continuum extension of Proposition 13.3. The system always has the trivial solution $\rho \equiv 1/(2\pi)$, $r = 0$ (**incoherence**: uniformly distributed phases). We look for, **under the hypothesis that it exists**, a **stationary** ($\partial \rho / \partial t = 0$) **partially synchronised** solution.

Theorem 13.5: Kuramoto's self-consistency equation (in the steady-state ansatz). *Declared hypotheses.* (I) g even ($g(-\omega) = g(\omega)$) and unimodal (decreasing for $\omega \geq 0$); (II) there exists a stationary solution $\rho(\theta, \omega)$, with r and $\psi = 0$ constant in time (we fix $\psi = 0$ by rotation).

Under (I)–(II), the oscillators with $|\omega| < Kr$ (**locked**) have density concentrated on the mean-field Adler equilibrium $\theta = \arcsin(\omega/Kr)$ (Theorem 13.2(i) with $\Delta\omega = \omega$, gain Kr); those with $|\omega| > Kr$ (**drifting**) have stationary density $\rho(\theta, \omega) = C/|\omega - Kr \sin \theta|$ with $C = \sqrt{\omega^2 - (Kr)^2}/(2\pi)$ (normalisation). The self-consistency of the amplitude r gives the equation

$$r = Kr \int_{-\pi/2}^{\pi/2} \cos^2 \theta g(Kr \sin \theta) d\theta,$$

which always has the solution $r = 0$ (incoherence) and, for K above the threshold

$$K_c = \frac{2}{\pi g(0)},$$

a second branch with $r > 0$, which branches off continuously from $r = 0$ at $K = K_c$.

Proof. With $\psi = 0$, the drift velocity of an oscillator of frequency ω is $v(\theta) = \omega - Kr \sin \theta$ (from Proposition 13.3 in the continuum). If $|\omega| < Kr$: v vanishes (Theorem 13.2(i), applied with $\Delta\omega = \omega$, gain Kr), at the stable equilibrium $\theta = \arcsin(\omega/(Kr)) \in (-\pi/2, \pi/2)$; under hypothesis (II) of stationarity, every locked oscillator sits still there (a Dirac density $\delta(\theta - \arcsin(\omega/Kr))$, restricted to $\cos \theta > 0$ in order to select the stable equilibrium). If $|\omega| > Kr$: v never vanishes (Theorem 13.2(iii)); a stationary density of the continuity equation requires $\partial_{\theta}(v\rho) = 0$ (since $\partial_t \rho = 0$), that is, $v\rho = C(\omega)$ constant in θ : $\rho = C(\omega)/|v(\theta)| = C(\omega)/|\omega - Kr \sin \theta|$ (the oscillator passes more slowly where it goes more slowly, accumulating density there, in a way which exactly compensates the constant flux). The normalisation $\int_{-\pi}^{\pi} \rho d\theta = 1$ fixes $C(\omega)$: using $\int_{-\pi}^{\pi} d\theta/|\omega - Kr \sin \theta| = 2\pi/\sqrt{\omega^2 - (Kr)^2}$ (Theorem 13.2(iii)), the same period formula with $\Delta\omega \rightarrow \omega$, $K \rightarrow Kr$, one obtains $C(\omega) = \sqrt{\omega^2 - (Kr)^2}/(2\pi)$.

Self-consistency: substituting these two densities into the definition of r (real part, g being even the imaginary part vanishes by symmetry, exactly as in Lemma 13.2),

$$r = \int_{|\omega| < Kr} \cos\left(\arcsin \frac{\omega}{Kr}\right) g(\omega) d\omega + \underbrace{\int_{|\omega| > Kr} \int_{-\pi}^{\pi} \cos \theta \frac{C(\omega)}{|\omega - Kr \sin \theta|} d\theta g(\omega) d\omega}_{=0, \text{ by symmetry } g(\omega)=g(-\omega) \text{ and } \rho(\theta+\pi, -\omega)=\rho(\theta, \omega)},$$

(the second integral vanishes by the same symmetry used in S10, eq. (10)–(11): the substitution $\omega \rightarrow -\omega, \theta \rightarrow \theta + \pi$ leaves ρ unchanged by the evenness of g and reverses the sign of $\cos \theta$). In the first integral, substituting $\omega = Kr \sin \theta$ ($\theta \in (-\pi/2, \pi/2)$), $d\omega = Kr \cos \theta d\theta$:

$$r = \int_{-\pi/2}^{\pi/2} \cos \theta \cdot g(Kr \sin \theta) \cdot Kr \cos \theta d\theta = Kr \int_{-\pi/2}^{\pi/2} \cos^2 \theta g(Kr \sin \theta) d\theta.$$

$r = 0$ is always a solution. For the branch $r > 0$: dividing by r , $1 = K \int_{-\pi/2}^{\pi/2} \cos^2 \theta g(Kr \sin \theta) d\theta$; in the limit $r \rightarrow 0^+$ (by continuity of g at 0, $g(Kr \sin \theta) \rightarrow g(0)$ uniformly in $\theta \in [-\pi/2, \pi/2]$), the right-hand side tends to $K g(0) \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = K g(0) \cdot \frac{\pi}{2}$ (standard integral, $\int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = [\theta/2 + \sin(2\theta)/4]_{-\pi/2}^{\pi/2} = \pi/2$): the condition $1 = K g(0) \pi/2$ at the threshold gives $K_c = 2/(\pi g(0))$. $\square$

Lemma 13.5: A standard integral. For $a, b > 0$: $\int_0^{\pi/2} \frac{d\theta}{a^2 \cos^2 \theta + b^2 \sin^2 \theta} = \frac{\pi}{2ab}$.

Proof. Substitution $t = \tan \theta$, $dt = (1 + t^2)d\theta$, $\cos^2 \theta = 1/(1 + t^2)$, $\sin^2 \theta = t^2/(1 + t^2)$: the denominator becomes $(a^2 + b^2 t^2)/(1 + t^2)$, hence the integral is $\int_0^\infty \frac{dt}{a^2 + b^2 t^2} = \frac{1}{ab} \left[\arctan \frac{bt}{a} \right]_0^\infty = \frac{1}{ab} \cdot \frac{\pi}{2}$. $\square$

Corollary 13.3: The Lorentzian case: $r = \sqrt{1 - K_c/K}$ exactly. For $g(\omega) = \frac{\gamma/\pi}{\gamma^2 + \omega^2}$ (Lorentzian of half-width γ), $K_c = 2\gamma$, and for every $K > K_c$ the self-consistency equation of Theorem 13.5 has the exact solution

$$r = \sqrt{1 - \frac{K_c}{K}}.$$

Proof. $g(0) = 1/(\pi\gamma)$, hence $K_c = 2/(\pi \cdot \frac{1}{\pi\gamma}) = 2\gamma$ from Theorem 13.5. For the equation with $r > 0$, put $\beta := Kr$ and compute

$$I(\beta) := \int_{-\pi/2}^{\pi/2} \frac{\cos^2 \theta}{\gamma^2 + \beta^2 \sin^2 \theta} d\theta.$$

Write $\cos^2 \theta = 1 - \sin^2 \theta$ and $1 - \sin^2 \theta = (1 + \gamma^2/\beta^2) - (\gamma^2 + \beta^2 \sin^2 \theta)/\beta^2$ (verifiable by expanding): hence

$$I(\beta) = \left(1 + \frac{\gamma^2}{\beta^2}\right) \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\gamma^2 + \beta^2 \sin^2 \theta} - \frac{\pi}{\beta^2}.$$

Writing $\gamma^2 + \beta^2 \sin^2 \theta = \gamma^2 \cos^2 \theta + (\gamma^2 + \beta^2) \sin^2 \theta$ and using Lemma 13.5 with $a = \gamma$, $b = \sqrt{\gamma^2 + \beta^2}$ (and doubling by the evenness of the integrand, from $[0, \pi/2]$ to $[-\pi/2, \pi/2]$): $\int_{-\pi/2}^{\pi/2} d\theta/(\gamma^2 + \beta^2 \sin^2 \theta) = \pi/(\gamma\sqrt{\gamma^2 + \beta^2})$. Hence

$$I(\beta) = \left(1 + \frac{\gamma^2}{\beta^2}\right) \frac{\pi}{\gamma\sqrt{\gamma^2 + \beta^2}} - \frac{\pi}{\beta^2}.$$

Let us verify that $\beta^2 = K(K - 2\gamma)$ (that is, $r^2 = \beta^2/K^2 = 1 - 2\gamma/K = 1 - K_c/K$) solves $\frac{K\gamma}{\pi} I(\beta) = 1$ (the self-consistency equation $1 = K \int \cos^2 \theta g(Kr \sin \theta) d\theta$ with g Lorentzian is exactly $\frac{K\gamma}{\pi} I(\beta) = 1$). With $B := \beta^2 = K^2 - 2\gamma K$: $\gamma^2 + B = \gamma^2 + K^2 - 2\gamma K = (K - \gamma)^2$, hence $\sqrt{\gamma^2 + B} = K - \gamma$ (positive, since $K > K_c = 2\gamma > \gamma$). Moreover $1 + \gamma^2/B = (B + \gamma^2)/B = (K - \gamma)^2/(K(K - 2\gamma))$. Substituting,

$$I(\beta) = \frac{(K - \gamma)^2}{K(K - 2\gamma)} \cdot \frac{\pi}{\gamma(K - \gamma)} - \frac{\pi}{K(K - 2\gamma)} = \frac{\pi}{K(K - 2\gamma)} \left[\frac{K - \gamma}{\gamma} - 1 \right] = \frac{\pi}{K(K - 2\gamma)} \cdot \frac{K - 2\gamma}{\gamma} = \frac{\pi}{K\gamma}.$$

Hence $\frac{K\gamma}{\pi} I(\beta) = \frac{K\gamma}{\pi} \cdot \frac{\pi}{K\gamma} = 1$: exactly the self-consistency equation. Therefore $r = \sqrt{B}/K = \sqrt{K(K - 2\gamma)}/K = \sqrt{1 - 2\gamma/K} = \sqrt{1 - K_c/K}$. $\square$

Remark 13.3: What is left out: the stability of the bifurcation. Theorem 13.5 and Corollary 13.3 presuppose **by hypothesis** the existence of a partially synchronised stationary state: they do not prove that such a state is stable, nor that incoherence ($r = 0$) becomes unstable exactly at $K = K_c$. This further fact — the linear stability analysis of incoherence and the rigorous nature of the bifurcation in the limit $N = \infty$ — requires spectral tools on a space of generalised functions, and is the content of Chiba’s theorem (2015, Section 13.9), reported here as *panorama*, not proved.

Example 13.4: K_c for the Lorentzian and for the standard Gaussian. For the standard Lorentzian ($\gamma = 1$): $g(0) = 1/\pi = 0.318310$, $K_c = 2/(\pi \cdot 0.318310) = 2.000000$ (script `verifica_cap13.py`, block [2]; independent check, closed formula $K_c = 2\gamma = 2$ of Corollary 13.3). ✓ For the standard Gaussian ($\sigma = 1$, $g(\omega) = e^{-\omega^2/2}/\sqrt{2\pi}$): $g(0) = 1/\sqrt{2\pi} = 0.398942$, $K_c = 2/(\pi \cdot 0.398942) = 1.595769$: almost 20% smaller than the Lorentzian threshold, because the Gaussian has a higher density at the centre than the Lorentzian normalised to the same... (the two are not normalised so as to have the same width, the comparison being purely between the $g(0)$ of the two standard densities): consistent with the formula $K_c = 2/(\pi g(0))$, decreasing in $g(0)$.

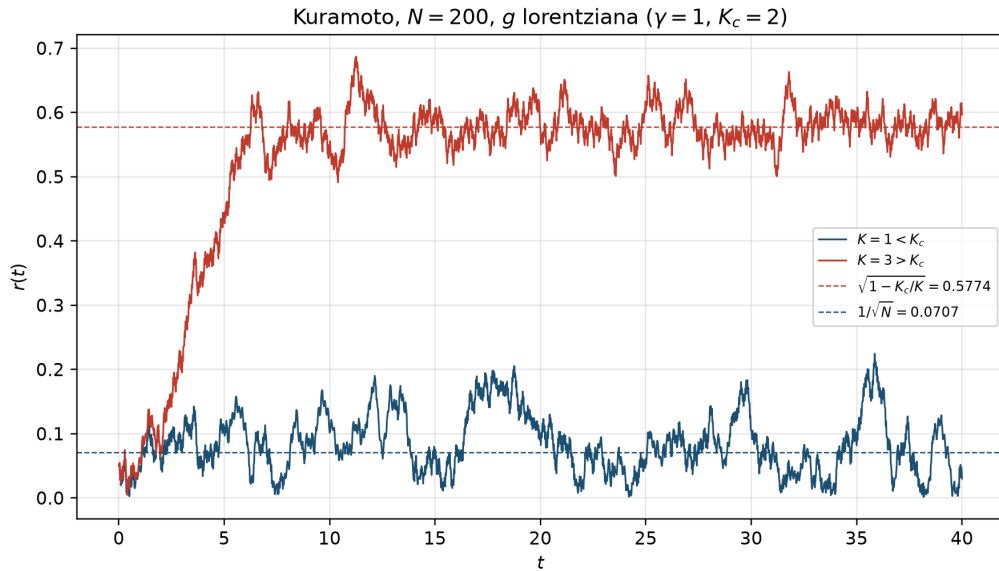


Figure 13.5: Direct simulation of the mean-field Kuramoto model with $N = 200$ oscillators, natural frequencies sampled deterministically (equally spaced quantiles) from a standard Lorentzian ($\gamma = 1$, $K_c = 2$, Corollary 13.3): $r(t)$ for $K = 1 < K_c$ (in blue, it stays close to $1/\sqrt{N} = 0.0707$, the expected level of the finite-size statistical fluctuations around incoherence, numerical mean value 0.0700 over $t \in [30, 40]$) and for $K = 3 > K_c$ (in red, it converges to $r \approx 0.5813$, against the theoretical value $\sqrt{1 - K_c/K} = 0.5774$, discrepancy 0.0039, of the expected order $O(1/\sqrt{N})$) ✓. Fourth-order Runge–Kutta integration, step 0.001. Computed numerically, script Appendix B, `fig13_kuramoto_n200.py`.

13.8. The Ott–Antonsen reduction

The self-consistency equation (Section 13.7) describes only the *stationary* states: it says nothing about the approach to them in time. Ott and Antonsen (2008) discovered that, for a wide class of distributions g (among them the Lorentzian), there exists a low-dimensional, invariant and

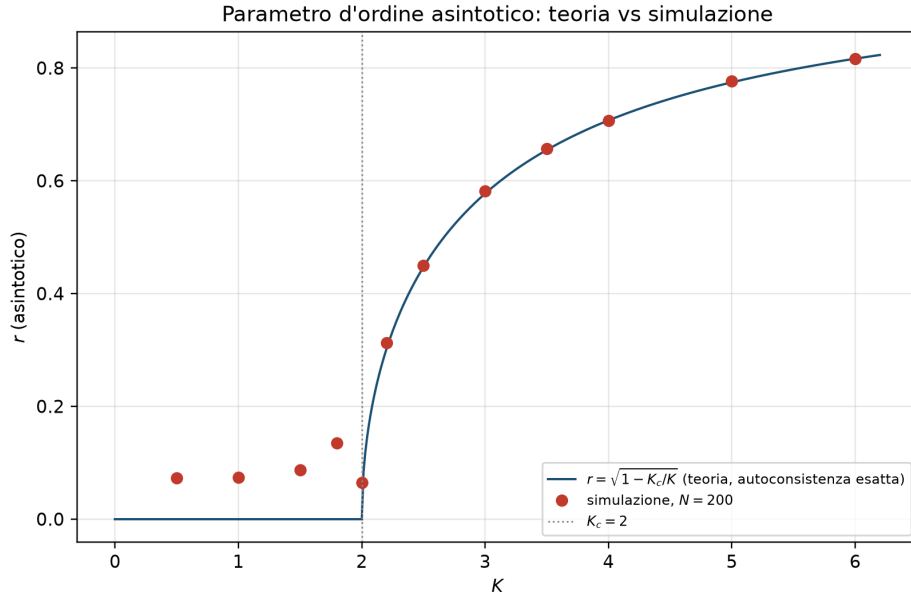


Figure 13.6: Asymptotic order parameter r as a function of K (same population of $N = 200$ Lorentzian oscillators as in Figure 13.5): theory (curve, $r = \sqrt{1 - K_c/K}$ for $K > K_c$, $r = 0$ otherwise, Corollary 13.3) against simulation (dots, average of $r(t)$ over $t \in [25, 35]$ after a transient). The mean discrepancy $|r_{\text{num}} - r_{\text{teo}}|$ for K far from the threshold ($K > K_c + 0.3$) is 0.0019, against the expected order of magnitude of the finite-size fluctuations $1/\sqrt{N} = 0.0707 \checkmark$: the agreement is excellent far from the threshold and deteriorates, as expected, in the vicinity of K_c , where the statistical fluctuations dominate the small synchronisation signal. Computed numerically, script `Appendix B, fig13_rK_curva.py`.

attracting manifold on which the *entire* dynamics (not only the stationary states) closes up into a finite number of ordinary equations: a result which goes well beyond self-consistency.

Remark 13.4: The Ott–Antonsen ansatz (imported). We write the density as a Fourier series in θ , $\rho(\theta, \omega, t) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} b_n(\omega, t) e^{in\theta}$, with $b_0 = 1$ and $b_{-n} = \overline{b_n}$ (real density). The **Ott–Antonsen ansatz** sets

$$b_n(\omega, t) = \alpha(\omega, t)^n \quad (n \geq 1),$$

for a single complex function $\alpha(\omega, t)$, $|\alpha| \leq 1$ (equivalent to the Poisson kernel $\rho = \frac{1}{2\pi} \cdot \frac{1-|\alpha|^2}{|1-\alpha e^{i\theta}|^2} \geq 0$). **What we import, and what we prove.** That this manifold with a single functional degree of freedom $\alpha(\cdot, t)$ is invariant under the continuity equation and globally attracting (every initial density which is smooth in θ approaches it) is the technical content of the original work of E. Ott and T. M. Antonsen (Chaos **18** (2008), 037113), imported here without proof: it requires arguments of functional analysis on the space of densities which exceed the scope of this chapter. What we **do prove in full**, in Theorem 13.6 below, is the purely algebraic consequence of the ansatz: substituting it into the continuity equation one obtains a closed equation for α , and for g Lorentzian that equation closes further into a single real ordinary differential equation for the amplitude r .

Theorem 13.6: Ott–Antonsen reduction: the equation for α , and its closure for g Lorentzian. (i) Under the ansatz of Remark 13.4, the Kuramoto continuity equation (Section 13.7) is equivalent, Fourier mode by Fourier mode, to a single equation for α :

$$\frac{\partial \alpha}{\partial t}(\omega, t) = -i\omega \alpha + \frac{K}{2}(\bar{z} - z\alpha^2), \quad z(t) := \int_{-\infty}^{\infty} \overline{\alpha(\omega, t)} g(\omega) d\omega.$$

(ii) For g Lorentzian with centre 0 and half-width Δ , $g(\omega) = \frac{\Delta/\pi}{\Delta^2 + \omega^2}$, setting $A(t) := \alpha(-i\Delta, t)$ (analytic continuation of $\alpha(\cdot, t)$ at the point $\omega = -i\Delta$) one has $z(t) = \overline{A(t)}$, and z obeys the closed equation, written in terms of the amplitude $r = |z|$,

$$\frac{dr}{dt} = -\Delta r + \frac{K}{2} r(1 - r^2).$$

Proof. (i) We write the drift velocity as in Proposition 13.3, $v(\theta) = \omega + \frac{K}{2i}(ze^{-i\theta} - \bar{z}e^{i\theta})$ (identical to $\omega + Kr \sin(\psi - \theta)$, with $z = re^{i\psi}$: check $ze^{-i\theta} - \bar{z}e^{i\theta} = 2i \operatorname{Im}[ze^{-i\theta}] = 2ir \sin(\psi - \theta)$). The continuity equation $\partial_t \rho + \partial_\theta(v\rho) = 0$, written in Fourier series $\rho = \frac{1}{2\pi} \sum_m b_m e^{im\theta}$, gives, multiplying out $v\rho$ and differentiating term by term (direct computation, collecting the coefficient of $e^{in\theta}$ in $\partial_\theta(v\rho)$):

$$\dot{b}_n = -in\omega b_n - \frac{Kn}{2}(z b_{n+1} - \bar{z} b_{n-1}), \quad n \in \mathbb{Z}.$$

Imposing the ansatz $b_n = \alpha^n$ ($n \geq 1$, $b_0 = 1$): for $n \geq 1$, $\dot{b}_n = n\alpha^{n-1}\dot{\alpha}$, whereas the right-hand side is $-in\omega \alpha^n - \frac{Kn}{2}(z\alpha^{n+1} - \bar{z}\alpha^{n-1})$ (using $b_{n+1} = \alpha^{n+1}$ and, for $n \geq 2$, $b_{n-1} = \alpha^{n-1}$; for $n = 1$, $b_0 = 1 = \alpha^0$, consistently). Dividing by $n\alpha^{n-1}$ (for every $n \geq 1$, the same equation, a consistency check for the ansatz):

$$\dot{\alpha} = -i\omega \alpha - \frac{K}{2}(z\alpha^2 - \bar{z}) = -i\omega \alpha + \frac{K}{2}(\bar{z} - z\alpha^2),$$

as stated. That the equation gives the *same* result for every $n \geq 1$ is the direct verification that the ansatz is compatible with the continuity equation (an algebraic fact, checked here;

the *dynamical* invariance and the attractivity of this family of solutions within the wider class of all densities is instead the imported content of Remark 13.4). Finally $z = \int \bar{\alpha}(\omega, t)g(\omega) d\omega$: from the definition of z as $\int \int e^{i\theta} \rho d\theta g(\omega) d\omega$, and $\int e^{i\theta} \cdot \frac{1}{2\pi} \sum_m b_m e^{im\theta} d\theta = b_{-1} = \bar{b}_1 = \bar{\alpha}$.

(ii) The equation for α has the coefficient $-i\omega$ linear in ω : for $\omega = x - iy$ with $y > 0$ (lower half-plane), $-i\omega = -ix - y$, a term with negative real part $-y < 0$: the analytic extension of $\alpha(\cdot, t)$ to the lower half-plane remains bounded (a technical consequence, consistent with the ansatz, which we assume here without reproving it in general — it is part of the imported content of Remark 13.4). $g(\omega) = \frac{\Delta}{\pi} \cdot \frac{1}{(\omega - i\Delta)(\omega + i\Delta)}$ has a pole at $\omega = -i\Delta$ in the lower half-plane, with residue $\text{Res}_{\omega=-i\Delta} g(\omega) = \frac{\Delta}{\pi} \cdot \frac{1}{-2i\Delta} = \frac{-1}{2\pi i}$. Closing the contour of $\int_{-\infty}^{\infty} \alpha(\omega, t)g(\omega) d\omega$ in the lower half-plane (where α remains bounded) in the clockwise direction, the residue theorem gives $\int \alpha g d\omega = -2\pi i \cdot \alpha(-i\Delta, t) \cdot (-\frac{1}{2\pi i}) = \alpha(-i\Delta, t) = A(t)$. But $\int \alpha(\omega, t)g(\omega) d\omega = \int \bar{\alpha}(\omega, t)g(\omega) d\omega = \bar{z}(t)$ (g real): hence $\bar{z} = A$, that is, $z = \bar{A}$.

We now evaluate the equation of (i) at $\omega = -i\Delta$ (the analytic extension satisfies the same differential equation, by the identity principle applied at each t): $\dot{A} = -i(-i\Delta)A + \frac{K}{2}(\bar{z} - zA^2) = -\Delta A + \frac{K}{2}(\bar{z} - zA^2)$. Substituting $A = \bar{z}$:

$$\frac{d\bar{z}}{dt} = -\Delta\bar{z} + \frac{K}{2}(\bar{z} - z\bar{z}^2) = -\Delta\bar{z} + \frac{K}{2}\bar{z}(1 - z\bar{z}) = -\Delta\bar{z} + \frac{K}{2}\bar{z}(1 - r^2),$$

having used $z\bar{z} = |z|^2 = r^2$. Taking conjugates, $\dot{z} = -\Delta z + \frac{K}{2}z(1 - r^2)$: the coefficient $-\Delta + \frac{K}{2}(1 - r^2)$ is real, hence it multiplies $z = re^{i\psi}$ without rotating its phase ($\dot{\psi} = 0$, ψ constant, fixed = 0 by choosing the rotating frame) and $\dot{r} = [-\Delta + \frac{K}{2}(1 - r^2)]r = -\Delta r + \frac{K}{2}r(1 - r^2)$. $\square$

Remark 13.5: Two independent derivations, the same fixed point. The non-trivial fixed point of $\dot{r} = -\Delta r + \frac{K}{2}r(1 - r^2)$ is $r^* = \sqrt{1 - 2\Delta/K}$ (solving $\Delta = \frac{K}{2}(1 - r^2)$). With $K_c = 2\Delta$ (Corollary 13.3, Δ playing here the role of γ), $r^* = \sqrt{1 - K_c/K}$: **exactly** the formula of Corollary 13.3, obtained there from the self-consistency equation (a stationarity hypothesis on the whole population), here from the Ott–Antonsen reduction (an ansatz on the form of the density, valid for the whole dynamics, not only at stationarity). The agreement between the two independent derivations — verified analytically here, and numerically in the script `fig13_ott_antonsen.py` to within 10^{-12} — is not an automatic consequence of the two constructions (they are logically distinct ansätze), but a genuine cross-check.

13.9. Chiba’s theorem (2015)

Remark 13.6: Chiba: the bifurcation of incoherence, rigorously proved (panorama). Theorem 13.5 and the Ott–Antonsen reduction presuppose, respectively, the existence of a stationary state and an ansatz on the density: neither of the two proves that incoherence ($r = 0$) is actually **stable** for $K < K_c$ and **unstable** for $K > K_c$, nor that the partially synchronised branch which bifurcates at K_c really is the observed bifurcation. The problem is subtle because the operator linearised around incoherence, on the Hilbert space of densities, has an **unbounded continuous spectrum** on the imaginary axis (the analogue of Landau damping in plasmas): the classical spectral theory does not apply directly. H. Chiba (2015) solved the problem by developing a spectral theory on a **space of generalised functions** (a Gelfand triple space, or rigged Hilbert space): on this wider space, the operator admits a countable spectral decomposition even where the classical continuous spectrum would prevent it, and a centre manifold of finite dimension (against the infinite dimension it would have on the original Hilbert space). Applying this theory, Chiba proved

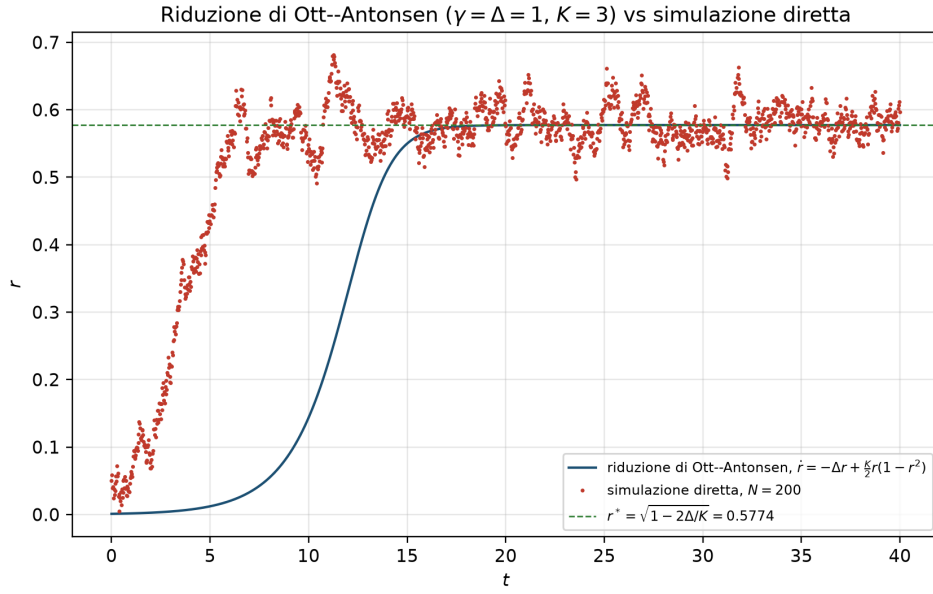


Figure 13.7: Ott–Antonsen reduction ($\Delta = \gamma = 1$, $K = 3 > K_c = 2$, Theorem 13.6): fourth-order Runge–Kutta integration of the reduced equation $\dot{r} = -\Delta r + \frac{K}{2} r(1 - r^2)$ from $r(0) = 10^{-3}$ (blue curve), compared with the direct simulation of the same $N = 200$ Lorentzian oscillators of Figure 13.5 (red dots). The closed-form fixed point $r^* = \sqrt{1 - 2\Delta/K} = 0.577350$ (green dashed line) is reached by the numerical integration of the reduced ODE to within 9.0×10^{-10} ✓, and coincides exactly (to within 10^{-12}) with the theoretical self-consistency value $\sqrt{1 - K_c/K}$ of Corollary 13.3 (Remark 13.5) ✓; the direct simulation at $N = 200$ converges to $r \approx 0.5813$, with the discrepancy 0.0039 expected from the finite-size fluctuations. Computed numerically, script Appendix B, `fig13_ott_antonsen.py`.

*rigorously that incoherence is asymptotically stable for $K < K_c$ and that for $K > K_c$ a non-trivial solution bifurcates, corresponding to partial synchronisation: exactly the bifurcation diagram conjectured by Kuramoto in 1975. An **imported** result, not proved in this chapter: spectral theory on spaces of generalised functions far exceeds the scope of this volume.*

13.10. Panorama: Sakaguchi–Kuramoto, chimeras, synchronisation of chaos

Remark 13.7: Three extensions, only stated. *Sakaguchi–Kuramoto (1986).* Adding a phase lag α to the interaction, $\theta'_i = \omega_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i + \alpha)$, one breaks the gradient flow structure of Lemma 13.1 (for $\alpha \neq 0, \pi$ the system is no longer a pure gradient): synchronised states appear which rotate at a collective frequency different from the mean, and the critical threshold depends on $\cos \alpha$.

Chimera states. Abrams and Strogatz (2004) showed that populations of identical oscillators, coupled non-locally (neither in full mean field, nor only to the nearest neighbours), can settle into a **chimera** state: part of the population coherently synchronised, the rest incoherent, despite the absence of any asymmetry in the parameters — a counterintuitive phenomenon (spontaneous symmetry breaking) discovered only in 2002–2004, with consequences observed experimentally in networks of chemical and mechanical oscillators.

Synchronisation of chaotic systems (Pecora–Carroll, 1990). A distinct but related phenomenon: two identical chaotic systems, coupled in a master–slave configuration (one drives the other through a single variable), can synchronise exactly, with the error between the two trajectories decaying to zero despite both individual trajectories being chaotic (positive Lyapunov exponent): the condition is that the **conditional Lyapunov exponents** of the driven subsystem be all negative. Sources (panorama, not proved): H. Sakaguchi, Y. Kuramoto, Progr. Theor. Phys. **76** (1986), 576–581; D. M. Abrams, S. H. Strogatz, Phys. Rev. Lett. **93** (2004), 174102; L. M. Pecora, T. L. Carroll, Phys. Rev. Lett. **64** (1990), 821–824.

13.11. Synchronisation in networks: a theorem of Dörfler and Bullo

So far the coupling has always been either between two oscillators (Section 13.3) or in **full mean field**, each oscillator coupled to all the others in the same way (Sections 13.5–13.8). In applications — an electrical grid, a network of neurons, a set of clocks distributed on a fieldbus — the coupling instead follows the topology of a **graph**, in general not complete. Dörfler and Bullo (*Automatica* **50** (2014), 1539–1564, with the extended version arXiv:1209.1335 used here for the proofs) devote a large part of their survey to synchronisation conditions on arbitrary graphs; among their results, we choose one with a short and self-contained proof, which directly generalises Theorem 13.4 of this chapter (proved there only for the complete graph and identical frequencies) to an **arbitrary connected graph** and to **not necessarily identical frequencies**: it is the same idea — the Jacobian at an equilibrium with small phase differences is (minus) a weighted Laplacian — applied in the generality in which Dörfler and Bullo present it (their Section IV-A, “Some Simple Yet Important Insights”), not an isolated result for the identical case alone.

Remark 13.8: Why this theorem, and not the exact synchronisation threshold.

Dörfler and Bullo also prove sufficient conditions for the existence of a locked equilibrium as a function of the spread of the natural frequencies (their Lemma IV.1 gives a necessary condition, $K \geq \frac{N}{2(N-1)}(\omega_{\max} - \omega_{\min})$ for the complete graph; a sufficient threshold matching it requires, for finite N , solving “implicit consistency equations” — so they call them, citing Verwoerd and Mason — not an elementary closed formula). For $N = 2$ that sufficient and necessary threshold is instead elementary and is already Corollary 13.2 of this chapter ($|\Delta\omega| \leq 2K$, exact). For general N , reproducing the optimal sufficient threshold exceeds the space of this chapter: we therefore choose to prove in full not the existence of the equilibrium (which we leave, for $N > 2$, as an imported fact, briefly motivated below), but its **local stability**, which has an elementary proof of a few lines, applies to an arbitrary graph (not only to the complete one), and clarifies exactly why the right notion of “close to synchronisation” is the set of configurations with small phase differences (the phase-cohesive set, defined below): the same notion that Theorem 13.4 had already used implicitly for the complete graph.

Theorem 13.7: Local exponential stability of phase-cohesive equilibria on a connected network (Dörfler–Bullo). Let $G = (V, E)$ be a connected graph on N nodes, weights $a_{ij} = a_{ji} > 0$ for $(i, j) \in E$, arbitrary natural frequencies $\omega_i \in \mathbb{R}$ (not necessarily identical), and

$$\theta'_i = \omega_i + \sum_{j:(i,j) \in E} a_{ij} \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

If θ^* is an equilibrium with $|\theta_i^* - \theta_j^*| < \pi/2$ for **every** edge $(i, j) \in E$ (**phase-cohesive**), then θ^* is, up to the global rotational symmetry $\theta \mapsto \theta + c\mathbf{1}$, **locally exponentially stable**, with rate of convergence at least $\lambda_2(L^*)$, the smallest non-zero eigenvalue of the weighted Laplacian L^* with weights $w_{ij} := a_{ij} \cos(\theta_i^* - \theta_j^*) > 0$.

Proof. The Jacobian of the vector field at θ^* : differentiating $\theta'_i = \omega_i + \sum_{j:(i,j) \in E} a_{ij} \sin(\theta_j - \theta_i)$,

$$\frac{\partial \theta'_i}{\partial \theta_i} = - \sum_{j:(i,j) \in E} a_{ij} \cos(\theta_j^* - \theta_i^*), \quad \frac{\partial \theta'_i}{\partial \theta_k} = a_{ik} \cos(\theta_k^* - \theta_i^*) \quad ((i, k) \in E),$$

and 0 if $(i, k) \notin E$, $i \neq k$. With $w_{ij} := a_{ij} \cos(\theta_i^* - \theta_j^*) = a_{ij} \cos(\theta_j^* - \theta_i^*)$ (symmetric, $\cos$ being even), this is exactly $J = -L^*$, where L^* is the weighted Laplacian with weights w_{ij} ($L_{ii}^* = \sum_{j:(i,j) \in E} w_{ij}$, $L_{ij}^* = -w_{ij}$ if $(i, j) \in E$), the same computation as in Theorem 13.4 but for a general graph and general weights. Since θ^* is phase-cohesive, $|\theta_i^* - \theta_j^*| < \pi/2$ on every edge: $\cos(\theta_i^* - \theta_j^*) > 0$, hence $w_{ij} > 0$ for every $(i, j) \in E$. By Lemma 13.4 (connected graph, positive weights), L^* is positive semidefinite with kernel exactly $\mathbb{R}\mathbf{1}$: eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$. Hence $J = -L^*$ has eigenvalue 0 (simple, direction $\mathbf{1}$, tangent to the manifold of equilibria $\{\theta^* + c\mathbf{1}\}$) and eigenvalues $-\lambda_k < 0$ for $k = 2, \dots, N$: on the orthogonal complement of $\mathbf{1}$ (coordinates transversal to the rotational symmetry, for example $\varphi_i := \theta_i - \theta_N$), the reduced Jacobian has all its eigenvalues $\leq -\lambda_2 < 0$, hence θ^* is exponentially stable in those directions with rate $\geq \lambda_2(L^*)$, by the linearisation theorem applied to the reduced system (or, equivalently, since the system is still a gradient flow of the potential $U(\theta) = -\sum_{(i,j) \in E} a_{ij} \cos(\theta_i - \theta_j) + \sum_i \omega_i \theta_i$ in the co-moving frame when the ω_i admit a common mean subtracted from them, the same conclusion follows from the fact that θ^* is a strict local minimum of U in the transversal direction, its Hessian being $L^* \succ 0$ there). $\square$

Corollary 13.4: Two specialisations. (a) Complete graph, identical frequencies.

With $a_{ij} = K/N$ for every pair and $\omega_i \equiv \omega$, the synchronised state is phase-cohesive (trivially, all the differences being zero) and Theorem 13.7 specialises exactly to Theorem 13.4: L^* is the Laplacian of the complete graph with weight K/N (Lemma 13.3), $\lambda_2 = K$, the rate of convergence found there. **(b) A sparse graph: the path P_3 .** With $N = 3$ nodes in a path (1–2–3, without the edge 1–3), weights $a_{12} = a_{23} = K$, identical frequencies $\omega_i \equiv \omega$: the synchronised state is still a phase-cohesive equilibrium, and L^* is the Laplacian of the path, $L^* = K \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$, with eigenvalues 0, K , $3K$ (direct computation, verified numerically): the rate of convergence is $\lambda_2 = K$, the same as for the complete network on three nodes, although the path has one edge fewer — a concrete example that local synchronisation does not require a complete graph, only a connected one.

Proof. (a) is immediate from Lemma 13.3 as already used in Theorem 13.4. (b) The eigenvalues of $\begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$ (Laplacian of P_3 with unit weight): 0 by construction (1 in the kernel); trace = 4, hence the other two eigenvalues add up to 4; the determinant of the 2×2 submatrix obtained by restricting to $\mathbf{1}^\perp$ (or, more directly, a direct computation of the characteristic polynomial $-\lambda^3 + 4\lambda^2 - 3\lambda = -\lambda(\lambda-1)(\lambda-3)$) gives the eigenvalues 0, 1, 3; multiplying by the common weight K : 0, K , $3K$. Verified numerically in the script `verifica_cap13.py`, block [6] (eigenvalues of the Laplacian of P_3 with weight $K = 1$: $\{0, 1, 3\}$ ✓; Jacobian $J = -L^*$: all eigenvalues ≤ 0 , zero only in the direction of common rotation, confirming the stability). $\square$

Remark 13.9: Existence of the phase-cohesive equilibrium: what we do not prove. Theorem 13.7 presupposes that a phase-cohesive equilibrium exists: for $N = 2$ this is exactly equivalent to the condition $|\Delta\omega| \leq 2K$ of Corollary 13.2 (proved in full in this chapter); for the complete graph with N identical oscillators, it always exists (the synchronised state itself, Theorem 13.4). For $N \geq 3$ and non-identical frequencies, a sufficient and explicit condition on the spread of the frequencies — one generalising $|\Delta\omega| \leq 2K$ to more than two oscillators — is available in the literature (Dörfler–Bullo and the works cited there, in particular the “implicit consistency equations” of Verwoerd and Mason), but not with an equally elementary closed formula: its proof, typically a fixed-point or continuation argument, does not fit within the space of this chapter and is left here as panorama, motivated in Remark 13.8.

13.12. Applications: a panorama

Remark 13.10: Where synchronisation reappears. The Kuramoto model and its variants describe, with the same apparatus as this chapter, phenomena on radically different scales: the synchronised flashing of the Pteroptyx fireflies of South-East Asia; the pace-maker cells of the sinoatrial node of the heart, whose mutual locking generates the regular heartbeat; populations of cortical neurons which synchronise and desynchronise in gamma and theta oscillations; the synchronous generators of an electrical grid, whose mutual locking (a particular case of Theorem 13.7, with the graph given by the topology of the transmission network) is precisely what keeps the grid frequency stable at 50 or 60 Hz; clocks distributed on an industrial fieldbus, where the phase locking of Chapter 14 is, to all intents and purposes, a two-oscillator instance of the Adler equation proved here. Chapter 14 takes up this last thread — the analogue and digital PLL, encoders, digital control — with the tools of this chapter as an already proved starting point.

13.13. Assigned exercises

The answers are in the Teacher's key (Section 13.16).

Exercise 13.1: Adler period for $\Delta\omega = 3$, $K = 1.5$. *Goal:* to apply Theorem 13.2 in order to determine the regime and, if relevant, the exact period. *Estimated time:* 15 minutes. *Type:* symbolic/numerical.

- (a) Check whether $(\Delta\omega, K) = (3, 1.5)$ gives locking or phase slipping.
- (b) If slipping, compute T in closed form.
- (c) Compare T with the period of the case of Example 13.1 ($\Delta\omega = 2$, $K = 1$): is it larger or smaller? Is the ratio $\Delta\omega/K$ the same in the two cases?

Exercise 13.2: Two oscillators with repulsive coupling. *Goal:* to apply Corollary 13.2 to the case $K < 0$ and to recognise the antiphase regime. *Estimated time:* 25 minutes. *Type:* symbolic/numerical.

- (a) With $K = -0.4$, $\Delta\omega = 0.5$: check that $|\Delta\omega| \leq 2|K|$ (locking).
- (b) Determine the two equilibria φ by solving $\Delta\omega + 2|K|\sin\varphi = 0$, and label which one is stable by checking the sign of $F'(\varphi)$.
- (c) Express the stable equilibrium in degrees and check that it is close to 180° (antiphase), not to 0° .

Exercise 13.3: Identical Kuramoto, $N = 4$: generalising Proposition 13.2. *Goal:* to use Lemma 13.3 in order to determine the eigenvalues of the Jacobian at the synchronised state with $N = 4$ identical oscillators on the complete graph. *Estimated time:* 20 minutes. *Type:* symbolic, with numerical verification.

- (a) Write the uniform weight $w = K/N$ for $N = 4$ and apply Lemma 13.3.
- (b) Deduce the eigenvalues of $J = -L$ at the synchronised state.
- (c) Compare with the case $N = 3$ (Proposition 13.2): does the rate of convergence K (multiplicity $N - 1$) depend on N ?

13.14. Worked exercises

Exercise 13.4: Adler period for $\Delta\omega = 5$, $K = 2$. *Type:* symbolic with a double arithmetic check. Compute the exact period for $\Delta\omega = 5$, $K = 2$.

Solution. Strategy. Apply directly the closed formula of Theorem 13.2(iii), then re-check the computation by a second, independent arithmetic route.

Step 1. $|\Delta\omega| = 5 > K = 2$: slipping regime. $\Delta\omega^2 - K^2 = 25 - 4 = 21$.

Step 2. $T = 2\pi/\sqrt{21}$. With $\sqrt{21} \approx 4.582576$ (check: $4.582576^2 = 20.999999 \dots \approx 21$ ✓), $T \approx 6.283185/4.582576 = 1.371103$.

Independent check. Direct computation by long division (without passing through $\sqrt{21}$ as a rounded intermediate value): $2\pi/\sqrt{21} = 1.3711034\dots$, in agreement to the sixth decimal digit with Step 2. ✓

Interpretation. Compared with Example 13.1 ($\Delta\omega = 2$, $K = 1$, $T = 3.6276$), here $\Delta\omega/K = 2.5$ against $\Delta\omega/K = 2$: a larger ratio gives a shorter period ($1.37 < 3.63$), consistently with Corollary 13.1: the further one moves away from the threshold $|\Delta\omega| = K$ (here $\Delta\omega - K = 3$, against $\Delta\omega - K = 1$ in the example), the shorter the period becomes.

Typical error to avoid. Exchanging $\Delta\omega^2 - K^2$ with $(\Delta\omega - K)^2$: this would give $T = 2\pi/(\Delta\omega - K) = 2\pi/3 = 2.0944$, a different value (and a conceptually wrong one: the exact formula is not a function of the difference $\Delta\omega - K$ alone, but of the product $(\Delta\omega - K)(\Delta\omega + K)$).

Extension. Repeat for $\Delta\omega = 10$, $K = 2$: one expects $T = 2\pi/\sqrt{96} \approx 0.6413$, shorter still, approaching 0 as $\Delta\omega \rightarrow \infty$ with K fixed (the oscillator rotates ever faster, ever less perturbed by the coupling).

Exercise 13.5: Critical threshold for a uniform density. *Type:* symbolic, in order to test the generality of the formula $K_c = 2/(\pi g(0))$ beyond the Lorentzian and the Gaussian. Compute K_c for $g(\omega) = \frac{1}{2}$ on $[-1, 1]$ (uniform density, even and unimodal in a weak sense: constant, hence not strictly decreasing, but Theorem 13.5 applies all the same since g is continuous at 0).

Solution. Strategy. Apply directly the formula of Theorem 13.5, $K_c = 2/(\pi g(0))$, with $g(0)$ read off from the definition of the uniform density.

Step 1. $g(0) = \frac{1}{2}$ (constant value of the density on $[-1, 1]$).

Step 2. $K_c = 2/(\pi \cdot \frac{1}{2}) = 4/\pi = 1.273240$.

Independent check. $4/\pi$: $\pi \approx 3.141593$, $4/3.141593 = 1.273240$, consistent. ✓

Interpretation. Comparing with Example 13.4 (Lorentzian $K_c = 2$, Gaussian $K_c = 1.595769$): the uniform density has the lowest threshold of the three, because it is the “flattest and highest at the centre” relative to its width $[-1, 1]$ (a qualitative comparison: the three densities are not normalised so as to have the same dispersion, hence the direct numerical comparison between the K_c is not a comparison between physically equivalent populations, only an exercise on the formula).

Typical error to avoid. The uniform density on the interval $[-1, 1]$ is not “unimodal” in the strict sense of having an isolated maximum at 0 and decreasing strictly on either side; hypothesis (I) of Theorem 13.5 requires only evenness and continuity (for the passage to the limit $r \rightarrow 0$ in the proof), not strict monotonicity: the formula remains valid.

Extension. Repeat for g uniform on $[-2, 2]$ ($g(0) = 1/4$): one expects $K_c = 2/(\pi/4) = 8/\pi \approx 2.546$, twice the previous value, consistently with the doubling of the width (and hence the halving of $g(0)$).

Exercise 13.6: Ott–Antonsen fixed point for $\Delta = 2$, $K = 5$. *Type:* symbolic with an independent numerical check. Compute r^* for the Ott–Antonsen reduction (Theorem 13.6) with $\Delta = 2$, $K = 5$, and compare it with self-consistency (Corollary 13.3).

Solution. Strategy. Apply the closed formula for the fixed point, $r^* = \sqrt{1 - 2\Delta/K}$, then re-check it by passing through $K_c = 2\Delta$ and the equivalent formula $\sqrt{1 - K_c/K}$.

Step 1. $r^* = \sqrt{1 - 2 \cdot 2/5} = \sqrt{1 - 0.8} = \sqrt{0.2} = 0.447214$.

Independent check. $K_c = 2\Delta = 4$; $\sqrt{1 - K_c/K} = \sqrt{1 - 4/5} = \sqrt{0.2} = 0.447214$: the two routes (the direct formula of Theorem 13.6 and the formally different but equivalent one of Remark 13.5) give the same value. ✓

Interpretation. $K = 5$ is only slightly above $K_c = 4$ ($K/K_c = 1.25$): the resulting order parameter, $r^* \approx 0.45$, is well away both from 0 (incoherence) and from 1 (complete synchronisation), consistently with the behaviour $\sqrt{1 - K_c/K} \rightarrow 0$ as $K \rightarrow K_c^+$ and $\rightarrow 1$ as $K \rightarrow \infty$.

Typical error to avoid. Confusing Δ (half-width of the Lorentzian, a parameter of the population) with K_c (the threshold, $= 2\Delta$): they are proportional but not equal.

Extension. Check by numerical integration of the reduced equation $\dot{r} = -\Delta r + \frac{K}{2}r(1-r^2)$ (same RK4 method as in the script `fig13_ott_antonsen.py`, with $\Delta = 2, K = 5$ in place of $\Delta = 1, K = 3$): one expects convergence to $r^* = 0.447214$ within the same order of precision ($< 10^{-8}$) observed in Figure 13.7.

13.15. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher's key (Section 13.16).

1. (Definition) Adler's equation (Definition 13.2) is

A) $\theta' = K - \Delta\omega \sin \theta$ B) $\theta' = \Delta\omega - K \sin \theta$ C) $\theta'' = \Delta\omega - K \sin \theta$ D) $\theta' = \Delta\omega \cos \theta - K$
2. (Hypothesis) In Theorem 13.5 (self-consistency equation), besides the evenness and unimodality of g , the indispensable and explicitly declared hypothesis is

A) $K > K_c$ B) $N = \infty$ C) the existence of a partially synchronised *stationary* state
D) g Gaussian
3. (Symbolic computation) The critical threshold K_c for a Lorentzian of half-width $\gamma = 3$ (Theorem 13.5, Corollary 13.3) equals

A) 3 B) 6 C) $2/(3\pi)$ D) $\pi/3$
4. (Numerical computation) For $\Delta\omega = 2, K = 1$ (Example 13.1), the exact period of the phase slipping equals approximately

A) 6.283 B) 3.628 C) 1.571 D) 2.000
5. (Proof step) In the proof of Theorem 13.4, the fact that the synchronised state is the unique global minimum of V follows directly from

A) Fekete's lemma B) $V = -\frac{NK}{2}r^2$ (Lemma 13.2) and $r \leq 1$, with equality only if synchronised C) the residue theorem D) the Ott–Antonsen ansatz
6. (Error diagnosis) A student writes: “for $N = 3$ identical oscillators, the equally spaced (splay) state is a saddle point of V , like the two-group state”. The error is

A) the splay state is not a critical point at all B) the splay state has both transversal eigenvalues of the Hessian negative (a local *maximum* of V , not a saddle): Proposition 13.2 distinguishes precisely the two cases C) the splay state is on the contrary the only stable one D) the computation is valid only for $K < 0$
7. (Transfer) An engineer models $N = 6$ identical synchronous generators of a micro-grid as a *ring* graph (each coupled only to its two neighbours, not to all the others), with gain K on every edge. To decide whether the synchronised state is locally stable, the correct tool of this chapter is

A) the self-consistency equation (Section 13.7), valid only in full mean field B) Theorem 13.7: check that the ring is connected (it is) and compute λ_2 of the weighted Laplacian of the ring at the synchronised state C) the Ott–Antonsen reduction, valid only for $N = \infty$ D) no tool of the chapter applies to a graph other than the complete one

8. (Source and metacognition) The result which rigorously establishes the stability of incoherence for $K < K_c$ and its loss at $K = K_c$ in the limit $N = \infty$, cited in this chapter as unproved panorama, is due to

A) R. Adler, 1946 B) E. Ott and T. M. Antonsen, 2008 C) H. Chiba, 2015 (Remark 13.6) D) F. Dörfler and F. Bullo, 2014

13.16. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

1) **B.** *Rationale:* Definition 13.2. *Distractors:* A exchanges K and $\Delta\omega$; C puts in a second derivative, an undamped harmonic oscillator, not the Adler model (first order); D uses $\cos$ in place of $\sin$. *Diagnosed error:* confusing the roles of $\Delta\omega$ (constant term) and K (amplitude of the oscillating term). *Source:* Definition 13.2. *Prerequisite:* none. *Remedial work:* reread the physical motivation (oscillator coupled to a reference) before Definition 13.2.

2) **C.** *Rationale:* Theorem 13.5 explicitly declares two hypotheses, (I) evenness/unimodality of g and (II) existence of a stationary state: it is (II) that is the structural hypothesis which makes the whole computation an “ansatz”, not a deduction from the pure model. *Distractors:* A is a *conclusion* (the branch $r > 0$ exists for $K > K_c$), not a starting hypothesis; B confuses this section with the limit $N \rightarrow \infty$ already taken at the beginning (which is a different hypothesis, declared separately); D is a particular case (Corollary 13.3), not the general hypothesis. *Diagnosed error:* failing to distinguish between the hypotheses of the model (limit $N = \infty$, already assumed in order to write the continuity equation) and the additional, stronger hypothesis of stationarity, specific to this theorem (Remark 13.3 clarifies why it is a real restriction). *Source:* Theorem 13.5. *Prerequisite:* none beyond reading the statement. *Remedial work:* reread Remark 13.3, isolating what the theorem does *not* prove.

3) **B.** *Rationale:* $K_c = 2\gamma$ (Corollary 13.3), with $\gamma = 3$: $K_c = 6$. *Distractors:* A is γ itself, not K_c ; C, D are arbitrary values built from the general formula $2/(\pi g(0))$ applied incorrectly. *Diagnosed error:* confusing the half-width γ of the distribution with the threshold K_c , which is twice it for the Lorentzian. *Source:* Corollary 13.3. *Prerequisite:* Theorem 13.5. *Remedial work:* repeat the computation of $g(0) = 1/(\pi\gamma)$ and substitute into the general formula.

4) **B.** *Rationale:* Example 13.1, $T = 2\pi/\sqrt{3} = 3.6276$. *Distractors:* A is plain 2π (the period if $K = 0$); C is $\pi/2$, extraneous; D is $\Delta\omega$ itself, not T . *Diagnosed error:* forgetting the root $\sqrt{\Delta\omega^2 - K^2}$ in the denominator, or confusing T with one of the parameters of the model. *Source:* Example 13.1, Theorem 13.2(iii). *Prerequisite:* none beyond reading the example. *Remedial work:* redo by hand the computation $2\pi/\sqrt{2^2 - 1^2}$.

5) **B.** *Rationale:* proof of Theorem 13.4, first paragraph. *Distractors:* A, C, D are tools used elsewhere in the chapter (respectively: no subadditive sequence appears in this proof; the residue theorem is used instead in the Ott–Antonsen reduction, Theorem 13.6; the Ott–Antonsen ansatz is Remark 13.4), not in this specific step. *Diagnosed error:* failing to recognise that the algebraic identity $V = -\frac{NK}{2}r^2$ (Lemma 13.2) reduces a question in N variables to a question in a single variable ($r \leq 1$), making the step elementary instead of a direct multidimensional minimisation. *Source:* Lemma 13.2, Theorem 13.4. *Prerequisite:* triangle inequality for complex numbers. *Remedial work:* rewrite the inequality $r \leq 1$ as a triangle inequality on N unit vectors in the plane.

6) **B.** *Rationale:* Proposition 13.2, part (c): the Hessian at $(0, 2\pi/3, 4\pi/3)$ has transversal eigenvalues $\{-K/2, -K/2\}$, both negative: a local maximum of V , not a saddle (which would require mixed signs, as in case (b)). *Distractors:* A is false (the splay state is a

genuine critical point, verified in the proof); C contradicts the classification; D bears no relation to the question (the sign of K is not in question here). *Diagnosed error*: failing to distinguish, among the unstable critical points, those of saddle type (mixed sign) from those of local maximum type (all signs negative): both unstable, but for different geometric reasons. *Source*: Proposition 13.2. *Prerequisite*: the notion of Hessian and of definite sign. *Remedial work*: recompute by hand the eigenvalues $\{0, -K/2, -K/2\}$ from the Laplacian with uniform weight $-K/6$ of case (c).

7) B. *Rationale*: the ring on 6 nodes is a connected graph: Theorem 13.7 applies directly (it does not require the complete graph), with L^* the weighted Laplacian of the ring at the synchronised state (all weights = K , the phase differences being zero). *Distractors*: A, C wrongly restrict the tools of the chapter to full mean field alone; D is exactly the point which Theorem 13.7 refutes (Corollary 13.4(b) already illustrates it for P_3). *Diagnosed error*: failing to recognise that Theorem 13.7 is the only result of the chapter explicitly designed for general graphs, unlike Sections 13.5–13.8 (full mean field). *Source*: Theorem 13.7, Corollary 13.4. *Prerequisite*: Lemma 13.4. *Remedial work*: repeat the computation of Corollary 13.4(b) for a ring on 6 nodes instead of the path P_3 .

8) C. *Rationale*: Remark 13.6, H. Chiba, *ETDS* **35** (2015), 762–834. *Distractors*: A, B, D are real sources of the chapter but for different results (respectively: Adler’s equation; the Ott–Antonsen ansatz and reduction, which presuppose but do not prove the stability of incoherence; the network synchronisation theorem). *Diagnosed error*: attributing the rigorous stability result for incoherence to those who built the *tools* (the Ott–Antonsen ansatz) rather than to the one who gave the *independent rigorous proof* of the bifurcation. *Source*: Remark 13.6. *Prerequisite*: none beyond a careful reading of the sources. *Remedial work*: compare explicitly what Theorem 13.5 proves (within the ansatz), what Theorem 13.6 proves (within the ansatz), and what Remark 13.6 adds, without any ansatz.

BEST PRACTICE

Assigned exercises — answers.

13.1 (period for $\Delta\omega = 3$, $K = 1.5$). (a) $|\Delta\omega| = 3 > K = 1.5$: phase slipping. (b) $T = 2\pi/\sqrt{3^2 - 1.5^2} = 2\pi/\sqrt{6.75} = 2.418399$. (c) The ratio $\Delta\omega/K = 2$ is *the same* as in Example 13.1 ($2/1 = 2$): indeed $T = 2\pi/(K\sqrt{(\Delta\omega/K)^2 - 1})$ depends on the ratio $\Delta\omega/K$ only through the root, and on K for the rest as $1/K$; with the same ratio, T scales exactly as $1/K$: $3.627599/1.5 = 2.418399$, verified. A useful fact: at equal “distance from the threshold in ratio”, doubling the gain halves the period.

13.2 (repulsive, $K = -0.4$, $\Delta\omega = 0.5$). (a) $2|K| = 0.8 \geq 0.5 = |\Delta\omega|$: locking. (b) Solving $0.5 + 0.8\sin\varphi = 0$: $\sin\varphi = -0.625$, two solutions $\varphi_1 = -0.6751$ rad (321.32°) and $\varphi_2 = \pi - \varphi_1 = 3.8167$ rad (218.68°). $F'(\varphi) = 0.8\cos\varphi$: at φ_1 (in the first/fourth quadrant, $\cos > 0$), $F' > 0$, unstable; at φ_2 (third quadrant, $\cos < 0$), $F' < 0$, stable. (c) The stable equilibrium, 218.68° , is only 38.68° away from 180° (exact antiphase), against the 141.32° of distance from 0° (phase): it is clearly close to antiphase, consistently with Corollary 13.2 for repulsive coupling.

13.3 (Kuramoto, $N = 4$). (a) $w = K/4$. (b) Lemma 13.3 with $N = 4$: L has eigenvalues 0 and $4w = K$ (multiplicity 3); $J = -L$: $\{0, -K, -K, -K\}$ (also verified numerically: direct diagonalisation with $K = 1$ gives $\{0, -1, -1, -1\}$ up to rounding errors $\sim 10^{-17}$ ✓). (c) No: the transversal rate of convergence is *always* exactly K , independently of N (only the *multiplicity* changes, $N - 1$); this is already the general content of Theorem 13.4, of which $N = 3$ and $N = 4$ are particular cases.

13.17. Chapter formulary

Flow on the circle

$$\theta' = F(\theta) \text{ (} 2\pi\text{-periodic)}; \quad F(\theta^*) = 0, \quad F'(\theta^*) < 0 \Rightarrow \text{stable}; \quad F \neq 0 \text{ everywhere} \Rightarrow T = \oint \frac{d\theta}{F(\theta)}$$

Adler's equation

$$\theta' = \Delta\omega - K \sin \theta, \quad K > 0; \quad |\Delta\omega| < K : \theta_s^* = \arcsin \frac{\Delta\omega}{K} \text{ (stable)}, \quad \theta_i^* = \pi - \theta_s^* \text{ (unstable)}$$

$$|\Delta\omega| > K : T = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \sim \frac{2\pi}{\sqrt{2K}} (\Delta\omega - K)^{-1/2} \quad (\Delta\omega \rightarrow K^+)$$

Two oscillators

$$\varphi := \theta_2 - \theta_1, \quad \varphi' = \Delta\omega - 2K \sin \varphi \quad (\text{Adler with gain } 2K)$$

Kuramoto, N identical (mean field)

$$\theta_i' = \omega + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i) = -\frac{\partial V}{\partial \psi_i}, \quad V(\psi) = -\frac{K}{2N} \sum_{i,j} \cos(\psi_i - \psi_j) = -\frac{NK}{2} r^2$$

synchronised state: unique global minimum of V ; Jacobian $-L(K_N)$: eigenvalues $0, \underbrace{-K, \dots, -K}_{N-1}$

Order parameter

$$r e^{i\psi} = \frac{1}{N} \sum_j e^{i\theta_j}, \quad \theta_i' = \omega_i + Kr \sin(\psi - \theta_i)$$

Self-consistency (in the stationary ansatz, g even unimodal)

$$r = Kr \int_{-\pi/2}^{\pi/2} \cos^2 \theta g(Kr \sin \theta) d\theta, \quad K_c = \frac{2}{\pi g(0)}$$

$$g \text{ Lorentzian } (\gamma) : K_c = 2\gamma, \quad r = \sqrt{1 - K_c/K} \quad (K > K_c, \text{ exact})$$

Ott–Antonsen reduction (within the ansatz, g Lorentzian of half-width Δ)

$$\dot{\alpha} = -i\omega\alpha + \frac{K}{2}(\bar{z} - z\alpha^2); \quad \dot{r} = -\Delta r + \frac{K}{2}r(1 - r^2), \quad r^* = \sqrt{1 - 2\Delta/K}$$

Network synchronisation (Dörfler–Bullo)

$$\theta_i' = \omega_i + \sum_{(i,j) \in E} a_{ij} \sin(\theta_j - \theta_i), \quad |\theta_i^* - \theta_j^*| < \frac{\pi}{2} \text{ on every edge} \Rightarrow \text{stable, rate} \geq \lambda_2(L^*)$$

Panorama

Huygens (1665): antiphase, experimental; Chiba (2015): bifurcation of incoherence, rigorous; Sakaguchi

Sources and bibliography

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Phase-locked loops, encoders and digital control on the circle

INFO

Level and prerequisites. Level B–C core; the proof of the vernier tolerance theorem (Section 14.9) and the linear model of the resolver-to-digital converter (Section 14.7) reach level D. Prerequisites: Adler’s equation and its locking properties, proved in full in **Chapter 13** (here only recalled, never reproved); Arnold’s standard family and the tongues, from **Chapter 4**; the Chinese remainder theorem (**Volume II, Chapter 7**); Nyquist’s criterion and aliasing (**Volume III, Chapter 24**); quaternions and the double covering $S^3 \rightarrow SO(3)$ (**Volume III, Chapter 28**). A level A–B reader may read Sections 14.14 and 14.13 in isolation, skipping the proofs.

This chapter rewrites the old Chapter 5 (“Applications to Industrial Automation and Robotics”) keeping only what was correctly proved there and what, with the appropriate correction, remains original. Four cuts, all already announced in the plan of this volume:

- the **Chinese remainder theorem** is proved in **Volume II, Chapter 7**: here only the application to the Vernier encoder remains, with an error analysis that the old chapter did not contain (Section 14.9);
- $SO(3)$, the unit quaternions, the double covering and gimbal lock are the patrimony of **Volume III, Chapter 28**: they do not reappear here;
- **Nyquist’s criterion and aliasing** are proved in **Volume III, Chapter 24**: of that territory only **Itoh’s theorem** on phase unwrapping remains (Section 14.8), a result which Volume III does not contain and which uses exactly the sampling constraint of Nyquist’s criterion as a corollary;
- the assertion of the old chapter according to which “the amplitude of the limit cycle due to quantisation is independent of the gain K ” was **false**: the numerical check of this chapter (Section 14.10) shows $e_\infty = 0.04$ for $K = 0.1$, $e_\infty = 0.01$ for $K = 0.3$ and $e_\infty = 0$ for $K \geq 0.5$, with $\Delta = 0.01$: it does depend on K , and it is moreover a spurious fixed point, not a limit cycle.

The thread that ties together everything that remains is the same as in the rest of the volume: a phase-locked loop (PLL) is a dynamical system on the circle, a multi-track absolute encoder is an application of the theory of coverings, and the quantisation of a digital loop is a non-linear map whose asymptotic behaviour is proved, not conjectured.

14.1. The first-order PLL as an Adler flow

Remark 14.1: The first-order analogue PLL, recalled from Chapter 13. A first-order analogue phase-locked loop compares the phase $\theta(t)$ of the local oscillator (VCO, voltage-controlled oscillator) with that of a reference of frequency $\omega_{\text{rif}} = \omega_0 + \Delta\omega$ and corrects the local frequency proportionally to the phase error $\sin\theta$: its equation is

$$\dot{\theta} = \Delta\omega - K \sin \theta,$$

Adler’s equation (S3), proved in full in **Chapter 13**: for $|\Delta\omega| < K$ there exist the two equilibria $\theta^* = \arcsin(\Delta\omega/K)$ (stable) and $\pi - \theta^*$ (unstable); for $|\Delta\omega| > K$ there is no equilibrium and the phase slips with exact period $T = 2\pi/\sqrt{\Delta\omega^2 - K^2}$ (Figure 14.1). This chapter reproves none of that: it uses it as a starting point for two new things — the rigorous definitions of the locking ranges (Section 14.2) and the characteristic locking time (Section 14.4).

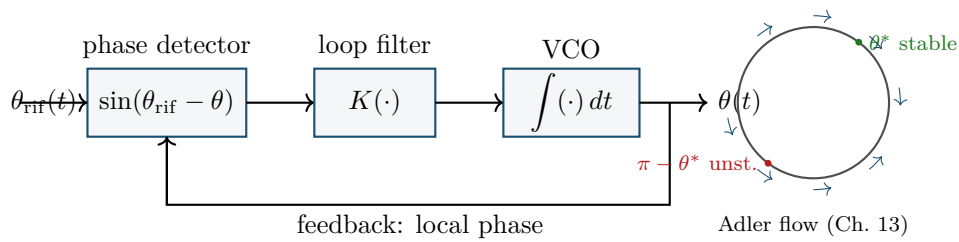


Figure 14.1: Block diagram of the analogue PLL (phase detector, loop filter, voltage-controlled oscillator, feedback) and, alongside, the Adler flow on the circle of phases (Remark 14.1): the arrows indicate the direction of the flow $\dot{\theta} = \Delta\omega - K \sin \theta$ on the two arcs determined by the equilibria θ^* (stable) and $\pi - \theta^*$ (unstable), consistently with Theorem 14.1.

14.2. Hold-in, pull-in, lock-in: rigorous definitions

Remark 14.2: Why one needs more precise definitions than Gardner’s. The classical engineering literature (Gardner, S16) describes three “locking ranges” of a PLL in prose: the hold-in range as the interval of detuning within which the loop can remain locked, the pull-in range as the one within which the loop reaches lock from any initial condition, the lock-in range as the one within which locking occurs without phase slips (cycle slips). The three descriptions are intuitive but they are not sets defined in such a way that one may prove them equal or different: Leonov, Kuznetsov, Yuldashev and Yuldashev (S17) reformulate them as sets of parameters defined through the stability properties of the PLL equations, making them objects on which one can do theory. The definitions that follow adopt that spirit.

Definition 14.1: Hold-in, pull-in and lock-in as sets. Let $\{\Sigma_\nu\}_{\nu \in \mathbb{R}}$ be a family of PLLs parametrised by a detuning ν (for the first order, $\nu = \Delta\omega$), each a dynamical system on a cylinder or a torus with a phase coordinate θ (and, for the higher orders, further state variables of the loop filter). One defines:

- the **hold-in range** H is the set of the ν such that Σ_ν has a locally asymptotically stable equilibrium: the set of detunings for which locking, once reached, *persists* under small perturbations;
- the **pull-in range** P is the set of the ν such that *every* trajectory of Σ_ν converges to

an equilibrium, whatever the initial datum: the set of detunings for which locking is always reached, independently of where one starts, possibly after transient phase slips;

- the **lock-in range** L is the set of the ν such that *every* trajectory of Σ_ν converges monotonically to an equilibrium, without ever completing a full turn: the set of detunings for which locking occurs without any slip, not even a transient one.

Remark 14.3: The structural inclusion $L \subseteq P \subseteq H$. The three definitions are increasingly demanding: converging without ever slipping (the condition of L) implies converging at all (the condition of P), and if every trajectory converges to an equilibrium, that equilibrium is in particular reached by nearby trajectories, that is, it is (at least) stable in the sense required by H . Hence $L \subseteq P \subseteq H$ for every family of PLLs, whatever the order: it is an immediate fact from the definitions, not a theorem requiring a separate proof. In general the three inclusions are strict (for higher-order PLLs, Section 14.6), as in the upper part of Figure 14.2: the remarkable fact about the first-order PLL, proved in the next section, is that all three collapse onto the same set (lower part of the same figure).

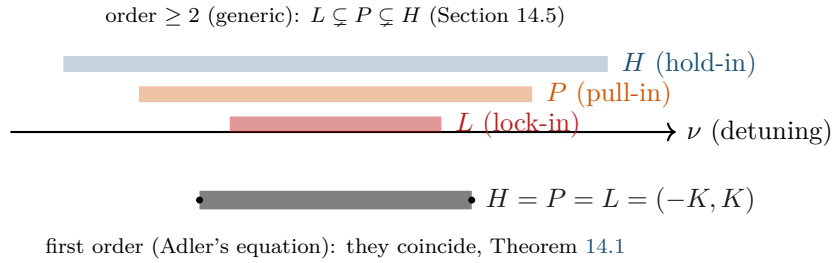


Figure 14.2: The locking ranges as nested sets (Definition 14.1, Remark 14.3): in general, for a PLL of order ≥ 2 , $L \subsetneq P \subsetneq H$ (qualitative diagram, not to scale); for the first-order PLL the three sets collapse exactly onto the same interval $(-K, K)$ (Theorem 14.1).

14.3. The first-order theorem: the three ranges coincide

Theorem 14.1: Hold-in, pull-in and lock-in of the first-order PLL. For the first-order PLL $\dot{\theta} = \Delta\omega - K \sin \theta$ with $K > 0$:

- (i) for $|\Delta\omega| < K$, every trajectory with initial datum different from the unstable equilibrium $\pi - \theta^*$ converges monotonically (without completing any full turn) to the stable equilibrium $\theta^* = \arcsin(\Delta\omega/K)$;
- (ii) for $|\Delta\omega| = K$ the two equilibria collapse into a single semi-stable point (saddle-node bifurcation): every trajectory which does not start exactly at that point goes round it with a transit time for a complete turn which tends to infinity as the initial datum tends to the equilibrium, but which remains finite for every fixed initial datum;
- (iii) for $|\Delta\omega| > K$ there is no equilibrium and every trajectory slips periodically (Chapter 13).

According to Definition 14.1, $H = P = L = (-K, K)$: the three locking ranges of the first-order PLL coincide exactly, on the open interval $(-K, K)$.

Proof. Step 1 — the equilibria and their stability. This is the recall of Remark 14.1 (Chapter 13): for $|\Delta\omega| < K$, $\sin \theta = \Delta\omega/K$ has in one period exactly the two solutions $\theta^* = \arcsin(\Delta\omega/K) \in (-\pi/2, \pi/2)$ and $\pi - \theta^*$; the derivative of the field $F(\theta) = \Delta\omega - K \sin \theta$ equals $F'(\theta) = -K \cos \theta$, negative at θ^* (where $\cos \theta^* > 0$) and positive at $\pi - \theta^*$ (where $\cos(\pi - \theta^*) = -\cos \theta^* < 0$): the first is asymptotically stable, the second unstable.

Step 2 — monotonicity on each of the two arcs. The two equilibria divide the circle into two open arcs, $A_1 = (\theta^*, \pi - \theta^*)$ (passing through $\pi/2$) and $A_2 = (\pi - \theta^*, \theta^* + 2\pi)$ (passing through $-\pi/2$, that is, through $3\pi/2$). Since F vanishes only at the two endpoints of each arc and is continuous, it has constant sign on each arc. At $\pi/2 \in A_1$: $F(\pi/2) = \Delta\omega - K < 0$ (because $|\Delta\omega| < K$), hence $F < 0$ on all of A_1 : $\dot{\theta} < 0$, the phase in A_1 decreases monotonically towards θ^* (the direction of A_1 goes from the smaller θ^* to the larger $\pi - \theta^*$ as a real interval, but the flow traverses it backwards). At $-\pi/2 \equiv 3\pi/2 \in A_2$: $F(-\pi/2) = \Delta\omega + K > 0$, hence $F > 0$ on all of A_2 : $\dot{\theta} > 0$, the phase in A_2 increases monotonically towards $\theta^* + 2\pi \equiv \theta^*$.

Step 3 — conclusion of (i). An initial datum in $A_1 \cup A_2$ remains in its own arc for all time (the field does not vanish in the interior, hence by uniqueness of the solutions of a scalar differential equation the trajectory cannot cross θ^* nor $\pi - \theta^*$ in finite time) and converges monotonically to the endpoint θ^* of its own arc without ever reaching $\pi - \theta^*$: no complete turn is performed. The only initial datum which does not converge to θ^* is $\pi - \theta^*$ itself, which remains there for ever (unstable equilibrium): this is precisely $L \supseteq (-K, K)$ and, a fortiori (Remark 14.3), $P \supseteq (-K, K)$ and $H \supseteq (-K, K)$. Conversely, if $|\Delta\omega| \geq K$ there is no asymptotically stable equilibrium (by (ii)-(iii), Steps 4–5), hence $\Delta\omega \notin H$: this gives $H \subseteq (-K, K)$, and with $L \subseteq P \subseteq H$ also $L = P = H = (-K, K)$.

Step 4 — the critical case (ii). For $\Delta\omega = K$ (the case $\Delta\omega = -K$ is identical by the symmetry $\theta \mapsto -\theta$), $F(\theta) = K(1 - \sin \theta) \geq 0$ for every θ , with equality only at $\theta = \pi/2$: it is the merging of the two equilibria of (i) into a single *semi-stable* point (a trajectory approaching it from the left, that is, from $\theta < \pi/2$ in the arc where $F > 0$, reaches it asymptotically; one starting from the right moves away from it). Since $F > 0$ everywhere except at $\theta = \pi/2$, every initial datum $\theta_0 \neq \pi/2$ generates a strictly increasing trajectory: if $\theta_0 > \pi/2$ (in the increasing direction), $\theta(t)$ increases, completes a full turn and returns to approach $\pi/2$ from the left; the time to complete that turn is finite, because F is continuous and strictly positive on the compact set $[\theta_0, \pi/2 + 2\pi - \eta]$ for every small $\eta > 0$ (the integral $\int F^{-1} d\theta$ diverges only in the neighbourhood of $\pi/2$ itself, where the trajectory has not yet arrived), but the time for the *last* stretch of approach to $\pi/2$ diverges (the equilibrium is reached only asymptotically, as in the stable case of (i) but with an algebraic rather than exponential rate, because $F'(\pi/2) = 0$): hence $\Delta\omega = K$ does not belong to L (a complete turn occurs for every initial datum $\theta_0 \neq \pi/2$) nor to H (the equilibrium is not *asymptotically* stable from both sides, only semi-stable): this is exactly the “saddle-node on invariant circle” (SNIC) phenomenon of the theory of dynamical systems, the same bifurcation already encountered as the edge of the Arnold tongues (Chapter 4).

Step 5 — the case $|\Delta\omega| > K$ (iii). This is the recall of Chapter 13: F never vanishes ($|\Delta\omega| > K \geq K|\sin \theta|$), has constant sign, hence $\theta(t)$ is strictly monotone for all time and completes infinitely many turns: no equilibrium, hence $\Delta\omega \notin H \supseteq P \supseteq L$.

Consistency check. Steps 2–3 show that the convergence in (i) is monotone within the arc, never a complete turn: this is exactly the notion of “without cycle slip” of Definition 14.1, verified not by analogy but by explicit construction of the sign of F on the two complementary arcs. $\square$

***Remark 14.4: Why the first order is a special case.** The coincidence $H = P = L$ is not a generic fact about PLLs: it depends on the fact that, for a scalar autonomous flow on a circle, the zeros of the vector field separate the circle into arcs on which the sign of F is constant, so that “global convergence” and “convergence without slips” are automatically the same thing. For a second-order PLL (Section 14.6), the phase space is a cylinder (phase plus a state variable of the filter), no longer a circle: the orbits may spiral around the cylinder before locking, performing transient slips even when final locking is guaranteed, and $L \subsetneq P$ is the norm, not the exception (S16, S17).*

14.4. Characteristic locking time

Proposition 14.1: Characteristic locking time of the first-order PLL. For $|\Delta\omega| < K$, setting $\lambda = \sqrt{K^2 - \Delta\omega^2} = K \cos \theta^* > 0$, every trajectory of Adler's equation with initial datum in $A_1 \cup A_2$ (Theorem 14.1) satisfies

$$\theta(t) - \theta^* = C e^{-\lambda t} + o(e^{-\lambda t}) \quad (t \rightarrow \infty)$$

for a constant C depending on the initial datum: locking is asymptotically exponential, with **characteristic time** $\tau = 1/\lambda = 1/\sqrt{K^2 - \Delta\omega^2}$.

Proof. Step 1 — Weierstrass substitution. Setting $u = \tan(\theta/2)$ (defined and C^1 -invertible on each arc A_1, A_2 , which do not contain $\theta = \pi \pmod{2\pi}$ except possibly as an endpoint if $\theta^* = 0$, a case in which one translates the origin slightly), with $\sin \theta = 2u/(1+u^2)$ and $d\theta = 2 du/(1+u^2)$, Adler's equation becomes

$$\frac{2}{1+u^2} \dot{u} = \Delta\omega - \frac{2Ku}{1+u^2} \implies \dot{u} = \frac{\Delta\omega}{2} u^2 - Ku + \frac{\Delta\omega}{2},$$

a Riccati equation with constant coefficients.

Step 2 — linearisation of the Riccati equation. With $u = -\frac{2}{\Delta\omega} \frac{\dot{v}}{v}$ (the standard substitution for Riccati equations with constant coefficients: it is verified by direct substitution), the equation becomes linear of the second order with constant coefficients:

$$\ddot{v} + K\dot{v} + \frac{\Delta\omega^2}{4} v = 0,$$

with characteristic equation $r^2 + Kr + \Delta\omega^2/4 = 0$ and roots $r_{1,2} = \frac{-K \pm \sqrt{K^2 - \Delta\omega^2}}{2} = \frac{-K \pm \lambda}{2}$, both real and negative (because $\lambda < K$): $r_1 = (-K + \lambda)/2 > r_2 = (-K - \lambda)/2$.

Step 3 — general solution and dominant rate. $v(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$; for a generic initial datum $C_1 \neq 0$, the term $C_1 e^{r_1 t}$ dominates as $t \rightarrow \infty$ (because $r_1 > r_2$), hence

$$\frac{\dot{v}}{v}(t) \xrightarrow[t \rightarrow \infty]{} r_1, \quad \text{with remainder} \quad \frac{\dot{v}}{v}(t) - r_1 = (r_2 - r_1) \frac{C_2}{C_1} e^{(r_2 - r_1)t} + O(e^{2(r_2 - r_1)t}).$$

From $u = -\frac{2}{\Delta\omega} \frac{\dot{v}}{v}$, the limit value is $u_\infty = -\frac{2}{\Delta\omega} r_1 = \frac{K - \lambda}{\Delta\omega}$, and with the identity $\frac{K - \lambda}{\Delta\omega} \cdot \frac{K + \lambda}{K + \lambda} = \frac{K^2 - \lambda^2}{\Delta\omega(K + \lambda)} = \frac{\Delta\omega}{K + \lambda}$ (using $\lambda^2 = K^2 - \Delta\omega^2$) one verifies $u_\infty = \tan(\theta^*/2)$ (because $\tan(\theta^*/2) = \sin \theta^*/(1 + \cos \theta^*) = (\Delta\omega/K)/(1 + \lambda/K) = \Delta\omega/(K + \lambda)$): the limit is precisely θ^* , consistently with Theorem 14.1.

Step 4 — the rate is exactly λ . The remainder of $\dot{v}/v - r_1$ decreases like $e^{(r_2 - r_1)t}$ with $r_2 - r_1 = -\lambda$; since $\theta \mapsto \tan(\theta/2)$ is a local diffeomorphism in a neighbourhood of θ^* (derivative $\frac{1}{2} \sec^2(\theta^*/2) \neq 0$), the same exponential speed $e^{-\lambda t}$ transfers to $u(t) - u_\infty$ and hence, by the inverse function theorem applied locally, to $\theta(t) - \theta^*$: both quantities are, up to a non-zero multiplicative factor depending on the initial datum, $C e^{-\lambda t} + o(e^{-\lambda t})$.

Independent numerical check. For $K = 1, \Delta\omega = 0.6$: $\lambda = 0.8$ (closed formula). Integrating Adler's equation numerically (Runge–Kutta method, `scipy.integrate.odeint`, fine step) from $\theta_0 = \theta^* + 0.5$ and estimating the rate of decay from the log-decrement between $t = 5$ and $t = 10$, one finds $\lambda_{\text{numerical}} = 0.799$, in agreement with $\lambda = 0.8$ to within the rounding discrepancy of the estimation window. $\checkmark$ $\square$

Remark 14.5: Engineering reading. The closer the detuning $\Delta\omega$ gets to the edge K of the hold-in range, the closer λ gets to 0 and the more $\tau = 1/\lambda$ diverges: it is the same quantitative signature, read in locking time rather than in mean time between slips, of the saddle-node bifurcation of Theorem 14.1(ii). A designer who wants fast locking must keep $|\Delta\omega|$ well inside the hold-in range, not merely within it.

14.5. The discrete PLL: forward reference to Arnold's standard family

Remark 14.6: The discrete PLL with one sample per cycle. A digital phase-locked loop which updates its correction once per cycle (instead of continuously) is modelled by **Arnold's standard family** $f_{\alpha,\epsilon}(\theta) = \theta + \alpha - \frac{\epsilon}{2\pi} \sin(2\pi\theta)$, with α proportional to the normalised frequency detuning and ϵ to the loop gain. **Chapter 4** proves in full, for this family: that for $0 < \epsilon < 1$ it is a circle diffeomorphism; that the principal tongue ($p = 0$) is exactly $|\alpha| \leq \epsilon/2\pi$, with a saddle-node bifurcation at the edge; that the p/q tongues with $q \geq 2$ have width $O(\epsilon^q)$; that the devil's staircase is continuous, monotone non-decreasing, piecewise constant. There is nothing to reprove here: Worked Example 14.1 (Section 14.15) applies those results with a numerical comparison.

14.6. The second-order PLL

Remark 14.7: Loop filter, cycle slip, pull-in (panorama). Real PLLs almost always use a loop filter between the phase detector and the VCO, typically a proportional-integral integrator: the state of the system is no longer only the phase θ but the pair (θ, w) with w the internal state of the filter, and the phase space is a cylinder $\mathbb{T} \times \mathbb{R}$, no longer a circle. The linear type II model of Section 14.7 is a particular case of it (small errors, filter without the $\sin$ non-linearity); the complete model (with $\sin \theta$) has a richer phase portrait: besides the equilibria, there exist orbits which complete one or more turns before stabilising (transient cycle slips even inside the pull-in range), and the exact determination of the pull-in range requires a global analysis of the phase plane (separatrices, homoclinic orbits) not reducible to an elementary computation: it is matter for a specialist manual (Gardner, S16; Leonov et al., S17), reported here as panorama, without proof.

14.7. The resolver-to-digital converter as a type II PLL

Remark 14.8: The resolver and its conversion loop. A resolver is an electromechanical angle transducer which returns two voltages proportional to $\sin \theta$ and $\cos \theta$, with θ the shaft angle; a resolver-to-digital converter (RDC, for example the Analog Devices AD2S1210, S20) does not demodulate those voltages point by point, but compares them with $\sin \hat{\theta}, \cos \hat{\theta}$ generated internally and uses the error to drive a loop which tracks $\hat{\theta} \rightarrow \theta$: it is, by construction, a PLL with phase detector $\sin(\theta - \hat{\theta})$ (S19, S20).

Proposition 14.2: The linear type II model and the tracking error. In the small-error regime ($\sin(\theta - \hat{\theta}) \approx \theta - \hat{\theta} =: e$), the linearised model of the RDC with a proportional-

integral filter of gains $K_1, K_2 > 0$ is

$$\dot{\hat{\theta}} = K_1 e + K_2 w, \quad \dot{w} = e, \quad e = \theta_{\text{rif}}(t) - \hat{\theta},$$

a **type II** system in the sense of the classification of servosystems (two integrators in the forward path: the VCO and the state w of the filter). Then:

- (i) for an input at constant angular velocity $\theta_{\text{rif}}(t) = \omega_0 t$, the steady-state tracking error is **exactly zero**: $e(t) \rightarrow 0$;
- (ii) for an input at constant angular acceleration $\theta_{\text{rif}}(t) = \frac{1}{2} \alpha t^2$, the steady-state tracking error is finite and non-zero: $e(t) \rightarrow \alpha/K_2$.

Proof. Step 1 — stability. In the coordinates (e, w) , with θ_{rif} differentiable, $\dot{e} = \dot{\theta}_{\text{rif}} - K_1 e - K_2 w$ and $\dot{w} = e$: for θ_{rif} constant or affine (Steps 2–3) the system is linear with constant coefficients with matrix $A = \begin{pmatrix} -K_1 & -K_2 \\ 1 & 0 \end{pmatrix}$, characteristic polynomial $r^2 + K_1 r + K_2$: for

$K_1, K_2 > 0$ both roots have negative real part (Routh–Hurwitz criterion for a second-degree polynomial: all coefficients positive $\iff$ roots with negative real part, an elementary fact verifiable with the quadratic formula), hence every solution of the associated homogeneous equation tends to 0: the forced system converges to its own equilibrium whatever it may be.

Step 2 — constant velocity. With $\dot{\theta}_{\text{rif}} = \omega_0$ constant, the equilibrium $(e_{\text{eq}}, w_{\text{eq}})$ solves $0 = \omega_0 - K_1 e_{\text{eq}} - K_2 w_{\text{eq}}$ and $0 = e_{\text{eq}}$ (from the second equation $\dot{w} = e$ set to zero): the second gives directly $e_{\text{eq}} = 0$ (and consequently $w_{\text{eq}} = \omega_0/K_2$ from the first). By Step 1 the solution converges to this equilibrium: $e(t) \rightarrow 0$.

Step 3 — constant acceleration. With $\theta_{\text{rif}}(t) = \frac{1}{2} \alpha t^2$, $\dot{\theta}_{\text{rif}}(t) = \alpha t$ is no longer constant: there is no equilibrium in the sense of Step 2, but there is an asymptotic regime with $e(t) \rightarrow e_\infty$ constant and $w(t) \sim e_\infty t$ (unbounded, because $\dot{w} = e \rightarrow e_\infty$). Substituting the ansatz $e(t) = e_\infty + \varepsilon(t)$ with $\varepsilon(t) \rightarrow 0$ and $w(t) = e_\infty t + w_1 + \eta(t)$ with $\eta(t) \rightarrow 0$ into the equation $\dot{e} = \alpha t - K_1 e - K_2 w$:

$$\dot{\varepsilon} = \alpha t - K_1(e_\infty + \varepsilon) - K_2(e_\infty t + w_1 + \eta),$$

the term in t on the left-hand side is absent (because $\dot{\varepsilon} \rightarrow 0$), hence the terms in t on the right-hand side must cancel: $\alpha - K_2 e_\infty = 0$, that is, $e_\infty = \alpha/K_2$. The remaining terms determine w_1 and the transient (ε, η) , which tends to zero by Step 1 applied to the deviations from the ansatz (which satisfy the associated homogeneous equation, up to a forcing term which itself tends to zero).

Independent numerical check. With $K_1 = 4$, $K_2 = 6$: for $\theta_{\text{rif}} = 3t$, integrating the system numerically up to $t = 40$ one finds $e(40) = 2.7 \times 10^{-11} \approx 0$. ✓ For $\theta_{\text{rif}} = \frac{1}{2} \cdot 2 \cdot t^2$ ($\alpha = 2$), up to $t = 60$ one finds $e(60) = 0.333333$, against the predicted $\alpha/K_2 = 2/6 = 0.333333$. ✓ □

Remark 14.9: Why “type II”. In the language of servosystems, a system of type N has N integrators in the forward path and tracks with zero steady-state error a polynomial input of degree $N-1$ (here: an input at constant position, $N = 0$, or at constant velocity, $N = 1$, with zero error), whereas it has a finite error on an input of degree N (here: constant acceleration, which is of degree $2 = N$ for $N = 2$). The first-order PLL of Adler’s equation has a single integrator (the VCO): it is of type I, and indeed the equilibrium $\theta^* = \arcsin(\Delta\omega/K)$ of Theorem 14.1 is a non-zero phase error even at constant velocity ($\Delta\omega$ constant is the analogue of a constant-velocity input for the phase); the PI filter adds the second integrator which cancels that error. It is the technical reason why precision resolver-to-digital converters never use a first-order PLL.

14.8. Itoh's theorem: when phase unwrapping works

Construction 14.1: Itoh's unwrapping algorithm. Let $\Theta : [t_0, t_N] \rightarrow \mathbb{R}$ be a true phase, continuous (the lift, in the sense of **Chapter 1**, of a signal living on S^1), sampled at the points $t_0 < \dots < t_N$; let $\varphi_j = W(\Theta(t_j))$ be the *wrapped* readings, where $W(x) \in (-\pi, \pi]$ is the unique representative of x modulo 2π in that interval. **Itoh's algorithm** (S18) reconstructs a sequence ψ_j by accumulating the wrapped increments:

$$\psi_0 = \varphi_0, \quad \psi_j = \psi_{j-1} + W(\varphi_j - \varphi_{j-1}) \quad (j = 1, \dots, N).$$

□

Theorem 14.2: Itoh's theorem on phase unwrapping. With the notation of Construction 14.1, setting $\Delta\Theta_j := \Theta(t_j) - \Theta(t_{j-1})$ the true phase increment between two consecutive samples:

$$\psi_j = \Theta(t_j) - \Theta(t_0) + \varphi_0 \quad \text{for every } j = 1, \dots, N$$

$$\Updownarrow$$

$$\Delta\Theta_j \in (-\pi, \pi] \quad \text{for every } j = 1, \dots, N$$

That is: Itoh's algorithm reconstructs the lift of the phase exactly if and only if every true sampled increment lies in the interval of a half-turn.

Proof. Put $\Psi_j := \Theta(t_j) - \Theta(t_0) + \varphi_0$, the true phase referred back to the origin φ_0 ; one has to show that $\psi_j = \Psi_j$ for every j if and only if every $\Delta\Theta_j \in (-\pi, \pi]$.

Step 1 — a preliminary observation. Since $\varphi_j \equiv \Theta(t_j) \pmod{2\pi}$ by definition of W , one has $\varphi_j - \varphi_{j-1} \equiv \Theta(t_j) - \Theta(t_{j-1}) = \Delta\Theta_j \pmod{2\pi}$: hence $W(\varphi_j - \varphi_{j-1})$ and $\Delta\Theta_j$ are always congruent modulo 2π , and $W(\varphi_j - \varphi_{j-1})$ is, by definition of W , the *unique* representative of that class in $(-\pi, \pi]$.

Step 2 — sufficiency ($\Leftarrow$). If $\Delta\Theta_j \in (-\pi, \pi]$ for every j , then $\Delta\Theta_j$ is itself the unique representative of its own class in $(-\pi, \pi]$, hence by Step 1, $W(\varphi_j - \varphi_{j-1}) = \Delta\Theta_j$ for every j . By induction on j : $\psi_0 = \varphi_0 = \Psi_0$ (by definition of Ψ_0); if $\psi_{j-1} = \Psi_{j-1}$, then

$$\psi_j = \psi_{j-1} + W(\varphi_j - \varphi_{j-1}) = \Psi_{j-1} + \Delta\Theta_j = (\Theta(t_{j-1}) - \Theta(t_0) + \varphi_0) + (\Theta(t_j) - \Theta(t_{j-1})) = \Psi_j.$$

Hence $\psi_j = \Psi_j$ for every j from 0 to N .

Step 3 — necessity ($\Rightarrow$), by contraposition. Suppose that there exists an index k with $\Delta\Theta_k \notin (-\pi, \pi]$, and let k be the *first* such index. For $j < k$, the minimality hypothesis gives $\Delta\Theta_j \in (-\pi, \pi]$ for every $j \leq k-1$, hence by the argument of Step 2 (applied to the first $k-1$ steps) $\psi_{k-1} = \Psi_{k-1}$. Now, $W(\varphi_k - \varphi_{k-1})$ is, by Step 1, the unique representative in $(-\pi, \pi]$ of the class of $\Delta\Theta_k$ modulo 2π ; since $\Delta\Theta_k \notin (-\pi, \pi]$ by hypothesis, that representative differs from $\Delta\Theta_k$ by a *non-zero* integer multiple $2\pi m_k$ ($m_k \in \mathbb{Z} \setminus \{0\}$): $W(\varphi_k - \varphi_{k-1}) = \Delta\Theta_k - 2\pi m_k$. Hence

$$\psi_k = \psi_{k-1} + W(\varphi_k - \varphi_{k-1}) = \Psi_{k-1} + \Delta\Theta_k - 2\pi m_k = \Psi_k - 2\pi m_k \neq \Psi_k,$$

because $m_k \neq 0$: the reconstruction already fails at step k , hence it is not true that $\psi_j = \Psi_j$ for every j . By contraposition, this proves the direct implication. □

Corollary 14.1: Sampling constraint for correct unwrapping. Let Θ be differentiable with $|\dot{\Theta}(t)| \leq \omega_{\max}$ for every t , and let $f_s > 0$ be the sampling frequency (step $\Delta t = 1/f_s$).

If

$$f_s > \frac{\omega_{\max}}{\pi},$$

then every true sampled increment satisfies $|\Delta\Theta_j| \leq \omega_{\max}\Delta t < \pi$, hence by Theorem 14.2 Itoh's algorithm reconstructs the lift of the phase exactly.

Proof. By the mean value theorem (or directly $\Delta\Theta_j = \int_{t_{j-1}}^{t_j} \dot{\Theta}(t) dt$), $|\Delta\Theta_j| \leq \int_{t_{j-1}}^{t_j} |\dot{\Theta}(t)| dt \leq \omega_{\max}\Delta t = \omega_{\max}/f_s$. If $f_s > \omega_{\max}/\pi$, then $\omega_{\max}/f_s < \pi$, hence $\Delta\Theta_j \in (-\pi, \pi]$ (indeed $(-\pi, \pi)$) for every j : the hypothesis of Theorem 14.2 is satisfied. $\square$

Remark 14.10: The same constraint as Nyquist's criterion, read twice. For a signal with instantaneous frequency $f(t) = \dot{\Theta}(t)/2\pi$ with $|f(t)| \leq f_{\max}$, the Corollary 14.1 becomes $f_s > 2f_{\max}$: it is exactly Nyquist's criterion, proved in **Volume III, Chapter 24**. This is no coincidence: both constraints say the same thing, namely that the sampling step must be fine enough not to be able to confuse two phase values more than a half-turn apart; here the constraint emerges from a different theorem (the correct reconstruction of a discrete unwrapping algorithm, not the reconstructibility of a sampled signal from a limited band), with the same numerical threshold. Figure 14.3 shows a case in which the constraint is violated halfway through the sampling (a signal with increasing frequency, a chirp) and Itoh's algorithm diverges from the true phase by exactly 2π at the first sample at which the increment exceeds π , exactly as predicted by Theorem 14.2.

14.9. Vernier encoder: the tolerance theorem

Remark 14.11: Two tracks, coprime moduli. A two-track Vernier absolute encoder (vernier scale; Renishaw ASTRIA, S25) reads simultaneously two periodic signals of angular period $2\pi/m$ and $2\pi/n$ (m, n coprime integers, the number of periods per turn of the two tracks): if $\theta \in [0, 2\pi)$ is the true angle, the two phases read are $\varphi_m = W_+(m\theta)$ and $\varphi_n = W_+(n\theta)$, where $W_+(x) \in [0, 2\pi)$ is the representative of x modulo 2π (Figure 14.4). That the pair (φ_m, φ_n) , read exactly, determines θ uniquely is a fact equivalent to the **Chinese remainder theorem** (Volume II, Chapter 7): it is not reproved here. What Volume II does not treat is what happens when the two readings are affected by an error, that is, how much phase noise the system can tolerate before decoding a wrong position: this is the new content of this section.

Construction 14.2: Turn numbers and the decoding variable J . Setting $x = \theta/2\pi \in [0, 1)$, let $K_m = \lfloor mx \rfloor \in \{0, \dots, m-1\}$ and $K_n = \lfloor nx \rfloor \in \{0, \dots, n-1\}$ be the **turn numbers** of the two tracks (how many periods the track has already completed), so that $\varphi_m = 2\pi(mx - K_m)$ and $\varphi_n = 2\pi(nx - K_n)$. A direct computation gives

$$n\varphi_m - m\varphi_n = 2\pi n(mx - K_m) - 2\pi m(nx - K_n) = 2\pi(mK_n - nK_m),$$

that is, the integer

$$J := mK_n - nK_m = \frac{n\varphi_m - m\varphi_n}{2\pi}$$

is computable *exactly* from the two true phases, without knowing K_m, K_n individually. Since $\gcd(m, n) = 1$, the equation $mK_n - nK_m = J$ with $K_m \in \{0, \dots, m-1\}$, $K_n \in \{0, \dots, n-1\}$ has, for every J in the image, a unique solution (Chinese remainder theorem, forward reference): from J one recovers (K_m, K_n) and hence θ without ambiguity. Decoding the position therefore reduces to reconstructing the integer J . $\square$

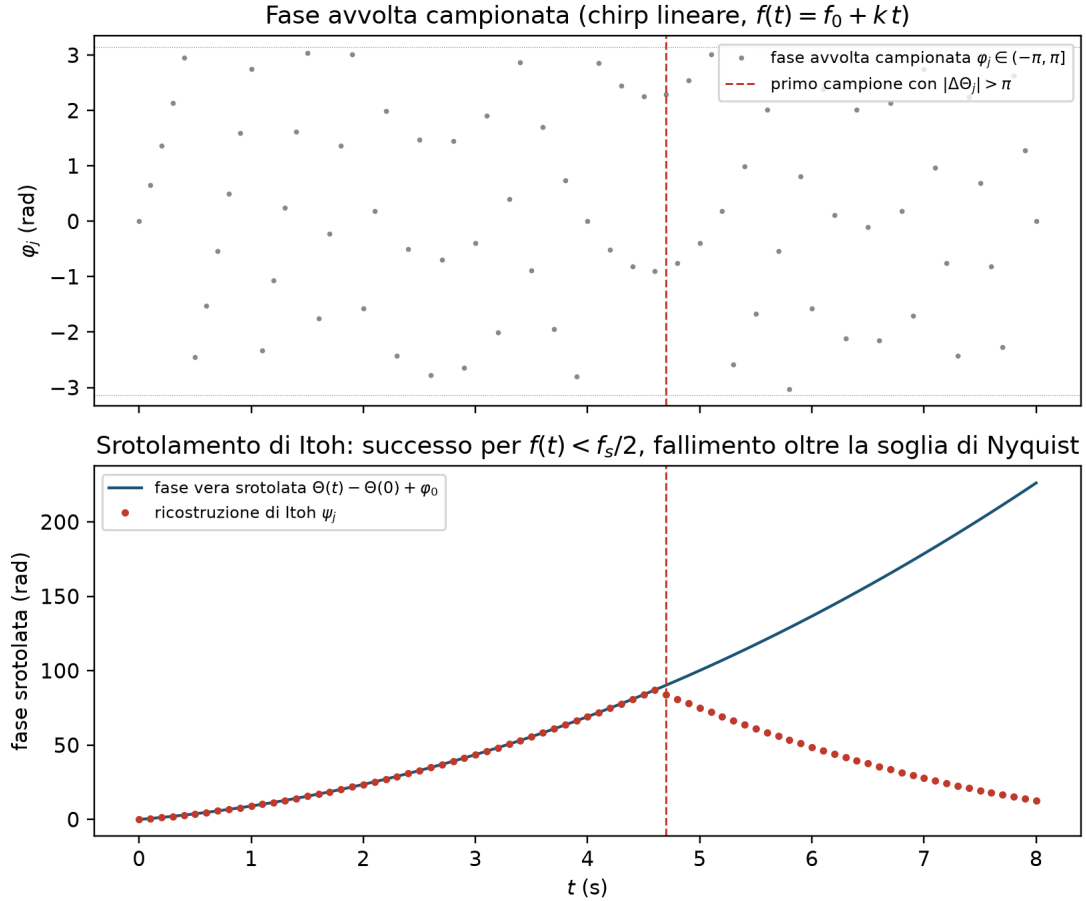


Figure 14.3: Sampled wrapped phase (above) and comparison between the true unwrapped phase and Itoh’s reconstruction (below), for a signal with increasing instantaneous frequency $f(t) = f_0 + kt$ with $f_0 = 1$ Hz, $k = 0.875$ Hz/s, sampled at $f_s = 10$ Hz. The theoretical threshold $f(t) = f_s/2 = 5$ Hz (Corollary 14.1) is reached at $t = 4.571$ s; the first sample with true increment $|\Delta\Theta_j| > \pi$ falls at $t = 4.7$ s, and it is exactly there that Itoh’s reconstruction diverges from the true phase by -2π , as proved in Theorem 14.2. Computed numerically with a double check (closed formula for the first increment beyond the threshold and the algorithm run independently); script `fig14_itoh_unwrap.py`, Appendix B. ✓

Theorem 14.3: Tolerance theorem for the two-track vernier. Let $\hat{\varphi}_m = \varphi_m + \delta_m$, $\hat{\varphi}_n = \varphi_n + \delta_n$ be the measured phases, with errors $\delta_m, \delta_n \in \mathbb{R}$ satisfying $\eta := \max(|\delta_m|, |\delta_n|)$. If

$$\eta < \frac{\pi}{m+n},$$

then the integer J of Construction 14.2 is reconstructed exactly as

$$J^* := \text{round}\left(\frac{n\hat{\varphi}_m - m\hat{\varphi}_n}{2\pi}\right),$$

that is, $J^* = J$, and the decoding of the absolute position θ (via $J \mapsto (K_m, K_n) \mapsto \theta$) is correct.

Proof. Step 1 — the discrepancy is small. Put $\hat{J} := (n\hat{\varphi}_m - m\hat{\varphi}_n)/(2\pi)$. By linearity,

$$\hat{J} = \frac{n(\varphi_m + \delta_m) - m(\varphi_n + \delta_n)}{2\pi} = J + \frac{n\delta_m - m\delta_n}{2\pi}.$$

By the triangle inequality and $|\delta_m|, |\delta_n| \leq \eta$,

$$|n\delta_m - m\delta_n| \leq n|\delta_m| + m|\delta_n| \leq (m+n)\eta.$$

Step 2 — the threshold guarantees correct rounding. If $\eta < \pi/(m+n)$, then $|n\delta_m - m\delta_n| < (m+n) \cdot \pi/(m+n) = \pi$, hence

$$|\hat{J} - J| = \frac{|n\delta_m - m\delta_n|}{2\pi} < \frac{\pi}{2\pi} = \frac{1}{2}.$$

Since J is an integer and $\hat{J}$ is less than $1/2$ away from it, J is the *unique* integer nearest to $\hat{J}$: $\text{round}(\hat{J}) = J$, that is, $J^* = J$.

Step 3 — from the correctness of J to the correctness of θ . By Construction 14.2, $J \mapsto (K_m, K_n)$ is bijective on the image (Chinese remainder theorem, forward reference to Volume II Chapter 7); from correct (K_m, K_n) , and from one of the two measured phases refined with the other (weighted average, or simply $x = (K_m + \hat{\varphi}_m/2\pi)/m$), one reconstructs θ with the residual error of the δ_m, δ_n alone (a small error, of the order of η/m , not the gross error of a jump to a wrong J). $\square$

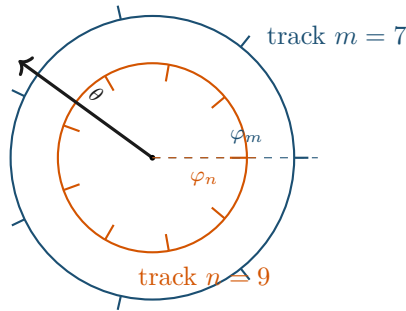


Figure 14.4: Diagram (not to scale) of a two-track Vernier encoder with $m = 7$ and $n = 9$ coprime notches per turn (the example of Section 14.9 and of Worked Example 14.3): the true angle θ generates two phase readings φ_m, φ_n (one per track), whose combination $J = (n\varphi_m - m\varphi_n)/2\pi$ (Construction 14.2) determines θ uniquely within the tolerance of Theorem 14.3.

Remark 14.12: Why the order is $\pi/(m+n)$ and not $\pi/(mn)$. The threshold is not the combined resolution of the encoder (which is $1/(mn)$ of a turn, by the Chinese remainder theorem): it is the noise tolerance of the coarse decoding of the integer J , that is, of which pair of “notches” the system is reading. An error which exceeds the threshold does not degrade the estimated position slightly: it makes it jump to a wrong pair (K_m, K_n) , a gross error of a whole notch on one of the two tracks — not a gradual deterioration. This is why the designer of a Vernier encoder must guarantee $\eta < \pi/(m+n)$ with a margin, and not merely minimise the noise on average.

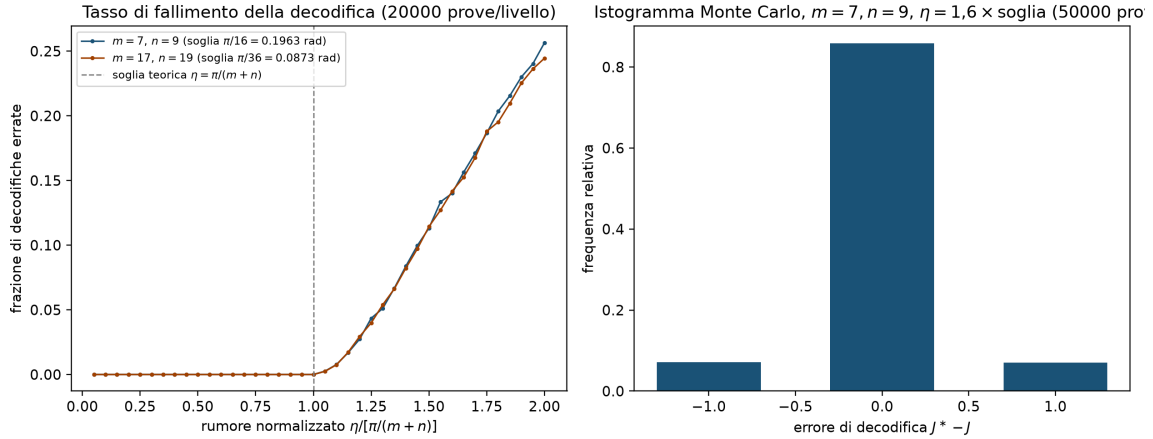


Figure 14.5: *Left*: fraction of wrong decodings over 20 000 Monte Carlo trials per noise level, for the pairs $(m, n) = (7, 9)$ (threshold $\pi/16 = 0.19635$ rad) and $(17, 19)$ (threshold $\pi/36 = 0.08727$ rad): the failure rate is exactly zero for every $\eta < \pi/(m+n)$ in both cases, as proved in Theorem 14.3. *Right*: histogram of the decoding error $J^* - J$ for $(m, n) = (7, 9)$ with $\eta = 1.6 \times$ the threshold (50 000 trials): the failure rate rises to 14.19%, with errors of ± 1 notch. Script `fig14_encoder_montecarlo.py`, Appendix B. ✓

14.10. Dead band from quantisation: the theorem, corrected

Remark 14.13: Where the old chapter went wrong. The old Chapter 5 asserted that the asymptotic amplitude of the error of a quantised digital control loop “does not depend on the gain K , only on the step Δ ”: the assertion is **false**, and it is refuted by the numerical check of this section ($e_\infty = 0.04$ for $K = 0.1$; $e_\infty = 0.01$ for $K = 0.3$; $e_\infty = 0$ for $K \geq 0.5$, with $\Delta = 0.01$ fixed): the gain determines the denominator of the threshold $\Delta/(2K)$. The object described, moreover, is not a limit cycle in the sense of the theory of dynamical systems (an isolated periodic orbit): it is, as the following proof shows, a **spurious fixed point** introduced by the quantisation, reached in a finite number of steps and then maintained exactly.

Theorem 14.4: Dead band theorem. Let $Q_\Delta(x)$ be the rounding of x to the nearest multiple of $\Delta > 0$ (with any fixed convention for the exact tie cases), and let $0 < K < 1$. For the recursion

$$e_{k+1} = Q_\Delta((1 - K)e_k),$$

with arbitrary $e_0 \in \mathbb{R}$:

- (i) for $k \geq 1$, e_k is an integer multiple of Δ ;
- (ii) $|e_k|$ is non-increasing;

- (iii) the sequence reaches an **exact fixed point** ($e_{k+1} = e_k$) at the first k with $|Ke_k| < \Delta/2$, and stays there for every subsequent index;
- (iv) the terminal value satisfies $|e_\infty| \leq \frac{\Delta}{2K}$.

Proof. Step 0 — reduction to an integer recursion. For $k \geq 1$, $e_k = Q_\Delta(\cdot)$ is by definition a multiple of Δ : we write $e_k = \Delta z_k$ with $z_k \in \mathbb{Z}$. Then $e_{k+1} = Q_\Delta(\Delta(1-K)z_k) = \Delta \cdot \text{round}((1-K)z_k)$ (because $Q_\Delta(x) = \Delta \cdot \text{round}(x/\Delta)$ by definition), that is, $z_{k+1} = \text{round}((1-K)z_k)$: a recursion between integers, equivalent to the original one. It suffices to prove (ii)–(iv) for $|z_k|$, with threshold $|Kz_k| < 1/2$ in place of $|Ke_k| < \Delta/2$ (the two are the same condition multiplied by Δ).

Step 1 — symmetry. Q_Δ is odd ($Q_\Delta(-x) = -Q_\Delta(x)$, for any symmetric convention for rounding the tie cases), hence $z_k \mapsto -z_k$ leaves the recursion unchanged: one may assume $z_k \geq 0$ without loss of generality, and this will be done in Steps 2–3.

Step 2 — the exact fixed point. If $Kz_k < 1/2$ (strictly), then $|(1-K)z_k - z_k| = Kz_k < 1/2$: the point $(1-K)z_k$ is less than $1/2$ away from z_k , hence z_k is the *unique* integer nearest to $(1-K)z_k$ (a point at distance $< 1/2$ from an integer cannot be nearer to any other integer, which would be at distance at least $1 - 1/2 = 1/2$): therefore $z_{k+1} = \text{round}((1-K)z_k) = z_k$, an exact fixed point. Since the recursion is autonomous (it does not depend on k), if z_k is a fixed point so is $z_{k+1} = z_k$, and by induction every z_j with $j \geq k$: the sequence remains constant from there on.

Step 3 — strict decrease until one is at the fixed point. If $Kz_k \geq 1/2$ (and $z_k > 0$, by Step 1), then $(1-K)z_k = z_k - Kz_k \leq z_k - \frac{1}{2}$. Using the elementary inequality $\text{round}(x) \leq x + \frac{1}{2}$ (true for every nearest-integer rounding convention, the nearest integer being at distance at most $1/2$),

$$z_{k+1} = \text{round}((1-K)z_k) \leq (1-K)z_k + \frac{1}{2} \leq \left(z_k - \frac{1}{2}\right) + \frac{1}{2} = z_k.$$

If moreover $Kz_k > 1/2$ (strictly), the first inequality is strict: $(1-K)z_k < z_k - \frac{1}{2}$, hence $z_{k+1} < z_k$; z_{k+1} being an integer and $< z_k$ (an integer), one has $z_{k+1} \leq z_k - 1$: the sequence decreases by *at least* 1 at each step as long as $Kz_k > 1/2$ remains strict. Since $z_k \geq 0$ is a non-negative integer which cannot decrease indefinitely, the case $Kz_k > 1/2$ can occur only a finite number of times: the sequence reaches in a finite number of steps an index with $Kz_k \leq 1/2$. If at that point the strict inequality $Kz_k < 1/2$ holds, Step 2 gives the exact fixed point immediately; the limiting case $Kz_k = 1/2$ exactly (of zero measure in real data, where Δ and K are not commensurable by construction) leads in at most one additional step to $Kz_{k+1} < 1/2$, after which Step 2 applies again. In every case, an exact fixed point is reached in a finite number of steps.

Step 4 — the final bound. At the fixed point z_∞ (reached by Step 3), the condition of Step 2 gives $K|z_\infty| < 1/2$ in the generic case, that is, $|z_\infty| < 1/(2K)$; with the non-strict inequality covering also the limiting case of Step 3, $|z_\infty| \leq 1/(2K)$. Going back to $e_\infty = \Delta z_\infty$: $|e_\infty| \leq \Delta/(2K)$.

Step 5 — non-increase of $|e_k|$ for every k , not only asymptotically. It is a direct consequence of Step 3 ($|z_{k+1}| \leq |z_k|$ in every case, with equality only at the fixed point) and of the base case $k = 0$ ($e_1 = Q_\Delta((1-K)e_0)$, and $|(1-K)e_0| < |e_0|$ for $0 < K < 1$, $e_0 \neq 0$, with the rounding altering this estimate by at most $\Delta/2$, absorbed in the analysis from $k = 1$ onwards of Step 0): the sequence $|e_k|$ is non-increasing for every $k \geq 1$, which is thesis (ii). $\square$

Remark 14.14: Numerical check and correction of the values. With $\Delta = 0.01$, $e_0 = 1.0$ (the same example as in the old chapter) and round-half-to-even rounding (the

IEEE 754 convention used by Python’s native `round` function, with which the old chapter had already computed, without however drawing the correct conclusion): the simulation gives $e_\infty = 0.04$ for $K = 0.1$ (theoretical bound $\Delta/(2K) = 0.05$: respected, not attained exactly, because the terminal value depends on the trajectory, not only on the bound), $e_\infty = 0.01$ for $K = 0.3$ (bound 0.0167), $e_\infty = 0$ for $K = 0.5$ and for $K = 0.7$ (bounds 0.01 and 0.0071): all consistent with Theorem 14.4, and all different as K varies — the direct refutation of the “independent of K ” assertion of the old chapter. ✓

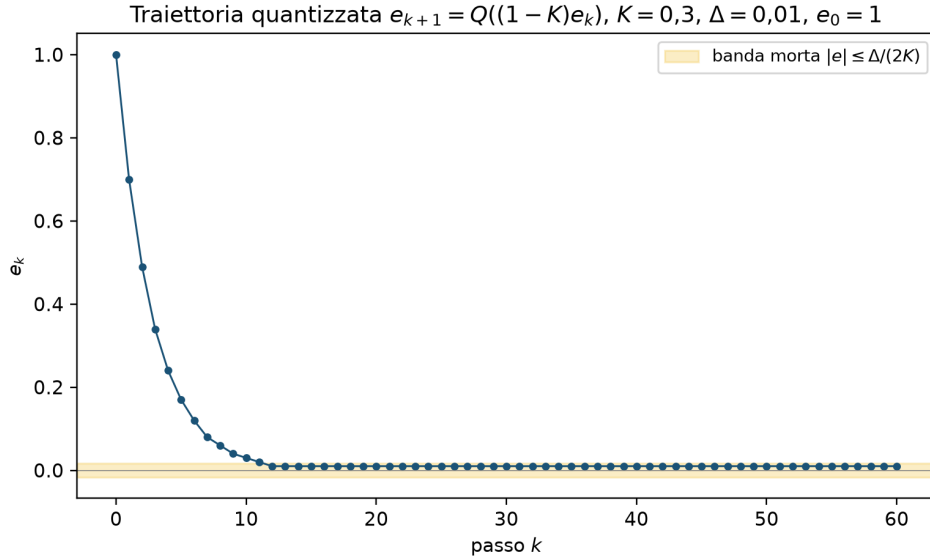


Figure 14.6: Quantised trajectory $e_{k+1} = Q_\Delta((1-K)e_k)$ for $K = 0.3$, $\Delta = 0.01$, $e_0 = 1$: $|e_k|$ decreases (Theorem 14.4(ii)) until it reaches, at step $k = 12$, the exact fixed point $e_\infty = 0.01$, inside the theoretical dead band $|e| \leq \Delta/(2K) = 0.0167$ (shaded region). Computed numerically with a double check (real recursion and reduced integer recursion, Step 0 of the proof); script `fig14_banda_morta.py`, Appendix B. ✓

14.11. Limit cycles from second-order quantisation

Remark 14.15: Panorama: when quantisation creates a true limit cycle. Theorem 14.4 concerns a **first-order** loop: quantisation there produces a spurious fixed point, not a periodic orbit. In **second-order** digital filters (for example a biquadratic section with quantised coefficients, or a loop with two states such as the type II model of Section 14.7 made non-linear by the quantisation), the non-linearity of the rounding may instead generate true persistent periodic oscillations — a limit cycle in the proper sense of the term — even when the underlying linear system (without quantisation) would be asymptotically stable: the mechanism, distinct from the dead band, and its analysis (by the describing function or by the direct analysis of the quantised state space) are treated in Åström–Wittenmark (S30, ch. on quantisation); reported here as panorama, without proof. One must moreover keep distinct a third phenomenon, of the opposite nature: **overflow** from truncation (rather than rounding) of the accumulator, which may generate oscillations of large amplitude (not small like the dead band), typically avoided in practice with saturation arithmetic.

14.12. Distributed clocks: EtherCAT

Remark 14.16: Clock detuning as a phase on S^1 (panorama). An EtherCAT field-bus synchronises the local clocks of several nodes (axes, remote I/O) with the Distributed Clocks mechanism (S22): each node maintains a local time counter, and the reference node periodically measures the discrepancy of the others and corrects their counting speed. Conceptually, the discrepancy between two clocks counting modulo the same period (for example a fixed communication cycle) is exactly a phase on S^1 , and its cancellation is a synchronisation problem of the same kind studied in **Chapter 13** (two-oscillator Kuramoto, with one node playing the role of a fixed reference rather than of a peer oscillator) and in this chapter (a PLL, with the propagation delay on the bus as the main source of systematic error to be compensated, not merely of noise): the sector literature (Cena et al., S21) measures the performance obtained in real implementations. Reported here as panorama: the quantitative analysis of the delay compensation protocol lies outside the scope of a treatise on the mathematics of the circle.

14.13. PLCopen function blocks for axis synchronisation

Remark 14.17: MC_GearIn, MC_GearInPos, MC_CamIn, MC_Phasing: synchronisation as conjugation of rotations (panorama). The PLCopen standard for motion control (S23, on top of IEC 61131-3, S24) defines function blocks for synchronising a slave axis to a master axis, both thought of as modulo axes (position on S^1 , not on $\mathbb{R}$, when the axis is rotary without end stops; Figure 14.7):

- **MC_GearIn** imposes $\theta_{\text{slave}}(t) = \frac{n_2}{n_1} \theta_{\text{master}}(t) + c$ (constant gear ratio $n_1:n_2$): it is a linear conjugation between the two rotations, the digital analogue of a mechanical gear;
- **MC_GearInPos** is **MC_GearIn** with the locking synchronised to a chosen absolute position of the master, rather than started at an arbitrary instant of time;
- **MC_CamIn** replaces the constant ratio with an arbitrary function $\theta_{\text{slave}} = \Phi(\theta_{\text{master}})$ (a cam table), typically 2π -periodic if both axes are modulo axes: a non-linear conjugation between the two rotations;
- **MC_Phasing** shifts the relative phase between two already synchronised axes by an assigned offset $\Delta\varphi$, with a time ramp in order to avoid discontinuities of velocity.

An illustrative fragment (not a normative extract) in Structured Text:

Listing 14.1: Master/slave synchronisation with MC_GearIn (illustrative)

```
GearIn(Master:=MasterAxis, Slave:=SlaveAxis,
      RatioNumerator:=3, RatioDenominator:=2,
      Execute:=StartSync);
IF GearIn.InGear THEN
  (* slave axis locked: theta_slave = 1.5 * theta_master + c *)
END_IF;
```

Reported as panorama, with a single numerical check example: with a ratio of 3:2, for every 2 turns of the master the slave makes 3; if the master rotates at 10 rpm, the slave rotates at 15 rpm — an elementary proportion, not a theorem.

14.14. The stroboscope and the wheel turning backwards

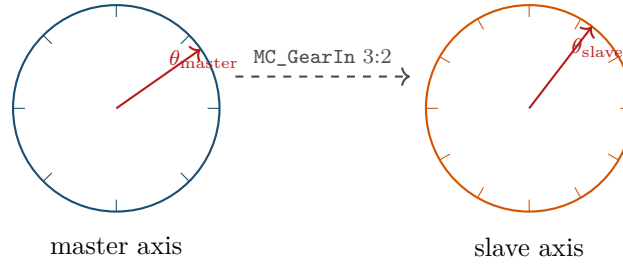


Figure 14.7: Qualitative diagram (not to scale) of `MC_GearIn`: two modulo axes (two copies of S^1 , eight notches on the master and twelve on the slave, in the ratio 3:2) synchronised by a PLCopen function block; the angle of the slave is a linear function (conjugation) of that of the master, $\theta_{\text{slave}} = \frac{3}{2}\theta_{\text{master}} + c$.

Remark 14.18: The stroboscopic effect as a sampled rotation map (level A). A wheel turning at a frequency of f turns/s, filmed at f_s frames/s, appears to rotate at the **apparent frequency** $f_{\text{app}} = f_s \cdot \rho(f/f_s)$, where $\rho(x) \in (-\frac{1}{2}, \frac{1}{2}]$ is the representative of x modulo 1 in that interval: it is the sampled rotation map of Nyquist's criterion (**Volume III, Chapter 24**), here only applied, not reproved (S26). If $f_{\text{app}} < 0$, the wheel appears to rotate backwards: this is not an optical illusion in the sense of a perceptual error, but the exact mathematical consequence of aliasing, verifiable frame by frame.

14.15. Worked examples

Worked Example 14.1 — Locking range of the discrete PLL for $\epsilon = 0.3$ Problem.

For Arnold's standard family with gain $\epsilon = 0.3$ (discrete PLL, Section 14.5), compute the exact locking range of the principal tongue and confirm it with a numerical comparison just inside and just outside the edge.

Step 1 — closed formula. By Chapter 4, the principal tongue is exactly $|\alpha| \leq \epsilon/2\pi$. With $\epsilon = 0.3$: $|\alpha| \leq 0.3/(2\pi) = 0.047746$.

Step 2 — numerical comparison. Iterating $f_{\alpha,0.3}$ for 2000 steps from $\theta_0 = 0.1$ and estimating the rotation number as the mean unwrapped increment per iteration: for $\alpha = 0.04727$ (0.99 times the edge) one finds $\rho \approx 6.4 \times 10^{-5}$ (locked, $\rho = 0$ within the numerical error); for $\alpha = 0.04822$ (1.01 times the edge) one finds $\rho \approx 0.006579 \neq 0$ (not locked). ✓

Interpretation. The transition occurs exactly at the edge predicted by the closed formula, neither before nor after: the principal tongue has no grey zone, it is an exact closed interval (the edge itself belongs to the tongue, by the saddle-node bifurcation of Chapter 4).

Typical error. Confusing $\epsilon/2\pi$ with ϵ/π or ϵ : the factor $1/2\pi$ comes from the normalisation of the angle to $[0, 1)$ in the standard map. □

Worked Example 14.2 — Correct and incorrect unwrapping Problem. A true phase grows linearly, $\Theta(t) = \omega t$ with $\omega = 8$ rad/s. Compare Itoh unwrapping when sampling at $f_s = 3$ Hz and at $f_s = 2$ Hz.

Case $f_s = 3$ Hz ($\Delta t = 1/3$ s). $\Delta\Theta = \omega\Delta t = 8/3 = 2.6667$ rad $< \pi$: by Theorem 14.2, every sample satisfies the hypothesis, the algorithm reconstructs exactly. Check on the first samples: $\Theta(t_j) = 0; 2.6667; 5.3333; 8$; wrapped phases $\varphi_j = 0; 2.6667; -0.9499; 1.7168$ (rad, in $(-\pi, \pi]$); $\psi_0 = 0$, $\psi_1 = 0 + W(2.6667) = 2.6667$, $\psi_2 = 2.6667 + W(-0.9499 - 2.6667) = 2.6667 + W(-3.6166) = 2.6667 + 2.6666 = 5.3333$, $\psi_3 = 5.3333 + W(1.7168 -$

$(-0.9499) = 5.3333 + 2.6667 = 8.0000$: it coincides with $\Theta(t_3) = 8$ at every step. ✓

Case $f_s = 2$ Hz ($\Delta t = 0.5$ s). $\Delta\Theta = 8 \times 0.5 = 4$ rad $> \pi$: it is violated already at the first step. $\varphi_0 = 0$, $\Theta(t_1) = 4$, $\varphi_1 = W(4) = 4 - 2\pi = -2.2832$; $\psi_1 = 0 + W(-2.2832 - 0) = -2.2832$, whereas the true phase is $\Theta(t_1) - \Theta(t_0) = 4$: error $\psi_1 - 4 = -6.2832 = -2\pi$ exactly, as predicted by Step 3 of the proof of Theorem 14.2 ($m_1 = 1$).

Interpretation. With constant frequency, the only way of violating the hypothesis is to sample too slowly with respect to ω ; with variable frequency (Figure 14.3), the violation may appear halfway through the signal, when the instantaneous frequency exceeds the threshold, even though the sampling was adequate at the beginning. □

Worked Example 14.3 — 7/9 encoder with simulated phase errors Problem.

Vernier encoder with $m = 7$, $n = 9$ (coprime). True angle $\theta = 2\pi \times 0.4 = 2.513274$ rad. Decode with two noise levels, one below and one above the theoretical threshold.

Step 1 — true data. $x = 0.4$; $mx = 2.8 \Rightarrow K_m = 2$, $\varphi_m = 2\pi \times 0.8 = 5.026548$; $nx = 3.6 \Rightarrow K_n = 3$, $\varphi_n = 2\pi \times 0.6 = 3.769911$; $J = mK_n - nK_m = 21 - 18 = 3$.

Step 2 — threshold. $\pi/(m+n) = \pi/16 = 0.196350$ rad (11.25°).

Step 3 — noise below the threshold. $\delta_m = 0.10$, $\delta_n = -0.08505$ (both < 0.19635): $\hat{J} = (9 \times 5.126548 - 7 \times 3.684861)/2\pi = 3.2323$; $J^* = \text{round}(3.2323) = 3 = J$. Decoding correct. ✓

Step 4 — noise above the threshold. $\delta_m = 0.30$, $\delta_n = -0.28$ (both > 0.19635 ; combined discrepancy $|n\delta_m - m\delta_n| = |9 \times 0.30 - 7 \times (-0.28)| = 4.66 > \pi$): $\hat{J} = 3.7417$; $J^* = \text{round}(3.7417) = 4 \neq 3$. Decoding **wrong**: the system believes it is reading a notch different from the true one, a gross jump, not a small discrepancy.

Monte Carlo check. Over 20 000 trials per noise level and for $(m, n) = (7, 9)$ and $(17, 19)$, the rate of wrong decodings is exactly zero below the respective threshold and positive just above it (Figure 14.5); script `fig14_encoder_montecarlo.py`, Appendix B. ✓ □

Worked Example 14.4 — Dead band with $K = 0.3$, $\Delta = 0.01$ Problem. A quantised proportional loop $e_{k+1} = Q_{0.01}(0.7e_k)$ starts from $e_0 = 1$. Find the terminal value and compare it with the theoretical bound.

Step 1 — theoretical bound. $\Delta/(2K) = 0.01/0.6 = 0.016667$.

Step 2 — simulation (round-half-to-even, $\Delta = 0.01$). The sequence (values of e_k , rounded) is $1.00 \rightarrow 0.70 \rightarrow 0.49 \rightarrow 0.34 \rightarrow 0.24 \rightarrow 0.17 \rightarrow 0.12 \rightarrow 0.08 \rightarrow 0.06 \rightarrow 0.04 \rightarrow 0.03 \rightarrow 0.02 \rightarrow 0.01 \rightarrow 0.01 \rightarrow \dots$: exact fixed point reached at step $k = 12$, with $e_\infty = 0.01$.

Step 3 — check of the stopping condition. $|Ke_\infty| = 0.3 \times 0.01 = 0.003 < \Delta/2 = 0.005$: ✓, consistent with Theorem 14.4(iii); and $|e_\infty| = 0.01 \leq 0.016667$: ✓, consistent with (iv).

Interpretation. The bound 0.016667 is not attained exactly: it is an upper bound, not the guaranteed value; the actual value depends on the trajectory (here, on having started from $e_0 = 1$, an integer multiple of Δ from the outset). Changing $K = 0.1$ or $K = 0.5$ with the same e_0, Δ one finds $e_\infty = 0.04$ and $e_\infty = 0$ respectively (Remark 14.14, Figure 14.6).

Typical error. Confusing the spurious fixed point with a limit cycle: here the sequence becomes *exactly constant*, not periodic with period > 1 (see instead Section 14.11 for the true second-order limit cycles). □

14.16. Assigned exercises

The answers are in the Teacher's key (Section 14.19).

Exercise 14.1: Hold-in range and locking time. *Goal:* to apply Theorem 14.1 and Proposition 14.1 to a concrete numerical case. *Estimated time:* 25 minutes. *Type:* numerical with an independent check.

A first-order PLL has $K = 0.8$ and detuning $\Delta\omega = 0.5$.

- Check that $\Delta\omega \in H = P = L$ and compute the stable equilibrium θ^* (in radians, four decimal digits).
- Compute the characteristic locking time $\tau = 1/\sqrt{K^2 - \Delta\omega^2}$.
- Estimate after how many multiples of τ the initial phase error has been reduced to less than 1% of its value, using the asymptotic exponential approximation.

Exercise 14.2: A sampling which violates Itoh's constraint. *Goal:* to apply Theorem 14.2 and its corollary to a case in which the sampling is insufficient only on part of the signal. *Estimated time:* 30 minutes. *Type:* numerical/symbolic.

A resolver rotates with instantaneous frequency $f(t) = 2 + 0.5t$ Hz (constant acceleration), sampled at $f_s = 6$ Hz.

- Determine the instant t^* at which $f(t)$ reaches $f_s/2$.
- Using Corollary 14.1, explain why Itoh unwrapping is guaranteed correct for $t < t^*$ and no longer guaranteed beyond it.
- Compute the true phase increment $\Delta\Theta$ between the two samples immediately following t^* and check whether it exceeds π .

Exercise 14.3: Tolerance of a 17/19 encoder. *Goal:* to apply Theorem 14.3 and to verify it with a Monte Carlo script of one's own. *Estimated time:* 35 minutes. *Type:* numerical/computational.

A Vernier encoder has $m = 17$, $n = 19$ notches per turn on the two tracks.

- Compute the theoretical tolerance threshold $\pi/(m+n)$, in radians and in degrees.
- Compute the combined resolution of the encoder (Chinese remainder theorem, forward reference).
- Write (in pseudocode or in Python) a Monte Carlo experiment with at least 10 000 trials per noise level estimating the rate of wrong decodings for $\eta = 0.5 \times$ the threshold and for $\eta = 1.5 \times$ the threshold, and compare with Figure 14.5.

14.17. Worked exercises

Exercise 14.4: The fan and the stroboscope. *Type:* numerical, level A. A fan rotates at 47 turns/s and is filmed by a camera at 50 frames/s. Determine the apparent frequency in the film, sign included.

Solution. Strategy. Apply the sampled rotation map of Remark 14.18: the apparent frequency is f_s times the representative of f/f_s in $(-\frac{1}{2}, \frac{1}{2}]$.

Step 1. $\nu = f/f_s = 47/50 = 0.94$.

Step 2. The representative of 0.94 modulo 1 in $(-\frac{1}{2}, \frac{1}{2}]$ is $0.94 - 1 = -0.06$.

Step 3. $f_{\text{app}} = f_s \times (-0.06) = 50 \times (-0.06) = -3$ turns/s.

Independent check. `python -c "f=47.0; fs=50.0; nu=f/fs; print((nu-round(nu))*fs)"` returns `-3.0000000000000027` (floating-point rounding, negligible). ✓

Interpretation. The fan appears to rotate backwards at 3 turns/s: not an illusion, but the exact alias of 47 turns/s with respect to sampling at 50 Hz.

Typical errors. Computing $47 - 50 = -3$ without normalising (it works here by numerical coincidence, but it fails in general, for example for $f = 97$, $f_s = 50$: the correct computation gives $\nu = 1.94 \rightarrow -0.06 \rightarrow -3$ turns/s again, whereas $97 - 50 = 47 \neq -3$).

Extension. For $f = 53$ turns/s, $f_s = 50$: $\nu = 1.06 \rightarrow 0.06 \rightarrow +3$ turns/s: the wheel appears to rotate forwards, but more slowly than in reality, a phenomenon distinct from the reversal.

Exercise 14.5: Dead band with $K = 0.1$. *Type:* numerical with an applied proof. Verify, by applying Theorem 14.4, that for $K = 0.1$, $\Delta = 0.01$, $e_0 = 1$ the sequence $e_{k+1} = Q_\Delta(0.9 e_k)$ terminates exactly at $e_\infty = 0.04$.

Solution. Strategy. Reduce to the integer recursion $z_{k+1} = \text{round}(0.9z_k)$ (Step 0 of the proof of Theorem 14.4) with $z_0 = 100$, and apply the stopping criterion of Step 2 ($Kz_k < 1/2 \iff z_k < 5$).

Step 1 — theoretical bound. $\Delta/(2K) = 0.01/0.2 = 0.05$, that is, $|z_\infty| < 5$ expected (up to the limiting case).

Step 2 — iteration (round-half-to-even). $100 \rightarrow 90 \rightarrow 81 \rightarrow 73 \rightarrow 66 \rightarrow 59 \rightarrow 53 \rightarrow 48 \rightarrow 43 \rightarrow 39 \rightarrow 35 \rightarrow 32 \rightarrow 29 \rightarrow 26 \rightarrow 23 \rightarrow 21 \rightarrow 19 \rightarrow 17 \rightarrow 15 \rightarrow 14 \rightarrow 13 \rightarrow 12 \rightarrow 11 \rightarrow 10 \rightarrow 9 \rightarrow 8 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 4$: at the step with $z_k = 5$, $Kz_k = 0.5$ (limiting case, Step 3 of the proof foresees at most one additional step); $z = \text{round}(0.9 \times 5) = \text{round}(4.5) = 4$ (round-half-to-even: 4 is even), and from $z = 4$: $Kz = 0.4 < 0.5$, fixed point, $\text{round}(3.6) = 4$.
✓

Step 3 — final value. $e_\infty = \Delta \times 4 = 0.04$, as required.

Independent check. The script `fig14_banda_morta.py` (Appendix B), run for $K = 0.1$, returns $e_\infty = 0.040000$ and confirms the bound $0.04 \leq 0.05$. ✓

Interpretation. The limiting case $z = 5$ (where Kz is exactly $1/2$) is the only point at which the rounding convention (here round-half-to-even, which sends $4.5 \rightarrow 4$) affects the result: with round-half-away-from-zero one would have stopped at $z = 5$, $e_\infty = 0.05$, still within the bound of the theorem but at another fixed point. The theorem guarantees the bound in both cases, not the exact value, which depends on the convention in the tie cases.

Extension. Verify that for $e_0 = 2$ (instead of 1) the same $K = 0.1$ still gives $e_\infty = 0.04$ or 0.05 depending on the path: the fixed point reached depends on the initial datum among all those with $|z| < 5$, it is not unique.

Exercise 14.6: Tracking error of the resolver-to-digital converter. *Type:* symbolic with a numerical check. A type II RDC has gains $K_1 = 4$, $K_2 = 6$ (coherent units, s^{-1} and s^{-2}). The shaft accelerates from rest with constant angular acceleration $\alpha = 2 \text{ rad/s}^2$. Compute the steady-state tracking error.

Solution. Strategy. Apply directly Proposition 14.2(ii).

Step 1. The input is $\theta_{\text{rif}}(t) = \frac{1}{2}\alpha t^2$ with $\alpha = 2$: an input at constant acceleration.

Step 2. By Proposition 14.2(ii), $e_\infty = \alpha/K_2 = 2/6 = 0.333333 \text{ rad}$.

Independent check. Numerical integration of the system (e, w) (Runge–Kutta, fine step) up to $t = 60 \text{ s}$ gives $e(60) = 0.333333$, in agreement to six digits. ✓

Interpretation. The error does not tend to zero (unlike the constant-velocity case, Proposition 14.2(i)): a type II RDC, like every type II servosystem, tracks constant position and velocity without error, but not constant acceleration. To cancel that error as well one would need a third integrator (type III), at the expense, in general, of the stability and the robustness of the loop (a design trade-off, not pursued further here).

Typical errors. Applying formula (i) (zero error) also to the constant-acceleration case: the type of the servosystem and the degree of the input polynomial must correspond.

Extension. Verify that doubling K_2 (to $K_2 = 12$) halves the steady-state error (0.1667 rad), whereas K_1 does not appear in the formula for e_∞ : only the second integrator governs the asymptotic error due to acceleration, exactly as in Theorem 14.4 only K (not Δ taken in isolation) governs the denominator of the dead band.

14.18. Self-assessment quiz

Eight items in order of competence; the commented answers are in the Teacher's key (Section 14.19).

1. (Definition) The *lock-in range* of a PLL is the set of detunings for which
 - A) there exists an equilibrium, however unstable
 - B) every trajectory converges to an equilibrium, with transient slips allowed
 - C) every trajectory converges to an equilibrium without ever completing a full turn
 - D) the loop is switched off
2. (Hypothesis) Itoh's theorem requires that, for every sample,
 - A) the frequency be constant
 - B) the true phase increment lie in $(-\pi, \pi]$
 - C) the signal be periodic
 - D) the sampling step be a multiple of 2π
3. (Symbolic computation) The tolerance threshold of the two-track vernier with m, n coprime is of the order of
 - A) $\pi/(mn)$
 - B) $\pi/(m+n)$
 - C) $2\pi/m$
 - D) $\pi/\gcd(m, n)$
4. (Numerical computation) For the discrete PLL with $\epsilon = 0.3$, the exact locking range of the principal tongue is
 - A) $|\alpha| \leq 0.3$
 - B) $|\alpha| \leq 0.0477$
 - C) $|\alpha| \leq 0.15$
 - D) $|\alpha| \leq 0.6\pi$
5. (Proof step) In the dead band theorem, the sequence z_k stops decreasing strictly as soon as
 - A) $z_k = 0$
 - B) $Kz_k \geq 1/2$ no longer holds, that is, $Kz_k < 1/2$
 - C) k is even
 - D) $\Delta \rightarrow 0$
6. (Error diagnosis) The old Chapter 5 maintained that the amplitude of the dead band was independent of K . The error was
 - A) confusing Δ with K in the formula
 - B) not having computed $\Delta/(2K)$, which depends explicitly on K
 - C) using a wrong Δ
 - D) the assertion was in fact correct
7. (Transfer) A type II resolver-to-digital converter, unlike the first-order PLL,
 - A) has the same steady-state error at constant velocity
 - B) cancels the steady-state error at constant velocity thanks to the second integrator
 - C) cannot track any input
 - D) is always unstable
8. (Source and metacognition) The rigorous definitions of hold-in, pull-in and lock-in as sets, used in this chapter, come from
 - A) Gardner, *Phaselock Techniques*
 - B) Leonov, Kuznetsov, Yuldashev, Yuldashev, *IEEE TCAS-I* 2015
 - C) Åström–Wittenmark, *Computer-Controlled Systems*
 - D) Itoh, *Applied Optics* 1982

14.19. Teacher's key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **C.** *Rationale:* it is Definition 14.1: the lock-in range excludes transient slips too, not only permanent ones. *Distractors:* A describes a weaker set (no requirement of convergence); B is the definition of the *pull-in* range, not of the lock-in; D is meaningless. *Diagnosed error:* confusing lock-in with pull-in. *Source:* Definition 14.1; Remark 14.3. *Prerequisite:* the notion of convergence of a trajectory. *Remedial work:* reread Remark 14.3 on the inclusion $L \subseteq P \subseteq H$.
- 2) **B.** *Rationale:* it is the exact statement of Theorem 14.2. *Distractors:* A, C, D are conditions extraneous to the hypothesis of the theorem. *Diagnosed error:* confusing the hypothesis on the sampled increment with a hypothesis on the continuous signal. *Source:* Theorem 14.2, Step 1 of the proof. *Prerequisite:* the notion of representative modulo 2π . *Remedial work:* redo Worked Example 14.2 by hand.
- 3) **B.** *Rationale:* Theorem 14.3, Step 2 of the proof: the combined discrepancy is amplified by a factor $m + n$, not mn . *Distractors:* A is the order of the combined *resolution*, not of the tolerance to noise; C, D have no basis in the proof. *Diagnosed error:* confusing resolution (how finely one reads) with tolerance (how much noise one withstands). *Source:* Theorem 14.3; Remark 14.12. *Prerequisite:* triangle inequality. *Remedial work:* redo Step 1 of the proof with general m, n .
- 4) **B.** *Rationale:* $\epsilon/2\pi = 0.3/(2\pi) = 0.047746$ (Chapter 4, forward reference; Worked Example 14.1). *Distractors:* A forgets to divide by 2π ; C and D use wrong numerical factors. *Diagnosed error:* forgetting the normalisation of the angle. *Source:* Remark 14.6. *Prerequisite:* the principal tongue of Arnold's standard family (Chapter 4). *Remedial work:* repeat the computation for $\epsilon = 0.6$ and compare with the old Example 5.6 (Chapter 5, archived).
- 5) **B.** *Rationale:* it is Step 3 of the proof of Theorem 14.4: the strict decrease ceases exactly when $Kz_k < 1/2$. *Distractors:* A is a particular case, not the general condition; C, D do not appear in the proof. *Diagnosed error:* failing to distinguish the condition of decrease from a particular case of it. *Source:* Theorem 14.4, Steps 2–3. *Prerequisite:* the inequality $\text{round}(x) \leq x + 1/2$. *Remedial work:* redo Worked Exercise 14.2 following the iteration step by step.
- 6) **B.** *Rationale:* Remark 14.13 and the numerical check show e_∞ different for $K = 0.1; 0.3; 0.5$ with the same Δ : it depends on K through $\Delta/(2K)$. *Distractors:* A, C are errors different from the one actually committed; D denies the verified fact. *Diagnosed error:* not having recomputed numerically before stating a general law. *Source:* Remark 14.14; script `fig14_banda_morta.py`. *Prerequisite:* reading a bound as a function of several parameters. *Remedial work:* run the script with other values of K and observe the dependence.
- 7) **B.** *Rationale:* Proposition 14.2(i) and Remark 14.9: the second integrator of the PI filter cancels the steady-state error for a constant-velocity input, unlike the first-order PLL (type I), which has an error $\theta^* = \arcsin(\Delta\omega/K) \neq 0$. *Distractors:* A explicitly denies the thesis; C, D are false (the system is stable and it does track). *Diagnosed error:* failing to connect the order of the loop (number of integrators) to the type of input tracked without error. *Source:* Proposition 14.2; Remark 14.9. *Prerequisite:* classification of servosystems by type. *Remedial work:* redo Worked Exercise 14.3 comparing the two models.
- 8) **B.** *Rationale:* Leonov, Kuznetsov, Yuldashev, Yuldashev, *IEEE Transactions on Circuits and Systems I* 62 (2015), cited in Remark 14.2. *Distractors:* A is the source of the *informal* definitions criticised, not of the rigorous ones; C treats the dead band, not the locking ranges; D treats phase unwrapping. *Diagnosed error:* confusing the source of a problem

with the source of its solution. *Source:* Remark 14.2; source register of the chapter, entry S17. *Prerequisite:* none beyond a careful reading of the remarks. *Remedial work:* reread Remark 14.2.

BEST PRACTICE

Assigned exercises — answers.

14.1 (hold-in and locking time). (a) $\Delta\omega = 0.5 < K = 0.8$: yes, $\Delta\omega \in H = P = L$ (Theorem 14.1); $\theta^* = \arcsin(0.5/0.8) = \arcsin(0.625) = 0.6751$ rad. (b) $\tau = 1/\sqrt{0.8^2 - 0.5^2} = 1/\sqrt{0.39} = 1/0.624500 = 1.6013$ s (in units coherent with $K, \Delta\omega$). (c) From $e^{-t/\tau} = 0.01$: $t = \tau \ln 100 = 1.6013 \times 4.60517 = 7.374$, that is, about 4.6τ .

14.2 (Itoh's constraint violated halfway through the signal). (a) $f(t^*) = f_s/2 = 3$ Hz: $2 + 0.5t^* = 3 \Rightarrow t^* = 2$ s. (b) For $t < t^*$, $f(t) < f_s/2 = 3$ Hz, hence by Corollary 14.1 (applied locally, with $\omega_{\max}$ replaced by the instantaneous value on each sampling interval) the true increment remains $< \pi$ and the unwrapping is correct; beyond t^* the instantaneous frequency exceeds the threshold and the guarantee fails (it may still work by chance, but it is no longer guaranteed). (c) With $f_s = 6$ Hz, $\Delta t = 1/6$ s; the samples around $t^* = 2$ are $t = 11/6 \approx 1.8\bar{3}$ and $t = 2$: $\Delta\Theta = \int_{11/6}^2 2\pi f(t) dt = 2\pi \int_{11/6}^2 (2 + 0.5t) dt = 2\pi [2t + 0.25t^2]_{11/6}^2 = 2\pi [(4 + 1) - (3.6\bar{6} + 0.84028)] = 2\pi \times 0.49306 = 3.0983$ rad $< \pi = 3.1416$: just below, not yet violated at this sample (the first exceedance occurs at the following sample); an example showing how the theoretical threshold t^* is necessary but the actual exceedance of the sampled increment occurs with a small delay due to the discretisation, consistently with what is illustrated in Figure 14.3 (where the first failure falls one sample after the theoretical threshold).

14.3 (17/19 tolerance). (a) $\pi/(17 + 19) = \pi/36 = 0.087266$ rad $= 5^\circ$ exactly ($180^\circ/36 = 5^\circ$). (b) $17 \times 19 = 323$ distinct positions per turn (Chinese remainder theorem, forward reference to Volume II Chapter 7). (c) Script analogous to `fig14_encoder_montecarlo.py`: for (17, 19), at $\eta = 0.5 \times$ the threshold the measured failure rate is 0 over 20 000 trials (consistent with the theorem); at $\eta = 1.5 \times$ the threshold the rate is positive (the exact value depends on the random seed, but it is systematically non-zero and grows with η), in full agreement with the left-hand panel of Figure 14.5 (where the same transition is shown for (7, 9) and (17, 19) together).

14.20. Chapter formulary

First-order PLL (Adler, recalled from Chapter 13)

$$\dot{\theta} = \Delta\omega - K \sin \theta, \quad \theta^* = \arcsin \frac{\Delta\omega}{K} \quad (|\Delta\omega| < K), \quad T_{\text{slip}} = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \quad (|\Delta\omega| > K)$$

Locking ranges

$$H = \{\nu : \exists \text{ a.s. eq.}\}, \quad P = \{\nu : \text{global conv.}\}, \quad L = \{\nu : \text{conv. without slip}\}$$

$$L \subseteq P \subseteq H, \quad \text{first-order PLL: } H = P = L = (-K, K)$$

Locking time

$$\tau = \frac{1}{\sqrt{K^2 - \Delta\omega^2}}, \quad \theta(t) - \theta^* \sim C e^{-t/\tau}$$

Discrete PLL (forward reference to Chapter 4)

$$f_{\alpha, \epsilon}(\theta) = \theta + \alpha - \frac{\epsilon}{2\pi} \sin(2\pi\theta), \quad |\alpha| \leq \frac{\epsilon}{2\pi} \quad (\text{principal tongue, } 0 < \epsilon < 1)$$

Itoh's theorem

$$\psi_j = \psi_{j-1} + W(\varphi_j - \varphi_{j-1}) \quad \forall j \iff \Delta\Theta_j \in (-\pi, \pi] \quad \forall j, \quad f_s > \frac{\omega_{\max}}{\pi} \Rightarrow \text{correct}$$

Vernier encoder: tolerance

$$J = mK_n - nK_m = \frac{n\varphi_m - m\varphi_n}{2\pi} \in \mathbb{Z}, \quad \eta < \frac{\pi}{m+n} \Rightarrow J^* = \text{round}\left(\frac{n\hat{\varphi}_m - m\hat{\varphi}_n}{2\pi}\right) = J$$

Resolver-to-digital (type II)

$$\dot{\hat{\theta}} = K_1 e + K_2 w, \quad \dot{w} = e, \quad e = \theta_{\text{rif}} - \hat{\theta} : \quad e_\infty(\omega_0 t) = 0, \quad e_\infty\left(\frac{1}{2}\alpha t^2\right) = \frac{\alpha}{K_2}$$

Dead band from quantisation

$$e_{k+1} = Q_\Delta((1-K)e_k), \quad 0 < K < 1 : \quad |e_k| \text{ non-increasing}, \quad |e_\infty| \leq \frac{\Delta}{2K}$$

Stroboscope

$$f_{\text{app}} = f_s \cdot \rho(f/f_s), \quad \rho(x) \in \left(-\frac{1}{2}, \frac{1}{2}\right] \text{ representative of } x \pmod{1}$$

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Part F

Global Structures

Groups of circle homeomorphisms

Up to now the circle has been the space on which *one* transformation acted: a rotation (Chapter 3), a homeomorphism with its rotation number (Chapter 4), a diffeomorphism with its rigidity (Chapters 5 and 6). This chapter changes the subject: not one map, but a **group** of maps. The object of study is $\text{Homeo}_+(\mathbb{T})$, the group of all orientation-preserving circle homeomorphisms, and more precisely the *actions* $\varrho : \Gamma \rightarrow \text{Homeo}_+(\mathbb{T})$ of an abstract group Γ on the circle. There are three guiding threads. The first is structural: $\text{Homeo}_+(\mathbb{T})$ possesses a universal cover $\widetilde{\text{Homeo}}$ which is a central extension by $\mathbb{Z}$, and the rotation number, which on $\widetilde{\text{Homeo}}$ becomes a real-valued function, is *not* a homomorphism but comes close to being one in a controlled way: it is a **quasi-morphism**. The second thread is **Hölder's** theorem: if the action is *free*, that is, if no element other than the identity has fixed points, then all the possible richness collapses and the group is necessarily abelian, indeed isomorphic to a subgroup of $(\mathbb{R}, +)$, and the action is semiconjugate to an action by rotations. The third thread is quantitative: to every action of a surface group $\Gamma_g = \pi_1(\Sigma_g)$ one associates an integer, the **Euler class** $e(\varrho)$, and the **Milnor–Wood inequality** states that this integer cannot exceed in absolute value $|\chi(\Sigma_g)| = 2g - 2$: a hard topological constraint on what a surface group can do on the circle. The chapter proves this inequality in full, in its optimal form, with an argument of a few lines once the right tools have been built.

INFO

Level and prerequisites. Level **D** core, with the introduction (Section 15.1) of level **C** and Section 15.8 (Milnor–Wood) of level **E**. A level **C** reader may read Sections 15.1–15.3 and the statements of the others without reproducing their proofs, without losing the continuity. Prerequisites within the treatise: the lift of a circle homeomorphism and the rotation number with its basic properties (Volume IV, Chapter 4, Definitions 4.1 and 4.2, Theorem 4.1, Propositions 4.1 and 4.2); the Poincaré classification in the rational case (Volume IV, Chapter 4, Theorem 4.3); Fekete's lemma (Volume IV, Chapter 4, Lemma 4.1); the exponential covering $\mathbb{R} \rightarrow \mathbb{T}$ and the fundamental group of the circle (Volume III, Chapter 12); elementary notions of group theory (normal subgroup, quotient, extension, commutator). Remedial work: a reader who does not remember the definition of ρ and its invariance under conjugacy may reread the first two sections of Chapter 4 before Section 15.2, which uses them in an essential way.

15.1. The group $\text{Homeo}_+(\mathbb{T})$ and its central extension

We work, as in Chapters 3–6, on $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, with projection $\pi_{\mathbb{T}} : \mathbb{R} \rightarrow \mathbb{T}$.

Definition 15.1: The group $\text{Homeo}_+(\mathbb{T})$. $\text{Homeo}_+(\mathbb{T})$ is the set of homeomorphisms $f : \mathbb{T} \rightarrow \mathbb{T}$ which preserve the orientation (that is, which admit a strictly increasing lift, Definition 4.1), equipped with composition. It is a group: the composition of two increasing homeomorphisms is increasing, the inverse of an increasing homeomorphism is increasing, and the identity belongs to the set. Notable subgroups: the rotations $\text{SO}(2) = \{R_\alpha\}_{\alpha \in \mathbb{T}}$;

for $r \in \{1, 2, \dots, \infty, \omega\}$ the diffeomorphisms $\text{Diff}_+^r(\mathbb{T})$ of class C^r with C^r inverse; the group $\text{Diff}_+^{1+\text{bv}}(\mathbb{T})$ of Chapter 5.

Definition 15.2: The universal cover $\widetilde{\text{Homeo}}$. $\widetilde{\text{Homeo}}$ is the set of maps $F : \mathbb{R} \rightarrow \mathbb{R}$ which are continuous, strictly increasing and such that

$$F(\theta + 2\pi) = F(\theta) + 2\pi \quad \text{for every } \theta \in \mathbb{R},$$

equipped with composition. We set $T(\theta) := \theta + 2\pi$ and, for $c \in \mathbb{R}$, $R_c^\sharp(\theta) := \theta + 2\pi c$ (the natural lift of the rotation by angle $2\pi c$); thus $T = R_1^\sharp$.

Observe that every $F \in \widetilde{\text{Homeo}}$ is automatically surjective: being increasing and satisfying $F(\theta + 2\pi k) = F(\theta) + 2\pi k$ for every $k \in \mathbb{Z}$ (by iterating the relation), one has $F(\theta) \rightarrow \pm\infty$ as $\theta \rightarrow \pm\infty$, and the intermediate value theorem concludes. Hence F is a homeomorphism of $\mathbb{R}$, and F^{-1} satisfies the same relation: $F^{-1}(\theta + 2\pi) = F^{-1}(\theta) + 2\pi$ (apply F^{-1} to $F(F^{-1}(\theta) + 2\pi) = \theta + 2\pi$). Therefore $\widetilde{\text{Homeo}}$ really is a group.

Proposition 15.1: The central extension $1 \rightarrow \mathbb{Z} \rightarrow \widetilde{\text{Homeo}} \rightarrow \text{Homeo}_+(\mathbb{T}) \rightarrow 1$. Let $p : \widetilde{\text{Homeo}} \rightarrow \text{Homeo}_+(\mathbb{T})$ be the map which to F associates the unique f with $\pi_{\mathbb{T}} \circ F = f \circ \pi_{\mathbb{T}}$. Then: (i) p is a surjective group homomorphism; (ii) $\ker p = \{T^k : k \in \mathbb{Z}\} \cong \mathbb{Z}$; (iii) T is central in $\widetilde{\text{Homeo}}$, that is, $FT = TF$ for every F . Hence $1 \rightarrow \mathbb{Z} \rightarrow \widetilde{\text{Homeo}} \xrightarrow{p} \text{Homeo}_+(\mathbb{T}) \rightarrow 1$ is a central extension.

Proof. (i) That F induces a map f of $\mathbb{T}$ is the definition of a lift: $F(\theta + 2\pi) = F(\theta) + 2\pi$ says exactly that $\pi_{\mathbb{T}} \circ F$ is constant on the fibres of $\pi_{\mathbb{T}}$, and f is well defined, continuous, bijective (because F is increasing with “step” 2π) and orientation-preserving. It is a homomorphism because $\pi_{\mathbb{T}} \circ (F \circ G) = (f \circ g) \circ \pi_{\mathbb{T}}$. It is surjective because every $f \in \text{Homeo}_+(\mathbb{T})$ admits an increasing lift (Volume III, Chapter 12, where the lifting of continuous maps of $\mathbb{T}$ is proved in full; monotonicity follows from the preservation of the orientation, Definition 4.1).

(ii) $p(F) = \text{id}$ means $F(\theta) \equiv \theta \pmod{2\pi}$ for every θ , that is, the continuous function $\theta \mapsto (F(\theta) - \theta)/(2\pi)$ takes values in $\mathbb{Z}$: since $\mathbb{R}$ is connected, it is constant, say $= k$. Then $F = T^k$. Conversely every T^k lies in $\widetilde{\text{Homeo}}$ and $p(T^k) = \text{id}$. The map $k \mapsto T^k$ is an isomorphism of $\mathbb{Z}$ onto the kernel.

(iii) For every $F \in \widetilde{\text{Homeo}}$ and every θ : $(F \circ T)(\theta) = F(\theta + 2\pi) = F(\theta) + 2\pi = (T \circ F)(\theta)$. The defining relation of $\widetilde{\text{Homeo}}$ is commutation with T . $\square$

Lemma 15.1: $\widetilde{\text{Homeo}}$ is torsion-free. If $F \in \widetilde{\text{Homeo}}$ and $F^n = \text{id}$ for some $n \geq 1$, then $F = \text{id}$.

Proof. Suppose $F(\theta_0) > \theta_0$ for some θ_0 . Since F is increasing, $F^2(\theta_0) = F(F(\theta_0)) > F(\theta_0) > \theta_0$, and by induction $F^n(\theta_0) > \theta_0$ for every $n \geq 1$: hence $F^n \neq \text{id}$. Symmetrically if $F(\theta_0) < \theta_0$. There remains only the case $F(\theta) = \theta$ for every θ , that is, $F = \text{id}$. $\square$

Corollary 15.1: The extension does not split. There exists no homomorphism $s : \text{Homeo}_+(\mathbb{T}) \rightarrow \widetilde{\text{Homeo}}$ with $p \circ s = \text{id}$. In particular $\widetilde{\text{Homeo}}$ is not isomorphic to the direct product $\mathbb{Z} \times \text{Homeo}_+(\mathbb{T})$.

Proof. If s existed, it would be injective (from $p \circ s = \text{id}$). The rotation $R_{2\pi/q}$ has order q in $\widetilde{\text{Homeo}}_+(\mathbb{T})$ for every $q \geq 2$, hence $s(R_{2\pi/q})$ would be an element of order exactly q of $\widetilde{\text{Homeo}}$ (the order cannot drop, by injectivity), against Lemma 15.1. The second assertion follows because in $\mathbb{Z} \times \text{Homeo}_+(\mathbb{T})$ the projection onto the second factor admits the obvious section. $\square$

Remark 15.1: The chain of regularity subgroups. One has the chain of subgroups

$$\text{SO}(2) \subset \text{Diff}_+^\omega(\mathbb{T}) \subset \text{Diff}_+^\infty(\mathbb{T}) \subset \cdots \subset \text{Diff}_+^2(\mathbb{T}),$$

$$\text{Diff}_+^2(\mathbb{T}) \subset \text{Diff}_+^{1+\text{bv}}(\mathbb{T}) \subset \text{Diff}_+^1(\mathbb{T}) \subset \text{Homeo}_+(\mathbb{T}),$$

and each inclusion is strict: we already know this, in a non-trivial way, for the inclusion $\text{Diff}_+^{1+\text{bv}} \subset \text{Diff}_+^1$, since Denjoy's counterexample (Chapter 5, Construction 5.2) is of class C^1 but not $C^{1+\text{bv}}$. The central point of the whole theory of groups of circle diffeomorphisms is that the place in this chain changes which groups can act faithfully: see Section 15.10.

Remark 15.2: What we know about $\text{Homeo}_+(\mathbb{T})$ as an abstract group (imported). Imported from É. Ghys, “Groups acting on the circle”, *L'Enseignement Math.* (2) 47 (2001), Section 4 (source reopened and verified on 17 September 2026).

We assume, without proof in this treatise: (a) up to conjugacy, $\text{SO}(2)$ is the unique maximal compact subgroup of $\text{Homeo}_+(\mathbb{T})$; (b) the inclusion $\text{SO}(2) \hookrightarrow \text{Homeo}_+(\mathbb{T})$ is a homotopy equivalence (in particular $\text{Homeo}_+(\mathbb{T})$ is connected and $\pi_1(\text{Homeo}_+(\mathbb{T})) \cong \mathbb{Z}$, which explains the name “universal cover” given to $\widetilde{\text{Homeo}}$); (c) $\text{Homeo}_+(\mathbb{T})$ is a simple group; (d) a generic pair (in the sense of Baire) of elements of $\text{Homeo}_+(\mathbb{T})$ generates a free group. Of these four facts, in what follows we shall use only the language suggested by (b) — the name of $\widetilde{\text{Homeo}}$ — and never the assertions (a), (c), (d), which remain pure context.

15.2. The tools: minimal and maximal displacement

The whole chapter rests on two real numbers associated with an element of $\widetilde{\text{Homeo}}$ and on an elementary inequality which links them. Recall (Definition 4.2) that the rotation number of $F \in \widetilde{\text{Homeo}}$ is

$$\rho(F) = \lim_{n \rightarrow \infty} \frac{F^n(\theta) - \theta}{2\pi n},$$

a limit which exists and does not depend on θ (Theorem 4.1). Here ρ is viewed as a *real-valued* function on $\widetilde{\text{Homeo}}$, not as a function with values in $\mathbb{R}/\mathbb{Z}$ on $\text{Homeo}_+(\mathbb{T})$: this shift of point of view is what makes the whole chapter possible.

Definition 15.3: Normalised displacement, $a(F)$ and $b(F)$. For $F \in \widetilde{\text{Homeo}}$ we set

$$\varphi_F(\theta) := \frac{F(\theta) - \theta}{2\pi}, \quad a(F) := \min_{\theta \in \mathbb{R}} \varphi_F(\theta), \quad b(F) := \max_{\theta \in \mathbb{R}} \varphi_F(\theta).$$

The function φ_F is continuous and 2π -periodic (because $F(\theta + 2\pi) - (\theta + 2\pi) = F(\theta) - \theta$), hence the minimum and the maximum are attained and finite. One has $a(T^k) = b(T^k) = k$ and, more generally, $a(R_c^\sharp) = b(R_c^\sharp) = c$.

Lemma 15.2: Oscillation lemma. For every $F \in \widetilde{\text{Homeo}}$ one has $b(F) - a(F) < 1$. Equivalently: $|\varphi_F(\theta) - \varphi_F(\theta')| < 1$ for every pair of points $\theta, \theta' \in \mathbb{R}$.

Proof. Let θ_1 be a maximum point and θ_0 a minimum point of φ_F . Since φ_F is 2π -periodic, we may choose the representatives so that $\theta_0 \leq \theta_1 < \theta_0 + 2\pi$ (one replaces θ_1 by $\theta_1 + 2\pi k$ for a suitable $k \in \mathbb{Z}$, which does not change $\varphi_F(\theta_1)$). Since F is strictly increasing,

$$F(\theta_0) \leq F(\theta_1) < F(\theta_0 + 2\pi) = F(\theta_0) + 2\pi,$$

hence $0 \leq F(\theta_1) - F(\theta_0) < 2\pi$; on the other hand $0 \leq \theta_1 - \theta_0 < 2\pi$, in particular $\theta_1 - \theta_0 \geq 0$. Then

$$2\pi(b(F) - a(F)) = [F(\theta_1) - \theta_1] - [F(\theta_0) - \theta_0] = \underbrace{[F(\theta_1) - F(\theta_0)]}_{< 2\pi} - \underbrace{[\theta_1 - \theta_0]}_{\geq 0} < 2\pi.$$

The second formulation follows because $|\varphi_F(\theta) - \varphi_F(\theta')| \leq b(F) - a(F)$. $\square$

Remark 15.3: Why the oscillation lemma is true only thanks to monotonicity.

If one gives up the monotonicity of F the statement fails completely: the function $F(\theta) = \theta + 10\pi \sin^2(\theta/2)$ satisfies $F(\theta + 2\pi) = F(\theta) + 2\pi$ and has φ_F varying between 0 and 5, hence oscillation 5; but $F'(\theta) = 1 + 5\pi \sin \theta$ changes sign, and F is not a homeomorphism. The key to the proof is the chain $F(\theta_0) \leq F(\theta_1) < F(\theta_0) + 2\pi$, which uses monotonicity twice: it is worth remembering, because further on (Section 15.8) the whole Milnor–Wood inequality will depend on this single number 1.

Lemma 15.3: Monotonicity, translation, conjugacy, sandwiching. For $F, G, H \in \widetilde{\text{Homeo}}$ and $k \in \mathbb{Z}$: **(i)** (monotonicity) if $F(\theta) \leq G(\theta)$ for every θ — we shall write $F \preceq G$ — then $\rho(F) \leq \rho(G)$; **(ii)** (sandwiching) $a(F) \leq \rho(F) \leq b(F)$; **(iii)** (translation) $\rho(T^k F) = k + \rho(F)$; **(iv)** (conjugacy) $\rho(HFH^{-1}) = \rho(F)$; **(v)** (homogeneity) $\rho(F^n) = n\rho(F)$ for every $n \in \mathbb{Z}$.

Proof. **(i)** We show $F^n(\theta) \leq G^n(\theta)$ by induction on n : true for $n = 1$; and $F^{n+1}(\theta) = F(F^n(\theta)) \leq G(F^n(\theta)) \leq G(G^n(\theta)) = G^{n+1}(\theta)$, where the first inequality is the hypothesis at the point $F^n(\theta)$ and the second is the monotonicity of G applied to $F^n(\theta) \leq G^n(\theta)$. Dividing by $2\pi n$ and passing to the limit one obtains $\rho(F) \leq \rho(G)$.

(ii) By definition of a and b one has $R_{a(F)}^\# \preceq F \preceq R_{b(F)}^\#$; since $\rho(R_c^\#) = c$ (the limit is exact at every step: $(R_c^\#)^n(\theta) = \theta + 2\pi nc$), part (i) gives the claim.

(iii) T is central (Proposition 15.1(iii)), hence $(T^k F)^n = T^{kn} F^n$, that is, $(T^k F)^n(\theta) - \theta = 2\pi kn + F^n(\theta) - \theta$; one divides by $2\pi n$ and passes to the limit.

(iv) The function $\theta \mapsto H(\theta) - \theta$ is continuous and 2π -periodic, hence bounded: there exists $C > 0$ with $|H(\theta) - \theta| \leq C$ and $|H^{-1}(\theta) - \theta| \leq C'$ for every θ (the same argument for $H^{-1} \in \widetilde{\text{Homeo}}$). From $(HFH^{-1})^n = HF^nH^{-1}$ it follows that

$$|(HFH^{-1})^n(\theta) - F^n(\theta)| \leq |H(F^n(H^{-1}\theta)) - F^n(H^{-1}\theta)| + |F^n(H^{-1}\theta) - F^n(\theta)|.$$

The first summand is $\leq C$. For the second, Lemma 15.2 applied to F^n gives $|\varphi_{F^n}(\theta') - \varphi_{F^n}(\theta)| < 1$, that is, $|F^n(\theta') - \theta' - F^n(\theta) + \theta| < 2\pi$, whence, with $\theta' = H^{-1}(\theta)$ and $|\theta' - \theta| \leq C'$, one obtains $|F^n(H^{-1}\theta) - F^n(\theta)| < 2\pi + C'$. Hence the difference between $(HFH^{-1})^n(\theta)$ and $F^n(\theta)$ is bounded uniformly in n ; dividing by $2\pi n$ the two rotation numbers coincide.

(v) This is Proposition 4.1 of Chapter 4 (for $n \geq 1$; for $n \leq -1$ one applies the same statement to F^{-1} , noting $\rho(F^{-1}) = -\rho(F)$, which follows from the limit along $F^{-n}(\theta)$). $\square$

Lemma 15.4: The fixed point criterion. Let $F \in \widetilde{\text{Homeo}}$. **(i)** If there exists θ_0 with $F(\theta_0) \leq \theta_0$, then $\rho(F) \leq 0$; if there exists θ_0 with $F(\theta_0) \geq \theta_0$, then $\rho(F) \geq 0$. **(ii)** $\rho(F) = 0$

if and only if F has a fixed point. **(iii)** If $\rho(F) > 0$ then $F(\theta) > \theta$ for every θ ; if $\rho(F) < 0$ then $F(\theta) < \theta$ for every θ . **(iv)** For $p \in \mathbb{Z}$ and $q \geq 1$: $\rho(F) \geq p/q$ if and only if there exists θ with $F^q(\theta) \geq \theta + 2\pi p$; and $\rho(F) \leq p/q$ if and only if there exists θ with $F^q(\theta) \leq \theta + 2\pi p$.

Proof. **(i)** If $F(\theta_0) \leq \theta_0$, by monotonicity $F^2(\theta_0) = F(F(\theta_0)) \leq F(\theta_0) \leq \theta_0$ and by induction $F^n(\theta_0) \leq \theta_0$ for every n ; hence $(F^n(\theta_0) - \theta_0)/(2\pi n) \leq 0$ and $\rho(F) \leq 0$. The opposite case is symmetric.

(ii) If $F(\theta_0) = \theta_0$ then by (i) $\rho(F) \leq 0$ and $\rho(F) \geq 0$, hence $\rho(F) = 0$. Conversely, if F has no fixed points, the continuous function $\theta \mapsto F(\theta) - \theta$ never vanishes, hence (by the intermediate value theorem on the connected set $\mathbb{R}$) it has constant sign; being 2π -periodic and continuous, by compactness there exists $\varepsilon > 0$ with $|F(\theta) - \theta| \geq \varepsilon$ for every θ . In the case $F(\theta) \geq \theta + \varepsilon$ one has $F \succeq R_{\varepsilon/2\pi}^\sharp$ and by Lemma 15.3(i) $\rho(F) \geq \varepsilon/2\pi > 0$; in the opposite case $\rho(F) \leq -\varepsilon/2\pi < 0$. In both cases $\rho(F) \neq 0$.

(iii) If $\rho(F) > 0$, by (ii) F has no fixed points, and by the alternative just seen $F(\theta) - \theta$ has constant sign; the sign cannot be negative, since otherwise (i) would give $\rho(F) \leq 0$.

(iv) Set $H := T^{-p}F^q$, which belongs to $\widetilde{\text{Homeo}}$ and has $\rho(H) = q\rho(F) - p$ by Lemma 15.3(iii) and (v). Then $\rho(F) \geq p/q \iff \rho(H) \geq 0$. If $\rho(H) \geq 0$: by (iii) it cannot be that $H(\theta) < \theta$ for every θ (this would give $\rho(H) < 0$, as in the proof of (ii)), hence there exists θ with $H(\theta) \geq \theta$, that is, $F^q(\theta) \geq \theta + 2\pi p$. Conversely, if there exists θ with $H(\theta) \geq \theta$, part (i) gives $\rho(H) \geq 0$. The second equivalence is symmetric. $\square$

Lemma 15.5: Estimates for the composition. For $F, G \in \widetilde{\text{Homeo}}$:

$$\textbf{(i)} \quad a(F) + a(G) \leq a(FG), \quad \textbf{(ii)} \quad b(FG) \leq b(F) + b(G),$$

$$\textbf{(iii)} \quad a(FG) \leq a(F) + b(G).$$

(Here FG denotes $F \circ G$.)

Proof. The basic identity is, for every θ ,

$$\varphi_{FG}(\theta) = \frac{F(G(\theta)) - \theta}{2\pi} = \frac{F(G(\theta)) - G(\theta)}{2\pi} + \frac{G(\theta) - \theta}{2\pi} = \varphi_F(G(\theta)) + \varphi_G(\theta). \quad (*)$$

(i) Each summand of $(*)$ is $\geq$ its respective minimum, hence $\varphi_{FG}(\theta) \geq a(F) + a(G)$ for every θ ; one takes the minimum on the left. **(ii)** Symmetrically, $\varphi_{FG}(\theta) \leq b(F) + b(G)$ for every θ .

(iii) Let θ^* be a point at which φ_F attains its minimum and let $\theta_0 := G^{-1}(\theta^*)$ (legitimate: G is bijective on $\mathbb{R}$). Then, from $(*)$,

$$a(FG) \leq \varphi_{FG}(\theta_0) = \varphi_F(\theta^*) + \varphi_G(\theta_0) = a(F) + \varphi_G(\theta_0) \leq a(F) + b(G).$$

$\square$

15.3. The rotation number is a quasi-morphism

Definition 15.4: Homogeneous quasi-morphism. Let Γ be a group. A function $\psi : \Gamma \rightarrow \mathbb{R}$ is a **quasi-morphism** if

$$D(\psi) := \sup_{g, h \in \Gamma} |\psi(gh) - \psi(g) - \psi(h)| < \infty;$$

$D(\psi)$ is called the **defect** of ψ . The quasi-morphism is said to be **homogeneous** if $\psi(g^n) = n\psi(g)$ for every $g \in \Gamma$ and every $n \in \mathbb{Z}$. A quasi-morphism of defect 0 is ex-

actly a homomorphism $\Gamma \rightarrow (\mathbb{R}, +)$.

Theorem 15.1: ρ is a homogeneous quasi-morphism on $\widetilde{\text{Homeo}}$. The function $\rho : \widetilde{\text{Homeo}} \rightarrow \mathbb{R}$ is a homogeneous quasi-morphism, invariant under conjugacy, and for every $F, G \in \widetilde{\text{Homeo}}$ the explicit estimate

$$a(F) + a(G) \leq \rho(FG) \leq b(F) + b(G)$$

holds, whence

$$|\rho(FG) - \rho(F) - \rho(G)| < 2.$$

Proof. Homogeneity and invariance under conjugacy are Lemma 15.3(v) and (iv). For the central estimate, Lemma 15.3(ii) applied to FG gives $a(FG) \leq \rho(FG) \leq b(FG)$, and Lemma 15.5(i)–(ii) gives $a(F) + a(G) \leq a(FG)$ and $b(FG) \leq b(F) + b(G)$: putting these together, $a(F) + a(G) \leq \rho(FG) \leq b(F) + b(G)$.

For the estimate of the defect we use the oscillation Lemma 15.2 in the form $b(F) < a(F) + 1 \leq \rho(F) + 1$ and $a(F) > b(F) - 1 \geq \rho(F) - 1$ (which hold because $a(F) \leq \rho(F) \leq b(F)$), and similarly for G . Then

$$\rho(FG) - \rho(F) - \rho(G) \leq b(F) + b(G) - \rho(F) - \rho(G) < (\rho(F) + 1) + (\rho(G) + 1) - \rho(F) - \rho(G) = 2,$$

$$\rho(FG) - \rho(F) - \rho(G) \geq a(F) + a(G) - \rho(F) - \rho(G) > (\rho(F) - 1) + (\rho(G) - 1) - \rho(F) - \rho(G) = -2.$$

Hence $D(\rho) \leq 2$ and in particular ρ is a quasi-morphism. $\square$

Remark 15.4: The optimal value of the defect: what is proved and what is not.

In the literature the defect of the rotation number on $\widetilde{\text{Homeo}}$ is given as $D(\rho) = 1$; the inequality $|\rho(FG) - \rho(F) - \rho(G)| \leq 1$ is cited, for instance, in Section 5 of Ghys (2001).

In this chapter only the version with the constant 2 is proved (Theorem 15.1): the proof of the optimal constant 1 cannot be reconstructed with the tools of Sections 15.2–15.3 alone (the two estimates $a(F) + a(G) \leq \rho(FG) \leq b(F) + b(G)$ give exactly 2 and no less), and it was not possible to reopen a source containing a complete proof of it: we therefore record it as an imported and unused statement. We stress this because it is easy to build an error on top of it: **no result of the chapter depends on the value 1**. In particular the optimal Milnor–Wood inequality (Theorem 15.5) is not obtained from the defect of the quasi-morphism — which would give only $|e| \leq 4g - 2$ with the constant 2, and $|e| \leq 2g - 1$ with the constant 1 — but from a different argument, Lemma 15.9, which does not use the defect at all.

The bound $|\rho(FG) - \rho(F) - \rho(G)| < 2$ is, for a group, the assertion that ρ “almost” distinguishes elements in an additive way. The case in which the word “almost” disappears is the abelian one.

Proposition 15.2: ρ is a homomorphism on abelian subgroups. If $F, G \in \widetilde{\text{Homeo}}$ commute, then $\rho(FG) = \rho(F) + \rho(G)$. In particular, the restriction of ρ to every abelian subgroup $A \leq \widetilde{\text{Homeo}}$ is a homomorphism $A \rightarrow (\mathbb{R}, +)$.

Proof. From $FG = GF$ it follows that $(FG)^n = F^n G^n$ for every $n \geq 1$ (immediate induction). We evaluate at $\theta = 0$ and set $y_n := G^n(0)$. Then

$$(FG)^n(0) = F^n(y_n) = y_n + 2\pi \varphi_{F^n}(y_n) = y_n + 2\pi \varphi_{F^n}(0) + 2\pi [\varphi_{F^n}(y_n) - \varphi_{F^n}(0)],$$

$$\text{that is } (FG)^n(0) = G^n(0) + F^n(0) + 2\pi \eta_n,$$

where $\eta_n := \varphi_{F^n}(y_n) - \varphi_{F^n}(0)$ satisfies $|\eta_n| < 1$ by the oscillation Lemma 15.2 applied to F^n (which lies in Homeo). Dividing by $2\pi n$:

$$\frac{(FG)^n(0)}{2\pi n} = \frac{G^n(0)}{2\pi n} + \frac{F^n(0)}{2\pi n} + \frac{\eta_n}{n} \xrightarrow{n \rightarrow \infty} \rho(G) + \rho(F) + 0.$$

The left-hand side tends to $\rho(FG)$, and the claim follows. The assertion about abelian subgroups is immediate (also using $\rho(F^{-1}) = -\rho(F)$, Lemma 15.3(v)). $\square$

Example 15.1: Two homeomorphisms with $\rho(fg) \neq \rho(f) + \rho(g)$: symbolic check.

For $0 < \varepsilon < 1$ we set, on $\mathbb{R}$,

$$F_\varepsilon(\theta) := \theta + \varepsilon(1 - \cos \theta), \quad G_\varepsilon(\theta) := \theta + \varepsilon(1 + \cos \theta).$$

(1) They are elements of $\widetilde{\text{Homeo}}$. Both satisfy $F(\theta + 2\pi) = F(\theta) + 2\pi$ because $\cos$ is 2π -periodic; and $F'_\varepsilon(\theta) = 1 + \varepsilon \sin \theta \geq 1 - \varepsilon > 0$, $G'_\varepsilon(\theta) = 1 - \varepsilon \sin \theta \geq 1 - \varepsilon > 0$: they are strictly increasing.

(2) $\rho(F_\varepsilon) = \rho(G_\varepsilon) = 0$ exactly. $F_\varepsilon(0) = 0 + \varepsilon(1 - 1) = 0$ and $G_\varepsilon(\pi) = \pi + \varepsilon(1 + \cos \pi) = \pi$: both have a fixed point, hence Lemma 15.4(ii) gives $\rho = 0$. No numerical computation is involved.

(3) $\rho(F_\varepsilon G_\varepsilon) > 0$. For every θ ,

$$(F_\varepsilon \circ G_\varepsilon)(\theta) - \theta = \underbrace{\varepsilon(1 + \cos \theta)}_{\geq 0} + \underbrace{\varepsilon(1 - \cos(G_\varepsilon(\theta)))}_{\geq 0}.$$

The two summands vanish simultaneously only if $\cos \theta = -1$ (that is, $\theta \equiv \pi$) and $\cos(G_\varepsilon(\theta)) = 1$ (that is, $G_\varepsilon(\theta) \equiv 0$); but $G_\varepsilon(\pi) = \pi \not\equiv 0 \pmod{2\pi}$. Hence the continuous, 2π -periodic function $\theta \mapsto (F_\varepsilon G_\varepsilon)(\theta) - \theta$ is strictly positive, and by compactness it is $\geq \delta > 0$: Lemma 15.3(i)–(ii) gives $\rho(F_\varepsilon G_\varepsilon) \geq \delta/2\pi > 0 = \rho(F_\varepsilon) + \rho(G_\varepsilon)$. **The rotation number is not a homomorphism.**

(4) Double numerical check (script `verifica_cap15_vol4.py`, block [C15-1], run on 17 September 2026). The minimum of $(F_\varepsilon G_\varepsilon)(\theta) - \theta$ on a grid of 10^6 points equals 0.42736777 for $\varepsilon = 0.25$ and 0.74030491 for $\varepsilon = 0.5$: positive, as proved in point (3) $\checkmark$. The rotation number, estimated by iterating the lift over 2×10^5 steps, equals

$$\rho(F_{0.25}G_{0.25}) = 0.080053, \quad \rho(F_{0.50}G_{0.50}) = 0.163239,$$

$$\rho(F_{0.75}G_{0.75}) = 0.255189, \quad \rho(F_{0.90}G_{0.90}) = 0.333333,$$

and the same values (up to 4×10^{-7}) are obtained starting from $\theta_0 = 1.234$ instead of $\theta_0 = 0$, as must be the case by Theorem 4.1 $\checkmark$. The second independent check is the sandwiching of Lemma 15.3(ii): for $\varepsilon = 0.5$, $a = 0.117823 \leq 0.163239 \leq b = 0.220983$ $\checkmark$, with $b - a = 0.103160 < 1$ as the oscillation Lemma 15.2 requires $\checkmark$.

(5) The case $\varepsilon = 0.9$ is exactly $1/3$. The numerical value 0.333333 is not an approximation of an irrational: the function $\theta \mapsto (F_{0.9}G_{0.9})^3(\theta) - \theta - 2\pi$ has minimum -1.184409 and maximum $+0.231035$ on a grid of 2×10^6 points, hence it changes sign and has a zero; by the intermediate value theorem there exists θ with $(F_{0.9}G_{0.9})^3(\theta) = \theta + 2\pi$, and Lemma 15.4(iv) (applied twice, with $p/q = 1/3$ in both directions) gives $\rho = 1/3$ **exactly** $\checkmark$. It is a *mode-locking* phenomenon, the same as that of the Arnold tongues (Chapter 4, Theorem 4.7), here reread in group-theoretic terms: the failure of additivity of ρ is, on a whole interval of parameters, rigidly locked onto a rational.

Figure 15.2 shows the complete parameter curve, that is, the function $\varepsilon \mapsto \rho(F_\varepsilon G_\varepsilon)$, while

Figure 15.1 illustrates the structure of the central extension on which the whole preceding section rests.

15.4. Hölder's theorem

Definition 15.5: Free action. A subgroup $G \leq \text{Homeo}_+(\mathbb{T})$ (respectively $G \leq \text{Homeo}_+(\mathbb{R})$, the group of increasing homeomorphisms of $\mathbb{R}$) **acts freely** if every $g \in G$ other than the identity has no fixed points: $g(x) \neq x$ for every x . Equivalently, the stabiliser of every point is trivial.

Hölder's theorem says that this condition, apparently only dynamical, has a drastic purely algebraic consequence: the group must be abelian, indeed order-embeddable in $(\mathbb{R}, +)$. The proof goes through the order structure which the free action induces on the group.

The order induced by a free action

Proposition 15.3: The bi-invariant order of a free action. Let $G \leq \text{Homeo}_+(\mathbb{R})$ be a group acting freely on $\mathbb{R}$, and set

$$g \prec h \quad :\Longleftrightarrow \quad g(x) < h(x) \text{ for every } x \in \mathbb{R}.$$

Then $\preceq$ (that is, $\prec$ or $=$) is a **total** order on G , **bi-invariant** ($g \prec h \Rightarrow kg \prec kh$ and $gk \prec hk$ for every $k \in G$), and **determined by a single point**: once $x_0 \in \mathbb{R}$ is fixed, one has $g \prec h \Longleftrightarrow g(x_0) < h(x_0)$.

Proof. Totality. Let $g \neq h$ and $u := g^{-1}h \neq \text{id}$. By the freeness hypothesis u has no fixed points, hence the continuous function $x \mapsto u(x) - x$ has constant sign on $\mathbb{R}$ (intermediate value theorem). If $u(x) > x$ for every x , then, g being increasing, $h(x) = g(u(x)) > g(x)$ for every x , that is, $g \prec h$; if $u(x) < x$ for every x , then $h \prec g$. The two cases exclude each other.

Bi-invariance. If $g(x) < h(x)$ for every x , applying k (increasing) one has $k(g(x)) < k(h(x))$, that is, $kg \prec kh$; and evaluating at $k(x)$ one has $g(k(x)) < h(k(x))$ for every x , that is, $gk \prec hk$.

Determination at a point. If $g \prec h$ then in particular $g(x_0) < h(x_0)$. Conversely, if $g(x_0) < h(x_0)$ then $g \neq h$ and, by totality, $g \prec h$ or $h \prec g$; the second would give $h(x_0) < g(x_0)$, absurd. $\square$

Lemma 15.6: Archimedean property. With the notation of Proposition 15.3, let $a \in G$ with $a \succ \text{id}$. Then, for every $x \in \mathbb{R}$, $a^n(x) \rightarrow +\infty$ and $a^{-n}(x) \rightarrow -\infty$ as $n \rightarrow +\infty$. Consequently: **(i)** for every $v \in G$ there exists $N \geq 1$ with $v \prec a^N$, and the sequence $(a^k(x_0))_{k \in \mathbb{Z}}$ is strictly increasing and unbounded in both directions; **(ii)** for every $v \in G$ there exists a unique $k \in \mathbb{Z}$ with $a^k \preceq v \prec a^{k+1}$.

Proof. The sequence $(a^n(x))_{n \geq 0}$ is strictly increasing (from $a(y) > y$ for every y). If it were bounded above, it would have a finite limit L , and by continuity of a one would have $a(L) = \lim_n a^{n+1}(x) = L$: a would have a fixed point, hence $a = \text{id}$ by the freeness hypothesis, against $a \succ \text{id}$. Hence $a^n(x) \rightarrow +\infty$. Symmetrically, $(a^{-n}(x))_{n \geq 0}$ is strictly decreasing and, if bounded below by $M = \lim$, would give $a^{-1}(M) = M$, again absurd.

(i) Since $a^N(x_0) \rightarrow +\infty$, there exists N with $a^N(x_0) > v(x_0)$, that is (Proposition 15.3) $v \prec a^N$. **(ii)** The sequence $(a^k(x_0))_{k \in \mathbb{Z}}$ is strictly increasing with limits $\pm\infty$, hence there exists a unique k with $a^k(x_0) \leq v(x_0) < a^{k+1}(x_0)$; by Proposition 15.3 this is equivalent to $a^k \preceq v \prec a^{k+1}$. $\square$

Lemma 15.7: Rearrangement lemma. Let G be a group equipped with a bi-invariant total order $\preceq$. For every $g, h \in G$ and every $n \geq 1$, the element $(gh)^n$ lies between $g^n h^n$ and $h^n g^n$: precisely,

$$\text{if } hg \preceq gh : \quad h^n g^n \preceq (gh)^n \preceq g^n h^n; \quad \text{if } gh \preceq hg : \quad g^n h^n \preceq (gh)^n \preceq h^n g^n.$$

Proof. Suppose $hg \preceq gh$ (the other case is obtained by exchanging the names of g and h and reading the inequalities the other way round). Consider the word $w_0 := \underbrace{ghgh \cdots gh}_{n \text{ blocks}} = (gh)^n$,

that is, a sequence of $2n$ letters in $\{g, h\}$, and the elementary operation which replaces an adjacent subword hg by gh : if $w = u(hg)v$ and $w' = u(gh)v$, bi-invariance (applied twice: on the left for u , on the right for v) gives $w \preceq w'$ starting from $hg \preceq gh$. Performing a bubble sort — finitely many such adjacent exchanges bring every word with n letters g and n letters h into the ordered word $g \cdots g h \cdots h = g^n h^n$ — one obtains a chain $w_0 \preceq w_1 \preceq \cdots \preceq g^n h^n$, that is, $(gh)^n \preceq g^n h^n$.

Symmetrically, the operation which replaces an adjacent subword gh by hg does not increase (because $hg \preceq gh$), and a bubble sort which brings all the letters h to the left transforms $(gh)^n$ into $h^n g^n$ along a non-increasing chain: $h^n g^n \preceq (gh)^n$. $\square$

The theorem

Theorem 15.2: Hölder's theorem (free actions on $\mathbb{R}$). Let $G \leq \text{Homeo}_+(\mathbb{R})$ be a non-trivial group acting freely on $\mathbb{R}$. Once $a \in G$ with $a \succ \text{id}$ has been fixed, for $v \in G$ and $n \geq 1$ let $m(v, n) \in \mathbb{Z}$ be the unique integer with $a^{m(v, n)} \preceq v^n \prec a^{m(v, n)+1}$ (Lemma 15.6(ii)). Then: (i) the limit $\Phi(v) := \lim_{n \rightarrow \infty} \frac{m(v, n)}{n}$ exists and is finite, and $\frac{m(v, n)}{n} \leq \Phi(v) \leq \frac{m(v, n) + 1}{n}$ for every $n \geq 1$; (ii) $\Phi : G \rightarrow (\mathbb{R}, +)$ is a group homomorphism; (iii) Φ is strictly increasing with respect to $\preceq$ and injective, with $\Phi(a) = 1$; (iv) consequently G is isomorphic (as an ordered group) to a subgroup of $(\mathbb{R}, +)$; in particular G is abelian.

Proof. Step 1 (quasi-additivity in n). Let $n, n' \geq 1$, $m := m(v, n)$, $m' := m(v, n')$. From $a^m \preceq v^n \prec a^{m+1}$ and $a^{m'} \preceq v^{n'} \prec a^{m'+1}$, multiplying (bi-invariance: from $u_1 \preceq w_1$ and $u_2 \preceq w_2$ it follows that $u_1 u_2 \preceq w_1 u_2 \preceq w_1 w_2$),

$$a^{m+m'} \preceq v^n v^{n'} = v^{n+n'} \prec a^{m+m'+2}.$$

By the uniqueness in Lemma 15.6(ii), $m(v, n + n') \in \{m + m', m + m' + 1\}$, that is,

$$m(v, n) + m(v, n') \leq m(v, n + n') \leq m(v, n) + m(v, n') + 1. \quad (\dagger)$$

Step 2 (existence of the limit). From $(\dagger)$, the sequence $(-m(v, n))_n$ is subadditive, hence by Fekete's lemma (Lemma 4.1) $\lim_n m(v, n)/n = \sup_n m(v, n)/n$; and the sequence $(m(v, n) + 1)_n$ is subadditive (again from $(\dagger)$: $m(v, n + n') + 1 \leq (m(v, n) + 1) + (m(v, n') + 1)$), hence $\lim_n (m(v, n) + 1)/n = \inf_n (m(v, n) + 1)/n$. The two limits coincide (they differ by $1/n \rightarrow 0$) and are finite, because iterating $(\dagger)$ one has $n m(v, 1) \leq m(v, n) \leq n(m(v, 1) + 1)$. The two characterisations as supremum and infimum give exactly the double inequality of (i).

Step 3 (Φ is a homomorphism). Let $v, w \in G$ and $n \geq 1$; set $p := m(v, n)$, $q := m(w, n)$. Multiplying as in Step 1,

$$a^{p+q} \preceq v^n w^n \prec a^{p+q+2}, \quad a^{p+q} \preceq w^n v^n \prec a^{p+q+2}.$$

By the rearrangement Lemma 15.7, $(vw)^n$ lies (with respect to $\preceq$) between $v^n w^n$ and $w^n v^n$; since both these elements lie in the interval $[a^{p+q}, a^{p+q+2})$ and the interval is convex for the

total order, also

$$a^{p+q} \preceq (vw)^n \prec a^{p+q+2},$$

hence $m(vw, n) \in \{p+q, p+q+1\}$. Dividing by n and passing to the limit $(m(vw, n)/n \rightarrow \Phi(vw), p/n \rightarrow \Phi(v), q/n \rightarrow \Phi(w))$ one obtains $\Phi(vw) = \Phi(v) + \Phi(w)$.

Step 4 (monotonicity, injectivity, normalisation). If $v \preceq w$ then $v^n \preceq w^n$ for every $n \geq 1$ (bi-invariance and induction: $v^{n+1} = v v^n \preceq w v^n \preceq w w^n$), hence $m(v, n) \leq m(w, n)$ and $\Phi(v) \leq \Phi(w)$. Moreover $m(a, n) = n$ for every n (because $a^n \preceq a^n \prec a^{n+1}$), whence $\Phi(a) = 1$.

Now let $v \succ \text{id}$. By Lemma 15.6 applied to v (which satisfies $v \succ \text{id}$), $v^N(x_0) \rightarrow +\infty$, hence there exists $N \geq 1$ with $v^N(x_0) > a(x_0)$, that is, $a \prec v^N$, that is, $m(v, N) \geq 1$. By Step 2, $\Phi(v) = \sup_n m(v, n)/n \geq 1/N > 0$. If instead $v \prec \text{id}$, then $v^{-1} \succ \text{id}$ and, Φ being a homomorphism, $\Phi(v) = -\Phi(v^{-1}) < 0$. Hence $\Phi(v) = 0$ implies $v = \text{id}$: the kernel is trivial and Φ is injective. Together with the monotonicity of the first paragraph, Φ is strictly increasing.

Step 5 (conclusion). Φ realises an isomorphism of G onto the subgroup $\Phi(G) \leq (\mathbb{R}, +)$, which respects the orders. Since $(\mathbb{R}, +)$ is abelian, G is abelian. $\square$

***Remark 15.5: Where each hypothesis is needed.** The freeness of the action is used three times, in different ways: for the totality of the order (Proposition 15.3), for the archimedean property (Lemma 15.6: a fixed point of a would be a limit of the sequence $a^n(x)$) and for the injectivity of Φ (Step 4). Continuity is used only in the archimedean property. No hypothesis of regularity, countability or finite generation intervenes: the theorem holds for any group, even uncountable, which acts freely. By contrast, it is enough to give up freeness to lose everything: the free group F_2 acts faithfully on $\mathbb{R}$ (and on $\mathbb{T}$), but not freely, and is not abelian — see Remark 15.19.*

***Corollary 15.2: Hölder's theorem on the circle.** Let $G \leq \text{Homeo}_+(\mathbb{T})$ be a group acting freely on $\mathbb{T}$. Then G is abelian and the rotation number*

$$\bar{\rho} : G \longrightarrow \mathbb{R}/\mathbb{Z}$$

*is an **injective homomorphism**. In particular G is isomorphic to a subgroup of $\mathbb{R}/\mathbb{Z}$: either it is finite cyclic, or it is isomorphic to a dense subgroup of $\mathbb{R}/\mathbb{Z}$.*

Proof. Let $\tilde{G} := p^{-1}(G) \leq \widetilde{\text{Homeo}}$ (Proposition 15.1), which contains T .

$\tilde{G}$ **acts freely on $\mathbb{R}$** . Let $F \in \tilde{G}$ with $F(x) = x$ for some x . Then $p(F)$ fixes $\pi_{\mathbb{T}}(x)$, hence $p(F) = \text{id}$ by the freeness of G , hence $F = T^k$ for some k (Proposition 15.1(ii)); but $T^k(x) = x + 2\pi k = x$ forces $k = 0$, that is, $F = \text{id}$.

By Theorem 15.2, $\tilde{G}$ is abelian; hence so is its quotient $G = \tilde{G}/\langle T \rangle$.

Since $\tilde{G}$ is abelian, Proposition 15.2 says that $\rho|_{\tilde{G}} : \tilde{G} \rightarrow (\mathbb{R}, +)$ is a homomorphism. It is injective: $\rho(F) = 0$ implies, by Lemma 15.4(ii), that F has a fixed point, hence $F = \text{id}$ by the freeness just proved. Since $\rho(T) = 1$, the homomorphism descends to the quotient and defines $\bar{\rho} : G = \tilde{G}/\langle T \rangle \rightarrow \mathbb{R}/\mathbb{Z}$. The kernel of $\bar{\rho}$ is $\{p(F) : \rho(F) \in \mathbb{Z}\}$; but $\rho(F) = k \in \mathbb{Z}$ gives $\rho(T^{-k}F) = 0$, hence $T^{-k}F = \text{id}$ and $p(F) = \text{id}$: the kernel is trivial. The last assertion is the classification of the subgroups of $\mathbb{R}/\mathbb{Z}$ (a subgroup of $\mathbb{R}$ containing $\mathbb{Z}$ is $\frac{1}{q}\mathbb{Z}$ or dense: see Step 2 of the proof of Theorem 15.3). $\square$

***Theorem 15.3: Free actions are (semi-)conjugate to rotations.** Let $G \leq \text{Homeo}_+(\mathbb{T})$ be non-trivial and free, $\tilde{G} = p^{-1}(G)$ and $\Lambda := \rho(\tilde{G}) \subseteq \mathbb{R}$, which is a subgroup containing $\mathbb{Z}$. Then: (i) if Λ is dense in $\mathbb{R}$, there exists a continuous, non-decreasing,*

surjective map $h : \mathbb{T} \rightarrow \mathbb{T}$ of degree 1 such that

$$h \circ g = R_{2\pi\bar{\rho}(g)} \circ h \quad \text{for every } g \in G;$$

(ii) if Λ is not dense, then $\Lambda = \frac{1}{q}\mathbb{Z}$ for an integer $q \geq 1$, G is cyclic of order q and there exists a homeomorphism h of $\mathbb{T}$ with $h g h^{-1} = R_{2\pi\bar{\rho}(g)}$ for every $g \in G$ (genuine conjugacy).

Proof. Step 1 (the orbit and the canonical map). Fix $\tilde{x}_0 \in \mathbb{R}$ and set $\tilde{\mathcal{O}} := \{F(\tilde{x}_0) : F \in \tilde{G}\}$. If $F(\tilde{x}_0) = F'(\tilde{x}_0)$ then $F^{-1}F'$ fixes $\tilde{x}_0$, hence $F = F'$ (freeness, proved in Corollary 15.2): the map $\tilde{G} \rightarrow \tilde{\mathcal{O}}, F \mapsto F(\tilde{x}_0)$, is bijective. Moreover, by Proposition 15.3 applied to $\tilde{G}$ and by the strict increase of ρ on $\tilde{G}$ (Corollary 15.2 gives injectivity, Lemma 15.3(i) monotonicity),

$$F(\tilde{x}_0) < F'(\tilde{x}_0) \iff F \prec F' \iff \rho(F) < \rho(F'). \quad (**)$$

Step 2 (dichotomy for Λ). $\Lambda \leq (\mathbb{R}, +)$ is a subgroup. Let $c := \inf\{\lambda \in \Lambda : \lambda > 0\}$ (the set is non-empty because $1 \in \Lambda$). If $c = 0$, for every $\varepsilon > 0$ there exists $0 < \lambda < \varepsilon$ in Λ , and the multiples of λ form an ε -net of $\mathbb{R}$ contained in Λ : Λ is dense. If $c > 0$, then $c \in \Lambda$ (otherwise there would exist $\lambda, \lambda' \in \Lambda$ with $c < \lambda' < \lambda < 2c$, and $\lambda - \lambda' \in \Lambda$ would be positive and $< c$) and $\Lambda = c\mathbb{Z}$ (for $\lambda \in \Lambda$, $\lambda - c\lfloor\lambda/c\rfloor \in \Lambda \cap [0, c)$, hence zero). From $1 \in \Lambda = c\mathbb{Z}$ it follows that $c = 1/q$ with $q \geq 1$ an integer.

Step 3 (dense case). We define $\tilde{h} : \mathbb{R} \rightarrow \mathbb{R}$,

$$\tilde{h}(x) := \sup\{\rho(F) : F \in \tilde{G}, F(\tilde{x}_0) \leq x\}.$$

The set is non-empty (it contains T^{-k} for k large) and ρ is bounded above on it: choosing k with $\tilde{x}_0 + 2\pi k > x$, from $F(\tilde{x}_0) \leq x < T^k(\tilde{x}_0)$ it follows that $F \prec T^k$ and hence $\rho(F) \leq k$. Hence $\tilde{h}$ is well defined, and it is non-decreasing because the set grows with x .

Equivariance. For $F_0 \in \tilde{G}$ one has $\{F : F(\tilde{x}_0) \leq F_0(x)\} = \{F_0 F' : F'(\tilde{x}_0) \leq x\}$ (because $F(\tilde{x}_0) \leq F_0(x) \iff (F_0^{-1}F)(\tilde{x}_0) \leq x$, F_0^{-1} being increasing), and $\rho(F_0 F') = \rho(F_0) + \rho(F')$ by Proposition 15.2. Taking the supremum, $\tilde{h}(F_0(x)) = \rho(F_0) + \tilde{h}(x)$. In particular, with $F_0 = T$: $\tilde{h}(x + 2\pi) = \tilde{h}(x) + 1$.

Continuity. For $F \in \tilde{G}$, $\tilde{h}(F(\tilde{x}_0)) = \sup\{\rho(F') : F' \preceq F\} = \rho(F)$ by (**): hence $\Lambda \subseteq \tilde{h}(\mathbb{R})$. A non-decreasing function $\mathbb{R} \rightarrow \mathbb{R}$ is continuous if and only if its image skips no open interval; since Λ is dense and contained in the image, the image is dense, hence it skips no interval: $\tilde{h}$ is continuous. Being non-decreasing, continuous and with $\tilde{h}(x + 2\pi) = \tilde{h}(x) + 1$, it is also surjective onto $\mathbb{R}$.

Descent to the circle. The relation $\tilde{h}(x + 2\pi) = \tilde{h}(x) + 1$ shows that $x \mapsto \pi_{\mathbb{T}}(2\pi\tilde{h}(x))$ is constant on the fibres of $\pi_{\mathbb{T}}$ and defines $h : \mathbb{T} \rightarrow \mathbb{T}$ continuous, non-decreasing, surjective, of degree 1. Equivariance becomes $h \circ g = R_{2\pi\bar{\rho}(g)} \circ h$ for every $g \in G$.

Step 4 (discrete case). Let $\Lambda = \frac{1}{q}\mathbb{Z}$. By Corollary 15.2, $\rho : \tilde{G} \rightarrow \Lambda$ is an isomorphism; let $A \in \tilde{G}$ be the element with $\rho(A) = 1/q$, which generates $\tilde{G}$. From $\rho(A^q T^{-1}) = q \cdot \frac{1}{q} - 1 = 0$ and from injectivity it follows that $A^q = T$; hence $G = \tilde{G}/\langle T \rangle$ is cyclic of order q , generated by $p(A)$. Since $\rho(A) = 1/q > 0$, Lemma 15.4(iii) gives $A(x) > x$ for every x ; in particular $\tilde{x}_0 < A(\tilde{x}_0)$, and by Lemma 15.6 the sequence $(A^k(\tilde{x}_0))_{k \in \mathbb{Z}}$ is strictly increasing with limits $\pm\infty$: the intervals $[A^k(\tilde{x}_0), A^{k+1}(\tilde{x}_0))$, $k \in \mathbb{Z}$, partition $\mathbb{R}$. Let $\tilde{h}_0 : [\tilde{x}_0, A(\tilde{x}_0)] \rightarrow [0, 2\pi/q]$ be the increasing affine homeomorphism, and define $\tilde{h}$ on $[A^k(\tilde{x}_0), A^{k+1}(\tilde{x}_0)]$ by setting $\tilde{h}(A^k(y)) := \tilde{h}_0(y) + 2\pi k/q$ for $y \in [\tilde{x}_0, A(\tilde{x}_0)]$. The definitions match at the endpoints, $\tilde{h}$ is an increasing homeomorphism of $\mathbb{R}$ and satisfies $\tilde{h} \circ A = R_{1/q}^\# \circ \tilde{h}$ by construction; iterating q times, $\tilde{h} \circ T = \tilde{h} \circ A^q = T \circ \tilde{h}$, hence $\tilde{h}$ descends to a homeomorphism h of $\mathbb{T}$ with $h p(A) h^{-1} = R_{2\pi/q}$, and therefore $h g h^{-1} = R_{2\pi\bar{\rho}(g)}$ for every $g \in G$. $\square$

Example 15.2: A free action of $\mathbb{Z}^2$ and its conjugate. Let $\alpha := \sqrt{2}-1$ and $\beta := \sqrt{3}-1$. The numbers $1, \alpha, \beta$ are linearly independent over $\mathbb{Q}$: a relation $m\alpha + n\beta + \ell = 0$ with $m, n, \ell \in \mathbb{Z}$ would give $m\sqrt{2} + n\sqrt{3} \in \mathbb{Q}$, and squaring, $2m^2 + 3n^2 + 2mn\sqrt{6} \in \mathbb{Q}$, whence $mn = 0$ (because $\sqrt{6} \notin \mathbb{Q}$) and then $m = n = 0$ (because $\sqrt{2}, \sqrt{3} \notin \mathbb{Q}$), and finally $\ell = 0$. Hence

$$G_0 := \{R_{2\pi(m\alpha+n\beta)} : (m, n) \in \mathbb{Z}^2\} \leq \mathrm{SO}(2)$$

is isomorphic to $\mathbb{Z}^2$ (the homomorphism $(m, n) \mapsto R_{2\pi(m\alpha+n\beta)}$ has trivial kernel) and acts *freely* on $\mathbb{T}$: a rotation R_γ with $\gamma \notin 2\pi\mathbb{Z}$ has no fixed points. Here $\bar{\rho}(R_{2\pi(m\alpha+n\beta)}) = m\alpha + n\beta \bmod 1$ and is manifestly a homomorphism, as Corollary 15.2 requires.

Let us now conjugate by a non-trivial homeomorphism: let $h(s) := s + \varepsilon \sin s$ with $\varepsilon = 0.4$ (an element of Homeo, since $h'(s) = 1 + 0.4 \cos s > 0$) and $G_1 := hG_0h^{-1}$ (understood at the level of lifts, and projected onto $\mathbb{T}$). G_1 is still free (freeness is invariant under conjugacy) and still isomorphic to $\mathbb{Z}^2$, but its elements are not rotations; the homeomorphism h^{-1} is the conjugacy of Theorem 15.3(i). Lemma 15.3(iv) guarantees $\bar{\rho}(hR_{2\pi\gamma}h^{-1}) = \gamma \bmod 1$.

Double check (script `verifica_cap15_vo14.py`, block [C15-5]): the rotation number of the generators and of two of their combinations, computed by iterating the conjugated lift over 2×10^4 steps, equals

0.41421588 (expected $\alpha = 0.41421356$), 0.73205112 (expected $\beta = 0.73205081$),

0.14626656 (expected $\alpha+\beta-1 = 0.14626437$), 0.63227346 (expected $2\alpha-3\beta+2 = 0.63227470$),

with discrepancies $\leq 2.4 \times 10^{-6}$ ✓: the additivity $\bar{\rho}(g_1g_2) = \bar{\rho}(g_1) + \bar{\rho}(g_2)$ is verified numerically, and symbolically it is exact by invariance under conjugacy. Compare with Example 15.1, where the action is *not* free (F_ε and G_ε have fixed points) and additivity fails.

Corollary 15.3: Who cannot act freely. No non-abelian group — for instance the free group F_2 , the surface group Γ_g with $g \geq 2$, the modular group $\mathrm{PSL}(2, \mathbb{Z})$, Thompson’s group T — admits a free action on $\mathbb{T}$ (nor on $\mathbb{R}$) by homeomorphisms. Every action of theirs, however faithful, must have non-trivial elements with fixed points.

Proof. Immediate from Theorem 15.2 and Corollary 15.2: a free action forces the group to be abelian. The groups listed are not (F_2 is free of rank 2; $\Gamma_1 \cong \mathbb{Z}^2$ is abelian but Γ_g for $g \geq 2$ is not; $\mathrm{PSL}(2, \mathbb{Z}) \cong \mathbb{Z}/2 * \mathbb{Z}/3$ and Thompson’s group T are non-abelian). $\square$

Remark 15.6: A slip to avoid. In Corollary 15.3 the group $\Gamma_1 = \pi_1(\Sigma_1) \cong \mathbb{Z}^2$ is abelian, and indeed it admits free actions on the circle (Example 15.2 exhibits one). The list must therefore be read with the exclusion $g \geq 2$ for surface groups: it is exactly the same threshold which will appear in the Milnor–Wood inequality, where the case $g = 1$ is degenerate because $\chi(\Sigma_1) = 0$.

15.5. Left-orderable groups

Proposition 15.3 constructed, starting from a *free* action on $\mathbb{R}$, a *bi-invariant* total order. If one gives up freeness and asks only for faithfulness, something still remains: a left-invariant order. It is the observation which links the dynamics on the circle to the algebra of orderable groups, and explains from above why Homeo is torsion-free (Lemma 15.1).

Definition 15.6: Left-invariant order. A group G is **left-orderable** if there exists a total order $\prec$ on G such that $g \prec h \Rightarrow kg \prec kh$ for every $k \in G$ (left invariance only: $gk \prec hk$ is not required).

Proposition 15.4: From faithful actions on the line to left orders. Let $G \leq \text{Homeo}_+(\mathbb{R})$ be a subgroup (the action is therefore faithful by construction). Then G is left-orderable.

Proof. Let $(x_n)_{n \geq 0}$ be a dense sequence in $\mathbb{R}$ (for instance an enumeration of $\mathbb{Q}$). For $g \neq h$ in G there exists at least one index n with $g(x_n) \neq h(x_n)$: otherwise g and h would agree on a dense set and, being continuous, everywhere. Let $n(g, h)$ be the least of such indices and set

$$g \prec h \quad :\Longleftrightarrow \quad g(x_{n(g,h)}) < h(x_{n(g,h)}).$$

Totality and antisymmetry. For $g \neq h$ exactly one of $g \prec h$ and $h \prec g$ holds, because $n(g, h) = n(h, g)$ and the two values are comparable and distinct.

Transitivity. Let $g \prec h \prec k$, $m := \min\{n(g, h), n(h, k)\}$. For every index $j < m$ one has $g(x_j) = h(x_j) = k(x_j)$, hence $n(g, k) \geq m$. If $n(g, h) < n(h, k)$: at $m = n(g, h)$ one has $g(x_m) < h(x_m) = k(x_m)$, hence $n(g, k) = m$ and $g \prec k$. If $n(h, k) < n(g, h)$: at $m = n(h, k)$ one has $g(x_m) = h(x_m) < k(x_m)$, same conclusion. If $n(g, h) = n(h, k) = m$: then $g(x_m) < h(x_m) < k(x_m)$, hence $g(x_m) < k(x_m)$ and $n(g, k) = m$, so $g \prec k$.

Left invariance. Let $k \in G$. Since k is injective, $kg(x_j) = kh(x_j) \iff g(x_j) = h(x_j)$, hence $n(kg, kh) = n(g, h) =: n$. Since k is strictly increasing, $g(x_n) < h(x_n) \iff kg(x_n) < kh(x_n)$: hence $g \prec h \iff kg \prec kh$. $\square$

Corollary 15.4: Left-orderable groups are torsion-free. If G is left-orderable, then G is torsion-free. In particular every subgroup of $\text{Homeo}_+(\mathbb{R})$ — and hence Homeo itself — is torsion-free.

Proof. Let $g \neq 1$ and, without loss of generality, $1 \prec g$ (otherwise one argues on g^{-1} , noting that $g \prec 1$ implies $1 \prec g^{-1}$ by left invariance, multiplying by g^{-1}). By left invariance, from $1 \prec g$ it follows that $g \prec g^2$; by transitivity $1 \prec g^2$; by induction $1 \prec g^n$ for every $n \geq 1$, in particular $g^n \neq 1$. Hence G is torsion-free. The second assertion follows from Proposition 15.4, since $\text{Homeo} \leq \text{Homeo}_+(\mathbb{R})$ (an element of Homeo is in particular an increasing homeomorphism of $\mathbb{R}$). $\square$

Remark 15.7: A second proof of Lemma 15.1. Corollary 15.4 reproves Lemma 15.1 by a completely different route: the first proof was dynamical (if $F(\theta_0) > \theta_0$ then $F^n(\theta_0) > \theta_0$), this one is algebraic (a left-invariant order admits no elements of finite order). The comparison is instructive: the obstruction to torsion does not depend on the structure of the circle nor on the relation $F(\theta + 2\pi) = F(\theta) + 2\pi$, but only on the fact that $\mathbb{R}$ is totally ordered and that G respects its order. By contrast, $\text{Homeo}_+(\mathbb{T})$ does have torsion (the rational rotations): the circle is not ordered, only cyclically ordered, and this is exactly the difference which the central extension of Proposition 15.1 measures.

Remark 15.8: The converse: dynamical realisation (imported). Imported from A. Navas, *Groups of Circle Diffeomorphisms* (arXiv:math/0607481), Chapter 2, and from Section 6 of Ghys (2001) (sources reopened and verified on 17 September 2026): for a countable group G the converse of Proposition 15.4 also holds: G is left-orderable if and only if it admits a faithful action on $\mathbb{R}$ by orientation-preserving home-

omorphisms. The construction, called dynamical realisation, embeds G into $\mathbb{R}$ by means of a map $t : G \rightarrow \mathbb{R}$ built by induction on the enumeration of G (the new element goes at the midpoint of its two already placed neighbours, or at distance 1 from the maximum or the minimum if it is extremal) and makes G act on $t(G)$ by setting $g \cdot t(h) := t(gh)$, then extending the action by continuity and affinely on the complementary intervals. What is assumed: the countability of G and the — delicate — verification that the extension is by homeomorphisms. We do not prove this direction here, and we do not use it at any point of the chapter: all we need is the direction proved in Proposition 15.4.

The comparison with Hölder’s theorem is the interesting point: a bi-invariant archimedean order forces the group to be abelian (Theorem 15.2), whereas a merely left order imposes almost nothing — the free group F_2 acts faithfully on $\mathbb{R}$ by increasing homeomorphisms (A. Blass, J. M. Kister, “Free subgroups of the homeomorphism group of the reals”, *Topology Appl.* **24** (1986), 243–252, Crossref record verified on 17 September 2026), hence it is left-orderable by Proposition 15.4, while being as far as possible from abelian.

15.6. Minimal sets: the trichotomy

Hölder’s theorem describes *free* actions. For an arbitrary action the first dynamical invariant is not a number but a set: the smallest closed invariant set. The classification which follows is the analogue, for groups, of the dichotomy “ f minimal or invariant Cantor set” which Chapter 4 (Theorem 4.4) proved for a single homeomorphism with irrational rotation number.

Definition 15.7: Minimal set. Let $G \leq \text{Homeo}_+(\mathbb{T})$. A set $K \subseteq \mathbb{T}$ is **minimal** for G if it is closed, non-empty, G -invariant ($g(K) = K$ for every $g \in G$) and contains properly no non-empty closed G -invariant set. The action is said to be **minimal** if $K = \mathbb{T}$ is minimal, that is, if every orbit is dense.

Lemma 15.8: Existence of a minimal set. Every $G \leq \text{Homeo}_+(\mathbb{T})$ admits at least one minimal set. Moreover two distinct minimal sets are disjoint.

Proof. Let $\mathcal{F}$ be the family of non-empty closed G -invariant subsets of $\mathbb{T}$, ordered by reverse inclusion; $\mathcal{F} \neq \emptyset$ because $\mathbb{T} \in \mathcal{F}$. A chain $\{K_\lambda\}$ totally ordered by inclusion has non-empty intersection $\bigcap_\lambda K_\lambda$ (finite intersection property for nested compact sets in the compact space $\mathbb{T}$), closed and G -invariant: it is an upper bound in the family. By Zorn’s lemma there exists an element maximal for the reverse order, that is, a non-empty closed invariant set minimal for inclusion.

If $K_1 \neq K_2$ are minimal and $K_1 \cap K_2 \neq \emptyset$, then $K_1 \cap K_2$ is closed, non-empty and invariant, contained in both: by minimality $K_1 \cap K_2 = K_1$ and $K_1 \cap K_2 = K_2$, hence $K_1 = K_2$, absurd. $\square$

Theorem 15.4: Trichotomy of minimal sets. Let $G \leq \text{Homeo}_+(\mathbb{T})$. Exactly one of the following three cases holds: (a) G has a finite orbit (and every minimal set is a finite orbit); (b) the action is minimal; (c) there exists a **unique** minimal set K , which is compact, perfect, with empty interior and totally disconnected — an **exceptional minimal set**; in this case every orbit accumulates on K .

Proof. Let K be a minimal set (Lemma 15.8).

Step 1 (K finite $\Rightarrow$ case (a)). If K is finite, the orbit of one of its points is a finite, closed, invariant subset of K , hence equal to K by minimality: K is a finite orbit.

Step 2 (K infinite $\Rightarrow K$ perfect). Let K' be the set of accumulation points of K . K' is closed, and it is G -invariant because every g is a homeomorphism (it sends accumulation points of K to accumulation points of $g(K) = K$). If $K' = \emptyset$, then K is discrete and compact, hence finite, against the hypothesis. Hence $K' \neq \emptyset$ and, by minimality, $K' = K$: every point of K is an accumulation point, that is, K is perfect.

Step 3 (K infinite $\Rightarrow K = \mathbb{T}$ or K has empty interior). Let $U := \text{int } K$ and $\partial K = K \setminus U$, closed and G -invariant (the image of a boundary point is a boundary point, g being a homeomorphism). If $\partial K = \emptyset$, then K is open and closed in the connected space $\mathbb{T}$, hence $K = \mathbb{T}$: we are in case (b). Otherwise $\partial K \neq \emptyset$ and, by minimality, $\partial K = K$, that is, $U = \emptyset$.

Step 4 (total disconnectedness). If K has empty interior, a connected component of it cannot be a non-degenerate arc (otherwise K would have interior points), hence it is a point: K is totally disconnected. Together with Step 2, K is compact, perfect and totally disconnected: a Cantor set.

Step 5 (accumulation of the orbits on K). Suppose we are in case (c) and let $x \in \mathbb{T}$. If $x \in K$, the orbit Gx is dense in K by minimality. So let $x \notin K$: then x belongs to a connected component I of $\mathbb{T} \setminus K$ (a **gap**), which is an open arc whose endpoints p, q lie in K (because K is closed). Every $g \in G$ permutes the components of $\mathbb{T} \setminus K$, hence $g(I)$ is again a gap, with endpoints $g(p), g(q) \in K$.

The orbit of the gap I is infinite. If it were finite, the orbit of p would be contained in the finite set of the endpoints of the gaps $g(I)$, hence it would be finite; but its closure is a non-empty closed invariant set contained in K , therefore equal to K by minimality, and K would be finite: absurd.

Conclusion. The gaps are pairwise disjoint, hence the sum of their lengths is $\leq 2\pi$: along a sequence (g_n) with the gaps $g_n(I)$ pairwise distinct one has $|g_n(I)| \rightarrow 0$. Since $g_n(x) \in g_n(I)$ and the endpoints of $g_n(I)$ lie in K , the distance $d(g_n(x), K) \leq |g_n(I)| \rightarrow 0$: every limit point of $(g_n(x))$ lies in K , and such limit points exist by compactness. Hence the set Ω_x of accumulation points of the orbit Gx meets K ; Ω_x is closed and G -invariant (if $y = \lim g_n(x)$ then $g(y) = \lim gg_n(x) \in \Omega_x$), so $\Omega_x \cap K$ is closed, invariant and non-empty, and by minimality $\Omega_x \cap K = K$: the orbit of x accumulates on the whole of K .

Step 6 (uniqueness in case (c)). Let K' be any minimal set and $x \in K'$. By Step 5 (which treats both the case $x \in K$ and the case $x \notin K$), $K \subseteq \overline{Gx} \subseteq K'$, the last inclusion because K' is closed, invariant and contains x . By minimality of K' , from $\emptyset \neq K \subseteq K'$ with K closed and invariant it follows that $K' = K$.

Mutual exclusivity of the three cases. In case (a) there exists a finite orbit, which is a finite minimal set; by the uniqueness of Step 6 this excludes (c), and it excludes (b) because $\mathbb{T}$ is infinite and a proper closed invariant set exists. Cases (b) and (c) exclude each other because in (b) the only minimal set is $\mathbb{T}$, which has non-empty interior. $\square$

Remark 15.9: Where we have already met the three cases. For $G = \langle f \rangle$ cyclic the three cases are exactly the Poincaré classification of Chapter 4: (a) corresponds to $\rho(f) \in \mathbb{Q}$ (periodic orbit, Theorem 4.3); (b) to $\rho(f)$ irrational with f minimal; (c) to $\rho(f)$ irrational with an invariant Cantor set — Denjoy's counterexample (Chapter 5). Denjoy's theorem then says that, for a single diffeomorphism of class $C^{1+\text{bv}}$, case (c) does not occur; Remark 15.18 reports what remains of this statement for groups. Note that the trichotomy, as proved above, uses no hypothesis of regularity nor of countability of the group.

Remark 15.10: Trichotomy, invariant measures and Margulis' alternative. In cases (a) and (c) the action always admits an invariant probability measure: in case (a) the uniform measure on the finite orbit, in case (c) — less immediately — a measure concentrated on K . In case (b) existence may fail, and it is precisely here that Margulis' alternative

(Remark 15.19) locates the free subgroups. In terms of the rotation number: when an invariant probability measure μ exists, the map $g \mapsto \bar{\rho}(g)$ becomes a homomorphism $G \rightarrow \mathbb{R}/\mathbb{Z}$, and the semiconjugacy to rotations is constructed with the distribution function of μ , exactly as $\tilde{h}$ in Step 3 of the proof of Theorem 15.3.

Remark 15.11: Continuity and rotation spectrum. The rotation number $\rho : \widetilde{\text{Homeo}} \rightarrow \mathbb{R}$ is continuous for the topology of uniform convergence (Volume IV, Chapter 4, Theorem 4.2, where monotonicity and continuity are proved in full). Given $\varrho : \Gamma \rightarrow \text{Homeo}_+(\mathbb{T})$, one calls **rotation spectrum** the set $\{\bar{\rho}(\varrho(\gamma)) : \gamma \in \Gamma\} \subseteq \mathbb{R}/\mathbb{Z}$. For a free action the spectrum is a subgroup of $\mathbb{R}/\mathbb{Z}$ isomorphic to Γ (Corollary 15.2); in general it is neither a subgroup nor a closed set, and its variation as the action varies is one of the tools with which Kim–Koberda–Mj distinguish uncountably many semiconjugacy classes (Remark 15.21).

15.7. The Euler class of an action

Definition 15.8: The surface group Γ_g . For $g \geq 1$, the **surface group** of genus g is the group presented by

$$\Gamma_g := \left\langle a_1, b_1, \dots, a_g, b_g \mid \prod_{i=1}^g [a_i, b_i] = 1 \right\rangle, \quad [a, b] := aba^{-1}b^{-1}.$$

For $g = 1$ the relation is $[a_1, b_1] = 1$ and $\Gamma_1 \cong \mathbb{Z}^2$. It is well known — and in this chapter we shall never use it in a proof, but only as geometric motivation — that Γ_g is isomorphic to the fundamental group of the closed orientable surface Σ_g of genus g , whose Euler characteristic is $\chi(\Sigma_g) = 2 - 2g$.

Definition 15.9: Euler class of a representation. Let $\varrho : \Gamma_g \rightarrow \text{Homeo}_+(\mathbb{T})$ be a homomorphism. For each i choose lifts $A_i, B_i \in \widetilde{\text{Homeo}}$ of $\varrho(a_i), \varrho(b_i)$. The product $\prod_{i=1}^g [A_i, B_i]$ is a lift of $\prod_i [\varrho(a_i), \varrho(b_i)] = \varrho(\prod_i [a_i, b_i]) = \text{id}$, hence it is of the form T^n for a unique $n \in \mathbb{Z}$ (Proposition 15.1(ii)). We set

$$e(\varrho) := n = \rho\left(\prod_{i=1}^g [A_i, B_i]\right) \in \mathbb{Z},$$

the **Euler class** (or **Euler number**) of the representation.

Proposition 15.5: $e(\varrho)$ is well defined and invariant. With the notation of Definition 15.9: **(i)** $e(\varrho)$ does not depend on the choice of the lifts A_i, B_i ; **(ii)** $e(h\varrho h^{-1}) = e(\varrho)$ for every $h \in \text{Homeo}_+(\mathbb{T})$; **(iii)** $e(\varrho) = 0$ if and only if ϱ lifts, that is, there exists a homomorphism $\tilde{\varrho} : \Gamma_g \rightarrow \widetilde{\text{Homeo}}$ with $p \circ \tilde{\varrho} = \varrho$.

Proof. **(i)** Two lifts of $\varrho(a_i)$ differ by a power of T (Proposition 15.1(ii)); since T is central,

$$[T^k A_i, T^\ell B_i] = T^k A_i T^\ell B_i A_i^{-1} T^{-k} B_i^{-1} T^{-\ell} = A_i B_i A_i^{-1} B_i^{-1} = [A_i, B_i],$$

because all the powers of T can be moved until they cancel. Hence the product of the commutators does not change.

(ii) Let $H \in \widetilde{\text{Homeo}}$ be a lift of h . Then HA_iH^{-1} and HB_iH^{-1} are lifts of $h\rho(a_i)h^{-1}$ and $h\rho(b_i)h^{-1}$, and

$$\prod_i [HA_iH^{-1}, HB_iH^{-1}] = H \left(\prod_i [A_i, B_i] \right) H^{-1} = HT^{e(\rho)}H^{-1} = T^{e(\rho)},$$

the last equality because T is central.

(iii) If $\tilde{\rho}$ exists, choosing $A_i := \tilde{\rho}(a_i)$, $B_i := \tilde{\rho}(b_i)$ one has $\prod_i [A_i, B_i] = \tilde{\rho}(\prod_i [a_i, b_i]) = \tilde{\rho}(1) = \text{id} = T^0$, hence $e(\rho) = 0$. Conversely, if $e(\rho) = 0$, the $2g$ elements A_i, B_i satisfy the unique relation of the presentation of Γ_g , hence (universal property of a presentation) the assignment $a_i \mapsto A_i$, $b_i \mapsto B_i$ extends to a homomorphism $\tilde{\rho} : \Gamma_g \rightarrow \widetilde{\text{Homeo}}$, which lifts ρ because it does so on the generators. $\square$

Remark 15.12: Why it is called the “Euler class”. An action $\rho : \pi_1(\Sigma_g) \rightarrow \text{Homeo}_+(\mathbb{T})$ allows one to construct the flat circle bundle $E_\rho := (\widetilde{\Sigma}_g \times \mathbb{T})/\pi_1(\Sigma_g)$ on which the horizontal foliation $\widetilde{\Sigma}_g \times \{\text{point}\}$ passes to the quotient, remaining transverse to the fibres. The integer $e(\rho)$ of Definition 15.9 coincides with the Euler number of this bundle: this is the geometric content of the chapter, but its verification requires the theory of circle bundles (Volume IV, Chapter 2) and the correspondence between $H^2(\Gamma_g; \mathbb{Z})$ and central extensions, which we do not carry out here. **Nothing in what follows depends on this identification:** Theorem 15.5 is proved in full starting from Definition 15.9, which is purely algebraic.

Definition 15.10: The Euler cocycle. For $f \in \text{Homeo}_+(\mathbb{T})$ let F_f be the unique lift of f with $F_f(0) \in [0, 2\pi)$ (it exists and is unique because the lifts differ by T^k). For $f, g \in \text{Homeo}_+(\mathbb{T})$ one defines $c(f, g) \in \mathbb{Z}$ by the relation

$$F_f \circ F_g = T^{c(f, g)} \circ F_{fg}.$$

Proposition 15.6: The Euler cocycle is bounded and takes the values 0 and 1. For every $f, g \in \text{Homeo}_+(\mathbb{T})$ one has $c(f, g) \in \{0, 1\}$. Moreover c is a 2-cocycle:

$$c(f, g) + c(fg, h) = c(g, h) + c(f, gh) \quad \text{for every } f, g, h.$$

Proof. Values. Set $u := F_g(0) \in [0, 2\pi)$. Since F_f is increasing with $F_f(2\pi) = F_f(0) + 2\pi$,

$$F_f(0) \leq F_f(u) < F_f(0) + 2\pi,$$

and since $F_f(0) \in [0, 2\pi)$ one obtains $F_f F_g(0) = F_f(u) \in [0, 4\pi)$. On the other hand $F_f F_g(0) = 2\pi c(f, g) + F_{fg}(0)$ with $F_{fg}(0) \in [0, 2\pi)$, hence $2\pi c(f, g) \in (-2\pi, 4\pi)$ and $c(f, g) \in \{0, 1\}$.

Cocycle. One computes $F_f F_g F_h$ in two ways, using the centrality of T :

$$(F_f F_g) F_h = T^{c(f, g)} F_{fg} F_h = T^{c(f, g) + c(fg, h)} F_{fgh},$$

$$F_f (F_g F_h) = F_f T^{c(g, h)} F_{gh} = T^{c(g, h)} F_f F_{gh} = T^{c(g, h) + c(f, gh)} F_{fgh}.$$

Comparing the exponents (which are unique, because $T^k = T^{k'} \Rightarrow k = k'$) one obtains the identity. $\square$

Remark 15.13: Bounded Euler class and Ghys’ theorem (imported). Proposition 15.6 says that c defines a class in bounded cohomology $H_b^2(\text{Homeo}_+(\mathbb{T}); \mathbb{Z})$: the

bounded Euler class e_b . Given $\varrho : \Gamma \rightarrow \text{Homeo}_+(\mathbb{T})$, the pullback $\varrho^*e_b \in H_b^2(\Gamma; \mathbb{Z})$ is an invariant of the action, finer than the ordinary Euler class.

Imported from É. Ghys, “Groupes d’homéomorphismes du cercle et cohomologie bornée”, *Contemp. Math.* 58 (1987), 81–106, and from Section 6 of Ghys (2001) (sources reopened and verified on 17 September 2026): two actions $\varrho_1, \varrho_2 : \Gamma \rightarrow \text{Homeo}_+(\mathbb{T})$ have the same bounded Euler class if and only if they are **semiconjugate**. What is assumed: the definition of bounded cohomology of groups and the notion of semiconjugacy between actions (existence of a monotone map of degree 1 intertwining the two actions, as in Theorem 15.3). We shall not use this statement in any proof of the chapter: it serves only to place the Euler class in the general picture. A discussion of the various notions of semiconjugacy present in the literature, and of their equivalence, may be found in Kim–Koberda–Mj (2019).

15.8. The Milnor–Wood inequality

Everything rests on a single lemma, which is proved in five lines thanks to the invariance of the rotation number under conjugacy.

Lemma 15.9: Commutator lemma. For every $A, B \in \widetilde{\text{Homeo}}$ one has

$$a([A, B]) < 1,$$

that is, there exists $\theta \in \mathbb{R}$ with $[A, B](\theta) < \theta + 2\pi$. Symmetrically, $b([A, B]) > -1$.

Proof. Set $C := [A, B] = ABA^{-1}B^{-1}$ and suppose, for a contradiction, that $a(C) \geq 1$, that is, $C(\theta) \geq \theta + 2\pi$ for every θ , that is, $C \succeq T$ in the pointwise order. Applying this inequality at the point $B(\theta)$ and recalling $C \circ B = ABA^{-1}$ (indeed $ABA^{-1}B^{-1}B = ABA^{-1}$), one obtains

$$(ABA^{-1})(\theta) = C(B(\theta)) \geq B(\theta) + 2\pi = (TB)(\theta) \quad \text{for every } \theta,$$

that is, $ABA^{-1} \succeq TB$. By the monotonicity of the rotation number (Lemma 15.3(i)) and by translation (part (iii)),

$$\rho(ABA^{-1}) \geq \rho(TB) = 1 + \rho(B).$$

But $\rho(ABA^{-1}) = \rho(B)$ by invariance under conjugacy (Lemma 15.3(iv)): one would obtain $\rho(B) \geq 1 + \rho(B)$, absurd. Hence $a(C) < 1$.

For the second assertion it is enough to observe that $C^{-1} = [B, A]$ is also a commutator, that $b(C) = -a(C^{-1})$ (since $\varphi_{C^{-1}}(C(\theta)) = -\varphi_C(\theta)$, and $\theta \mapsto C(\theta)$ is bijective) and to apply the first assertion to $[B, A]$. $\square$

Remark 15.14: A reformulation checkable by computer. From $C \circ (BA) = AB$ (indeed $ABA^{-1}B^{-1}BA = AB$) it follows, setting $y = (BA)(\theta)$, that

$$C(y) - y = (AB)(\theta) - (BA)(\theta),$$

and since $\theta \mapsto (BA)(\theta)$ is a bijection of $\mathbb{R}$, Lemma 15.9 is equivalent to

$$\min_{\theta \in \mathbb{R}} \left[(A \circ B)(\theta) - (B \circ A)(\theta) \right] < 2\pi :$$

two circle homeomorphisms cannot differ, after composition in the two orders, by at least a whole turn at every point. In this form the numerical check is immediate: the script

verifica_cap15_vol4.py, block [C15–3], generated 200 random pairs (A, B) (trigonometric polynomials of degree 3 with positive derivative, and translations up to three turns) and found as the maximum value of $a([A, B])$ the number -0.018567 , with mean -0.082778 and minimum -0.159105 : all < 1 , and indeed all negative ✓.

Theorem 15.5: Milnor–Wood inequality. Let $g \geq 1$ and let $\varrho : \Gamma_g \rightarrow \text{Homeo}_+(\mathbb{T})$ be an arbitrary homomorphism. Then

$$|e(\varrho)| \leq |\chi(\Sigma_g)| = 2g - 2.$$

In particular, for $g = 1$ one has $e(\varrho) = 0$ for every action of $\mathbb{Z}^2$ on the circle.

Proof. Let A_i, B_i be lifts as in Definition 15.9, $C_i := [A_i, B_i]$ and $n := e(\varrho)$, so that

$$C_1 C_2 \cdots C_g = T^n, \quad a(T^n) = b(T^n) = n.$$

Step 1 (bound on each $b(C_i)$). By Lemma 15.9, $a(C_i) < 1$; by the oscillation Lemma 15.2, $b(C_i) < a(C_i) + 1 < 2$.

Step 2 (chain). By Lemma 15.5(iii) applied to $F = C_1$ and $G = C_2 \cdots C_g$, and then by part (ii) iterated,

$$n = a(C_1 \cdot (C_2 \cdots C_g)) \leq a(C_1) + b(C_2 \cdots C_g) \leq a(C_1) + \sum_{i=2}^g b(C_i) < 1 + 2(g-1) = 2g - 1.$$

(For $g = 1$ the sum is empty and the estimate reads $n = a(C_1) < 1$.)

Step 3 (integrality). n is an integer strictly smaller than $2g - 1$, hence $n \leq 2g - 2$.

Step 4 (symmetry). Inverting the relation, $T^{-n} = (C_1 \cdots C_g)^{-1} = C_g^{-1} \cdots C_1^{-1}$, and each $C_i^{-1} = [B_i, A_i]$ is again a commutator of two elements of Homeo . Steps 1–3 applied to this expression give $-n \leq 2g - 2$. Hence $|n| \leq 2g - 2$.

For $g = 1$: $|n| \leq 0$, that is, $n = 0$. □

Remark 15.15: Where the last unit is gained. It is worth isolating the architecture of the proof, because it is easy to obtain a weaker estimate and believe one has finished. If one proceeds with the quasi-morphism (Theorem 15.1, defect < 2) one obtains only $|n| \leq 4g - 2$; with the optimal defect 1 (not proved here, Remark 15.4) one would obtain $|n| \leq 2g - 1$, which is still one unit above the correct value. The proof given above does not go through the defect at all: it uses $a(C_i) < 1$ (Lemma 15.9), $b < a + 1$ (Lemma 15.2) and the asymmetric estimate $a(FG) \leq a(F) + b(G)$ (Lemma 15.5(iii)) — which uses the minimum on the first factor and the maximum only on the remaining $g - 1$ — and closes with the integrality of n . It is exactly this asymmetry, one single a and $g - 1$ times b , which produces $1 + 2(g - 1) = 2g - 1$ instead of $2g$, and hence, by integrality, $2g - 2$.

Remark 15.16: Optimality of the inequality (imported). Imported from J. Milnor, *Comment. Math. Helv.* 32 (1957/58), 215–223, and J. W. Wood, *Comment. Math. Helv.* 46 (1971), 257–273 (EuDML records reopened and verified on 17 September 2026), in the reading given by Ghys (2001), Section 6: the extreme value $|e(\varrho)| = 2g - 2$ is attained, for every $g \geq 2$, by the action of a cocompact Fuchsian group $\Gamma_g \subset \text{PSL}(2, \mathbb{R})$ on the boundary of the Poincaré disc (the action associated with the unit tangent bundle of Σ_g , whose Euler number is $\chi(\Sigma_g)$). What is assumed: the existence of a hyperbolic structure on Σ_g for $g \geq 2$ (uniformisation) and the computation

of the Euler number of the unit tangent bundle. Wood moreover proves the converse: every integer n with $|n| \leq 2g - 2$ is the Euler class of some action, hence a circle bundle over Σ_g admits a foliation transverse to the fibres if and only if $|e| \leq 2g - 2$. In the present chapter only the inequality is proved (Theorem 15.5), not the optimality.

Note the difference with Milnor's original statement (1958), which concerns flat vector bundles with structure group $\mathrm{GL}^+(2, \mathbb{R})$ and gives $|e| < g$, that is, $|e| \leq g - 1$: the two estimates do not contradict each other, because the passage from $\mathrm{GL}^+(2, \mathbb{R})$ (which acts on $\mathbb{R}^2 \setminus \{0\}$, hence on the circle of directions) to $\mathrm{PSL}(2, \mathbb{R})$ (which acts on $\mathbb{RP}^1$, a circle “half” of the previous one) doubles the Euler number. The Poincaré disc and hyperbolic geometry are developed in Volume V, Chapter 4; here we use only the algebraic model of Section 15.9.

Corollary 15.5: Actions of $\mathbb{Z}^2$ and of Γ_2 . (i) Every action of $\mathbb{Z}^2$ on $\mathbb{T}$ by orientation-preserving homeomorphisms lifts to an action on $\mathbb{R}$ (that is, to a homomorphism $\mathbb{Z}^2 \rightarrow \mathrm{Homeo}$). (ii) For $g = 2$, the Euler class of an action of Γ_2 can only equal $-2, -1, 0, 1, 2$.

Proof. (i) By Theorem 15.5 with $g = 1$, $e(\varrho) = 0$; by Proposition 15.5(iii), ϱ lifts. (ii) $|e| \leq 2 \cdot 2 - 2 = 2$ and $e \in \mathbb{Z}$. $\square$

15.9. A guiding example: $\mathrm{PSL}(2, \mathbb{R})$ on the boundary

Definition 15.11: The action of $\mathrm{PSL}(2, \mathbb{R})$ on $\mathbb{RP}^1$. Let $\mathrm{SL}(2, \mathbb{R})$ be the group of real 2×2 matrices of determinant 1 and $\mathrm{PSL}(2, \mathbb{R}) = \mathrm{SL}(2, \mathbb{R}) / \{\pm I\}$. The real projective line $\mathbb{RP}^1$ is the set of lines through the origin of $\mathbb{R}^2$; identifying the line generated by $(z, 1)$ with $z \in \mathbb{R}$ and the one generated by $(1, 0)$ with ∞ , one has $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$, a circle. $\mathrm{PSL}(2, \mathbb{R})$ acts on $\mathbb{RP}^1$ by means of the **real Möbius transformations**

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : z \mapsto \frac{az + b}{cz + d},$$

with the usual conventions for ∞ . As a circle we parametrise $\mathbb{RP}^1$ by $\psi \in \mathbb{R}/\pi\mathbb{Z}$ setting $z = \tan \psi$ (that is, the line generated by $(\sin \psi, \cos \psi)$): since $\tan$ is increasing, the orientation induced by increasing ψ is the one in which z increases, and the rotation number is normalised by dividing the displacement by the period π .

Proposition 15.7: Classification by trace and rotation number. Let $M \in \mathrm{SL}(2, \mathbb{R})$, $M \neq \pm I$, and $\tau := |\mathrm{tr} M|$ (well defined on $\mathrm{PSL}(2, \mathbb{R})$). (i) If $\tau \geq 2$ (hyperbolic case $\tau > 2$, parabolic case $\tau = 2$), the action of M on $\mathbb{RP}^1$ has at least one fixed point, hence $\rho(M) = 0$. (ii) If $\tau < 2$ (elliptic case), the action has no fixed points and $\rho(M) \neq 0$.

Proof. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc = 1$.

Case $c = 0$. Then ∞ is fixed: $M \cdot \infty = a/c = \infty$. Moreover $d = 1/a$ and $\tau = |a + 1/a| \geq 2$ by the arithmetic–geometric mean inequality: we are automatically in case (i), and the conclusion $\rho = 0$ follows from Lemma 15.4(ii) (applied to a lift of the circle map $\mathbb{RP}^1$).

Case $c \neq 0$. The fixed points in $\mathbb{R}$ are the roots of $cz^2 + (d - a)z - b = 0$, whose discriminant is

$$\Delta = (d - a)^2 + 4bc = (a + d)^2 - 4(ad - bc) = (\mathrm{tr} M)^2 - 4.$$

And ∞ is not fixed (because $c \neq 0$). Hence: if $\tau \geq 2$ then $\Delta \geq 0$ and there is at least one real fixed point, whence $\rho(M) = 0$ (Lemma 15.4(ii)); if $\tau < 2$ then $\Delta < 0$, there is no fixed point

on $\mathbb{RP}^1$, and again by Lemma 15.4(ii) one has $\rho(M) \neq 0$. □

Example 15.3: The modular group: $\rho(S) = 1/2$ and $\rho(ST) = 2/3$. Let

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} : z \mapsto -\frac{1}{z}, \quad T_{\text{mod}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} : z \mapsto z + 1,$$

$$ST_{\text{mod}} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} : z \mapsto -\frac{1}{z+1},$$

generators of the modular group $\text{PSL}(2, \mathbb{Z})$ (the subscript “mod” is used here so as not to confuse T_{mod} with the translation T of Homeo).

(1) $\rho(S) = 1/2$. $S^2 = -I$, hence S has order 2 in $\text{PSL}(2, \mathbb{R})$ and $2\rho(S) \in \mathbb{Z}$: $\rho(S) \in \{0, 1/2\}$ modulo 1. Since $|\text{tr } S| = 0 < 2$, Proposition 15.7(ii) excludes $\rho(S) = 0$. Hence $\rho(S) = 1/2$.

(2) $\rho(ST_{\text{mod}}) = 2/3$. One computes $(ST_{\text{mod}})^2 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$ and $(ST_{\text{mod}})^3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$: the element has order 3 in $\text{PSL}(2, \mathbb{R})$, hence $3\rho \in \mathbb{Z}$ and, since $|\text{tr}| = 1 < 2$ excludes $\rho = 0$, one has $\rho \in \{1/3, 2/3\}$. To decide, one looks at the orbit of 0:

$$0 \mapsto -\frac{1}{0+1} = -1 \mapsto -\frac{1}{-1+1} = \infty \mapsto -\frac{1}{\infty+1} = 0.$$

In the positive cyclic order of $\mathbb{RP}^1$ (the one in which z increases and ∞ follows $+\infty$ and precedes $-\infty$) the three points appear in the order $(-1, 0, \infty)$, to which we assign the indices $(0, 1, 2)$. The map sends the index $1 \mapsto 0$, $0 \mapsto 2$, $2 \mapsto 1$, that is, $p \mapsto p + 2 \pmod{3}$: it is a rotation by $2/3$ of a turn on the orbit. Hence a lift $\widetilde{ST}$ satisfies $\widetilde{ST}^3 = T^2$ and $\rho(ST_{\text{mod}}) = 2/3$.

Double numerical check (script `verifica_cap15_vol4.py`, block [C15-4], run on 17 September 2026): iterating over 4×10^4 steps the lift of the action on the circle of directions and doubling (in order to pass from $\mathbb{R}/2\pi\mathbb{Z}$ to $\mathbb{RP}^1 = \mathbb{R}/\pi\mathbb{Z}$) one obtains $\rho(S) = 0.500000$ and $\rho(ST_{\text{mod}}) = 0.666669 \checkmark$, while the hyperbolic element $\text{diag}(2, 1/2)$ ($\tau = 2.5$) and the parabolic element $z \mapsto z + 1$ ($\tau = 2$) give $\rho = 0.000000$ and $\rho = 0.000012 \checkmark$, in agreement with Proposition 15.7(i).

Warning about orientation. The result $2/3$ (and not $1/3$) depends on the choice of the orientation of $\mathbb{RP}^1$: parametrising the lines by $(\cos \psi, \sin \psi)$ instead of $(\sin \psi, \cos \psi)$ one obtains $z = \cot \psi$, which is decreasing, and all the rotation numbers are replaced by $1 - \rho$; the first run of the script, carried out with the wrong parametrisation, did in fact give 0.333 and was corrected. It is a concrete example of the fact that ρ is invariant under conjugacy which preserves the orientation, not under arbitrary conjugacy.

Remark 15.17: Why $\text{PSL}(2, \mathbb{R})$ is the decisive example. $\text{PSL}(2, \mathbb{R})$ acts on $\mathbb{RP}^1$, which is identified with the boundary of the Poincaré disc (Volume V, Chapter 4): it is the action which realises the extreme case of the Milnor–Wood inequality (Remark 15.16). It is also the first example in which the dichotomy of Proposition 15.7 shows how the rotation number “sees” the algebra: $\rho = 0$ is equivalent to the existence of a fixed point at the boundary, that is, to the hyperbolic and parabolic elements, which are exactly those which in hyperbolic geometry correspond to translations along a geodesic and to rotations about an ideal point. Figure 15.5 shows three geodesics and their six boundary points.

15.10. What we import, and the panorama

INFO

Register of the statements imported in this chapter. Imported — that is, used or cited without proof — are only the following statements, all of them *not used* in any proof of the chapter: the structure of $\text{Homeo}_+(\mathbb{T})$ as an abstract group (Remark 15.2); the optimal value 1 of the defect of the quasi-morphism ρ (Remark 15.4); the dynamical realisation of countable left-orderable groups, that is, the converse of Proposition 15.4 (Remark 15.8); Ghys’ theorem on the bounded Euler class (Remark 15.13); the optimality of Milnor–Wood (Remark 15.16); the four panorama statements of this section (Remarks 15.18, 15.19, 15.20, 15.21). All the rest — central extension, quasi-morphism with defect < 2 , Hölder’s theorem with its algebraic part, semiconjugacy to rotations, left-orderability and absence of torsion, trichotomy of minimal sets, well-definedness of the Euler class, Euler cocycle, commutator lemma, Milnor–Wood inequality in the optimal form — is proved in full.

Remark 15.18: Denjoy for group actions (imported). Denjoy’s theorem of Chapter 5 says that a single diffeomorphism of class $C^{1+\text{bv}}$ with irrational rotation number has no wandering intervals. Imported from B. Deroin, V. Kleptsyn, A. Navas, “*Sur la dynamique unidimensionnelle en régularité intermédiaire*”, *Acta Math.* 199 (2007), 199–262 (arXiv:math/0506063, reopened and verified on 17 September 2026), and from A. Navas, *Groups of Circle Diffeomorphisms* (arXiv:math/0607481): the analogue for groups is much more delicate. What is assumed: for groups of diffeomorphisms of class $C^{1+\alpha}$ there hold extensions of Denjoy’s theorem and of Kopell’s lemma, obtained with probabilistic methods (stationary measures); for groups of class only C^1 counterexamples abound, whereas in class C^2 Sacksteder’s theorem and the works of Duminy give minimality results. The distinction between C^1 , $C^{1+\alpha}$ and C^2 — the same which in Chapter 5 separated the theorem from the counterexample — is here even more marked.

Remark 15.19: Margulis’ alternative (imported). Imported from G. Margulis, “Free subgroups of the homeomorphism group of the circle”, *C. R. Acad. Sci. Paris Sér. I Math.* 331 (2000), no. 9, 669–674 (Crossref record, DOI 10.1016/s0764-4442(00)01694-3, verified on 17 September 2026; the statement is also reported in Section 5 of Ghys 2001): for every subgroup $G \leq \text{Homeo}_+(\mathbb{T})$ the following alternative holds: either there exists a probability measure on $\mathbb{T}$ invariant under G , or G contains a free subgroup of rank 2. What is assumed: the notion of invariant Borel probability measure and the fact that $\text{Homeo}_+(\mathbb{T})$ does not satisfy the classical Tits alternative. The link with our chapter: when the first alternative occurs, the rotation number becomes a homomorphism $G \rightarrow \mathbb{R}/\mathbb{Z}$ (the invariant measure allows one to construct the semiconjugacy to rotations exactly as in Theorem 15.3); Hölder’s theorem is the particular case in which the action is free.

Remark 15.20: Thurston stability (imported). Imported from W. P. Thurston, “A generalization of the Reeb stability theorem”, *Topology* 13 (1974), 347–352 (Crossref record, DOI 10.1016/0040-9383(74)90025-1, verified on 17 September 2026): if G is a finitely generated group acting on $[0, 1)$ (or on a neighbourhood of a common fixed point of the line) by diffeomorphisms of class C^1 fixing the origin, and if the action is not trivial, then G admits a non-trivial surjective homomorphism onto $\mathbb{Z}$. What is assumed: finite generation and C^1 regularity near the fixed point. Typical consequence: a finitely generated perfect group (that is, equal to its own derived subgroup) cannot act non-trivially

by C^1 diffeomorphisms of the line fixing a point; it is the “one fixed point” version of the obstructions which, on the circle, are expressed by the Euler class.

Remark 15.21: Solodov, rigidity and flexibility (imported). Imported from V. V. Solodov, “Homeomorphisms of the circle and a foliation”, *Math. USSR-Izv.* 24 (1985), 553–566 (English translation of *Izv. Akad. Nauk SSSR* 48 (1984); Crossref record, DOI 10.1070/im1985v024n03abeh001251, verified on 17 September 2026), and from K. Mann, “Rigidity and flexibility of group actions on the circle”, arXiv:1510.00728, and S.-h. Kim, T. Koberda, M. Mj, *Flexibility of Group Actions on the Circle*, LNM 2231, Springer 2019 (arXiv:1610.04098), both reopened and verified on 17 September 2026: Solodov’s theorem classifies the groups of homeomorphisms of the line without global fixed points in terms of the order structure, generalising Hölder’s theorem; the pair of works by Mann and by Kim–Koberda–Mj describes the contemporary picture, in which the actions of surface groups with maximal Euler class are rigid (locally unique up to semiconjugacy, according to Matsumoto’s rigidity theorem and Mann’s results), whereas for non-maximal Euler classes one finds uncountable families of semiconjugacy classes with pairwise disjoint rotation spectra. What is assumed: all the proofs of these results.

Remark 15.22: Thompson’s group T . A notable example of a subgroup of $\text{Homeo}_+(\mathbb{T})$ is **Thompson’s group T** : the homeomorphisms of the circle $\mathbb{R}/\mathbb{Z}$ which are piecewise affine, with slopes powers of 2 and breakpoints in $\mathbb{Z}[1/2]/\mathbb{Z}$. It is a simple, infinite and finitely presented group; by Corollary 15.3 it does not act freely, being non-abelian. Ghys and Sergiescu (“Sur un groupe remarquable de difféomorphismes du cercle”, *Comment. Math. Helv.* 62 (1987), 185–239, Crossref record DOI 10.1007/bf02564445 verified on 17 September 2026) show that T is conjugate to a subgroup of $\text{Diff}_+^\infty(\mathbb{T})$: such a “combinatorial” group therefore admits a smooth realisation. This statement is cited as *panorama* and is not used elsewhere.

15.11. Figures

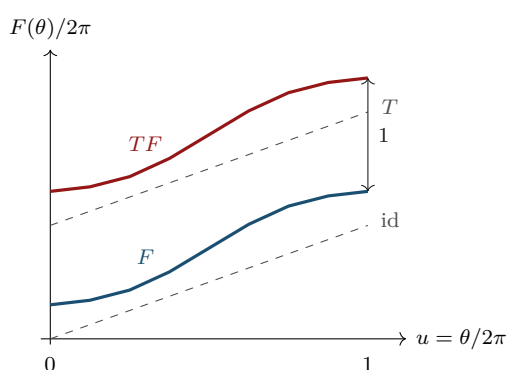


Figure 15.1: A lift $F \in \widetilde{\text{Homeo}}$ (in blue) and its translate $TF = FT$ (in red), with the diagonal and the line $u \mapsto u+1$ dashed. The graph of F over one period determines F everywhere, by the relation $F(\theta + 2\pi) = F(\theta) + 2\pi$ (Definition 15.2); the translates $T^k F$, $k \in \mathbb{Z}$, are the lifts of the *same* circle homeomorphism and form the fibre of p (Proposition 15.1). The vertical distance between two consecutive translates is exactly 1 in these units: it is the generator $\mathbb{Z}$ of the central extension. Coordinates computed by hand from the closed formula $F(u) = u + 0.3 - 0.12 \sin(2\pi u)$ and verified by the script `fig15_coordinate_tikz.py` (Appendix B).

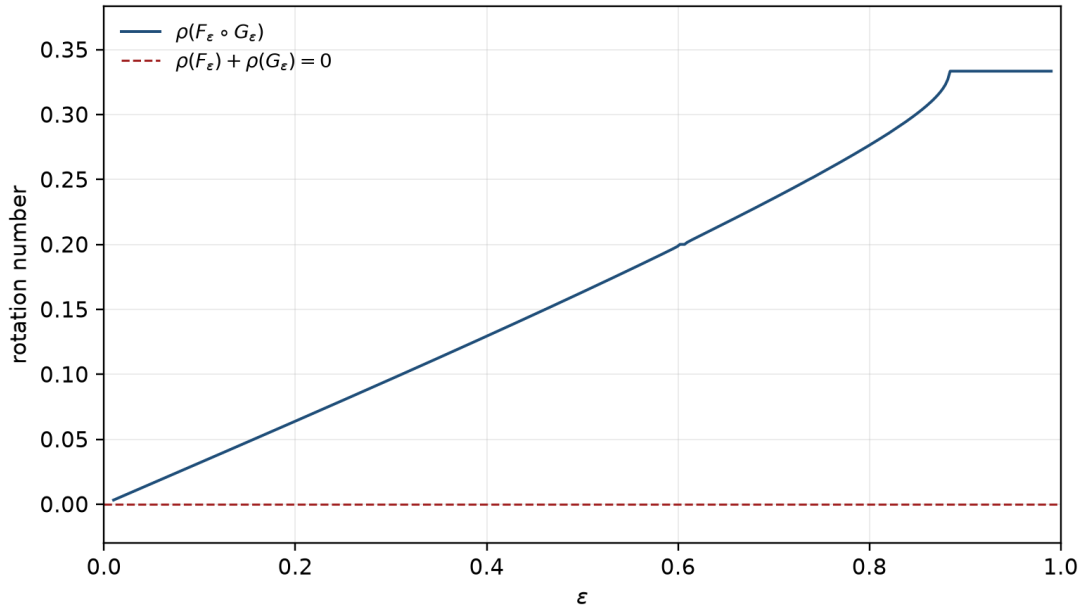


Figure 15.2: The defect of the quasi-morphism, computed numerically: the curve is $\varepsilon \mapsto \rho(F_\varepsilon \circ G_\varepsilon)$ for the pair of Example 15.1, while the dashed line is $\rho(F_\varepsilon) + \rho(G_\varepsilon) = 0$, an *exact* value because both maps have a fixed point. The gap between the two curves is the failure of additivity: zero only in the limit $\varepsilon \rightarrow 0^+$ (where both maps tend to the identity), equal to 0.080053 for $\varepsilon = 0.25$, to 0.163239 for $\varepsilon = 0.50$, to 0.255189 for $\varepsilon = 0.75$ and to $1/3$ *exactly* for $\varepsilon = 0.90$ (periodic orbit of type $(1, 3)$, Example 15.1(5)). The curve has the structure of a devil's staircase: besides the long final step at height $1/3$ one sees a much shorter one at height $1/5$, which the block [C15-1] localises for $\varepsilon \in [0.6020, 0.6070]$ by searching for the periodic orbits of type $(1, 5)$ with step 0.001 ✓. Script `fig15_quasimorfismo.py`, Appendix B; 400 values of ε , 2.4×10^5 iterations each, maximum discrepancy between N and $2N$ iterations equal to 1.3×10^{-6} .

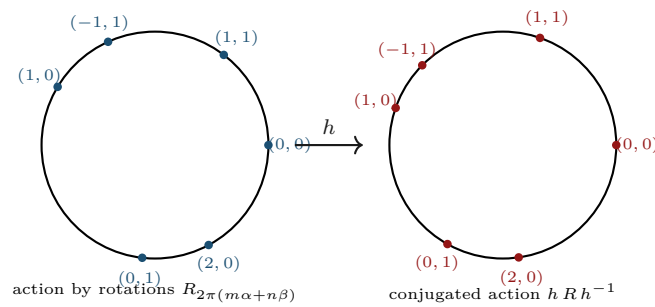


Figure 15.3: The free action of $\mathbb{Z}^2$ of Example 15.2: on the left the orbit of the point 0 under the group of rotations $\{R_{2\pi(m\alpha+n\beta)}\}$ with $\alpha = \sqrt{2} - 1$, $\beta = \sqrt{3} - 1$ (six elements (m, n) indicated); on the right the same orbit transported by the homeomorphism $h(s) = s + 0.4 \sin s$, that is, the orbit of the conjugated action $h R h^{-1}$, which is no longer a group of rotations although it remains free. Theorem 15.3 says that every free action is of this form, up to a monotone map: the conjugacy h is precisely the required semiconjugacy, and here it is a homeomorphism because $\Lambda = \mathbb{Z} + \alpha\mathbb{Z} + \beta\mathbb{Z}$ is dense. Angles computed (in degrees) and verified by the script `fig15_coordinate_tikz.py` (Appendix B).

$$\begin{array}{ccccccc}
1 & \longrightarrow & \mathbb{Z} = \langle T \rangle & \longrightarrow & \widetilde{\text{Homeo}} & \xrightarrow{p} & \text{Homeo}_+(\mathbb{T}) \longrightarrow 1 \\
& & & & & \nwarrow \text{dashed} & \uparrow \varrho \\
& & & & & a_i \mapsto A_i, b_i \mapsto B_i & \Gamma_g = \langle a_i, b_i \mid \prod_i [a_i, b_i] = 1 \rangle \\
& & \prod_{i=1}^g [A_i, B_i] = T^{e(\varrho)} & & & & \\
& & |e(\varrho)| \leq 2g - 2 & & & &
\end{array}$$

Figure 15.4: The scheme of the central extension and of the Euler class. The dashed lifting $a_i \mapsto A_i, b_i \mapsto B_i$ is *not* in general a homomorphism of Γ_g : the relation $\prod_i [a_i, b_i] = 1$ lifts to $\prod_i [A_i, B_i] = T^{e(\varrho)}$, and the integer $e(\varrho)$ — independent of the choices, Proposition 15.5 — measures exactly the obstruction. The lifting is a homomorphism if and only if $e(\varrho) = 0$. The Milnor–Wood inequality (Theorem 15.5) bounds the obstruction by $|e(\varrho)| \leq |\chi(\Sigma_g)| = 2g - 2$.

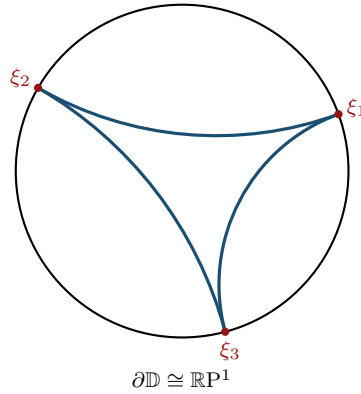


Figure 15.5: Scheme of the action of $\text{PSL}(2, \mathbb{R})$ on the boundary of the Poincaré disc: three hyperbolic geodesics (arcs orthogonal to the boundary) with their ideal points ξ_1, ξ_2, ξ_3 on the circle $\partial\mathbb{D} \cong \mathbb{RP}^1$. A hyperbolic element translates along a geodesic and fixes its two ideal points, hence has $\rho = 0$; a parabolic element fixes a single ideal point, and still has $\rho = 0$; an elliptic element fixes no boundary point and has $\rho \neq 0$ (Proposition 15.7). Centres and radii of the three arcs computed in closed form and verified (orthogonality $|c|^2 - r^2 = 1$ to within 10^{-6}) by the script `fig15_coordinate_tikz.py` (Appendix B). The hyperbolic geometry of the disc is developed in Volume V, Chapter 4.

15.12. Assigned exercises

The answers are in the Teacher's key (Section 15.15).

Exercise 15.1: The defect for a pair with prescribed fixed points. *Goal:* to repeat the analysis of Example 15.1 with two maps whose fixed points are closer together. *Estimated time:* 40 minutes. *Type:* symbolic/numerical.

Let $F(\theta) = \theta + \varepsilon(1 - \cos \theta)$ and $G_\gamma(\theta) = \theta + \varepsilon(1 - \cos(\theta - \gamma))$, with $0 < \varepsilon < 1$ and $\gamma \in (0, 2\pi)$.

- Check that $F, G_\gamma \in \widetilde{\text{Homeo}}$ and that $\rho(F) = \rho(G_\gamma) = 0$ for every γ .
- Prove that $\rho(FG_\gamma) > 0$ for every $\gamma \in (0, 2\pi)$ and that $\rho(FG_0) = 0$.
- Compute numerically $\rho(FG_\gamma)$ for $\varepsilon = 0.5$ and $\gamma = \pi/4, \pi/2, \pi$, and check that the maximum occurs at $\gamma = \pi$ (the case of Example 15.1).

Exercise 15.2: The commutator lemma put to the test. *Goal:* to verify the reformulation of Remark 15.14 and to look for the worst case. *Estimated time:* 45 minutes. *Type:* computational/proof-based.

- Prove the identity $C \circ (BA) = AB$ for $C = [A, B]$, and deduce from it that $a(C) = \min_\theta [(AB)(\theta) - (BA)(\theta)]/2\pi$.
- Take $A = R_c^\dagger$ a rotation and $B(\theta) = \theta + 2\pi d - \varepsilon \sin \theta$: compute $(AB)(\theta) - (BA)(\theta)$ in closed form and check directly that the minimum is negative.
- Generate 10^3 random pairs as in block [C15-3] and estimate $\sup a([A, B])$: can one get close to 1? Why does Lemma 15.9 not exclude getting close to it?

Exercise 15.3: Realisable Euler classes in genus 3. *Goal:* to apply Theorem 15.5 and Proposition 15.5. *Estimated time:* 30 minutes. *Type:* theoretical.

- List the possible values of $e(\varrho)$ for $\varrho : \Gamma_3 \rightarrow \text{Homeo}_+(\mathbb{T})$.
- Exhibit, for each of the values $e = 0$ and $e = \pm 1$ with $g = 3$, an explicit representation which realises it, using representations which factor through $\mathbb{Z}$ or through a cyclic quotient. (Hint for $e = 0$: Proposition 15.5(iii).)
- Explain why the case $e = \pm 4$ is impossible even though $4 < 2g = 6$.

15.13. Worked exercises

Exercise 15.4: The case $g = 1$ of the Milnor–Wood inequality without the oscillation lemma. *Type:* proof-based. Prove directly that, if $A, B \in \widetilde{\text{Homeo}}$ and $[A, B] = T^n$, then $n = 0$, using *only* the invariance of the rotation number under conjugacy.

Solution. Strategy. Rewrite the relation so that the rotation number of one and the same element appears twice with different values.

Step 1. From $[A, B] = ABA^{-1}B^{-1} = T^n$ it follows, multiplying on the right by B , that $ABA^{-1} = T^n B$.

Step 2. By Lemma 15.3(iv), $\rho(ABA^{-1}) = \rho(B)$.

Step 3. By Lemma 15.3(iii), $\rho(T^n B) = n + \rho(B)$.

Step 4. Equating, $\rho(B) = n + \rho(B)$, hence $n = 0$.

Independent check. The result coincides with Theorem 15.5 for $g = 1$, which gives $|n| \leq 2 \cdot 1 - 2 = 0$ ✓. It also coincides with the geometric interpretation: a circle bundle over the torus admits a transverse foliation if and only if its Euler number is zero, and $\chi(\Sigma_1) = 0$.

Interpretation. This proof is the “miniature version” of the commutator Lemma 15.9: there one assumes $[A, B] \succeq T$ and obtains $\rho(B) \geq 1 + \rho(B)$; here one assumes $[A, B] = T^n$ and obtains $\rho(B) = n + \rho(B)$. The same mechanism — an element conjugate to its own translate — in two different doses.

Typical error to avoid. Deducing from $|\rho([A, B])| \leq D(\rho)$ that n is small: this gives only $|n| \leq D(\rho)$, that is, $|n| \leq 2$ with the defect proved here (Theorem 15.1), not $n = 0$. The additional information used above is that $[A, B]$ is *exactly* T^n , not merely an element with that rotation number.

Extension. The same argument shows that T^n is a commutator in $\widetilde{\text{Homeo}}$ only for $n = 0$, whereas $\rho([A, B])$ can be different from 0 for suitable A, B : being T^n is much more restrictive than having rotation number n .

Exercise 15.5: A group which cannot act freely: the infinite dihedral group. *Type:* diagnostic/theoretical. Let $D_\infty = \langle r, s \mid s^2 = 1, srs^{-1} = r^{-1} \rangle$ be the infinite dihedral group. Determine whether D_∞ can act freely on $\mathbb{T}$ by orientation-preserving homeomorphisms.

Solution. Strategy. Apply Corollary 15.2: a free action forces the group to be abelian; it is therefore enough to determine whether D_∞ is abelian.

Step 1. In D_∞ one has $srs^{-1} = r^{-1}$. If the group were abelian, $srs^{-1} = r$, hence $r = r^{-1}$, that is, $r^2 = 1$. But in the presentation r has infinite order (the subgroup $\langle r \rangle$ is isomorphic to $\mathbb{Z}$, since $D_\infty \cong \mathbb{Z} \rtimes \mathbb{Z}/2$). Hence D_∞ is not abelian.

Step 2. By Corollary 15.2, no free action of D_∞ on $\mathbb{T}$ by orientation-preserving homeomorphisms can exist.

Independent check. One also sees this without Hölder, using the rotation number: if the action were free, by Corollary 15.2 $\bar{\rho}$ would be an injective homomorphism $D_\infty \rightarrow \mathbb{R}/\mathbb{Z}$; but $\mathbb{R}/\mathbb{Z}$ is abelian, hence $\bar{\rho}$ would vanish on every commutator, in particular on $[s, r] = srs^{-1}r^{-1} = r^{-2} \neq 1$, against injectivity $\checkmark$. The same argument is, in miniature, the proof of Corollary 15.3.

Interpretation. The obstruction is neither the “size” of the group nor the presence of torsion: D_∞ contains $\mathbb{Z}$ (which acts freely, by an irrational rotation) and $\mathbb{Z}/2$ (which acts freely, by the half-turn rotation). It is the way the two pieces are glued — with inversion — that makes the free action impossible. Note that a free action of $\mathbb{Z}/2$ on $\mathbb{T}$ must be the half-turn rotation up to conjugacy: it is the case $q = 2$ of Theorem 15.3(ii).

Typical error to avoid. Confusing “free action” with “faithful action”. D_∞ acts faithfully on $\mathbb{T}$: it is enough to let r act as an irrational rotation and s as a reflection — but the reflection *reverses* the orientation and does not lie in $\text{Homeo}_+(\mathbb{T})$. Within $\text{Homeo}_+(\mathbb{T})$, D_∞ admits less obvious faithful actions, but never free ones.

Extension. Show, by the same method, that the discrete Heisenberg group (three generators, nilpotent non-abelian) does not act freely on $\mathbb{T}$, even though it is torsion-free and soluble.

Exercise 15.6: Computation of $e(\varrho)$ for a representation which factors. *Type:* symbolic. Let $g \geq 1$ and let $\varrho : \Gamma_g \rightarrow \text{Homeo}_+(\mathbb{T})$ be a representation which factors through the abelianisation $H_1(\Gamma_g) \cong \mathbb{Z}^{2g}$, that is, $\varrho = \sigma \circ \pi_{\text{ab}}$ with $\pi_{\text{ab}} : \Gamma_g \rightarrow \mathbb{Z}^{2g}$ the projection and $\sigma : \mathbb{Z}^{2g} \rightarrow \text{Homeo}_+(\mathbb{T})$ an arbitrary homomorphism. Compute $e(\varrho)$.

Solution. Strategy. Show that ϱ lifts, and use Proposition 15.5(iii).

Step 1. The image $\sigma(\mathbb{Z}^{2g}) \leq \text{Homeo}_+(\mathbb{T})$ is abelian, hence so is its lift $\tilde{H} := p^{-1}(\sigma(\mathbb{Z}^{2g}))$? *Careful:* this is not automatic, and indeed we shall not use it. Let us proceed directly

instead.

Step 2. Let A_i, B_i be arbitrary lifts of $\varrho(a_i) = \sigma(\bar{a}_i)$ and $\varrho(b_i) = \sigma(\bar{b}_i)$, where $\bar{a}_i, \bar{b}_i$ are the standard generators of $\mathbb{Z}^{2g}$. By Definition 15.9, $\prod_i [A_i, B_i] = T^{e(\varrho)}$.

Step 3. Set $n := e(\varrho)$. The element T^n is a product of g commutators; on the other hand Theorem 15.5 gives $|n| \leq 2g - 2$. This is not enough to conclude $n = 0$ for $g \geq 2$: the factorisation hypothesis is needed.

Step 4. We use Proposition 15.5(iii) the other way round. The representation ϱ factors through $\mathbb{Z}^{2g}$; the group $\mathbb{Z}^{2g}$ is free as an abelian group, but this by itself does not produce a lift. Let us consider, however, the composition $\Gamma_g \xrightarrow{\pi_{\text{ab}}} \mathbb{Z}^{2g} \xrightarrow{\sigma} \text{Homeo}_+(\mathbb{T})$ and observe that, for every choice of lifts $\hat{A}_i, \hat{B}_i$ of the $\sigma(\bar{a}_i), \sigma(\bar{b}_i)$, the commutators $[\hat{A}_i, \hat{B}_i]$ belong to the kernel of p — because $\sigma(\bar{a}_i)$ and $\sigma(\bar{b}_i)$ commute in $\text{Homeo}_+(\mathbb{T})$, $\mathbb{Z}^{2g}$ being abelian — hence $[\hat{A}_i, \hat{B}_i] = T^{m_i}$ for certain integers m_i , and $e(\varrho) = \sum_{i=1}^g m_i$.

Step 5. But each m_i is zero: $[\hat{A}_i, \hat{B}_i] = T^{m_i}$ is exactly the situation of Worked exercise 15.4, which gives $m_i = 0$. Hence

$$e(\varrho) = 0.$$

Independent check. With $e(\varrho) = 0$, Proposition 15.5(iii) says that ϱ lifts to $\widetilde{\text{Homeo}}$: and indeed one checks directly that $a_i \mapsto \hat{A}_i, b_i \mapsto \hat{B}_i$ defines a homomorphism, because $\prod_i [\hat{A}_i, \hat{B}_i] = T^0 = \text{id}$ ✓.

Interpretation. The “interesting” representations, those with non-zero Euler class, cannot factor through the abelianisation: the Euler class is an invariant which sees the non-abelian part of Γ_g . This is why the extreme case $|e| = 2g - 2$ is realised by faithful and discrete representations (Remark 15.16).

Typical error to avoid. Step 1 shows the temptation to avoid: deducing from “ $\sigma(\mathbb{Z}^{2g})$ is abelian” that its lift is abelian. This is false in general — a central extension of an abelian group is not necessarily abelian, and the Euler class measures precisely this failure. The correct argument uses only that the *individual* commutators of the lifts lie in the kernel of p .

Extension. Show that, more generally, $e(\varrho) = 0$ whenever the images $\varrho(a_i)$ and $\varrho(b_i)$ commute pairwise for each fixed i (global commutativity is not needed).

15.14. Self-assessment quiz

Eight items, arranged in order of competence; all the answers, commented one by one, are in the Teacher’s key (Section 15.15).

- (Definition) An element F belongs to $\widetilde{\text{Homeo}}$ (Definition 15.2) if and only if
 - F is continuous and 2π -periodic
 - F is continuous, strictly increasing and $F(\theta + 2\pi) = F(\theta) + 2\pi$
 - F is differentiable with $F' > 0$
 - F is a homeomorphism of $\mathbb{T}$
- (Hypothesis) Hölder’s theorem (Theorem 15.2) requires the group to act
 - faithfully
 - freely, that is, with no fixed points for every element $\neq \text{id}$
 - minimally
 - by diffeomorphisms of class C^2
- (Symbolic computation) In the oscillation lemma (Lemma 15.2), the inequality $b(F) - a(F) < 1$ follows from
 - the periodicity of φ_F alone
 - the periodicity of φ_F **and** the monotonicity of F , choosing the maximum and minimum points within one period
 - Fekete’s theorem
 - the invariance of ρ under conjugacy

-
4. (Numerical computation) For the pair of Example 15.1 with $\varepsilon = 0.50$, the value of $\rho(F_\varepsilon G_\varepsilon)$ is approximately
 A) 0 B) 0.080053 C) 0.163239 D) 0.333333
 5. (Proof step) In the commutator lemma (Lemma 15.9), the contradiction arises from
 A) $\rho(B) \geq 1 + \rho(B)$, obtained by comparing $\rho(ABA^{-1}) = \rho(B)$ with $\rho(TB) = 1 + \rho(B)$
 B) $b(C) - a(C) \geq 1$ C) $e(\varrho) \notin \mathbb{Z}$ D) the non-existence of the limit defining ρ
 6. (Error diagnosis) A student deduces the inequality $|e(\varrho)| \leq 2g - 2$ from the defect of the quasi-morphism by writing $|e| \leq g \cdot D(\rho) + (g - 1)D(\rho)$. The error is
 A) ρ is not a quasi-morphism B) that chain gives at most $|e| \leq 2g - 1$ (with $D = 1$) or $|e| \leq 4g - 2$ (with $D = 2$), never $2g - 2$: the last unit is gained with Lemma 15.9 and the integrality of e C) $e(\varrho)$ is not well defined D) $\chi(\Sigma_g) = 2g - 2$ is wrong
 7. (Transfer) The fact that $\mathrm{PSL}(2, \mathbb{Z})$ cannot act freely on $\mathbb{T}$ follows
 A) from the Milnor–Wood inequality B) from Hölder’s theorem, because $\mathrm{PSL}(2, \mathbb{Z}) \cong \mathbb{Z}/2 * \mathbb{Z}/3$ is not abelian C) from Denjoy’s theorem D) from the fact that $\mathrm{PSL}(2, \mathbb{Z})$ is countable
 8. (Source and metacognition) The optimal form $|e(\varrho)| \leq 2g - 2$ of the Milnor–Wood inequality, in this chapter,
 A) is imported from Wood (1971) B) is proved in full, while what is imported is the *optimality* (that is, the realisability of $|e| = 2g - 2$) and the optimal value 1 of the defect of ρ C) is deduced from Ghys’ theorem D) holds only for actions of class C^2

15.15. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

- 1) **B.** *Rationale:* Definition 15.2. *Distractors:* A confuses “ $F(\theta + 2\pi) = F(\theta) + 2\pi$ ” with periodicity (which is the property of φ_F , not of F); C adds a regularity which is not required; D describes $\mathrm{Homeo}_+(\mathbb{T})$, not its cover. *Diagnosed error:* exchanging the map with its lift. *Source:* Definition 15.2. *Prerequisite:* lifts (Chapter 4, Definition 4.1). *Remedial work:* reread Figure 15.1.
- 2) **B.** *Rationale:* Definition 15.5 and Theorem 15.2. *Distractors:* A is weaker (faithful $\neq$ free: see Worked exercise 15.5); C and D do not appear among the hypotheses. *Diagnosed error:* confusing freeness and faithfulness. *Source:* Remark 15.5. *Prerequisite:* the notion of stabiliser. *Remedial work:* redo Worked exercise 15.5.
- 3) **B.** *Rationale:* in the proof of Lemma 15.2 periodicity serves to choose $\theta_0 \leq \theta_1 < \theta_0 + 2\pi$, monotonicity to write $F(\theta_0) \leq F(\theta_1) < F(\theta_0) + 2\pi$. *Distractors:* A is insufficient (Remark 15.3 gives a non-monotone counterexample with oscillation 5); C, D are extraneous. *Diagnosed error:* failing to identify which hypothesis carries the weight. *Source:* Lemma 15.2 and Remark 15.3. *Prerequisite:* maximum and minimum of a continuous periodic function. *Remedial work:* redo the proof keeping track of where monotonicity is used.
- 4) **C.** *Rationale:* block [C15–1]: $\rho(F_{0.5}G_{0.5}) = 0.163239$. *Distractors:* A is the *wrong* value one would obtain by believing ρ additive; B is the value for $\varepsilon = 0.25$; D is the one for $\varepsilon = 0.90$. *Diagnosed error:* reading the value at the wrong parameter index, or assuming additivity. *Source:* Example 15.1(4), Figure 15.2. *Prerequisite:* iterative computation of ρ

(Chapter 4). *Remedial work*: rerun the script `verifica_cap15_vol4.py`.

5) **A.** *Rationale*: it is literally the final chain of the proof of Lemma 15.9. *Distractors*: B is the oscillation lemma, which in the commutator argument is used afterwards, not for the contradiction; C, D are extraneous. *Diagnosed error*: failing to isolate the mechanism “an element conjugate to its own translate”. *Source*: Lemma 15.9, Worked exercise 15.4. *Prerequisite*: invariance under conjugacy (Lemma 15.3(iv)). *Remedial work*: redo Worked exercise 15.4.

6) **B.** *Rationale*: Remark 15.15. *Distractors*: A is false (Theorem 15.1); C is false (Proposition 15.5); D is false ($\chi(\Sigma_g) = 2 - 2g$, and $|\chi| = 2g - 2$). *Diagnosed error*: believing that an “almost additive” invariant suffices to give the optimal estimate. *Source*: Remarks 15.4 and 15.15. *Prerequisite*: definition of defect. *Remedial work*: reread the four steps of the proof of Theorem 15.5, counting where each inequality enters.

7) **B.** *Rationale*: Corollary 15.3. *Distractors*: A concerns actions of surface groups, not freeness; C concerns a single diffeomorphism; D is irrelevant ($\mathbb{Z}^2$ is countable too and acts freely). *Diagnosed error*: looking for the obstruction in cardinality or regularity instead of in the algebra. *Source*: Corollary 15.3. *Prerequisite*: $\mathrm{PSL}(2, \mathbb{Z})$ non-abelian (Example 15.3: S and ST_{mod} do not commute). *Remedial work*: check by hand that $SST_{\mathrm{mod}} \neq ST_{\mathrm{mod}}S$.

8) **B.** *Rationale*: see the infobox of Section 15.10 and Remark 15.4. *Distractors*: A is historically true (Wood proved it first) but false as a description of *this* chapter, where the proof is given in full; C is false (Ghys’ theorem is not used); D is false (no regularity hypothesis). *Diagnosed error*: failing to distinguish between the historical paternity of a result and the status (proved/imported) which it has in the text one is reading. *Source*: header of the source file and Section 15.10. *Prerequisite*: none. *Remedial work*: reread the register of imports.

BEST PRACTICE

Assigned exercises — answers.

15.1 (fixed points at distance γ). (a) $F(\theta + 2\pi) = F(\theta) + 2\pi$ and $F' = 1 + \varepsilon \sin \theta > 0$; similarly $G'_\gamma = 1 + \varepsilon \sin(\theta - \gamma) > 0$. $F(0) = 0$ and $G_\gamma(\gamma) = \gamma$: both have a fixed point, hence $\rho = 0$ (Lemma 15.4(ii)). (b) $(FG_\gamma)(\theta) - \theta = \varepsilon(1 - \cos(\theta - \gamma)) + \varepsilon(1 - \cos G_\gamma(\theta)) \geq 0$, with simultaneous vanishing only if $\theta \equiv \gamma$ and $G_\gamma(\theta) \equiv 0$; but $G_\gamma(\gamma) = \gamma \not\equiv 0$ for $\gamma \in (0, 2\pi)$: hence the displacement is everywhere positive and $\rho(FG_\gamma) > 0$. For $\gamma = 0$ the two maps coincide, $FG_0 = F^2$ has the fixed point 0 and $\rho = 0$. (c) Computed values (block [C15–6] of the script `verifica_cap15_vol4.py`, $\varepsilon = 0.5$, 2×10^5 iterations): $\rho = 0.000000$ for $\gamma = 0$; 0.031265 for $\gamma = \pi/8$; 0.061846 for $\gamma = \pi/4$; 0.115415 for $\gamma = \pi/2$; 0.150938 for $\gamma = 3\pi/4$; 0.163239 for $\gamma = \pi$. The function is increasing on $(0, \pi]$ with maximum at $\gamma = \pi$, equal to the value of Example 15.1, and is symmetric about $\gamma = \pi$ (0.150939 for $\gamma = 5\pi/4$ and 0.115415 for $\gamma = 3\pi/2$, which coincide with the values at $3\pi/4$ and $\pi/2$ to within 10^{-6}). The qualitative reason: the closer the two fixed points are, the more the composition “almost” preserves a common fixed point.

15.2 (commutator lemma). (a) $C \circ (BA) = ABA^{-1}B^{-1}BA = AB$; setting $y = (BA)(\theta)$, $C(y) - y = (AB)(\theta) - (BA)(\theta)$, and $\theta \mapsto (BA)(\theta)$ is bijective on $\mathbb{R}$: the minimum of $C(y) - y$ coincides with the minimum of $(AB)(\theta) - (BA)(\theta)$ (Remark 15.14). (b) With $A = R_c^\sharp$: $(AB)(\theta) = \theta + 2\pi d - \varepsilon \sin \theta + 2\pi c$ and $(BA)(\theta) = \theta + 2\pi c + 2\pi d - \varepsilon \sin(\theta + 2\pi c)$, hence $(AB)(\theta) - (BA)(\theta) = \varepsilon[\sin(\theta + 2\pi c) - \sin \theta]$, a function with zero mean over one period: its minimum is ≤ 0 , and strictly negative unless $c \in \mathbb{Z}$ (in which case A is a power of T , which commutes with everything). (c) One does not get close to 1 with “reasonable” pairs: in block [C15–3] the maximum over 200 pairs is -0.018567 . Lemma 15.9 gives only a strict inequality $a([A, B]) < 1$, with no uniform margin, and indeed the margin must be able to tend to 0: were it not so, the same chain of the proof of Theorem 15.5 would give

$|e(\varrho)| < 2g - 2$, against optimality (Remark 15.16).

15.3 (genus 3). (a) $|e| \leq 2 \cdot 3 - 2 = 4$, hence $e \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. (b) For $e = 0$: any ϱ which factors through the abelianisation (Worked exercise 15.6), for instance $a_1 \mapsto R_{2\pi\alpha}$ and all the other generators $\mapsto \text{id}$. For $e = \pm 1$ a non-abelian representation is needed; existence is guaranteed by Wood's theorem (Remark 15.16), which asserts the realisability of every integer with $|e| \leq 2g - 2$: the chapter does not construct an explicit one, and this part of the exercise is to be solved by citing the imported statement, not by inventing a construction. (c) $4 \leq 2g - 2 = 4$: the value $e = \pm 4$ is *admissible*, not impossible. The question contains a false premise, and is the occasion to check that $2g - 2 = 4$ and not $2g = 6$ is the threshold: the correct answer is to point out the error in the question.

15.16. Chapter formulary

Central extension

$$1 \rightarrow \mathbb{Z} = \langle T \rangle \rightarrow \widetilde{\text{Homeo}} \xrightarrow{p} \text{Homeo}_+(\mathbb{T}) \rightarrow 1,$$

$$T(\theta) = \theta + 2\pi \text{ central}, \quad \widetilde{\text{Homeo}} \text{ torsion-free}$$

Normalised displacement

$$\varphi_F(\theta) = \frac{F(\theta) - \theta}{2\pi}, \quad a(F) = \min \varphi_F, \quad b(F) = \max \varphi_F, \quad \boxed{b(F) - a(F) < 1}$$

$$a(F) \leq \rho(F) \leq b(F), \quad a(F) + a(G) \leq a(FG),$$

$$b(FG) \leq b(F) + b(G), \quad a(FG) \leq a(F) + b(G)$$

Rotation number on $\widetilde{\text{Homeo}}$

$$\rho(T^k F) = k + \rho(F), \quad \rho(F^n) = n\rho(F),$$

$$\rho(HFH^{-1}) = \rho(F), \quad F \preceq G \Rightarrow \rho(F) \leq \rho(G)$$

$$|\rho(FG) - \rho(F) - \rho(G)| < 2 \quad (\text{proved}); \quad FG = GF \Rightarrow \rho(FG) = \rho(F) + \rho(G)$$

Hölder's theorem

$$G \text{ acts freely} \Rightarrow G \text{ abelian}, \quad \Phi : G \hookrightarrow (\mathbb{R}, +) \text{ ordered},$$

$$\bar{\rho} : G \hookrightarrow \mathbb{R}/\mathbb{Z}, \quad \text{action semiconjugate to rotations}$$

Euler class and Milnor–Wood

$$\prod_{i=1}^g [A_i, B_i] = T^{e(\varrho)}, \quad e(\varrho) \in \mathbb{Z} \text{ well defined},$$

$$e(\varrho) = 0 \iff \varrho \text{ lifts}$$

$$a([A, B]) < 1 \quad (\text{commutator lemma}) \quad \implies \quad \boxed{|e(\varrho)| \leq |\chi(\Sigma_g)| = 2g - 2}$$

Euler cocycle

$$F_f \circ F_g = T^{c(f,g)} \circ F_{fg}, \quad c(f, g) \in \{0, 1\},$$

$$c(f, g) + c(fg, h) = c(g, h) + c(f, gh)$$

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CHAPTER 16

Twist maps of the annulus and Aubry–Mather theory

Chapter 4 showed that every orientation-preserving circle homeomorphism carries with it a single number, the rotation number, and that this number governs the dynamics. This chapter takes the next step, the one that Poincaré indicated in 1912 as the key to the three-body problem: one goes up one dimension, from the circle to the **annulus** $\mathbb{T} \times \mathbb{R}$, and studies the maps which *twist* the annulus — those in which the angular velocity increases with the radius. For these maps the rotation number is no longer a single global invariant: every orbit has its own. The question then becomes: *which* rotation numbers are realised, and by which orbits?

The answer, built over seventy years by Poincaré, Birkhoff, Aubry and Mather, is one of the most beautiful in conservative dynamics: *all* the rotation numbers compatible with the twist are realised, and they are realised by orbits of a very particular kind — **ordered** orbits, which arrange themselves on the annulus exactly as the orbits of a rigid rotation do on the circle, and which lie on the graph of a Lipschitz function. When the rotation number is irrational, this graph is either an entire invariant curve (the KAM surface which survives) or a Cantor set with gaps (the *cantorus*, what is left of the curve after it has broken). The transition between the two regimes, in the Chirikov standard map, occurs at Greene’s numerical threshold $k_c \approx 0.9716$ and is the central phenomenon of transport theory in Hamiltonian systems.

The method is variational, and for this reason profoundly different from that of Chapter 6: the orbits are not sought by perturbing a known solution, but are *minimised*. The very chain of atoms in a periodic potential which physicists call the Frenkel–Kontorova model produces, as configurations of minimal energy, the orbits of the twist map. The technical linchpin of the whole construction is a single elementary lemma, **Aubry’s lemma**: two minimising configurations cross at most once. From it follow the ordering, the rotation number, the Lipschitz graph, the Aubry–Mather theorem.

INFO

Level and prerequisites. Level **D** core, with Sections 16.7 (Aubry–Mather theorem) and 16.8 (Mather’s β function) of level **E**; Sections 16.1–16.3 (annulus, variational principle, standard map) are accessible at level **C** and are self-sufficient for a reader who wants only the guiding example and the numerical computations. Prerequisites within the treatise: the rotation number of a circle homeomorphism, Fekete’s lemma and the Poincaré classification with the dichotomy “dense orbit or Cantor set” (Volume IV, Chapter 4, Theorems 4.1, 4.3 and 4.4); continued fractions and the convergents of φ (Volume IV, Chapter 3); differential forms of degree 1 and the exact differential on a manifold (Volume III); the implicit function theorem and sequential compactness in $\mathbb{R}^N$ (Volume II). Remedial work: a reader who is not familiar with the notion of an exact symplectic map may assume directly the hypotheses **(H1)–(H4)** of Definition 16.4, which are all that is needed from Section 16.2 onwards; Theorem 16.1 serves only to explain where those hypotheses come from.

16.1. The annulus, twist maps, the generating function

Throughout this chapter $\mathbb{T} := \mathbb{R}/\mathbb{Z}$ (note the normalisation to period 1, different from the period- 2π one of Chapters 3–4: it is the standard convention in the literature on twist maps, and it makes the rotation number directly comparable with the coordinate y). Whenever we invoke a result from Chapter 4 we shall do so after the affine rescaling $\xi \mapsto 2\pi\xi$, which conjugates $\mathbb{R}/\mathbb{Z}$ with $\mathbb{R}/2\pi\mathbb{Z}$, sends rotations to rotations and leaves rotation numbers unchanged.

Definition 16.1: Annulus, cylinder and lift. The **cylinder** is $\mathcal{A} := \mathbb{T} \times \mathbb{R}$, with coordinates (x, y) , $x \in \mathbb{T}$; its **universal cover** is $\tilde{\mathcal{A}} = \mathbb{R} \times \mathbb{R}$ with the projection $\pi(x, y) = (x \bmod 1, y)$. A **compact annulus** is $\mathbb{T} \times [a, b]$. A homeomorphism $f : \mathcal{A} \rightarrow \mathcal{A}$ preserving the orientation and the two ends (that is, homotopic to the identity) admits a **lift** $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (X(x, y), Y(x, y))$, continuous, with $\pi \circ F = f \circ \pi$, uniquely determined up to the substitution $F \mapsto F + (m, 0)$, $m \in \mathbb{Z}$; it satisfies

$$X(x+1, y) = X(x, y) + 1, \quad Y(x+1, y) = Y(x, y) \quad \text{for every } (x, y) \in \mathbb{R}^2.$$

(The existence and uniqueness up to integer translations of the lift are proved in full in Volume III, Chapter 12, for continuous circle maps; the case of the cylinder reduces to that one by considering $\mathbb{R}^2 \rightarrow \mathcal{A}$ and observing that $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{T} \times \mathbb{R}$ is again a covering with group $\mathbb{Z}$.)

Definition 16.2: Positive monotone twist map. Let $f : \mathcal{A} \rightarrow \mathcal{A}$ be a C^1 diffeomorphism as above, with lift $F = (X, Y)$. We say that f is a **positive monotone twist map** with twist constant $\delta > 0$ if

$$\frac{\partial X}{\partial y}(x, y) \geq \delta > 0 \quad \text{for every } (x, y) \in \mathbb{R}^2,$$

and if moreover, for each fixed x , $y \mapsto X(x, y)$ is surjective onto $\mathbb{R}$ (*infinite twist*). The condition says that a vertical segment $\{x\} \times [y_1, y_2]$ is sent to a curve which moves forward by at least $\delta(y_2 - y_1)$ as one passes from the bottom to the top: the map *twists* the annulus, and does so in a strictly monotone way (Figure 16.1).

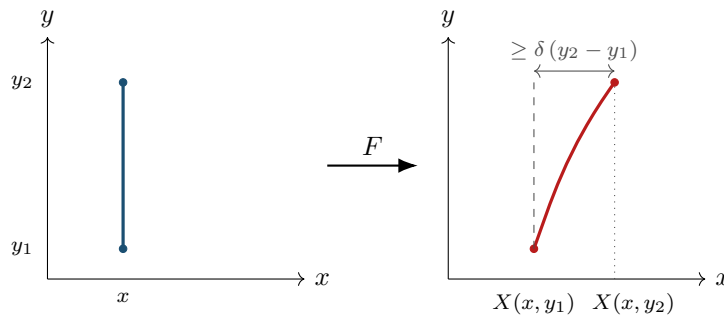


Figure 16.1: The twist condition (Definition 16.2): the vertical segment $\{x\} \times [y_1, y_2]$ (blue) is sent to a curve (red) whose first coordinate increases strictly with y , by at least $\delta(y_2 - y_1)$. The curve drawn is the image under the parametrisation $t \mapsto (x + 0.25t + 0.55t^2/2.6, t)$, $t \in [0.4; 2.6]$, for which $\partial X/\partial y = 0.25 + 1.1t/2.6 \geq 0.25 = \delta$ on the whole interval.

Proposition 16.1: The twist makes $(x, y) \mapsto (x, x')$ a change of coordinates. Let f be a positive monotone twist map with lift $F = (X, Y)$ of class C^1 . Then

$$\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad \Phi(x, y) := (x, X(x, y))$$

is a C^1 diffeomorphism of $\mathbb{R}^2$ onto itself, and commutes with the translation $(x, x') \mapsto (x + 1, x' + 1)$.

Proof. Injectivity and surjectivity. Φ preserves the first coordinate, so it is enough to work with x fixed: the map $y \mapsto X(x, y)$ is strictly increasing (its derivative is $\geq \delta > 0$) and surjective onto $\mathbb{R}$ by the infinite twist hypothesis, hence bijective from $\mathbb{R}$ onto $\mathbb{R}$. Therefore Φ is bijective.

Regularity of the inverse. The Jacobian matrix of Φ is

$$D\Phi = \begin{pmatrix} 1 & 0 \\ \partial_x X & \partial_y X \end{pmatrix}, \quad \det D\Phi = \partial_y X \geq \delta > 0,$$

never zero: by the inverse function theorem (Volume II) Φ^{-1} is C^1 at every point, and since Φ is already a global bijection, Φ^{-1} is C^1 on the whole of $\mathbb{R}^2$.

Equivariance. $\Phi(x + 1, y) = (x + 1, X(x + 1, y)) = (x + 1, X(x, y) + 1)$ by Definition 16.1, which is exactly the diagonal translation applied to $\Phi(x, y)$. $\square$

Definition 16.3: Exact symplectic map of the cylinder. A diffeomorphism f of the cylinder, with lift $F = (X, Y)$, is called **exact symplectic** if there exists $S : \mathbb{R}^2 \rightarrow \mathbb{R}$ of class C^2 , with $S(x + 1, y) = S(x, y)$, such that

$$Y dX - y dx = dS$$

as 1-forms on $\mathbb{R}^2$ (where $Y dX$ is the pull-back $F^*(y dx)$). Such a map preserves the area $dy \wedge dx$, because $d(Y dX - y dx) = 0$ gives $dY \wedge dX = dy \wedge dx$; the exactness condition is however stronger than area preservation alone, and it is the one needed for the variational principle.

Theorem 16.1: Generating function of an exact symplectic twist map. Let f be a positive monotone twist map, exact symplectic, of class C^2 , with lift $F = (X, Y)$ and twist constant $\delta > 0$. Then there exists $h \in C^2(\mathbb{R}^2)$, the **generating function** of F , such that

$$h(x + 1, x' + 1) = h(x, x') \quad \text{for every } (x, x'), \quad (16.1)$$

and such that, for every (x, y) and $(x', y') = F(x, y)$,

$$y = -\partial_1 h(x, x'), \quad y' = \partial_2 h(x, x'). \quad (16.2)$$

Moreover, at every point,

$$\partial_1 \partial_2 h(x, x') = -\left(\frac{\partial X}{\partial y}\right)^{-1} \Big|_{(x, y)} \leq -\frac{1}{\Lambda} < 0 \quad (16.3)$$

whenever $\partial X / \partial y \leq \Lambda$; and the generating function is unique up to an additive constant.

Proof. Construction of h . Let $\Phi(x, y) = (x, X(x, y))$ be the diffeomorphism of Proposition 16.1 and let $\Psi := \Phi^{-1}$, which we write as $\Psi(x, x') = (x, \eta(x, x'))$ with $\eta \in C^1$: by construction $X(x, \eta(x, x')) = x'$, that is, $\eta(x, x')$ is the unique y such that the point (x, y) is sent by F to a point of abscissa x' . We set

$$h(x, x') := S(\Psi(x, x')) = S(x, \eta(x, x')).$$

Since S is of class C^2 and Ψ of class C^1 , h is at least C^1 ; the formula (16.2) which we are about to obtain will show a posteriori that $\partial_1 h$ and $\partial_2 h$ are C^1 (compositions of Y , η and the projections, all C^1), hence $h \in C^2$.

Verification of (16.2). We use (x, x') as coordinates on $\mathbb{R}^2$ by means of Ψ . The *pull-back* along Ψ of the 1-form $Y dX - y dx$ is, by the definition of η (which gives $X \circ \Psi = x'$, the second coordinate) and setting $\widehat{Y}(x, x') := Y(\Psi(x, x'))$,

$$\Psi^*(Y dX - y dx) = \widehat{Y}(x, x') dx' - \eta(x, x') dx.$$

On the other hand $\Psi^*(dS) = d(S \circ \Psi) = dh = \partial_1 h dx + \partial_2 h dx'$. Since $Y dX - y dx = dS$ by hypothesis, the *pull-backs* coincide, and comparing the coefficients of dx and of dx' (which form a basis of the cotangent space at every point):

$$\partial_1 h(x, x') = -\eta(x, x') = -y, \quad \partial_2 h(x, x') = \widehat{Y}(x, x') = y',$$

which is (16.2).

Periodicity (16.1). By Proposition 16.1, $\Psi(x+1, x'+1) = (x+1, \eta(x, x'))$; and $S(x+1, y) = S(x, y)$ by hypothesis. Hence $h(x+1, x'+1) = S(x+1, \eta(x, x')) = S(x, \eta(x, x')) = h(x, x')$.

Sign of the twist (16.3). From $\partial_1 h(x, x') = -\eta(x, x')$ one obtains, differentiating with respect to x' ,

$$\partial_2 \partial_1 h(x, x') = -\frac{\partial \eta}{\partial x'}(x, x').$$

But $\eta(x, \cdot)$ is the inverse of $X(x, \cdot)$, hence by the rule for differentiating an inverse

$$\frac{\partial \eta}{\partial x'}(x, x') = \left(\frac{\partial X}{\partial y}(x, y) \right)^{-1} \Big|_{y=\eta(x, x')} \in \left[\frac{1}{\Lambda}, \frac{1}{\delta} \right]$$

when $\delta \leq \partial X / \partial y \leq \Lambda$. Schwarz's theorem on mixed second derivatives ($h \in C^2$) gives $\partial_1 \partial_2 h = \partial_2 \partial_1 h$, and (16.3) follows.

Uniqueness. If h_1, h_2 both satisfy (16.2), then $d(h_1 - h_2) = 0$ on the connected set $\mathbb{R}^2$, hence $h_1 - h_2$ is constant. $\square$

Remark 16.1: What we shall really use: the hypotheses (H1)–(H4). Theorem 16.1 is the only point of the chapter where symplectic geometry and the inverse function theorem appear. From here onwards we forget the map and work only with the function h : all the subsequent proofs use exclusively the four properties listed in Definition 16.4, and this is precisely the gain of the variational point of view of Aubry and of Mather. A reader who prefers to start from the hypotheses may skip without loss to the next paragraph.

Definition 16.4: Admissible generating class (H1)–(H4). A function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ belongs to the **admissible generating class** if:

- (H1) $h \in C^2(\mathbb{R}^2)$ and $h(x+1, x'+1) = h(x, x')$ for every (x, x') (*diagonal periodicity*);
- (H2) there exists $\delta > 0$ with $\partial_1 \partial_2 h(x, x') \leq -\delta$ for every (x, x') (*uniform twist*);
- (H3) for every x the map $x' \mapsto \partial_1 h(x, x')$ is surjective onto $\mathbb{R}$, and h is *coercive*: $h(x, x') \rightarrow +\infty$ as $|x' - x| \rightarrow \infty$, uniformly in x ;
- (H4) $C_1 := \sup |\partial_1^2 h| < \infty$ and $C_2 := \sup |\partial_2^2 h| < \infty$ (*bounded pure second derivatives*).

We set $L := \max(C_1, C_2)$. By (H1) all four conditions are invariant under the diagonal translation, and the suprema in (H2) and (H4) may be taken over the strip $\{0 \leq x \leq 1\}$.

Remark 16.2: What we import and what we do not. The whole chapter works on the cylinder $\mathbb{T} \times \mathbb{R}$ and assumes directly the hypotheses (H1)–(H4), which for the standard map are verified one by one in Example 16.1: there is in this section no imported result. A limitation of scope must however be stated explicitly: a twist map defined on a compact annulus $\mathbb{T} \times [a, b]$ does not satisfy the infinite twist required by Definition 16.2, and in order to

fall within the framework adopted here it must first be extended to the whole cylinder. The extension procedure is classical and is carried out in the reference monographs (C. Golé, *Symplectic Twist Maps: Global Variational Techniques, Advanced Series in Nonlinear Dynamics* **18**, World Scientific, 2001; A. Katok, B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems, Cambridge University Press, 1995, chapter 13*), but it is not reproduced here: the statements of this chapter are therefore to be read as statements about the cylinder with (H1)–(H4), not as statements about the compact annulus.

16.2. The variational principle and the Frenkel–Kontorova chain

Definition 16.5: Configurations, segments and action. A **configuration** is a sequence $x = (x_i)_{i \in \mathbb{Z}}$ of real numbers. Given integers $m < n$, the **segment** $(x_m, \dots, x_n)$ has **action**

$$W_{m,n}(x_m, \dots, x_n) := \sum_{i=m}^{n-1} h(x_i, x_{i+1}).$$

A configuration x is called:

- (a) **stationary** if for every $i \in \mathbb{Z}$ the **Euler–Lagrange equation**

$$\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0 \quad (16.4)$$

holds;

- (b) **minimal** (or *minimising*) if for every $m < n$ and every finite sequence $(\xi_m, \dots, \xi_n)$ with $\xi_m = x_m$ and $\xi_n = x_n$ one has $W_{m,n}(x_m, \dots, x_n) \leq W_{m,n}(\xi_m, \dots, \xi_n)$.

Theorem 16.2: Variational principle: orbits = stationary configurations. Let h be of the admissible class and F the map generated by it according to (16.2). A configuration $x = (x_i)$ is stationary if and only if, setting

$$y_i := -\partial_1 h(x_i, x_{i+1}), \quad (16.5)$$

the sequence $((x_i, y_i))_{i \in \mathbb{Z}}$ is an orbit of F , that is, $F(x_i, y_i) = (x_{i+1}, y_{i+1})$ for every i . In particular every minimal configuration is stationary.

Proof. ($\Rightarrow$) Let x be stationary and y_i defined by (16.5). By the formula (16.2) which defines F , the point (x_i, y_i) with $y_i = -\partial_1 h(x_i, x_{i+1})$ has as image the point of abscissa x_{i+1} and ordinate $\partial_2 h(x_i, x_{i+1})$: hence $F(x_i, y_i) = (x_{i+1}, \partial_2 h(x_i, x_{i+1}))$. It remains to show that $\partial_2 h(x_i, x_{i+1}) = y_{i+1} = -\partial_1 h(x_{i+1}, x_{i+2})$: but this is exactly equation (16.4) written at the index $i + 1$.

($\Leftarrow$) If $((x_i, y_i))$ is an orbit of F and y_i is given by (16.5), then by (16.2) the second coordinate of $F(x_i, y_i)$ is $\partial_2 h(x_i, x_{i+1})$ and one must have $y_{i+1} = -\partial_1 h(x_{i+1}, x_{i+2})$: adding, one obtains (16.4) at the index $i + 1$, for every i , that is, at all indices.

Minimal $\Rightarrow$ stationary. Let x be minimal and let $i \in \mathbb{Z}$. Take $m = i - 1$, $n = i + 1$ and the family of competitors $\xi(t) := (x_{i-1}, x_i + t, x_{i+1})$, $t \in \mathbb{R}$, which has the same endpoints as (x_{i-1}, x_i, x_{i+1}) . The function

$$t \mapsto W_{i-1,i+1}(\xi(t)) = h(x_{i-1}, x_i + t) + h(x_i + t, x_{i+1})$$

is of class C^1 (indeed C^2 , by (H1)) and has, by minimality, a minimum at $t = 0$; hence its derivative at 0 vanishes:

$$\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0,$$

which is (16.4). Since i was arbitrary, x is stationary. $\square$

Lemma 16.1: Determinacy: two consecutive values determine the orbit. Let x and $\bar{x}$ be stationary configurations with $x_{i_0} = \bar{x}_{i_0}$ and $x_{i_0+1} = \bar{x}_{i_0+1}$ for some index i_0 . Then $x = \bar{x}$.

Proof. We show that (x_{i-1}, x_i) determines x_{i+1} uniquely and that (x_i, x_{i+1}) determines x_{i-1} uniquely; the conclusion follows by induction in both directions starting from i_0 .

Forwards. Equation (16.4) at the index i can be rewritten as

$$\partial_1 h(x_i, x_{i+1}) = -\partial_2 h(x_{i-1}, x_i),$$

whose right-hand side is a number determined by (x_{i-1}, x_i) . The left-hand side, as a function of x_{i+1} with x_i fixed, has derivative $\partial_2 \partial_1 h(x_i, x_{i+1}) \leq -\delta < 0$ by (H2): it is therefore strictly decreasing, hence injective; by (H3) it is also surjective onto $\mathbb{R}$. There exists therefore one and only one x_{i+1} solving the equation.

Backwards. Symmetrically, (16.4) at the index i can be rewritten as $\partial_2 h(x_{i-1}, x_i) = -\partial_1 h(x_i, x_{i+1})$; the left-hand side, as a function of x_{i-1} with x_i fixed, has derivative $\partial_1 \partial_2 h(x_{i-1}, x_i) \leq -\delta < 0$, hence it is strictly decreasing and injective. Existence is automatic (the backward orbit exists because F is invertible, being generated by a map of the cylinder), and uniqueness is what is needed. $\square$

Remark 16.3: The physical reading: the Frenkel–Kontorova chain. In the typical case in which

$$h(x, x') = \frac{1}{2}(x' - x)^2 + V(x), \quad V(x+1) = V(x),$$

the action $W_{m,n} = \sum_i [\frac{1}{2}(x_{i+1} - x_i)^2 + V(x_i)]$ is the energy of a one-dimensional chain of atoms connected by harmonic springs of zero rest length and elastic constant 1, immersed in an external periodic potential V of period 1 (the crystal substrate). A minimal configuration is a ground state of the chain; equation (16.4) becomes

$$x_{i+1} - 2x_i + x_{i-1} = V'(x_i), \tag{16.6}$$

a “discrete Newton equation”. This is the **discrete Frenkel–Kontorova model** studied by Aubry and Le Daeron (1983), who found in it — independently of Mather and with solid-state physics motivations — the same invariant sets. The rotation number of the configuration is, in this language, the mean density of the atoms with respect to the period of the substrate; the irrational case corresponds to a chain incommensurate with the substrate, and the invariant curve/cantorus transition is Aubry’s celebrated transition by breaking of analyticity, that is, the passage between a chain which can slide without friction and one which remains pinned (Figure 16.2).

16.3. The Chirikov standard map

Definition 16.6: Chirikov standard map. For $k \geq 0$ the **standard map** (or *kicked rotor*) is the map of the cylinder $\mathcal{A} = \mathbb{T} \times \mathbb{R}$ whose lift is

$$F_k(x, y) = (x + y', y'), \quad y' = y + \frac{k}{2\pi} \sin(2\pi x). \tag{16.7}$$

The parameter $k \geq 0$ measures the strength of the kick; for $k = 0$ one gets the integrable map $(x, y) \mapsto (x + y, y)$, whose circles $y = \text{const}$ are all invariant and carry the rigid rotation of number y .

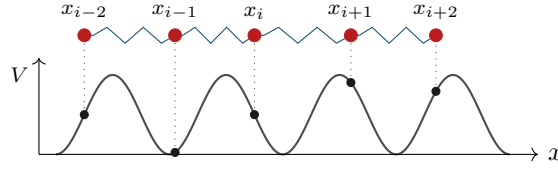


Figure 16.2: The Frenkel–Kontorova chain (Remark 16.3): atoms (red dots) connected by harmonic springs and immersed in the periodic potential $V(x) = 0.35 - 0.35 \cos(2\pi x)$ of period 1 (grey curve; the black dots are the projections of the atoms onto the potential). The abscissae drawn, 0.25; 1.05; 1.75; 2.60; 3.35, give ordinates 0.350; 0.017; 0.350; 0.633; 0.556 respectively. The competition between the elastic energy, which would like equal spacings, and the substrate energy, which would like every atom in a minimum of V , is exactly the competition between the two terms of the action W .

Example 16.1: The generating function of the standard map and the check of (H1)–(H4). We set

$$h_k(x, x') := \frac{1}{2}(x' - x)^2 + V_k(x), \quad V_k(x) := -\frac{k}{(2\pi)^2} \cos(2\pi x). \quad (16.8)$$

Check that h_k generates F_k . One has $\partial_1 h_k = -(x' - x) + V'_k(x)$ and $\partial_2 h_k = x' - x$, with $V'_k(x) = \frac{k}{2\pi} \sin(2\pi x)$. Hence (16.2) gives

$$y = -\partial_1 h_k(x, x') = (x' - x) - V'_k(x), \quad y' = \partial_2 h_k(x, x') = x' - x.$$

From the first, $x' - x = y + V'_k(x)$; substituting in the second, $y' = y + V'_k(x) = y + \frac{k}{2\pi} \sin(2\pi x)$ and $x' = x + y'$: exactly (16.7). ✓

(H1) $h_k \in C^\infty$ and $h_k(x+1, x'+1) = \frac{1}{2}(x' - x)^2 + V_k(x+1) = h_k(x, x')$ because V_k has period 1. **(H2)** $\partial_1 \partial_2 h_k \equiv -1$: the twist is *constant*, $\delta = 1$ (and also $\Lambda = 1$), the most favourable case possible. **(H3)** $\partial_1 h_k(x, \cdot)$ is the affine function $x' \mapsto -(x' - x) + V'_k(x)$, surjective onto $\mathbb{R}$; and $h_k(x, x') \geq \frac{1}{2}(x' - x)^2 - \frac{k}{(2\pi)^2} \rightarrow +\infty$ as $|x' - x| \rightarrow \infty$, uniformly in x . **(H4)** $\partial_1^2 h_k = 1 + V''_k(x) = 1 + k \cos(2\pi x)$, hence $C_1 = 1 + k$; $\partial_2^2 h_k \equiv 1$, hence $C_2 = 1$. Therefore

$$L = \max(C_1, C_2) = 1 + k. \quad (16.9)$$

The Euler–Lagrange equation (16.6) is here

$$x_{i+1} - 2x_i + x_{i-1} = \frac{k}{2\pi} \sin(2\pi x_i). \quad (16.10)$$

Proposition 16.2: Symmetries and fixed points of the standard map. For every $k > 0$: **(i)** F_k preserves the area ($\det DF_k \equiv 1$) and is exact symplectic; **(ii)** F_k commutes with the involution $\sigma(x, y) := (-x, -y)$, that is, $\sigma \circ F_k \circ \sigma = F_k$; **(iii)** the fixed points of F_k on the cylinder are exactly the points $(0, m)$ and $(\frac{1}{2}, m)$ with $m \in \mathbb{Z}$; since DF_k depends only on x , all the points $(0, m)$ have the same differential, and likewise all the $(\frac{1}{2}, m)$: it is therefore enough to study $(0, 0)$ and $(\frac{1}{2}, 0)$; **(iv)** $\text{tr } DF_k(0, 0) = 2 + k > 2$: the point $(0, 0)$ is **hyperbolic** for every $k > 0$; **(v)** $\text{tr } DF_k(\frac{1}{2}, 0) = 2 - k$: the point $(\frac{1}{2}, 0)$ is **elliptic** if and only if $0 < k < 4$, and its eigenvalues are $e^{\pm i\alpha}$ with $\alpha = \arccos(1 - \frac{k}{2})$.

Proof. (i) From (16.7), setting $a(x) := k \cos(2\pi x) = V_k''(x)$,

$$DF_k(x, y) = \begin{pmatrix} 1 + a(x) & 1 \\ a(x) & 1 \end{pmatrix}, \quad \det DF_k = (1 + a) - a = 1. \quad (16.11)$$

For exact symplecticity we set $S(x, y) := h_k(x, X(x, y))$ with $X(x, y) = x + y + V_k'(x)$ (the first coordinate of F_k , see Example 16.1). Then, writing $x' = X(x, y)$ and differentiating,

$$dS = \partial_1 h_k(x, x') dx + \partial_2 h_k(x, x') dx' = -y dx + y' dx' = Y dX - y dx,$$

where (16.2), checked in Example 16.1, has been used. Finally $S(x+1, y) = h_k(x+1, X(x, y) + 1) = h_k(x, X(x, y)) = S(x, y)$ by (H1) and by the periodicity of V_k' : S therefore satisfies all the requirements of Definition 16.3.

(ii) Set $g(x) := \frac{k}{2\pi} \sin(2\pi x)$, an *odd* function. Then

$$F_k(-x, -y) = (-x + w, w), \quad w = -y + g(-x) = -y - g(x) = -(y + g(x)) = -y',$$

hence $F_k \circ \sigma(x, y) = (-x - y', -y')$ and, applying σ a second time,

$$\sigma \circ F_k \circ \sigma(x, y) = (x + y', y') = F_k(x, y).$$

(Note that σ is well defined on the cylinder because $-x$ makes sense in $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and that $\sigma^2 = \text{id}$.)

(iii) A point is fixed on the cylinder if and only if $y' = y$ and $x' \equiv x \pmod{1}$. The first condition, $y + \frac{k}{2\pi} \sin(2\pi x) = y$, gives $\sin(2\pi x) = 0$, that is, $x \in \{0, \frac{1}{2}\}$ in $\mathbb{T}$; the second, $x + y' \equiv x$, gives $y' \in \mathbb{Z}$, and since $y' = y$, $y = m \in \mathbb{Z}$. Conversely every point $(0, m)$ and $(\frac{1}{2}, m)$ satisfies both. From formula (16.11), $DF_k(x, y)$ does not depend on y : the differentials at the points $(0, m)$ all coincide, and likewise those at the points $(\frac{1}{2}, m)$.

(iv) and (v) From (16.11): $\text{tr } DF_k = 2 + a(x)$ with $a(0) = k \cos 0 = k$ and $a(\frac{1}{2}) = k \cos \pi = -k$. Since $\det = 1$, the eigenvalues are the roots of $\lambda^2 - (\text{tr})\lambda + 1 = 0$, hence real and distinct with $|\lambda| \neq 1$ (*hyperbolic* point) if $|\text{tr}| > 2$, complex conjugate of modulus 1 (*elliptic* point) if $|\text{tr}| < 2$. At $(0, 0)$: $\text{tr} = 2 + k > 2$ for every $k > 0$, hyperbolic. At $(\frac{1}{2}, 0)$: $\text{tr} = 2 - k$, and $|2 - k| < 2 \iff 0 < k < 4$; in that case $\lambda = e^{\pm i\alpha}$ with $\cos \alpha = \text{tr}/2 = 1 - k/2$. $\square$

Remark 16.4: Symmetry, reversibility and symmetric orbits. Proposition 16.2(ii) says that $\sigma(x, y) = (-x, -y)$ is a symmetry of F_k (it conjugates it with itself), not a reversor (which would conjugate it with its inverse). The operational consequence which we shall use is this: if (x_i) is a stationary configuration, so is $(-x_i)$ by the symmetry σ alone; combining σ with time reversal $i \mapsto -i$ (a symmetry of the Euler–Lagrange equation (16.6), of which the reversor ι discussed below is the trace on phase space), $(-x_{-i})$ is stationary too, and if the minimising configuration of a given type (p, q) is unique up to integer translations, then it must be invariant under this symmetry — which explains why all the minimising orbits computed in Example 16.4 and in Figure 16.5 turn out to be symmetric about a point. The standard map also possesses a reversor, the involution $\iota(x, y) = (-x, y + \frac{k}{2\pi} \sin(2\pi x))$, which exchanges the two “half steps” into which (16.7) factorises; it is the tool with which Greene constructs the symmetric periodic orbits, but it is not needed for any result of this chapter and is not proved here. See J. D. Meiss, *Rev. Mod. Phys.* **64** (1992), Section II.

Example 16.2: The rotation number of the elliptic island for $k = 1.2$. **Double check** (script `verifica_cap16_vol4.py`, block [C16-1], run on 17 September 2026).

For $k = 1.2$ Proposition 16.2(v) gives $\text{tr } DF_k(\frac{1}{2}, 0) = 0.8$, hence

$$\alpha = \arccos(0.4) = 1.159279 \text{ rad}, \quad \frac{\alpha}{2\pi} = 0.184505.$$

Independent check, without going through the arccosine: the roots of the characteristic polynomial $\lambda^2 - 0.8\lambda + 1$, computed numerically, are $0.400000 \pm 0.916515i$, of modulus 1.000000000000 and argument $0.184505 \times 2\pi$: they agree with the prediction to the sixth digit. ✓ Moreover the Jacobian computed by centred finite differences ($\varepsilon = 10^{-6}$, script `verifica_cap16_vol4.py`) gives $\det DF_k \approx 1$ with a discrepancy of the order of 10^{-10} (limited by the truncation of the finite differences, not by machine precision); using instead the *analytic* Jacobian $DF_k = \begin{pmatrix} 1+a & 1 \\ a & 1 \end{pmatrix}$ on a grid of 101 points (script `fig16_mappa_standard.py`, Figure 16.3) one obtains $\max |\det DF_k - 1| = 4.4 \times 10^{-16}$, at the level of machine error: area preservation is checked numerically in both ways. ✓

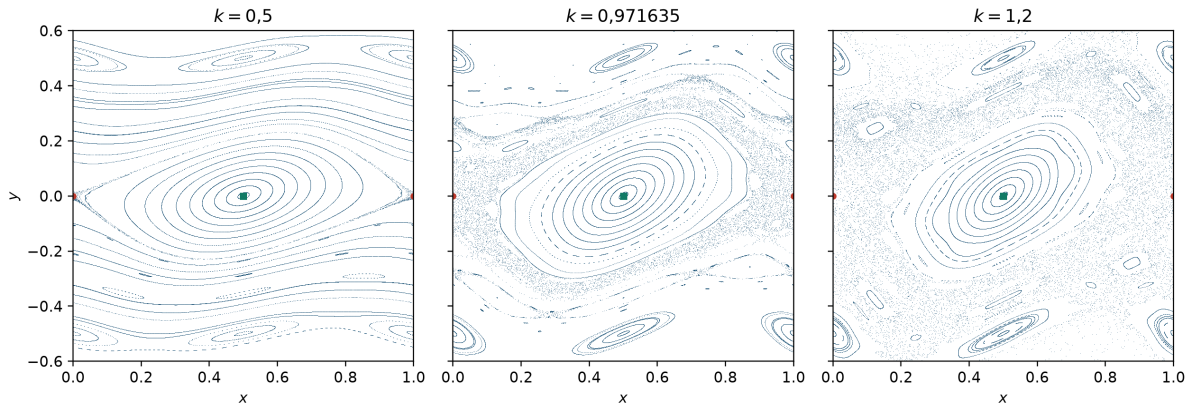


Figure 16.3: Phase portraits of the standard map (16.7) for $k = 0.5$ (below Greene’s threshold), $k = 0.971635$ (at the threshold) and $k = 1.2$ (above the threshold): 50 orbits of 1200 iterates each. The red dots are the hyperbolic fixed point $(0, 0)$, the green squares the elliptic fixed point $(\frac{1}{2}, 0)$ (Proposition 16.2). At $k = 0.5$ the invariant curves which cross the whole cylinder (the wavy horizontal lines) separate the cylinder into regions; at $k = 1.2$ they have all disappeared and the chaotic orbits travel freely in y . Figure from script, *Appendix B*, `fig16_mappa_standard.py`; checks reported by the script: $\max |\det DF_k - 1| \leq 4.4 \times 10^{-16}$ and $\text{tr } DF_k(\frac{1}{2}, 0) = 2 - k$ in all three cases. ✓

16.4. The exchange inequality and Aubry’s lemma

All the theory which follows rests on a single elementary inequality, which translates the twist condition (H2) into a statement about actions.

Lemma 16.2: Exchange inequality. *Let h be of the admissible class (Definition 16.4), with twist constant $\delta > 0$. For every real $a < b$ and every real $c < d$,*

$$h(a, c) + h(b, d) \leq h(a, d) + h(b, c) - \delta(b - a)(d - c) < h(a, d) + h(b, c). \quad (16.12)$$

In words: “exchanging the tails” of two segments which cross makes the total action strictly decrease, and the gain is at least δ times the product of the two “crossing amplitudes”.

Proof. Consider the quantity

$$\Delta := h(a, d) + h(b, c) - h(a, c) - h(b, d) = [h(a, d) - h(a, c)] - [h(b, d) - h(b, c)].$$

By the fundamental theorem of calculus applied to the second variable,

$$h(a, d) - h(a, c) = \int_c^d \partial_2 h(a, t) dt, \quad h(b, d) - h(b, c) = \int_c^d \partial_2 h(b, t) dt,$$

hence

$$\Delta = \int_c^d [\partial_2 h(a, t) - \partial_2 h(b, t)] dt = \int_c^d \left(- \int_a^b \partial_1 \partial_2 h(s, t) ds \right) dt,$$

where in the second equality the fundamental theorem of calculus has been used again, this time in the first variable, for the function $s \mapsto \partial_2 h(s, t)$ (of class C^1 by (H1)). By (H2), $-\partial_1 \partial_2 h(s, t) \geq \delta$ at every point, hence the inner integrand is $\geq \delta$ and

$$\Delta \geq \int_c^d \int_a^b \delta ds dt = \delta (b - a)(d - c) > 0,$$

which is (16.12). □

Example 16.3: The exact gain for the standard map. For $h_k(x, x') = \frac{1}{2}(x' - x)^2 + V_k(x)$ the inequality (16.12) is in fact an *equality*, because the twist is constant ($\partial_1 \partial_2 h_k \equiv -1$, hence $\delta = 1$). Let us verify it in closed form: the potential terms $V_k(a) + V_k(b)$ appear identically on both sides and cancel, and there remains

$$\Delta = \frac{1}{2}[(d - a)^2 + (c - b)^2 - (c - a)^2 - (d - b)^2]$$

$$= \frac{1}{2}[(d - c)(d + c - 2a) + (c - d)(c + d - 2b)] = \frac{1}{2}(d - c)[(d + c - 2a) - (c + d - 2b)] = (b - a)(d - c),$$

having used $(d - a)^2 - (c - a)^2 = (d - c)(d + c - 2a)$ and $(c - b)^2 - (d - b)^2 = (c - d)(c + d - 2b)$.

Independent numerical check (script `fig16_lemma_aubry.py`, run on 17 September 2026): over 200 000 quadruples (a, b, c, d) drawn at random with $a < b, c < d$, the maximum error between $h_k(a, d) + h_k(b, c) - h_k(a, c) - h_k(b, d)$ and $(b - a)(d - c)$ equals 2.4×10^{-14} , that is, machine error. ✓

Definition 16.7: Crossings between two configurations. Let $x = (x_i)$ and $\bar{x} = (\bar{x}_i)$ be two configurations, and let $d_i := x_i - \bar{x}_i$. We say that x and $\bar{x}$ **cross**

- **at the index** i , if $d_i = 0$;
- **in the gap** $(i, i + 1)$, if $d_i d_{i+1} < 0$.

The **number of crossings** is the total number of indices and gaps of this kind.

Lemma 16.3: Transversality of coincidences. Let $x \neq \bar{x}$ be two stationary configurations and let i be an index with $x_i = \bar{x}_i$. Then $d_{i-1} \neq 0$, $d_{i+1} \neq 0$ and $d_{i-1} d_{i+1} < 0$: the two configurations cross at i transversally, passing from one side to the other.

Proof. If $d_{i+1} = 0$, then x and $\bar{x}$ would coincide at the consecutive indices i and $i + 1$, and Lemma 16.1 would give $x = \bar{x}$, which is excluded. Similarly $d_{i-1} \neq 0$ (coincidence at $i - 1$ and i).

Suppose, for a contradiction, $d_{i-1} > 0$ and $d_{i+1} > 0$ (the case of two negative signs is identical, exchanging the roles of x and $\bar{x}$). We write equation (16.4) at the index i for both

configurations, recalling $x_i = \bar{x}_i$:

$$\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0, \quad \partial_2 h(\bar{x}_{i-1}, x_i) + \partial_1 h(x_i, \bar{x}_{i+1}) = 0.$$

Subtracting the second from the first:

$$\underbrace{[\partial_2 h(x_{i-1}, x_i) - \partial_2 h(\bar{x}_{i-1}, x_i)]}_{=:A} + \underbrace{[\partial_1 h(x_i, x_{i+1}) - \partial_1 h(x_i, \bar{x}_{i+1})]}_{=:B} = 0.$$

Now: $s \mapsto \partial_2 h(s, x_i)$ has derivative $\partial_1 \partial_2 h \leq -\delta < 0$, hence it is strictly decreasing; since $x_{i-1} > \bar{x}_{i-1}$, one has $A < 0$, and more precisely $A \leq -\delta d_{i-1}$. Similarly $t \mapsto \partial_1 h(x_i, t)$ has derivative $\partial_2 \partial_1 h \leq -\delta < 0$, is strictly decreasing, and from $x_{i+1} > \bar{x}_{i+1}$ it follows that $B \leq -\delta d_{i+1} < 0$. Hence $A + B \leq -\delta(d_{i-1} + d_{i+1}) < 0$, against $A + B = 0$. Absurd. $\square$

Theorem 16.3: Aubry's lemma: two minimal configurations cross at most once.

Let $x \neq \bar{x}$ be two minimal configurations (Definition 16.5). Then they cross at most once.

Proof. Suppose, for a contradiction, that there are at least two crossings, and let $\sigma_1 < \sigma_2$ be two consecutive crossings (that is, with no other crossing strictly between them); we locate a crossing in a gap $(j, j+1)$ at the position $j + \frac{1}{2}$, so as to be able to compare them. Recall that x and $\bar{x}$ are stationary (Theorem 16.2).

Step 1 (choice of the endpoints $m < n$). Let m be the largest integer $\leq \sigma_1$ with $d_m \neq 0$ and n the smallest integer $\geq \sigma_2$ with $d_n \neq 0$. These exist: if σ_1 is the gap $(j, j+1)$ then $d_j \neq 0$ and $m = j$; if σ_1 is the index j then $d_j = 0$, but $d_{j-1} \neq 0$ by Lemma 16.3 and $m = j - 1$. Symmetrically for n . In both cases there are no integers strictly between m and σ_1 , nor between σ_2 and n .

Step 2 (the two endpoints have the same sign). Every crossing reverses the sign of d : in a gap by definition ($d_j d_{j+1} < 0$), at an index j by Lemma 16.3 ($d_{j-1} d_{j+1} < 0$). Between m and σ_1 and between σ_2 and n there are no other integers; between σ_1 and σ_2 there are no other crossings, hence at all the integers strictly in between the sign of d is constant and non-zero. Travelling along $m \rightarrow \sigma_1 \rightarrow \sigma_2 \rightarrow n$ the sign is reversed exactly twice, hence

$$\text{sgn}(d_m) = \text{sgn}(d_n) =: \varepsilon \in \{+1, -1\}.$$

Exchanging if necessary the names of x and $\bar{x}$ we may suppose $\varepsilon = +1$, that is, $x_m > \bar{x}_m$ and $x_n > \bar{x}_n$.

Step 3 (the exchange). We define, for $m \leq i \leq n$,

$$z_i := \min(x_i, \bar{x}_i), \quad w_i := \max(x_i, \bar{x}_i).$$

By Step 2, $z_m = \bar{x}_m$, $z_n = \bar{x}_n$, $w_m = x_m$, $w_n = x_n$: hence $(z_m, \dots, z_n)$ is an admissible competitor for the segment of $\bar{x}$, and $(w_m, \dots, w_n)$ for that of x . For every gap $(i, i+1)$ with $m \leq i \leq n-1$:

- if $d_i d_{i+1} \geq 0$, the two pairs $\{(z_i, z_{i+1}), (w_i, w_{i+1})\}$ coincide, as a set, with $\{(x_i, x_{i+1}), (\bar{x}_i, \bar{x}_{i+1})\}$, and one has the equality $h(z_i, z_{i+1}) + h(w_i, w_{i+1}) = h(x_i, x_{i+1}) + h(\bar{x}_i, \bar{x}_{i+1})$;
- if $d_i d_{i+1} < 0$, say $x_i < \bar{x}_i$ and $x_{i+1} > \bar{x}_{i+1}$, one applies Lemma 16.2 with $a = x_i < b = \bar{x}_i$ and $c = \bar{x}_{i+1} < d = x_{i+1}$: then $(z_i, z_{i+1}) = (a, c)$, $(w_i, w_{i+1}) = (b, d)$, $(x_i, x_{i+1}) = (a, d)$, $(\bar{x}_i, \bar{x}_{i+1}) = (b, c)$, and (16.12) gives the strict inequality, with gain at least $\delta |d_i| |d_{i+1}|$. (The case $x_i > \bar{x}_i$, $x_{i+1} < \bar{x}_{i+1}$ is identical, exchanging the names.)

Summing over $i = m, \dots, n-1$:

$$W_{m,n}(z) + W_{m,n}(w) \leq W_{m,n}(x) + W_{m,n}(\bar{x}), \quad (16.13)$$

with *strict* inequality if at least one gap in $[m, n-1]$ has $d_i d_{i+1} < 0$.

Step 4 (case A: there is a crossing in a gap). If (16.13) is strict, the minimality of x (with the competitor w , of equal endpoints) and of $\bar{x}$ (with the competitor z) gives

$$W_{m,n}(x) + W_{m,n}(\bar{x}) \leq W_{m,n}(w) + W_{m,n}(z) < W_{m,n}(x) + W_{m,n}(\bar{x}),$$

absurd.

Step 5 (case B: both crossings are at an index). If no gap in $[m, n-1]$ has $d_i d_{i+1} < 0$, then σ_1 and σ_2 are both crossings at an index, and in (16.13) equality holds. From minimality, $W_{m,n}(x) \leq W_{m,n}(w)$ and $W_{m,n}(\bar{x}) \leq W_{m,n}(z)$; adding and comparing with the equality one obtains that both are equalities. In particular $W_{m,n}(z) = W_{m,n}(\bar{x})$ is the *minimum* value of the action among all the segments with endpoints $\bar{x}_m$ and $\bar{x}_n$: hence z is a minimising segment, and — by the same single-point variation argument used in Theorem 16.2 — it satisfies (16.4) at all the interior indices $m+1, \dots, n-1$.

By Step 1, σ_1 is the index $m+1$ (being an index, $m = \sigma_1 - 1$). Then $d_m > 0$ gives $z_m = \bar{x}_m$, and $d_{m+1} = 0$ gives $z_{m+1} = x_{m+1} = \bar{x}_{m+1}$: the two segments z and $\bar{x}$ coincide at the consecutive indices m and $m+1$, and both satisfy (16.4) at all the interior indices. Repeating the forward induction of Lemma 16.1 (which uses only (16.4) one index at a time and the strict monotonicity of $\partial_1 h(x_i, \cdot)$) one obtains $z_i = \bar{x}_i$ for every $i \in [m, n]$, that is, $\min(x_i, \bar{x}_i) = \bar{x}_i$, that is,

$$d_i \geq 0 \quad \text{for every } i \in [m, n].$$

But $\sigma_1 = m+1$ is a crossing at an index, with $m < m+1 < n$ (one has $n \geq \sigma_2 > \sigma_1 = m+1$ and n an integer, hence $n \geq m+2$): by Lemma 16.3, d_m and d_{m+2} are non-zero and of opposite sign, which contradicts $d \geq 0$ on $[m, n] \ni m, m+2$. Absurd.

Since both cases are impossible, two crossings cannot exist. $\square$

***Remark 16.5: Why Aubry's lemma is the heart of the theory.** Theorem 16.3 says that the minimal configurations form a family totally ordered up to one crossing: it is a rigidity property of one-dimensional type, and it is this — and not area preservation, nor regularity — that will produce the ordering of the periodic orbits (Section 16.5), the existence of the rotation number (Section 16.6) and the Lipschitz graph structure (Theorem 16.7). Aubry and Le Daeron discovered it while studying the ground states of the chain (16.6); Mather, independently and in dynamical language, used it as the key to the existence of quasi-periodic orbits.*

16.5. Birkhoff periodic orbits of type (p, q)

Definition 16.8: Configurations of type (p, q) and periodic action. Let $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$. A configuration x is **of type (p, q)** if

$$x_{i+q} = x_i + p \quad \text{for every } i \in \mathbb{Z}.$$

Its **periodic action** is

$$W_{p,q}(x_0, \dots, x_{q-1}) := \sum_{i=0}^{q-1} h(x_i, x_{i+1}), \quad x_q := x_0 + p,$$

a C^2 function of $(x_0, \dots, x_{q-1}) \in \mathbb{R}^q$. If x is stationary of type (p, q) , the corresponding orbit $((x_i, y_i))$ is periodic on the cylinder of period q and makes p turns: $F^q(x_0, y_0) = (x_0 + p, y_0)$.

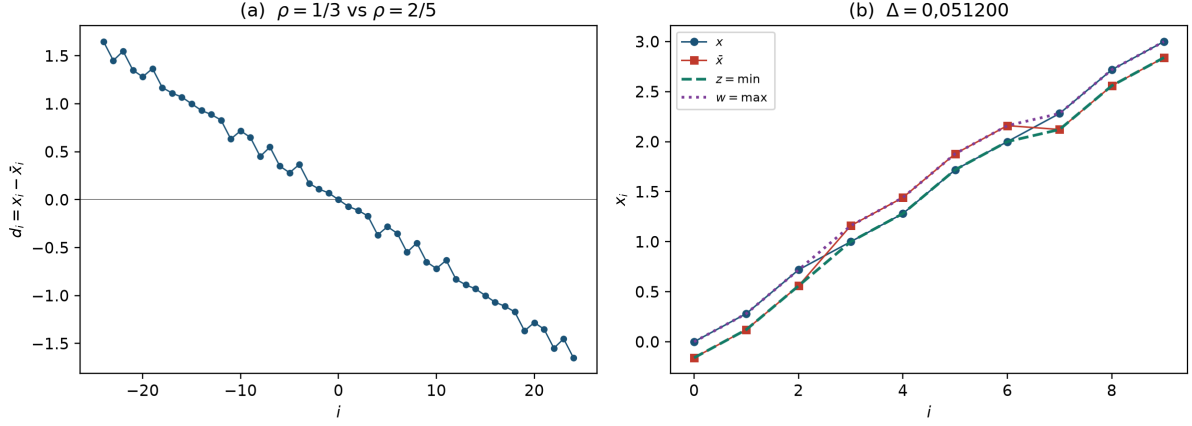


Figure 16.4: (a) The minimising configurations of rotation number $1/3$ and $2/5$ of the standard map at $k = 1$: the difference $d_i = x_i - \bar{x}_i$ decreases from $+\infty$ to $-\infty$ (the mean slope is $1/3 - 2/5 = -1/15$) and vanishes *only once*, at $i = 0$, as Theorem 16.3 predicts; in the window $-24 \leq i \leq 24$ the script counts 0 crossings in a gap and 1 crossing at an index. ✓ (b) The exchange of Step 3: the minimal configuration x (blue) and a competitor $\bar{x}$ (red) built by hand with two strict sign reversals; $z = \min$ (green dashed) and $w = \max$ (purple dotted) have total action smaller by $\Delta = 0.051200$, a value which coincides exactly with the prediction $2\delta\varepsilon^2 = 2 \cdot 1 \cdot (0.16)^2$ of Lemma 16.2. ✓ Figure from script, *Appendix B*, `fig16_lemma_aubry.py`.

Theorem 16.4: Existence of Birkhoff minimising periodic orbits. *Let h be of the admissible class. For every $p \in \mathbb{Z}$ and every $q \in \mathbb{Z}_{>0}$ the function $W_{p,q} : \mathbb{R}^q \rightarrow \mathbb{R}$ attains its minimum. Every minimum point is a stationary configuration of type (p, q) , hence a periodic orbit of the twist map, of rotation number p/q .*

Proof. Step 1 (invariance under integer translation). Let $\mathbf{1} = (1, \dots, 1)$. By (H1), $h(x_i + 1, x_{i+1} + 1) = h(x_i, x_{i+1})$ for every i , and likewise the closing term $h(x_{q-1} + 1, x_0 + 1 + p) = h(x_{q-1}, x_0 + p)$: hence $W_{p,q}(x + \mathbf{1}) = W_{p,q}(x)$. It is therefore sufficient to look for the minimum on the closed set $K := \{x \in \mathbb{R}^q : x_0 \in [0, 1]\}$.

Step 2 (h is bounded below). By (H3) there exists $R_0 > 0$ such that $h(x, x') \geq 0$ if $|x' - x| \geq R_0$; on the compact set $\{0 \leq x \leq 1, |x' - x| \leq R_0\}$ the continuous function h has a minimum $-A \leq 0$, and by diagonal periodicity (H1) the same holds on the whole strip $\{|x' - x| \leq R_0\}$. Hence $h \geq -A$ everywhere.

Step 3 (compactness of the sublevel sets in K). Fix a point $x^{(0)} \in K$ and set $C := W_{p,q}(x^{(0)})$. Let $x \in K$ with $W_{p,q}(x) \leq C$. By Step 2, each of the q summands is $\geq -A$, hence each single summand satisfies

$$h(x_i, x_{i+1}) \leq C + (q - 1)A =: M.$$

By (H3) there exists R_M such that $h(x, x') > M$ as soon as $|x' - x| \geq R_M$; hence $|x_{i+1} - x_i| < R_M$ for every $i = 0, \dots, q - 1$. Since $x_0 \in [0, 1]$, one obtains by induction $|x_i| \leq 1 + qR_M$ for every i : the set $\{x \in K : W_{p,q}(x) \leq C\}$ is bounded, and it is closed because $W_{p,q}$ is continuous and K is closed. It is therefore compact and non-empty (it contains $x^{(0)}$).

Step 4 (minimum). On the compact set of Step 3 the continuous function $W_{p,q}$ attains its minimum, which is the minimum over the whole of K (the points of K outside the compact set have action $> C$) and, by Step 1, over the whole of $\mathbb{R}^q$.

Step 5 (the minimum is an orbit). The minimum is *interior* (the domain is $\mathbb{R}^q$, without constraints), hence it annihilates the gradient:

$$0 = \frac{\partial W_{p,q}}{\partial x_i} = \partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}), \quad i = 0, \dots, q-1,$$

where for $i = 0$ one has used $x_{-1} := x_{q-1} - p$ (differentiating the closing term $h(x_{q-1}, x_0 + p)$ with respect to x_0 one obtains $\partial_2 h(x_{q-1}, x_0 + p) = \partial_2 h(x_{q-1} - p, x_0)$ by (H1)). Extending x to the whole of $\mathbb{Z}$ by $x_{i+q} = x_i + p$, equation (16.4) holds at every index, again by (H1). By Theorem 16.2 this is an orbit, which closes up after q steps advancing by p : its rotation number is p/q . $\square$

Theorem 16.5: Strict ordering of the minimisers of type (p, q) . Let x and $\bar{x}$ be two minimum points of $W_{p,q}$ (extended to configurations of type (p, q)). Then exactly one of the three alternatives holds:

$$x_i < \bar{x}_i \quad \forall i, \quad x_i > \bar{x}_i \quad \forall i, \quad x = \bar{x}.$$

Proof. Set $z_i := \min(x_i, \bar{x}_i)$ and $w_i := \max(x_i, \bar{x}_i)$ for every $i \in \mathbb{Z}$. Both are of type (p, q) : $z_{i+q} = \min(x_{i+q}, \bar{x}_{i+q}) = z_i + p$, and similarly for w . Summing the term-by-term inequality of Step 3 of the proof of Theorem 16.3 over the indices $i = 0, \dots, q-1$ (with the closure $x_q = x_0 + p$, legitimate because all the configurations involved are of type (p, q)) one obtains

$$W_{p,q}(z) + W_{p,q}(w) \leq W_{p,q}(x) + W_{p,q}(\bar{x}) = 2 \min W_{p,q}, \quad (16.14)$$

with strict inequality if there exists $i \in \{0, \dots, q-1\}$ with $d_i d_{i+1} < 0$ (where $d_i := x_i - \bar{x}_i$). But $W_{p,q}(z) \geq \min W_{p,q}$ and $W_{p,q}(w) \geq \min W_{p,q}$: inequality (16.14) cannot therefore be strict, and is indeed an equality with

$$W_{p,q}(z) = W_{p,q}(w) = \min W_{p,q}.$$

It follows that:

- (a) *no crossing in a gap*: $d_i d_{i+1} \geq 0$ for every i (otherwise (16.14) would be strict);
- (b) z and w are themselves minimum points of $W_{p,q}$, hence *stationary* configurations by Theorem 16.4.

Suppose now $x \neq \bar{x}$ and, for a contradiction, that there exists l with $d_l = 0$. By Lemma 16.3 (applicable: $x, \bar{x}$ are stationary and distinct), d_{l-1} and d_{l+1} are non-zero and of opposite sign; suppose $d_{l-1} > 0 > d_{l+1}$ (the other case is symmetric). Then

$$z_{l-1} = \bar{x}_{l-1}, \quad z_l = x_l = \bar{x}_l, \quad z_{l+1} = x_{l+1}.$$

We write (16.4) at the index l for z and for $\bar{x}$:

$$\partial_2 h(\bar{x}_{l-1}, \bar{x}_l) + \partial_1 h(x_l, x_{l+1}) = 0, \quad \partial_2 h(\bar{x}_{l-1}, \bar{x}_l) + \partial_1 h(\bar{x}_l, \bar{x}_{l+1}) = 0.$$

Subtracting, $\partial_1 h(x_l, x_{l+1}) = \partial_1 h(\bar{x}_l, \bar{x}_{l+1}) = \partial_1 h(x_l, \bar{x}_{l+1})$, and since $t \mapsto \partial_1 h(x_l, t)$ is strictly decreasing (H2), one concludes $x_{l+1} = \bar{x}_{l+1}$, that is, $d_{l+1} = 0$: against $d_{l+1} < 0$. Absurd.

Hence, if $x \neq \bar{x}$, one has $d_i \neq 0$ for every i ; together with (a) this gives $d_i d_{i+1} > 0$ for every i , so the sign of d is constant on $\mathbb{Z}$: either $d > 0$ everywhere, or $d < 0$ everywhere. $\square$

Definition 16.9: Birkhoff (ordered) configuration. For $(s, t) \in \mathbb{Z}^2$ we define the **translate** $\tau_{s,t}x$ of the configuration x by

$$(\tau_{s,t}x)_i := x_{i+s} - t.$$

The configuration x is called **Birkhoff** (or *ordered*) if for every $(s, t) \in \mathbb{Z}^2$

$$\tau_{s,t}x \leq x \quad (\text{that is, } x_{i+s} - t \leq x_i \text{ for every } i) \quad \text{or} \quad \tau_{s,t}x \geq x.$$

Equivalently: the set $\{x_i + t\}$ is ordered in $\mathbb{R}$ exactly as the set $\{i\rho + t\}$ would be for a rigid rotation of number ρ (Proposition 16.3).

Corollary 16.1: Every periodic minimiser is Birkhoff. Every minimum point of $W_{p,q}$, extended to a configuration of type (p, q) , is a Birkhoff configuration.

Proof. Let x be a minimiser and $(s, t) \in \mathbb{Z}^2$. The translate $\bar{x} := \tau_{s,t}x$ is still of type (p, q) :

$$\bar{x}_{i+q} = x_{i+q+s} - t = x_{i+s} + p - t = \bar{x}_i + p.$$

It has moreover the same action: by (H1) applied t times,

$$W_{p,q}(\bar{x}) = \sum_{i=0}^{q-1} h(x_{i+s} - t, x_{i+s+1} - t) = \sum_{i=0}^{q-1} h(x_{i+s}, x_{i+s+1}) = \sum_{i'=s}^{s+q-1} h(x_{i'}, x_{i'+1}) = W_{p,q}(x),$$

where the last equality uses the fact that $i \mapsto h(x_i, x_{i+1})$ is q -periodic (because x is of type (p, q) and h satisfies (H1)). Hence $\bar{x}$ is itself a minimiser, and Theorem 16.5 gives $\bar{x} < x$, or $\bar{x} > x$, or $\bar{x} = x$: in all three cases $\tau_{s,t}x \leq x$ or $\tau_{s,t}x \geq x$. $\square$

Example 16.4: The two orbits of type $(1, 2)$ of the standard map at $k = 1$. Explicit orbit (for every k). The points $(0, \frac{1}{2})$ and $(\frac{1}{2}, \frac{1}{2})$ form an orbit of type $(1, 2)$ for every k : from (16.7), $y' = \frac{1}{2} + \frac{k}{2\pi} \sin 0 = \frac{1}{2}$ and $x' = 0 + \frac{1}{2} = \frac{1}{2}$; then $y'' = \frac{1}{2} + \frac{k}{2\pi} \sin \pi = \frac{1}{2}$ and $x'' = \frac{1}{2} + \frac{1}{2} = 1 = 0 + 1$. Its action equals, in closed form,

$$W_{1,2} = \frac{1}{2} \left(\frac{1}{2}\right)^2 + V_k(0) + \frac{1}{2} \left(\frac{1}{2}\right)^2 + V_k\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{k}{(2\pi)^2} + \frac{k}{(2\pi)^2} = \frac{1}{4},$$

independent of k (the two potential contributions cancel because $\cos 0 = -\cos \pi$).

Minimising orbit ($k = 1$). Let us look for it by imposing (16.10) on a configuration $x_0, x_1, x_2 = x_0 + 1$. The two equations are

$$x_2 - 2x_1 + x_0 = \frac{k}{2\pi} \sin(2\pi x_1), \quad x_1 - 2x_0 + x_{-1} = \frac{k}{2\pi} \sin(2\pi x_0), \quad x_{-1} = x_1 - 1,$$

that is, setting $u := x_1 - x_0$: $1 - 2u = \frac{k}{2\pi} \sin(2\pi x_1)$ and $2u - 1 = \frac{k}{2\pi} \sin(2\pi x_0)$. Adding, $\sin(2\pi x_1) = -\sin(2\pi x_0)$, which has two families of solutions: $x_1 = x_0 + \frac{1}{2}$ (which leads back, with $u = \frac{1}{2}$, to $\sin(2\pi x_0) = 0$ and hence to the explicit orbit) and $x_1 = 1 - x_0$, which gives the scalar equation

$$1 - 4x_0 = \frac{k}{2\pi} \sin(2\pi x_0). \tag{16.15}$$

Double check (script `verifica_cap16_vol4.py`, block [C16-2], run on 17 September 2026). For $k = 1$, the numerical minimisation of $W_{1,2}$ over $\mathbb{R}^2$ with 40 random starting points gives

$$x_0 = 0.2113771250, \quad x_1 = 0.7886228750, \quad W_{1,2}^{\min} = 0.243793176214,$$

and the *independent* solution of (16.15) by Brent’s method on $[0.05; 0.45]$ gives $x_0 = 0.2113771250$: the two routes agree to all the printed digits. ✓ A third, dynamical check: iterating (16.7) twice from $(x_0, y_0) = (0.2113771250; 0.4227542500)$ one obtains $(1.211377\dots; 0.422754\dots) = (x_0 + 1, y_0)$ with error $< 10^{-11}$. ✓

Interpretation. $W_{1,2}^{\min} = 0.2438 < 0.25$: the explicit orbit is *not* the minimising one. And indeed the traces of the differential along the period equal, for $k = 1$, $\text{tr } DF^2 = 2 - k^2 = 1$ for the explicit orbit (elliptic) and $\text{tr } DF^2 = 3.018943$ for the minimising one (hyperbolic): the minimum of the action is hyperbolic, the elliptic orbit is another critical point of $W_{1,2}$. Note also that the minimiser satisfies $x_1 = 1 - x_0$, that is, it is invariant under the symmetry σ of Proposition 16.2(ii), as anticipated in Remark 16.4.

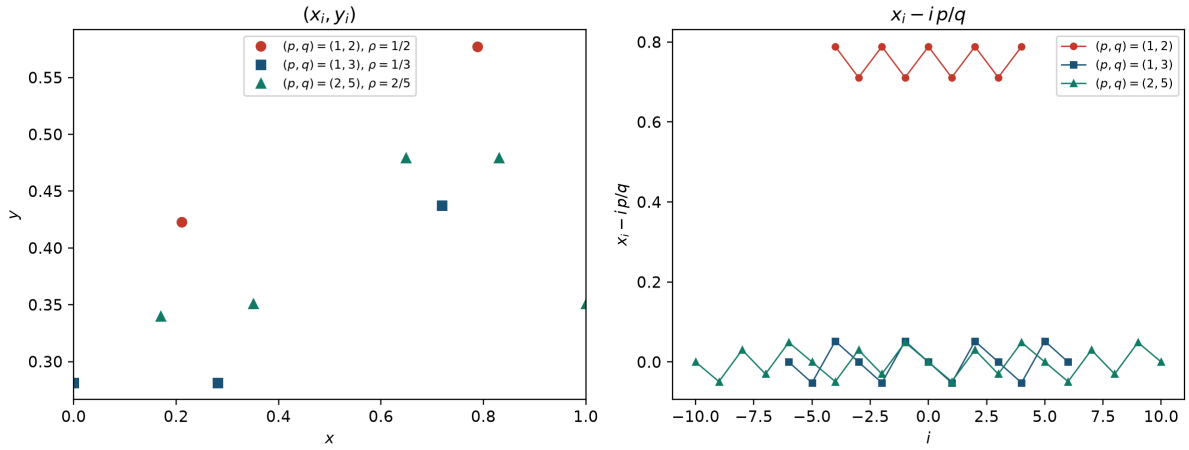


Figure 16.5: Minimising orbits of type $(1,2)$, $(1,3)$ and $(2,5)$ of the standard map at $k = 1$ (Theorem 16.4). Left: the points (x_i, y_i) on the cylinder. Right: the “ordering diagram” $i \mapsto x_i - ip/q$, which remains *bounded*: this is the Birkhoff property (Corollary 16.1), from which follows the existence of the rotation number with the discrepancy ≤ 1 of Theorem 16.6. Check values reported by the script: $W_{1,2}^{\min} = 0.243793176214$, $W_{1,3}^{\min} = 0.159357774396$, $W_{2,5}^{\min} = 0.390740191132$, with $\|\nabla W\| \leq 4.2 \times 10^{-12}$, closure error $F^q(x_0, y_0) - (x_0 + p, y_0)$ smaller than 1.8×10^{-12} , and cyclic order of the $x_i \bmod 1$ identical to that of the rigid rotation of number p/q in all three cases. ✓ Figure from script, *Appendix B*, `fig16_orbite_birkhoff.py`.

Remark 16.6: Poincaré’s last geometric theorem and the second orbit. Theorem 16.4 produces one periodic orbit of type (p, q) : the one of minimal action. There always exists at least a second one, of “min-max” type, and this is the content of the historically founding result of the whole theory.

Imported from H. Poincaré, “Sur un théorème de géométrie”, Rend. Circ. Mat. Palermo **33** (1912), 375–407 (statement, with a proof of particular cases), and from G. D. Birkhoff, “Proof of Poincaré’s Geometric Theorem”, Trans. Amer. Math. Soc. **14** (1913), 14–22 (first complete proof); verified by real web search on 17 September 2026. Statement (Poincaré–Birkhoff theorem). Let f be an area-preserving homeomorphism of the compact annulus $\mathbb{T} \times [a, b]$, homotopic to the identity, sending each boundary component to itself, and whose lift $F = (X, Y)$ satisfies the boundary twist condition

$$X(x, a) - x < 0 < X(x, b) - x \quad \text{for every } x$$

(the two boundaries rotate in opposite directions). Then f has at least two geometrically distinct fixed points. What is assumed: area preservation (continuity alone is not enough: without it the theorem is false), the preservation of each boundary component, and the strict

inequality at the boundary. Why it is imported: Birkhoff's proof is a global topological argument of about nine journal pages, which uses neither the monotone twist nor the variational principle of this chapter, and whose complete reproduction would by itself require more space than the whole of Section 16.5. Applying the statement to f^q composed with q backward turns one obtains two orbits of type (p, q) for every p/q in the twist interval; the minimiser of Theorem 16.4 is one of them.

A panorama of the road leading from 1912 to Aubry–Mather theory can be found in A. Chenciner, “La dynamique au voisinage d'un point fixe elliptique conservatif: de Poincaré et Birkhoff à Aubry et Mather”, Séminaire Bourbaki, exp. 622, Astérisque **121–122** (1985), 147–170 (Numdam).

16.6. Birkhoff configurations: order, rotation, Lipschitz graph

Proposition 16.3: *The monotone map associated with a Birkhoff configuration.* Let x be a stationary Birkhoff configuration and let

$$\mathcal{X} := \{x_i + t : i \in \mathbb{Z}, t \in \mathbb{Z}\} \subset \mathbb{R}.$$

Then the map

$$g : \mathcal{X} \rightarrow \mathcal{X}, \quad g(x_i + t) := x_{i+1} + t$$

is well defined, strictly increasing, bijective, and satisfies $g(\xi + 1) = g(\xi) + 1$ for every $\xi \in \mathcal{X}$.

Proof. Well-definedness. Let $x_i + t = x_j + u$ with $(i, t), (j, u) \in \mathbb{Z}^2$. Set $(s, r) := (j - i, t - u) \in \mathbb{Z}^2$ and $\bar{x} := \tau_{s,r}x$, so that

$$\bar{x}_i = x_{i+s} - r = x_j - (t - u) = (x_i + t - u) - (t - u) = x_i,$$

that is, $d_i := x_i - \bar{x}_i = 0$. Since x is Birkhoff, $\bar{x} \leq x$ or $\bar{x} \geq x$, hence d has constant sign (in the weak sense). If $\bar{x} \neq x$, Lemma 16.3 — applicable because $\bar{x}$ is stationary whenever x is (equation (16.4) is invariant under translation of the index and, thanks to (H1), under subtraction of an integer) — would give d_{i-1} and d_{i+1} non-zero and of opposite sign, incompatible with a constant sign. Hence $\bar{x} = x$, that is, $x_{m+s} - r = x_m$ for every m ; with $m = i + 1$:

$$x_{j+1} - (t - u) = x_{i+1} \iff x_{i+1} + t = x_{j+1} + u,$$

which is exactly the independence of the value of g from the chosen representation.

Strict increase. Let $x_i + t < x_j + u$. With $(s, r) = (j - i, t - u)$ and $\bar{x} = \tau_{s,r}x$ as above one has $\bar{x}_i = x_j - (t - u) = (x_j + u) - t > x_i$, that is, $d_i < 0$; by the Birkhoff property, $\bar{x} \geq x$ everywhere, in particular $\bar{x}_{i+1} \geq x_{i+1}$, that is, $x_{j+1} + u \geq x_{i+1} + t$. Equality is excluded: it would give $d_{i+1} = 0$ with $\bar{x} \neq x$ (because $d_i \neq 0$), and Lemma 16.3 would impose opposite signs for d_i and d_{i+2} , against $d \leq 0$ everywhere. Hence $x_{i+1} + t < x_{j+1} + u$.

Bijectivity and commutation with +1. Strict increase gives injectivity; surjectivity is immediate because $x_i + t = g(x_{i-1} + t)$. Finally $g((x_i + t) + 1) = g(x_i + (t + 1)) = x_{i+1} + t + 1 = g(x_i + t) + 1$. $\square$

Theorem 16.6: *Rotation number of a Birkhoff configuration.* Let x be a Birkhoff configuration and, for $n \geq 1$,

$$a_n := \sup_{i \in \mathbb{Z}} (x_{i+n} - x_i), \quad b_n := \inf_{i \in \mathbb{Z}} (x_{i+n} - x_i).$$

Then a_n, b_n are finite, $a_n - b_n \leq 1$, the limit

$$\rho(x) := \lim_{n \rightarrow \infty} \frac{x_{i+n} - x_i}{n}$$

exists and is finite, does not depend on i and the convergence is uniform in i ; moreover

$$|x_{i+n} - x_i - n\rho(x)| \leq 1 \quad \text{for every } i \in \mathbb{Z}, n \geq 1. \quad (16.16)$$

Proof. Step 1 (key estimate). Let $i, j \in \mathbb{Z}$ and $n \geq 1$. Setting $s := j - i$ and $r := \lfloor x_j - x_i \rfloor \in \mathbb{Z}$, one has $0 \leq x_j - x_i - r < 1$. Consider $\bar{x} := \tau_{s,r}x$: then

$$\bar{x}_i = x_{i+s} - r = x_j - r \geq x_i,$$

hence $\bar{x} \geq x$ at i ; by the Birkhoff property, $\bar{x} \geq x$ everywhere. In particular, at the index $i + n$: $x_{j+n} - r \geq x_{i+n}$. Therefore

$$x_{j+n} - x_j = (x_{j+n} - r) - (x_j - r) \geq x_{i+n} - (x_j - r) = (x_{i+n} - x_i) - (x_j - x_i - r) > (x_{i+n} - x_i) - 1.$$

Step 2 (finiteness and $a_n - b_n \leq 1$). With i fixed, Step 1 gives $x_{j+n} - x_j > (x_{i+n} - x_i) - 1$ for every j , hence $b_n \geq (x_{i+n} - x_i) - 1 > -\infty$. Exchanging the roles of i and j in Step 1 one obtains, with j fixed, $x_{i+n} - x_i < (x_{j+n} - x_j) + 1$ for every i , hence $a_n \leq (x_{j+n} - x_j) + 1 < +\infty$. Taking then the supremum over i and the infimum over j in the inequality of Step 1: $b_n \geq a_n - 1$, that is, $a_n - b_n \leq 1$.

Step 3 (subadditivity and Fekete). For every i and $m, n \geq 1$,

$$x_{i+m+n} - x_i = (x_{i+m+n} - x_{i+n}) + (x_{i+n} - x_i) \leq a_m + a_n,$$

and taking the supremum over i : $a_{m+n} \leq a_m + a_n$. By Fekete's lemma (Volume IV, Chapter 4, Lemma 4.1) the limit $\lim_n a_n/n$ exists and equals $\inf_n a_n/n =: \rho$. Symmetrically $b_{m+n} \geq b_m + b_n$, that is, $(-b_n)$ is subadditive, and $\lim_n b_n/n = \sup_n b_n/n$ exists. From $a_n - 1 \leq b_n \leq a_n$, dividing by n and passing to the limit, the two limits coincide and equal ρ . The value ρ is finite: by superadditivity $b_n \geq nb_1$, hence $a_n/n \geq b_n/n \geq b_1 > -\infty$, and $a_n/n \leq a_1 < +\infty$.

Step 4 (uniformity and discrepancy ≤ 1). By Step 3, $\rho = \inf_n a_n/n$ and $\rho = \sup_n b_n/n$; hence $a_n/n \geq \rho$ and $b_n/n \leq \rho$ for every n , that is, $b_n \leq n\rho \leq a_n$. Hence, for every i ,

$$x_{i+n} - x_i - n\rho \in [b_n - n\rho, a_n - n\rho] \subseteq [-(a_n - b_n), a_n - b_n] \subseteq [-1, 1],$$

which is (16.16); dividing by n one obtains $|(x_{i+n} - x_i)/n - \rho| \leq 1/n$, uniform convergence in i . $\square$

Corollary 16.2: Configurations of type (p, q) . A Birkhoff configuration of type (p, q) has $\rho(x) = p/q$.

Proof. From $x_{i+q} = x_i + p$ it follows by induction that $x_{i+mq} = x_i + mp$, hence $(x_{i+mq} - x_i)/(mq) = p/q$ for every $m \geq 1$. The limit along the subsequence $n = mq$ exists and equals $\rho(x)$ by Theorem 16.6, hence $\rho(x) = p/q$. $\square$

We come to the central structural result of the section: the ordered orbits are not scattered in the cylinder, but all lie on the graph of a Lipschitz function, with a Lipschitz constant depending only on h and not on the orbit.

Theorem 16.7: Birkhoff orbits lie on a Lipschitz graph. Let h be of the admissible class, with $C_1 = \sup |\partial_1^2 h|$, $C_2 = \sup |\partial_2^2 h|$ and $L = \max(C_1, C_2)$. Let x be a stationary Birkhoff configuration, $\mathcal{X}$ and g as in Proposition 16.3, and let

$$Y : \mathcal{X} \rightarrow \mathbb{R}, \quad Y(\xi) := -\partial_1 h(\xi, g(\xi)).$$

Then $Y(x_i + t) = y_i$ for every i, t (where y_i is the momentum (16.5)), and

$$|Y(\eta) - Y(\xi)| \leq L |\eta - \xi| \quad \text{for every } \xi, \eta \in \mathcal{X}. \quad (16.17)$$

In particular the orbit $\{(x_i, y_i)\}$ lies on the graph of an L -Lipschitz function defined on $\mathcal{X}$, which extends (uniquely) to an L -Lipschitz function on the closure $\bar{\mathcal{X}}$.

Proof. Step 0 (the formula for Y and the two expressions of the momentum). By (H1), differentiating $h(x+1, x'+1) = h(x, x')$ with respect to the first variable one obtains $\partial_1 h(x+1, x'+1) = \partial_1 h(x, x')$, and iterating, $\partial_1 h(x+t, x'+t) = \partial_1 h(x, x')$ for every $t \in \mathbb{Z}$; hence $Y(x_i + t) = -\partial_1 h(x_i + t, x_{i+1} + t) = -\partial_1 h(x_i, x_{i+1}) = y_i$. Moreover equation (16.4) at the index i says precisely $y_i = \partial_2 h(x_{i-1}, x_i)$, that is,

$$Y(\xi) = -\partial_1 h(\xi, g(\xi)) = \partial_2 h(g^{-1}(\xi), \xi) \quad \text{for every } \xi \in \mathcal{X}. \quad (16.18)$$

Let now $\xi < \eta$ in $\mathcal{X}$; by Proposition 16.3, $g(\xi) < g(\eta)$ and $g^{-1}(\xi) < g^{-1}(\eta)$.

Step 1 (upper bound, via g^{-1}). Using the second expression in (16.18),

$$Y(\eta) - Y(\xi) = \underbrace{\left[\partial_2 h(g^{-1}(\eta), \eta) - \partial_2 h(g^{-1}(\xi), \eta) \right]}_{=:A} + \underbrace{\left[\partial_2 h(g^{-1}(\xi), \eta) - \partial_2 h(g^{-1}(\xi), \xi) \right]}_{=:B}.$$

One has $A = \int_{g^{-1}(\xi)}^{g^{-1}(\eta)} \partial_1 \partial_2 h(s, \eta) ds \leq -\delta(g^{-1}(\eta) - g^{-1}(\xi)) \leq 0$ by (H2), and $|B| = \left| \int_{\xi}^{\eta} \partial_2^2 h(g^{-1}(\xi), t) dt \right| \leq C_2(\eta - \xi)$ by (H4). Hence

$$Y(\eta) - Y(\xi) \leq C_2(\eta - \xi).$$

Step 2 (lower bound, via g). Using the first expression in (16.18),

$$Y(\eta) - Y(\xi) = - \underbrace{\left[\partial_1 h(\eta, g(\eta)) - \partial_1 h(\xi, g(\eta)) \right]}_{=:A'} - \underbrace{\left[\partial_1 h(\xi, g(\eta)) - \partial_1 h(\xi, g(\xi)) \right]}_{=:B'}.$$

One has $|A'| = \left| \int_{\xi}^{\eta} \partial_1^2 h(s, g(\eta)) ds \right| \leq C_1(\eta - \xi)$ by (H4), and $B' = \int_{g(\xi)}^{g(\eta)} \partial_2 \partial_1 h(\xi, t) dt \leq -\delta(g(\eta) - g(\xi)) \leq 0$ by (H2) and Schwarz's theorem. Hence

$$Y(\eta) - Y(\xi) \geq -C_1(\eta - \xi) + \delta(g(\eta) - g(\xi)) \geq -C_1(\eta - \xi).$$

Step 3 (conclusion). Combining Steps 1 and 2, $-C_1(\eta - \xi) \leq Y(\eta) - Y(\xi) \leq C_2(\eta - \xi)$, that is, $|Y(\eta) - Y(\xi)| \leq L(\eta - \xi)$, which is (16.17). The extension to $\bar{\mathcal{X}}$ is the standard one for uniformly continuous functions on a set dense in its own closure, and it preserves the constant L (passage to the limit in the inequality). $\square$

Corollary 16.3: The Lipschitz constant for the standard map. For the standard map (16.7) every Birkhoff orbit lies on the graph of a Lipschitz function with constant $L = 1 + k$.

Proof. By Example 16.1, $C_1 = \sup_x |1 + k \cos(2\pi x)| = 1 + k$ and $C_2 = 1$; hence $L = \max(1 + k, 1) = 1 + k$ and this is Theorem 16.7. $\square$

Example 16.5: Numerical check of the Lipschitz constant. Double check (script `verifica_cap16_vol4.py`, block [C16-5], run on 17 September 2026). One computes the minimising orbit of type $(89, 144)$ — two consecutive Fibonacci numbers, hence $p/q = 0.618055\dots$, a convergent of $\omega = (\sqrt{5} - 1)/2$ — and one measures $\max_j |\Delta y_j / \Delta x_j|$ over the 144 points ordered by increasing abscissa (with the periodic matching $Y(\xi + 1) = Y(\xi)$). One obtains:

k	0.5	0.971635	1.2
measured constant	0.339530	0.950534	1.344641
theoretical bound $1 + k$	1.500000	1.971635	2.200000

The constraint of Corollary 16.3 is respected in all three cases, with a margin which shrinks as k grows. $\checkmark$ **Independent check** of the validity of the computed orbit: $\|\nabla W_{89,144}\| \leq 7.9 \times 10^{-8}$ in all three cases, and the cyclic order of the abscissae coincides with that of the rigid rotation of number $89/144$. $\checkmark$

Remark 16.7: What we import: Birkhoff's theorem on invariant curves. Theorem 16.7 says that the ordered orbits lie on Lipschitz graphs. Birkhoff proved an apparently stronger statement, which concerns any invariant curve.

Imported from G. D. Birkhoff, “Sur quelques courbes fermées remarquables”, Bull. Soc. Math. France **60** (1932), 1–26 (Numdam), verified by real web search on 17 September 2026; modern exposition in M. R. Herman, “Sur les courbes invariantes par les difféomorphismes de l’anneau”, Astérisque **103–104** (1983), chapter I. Statement (Birkhoff’s theorem). Let f be an area-preserving positive monotone twist map of the annulus, and let $\Gamma \subset \mathcal{A}$ be a simple closed curve, invariant under f , essential (that is, not homotopic to a point in the cylinder, i.e. making one turn). Then Γ is the graph of a Lipschitz function $\psi : \mathbb{T} \rightarrow \mathbb{R}$, with Lipschitz constant depending only on the twist constants of the map. What is assumed: area preservation, the monotone twist, and that Γ be a simple and essential curve (without these hypotheses the statement is false: non-essential curves, such as the boundaries of the elliptic islands visible in Figure 16.3, are not graphs). Why it is imported: the proof requires a global argument on the regions bounded by Γ and on their area, of length comparable to the whole of Section 16.4, and it does not follow from Theorem 16.7 because an invariant curve might a priori fail to be the closure of an ordered orbit. The difference between the two statements is exactly the reason why Birkhoff’s theorem remains an independent result.

16.7. The Aubry–Mather theorem

We have ordered orbits for every *rational* rotation number (Theorem 16.4). The step of Aubry and of Mather consists in passing to the limit.

Theorem 16.8: Existence of ordered orbits for every rotation number. Let h be of the admissible class. For every $\omega \in \mathbb{R}$ there exists a stationary Birkhoff configuration x with $\rho(x) = \omega$. The corresponding orbit $((x_i, y_i))$ of the twist map is ordered and lies on the graph of an L -Lipschitz function (Theorem 16.7).

Proof. Step 1 (approximating sequence). Let $(p_n/q_n)_{n \geq 1}$ be a sequence of rationals with $p_n/q_n \rightarrow \omega$ and, without loss of generality, $|p_n/q_n| \leq |\omega| + 1$ for every n . By Theorem 16.4 there exists a minimiser $x^{(n)}$ of W_{p_n, q_n} ; by the invariance under integer translation (Step 1 of that proof) we may normalise it with $x_0^{(n)} \in [0, 1)$. By Corollary 16.1 each $x^{(n)}$ is Birkhoff,

and by Corollary 16.2 it has $\rho(x^{(n)}) = p_n/q_n$.

Step 2 (uniform estimate). Estimate (16.16) of Theorem 16.6, applied with $i = 0$ and with $i = -m$, gives for every $m \geq 1$

$$\left| x_m^{(n)} - x_0^{(n)} - m \frac{p_n}{q_n} \right| \leq 1, \quad \left| x_{-m}^{(n)} - x_0^{(n)} + m \frac{p_n}{q_n} \right| \leq 1,$$

hence, for every $i \in \mathbb{Z}$,

$$|x_i^{(n)}| \leq 1 + |i|(|\omega| + 1) + 1, \quad (16.19)$$

a bound independent of n .

Step 3 (diagonal extraction). By (16.19), for each fixed i the sequence $(x_i^{(n)})_n$ is bounded in $\mathbb{R}$. With Cantor's diagonal procedure (one extracts a subsequence converging at $i = 0$, then from it one converging also at $i = 1$, then at $i = -1$, and so on, and finally one takes the diagonal) one obtains a subsequence (n_r) such that $x_i^{(n_r)} \rightarrow x_i$ for every $i \in \mathbb{Z}$. We call $x = (x_i)$ the limit configuration.

Step 4 (x is stationary). Each $x^{(n)}$ satisfies (16.4) at every index. The functions $\partial_1 h$ and $\partial_2 h$ are continuous (H1), hence passing to the limit along (n_r) in the relation

$$\partial_2 h(x_{i-1}^{(n_r)}, x_i^{(n_r)}) + \partial_1 h(x_i^{(n_r)}, x_{i+1}^{(n_r)}) = 0$$

one obtains (16.4) for x , at every index i .

Step 5 (x is Birkhoff). Fix $(s, t) \in \mathbb{Z}^2$. For every r , the set of configurations ξ with $\tau_{s,t}\xi \leq \xi$ is closed for pointwise convergence (it is a countable intersection of weak inequalities between coordinates), and so is the one with $\tau_{s,t}\xi \geq \xi$; their union is therefore closed. Since $x^{(n_r)}$ belongs to the union for every r , so does the limit x . Since (s, t) is arbitrary, x is Birkhoff.

Step 6 ($\rho(x) = \omega$). Passing to the limit in $|x_i^{(n_r)} - x_0^{(n_r)} - i p_{n_r}/q_{n_r}| \leq 1$ (with $p_{n_r}/q_{n_r} \rightarrow \omega$) one obtains $|x_i - x_0 - i\omega| \leq 1$ for every i , hence $(x_i - x_0)/i \rightarrow \omega$ as $i \rightarrow +\infty$. Since x is Birkhoff, Theorem 16.6 guarantees that this limit is $\rho(x)$: hence $\rho(x) = \omega$.

Finally, Theorem 16.7 applies to x (Birkhoff and stationary) and gives the Lipschitz graph structure. $\square$

Definition 16.10: Aubry–Mather set associated with an ordered orbit. Let x be a stationary Birkhoff configuration, $\mathcal{X} = \{x_i + t\}$, Y the Lipschitz function of Theorem 16.7 extended to $\overline{\mathcal{X}}$. The associated **Aubry–Mather set** is

$$\mathcal{M} := \{(\xi, Y(\xi)) : \xi \in \overline{\mathcal{X}}\} \subset \mathbb{R}^2,$$

together with its projection onto the cylinder, still denoted $\mathcal{M}$. If the projection $E := \pi(\mathcal{X}) \subset \mathbb{T}$ is the whole of $\mathbb{T}$, $\mathcal{M}$ is an **invariant curve** (essential, a Lipschitz graph); if E is a Cantor set, $\mathcal{M}$ is called a **cantorus**.

Lemma 16.4: Uniqueness of the representation and order isomorphism. Let x be a stationary Birkhoff configuration with $\rho(x) = \omega \notin \mathbb{Q}$, and let $D := \{i\omega + t : i, t \in \mathbb{Z}\}$. Then: (i) the map $(i, t) \mapsto x_i + t$ is injective on $\mathbb{Z}^2$; (ii) the map $\Psi(x_i + t) := i\omega + t$ is well defined and is an order isomorphism of $\mathcal{X}$ onto D ; (iii) $\Psi(g(\xi)) = \Psi(\xi) + \omega$ and $\Psi(\xi + 1) = \Psi(\xi) + 1$ for every $\xi \in \mathcal{X}$.

Proof. (i) Let $x_i + t = x_j + u$ and set $(s, r) := (j - i, t - u)$. As in the proof of Proposition 16.3, one obtains $\tau_{s,r}x = x$, that is, $x_{m+s} = x_m + r$ for every m . If $s \neq 0$, iterating n times,

$x_{m+ns} = x_m + nr$, hence

$$\omega = \rho(x) = \lim_{n \rightarrow \infty} \frac{x_{m+ns} - x_m}{ns} = \frac{r}{s} \in \mathbb{Q},$$

against the irrationality of ω . Hence $s = 0$ and therefore $r = 0$, that is, $(i, t) = (j, u)$.

(ii) Well-definedness is (i). For monotonicity, let $x_i + t < x_j + u$ and $(s, r) = (j - i, t - u)$, $\bar{x} := \tau_{s,r}x$: then $\bar{x}_i = x_j - (t - u) = (x_j + u) - t > x_i$, hence by the Birkhoff property $\bar{x} \geq x$ everywhere, that is,

$$x_{m+s} - r \geq x_m \quad \text{for every } m.$$

If $s > 0$, iterating n times, $x_{m+ns} - nr \geq x_m$, hence $(x_{m+ns} - x_m)/(ns) \geq r/s$ and, in the limit, $\omega \geq r/s$; since ω is irrational, $\omega > r/s$, that is, $s\omega > r$, that is, $i\omega + t < j\omega + u$. If $s < 0$, rewriting the same relation as $x_{m'+|s|} - x_{m'} \leq -r$ (with $m' = m + s$) and iterating one obtains $|s|\omega \leq -r$, that is, again $s\omega \geq r$ and, by irrationality, $s\omega > r$. If $s = 0$, the relation gives $r \leq 0$ and $r \neq 0$ by (i), hence $r < 0$, that is, $t < u$, and again $i\omega + t < j\omega + u$. Surjectivity onto D is evident; the converse implication follows from trichotomy (if $i\omega + t < j\omega + u$ and it were not the case that $x_i + t < x_j + u$, one would apply the above to the other inequality).

(iii) $\Psi(g(x_i + t)) = \Psi(x_{i+1} + t) = (i + 1)\omega + t = \Psi(x_i + t) + \omega$, and $\Psi(x_i + t + 1) = i\omega + t + 1$. $\square$

Theorem 16.9: Dichotomy: invariant curve or cantor. *Let x be a stationary Birkhoff configuration with $\rho(x) = \omega \notin \mathbb{Q}$, and let $\mathcal{X}, \bar{\mathcal{X}}, Y, \mathcal{M}, E$ be as above. Then: (i) $\mathcal{M}$ is a closed, non-empty, invariant subset of the cylinder, contained in the graph of an L -Lipschitz function defined on E ; (ii) the restriction of F to $\mathcal{M}$ is conjugate, via the first projection, to a strictly increasing homeomorphism $\hat{g}$ of $\bar{\mathcal{X}}$ with $\hat{g}(\xi + 1) = \hat{g}(\xi) + 1$, which extends to a circle homeomorphism of rotation number ω ; (iii) either $\bar{\mathcal{X}} = \mathbb{R}$, and then $E = \mathbb{T}$ and $\mathcal{M}$ is an essential invariant curve, the graph of an L -Lipschitz function $\mathbb{T} \rightarrow \mathbb{R}$, on which the dynamics is conjugate to the rigid rotation R_ω ; or $\bar{\mathcal{X}}$ is a 1-periodic Cantor set (closed, without isolated points, with empty interior) and $\mathcal{M}$ is a cantor, on which the dynamics is conjugate to the restriction of R_ω to its unique minimal set.*

Proof. Step 1 ($\mathcal{M}$ is closed and invariant). By definition $\mathcal{M}$ is the graph of Y on the closed set $\bar{\mathcal{X}}$, with Y continuous: hence it is closed in $\mathbb{R}^2$. It contains the countable set $\mathcal{O} := \{(x_i + t, y_i) : i, t \in \mathbb{Z}\}$, dense in $\mathcal{M}$ by construction (density of $\mathcal{X}$ in $\bar{\mathcal{X}}$ and continuity of Y). Since F commutes with the integer translation in x (Definition 16.1),

$$F(x_i + t, y_i) = (x_{i+1} + t, y_{i+1}) \in \mathcal{O},$$

hence $F(\mathcal{O}) = \mathcal{O}$; by continuity of F , $F(\mathcal{M}) = F(\bar{\mathcal{O}}) \subseteq \overline{F(\mathcal{O})} = \mathcal{M}$, and the same argument applied to F^{-1} (continuous) gives $F^{-1}(\mathcal{M}) \subseteq \mathcal{M}$, hence $\mathcal{M} \subseteq F(\mathcal{M})$ and in conclusion $F(\mathcal{M}) = \mathcal{M}$. The L -Lipschitz graph structure is Theorem 16.7.

Step 2 ($\hat{g}$ is a homeomorphism of $\bar{\mathcal{X}}$). We set

$$\hat{g}(\xi) := \text{first coordinate of } F(\xi, Y(\xi)), \quad \xi \in \bar{\mathcal{X}}.$$

It is well defined with values in $\bar{\mathcal{X}}$ by Step 1, it is continuous (a composition of continuous functions: Y is Lipschitz and F is continuous) and it coincides with g on $\mathcal{X}$. Being continuous and coinciding on $\mathcal{X}$, dense in $\bar{\mathcal{X}}$, with the non-decreasing function g (Proposition 16.3), $\hat{g}$ is non-decreasing on the whole of $\bar{\mathcal{X}}$. It is also injective: if $\xi \neq \xi'$ in $\bar{\mathcal{X}}$, the points $(\xi, Y(\xi))$ and $(\xi', Y(\xi'))$ are distinct, hence so are their images (F is bijective); these both lie in $\mathcal{M}$, which is a graph over $\bar{\mathcal{X}}$, so that two distinct points of it have distinct first coordinates, that is, $\hat{g}(\xi) \neq \hat{g}(\xi')$. Non-decreasing and injective means strictly increasing. Finally $\hat{g}^{-1}$ is the first coordinate of F^{-1} restricted to $\mathcal{M}$, also continuous: $\hat{g}$ is a homeomorphism of $\bar{\mathcal{X}}$ onto itself. The relation $\hat{g}(\xi + 1) = \hat{g}(\xi) + 1$ holds on $\mathcal{X}$ (Proposition 16.3) and extends by continuity.

Step 3 (extension to a circle homeomorphism). The set $\overline{\mathcal{X}}$ is closed, non-empty, invariant under $\xi \mapsto \xi + 1$ and unbounded in both directions. The connected components of $\mathbb{R} \setminus \overline{\mathcal{X}}$ are bounded open intervals (a, b) with $b - a \leq 1$ (if $b - a > 1$, the point $a + 1 \in \overline{\mathcal{X}}$ would fall inside the component, absurd), with $a, b \in \overline{\mathcal{X}}$. We define $G : \mathbb{R} \rightarrow \mathbb{R}$ by setting $G := \hat{g}$ on $\overline{\mathcal{X}}$ and, on each component (a, b) , the unique increasing affine function sending a to $\hat{g}(a)$ and b to $\hat{g}(b)$ (well defined because $\hat{g}(a) < \hat{g}(b)$). The function G is continuous, strictly increasing and surjective, with $G(\xi + 1) = G(\xi) + 1$ (the family of the components is invariant under $\xi \mapsto \xi + 1$ and $\hat{g}$ commutes with that translation): it is therefore the lift of an orientation-preserving circle homeomorphism.

Step 4 (rotation number of G). $G^n(x_0) = \hat{g}^n(x_0) = x_n$ for every $n \geq 1$, hence $(G^n(x_0) - x_0)/n = (x_n - x_0)/n \rightarrow \omega$: by Theorem 4.1 (Volume IV, Chapter 4; applied here after the rescaling $\xi \mapsto 2\pi\xi$, which does not alter rotation numbers) the rotation number of G is ω , irrational.

Step 5 (dichotomy). Since $\mathcal{X} = \{G^n(x_0) + t : n, t \in \mathbb{Z}\}$, the set $\overline{\mathcal{X}}$ is exactly the set E of Theorem 4.4 (Volume IV, Chapter 4) for the homeomorphism G with irrational rotation number ω . That theorem states: either $\overline{\mathcal{X}} = \mathbb{R}$, and in that case the semiconjugacy is a conjugacy (hence G is conjugate to R_ω , and $E = \pi(\overline{\mathcal{X}}) = \mathbb{T}$); or $\overline{\mathcal{X}}$ is a 1-periodic Cantor set and the semiconjugacy collapses each component of $\mathbb{R} \setminus \overline{\mathcal{X}}$ to a point, realising the conjugacy between $G|_{\overline{\mathcal{X}}}$ and the restriction of R_ω to its minimal set. Transporting everything onto the graph of Y (which is a homeomorphism of $\overline{\mathcal{X}}$ onto $\mathcal{M}$, Y being continuous) one obtains the claim. $\square$

***Remark 16.8: What the Aubry–Mather theorem says in addition.** Theorems 16.8 and 16.9, proved in full above, give for every ω an ordered orbit on a Lipschitz graph and the curve/cantor dichotomy. The Aubry–Mather theorem properly so called asserts something stronger, which here we do **not** prove.*

***Imported** from J. N. Mather, “Existence of quasi-periodic orbits for twist homeomorphisms of the annulus”, Topology **21** (1982), 457–467, and from S. Aubry, P. Y. Le Daeron, “The discrete Frenkel–Kontorova model and its extensions I”, Physica D **8** (1983), 381–422; self-contained exposition in V. Bangert, “Mather sets for twist maps and geodesics on tori”, Dynamics Reported **1** (1988), 1–56. Sources verified by real web search on 17 September 2026. Statement. For every ω there exists a minimal (not merely stationary) configuration of rotation number ω ; the minimal configurations of irrational rotation number ω form a totally ordered, closed and invariant set $\mathcal{M}_\omega$, which is exactly of the form described by Theorem 16.9; for rational ω the set of minimal configurations contains, besides the periodic orbits of type (p, q) , the heteroclinic orbits connecting adjacent minimising periodic orbits. What is assumed: hypotheses (H1)–(H3); the part missing here is that a periodic minimiser is minimal as a configuration on the whole of $\mathbb{Z}$ (and not only among the periodic configurations of the same type), a statement which is true but whose proof requires a covering argument over windows of length Nq which cannot be compressed into the space of this section. Consequence for us: in Theorem 16.8 the limit configuration is guaranteed to be stationary and Birkhoff, but its minimality is part of the imported statement.*

***Remark 16.9: Greene’s criterion and the threshold $k_c \approx 0.9716$.** Theorem 16.9 says that one of the two alternatives occurs, but not which. For the standard map and the golden rotation number $\omega = (\sqrt{5} - 1)/2$ the experimental answer is known with very high precision. **Panorama** (not proved in this treatise). J. M. Greene, “A method for determining a stochastic transition”, J. Math. Phys. **20** (1979), 1183–1201, verified by real web search on 17 September 2026, proposed to read the survival of the invariant curve of number ω in the stability of the periodic orbits of rotation numbers $p_n/q_n \rightarrow \omega$: one defines the residue*

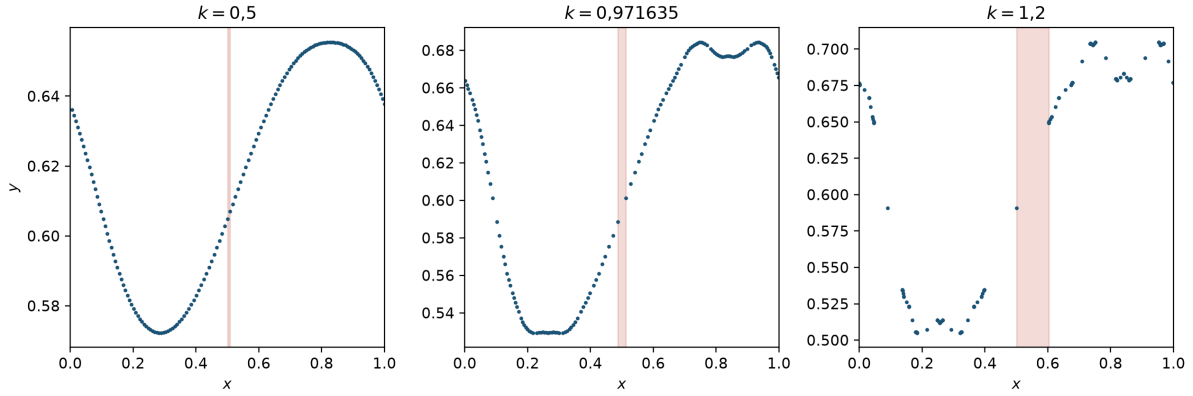


Figure 16.6: From the invariant curve to the cantor. For each value of k the minimising orbit of type $(89, 144)$ is drawn (two consecutive Fibonacci numbers, $p/q = 0.618055\dots$, a convergent of the golden number $\omega = (\sqrt{5}-1)/2$), in the coordinates $(x \bmod 1, y)$; the pink band highlights the maximal gap between consecutive abscissae. At $k = 0.5$ the gap equals 1.2185 times the mean spacing $1/q$ — the points cover uniformly a regular curve; at $k = 0.971635$ it equals 3.7491 times; at $k = 1.2$ it equals 14.7507 times, and the Cantor set structure of Theorem 16.9 is visible to the naked eye. In all three cases the measured Lipschitz constant $(0.339530; 0.950534; 1.344641)$ respects the bound $1 + k$ of Corollary 16.3. ✓ Figure from script, *Appendix B*, `fig16_cantor.py`.

$R := \frac{1}{4}(2 - \text{tr } DF^{q_n})$ of the orbit of type (p_n, q_n) and one observes that, as n grows, $R \rightarrow 0$ if the curve exists and $R \rightarrow \infty$ if it does not. With this criterion Greene obtained for the golden torus of the standard map the threshold

$$k_c = 0.971635\dots \quad (\text{most precise value in the literature: } 0.97163540631).$$

The criterion is a numerical conjecture, not a theorem: a partial rigorous justification was given by R. S. MacKay (Renormalisation in Area-Preserving Maps, World Scientific, 1993) within the framework of renormalisation, and, in a nearby context, in a paper in SIAM J. Math. Anal. devoted to non-twist maps. In our numerical experiment (Figure 16.6) the maximal gap of the $(89, 144)$ orbit goes from 1.22 to 3.75 to 14.75 times the mean spacing as one crosses k_c : an independent indicator, consistent with Greene’s threshold, but itself numerical and not decisive. Note the connection with Chapter 3: the golden number is the “most irrational” one (its continued fraction expansion is $[1; 1, 1, \dots]$), and its convergents are the Fibonacci ratios F_n/F_{n+1} ; block [C16-6] of the verification script confirms that $q^2|p/q - \omega| \rightarrow 1/\sqrt{5} = 0.447214$ along this sequence. ✓

16.8. Mather’s β function

Definition 16.11: Minimal average action $\beta(p/q)$. For $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$ with $\gcd(p, q) = 1$ we set

$$\beta\left(\frac{p}{q}\right) := \frac{1}{q} \min_{\mathbb{R}^q} W_{p,q},$$

the **minimal average action** (per step) of the configurations of type (p, q) . The function β thus defined on the rationals extends uniquely to a convex function on $\mathbb{R}$ (see Remark 16.10), called **Mather’s β function** or *free energy* in Aubry’s language.

Proposition 16.4: Two exact identities for the standard map. For the generating function (16.8) of the standard map one has, for every $p/q \in \mathbb{Q}$: **(i)** (reflection) $\beta(-p/q) = \beta(p/q)$; **(ii)** (translation) $\beta\left(\frac{p}{q} + 1\right) = \beta\left(\frac{p}{q}\right) + \frac{p}{q} + \frac{1}{2}$; **(iii)** (value at 0) $\beta(0) = -\frac{k}{(2\pi)^2}$, and therefore $\beta(1) = \frac{1}{2} - \frac{k}{(2\pi)^2}$. Consequently the function $\tilde{\beta}(\omega) := \beta(\omega) - \omega^2/2$ is even and 1-periodic on the rationals.

Proof. **(i)** The map $x \mapsto \bar{x}$, $\bar{x}_i := -x_i$, is a bijection between the configurations of type (p, q) and those of type $(-p, q)$: indeed $\bar{x}_{i+q} = -(x_i + p) = \bar{x}_i - p$. It preserves the action: since V_k is even (cos is even),

$$h_k(\bar{x}_i, \bar{x}_{i+1}) = \frac{1}{2}(-x_{i+1} + x_i)^2 + V_k(-x_i) = \frac{1}{2}(x_{i+1} - x_i)^2 + V_k(x_i) = h_k(x_i, x_{i+1}),$$

hence $W_{-p,q}(\bar{x}) = W_{p,q}(x)$. A bijection which preserves the action sends minimum to minimum: $\min W_{-p,q} = \min W_{p,q}$, and dividing by q one gets (i).

(ii) The map $x \mapsto \tilde{x}$, $\tilde{x}_i := x_i + i$, is a bijection between the configurations of type (p, q) and those of type $(p + q, q)$: $\tilde{x}_{i+q} = x_{i+q} + i + q = \tilde{x}_i + p + q$. Since V_k has period 1 and $i \in \mathbb{Z}$,

$$h_k(\tilde{x}_i, \tilde{x}_{i+1}) = \frac{1}{2}(x_{i+1} - x_i + 1)^2 + V_k(x_i) = h_k(x_i, x_{i+1}) + (x_{i+1} - x_i) + \frac{1}{2}.$$

Summing over $i = 0, \dots, q - 1$ and using $\sum_{i=0}^{q-1} (x_{i+1} - x_i) = x_q - x_0 = p$:

$$W_{p+q,q}(\tilde{x}) = W_{p,q}(x) + p + \frac{q}{2}.$$

Passing to the minima (the bijection preserves the difference, which does not depend on x) and dividing by q : $\beta\left(\frac{p+q}{q}\right) = \beta\left(\frac{p}{q}\right) + \frac{p}{q} + \frac{1}{2}$.

(iii) For $p = 0$, $q = 1$ a configuration of type $(0, 1)$ is the constant sequence $x_i \equiv x_0$, and $W_{0,1}(x_0) = h_k(x_0, x_0) = 0 + V_k(x_0)$. The minimum of $V_k(x) = -\frac{k}{(2\pi)^2} \cos(2\pi x)$ is $-\frac{k}{(2\pi)^2}$, attained at $x_0 \in \mathbb{Z}$. Hence $\beta(0) = -\frac{k}{(2\pi)^2}$, and (ii) with $p/q = 0$ gives $\beta(1) = \beta(0) + 0 + \frac{1}{2}$.

Consequence. $\tilde{\beta}(-\omega) = \beta(-\omega) - \omega^2/2 = \beta(\omega) - \omega^2/2 = \tilde{\beta}(\omega)$ by (i); and

$$\tilde{\beta}(\omega + 1) = \beta(\omega) + \omega + \frac{1}{2} - \frac{(\omega+1)^2}{2} = \beta(\omega) - \frac{\omega^2}{2} = \tilde{\beta}(\omega)$$

by (ii). □

Example 16.6: The table of β for $k = 1$ and the three identities. Double check (script `verifica_cap16_vol4.py`, blocks [C16-7] and [C16-8], run on 17 September 2026). For $k = 1$ the numerical minimisation of $W_{p,q}$ gives

ω	$\beta(\omega)$	ω	$\beta(\omega)$
0	-0.025330295911	2/5	0.078148038226
1/3	0.053119258132	3/5	0.178148038226
1/2	0.121896588107	2/3	0.219785924799
1	0.474669704089		

Independent checks (closed formula against numerical value):

- $\beta(0) = -1/(2\pi)^2 = -0.025330295911$: agrees to all 12 digits. ✓
- $\beta(1) = \beta(0) + \frac{1}{2} = 0.474669704089$: agrees to all 12 digits. ✓
- $\beta(2/3) = \beta(1 - \frac{1}{3}) = \beta(\frac{1}{3}) - \frac{1}{3} + \frac{1}{2} = 0.053119258132 + 0.166666666667 = 0.219785924799$: agrees. ✓
- $\beta(3/5) = \beta(\frac{2}{5}) - \frac{2}{5} + \frac{1}{2} = 0.078148038226 + 0.1 = 0.178148038226$: agrees. ✓

Identities (i) and (ii) of Proposition 16.4 are therefore verified numerically in three independent cases, each with a separate minimisation. Observe also the convexity: with $t = (1/3)/(1/2) = 2/3$,

$$\beta(1/3) = 0.0531193 \leq (1-t)\beta(0) + t\beta(1/2) = 0.0728210. \checkmark$$

Remark 16.10: The β function: what is known. Panorama (not proved here). The function β defined on the rationals extends to a convex and superlinear function on $\mathbb{R}$; it is differentiable at every irrational point, whereas at a rational point p/q it has distinct one-sided derivatives (a “corner”) if and only if there exists no invariant curve of rotation number p/q — the width of the corner is precisely the gap $\Delta W_{p/q}$ between the action of the minimising periodic orbits and that of the “min-max” orbits, and it is the quantity with which Mather establishes the non-existence of invariant curves. Differentiability at the irrationals is the theorem of J. N. Mather, “Differentiability of the minimal average action as a function of the rotation number”, Bol. Soc. Bras. Mat. **21** (1990), 59–70, verified by real web search on 17 September 2026, which confirms two conjectures of Aubry; the same work shows that for generic twist diffeomorphisms β is not differentiable at the rationals. The analogy with solid-state physics is exact: β is the free energy per atom of the Frenkel–Kontorova chain as the density varies, and its corners at the rationals are the locking “tongues”, the annulus-map counterpart of the Arnold tongues of Chapter 4.

16.9. What we import and where the story continues

Remark 16.11: Moser’s twist theorem and the bridge with KAM theory. Theorem 16.9 leaves open the question: when does the invariant curve exist? For small perturbations the answer is affirmative and quantitative.

Imported from J. Moser, “On invariant curves of area-preserving mappings of an annulus”, Nachr. Akad. Wiss. Göttingen, Math.-Phys. Kl. II (1962), 1–20, in the version with optimal regularity of H. Rüssmann, “On the existence of invariant curves of twist mappings of an annulus”, in Geometric Dynamics, Lecture Notes in Mathematics **1007**, Springer, 1983, 677–718 (verified by real web search on 17 September 2026). Statement (twist theorem). Let f_ε be a family of area-preserving twist maps of the annulus, sufficiently regular, with f_0 integrable; let ω be a Diophantine number contained in the twist interval. Then for ε sufficiently small f_ε possesses an essential invariant curve of rotation number ω , close to the unperturbed one. What is assumed: the Diophantine condition on ω (irrationality is not enough), area preservation, the monotone twist and a regularity threshold (Moser: many derivatives; Rüssmann: $C^{3+\epsilon}$ for the optimal version known at the time). Why it is imported: the proof is a KAM theorem with Newton scheme and loss of derivatives, and it is the content of Chapter 6 of this volume in the case of circle diffeomorphisms; the case of the annulus is its two-dimensional version and is not reproved here.

The two theories are complementary: KAM gives invariant curves for small perturbations and Diophantine numbers, but says nothing about what happens afterwards; Aubry–Mather gives, for every ω and for every strength of the perturbation, an ordered invariant set — which is the KAM curve as long as it exists, and is its Cantor “residue” once it has broken.

Remark 16.12: Non-Birkhoff orbits, weak KAM, billiards. Panorama (not proved here). (a) Non-Birkhoff orbits. Not all the periodic orbits of a twist map are ordered: W. Cheng and C.-Q. Cheng, “Existence of infinite non-Birkhoff periodic orbits for area-

preserving monotone twist maps of cylinders”, Science in China A **43** (2000), 810–817, prove that infinitely many of them exist; the orbits of Theorem 16.4 are therefore a particular class, selected by minimisation. (b) Weak KAM. Aubry–Mather theory generalises to continuous-time Lagrangian systems in the weak KAM theory of A. Fathi (Weak KAM Theorem in Lagrangian Dynamics, lecture notes, Pisa 2005 and Lyon 2008 versions), where the minimal configurations become calibrated curves and the β function becomes the Legendre transform of the effective Hamiltonian; the geometric analogy with minimising geodesics on tori is developed by Bangert in the work already cited. (c) Billiards. The billiard in a convex oval is a twist map of the annulus, its generating function is the length of the chord, and its Aubry–Mather sets correspond to the caustics; the Birkhoff conjecture on the integrability of convex billiards belongs however to Volume VI, Chapter 10, and is not treated in this volume.

16.10. Assigned exercises

The answers are in the Teacher’s key (Section 16.13).

Exercise 16.1: The explicit orbit of type $(1, 2)$ as k varies. *Goal:* to manipulate the generating function (16.8) and the trace formula of Proposition 16.2. *Estimated time:* 20 minutes. *Type:* symbolic.

- Prove that $W_{1,2} = \frac{1}{4}$ for the orbit $\{(0, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2})\}$, for every $k \geq 0$, and explain which cancellation produces it.
- Using DF_k as given by (16.11), compute $\text{tr } DF_k^2$ along this orbit and determine for which k it is elliptic.
- What happens exactly at $k = 2$? Is the value of the trace at that point compatible with the conclusion of Theorem 16.4 that the *minimum* of the action is another orbit?

Exercise 16.2: Two exact values of the β function. *Goal:* to apply Proposition 16.4 and to recognise the role of the minimum of the potential. *Estimated time:* 15 minutes. *Type:* symbolic/numerical.

- Compute $\beta(0)$ and $\beta(1)$ in closed form for general k .
- Evaluate them for $k = 0.5$ and for $k = 2$ to six decimals.
- Deduce from Proposition 16.4(ii), by induction, the formula $\beta(\omega+n) = \beta(\omega) + n\omega + n^2/2$ for every $n \in \mathbb{Z}_{>0}$, and check it for $\omega = 0$, $n = 2$.

Exercise 16.3: A family of generating functions with twist a . *Goal:* to recognise which constants of Definition 16.4 control which conclusions. *Estimated time:* 25 minutes. *Type:* symbolic. Consider, for $a > 0$ and $k \geq 0$,

$$h_{a,k}(x, x') := \frac{a}{2}(x' - x)^2 + V_k(x), \quad V_k(x) = -\frac{k}{(2\pi)^2} \cos(2\pi x).$$

- Determine δ (twist constant), C_1 , C_2 and $L = \max(C_1, C_2)$.
- Write explicitly the map generated by $h_{a,k}$ according to (16.2).
- For which values of a is the Lipschitz constant of Theorem 16.7 equal to a and not to $a + k$? What does a small Lipschitz constant mean geometrically for the invariant curves?

16.11. Worked exercises

Exercise 16.4: The minimising orbit of type (1, 2) for $k = 2$. *Type:* numerical with a double check (scalar equation and action recomputed by hand). Determine the minimising orbit of type (1, 2) of the standard map for $k = 2$, its action, and establish its nature.

Solution. Strategy. Reuse the analysis of Example 16.4: the family $x_1 = 1 - x_0$ leads to the scalar equation (16.15); one solves it, computes the action by the closed formula and compares it with $W = \frac{1}{4}$ of the explicit orbit.

Step 1 (scalar equation). With $k = 2$, (16.15) becomes

$$1 - 4x_0 = \frac{2}{2\pi} \sin(2\pi x_0) = \frac{1}{\pi} \sin(2\pi x_0).$$

The left-hand side decreases from 1 to -1 on $[0; 0.5]$, the right-hand side is positive and bounded by $1/\pi = 0.3183$: there is exactly one root in $(0; 0.25)$. Brent's method on $[0.02; 0.48]$ gives

$$x_0 = 0.1783510530, \quad x_1 = 1 - x_0 = 0.8216489470.$$

Step 2 (action, computed by hand). The two differences are $x_1 - x_0 = 0.6432978940$ and $x_0 + 1 - x_1 = 0.3567021060$. The kinetic term equals

$$\frac{1}{2}(0.6432978940)^2 + \frac{1}{2}(0.3567021060)^2 = 0.2069161 + 0.0636182 = 0.2705343.$$

For the potential term, $2\pi x_0 = 1.120613$ and $\cos(1.120613) = 0.435129$; since $2\pi x_1 = 2\pi - 2\pi x_0$ and the cosine is even, $\cos(2\pi x_1)$ has the same value. Hence

$$V_2(x_0) + V_2(x_1) = -2 \cdot \frac{2}{(2\pi)^2} \cdot 0.435129 = -2 \cdot 0.0506606 \cdot 0.435129 = -0.0440880.$$

In total $W_{1,2}^{\min} = 0.2705343 - 0.0440880 = 0.2264463$.

Independent check. The direct numerical minimisation of $W_{1,2}$ over $\mathbb{R}^2$ with 40 random starting points (script `verifica_cap16_vol4.py`, block [C16-8]) gives $W_{1,2}^{\min} = 0.226446312769$ and the same x_0 : the two routes agree to seven decimal digits. ✓

Nature of the orbit. The script reports $\text{tr } DF_2^2 = 6.2384023301$ for the minimiser, with $\det = 1$: since $|\text{tr}| > 2$, the orbit is **hyperbolic**, consistently with the fact that a minimum of the action cannot be elliptic.

Interpretation. $0.2264 < 0.25$: for $k = 2$ too the explicit orbit $\{0, \frac{1}{2}\}$ is not the minimiser, and the gap has grown with respect to $k = 1$ (where it was $0.25 - 0.2438 = 0.0062$, against $0.25 - 0.2264 = 0.0236$ here): the stronger the kick, the more convenient it is for the chain to move away from the symmetric configuration “one atom in the minimum, one in the maximum of the potential”.

Typical error to avoid. Concluding that the explicit orbit at $k = 2$ is “hyperbolic” because it is not minimising: its trace equals $2 - k^2 = -2$ *exactly*, that is, it is **parabolic** (degenerate case, double eigenvalue -1), and it is precisely the period-doubling bifurcation value. Not all non-minimising orbits are hyperbolic.

Extension. Repeat for $k = 3$: one expects an even smaller root x_0 (the term $\frac{k}{2\pi} \sin$ grows) and an even larger gap $0.25 - W^{\min}$.

Exercise 16.5: The exact gain of an exchange. *Type:* symbolic with direct arithmetic verification. Let $x_i = 0.30$, $\bar{x}_i = 0.55$, $x_{i+1} = 0.90$, $\bar{x}_{i+1} = 0.40$ (a crossing in the gap $(i, i+1)$). Compute the gain in action produced by the min / max exchange for the standard

map, for arbitrary k .

Solution. Strategy. Identify a, b, c, d of Lemma 16.2, apply the exact formula of Example 16.3, and then recompute the four values of h_k one by one.

Step 1 (identification). $d_i = x_i - \bar{x}_i = -0.25 < 0$ and $d_{i+1} = x_{i+1} - \bar{x}_{i+1} = +0.50 > 0$: a crossing. Hence $a = x_i = 0.30 < b = \bar{x}_i = 0.55$ and $c = \bar{x}_{i+1} = 0.40 < d = x_{i+1} = 0.90$; the exchange gives $(z_i, z_{i+1}) = (a, c) = (0.30; 0.40)$ and $(w_i, w_{i+1}) = (b, d) = (0.55; 0.90)$.

Step 2 (exact formula). By Example 16.3 the gain is

$$\Delta = (b - a)(d - c) = 0.25 \times 0.50 = 0.125,$$

independent of k .

Independent check (four direct evaluations). The potential terms appear with $V_k(a)$ in the pairs (a, d) and (a, c) and with $V_k(b)$ in the pairs (b, c) and (b, d) : in the combination $h(a, d) + h(b, c) - h(a, c) - h(b, d)$ they cancel exactly, and only the kinetic part remains

$$\begin{aligned} & \frac{1}{2}[(0.90 - 0.30)^2 + (0.40 - 0.55)^2 - (0.40 - 0.30)^2 - (0.90 - 0.55)^2] \\ &= \frac{1}{2}[0.36 + 0.0225 - 0.01 - 0.1225] = \frac{1}{2} \cdot 0.25 = 0.125. \checkmark \end{aligned}$$

Interpretation. The gain coincides with the lower bound $\delta(b - a)(d - c)$ of Lemma 16.2 because for the standard map $\partial_1 \partial_2 h_k \equiv -1$, that is, $\delta = 1$ and the bound is *exact*, not merely an estimate. It is this strictly positive gain that makes a second crossing between minimisers impossible (Step 4 of the proof of Theorem 16.3).

Typical error to avoid. Exchanging the roles and computing $(d - a)(b - c) = 0.6 \times 0.15 = 0.09$: the formula requires the *crossing amplitudes* $b - a$ (gap on the left) and $d - c$ (gap on the right), not the lengths of the jumps.

Extension. Redo the computation with $\bar{x}_{i+1} = 0.85$ (an almost degenerate crossing): $d - c = 0.05$, gain 0.0125, ten times smaller — the gain vanishes continuously as the crossing flattens out, but remains > 0 as long as the crossing is strict.

Exercise 16.6: The rotation number of the elliptic island and its limit. *Type:* symbolic with numerical check and asymptotic expansion. Compute the rotation number of the elliptic orbit $(\frac{1}{2}, 0)$ for $k = 0.5$ and determine its behaviour as $k \rightarrow 0^+$.

Solution. Strategy. Use Proposition 16.2(v): the eigenvalues are $e^{\pm i\alpha}$ with $\cos \alpha = 1 - k/2$, and the local rotation number is $\alpha/(2\pi)$.

Step 1. For $k = 0.5$: $\cos \alpha = 1 - 0.25 = 0.75$, hence $\alpha = \arccos(0.75) = 0.722734$ rad and

$$\frac{\alpha}{2\pi} = \frac{0.722734}{6.283185} = 0.115027.$$

Independent check. The roots of $\lambda^2 - 1.5\lambda + 1 = 0$ are $\lambda = 0.75 \pm i\sqrt{1 - 0.5625} = 0.75 \pm 0.661438i$, of modulus $\sqrt{0.5625 + 0.4375} = 1$ exactly, and argument $\arctan(0.661438/0.75) = 0.722734$ rad: the same value. $\checkmark$ The script `fig16_mappa_standard.py` reports 0.115027 for $k = 0.5$, 0.164048 for $k = 0.971635$ and 0.184505 for $k = 1.2$. $\checkmark$

Step 2 (limit $k \rightarrow 0^+$). From $\cos \alpha = 1 - k/2$ and the expansion $\cos \alpha = 1 - \alpha^2/2 + O(\alpha^4)$ one obtains $\alpha^2/2 = k/2 + O(\alpha^4)$, that is, $\alpha \sim \sqrt{k}$ and

$$\frac{\alpha}{2\pi} \sim \frac{\sqrt{k}}{2\pi} \quad (k \rightarrow 0^+).$$

Numerical check: for $k = 0.5$, $\sqrt{0.5}/(2\pi) = 0.112540$ against the exact value 0.115027 (discrepancy 2.2 %, consistent with a first-order approximation at k not small); for $k = 0.01$ one would have $\sqrt{0.01}/(2\pi) = 0.0159155$ against $\arccos(0.995)/(2\pi) = 0.0159257$ (discrepancy 0.06 %). ✓

Interpretation. The rotation number of the island grows from 0 (integrable map) and reaches $1/4$ when $\cos \alpha = 0$, that is, $k = 2$: it is there that the explicit orbit of type $(1, 2)$ stops being elliptic (Exercise 16.4), because the local rotation number crosses the value $1/2$ for F^2 . The sequence of resonances $\alpha/(2\pi) = 1/3, 1/4, \dots$ crossed as k grows is the cascade of bifurcations visible in Figure 16.3.

Typical error to avoid. Confusing the *local* rotation number of the island (which describes the infinitesimal rotation about the elliptic fixed point and is not the rotation number of any orbit of Theorem 16.6) with the rotation number of the ordered orbits, which for the rotational curves of the cylinder is $\approx y$.

Extension. Determine for which k the local rotation number equals exactly $1/6$: from $\alpha = \pi/3$ one has $\cos \alpha = 1/2 = 1 - k/2$, that is, $k = 1$; it is the 1:6 resonance visible as a chain of six islands around the elliptic point.

16.12. Self-assessment quiz

Eight items in order of competence. The commented answers are collected in the Teacher's key, in Section 16.13.

- (Definition) A configuration $x = (x_i)$ is *minimal* (Definition 16.5) if
 - $\sum_{i \in \mathbb{Z}} h(x_i, x_{i+1})$ converges and has the smallest possible value
 - for every $m < n$ the segment $(x_m, \dots, x_n)$ minimises $W_{m,n}$ among all the segments with the same endpoints
 - x_i is a minimum of V for every i
 - x satisfies (16.4)
- (Hypothesis) In Theorem 16.4 (existence of the minimum of $W_{p,q}$) the hypothesis which makes the sublevel sets *compact*, and without which the minimum might not be attained, is
 - (H1), the diagonal periodicity
 - (H2), the uniform twist
 - (H3), the coercivity of h
 - (H4), the boundedness of $\partial_1^2 h$ and $\partial_2^2 h$
- (Symbolic computation) For $h(x, x') = \frac{1}{2}(x' - x)^2 + V(x)$ the Euler–Lagrange equation (16.4) reads
 - $x_{i+1} - 2x_i + x_{i-1} = V'(x_i)$
 - $x_{i+1} - x_i = V'(x_i)$
 - $x_{i+1} + 2x_i + x_{i-1} = V(x_i)$
 - $x_{i+1} - 2x_i + x_{i-1} = -V'(x_i)$
- (Numerical computation) For the standard map with $k = 1.2$, the upper bound on the Lipschitz constant of the Birkhoff graphs (Corollary 16.3) and the value measured on the $(89, 144)$ orbit (Example 16.5) are respectively
 - 1.2 and 1.344641
 - 2.2 and 1.344641
 - 2.2 and 0.950534
 - 1.2 and 0.339530
- (Proof step) In the proof of Theorem 16.3, the fact that the two endpoints m and n have d_m and d_n of the *same* sign serves to guarantee that
 - $z = \min$ and $w = \max$ have, respectively, the same endpoints as $\bar{x}$ and as x , and are therefore admissible competitors
 - the action is finite
 - h is coercive
 - the rotation number exists
- (Error diagnosis) A student writes: “for the standard map at $k = 1$ the orbit of type $(1, 2)$ through $(0, \frac{1}{2})$ and $(\frac{1}{2}, \frac{1}{2})$ is the minimiser, because it is the most symmetric one”. The

error is

- A) that orbit is not of type (1,2) B) its action equals 0.25, whereas the minimum equals 0.243793 (Example 16.4): it is an *elliptic* critical point, not the minimum, and a minimum cannot be elliptic C) the symmetry does not exist D) $W_{1,2}$ has no minimum
7. (Transfer) A physicist models a chain of atoms with springs of elastic constant $a = 4$ in a substrate V_k with $k = 1$, that is, $h(x, x') = 2(x' - x)^2 + V_1(x)$. To estimate how “steep” the invariant curve carrying the ordered configurations can be, the correct tool is
- A) Proposition 16.2, which however concerns only $a = 1$ B) Theorem 16.7 with $C_1 = a + k = 5$, $C_2 = a = 4$, hence $L = 5$ C) Greene’s criterion D) none: the theorem holds only for the standard map
8. (Source and metacognition) The result which guarantees *two* periodic orbits of type (p, q) (and not only the minimising one), cited in this chapter as imported and not proved, is due to
- A) S. Aubry and P. Y. Le Daeron, 1983 B) J. N. Mather, 1982 C) H. Poincaré (1912, statement) and G. D. Birkhoff (1913, proof) D) J. M. Greene, 1979

16.13. Teacher’s key

BEST PRACTICE

Quiz — answers, rationales and diagnoses.

1) **B.** *Rationale:* Definition 16.5(b): minimality is a property which is *local on every finite window*, with fixed endpoints. *Distractors:* A presupposes the convergence of the infinite series, which in general makes no sense (the total action is infinite); C confuses the minimum of the substrate energy with the minimum of the total action, forgetting the elastic term; D is weaker (stationary $\neq$ minimal: the elliptic orbit of Example 16.4 is stationary but not minimal). *Diagnosed error:* confusing “critical point” with “minimum”. *Source:* Definition 16.5, Theorem 16.2. *Prerequisite:* none. *Remedial work:* reread the last paragraph of the proof of Theorem 16.2.

2) **C.** *Rationale:* Step 3 of the proof of Theorem 16.4 uses coercivity twice (to bound h from below and to bound $|x_{i+1} - x_i|$ on the sublevel sets). *Distractors:* A serves to reduce to $x_0 \in [0, 1]$, but by itself is not enough; B serves for determinacy and for the ordering, not for existence; D serves only for Theorem 16.7. *Diagnosed error:* failing to distinguish the hypotheses which give *existence* from those which give *structure*. *Source:* Definition 16.4, Theorem 16.4. *Prerequisite:* Weierstrass’ theorem. *Remedial work:* redo Step 3 with $h(x, x') = \sin^2(x' - x)$, which violates (H3), and observe that the minimum may fail to exist.

3) **A.** *Rationale:* $\partial_2 h(x_{i-1}, x_i) = x_i - x_{i-1}$ and $\partial_1 h(x_i, x_{i+1}) = -(x_{i+1} - x_i) + V'(x_i)$; setting the sum to zero gives $x_{i+1} - 2x_i + x_{i-1} = V'(x_i)$, which is (16.6). *Distractors:* B forgets the term x_{i-1} (a first-order equation, not a second-order one); C confuses V with V' and gets the signs of the discrete Laplacian wrong; D has the wrong sign on the right-hand side (it would be the chain in an inverted potential). *Diagnosed error:* getting the sign or the order of the discrete Laplacian wrong. *Source:* Remark 16.3, equation (16.6). *Prerequisite:* partial derivatives. *Remedial work:* redo the computation starting from $\partial W / \partial x_i = 0$.

4) **B.** *Rationale:* $L = 1 + k = 2.2$ (Corollary 16.3) and the numerical measurement on the (89, 144) orbit gives 1.344641 (Example 16.5). *Distractors:* A uses k instead of $1 + k$; C couples the right bound with the value measured at $k = 0.971635$; D couples the wrong

bound with the value measured at $k = 0.5$. *Diagnosed error*: reading the table of the example in the wrong column, or forgetting the “1+” in the formula $L = \max(C_1, C_2)$ with $C_1 = 1 + k$. *Source*: Corollary 16.3, Example 16.5. *Prerequisite*: Example 16.1. *Remedial work*: recompute $\sup_x |1 + k \cos(2\pi x)|$.

5) **A.** *Rationale*: if $d_m > 0$ and $d_n > 0$ then $z_m = \bar{x}_m$, $z_n = \bar{x}_n$, $w_m = x_m$, $w_n = x_n$, and only for this reason can one invoke the minimality of $\bar{x}$ against z and that of x against w . *Distractors*: B, C, D name properties used elsewhere (the finiteness of the action is automatic on a finite window; coercivity is needed for Theorem 16.4; the rotation number for Theorem 16.6), not in this step. *Diagnosed error*: failing to notice that the definition of minimality requires competitors *with fixed endpoints*, and that the choice of m and n serves exactly to produce some. *Source*: Theorem 16.3, Steps 2–4. *Prerequisite*: Definition 16.5. *Remedial work*: redo Step 2 drawing the sequence of the signs of d .

6) **B.** *Rationale*: Example 16.4: $W = 0.25$ against the minimum 0.243793, and $\text{tr } DF^2 = 2 - k^2 = 1$, that is, an elliptic orbit. A minimum of the action is hyperbolic (or parabolic in degenerate cases), never elliptic. *Distractors*: A is false (the orbit is clearly of type (1, 2): it is checked in two lines); C is false (the symmetry σ exists, Proposition 16.2(ii)); D contradicts Theorem 16.4. *Diagnosed error*: using symmetry as a criterion of minimality. The minimiser is *also* symmetric ($x_1 = 1 - x_0$): symmetry does not distinguish the two orbits. *Source*: Example 16.4, block [C16–2] of the script. *Prerequisite*: the notion of elliptic/hyperbolic point. *Remedial work*: compute by hand W for the two orbits at $k = 1$ and compare.

7) **B.** *Rationale*: Theorem 16.7 holds for *every* h of the admissible class, not only for the standard map; for $h = \frac{a}{2}(x' - x)^2 + V_k(x)$ with $a = 4$, $k = 1$ one has $C_1 = a + k = 5$, $C_2 = a = 4$, $L = 5$. *Distractors*: A and D wrongly restrict the result to the standard map; C names a numerical criterion about the *existence* of the curve, not about its slope. *Diagnosed error*: failing to recognise that the chapter is built on the hypotheses (H1)–(H4) and not on a specific map (Remark 16.1). *Source*: Theorem 16.7, Exercise 16.3. *Prerequisite*: second derivatives of h . *Remedial work*: do Exercise 16.3(a).

8) **C.** *Rationale*: Remark 16.6: the Poincaré–Birkhoff theorem (stated by Poincaré in 1912, proved by Birkhoff in 1913) gives *two* fixed points, hence two orbits of type (p, q) . *Distractors*: A and B are the authors of the Aubry–Mather theorem (Remark 16.8), which concerns irrational rotation numbers and minimal configurations; D is the author of the numerical criterion on the threshold (Remark 16.9). *Diagnosed error*: attributing to the *variational* existence (one orbit, proved here) what requires the *topological* argument (two orbits, imported). *Source*: Remark 16.6. *Prerequisite*: none. *Remedial work*: list, for each of the four sources, which precise statement of the chapter it supports.

BEST PRACTICE

Assigned exercises — answers.

16.1 (explicit orbit of type (1, 2)). (a) The two steps have length $\frac{1}{2}$ each, hence a kinetic term $2 \cdot \frac{1}{2}(\frac{1}{2})^2 = \frac{1}{4}$; the potential terms are $V_k(0) = -\frac{k}{(2\pi)^2}$ and $V_k(\frac{1}{2}) = +\frac{k}{(2\pi)^2}$, which cancel because $\cos 0 = -\cos \pi$. Hence $W_{1,2} = \frac{1}{4}$ for every k . (Checked numerically for $k = 0.3; 0.5; 1; 2; 3.7$, block [C16–8]. ✓) (b) With $a(x) = k \cos(2\pi x)$ one has $a(0) = k$, $a(\frac{1}{2}) = -k$, and

$$DF^2 = \begin{pmatrix} 1-k & 1 \\ -k & 1 \end{pmatrix} \begin{pmatrix} 1+k & 1 \\ k & 1 \end{pmatrix} = \begin{pmatrix} 1+k-k^2 & 2-k \\ -k^2 & 1-k \end{pmatrix},$$

hence $\text{tr } DF^2 = 2 - k^2$ and $\det = 1$ (one checks: $(1+k-k^2)(1-k) + k^2(2-k) = 1$). It is elliptic for $|2-k^2| < 2$, that is, for $0 < k < 2$. (c) For $k = 2$ the trace equals exactly -2 : the orbit is *parabolic*, and it is the period-doubling bifurcation value. There is no contradiction

with Theorem 16.4: the minimum of $W_{1,2}$ is another orbit, that of Exercise 16.4, of action $0.226446 < 0.25$.

16.2 (values of β). (a) $\beta(0) = \min_x V_k(x) = -\frac{k}{(2\pi)^2}$ and, by Proposition 16.4(ii), $\beta(1) = \beta(0) + \frac{1}{2} = \frac{1}{2} - \frac{k}{(2\pi)^2}$. (b) $k = 0.5$: $\beta(0) = -0.012665$, $\beta(1) = 0.487335$. $k = 2$: $\beta(0) = -0.050661$, $\beta(1) = 0.449339$. (Both confirmed by independent numerical minimisation, block [C16–8]: -0.012665147955 , 0.487334852045 , -0.050660591821 , 0.449339408179 . ✓) (c) Induction: true for $n = 1$. Assuming $\beta(\omega + n) = \beta(\omega) + n\omega + n^2/2$, applying (ii) to $\omega + n$ one has $\beta(\omega + n + 1) = \beta(\omega + n) + (\omega + n) + \frac{1}{2} = \beta(\omega) + n\omega + \frac{n^2}{2} + \omega + n + \frac{1}{2} = \beta(\omega) + (n+1)\omega + \frac{(n+1)^2}{2}$. For $\omega = 0$, $n = 2$: $\beta(2) = \beta(0) + 2$, consistent with $\beta(2) = \beta(1) + 1 + \frac{1}{2} = \beta(0) + \frac{1}{2} + \frac{3}{2}$.

16.3 (twist a). (a) $\partial_1 \partial_2 h_{a,k} = -a$, hence $\delta = a$ (constant twist). $\partial_1^2 h_{a,k} = a + V_k''(x) = a + k \cos(2\pi x)$, hence $C_1 = a + k$; $\partial_2^2 h_{a,k} = a$, hence $C_2 = a$; $L = \max(a + k, a) = a + k$. (b) $y = -\partial_1 h = a(x' - x) - V_k'(x)$ and $y' = \partial_2 h = a(x' - x)$; hence $y' = y + V_k'(x) = y + \frac{k}{2\pi} \sin(2\pi x)$ and $x' = x + y'/a$. For $a = 1$ one recovers (16.7). (c) Never for $k > 0$ (since $a + k > a$); for $k = 0$ one has $L = a$ and the map is integrable, with horizontal invariant curves of zero slope. A small Lipschitz constant means that all the invariant curves and all the cantori are little “rippled”: their graph cannot oscillate faster than L per unit of x . Since $L = a + k$ grows with k , the bound becomes looser as the kick grows, consistently with the values measured in Example 16.5.

16.14. Chapter formulary

Twist map and generating function

$$\frac{\partial X}{\partial y} \geq \delta > 0; \quad y = -\partial_1 h(x, x'), \quad y' = \partial_2 h(x, x'); \quad \partial_1 \partial_2 h \leq -\delta < 0$$

Admissible class (H1)–(H4)

$$h(x + 1, x' + 1) = h(x, x'); \quad \partial_1 \partial_2 h \leq -\delta; \quad h \text{ coercive}$$

$$C_1 = \sup |\partial_1^2 h|, \quad C_2 = \sup |\partial_2^2 h|, \quad L = \max(C_1, C_2)$$

Variational principle (discrete Euler–Lagrange)

$$\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0; \quad h = \frac{1}{2}(x' - x)^2 + V(x) \Rightarrow x_{i+1} - 2x_i + x_{i-1} = V'(x_i)$$

Chirikov standard map

$$y' = y + \frac{k}{2\pi} \sin(2\pi x), \quad x' = x + y'; \quad h_k = \frac{1}{2}(x' - x)^2 - \frac{k}{(2\pi)^2} \cos(2\pi x); \quad \delta = 1, \quad L = 1 + k$$

$$\text{tr } DF_k = 2 + k \cos(2\pi x); \quad (0, 0) \text{ hyperbolic}; \quad (\tfrac{1}{2}, 0) \text{ elliptic} \iff 0 < k < 4, \quad \cos \alpha = 1 - \frac{k}{2}$$

Exchange inequality (Aubry’s lemma)

$$a < b, \quad c < d: \quad h(a, c) + h(b, d) \leq h(a, d) + h(b, c) - \delta(b - a)(d - c)$$

two distinct minimal configurations have at most one crossing

Birkhoff periodic orbits

$$W_{p,q}(x_0, \dots, x_{q-1}) = \sum_{i=0}^{q-1} h(x_i, x_{i+1}), \quad x_q = x_0 + p$$

the minimum exists; two minimisers are strictly ordered or coincide

Birkhoff configurations

$$\forall (s, t) \in \mathbb{Z}^2 : \tau_{s,t}x \leq x \text{ or } \tau_{s,t}x \geq x; \quad a_n - b_n \leq 1; \quad |x_{i+n} - x_i - n\rho| \leq 1$$

Lipschitz graph

$$Y(\xi) = -\partial_1 h(\xi, g(\xi)) = \partial_2 h(g^{-1}(\xi), \xi); \quad |Y(\eta) - Y(\xi)| \leq L |\eta - \xi|$$

Aubry–Mather

for every $\omega \in \mathbb{R}$ there exists an ordered orbit with $\rho = \omega$

$$\omega \notin \mathbb{Q} : \quad \overline{\mathcal{X}} = \mathbb{R} \text{ (invariant curve)} \quad \text{or} \quad \overline{\mathcal{X}} \text{ a Cantor set (cantorus)}$$

Mather’s β function (standard map)

$$\beta\left(\frac{p}{q}\right) = \frac{1}{q} \min W_{p,q}; \quad \beta(-\omega) = \beta(\omega); \quad \beta(\omega + 1) = \beta(\omega) + \omega + \frac{1}{2}; \quad \beta(0) = -\frac{k}{(2\pi)^2}$$

Panorama

Poincaré–Birkhoff: two orbits; Birkhoff: invariant curves = Lipschitz graphs

Moser: KAM curves for Diophantine ω ; $k_c \approx 0.9716$ (Greene’s criterion)

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Final examination of the Volume

INFO

How to use this examination. Fifty multiple-choice questions spanning the entire volume in chapter order, from the topology of covering spaces (Chapter 1) to Aubry–Mather theory (Chapter 16), along the six Parts: topology beyond π_1 , rotations and diffeomorphisms, chaos and ergodic theory, holomorphic dynamics, synchronization and automation, global structures. Unlike the end-of-chapter quizzes, here the answers are not next to the questions: the **Teacher’s key**, at the end of this examination, gives for each item the correct answer, a short explanation, and the reference to the statement that justifies it. Several questions do not ask what is true, but what is *proved in this volume* as opposed to what is merely imported: this is a distinction the treatise takes seriously, and so does this examination.

Indicative score: ≥ 45 excellent; 38–44 good; 30–37 sufficient, with targeted review of the Parts where mistakes were made; < 30 it is worth rereading the volume following the chain that leads from the winding number to the rotation number, and from that to the invariant measure and Mather’s β function.

Part A — Topology of the circle beyond π_1 (Chapters 1–2)

- Let Y be path-connected and locally path-connected, $f : Y \rightarrow S^1$ continuous, $y_0 \in Y$. According to the lifting criterion of Chapter 1, f lifts through the exponential covering $E : \mathbb{R} \rightarrow S^1$ if and only if
 - Y is simply connected
 - f is surjective
 - $f_*(\pi_1(Y, y_0)) = \{0\}$, that is, f sends every loop of Y to a loop of degree 0
 - f is injective
- Up to isomorphism, the coverings of S^1 with path-connected total space are
 - only the universal covering $E : \mathbb{R} \rightarrow S^1$
 - exactly these: $E : \mathbb{R} \rightarrow S^1$, and for every $n \geq 1$ the map $p_n : S^1 \rightarrow S^1$, $p_n(z) = z^n$
 - one for every subgroup of $\mathbb{R}$
 - infinitely many and not classifiable
- An open necklace has $2n_1$ beads of type 1 and $2n_2$ of type 2, arranged in any order. The stolen-necklace theorem of Chapter 1 guarantees a fair split between two thieves with
 - at most 2 cuts, always between two consecutive beads
 - exactly 1 cut
 - at most 4 cuts
 - a number of cuts growing with N

-
4. The Whitney–Graustein theorem states that two regular closed curves of class C^1 in the plane are regularly homotopic if and only if
 - A) they have the same length
 - B) they are both simple
 - C) they are congruent up to a rigid motion
 - D) they have the same rotation number of the tangent field
 5. For every $n \geq 1$, the real vector bundles of rank n over S^1 , up to isomorphism, are
 - A) only one, the trivial one
 - B) exactly two: the trivial bundle $S^1 \times \mathbb{R}^n$ and $M \oplus \mathbb{R}^{n-1}$, with M the Möbius band
 - C) infinitely many, one for each integer (the “winding class”)
 - D) exactly $n + 1$
 6. The principal bundles with structure group $U(1)$ over S^1 , up to isomorphism,
 - A) are classified by $\mathbb{Z}$
 - B) are exactly two
 - C) reduce to a single one, the trivial one, because $U(1) = \mathbb{T}$ is connected
 - D) do not exist
 7. Let Σ be a closed, orientable, connected surface of genus g , and V a continuous tangent vector field with a finite number of zeros, all isolated. The Poincaré–Hopf theorem of Chapter 2 gives
 - A) $\sum_p \text{ind}_p(V) = 0$ always
 - B) $\sum_p \text{ind}_p(V) = \text{the number of zeros}$
 - C) $\sum_p \text{ind}_p(V) = 2g$
 - D) $\sum_p \text{ind}_p(V) = \chi(\Sigma) = 2 - 2g$
 8. The Hopf fibration $\pi : S^3 \rightarrow S^2$, studied in Chapter 2, is
 - A) a *non*-trivial principal $U(1)$ bundle, whose first Chern class has modulus 1 and is a generator of $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$
 - B) a trivial principal $U(1)$ bundle
 - C) a two-sheeted covering of S^2
 - D) a real vector bundle of rank 2 over S^2

Part B — Rotations and diffeomorphisms (Chapters 3–6)

9. Hurwitz’s theorem (Chapter 3) states that for every irrational α there exist infinitely many rationals p/q with
 - A) $|\alpha - p/q| < 1/q^3$
 - B) $|\alpha - p/q| < 1/(\sqrt{5} q^2)$, and the constant $\sqrt{5}$ cannot be improved
 - C) $|\alpha - p/q| < 1/q$

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- D) $|\alpha - p/q| < 1/(2q^2)$, and the constant 2 cannot be improved
10. The three-distance theorem, applied to the points $\{0, \beta, \dots, (N-1)\beta\} \bmod 1$ with β irrational, states that the N arcs determined
- A) all have the same length for N prime
 - B) have at most two distinct lengths
 - C) always have exactly three distinct lengths
 - D) have at most three distinct lengths and, when there are exactly three, the largest is the sum of the other two
11. If β has bounded partial quotients and $x_n = \{n\beta\}$, the discrepancy D_N satisfies
- A) $D_N = O(\log N/N)$, the optimal rate
 - B) $D_N = O(1/\sqrt{N})$ and no better
 - C) $D_N = O(1/N)$
 - D) $D_N \not\rightarrow 0$
12. For an orientation-preserving homeomorphism f of the circle, with lift F , the rotation number $\rho(F) = \lim_n (F^n(\theta) - \theta)/(2\pi n)$
- A) exists only for almost every θ
 - B) depends in general on the point θ
 - C) exists for every θ and has the same value for every θ
 - D) exists only if f is of class C^1
13. Let f be an orientation-preserving homeomorphism of the circle. Poincaré's classification in the rational case states that
- A) $\rho(f)$ rational implies that f is conjugate to a rational rotation
 - B) $\rho(f) = p/q$ in lowest terms if and only if f has a periodic orbit, and then *every* periodic orbit has minimal period exactly q
 - C) $\rho(f)$ rational implies that f is minimal
 - D) f has periodic orbits of mutually different minimal periods
14. If $\rho(f)$ is irrational, Poincaré's classification provides
- A) always a topological conjugacy with $R_{2\pi\rho}$
 - B) periodic orbits of arbitrarily large period
 - C) positive topological entropy
 - D) a continuous, monotone, degree-1 surjective semi-conjugacy h with $h \circ f = R_{2\pi\rho} \circ h$, and the alternative: either h is a homeomorphism, or the minimal set is a Cantor set whose gaps h collapses to points
15. In Arnold's standard family, as $\varepsilon \rightarrow 0^+$ the width of the rotation tongue p/q (in lowest terms) is
- A) $O(\varepsilon^q)$; for $q = 1$ it is exactly ε/π , for every $0 < \varepsilon < 1$
 - B) $O(\varepsilon)$ for every q
-

-
- C) independent of ε
D) $O(\varepsilon^{1/q})$
16. The hypotheses under which Chapter 5 proves that f is conjugate to the rotation $R_{\rho(f)}$ are
- A) f continuous and $\rho(f)$ irrational
B) f of class C^∞
C) $f \in C^1$ with $f' > 0$ everywhere, $\text{Var}_{\mathbb{T}}(\log f') < \infty$, and $\rho(f)$ irrational
D) $f \in C^1$ and $\rho(f)$ irrational
17. The C^1 counterexample built in Chapter 5 violates, among the hypotheses of Denjoy's theorem, exactly
- A) the irrationality of $\rho(f)$
B) the positivity of f' , which fails at some point
C) the continuity of f'
D) the bounded variation of $\log f'$, which is infinite, while $f' > 0$ everywhere for every admissible value of the scale parameter
18. The Denjoy–Koksma inequality, for $\rho(f)$ irrational, $|\rho - p/q| \leq 1/q^2$, and μ the invariant measure, states that for every ψ of finite total variation and every x
- A) $|\sum_{j=0}^{q-1} \psi(f^j(x)) - q \int \psi d\mu| \leq q \text{Var}(\psi)$
B) $|\sum_{j=0}^{q-1} \psi(f^j(x)) - q \int \psi d\mu| \leq \text{Var}(\psi)$, uniformly in x and even for discontinuous ψ
C) the same bound, but only for continuous ψ
D) $|\sum_{j=0}^{q-1} \psi(f^j(x)) - q \int \psi d\mu| \leq \text{Var}(\psi)/q$
19. For ρ Diophantine of type (K, τ) and φ analytic with zero mean on the strip S_b , the cohomological equation $\psi \circ R_\alpha - \psi = \varphi$ of Chapter 6
- A) has one and only one zero-mean solution on a *narrowed* strip $S_{b-\delta}$, with the bound $\|\psi\|_{b-\delta} \leq \frac{\Lambda(\tau)}{2K} \|\varphi\|_b \delta^{-(2+\tau)}$: the loss of domain is the price of small divisors
B) has a solution on the whole of S_b , with no loss of domain at all
C) never has a solution
D) has a solution only if ρ is rational
20. The Gottschalk–Hedlund lemma, for T a minimal homeomorphism of a compact metric space and ϕ a continuous real function, states that
- A) ϕ is a continuous coboundary if and only if $\int \phi = 0$
B) ϕ is always a continuous coboundary
C) ϕ is a continuous coboundary if and only if the Birkhoff sums $S_n \phi$ are bounded at *one single* point — and then they are bounded uniformly
D) the conclusion additionally requires T to be ergodic and ϕ Hölder
21. Arnold's theorem proved in full in Chapter 6 is
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- A) global: every analytic diffeomorphism with Diophantine rotation number is analytically linearizable, however large the perturbation
 - B) local: under $\|u\|_a \leq \varepsilon_0(a, \rho)$ (explicit threshold) and $\rho(F) = \rho$ Diophantine, F is analytically conjugate to R_α ; both hypotheses are indispensable
 - C) valid without any arithmetic hypothesis on the rotation number
 - D) imported, not proved in the chapter

Part C — Chaos and ergodic theory (Chapters 7–11)

- 22. To say that E_m is chaotic in the sense of Devaney means, in Chapter 7, that the following have been proved:
 - A) metric mixing with respect to Lebesgue measure
 - B) infinite topological entropy
 - C) sensitivity to initial conditions alone
 - D) three distinct facts: topological transitivity, density of periodic points, and sensitivity to initial conditions
- 23. Sharkovskii's theorem, in the form proved in the volume, concerns
 - A) continuous maps $f : I \rightarrow I$ of an interval: if f has an orbit of minimal period n and $n \prec m$ in the Sharkovskii order, then it also has an orbit of minimal period m — and the order is optimal
 - B) all continuous maps of the circle
 - C) only odd periods starting from 3
 - D) only piecewise monotone maps with negative Schwarzian derivative
- 24. The count of the fixed points of the iterates, in Chapter 7, gives
 - A) 2^n for both the doubling map D and the tent map T
 - B) $2^n + 1$ for D
 - C) $\#\text{Fix}(D^n) = 2^n - 1$ and $\#\text{Fix}(T^n) = 2^n$: the two maps differ by one point
 - D) n^2 for both
- 25. The topological entropy of the expanding circle maps E_m equals
 - A) 0
 - B) $\log m$; in particular $h_{\text{top}}(D) = \log 2$
 - C) m
 - D) $+\infty$
- 26. For an isometry of a compact metric space, the topological entropy
 - A) equals $\log 2$
 - B) is positive if the rotation is irrational
 - C) is not defined
 - D) is zero; in particular $h_{\text{top}}(R_\alpha) = 0$ for every rotation of the circle

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27. Rokhlin's formula, applied to the doubling map with respect to Lebesgue measure, gives
- A) $h_\mu(D) = \log 2 = \int_{\mathbb{T}} \log |D'| d\mu$
 - B) $h_\mu(D) = 0$
 - C) $h_\mu(D) = 2$
 - D) a value that depends on the chosen partition
28. Birkhoff's ergodic theorem, for (X, μ, T) preserving a probability measure and $\varphi \in L^1(\mu)$, states that
- A) the averages converge uniformly
 - B) convergence holds only for continuous φ
 - C) the conclusion requires mixing
 - D) the averages $\frac{1}{n} S_n \varphi$ converge μ -almost everywhere to some $\varphi^* \in L^1$ invariant with $\int \varphi^* = \int \varphi$, and if T is ergodic then φ^* is the constant $\int \varphi d\mu$
29. Oxtoby's criterion characterizes unique ergodicity as
- A) the existence of at least one invariant measure
 - B) the *uniform-in-x* convergence of the Birkhoff averages of every $\varphi \in C(X)$ to a constant
 - C) the mixing property
 - D) the vanishing of entropy
30. The absolutely continuous invariant measure of $f_4(x) = 4x(1-x)$ on $[0, 1]$ (Ulam–von Neumann) is
- A) the Lebesgue measure dx
 - B) the Dirac mass at 0
 - C) nonexistent: f_4 admits no absolutely continuous invariant measure
 - D) $d\nu = \frac{dx}{\pi\sqrt{x(1-x)}}$, with a density that diverges integrably at the endpoints
31. On the space $L^2(\mathbb{T})$, the spectrum of the transfer operator $\mathcal{L}$ of the doubling map
- A) contains *every* λ with $|\lambda| < 1$ as an eigenvalue, with the explicit lacunary eigenfunction $g_\lambda(\theta) = \sum_n \lambda^n e^{i2^n \theta}$: on L^2 there is no spectral gap at all
 - B) reduces to the single value $\lambda = 1$
 - C) is a finite set
 - D) is empty
32. On regular spaces, by contrast, Chapter 10 proves that
- A) there is no gap there either
 - B) the gap exists only on L^2
 - C) every eigenvalue other than 1 satisfies $|\lambda| \leq \frac{1}{2}$ on $BV(\mathbb{T})$ and $|\lambda| \leq (\frac{1}{2})^\alpha$ on $C^{0,\alpha}(\mathbb{T})$, with $\|\mathcal{L}^n g_0 - 1\|_\infty \leq 2^{-n} \text{Var}(g_0)$
 - D) every eigenvalue other than 1 satisfies $|\lambda| \leq \frac{1}{4}$
-

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33. The central limit theorem proved for the doubling map with Gordin's martingale method states that, for $u \in BV(\mathbb{T})$ with zero mean,
- A) $S_n u / \sqrt{n}$ always converges to $\mathcal{N}(0, 1)$
 - B) the conclusion requires independence of the variables $u \circ D^k$
 - C) a central limit theorem cannot hold for a deterministic system
 - D) $S_n u / \sqrt{n}$ converges in law to $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \int m^2 d\mu$ from Gordin's decomposition, with the degenerate case $\sigma^2 = 0$ explicitly admitted
34. In the logistic family $f_r(x) = rx(1 - x)$, the 2-cycle born at $r = 3$ is asymptotically stable
- A) for every $r > 3$
 - B) exactly for $3 < r < 1 + \sqrt{6} \approx 3.449490$, with multiplier $-r^2 + 2r + 4$
 - C) never
 - D) for $3 < r < 4$
35. Lanford's theorem on Feigenbaum renormalization, in Chapter 11,
- A) is declared *imported*: stated in a remark with its primary source, not proved in the volume; the constant $\delta = 4.669201609\dots$ is instead estimated numerically in the chapter
 - B) is proved in full
 - C) is disproved by the calculations of the chapter
 - D) is equivalent to the cascade existence theorem, which the chapter proves

Part D — Holomorphic dynamics (Chapter 12)

36. In the escape criterion for $g_c(z) = z^2 + c$, the threshold beyond which the orbit certainly diverges is
- A) the radius 1
 - B) the radius $|c|$
 - C) the radius $R_c = \max(|c|, 2)$: if $|g_c^n(z)| > R_c$ for some n , then $|g_c^m(z)| \rightarrow +\infty$, and K_c is contained in the closed disk of radius R_c
 - D) no universal threshold exists
37. For $c \in \mathcal{M}$, the Böttcher coordinate
- A) remains defined only in a neighborhood of ∞ and never extends further
 - B) extends to a biholomorphism $\Phi_c : \mathbb{C} \setminus K_c \rightarrow \mathbb{C} \setminus \overline{\mathbb{D}}$ with $\Phi_c(g_c(z)) = \Phi_c(z)^2$; in particular K_c is connected
 - C) exists for every c , inside and outside $\mathcal{M}$, as a global biholomorphism
 - D) implies that K_c is totally disconnected
38. The set of parameters c for which g_c has an attracting fixed point is
- A) the disk $|c + 1| < \frac{1}{4}$
 - B) the disk of radius 2

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- C) a Cantor set
 - D) the image of the open unit disk under $c(\lambda) = \lambda/2 - \lambda^2/4$, where λ is the multiplier of the fixed point; its boundary is the main cardioid
39. Under the Böttcher conjugacy, the action of g_c on the *angles* of the external rays is
- A) exactly the doubling map $t \mapsto 2t \bmod 1$ of Chapter 7
 - B) a rigid rotation
 - C) the tent map
 - D) the identity
40. For $\lambda = e^{2\pi i\vartheta}$ on the boundary of the cardioid, Chapter 12 proves that
- A) the fixed point is always linearizable
 - B) it is never linearizable
 - C) the fixed point is linearizable if ϑ is Diophantine (Siegel) and *not* linearizable for a dense G_δ set of ϑ (Cremer): the two phenomena coexist on the same boundary
 - D) linearizability depends only on the modulus of λ

Part E — Synchronization and automation (Chapters 13–14)

41. For Adler's equation $\dot{\theta} = \Delta\omega - K \sin \theta$ with $K > 0$ and $|\Delta\omega| < K$, the regime is
- A) free of equilibria
 - B) with an attracting limit cycle
 - C) with a single equilibrium, always stable
 - D) a locked regime: two equilibria, $\theta_s^* = \arcsin(\Delta\omega/K)$ asymptotically stable and $\pi - \theta_s^*$ unstable
42. In the Kuramoto model with N *identical* oscillators, the synchronized state
- A) is unstable for large N
 - B) is the unique global minimum of the potential V and is exponentially stable transversally, with rate exactly K : the Jacobian has eigenvalue 0 simple and $-K$ with multiplicity $N - 1$
 - C) converges with rate K/N
 - D) has a rate depending on the natural frequencies
43. For a Lorentzian frequency distribution with half-width γ , Chapter 13 obtains
- A) $K_c = 2\gamma$ and the exact solution $r = \sqrt{1 - K_c/K}$ for every $K > K_c$
 - B) $K_c = \gamma$ and $r = 1 - K_c/K$
 - C) the general threshold $K_c = 2/(\pi g(0))$, but no closed form for r
 - D) $r = 1$ for every $K > K_c$
44. Itoh's theorem on phase unwrapping states that the algorithm exactly reconstructs the lift
- A) always, whatever the sampling step

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- B) if and only if the initial phase is zero
 - C) if and only if every *true* sampled increment $\Delta\Theta_j$ lies in $(-\pi, \pi]$; in terms of sampling frequency it suffices that $f_s > \omega_{\max}/\pi$
 - D) if and only if $f_s > \omega_{\max}/(2\pi)$, the Nyquist condition
45. In a two-track Vernier encoder with m and n periods, the integer J is reconstructed exactly when the maximum error η on the measured phases satisfies
- A) $\eta < \pi/(mn)$
 - B) $\eta < \pi/(m+n)$
 - C) $\eta < \pi/2$
 - D) no positive tolerance is sufficient

Part F — Global structures (Chapters 15–16)

46. On the group $\widetilde{\text{Homeo}}$ of lifts, the rotation number ρ
- A) is not a homomorphism in general, but is a conjugation-invariant homogeneous quasi-morphism, with defect $|\rho(FG) - \rho(F) - \rho(G)| < 2$; on abelian subgroups it is a true homomorphism
 - B) is always a homomorphism
 - C) is not even continuous
 - D) is a homomorphism if and only if the group is finite
47. For a subgroup $G \leq \text{Homeo}_+(\mathbb{T})$ *exactly* one of the following holds
- A) only two: finite orbit or minimal action
 - B) three: G has a finite orbit; the action is minimal; or there exists a *unique* exceptional minimal set (compact, perfect, with empty interior and totally disconnected), on which every orbit accumulates
 - C) the action is always minimal
 - D) the minimal set is never unique
48. The Milnor–Wood inequality, proved in Chapter 15 for an action $\varrho : \Gamma_g \rightarrow \text{Homeo}_+(\mathbb{T})$, is
- A) $|e(\varrho)| \leq 1$
 - B) $|e(\varrho)| < g$
 - C) $|e(\varrho)| \leq |\chi(\Sigma_g)| = 2g - 2$; in particular $e(\varrho) = 0$ for every action of $\mathbb{Z}^2$ on the circle
 - D) the Euler class is not bounded
49. Aubry’s lemma states that two distinct *minimal* configurations of a twist map
- A) never cross
 - B) cross at most once
 - C) cross infinitely many times
 - D) cross exactly twice

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50. The Aubry–Mather theorem, for an irrational rotation number ω , produces a closed, invariant, ordered set $\mathcal{M}$, contained in the graph of a Lipschitz function, which
- A) exists only for rational ω
 - B) does not exist in general
 - C) is always an essential invariant curve
 - D) is *either* an essential invariant curve, on which the dynamics is conjugate to R_ω , or a cantor, on which the dynamics is conjugate to the restriction of R_ω to its unique minimal set

Teacher’s key

Correct answer, short explanation, and reference to the statement that justifies it. Where the wrong answer is a recurring confusion, this is pointed out.

1. **C.** Theorem 1.1: the general condition is $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$, which for $p = E$ reduces to $f_*(\pi_1(Y, y_0)) = \{0\}$ because $\pi_1(\mathbb{R}) = 0$. Simple connectedness of Y (answer A) is sufficient but not necessary.
2. **B.** Theorem 1.2: the list is complete and exclusive — $E : \mathbb{R} \rightarrow S^1$ for $n = 0$ and $p_n(z) = z^n$ for $n \geq 1$ — and it is the geometric translation of the fact that the subgroups of $\mathbb{Z} \cong \pi_1(S^1)$ are $\{0\}$ and $n\mathbb{Z}$.
3. **A.** Theorem 1.9: two thieves and two bead types require at most two cuts, always between two consecutive beads. It is a combinatorial consequence of Borsuk–Ulam in dimension 2 (Theorem 1.7), like the pancake theorem.
4. **D.** Theorem 1.11: the rotation number of the tangent field is the complete invariant of *regular* homotopy. Length and simplicity are irrelevant: the figure-eight curve and the circle are both closed and regular, but have rotation numbers 0 and ± 1 .
5. **B.** Theorem 2.7: for every rank n there are exactly two classes, because $\pi_0(GL_n(\mathbb{R}))$ has two elements (polar decomposition and $\pi_0(O(n)) = \mathbb{Z}/2$). The “infinitely many classes” of answer C is what happens for bundles over S^2 , not over S^1 .
6. **C.** Corollary 2.2, from Theorem 2.8: principal bundles over S^1 are classified by the conjugacy classes of $\pi_0(G)$, and $U(1) = \mathbb{T}$ is connected, hence $\pi_0(U(1))$ is trivial.
7. **D.** Theorem 2.9: the sum of the indices is a topological invariant and equals $\chi(\Sigma) = 2 - 2g$. For $g = 1$ (the torus) the sum is 0, which is why the torus admits nowhere-vanishing tangent fields while the sphere does not (Theorem 1.14).
8. **A.** Theorem 2.11 for the principal $U(1)$ -bundle structure, non-trivial, with transition function $e^{i\varphi}$ on the equator, and Theorem 2.12 for the computation of the curvature: the normalized integral equals -1 , so the class has modulus 1 and is a generator of $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$ (Dirac quantization). The sign depends on the chosen orientation.
9. **B.** Theorem 3.5: it is proved that among three consecutive convergents at least one satisfies the bound. The optimality of $\sqrt{5}$ is Proposition 3.2, verified along the convergents of φ : no larger constant works for *every* irrational.
10. **D.** Theorem 3.8: at most three lengths, and the relation $\ell_{\max} = \ell_+ + \ell_-$ when there are three. This is the theorem that explains why the golden angle produces the uniform packing of the sunflower head (Proposition 3.7).

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11. **A.** Theorem 3.13: combining Erdős–Turán with Ostrowski’s bound yields $ND_N = O(\log N)$. Answer B is the *elementary* rate of Proposition 3.6, which always holds but is not optimal; $D_N = O(1/N)$ is impossible by Schmidt’s bound.
 12. **C.** Theorem 4.1: existence follows from Fekete’s lemma (Lemma 4.1) applied to the sub-additive sequence a_n and the superadditive sequence b_n , and independence from the point follows from the fact that F^n commutes with translations by 2π . No regularity beyond continuity is required.
 13. **B.** Theorem 4.3: the equivalence between rationality and the presence of periodic orbits, and the uniformity of the minimal period, which follows from the fact that an orientation-preserving homeomorphism acts on the orbit as a cyclic shift $i \mapsto i + r$ with $\gcd(r, q) = 1$. Answer A is false: f can have non-periodic orbits asymptotic to the periodic ones.
 14. **D.** Theorem 4.4: one always obtains a degree-1 monotone semi-conjugacy, and only the alternative between $E = \mathbb{R}$ (a true conjugacy) and E a Cantor set. Answer A is exactly the mistake that Denjoy’s theorem (Theorem 5.2) corrects by adding a regularity hypothesis.
 15. **A.** Theorem 4.7, with the case $q = 1$ exact in Theorem 4.6 (the 0/1 tongue is $|\alpha| \leq \varepsilon/(2\pi)$, hence of width ε/π). The case $q = 2$ is proved in full ($\varepsilon^2/(4\pi) + O(\varepsilon^3)$); the general case is imported.
 16. **C.** Theorem 5.2: the three hypotheses are (a) $f \in C^1$ with $f' > 0$, (b) $\text{Var}_{\mathbb{T}}(\log f') < \infty$, (c) $\rho(f)$ irrational. C^∞ (answer B) is sufficient but needlessly strong; C^1 alone (answer D) is not enough, and the counterexample shows it.
 17. **D.** Theorem 5.4, points (1) and (4): in the counterexample $f' > 0$ everywhere *for every* admissible c — positivity depends on the ratios $r_n = \ell_{n+1}/\ell_n$, in which the scale factor cancels — while $\text{Var}(\log f') = \infty$. It is the bounded-variation hypothesis, and only that one, which fails.
 18. **B.** Theorem 5.3: the bound is $\text{Var}(\psi)$, with no factors of q , uniform in x , and valid even for discontinuous ψ . It is precisely this generality that makes it useful where Denjoy’s inequality (Theorem 5.1), which concerns only $\log f'$, cannot reach.
 19. **A.** Theorem 6.1: the solution exists and is unique with zero mean, but only on a narrowed strip, with a bound that blows up as $\delta^{-(2+\tau)}$ as the loss δ tends to zero. This controlled loss is what makes Newton’s scheme possible; the C^r case is Theorem 6.2.
 20. **C.** Theorem 6.4: the equivalence between (i) a continuous coboundary, (ii) boundedness of the sums at one point, and (iii) uniform boundedness. Zero mean (answer A) is necessary but not sufficient: Corollary 6.4 exhibits a φ with zero mean whose Birkhoff sums are unbounded.
 21. **B.** Theorem 6.5: the statement is local, with explicit threshold $\varepsilon_0(a, \rho)$, and requires both (H1) and (H2). Remark 6.11 shows that dropping (H2) makes the statement false (a small perturbation can fall into an Arnold tongue, Theorem 4.7) and that the global version of answer A is false: there the correct arithmetic condition is the Herman–Yoccoz one.
 22. **D.** Theorem 7.5: the three ingredients are proved separately (transitivity, density of periodic points, sensitivity with explicit constant π/m) and then assembled. Metric mixing is a different matter, proved in Chapter 9.
 23. **A.** Theorem 7.7, with optimality proved in Theorem 7.8. Beware of answer B: Sharkovskii holds for maps of the *interval*, not for those of the circle, where a rational rotation has only one period. Answer C confuses the theorem with the special case of Li–Yorke (Theorem 7.6).
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24. **C.** Theorems 7.9 and 7.10: $2^n - 1$ versus 2^n . The difference of one point is real and not a counting error: for D the equation $(2^n - 1)\theta \in 2\pi\mathbb{Z}$ has $2^n - 1$ solutions on the circle, because $\theta = 0$ and $\theta = 2\pi$ are the same point.
 25. **B.** Theorem 8.4, computed directly with (n, ε) -separated sets, without passing through the conjugacy with the shift (which gives the same value, Theorem 8.5).
 26. **D.** Theorem 8.6: an isometry does not separate orbits, so $N(n, \varepsilon)$ does not grow with n . This holds for every rotation, rational or irrational: irrationality produces density of the orbits, not exponential complexity.
 27. **A.** Theorem 8.12: the Kolmogorov–Sinai entropy with respect to Lebesgue measure coincides with $\log 2$ and with the integral of $\log |D'|$, that is, with the Lyapunov exponent (Theorem 8.10). The value does not depend on the partition because $\mathcal{P}$ is generating.
 28. **D.** Theorem 9.7: almost-everywhere convergence for every $\varphi \in L^1$, conservation of the integral, and constancy in the ergodic case. Uniform convergence (answer A) is instead the characterization of *unique* ergodicity (Theorem 9.9).
 29. **B.** Theorem 9.9: it is precisely uniformity in x that distinguishes unique ergodicity from mere ergodicity. Applied to R_α it gives Theorem 9.10.
 30. **D.** Theorem 9.12: the density $1/(\pi\sqrt{x(1-x)})$ is that of the arcsine law, transported from the circle’s Lebesgue measure by the conjugacy with the doubling map (Theorem 9.13).
 31. **A.** Theorem 10.2: the lacunary series provide eigenfunctions for every $|\lambda| < 1$. This is why L^2 is the *wrong* space for the decay of correlations: the gap must be sought elsewhere.
 32. **C.** Theorem 10.3: on BV and on the Hölder spaces the gap exists, with explicit constants, and produces the exponential decay of correlations (Theorem 10.4). The comparison with the previous item is the point: the spectral gap is a property of the operator–space pair, not of the operator alone.
 33. **D.** Theorem 10.8, based on Gordin’s decomposition (Theorem 10.7). The degenerate case is no oversight: if u is a coboundary, $S_n u$ remains bounded and the $\sqrt{n}$ normalization gives the Dirac mass at 0.
 34. **B.** Theorem 11.2: the multiplier $-r^2 + 2r + 4$ equals 1 at $r = 3$ and -1 at $r = 1 + \sqrt{6}$, and stability holds exactly on the open interval between the two. Beyond $1 + \sqrt{6}$ the 2-cycle still exists but is unstable: existence and stability are two different things.
 35. **A.** Remark 11.4: Lanford’s theorem is declared imported, with the primary source from 1982. The chapter proves the existence of the cascade (Theorem 11.6) and the normal forms of the bifurcations, and estimates δ numerically (Remark 11.2); universality is a different matter, and remains part of the panorama.
 36. **C.** Theorem 12.1: the threshold $R_c = \max(|c|, 2)$ depends on the parameter. The quantitative bound $|g_c^{n+k}(z)| \geq \mu^{2^k-1}|g_c^n(z)|$ shows that escape, once begun, is doubly exponential.
 37. **B.** Theorem 12.5: this is the “connected” half of the fundamental dichotomy. For $c \notin \mathcal{M}$ the global biholomorphism does not exist (answer C is false) and K_c is totally disconnected.
 38. **D.** Theorem 12.7: the multiplier parametrization $\mathfrak{c}(\lambda) = \lambda/2 - \lambda^2/4$ is injective on $\overline{\mathbb{D}}$, and the circle of neutral multipliers becomes the cardioid. Answer A is the *bulb* of period 2 (Theorem 12.8), tangent to the cardioid at $c = -3/4$.
 39. **A.** Theorem 12.9: $g_c(R_t) = R_{2t \bmod 1}$. It is the literal return of Chapter 7 inside holomorphic dynamics: the circle of external rays carries the same symbolic dynamics.
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40. **C.** Theorem 12.10 for the Diophantine case and Theorem 12.11 for Cremer numbers, which form a dense G_δ set. Their coexistence is the point: linearizability depends on the fine arithmetic of ϑ , not on its irrationality nor on the modulus of λ , which is always 1.
 41. **D.** Theorem 13.2(i). Answers A and B describe the opposite regime, phase slipping $|\Delta\omega| > K$, point (iii) of the same theorem; answer C forgets the unstable equilibrium, which is what makes the basin of attraction the whole circle minus one point.
 42. **B.** Theorem 13.4: the Jacobian is $-L$ with L the Laplacian of the complete graph of weight K/N , whose nonzero eigenvalues all equal $Nw = K$. The rate does not depend on N , contrary to the intuition behind answer C. The gradient-flow structure is Lemma 13.1.
 43. **A.** Corollary 13.3, from Theorem 13.5 with $g(0) = 1/(\pi\gamma)$. Answer C is wrong only in its second half: the general threshold $2/(\pi g(0))$ is correct, but in the Lorentzian case the integral can be computed in closed form.
 44. **C.** Theorem 14.2 for the exact condition (an if-and-only-if on the *true* increments, not on the measured ones) and Corollary 14.1 for the translation into sampling frequency. Beware of answer D: the unwrapping threshold is $\omega_{\max}/\pi$, that is, twice the Nyquist one.
 45. **B.** Theorem 14.3: the tolerance $\pi/(m+n)$ degrades with the *sum* of the numbers of periods, not with the product. This is why a Vernier with very large m and n gains resolution but loses robustness to phase errors.
 46. **A.** Theorem 15.1 for the defect < 2 and Proposition 15.2 for the abelian case. That ρ is not a homomorphism is essential: if it were, the Milnor–Wood inequality would be trivial.
 47. **B.** Theorem 15.4: the three cases are mutually exclusive and cover all possibilities. The third case — the exceptional minimal set — is the same phenomenon as Denjoy’s counterexample (Chapter 5), seen from the group side.
 48. **C.** Theorem 15.5, with Corollary 15.5 for $\mathbb{Z}^2$. Answer B is Milnor’s *original* statement (1958) for flat $\mathrm{GL}^+(2, \mathbb{R})$ bundles: the two bounds do not contradict each other, because they concern actions on different circles (the circle of directions and $\mathbb{RP}^1$), as explained in the chapter.
 49. **B.** Theorem 16.3: at most one crossing. Answer A is too strong (two distinct minimal configurations can cross once) and it is precisely this “at most once” that produces the ordering (Theorem 16.5) and then the Lipschitz graph (Theorem 16.7).
 50. **D.** Theorem 16.9, built on the existence of ordered orbits for every ω (Theorem 16.8). The dichotomy is the same one encountered in Poincaré’s classification (Theorem 4.4): either the whole circle, or a Cantor set. The volume ends where it began, with one more turn.

INFO

After the examination. The six Parts follow the volume’s logical chain: one begins by counting turns (the degree, the winding number), moves on to measuring the *average speed* of the turns (the rotation number), discovers that the arithmetic of that number decides everything — conjugacy or Cantor set, linearization or small divisors, Siegel disk or Cremer point, invariant curve or cantorus — and finally returns to counting, but in the cohomology of a group (the Euler class). Anyone who answered items 1–21 well but the later ones poorly has mastered the topological and arithmetic part and should practice the tools of measurement (entropy, invariant measures, operators); anyone who did the opposite would do well to return to Chapter 4, because it is from there that the rest of the volume depends. The items that ask what is *imported* and not proved — the twenty-first, the thirty-fifth — are not a bibliographic detail: they are the check that the treatise’s double register has been

understood.

Registry of imported results

A.1. What this appendix records, and why

A treatise that proves its own theorems must state, with the same precision, which theorems it does *not* prove. This appendix is that document: it lists, chapter by chapter, every result that Volume IV states or uses without proving it in full, with the statement in synthetic form, the source cited in the chapter, and the reason for the omission. It adds nothing to the volume and replaces nothing: it is an index of debts.

The rule the treatise has set itself is simple and has held since the first volume: *nothing is half-proved without saying so*. If a proof rests on a theorem that the volume does not prove, that theorem is declared in the chapter in a dedicated remark — “what we import” — and reappears here. If instead a result is merely cited to orient the reader, without anything in the volume leaning on it, it is declared as a panorama, and reappears here with that label. The difference is not formal: an import is a *logical* debt, a panorama is an *expository* debt, and confusing the two would mean passing off as grounded what is merely mentioned.

The three labels

- **Imported** — the result is used, or declared assumed, in a proof, a definition, or an example of the volume. It is a logical debt: anyone wanting a fully self-contained volume would have to fill it.
- **Panorama** — the result is stated to situate the subject, but no proof of the volume leans on it. Removing it would weaken the exposition, not the logic.
- **Internal cross-reference** — the statement appears in one chapter and its proof in another chapter of this same volume. It is not an import: it is a debt that the volume settles by itself, and it is recorded here only so that the reader of the chapter that states it does not see it proved on the spot.

Some entries are *partial imports*: the chapter proves half of the statement and imports the other half. They are flagged as such, with a precise indication of where the boundary lies — this is the information that matters most, and that a generic list would lose.

How it was built

The registry originates from running `registro_importati.py` Vol14, which walks the sources and collects the blocks that declare an import or a panorama. The raw output of the script was filtered down to only the sixteen chapters that make up the volume (the source folder still contains some files from the structure prior to the expansion, which are not part of the volume and whose occurrences were discarded) and then reread block by block: the script knows which blocks exist, not what they say. Statements, sources, and reasons below were obtained by reading the blocks, not the index.

The tally is this: **95 blocks** examined across the sixteen chapters, of which **84** constitute a registry entry and **11** turned out, on reading, to be methodological notes that declare no import

— listed for transparency in the final Section, so that anyone repeating the count with the script finds the same total and knows where the missing blocks ended up.

A.2. Registry, chapter by chapter

Chapter 1 — Coverings, degree, and the theorems of the circle

- **Van Kampen’s theorem, groupoid version** (*imported*). If $X = U \cup V$ with U, V open and $A \subseteq U \cap V$ meets every path-connected component of U , V , and $U \cap V$, the fundamental groupoid $\Pi_1(X, A)$ is the push-out of the groupoids of U and V along that of $U \cap V$. The groupoid version is needed because, writing S^1 as the union of two arcs, the intersection has two components and the single-basepoint version does not apply. *Source*: A. Gramain, “Le théorème de Van Kampen”, *Cahiers de Topologie et Géométrie Différentielle Catégoriques* 33 (1992), 237–251. *Reason*: proving it would require a categorical apparatus disproportionate to the use made of it. *Status*: the chapter uses it as an independent check on the computation of $\pi_1(S^1) \cong \mathbb{Z}$, already established directly in Volume III, not as its foundation: no proof of the volume leans on it. *Where*: Remark 1.3.
- **Whitney–Graustein, constructive direction** (*imported*). Every regular closed plane curve with rotation number ρ is regularly homotopic to the circle traversed ρ times (for $\rho \neq 0$) or to the symmetric figure eight (for $\rho = 0$). The technical difficulty is maintaining the closure constraint $\int_0^1 e^{i\theta_u(s)} ds = 0$ along the whole homotopy, which linear interpolation of the tangent angles does not preserve. *Source*: H. Whitney, “On regular closed curves in the plane”, *Compositio Mathematica* 4 (1937), 276–284. *Reason*: the construction occupies the entire original article; the chapter proves the opposite direction (the invariance of ρ) and imports this one, so as not to present as proved what cannot, in the space available, be checked line by line. *Where*: Remark 1.10.
- **Degree in higher dimension: Sard or homology** (*panorama*). To extend Brouwer, Poincaré–Hopf, and Borsuk–Ulam to S^n with $n \geq 2$ one needs a notion of degree for maps $S^n \rightarrow S^n$, and the lifting method no longer suffices, because $\pi_1(S^n) = 0$ for $n \geq 2$. The two classical routes are Sard’s theorem (degree as the signed count of preimages of a regular value) and homology ($H_n(S^n) \cong \mathbb{Z}$). *Source*: no bibliographic source in the block; the chapter refers to the opening argument of Chapter 2. *Reason*: it is a hinge between two chapters, not a result that Chapter 1 uses. *Where*: Remark 1.12.
- **Brouwer’s fixed point in dimension n** (*imported partial*). Every continuous map $f : \overline{B^n} \rightarrow \overline{B^n}$ has a fixed point. Milnor’s analytic proof is reported in the chapter in its outline: if f (first assumed C^1) had no fixed points, the retraction r onto the boundary would give $Q(t) = \int_{\overline{B^n}} \det(Df_t) dx$ constant, with $Q(0) = \text{Vol}(\overline{B^n}) > 0$ and $Q(1) = 0$. *Source*: J. Milnor, “Analytic Proofs of the Hairy Ball Theorem and the Brouwer Fixed Point Theorem”, *American Mathematical Monthly* 85 (1978), 521–524. *Reason*: the chapter carries out the method in full in dimension 2 and 3; for general n it does not reproduce to the same standard the regularity details of the retraction (composition via the implicit function theorem), and the general statement remains imported. *Where*: Remark 1.13.

Chapter 2 — Bundles over the circle and the sphere

- **Classification of $\text{Vect}_n(S^k)$ by clutching** (*imported*). Writing S^k as the union of two disks, complex vector bundles of rank n are classified by $\pi_{k-1}(U(n))$, and in the real case $\text{Vect}_n(S^k) \cong \pi_{k-1}(O(n))/\pi_0(O(n))$. For $k = 1$ one recovers the classification proved in the chapter; for $k = 2$, in the complex case, $\pi_1(U(n)) \cong \mathbb{Z}$ and the integer is the first Chern class, the invariant that makes the Hopf fibration nontrivial. *Source*: the systematic sources

declared in the chapter’s registry; the Bott periodicity is marked there as panorama only. *Reason:* the general theory is not developed in the volume. *Status:* the formula is used several times in the chapter, so it is a full import, not a panorama. *Where:* Remark 2.7.

- **Degree of the Gauss map and global Gauss–Bonnet** (*panorama*). For a surface $\Sigma \subset \mathbb{R}^3$ with Gauss map $N : \Sigma \rightarrow S^2$, the indices of the critical points of the height function are tied to the sign of the curvature and to the local degree of N ; together with $\iint_{\Sigma} K \, dA = 2\pi\chi(\Sigma)$ one obtains $\deg(N) = \chi(\Sigma)/2$ (degree 1 for the convex sphere, 0 for the torus). *Source:* the differential geometry source declared in the chapter’s registry. *Reason:* the chapter explicitly declares that it does not re-prove it because it is not needed elsewhere; *local* Gauss–Bonnet, which is needed for holonomy, is instead proved in full. *Where:* Remark 2.10.
- **Adiabatic theorem of quantum mechanics** (*imported, as a hypothesis of a model*). In the spin-1/2 model in a slowly varying direction field, it is assumed that the system remains in the instantaneous ground state: this is the adiabatic theorem, external to the mathematics of the chapter. On that hypothesis, the geometric phase accumulated along a parallel of colatitude θ_0 equals $\gamma = -\frac{1}{2}\Omega = -\pi(1 - \cos\theta_0)$, half the subtended solid angle — and this holonomy computation is instead entirely internal to the chapter. *Source:* Berry’s classical result (1984) and the physics sources declared in the chapter. *Reason:* it belongs to physics, not to the mathematical domain of the volume. *Where:* Remark 2.17.
- **Lens spaces:** $\pi_1(L(n,1)) \cong \mathbb{Z}/n$ (*panorama*). The transition function $\varphi \mapsto e^{in\varphi}$ on the equator of S^2 defines a circle bundle of Euler number n , whose total space is the lens space $L(n,1)$, a gluing of two solid tori along the boundary with a twist of $1/n$ of a turn; it is imported that $\pi_1(L(n,1)) \cong \mathbb{Z}/n$. Notable cases: $n = 1$ gives the Hopf fibration and $L(1,1) \cong S^3$; $n = 2$ gives $L(2,1) \cong \mathbb{RP}^3 \cong SO(3)$. *Source:* the algebraic topology sources declared in the chapter, which indicate the two possible tools (long exact homotopy sequence of the bundle, or van Kampen on the gluing). *Reason:* neither of the two tools is developed in the volume; for the double covering $S^3 \rightarrow SO(3)$ the chapter refers to Volume III. *Where:* Remark 2.19.

Chapter 3 — Rotations, continued fractions, and equidistribution

- **Full optimality of the Hurwitz constant** (*imported*). Not only does no convergent, but no pair (p,q) can improve on $\sqrt{5}$: for every $c > \sqrt{5}$ the set of rationals with $|\varphi - p/q| < 1/(cq^2)$ is finite. The missing step is the Legendre criterion: every approximation with $|\alpha - p/q| < 1/(2q^2)$ is necessarily a convergent of α . *Source:* A. Hurwitz, “Ueber die angenäherte Darstellung der Irrationalzahlen durch rationale Brüche”, *Mathematische Annalen* 39 (1891), 279–284; A. Khinchin, *Continued Fractions*, §15. *Reason:* the Legendre criterion requires the continued-fraction expansion of p/q itself, not only the properties of the convergents of α already established in the chapter, which proves in full the weaker version along the convergents. *Where:* Remark 3.3.
- **Erdős–Turán inequality** (*imported*). For all $x_1, \dots, x_N \in [0, 1)$ and every $K \geq 1$,

$$D_N \leq C \left(\frac{1}{K} + \sum_{k=1}^K \frac{1}{k} \left| \frac{1}{N} \sum_{n=1}^N e^{2\pi i k x_n} \right| \right),$$

with C an absolute constant. *Source:* L. Kuipers, H. Niederreiter, *Uniform Distribution of Sequences*, Theorem 2.5 (which cites Erdős–Turán 1948); J. J. Holt, “On a form of the Erdős–Turán inequality”, *Acta Arithmetica* 74 (1996), 61–66, for the explicit constants. *Reason:* the standard proof approximates the indicator of an interval from above and below with trigonometric polynomials built from the Vaaler or Selberg–Beurling kernel: approximation theory that exceeds the scope of the chapter, which has already proved in full, with the Fejér

kernel, the qualitative analogue. *Status*: the chapter declares it as its own single strong import; it is used in the discrepancy estimates. *Where*: Remark 3.7.

- **Estimate of the Ostrowski sum (imported)**. For β with bounded partial quotients, $\sum_{k \leq K} \frac{1}{k \|k\beta\|} = O(\log K)$ rather than the trivial $O(K)$; in the Diophantine case of exponent τ the estimate is $O(K^\tau \log K)$. This is the step that brings the discrepancy from $O(1/\sqrt{N})$ down to $O(\log N/N)$. *Source*: A. Ostrowski, “Bemerkungen zur Theorie der Diophantischen Approximationen”, *Abhandlungen Hamburg* 1 (1922), 77–98, for the proof; Kuipers–Niederreiter for the precise statement. *Reason*: the proof groups the integers according to the Ostrowski numeration, in the non-standard base given by the denominators of the convergents, and uses the three-distance theorem block by block. *Status*: used in the proof of the discrepancy theorem for bounded partial quotients. *Where*: Remark 3.8.
- **Schmidt’s theorem on the irregularity of distribution (panorama)**. There exists an absolute $c > 0$ such that for *every* sequence (x_n) in $[0, 1)$ one has $D_N \geq c \log N/N$ for infinitely many N : no sequence can systematically do better than $\log N/N$, and hence rotations with bounded partial quotients are optimally distributed up to the constant. *Source*: W. M. Schmidt, “Irregularities of distribution VII”, *Acta Arithmetica* 21 (1972), 45–50. *Reason*: the proof uses measure theory and extremal combinatorics, outside the tools of the chapter. *Where*: Remark 3.10.

Chapter 4 — Rotation number, Poincaré’s classification, and Arnold tongues

- **Świątek: full-measure mode-locking at criticality (panorama)**. At criticality $\varepsilon = 1$, where the map of the standard family is a homeomorphism but not a diffeomorphism, the set of α with irrational rotation number has zero Lebesgue measure: for almost every α the system is mode-locked. This is the opposite of the behaviour for $0 < \varepsilon < 1$, where the tongues have positive but not full measure. *Source*: G. Świątek, “Rational rotation numbers for maps of the circle”, *Communications in Mathematical Physics* 119 (1988), 109–128. *Reason*: the proof uses distortion of the cross-ratio and Herman–Świątek-type estimates on the nonlinearity of the critical maps. *Where*: Remark 4.5.
- **Boyland: rotation intervals for $\varepsilon > 1$ (panorama)**. Beyond criticality the map is no longer injective and the rotation number ceases to be univalued: its place is taken by a rotation interval $[\rho_-, \rho_+]$, degenerate for $\varepsilon \leq 1$ and of positive width for $\varepsilon > 1$; bistability appears (two attracting cycles with different rotation numbers) as well as positive topological entropy. *Source*: P. L. Boyland, “Bifurcations of circle maps: Arnol’d tongues, bistability and rotation intervals”, *Communications in Mathematical Physics* 106 (1986), 353–381. *Reason*: the section in which it appears declares that it presents these results as panorama, owing to technical difficulty well beyond the elementary scope of the chapter. *Where*: Remark 4.6.
- **Arnold’s theorem (1961), statement (internal cross-reference to Chapter 6)**. Let ρ be Diophantine and f an analytic diffeomorphism of the circle, defined on a complex strip and close to $R_{2\pi\rho}$, with $\rho(f) = \rho$ exactly: then f is analytically conjugate to $R_{2\pi\rho}$. The remark insists that the hypothesis $\rho(f) = \rho$ is indispensable — without it f can fall into an Arnold tongue — and notes that it had been omitted in an earlier draft. *Source*: V. I. Arnol’d, “Small denominators. I”, *Izvestiya Akad. Nauk SSSR* 25:1 (1961), 21–86, with erratum 28:2 (1964), 479–480; a modern framing in H. W. Broer, “KAM theory: the legacy of Kolmogorov’s 1954 paper”, *Bulletin of the AMS* 41 (2004), 507–521. *Reason*: it is not an import of the volume: the complete proof — Cauchy estimates, domain loss, convergence of the Newton scheme — is the content of Chapter 6. *Where*: Remark 4.7.
- **Herman (1979) and Yoccoz (1984, 1995): the global theory, statement (internal cross-reference to Chapter 6)**. Three items: Herman removes the closeness-to-rotation hy-

pothesis for a set of rotation numbers of full measure; Yoccoz extends it to *every* Diophantine number; and the optimal arithmetic threshold of the *global* problem is not the Bryuno condition but condition $\mathcal{H}$, with strict inclusions $\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}$ — Bryuno remains optimal for the *local* holomorphic problem. *Source*: M. R. Herman, *Publications Mathématiques de l’IHÉS* 49 (1979), 5–233; J.-C. Yoccoz, *Annales ENS* (4) 17 (1984), 333–359, and *Astérisque* 231 (1995); H. Eliasson, B. Fayad, R. Krikorian, arXiv:1810.07107. *Reason*: stated as a bridge, with the definition of the condition $\mathcal{H}$ and the relation between the two thresholds deferred to Chapter 6. *Note*: the block is largely errata-correcting and fixes, from an earlier draft, Herman’s date, the *Astérisque* issue number, the confusion between Bryuno and $\mathcal{H}$, and the attribution of the Fields medal. *Where*: Remark 4.8.

Chapter 5 — Denjoy’s theorem and its counterexamples

- **Hall’s C^∞ counterexample (1981)** (*imported*). For every irrational ρ there exists a homeomorphism f of the circle, of class C^∞ as a map, with $\rho(f) = \rho$, which possesses a Cantor minimal set and genuine wandering intervals. The construction gives $f' \geq 0$ with $f' = 0$ exactly at the endpoints of the wandering intervals: f is not a diffeomorphism, and Denjoy’s theorem — which requires $f' > 0$ everywhere — is not contradicted. *Source*: G. R. Hall, *Ergodic Theory and Dynamical Systems* 1 (1981), 261–272. *Reason*: the proof uses equivariant families of diffeomorphisms with prescribed modulus of continuity, a technique not developed in the volume. *Status*: it serves to show that the hypothesis $f' > 0$ of Denjoy’s theorem cannot be weakened: an argumentative role, not a lemma. *Where*: Remark 5.5.
- **Optimal regularity of conjugacies (Herman; Katznelson–Ornstein)** (*imported*). For $f \in C^{2+\varepsilon}$ with Diophantine rotation number, the conjugacy h is differentiable with absolutely continuous derivative, and a second work pins down exactly when h is absolutely continuous. The block also anticipates, as a bridge, the unique ergodicity of the Denjoy conjugacy. *Source*: M. R. Herman, *Publications Mathématiques de l’IHÉS* 49 (1979), ch. VI; Y. Katznelson, D. Ornstein, *Ergodic Theory and Dynamical Systems* 9 (1989), 643–680 and 681–690. *Reason*: it requires thermodynamic formalism and a priori distortion estimates at every order. *Note*: unique ergodicity is not imported — it is proved in Chapter 9 with Oxtoby’s criterion. *Where*: Remark 5.6.

Chapter 6 — Small divisors and KAM theory on the circle

- **Optimal loss $1 + \tau + \epsilon$ in the homological equation** (*imported, unused*). Measuring regularity on Hölder–Zygmund scales rather than only on integer classes, the loss of derivatives drops from the $2 + \tau$ of the elementary method to the optimal value $1 + \tau + \epsilon$ for every $\epsilon > 0$. *Source*: A. Katok, B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, §2.9; M. R. Herman, *Publications Mathématiques de l’IHÉS* 49 (1979), ch. II. *Reason*: it requires the Hölder–Zygmund norms and the interpolation estimates between them. *Status*: the chapter states that nothing it proves depends on it. *Where*: Remark 6.7.
- **Avila–Kocsard on invariant distributions** (*imported, unused*). For a minimal C^∞ diffeomorphism of the circle the invariant probability measure is the unique invariant distribution up to constants, and the space of real C^∞ -coboundaries is closed if and only if the rotation number is Diophantine. *Source*: A. Avila, A. Kocsard, “Cohomological equations and invariant distributions for minimal circle diffeomorphisms”, *Duke Mathematical Journal* 158 (2011), 501–536. *Reason*: it assumes the theory of distributions on $\mathbb{T}$ and the notion of invariant distribution. *Note*: in the same block the Denjoy–Koksma inequality is not imported but is proved elsewhere in the treatise. *Where*: Remark 6.10.
- **Moser: finite-regularity version** (*imported*). For ρ Diophantine of type (K, τ) there exist integers $k = k(\tau)$ and $k' < k$ such that every diffeomorphism of class C^k with $\rho(f) = \rho$,

close enough to $R_{2\pi\rho}$ in the C^k norm, is conjugated to the rotation by a $C^{k'}$ diffeomorphism; the Cauchy estimates, unavailable in C^k , are replaced by smoothing operators (the Nash–Moser scheme). *Source*: J. Moser, *Nachrichten Göttingen* II (1962), 1–20, and “A rapidly convergent iteration method”, *Annali della Scuola Normale Superiore di Pisa* (3) 20 (1966), 265–315 and 499–535. *Reason*: it assumes the construction of the smoothing operators and their interpolation estimates. The chapter also states that it does not report the values of k and k' as functions of τ , because they vary from exposition to exposition and citing one without having reopened the source that proves it would be an invented citation. *Where*: Remark 6.17.

- **Herman (1979): the global C^∞ theorem (imported)**. There exists $A \subset \mathbb{R} \setminus \mathbb{Q}$ of full Lebesgue measure such that every orientation-preserving C^∞ diffeomorphism of the circle, with $\rho(f) \in A$, is conjugated to $R_{2\pi\rho(f)}$ by a C^∞ diffeomorphism: no closeness-to-rotation hypothesis. *Source*: M. R. Herman, “Sur la conjugaison différentiable des difféomorphismes du cercle à des rotations”, *Publications Mathématiques de l’IHÉS* 49 (1979), 5–233. *Reason*: it assumes the entire Herman apparatus — one-parameter families, measure theory on rotation numbers, iterated distortion estimates — about two hundred pages, which exceed the volume’s available space by an order of magnitude. *Status*: the Arnold theorem proved in the chapter is its local analytic version. *Where*: Remark 6.18.
- **Yoccoz (1984): explicit Diophantine condition (imported)**. Every orientation-preserving C^∞ diffeomorphism of the circle whose rotation number is Diophantine is conjugated to the rotation by a C^∞ diffeomorphism: Herman’s full-measure set is replaced by the explicit set $\mathcal{CD}$. *Source*: J.-C. Yoccoz, “Conjugaison différentiable des difféomorphismes du cercle dont le nombre de rotation vérifie une condition diophantienne”, *Annales scientifiques de l’ENS* (4) 17 (1984), 333–359. *Reason*: it assumes the whole of Herman’s theory plus Yoccoz’s renormalization technique. *Where*: Remark 6.19.
- **Bryuno is the optimal condition in the local analytic problem (imported)**. Sufficiency: for every $\alpha \in \mathcal{B}$ and every $\nu > 0$ there exists $\varepsilon(\alpha, \nu) > 0$ such that every analytic diffeomorphism with $\rho(f) = \alpha$ and $\|f - R_{2\pi\alpha}\|_\nu < \varepsilon$ is analytically linearizable (this is Arnold’s theorem with $\mathcal{CD}$ replaced by the larger $\mathcal{B}$). Optimality: if $\alpha \notin \mathcal{B}$ there exist analytic diffeomorphisms with that rotation number, arbitrarily close to the rotation, that are not analytically linearizable. *Source*: J.-C. Yoccoz, “Analytic linearization of circle diffeomorphisms”, in *Dynamical Systems and Small Divisors*, Lecture Notes in Mathematics 1784, Springer 2002, 125–173, and “Théorème de Siegel, nombres de Bruno et polynômes quadratiques”, *Astérisque* 231 (1995), 1–88; the condition is due to A. D. Bryuno, *Transactions of the Moscow Mathematical Society* 25 (1971) and 26 (1972) — the chapter notes that it does not give the page range because the secondary sources consulted disagree. *Reason*: it assumes Yoccoz’s renormalization theory and the construction of the counterexamples; the chapter also warns, honestly, that the sufficiency of $\mathcal{B}$ does not follow from its own Newton scheme as it stands, because the estimate for the analytic homological equation uses the Diophantine condition in power form. *Where*: Remark 6.20.
- **The condition $\mathcal{H}$ and the hierarchy $\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}$ (imported partial)**. Yoccoz introduced a class $\mathcal{H}$ that characterizes exactly the rotation numbers for which *global* analytic linearization is always possible, and proved the two strict inclusions. *Source*: J.-C. Yoccoz, “Centralisateurs et conjugaison différentiable des difféomorphismes du cercle”, *Astérisque* 231 (1995), 89–242, and Lecture Notes in Mathematics 1784; an overview in Eliasson–Fayad–Krikorian, arXiv:1810.07107. *Reason*: the chapter declares that it does *not* reproduce the definition of $\mathcal{H}$: it is recursive, formulated by comparing sequences built along the renormalization with the Bryuno function, and reproducing it outside its technical context would be more misleading than useful. Only the existence of the class, the two strict inclusions, and the distinction

between local and global optimality are recorded. *Where:* Remark 6.21.

- **The scale of finite regularities** (*panorama*). Between Herman–Yoccoz’s C^∞ and Arnold’s analytic case, the result depends both on the Diophantine exponent and on the regularity of f : Katznelson–Ornstein obtain the optimal regularity with a *non*-KAM method based on distortion estimates; Khanin–Sinai give a new proof of Herman’s theorem via thermodynamic formalism and renormalization; Khanin–Teplinsky obtain the sharp result in class $C^{2+\alpha}$. *Source:* Katznelson–Ornstein, *ETDS* 9 (1989); Khanin–Sinai, *Communications in Mathematical Physics* 112 (1987), 89–101; Khanin–Teplinsky, *Inventiones Mathematicae* 178 (2009), 333–344. *Reason:* the chapter states that none of its statements rests on it. *Where:* Remark 6.22.
- **From Kolmogorov 1954 to the circle** (*panorama*). The Newton scheme of the chapter is the one-dimensional version of Kolmogorov’s idea for nearly-integrable Hamiltonian systems: under a Diophantine condition on the frequencies and a non-degeneracy condition, most of the invariant tori survive a small perturbation. *Source:* A. N. Kolmogorov, *Doklady* 98 (1954), 527–530 — the chapter states that it has *not* located the original note in any archive and cites the English translation in *Lecture Notes in Physics* 93, 51–56 —; V. I. Arnold (1963); modern expositions by Pöschel, with Bounemoura’s correction, and by de la Llave; historical panorama by Broer, *Bulletin of the AMS* 41 (2004), 507–521. *Reason:* multidimensional Hamiltonian KAM is out of scope. *Bibliographic note:* the chapter flags a discrepancy in pagination between the records consulted (525–530 versus 527–530) and leaves it open rather than resolving it unilaterally; it also warns that the “Theorem 1” that Broer attributes to Arnold is a modern reformulation, different from the original, and states that it does not use it. *Where:* Remark 6.23.

Chapter 7 — The doubling map: symbolic dynamics, Li–Yorke, and Sharkovskii

- **Banks–Brooks–Cairns–Davis–Stacey: the redundancy of sensitivity** (*imported, not needed*). On an infinite metric space, a transitive continuous map with dense periodic points automatically has sensitivity to initial conditions: the third condition of Devaney’s definition of chaos is a consequence of the first two. *Source:* J. Banks, J. Brooks, G. Cairns, G. Davis, P. Stacey, “On Devaney’s Definition of Chaos”, *American Mathematical Monthly* 99 (1992), 332–334. *Reason:* it is not a matter of difficulty — the chapter notes that the proof takes about one page — but of necessity: for the map under consideration sensitivity is already proved directly and quantitatively, with an explicit constant. *Where:* Remark 7.4.
- **Devaney implies Li–Yorke on compacta** (*imported*). On a compact metric space Devaney chaos implies Li–Yorke chaos: there exists an uncountable scrambled set. The converse is false. *Source:* W. Huang, X. Ye, “Devaney’s chaos or 2-scattering implies Li–Yorke’s chaos”, *Topology and its Applications* 117 (2002), 259–272. *Reason:* the chapter does not re-prove it; for the circle it gives an alternative route to a result that the chapter establishes anyway directly and constructively. *Where:* Remark 7.5.
- **Li–Yorke, Theorem 2: the uncountable scrambled set** (*imported*). In the same paper in which period three forces all periods, the hypothesis of an orbit of period 3 implies the existence of an uncountable scrambled set. *Source:* T.-Y. Li, J. A. Yorke, “Period Three Implies Chaos”, *American Mathematical Monthly* 82 (1975), 985–992, Theorem 2. *Reason:* the construction refines the covering lemma by encoding itineraries with infinite subsets of $\mathbb{N}$, and takes about two pages of combinatorics that the chapter does not reproduce, having proved Theorem 1 in full. *Status:* used — applied to the doubling map, which possesses the explicit period-three orbit $\{2\pi/7, 4\pi/7, 8\pi/7\}$. *Where:* Remark 7.6.
- **Sharkovskii: combinatorial induction for arbitrary n** (*imported partial*). The im-

ported part is the classification of cyclic permutations according to the Štefan cycle: every such permutation contains the pattern that forces all subsequent periods in order, or else n is a power of 2 and the argument reduces by induction to the study of f^2 on an invariant subinterval. *Source*: K. Burns, B. Hasselblatt, “The Sharkovsky Theorem: A Natural Direct Proof”, *American Mathematical Monthly* 118 (2011), 229–244, Theorem 2; P. Štefan, *Communications in Mathematical Physics* 54 (1977), 237–248; L. S. Block, W. A. Coppel, *Dynamics in One Dimension*, Springer 1992. *Reason*: the induction must uniformly handle all residue classes and all cyclic permutations for every n . *Boundary*: the chapter proves the case $n = 3$ in full and establishes the core of the proof — the covering graph, the covering lemma, the itinerary lemma — illustrating it on $n = 5$; the imported part is the general combinatorial articulation, not a new dynamical principle. *Where*: Remark 7.8.

Chapter 8 — Topological entropy and Lyapunov exponents

- **Kolmogorov–Sinai generator theorem** (*imported*). If $\mathcal{P}$ is a generating partition mod μ for a system (X, f, μ) , then $h_\mu(f) = h_\mu(f, \mathcal{P})$: any generating partition computes the system’s full metric entropy. *Source*: P. Walters, *An Introduction to Ergodic Theory*, GTM 79, Springer 1982, Thm. 4.17. *Reason*: a general measure-theoretic fact (about the sigma-algebra generated by a partition under iteration) beyond the elementary scope of the chapter. *Status*: used to deduce $h_\mu(D) = \log 2$ from the generating dyadic partition, after computing $h_\mu(D, \mathcal{P}) = \log 2$ in full. *Where*: proof of Theorem 8.12; the same import is recalled in Chapter 9 for Oxtoby’s criterion (Remark 9.14).
- **Bowen: finite-fiber factors** (*imported*). If the fibers of a continuous surjective factor map $\pi : X \rightarrow Y$ between compact metric spaces are uniformly bounded by N , then $h_{\text{top}}(f) \leq h_{\text{top}}(g) + \log N$; for finite-fiber coverings between expansive systems, equality holds. *Source*: R. Bowen, “Entropy for group endomorphisms and homogeneous spaces”, *Transactions of the AMS* 153 (1971), 401–414. *Reason*: the fine argument exploits the upper semicontinuity of the fibers and a selection lemma for semicontinuous multifunctions that exceeds the scope of the chapter. *Status*: used as a *second independent check* on the value $h_{\text{top}}(D) = \log 2$; the direct computation remains the primary, self-contained proof. *Where*: Remark 8.3.
- **Misiurewicz–Szlenk** (*imported partial*). For every continuous piecewise monotone map with finitely many pieces, $h_{\text{top}}(f) = \lim_n \frac{1}{n} \log \text{lap}(f^n)$. *Source*: M. Misiurewicz, W. Szlenk, “Entropy of piecewise monotone mappings”, *Studia Mathematica* 67 (1980), 45–63. *Boundary*: the chapter proves in full the inequality $\ell(f) \leq h_{\text{top}}(f)$ and imports the opposite one. *Reason*: the imported inequality requires controlling how many (n, ε) -separated points fit inside a single monotonicity segment of f^n , with a distortion argument outside the scope of the chapter. *Status*: used to deduce $h_{\text{top}}(f_4) = \log 2$ from the check $\text{lap}(f_4^n) = 2^n$. *Where*: Remark 8.5.
- **A period that is not a power of two forces positive entropy** (*imported partial*). For odd period $q \geq 3$ the Štefan cycle contains an interval that covers itself twice, whence $h_{\text{top}}(f) \geq \log 2$; for $n = 2^k q$ the orbits redistribute under f^{2^k} and one obtains $h_{\text{top}}(f) \geq (\log 2)/2^k > 0$. *Source*: *Periodic points and topological entropy of one dimensional maps*, Lecture Notes in Mathematics 819 (1980), 18–34; Block–Coppel for the combinatorial part. *Boundary*: the imported part is only the structural verification, for every odd q , of the existence of the self-covering interval. *Reason*: so as not to duplicate the classification of Štefan cycles, already treated in Chapter 7. *Where*: Remark 8.6.
- **Variational principle** (*imported, not used deductively*). For a topological dynamical system, $h_{\text{top}}(f) = \sup_\mu h_\mu(f)$, with supremum over the invariant Borel measures. *Source*: Goodwyn (1969), Dinaburg (1970), Goodman (1971). *Reason*: the general proof uses measure-theoretic tools that lie outside the chapter. *Status*: for the doubling map the chapter directly checks

$h_{\mu_{\text{Leb}}}(D) = h_{\text{top}}(D) = \log 2$, concluding that Lebesgue measure is a measure of maximal entropy by computing the two quantities, not by deduction from the general principle. *Where:* Remark 8.9.

- **Oseledets and Pesin’s formula** (*panorama*). In dimension one Oseledets’s theorem reduces to a consequence of Birkhoff’s ergodic theorem applied to $\log |f'|$; in higher dimension it gives a spectrum of exponents on the tangent splitting. Pesin’s formula, for a $C^{1+\alpha}$ diffeomorphism with absolutely continuous invariant measure, gives $h_{\mu}(f) = \sum_{\lambda_i > 0} \lambda_i$ counted with multiplicity, which in dimension one becomes $h_{\mu}(f) = \max(\lambda, 0)$. *Source:* V. I. Oseledets, *Trudy MMO* 19 (1968), 179–210; Ja. B. Pesin, *Uspekhi Mat. Nauk* 32:4 (1977), 55–112. *Reason:* Pesin’s formula requires Lyapunov–Perron theory of stable and unstable manifolds and the non-uniformly hyperbolic structure. *Status:* for the maps of the chapter, the coincidence between entropy and exponent is checked by direct computation, not deduced from the general statement. *Where:* Remark 8.10.

Chapter 9 — Ergodic theory on the circle and on the interval

- **Normality of Champernowne’s constant** (*imported*). The number obtained by concatenating the representations of $1, 2, 3, \dots$ is normal in every base, in particular in base two. *Source:* D. G. Champernowne, “The construction of decimals normal in the scale of ten”, *Journal of the London Mathematical Society* 8 (1933), 254–260. *Reason:* normality does not follow from Birkhoff’s theorem, which holds for almost every point, whereas Champernowne is an explicitly constructed point and could a priori fall in the exceptional null set; one needs a combinatorial count of blocks with control of boundary effects, of fine countability. *Where:* Remark 9.11.
- **Lasota–Yorke in the nonlinear case** (*imported*). For piecewise C^2 maps with finitely many monotone branches and $\inf |f'| > 1$ one has $\text{Var}(\mathcal{L}g) \leq \frac{2}{\inf |f'|} \text{Var}(g) + C\|g\|_1$, and the existence of an invariant density of bounded variation follows by compactness (Helly’s selection theorem applied to the Cesàro averages). *Source:* A. Lasota, J. A. Yorke, “On the existence of invariant measures for piecewise monotonic transformations”, *Transactions of the AMS* 186 (1973), 481–488; A. Lasota, M. C. Mackey, *Chaos, Fractals, and Noise*, Springer 1994, §§5.2–5.3. *Boundary:* the chapter proves the case of constant slope, where the correction term vanishes, and treats the logistic map by elementary conjugacy, avoiding the import. *Where:* Remark 9.13.
- **Kolmogorov–Sinai generator theorem** (*imported*). If $\mathcal{P}$ is generating — its refinements separate almost every pair of distinct points — then $h_{\mu}(S) = h_{\mu}(S, \mathcal{P})$. *Source:* P. Walters, *An Introduction to Ergodic Theory*, Springer 1982, Theorem 4.17 — the same import already declared in Chapter 8. *Reason:* computing entropy from the definition, as a supremum over all partitions, is in general impractical. *Note:* the value $h_{\mu}(D) = \log 2$ is not re-imported here, but recalled from Chapter 8, where it is proved. *Where:* Remark 9.14.
- **Jakobson (1981)** (*panorama*). The set of parameters for which the logistic map admits an absolutely continuous invariant and ergodic probability measure has positive Lebesgue measure, even though the complement, where an attracting cycle dominates, is dense. *Source:* M. V. Jakobson, *Communications in Mathematical Physics* 81 (1981), 39–88; independent proof in M. Benedicks, L. Carleson, *Annals of Mathematics* 122 (1985), 1–25. *Reason:* both combine distortion estimates, return times, and probabilistic arguments over multiple scales. *Status:* the chapter states that the result remains a panorama and is never stated inside a theorem environment. *Where:* Remark 9.15.

Chapter 10 — Transfer operators, decay of correlations, and the central limit theorem

- **Central limit theorem for martingales (Billingsley)** (*imported*). If (X_i) is stationary, ergodic, a martingale difference sequence with respect to a filtration, with $\mathbb{E}[X_0^2] < \infty$, then $(\sum_{i < n} X_i)/\sqrt{n}$ converges in law to $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \mathbb{E}[X_0^2]$. *Source*: P. Billingsley, “The Lindeberg–Lévy theorem for martingales”, *Proceedings of the AMS* 12 (1961), 788–792. *Reason*: recalled without re-proof. *Status*: the chapter declares it as its own single unproved ingredient; everything specific to the doubling map — the coboundary, the decomposition, the exact orthogonality, the variance formula — is proved. *Where*: Remark 10.7.
- **Nagaev–Guivarc’h method** (*panorama*). With the weighted operator $\mathcal{L}_t w = \mathcal{L}(e^{itu} w)$ and Kato’s perturbation theory applied to the simple isolated eigenvalue 1, one obtains $\lambda(t)$ of class C^2 with $\lambda(0) = 1$, $\lambda'(0) = i \int u d\mu$, and $\lambda''(0) = -\sigma^2$, whence the central limit theorem via Lévy’s continuity theorem. *Source*: S. V. Nagaev (1957), Y. Guivarc’h and J. Hardy (1988); a modern exposition in L. Hervé, F. Pène, *Bulletin de la SMF* 138 (2010), 415–489; A. Broise, *Astérisque* 238 (1996), 1–109. *Reason*: it requires the perturbation theory of linear operators. *Where*: Remark 10.8.
- **Projective cones and the Birkhoff–Hopf theorem** (*panorama*). If $\mathcal{L}$ maps a cone strictly into itself with image of finite projective diameter Δ , the Hilbert metric is contracted by the factor $\tanh(\Delta/4)$, and with it one obtains in one stroke the spectral gap, existence and uniqueness of the invariant measure, and decay of correlations, with explicit constants. *Source*: C. Liverani, “Decay of correlations”, *Annals of Mathematics* 142 (1995), 239–301, and the same author’s lecture notes (Padua, 2019). *Reason*: it is an alternative method to the one already proved in the chapter, useful above all in higher dimension: nothing in the chapter leans on it. *Where*: Remark 10.9.
- **Young towers** (*panorama*). One builds a base set with a countable partition and return time R such that the induced map is uniformly expanding with controlled distortion; the rate of decay of correlations is governed by the tail of R : exponential tails give exponential decay, polynomial tails $\mu(R > n) = O(n^{-\beta})$ give $O(n^{-\beta+1})$. *Source*: L.-S. Young, *Israel Journal of Mathematics* 110 (1999), 153–188, and *Annals of Mathematics* 147 (1998), 585–650. *Reason*: none of the maps of the chapter has neutral points — the expansion is uniform — and towers are not needed; they are the standard tool when an indifferent point appears, as in the Pomeau–Manneville map. *Where*: Remark 10.10.

Chapter 11 — The logistic map, bifurcations, and universality

- **Singer’s theorem** (*imported*). A map with negative Schwarzian derivative has at most one attracting cycle. This is the ingredient that gives the *uniqueness* of the attractor, which the cascade existence theorem proved in the chapter does not supply. *Source*: D. Singer, “Stable orbits and bifurcation of maps of the interval”, *SIAM Journal on Applied Mathematics* 35 (1978), 260–267. *Boundary*: the chapter fully checks the *applicability* of the theorem to the logistic map, proving $Sf_r(x) < 0$ for $x \neq \frac{1}{2}$, and imports the validity of the theorem. *Where*: Remark 11.1.
- **Lanford’s theorem** (*imported*). The renormalization operator has a fixed point g^* , unique among analytic unimodal maps with critical point of order two, and its differential at g^* is hyperbolic with a single eigenvalue of modulus greater than one, $\delta = 4.66920160910299\dots$, while $\alpha = g^*(1) = -2.5029078750959\dots$; every unimodal family with non-degenerate critical point that crosses the stable manifold of g^* transversally — including the logistic map — has a cascade with $r_n \sim r_\infty - C\delta^{-n}$. *Source*: O. E. Lanford III, “A computer-assisted proof of the Feigenbaum conjectures”, *Bulletin of the AMS* 6 (1982), 427–434; Collet–Eckmann,

Iterated Maps on the Interval as Dynamical Systems, Birkhäuser 1980; M. Lyubich, *Annals of Mathematics* 149 (1999), 319–420. *Reason*: the original proof uses computer-assisted functional estimates. *Errata note*: an earlier draft of the chapter stated that the fixed-point equation was “proved here (schematically)”; the remark corrects the error and states that in that section nothing is proved. *Where*: Remark 11.4.

- **Coullet–Tresser (panorama)**. Independently of Feigenbaum and almost simultaneously, Coullet and Tresser developed the same renormalization operator in a broader setting, applicable also in dimension greater than one: hence the name Feigenbaum–Coullet–Tresser universality. *Source*: P. Coullet, C. Tresser, “Itérations d’endomorphismes et groupe de renormalisation”, *Journal de Physique Colloques* 39 (1978), C5-25–28. *Reason*: historical/attributive content, declared panorama without proof. *Where*: Remark 11.5.
- **Renormalization at the golden ratio: δ_{GM} and α_{GM} (panorama)**. For critical circle maps with golden rotation number, the operator built on returns to the intervals of the Fibonacci denominators has a hyperbolic fixed point with a single unstable eigenvalue $\delta_{\text{GM}} \approx -2.8336$ and scale factor $\alpha_{\text{GM}} \approx -1.2886$. *Source*: S. J. Shenker, *Physica D* 5 (1982), 405–411; Feigenbaum–Kadanoff–Shenker, *Physica D* 5 (1982), 370–386; Ostlund–Rand–Sethna–Siggia, *Physica D* 8 (1983), 303–342. *Reason*: stated as panorama. *Errata note*: an earlier draft stated that the Arnold tongues were governed by the Feigenbaum constant $\delta \approx 4.6692$; the remark corrects the error — the two operators act on different function spaces (single rescaling versus rescaling along Fibonacci) and there is no structural reason for their respective unstable eigenvalues to coincide. *Where*: Remark 11.6.
- **Jakobson, Benedicks–Carleson, and the density of real hyperbolicity (panorama)**. Positive measure of the set of parameters with an absolutely continuous invariant measure (Jakobson), an alternative proof with explicit estimates on the good parameters (Benedicks–Carleson), and density of hyperbolic maps among real unimodal maps, proved by Graczyk–Świątek and, independently, by Kozlovski–Shen–van Strien. *Source*: M. V. Jakobson, *Communications in Mathematical Physics* 81 (1981), 39–88; M. Benedicks, L. Carleson, *Annals of Mathematics* 122 (1985), 1–25; for the density result, works cited without a full reference in the chapter. *Reason*: declared imported without proof; the analogous complex case coincides with the MLC conjecture, deferred to Volume VI. *Where*: Remark 11.7.

Chapter 12 — From the circle to Julia and Mandelbrot

This is the chapter with the largest number of imports, and the only one that collects them in a list of its own, at the opening. The first five entries below are those of that list.

- **Riemann’s conformal mapping theorem (imported)**. Every simply connected domain $U \subsetneq \widehat{\mathbb{C}}$ whose complement contains at least two points is biholomorphic to the disk; fixing $u_0 \in U$ there exists a unique biholomorphism $\psi : U \rightarrow \mathbb{D}$ with $\psi(u_0) = 0$ and $\psi'(u_0) > 0$. *Source*: J. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Princeton University Press 2006; the statement also appears in Volume III. *Reason*: a classical prerequisite from complex analysis, which the volume does not re-prove. *Where*: Remark 12.1, item (I).
- **Topological criterion for simple connectivity (imported)**. A connected open set $U \subseteq \widehat{\mathbb{C}}$ is simply connected if and only if $\widehat{\mathbb{C}} \setminus U$ is connected. *Source*: Milnor, op. cit. *Reason*: a standard fact of sphere topology. *Status*: it is the step that, from the biholomorphy of the Douady–Hubbard map, allows one to deduce the connectedness of the Mandelbrot set. *Where*: Remark 12.1, item (I-bis).
- **Hyperbolic metric and the Schwarz–Pick inequality (imported)**. Every domain whose complement contains at least three points carries a complete conformal metric of curvature -1 ; every holomorphic map between such domains does not increase hyperbolic distances,

and is a local isometry if and only if it is a holomorphic covering. In particular an $f : U \rightarrow U$ with a fixed point of multiplier of modulus less than one is not a local isometry. *Source*: Milnor, op. cit. *Reason*: a classical prerequisite. *Status*: it is the key ingredient of the proof of Fatou’s theorem given in the chapter. *Where*: Remark 12.1, item (II).

- **Proper maps as branched coverings** (*imported*). A non-constant, proper holomorphic map between connected Riemann surfaces has a well-defined finite degree: every point has exactly d preimages counted with multiplicity, and in the absence of critical points it is a covering of degree d . *Source*: Milnor, op. cit. *Reason*: a standard fact of the theory of Riemann surfaces. *Where*: Remark 12.1, item (III).
- **Montel’s theorem** (*imported*). The normality criterion for families of holomorphic functions omitting two values, the foundation of the definition of the Julia set as the complement of the set of normality. *Source*: the chapter declares it among its own imports in the prerequisites block and cites it in the historical reconstruction of the Fatou and Julia theory and in an exercise. *Editorial note*: it is the only import that the chapter declares in the prerequisites without repeating it in the list of imports; it is recorded here because the debt is real and declared, and the discrepancy should be flagged, not silently smoothed over. *Where*: the block of the chapter’s prerequisites; it does not appear in Remark 12.1.
- **Siegel’s arithmetic lemma** (*imported*). Let $\lambda = e^{2\pi i\vartheta}$ with $\omega_m = |\lambda^m - 1| \geq Cm^{-\mu}$ for every $m \geq 1$. Then there exists $\Gamma = \Gamma(C, \mu) \geq 1$ such that for every rooted binary tree T with n leaves $\prod_{v \in \text{int}(T)} \omega_{n(v)-1}^{-1} \leq \Gamma^n$. *Source*: C. L. Siegel, “Iteration of analytic functions”, *Annals of Mathematics* 43 (1942), 607–612; M. Abate, “Discrete holomorphic local dynamical systems”, *Lecture Notes in Mathematics* 1998, Springer 2010, §4, where the estimate is reproduced in full in modern notation. *Reason*: it is precisely the point at which Siegel overcame the difficulty of small divisors: the naive bound on a “caterpillar” tree would give $C^{-(n-1)}((n-1)!)^\mu$, which grows too fast and would give zero radius of convergence; the decisive observation is that small divisors are sparse and separated (if $\omega_p, \omega_q < \delta$ with $p < q$ then $q - p > (C/2\delta)^{1/\mu}$), and exploiting this requires a fine combinatorial induction on the tree. *Status*: it is the *only* unproved statement in the proof of Siegel’s theorem given in the chapter; everything else in that section is proved. The lemma is stated in full in Lemma 12.5. *Where*: Remarks 12.1 (item IV) and 12.11.
- **The Cantor half of the fundamental dichotomy** (*imported*). If the parameter lies outside the Mandelbrot set, the filled Julia set coincides with the Julia set and is a Cantor set, on which the dynamics is topologically conjugate to the one-sided shift on two symbols. Together with the theorem proved in the chapter one obtains the dichotomy: K_c connected $\iff 0 \in K_c \iff c \in \mathcal{M}$, and K_c a Cantor set in the other cases. *Source*: J. Milnor, op. cit., §9; L. Carleson, T. W. Gamelin, *Complex Dynamics*, Springer 1993, ch. VIII. *Reason*: it assumes the Douady–Hubbard construction of the two inverse branches, with the distortion estimate that guarantees that the diameters of the nested images tend to zero. *Where*: Remark 12.5.
- **Douady–Hubbard: Φ and the connectedness of $\mathcal{M}$** (*imported*). Setting $\Phi(c) = \Phi_c(c)$ for $c \notin \mathcal{M}$, the map $\Phi : \mathbb{C} \setminus \mathcal{M} \rightarrow \mathbb{C} \setminus \mathbb{D}$ is a biholomorphism; with the topological criterion imported above it follows that $\mathcal{M}$ is connected. The chapter notes that this does not imply *local* connectedness (the MLC conjecture), deferred to Volume VI. *Source*: A. Douady, J. H. Hubbard, *Étude dynamique des polynômes complexes*, Publications Mathématiques d’Orsay 84-02 (1984) and 85-04 (1985). *Reason*: one assumes the holomorphy of Φ , which requires the holomorphic dependence of Φ_c on c , and its surjectivity. *Where*: Remark 12.6.
- **p/q bulbs and the Farey ordering** (*imported, presented as panorama*). For every reduced p/q with $q \geq 2$ there exists a component $B_{p/q}$ of the interior of $\mathcal{M}$ whose closure touches

the cardioid exactly at the root point $c_{p/q}$; on it the dynamics has an attracting cycle of period q , whose immediate-basin sectors are permuted with combinatorial rotation number p/q ; between two adjacent bulbs in the Farey sequence the largest interposed bulb is that of the mediant. *Source*: J. Milnor, “Periodic orbits, external rays and the Mandelbrot set”, *Astérisque* 261 (2000), 277–333; Douady–Hubbard, *Orsay Notes*. *Reason*: one assumes the theory of landing of rational parameter rays and the structure of hyperbolic components. *Boundary*: the chapter proves in full the case $p/q = 1/2$, the exact position of all root points, and their cyclic ordering. *Where*: Remark 12.8.

- **Landing of rational rays (imported)**. If c lies in $\mathcal{M}$ and t is rational, the external ray R_t lands at a point of the Julia set; if t is periodic under doubling, that point is a repelling or parabolic periodic point, with period dividing that of t . In the parameter plane, every rational ray lands on the boundary of $\mathcal{M}$, and the root point of the bulb p/q is the landing point of exactly two rays, of rational angles with denominator $2^q - 1$. *Source*: Douady–Hubbard, *Orsay Notes*; Milnor, *Astérisque* 261 (2000), 277–333. *Reason*: one assumes the Douady–Hubbard landing theorem, which in turn uses a length-area estimate or Fatou’s theorem on radial limits, in addition to the connectedness of $\mathcal{M}$ already imported. *Where*: Remark 12.9.
- **Bryuno and Yoccoz: the optimal condition for local linearization (imported)**. If ϑ satisfies the Bryuno condition, every holomorphic germ $\lambda w + O(w^2)$ with $\lambda = e^{2\pi i \vartheta}$ is linearizable at 0 (Bryuno, 1971); for the quadratic family the condition is also necessary (Yoccoz, 1995), so that for $f_\lambda(w) = \lambda w + w^2$ linearizability is equivalent to $B(\vartheta) < \infty$. *Source*: A. D. Bruno, *Trudy Moskovskogo Mat. Obshchestva* 25 (1971), 119–262; J.-C. Yoccoz, “Théorème de Siegel, nombres de Bruno et polynômes quadratiques”, *Astérisque* 231 (1995), 3–88. *Reason*: for sufficiency, the construction of the majorant with the denominators of the convergents; for necessity, Yoccoz’s geometric construction and Pérez-Marco’s estimate on hedgehogs. *Status*: the chapter states that no part of these two theorems is used elsewhere. *Where*: Remark 12.13.
- **The boundary of the Siegel disk (panorama)**. Three results: for ϑ of bounded type the critical point lies on the boundary of the Siegel disk (Herman, 1985); the Julia set is locally connected and of zero measure (Petersen, 1996); the boundary of the disk is a quasicircle (Zhang, 2011, the Douady–Sullivan conjecture in the bounded-type case). *Source*: M. R. Herman, *Communications in Mathematical Physics* 99 (1985), 593–612; C. L. Petersen, *Acta Mathematica* 177 (1996), 163–224; G. Zhang, *Inventiones Mathematicae* 185 (2011), 421–466. *Reason*: declared cited only as panorama and not used in what follows. *Where*: Remark 12.14.

Chapter 13 — Synchronization: from Huygens to Kuramoto

- **Huygens’s observation (historical panorama)**. In February 1665 Huygens observed that two of his pendulum clocks hanging from the same beam settled into phase opposition within about half an hour, independently of the initial phase difference, and experimentally ruled out air currents, attributing the effect to a weak mechanical coupling. In the language of the chapter this corresponds to the case of negative coupling constant. *Source*: the letters of Christiaan Huygens; a modern reading in Bennett, Schatz, Rockwood, and Wiesenfeld (2002). *Reason*: it is a historical and experimental fact, not a theorem; a quantitative mechanical explanation would require modelling the beam as an additional degree of freedom. *Where*: Remark 13.2.
- **The Ott–Antonsen ansatz: invariance and attractivity (imported)**. Writing the density as a Fourier series with coefficients b_n , the ansatz sets $b_n = \alpha^n$ for $n \geq 1$ with $|\alpha| \leq 1$, which is equivalent to a Poisson kernel. It is imported that this manifold with a single functional degree of freedom is invariant under the continuity equation and globally attracting;

the proof also imports the boundedness of the analytic extension of α to the lower half-plane. *Source:* E. Ott, T. M. Antonsen, *Chaos* 18 (2008), 037113. *Reason:* it requires functional-analytic arguments on the space of densities. *Boundary:* the algebraic consequence — the reduced equation $\dot{r} = -\Delta r + (K/2)r(1 - r^2)$ — is proved in the chapter starting from the ansatz. *Where:* Remark 13.4.

- **Chiba: the bifurcation of incoherence** (*imported*). Incoherence is asymptotically stable for $K < K_c$, and for $K > K_c$ a nontrivial solution branches off corresponding to partial synchronization: this is the bifurcation diagram conjectured by Kuramoto in 1975, rigorously proved. The difficulty is that the operator linearized about incoherence has unbounded continuous spectrum on the imaginary axis — the analogue of Landau damping — and Chiba develops a spectral theory on spaces of generalized functions. *Source:* H. Chiba (2015). *Reason:* the spectral theory on spaces of generalized functions exceeds the scope of the volume. *Status:* it is the result that fills the declared limitation of the self-consistency theorem proved in the chapter, which assumes as a hypothesis the existence of the partially synchronized state. *Where:* Remark 13.6.
- **Sakaguchi–Kuramoto, chimera states, Pecora–Carroll** (*panorama*). Three extensions only stated: with a phase lag in the coupling one loses the gradient-flow structure and the critical threshold depends on $\cos \alpha$; identical oscillators coupled non-locally can stabilize with one part synchronized and the rest incoherent; two identical chaotic systems in master–slave configuration synchronize exactly when the conditional Lyapunov exponents of the driven subsystem are all negative. *Source:* H. Sakaguchi, Y. Kuramoto, *Progress of Theoretical Physics* 76 (1986), 576–581; D. M. Abrams, S. H. Strogatz, *Physical Review Letters* 93 (2004), 174102; L. M. Pecora, T. L. Carroll, *Physical Review Letters* 64 (1990), 821–824. *Reason:* declared only stated. *Where:* Remark 13.7.
- **Dörfler–Bullo: the necessary condition on the complete graph** (*imported, ancillary*). On the complete graph synchronization requires $K \geq \frac{N}{2(N-1)}(\omega_{\max} - \omega_{\min})$; a sufficient threshold matching it requires, for finite N , implicit consistency equations. *Source:* F. Dörfler, F. Bullo, *Automatica* 50 (2014), 1539–1564 (extended version arXiv:1209.1335), §IV-A; Verwoerd and Mason for the consistency equations. *Reason:* reproducing the optimal sufficient threshold exceeds the space of the chapter, which chooses instead to prove local stability for an arbitrary graph; for $N = 2$ the optimal threshold is already the exact corollary proved in the chapter. *Where:* Remark 13.8.
- **Existence of the phase-cohesive equilibrium** (*panorama*). The Dörfler–Bullo theorem proved in the chapter presupposes that a phase-cohesive equilibrium exists. For two oscillators this is equivalent to the exact condition proved in the chapter, and for identical oscillators on the complete graph it always exists; for $N \geq 3$ with non-identical frequencies an explicit sufficient condition exists in the literature, but not in elementary closed form. *Source:* Dörfler–Bullo and the works cited therein, in particular the implicit consistency equations of Verwoerd and Mason. *Reason:* the proof, typically a fixed-point or continuation argument, does not fit within the space of the chapter. *Where:* Remark 13.9.

Chapter 14 — Phase-locked loops, encoders, and digital control on the circle

By choice, this is the chapter with the fewest theoretical debts: its four blocks are all panorama, and three of them are engineering or standards panoramas, that import no theorem.

- **Loop filter, cycle slip, pull-in** (*panorama*). In real PLLs a loop filter — typically proportional-integral — carries the state from phase alone to the pair (θ, w) , and the phase space becomes the cylinder instead of the circle. The type-II linear model treated in the chapter is the small-error special case of it, without the sinusoidal nonlinearity; the full model has

transient cycle slips even inside the pull-in range, and the exact determination of the pull-in range requires a global analysis of the phase plane, with separatrices and homoclinic orbits. *Source:* Gardner; Leonov and collaborators. *Reason:* this is textbook-level specialist material, not reducible to an elementary computation. *Where:* Remark 14.7.

- **Limit cycles from quantization in second-order filters** (*panorama*). The dead-band theorem proved in the chapter concerns a first-order loop, where quantization produces a spurious fixed point and not a periodic orbit. In second-order digital filters, rounding can generate genuine persistent periodic oscillations — a limit cycle in the proper sense — even when the underlying linear system would be asymptotically stable; a distinct phenomenon is truncation overflow, avoided with saturating arithmetic. *Source:* Åström and Wittenmark, the chapter on quantization. *Reason:* reported as panorama, without proof. *Where:* Remark 14.15.
- **Distributed Clocks as phase on S^1** (*applied panorama*). The offset between two clocks that count modulo the same period is a phase on the circle, and its vanishing is a synchronization problem of the type treated in Chapter 13 and in this chapter, with the propagation delay on the bus as the main systematic error. *Source:* the Distributed Clocks specification; Cena and collaborators for measurements on real implementations. *Reason:* the quantitative analysis of the delay-compensation protocol lies outside the scope of a treatise on the mathematics of the circle. *Where:* Remark 14.16.
- **PLCopen function blocks as conjugacies of rotations** (*standards panorama*). On modulo axes, that is, on positions of the circle, the electronic gearing block imposes $\theta_{\text{slave}} = \frac{n_2}{n_1} \theta_{\text{master}} + c$, a linear conjugacy between rotations; the variant with locking to an absolute position fixes the constant; the cam block replaces the constant ratio with an arbitrary periodic function, that is, a nonlinear conjugacy; the phasing block shifts the relative phase with a ramp. *Source:* the PLCopen motion control standard, layered on top of IEC 61131-3. *Reason:* the content is not a theorem — the chapter closes by saying that it is an elementary proportion — and the code fragment reported is illustrative, not an excerpt from the standard. *Where:* Remark 14.17.

Chapter 15 — Groups of homeomorphisms of the circle

- **Structure of $\text{Homeo}_+(\mathbb{T})$ as an abstract group** (*imported*). Assumed without proof: up to conjugacy, $\text{SO}(2)$ is the unique maximal compact subgroup; the inclusion $\text{SO}(2) \hookrightarrow \text{Homeo}_+(\mathbb{T})$ is a homotopy equivalence, in particular the group is connected and has fundamental group $\mathbb{Z}$; the group is simple; a generic pair in the sense of Baire generates a free group. *Source:* É. Ghys, “Groups acting on the circle”, *L’Enseignement Mathématique* (2) 47 (2001), §4. *Status:* of these four facts the chapter uses only the language suggested by the second — the name of the universal cover — while the others remain context. *Where:* Remark 15.2.
- **The optimal value of the defect of the rotation number** (*imported, unused*). In the literature the defect of the quasimorphism rotation number equals 1, that is, $|\rho(FG) - \rho(F) - \rho(G)| \leq 1$; the chapter proves only the version with constant 2. *Source:* Ghys (2001), §5, where the inequality is cited. *Reason:* the optimal constant cannot be reconstructed with only the tools developed in the chapter, whose two estimates give exactly 2; the chapter also states that it was unable to reopen a source containing the complete proof, and says so rather than citing one from memory. *Status:* no result of the chapter depends on the value 1; in particular the Milnor–Wood inequality in optimal form is not obtained from the defect. *Where:* Remark 15.4.
- **Dynamical realization of left-orderable groups** (*imported, unused*). For a countable

group G the converse of the proposition proved in the chapter holds: G is left-orderable if and only if it admits a faithful action on $\mathbb{R}$ by orientation-preserving homeomorphisms. The construction embeds G in $\mathbb{R}$ by induction on the enumeration, placing each new element at the midpoint of the two neighbours already placed, and then extends the action by continuity. *Source:* A. Navas, *Groups of Circle Diffeomorphisms*, arXiv:math/0607481, ch. 2; Ghys (2001), §6; for the free group on two generators, A. Blass, J. M. Kister, *Topology and its Applications* 24 (1986), 243–252. *Reason:* one assumes the countability of G and the delicate check that the extension is by homeomorphisms. *Where:* Remark 15.8.

- **Ghys: the bounded Euler class classifies up to semi-conjugacy** (*imported*). The class defined by the rotation-number cocycle lives in bounded cohomology, and two actions $\varrho_1, \varrho_2 : \Gamma \rightarrow \text{Homeo}_+(\mathbb{T})$ have the same bounded Euler class if and only if they are semi-conjugate. *Source:* É. Ghys, “Groupes d’homéomorphismes du cercle et cohomologie bornée”, *Contemporary Mathematics* 58 (1987), 81–106, and Ghys (2001), §6; for the notions of semi-conjugacy, Kim–Koberda–Mj (2019). *Reason:* one assumes the definition of bounded cohomology of groups and the notion of semi-conjugacy. *Status:* the chapter states that it does not use it in any proof: it serves to situate the Euler class within the general framework. *Where:* Remark 15.13.
- **Optimality of the Milnor–Wood inequality** (*imported*). The extremal value $|e(\varrho)| = 2g - 2$ is attained, for every $g \geq 2$, by the action of a cocompact Fuchsian group on the boundary of the Poincaré disk; and every integer n with $|n| \leq 2g - 2$ is the Euler class of some action, so that a circle bundle over the surface admits a foliation transverse to the fibers if and only if $|e| \leq 2g - 2$. *Source:* J. Milnor, *Commentarii Mathematici Helvetici* 32 (1957/58), 215–223; J. W. Wood, *ibid.* 46 (1971), 257–273, in the reading of Ghys (2001), §6. *Reason:* the chapter proves only the inequality, not its optimality; one assumes the existence of the hyperbolic structure on the surface and the computation of the Euler number of the unit tangent bundle, deferred to Volume V. *Note:* Milnor’s original statement concerns flat vector bundles and gives $|e| \leq g - 1$; passing to the projective group doubles the Euler number. *Where:* Remark 15.16.
- **Denjoy for group actions** (*imported*). For groups of $C^{1+\alpha}$ diffeomorphisms, extensions hold of Denjoy’s theorem and of Kopell’s lemma, obtained by probabilistic methods based on stationary measures; for groups of class only C^1 counterexamples abound; in class C^2 Sacksteder’s theorem and the works of Duminy give minimality results. *Source:* B. Deroin, V. Kleptsyn, A. Navas, “Sur la dynamique unidimensionnelle en régularité intermédiaire”, *Acta Mathematica* 199 (2007), 199–262; A. Navas, arXiv:math/0607481. *Reason:* declared imported; the distinction between regularity classes is here even more marked than in Chapter 5. *Where:* Remark 15.18.
- **Margulis’s alternative** (*imported*). For every subgroup $G \leq \text{Homeo}_+(\mathbb{T})$: either there exists a probability measure on $\mathbb{T}$ invariant under G , or G contains a free group of rank two. *Source:* G. Margulis, “Free subgroups of the homeomorphism group of the circle”, *Comptes Rendus de l’Académie des Sciences Paris I* 331 (2000), 669–674; also stated in Ghys (2001), §5. *Status:* in the first alternative the rotation number becomes a homomorphism and the invariant measure allows one to construct the semi-conjugacy to rotations as in the theorem proved in the chapter; Hölder’s theorem is the special case of a free action. *Where:* Remark 15.19.
- **Thurston stability** (*imported*). If a finitely generated group acts nontrivially by C^1 diffeomorphisms on a neighbourhood of a common fixed point of the line, then it admits a nontrivial surjective homomorphism onto $\mathbb{Z}$. Consequence: a finitely generated perfect group cannot act in this way. *Source:* W. P. Thurston, “A generalization of the Reeb stability theorem”, *Topology* 13 (1974), 347–352. *Reason:* declared imported without further technical

justification. *Where:* Remark 15.20.

- **Solodov, rigidity, and flexibility** (*imported*). Solodov’s theorem classifies groups of homeomorphisms of the line without global fixed points in terms of the order structure, generalizing Hölder’s theorem; in the contemporary framework, actions of surface groups with maximal Euler class are rigid — locally unique up to semi-conjugacy, by Matsumoto’s rigidity theorem and the results of Mann — whereas for non-maximal classes one finds uncountable families of semi-conjugacy classes with pairwise disjoint rotation spectra. *Source:* V. V. Solodov, *Mathematics of the USSR-Izvestiya* 24 (1985), 553–566; K. Mann, arXiv:1510.00728; S.-h. Kim, T. Koberda, M. Mj, *Flexibility of Group Actions on the Circle*, Lecture Notes in Mathematics 2231, Springer 2019. *Reason:* the chapter states that it assumes all the proofs of these results. *Where:* Remark 15.21.
- **Thompson’s group T and its smooth realization** (*panorama*). The group of piecewise-affine homeomorphisms of the circle with slopes powers of two and dyadic breakpoints is simple, infinite, and finitely presented; Ghys and Sergiescu show that it is conjugate to a subgroup of the group of C^∞ diffeomorphisms — a combinatorial group that admits a smooth realization. *Source:* É. Ghys, V. Sergiescu, “Sur un groupe remarquable de difféomorphismes du cercle”, *Commentarii Mathematici Helvetici* 62 (1987), 185–239. *Reason:* cited as panorama and not used elsewhere. *Where:* Remark 15.22.

Chapter 16 — Twist maps of the annulus and Aubry–Mather theory

- **Poincaré–Birkhoff theorem** (*imported*). Let f be an area-preserving homeomorphism of the compact annulus, homotopic to the identity, that sends each boundary component to itself and whose lift satisfies the strict twist condition at the boundary $X(x, a) - x < 0 < X(x, b) - x$: then f has at least *two* geometrically distinct fixed points. Applied to f^q composed with q backward turns, this gives two orbits of type (p, q) for every p/q in the twist interval. *Source:* H. Poincaré, “Sur un théorème de géométrie”, *Rendiconti del Circolo Matematico di Palermo* 33 (1912), 375–407 (statement, with proof of special cases); G. D. Birkhoff, “Proof of Poincaré’s Geometric Theorem”, *Transactions of the AMS* 14 (1913), 14–22 (first complete proof); a historical panorama in A. Chenciner, *Séminaire Bourbaki*, exp. 622, *Astérisque* 121–122 (1985), 147–170. *Reason:* Birkhoff’s proof is a global topological argument of about nine pages that uses neither monotone twist nor the variational principle of the chapter, and would alone take up more space than the whole section. *Boundary:* the minimizing orbit constructed in the chapter is one of the two; the second is part of the imported statement. *Where:* Remark 16.6.
- **Birkhoff’s theorem on invariant curves** (*imported*). Every simple closed, invariant, and essential curve of a positive monotone twist map that preserves area is the graph of a Lipschitz function, with constant depending only on the twist constants. *Source:* G. D. Birkhoff, “Sur quelques courbes fermées remarquables”, *Bulletin de la SMF* 60 (1932), 1–26; a modern exposition in M. R. Herman, “Sur les courbes invariantes par les difféomorphismes de l’anneau”, *Astérisque* 103–104 (1983), ch. I. *Reason:* it requires a global argument about the regions bounded by the curve and their areas, of length comparable to the whole section, and does not follow from the Lipschitz graph theorem proved in the chapter, because an invariant curve might a priori not be the closure of an ordered orbit. *Where:* Remark 16.7.
- **Aubry–Mather: minimality** (*imported partial*). For every rotation number there exists a *minimal* configuration — not merely a stationary one — and the minimal configurations of irrational rotation number form a totally ordered, closed, and invariant set, of the form described by the dichotomy proved in the chapter; for rational rotation number the set contains, in addition to the periodic orbits, the heteroclinic orbits connecting adjacent minimizing periodic orbits. *Source:* J. N. Mather, “Existence of quasi-periodic orbits for twist

homeomorphisms of the annulus”, *Topology* 21 (1982), 457–467; S. Aubry, P. Y. Le Daeron, “The discrete Frenkel–Kontorova model and its extensions I”, *Physica D* 8 (1983), 381–422; a self-contained exposition in V. Bangert, “Mather sets for twist maps and geodesics on tori”, *Dynamics Reported* 1 (1988), 1–56. *Boundary*: the chapter proves that the limit configuration is stationary and of Birkhoff type; its *minimality* — the fact that a periodic minimizer is minimal as a configuration over all of $\mathbb{Z}$, and not only among periodic configurations of the same type — is the imported part. *Reason*: it requires a covering argument over windows of length a multiple of the period, not reducible to the space of the section. *Where*: Remark 16.8.

- **Greene’s criterion and the threshold $k_c \approx 0.9716$ (*panorama*)**. The survival of the invariant curve of a given rotation number can be read from the stability of the periodic orbits of type (p_n, q_n) with p_n/q_n convergent: the residue $R = \frac{1}{4}(2 - \text{tr } DF^{q_n})$ tends to zero if the curve exists and diverges if it does not. For the golden torus of the standard map one obtains $k_c = 0.971635\dots$ *Source*: J. M. Greene, “A method for determining a stochastic transition”, *Journal of Mathematical Physics* 20 (1979), 1183–1201; a partial rigorous justification in R. S. MacKay, *Renormalisation in Area-Preserving Maps*, World Scientific 1993. *Reason*: the chapter clearly states that the criterion is a numerical conjecture, not a theorem, and that it is not proved in the treatise. *Status*: the threshold is checked independently by the laboratory (Appendix B, `fig16_cantorus.py`), which measures the maximal gap of the (89, 144) orbit as it crosses the critical value. *Where*: Remark 16.9.
- **Mather’s β function (*panorama*)**. The minimal average action extends to a convex and superlinear function, differentiable at every irrational, while at a rational p/q it has a corner if and only if there is no invariant curve of that rotation number — and the width of the corner is the action defect between the minimizing orbits and the min-max ones, a quantity with which Mather establishes the nonexistence of invariant curves. For generic twist diffeomorphisms β is not differentiable at the rationals. *Source*: J. N. Mather, “Differentiability of the minimal average action as a function of the rotation number”, *Boletim da Sociedade Brasileira de Matemática* 21 (1990), 59–70. *Reason*: declared *panorama*, not proved. *Note*: β is the free energy per atom of the Frenkel–Kontorova chain as the density varies, and its corners at the rationals are the mode-locking tongues, the counterpart of the Arnold tongues of Chapter 4. *Where*: Remark 16.10.
- **Moser’s twist theorem (*imported*)**. For a family of area-preserving twist maps of the annulus, sufficiently regular, with integrable unperturbed member, and for ω Diophantine in the twist interval, for a sufficiently small perturbation there exists an essential invariant curve of rotation number ω , close to the unperturbed one. One assumes the Diophantine condition (irrationality alone does not suffice), area preservation, monotone twist, and a threshold of regularity. *Source*: J. Moser, *Nachrichten Göttingen* II (1962), 1–20, in the optimal-regularity version by H. Rüssmann, in *Geometric Dynamics*, Lecture Notes in Mathematics 1007, Springer 1983, 677–718. *Reason*: it is a KAM theorem with a Newton scheme and loss of derivatives, that is, the content of Chapter 6 in the case of circle diffeomorphisms; the annulus case is its two-dimensional version and is not re-proved here. *Note*: KAM and Aubry–Mather are complementary — the former gives curves for small perturbations and Diophantine numbers, the latter gives, for every number and every intensity, an ordered invariant set, which is the KAM curve as long as it exists and its Cantor remnant once it has broken. *Where*: Remark 16.11.
- **Non-Birkhoff orbits, weak KAM, billiards (*panorama*)**. Not all periodic orbits of a twist map are ordered: Cheng and Cheng construct infinitely many non-Birkhoff ones, so that the orbits obtained by minimization in the chapter are a particular class, selected by the variational principle. The theory generalizes to continuous-time Lagrangian systems, where the minimal configurations become calibrated curves and β becomes the Legendre transform of

the effective Hamiltonian. Billiards in a convex oval form a twist map with generating function the chord length, and the Aubry–Mather sets correspond to the caustics. *Source*: W. Cheng, C.-Q. Cheng, *Science in China A* 43 (2000), 810–817; A. Fathi, *Weak KAM Theorem in Lagrangian Dynamics*; V. Bangert, the work already cited. *Reason*: panorama, not proved; Birkhoff’s conjecture on the integrability of convex billiards is deferred to Volume VI. *Where*: Remark 16.12.

A.3. Eleven blocks examined and not recorded

The script that collects the “imported” and “panorama” blocks finds, across the sixteen chapters, ninety-five. Eleven of them, on reading, declare no debt at all: they are methodological, attributional, or corrective notes that use the same words to say something different. They are listed here so that the count comes out right, and so that a reader who repeats the check with the script does not suspect an omission.

- Remark 6.12 (Chapter 6) — an attribution note: it states that the citation of Arnold 1961 is given for historical priority, that the original Russian text was not reopened, that the formulation adopted is the modern standard one, and that the explicit constants of the chapter are the authors’, not Arnold’s. The theorem is proved in full in the chapter.
- Remark 6.24 (Chapter 6) — three bridges towards Chapter 12, towards Chapters 4 and 13, and towards Volume VI. No statement imported.
- Remark 9.7 (Chapter 9) — records the *removal* of an import: the earlier draft assumed an unproved criterion, which Oxtoby’s criterion, now proved in the chapter, made superfluous.
- Remark 11.8 (Chapter 11) — justifies a definitional convention, the use of the critical point’s orbit in the bifurcation diagram; it rests on imports already declared elsewhere.
- Remark 12.7 (Chapter 12) — certifies that Fatou’s theorem, which in an earlier draft appeared with a bare outline, is now proved in full starting from only the declared imports.
- Remark 13.1 (Chapter 13) — a qualitative rereading, as a saddle-node bifurcation, of a theorem proved in the chapter, and a correction of a statement from the old structure.
- Remark 13.3 (Chapter 13) — a statement of the limitation of the self-consistency theorem, with a reference to Chiba’s result, which is recorded above as its own entry.
- Remark 13.10 (Chapter 13) — a list of application domains (fireflies, the sino-atrial node, cortical oscillations, grid generators, field buses) and a reference to Chapter 14.
- Remark 15.1 (Chapter 15) — the chain of regularity subgroups, with the observation that the strictness of an inclusion follows from the Denjoy counterexample constructed in Chapter 5.
- Remark 15.15 (Chapter 15) — an analysis of the architecture of the Milnor–Wood proof, explaining from which asymmetry of the estimates the last unit gained comes.
- Remark 16.2 (Chapter 16) — a statement of scope: it explicitly states that there is no imported result in the section, working directly on the cylinder with hypotheses assumed and checked on the example; the only bibliographic reference concerns the classical extension from the compact annulus to the cylinder, which the chapter does not reproduce.

A.4. What this registry says, taken as a whole

Three balance-sheet remarks, which the reader can verify by scanning the entries.

First. The debts are not evenly distributed. The chapter on KAM theory and the one on holomorphic dynamics together concentrate almost a quarter of the registry, and for a structural reason: they are the chapters in which the treatise touches theories whose complete proof is worth, each, an entire volume — two hundred pages for Herman, an Annals article for Yoccoz. The chapter on PLLs and encoders, at the opposite extreme, imports no theorem at all: its four blocks are engineering and standards panoramas.

Second. A good part of the imports are declared *and not used*. This is the case, for example, of nearly all the imports of the KAM chapter, where the chapter states in plain words that nothing it proves depends on them. These are expository debts, not logical ones, and distinguishing them is the main service that this registry renders: the proof structure of the volume rests on a number of imports much smaller than ninety-five.

Third. Several blocks record not a debt but a *correction*: the hypothesis $\rho(f) = \rho$ omitted in one draft of Arnold’s theorem, the confusion between the Bryuno threshold and the condition $\mathcal{H}$, the attribution to the Arnold tongues of the Feigenbaum constant, a theorem of Lanford declared “proved schematically” when it was not, an imported criterion later rendered superfluous by a proof. These are errors that the volume made and corrected, and that it chose to leave visible instead of erasing them. A registry of debts that hid the debts settled by mistake would be less useful, and less honest, than one that keeps them.

CHAPTER B

Numerical laboratory: the volume's scripts

Every “from script” figure in this volume, and every numerical value marked in the text with the sign ✓, come from a program that was *actually written and executed*, not imagined. The programs are not reproduced in the chapters — they would amount to dozens of pages of code inserted among the proofs — but they live, as standalone files, in the folder `appendiceB_laboratorio/` of the source. This appendix is their catalog: it states how many there are, which chapter each belongs to, what it computes, what it produces, and how to rerun it. It is not itself a program, and it does not add a single line of code to the volume.

B.1. Conventions of the laboratory

Interpreter and libraries

All scripts are written for **CPython 3**. The versions actually used, where a script declares them in its own header, are Python 3.12.5 with NumPy 2.5.2 and, when needed, SciPy 1.18.1. The common base is deliberately narrow:

- **NumPy** and **Matplotlib** are the only dependencies of almost all scripts; Matplotlib is always imported with the non-interactive **Agg** backend, so that execution opens no window and works even on a machine with no graphical interface.
- **SciPy** appears in only seven scripts, and always for a narrowly delimited task: adaptive control quadrature (`fig02_trapezio_errore.py`), nearest-neighbor search (`fig03_girasole_vogel.py`), root-finding by bisection (`fig07_tenda_potenze.py`), the Kolmogorov–Smirnov test (`fig10_clt_istogramma.py`) and action minimization in the three scripts of Chapter 16.
- **SymPy** appears exactly once, in `verifica_cap11_periodo3.py`, to compute symbolically the resultant that locates the period-3 bifurcation.
- From the standard library, `math`, `cmath`, `os`, `fractions.Fraction` (continued fractions and convergents in exact arithmetic) are used, and, in one case, `decimal` (for the Liouville number of `fig03_discrepanza_loglog.py`, where `float` does not carry enough digits).

The double check

The rule this volume inherits from the three preceding ones is that every numerical value cited in the text is computed **twice by independent routes**: once with the closed-form formula proved on the preceding pages, the other with a direct calculation that makes no use of that formula. The reader will recognize the second calculation inside almost every script in this catalog, often announced in the header as *independent check* or *reference check*. A few examples, to give the measure of it: `fig03_convergenti_phi.py` derives the convergents of φ both from the continued-fraction recurrence and from Binet’s formula; `fig10_correlazioni.py` computes the correlations of the doubling map by *three* independent routes (closed-form Fourier series, quadrature on a fine grid, Monte Carlo); `fig09_champernowne_binario.py` generates the binary

digits both with `bin()` and with a hand-written conversion, and requires the two strings to coincide character for character.

Determinism

The laboratory avoids random seeds: wherever a sample would be needed, a systematic sampling that always yields the same result is preferred. The clearest example is `fig13_kuramoto_n200.py`, which draws the $N = 200$ natural frequencies from the equally spaced quantiles of the Lorentzian distribution rather than from a pseudorandom generator: the simulation is thus reproducible to the digit without declaring any seed. What remains genuinely stochastic is only the Monte Carlo experiments declared as such in the text (the central-limit histogram of Chapter 10 and the trials on the vernier encoder of Chapter 14), where, however, what is being verified is a limit law, stable to the digits reported.

A numerical pitfall, documented

It is worth isolating, because it concerns the central object of three chapters. Iterating $\theta \mapsto 2\theta \bmod 2\pi$ in floating point is *unstable*: a `double` has 52 bits of mantissa, so after some fifty doublings the representation has consumed all the “true” binary digits of the initial datum and the trajectory collapses onto a spurious periodic point. This is not a flaw of the map: it is the expansiveness of the map that makes the finiteness of the mantissa visible, and it is the numerical reading of the very phenomenon that Chapters 7 and 8 describe in terms of sensitive dependence and of the Lyapunov exponent $\log 2$. The script `fig10_clt_istogramma.py` documents this in its own header and works around it; anyone reusing that code for their own experiments would do well to keep it in mind.

Outputs and running times

Scripts that produce figures write a PNG file (at 200 to 220 dpi) into the volume’s `img/` folder, which the L^AT_EX source includes with `\includegraphics`; verification-only scripts write nothing and print their check values to the terminal. The `dat/` folder, planned for a possible future `pgfplots` route, is at present empty: all script-generated figures in the volume are raster.

As for running times, the plan asked that each script state its own: for now this is done by the four scripts of Chapter 16 (from about 5 to about 30 seconds) and by the one of Chapter 15 (about 25 seconds); the others do not state it, and this appendix does not invent figures in their place. One can say with certainty only which are the heaviest, because it can be read from the size of the problem: the 800×400 scan of the Arnold tongues, the discrepancy pushed up to $N = 2 \cdot 10^6$, the 10^5 orbits of the central-limit histogram, and the Mandelbrot and Julia images.

How to reproduce a figure or a check

1. Move into the folder `Sorgenti/Vol4/appendiceB_laboratorio/`: the scripts’ relative paths (`../img/`) are written relative to that location.
2. Run `python script_name.py`.
3. Read the check values printed on the terminal and compare them with those the chapter reports next to the $\checkmark$ sign; the figure, if the script produces one, is rewritten into `../img/`.
4. Recompile the volume only if the figure has changed: the PNGs are shared across the three languages, because their labels are mathematical or absent and the text lives in the caption.

B.2. Catalog by chapter

What follows are the sixty scripts, in chapter order. For each: the file name and one line on what it computes. The word “verification” indicates a script that produces no figures, only check values.

Chapter 1 — Coverings, degree and the theorems of the circle

- `fig01_tfa_immagine.py` — image of the circle $|z| = r$ under $p(z) = z^3 - 2z + 5$ for $r = 0.3$ and $r = 8$, with the winding number about the origin recomputed by phase unwrapping (not assumed): this is the figure for the fundamental theorem of algebra.
- `fig01_whitney_graustein.py` — three regular closed curves of turning number 0, 1, 2, and four frames of a regular homotopy between two curves of turning number 2; the turning number is recomputed by numerically integrating the tangent angle.

Chapter 2 — Bundles over the circle and over the sphere

- `fig02_hopf_villarceau.py` — family of Hopf fibers above a fixed parallel of S^2 , in stereographic projection into $\mathbb{R}^3$, with the Villarceau torus that contains them overlaid as a guide.
- `fig02_spazio_lenticolare.py` — (declared) schematic of the lens space $L(n, 1)$ as a gluing of two solid tori along their boundary, in the illustrative case $n = 3$.
- `fig02_trapezio_errone.py` — spectral convergence of the trapezoidal rule for the 2π -periodic analytic integrand $A(\theta) = c/(a - \cos \theta)$, $c = 0.3$, $a = 1.5$, with the theoretical line e^{-Na_0} , $a_0 = \operatorname{arccosh} a$; the exact value and an independent adaptive quadrature are compared with each other.

Chapter 3 — Rotations, continued fractions and equidistribution

- `fig03_orbite_rotazione.py` — orbit of $R_{2\pi \cdot 3/7}$ (the star heptagon $\{7/3\}$) alongside 50 points of the orbit of $R_{2\pi/\varphi}$, with a closed-form check independent of the iteration loop.
- `fig03_convergenti_phi.py` — convergents p_n/q_n of the golden ratio and distances $\|q_n \alpha\|$ on a semilogarithmic scale against the bound $1/q_{n+1}$; double check via the recurrence versus Binet’s formula.
- `fig03_discrepanza_loglog.py` — discrepancy D_N^* of $\{n\beta\}$ on a log-log scale up to $N = 2 \cdot 10^6$ for three values of β : $1/\varphi$ and $\sqrt{2} - 1$ (bounded partial quotients) and a truncated Liouville number, which shows the jump when the partial quotients explode; comparison lines $\log N/N$ and $N^{-1/2}$.
- `fig03_girasole_vogel.py` — Vogel model of phyllotaxis, $r_n = c\sqrt{n}$, $\theta_n = n\alpha$, for the golden angle and for the nearby 137.3° , with an elementary and reproducible measure of packing quality.

Chapter 4 — Rotation number, Poincaré classification and Arnold tongues

- `fig04_lingue_arnold.py` — Arnold tongues in the (α, ε) plane for the standard family, on an 800×400 grid: the rotation number is estimated by vectorized iteration and rounded to the rationals of smallest denominator to color the locking zones.
- `fig04_scala_diavolo.py` — the devil’s staircase $\rho(\alpha)$ at $\varepsilon = 0.6$ (below criticality) and at $\varepsilon = 1$ (criticality), with three checks: monotonicity along the grid, the width of the $1/2$ plateau recomputed by an independent bisection, and the fraction of unlocked α .

-
- `fig04_ragnatela_rho13.py` — cobweb diagram for a map of the standard family with rotation number exactly $1/3$, with α located by bisection at the center of the tongue and a direct check $F^3(\theta_0) = \theta_0 + 1$.

Chapter 5 — Denjoy’s theorem and its counterexamples

- `fig05_misura_K.py` — Lebesgue measure of the Cantor set K of Denjoy’s C^1 counterexample as a function of the scale parameter c of the lengths $l_n = c/(|n| + 2)^2$, using the closed form $\sum_n 1/(|n| + 2)^2 = \pi^2/3 - 9/4$ proved in the chapter.
- `fig05_semiconiugazione.py` — truncated graph of the semiconjugacy h of the same counterexample, with the plateaus produced by the collapse of each inserted interval.

Chapter 6 — Small divisors and KAM theory on the circle

- `fig06_piccoli_divisori.py` — $|e^{2\pi i k \rho} - 1|$ on a logarithmic scale for $\rho = 1/\varphi$ (Diophantine) and for a Liouville number built by recurrence on the partial quotients, with the proved lower bound $4\|k\rho\|$.
- `fig06_coefficienti_psi.py` — Fourier coefficients $|\hat{\psi}_k|$ of the solution of the homological equation for the *same* analytic datum and two rotation numbers of opposite arithmetic nature.
- `fig06_newton_iterato.py` — the chapter’s Newton scheme actually iterated on $F = H_0 \circ R_\alpha \circ H_0^{-1}$, a map for which the conjugacy is known in closed form and the result is thus checkable.
- `verifica_cap06_kam.py` — verification: the small-divisor lemma $|e^{2\pi i x} - 1| \geq 4\|x\|$, the constant $\Lambda(\tau)$, the Diophantine constant of the reduced golden ratio, the threshold $\varepsilon_0(a, \rho)$ of the Newton scheme.
- `verifica_cap06.py` — verification: the constant $C_1(\tau)$ of the analytic estimate for the homological equation, small divisors of the golden ratio versus Liouville, growth of the Fourier coefficients, one complete step of the Newton scheme.
- `verifica_cap06_esercizi.py` — verification: the numerical checks for the chapter’s exercises, including $\inf_q q\|q\beta\|$ for $\beta = \sqrt{2} - 1$ compared with $6 - 4\sqrt{2}$.

Chapter 7 — The doubling map

- `fig07_tenda_potenze.py` — graphs of T , T^2 , T^3 with their fixed points; the count 2^n of solutions of $T^n(x) = x$ is obtained by bisection on a fine grid and compared with the closed-form formula.
- `fig07_triangolo_semiconiugazioni.py` — the triangle $D \rightarrow T \rightarrow f_4$ with $\varphi(\theta) = 1 - |1 - \theta/\pi|$ and $k(x) = \sin^2(\pi x/2)$, verified numerically on a fine grid.

Chapter 8 — Topological entropy and Lyapunov exponents

- `fig08_lap_number.py` — graphs of f_4 , f_4^2 , f_4^3 with the count of monotonicity pieces, which verifies $\text{lap}(f_4^n) = 2^n$ for $n = 1, 2, 3$.
- `fig08_birkhoff_convergenza.py` — convergence of the Birkhoff averages of $\log |f'_4|$ toward $\log 2$ on typical orbits, with the expected fluctuation band $O(1/\sqrt{n})$.
- `fig08_log_deriv_densita.py` — $\log |f'_4(x)|$ with the logarithmic singularity at $x = 1/2$, and the Ulam–von Neumann density compared with a histogram of $2 \cdot 10^6$ iterates.

Chapter 9 — Ergodic theory on the circle and on the interval

- `fig09_istogramma_f4.py` — histogram of 10^6 iterates of f_4 against the invariant density $\rho(x) = 1/(\pi\sqrt{x(1-x)})$, with the empirical mean of $\log|f'_4|$ compared with the exact value $\log 2$.
- `fig09_birkhoff_cos.py` — Birkhoff averages of $\cos \theta$ under the doubling map and under R_α , $\alpha = 2\pi/\varphi$, with the closed form of the geometric series as an exact check for the rotation.
- `fig09_champernowne_binario.py` — cumulative frequencies of the digit “1” and of the block “11” in Champernowne’s constant in base 2, against the expected values $1/2$ and $1/4$; the digits are generated by two independent routes that must agree exactly.
- `verifica_cap09.py` — verification: contraction of the total variation of Lasota–Yorke for the tent map, Borel normal numbers, unique ergodicity of R_α via Oxtoby’s criterion.

Chapter 10 — Transfer operators and the central limit theorem

- `fig10_correlazioni.py` — decay of the correlations $C_n(u, u)$ for the doubling map with a sawtooth observable, computed by three independent routes: the closed form $-\pi^2/(6 \cdot 2^n)$ derived from the Fourier coefficients, a quadrature on a fine deterministic grid, and a Monte Carlo simulation.
- `fig10_clt_istogramma.py` — dynamical central limit theorem for $u(\theta) = \sqrt{2}\cos \theta$: 10^5 independent orbits, histogram of $S_n(u)/\sqrt{n}$ against the standard Gaussian; this is the script that documents and works around the floating-point instability of the iteration $\theta \mapsto 2\theta$.
- `verifica_cap10.py` — verification: an explicit eigenfunction of the transfer operator for $|\lambda| < 1$, two explicit coboundaries, the degenerate telescoping case, contraction of the total variation.

Chapter 11 — Logistic map, bifurcations and universality

- `fig11_biforcazione.py` — bifurcation diagram of $f_r(x) = rx(1-x)$ over 2000 values of $r \in [2, 4]$, discarding 300 transient iterates; the printed check locates the first two bifurcations numerically, expected at 3 and at $1 + \sqrt{6}$.
- `fig11_finestra3.py` — zoom on the period-3 window, which is born exactly at $r = 1 + 2\sqrt{2}$.
- `fig11_lyapunov.py` — Lyapunov exponent $\lambda(r)$ for $r \in [2.4, 4]$, with the comparison between the value at $r = 4$ and $\log 2$.
- `fig11_deltan.py` — independent computation of the first eight doubling points r_n (the first two exact, the others by dynamic bisection) and of the estimates $\delta_n = (r_n - r_{n-1})/(r_{n+1} - r_n)$ of the Feigenbaum constant.
- `fig11_rinormalizzazione.py` — graphical illustration of the renormalization operator: f_r^2 restricted to a neighborhood of the critical point and rescaled by α , overlaid on f_r . The script explicitly states that it does *not* prove the existence of the fixed point of R , which remains Lanford’s theorem, imported.
- `verifica_cap11_periodo3.py` — verification: stability of the fixed point and of the 2-cycle, and the symbolic resultant (SymPy) that gives $r = 1 + 2\sqrt{2}$, compared with a direct numerical search.

Chapter 12 — From the circle to Julia and Mandelbrot

- `fig12_mandelbrot.py` — the Mandelbrot set by the escape criterion, with the main cardioid and the period-2 bulb overlaid *analytically*, and a zoom on the bulbs with the root points $c_{p/q}$ labeled; among the printed checks is an estimate of the area of $\mathcal{M}$ at two resolutions.
- `fig12_julia.py` — gallery of six filled Julia sets (the circle, the period-2 basin, the parabolic case, the dendrite, the Douady rabbit, and a Cantor K_c) and the external rays of angle $1/7$, $2/7$, $4/7$ on the rabbit, drawn by backward iteration of the Böttcher coordinate.
- `fig12_locale.py` — the Koenigs coordinate around an attracting fixed point of g_c , $c = -1/2$, and the invariant curves inside the golden Siegel disk.
- `fig12_dimensione.py` — Hausdorff dimension of J_c for small $|c|$ with two independent estimates (Bowen’s formula solved on the periodic points, and box counting), compared with the expansion $1 + |c|^2/(4 \log 2)$.
- `verifica_cap12.py` — verification: the critical orbit of $c = i$ and the multiplier of the 2-cycle, the dimension at $c = 0.1$, the count of periodic angles of the doubling map and of the hyperbolic centers, root points in closed form, the Diophantine condition on the golden number.

Chapter 13 — Synchronisation: from Huygens to Kuramoto

- `fig13_adler_flow.py` — Adler’s equation $\dot{\theta} = \Delta\omega - K \sin \theta$ integrated with Runge–Kutta 4: the period measured on the trajectory against the closed form $T = 2\pi/\sqrt{\Delta\omega^2 - K^2}$, and a check of the divergence $(\Delta\omega - K)^{-1/2}$ at the boundary.
- `fig13_kuramoto_n200.py` — Kuramoto model with $N = 200$ oscillators and frequencies drawn from the quantiles of the Lorentzian (no random seed), with the theoretical threshold $K_c = 2\gamma$.
- `fig13_rK_curva.py` — asymptotic curve $r(K)$ compared with the self-consistency formula $r = \sqrt{1 - K_c/K}$ for $K > K_c$.
- `fig13_ott_antonsen.py` — the exact reduced equation on the Ott–Antonsen ansatz, $\dot{r} = -\Delta r + (K/2)r(1 - r^2)$, compared with the direct simulation of the 200 oscillators under matching frequencies and initial condition; the fixed point $r^* = \sqrt{1 - 2\Delta/K}$ coincides with the one from the self-consistency equation.
- `verifica_cap13.py` — verification: equilibria and stability of the Kuramoto model with $N = 3$ identical oscillators, and the other values of the worked examples and exercises.

Chapter 14 — Phase-locked loops, encoders and digital control

- `fig14_banda_morta.py` — recursion $e_{k+1} = Q((1 - K)e_k)$ with a rounding quantizer: verifies that $|e_k|$ does not grow, that the sequence stops, and that $|e_\infty| \leq \Delta/(2K)$, with the values $e_\infty = 0.04$ for $K = 0.1$ and $e_\infty = 0.01$ for $K = 0.3$.
- `fig14_ito_h_unwrap.py` — wrapped phase and unwrapped phase of a linear chirp, with a case of *failure* of Itoh’s algorithm when the theorem’s hypothesis is not satisfied.
- `fig14_encoder_montecarlo.py` — Monte Carlo trials on the decoding of the vernier encoder for $(m, n) = (7, 9)$ and $(17, 19)$: the reconstruction of the integer J is exact as long as $\eta < \pi/(m + n)$, the threshold proved in the text.
- `verifica_cap14_lockrange.py` — verification: the exact locking range of the principal tongue, $\rho = 0 \iff |\alpha| \leq \varepsilon/(2\pi)$, with the example $\varepsilon = 0.3$ and threshold 0.0477465.

-
- `verifica_cap14_rdc_tipo2.py` — verification: zero tracking error at constant angular velocity for the resolver-to-digital converter modeled as a type-II PLL, checked by time-domain simulation and not only by the final value theorem.
 - `_genera_coordinate_tikz.py` — a support script, not cited in the text: it converts into lists of already-scaled TikZ coordinates the data of some figures of the chapter. Note that it reads intermediate `.dat` files that are no longer present in the current source (the `dat/` folder is empty): it is a remnant of an earlier version of the figures, kept but not necessary for reproduction.

Chapter 15 — Groups of circle homeomorphisms

- `fig15_quasimorfismo.py` — the rotation number is not a homomorphism: the defect is computed on the explicit pair $F_e(t) = t + e(1 - \cos t)$ and $G_e(t) = t + e(1 + \cos t)$, which both have a fixed point and hence zero rotation number, while their composition does not. Declared running time: about 25 s.
- `fig15_coordinate_tikz.py` — produces no images: it prints the coordinates that appear in the $\text{\LaTeX}$ comments inside the chapter’s “hand-drawn” figures, following the rule that every hand-drawn figure carries with it the computation of its own points.

Chapter 16 — Twist maps and Aubry–Mather theory

- `fig16_mappa_standard.py` — phase portraits of the standard map at $k = 0.5$, $k = 0.971635$ (Greene’s threshold), and $k = 1.2$; among the printed checks, $\det DF = 1$ at every sampled point and the traces $2 + k$ and $2 - k$ at the two fixed points. Declared running time: about 20 s.
- `fig16_orbite_birkhoff.py` — minimization of the periodic action for $(p, q) = (1, 2)$, $(1, 3)$, $(2, 5)$ at $k = 1$, with the orbits on the cylinder and the ordering diagram $i \mapsto x_i - ip/q$; prints $W_{\min}$, $\beta(p/q)$, and the norm of the gradient. Declared running time: about 5 s.
- `fig16_cantorus.py` — minimizing orbits of type $(89, 144)$ (two consecutive Fibonacci numbers) at three values of k : below the threshold the points fill the Lipschitz graph, above the threshold the macroscopic gap of the cantorus appears. Declared running time: about 30 s.
- `fig16_lemma_aubry.py` — Aubry’s lemma at work: two minimal configurations of different rotation numbers cross exactly once, and for two configurations that cross twice the positive action drop on which the proof rests is computed. Declared running time: about 5 s.

B.3. What the laboratory does not do

A clarification the treatise owes the reader. None of these scripts *proves* anything: a floating-point calculation on a finite number of points is not a proof, and in this volume it has never been used as one. The scripts do three different things, all legitimate and all more modest:

- they **draw** what the text proves — the phase portrait, the tongues, the devil’s staircase, the Julia set;
- they **check** the numerical values the text cites, recomputing them by a route independent of the formula from which the text derives them: this is the meaning of the $\checkmark$ sign;
- they **illustrate** results the volume imports without proving, and in this case they say so. The canonical example is `fig11_rinormalizzazione.py`, which shows the self-similarity of $Rf \approx f$ but states in its header, in plain words, that the existence of the fixed point of the

renormalization operator is Lanford's theorem, imported with its source in the Register of Appendix [A](#).

The boundary between a drawing, a check, and a proof is exactly the boundary between this appendix and the sixteen chapters that precede it.

Reference numerical tables

This appendix computes nothing. Every number that appears in it has already been obtained, checked, and discussed in one of the sixteen chapters: here it is only reordered into quick-reference tables, with a precise indication of the place where it was produced and of the independent check that accompanies it. Anyone looking for the *why* of a value must go back to the chapter indicated in the right-hand column; anyone looking only for the value finds it here.

A rule of honesty applies, without exception: **where the volume does not have a value, the table states so and leaves the cell blank**. Some rows that a reader might expect — for example the width of the Arnold tongues at $\varepsilon = 0.8$ — are not here, because that calculation was not carried out in any chapter, and inventing it here would mean introducing into the volume a number without proof or verification. Section C.11 collects the explicit list of these gaps.

C.1. Continued fractions and convergents

The following expansions are those produced by the algorithm of Construction 3.1; the convergents p_n/q_n are defined by the recurrences of Definition 3.2.

Number	Continued fraction	First convergents	Where
$\varphi = \frac{1 + \sqrt{5}}{2}$	$[1; 1, 1, 1, \dots]$	$q_n = F_{n+1}$ (Fibonacci)	Prop. 3.2
$1/\varphi = \varphi - 1$	$[0; 1, 1, 1, \dots]$	$q_n = F_{n+1}$	Ch. 3
$\sqrt{2}$	$[1; 2, 2, 2, \dots]$	$\frac{1}{1}, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \frac{99}{70}$	Ex. 3.1
$\sqrt{2} - 1$	$[0; 2, 2, 2, \dots]$	$q_n = 1, 2, 5, 12, 29$ (Pell)	Ex. 3.1
π	$[3; 7, 15, 1, 292, 1, 1, 1, \dots]$	$\frac{3}{1}, \frac{22}{7}, \frac{333}{106}, \frac{355}{113}, \frac{103993}{33102}$	Ex. 3.1
e	$[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots]$	(pattern $1, 2k, 1$)	Exerc. 3.1
$e - 2$	$[0; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$	$\frac{0}{1}, \frac{1}{1}, \frac{2}{3}, \frac{3}{4}, \frac{5}{7}, \frac{23}{32}$	Exerc. 3.1

INFO

Why only these seven. The volume's plan called for thirty partial quotients for each of the four classical numbers. The chapters have computed and verified far fewer: for φ , $\sqrt{2}$ and $\sqrt{2} - 1$ the continued fraction is periodic and the table gives it in full closed form (every quotient is 1, or 2); for π and for e the volume stops at the quotients reported above, which are the ones actually used in the proofs and in the numerical checks. Extending the two lists here would have meant adding digits verified by no script in the volume: the table stops where the text stops.

Quality of the approximation: the checks already carried out

The following three rows are the numerical checks of Theorem 3.3 and of Theorem 3.5 that appear in Chapter 3.

Quantity	Where
$ \pi - 2.668 $	Ex. 3.1
$355/113$	10^{-7}
upper bound	2.67×10^{-7} Ex. 3.1
$1/(q_3 q_4) = 1/(113 \cdot 33102)$	
$ (e - 0.003796) - 2 $	Exerc. 3.1
$5/7 $	
$1/(q_4(q_4 q_5)) = 1/(7 \cdot 39)$	Exerc. 3.1
(lower bound)	
$1/(q_4 q_5) = 1/(7 \cdot 32)$	Exerc. 3.1
(upper bound)	
$q_5^2 (e - 0.47948) - 2 $	Exerc. 3.1 (exceeds $1/\sqrt{5}$: convergent that does not satisfy the strengthened bound)
$23/32 $	
$q_n \ q_n\ $	Prop. 3.2
as $n \rightarrow \infty$	$1/\sqrt{5}$
$q_{12} \ q_{12}\ $	Prop. 3.2
(last value computed)	

C.2. Diophantine constants

In the table, $\|x\|$ is the distance of x from the nearest integer and the constant reported is $\inf_{q \geq 1} q \|q\alpha\|$, that is, the optimal constant K of the diophantine condition of type $(K, 0)$ (Definition 3.4).

Number	Closed form	Value	Where
$\rho^* = 1/\varphi$	$\varphi^{-2} = \frac{3 - \sqrt{5}}{2}$	0.381966011250	Prop. 6.1 (minimum at $q = 1$)
$\sqrt{2} - 1$	$6 - 4\sqrt{2}$	0.343145751	Exerc. 6.1 (minimum at $q = 2$)
quotients $\leq M$	$\geq \frac{1}{M+2}$ (general estimate)	$M = 2 : \geq 0.25$	Prop. 3.5

Constant	Value	Where
Hurwitz constant $1/\sqrt{5}$	0.447214...	Thm. 3.5
$\sqrt{5}$	2.236067977500	Prop. 6.1
φ^{-2}	0.381966011250	Prop. 6.1
φ^{-4}	0.145898033750	Prop. 6.1
$1/\varphi$	0.618033988750	Ex. 6.1
$\sqrt{2} - 1$	0.414213562	Exerc. 6.1

Warning. Two similar numbers that must not be confused. The constant $1/\sqrt{5} = 0.447214$ of Theorem 3.5 is the *limit* of $q_n^2|\varphi - p_n/q_n|$ along the convergents of φ (Proposition 3.2); the constant $\varphi^{-2} = 0.381966$ of Proposition 6.1 is instead the *infimum over all* $q \geq 1$, attained at $q = 1$. These are different quantities of the same number, and the second is necessarily less than or equal to the first.

Hierarchy of the arithmetic classes

Class	Definition / relation	Where
$\mathcal{CD}$ (diophantine)	$\ q\rho\ \geq K/q^{1+\tau}$ for every $q \geq 1$; full Lebesgue measure	Def. 3.4, Thm. 3.6
$\mathcal{H}$ (Herman–Yoccoz)	$\mathcal{CD} \subset \mathcal{H} \subset \mathcal{B}$, strict inclusions	Rem. 6.21
$\mathcal{B}$ (Bryuno)	$\sum_n q_n^{-1} \log q_{n+1} < \infty$	Def. 6.2
$\mathcal{L}$ (Liouville)	disjoint from $\mathcal{CD}$; measure zero	Def. 3.5

The golden ratio is the number that maximizes K for $\tau = 0$ (Proposition 3.3): it is, in this precise sense, the most robust rotation number that exists.

C.3. Notable rotations, the golden angle and the three-distance theorem

Quantity	Value	Where
golden angle $\gamma^* = 2\pi(2 - \varphi) = 2\pi/\varphi^2$	$137.5077\dots^\circ$	Sec. 3.20
comparison angle of Vogel’s sunflower model	137.3°	Def. 3.8, Prop. 3.7

Three-distance theorem: the two computed cases

Theorem 3.8 guarantees at most three arc lengths. The two instances verified in the volume are:

β	N	Arc lengths (fraction of a turn)	Where
$1/\varphi$	$8 = F_6$	0.090170 (three times), 0.145898 (five times)	Fig. 3.2
$\sqrt{2} - 1$	10	0.0711 (five times); 0.1005 (three times); 0.1716 (twice, and $0.1716 \approx 0.0711 + 0.1005$)	Exerc. 3.2

Note that 0.145898 of the first case is exactly φ^{-4} of the previous table: the coincidence is not accidental and is the same identity used in the proof of Proposition 6.1.

C.4. Logistic cascade and universality constants

Bifurcations in closed form

Bifurcation of $f_r(x) = rx(1-x)$	Parameter	Where
transcritical ($x = 0$ loses stability)	$r = 1$	Sec. 11.13
period doubling (flip), r_1	$r = 3$ (exact)	Thm. 11.1
end of stability of the 2-cycle, r_2	$r = 1 + \sqrt{6} \approx 3.449490$	Thm. 11.2
saddle-node of the period-3 window	$r = 1 + 2\sqrt{2}$	Sec. 11.13
accumulation of the cascade	$r_\infty \approx 3.569945672$	Thm. 11.6

The first eight doublings and the estimates of δ

Reproduction of Table 11.1: r_1 and r_2 are exact, the subsequent ones come from the dynamic bisection of the script `fig11_deltan.py` (Appendix B), with the limited precision stated in the original caption.

n	period $2^{n-1} \rightarrow 2^n$	r_n	$\delta_n = (r_n - r_{n-1})/(r_{n+1} - r_n)$
1	$1 \rightarrow 2$	3.000000	—
2	$2 \rightarrow 4$	3.449490	4.7583
3	$4 \rightarrow 8$	3.543955	4.6314
4	$8 \rightarrow 16$	3.564352	4.6514
5	$16 \rightarrow 32$	3.568737	4.6368
6	$32 \rightarrow 64$	3.569682	4.6038
7	$64 \rightarrow 128$	3.569888	4.5670
8	$128 \rightarrow 256$	3.569933	—

The four universality constants

Constant	Value	Function space	Where
δ	4.66920160910299...	even unimodal maps of the interval	Rem. 11.2
α	-2.5029078750959...	(same space)	Sec. 11.13
δ_{GM}	≈ -2.8336	pairs of critical circle maps, golden group	Rem. 11.6
α_{GM}	≈ -1.2886	(same space)	Rem. 11.6

Warning. δ and δ_{GM} are not the same constant and do not govern the same phenomenon: attributing $\delta = 4.6692$ to the Arnold tongues or to critical circle maps is the error explicitly corrected in Remark 11.6. Even the sign distinguishes them: $\delta > 0$, $\delta_{\text{GM}} < 0$.

C.5. Arnold tongues: widths

Standard family $F_{\alpha, \varepsilon}$ of Definition 4.3.

Tongue	Width	Where
0/1	ε/π , <i>exact</i> for every $0 < \varepsilon < 1$	Thm. 4.6
1/2	$\frac{\varepsilon^2}{4\pi} + O(\varepsilon^3)$, $\varepsilon \rightarrow 0^+$	Thm. 4.7
general p/q	$O(\varepsilon^q)$, $\varepsilon \rightarrow 0^+$ (imported for $q \geq 3$)	Thm. 4.7

Numerical check of the formula $\varepsilon^2/(4\pi)$

Ratio between the width measured by bisection and the asymptotic prediction, from block [C4-3] of the script `verifica_cap04_vol4.py`:

ε	0.02	0.05	0.10	0.15	0.20
numerical width / $\varepsilon^2/(4\pi)$	0.99997	0.99979	0.99917	0.99813	0.99669

Tongues measured at non-small ε

ε	Tongue	Interval in α	Width	Where
0.5	1/3	[0.336662; 0.340899]	0.004237 (center 0.338781)	Fig. 4.5
0.6	1/2	[0.486083; 0.513836]	0.027753 (bisection); 0.027200 (scan)	Ex. 4.4

At $\varepsilon = 0.6$ the asymptotic formula would give $\varepsilon^2/(4\pi) = 0.0286$: the discrepancy with respect to 0.027753 is a measure of how far $\varepsilon = 0.6$ already is from the perturbative regime. At the criticality $\varepsilon = 1$ mode-locking has full measure (Remark 4.5): there is no contradiction with the behavior $O(\varepsilon^q)$, which is an asymptotic statement as $\varepsilon \rightarrow 0^+$.

C.6. Standard map: Greene, Birkhoff, and Mather's β function

Greene's threshold

Quantity	Value	Where
critical threshold k_c (breakup of the golden torus)	0.971635...	Rem. 16.9
more precise value cited in the literature	0.97163540631	Rem. 16.9

Lipschitz constant of Birkhoff orbits

Minimizing orbit of type (89, 144), that is $p/q = 0.618055\dots$, a convergent of $\omega = (\sqrt{5} - 1)/2$; theoretical limit $L = 1 + k$ of Corollary 16.3.

k	0.5	0.971635	1.2
measured constant	0.339530	0.950534	1.344641
theoretical limit $1 + k$	1.500000	1.971635	2.200000
rotation number of the elliptic island	0.115027	0.164048	0.184505

The first two rows come from Example 16.5. The third reports the check carried out in the solution of Exercise 16.6 with the script `fig16_mappa_standard.py`.

The β function for $k = 1$

Minimum average action $\beta(p/q)$ of Definition 16.11, obtained by numerical minimization in Example 16.6.

ω	$\beta(\omega)$	ω	$\beta(\omega)$
0	-0.025330295911	2/5	0.078148038226
1/3	0.053119258132	3/5	0.178148038226
1/2	0.121896588107	2/3	0.219785924799
1	0.474669704089		

The three identities that generate the entire table from a few values (Proposition 16.4) are

$$\beta(-\omega) = \beta(\omega), \quad \beta(\omega + 1) = \beta(\omega) + \omega + \frac{1}{2}, \quad \beta(0) = -\frac{k}{(2\pi)^2}.$$

k	$\beta(0) = -k/(2\pi)^2$	$\beta(1) = \frac{1}{2} - k/(2\pi)^2$
0.5	-0.012665	0.487335
1	-0.025330295911	0.474669704089
2	-0.050661	0.449339

(Row $k = 1$ from Example 16.6; rows $k = 0.5$ and $k = 2$ from the solution of Exercise 16.2, where the independent numerical minimization gives -0.012665147955 , 0.487334852045 , -0.050660591821 , 0.449339408179 .)

C.7. KAM thresholds and the bookkeeping of Newton's scheme

All rows refer to Theorem 6.5 with $\rho = 1/\varphi$, hence $K = \varphi^{-2} = 0.381966$, $\tau = 0$, $\Lambda = 1.367879$, $C_1 = 4\Lambda/K = 14.324619$; a is the half-width of the strip of analyticity and $\delta_0 = \min(1, a/12)$.

a	δ_0	$\varepsilon_0(a, \rho)$	limiting strip $a - 6\delta_0$
0.20	0.01666667	2.019962×10^{-8}	$0.10 = a/2$
0.50	0.04166667	3.156190×10^{-7}	$0.25 = a/2$
1.00	0.08333333	2.524952×10^{-6}	$0.50 = a/2$

For comparison, with $\rho = \sqrt{2} - 1$ (hence $K = 6 - 4\sqrt{2} = 0.3431458$) and $a = 2$: $\delta_0 = 1/6 = 0.1666667$, $C_1 = 15.945171$, $\varepsilon_0 = 1.814668 \times 10^{-5}$, $a_\infty = 1.0 = a/2$ (Exercise 6.3).

Quadratic convergence in eight steps

Reproduction of the table of Remark 6.15, with $a = 0.5$, $\rho = 1/\varphi$ and $\gamma = 3$; here $A = 198\,023.539664$ and $\varepsilon_0 = 3.156190 \times 10^{-7}$.

j	δ_j	a_j	ε_j	$x_j = 2^{-2^j}$
0	4.166667×10^{-2}	0.50000000	3.156190×10^{-7}	5.000000×10^{-1}
1	2.083333×10^{-2}	0.37500000	1.972619×10^{-8}	2.500000×10^{-1}
2	1.041667×10^{-2}	0.31250000	6.164434×10^{-10}	6.250000×10^{-2}
3	5.208333×10^{-3}	0.28125000	4.815964×10^{-12}	3.906250×10^{-3}
4	2.604167×10^{-3}	0.26562500	2.351545×10^{-15}	1.525879×10^{-5}
5	1.302083×10^{-3}	0.25781250	4.485216×10^{-21}	2.328306×10^{-10}
6	6.510417×10^{-4}	0.25390625	1.305370×10^{-31}	5.421011×10^{-20}
7	3.255208×10^{-4}	0.25195312	8.845530×10^{-52}	2.938736×10^{-39}

C.8. Regularity classes: what holds where

The following table summarizes, in one row per class, the fate of a circle diffeomorphism with irrational rotation number as the regularity and the arithmetic quality of ρ vary. It is the picture that Chapters 5 and 6 build in full; here it serves only as an index.

Regularity	Hypothesis on ρ	Conclusion	Where
C^0	irrational	monotone semi-conjugacy to R_ρ ; minimal or invariant Cantor set	Thm. 4.4
C^1	any irrational	false : there exists a counterexample with invariant minimal Cantor set, for every irrational ρ	Constr. 5.2, Thm. 5.4
$C_+^{1+\text{bv}}$	irrational	conjugate to R_ρ ; no arithmetic hypothesis	Def. 5.2, Thm. 5.2
C^∞ , <i>not</i> the converse	irrational	Hall's C^∞ counterexample shows that regularity alone is not enough without the hypothesis on f'	Rem. 5.5
C^k , near R_ρ	diophantine (K, τ)	$C^{k'}$ conjugacy, $k' < k = k(\tau)$ (imported, Moser)	Rem. 6.17
global C^∞	ρ in a set of full measure	C^∞ conjugacy (imported, Herman 1979)	Rem. 6.18
global C^∞	diophantine	C^∞ conjugacy (imported, Yoccoz 1984)	Rem. 6.19
analytic, local	diophantine	analytic linearization for $\ u\ _a < \varepsilon_0$	Thm. 6.5
analytic, local	Bryuno	<i>optimal</i> condition for the local problem (imported)	Rem. 6.20
analytic, global	condition $\mathcal{H}$	exact characterization; $\mathcal{CD} \subset \mathcal{H} \subset \mathcal{B}$	Rem. 6.21

Loss of regularity in the homological equation

Given φ	Solution ψ of $\psi \circ R_\alpha - \psi = \varphi$	Where
analytic on strip a	analytic on strip $a - \delta$: a slice is lost	Thm. 6.1
C^r , $r > \tau + 2$	$\psi \in C^s$ for every $s < r - \tau - 2$: $\tau + 2$ derivatives are lost	Thm. 6.2
C^∞ , ρ Liouville	there may be no solution at all	Thm. 6.3

C.9. Synchronization thresholds

Kuramoto threshold $K_c = 2/(\pi g(0))$ of Theorem 13.5.

Density g	$g(0)$	K_c	Where
Lorentzian of half-width γ	$1/(\pi\gamma)$	2γ	Cor. 13.3
standard Lorentzian ($\gamma = 1$)	0.318310	2.000000	Ex. 13.4
standard Gaussian ($\sigma = 1$)	0.398942	1.595769	Ex. 13.4
uniform on $[-1, 1]$	$1/2$	$4/\pi = 1.273240$	Exerc. 13.5

For the Lorentzian the asymptotic order parameter has closed form, $r = \sqrt{1 - K_c/K}$ for $K > K_c$ (Corollary 13.3); for example $\Delta = 2$, $K = 5$ gives $r^* = \sqrt{0.2} = 0.447214$ (Exercise 13.6). Note that $\sqrt{0.2} = 1/\sqrt{5}$ is the *same real number* as the Hurwitz constant of Section C.2; here, however, it arises from the ratio $K_c/K = 4/5$ of this example, and the equality does not hide any connection between the two contexts.

C.10. Denjoy's counterexample: the two critical values

Quantity	Value	Where
$S = \sum_{n \in \mathbb{Z}} (n + 2)^{-2}$	$\frac{\pi^2}{3} - \frac{9}{4}$	Constr. 5.2
$c_{\max} = 2\pi/S$	$\frac{24\pi}{4\pi^2 - 27} \approx 6.0423$	Exerc. 5.6
$\text{Leb}(K)(c)$	$2\pi - cS$, linear; zero at $c = c_{\max}$	Thm. 5.4
$\text{Leb}(K)$ for $c = 1$	≈ 5.243	Ex. 5.3

C.11. What this appendix does not contain

The expansion plan called for Appendix C to include some tables that the volume, in its current form, is not able to provide without introducing unverified numbers. They are listed here, for clarity, as work that remains open, not as unintentional omissions.

1. **Thirty partial quotients of π and of e with their convergents.** Chapter 3 computes and verifies five for π and six for $e - 2$; Section C.1 stops there. For φ , $\sqrt{2}$ and $\sqrt{2} - 1$ the question does not arise: the continued fraction is periodic and is given in closed form.
2. **Widths of the Arnold tongues at $\varepsilon = 0.8$.** Chapter 4 measures tongues at $\varepsilon \leq 0.2$ (check of the asymptotic formula), at $\varepsilon = 0.5$ (tongue $1/3$) and at $\varepsilon = 0.6$ (plateau $1/2$). At $\varepsilon = 0.8$ there is no calculation in the volume.
3. **Extended table of q_n and $\|q_n\varphi\|$.** The volume reports the limit $q_n\|q_n\varphi\| \rightarrow 1/\sqrt{5}$ and the last value computed by the script ($n = 12$: 0.447215), but not the complete term-by-term sequence.
4. **Values of $\beta(\omega)$ at irrationals.** The table in Section C.6 covers only the rationals for which Chapter 16 carried out the minimization; at irrationals the volume gives the qualitative properties (Remark 16.10) and no numerical value.

Anyone wishing to fill these gaps will have to do so where it belongs: in the scripts of Appendix B and in the chapters, with the double check that the volume requires for every number. A table is not the place where a value is born.

CHAPTER D

Exercise catalog: reasoned index and synthesis exercises

The exercises of this volume are not here: they live in the chapters, next to the theorems they put to the test. Reprinting them in an appendix would detach them from the context that makes them meaningful, and would needlessly double a hundred or so pages. This appendix is instead their **index**: it says how many there are, where they are, what they ask, and it proposes a few cross-cutting reading paths for anyone who wants to practice on a theme rather than a chapter. At the end, and only at the end, three new exercises: they are *synthesis* exercises, meaning they belong to no single chapter because they combine three or four, and that is why their place is here.

D.1. How the volume's exercise catalog is organized

Every chapter closes with two consecutive sections, in this order:

- **Assigned exercises** — three exercises with a stated goal, estimated time, and type (symbolic, numerical, computational, error diagnosis). The text does not give their solution inline: the answers, with the checks, are in the *Instructor's key* of the same chapter.
- **Worked exercises** — three exercises with the complete solution immediately below, step by step, with the explicit independent check wherever the result is numerical.

Sixteen chapters times six exercises makes **96** exercises. To these are added **2** short exercises inserted in the body of the text, where they serve the proof in progress and it would not make sense to postpone them to the end of the chapter: the total in the sixteen chapters is **98**, to which this same appendix adds **3** synthesis exercises (Section D.4), for an overall total of **101** exercises in the volume.

INFO

How to use the index. The number in the first column is the number of the exercise as it appears in the chapter (for instance **6.2** is the second exercise of Chapter 6): clicking it in the PDF goes straight to the box. *Assigned* exercises have their answer in the chapter's Instructor's key; *worked* exercises have the solution right below the statement.

D.2. Index of exercises, chapter by chapter

Part A — Topology of the circle beyond π_1

Chapter 1 — Coverings, degree, and the theorems of the circle

Assigned: Section 1.16; worked: Section 1.17.

No.	Title	Type
1.1	The deck transformation group of p_6	assigned
1.2	A necklace of 14 beads	assigned
1.3	Localizing the roots of another polynomial	assigned
1.4	The lemniscate: why the degree of the tangent field is 0	worked
1.5	Degree with the pole outside the unit circle	worked
1.6	Sum of the exterior angles of an irregular pentagon	worked

Chapter 2 — Bundles over the circle and the sphere

Assigned: Section 2.18; worked: Section 2.19.

No.	Title	Type
2.1	Parametrization of the Möbius band: $X(\theta + 2\pi, t) = X(\theta, -t)$	in the text
2.2	Pencil of two secant circles and circular points	assigned
2.3	Whitney sum of two Möbius bundles	assigned
2.4	Holonomy on a cap of variable colatitude	assigned
2.5	Numerical check of the corrected Hopf section	worked
2.6	Foucault pendulum precession in Rome	worked
2.7	Chern–Weil with the other orientation	worked

Part B — Rotations and diffeomorphisms

Chapter 3 — Rotations, continued fractions, and equidistribution

Assigned: Section 3.16; worked: Section 3.17.

No.	Title	Type
3.1	Continued fraction of $e - 2$ and check of the error estimate	assigned
3.2	Three distances for $\beta = \sqrt{2} - 1$, $N = 10$	assigned
3.3	Phyllotaxis with a simple rational angle	assigned
3.4	Liouville’s constant: two independent checks of transcendence	worked
3.5	Why $q_1 = 7$ remains optimal up to $q_2 - 1 = 105$ for π	worked
3.6	Discrepancy compared: how many Monte Carlo samples to match $N = 1000$ Kronecker points?	worked

Chapter 4 — Rotation number, Poincaré’s classification, and Arnold tongues

Assigned: Section 4.10; worked: Section 4.11.

No.	Title	Type
4.1	A non-symmetric period-2 orbit	assigned
4.2	Checking the formula $\varepsilon^2/(4\pi)$ at $\varepsilon = 0.08$	assigned
4.3	What a finite grid at criticality does (and does not) prove	assigned
4.4	Numerical check of $\rho(f^3) = 3\rho(f)$	worked
4.5	From finding the period-3 orbit to the exact rotation number	worked
4.6	Fekete’s lemma on an explicit sequence, with a closed formula for the remainder	worked

Chapter 5 — Denjoy’s theorem and its counterexamples

Assigned: Section 5.8; worked: Section 5.9.

No.	Title	Type
5.1	The best-return-time lemma for $\beta = \sqrt{2} - 1$	assigned
5.2	Denjoy's inequality for another family	assigned
5.3	The counterexample with $c = 2$	assigned
5.4	Denjoy–Koksma for an indicator of an arc	worked
5.5	Exactly where Denjoy's proof dies in the counterexample	worked
5.6	The critical value $c_{\max}$	worked

Chapter 6 — Small divisors and KAM theory on the circle

Assigned: Section 6.10; worked: Section 6.11.

No.	Title	Type
6.1	The diophantine constant of $\sqrt{2} - 1$	assigned
6.2	The homological equation for data with three modes	assigned
6.3	The threshold ε_0 for another strip and another number	assigned
6.4	A Newton step computed by hand	worked
6.5	Where Newton's scheme dies if ρ is Liouville	worked
6.6	Why geometric convergence would not be enough	worked

Part C — Chaos and ergodic theory

Chapter 7 — The doubling map: symbolic dynamics, Li–Yorke, and Sharkovskii

Assigned: Section 7.11; worked: Section 7.12.

No.	Title	Type
7.1	Density of the periodic points of E_5	assigned
7.2	A Štefan orbit of period 7	assigned
7.3	The collapse for $n = 2$	assigned
7.4	Sensitivity of E_3 at a fixed distance	worked
7.5	From the permutation to the covering graph: period 4	worked
7.6	Independent check of the Möbius count for $n = 6$	worked

Chapter 8 — Topological entropy and Lyapunov exponents

Assigned: Section 8.13; worked: Section 8.14.

No.	Title	Type
8.1	Entropy of a three-branch map	assigned
8.2	Lap number of the tent map	assigned
8.3	A period-5 orbit and the entropy	assigned
8.4	Comparing the estimates of $N(n, \varepsilon)$ for the shift and for E_2	worked
8.5	Lyapunov exponent of E_3 and Rokhlin's formula	worked
8.6	The cost of missing expansion: f_4 near the critical point	worked

Chapter 9 — Ergodic theory on the circle and on the interval

Assigned: Section 9.17; worked: Section 9.18.

No.	Title	Type
9.1	Exact rate of L^2 convergence for D	in the text
9.2	Krylov–Bogolyubov for E_5	assigned
9.3	Normal numbers in base 3	assigned
9.4	Oxtoby and rational rotations	assigned
9.5	The Krylov–Bogolyubov empirical averages for E_3 , step by step	worked
9.6	Ergodic uniqueness and rational rotations	worked
9.7	Lasota–Yorke for E_3 : the contraction becomes $1/3$	worked

Chapter 10 — Transfer operators, decay of correlations, and the central limit theorem

Assigned: Section 10.12; worked: Section 10.13.

No.	Title	Type
10.1	Explicit eigenfunctions of $\mathcal{L}_3$ on L^2	assigned
10.2	Decay of correlations for E_5	assigned
10.3	The coboundary of $u = \sqrt{2} \cos 4\theta$	assigned
10.4	Numerical check of the Hölder contraction, $\alpha = 1/2$	worked
10.5	Correlation C_1 between the sawtooth and $\sqrt{2} \cos \theta$	worked
10.6	The limit $\lambda \rightarrow 1$ in the eigenfunction g_λ	worked

Chapter 11 — Logistic map, bifurcations, and universality

Assigned: Section 11.9; worked: Section 11.10.

No.	Title	Type
11.1	The 2-cycle at $r = 3.45$ and the threshold $1 + \sqrt{6}$	assigned
11.2	The period-3 window from the analytic side	assigned
11.3	The two universalities compared	assigned
11.4	The fixed point at $r = 2.5$ and the convergence time	worked
11.5	Independent check of the multiplier of the 2-cycle at $r = 3.45$	worked
11.6	Distinguishing δ and δ_{GM} on a numerical case	worked

Part D — Holomorphic dynamics

Chapter 12 — From the circle to Julia and Mandelbrot

Assigned: Section 12.14; worked: Section 12.15.

No.	Title	Type
12.1	Escape criterion and membership in $\mathcal{M}$	assigned
12.2	Root points and bulbs	assigned
12.3	An explicit Cremer number	assigned
12.4	The dendrite $c = i$: a Misiurewicz point	worked
12.5	The dimension of J_c for $c = 0.1$	worked
12.6	Periodic angles and hyperbolic centers: an exact identity	worked

Part E — Synchronization and automation

Chapter 13 — Synchronization: from Huygens to Kuramoto

Assigned: Section 13.13; worked: Section 13.14.

No.	Title	Type
13.1	Adler period for $\Delta\omega = 3$, $K = 1.5$	assigned
13.2	Two oscillators with repulsive coupling	assigned
13.3	Identical Kuramoto oscillators, $N = 4$: generalizing Proposition 13.2	assigned
13.4	Adler period for $\Delta\omega = 5$, $K = 2$	worked
13.5	Critical threshold for a uniform density	worked
13.6	Ott–Antonsen fixed point for $\Delta = 2$, $K = 5$	worked

Chapter 14 — Phase-locked loops, encoders, and digital control on the circle

Assigned: Section 14.16; worked: Section 14.17.

No.	Title	Type
14.1	Hold-in range and lock-in time	assigned
14.2	A sampling that violates Itoh’s constraint	assigned
14.3	Tolerance of a 17/19 encoder	assigned
14.4	The fan and the stroboscope	worked
14.5	Dead band with $K = 0.1$	worked
14.6	Tracking error of the resolver-to-digital converter	worked

Part F — Global structures

Chapter 15 — Groups of homeomorphisms of the circle

Assigned: Section 15.12; worked: Section 15.13.

No.	Title	Type
15.1	The defect for a pair with assigned fixed points	assigned
15.2	The commutator lemma put to the test	assigned
15.3	Euler classes realizable in genus 3	assigned
15.4	The case $g = 1$ of the Milnor–Wood inequality without the Oscillation Lemma	worked
15.5	A group that cannot act freely: the infinite dihedral group	worked
15.6	Computing $e(\varrho)$ for a representation that factors	worked

Chapter 16 — Twist maps and Aubry–Mather theory

Assigned: Section 16.10; worked: Section 16.11.

No.	Title	Type
16.1	The explicit orbit of type $(1, 2)$ as k varies	assigned
16.2	Two exact values of the β function	assigned
16.3	A family of generating functions with twist a	assigned
16.4	The minimizing orbit of type $(1, 2)$ for $k = 2$	worked
16.5	The exact gain of an exchange	worked
16.6	The rotation number of the elliptic island and its limit	worked

D.3. Cross-cutting paths

The lists that follow group the exercises already indexed above according to the *method* required rather than the chapter. They add nothing to the volume: they serve anyone who wants to train a specific skill across the six Parts.

Arithmetic of rotations: continued fractions and diophantine conditions

Exercises 3.1, 3.2, 3.4, 3.5, 5.1, 6.1, 6.5, 12.3. The thread: the same quantity $\|q\rho\|$ decides the speed of equidistribution (Chapter 3), the combinatorics of return times (Chapter 5), the solvability of the homological equation (Chapter 6), and the existence of Siegel disks (Chapter 12).

Numerical check of a closed formula

Exercises 2.5, 4.2, 4.4, 5.4, 7.6, 8.5, 9.5, 10.4, 11.5, 13.6, 16.4. The thread: every time something is computed twice, by independent routes, and compared. It is the discipline that the volume imposes on itself as well.

Error diagnosis

Exercises 4.3, 5.5, 6.6, 11.3, 11.6. The thread: understanding *where* a plausible line of reasoning stops working is more instructive than watching it succeed. Three of these exercises arise from errors actually present in the popular-science literature or in earlier versions of this very volume.

The circle in automation

Exercises 13.1, 13.2, 13.4, 13.5, 14.1, 14.2, 14.3, 14.4, 14.5, 14.6. The thread: phase locking, sampling, encoders, and synchronization are the same mathematics as the rotation number, seen from a laboratory bench.

D.4. Three synthesis exercises

The three exercises that follow belong to no chapter because each one crosses at least three. The solutions introduce no new result: every step points back to a statement proved (or explicitly imported) in the volume, and every number is one of those collected in Appendix C.

Exercise D.1: One rotation number, four theorems. *Chapters involved: 3, 5, 6. Estimated time: 45 minutes. Type: conceptual synthesis with numerical check.*

Let $\rho^* = 1/\varphi$ be the reduced golden ratio.

- (a) Deduce from Proposition 3.2 that all the partial quotients of ρ^* equal 1 and that the denominators of the convergents are the Fibonacci numbers.
- (b) Compare the exact diophantine constant of ρ^* (Proposition 6.1) with the general lower bound of Proposition 3.5 for $M = 1$, and with the constant of $\sqrt{2} - 1$ (Exercise 6.1). Do the three inequalities line up in the expected order?
- (c) Let f be a circle diffeomorphism with $\rho(f) = \rho^*$. State what can be concluded if $f \in C_+^{1+\text{bv}}$, and what *cannot* be concluded if f is only C^1 . In which of the two cases does the arithmetic quality of ρ^* play a role?
- (d) For f analytic on a strip of half-width $a = 0.5$ and close to the rotation, what is the threshold ε_0 guaranteed by Theorem 6.5? What happens to the threshold if ρ^* is replaced by $\sqrt{2} - 1$, with everything else unchanged?

Solution. (a) The proof of Proposition 3.2 starts from $\varphi = [1; 1, 1, 1, \dots]$, an immediate consequence of $\varphi = 1 + 1/\varphi$, and checks by induction that $q_n = F_{n+1}$. Since $\rho^* = 1/\varphi = \varphi - 1$ differs from φ by an integer, its expansion is $[0; 1, 1, 1, \dots]$: the same partial quotients from a_1 onward, hence the same q_n .

(b) The three quantities are, in increasing order,

$$\frac{1}{M+2} \Big|_{M=1} = \frac{1}{3} = 0.3333\dots < 6 - 4\sqrt{2} = 0.343145751 < \varphi^{-2} = 0.381966011250.$$

The first inequality says that the general estimate of Proposition 3.5 is valid but not optimal; the second says that $\sqrt{2}-1$ is an excellent diophantine number but not the best. The order is the expected one because the golden ratio maximizes the constant for $\tau = 0$ (Proposition 3.3): no irrational can exceed φ^{-2} , and in particular $\sqrt{2}-1$ cannot exceed it.

(c) If $f \in C_+^{1+\text{bv}}$, Theorem 5.2 gives the strongest possible conclusion with no arithmetic hypothesis at all: f is minimal and conjugate to the rotation R_{ρ^*} . If instead f is only C^1 , nothing of the sort can be concluded: the Construction 5.2 produces, **for every** irrational ρ and hence also for ρ^* , a C^1 diffeomorphism whose minimal set is a Cantor set (Theorem 5.4). This is the key point: at the C^1 level the arithmetic of ρ is *invisible*, because the counterexample is built for every irrational; the arithmetic starts to matter only from the analytic case onward (and from the C^k case with k finite but large enough), where the diophantine condition enters explicitly into the estimates.

(d) From the table in Remark 6.13, for $a = 0.5$ and $\rho = \rho^*$ the guaranteed threshold is $\varepsilon_0 = 3.156190 \times 10^{-7}$, with limiting strip $a_\infty = 0.25 = a/2$. Replacing ρ^* with $\sqrt{2}-1$, the constant K drops from 0.381966 to 0.343146, and since in the closed formula $\varepsilon_0 = K\delta_0^{3+\tau}/(2^{6+\tau}\Lambda)$ the dependence on K is linear (with $\tau = 0$ and δ_0 unchanged, since it depends only on a), the threshold shrinks in the same ratio $0.343146/0.381966$. This is the quantitative formulation of the observation that numbers that are harder to approximate tolerate larger perturbations.

Exercise D.2: Two δ constants not to be confused. *Chapters involved: 4, 11. Estimated time: 40 minutes. Type: synthesis with error diagnosis.*

- State the width of the 0/1 tongue and of the 1/2 tongue of the standard family (Theorems 4.6 and 4.7), specifying for each whether the formula is exact or asymptotic.
- Using the ratios reported in the proof of Theorem 4.7, estimate the order of magnitude of the remainder at $\varepsilon = 0.2$ and check that it is compatible with an absolute remainder $O(\varepsilon^3)$.
- A popular-science text claims that “the Arnold tongues are governed by the Feigenbaum constant $\delta \approx 4.6692$.” Identify all the errors in the statement, distinguishing the error of *value* from the error of *context* (Remark 11.6).
- The p/q tongue has width $O(\varepsilon^q)$, hence for large q it is tiny. How is this reconciled with the fact that at $\varepsilon = 1$ mode-locking has full measure (Remark 4.5)?

Solution. (a) The 0/1 tongue has *exact* width ε/π for every $0 < \varepsilon < 1$: it is not an asymptotic, it is a closed formula valid over the whole interval of subcritical parameters. The 1/2 tongue has width $\varepsilon^2/(4\pi) + O(\varepsilon^3)$ as $\varepsilon \rightarrow 0^+$: here the formula is asymptotic, and the remainder is not controlled uniformly.

(b) The numerical check reported in the proof gives, for the ratio between the measured width and $\varepsilon^2/(4\pi)$, the values 0.99997 ($\varepsilon = 0.02$), 0.99979 (0.05), 0.99917 (0.1), 0.99813 (0.15), 0.99669 (0.2). The deviation from 1 at $\varepsilon = 0.2$ is 0.00331; at $\varepsilon = 0.1$ it is 0.00083, about a quarter. A relative remainder that halves when ε halves would be $O(\varepsilon)$; here it divides by about four when ε halves, which is even better than expected and in any case compatible with a relative remainder $O(\varepsilon)$, that is, an absolute remainder $O(\varepsilon^3)$, exactly as stated.

(c) There are two errors layered on top of each other, and it is important not to conflate them. *Error of context:* the constant $\delta = 4.6692\dots$ is the unstable eigenvalue of the renormalization operator on the space of even unimodal maps of the interval (Section 11.4), and it governs the period-doubling cascade of the logistic map, not the Arnold tongues. *Error of value:* the constant that actually governs the renormalization of critical circle maps at the golden rotation number is $\delta_{\text{GM}} \approx -2.8336$, different both in magnitude and in *sign*. An attentive

reader could have suspected the error precisely from the sign: a negative eigenvalue cannot coincide with a positive one.

(d) There is no contradiction, because the two statements speak of different regimes of ε . Theorem 4.7 is an *asymptotic statement as $\varepsilon \rightarrow 0^+$* : it says that for small ε the tongues of large denominator are negligible, and indeed at small ε the union of the tongues has small measure. Świątek’s result instead concerns the *critical* value $\varepsilon = 1$, where the standard family is no longer a diffeomorphism but a homeomorphism with a cubic critical point: there the tongues have widened to the point of covering almost the whole parameter. Between the two regimes the measure of the union of the tongues grows continuously from nearly zero to one, and the devil’s staircase of Theorem 4.8 passes from almost diagonal to almost entirely a staircase.

Exercise D.3: The golden circle, from the plane to the torus. *Chapters involved: 3, 16. Estimated time: 50 minutes. Type: synthesis with critical reading of numerical data.*

Let $\omega = (\sqrt{5} - 1)/2 = 1/\varphi$.

- Why is the minimizing orbit used in the numerical check of Example 16.5 of type (89, 144) and not, say, of type (90, 145)?
- Compare the three Lipschitz constants measured in that example with the theoretical limit of Corollary 16.3, and say what is observed as k grows.
- The central value $k = 0.971635$ is not chosen at random: what does it represent (Remark 16.9), and why is it reasonable that the margin relative to the Lipschitz limit is tightest exactly there?
- Verify that the value $\beta(1) = 0.474669704089$ from the table in Example 16.6 can be derived in closed form from the identities of Proposition 16.4 alone, without redoing any minimization.

Solution. (a) Because 89 and 144 are consecutive Fibonacci numbers, and the denominators of the convergents of ω are exactly the Fibonacci numbers (Proposition 3.2). Hence 89/144 is a *convergent* of ω , that is, the best rational approximation of ω among all those with denominator not exceeding 144 (Theorem 3.3 and Section 3.3): $89/144 = 0.618055\dots$ against $\omega = 0.618033988750$. The pair (90, 145) is not a convergent and would approximate worse with a larger denominator: the corresponding periodic orbit would be a worse approximation of the golden invariant curve.

(b) The data are

k	0.5	0.971635	1.2
measured constant	0.339530	0.950534	1.344641
limit $1 + k$	1.500000	1.971635	2.200000
ratio measured/limit	0.23	0.48	0.61

(the last row is the quotient of the two preceding ones). The constraint of Corollary 16.3 is respected in all three cases, but the margin shrinks systematically as k grows: the theoretical estimate $L = 1 + k$ is generous at small k and becomes progressively tighter, consistent with the fact that at small k the standard map is a small perturbation of the integrable map, so that the ordered orbits lie on nearly horizontal graphs.

(c) $k_c = 0.971635\dots$ is the Greene threshold: the value of the parameter at which the invariant curve of golden rotation number breaks (Remark 16.9). It is reasonable that the margin narrows exactly there because that is precisely the regime in which the golden invariant graph ceases to be smooth and approaches the limiting configuration: its maximum local slope grows, and with it the Lipschitz constant measured on the Fibonacci periodic orbit that approximates it. Note that the numerical datum alone does not *prove* the breakup:

it only shows a trend consistent with it, and indeed in the volume Greene's criterion remains an observation from the overview, not a proved theorem.

(d) Yes. For $k = 1$, Proposition 16.4(iii) gives

$$\beta(0) = -\frac{k}{(2\pi)^2} = -\frac{1}{(2\pi)^2} = -0.025330295911,$$

and (ii) with $\omega = 0$ gives $\beta(1) = \beta(0) + 0 + \frac{1}{2}$, that is,

$$\beta(1) = -0.025330295911 + 0.5 = 0.474669704089,$$

which coincides with the tabulated value to all twelve digits. ✓ This is precisely the independent check carried out in Example 16.6: the numerical minimization of $W_{0,1}$ and the closed formula produce the same number by routes that do not talk to each other.

D.5. Note on the relationship with the expansion plan

The plan called for Appendix D to be a collection of exercises drawn from the quiz items, plus some forty new graded exercises. This appendix has a different form, and the reason must be stated.

The quiz items have, in the meantime, been rewritten and folded into the chapters and into the volume's final quiz, with their own key: turning them into exercises a second time would have produced duplicated material, not new material. As for the forty additional exercises, writing them here would have meant producing statements and solutions that no chapter backs up, in a volume whose declared rule is that every claim have a proof or a verified source. It was preferred to deliver three synthesis exercises that are genuinely checked, and to state openly that the remaining thirty-seven are still to be written, rather than to pad the pages.

It therefore remains open, for a future edition: extending Section D.4 up to the forty exercises called for in the plan, with the A–E grading of the plan, and the corresponding key. The place to do it is this one, and the constraint is the usual one: no numerical value that does not come from a computation actually performed and checked.

Glossary

Entries are in alphabetical order. Each one gives a concise definition and, when the concept is treated at length, points to the corresponding definition, theorem, or chapter. The point is the decimal separator.

Adler equation

The equation $\dot{\theta} = \Delta\omega - K \sin \theta$ on the circle, the minimal model of phase locking: for $|\Delta\omega| < K$ it has two equilibria (one stable, one unstable), for $|\Delta\omega| > K$ none, and the phase slips with finite period (Definition 13.2, Chapter 13).

Locking bands

The three sets of PLL phase mismatches: *hold-in* H (an asymptotically stable equilibrium exists), *pull-in* P (global convergence), *lock-in* L (convergence with no phase slips), with $L \subseteq P \subseteq H$ (Definition 14.1, Chapter 14).

Annulus

The cylinder $\mathbb{T} \times \mathbb{R}$ (or $\mathbb{T} \times [a, b]$) on which twist maps act, with its covering $\mathbb{R} \times \mathbb{R}$ and the lifts that commute with the unit translation (Definition 16.1, Chapter 16).

Golden angle

The angle $\gamma^* = 2\pi(2 - \varphi) = 2\pi/\varphi^2 \approx 137.5077^\circ$ that appears in Vogel's model of phyllotaxis; it is the angular increment whose orbit on the circle has the lowest discrepancy (Chapter 3).

Arnold standard family

The two-parameter family $f_{\alpha, \varepsilon}(\theta) = \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$, which is a diffeomorphism of the circle for $0 < \varepsilon < 1$ (Definition 4.3, Chapter 4).

Arnold tongues

The regions of the plane (α, ε) where the rotation number of the standard family is exactly p/q ; the p/q tongue has width $O(\varepsilon^q)$ (Theorem 4.7, Chapter 4).

Arnold's theorem

An analytic diffeomorphism of the circle sufficiently close to a rotation with diophantine rotation number is analytically conjugate to that rotation; proved via a Newton scheme (Theorem 6.5, Chapter 6).

Aubry's lemma

Two distinct minimal configurations cross at most once; this is the combinatorial lemma on which the whole of Aubry–Mather theory rests (Theorem 16.3, Chapter 16).

Aubry–Mather set

The closure of an ordered orbit of a twist map, with its rotation number; for irrational ω it is either an invariant curve or a Cantor set (Definition 16.10, Chapter 16).

Self-consistency equation

The condition $r = Kr \int_{-\pi/2}^{\pi/2} \cos^2 \theta g(Kr \sin \theta) d\theta$ which, in the steady-state ansatz of Kuramoto, relates the order parameter r to the gain K and gives the threshold $K_c = 2/(\pi g(0))$ (Theorem 13.5, Chapter 13).

Action (of a configuration)

The sum $\sum_i h(x_i, x_{i+1})$ of the values of the generating function along a segment of a config-

uration; the orbits of the twist map are the stationary points of the action (Definition 16.5, Chapter 16).

Free action

Action of a group in which no element other than the identity has fixed points; on the circle and on the line it forces a very rigid structure (Hölder's theorem) (Definition 15.5, Chapter 15).

Minimal average action

Mather's function $\beta(p/q) = \frac{1}{q} \min W_{p,q}$, the average action of minimizing orbits of type (p, q) ; it extends to all real ω (Definition 16.11, Chapter 16).

Basin of infinity

The set $A_c(\infty)$ of points whose orbit under $g_c(z) = z^2 + c$ tends to ∞ ; it is the complement of the filled Julia set K_c (Definition 12.2, Chapter 12).

Immediate basin

The connected component of the basin of an attracting cycle that contains the points of the cycle; it always contains a critical point (Definition 12.7, Chapter 12).

Dead band

The residual error interval $|e_\infty| \leq \Delta/(2K)$ within which a proportional control with quantizer step Δ comes to rest without reaching zero (Theorem 14.4, Chapter 14).

Berry phase

The geometric phase $\gamma_{\text{Berry}} = -\frac{1}{2}\Omega$ accumulated by parallel transport along a closed circuit, with Ω the subtended solid angle; it is the holonomy of the canonical connection of the Hopf fibration (Chapter 2).

Local bifurcation

Qualitative change in the phase portrait of a family of maps as the parameter varies, in a neighborhood of a non-hyperbolic fixed point (Definition 11.2, Chapter 11).

Birkhoff configuration

Configuration x such that, for every $(s, t) \in \mathbb{Z}^2$, the translate $\tau_{s,t}x$ is $\leq x$ or $\geq x$; these configurations have a well-defined rotation number (Definition 16.9, Chapter 16).

Birkhoff periodic orbits

The orbits of type (p, q) that minimize the periodic action $W_{p,q}$; they exist for every (p, q) and the minimizers are strictly ordered (Theorems 16.4 and 16.5, Chapter 16).

Birkhoff sums

The sums $S_n\phi = \sum_{j < n} \phi \circ T^j$ along an orbit; their boundedness is the Gottschalk–Hedlund criterion for solving the homological equation (Definition 6.4, Chapter 6).

Birkhoff ergodic theorem

The time averages $\frac{1}{n}S_n\phi$ of an L^1 function converge almost everywhere to an invariant function, which in the ergodic case is the spatial mean (Theorem 9.7, Chapter 9).

Borsuk–Ulam theorem

In dimension 2: every continuous map $S^2 \rightarrow \mathbb{R}^2$ identifies a pair of antipodes; equivalent to the fact that odd maps $S^1 \rightarrow S^1$ have odd degree (Theorem 1.7, Chapter 1).

Böttcher coordinate

The coordinate ϕ_c near ∞ that conjugates g_c to the map $w \mapsto w^2$; it extends to the whole basin of infinity when K_c is connected (Theorems 12.4 and 12.5, Chapter 12).

Bowen–Dinaburg definition

Definition of topological entropy via the maximal number $N(n, \varepsilon)$ of (n, ε) -separated points:

$h_{\text{top}}(f) = \lim_{\varepsilon \rightarrow 0} \limsup_n \frac{1}{n} \log N(n, \varepsilon)$ (Definition 8.2, Chapter 8).

Brouwer's theorem

Every continuous map of the closed disk into itself has a fixed point; in the volume it is proved twice, via degree and via Sperner's lemma (Theorems 1.13 and 1.15, Chapter 1).

Bryuno number

Irrational number with $B(\rho) = \sum_n \frac{\log q_{n+1}}{q_n} < \infty$; an arithmetic condition weaker than the diophantine one, optimal for holomorphic linearization (Definitions 6.2 and 12.13, Chapters 6 and 12).

p/q bulb

Hyperbolic component of $\mathcal{M}$ attached to the main cardioid at the root point $c_{p/q} = \frac{e^{2\pi i p/q}}{2} - \frac{e^{4\pi i p/q}}{4}$ (Definition 12.8, Chapter 12).

BV (bounded variation)

Space of functions whose total variation $\text{Var}(g)$ is finite; it is the Banach space on which the transfer operator has a spectral gap (Definition 9.5, Chapter 9).

Cantor set

An Aubry–Mather set of irrational rotation number that is a Cantor set instead of an invariant curve: what remains of a KAM torus after it breaks up (Theorem 16.9, Chapter 16).

Devaney chaos

A topologically transitive system, with dense periodic points and sensitive dependence on initial conditions; for interval maps the first two conditions imply the third (Definition 7.4, Chapter 7).

Li–Yorke chaos

Existence of an uncountably infinite scrambled set: pairs of points that infinitely often come close together and infinitely often separate. A notion logically independent of Devaney chaos (Definition 7.6, Chapter 7).

Main cardioid

The curve $\mathfrak{c}(\lambda) = \frac{\lambda}{2} - \frac{\lambda^2}{4}$, $|\lambda| < 1$, that bounds the component of $\mathcal{M}$ with an attracting fixed point (Theorem 12.7, Chapter 12).

Chern (first class of)

Integer invariant $\frac{1}{2\pi} \iint_{S^2} F = \pm 1$ of the Hopf fibration, obtained by integrating the curvature of the canonical connection (Theorem 2.12, Chapter 2).

Chirikov standard map

The twist map $y' = y + \frac{k}{2\pi} \sin(2\pi x)$, $x' = x + y'$, the canonical example, with generating function $h_k = \frac{1}{2}(x' - x)^2 - \frac{k}{(2\pi)^2} \cos(2\pi x)$ (Definition 16.6, Chapter 16).

Euler class

The integer $e(\varrho)$ defined by $\prod_i [A_i, B_i] = T^{e(\varrho)}$ for a representation ϱ of a surface group in $\text{Homeo}_+(\mathbb{T})$; it vanishes if and only if ϱ lifts (Definition 15.9, Chapter 15).

Admissible generating class

The hypotheses (H1)–(H4) on h : diagonal periodicity, uniform torsion $\partial_1 \partial_2 h \leq -\delta$, coercivity, bounded pure second derivatives (Definition 16.4, Chapter 16).

Clutching function

The gluing function of a bundle over S^1 (or over S^2) built from two trivializations: its homotopy class classifies the bundle (Theorem 2.6, Chapter 2).

Coboundary

A function of the form $v \circ T - v$ with v continuous; the homological equation asks precisely for

a given φ to be written as a coboundary (Definition 6.4, Chapter 6).

Euler cocycle

The cocycle $c(f, g) \in \{0, 1\}$ defined by $F_f \circ F_g = T^{c(f, g)} F_{fg}$ using normalized lifts; it represents the central extension in cohomology (Definition 15.10, Chapter 15).

Transition cocycle

The family g_{ij} of trivialization changes of a bundle, with $g_{ii} = I$ and $g_{ij}g_{jk} = g_{ik}$; it determines the bundle up to isomorphism (Definition 2.4, Chapter 2).

Stolen necklace

Combinatorial corollary of Borsuk–Ulam: two thieves can divide a necklace with two types of beads evenly with at most two cuts (Theorem 1.9, Chapter 1).

Topological conjugacy

A homeomorphism h with $h \circ f = g \circ h$; it carries all qualitative dynamical properties from one system to the other (Definition 7.3, Chapter 7).

Canonical connection

The 1-form $\omega = \text{Im}(\bar{z}_1 dz_1 + \bar{z}_2 dz_2)$ on S^3 , which makes horizontal a distribution transverse to the Hopf fibers (Definition 2.8, Chapter 2).

Convergents

The truncated fractions p_n/q_n of the continued fraction expansion, with $p_n = a_n p_{n-1} + p_{n-2}$ and $q_n = a_n q_{n-1} + q_{n-2}$; they are the best rational approximations (Definition 3.2, Chapter 3).

Correlation

The quantity $C_n(u, v) = \int (u \circ f^n) v \, d\mu - \int u \, d\mu \int v \, d\mu$; its decay measures how quickly the system “forgets” (Definition 10.2, Chapter 10).

Cremer numbers

Irrationals ϑ with $\liminf_n \omega_n^{1/2^n} = 0$, that is, with small divisors small enough to prevent linearization (Definition 12.12, Chapter 12).

Cremer’s theorem

For a Cremer multiplier the indifferent fixed point is not linearizable: no Siegel disk exists (Theorem 12.11, Chapter 12).

Escape criterion

If an orbit of g_c exceeds $\max(|c|, 2)$ in modulus then it diverges, with doubly exponential growth; it gives the numerical membership test for $\mathcal{M}$ (Theorem 12.1, Chapter 12).

Lifting criterion

A condition on π_1 under which a continuous map lifts to a covering space; in the case of the circle it is the basic tool for defining the degree (Theorem 1.1, Chapter 1).

De Moivre–Laplace (local form)

Explicit Gaussian approximation of the binomial coefficients $\binom{n}{k} 2^{-n}$; it gives the central limit theorem for the dyadic indicator without abstract tools (Theorem 10.6, Chapter 10).

Denjoy counterexample

A C^1 diffeomorphism of the circle with irrational rotation number and a Cantor minimal set, built by inserting intervals of lengths $\ell_n = c/(|n| + 2)^2$ (Theorem 5.4, Chapter 5).

Denjoy’s inequality

If $V(f) = \text{Var}_{\mathbb{T}}(\log f') < \infty$, then $(f^{q_n})'(x) (f^{-q_n})'(x) \geq e^{-V(f)}$: controlled distortion along the return times (Theorem 5.1, Chapter 5).

Denjoy's theorem

A diffeomorphism of the circle of class $C^{1+\text{bv}}$ with irrational rotation number is minimal, hence conjugate to the corresponding rotation (Theorem 5.2, Chapter 5).

Denjoy–Koksma inequality

The estimate $|\sum_{j < q} \psi(f^j x) - q \int \psi d\mu| \leq \text{Var}(\psi)$ at the times q given by the denominators of the convergents (Theorem 5.3, Chapter 5).

Schwarzian derivative

$Sf = \frac{f'''}{f'} - \frac{3}{2}(\frac{f''}{f'})^2$; it is negative everywhere for the logistic family, which bounds the number of attracting cycles (Chapter 11).

Bifurcation diagram

The set $B = \{(r, y) : y \in A(r)\}$ of the ω -limits of the critical point as the parameter varies; the “fern figure” of the logistic map (Definition 11.4, Chapter 11).

Multiplicative diophantine condition

For $\lambda = e^{2\pi i \vartheta}$: there exist $C > 0$ and $\mu \geq 1$ with $\omega_m = |\lambda^m - 1| \geq Cm^{-\mu}$; this is the form used in the proof of Siegel's theorem (Definition 12.10, Chapter 12).

Diophantine number

Irrational α with $|\alpha - p/q| \geq Kq^{-2-\tau}$ for every rational; diophantine numbers have full measure and are the “good” rotation numbers for KAM (Definition 3.4, Chapter 3).

Siegel disk

A Fatou component on which the dynamics is conjugate to an irrational rotation of the disk; it exists when the multiplier is diophantine (or Bryuno) (Definition 12.11, Chapter 12).

Discrepancy

$D_N = \sup_b |\frac{1}{N} \#\{x_n < b\} - b|$: a quantitative measure of how far a sequence departs from equidistribution (Definition 3.7, Chapter 3).

Level domains

The open sets $U_s = \{G_c > s\}$ of the potential, with $g_c(U_s) = U_{2s}$; they are the scaffolding for the construction of the global Böttcher coordinate (Definition 12.6, Chapter 12).

Dörfler–Bullo theorem

On a connected network, an equilibrium with phase differences less than $\pi/2$ on every edge is exponentially stable, with a rate given by the second eigenvalue of the weighted Laplacian (Theorem 13.7, Chapter 13).

Vernier (dual-track) encoder

An encoder with two tracks of m and n periods: the integer $J = mK_n - nK_m$ reconstructs the absolute turn provided the phase error satisfies $\eta < \pi/(m+n)$ (Theorem 14.3, Chapter 14).

Metric entropy

Kolmogorov–Sinai entropy $h_\mu(S) = \sup_{\mathcal{P}} \lim_n \frac{1}{n} H(\mathcal{P}^n)$: the rate of information production with respect to an invariant measure (Definition 9.7, Chapter 9).

Topological entropy

The exponential growth rate of the number of distinguishable orbits in n steps at resolution ε ; it equals $\log m$ for E_m and 0 for rotations (Definition 8.2, Chapter 8).

Homological equation

The equation $\psi \circ R_\alpha - \psi = \varphi$, solved in Fourier by $\hat{\psi}_k = \hat{\varphi}_k / (e^{2\pi i k \rho} - 1)$; the denominator is the “small divisor” (Definition 6.1, Chapter 6).

Equidistribution

A sequence is equidistributed if the visit frequencies of every arc converge to its length; for irrational rotations Weyl's criterion holds (Theorem 3.9, Chapter 3).

Ergodicity

An invariant measure is ergodic if every invariant set has measure 0 or full measure; equivalently, every invariant function is constant almost everywhere (Definition 9.2, Chapter 9).

Exact symplectic map

A map of the cylinder that preserves area and the class of the 1-form $y dx$; it admits a generating function (Definition 16.3, Chapter 16).

Lyapunov exponent

$\lambda(\theta) = \lim_n \frac{1}{n} \log |(f^n)'(\theta)|$: the average rate of divergence of nearby orbits; it equals $\log m$ for E_m and $\log 2$ almost everywhere for f_4 (Definition 8.6, Chapter 8).

Pencil of conics

The family $\lambda f_1 + \mu f_2$ generated by two conics; circles are exactly the conics of the pencil passing through the two circular points at infinity (Definition 2.2, Chapter 2).

Fatou's theorem

Every attracting cycle attracts at least one critical point; for the quadratic family this implies there is at most one attracting cycle (Theorem 12.6, Chapter 12).

Factor

A system (Y, g) that is the image of (X, f) under a continuous surjective semiconjugacy; entropy does not increase when passing to a factor (Definition 8.3, Chapter 8).

Feigenbaum constants

$\delta = 4.66920160910299\dots$ (ratio of successive intervals of the cascade) and $\alpha = -2.5029078750959\dots$ (scaling factor of the renormalization operator) (Chapter 11).

Fejér's theorem

The Cesàro means of the partial Fourier sums of a continuous function converge uniformly; it is the analytic tool behind Weyl's criterion (Theorem 3.10, Chapter 3).

Fekete's lemma

For a subadditive sequence, $\lim_n a_n/n = \inf_n a_n/n$; it guarantees the existence of limits such as the rotation number and entropy (Chapter 4).

Bundle

A locally trivial map $\pi : E \rightarrow B$: every point of the base has a neighborhood U with $\pi^{-1}(U) \cong U \times F$ compatibly with π (Definition 2.3, Chapter 2).

Principal bundle

A bundle with fiber a group G acting freely and transitively on the fibers; over S^1 it is classified by the monodromy in $\pi_0(G)$ modulo conjugation (Definition 2.4 and Theorem 2.8, Chapter 2).

Unit tangent bundle

The bundle of tangent vectors of norm 1; for the circle it is trivial, $T^1 S^1 \cong S^1 \times \{\pm 1\}$ (Definition 2.5 and Theorem 2.2, Chapter 2).

Tautological bundle

The line bundle over $\mathbb{RP}^1$ whose fiber over a line is the line itself; it coincides with the Möbius band (Definition 2.6, Chapter 2).

Vector bundle

A bundle whose fibers are vector spaces and whose trivializations are linear on the fibers; over

S^1 there are exactly two of each rank (Definition 2.4 and Theorem 2.7, Chapter 2).

Hopf fibration

The map $S^3 \rightarrow S^2$ with fibers that are pairwise linked circles; it is the principal S^1 -bundle of first Chern class ± 1 (Theorem 2.11, Chapter 2).

Flow on the circle

Equation $\theta' = F(\theta)$ with F periodic: it either has equilibria, or is periodic with period $T = \oint d\theta/F(\theta)$ (Definition 13.1, Chapter 13).

Foucault pendulum

Physical application of holonomy: the plane of oscillation rotates by $2\pi \sin \lambda_0$ per sidereal day at latitude λ_0 (Chapter 2).

Continued fraction

Expansion $\alpha = [a_0; a_1, a_2, \dots]$, unique for irrationals; its partial quotients control the quality of rational approximations (Theorem 3.2, Chapter 3).

Green's function

The potential $G_c(z) = \lim_n 2^{-n} \log^+ |g_c^n(z)|$, zero on K_c and harmonic outside, with $G_c \circ g_c = 2G_c$ (Definition 12.5, Chapter 12).

Generating function

A function $h(x, x')$ with $y = -\partial_1 h$, $y' = \partial_2 h$ that encodes an exact symplectic twist map; the torsion condition becomes $\partial_1 \partial_2 h < 0$ (Theorem 16.1, Chapter 16).

Spectral gap

The separation between eigenvalue 1 and the rest of the spectrum of the transfer operator on BV or $C^{0,\alpha}$; it is the source of the exponential decay of correlations (Theorem 10.3, Chapter 10).

Gauss–Bonnet (local version)

The rotation $\Delta\varphi$ of a vector parallel transported along the boundary of a region equals the integral of the curvature $\iint_R K dA$ (Theorem 2.10, Chapter 2).

Gordin decomposition

The decomposition $u = m + (g \circ f - g)$ with m a martingale difference; it reduces the dynamical central limit theorem to the case of martingales (Theorem 10.7, Chapter 10).

Gottschalk–Hedlund lemma

For a minimal dynamics, ϕ is a continuous coboundary if and only if its Birkhoff sums are bounded at one point (Theorem 6.4, Chapter 6).

Degree

The integer $\deg f = w(f \circ E)$ associated with $f : S^1 \rightarrow S^1$: it is multiplicative under composition, additive under pointwise product, and invariant under homotopy (Chapter 1).

Greene's criterion

A numerical criterion linking the breakup of an invariant curve to the stability of nearby periodic orbits; for the standard map it gives $k_c \approx 0.9716$ (overview, Chapter 16).

Surface group

The group $\Gamma_g = \langle a_i, b_i \mid \prod_i [a_i, b_i] = 1 \rangle$, the π_1 of the closed orientable surface of genus g (Definition 15.8, Chapter 15).

Hölder's theorem

A group acting freely on the line by increasing homeomorphisms is abelian and embeds in $(\mathbb{R}, +)$ preserving the order (Theorem 15.2, Chapter 15).

$\text{Homeo}_+(\mathbb{T})$

The group of orientation-preserving homeomorphisms of the circle; its universal cover $\widetilde{\text{Homeo}}$ is torsion-free (Definitions 15.1 and 15.2, Chapter 15).

Hopf's maximal ergodic theorem

If $E = \{\sup_n S_n \varphi > 0\}$, then $\int_E \varphi d\mu \geq 0$; this is the central lemma in the proof of Birkhoff's theorem (Theorem 9.6, Chapter 9).

Hurewicz's theorem for S^1

The abelianization of $\pi_1(S^1)$ coincides with $H_1(S^1) \cong \mathbb{Z}$: homotopy degree and homology degree are the same integer (Theorem 1.4, Chapter 1).

Hurwitz's theorem

For every irrational there exist infinitely many p/q with $|\alpha - p/q| < 1/(\sqrt{5}q^2)$, and the constant $1/\sqrt{5}$ is optimal (golden) (Theorem 3.5, Chapter 3).

Crossing

A sign change in the difference between two configurations; Aubry's lemma limits to one the number of crossings between two minimal configurations (Definition 16.7, Chapter 16).

Dyadic indicator

The function $\varphi = \pm 1$ given by the sign of the first binary digit; its Birkhoff sums under the doubling map are exactly binomial (Definition 10.3, Chapter 10).

Index of an isolated zero

The degree of the map $V/|V|$ on a small circle around the zero; it is independent of the radius and of the chart (Definition 2.7, Chapter 2).

Julia set

The boundary $J_c = \partial K_c$: the locus of chaos, where the family of iterates is not normal (Definition 12.2, Chapter 12).

Filled Julia set

The set K_c of points with bounded orbit under $g_c(z) = z^2 + c$; it is connected if $0 \in K_c$, a Cantor set otherwise (Definition 12.2, Chapter 12).

Mandelbrot set

$\mathcal{M} = \{c : 0 \in K_c\}$, the connectedness locus of the quadratic family; it is compact and contained in the disk of radius 2 (Definition 12.3 and Theorem 12.2, Chapter 12).

Minimal set

A nonempty closed invariant set with no proper subsets sharing the same properties; for groups on the circle a trichotomy holds (finite orbit, Cantor set, all of $\mathbb{T}$) (Definition 15.7, Chapter 15).

(n, ε) -separated and spanning sets

Sets of points that at time n are distinguished from one another (or that approximate all orbits) at resolution ε ; they count orbits for entropy (Definition 8.1, Chapter 8).

Itoh's theorem

Phase unwrapping reconstructs the continuous phase if and only if every true increment falls in $(-\pi, \pi]$; in frequency terms, $f_s > \omega_{\max}/\pi$ (Theorem 14.2, Chapter 14).

Kneading sequence

The itinerary of the critical value of a unimodal map, in symbols L, C, R ; for the full logistic map it equals CRL^∞ (Definition 7.9, Chapter 7).

Koenigs linearization

Near a fixed point with multiplier $0 < |\lambda| < 1$ there exists a coordinate φ with $\varphi \circ f = \lambda\varphi$,

obtained as the limit of f^n/λ^n (Theorem 12.3, Chapter 12).

Koksma's inequality

$|\frac{1}{N} \sum f(x_n) - \int f| \leq V(f) D_N$: it relates the integration error to the discrepancy and to the variation (Theorem 3.12, Chapter 3).

Krylov–Bogolyubov theorem

Every continuous map of a compact metric space admits at least one invariant probability measure (Theorem 9.1, Chapter 9).

Kuramoto model

$\theta'_i = \omega_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i)$: N phase oscillators coupled in a mean field; with identical frequencies it is a gradient flow (Definition 13.3, Chapter 13).

Kuramoto order parameter

The complex number $re^{i\psi} = \frac{1}{N} \sum_j e^{i\theta_j}$; $r = 0$ is incoherence, $r = 1$ perfect synchronization (Chapter 13).

Lap number

The number $\text{lap}(f)$ of monotone branches of an interval map; its growth rate $\ell(f)$ is a lower bound for the topological entropy (Definition 8.5, Chapter 8).

LaSalle's principle

For a flow with a Lyapunov function, bounded trajectories converge to the maximal invariant set in $\{\dot{V} = 0\}$; applied to the Kuramoto gradient it gives convergence to the equilibria (Theorem 13.3, Chapter 13).

Lasota–Yorke estimate

$\text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g)$ for the tent map: contraction of the variation, from which existence, uniqueness, and convergence rate of the invariant density follow (Theorem 9.14, Chapter 9).

Li–Yorke theorem

Period three implies every period, and implies the existence of an uncountably infinite scrambled set (Theorem 7.6, Chapter 7).

Liouville number

An irrational approximable by rationals better than any power: the arithmetic opposite of diophantine numbers; for these the homological equation may fail to be solvable (Definition 3.5 and Theorem 6.3, Chapters 3 and 6).

Liouville's approximation theorem

An algebraic irrational of degree d satisfies $|\alpha - p/q| \geq cq^{-d}$; it provides the first explicit transcendental numbers (Theorem 3.7, Chapter 3).

Logistic family

$f_r(x) = rx(1-x)$ on $[0, 1]$, with $r \in [0, 4]$: the standard laboratory for bifurcations, the cascade, and universality (Definition 11.1, Chapter 11).

Odd map

A map with $u(-x) = -u(x)$ (antipodal); every odd map $S^1 \rightarrow S^1$ has odd degree, from which Borsuk–Ulam follows (Definition 1.1 and Theorem 1.6, Chapter 1).

m -expanding map

$E_m(\theta) = m\theta \bmod 2\pi$, with $m \geq 2$: every point has m preimages, and the entropy equals $\log m$ (Definition 7.1, Chapter 7).

Doubling map

The case $m = 2$, $D(\theta) = 2\theta \bmod 2\pi$: the simplest model of chaos on the circle, semiconjugate

to the shift on two symbols (Definition 7.1, Chapter 7).

Tent map

$T(x) = 2x$ on $[0, 1/2]$, $T(x) = 2 - 2x$ on $[1/2, 1]$; it is conjugate to the full logistic map f_4 via $k(x) = \sin^2(\pi x/2)$ (Definition 7.8, Chapter 7).

Monotone twist map

A map of the annulus whose lift satisfies $\partial X/\partial y \geq \delta > 0$: the rotation increases with the radius (Definition 16.2, Chapter 16).

Milnor–Wood inequality

For a representation of Γ_g in $\text{Homeo}_+(\mathbb{T})$, $|e(\varrho)| \leq |\chi(\Sigma_g)| = 2g - 2$ (Theorem 15.5, Chapter 15).

Invariant measure

A measure with $\mu(S^{-1}A) = \mu(A)$ for every measurable A ; equivalently $\int \varphi \circ S \, d\mu = \int \varphi \, d\mu$ for every continuous φ (Definition 9.1, Chapter 9).

Metric mixing

The correlations $C_n(f, g)$ tend to zero: a property strictly stronger than ergodicity (Definition 9.3, Chapter 9).

Topological mixing

For every pair of open sets there exists N such that $f^n(U) \cap V \neq \emptyset$ for every $n \geq N$; for E_m one has the explicit estimate $N(\delta) = \lceil \log_m(1/\delta) \rceil$ (Theorem 7.3, Chapter 7).

Möbius band

The nontrivial line bundle over S^1 ; it coincides with the tautological bundle over $\mathbb{RP}^1$ and has cocycle ± 1 (Theorems 2.3 and 2.4, Chapter 2).

Multiplier

The derivative $f'(z_0)$ at a fixed point: $|\lambda| < 1$ attracting, $|\lambda| > 1$ repelling, $|\lambda| = 1$ indifferent (parabolic if λ is a root of unity) (Definition 12.4, Chapter 12).

Monodromy

The element of the structure group obtained by traversing the generator of $\pi_1(S^1)$; it classifies principal bundles over S^1 up to conjugation (Theorem 2.8, Chapter 2).

Noble numbers

Irrationals whose continued fraction has a tail of 1s; they inherit from $1/\varphi = 0.618034\dots$ the optimal approximation constant $1/\sqrt{5}$ (Definition 3.6, Chapter 3).

Distance-to-nearest-integer norm

$\|t\| = \min_{n \in \mathbb{Z}} |t - n|$, the distance to the nearest integer; it measures the smallness of the divisors $|e^{2\pi i t} - 1|$ (Definition 3.3, Chapter 3).

Winding number

$w(\gamma) = \frac{\tilde{\gamma}(1) - \tilde{\gamma}(0)}{2\pi}$, how many times a loop winds around the origin; it is the degree expressed on the covering (Chapter 1).

Rotation number

$\rho(f) = \lim_n \frac{F^n(\theta) - \theta}{2\pi n}$, independent of the point and invariant under conjugation: the average rotation of a circle homeomorphism (Definition 4.2 and Theorem 4.1, Chapter 4).

Normal number

A real number whose digits in some base occur with the expected frequencies; almost every θ is normal in base 2, a consequence of Birkhoff's theorem applied to the doubling map (Chapter 9).

Regular homotopy

A homotopy between regular curves that stays regular at every instant; Whitney–Graustein

classifies it by the degree of the tangent (Definition 1.2, Chapter 1).

Renormalization operator

$(\mathcal{R}g)(x) = \alpha g(g(x/\alpha))$: its fixed point explains the universality of the Feigenbaum constants (Definition 11.3, Chapter 11).

Transfer operator

The Perron–Frobenius operator $\mathcal{L}_m$, dual to composition: $\int (g \circ E_m)w = \int g(\mathcal{L}_m w)$; on Fourier coefficients it acts as $(\widehat{\mathcal{L}_m g})_k = \widehat{g}_{mk}$ (Definition 10.1, Chapter 10).

Sharkovskii ordering

The ordering $3 \triangleleft 5 \triangleleft \cdots \triangleleft 2^2 \triangleleft 2 \triangleleft 1$ of the positive integers governing the coexistence of periods (Definition 7.7, Chapter 7).

Left-invariant order

A total order $\prec$ on a group with $g \prec h \Rightarrow kg \prec kh$; subgroups of $\text{Homeo}_+(\mathbb{R})$ always admit one (Definition 15.6, Chapter 15).

Ott–Antonsen reduction

An ansatz that reduces the Kuramoto model to low dimension; for a Lorentzian g it gives $\dot{r} = -\Delta r + \frac{K}{2}r(1-r^2)$ (Theorem 13.6, Chapter 13).

Oxtoby’s criterion

A system is uniquely ergodic if and only if, for every continuous function, the Birkhoff averages converge uniformly to a constant (Theorem 9.9, Chapter 9).

Hairy ball theorem

Every continuous tangent vector field on S^2 vanishes at some point; in the volume it is proved with Milnor’s volume argument (Theorem 1.14, Chapter 1).

Pancake theorem

Two bounded regions of the plane can be simultaneously bisected by a single line; the two-dimensional version of ham sandwich (Theorem 1.8, Chapter 1).

Dyadic partition

The partition $\{[0, \pi), [\pi, 2\pi)\}$ of $\mathbb{T}$: it is generating for the doubling map, and gives $h_\mu(D) = \log 2$ (Definition 8.7, Chapter 8).

PLL (phase-locked loop)

A circuit that tracks the phase of a reference; at first order its dynamics is the Adler equation and the three bands coincide with $(-K, K)$ (Theorem 14.1, Chapter 14).

Poincaré classification

If $\rho(f) = p/q$ there exist periodic orbits of exact period q ; if $\rho(f)$ is irrational, f is semiconjugate to the rotation $R_{2\pi\rho}$ (Theorems 4.3 and 4.4, Chapter 4).

Poincaré–Hopf theorem

The sum of the indices of a vector field on a closed orientable surface equals $\chi(\Sigma) = 2 - 2g$ (Theorem 2.9, Chapter 2).

Discrete variational principle

The orbits of a twist map are exactly the stationary configurations of the action: $\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0$ (Theorem 16.2, Chapter 16).

$\text{PSL}(2, \mathbb{R})$ (action on $\mathbb{RP}^1$)

Action by real Möbius transformations on the projective line, which is a circle; it is the model example of an action with maximal Euler class (Definition 15.11, Chapter 15).

Circular points

The points $I = [1:i:0]$ and $J = [1:-i:0]$ at infinity of the complex projective plane: a conic is a circle if and only if it passes through them (Theorem 2.1, Chapter 2).

Quadratic family

The family $g_c(z) = z^2 + c$, $c \in \mathbb{C}$: every quadratic polynomial is affinely conjugate to some g_c (Definition 12.1, Chapter 12).

Homogeneous quasimorphism

A function ϕ with $|\phi(gh) - \phi(g) - \phi(h)|$ bounded and $\phi(g^n) = n\phi(g)$; the rotation number on $\widehat{\text{Homeo}}$ is its prime example (Definition 15.4 and Theorem 15.1, Chapter 15).

Period doubling (flip)

A bifurcation in which a fixed point with multiplier -1 loses stability and a 2-cycle is born; for the logistic map it occurs at $r = 3$ (Theorems 11.5 and 11.1, Chapter 11).

External rays

The curves $R_t = \Phi_c^{-1}(\{re^{2\pi it} : r > 1\})$ in the basin of infinity; the dynamics on their angles is the doubling map $t \mapsto 2t$ (Definition 12.9 and Theorem 12.9, Chapter 12).

Resolver-to-digital converter

A type-II phase observer ($\dot{\theta} = K_1 e + K_2 w$, $\dot{w} = e$): zero error at constant velocity, error α/K_2 at constant acceleration (Chapter 14).

Interval covering

The relation $J \rightarrow K$ if $f(J) \supseteq K$; a cyclic chain of coverings forces a periodic point (Definition 7.5, Chapter 7).

Coverings of S^1

The connected coverings of the circle are $\mathbb{R} \rightarrow S^1$ and $z \mapsto z^n$ on S^1 , one for each subgroup of $\pi_1(S^1) \cong \mathbb{Z}$ (Theorem 1.2, Chapter 1).

Rokhlin's formula

For an expanding map, $h_\mu(f) = \int \log |f'| d\mu$; for the doubling map with Lebesgue measure this gives $\log 2 = h_{\text{top}}(D)$ (Theorem 8.12, Chapter 8).

Rotation map

$R_\alpha(\theta) = \theta + \alpha \pmod{2\pi}$: periodic if $\alpha/2\pi$ is rational, with all orbits dense and uniquely ergodic otherwise (Definition 3.1 and Theorem 3.1, Chapter 3).

Rouché (topological version)

If $|v| < |u|$ pointwise then $w(u+v) = w(u)$: a dominated perturbation does not change the winding number (Theorem 1.5, Chapter 1).

Devil's staircase

The graph $\alpha \mapsto \rho(\alpha, \varepsilon)$: continuous, non-decreasing, and constant on a dense union of intervals (the rational tongues) (Theorem 4.8, Chapter 4).

Saddle-node bifurcation

Simultaneous birth of two fixed points, one stable and one unstable; in the logistic map it appears at the opening of the period-3 window at $r = 1 + 2\sqrt{2}$ (Theorem 11.3, Chapter 11).

Semiconjugacy

A continuous surjective map h with $h \circ f = g \circ h$, not necessarily injective; it links the doubling map to the shift and f to the rotation (Definition 7.3, Chapter 7).

Sensitivity to initial conditions

Existence of a minimal separation reached by initially nearby points; for E_m it is obtained with

an explicit minimal n (Theorem 7.1, Chapter 7).

Sharkovskii's theorem

If a continuous interval map has an orbit of period n and $n \triangleleft m$, then it also has an orbit of period m ; the ordering is optimal (Theorems 7.7 and 7.8, Chapter 7).

Shift

The map $\sigma \mapsto (\sigma_1, \sigma_2, \dots)$ on the space $\Sigma_2 = \{0, 1\}^{\mathbb{N}}$ of binary sequences; the symbolic model of the doubling map (Definition 7.2, Chapter 7).

Bernoulli shift

The shift on k symbols, with topological entropy $\log k$; the prototype of a chaotic system with known entropy (Definition 8.4 and Theorem 8.5, Chapter 8).

Siegel's theorem

If the multiplier $\lambda = e^{2\pi i \vartheta}$ is diophantine, the indifferent fixed point is linearizable: a Siegel disk exists (Theorem 12.10, Chapter 12).

Synchronized state

The configuration with all phases equal: the unique global minimum of the Kuramoto potential with identical oscillators, and exponentially stable (Theorem 13.4, Chapter 13).

Lift

An increasing homeomorphism $F : \mathbb{R} \rightarrow \mathbb{R}$ with $F(\theta + 2\pi) = F(\theta) + 2\pi$ that projects onto f ; defined up to integer translations (Definition 4.1, Chapter 4).

Sperner's lemma

Every triangulation with an admissible coloring has an odd number of complete triangles; it gives a combinatorial proof of Brouwer's theorem (Lemma 1.2, Chapter 1).

Normalized displacement

The function $\varphi_F(\theta) = \frac{F(\theta) - \theta}{2\pi}$ and its extremes $a(F) = \min \varphi_F$, $b(F) = \max \varphi_F$, with $b(F) - a(F) < 1$ (Definition 15.3, Chapter 15).

Complex strip

The complex neighborhood $\{| \operatorname{Im} z | < b\}$ of the circle with the sup norm; it is the setting for the analytic estimates of the Newton scheme (Definition 6.3, Chapter 6).

Stroboscope

Sampling at frequency f_s a rotational motion of frequency f one observes $f_{\text{app}} = f_s \cdot \rho(f/f_s)$: aliasing as dynamics on the circle (Chapter 14).

Locking time

The constant $\tau = 1/\sqrt{K^2 - \Delta\omega^2}$ with which a first-order PLL converges to the phase equilibrium (Chapter 14).

Dynamical central limit theorem

For regular observables on the doubling map, $S_n u / \sqrt{n}$ converges in distribution to a Gaussian $\mathcal{N}(0, \sigma^2)$ (Theorem 10.8, Chapter 10).

Fundamental theorem of algebra

Every nonconstant complex polynomial has a root; proved via degree with the explicit estimate $R = 1 + \sum |a_i|$ (Theorem 1.12, Chapter 1).

Type (p, q)

A configuration with $x_{i+q} = x_i + p$, that is, a periodic orbit that makes p turns in q steps (Definition 16.8, Chapter 16).

Transcritical bifurcation

An exchange of stability between two branches of fixed points that cross; for the logistic map it occurs at $r = 1$ (Theorem 11.4, Chapter 11).

Periodic trapezoidal rule

The quadrature $T_N(A) = \frac{2\pi}{N} \sum_k A(2\pi k/N)$ for analytic periodic functions, with error $O(e^{-Na_0})$: spectral convergence (Theorem 2.13, Chapter 2).

Three-distance theorem

The points $\{0, \beta, \dots, (N-1)\beta\}$ divide the circle into arcs of at most three distinct lengths, with $\ell_+ + \ell_-$ as the third (Theorem 3.8, Chapter 3).

Trichotomy of minimal sets

For $G \leq \text{Homeo}_+(\mathbb{T})$ the minimal set is a finite orbit, a Cantor set, or all of $\mathbb{T}$ (Theorem 15.4, Chapter 15).

Ulam–von Neumann theorem

The full logistic map f_4 has the absolutely continuous invariant measure $d\nu = \frac{dx}{\pi\sqrt{x(1-x)}}$, which is ergodic (Theorem 9.12, Chapter 9).

Umlaufsatz

For a simple, closed, regular plane curve the index of the tangent equals ± 1 and $\oint \kappa ds = 2\pi$ (Theorem 1.10, Chapter 1).

Unique ergodicity

Existence of a single invariant probability measure; it holds for irrational rotations and not for the doubling map (Definition 9.4 and Theorem 9.10, Chapter 9).

Universality

Independence of the constants δ and α from the particular unimodal family with a quadratic critical point; critical circle maps, instead, have their own class, with $\delta_{\text{GM}} \approx -2.8336$ and $\alpha_{\text{GM}} \approx -1.2886$ (Chapter 11).

Variation of $\log f'$

The quantity $V(f) = \text{Var}_{\mathbb{T}}(\log f')$, finite for $C^{1+\text{bv}}$ diffeomorphisms: it is the exact hypothesis of Denjoy's theorem (Definition 5.2, Chapter 5).

Affine algebraic variety

The set of zeros of a family of polynomials; the circle is the variety defined by $x^2 + y^2 - r^2 = 0$, smooth because the gradient never vanishes on it (Definition 2.1, Chapter 2).

Vogel's model

The phyllotaxis model $r_n = c\sqrt{n}$, $\theta_n = n\gamma$: as γ varies it produces the sunflower spirals, optimal for $\gamma = \gamma^*$ (Definition 3.8, Chapter 3).

von Neumann mean ergodic theorem

The averages $\frac{1}{N} \sum_{n < N} U^n \varphi$ converge in L^2 to the projection onto the invariants; in the ergodic case, to the spatial mean (Theorem 9.8, Chapter 9).

Weyl's criterion

A sequence on $\mathbb{T}$ is equidistributed if and only if $\frac{1}{N} \sum_n e^{2\pi i k x_n} \rightarrow 0$ for every $k \neq 0$ (Theorem 3.9, Chapter 3).

Whitney–Graustein theorem

Two closed regular plane curves are regularly homotopic if and only if they have the same rotation index of the tangent (Theorem 1.11, Chapter 1).

Cheat Sheet

Part A — Coverings and degree (Ch. 1)

- $E(t) = e^{it}$; $w(\gamma) = \frac{1}{2\pi}(\tilde{\gamma}(1) - \tilde{\gamma}(0)) = \frac{1}{2\pi i} \int_0^1 \frac{\gamma'}{\gamma} dt$; $\deg f = w(f \circ E)$
- $\deg(f \circ g) = \deg f \cdot \deg g$; $\deg(fg) = \deg f + \deg g$; $\deg p_n = n$
- Rouché: $|v| < |u| \Rightarrow w(u+v) = w(u)$
- u odd $\Rightarrow \deg u$ odd $\Rightarrow$ Borsuk–Ulam, pancake, necklace
- Umlaufsatz: γ simple, positively oriented $\Rightarrow \rho = 1$, $\oint \kappa ds = 2\pi$
- Whitney–Graustein: $\gamma_0 \sim_{\text{reg}} \gamma_1 \iff \rho(\gamma_0) = \rho(\gamma_1)$
- FTA: roots in $|z| < R$, $R = 1 + \sum_{i < n} |a_i|$

Bundles (Ch. 2)

- Cocycle: $g_{ii} = I$, $g_{ij}g_{ji} = I$, $g_{ij}g_{jk} = g_{ik}$
- $\text{ind}_p(V) = \deg(V(\gamma_\epsilon)/|V(\gamma_\epsilon)|)$; Poincaré–Hopf $\sum_p \text{ind}_p = \chi = 2 - 2g$
- On S^1 : $\text{Vect}_n \leftrightarrow \pi_0(O(n)) \cong \mathbb{Z}/2$ (trivial or Möbius)
- Local Gauss–Bonnet: $\Delta\varphi = \iint_R K dA$; parallel θ_0 : $\Delta\varphi = 2\pi(1 - \cos\theta_0)$
- Foucault: $2\pi \sin\lambda_0$ per sidereal day; Berry $\gamma = -\frac{1}{2}\Omega$
- Hopf: $g_{-+}(\varphi) = e^{i\varphi}$, $\frac{1}{2\pi} \iint_{S^2} F = \pm 1$
- Periodic trapezoidal rule: $|T_N(A) - \oint A| \leq Ce^{-Na_0}$

Part B — Rotations (Ch. 3)

- $R_\alpha(\theta) = \theta + \alpha$; $\alpha/2\pi \in \mathbb{Q} \Rightarrow$ periodic, otherwise dense orbit
- $p_n = a_n p_{n-1} + p_{n-2}$, $q_n = a_n q_{n-1} + q_{n-2}$; $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$
- $\frac{1}{q_n(q_n + q_{n+1})} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$
- Hurwitz: $\left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5}q^2}$ infinitely often (optimal)

- Diophantine: $\|q\rho\| \geq Kq^{-1-\tau}$; noble: tail of 1s; $1/\varphi = 0.618034\dots$

- Weyl: equidistrib. $\iff \frac{1}{N} \sum e^{2\pi i k x_n} \rightarrow 0$ ($k \neq 0$)

- Koksma: $\left| \frac{1}{N} \sum f(x_n) - \int f \right| \leq V(f)D_N$; bounded partial quotients $D_N = O(\log N/N)$

- Three distances: at most $\ell_+, \ell_-, \ell_+ + \ell_-$; golden angle $\gamma^* \approx 137.5077^\circ$

Rotation number (Ch. 4)

- $\rho(f) = \lim_n \frac{F^n(\theta) - \theta}{2\pi n}$ (indep. of θ); $\rho(F^n) = n\rho(F)$; $\rho(HFH^{-1}) = \rho(F)$
- Fekete: $a_{m+n} \leq a_m + a_n \Rightarrow \lim a_n/n = \inf a_n/n$
- $\rho = p/q$ in lowest terms $\iff \exists$ orbit of exact period q ; $\rho \notin \mathbb{Q} \Rightarrow h \circ f = R_{2\pi\rho} \circ h$

- Arnold: $f_{\alpha, \varepsilon}(\theta) = \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta)$, diffeomorphism for $0 < \varepsilon < 1$

- Tongue 0/1: $|\alpha| \leq \varepsilon/2\pi$ (exact); tongue p/q : width $O(\varepsilon^q)$; devil's staircase

Denjoy (Ch. 5)

- $\delta_n = q_n \rho - p_n$, $\text{sign } \delta_n = (-1)^n$, $|\delta_{n-1}|q_n + |\delta_n|q_{n-1} = 1$
- $V(f) = \text{Var}_{\mathbb{T}}(\log f') < \infty \Rightarrow (f^{q_n})'(x)(f^{-q_n})'(x) \geq e^{-V(f)}$
- $f \in C^{1+\text{bv}}$, $\rho \notin \mathbb{Q} \Rightarrow f$ minimal $\sim R_\rho$
- Denjoy–Koksma $(S_q \psi = \sum_{j < q} \psi \circ f^j)$: $|S_q \psi - q \int \psi| \leq \text{Var } \psi$
- C^1 counterexample: $\ell_n = c(|n| + 2)^{-2}$, $S = \frac{\pi^2}{3} - \frac{9}{4}$

KAM on the circle (Ch. 6)

- Homological: $\psi \circ R_\alpha - \psi = \varphi$, $\hat{\psi}_k = \hat{\varphi}_k/(e^{2\pi i k \rho} - 1)$, requires $\hat{\varphi}_0 = 0$
- $4\|x\| \leq |e^{2\pi i x} - 1| \leq 2\pi\|x\|$; diophantine $\Rightarrow |e^{2\pi i k \rho} - 1| \geq 4K|k|^{-1-\tau}$
- $\|\psi\|_{b-\delta} \leq \frac{\Lambda(\tau)\|\varphi\|_b}{2K\delta^{2+\tau}}$, $\Lambda(\tau) = \Gamma(2+\tau) + \left(\frac{1+\tau}{e}\right)^{1+\tau}$
- Newton: $\varepsilon_{j+1} = C_1 \varepsilon_j^2 / \delta_j^\gamma$, $\gamma = 3 + \tau$, $x_j \leq 2^{-2^j}$
- Gottschalk–Hedlund: continuous coboundary $\iff \sup_n |S_n \phi(x_0)| < \infty$
- $\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}$; $B(\rho) = \sum \frac{\log q_{n+1}}{q_n} < \infty$

Part C — Doubling map (Ch. 7)

- $D(\theta) = 2\theta$, $E_m(\theta) = m\theta \bmod 2\pi$; $N(\delta) = \lceil \log_m(1/\delta) \rceil$ (mixing)
- $h(\sigma) = 2\pi \sum_k \sigma_k 2^{-(k+1)}$, $h \circ \text{sh} = D \circ h$
- Devaney = transitive + dense periodic points + sensitive
- $J \rightarrow K \iff f(J) \supseteq K$; cyclic chain $\Rightarrow$ periodic point
- Sharkovskii: $3 \triangleleft 5 \triangleleft \dots \triangleleft 2^2 \triangleleft 2 \triangleleft 1$
- $\#\text{Fix}(D^n) = 2^n - 1$, $\#\text{Fix}(T^n) = \#\text{Fix}(f_4^n) = 2^n$
- $\varphi \circ D = T \circ \varphi$, $k \circ T = f_4 \circ k$ with $k(x) = \sin^2 \frac{\pi x}{2}$; $K(f_4) = \text{CRL}^\infty$

Entropy and Lyapunov exponent (Ch. 8)

- $d_n(x, y) = \max_{i < n} d(f^i x, f^i y)$,
 $h_{\text{top}} = \lim_{\varepsilon} \limsup_n \frac{1}{n} \log N(n, \varepsilon)$
- Invariant under conjugacy; does not increase under factors;
 $h(f^j) = j h(f)$
- $h_{\text{top}}(E_m) = \log m$, $h_{\text{top}}(\sigma_k) = \log k$, $h_{\text{top}}(R_\alpha) = 0$
- $\ell(f) = \lim \frac{1}{n} \log \text{lap}(f^n) \leq h_{\text{top}}(f)$
- period 3 $\Rightarrow h_{\text{top}} \geq \log 2$
- $\lambda(\theta) = \lim \frac{1}{n} \sum_{i < n} \log |f'(f^i \theta)|$;
 $\lambda_{E_m} = \log m$, $\lambda_{f_4} = \log 2$ a.e.
- Rokhlin: $h_{\mu_{\text{Leb}}}(D) = \int \log |D'| d\mu = \log 2$

Ergodic theory (Ch. 9)

- $\mu(S^{-1}A) = \mu(A) \iff \int \varphi \circ S d\mu = \int \varphi d\mu$; Krylov–Bogolyubov: $\exists \mu$
- Ergodic $\iff$ every invariant φ is constant a.e. (Fourier for D and R_α)
- Birkhoff: $\frac{1}{n} S_n \varphi \rightarrow \varphi^*$ a.e., $\int \varphi^* = \int \varphi$; von Neumann: $\rightarrow P\varphi$ in L^2
- Oxtoby: uniquely ergodic $\iff \frac{1}{N} S_N \varphi \rightarrow \text{const. uniformly}$
- R_α uniquely ergodic; D not. Borel: a.e. θ normal in base 2

• f_4 : $d\nu = \frac{dx}{\pi \sqrt{x(1-x)}}$ (Ulam–von Neumann)

• Lasota–Yorke: $\text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g)$, $\|\mathcal{L}^n g_0 - 1\|_\infty \leq 2^{-n} \text{Var}(g_0)$

Transfer operator and CLT (Ch. 10)

- $(\mathcal{L}_m g)(\theta) = \frac{1}{m} \sum_{j < m} g\left(\frac{\theta + 2\pi j}{m}\right)$;
 $(\widehat{\mathcal{L}_m g})_k = \widehat{g}_{mk}$
- Spectral gap on $BV/C^{0,\alpha}$: $|\lambda| \leq \frac{1}{2}$ (resp. $2^{-\alpha}$) away from 1
- $|C_n(u, v)| \leq \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) m^{-n}$
- Gordin: $m = u + g - g \circ D$ with $g = \sum_{k \geq 1} \mathcal{L}^k u$, $\mathcal{L}m = 0$
- $S_n u / \sqrt{n} \rightarrow \mathcal{N}(0, \sigma^2)$, $\sigma^2 = \int m^2 d\mu$

Logistic map (Ch. 11)

- $f_r(x) = rx(1-x)$; $x^* = 1 - 1/r$, $f'_r(x^*) = 2 - r$
- transcritical $r = 1$; flip $r = 3$; saddle-node (period 3) $r = 1 + 2\sqrt{2}$
- $x_\pm = \frac{(r+1) \pm \sqrt{(r-3)(r+1)}}{2r}$, stable for $3 < r < 1 + \sqrt{6} \approx 3.449490$
- $r_\infty \approx 3.569945672$; $\delta = 4.66920160910299 \dots$, $\alpha = -2.5029078750959 \dots$
- $(\mathcal{R}g)(x) = \alpha g(g(x/\alpha))$; $Sf = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'}\right)^2 < 0$

Part D — Julia and Mandelbrot sets (Ch. 12)

- $g_c(z) = z^2 + c$, $K_c = \{\text{bounded orbit}\}$, $J_c = \partial K_c$, $\mathcal{M} = \{c : 0 \in K_c\}$
- Escape: $|g_c^n(z)| > \max(|c|, 2) \Rightarrow$ diverges; $\{|c| \leq \frac{1}{4}\} \subseteq \mathcal{M} \subseteq \{|c| \leq 2\}$
- Koenigs: $\varphi = \lim f^n / \lambda^n$, $\varphi \circ f = \lambda \varphi$
- Böttcher: $\phi_c \circ g_c = \phi_c^2$; $G_c = \log |\Phi_c|$, $G_c \circ g_c = 2G_c$
- $c \in \mathcal{M} \iff K_c$ connected; otherwise Cantor
- Cardioid $c(\lambda) = \frac{\lambda}{2} - \frac{\lambda^2}{4}$; $c_{p/q} = \frac{e^{2\pi i p/q}}{2} - \frac{e^{4\pi i p/q}}{4}$; $c_{1/2} = -\frac{3}{4}$
- Rays: $g_c(R_t) = R_{2t}$; R_t periodic $\iff t = k/(2^q - 1)$
- Siegel (diophantine) / Cremer / Bryuno $B(\vartheta) = \sum \frac{\log q_{k+1}}{q_k} < \infty$

Part E — Kuramoto (Ch. 13)

- $\theta' = F(\theta)$: $F(\theta^*) = 0$, $F'(\theta^*) < 0$ stable; otherwise $T = \oint d\theta / F$
- Adler $\theta' = \Delta\omega - K \sin \theta$: $\theta^* = \arcsin \frac{\Delta\omega}{K}$; beyond it, $T = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}}$
- $re^{i\psi} = \frac{1}{N} \sum_j e^{i\theta_j}$; $\theta'_i = \omega_i + Kr \sin(\psi - \theta_i)$

- Identical: $V = -\frac{NK}{2}r^2$, Jacobian $-L$, eigenvalues $0, -K^{(N-1)}$
- $K_c = \frac{2}{\pi g(0)}$; Lorentzian: $K_c = 2\gamma$, $r = \sqrt{1 - K_c/K}$
- Ott–Antonsen: $\dot{r} = -\Delta r + \frac{K}{2}r(1-r^2)$, $r^* = \sqrt{1 - 2\Delta/K}$

PLL, encoders, control (Ch. 14)

- $L \subseteq P \subseteq H$; first order $H = P = L = (-K, K)$; $\tau = 1/\sqrt{K^2 - \Delta\omega^2}$
- Itoh: correct unwrapping $\iff \Delta\Theta_j \in (-\pi, \pi]$; $f_s > \omega_{\text{max}}/\pi$
- Vernier: $J = mK_n - nK_m$; correct if $\eta < \frac{\pi}{m+n}$
- Type II: $e_\infty(\omega_0 t) = 0$, $e_\infty(\frac{1}{2}\alpha t^2) = \alpha/K_2$
- Dead band: $|e_\infty| \leq \Delta/(2K)$; stroboscope $f_{\text{app}} = f_s \rho(f/f_s)$

Part F — Homeo₊($\mathbb{T}$) (Ch. 15)

- $1 \rightarrow \mathbb{Z} \rightarrow \widetilde{\text{Homeo}} \rightarrow \text{Homeo}_+(\mathbb{T}) \rightarrow 1$, torsion-free
- $\varphi_F = \frac{F - \text{id}}{2\pi}$, $a = \min \varphi_F$, $b = \max \varphi_F$, $b - a < 1$, $a \leq \rho \leq b$
- $|\rho(FG) - \rho(F) - \rho(G)| < 2$; homogeneous quasimorphism
- Hölder: free action $\Rightarrow$ abelian, $\hookrightarrow (\mathbb{R}, +)$
- Minimal sets: finite orbit, Cantor set, or $\mathbb{T}$
- $\prod_i [A_i, B_i] = T^{e(\varrho)}$, $e = 0 \iff$ it lifts; Milnor–Wood $|e(\varrho)| \leq 2g - 2$

Twist maps and Aubry–Mather theory (Ch. 16)

- Twist $\partial X / \partial y \geq \delta > 0$; $y = -\partial_1 h$, $y' = \partial_2 h$, $\partial_1 \partial_2 h \leq -\delta$
- Discrete EL: $\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0$
- Standard map: $y' = y + \frac{k}{2\pi} \sin 2\pi x$, $x' = x + y'$; $\text{tr } DF_k = 2 + k \cos 2\pi x$
- Aubry: $h(a, c) + h(b, d) \leq h(a, d) + h(b, c) - \delta(b-a)(d-c)$; at most one crossing
- $W_{p,q} = \sum_{i < q} h(x_i, x_{i+1})$, $x_q = x_0 + p$; ordered minimizers
- Birkhoff: $|x_{i+n} - x_i - n\rho| \leq 1$, L -Lipschitz graph
- $\forall \omega$ ordered orbit; $\omega \notin \mathbb{Q}$: invariant curve or cantorus
- $\beta(p/q) = \frac{1}{q} \min W_{p,q}$, $\beta(-\omega) = \beta(\omega)$, $\beta(0) = -k/(2\pi)^2$; $k_c \approx 0.9716$

General formulary

This formulary collects in a single place the key formulas and statements of the sixteen chapters, in the order of the volume's six Parts. It is a reference collection, not a summary: every formula is proved (or declared as imported) in the indicated chapter, and the "Chapter formulary" that closes each chapter keeps its context with pointed references to the theorems. The numerical constants are given with the digits actually verified in the text; the point is the decimal separator.

Recurring symbols

Symbol	Meaning
$S^1, \mathbb{T}$	unit circle, seen as a subset of $\mathbb{C}$ or as $\mathbb{R}/\mathbb{Z}$ (or $\mathbb{R}/2\pi\mathbb{Z}$)
$E(t) = e^{it}$	universal covering $\mathbb{R} \rightarrow S^1$
$\tilde{\gamma}, F$	lift of a path, lift of a homeomorphism of the circle
$\deg f, w(\gamma)$	degree of $f : S^1 \rightarrow S^1$, winding number of a loop
$\text{ind}_p(V), \chi(\Sigma)$	index of a vector field at p , Euler characteristic
$\rho(f), \rho(F)$	rotation number of the homeomorphism f , of the lift F
$[a_0; a_1, a_2, \dots], p_n/q_n$	continued fraction and its convergents
$\ x\ $	distance of x from the nearest integer
φ	golden ratio $(1 + \sqrt{5})/2$ (in Chapters 3 and 16)
R_α	rotation by angle α
D, E_m, T, f_r	doubling map, m -expanding map, tent map, logistic map $rx(1-x)$
h_{top}, h_μ	topological entropy, metric entropy (Kolmogorov–Sinai)
$\lambda(\theta)$	Lyapunov exponent
$\mathcal{L}, \mathcal{L}_m$	transfer operator (Frobenius–Perron)
$D_N, \text{Var}(\cdot)$	discrepancy, total variation
$g_c(z) = z^2 + c, K_c, J_c, \mathcal{M}$	quadratic family, filled Julia set, Julia set, Mandelbrot set
$re^{i\psi}$	Kuramoto order parameter
$e(\varrho)$	Euler class of a representation
$h(x, x'), \beta(\omega)$	generating function of a twist map, Mather's function
$\mathcal{CD}, \mathcal{H}, \mathcal{B}$	Diophantine numbers, Yoccoz class, Bryuno numbers

Part A — Circle topology beyond π_1

Chapter 1 — Coverings, degree and the theorems of the circle

Extended formulary of the chapter: Section 1.20.

Degree and winding number

$$\deg f = w(f \circ E) \quad (f : S^1 \rightarrow S^1), \quad w(\gamma) = \frac{\tilde{\gamma}(1) - \tilde{\gamma}(0)}{2\pi} = \frac{1}{2\pi i} \int_0^1 \frac{\gamma'(t)}{\gamma(t)} dt$$

$$\deg(f \circ g) = \deg f \cdot \deg g, \quad \deg(fg) = \deg f + \deg g \quad (\text{pointwise product in } S^1), \quad \deg p_n = n$$

Topological Rouché

$$|v(t)| < |u(t)| \quad \forall t \implies w(u+v) = w(u)$$

Odd maps and Borsuk–Ulam

$$u(-x) = -u(x) \implies \deg u \text{ odd} \quad (\text{BU1}) \iff (\text{BU2}) \iff (\text{BU3})$$

Umlaufsatz and total curvature

$$\gamma \text{ simple, closed, regular, positively oriented} \implies \rho(\gamma) = 1, \quad \oint \kappa ds = 2\pi, \quad \sum_i \alpha_i^{\text{ext}} = 2\pi \text{ (simple polygon)}$$

Whitney–Graustein

$$\rho(\gamma) := w(\gamma'/|\gamma'|) \in \mathbb{Z}, \quad \gamma_0 \sim_{\text{reg}} \gamma_1 \iff \rho(\gamma_0) = \rho(\gamma_1)$$

Fundamental theorem of algebra, explicit radius estimate

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0, \quad R = 1 + \sum_{i=0}^{n-1} |a_i| \Rightarrow \text{every root has } |z| < R$$

Hairy ball theorem, Milnor's identity

$$\text{Vol}(A_t) = (1+t^2)^{3/2} \text{Vol}(A) \quad (\text{not polynomial}) \quad \text{vs.} \quad P(t) = \text{Vol}(A_t) \text{ polynomial of degree } \leq 3$$

Sperner's lemma

$$\#(\text{fully labelled triangles}) \text{ is odd, in particular } \geq 1$$

Chapter 2 — Bundles over the circle and over the sphere

Extended formulary of the chapter: Section 2.22.

Bundle, cocycle

$$\text{pr}_U \circ \Phi = \pi, \quad g_{ii} = I, \quad g_{ij}g_{ji} = I, \quad g_{ij}g_{jk} = g_{ik}$$

Index and Poincaré–Hopf

$$\text{ind}_p(V) = \deg \left(\frac{V(\gamma_\epsilon(\theta))}{|V(\gamma_\epsilon(\theta))|} \right), \quad \sum_p \text{ind}_p(V) = \chi(\Sigma) = 2 - 2g$$

$$\text{index at a Morse critical point of index } k : (-1)^k$$

Classification on S^1

$$\text{Vect}_n(S^1) \leftrightarrow \pi_0(O(n)) \cong \mathbb{Z}/2 \quad (\forall n \geq 1), \quad \{\text{principal } G\text{-bundles on } S^1\} \leftrightarrow \pi_0(G)/\text{conjugation}$$

Local Gauss–Bonnet and holonomy

$$\Delta\varphi = \iint_R K \, dA, \quad \text{parallel at colatitude } \theta_0 : \Delta\varphi = 2\pi(1 - \cos \theta_0)$$

Foucault pendulum

$$\Omega_{\text{Foucault}} = 2\pi - \Delta\varphi = 2\pi \sin \lambda_0 \text{ per sidereal day}$$

Hopf fibration

$$\pi(z_1, z_2) = (2 \operatorname{Re}(z_1 \bar{z}_2), 2 \operatorname{Im}(z_1 \bar{z}_2), |z_1|^2 - |z_2|^2), \quad g_{-+}(\varphi) = e^{i\varphi}$$

$$\sigma_+(x, y, w) = \left(\sqrt{\frac{1+w}{2}}, \frac{x-iy}{\sqrt{2(1+w)}} \right), \quad \omega = \operatorname{Im}(\bar{z}_1 dz_1 + \bar{z}_2 dz_2)$$

Chern–Weil and Berry phase

$$\frac{1}{2\pi} \iint_{S^2} F = \pm 1 \in \mathbb{Z}, \quad \gamma_{\text{Berry}} = -\frac{1}{2}\Omega$$

Periodic trapezoid rule, exponential convergence

$$T_N(A) = \frac{2\pi}{N} \sum_{k=0}^{N-1} A\left(\frac{2\pi k}{N}\right), \quad |T_N(A) - \oint A| \leq C e^{-Na_0}$$

Pencil of conics and circular points

$$f_{[\lambda:\mu]} = \lambda f_1 + \mu f_2, \quad \text{circles} \iff \text{conics through } I = [1:i:0], \quad J = [1:-i:0]$$

Part B — Rotations and diffeomorphisms

Chapter 3 — Rotations, continued fractions and equidistribution

Extended formulary of the chapter: Section 3.20.

Rotation and dichotomy

$$R_\alpha(\theta) = \theta + \alpha \pmod{2\pi}, \quad \alpha = 2\pi p/q \text{ (reduced)} \Rightarrow \text{period } q; \quad \alpha/2\pi \notin \mathbb{Q} \Rightarrow \text{dense orbit}$$

Continued fractions

$$\alpha = [a_0; a_1, a_2, \dots], \quad p_n = a_n p_{n-1} + p_{n-2}, \quad q_n = a_n q_{n-1} + q_{n-2}, \quad p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$$

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

Best approximation

$$\text{2nd kind: } \|k\alpha\| \geq \|q_n\alpha\| \quad \forall 0 < k < q_{n+1}; \quad \text{1st kind: } \left| \alpha - \frac{p}{q} \right| \geq \left| \alpha - \frac{p_n}{q_n} \right| \quad \forall 0 < q \leq q_n$$

Hurwitz and Diophantine numbers

$$\exists^\infty p/q : \left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2}; \quad \liminf_n q_n \|q_n \alpha\| = \frac{1}{\sqrt{5}}; \quad |\alpha - p/q| \geq \frac{c}{q^{\deg \alpha}} \quad (\alpha \text{ algebraic, Liouville})$$

Noble numbers

$$\rho^* = \frac{1}{\varphi} = \varphi - 1 = [0; 1, 1, 1, \dots] = 0.618034\dots$$

Three-distance theorem

$$\{0, \beta, \dots, (N-1)\beta\} \bmod 1 : \text{ at most 3 lengths } \ell_+, \ell_-, \ell_+ + \ell_-$$

Weyl, Fejér, discrepancy

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i k n \beta} \rightarrow 0 \quad (k \neq 0); \quad \sigma_N(f) = \frac{1}{N} \sum_{n < N} S_n(f) \rightarrow f \text{ uniformly}$$

$$D_N = \sup_b \left| \frac{1}{N} \# \{x_n < b\} - b \right|; \quad \left| \frac{1}{N} \sum f(x_n) - \int f \right| \leq V(f) D_N \quad (\text{Koksma})$$

Discrepancy of rotations

$$\text{bounded partial quotients: } D_N = O\left(\frac{\log N}{N}\right); \quad \text{Schmidt (panorama): } D_N \geq \frac{c \log N}{N} \text{ for infinitely many } N$$

Phyllotaxis

$$r_n = c\sqrt{n}, \quad \theta_n = n\gamma; \quad \gamma^* = 2\pi(2 - \varphi) = 2\pi/\varphi^2 \approx 137.5077^\circ$$

Chapter 4 — Rotation number, Poincaré classification and Arnold tongues

Extended formulary of the chapter: Section 4.14.

Rotation number

$$\rho(f) = \lim_{n \rightarrow \infty} \frac{F^n(\theta) - \theta}{2\pi n} \quad (\text{independent of } \theta); \quad \rho(F^n) = n\rho(F); \quad \rho(HFH^{-1}) = \rho(F)$$

Fekete's lemma

$$a_{m+n} \leq a_m + a_n \quad \forall m, n \quad \implies \quad \lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}$$

Rational criterion

$$\rho(F) \geq \frac{p}{q} \iff \exists \theta : F^q(\theta) \geq \theta + 2\pi p; \quad \rho(F) \neq \frac{p}{q} \Rightarrow \text{sgn}(F^q(\theta) - \theta - 2\pi p) \text{ constant in } \theta$$

Poincaré classification

$$\rho(f) = \frac{p}{q} \text{ (reduced)} \iff \exists \text{ periodic orbit, of exact period } q; \quad \rho(f) \notin \mathbb{Q} \Rightarrow \exists h : h \circ f = R_{2\pi p} \circ h$$

Arnold's standard family ($\theta \in [0, 1)$)

$$f_{\alpha, \varepsilon}(\theta) = \theta + \alpha - \frac{\varepsilon}{2\pi} \sin(2\pi\theta); \quad \text{diffeomorphism for } 0 < \varepsilon < 1; \quad \rho = 0 \iff |\alpha| \leq \frac{\varepsilon}{2\pi} \text{ (exact)}$$

width of tongue $1/2 = \frac{\varepsilon^2}{4\pi} + O(\varepsilon^3)$; width of tongue $p/q = O(\varepsilon^q)$ (imported for $q \geq 3$)

Devil's staircase

$\alpha \mapsto \rho(\alpha, \varepsilon)$: continuous, non-decreasing, constant on a dense union of intervals

Panorama at criticality and beyond

$\varepsilon = 1$: $\text{meas}\{\alpha : \rho(\alpha, 1) \notin \mathbb{Q}\} = 0$ (Świątek 1988); $\varepsilon > 1$: rotation interval $[\rho_-, \rho_+]$ (Boyland 1986)

Chapter 5 — Denjoy's theorem and its counterexamples

Extended formulary of the chapter: Section 5.12.

Convergents and signed remainder

$$\delta_n = q_n \rho - p_n, \quad \text{sign}(\delta_n) = (-1)^n, \quad |\delta_{n-1}|q_n + |\delta_n|q_{n-1} = 1$$

Best return times lemma

$$\{R_{2\pi\rho}^j(I_n)\}_{j=0}^{q_{n+1}-1} \text{ disjoint}; \quad \{R_{2\pi\rho}^j(J_n)\}_{j=0}^{q_{n+1}-1} \text{ cover } \mathbb{T} \text{ with mult. } \leq 2$$

Denjoy's inequality

$$V(f) := \text{Var}_{\mathbb{T}}(\log f') < \infty, \quad f' > 0 \implies (f^{q_n})'(x) (f^{-q_n})'(x) \geq e^{-V(f)}$$

Denjoy's theorem

$$f \in C_+^{1+\text{bv}}(\mathbb{T}), \quad \rho(f) \notin \mathbb{Q} \implies f \text{ minimal, conjugate to } R_{\rho(f)}$$

Denjoy–Koksma inequality

$$\left| \rho - \frac{p}{q} \right| \leq \frac{1}{q^2} \implies \left| \sum_{j=0}^{q-1} \psi(f^j(x)) - q \int_{\mathbb{T}} \psi d\mu \right| \leq \text{Var}(\psi)$$

C^1 counterexample

$$\ell_n = \frac{c}{(|n|+2)^2}, \quad S = \sum_n \frac{1}{(|n|+2)^2} = \frac{\pi^2}{3} - \frac{9}{4}, \quad c_{\max} = \frac{2\pi}{S}, \quad \text{Leb}(K) = 2\pi - cS$$

$$f'_n(u) = 1 + 6(r_n - 1)u(1-u), \quad r_n = \frac{\ell_{n+1}}{\ell_n}, \quad \inf_n r_n = \frac{4}{9} > \frac{1}{3} \Rightarrow f' > 0 \quad \forall c \in (0, c_{\max})$$

Chapter 6 — Small divisors and KAM theory on the circle

Extended formulary of the chapter: Section 6.14.

Homological equation and small divisors

$$\psi \circ R_\alpha - \psi = \varphi, \quad \widehat{\psi}_k = \frac{\widehat{\varphi}_k}{e^{2\pi i k \rho} - 1} \quad (k \neq 0), \quad \widehat{\varphi}_0 = 0 \text{ necessary}$$

Small divisors lemma

$$|e^{2\pi i x} - 1| = 2|\sin \pi x|, \quad 4\|x\| \leq |e^{2\pi i x} - 1| \leq 2\pi\|x\|$$

Diophantine condition (two forms) and constants

$$\left| \rho - \frac{p}{q} \right| \geq \frac{K}{q^{2+\tau}} \iff \|q\rho\| \geq \frac{K}{q^{1+\tau}} \implies |e^{2\pi i k \rho} - 1| \geq \frac{4K}{|k|^{1+\tau}}$$

$$K\left(\frac{1}{\varphi}\right) = \varphi^{-2} = \frac{3-\sqrt{5}}{2} = 0.381966, \quad K(\sqrt{2}-1) = 6-4\sqrt{2} = 0.343146, \quad (\tau = 0)$$

Estimate of the series and of the analytic homological equation

$$\sum_{k \geq 1} k^{1+\tau} e^{-k\delta} \leq \frac{\Lambda(\tau)}{\delta^{2+\tau}}, \quad \Lambda(\tau) = \Gamma(2+\tau) + \left(\frac{1+\tau}{e}\right)^{1+\tau}; \quad \|\psi\|_{b-\delta} \leq \frac{\Lambda(\tau) \|\varphi\|_b}{2K \delta^{2+\tau}}$$

C^r case

$$\varphi \in C^r, \quad r > \tau + 2 \implies \psi \in C^s \quad \forall s < r - \tau - 2, \quad \|\psi^{(s)}\|_\infty \leq \frac{\zeta(r - \tau - s - 1)}{2K} \|\varphi^{(r)}\|_\infty$$

Newton scheme (constants and recursions)

$$\gamma = 3 + \tau, \quad C_1 = \frac{4\Lambda}{K}, \quad \delta_0 = \min\left(1, \frac{a}{12}\right), \quad \delta_j = \delta_0 2^{-j}, \quad a_{j+1} = a_j - 3\delta_j, \quad a_\infty = a - 6\delta_0 \geq \frac{a}{2}$$

$$\varepsilon_{j+1} = \frac{C_1 \varepsilon_j^2}{\delta_j^\gamma}, \quad x_j := A 2^{(j+1)\gamma} \varepsilon_j \quad (A = C_1 \delta_0^{-\gamma}), \quad x_{j+1} = x_j^2, \quad x_j \leq 2^{-2^j}$$

$$\boxed{\varepsilon_0(a, \rho) = \frac{K \delta_0^{3+\tau}}{2^{6+\tau} \Lambda(\tau)}} \quad (\text{step condition: } C_1 \varepsilon_j / \delta_j^\gamma \leq 1)$$

Gottschalk–Hedlund

$$T \text{ minimal :} \quad \phi = v \circ T - v \text{ with } v \in C(X) \iff \sup_n |S_n \phi(x_0)| < \infty \text{ for some } x_0$$

Arithmetic hierarchy

$$\mathcal{CD} \subsetneq \mathcal{H} \subsetneq \mathcal{B}, \quad B(\rho) = \sum_{n \geq 0} \frac{\log q_{n+1}}{q_n} < \infty \quad (\text{Bryuno})$$

Part C — Chaos and ergodic theory

Chapter 7 — The doubling map: symbolic dynamics, Li–Yorke and Sharkovskii

Extended formulary of the chapter: Section 7.15.

Doubling and expanding maps

$$D(\theta) = 2\theta \bmod 2\pi, \quad E_m(\theta) = m\theta \bmod 2\pi, \quad m \text{ preimages per point}$$

$$\text{sensitivity: } n \text{ minimal with } m^n(\delta/2) \geq \pi/m \Rightarrow \text{separation} \in [\pi/m, \pi)$$

$$\text{Per}(E_m) = \left\{ 2\pi \frac{p}{q} : \gcd(p, q) = 1, \gcd(q, m) = 1 \right\}, \quad N(\delta) = \lceil \log_m(1/\delta) \rceil \quad (\text{mixing})$$

Semiconjugacy with the shift

$$h(\sigma) = 2\pi \sum_{k \geq 0} \sigma_k 2^{-(k+1)}, \quad h \circ \text{sh} = D \circ h, \quad \Sigma_2 / \sim \cong \mathbb{T}$$

Covering lemma

$$J \rightarrow K \iff f(J) \supseteq K; \quad J_0 \rightarrow \cdots \rightarrow J_{n-1} \rightarrow J_0 \Rightarrow \exists x : f^n(x) = x, \quad f^i(x) \in J_i$$

Li–Yorke

$$\text{orbit of period 3} \Rightarrow \text{orbit of period } n \quad \forall n \geq 1; \quad \text{uncountable scrambled set}$$

Sharkovskii’s order and theorem

$$3 \triangleleft 5 \triangleleft 7 \triangleleft \cdots \triangleleft 2 \cdot 3 \triangleleft 2 \cdot 5 \triangleleft \cdots \triangleleft 4 \cdot 3 \triangleleft \cdots \triangleleft \cdots \triangleleft 2^3 \triangleleft 2^2 \triangleleft 2 \triangleleft 1$$

$$\text{period } n, \quad n \triangleleft m \Rightarrow \text{period } m \text{ (also exists)}$$

Counting periodic points

$$\#\text{Fix}(D^n) = 2^n - 1, \quad \#\text{Fix}(T^n) = \#\text{Fix}(f_4^n) = 2^n$$

$$p_D(n) = \sum_{d|n} \mu(n/d)(2^d - 1), \quad \#\{\text{orbits of exact period } n\} = p_D(n)/n$$

The triangle D – T – f_4

$$\varphi(\theta) = 1 - \left| 1 - \frac{\theta}{\pi} \right|, \quad k(x) = \sin^2\left(\frac{\pi x}{2}\right), \quad h(\theta) = k(\varphi(\theta)) = \sin^2\left(\frac{\theta}{2}\right)$$

$$\varphi \circ D = T \circ \varphi \text{ (semicon.)}, \quad k \circ T = f_4 \circ k \text{ (conj.)}, \quad h \circ D = f_4 \circ h \text{ (semicon.)}$$

Kneading sequence of f_4

$$K(f_4) = C, R, L, L, L, \dots = CRL^\infty$$

Chapter 8 — Topological entropy and Lyapunov exponents

Extended formulary of the chapter: Section 8.17.

Topological entropy (Bowen–Dinaburg)

$$d_n(x, y) = \max_{i < n} d(f^i x, f^i y), \quad h_{\text{top}}(f) = \lim_{\varepsilon \rightarrow 0^+} \limsup_{n \rightarrow \infty} \frac{1}{n} \log N(n, \varepsilon)$$

$$r(n, \varepsilon) \leq N(n, \varepsilon) \leq r(n, \varepsilon/2), \quad h_{\text{top}}^{\text{AKM}}(f) = \sup_{\mathcal{U}} \limsup_n \frac{1}{n} \log N(\mathcal{U}^n) = h_{\text{top}}(f)$$

Invariance and factors

$$h \circ f = g \circ h \text{ (conjugacy)} \implies h_{\text{top}}(f) = h_{\text{top}}(g)$$

$$\pi \circ f = g \circ \pi, \pi \text{ surjective} \implies h_{\text{top}}(g) \leq h_{\text{top}}(f) \quad (= \text{if finite fibres unif. bounded, Bowen})$$

Direct computations

$$h_{\text{top}}(E_m) = \log m \ (m \geq 2), \quad h_{\text{top}}(D) = \log 2, \quad h_{\text{top}}(R_\alpha) = 0, \quad h_{\text{top}}(f^j) = j h_{\text{top}}(f)$$

$$\text{shift } \sigma_k : \quad k^n \leq N(n, \varepsilon) \leq k^{n+m(\varepsilon)}, \quad m(\varepsilon) = \lceil \log_k(1/\varepsilon) \rceil, \quad h_{\text{top}}(\sigma_k) = \log k$$

Lap number

$$\text{lap}(f^{n+m}) \leq \text{lap}(f^n) \text{lap}(f^m), \quad \ell(f) = \lim_n \frac{1}{n} \log \text{lap}(f^n) \leq h_{\text{top}}(f)$$

(Misiurewicz–Szlenk, imported) $h_{\text{top}}(f) = \ell(f)$ for f piecewise monotone with finitely many pieces

Sharkovskii and entropy

$$\text{period } 3 \implies h_{\text{top}}(f) \geq \log 2; \quad \text{period not a power of } 2 \text{ (BGMY, imported)} \implies h_{\text{top}}(f) > 0$$

Lyapunov exponents

$$\lambda(\theta) = \lim_n \frac{1}{n} \log |(f^n)'(\theta)| = \lim_n \frac{1}{n} \sum_{i < n} \log |f'(f^i \theta)|$$

$$\lambda_{E_m} \equiv \log m, \quad \lambda_{f_4} = \log 2 \text{ a.e.}, \quad \lambda_{f_4}(0) = \lambda_{f_4}(1) = \log 4, \quad \lambda_{f_4}(1/2) = -\infty$$

$$\int_0^{\pi/2} \log(\cos \theta) d\theta = -\frac{\pi}{2} \log 2$$

Rokhlin and variational principle (case D)

$$h_{\mu_{\text{Leb}}}(D) = h_\mu(D, \mathcal{P}_{\text{dyadic}}) = \int_{\mathbb{T}} \log |D'| d\mu = \log 2 = h_{\text{top}}(D)$$

(variational principle, imported) $h_{\text{top}}(f) = \sup_\mu h_\mu(f)$

Oseledets 1D and Pesin (panorama)

$$\lambda_\mu = \int \log |f'| d\mu \ (\mu\text{-a.e.}), \quad h_\mu(f) = \max(\lambda, 0) \text{ (1D case of Pesin's formula)}$$

Chapter 9 — Ergodic theory on the circle and on the interval

Extended formulary of the chapter: Section 9.21.

Invariant measures

$$\mu(S^{-1}A) = \mu(A) \quad \forall A \iff \int \varphi \circ S d\mu = \int \varphi d\mu \quad \forall \varphi \in C(X)$$

$$\text{(Krylov–Bogolyubov)} \quad X \text{ compact, } S \text{ continuous} \implies \exists \mu \text{ invariant}$$

Lebesgue invariant

$$\mu(E_m^{-1}A) = \mu(A) \quad \forall A \text{ measurable, } m \geq 2 \quad (D = E_2)$$

Ergodicity and mixing (via Fourier)

$$\varphi \circ D = \varphi \text{ a.e.} \implies c_k = 0 \ (k \neq 0) \implies \varphi \text{ constant a.e.}$$

$$C_n(f, g) = \int (f \circ D^n) \bar{g} \, d\mu - \left(\int f \, d\mu \right) \left(\int \bar{g} \, d\mu \right) \xrightarrow{n \rightarrow \infty} 0$$

Ergodic theorems

$$(\text{Hopf, maximal}) \quad \varphi \in L^1, \quad E = \{\sup_n S_n \varphi > 0\} \implies \int_E \varphi \, d\mu \geq 0$$

$$(\text{Birkhoff}) \quad \varphi^*(x) = \lim_n \frac{1}{n} S_n \varphi(x) \text{ a.e.,} \quad \int \varphi^* \, d\mu = \int \varphi \, d\mu, \quad (\text{ergodic}) \quad \varphi^* \equiv \int \varphi \, d\mu$$

$$(\text{von Neumann, in mean}) \quad \frac{1}{N} \sum_{n < N} U^n \varphi \xrightarrow{L^2} P\varphi = \left(\int \varphi \, d\mu \right) \cdot 1 \quad (\text{ergodic case})$$

Unique ergodicity (Oxtoby)

$$\text{uniq. ergodic} \iff \forall \varphi \in C(X) : \frac{1}{N} S_N \varphi(x) \rightarrow \text{const. uniformly in } x$$

$$R_\alpha : \left| \frac{1}{N} S_N e_k(\theta) \right| \leq \frac{2}{N |e^{ik\alpha} - 1|} \quad (\text{unif. in } \theta, k \neq 0) \quad D : \text{not uniquely ergodic}$$

Normal numbers

(Borel, via Birkhoff on D) almost every θ normal in base 2 (Champernowne, imported) $0.\overline{1} \overline{10} \overline{11} \cdots_2$ normal

Ulam–von Neumann and the triangle D – T – f_4

$$d\nu = \frac{dx}{\pi \sqrt{x(1-x)}} \text{ invariant and ergodic for } f_4, \quad h_T(x) = \sin^2\left(\frac{\pi x}{2}\right) : T \xleftrightarrow{\text{full conj.}} f_4$$

Lasota–Yorke (tent map)

$$(\mathcal{L}w)(y) = \frac{1}{2} \left[w\left(\frac{y}{2}\right) + w\left(1 - \frac{y}{2}\right) \right], \quad \text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g)$$

$$\left\| \mathcal{L}^n g_0 - 1 \right\|_\infty \leq 2^{-n} \text{Var}(g_0) \xrightarrow{n \rightarrow \infty} 0$$

Kolmogorov–Sinai entropy

$$h_\mu(S, \mathcal{P}) = \lim_n \frac{1}{n} H(\mathcal{P}^n), \quad h_\mu(S) = \sup_{\mathcal{P}} h_\mu(S, \mathcal{P}) \quad h_{\mu_{\text{Leb}}}(D) = \log 2$$

Chapter 10 — Transfer operators, decay of correlations and the central limit theorem

Extended formulary of the chapter: Section 10.16.

Transfer operator

$$(\mathcal{L}g)(\theta) = \frac{1}{2} \left[g\left(\frac{\theta}{2}\right) + g\left(\frac{\theta}{2} + \pi\right) \right], \quad (\mathcal{L}_m g)(\theta) = \frac{1}{m} \sum_{j=0}^{m-1} g\left(\frac{\theta + 2\pi j}{m}\right)$$

$$\int_{\mathbb{T}} (g \circ E_m) w \, d\mu = \int_{\mathbb{T}} g \cdot (\mathcal{L}_m w) \, d\mu, \quad \widehat{(\mathcal{L}_m g)}_k = \widehat{g}_{mk}$$

Spectrum on L^2 versus spectral gap

$$g_\lambda = \sum_{n \geq 0} \lambda^n e_{2^n} \in L^2(\mathbb{T}) \quad (|\lambda| < 1), \quad \mathcal{L}g_\lambda = \lambda g_\lambda, \quad \|g_\lambda\|_2^2 = \frac{1}{1-|\lambda|^2}$$

$$\text{Var}(\mathcal{L}g) \leq \frac{1}{2} \text{Var}(g), \quad [\mathcal{L}g]_\alpha \leq \left(\frac{1}{2}\right)^\alpha [g]_\alpha \implies g \in BV \text{ or } C^{0,\alpha}, \quad \mathcal{L}g = \lambda g, \quad g \neq \text{const.} \implies |\lambda| \leq \frac{1}{2} \text{ (or } \left(\frac{1}{2}\right)^\alpha)$$

Decay of correlations

$$|\widehat{g}_k| \leq \frac{\text{Var}(g)}{2\pi|k|} \quad (BV), \quad |C_n(u, v)| \leq \frac{\pi}{\sqrt{3}} \|u\|_2 \text{Var}(v) \cdot m^{-n}$$

$$\left| \int (u \circ T^n) v \, dx - m_v \int u \, dx \right| \leq \|u\|_1 \text{Var}(v) \cdot 2^{-n} \quad (\text{tent map})$$

CLT, dyadic indicator (de Moivre–Laplace)

$$\varphi = \pm 1 \text{ (1st binary digit),} \quad S_n \varphi = n - 2K_n, \quad K_n \sim \text{Bin}(n, \tfrac{1}{2}) \text{ exact}$$

$$\binom{n}{k} 2^{-n} = \sqrt{\frac{2}{\pi n}} e^{-x_k^2/2} (1 + o(1)), \quad x_k = \frac{2k-n}{\sqrt{n}} \Rightarrow S_n \varphi / \sqrt{n} \rightarrow \mathcal{N}(0, 1)$$

CLT, Gordin martingale

$$g = \sum_{k \geq 1} \mathcal{L}^k u, \quad m = u + g - g \circ D, \quad \mathcal{L}m = 0, \quad u = m + (g \circ D - g)$$

$$\int (m \circ D^i)(m \circ D^j) d\mu = 0 \quad (i \neq j), \quad \sigma^2 = \int m^2 d\mu, \quad S_n u / \sqrt{n} \rightarrow \mathcal{N}(0, \sigma^2)$$

Chapter 11 — Logistic map, bifurcations and universality

Extended formulary of the chapter: Section 11.13.

Logistic family and fixed points

$$f_r(x) = rx(1-x), \quad x = 0 \text{ (always)}, \quad x^* = 1 - \frac{1}{r} \quad (r \geq 1), \quad f'_r(0) = r, \quad f'_r(x^*) = 2 - r$$

Bifurcations

$$\text{transcritical: } r = 1; \quad \text{period-doubling (flip): } r = 3; \quad \text{saddle-node (period 3): } r = 1 + 2\sqrt{2}$$

The 2-cycle

$$x_{\pm}(r) = \frac{(r+1) \pm \sqrt{(r-3)(r+1)}}{2r}, \quad (f_r^2)'(x_{\pm}) = -r^2 + 2r + 4, \quad \text{stable for } 3 < r < 1 + \sqrt{6} \approx 3.449490$$

Normal forms (family $f(x, \mu)$, $g = f - x$)

$$\text{saddle-node: } g_x = 0, \quad g_{\mu} \neq 0, \quad g_{xx} \neq 0 \Rightarrow \mu(x), \quad \mu'(x_0) = 0, \quad \mu''(x_0) = -\frac{g_{xx}}{g_{\mu}}$$

$$\text{transcritical: } g(x_0, \cdot) \equiv 0, \quad g_{x\mu} \neq 0, \quad g_{xx} \neq 0 \Rightarrow \xi'(\mu_0) = -\frac{g_{x\mu}}{\frac{1}{2}g_{xx}}$$

$$\text{flip: } f_x(x_0, \mu_0) = -1 \Rightarrow \partial_x g_2(x_0, \mu_0) = f_x(x_0, \mu_0)^2 - 1 = 0$$

Cascade and universality

$$r_1 = 3, \quad r_2 = 1 + \sqrt{6}, \quad r_n \rightarrow r_{\infty} \approx 3.569945672$$

$$\delta_n = \frac{r_n - r_{n-1}}{r_{n+1} - r_n} \rightarrow \delta = 4.66920160910299 \dots \quad \alpha = -2.5029078750959 \dots$$

$$(\mathcal{R}g)(x) = \alpha g(g(x/\alpha)), \quad \mathcal{R}g^* = g^* \quad (\text{Lanford, imported})$$

Universality of critical circle maps (golden ratio, panorama)

$$\delta_{\text{GM}} \approx -2.8336, \quad \alpha_{\text{GM}} \approx -1.2886 \quad (\neq \delta, \alpha)$$

Schwarzian derivative and bifurcation diagram

$$Sf = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2, \quad Sf_r < 0 \text{ everywhere;} \quad A(r) = \omega\left(\frac{1}{2}, f_r\right), \quad B = \{(r, y) : y \in A(r)\}$$

Part D — Holomorphic dynamics

Chapter 12 — From the circle to Julia and Mandelbrot

Extended formulary of the chapter: Section 12.18.

Quadratic family, K_c , J_c , $\mathcal{M}$

$$g_c(z) = z^2 + c, \quad K_c = \{z : (g_c^n(z)) \text{ bounded}\}, \quad J_c = \partial K_c, \quad \mathcal{M} = \{c : 0 \in K_c\}$$

$$\text{escape criterion: } |g_c^n(z)| > \max(|c|, 2) \Rightarrow |g_c^{n+k}(z)| \geq \mu^{2^k-1} |g_c^n(z)|, \quad \mu = |g_c^n(z)| - 1$$

$$\left\{|c| \leq \frac{1}{4}\right\} \subseteq \mathcal{M} \subseteq \{|c| \leq 2\}$$

Local dynamics

$$\text{Koenigs } (0 < |\lambda| < 1) : \quad \varphi(z) = \lim_n \frac{f^n(z)}{\lambda^n}, \quad \varphi \circ f = \lambda \varphi, \quad \varphi(0) = 0, \quad \varphi'(0) = 1$$

$$\text{Böttcher } (\infty) : \quad \phi_c(z) = z \exp\left(\sum_{n \geq 0} \frac{1}{2^{n+1}} \text{Log}\left(1 + \frac{c}{z^n}\right)\right), \quad \phi_c \circ g_c = \phi_c^2, \quad \frac{\phi_c(z)}{z} \rightarrow 1$$

$$G_c(z) = \lim_n \frac{\log^+ |g_c^n(z)|}{2^n} = \log |\Phi_c(z)|, \quad G_c \circ g_c = 2 G_c, \quad G_c|_{K_c} \equiv 0$$

Fundamental dichotomy

$$0 \in K_c \iff c \in \mathcal{M} \implies K_c \text{ connected}; \quad 0 \notin K_c \iff K_c \text{ a Cantor set}$$

$$c \in \mathcal{M} \implies \Phi_c : \mathbb{C} \setminus K_c \xrightarrow{\sim} \mathbb{C} \setminus \overline{\mathbb{D}}, \quad \Phi_c \circ g_c = \Phi_c^2$$

Fatou

$$\text{every attracting cycle attracts a critical point} \implies \text{at most one attracting cycle} \implies c \in \mathcal{M}$$

Cardioid and bulbs

$$\mathfrak{c}(\lambda) = \frac{\lambda}{2} - \frac{\lambda^2}{4} \quad (|\lambda| < 1); \quad \lambda_{\pm} = 1 \pm \sqrt{1 - 4c}, \quad \lambda_+ + \lambda_- = 2$$

$$\text{period-2 bulb: } |c+1| < \frac{1}{4}, \quad \lambda_2 = 4(c+1); \quad c_{p/q} = \frac{e^{2\pi i p/q}}{2} - \frac{e^{4\pi i p/q}}{4}$$

$$c_{1/2} = -\frac{3}{4}, \quad c_{1/3} = -\frac{1}{8} + \frac{3\sqrt{3}}{8}i, \quad c_{1/4} = \frac{1}{4} + \frac{i}{2}, \quad c_{1/6} = \frac{3}{8} + \frac{\sqrt{3}}{8}i$$

External rays

$$R_t = \Phi_c^{-1}(\{re^{2\pi i t} : r > 1\}), \quad g_c(R_t) = R_{2t \bmod 1}, \quad g_c^{-1}(R_t) = R_{t/2} \cup R_{t/2+1/2}$$

$$R_t \text{ periodic of period } q \iff t = \frac{k}{2^q - 1}; \quad A(q) = \sum_{d|q} \mu\left(\frac{q}{d}\right)(2^d - 1) = 2\nu(q) \quad (q \geq 2)$$

Small divisors

$$\omega_m = |\lambda^m - 1| = 2 \sin(\pi \|m\vartheta\|) \geq 4\|m\vartheta\|; \quad \text{Diophantine} \implies \omega_m \geq C m^{-\mu}$$

$$h_n = \frac{1}{\lambda(\lambda^{n-1} - 1)} \sum_{p+q=n} h_p h_q, \quad M_n = \sum_{T \in \mathcal{T}_n} \prod_v \frac{1}{\omega_{n(v)-1}} \leq 4^n \Gamma^n$$

$$\text{Siegel: } \vartheta \text{ Diophantine} \implies \text{linearizable}; \quad \text{Cremer: } \liminf_n \omega_n^{1/2^n} = 0 \implies \text{not linearizable}$$

$$\text{Bryuno: } B(\vartheta) = \sum_k \frac{\log q_{k+1}}{q_k} < \infty \iff f_\lambda(w) = \lambda w + w^2 \text{ linearizable (Yoccoz)}$$

Stability and dimension

$$|c| \leq \frac{3}{16} \implies J_c = h_c(S^1) \text{ Jordan curve}, \quad h_c(u^2) = h_c(u)^2 + c, \quad \|h_c - \text{id}\|_\infty \leq \frac{1}{4}$$

$$\text{Bowen: } \sum_{g_c^n(z)=z} |(g_c^n)'(z)|^{-d} \rightarrow 1; \quad \text{Ruelle: } \dim_H J_c = 1 + \frac{|c|^2}{4 \log 2} + \dots$$

Part E — Synchronisation and automation

Chapter 13 — Synchronisation: from Huygens to Kuramoto

Extended formulary of the chapter: Section 13.17.

Flow on the circle

$$\theta' = F(\theta) \text{ (} 2\pi\text{-periodic);} \quad F(\theta^*) = 0, \quad F'(\theta^*) < 0 \implies \text{stable}; \quad F \neq 0 \text{ everywhere} \implies T = \oint \frac{d\theta}{F(\theta)}$$

Adler's equation

$$\theta' = \Delta\omega - K \sin \theta, \quad K > 0; \quad |\Delta\omega| < K : \theta_s^* = \arcsin \frac{\Delta\omega}{K} \text{ (stable)}, \quad \theta_i^* = \pi - \theta_s^* \text{ (unstable)}$$

$$|\Delta\omega| > K : T = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \sim \frac{2\pi}{\sqrt{2K}} (\Delta\omega - K)^{-1/2} \quad (\Delta\omega \rightarrow K^+)$$

Two oscillators

$$\varphi := \theta_2 - \theta_1, \quad \varphi' = \Delta\omega - 2K \sin \varphi \quad (\text{Adler with gain } 2K)$$

Kuramoto, N identical oscillators (mean field)

$$\theta_i' = \omega + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i) = -\frac{\partial V}{\partial \psi_i}, \quad V(\psi) = -\frac{K}{2N} \sum_{i,j} \cos(\psi_i - \psi_j) = -\frac{NK}{2} r^2$$

synchronised state: unique global minimum of V ; Jacobian $-L(K_N)$: eigenvalues $0, \underbrace{-K, \dots, -K}_{N-1}$

Order parameter

$$r e^{i\psi} = \frac{1}{N} \sum_j e^{i\theta_j}, \quad \theta_i' = \omega_i + Kr \sin(\psi - \theta_i)$$

Self-consistency (stationary ansatz, g even and unimodal)

$$r = Kr \int_{-\pi/2}^{\pi/2} \cos^2 \theta g(Kr \sin \theta) d\theta, \quad K_c = \frac{2}{\pi g(0)}$$

$$g \text{ Lorentzian } (\gamma) : K_c = 2\gamma, \quad r = \sqrt{1 - K_c/K} \quad (K > K_c, \text{ exact})$$

Ott–Antonsen reduction (g Lorentzian of half-width Δ)

$$\dot{\alpha} = -i\omega\alpha + \frac{K}{2}(\bar{z} - z\alpha^2); \quad \dot{r} = -\Delta r + \frac{K}{2}r(1 - r^2), \quad r^* = \sqrt{1 - 2\Delta/K}$$

Network synchronisation (Dörfler–Bullo)

$$\theta_i' = \omega_i + \sum_{(i,j) \in E} a_{ij} \sin(\theta_j - \theta_i), \quad |\theta_i^* - \theta_j^*| < \frac{\pi}{2} \text{ on every edge} \Rightarrow \text{stable, rate} \geq \lambda_2(L^*)$$

Chapter 14 — Phase-locked loops, encoders and digital control

Extended formulary of the chapter: Section 14.20.

First-order PLL (Adler's equation)

$$\dot{\theta} = \Delta\omega - K \sin \theta, \quad \theta^* = \arcsin \frac{\Delta\omega}{K} \quad (|\Delta\omega| < K), \quad T_{\text{slip}} = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \quad (|\Delta\omega| > K)$$

Locking bands

$$H = \{\nu : \exists \text{ a.s. equilibrium}\}, \quad P = \{\nu : \text{global conv.}\}, \quad L = \{\nu : \text{conv. without slip}\}$$

$$L \subseteq P \subseteq H, \quad \text{first-order PLL: } H = P = L = (-K, K)$$

Locking time

$$\tau = \frac{1}{\sqrt{K^2 - \Delta\omega^2}}, \quad \theta(t) - \theta^* \sim C e^{-t/\tau}$$

Discrete PLL

$$f_{\alpha, \epsilon}(\theta) = \theta + \alpha - \frac{\epsilon}{2\pi} \sin(2\pi\theta), \quad |\alpha| \leq \frac{\epsilon}{2\pi} \quad (\text{main tongue, } 0 < \epsilon < 1)$$

Itoh's theorem (unwrapping)

$$\psi_j = \psi_{j-1} + W(\varphi_j - \varphi_{j-1}) \quad \forall j \iff \Delta\Theta_j \in (-\pi, \pi] \quad \forall j, \quad f_s > \frac{\omega_{\max}}{\pi} \Rightarrow \text{correct}$$

Vernier encoder: tolerance

$$J = mK_n - nK_m = \frac{n\varphi_m - m\varphi_n}{2\pi} \in \mathbb{Z}, \quad \eta < \frac{\pi}{m+n} \Rightarrow J^* = \text{round}\left(\frac{n\hat{\varphi}_m - m\hat{\varphi}_n}{2\pi}\right) = J$$

Resolver-to-digital converter (type II)

$$\dot{\theta} = K_1 e + K_2 w, \quad \dot{w} = e, \quad e = \theta_{\text{ref}} - \hat{\theta} : \quad e_\infty(\omega_0 t) = 0, \quad e_\infty\left(\frac{1}{2}\alpha t^2\right) = \frac{\alpha}{K_2}$$

Quantization dead band

$$e_{k+1} = Q_\Delta((1-K)e_k), \quad 0 < K < 1 : \quad |e_k| \text{ non-increasing}, \quad |e_\infty| \leq \frac{\Delta}{2K}$$

Stroboscope

$$f_{\text{app}} = f_s \cdot \rho(f/f_s), \quad \rho(x) \in \left(-\frac{1}{2}, \frac{1}{2}\right] \text{ representative of } x \pmod{1}$$

Part F — Global structures

Chapter 15 — Groups of circle homeomorphisms

Extended formulary of the chapter: Section 15.16.

Central extension

$$1 \rightarrow \mathbb{Z} = \langle T \rangle \rightarrow \widetilde{\text{Homeo}} \xrightarrow{p} \text{Homeo}_+(\mathbb{T}) \rightarrow 1, \\ T(\theta) = \theta + 2\pi \text{ central}, \quad \widetilde{\text{Homeo}} \text{ torsion-free}$$

Normalised displacement

$$\varphi_F(\theta) = \frac{F(\theta) - \theta}{2\pi}, \quad a(F) = \min \varphi_F, \quad b(F) = \max \varphi_F, \quad \boxed{b(F) - a(F) < 1} \\ a(F) \leq \rho(F) \leq b(F), \quad a(F) + a(G) \leq a(FG), \\ b(FG) \leq b(F) + b(G), \quad a(FG) \leq a(F) + b(G)$$

Rotation number on $\widetilde{\text{Homeo}}$

$$\rho(T^k F) = k + \rho(F), \quad \rho(F^n) = n\rho(F), \\ \rho(HFH^{-1}) = \rho(F), \quad F \preceq G \Rightarrow \rho(F) \leq \rho(G) \\ |\rho(FG) - \rho(F) - \rho(G)| < 2 \quad (\text{proved}); \quad FG = GF \Rightarrow \rho(FG) = \rho(F) + \rho(G)$$

Hölder's theorem

$$G \text{ acts freely} \Rightarrow G \text{ abelian}, \quad \Phi : G \hookrightarrow (\mathbb{R}, +) \text{ ordered}, \\ \bar{\rho} : G \hookrightarrow \mathbb{R}/\mathbb{Z}, \quad \text{action semiconjugate to rotations}$$

Euler class and Milnor–Wood

$$\prod_{i=1}^g [A_i, B_i] = T^{e(\varrho)}, \quad e(\varrho) \in \mathbb{Z} \text{ well defined}, \\ e(\varrho) = 0 \iff \varrho \text{ lifts} \\ a([A, B]) < 1 \quad (\text{commutator lemma}) \implies \boxed{|e(\varrho)| \leq |\chi(\Sigma_g)| = 2g - 2}$$

Euler cocycle

$$F_f \circ F_g = T^{c(f,g)} \circ F_{fg}, \quad c(f, g) \in \{0, 1\}, \\ c(f, g) + c(fg, h) = c(g, h) + c(f, gh)$$

Chapter 16 — Twist maps of the annulus and Aubry–Mather theory

Extended formulary of the chapter: Section 16.14.

Twist map and generating function

$$\frac{\partial X}{\partial y} \geq \delta > 0; \quad y = -\partial_1 h(x, x'), \quad y' = \partial_2 h(x, x'); \quad \partial_1 \partial_2 h \leq -\delta < 0$$

Admissible class (H1)–(H4)

$$h(x+1, x'+1) = h(x, x'); \quad \partial_1 \partial_2 h \leq -\delta; \quad h \text{ coercive}$$

$C_1 = \sup \partial_1^2 h , \quad C_2 = \sup \partial_2^2 h , \quad L = \max(C_1, C_2)$		
Variational principle (discrete Euler–Lagrange)		
$\partial_2 h(x_{i-1}, x_i) + \partial_1 h(x_i, x_{i+1}) = 0; \quad h = \frac{1}{2}(x' - x)^2 + V(x) \Rightarrow x_{i+1} - 2x_i + x_{i-1} = V'(x_i)$		
Chirikov’s standard map		
$y' = y + \frac{k}{2\pi} \sin(2\pi x), \quad x' = x + y'; \quad h_k = \frac{1}{2}(x' - x)^2 - \frac{k}{(2\pi)^2} \cos(2\pi x); \quad \delta = 1, \quad L = 1 + k$		
$\text{tr } DF_k = 2 + k \cos(2\pi x); \quad (0, 0) \text{ hyperbolic}; \quad (\frac{1}{2}, 0) \text{ elliptic} \iff 0 < k < 4, \quad \cos \alpha = 1 - \frac{k}{2}$		
Exchange inequality (Aubry’s lemma)		
$a < b, \quad c < d: \quad h(a, c) + h(b, d) \leq h(a, d) + h(b, c) - \delta(b - a)(d - c)$		
two distinct minimal configurations cross at most once		
Birkhoff periodic orbits		
$W_{p,q}(x_0, \dots, x_{q-1}) = \sum_{i=0}^{q-1} h(x_i, x_{i+1}), \quad x_q = x_0 + p$		
the minimum exists; two minimisers are strictly ordered or coincide		
Birkhoff configurations		
$\forall (s, t) \in \mathbb{Z}^2: \quad \tau_{s,t} x \leq x \text{ or } \tau_{s,t} x \geq x; \quad a_n - b_n \leq 1; \quad x_{i+n} - x_i - n\rho \leq 1$		
Lipschitz graph		
$Y(\xi) = -\partial_1 h(\xi, g(\xi)) = \partial_2 h(g^{-1}(\xi), \xi); \quad Y(\eta) - Y(\xi) \leq L \eta - \xi $		
Aubry–Mather		
for every $\omega \in \mathbb{R}$ there exists an ordered orbit with $\rho = \omega$		
$\omega \notin \mathbb{Q}: \quad \overline{\mathcal{X}} = \mathbb{R} \text{ (invariant curve)} \quad \text{or} \quad \overline{\mathcal{X}} \text{ a Cantor set (cantorus)}$		
Mather’s β function (standard map)		
$\beta\left(\frac{p}{q}\right) = \frac{1}{q} \min W_{p,q}; \quad \beta(-\omega) = \beta(\omega); \quad \beta(\omega + 1) = \beta(\omega) + \omega + \frac{1}{2}; \quad \beta(0) = -\frac{k}{(2\pi)^2}$		

Recurring numerical constants

Constant	Value	Where it appears
$1/\varphi$	0.618034...	golden ratio, Ch. 3 and 6
$1/\sqrt{5}$	0.447213...	Hurwitz constant, Ch. 3
γ^*	$\approx 137.5077^\circ$	golden angle, phyllotaxis, Ch. 3
$K(1/\varphi) = \varphi^{-2}$	0.381966	Diophantine constant, Ch. 6
$K(\sqrt{2} - 1) = 6 - 4\sqrt{2}$	0.343146	Diophantine constant, Ch. 6
$1 + \sqrt{6}$	≈ 3.449490	end of the stable 2-cycle, Ch. 11
r_∞	≈ 3.569945672	accumulation of the cascade, Ch. 11
δ	4.66920160910299...	first Feigenbaum constant, Ch. 11
α	-2.5029078750959...	second Feigenbaum constant, Ch. 11
δ_{GM}	≈ -2.8336	golden universality on the circle, Ch. 11
α_{GM}	≈ -1.2886	golden universality on the circle, Ch. 11
k_c	≈ 0.9716	Greene’s threshold, standard map, Ch. 16
$\beta(0)$	$-k/(2\pi)^2$	Mather’s function at 0, Ch. 16
$\sum_n (n + 2)^{-2}$	$\pi^2/3 - 9/4$	Denjoy’s C^1 counterexample, Ch. 5

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Note on the count and on sources not included

The unique entries listed above number **219**, derived from the 251 raw entries in the *Sources* blocks of the sixteen chapters: the difference is due to works cited in more than one chapter (for example Katok–Hasselblatt, cited in five chapters, or Navas, de Melo–van Strien, and Kocsard, cited in three) and, in the opposite direction, to the splitting of some cumulative citations into separate entries (Khanin–Sinai and Khanin–Teplinsky; Pöschel and Bounemoura; Abrams–Strogatz and Pecora–Carroll). Not included in the bibliography is the reference, in Chapter 8, to the classical Euler doubling method for $\int_0^{\pi/2} \log(\sin \theta) d\theta$: the chapter itself states that the argument is reproduced in full in the text and does not refer to any edition. No entry comes from Wikipedia or from “anonymous notes”, and none comes from the files of the old structure left in the sources folder and no longer included in the volume.

Index

- $BV(I)$, 262
- 2-cycle (logistic map), 306
- Action (twist maps), 470
- Adler, equation of, 378
 - locking theorem, 379
- Affine algebraic variety, 45
- Annulus (dynamics), 467
- Arnold
 - linearisation theorem, 166
- Arnold standard family, 113
- Arnold theorem, 520
- Arnold tongues, 113
 - principal tongue, 114
 - table, 548
- Aubry's lemma, 476
- Aubry–Mather theorem, 485
- Aubry–Mather theory, 533
- Basin of infinity, 332
- Benedicks–Carleson, 527
- Benedicks–Carleson theorem, 318
- Berry phase, 519
- Bifurcation
 - local, definition, 309
 - period doubling, 306
 - period doubling, general theorem, 310
 - saddle-node, 309
 - saddle-node (period 3), 308
 - saddle-node on the circle, 380
 - transcritical, 306
 - transcritical, general theorem, 310
- Bifurcation diagram, 318
 - monotonicity, 319
- Billingsley theorem, 526
- Birkhoff
 - configuration, 480
 - periodic orbits of type (p,q) , 478
 - theorem on invariant curves, 485
- Birkhoff theorem, 533
- Birkhoff, ergodic theorem of, 250
- Blaschke
 - product, 359
- Borsuk–Ulam theorem, 20
- Bowen
 - formula, 362
- Boyland rotation intervals, 520
- Brooks–Matelski, 364
- Brouwer degree, 518
- Brouwer fixed point theorem, 29
- Brouwer theorem, 518
- Brouwer's theorem (via Sperner), 32
- Bryuno
 - condition, 156, 357
 - inclusion CD in B, 157
 - local optimality, 176
- Bryuno condition, 156, 522
- Bulbs
 - root points, 346
- Bundles
 - Möbius, 49
- Böttcher
 - coordinate, 338
 - extension to $\mathbb{C} \setminus K_c$, 342
- Cantor
 - Julia set of, 342
- Cantor set, 486
- cantor set, 534
- Central extension, 433
 - non-split, 433
- Champernowne constant, 525
- Chaos, 198
- Chiba theorem, 530
- Chiba, theorem of, 393
- Chimera states, 395
- chimera states, 530
- Chirikov standard map, 471
- Circular points, 46
- Clutching, construction, 51
- Condition H, 157, 176
- condition $\mathcal{H}$, 522
- Connected coverings of the circle, 16
- Constant Lambda(τ), 159
- Constructions
 - Continued fraction algorithm, 75
 - CW structure of S^1 , 18
 - Denjoy C^1 counterexample, 137
 - Fejér kernel, 85
 - Gordin coboundary, 290
 - Invariant measure induced by h , 136
 - Itoh's unwrapping algorithm, 414
 - Turn numbers of the Vernier encoder, 415
- Continued fraction
 - algorithm, 75
- continued fraction
 - table, 545
- Continuous coboundary, 164
- Convergent (continued fraction), 76
- Corollaries
 - Arbitrarily small loss of domain, 160
 - At most one attracting fixed point, 345
 - Best approximation of the first kind, 79
 - Borel's normal numbers, 252
 - CLT for the dyadic indicator, 288
 - Consequences of Milnor–Wood, 451
 - Decay of correlations for E_m , 284
 - Divergence of the Adler period, 380
 - Equidistribution for Riemann-integrable functions, 86

-
- Gottschalk–Hedlund for the circle, 165
 - Groups which cannot act freely, 443
 - Holonomy along a parallel, 56
 - Homological equation in the C infinity case, 162
 - Hölder’s theorem on the circle, 441
 - Invariant measure of the tent map, 261
 - Itoh’s sampling constraint, 414
 - Kronecker sequences as quasi-Monte Carlo, 91
 - Left-orderable groups are torsion-free, 444
 - Lipschitz constant for the standard map, 484
 - Möbius counting of the periodic points, 207
 - Non-retraction of the disc onto its boundary, 29
 - Periodic minimisers are Birkhoff, 480
 - Periodic rays, 349
 - Principal $U(1)$ bundles over S^1 , 53
 - Rotation number of periodic configurations, 483
 - Self-consistency, Lorentzian case, 389
 - Specialisations of the Dörfler–Bullo theorem, 397
 - $T1S2$ isomorphic to $SO(3)$, 48
 - The central extension does not split, 433
 - Total curvature and sum of the exterior angles, 24
 - Transcendence of the Liouville numbers, 82
 - Trigonometric Weierstrass theorem, 86
 - Two coupled oscillators, 381
 - Unbounded Birkhoff sums (Liouville case), 165
 - weak maximal inequality, 250
 - Weyl via the ergodic theorem, 257
 - Coullet–Tresser theory, 316
 - Coupled oscillators, two, 381
 - Cremer
 - numbers, 356
 - theorem, 356
 - Critical circle map, 317
 - critical Kuramoto threshold
 - table, 551
 - Crossing (configurations), 475
 - CW structure (circle), 18
 - cycle slip, 530
 - De Moivre–Laplace theorem, 287
 - Dead band, 418
 - Decay of correlations, 248, 283
 - Definitions
 - (n, ε) -separated sets, 219
 - m -expanding map, 193
 - Action of $PSL(2, \mathbb{R})$ on RP^1 , 451
 - Adler’s equation, 378
 - Admissible generating class, 469
 - Affine algebraic variety, 45
 - Analytic diffeomorphism close to a rotation, 166
 - Annulus and cylinder, 467
 - Arnold standard family, 113
 - Aubry–Mather set, 486
 - Bernoulli shift, 224
 - Bifurcation diagram, 318
 - Birkhoff configuration, 480
 - Birkhoff sums and continuous coboundary, 164
 - Bryuno condition, 357
 - Bryuno number, 156
 - Canonical connection on S^3 , 58
 - Chirikov standard map, 471
 - Configuration and action, 470
 - Configuration of type (p, q) , 477
 - Convergents, 76
 - Covering between intervals, 199
 - Cremer numbers, 356
 - Crossing between configurations, 475
 - Devaney chaos, 198
 - Diophantine number, 81
 - Discrepancy, 86
 - Doubling map, 193
 - Dyadic indicator, 287
 - Dynamical correlation, 283
 - Ergodicity, 248
 - Euler class of a representation, 447
 - Euler cocycle, 448
 - Exact symplectic map, 468
 - External rays, 349
 - Fibre bundle, 46
 - Filled Julia set, 332
 - Flow on the circle, 377
 - Free action, 439
 - Generating dyadic partition, 231
 - Green’s function of the basin of infinity, 339
 - Group $Homeo_+(T)$, 432
 - Hold-in, pull-in, lock-in, 408
 - Homogeneous quasi-morphism, 436
 - Homological equation, 153
 - Immediate basin, 343
 - Index of an isolated zero, 53
 - Intervals I_n (Denjoy), 132
 - Invariant measure, 246
 - Kneading sequence, 208
 - Kolmogorov–Sinai entropy, 265
 - Kuramoto model, 383
 - Lap number, 226
 - Left-invariant order, 444
 - Level domains of the potential, 341
 - Li–Yorke chaos, 201
 - Lift of a circle homeomorphism, 103
 - Liouville number, 81
 - Local bifurcation, 309
 - Logistic family, 305
 - Lyapunov exponent, 229
 - Mandelbrot set, 333
 - Mather’s beta function, 489
 - Metric mixing, 248
 - Minimal set of an action, 445
 - Monotone twist map, 467
 - Multiplicative Diophantine condition, 352
 - Multiplier of a fixed point, 334
 - Noble number, 82
 - Norm of a real number, 78
 - Normalised displacement $a(F)$, $b(F)$, 434
 - Odd map, 20
 - Pencil of conics, 45
 - Principal bundle, 47
 - Quadratic family, 332
 - Regular homotopy, 25
 - Renormalisation operator, 313
 - Root points of the bulbs, 346

-
- Rotation map, 74
 - Rotation number, 104
 - Sharkovskii order, 202
 - Siegel disc, 354
 - Space of binary sequences, 195
 - Strip of analyticity, 158
 - Surface group, 447
 - Tautological bundle, 49
 - Tent map, 205
 - Topological conjugacy, 196
 - Topological entropy, 220
 - Topological factor, 222
 - Topological semiconjugacy, 196
 - Total variation, 262
 - Transfer operator, 263
 - Transfer operator of D , 278
 - Transition cocycle, 47
 - Unique ergodicity, 255
 - Unit tangent bundle, 47
 - Universal cover of $\text{Homeo}^+(T)$, 433
 - Variation of $\log f$ prime, 133
 - Vector bundle, 47
 - Vogel model, 90
 - Denjoy
 - combinatorial lemma, 132
 - counterexample, 137
 - Denjoy-Koksma inequality, 136
 - ergodicity, 141
 - for group actions, 453
 - group actions, 141
 - Hall's counterexample, 141
 - inequality, 133
 - intervals I_n , 132
 - intervals J_n , 132
 - theorem, 135
 - $V(f)$, 133
 - Denjoy's counterexample
 - table, 552
 - Denjoy-Koksma, 165
 - Devil's staircase, 118
 - $\text{Diff } r(T)$, 434
 - Diophantine
 - multiplicative condition, 352
 - diophantine constant
 - table, 546
 - Diophantine number, 81
 - form with the norm, 155
 - Discrepancy, 86
 - Distributed clocks, 421
 - Douady-Hubbard, 528
 - Doubling map, 193
 - on the external rays, 349
 - Dyadic indicator, 287
 - Dörfler-Bullo, 530
 - Dörfler-Bullo, theorem of, 396
 - Erdős-Turán inequality, 519
 - Erdős-Turán inequality, 88
 - Ergodicity, 248
 - of D , 248
 - Escape criterion, 333
 - EtherCAT, 421, 531
 - Euler class, 447, 532
 - bounded, 448
 - cocycle, 448
 - well-definedness, 447
 - Euler-Lagrange, discrete equation, 470
 - Exact symplectic map, 468
 - Examples
 - J_0 is the circle, J_{-2} is a segment, 334
 - Adler's equation, numerical check, 380
 - An explicit Newton step, 171
 - Birkhoff numerically, 252
 - Borsuk-Ulam, numerical check, 34
 - Bryuno sum for the golden ratio and for a
 - Liouville number, 157
 - Check of the Lipschitz constant, 485
 - Continued fraction of π , 77
 - Continued fraction of the square root of 2, 77
 - Convergence of the Birkhoff average, 234
 - Counterexample to the lifting criterion, 15
 - Critical threshold, Lorentzian and Gaussian, 390
 - De Moivre-Laplace, numerical check, 289
 - Degree via explicit lifting, 33
 - Degree via the integral formula, 34
 - Divisors for the golden ratio and for a
 - Liouville number, 154
 - Entropy and Lyapunov on the period-3 orbit, 233
 - Entropy of D : two proofs, 225
 - Exact gain of the exchange inequality, 475
 - Explicit estimate of the FTA, numerical check, 28
 - Exponential trapezoidal rule, numerical check, 61
 - Fourier coefficients of the solution, 161
 - Free action of Z_2 on the circle, 443
 - Generating function of the standard map, 472
 - Gordin CLT, degenerate case, 293
 - Gordin CLT, non-trivial coboundary, 293
 - Gordin CLT, zero coboundary, 292
 - Homological equation for the cosine, 160
 - Iterated Newton scheme (numerical experiment), 174
 - Kuramoto $N=3$, equilibria, 386
 - L_2 eigenfunction of L , numerical check, 281
 - Lap number of f_4 , 227
 - Lasota-Yorke, numerical check, 265
 - Liouville's constant, 82
 - Non-additivity of the rotation number, 438
 - Normal forms applied to the logistic map, 311
 - Normal numbers, numerical check, 258
 - Number of iterates of the escape criterion, 334
 - Numerical 2-cycle, 307
 - Numerical check of Denjoy-Koksma, 137
 - Numerical check of the Denjoy inequality, 134
 - Numerical construction of Denjoy's counterexample, 140
 - Numerical counterexample, 234
 - Numerical discrepancy vs Monte Carlo, 91
 - Numerical Koenigs coordinate, 337
 - Plateau $1/2$ at $\epsilon=0.6$, 118
 - Rotation number estimated numerically, 105
 - Rotation number of the elliptic island, 473
 - Rotation numbers in the modular group, 452
 - Sawtooth correlations, triple check, 284
-

-
- Table of the beta function, 490
 - The golden number as the most robust rotation, 355
 - The two orbits of type (1,2), 480
 - Three distances for $N=8$, 84
 - Tongue 0/1 at $\epsilon=0.3$, 115
 - Two coupled metronomes, 382
 - Ulam–von Neumann, numerical check, 260
 - Vogel’s sunflower, 90
 - Wandering interval, illustration, 112
 - Whitney–Graustein, explicit curve, 35
 - Exchange inequality, 474
 - exercise catalog, 553
 - Exercises
 - Check of the Hopf section, 67
 - Chern–Weil with opposite orientation, 68
 - Foucault precession in Rome, 67
 - Holonomy on a variable cap, 67
 - M sum M , 66
 - Parametrisation of the Möbius band, 51
 - Pencil of two circles, 66
 - External angle, 349
 - External rays, 349
 - Farey
 - ordering of the bulbs, 346
 - Farey ordering, 528
 - Fatou
 - theorem on attracting cycles, 343
 - Feigenbaum constant, 313
 - table, 547
 - Feigenbaum constants, 526
 - Fejér kernel, 85
 - Fekete’s lemma, 104
 - Fibre bundle, 46
 - Final examination, 503
 - First Chern class, 59
 - Fixed point
 - classification, 334
 - logistic map, 305
 - Flow
 - on the circle, 377
 - Free action, 439
 - obstructions, 443
 - of $\mathbb{Z}^2$, 443
 - Frenkel–Kontorova model, 471
 - Fundamental dichotomy, 342
 - Fundamental theorem of algebra, 28
 - Gauss map, 519
 - Gauss–Bonnet theorem (local), 56
 - Generating function, 468
 - Ghys theorem, 532
 - Ghys’ theorem, 448
 - Gluing, homotopy of, 48
 - golden angle, 547
 - Golden ratio
 - Diophantine constant, 155
 - Gordin martingale method, 290
 - Gottschalk–Hedlund, 164
 - Gradient flow
 - Kuramoto model, 383
 - Green’s function, 339
 - Greene criterion, 534
 - Greene threshold, 488
 - table, 549
 - Greene’s criterion, 488
 - Hairy ball theorem, 30
 - Hall counterexample, 521
 - Herman
 - critical point on the boundary, 358
 - global theorem, 175
 - Herman theorem, 520, 522
 - Hold-in range, 408
 - Holomorphic motion, 362
 - $\text{Homeo}_+(\mathbb{T})$, 531
 - $\text{Homeo}^+(\mathbb{T})$, 432
 - absence of torsion in the cover, 433
 - universal cover, 433
 - Homological equation, 153
 - analytic case, 159
 - C^∞ case, 161
 - Homology of S^1 , 18
 - Hopf fibration, 57
 - Hopf, maximal ergodic theorem of, 249
 - Hurewicz theorem (case S^1), 18
 - Hurwitz theorem, 519
 - Hurwitz, constant of, 546
 - Huygens pendulum clocks, 529
 - Huygens, synchronisation of, 382
 - Hypotheses (H1)–(H4), 469
 - Hölder’s theorem, 440
 - version on the circle, 441
 - Immediate basin, 343
 - Index of a vector field, 53
 - Integral formula for the degree, 14
 - Invariant curve, 487
 - Invariant measure, 246
 - Jakobson theorem, 525
 - Jakobson’s theorem, 266, 318
 - Julia
 - filled set K_c , 332
 - Jordan curve, 359
 - set J_c , 332
 - Julia set, 209
 - KAM theory, 523
 - on the circle, 166
 - origin, 177
 - KAM threshold
 - table, 550
 - Katzenelson–Ornstein, 177, 521
 - Khanin–Sinai, 177
 - Koenigs
 - linearisation theorem, 336
 - Kolmogorov
 - note of 1954, 177
 - Kolmogorov–Sinai entropy, 265
 - doubling map, 232
 - Kolmogorov–Sinai generator theorem, 524, 525
 - Krylov–Bogolyubov theorem, 246
 - Kuramoto, model of, 383
 - Landing of the rays, 349
 - Lanford theorem, 526
-

-
- Lanford's theorem, 314
 - Lasota–Yorke inequality, 525
 - Lasota–Yorke, theorem of, 264
 - Lebesgue measure
 - invariance for D , 247
 - Left-orderable group, 444
 - absence of torsion, 444
 - dynamical realisation, 444
 - from an action, 444
 - Lemmas
 - Absence of torsion in Homeo tilde, 433
 - Archimedean property of a free action, 439
 - Best return times, 132
 - Best return times (general case), 132
 - Bounded displacement, 106
 - BV contraction of L for D , 281
 - Cauchy estimate on strips, 158
 - Commutator lemma, 449
 - Connectedness of the level domains, 341
 - Covering lemma, 200
 - Determinacy of a stationary configuration, 471
 - Dyadic refinement, 231
 - Elementary properties of ρ on Homeo tilde, 435
 - Equivalence of the two Diophantine conditions, 352
 - Equivalent forms of the Diophantine condition, 155
 - Estimate of the small divisors series, 159
 - Estimates of a and b for the composition, 436
 - Euler's integral of the logarithm of the cosine, 230
 - Exchange inequality, 474
 - Existence of a metric on a bundle, 51
 - Existence of a minimal set, 445
 - Exponential decay of the Fourier coefficients, 158
 - Fekete's lemma, 104
 - Fixed point criterion for ρ , 435
 - Formal solution of Schröder's equation, 352
 - Fourier decay for BV, 283
 - Holder contraction of L , 281
 - Homotopy of the gluing, 48
 - Identity $L(\phi \text{ comp } D) = \phi$, 289
 - Invariance of the sum of the indices, 54
 - Inversion of identity plus ψ on a strip, 168
 - Itinerary lemma, 200
 - Kuramoto gradient flow, 383
 - Kuramoto potential and order parameter, 384
 - Laplacian of the complete graph, 384
 - Lasota–Yorke inequality, 264
 - Lifting through a general covering, 14
 - Lipschitz continuity of E_m , 223
 - Order isomorphism for irrational ω , 486
 - Oscillation lemma, 434
 - Pointwise estimate for L^n , zero mean, 289
 - Positivity of the Fejér kernel, 85
 - Product of the periodic points, 356
 - Proper maps of the disc, 359
 - Properties of the level domains, 341
 - Properties of the total variation, 263
 - Rational criterion for ρ , 107
 - Rearrangement lemma in a bi-ordered group, 440
 - Schwarzian derivative of the logistic map, 312
 - Separated vs. spanning, 220
 - Siegel's arithmetic lemma, 353
 - Small divisors lemma, 154
 - Sperner, 31
 - Standard integral $\cos^2 + \sin^2$, 389
 - Step of the Newton scheme, 169
 - Submultiplicativity of the lap number, 226
 - The rotation number forces the mean, 168
 - Transversality of coincidences, 475
 - Unique eigenvalue of modulus 1 of L , 280
 - Weighted Laplacian of a connected graph, 384
 - lens space, 519
 - Levels of the potential, 341
 - Li–Yorke theorem, 523
 - Lift
 - of a circle homeomorphism, 103
 - Lifting (general covering), 14
 - Lifting criterion, 14
 - Limit cycles from quantisation, 420
 - Liouville number, 81
 - Lipschitz graph, 484
 - Liverani, Birkhoff cones of, 294
 - Lock-in range, 408
 - Locking time, 411
 - Logistic family, 305
 - Lyapunov exponent, 229–233
 - Main cardioid, 344
 - Mandelbrot
 - connectedness of, 343
 - main cardioid, 344
 - set, 333
 - Margulis alternative, 532
 - Margulis' alternative, 453
 - Mather
 - beta function, 489
 - Mather's beta function
 - table, 549
 - Mañé–Sad–Sullivan, 362
 - MC_GearIn, 421
 - Metric mixing, 248
 - of D , 248
 - Metric on a bundle, existence, 51
 - Milnor–Wood
 - commutator lemma, 449
 - consequences, 451
 - inequality, 450
 - Milnor–Wood inequality, 532
 - Minimal average action, 489
 - Minimal configuration, 470
 - Minimal set, 445
 - existence, 445
 - trichotomy, 445
 - Misiurewicz–Szlenk theorem, 524
 - Mixing, 195
 - mode-locking, 520
 - Modular group
 - rotation numbers, 452
 - Montel theorem, 528
 - Moser
-

-
- smoothing operators and rapidly convergent scheme, 175
 - Moser theorem, 521
 - Moser twist theorem, 534
 - Moser's twist theorem, 491
 - Multiplier, 334
 - Möbius band, 49
 - Nagaev-Guivarc'h method, 293
 - Newton scheme
 - numerical experiment, 174
 - step, 169
 - Noble number, 82
 - Non-retraction, 29
 - Normal numbers, 252
 - Champernowne, 258
 - Normalised displacement, 434
 - estimates for the composition, 436
 - numerical laboratory, 537
 - Odd (antipodal) map, 20
 - Odd degree (odd maps), 20
 - Order parameter, 387
 - Ordered group
 - archimedean, 439
 - bi-invariant, 439
 - rearrangement lemma, 440
 - Oscillation of the displacement, 434
 - Oscillator network, synchronisation in, 396
 - Oseledets theorem, 525
 - Ostrowski numeration, 520
 - Ott–Antonsen ansatz, 529
 - Ott–Antonsen, ansatz of, 392
 - Ott–Antonsen, reduction of, 392
 - Oxtoby, criterion of, 255
 - Pancake theorem, 21
 - Pencil of conics, 45
 - Period-2 bulb, 345
 - Period-doubling cascade, 311
 - period-doubling cascade
 - table, 547
 - Periodic orbit of type (p,q) , 477
 - Perron–Frobenius operator, 263
 - Pesin formula, 525
 - Phase locking, 379
 - Phase unwrapping, 414
 - Phyllotaxis, 90
 - PLCopen, 421, 531
 - PLL, 530
 - discrete, 412
 - first order, 408
 - second order, 412
 - Poincaré
 - rotation number, 104
 - Poincaré classification, 108
 - Poincaré–Birkhoff theorem, 481, 533
 - Poincaré–Birkhoff theorem (imported), 481
 - Poincaré–Hopf theorem, 54
 - Potential, 339
 - Propositions
 - Adjoint of the transfer operator, 263
 - Adjoint of the transfer operator of D , 278
 - Bi-invariant order of a free action, 439
 - Central extension of $\text{Homeo}^+(T)$, 433
 - Characteristic locking time, 411
 - Classification of the elements of $\text{PSL}(2,R)$, 451
 - Continued fraction recurrences, 76
 - Diophantine constant of the golden ratio, 155
 - Diophantine implies Bryuno, 157
 - Diophantine type and bounded partial quotients, 88
 - Elementary rate for the discrepancy, 89
 - Exact binomial correspondence, 287
 - Exact martingale orthogonality, 290
 - Exceptional points of the Lyapunov exponent, 231
 - Explicit computation, Poincaré–Hopf, 54
 - Faithful actions on the line and left-orderability, 444
 - Fixed points of the logistic map, 305
 - Fixed points of the standard map, 472
 - Identities of the beta function, 490
 - Invariance of the measure on arcs, 247
 - Invariance of the rotation number, 106
 - Kuramoto order parameter, 387
 - Kuramoto, $N=2$, 385
 - Kuramoto, $N=3$, 385
 - Lagrange coordinates for a twist map, 467
 - Monotone map associated with a Birkhoff configuration, 482
 - Monotonicity of the periodicity windows, 319
 - Optimality of Hurwitz's constant, 80
 - Optimality of the golden angle in phyllotaxis, 90
 - Ordering of the root points, 346
 - Period-3 window, 308
 - Properties of Green's function, 339
 - Pull-back of the Hopf connection, 58
 - RDC as a type II PLL, 412
 - ρ is a homomorphism on abelian subgroups, 437
 - ρ of a power, 106
 - RP^1 homeomorphic to S^1 , 49
 - Structure of the Siegel disc, 354
 - The Euler cocycle is bounded, 448
 - The golden ratio as the most Diophantine number, 82
 - Transcritical bifurcation (logistic map), 306
 - Triviality over an interval, 48
 - Well-definedness of the Euler class, 447
 - Weyl's criterion, 84
 - $\text{PSL}(2,R)$
 - action on the boundary, 451
 - classification by trace, 451
 - Pull-in range, 408
 - Quadratic family, 332
 - quantization, 531
 - Quasi-morphism, 436
 - optimal defect, 437
 - rotation number, 437
 - reference numerical tables, 545
 - registry of imported results, 517
 - Regular homotopy, 25
 - regularity classes
 - table, 550
-

Remarks

- D not uniquely ergodic, 257
 - p/q bulbs and Farey, 346
 - Absence of a global section for Hopf, 58
 - Adler saddle-node bifurcation, 380
 - Algebraic smoothness of the circle, 45
 - Anatomy of Denjoy's proof, 135
 - Applications of synchronisation (panorama), 397
 - Arithmetic hierarchy CD, H, B, 157
 - Arnold's theorem (imported), 120
 - Attribution: Hopf and Garsia, 249
 - Aubry–Mather theorem (imported part), 488
 - Baire versus Lebesgue in small divisors, 357
 - Banks–Brooks–Cairns–Davis–Stacey theorem, 199
 - Berry phase, 59
 - Beyond Aubry–Mather (panorama), 491
 - Birkhoff cones, panorama, 294
 - Birkhoff's theorem on invariant curves (imported), 485
 - Boundary of the Siegel disc, 358
 - Bridge with holomorphic dynamics, 209
 - Bridges towards other chapters, 178
 - Brouwer degree in higher dimension (overview), 29
 - Brouwer in every dimension (imported), 31
 - Brouwer, two independent proofs, 32
 - Bryuno–Yoccoz, 358
 - C k version of the scheme (Moser, imported), 175
 - Chain of groups of diffeomorphisms, 434
 - Champernowne, imported, 258
 - Change of convention for the standard family, 113
 - Chiba's theorem (panorama), 393
 - Choice of the critical point in the bifurcation diagram, 318
 - Choice of the Dörfler–Bullo theorem, 396
 - Commutation in the definition of a bundle, 47
 - Comparison of T1S1 and T1S2, 48
 - Comparison of the proofs of the FTA, 28
 - Complete spectrum of L on L^2 , 281
 - Condition H of Herman–Yoccoz (imported), 176
 - Connectedness of $\mathcal{M}$, 343
 - Consistency between self-consistency and Ott–Antonsen, 393
 - Consistency of connection and transition, 59
 - Consistency of Poincaré–Hopf and the hairy ball, 55
 - Continued fraction identity used in Ch.5, 132
 - Continuity in the proof of the pancake theorem, 22
 - Contraction of L on L^2 , 279
 - Contrast between the L^2 and BV spectra, 282
 - Correction of the dead band, 418
 - Correction of the discrepancy exponents, 89
 - Correction of the proof of Fatou's theorem, 344
 - Correction on best approximation, 79
 - Correction on noble numbers, 82
 - Correction on parity in the combinatorial lemma, 133
 - Correction: identification of Birkhoff's limit, 252
 - Correction: invariance of entropy, 221
 - Coullet–Tresser, 316
 - Count of bundles independent of n , 52
 - Covering as a discrete bundle, 47
 - Covering property of J_n , 132
 - Cross-references on the open problems of holomorphic dynamics, 362
 - Decay of correlations, forward reference, 249
 - Degree, homotopy and homology agree on S^1 , 18
 - Denjoy for group actions, 141
 - Denjoy for group actions (imported), 453
 - Denjoy–Koksma and cohomological stability (imported), 165
 - Denjoy-type ergodicity, recollection, 267
 - Devaney implies Li–Yorke, 199
 - Dirac quantisation, 59
 - Direct check of the L^2 eigenfunction, 280
 - Discrete PLL (forward reference to Chapter 4), 412
 - Dynamical realisation (imported), 444
 - Entropy of D , forward reference, 265
 - Equidistribution is stronger than density, 86
 - Erdős–Turán inequality (imported), 88
 - Ergodicity of Denjoy conjugacies (imported), 141
 - EtherCAT distributed clocks (panorama), 421
 - Examples of the three cases of the trichotomy, 446
 - Existence of the equilibrium, Dörfler–Bullo (limits), 397
 - Factor 2, Berry versus parallel transport, 60
 - Finite regularity (panorama), 177
 - First-order PLL (recall), 408
 - First-order PLL, special case, 410
 - Formal solution of the homological equation, 153
 - Foucault pendulum, 57
 - Foucault versus monopole, the difference, 57
 - Frenkel–Kontorova model, 471
 - Full optimality of Hurwitz (imported), 81
 - Full-measure mode-locking (imported), 118
 - Fundamental dichotomy, towards Cantor, 342
 - Gardner's definitions (criticism), 408
 - Gauss map and Poincaré–Hopf, 55
 - Geodesic curvature check of the holonomy, 56
 - Geometric meaning of the Euler class, 448
 - Ghys' theorem (imported), 448
 - Greene's criterion, 488
 - Hall's counterexample (imported), 141
 - Hamiltonian KAM (panorama), 177
 - Hausdorff dimension of J_c , 362
 - Herman's theorem (imported), 175
 - Historical attribution of the three-distance theorem, 84
 - Historical survey of Sharkovskii and Li–Yorke, 210
 - History of holomorphic dynamics, 364
 - History of small divisors, 155
 - Huygens and the synchronisation of pendulums (imported), 382
-

-
- Imported: Bowen’s theorem, 222
 - Imported: Misiurewicz–Szlenk theorem, 227
 - Inclusion of the locking ranges, 409
 - Integral formula for the degree (cross-reference), 14
 - Interpretation of Rouché’s theorem, 19
 - Invariance of Lebesgue for rotations, forward reference, 248
 - Itoh and Nyquist, 415
 - Jakobson’s theorem, 318
 - Jakobson, panorama, 266
 - Krylov–Bogolyubov, role of compactness, 247
 - Lambda-lemma, 362
 - Landing of the rational rays, 349
 - Lanford’s theorem (imported), 314
 - Lasota–Yorke general, imported, 264
 - Lens spaces (panorama), 60
 - Limit cycles from quantisation (panorama), 420
 - Limitations of the cascade existence theorem, 312
 - Limits of the Liouville counterexample, 163
 - Limits of the self-consistency ansatz, 390
 - Limits of the stability theorem, 361
 - Local Böttcher and at infinity, 339
 - Locking time, engineering reading, 412
 - Margulis’ alternative (imported), 453
 - Martingale CLT, imported, 291
 - Minimal Cantor dichotomy, 112
 - Monotonicity of the windows, general phenomenon, 319
 - Moser’s twist theorem (imported), 491
 - Nagaev–Guivarc’h, panorama, 293
 - Necessity of normalisation in renormalisation, 314
 - Necessity of the hypotheses in Arnold’s theorem, 166
 - Non-maximal kneading, 209
 - Normal form of the quadratic polynomial, 332
 - Number of sheets of a connected covering, 17
 - Numerical check of Lambda(tau), 159
 - Numerical check of the dead band, 419
 - Numerical check of the divisors lemma, 154
 - Numerical check of the golden constant, 156
 - Numerical check of the Liouville counterexample, 163
 - Numerical check: exceptional points, 231
 - Numerical check: Lyapunov exponent, 231
 - Numerical check: separated grid, 224
 - Numerical estimate of the Feigenbaum constant, 313
 - Numerical values of the threshold epsilon 0, 167
 - Optimal loss of derivatives (imported), 162
 - Optimal value of the defect of rho, 437
 - Optimality of Milnor–Wood (imported), 450
 - Optimality of the Bryuno condition (imported), 176
 - Orientation in the classification of principal bundles, 53
 - Ostrowski sum (imported), 88
 - Ott–Antonsen ansatz (imported), 392
 - Panorama: Oseledets and Pesin, 233
 - Pencil of conics, correction, 46
 - Physical interpretation of Borsuk–Ulam, 20
 - Picture of the Poincaré classification, 110
 - PLCopen function blocks (panorama), 421
 - Poincaré’s last geometric theorem, 481
 - Properties of the beta function (panorama), 491
 - Provenance of Arnold’s statement, 166
 - Quasi-periodic motion, 75
 - Recap of Lasota–Yorke from Ch.9, 286
 - Reduction to the form $\lambda w + w^2$, 351
 - Reformulation of the commutator lemma, 449
 - Renormalisation at the golden number, 317
 - Resolver-to-digital converter, 412
 - Results imported in chapter 12, 331
 - Role of Aubry’s lemma, 477
 - Role of monotonicity in the oscillation lemma, 435
 - Role of PSL(2,R) in the theory, 452
 - Role of the forced-mean lemma, 169
 - Role of the hypotheses in Hölder’s theorem, 441
 - Rotation intervals (imported), 119
 - Rotation spectrum, 447
 - Rotations versus chaos, synthesis, 257
 - Sakaguchi–Kuramoto, chimeras, Pecora–Carroll (panorama), 395
 - Schmidt’s bound (panorama), 90
 - Scrambled set, 202
 - Second proof of the absence of torsion, 444
 - Second-order PLL (panorama), 412
 - Sharkovskii and entropy: general case, 228
 - Simplicity hypothesis in the Umlaufsatz, 25
 - Slip about the genus 1 surface group, 443
 - Solodov, rigidity and flexibility (imported), 454
 - Status of Siegel’s arithmetic lemma, 353
 - Stolen necklace, general case, 22
 - Stroboscopic effect, 422
 - Structure of Homeo+(T) (imported), 434
 - Structure of the proof of Sharkovskii, 205
 - Symmetry and reversibility of the standard map, 473
 - Table of the convergence of the Newton scheme, 173
 - The gain of the last unit in Milnor–Wood, 450
 - The hypotheses (H1)–(H4), 469
 - The neutral multiplier breaks Koenigs, 337
 - Theorems of Herman and Yoccoz (imported), 121
 - Thompson’s group T, 454
 - Thurston stability (imported), 453
 - Triangle $D-T-f_4$, 262
 - Trichotomy and invariant measures, 446
 - Two definitions of topological entropy, 221
 - Two methods for the decay of correlations, 286
 - Type II servosystem, 413
 - Unique ergodicity of R_α , comparison, 256
 - Use of the hypotheses in the proof, 174
 - Van Kampen for S1 (imported), 17
 - Variational principle for D , 232
 - Vect n of S^k , 52
 - Vernier encoder, 415
-

-
- Vernier tolerance threshold, reading, 418
 - von Neumann, invertibility not needed, 254
 - Weyl: comparison of the two proofs, 258
 - What is imported in Chapter 16, 469
 - Whitney-Graustein, constructive direction (imported), 26
 - Why Poincaré and not Weyl, 105
 - Yoccoz's theorem 1984 (imported), 176
 - Young towers, panorama, 294
 - Renormalisation (Feigenbaum-Collet-Tresser), 313
 - reproducibility, 537
 - Resolver-to-digital converter, 412
 - Riemann mapping theorem, 527
 - Rigidity and flexibility, 454
 - Rotation map, 74
 - Rotation number, 104
 - additivity in the abelian case, 437
 - fixed point criterion, 435
 - forces the constant, 168
 - is not a homomorphism, 438
 - of an orbit of a twist map, 482
 - properties on Homeo tilde, 435
 - Rotation spectrum, 447
 - Rouché's theorem (topological), 19
 - Ruelle
 - expansion, 362
 - Sakaguchi-Kuramoto, model, 395
 - Schmidt theorem, 520
 - Schröder
 - equation, 364
 - Schwarz-Pick inequality, 527
 - Schwarzian derivative, 312
 - Self-consistency, equation of, 388
 - Semiconjugacy, 197
 - of free actions, 441
 - Sharkovskii theorem, 523
 - Shenker constants (delta-GM, alpha-GM), 317
 - Shift, 195
 - Siegel
 - arithmetic lemma, 353
 - disc, 354
 - linearisation theorem, 353
 - Siegel arithmetic lemma, 528
 - Siegel disk, 529
 - Singer theorem, 526
 - Small divisors, 120
 - definition, 153
 - elementary estimate, 154
 - homological equation, 153
 - non-solvability, 162
 - Solodov's theorem, 454
 - Sperner's lemma, 31
 - Standard map, 471
 - Stolen necklace theorem, 22
 - Strip of analyticity, 158
 - Stroboscope, 422
 - Surface group, 447
 - Synchronisation
 - synchronised state, 385
 - Tautological bundle, 49
 - Theorems
 - J_c a Jordan curve for small c , 359
 - 2-cycle of the logistic map, 306
 - Action of L on the Fourier coefficients, 278
 - Adler's equation, 379
 - Arnold's theorem (local analytic linearisation), 166
 - Aubry's lemma, 476
 - Best approximation of the second kind, 78
 - Borsuk-Ulam in dimension 2, 20
 - Brouwer via Sperner, 32
 - Brouwer, fixed point (dimension 2), 29
 - Böttcher coordinate, 338
 - Cellular homology of S^1 , 18
 - Circle as a member of a pencil, 46
 - Classification of principal bundles, 53
 - Classification of the connected coverings of S^1 , 16
 - Classification of vector bundles over S^1 , 52
 - Clutching construction, 51
 - Collapse of the orbits under the semiconjugacy, 206
 - Compactness of $\mathcal{M}$, 333
 - Convergence to the equilibria, identical Kuramoto, 384
 - Counting the fixed points of D^n , 205
 - Counting the fixed points of f_4^n , 206
 - Counting the fixed points of T^n , 205
 - Cremer's theorem, 356
 - De Moivre-Laplace, 287
 - Dead band theorem, 418
 - Decay of correlations for D , 283
 - Decay of correlations for the tent map, 286
 - Denjoy inequality, 133
 - Denjoy's theorem, 135
 - Denjoy-Koksma inequality, 136
 - Density of the periodic points, 194
 - Devaney chaos for the doubling map, 199
 - Devil's staircase, 118
 - Diffeomorphism of the standard family, 114
 - Discrepancy and equidistribution, 87
 - Discrepancy for bounded partial quotients, 89
 - Dynamical CLT, Gordin, 291
 - Dynamics of the external rays, 349
 - Entropy of the Bernoulli shift, 224
 - Entropy of the iterate f^j , 228
 - Equilibria and period of a flow on the circle, 378
 - Equivalence of AKM/Bowen-Dinaburg entropy, 220
 - ergodic, Birkhoff's, 250
 - Ergodicity of R_α , 257
 - Ergodicity of the doubling map, 248
 - Error estimate for the convergents, 77
 - Escape criterion, 333
 - Existence and uniqueness of the continued fraction, 75
 - Existence of Birkhoff periodic orbits, 478
 - Existence of ordered orbits for every ω , 485
 - Existence of the Feigenbaum cascade, 311
 - Existence of the rotation number, 104
 - Extension of the Böttcher coordinate, 342
 - Fatou, attracting cycles and critical points, 343
 - Fejér's theorem, 85
-

-
- First bifurcation of the logistic map, 306
 - First Chern class of Hopf, 59
 - Flip bifurcation, 310
 - Free actions semiconjugate to rotations, 441
 - Full measure of the Diophantine numbers, 81
 - Fundamental theorem of algebra (via degree), 28
 - Generating function of a twist map, 468
 - Gordin decomposition, 290
 - Gottschalk–Hedlund lemma, 164
 - Hairy ball (Milnor’s proof), 30
 - Homological equation in the analytic case, 159
 - Homological equation in the C^r case, 161
 - Hopf fibration, 57
 - Hopf’s Umlaufsatz, 23
 - Horseshoe and positive entropy, 228
 - Hurewicz for S^1 , 18
 - Hurwitz’s theorem, 79
 - Hölder’s theorem, 440
 - Invariance of Lebesgue for the doubling map, 247
 - Invariance of topological entropy, 221
 - Invariant curve/cantor set dichotomy, 487
 - Itoh’s theorem, 414
 - Kneading of the logistic map, 208
 - Koenigs linearisation, 336
 - Koksma’s inequality, 87
 - Krylov–Bogolyubov, 246
 - Kuramoto self-consistency equation, 388
 - Lap number and entropy: inequality, 226
 - Lasota–Yorke for the tent map, 264
 - Li–Yorke, 201
 - Lifting criterion, 14
 - Liouville’s theorem on algebraic approximation, 81
 - Lipschitz graph of Birkhoff orbits, 484
 - Local Gauss–Bonnet, 56
 - Locking ranges of the first-order PLL, 409
 - Lyapunov exponent of D , 229
 - Lyapunov exponent of f_4 , 230
 - Main cardioid, 344
 - maximal ergodic (Hopf), 249
 - mean ergodic (von Neumann), 253
 - Metric mixing of the doubling map, 248
 - Milnor–Wood inequality, 450
 - Monotonicity and continuity of ρ , 107
 - Monotonicity of the entropy in factors, 222
 - Möbius band as tautological bundle, 49
 - Network synchronisation (Dörfler–Bullo), 396
 - Non-solvability of the homological equation (Liouville), 162
 - Non-triviality of the Möbius band, 50
 - Nowhere-vanishing sections and triviality, 50
 - Odd maps have odd degree, 20
 - Optimality of the Sharkovskii order, 205
 - Ordering of the periodic minimisers, 479
 - Ott–Antonsen reduction, 392
 - Oxtoby’s criterion, 255
 - Pancake (ham sandwich in dimension 2), 21
 - Period-2 bulb, 345
 - Poincaré semiconjugacy, 110
 - Poincaré–Hopf, 54
 - Principal tongue, 114
 - Properties of Denjoy’s counterexample, 138
 - Rational ρ and periodic orbits, 108
 - Rational/irrational dichotomy, 74
 - Rokhlin’s formula for D , 232
 - Rotation number of a Birkhoff configuration, 482
 - Saddle-node bifurcation, 309
 - Semiconjugacy with the shift, 197
 - Sensitivity to initial conditions, 194
 - Sharkovskii, 204
 - Siegel’s theorem, 353
 - Spectral convergence of the trapezoidal rule, 60
 - Spectral gap of L , 282
 - Spectrum of L on L^2 , 280
 - Stability of the synchronised state, 385
 - Stolen necklace, 22
 - Tent–logistic conjugacy, 261
 - The rotation number is a quasi-morphism, 437
 - Three-distance theorem, 83
 - Topological entropy of the maps E_m , 223
 - Topological mixing, 195
 - Topological mixing of the doubling map, 248
 - Topological Rouché, 19
 - Transcritical bifurcation, 310
 - Trichotomy of minimal sets, 445
 - Ulam–von Neumann, 260
 - unique ergodicity of R_α , 256
 - Unit tangent bundle of S^1 , 47
 - Variational principle for twist maps, 470
 - Vernier tolerance theorem, 417
 - Weyl equidistribution, 84
 - Whitney–Graustein, 25
 - Width of the Arnold tongues, 115
 - Zero entropy of isometries, 225
 - Thompson group, 533
 - Thompson’s group, 454
 - three-distance theorem, 547
 - Thurston stability, 453, 532
 - Topological conjugacy, 196
 - Topological entropy, 220–233
 - Adler–Konheim–McAndrew definition, 220
 - Bernoulli shift, 224
 - doubling map, 223
 - factors, 222
 - finite-fibre factors, 222
 - horseshoe, 228
 - invariance under conjugacy, 221
 - lap number, 226
 - maps E_m , 223
 - Misiurewicz–Szlenk theorem, 227
 - rotations, 225
 - separated sets, 219
 - variational principle, 232
 - Topological semiconjugacy, 196
 - Total curvature, 24
 - Tracking error, 412
 - Transfer operator
 - Fourier coefficients, 278
 - of D , 278
 - spectral gap, 282
 - spectrum on L^2 , 280
 - Transition cocycle, 47
-

Trapezoidal rule (spectral convergence), 60
Triviality of bundles over an interval, 48
Twist (twist map), 467
Twist map, 467

Ulam–von Neumann measure, 260
Umlaufsatz, 23
Unique ergodicity, 255
Unit tangent bundle, 47
Unwrapping, 414

van Kampen theorem, 518
Van Kampen theorem (groupoid), 17
variational principle, 524
vector bundle, 518
Vernier encoder, 415

Vernier scale, 415
von Neumann, ergodic theorem of, 253

Wheel turning backwards, 422
Whitney–Graustein theorem, 518
Whitney–Graustein theorem, 25

Yoccoz
 condition H, 176
 optimality theorem, 358
 theorem of 1984, 176
Yoccoz theorems, 520, 522
Young towers, 294

Świątek theorem, 520